

LeetMastery

Study-Order Edition

Python Only · Priority-Grouped ·

Difficulty-First · 1×1 Portrait

331 questions · 9 rounds · Interview Approach
High Easy → High Med → High Hard → Mid Easy
→ Mid Med → Mid Hard → Low Easy → Low Med
→ Low Hard

Contents

★ = LeetCode Premium (Premium 98 list)

Round 1 | High | E Easy (43) ↗

Arrays & Hashing (6) ↗

☐ ● #1 Two Sum ↗

☐ ● ★ #163 Missing Ranges ↗

☐ ● ★ #346 Moving Average from Data Stream ↗

☐ ● ★ #760 Find Anagram Mappings ↗

☐ ● ★ #1426 Counting Elements ↗

☐ ● #1534 Count Good Triplets ↗

String (3) ↗

☐ ● #383 Ransom Note ↗

☐ ● ★ #734 Sentence Similarity ↗

☐ ● ★ #1165 Single Row Keyboard ↗

Two Pointers (5) ↗

☐ ● #88 Merge Sorted Array ↗

☐ ● #125 Valid Palindrome ↗

☐ ● #283 Move Zeroes ↗

☐ ● ★ #408 Valid Word Abbreviation ↗

☐ ● #977 Squares of a Sorted Array ↗

Sorting (6) ↗

☐ ● #169 Majority Element ↗

☐ ● #217 Contains Duplicate ↗

☐ ● #242 Valid Anagram ↗

☐ ● ★ #252 Meeting Rooms ↗

☐ ● ★ #1086 Largest Unique Number ↗

☐ ● #1122 Relative Sort Array ↗

Binary Search (5) ↗

☐ ● #35 Search Insert Position ↗

☐ ● #69 Sqrt(x) ↗

☐ ● #278 First Bad Version ↗

☐ ● #374 Guess Number Higher or Lower ↗

☐ ● #704 Binary Search ↗

Matrix (5) ↗

☐ ● ★ #422 Valid Word Square ↗

☐ ● #566 Reshape the Matrix ↗

- ☐ ● #766 Toeplitz Matrix ↗
- ☐ ● #867 Transpose Matrix ↗
- ☐ ● #2373 Largest Local Values in a Matrix ↗
- Trees & BST (11)** ↗
- ☐ ● #100 Same Tree ↗
- ☐ ● #101 Symmetric Tree ↗
- ☐ ● #104 Maximum Depth of Binary Tree ↗
- ☐ ● #108 Convert Sorted Array to Binary Search Tree ↗
- ☐ ● #110 Balanced Binary Tree ↗
- ☐ ● #112 Path Sum ↗
- ☐ ● #226 Invert Binary Tree ↗
- ☐ ● ★ #270 Closest Binary Search Tree Value ↗
- ☐ ● #501 Find Mode in Binary Search Tree ↗
- ☐ ● #543 Diameter of Binary Tree ↗
- ☐ ● #572 Subtree of Another Tree ↗
- DFS (2)** ↗
- ☐ ● #463 Island Perimeter ↗

☐ ● #733 Flood Fill ↗

Round 2 | High | M Medium (116) ↗

Arrays & Hashing (11) ↗

☐ ● #57 Insert Interval ↗

☐ ● #238 Product of Array Except Self ↗

☐ ● #281 Zigzag Iterator ↗

☐ ● #307 Range Sum Query – Mutable ↗

☐ ● #454 4Sum II ↗

☐ ● #525 Contiguous Array ↗

☐ ● #560 Subarray Sum Equals K ↗

☐ ● #1010 Pairs of Songs With Total Durations
Divisible by 60 ↗

☐ ● ★ #1272 Remove Interval ↗

☐ ● ★ #1429 First Unique Number ↗

☐ ● #3159 Find Occurrences of an Element in
an Array ↗

String (5) ↗

☐ ● #8 String to Integer (atoi) ↗

☐ ● ★ #249 Group Shifted Strings ↗

☐ ● ★ #271 Encode and Decode Strings ↗

☐ ● #635 Design Log Storage System ↗

☐ ● #1138 Alphabet Board Path ↗

Two Pointers (14) ↗

☐ ● #11 Container With Most Water ↗

☐ ● #15 3Sum ↗

☐ ● #16 3Sum Closest ↗

☐ ● #31 Next Permutation ↗

☐ ● #75 Sort Colors ↗

☐ ● ★ #161 One Edit Distance ↗

☐ ● #167 Two Sum II – Input Array Is Sorted ↗

☐ ● ★ #186 Reverse Words in a String II ↗

☐ ● #189 Rotate Array ↗

☐ ● #532 K-diff Pairs in an Array ↗

☐ ● #923 3Sum With Multiplicity ↗

☐ ● #986 Interval List Intersections ↗

☐ ● ★ #1055 Shortest Way to Form String ↗

☐ ● #2563 Count the Number of Fair Pairs ↗

Sliding Window (8) ↗

- ☐ ● #3 Longest Substring Without Repeating Characters ↗
- ☐ ● ★ #159 Longest Substring with At Most Two Distinct Characters ↗
- ☐ ● ★ #340 Longest Substring with At Most K Distinct Characters ↗
- ☐ ● #424 Longest Repeating Character Replacement ↗
- ☐ ● #438 Find All Anagrams in a String ↗
- ☐ ● ★ #487 Max Consecutive Ones II ↗
- ☐ ● ★ #1100 Find K-Length Substrings with No Repeated Characters ↗
- ☐ ● #1248 Count Number of Nice Subarrays ↗

Sorting (3) ↗

- ☐ ● #49 Group Anagrams ↗
- ☐ ● #56 Merge Intervals ↗
- ☐ ● ★ #1152 Analyze User Website Visit Pattern ↗

Binary Search (12) ↗

- ☐ ● #33 Search in Rotated Sorted Array ↗

- ☐ ● #34 Find First and Last Position of Element in Sorted Array ↗
- ☐ ● #153 Find Minimum in Rotated Sorted Array ↗
- ☐ ● #300 Longest Increasing Subsequence ↗
- ☐ ● #362 Design Hit Counter ↗
- ☐ ● #528 Random Pick with Weight ↗
- ☐ ● #852 Peak Index in a Mountain Array ↗
- ☐ ● #981 Time Based Key-Value Store ↗
- ☐ ● #1011 Capacity to Ship Packages Within D Days ↗
- ☐ ● ★ #1060 Missing Element in Sorted Array ↗
- ☐ ● #1062 Longest Repeating Substring ↗
- ☐ ● ★ #1533 Find the Index of the Large Integer ↗
- Matrix (14)** ↗
- ☐ ● #36 Valid Sudoku ↗
- ☐ ● #48 Rotate Image ↗
- ☐ ● #54 Spiral Matrix ↗
- ☐ ● #59 Spiral Matrix II ↗

- ☐ ● #64 Minimum Path Sum ↗
- ☐ ● #73 Set Matrix Zeroes ↗
- ☐ ● #74 Search a 2D Matrix ↗
- ☐ ● #221 Maximal Square ↗
- ☐ ● ★ #311 Sparse Matrix Multiplication ↗
- ☐ ● #498 Diagonal Traverse ↗
- ☐ ● ★ #531 Lonely Pixel I ↗
- ☐ ● ★ #723 Candy Crush ↗
- ☐ ● #835 Image Overlap ↗
- ☐ ● ★ #1198 Find Smallest Common Element in All Rows ↗

Trees & BST (24) ↗

- ☐ ● #98 Validate Binary Search Tree ↗
- ☐ ● #102 Binary Tree Level Order Traversal ↗
- ☐ ● #103 Binary Tree Zigzag Level Order Traversal ↗
- ☐ ● #105 Construct Binary Tree from Preorder and Inorder Traversal ↗
- ☐ ● #199 Binary Tree Right Side View ↗

- ☐ ● #230 Kth Smallest Element in a BST ↗
- ☐ ● #235 Lowest Common Ancestor of a Binary Search Tree ↗
- ☐ ● #236 Lowest Common Ancestor of a Binary Tree ↗
- ☐ ● ★ #250 Count Univalue Subtrees ↗
- ☐ ● #285 Inorder Successor in BST ↗
- ☐ ● ★ #298 Binary Tree Longest Consecutive Sequence ↗
- ☐ ● ★ #314 Binary Tree Vertical Order Traversal ↗
- ☐ ● ★ #333 Largest BST Subtree ↗
- ☐ ● ★ #366 Find Leaves of Binary Tree ↗
- ☐ ● #437 Path Sum III ↗
- ☐ ● ★ #545 Boundary of Binary Tree ↗
- ☐ ● ★ #549 Binary Tree Longest Consecutive Sequence II ↗
- ☐ ● ★ #582 Kill Process ↗
- ☐ ● #662 Maximum Width of Binary Tree ↗

- ☐ ● #863 All Nodes Distance K in Binary Tree ↗
- ☐ ● ★ #1120 Maximum Average Subtree ↗
- ☐ ● ★ #1490 Clone N-ary Tree ↗
- ☐ ● ★ #1522 Diameter of N-Ary Tree ↗
- ☐ ● ★ #1650 Lowest Common Ancestor of a Binary Tree III ↗

DFS (16) ↗

- ☐ ● #130 Surrounded Regions ↗
- ☐ ● #133 Clone Graph ↗
- ☐ ● #200 Number of Islands ↗
- ☐ ● #207 Course Schedule ↗
- ☐ ● #210 Course Schedule II ↗
- ☐ ● ★ #261 Graph Valid Tree ↗
- ☐ ● #310 Minimum Height Trees ↗
- ☐ ● ★ #323 Number of Connected Components in an Undirected Graph ↗
- ☐ ● #417 Pacific Atlantic Water Flow ↗
- ☐ ● ★ #490 The Maze ↗

☐ ● ★ #694 Number of Distinct Islands ↗

☐ ● #695 Max Area of Island ↗

☐ ● #721 Accounts Merge ↗

☐ ● #785 Is Graph Bipartite? ↗

☐ ● ★ #1236 Web Crawler ↗

☐ ● #1242 Web Crawler Multithreaded ↗

Graphs (3) ↗

☐ ● #128 Longest Consecutive Sequence ↗

☐ ● ★ #277 Find the Celebrity ↗

☐ ● ★ #1136 Parallel Courses ↗

BFS (6) ↗

☐ ● ★ #286 Walls and Gates ↗

☐ ● #322 Coin Change ↗

☐ ● #542 01 Matrix ↗

☐ ● #994 Rotting Oranges ↗

☐ ● ★ #1197 Minimum Knight Moves ↗

☐ ● #1730 Shortest Path to Get Food ↗

Round 3 | High | H Hard (23)



Arrays & Hashing (1) ↗

☐ ● #41 First Missing Positive ↗

Sliding Window (1) ↗

☐ ● #76 Minimum Window Substring ↗

Binary Search (7) ↗

☐ ● #4 Median of Two Sorted Arrays ↗

☐ ● #315 Count of Smaller Numbers After Self ↗

☐ ● #410 Split Array Largest Sum ↗

☐ ● #668 Kth Smallest Number in
Multiplication Table ↗

☐ ● #710 Random Pick with Blacklist ↗

☐ ● #774 Minimize Max Distance to Gas Station ↗

☐ ● #1235 Maximum Profit in Job Scheduling ↗

Matrix (1) ↗

☐ ● #1074 Number of Submatrices That Sum to
Target ↗

Trees & BST (2) ↗

☐ ● #124 Binary Tree Maximum Path Sum ↗

☐ ● #297 Serialize and Deserialize Binary Tree ↗

DFS (5) ↗

- ☐ ● ★ #269 Alien Dictionary ↗
- ☐ ● #329 Longest Increasing Path in a Matrix ↗
- ☐ ● #685 Redundant Connection II ↗
- ☐ ● #1036 Escape a Large Maze ↗
- ☐ ● #1192 Critical Connections in a Network ↗

Graphs (2) ↗

- ☐ ● ★ #305 Number of Islands II ↗
- ☐ ● #1153 String Transforms Into Another String ↗

BFS (4) ↗

- ☐ ● #127 Word Ladder ↗
- ☐ ● ★ #317 Shortest Distance from All Buildings ↗
- ☐ ● #773 Sliding Puzzle ↗
- ☐ ● #815 Bus Routes ↗

Round 4 | Mid | E Easy (12) ↗

Linked List (7) ↗

- ☐ ● #21 Merge Two Sorted Lists ↗

- ☐ ● #141 Linked List Cycle ↗
- ☐ ● #206 Reverse Linked List ↗
- ☐ ● #234 Palindrome Linked List ↗
- ☐ ● #706 Design HashMap ↗
- ☐ ● #876 Middle of the Linked List ↗
- ☐ ● ★ #1474 Delete N Nodes After M Nodes of Linked List ↗

Stack (3) ↗

- ☐ ● #20 Valid Parentheses ↗
- ☐ ● #232 Implement Queue using Stacks ↗
- ☐ ● #844 Backspace String Compare ↗

Trie (1) ↗

- ☐ ● #14 Longest Common Prefix ↗

Greedy (1) ↗

- ☐ ● #409 Longest Palindrome ↗

Round 5 | Mid | M Medium (56) ↗

Linked List (13) ↗

- ☐ ● #2 Add Two Numbers ↗
- ☐ ● #19 Remove Nth Node From End of List ↗

- ☐ ● #24 Swap Nodes in Pairs ↗
- ☐ ● #61 Rotate List ↗
- ☐ ● #146 LRU Cache ↗
- ☐ ● #148 Sort List ↗
- ☐ ● #328 Odd Even Linked List ↗
- ☐ ● ★ #369 Plus One Linked List ↗
- ☐ ● ★ #430 Flatten a Multilevel Doubly Linked List ↗
- ☐ ● #622 Design Circular Queue ↗
- ☐ ● #707 Design Linked List ↗
- ☐ ● ★ #708 Insert into a Sorted Circular Linked List ↗
- ☐ ● #1171 Remove Zero Sum Consecutive Nodes from Linked List ↗

Stack (14) ↗

- ☐ ● #143 Reorder List ↗
- ☐ ● #150 Evaluate Reverse Polish Notation ↗
- ☐ ● #155 Min Stack ↗
- ☐ ● #173 Binary Search Tree Iterator ↗

- ☐ ● #227 Basic Calculator II ↗
- ☐ ● ★ #255 Verify Preorder Sequence in Binary Search Tree ↗
- ☐ ● #394 Decode String ↗
- ☐ ● ★ #439 Ternary Expression Parser ↗
- ☐ ● ★ #484 Find Permutation ↗
- ☐ ● #735 Asteroid Collision ↗
- ☐ ● #739 Daily Temperatures ↗
- ☐ ● #962 Maximum Width Ramp ↗
- ☐ ● ★ #1214 Two Sum BSTs ↗
- ☐ ● ★ #1762 Buildings With an Ocean View ↗
- ☐ ● #Heap (11) ↗
- ☐ ● #215 Kth Largest Element in an Array ↗
- ☐ ● ★ #253 Meeting Rooms II ↗
- ☐ ● #347 Top K Frequent Elements ↗
- ☐ ● ★ #505 The Maze II ↗
- ☐ ● #621 Task Scheduler ↗
- ☐ ● #658 Find K Closest Elements ↗

- ☐ **● #787 Cheapest Flights Within K Stops** ↗
- ☐ **● #973 K Closest Points to Origin** ↗
- ☐ **● ★ #1057 Campus Bikes** ↗
- ☐ **● ★ #1167 Minimum Cost to Connect Sticks** ↗
- ☐ **● #1424 Diagonal Traverse II** ↗

Trie (7) ↗

- ☐ **● #139 Word Break** ↗
- ☐ **● #208 Implement Trie (Prefix Tree)** ↗
- ☐ **● #211 Design Add and Search Words Data Structure** ↗
- ☐ **● ★ #616 Add Bold Tag in String** ↗
- ☐ **● #692 Top K Frequent Words** ↗
- ☐ **● #720 Longest Word in Dictionary** ↗
- ☐ **● #1166 Design File System** ↗

Backtracking (6) ↗

- ☐ **● #17 Letter Combinations of a Phone Number** ↗
- ☐ **● #22 Generate Parentheses** ↗
- ☐ **● #39 Combination Sum** ↗

☐ ● #46 Permutations ↗

☐ ● #79 Word Search ↗

☐ ● #113 Path Sum II ↗

Greedy (5) ↗

☐ ● #134 Gas Station ↗

☐ ● #179 Largest Number ↗

☐ ● ★ #280 Wiggle Sort ↗

☐ ● #954 Array of Doubled Pairs ↗

☐ ● #2131 Longest Palindrome by
Concatenating Two Letter Words

Round 6 | Mid | H Hard (28) ↗

Linked List (3) ↗

☐ ● #25 Reverse Nodes in k-Group ↗

☐ ● #432 All O'one Data Structure ↗

☐ ● #460 LFU Cache ↗

Stack (8) ↗

☐ ● #32 Longest Valid Parentheses ↗

☐ ● #42 Trapping Rain Water ↗

☐ ● #84 Largest Rectangle in Histogram ↗

☐ ● #224 Basic Calculator ↗

☐ ● #239 Sliding Window Maximum ↗

☐ ● #716 Max Stack ↗

☐ ● ★ #772 Basic Calculator III ↗

☐ ● #895 Maximum Frequency Stack ↗

Heap (9) ↗

☐ ● #23 Merge k Sorted Lists ↗

☐ ● #218 The Skyline Problem ↗

☐ ● ★ #272 Closest Binary Search Tree Value II ↗

☐ ● #295 Find Median from Data Stream ↗

☐ ● ★ #358 Rearrange String k Distance Apart ↗

☐ ● ★ #499 The Maze III ↗

☐ ● #632 Smallest Range Covering Elements
from K Lists ↗

☐ ● #759 Employee Free Time ↗

☐ ● #1425 Constrained Subsequence Sum ↗

Trie (4) ↗

- ☐ ● #212 Word Search II ↗
- ☐ ● #336 Palindrome Pairs ↗
- ☐ ● ★ #588 Design In-Memory File System ↗
- ☐ ● ★ #642 Design Search Autocomplete System ↗

Backtracking (4) ↗

- ☐ ● #37 Sudoku Solver ↗
- ☐ ● #51 N-Queens ↗
- ☐ ● #126 Word Ladder II ↗
- ☐ ● #996 Number of Squareful Arrays ↗

Round 7 | Low | E Easy (14) ↗

Dynamic Programming (2) ↗

- ☐ ● #70 Climbing Stairs ↗
- ☐ ● #121 Best Time to Buy and Sell Stock ↗

Bit Manipulation (6) ↗

- ☐ ● #67 Add Binary ↗
- ☐ ● #136 Single Number ↗
- ☐ ● #190 Reverse Bits ↗

☐ ● #191 Number of 1 Bits ↗

☐ ● #268 Missing Number ↗

☐ ● #338 Counting Bits ↗

Math (6) ↗

☐ ● #9 Palindrome Number ↗

☐ ● #13 Roman to Integer ↗

☐ ● #504 Base 7 ↗

☐ ● ★ #1056 Confusing Number ↗

☐ ● ★ #1228 Missing Number in Arithmetic Progression ↗

☐ ● ★ #1427 Perform String Shifts ↗

Round 8 | Low | M Medium (32) ↗

Dynamic Programming (16) ↗

☐ ● #5 Longest Palindromic Substring ↗

☐ ● #53 Maximum Subarray ↗

☐ ● #55 Jump Game ↗

☐ ● #62 Unique Paths ↗

☐ ● #72 Edit Distance ↗

- ☐ ● #91 Decode Ways ↗
- ☐ ● #152 Maximum Product Subarray ↗
- ☐ ● #198 House Robber ↗
- ☐ ● ★ #256 Paint House ↗
- ☐ ● #309 Best Time to Buy and Sell Stock with ↗
Cooldown
- ☐ ● #377 Combination Sum IV ↗
- ☐ ● #416 Partition Equal Subset Sum ↗
- ☐ ● #435 Non-overlapping Intervals ↗
- ☐ ● #516 Longest Palindromic Subsequence ↗
- ☐ ● #576 Out of Boundary Paths ↗
- ☐ ● #1143 Longest Common Subsequence ↗

Bit Manipulation (4) ↗

- ☐ ● #78 Subsets ↗
- ☐ ● #90 Subsets II ↗
- ☐ ● #287 Find the Duplicate Number ↗
- ☐ ● ★ #1506 Find Root of N-Ary Tree ↗

Math (5) ↗

- ☐ **● #7 Reverse Integer** ↗
- ☐ **● #50 Pow(x, n)** ↗
- ☐ **● #380 Insert Delete GetRandom O(1)** ↗
- ☐ **● #1017 Convert to Base -2** ↗
- ☐ **● #3160 Find the Number of Distinct Colors Among the Balls** ↗

JavaScript (7) ↗

- ☐ **● ★ #2627 Debounce** ↗
- ☐ **● ★ #2628 JSON Deep Equal** ↗
- ☐ **● ★ #2630 Memoize** ↗
- ☐ **● ★ #2633 Convert Object to JSON String** ↗
- ☐ **● ★ #2636 Promise Pool** ↗
- ☐ **● ★ #2675 Array of Objects to Matrix** ↗
- ☐ **● ★ #2700 Differences Between Two Objects** ↗

Round 9 | Low | H Hard (7) ↗

Dynamic Programming (6) ↗

- ☐ **● #10 Regular Expression Matching** ↗
- ☐ **● #115 Distinct Subsequences** ↗

☐ ● #123 Best Time to Buy and Sell Stock III ↗

☐ ● #132 Palindrome Partitioning II ↗

☐ ● #312 Burst Balloons ↗

☐ ● #1000 Minimum Cost to Merge Stones ↗

Math (1) ↗

☐ ● #68 Text Justification ↗

Round 1 · High · Easy

Round 1

High Priority · Easy

43 questions · 8 patterns

**Arrays & Hashing #1 #163 #346 #760 #1426
#1534**

String #383 #734 #1165

Two Pointers #88 #125 #283 #408 #977

Sorting #169 #217 #242 #252 #1086 #1122

Binary Search #35 #69 #278 #374 #704

Matrix #422 #566 #766 #867 #2373

**Trees & BST #100 #101 #104 #108 #110 #112
#226 #270 #501 #543 #572**

DFS #463 #733

Arrays & Hashing

Round 1 · High · Easy · 6 questions

Arrays & Hashing

□ #1 Two Sum

EAS

[Open on LeetCode](#)

Array · Hash Table

Lists: Grind 169

Problem

Given an array of integers `nums` and an integer `target`, return indices of the two numbers such that they add up to `target`.

You may assume that each input would have exactly one solution, and you may not use the same element twice.

You can return the answer in any order.

Example 1:

```
Input: nums = [2,7,11,15], target = 9  
Output: [0,1]  
Explanation: Because nums[0] + nums[1] == 9, we return [0, 1].
```

Example 2:

```
Input: nums = [3,2,4], target = 6  
Output: [1,2]
```

Example 3:

```
Input: nums = [3,3], target = 6  
Output: [0,1]
```

Constraints:

- $2 \leq \text{nums.length} \leq 104$
- $-109 \leq \text{nums}[i] \leq 109$
- $-109 \leq \text{target} \leq 109$
- Only one valid answer exists.

Follow-up: Can you come up with an algorithm that is less than $O(n^2)$ time complexity?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Let me make sure I understand the problem correctly.
# We're given an array of integers and a target value,
# and we need to return the indices of the two numbers
# that add up to the target – not the values, the indices. Right?"
#
# "A few quick questions:
# Can the same element be used twice? I'm guessing no.
# Is there always exactly one valid answer?
# Can the array be empty or have only one element?
# Are there negative numbers or duplicates?"
#
# "Let me walk the example: nums = [2, 7, 11, 15], target = 9.
# 2 + 7 = 9, so we return [0, 1]. Got it."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# For every pair (i, j) where j > i, check if nums[i] + nums[j] equals target.
# Time:  $O(n^2)$  from the nested loops. Space:  $O(1)$ , no extra storage.
# Should I go ahead and optimize?"
```

```
class _1_TwoSum_Brute:
    def twoSum(self, nums: List[int], target: int) -> List[int]:
        for i in range(len(nums)):
            for j in range(i + 1, len(nums)):
                if nums[i] + nums[j] == target:
                    return [i, j]
        return []
```

PHASE 3 — OPTIMIZE

```
# "The bottleneck is searching the rest of the array for a complement each time."
```

```

# I notice we're repeatedly asking: does the number I need exist?
# A hash map answers that in O(1).
#
# Walk forward through the array. For each number, compute complement = target -
num.
# If complement is already in the map, return both indices.
# Otherwise, store the current number and index and keep going.
#
# Pseudocode:
# seen = {}
# for index, num in nums:
#     complement = target - num
#     if complement in seen: return [seen[complement], index]
#     seen[num] = index
#
# Time: O(n). Space: O(n). Does that make sense before I code it?"

# PHASE 4 — CLEAN CODE
# "Now I'll implement the optimized approach with meaningful names."
class _1_TwoSum:
    def twoSum(self, nums: List[int], target: int) -> List[int]:
        seen = {} # maps value -> index
        for index, num in enumerate(nums):
            complement = target - num
            if complement in seen:
                return [seen[complement], index]
            seen[num] = index
        return []

# PHASE 5 — TEST & DEBUG
# "Let me trace through a few cases."
#
# nums=[2,7,11,15], target=9
# index=0: complement=7, not in seen. seen={2:0}
# index=1: complement=2, found at 0. return [0,1] ✓
#
# nums=[3,2,4], target=6
# index=0: complement=3, not in seen. seen={3:0}
# index=1: complement=4, not in seen. seen={3:0,2:1}
# index=2: complement=2, found at 1. return [1,2] ✓
#
# nums=[3,3], target=6
# index=0: complement=3, not in seen. seen={3:0}
# index=1: complement=3, found at 0. return [0,1] ✓
#
# "No bugs. Holds across all cases."

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(n) — one pass. Space O(n) — worst case every element enters the map.
# Trade-off: O(1) space would require O(n²) time — no free lunch here.

```

```
# If the array were sorted I'd use two pointers: O(n) time, O(1) space.  
# But we need original indices and the input is unsorted, so hash map wins.  
# Given more time I'd ask: what if there are multiple valid pairs?  
# We'd collect all of them rather than returning on the first hit."
```

Arrays & Hashing

★ #163 Missing Ranges ★

EAS

[Open on LeetCode](#)

Array

Lists: ★ Premium | Premium 98

Problem

You are given an inclusive range [lower, upper] and a sorted unique integer array nums, where all elements are within the inclusive range.

A number x is considered missing if x is in the range [lower, upper] and x is not in nums.

Return the shortest sorted list of ranges that exactly covers all the missing numbers. That is, no element of nums is included in any of the ranges, and each missing number is covered by one of the ranges.

Example 1:

Input: nums = [0,1,3,50,75], lower = 0, upper = 99

Output: [[2,2],[4,49],[51,74],[76,99]]

Explanation: The ranges are:

[2,2]

[4,49]

[51,74]

[76,99]

Example 2:

Input: nums = [-1], lower = -1, upper = -1

Output: []

Explanation: There are no missing ranges since there are no missing numbers.

Constraints:

- $-109 \leq \text{lower} \leq \text{upper} \leq 109$
- $0 \leq \text{nums.length} \leq 100$
- $\text{lower} \leq \text{nums}[i] \leq \text{upper}$
- All the values of nums are unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Let me make sure I understand the problem correctly.

We have a sorted array of unique integers, and an inclusive range [lower, upper].

We need to return the list of ranges that cover every number

inside [lower, upper] that does NOT appear in nums. Right?"

#

"A few quick questions:

Can nums be empty? If so I assume the whole [lower, upper] is one missing range.

Are all elements in nums guaranteed to be within [lower, upper]? Good.

The output ranges should be as compact as possible – so [2,2] not just [2]?"

#

"Let me walk the example:

nums=[0,1,3,50,75], lower=0, upper=99.

0 and 1 are present so no gap at the start.

3 is present but 2 is missing → [2,2].

50 is present but 4–49 are missing → [4,49].

75 is present but 51–74 are missing → [51,74].

76–99 are all missing → [76,99].

Output: [[2,2],[4,49],[51,74],[76,99]]. Got it."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

The simplest approach: build a set from nums for O(1) lookups,

then walk every integer from lower to upper.

Whenever I hit a missing number, I extend the current gap range.

When I hit a number that IS in the set, I close the gap and record it.

#

Time: O(upper – lower) which can be up to 2 billion – way too slow.

Space: O(n) for the set.

Should I optimize?"

```

class _163_MissingRanges_Brute:
    def findMissingRanges(self, nums: List[int], lower: int, upper: int) ->
List[List[int]]:
    present = set(nums)
    result = []
    gap_start = None
    for x in range(lower, upper + 1):
        if x not in present:
            if gap_start is None:
                gap_start = x
            else:
                if gap_start is not None:
                    result.append([gap_start, x - 1])
                    gap_start = None
    if gap_start is not None:
        result.append([gap_start, upper])
    return result

```

PHASE 3 — OPTIMIZE

```

# "The brute force iterates over the entire numerical range – up to 2 billion steps.
# But nums has at most 100 elements. The gaps I care about
# only exist between consecutive numbers in nums, before the first, and after the
# last.
# So I only need to check those boundaries – that's O(n), not O(range).
#
# I think of nums as a set of walls. The missing ranges are the gaps between walls.
# There are at most n+1 gaps: one before nums[0], one between each pair, one after
# nums[-1].
#
# Pseudocode:
# if nums is empty: return [[lower, upper]]
# if lower < nums[0]: add [lower, nums[0]-1]
# for each consecutive pair (prev, curr) in nums:
# if prev+1 < curr: add [prev+1, curr-1]
# if nums[-1] < upper: add [nums[-1]+1, upper]
#
# Time: O(n). Space: O(1) extra. Does that make sense before I code it?"

```

PHASE 4 — CLEAN CODE

```

# "Now I'll implement the optimized approach.
# I'll use pairwise from itertools to cleanly iterate consecutive pairs."
class _163_MissingRanges:
    def findMissingRanges(self, nums: List[int], lower: int, upper: int) ->
List[List[int]]:
    if not nums:
        return [[lower, upper]]
    result = []
    first, last = nums[0], nums[-1]
    if lower < first:

```

```

        result.append([lower, first - 1])
    for prev, curr in pairwise(nums):
        if prev + 1 < curr:
            result.append([prev + 1, curr - 1])
    if last < upper:
        result.append([last + 1, upper])
    return result

```

PHASE 5 — TEST & DEBUG

"Let me trace through a few cases."

#

nums=[0,1,3,50,75], lower=0, upper=99

nums not empty. first=0, last=75.

lower(0) < first(0)? No. No leading gap.

pairwise: (0,1) → 1 < 1? No. (1,3) → 2 < 3? Yes → [2,2]. (3,50) → 4 < 50? Yes → [4,49]. (50,75) → 51 < 75? Yes → [51,74].

last(75) < upper(99)? Yes → [76,99].

result = [[2,2],[4,49],[51,74],[76,99]] ✓

#

nums=[-1], lower=-1, upper=-1

first=-1, last=-1.

lower(-1) < first(-1)? No.

pairwise: empty (only one element).

last(-1) < upper(-1)? No.

result = [] ✓

#

nums=[], lower=1, upper=5

Empty → return [[1,5]] ✓

#

"No bugs. All cases pass."

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$ – one pass through nums plus constant work per gap.

Space $O(1)$ extra – result list aside, no auxiliary storage.

This is a massive improvement over $O(\text{upper} - \text{lower})$ brute force

given the constraint that nums has at most 100 elements.

#

Trade-off: if nums were not sorted I'd need to sort first – $O(n \log n)$.

The problem guarantees sorted input so we skip that.

#

pairwise is Python 3.10+. If the interviewer is on an older version

I'd write: for i in range(1, len(nums)): prev, curr = nums[i-1], nums[i].

#

Given more time I'd ask: what if the output should be strings like '2->2'

instead of lists? Just a formatting change at the append step."

Arrays & Hashing

□ ★ #346 Moving Average from Data Stream ★

EAS

[Open on LeetCode](#)

Array · Design · Queue · Data Stream

Lists: ★ Premium | Premium 98

Problem

Given a stream of integers and a window size, calculate the moving average of all integers in the sliding window.

Implement the `MovingAverage` class:

- `MovingAverage(int size)` Initializes the object with the size of the window size.
- `double next(int val)` Returns the moving average of the last size values of the stream.

Example 1:

```
Input
["MovingAverage", "next", "next", "next", "next"]
[[3], [1], [10], [3], [5]]
Output
[null, 1.0, 5.5, 4.66667, 6.0]

Explanation
MovingAverage movingAverage = new MovingAverage(3);
```

Constraints:

- $1 \leq \text{size} \leq 1000$
- $-105 \leq \text{val} \leq 105$
- At most 104 calls will be made to next.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Let me make sure I understand the problem correctly.

We're designing a class that receives a stream of integers one at a time.

Each call to next() should return the average of the last 'size' values seen so far.

If fewer than 'size' values have arrived, we average however many we have. Right?"

#

"A few quick questions:

Can size be 1? Constraints say yes – that's just returning the latest value.

Can values be negative? Yes, constraints say down to -10^5 .

Is there a limit on how many next() calls we get? Up to 10^4 – good to know for space."

#

"Let me walk the example:

size=3. next(1)=1.0 (only 1 value). next(10)=5.5 (avg of 1,10).

next(3)=4.667 (avg of 1,10,3 – window full now).

next(5)=6.0 (avg of 10,3,5 – 1 falls out). Got it."

PHASE 2 — BRUTE FORCE

"The simplest approach: store every value in a plain list.

On each next() call, slice the last 'size' elements and compute the average.

#

Time: $O(\text{size})$ per next() call due to the slice and sum.

Space: $O(n)$ where n is total calls – the list grows forever.

Should I optimize?"

```
class _346_MovingAverage_Brute:
```

```
    def __init__(self, size: int):
```

```
        self.size = size
```

```
        self.stream = []
```

```
    def next(self, val: int) -> float:
```

```
        self.stream.append(val)
```

```
        window = self.stream[-self.size:]
```

```
        return sum(window) / len(window)
```

PHASE 3 — OPTIMIZE

"Two inefficiencies in the brute force:

First, we recompute the sum from scratch every call – that's $O(\text{size})$ work.

Second, we store the entire history but only ever need the last 'size' values.

```
#
# I notice we're recomputing the same sum repeatedly. What if we cache it?
# I'll keep a running sum and update it incrementally:
# when a new value comes in, add it. When the window is full,
# subtract the oldest value before adding the new one.
#
# For the window itself, a deque is perfect – O(1) append on the right,
# O(1) popleft when the oldest value falls out. The deque never exceeds 'size'
# elements.
#
# Pseudocode:
# __init__: q = deque(), running_sum = 0, self.size = size
# next(val):
# if len(q) == size: running_sum -= q.popleft()
# running_sum += val
# q.append(val)
# return running_sum / len(q)
#
# Time: O(1) per next() call. Space: O(size) – the deque is capped.
# Does that make sense before I write it out?"
```

PHASE 4 — CLEAN CODE

"Now I'll implement the optimized approach with meaningful names."

```
class _346_MovingAverage:
    def __init__(self, size: int):
        self.size = size
        self.window = deque()
        self.running_sum = 0

    def next(self, val: int) -> float:
        if len(self.window) == self.size:
            self.running_sum -= self.window.popleft() # evict oldest
        self.running_sum += val
        self.window.append(val)
        return self.running_sum / len(self.window)
```

PHASE 5 — TEST & DEBUG

"Let me trace through the example: size=3."

```
#
# next(1): window not full. sum=1, window=[1]. return 1/1 = 1.0 ✓
# next(10): window not full. sum=11, window=[1,10]. return 11/2 = 5.5 ✓
# next(3): window not full. sum=14, window=[1,10,3]. return 14/3 = 4.667 ✓
# next(5): window full → popleft() removes 1, sum=14-1=13.
# sum=13+5=18, window=[10,3,5]. return 18/3 = 6.0 ✓
#
# Edge – size=1:
# next(7): window full immediately after first call? No – starts empty.
# sum=7, window=[7]. return 7/1 = 7.0 ✓
# next(4): window full (size=1). popleft removes 7, sum=0+4=4, window=[4].
# return 4/1 = 4.0 ✓
#
```

"No bugs. Running sum stays consistent across evictions."

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(1)$ per `next()` call – all operations are constant: deque append, popleft, addition, division.

Space $O(\text{size})$ – the deque holds at most 'size' elements regardless of how many total calls are made. Much better than the $O(n)$ brute force.

#

Trade-off worth mentioning: floating point precision. If values are large and size is large, the running sum can accumulate rounding errors over time. A production system might use Python's Decimal or periodically recompute the sum from scratch to correct drift.

#

Alternative: a circular buffer (fixed-size list with a pointer) does the same thing with slightly better cache locality than a deque, but the deque is cleaner to read and interview-safe.

#

Given more time I'd ask: what if we need to support a variable window size that can change between calls? We'd need to resize the deque dynamically."

Arrays & Hashing

★ #760 Find Anagram Mappings ★

EAS

[Open on LeetCode](#)

Array · Hash Table

Lists: ★ Premium | Premium 98

Problem

You are given two integer arrays `nums1` and `nums2` where `nums2` is an anagram of `nums1`. Both arrays may contain duplicates.

Return an index mapping array mapping from `nums1` to `nums2` where `mapping[i] = j` means the *i*th element in `nums1` appears in `nums2` at index *j*. If there are multiple answers, return any of them.

An array *a* is an anagram of an array *b* means *b* is made by randomizing the order of the elements in *a*.

Example 1:

Input: `nums1 = [12,28,46,32,50]`, `nums2 = [50,12,32,46,28]`

Output: `[1,4,3,2,0]`

Explanation: As `mapping[0] = 1` because the 0th element of `nums1` appears at `nums2[1]`, and `mapping[1] = 4` because the 1st element of `nums1` appears at `nums2[4]`, and so on.

Example 2:

Input: nums1 = [84,46], nums2 = [84,46]
 Output: [0,1]

Constraints:

- $1 \leq \text{nums1.length} \leq 100$
- $\text{nums2.length} == \text{nums1.length}$
- $0 \leq \text{nums1}[i], \text{nums2}[i] \leq 105$
- nums2 is an anagram of nums1 .

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Let me make sure I understand the problem correctly.
# We have two arrays nums1 and nums2. nums2 is an anagram of nums1 –
# same elements, different order. We need to return a mapping array
# where mapping[i] is the index in nums2 where nums1[i] appears. Right?"
#
# "A few quick questions:
# Can there be duplicate values? The problem says yes.
# If a value appears multiple times in nums2, which index do we return?
# The problem says any valid answer is accepted – good, that simplifies things.
# Both arrays are the same length and nums2 is guaranteed to be an anagram,
# so every element in nums1 is guaranteed to exist in nums2?"
#
# "Let me walk the example:
# nums1=[12,28,46,32,50], nums2=[50,12,32,46,28].
# 12 is at index 1 in nums2 → mapping[0]=1.
# 28 is at index 4 → mapping[1]=4.
# 46 is at index 3 → mapping[2]=3.
# 32 is at index 2 → mapping[3]=2.
# 50 is at index 0 → mapping[4]=0.
# Output: [1,4,3,2,0]. Got it."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# For each element in nums1, scan nums2 left to right until I find a match
# and record that index.
#
# Time:  $O(n^2)$  – for each of  $n$  elements in  $\text{nums1}$  we do a linear scan of  $\text{nums2}$ .
# Space:  $O(1)$  extra beyond the output array.
# Should I optimize?"
```

```
class _760_AnagramMappings_Brute:
```

```
def anagramMappings(self, nums1: List[int], nums2: List[int]) -> List[int]:
    mapping = []
    for num in nums1:
        for j, val in enumerate(nums2):
            if val == num:
                mapping.append(j)
                break
    return mapping
```

PHASE 3 — OPTIMIZE

```
# "The bottleneck is scanning nums2 from scratch for every element in nums1.
# We're asking the same question repeatedly: where does this value live in nums2?
# A hash map lets us answer that in O(1).
#
# I'll preprocess nums2 once: build a map from value -> index.
# Then for each element in nums1, look it up directly.
#
# One thing to flag: if there are duplicates, building the map will overwrite
# earlier indices with later ones. Since the problem allows any valid answer,
# that's fine - we just return whichever index was stored last.
#
# Pseudocode:
# index_in_nums2 = {val: i for i, val in enumerate(nums2)}
# return [index_in_nums2[num] for num in nums1]
#
# Time: O(n). Space: O(n) for the hash map.
# Does that make sense before I code it?"
```

PHASE 4 — CLEAN CODE

```
# "Now I'll implement the optimized approach with meaningful names."
```

```
class _760_AnagramMappings:
    def anagramMappings(self, nums1: List[int], nums2: List[int]) -> List[int]:
        index_in_nums2 = {val: i for i, val in enumerate(nums2)}
        return [index_in_nums2[num] for num in nums1]
```

PHASE 5 — TEST & DEBUG

```
# "Let me trace through both examples."
#
# nums1=[12,28,46,32,50], nums2=[50,12,32,46,28]
# index_in_nums2 = {50:0, 12:1, 32:2, 46:3, 28:4}
# nums1[0]=12 -> 1
# nums1[1]=28 -> 4
# nums1[2]=46 -> 3
# nums1[3]=32 -> 2
# nums1[4]=50 -> 0
# Output: [1,4,3,2,0] ✓
#
# nums1=[84,46], nums2=[84,46]
# index_in_nums2 = {84:0, 46:1}
# Output: [0,1] ✓
```

```
#
# Duplicate case – nums1=[3,3], nums2=[3,3]
# enumerate(nums2): (0,3) → map={3:0}, then (1,3) → map={3:1} (overwritten).
# nums1[0]=3 → 1, nums1[1]=3 → 1. Output: [1,1].
# Both point to the same index – valid since any answer is accepted ✓
#
# "No bugs. Duplicate overwrite is intentional and within spec."

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(n)$  – one pass to build the map, one pass to build the result.
# Space  $O(n)$  for the hash map.
#
# The duplicate handling is worth calling out explicitly:
# by iterating enumerate(nums2) forward, later indices overwrite earlier ones.
# If the interviewer wanted ALL valid indices for duplicates returned,
# I'd change the map to store a list of indices per value and pop one per lookup.
#
# Trade-off: if n is tiny (say n=5 like in the example), the brute force  $O(n^2)$ 
# is barely slower and uses no extra space. The hash map pays off as n grows.
#
# Given more time I'd ask: what if we need to return the mapping in sorted order
# by nums2 index? We'd sort the output –  $O(n \log n)$  – but that changes the spec."
```


Arrays & Hashing

□ ★ #1426 Counting Elements ★

EAS

[Open on LeetCode](#)

Array · Hash Table

Lists: ★ Premium | Premium 98

Problem

Given an integer array `arr`, count how many elements `x` there are, such that `x + 1` is also in `arr`. If there are duplicates in `arr`, count them separately.

Example 1:

```
Input: arr = [1,2,3]
Output: 2
Explanation: 1 and 2 are counted cause 2 and 3 are in arr.
```

Example 2:

```
Input: arr = [1,1,3,3,5,5,7,7]
Output: 0
Explanation: No numbers are counted, cause there is no 2, 4, 6, or 8 in arr.
```

Constraints:

- $1 \leq \text{arr.length} \leq 1000$
- $0 \leq \text{arr}[i] \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Let me make sure I understand the problem correctly.

For each element x in the array, we check whether $x+1$ also exists in the array.

If it does, we count x . And if x appears multiple times, we count each occurrence separately – so duplicates all contribute individually. Right?"

#

"A few quick questions:

Can the array be empty? Constraints say $\text{length} \geq 1$, so no.

Can values be 0? Yes – so $x=0$ counts if 1 is present.

Can values be negative? Constraints say $0 \leq \text{arr}[i] \leq 1000$, so no.

Are there duplicates? Yes, and the problem explicitly says to count them separately."

#

"Let me walk the examples:

$\text{arr}=[1,2,3]$: $1+1=2$ is in $\text{arr} \rightarrow$ count 1. $2+1=3$ is in $\text{arr} \rightarrow$ count 2. $3+1=4$ not in arr .

Output: 2 ✓

$\text{arr}=[1,1,3,3,5,5,7,7]$: $1+1=2$ not in arr . $3+1=4$ not in arr . Same for 5,7.

Output: 0 ✓ Got it."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

For each element x in arr , scan the entire array to see if $x+1$ exists.

If it does, increment the count.

#

Time: $O(n^2)$ – nested scan for every element.

Space: $O(1)$ – no extra storage.

Should I optimize?"

class _1426_CountingElements_Brute:

def countElements(self, arr: List[int]) -> int:

count = 0

for x in arr:

for y in arr:

if y == x + 1:

count += 1

break

return count

PHASE 3 — OPTIMIZE

"The bottleneck is the inner scan – we're asking 'does $x+1$ exist?' n times, each taking $O(n)$. A hash set answers that in $O(1)$.

#

But there's a subtlety: duplicates. If $\text{arr}=[1,1,2]$, both 1s should be counted.

A plain set tells us $x+1$ exists but not how many times x appears.

So I'll use a Counter instead – it gives me $O(1)$ existence checks

AND stores the frequency of each value.

#

Walk over the unique keys in the counter. If $\text{key}+1$ is also in the counter,

add $\text{counter}[\text{key}]$ to the answer – that accounts for all duplicates of key at once.

#

```
# Pseudocode:
# freq = Counter(arr)
# for x in freq:
# if x+1 in freq: ans += freq[x]
# return ans
#
# Time: O(n) to build the Counter, O(k) to iterate unique keys where k <= n.
# Overall O(n). Space: O(k) for the Counter. Does that make sense?"
```

PHASE 4 — CLEAN CODE

"Now I'll implement the optimized approach with meaningful names."

```
class _1426_CountingElements:
    def countElements(self, arr: List[int]) -> int:
        freq = Counter(arr)
        count = 0
        for x in freq:
            if x + 1 in freq:
                count += freq[x]
        return count
```

PHASE 5 — TEST & DEBUG

"Let me trace through a few cases."

```
#
# arr=[1,2,3]
# freq={1:1, 2:1, 3:1}
# x=1: 2 in freq → count+=1. count=1.
# x=2: 3 in freq → count+=1. count=2.
# x=3: 4 not in freq.
# return 2 ✓
#
# arr=[1,1,3,3,5,5,7,7]
# freq={1:2, 3:2, 5:2, 7:2}
# x=1: 2 not in freq.
# x=3: 4 not in freq.
# x=5: 6 not in freq.
# x=7: 8 not in freq.
# return 0 ✓
#
# Duplicate counting – arr=[1,1,2]
# freq={1:2, 2:1}
# x=1: 2 in freq → count+=freq[1]=2. count=2.
# x=2: 3 not in freq.
# return 2 ✓ (both 1s counted, as the problem requires)
#
# "No bugs. Duplicate handling confirmed."
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n) – one pass to build the Counter, one pass over unique keys.
# Space O(k) where k is the number of distinct values, at most O(n).
#
```

```
# Alternative: use a plain set instead of a Counter, then iterate the original arr.
# present = set(arr)
# return sum(1 for x in arr if x+1 in present)
# Same O(n) time and O(n) space – slightly simpler but both are valid.
# The Counter approach avoids re-iterating arr which is a minor readability win.
#
# Trade-off: if the value range is small and fixed (like 0-1000 here),
# a boolean array of size 1001 works as a lookup in O(1) with better cache
# performance than a hash map, and O(1001) ~ O(1) space in practice.
#
# Given more time I'd ask: what if we needed to return the actual elements
# that qualify rather than a count? We'd collect them into a list instead."
```

Arrays & Hashing

□ #1534 Count Good Triplets

EAS

[Open on LeetCode](#)

Array · Enumeration

Lists: CodeSignal

Problem

Given an array of integers `arr`, and three integers `a`, `b` and `c`. You need to find the number of good triplets.

A triplet $(arr[i], arr[j], arr[k])$ is good if the following conditions are true:

- $0 \leq i < j < k < arr.length$
- $|arr[i] - arr[j]| \leq a$
- $|arr[j] - arr[k]| \leq b$
- $|arr[i] - arr[k]| \leq c$

Where $|x|$ denotes the absolute value of x .

Return the number of good triplets.

Example 1:

Input: `arr = [3,0,1,1,9,7]`, `a = 7`, `b = 2`, `c = 3`

Output: 4

Explanation: There are 4 good triplets: $[(3,0,1), (3,0,1), (3,1,1), (0,1,1)]$.

Example 2:

Input: arr = [1,1,2,2,3], a = 0, b = 0, c = 1

Output: 0

Explanation: No triplet satisfies all conditions.

Constraints:

- $3 \leq \text{arr.length} \leq 100$
- $0 \leq \text{arr}[i] \leq 1000$
- $0 \leq a, b, c \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Let me make sure I understand the problem correctly.

We need to count all index triplets (i, j, k) where $i < j < k$,

and all three pairwise conditions hold simultaneously:

the first pair must differ by at most a,

the middle-to-last pair by at most b,

and the first-to-last pair by at most c.

We're counting triplets, not returning them. Right?"

#

"A few quick questions:

Can values in arr be 0? Yes, constraints allow it.

Can a, b, c be 0? Yes – that means exact equality required for that pair.

Array length is at most 100 – good to know for complexity decisions."

#

"Let me walk the example:

arr=[3,0,1,1,9,7], a=7, b=2, c=3.

(3,0,1) at indices (0,1,2): $|3-0|=3 \leq 7$, $|0-1|=1 \leq 2$, $|3-1|=2 \leq 3$ ✓

(3,0,1) at indices (0,1,3): $|3-0|=3 \leq 7$, $|0-1|=1 \leq 2$, $|3-1|=2 \leq 3$ ✓

(3,1,1) at indices (0,2,3): $|3-1|=2 \leq 7$, $|1-1|=0 \leq 2$, $|3-1|=2 \leq 3$ ✓

(0,1,1) at indices (1,2,3): $|0-1|=1 \leq 7$, $|1-1|=0 \leq 2$, $|0-1|=1 \leq 3$ ✓

Output: 4. Got it."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Three nested loops over indices (i, j, k) with $i < j < k$.

Count the triplet when $\text{arr}[i] + \text{arr}[j] + \text{arr}[k] \leq a + b + c$.

Correct, but $O(n^3)$ is too slow when n is up to 100.

Time: $O(n^3)$. Space: $O(1)$.

Should I go ahead and optimize?"

```
class _1534_CountGoodTriplets_Brute:
    def countGoodTriplets(self, arr: List[int], a: int, b: int, c: int) -> int:
        count = 0
        n = len(arr)
        for i in range(n):
            for j in range(i + 1, n):
                for k in range(j + 1, n):
                    if (abs(arr[i] - arr[j]) <= a and
                        abs(arr[j] - arr[k]) <= b and
                        abs(arr[i] - arr[k]) <= c):
                        count += 1
        return count
```

PHASE 3 — OPTIMIZE

"The algorithm stays $O(n^3)$ – there's no known sub-cubic solution for this specific three-condition problem, and with $n=100$ we don't need one.
 # But I can make two practical improvements:
 #
 # First: early pruning. The three conditions are checked in sequence.
 # If $|arr[i] - arr[j]| > a$, there's no point checking any k for this (i,j) pair.
 # We skip the entire inner k loop, reducing the constant factor significantly.
 #
 # Second: use `itertools.combinations` for cleaner, more readable code.
 # It generates all (i,j,k) triples with $i < j < k$ in one line.
 # Note that `combinations` doesn't support early pruning on the outer pair,
 # so for readability I'll use it here since n is small enough that it doesn't matter.
 #
 # I'll go with the early-pruning nested loop to show I understand the trade-off,
 # then mention the `itertools` alternative.
 #
 # Time: $O(n^3)$ worst case, faster in practice due to pruning.
 # Space: $O(1)$."

PHASE 4 — CLEAN CODE

"Now I'll implement with early pruning and meaningful names."

```
class _1534_CountGoodTriplets:
    def countGoodTriplets(self, arr: List[int], a: int, b: int, c: int) -> int:
        count = 0
        n = len(arr)
        for i in range(n):
            for j in range(i + 1, n):
                if abs(arr[i] - arr[j]) > a: # prune: skip all k for this pair
                    continue
                for k in range(j + 1, n):
                    if (abs(arr[j] - arr[k]) <= b and
                        abs(arr[i] - arr[k]) <= c):
                        count += 1
```

```

        return count

# itertools one-liner alternative (interview-clean, no pruning):
class _1534_CountGoodTriplets_Itertools:
    def countGoodTriplets(self, arr: List[int], a: int, b: int, c: int) -> int:
        return sum(
            abs(arr[i] - arr[j]) <= a and
            abs(arr[j] - arr[k]) <= b and
            abs(arr[i] - arr[k]) <= c
            for i, j, k in itertools.combinations(range(len(arr)), 3)
        )

```

PHASE 5 — TEST & DEBUG

"Let me trace the pruning version on example 1."

```

#
# arr=[3,0,1,1,9,7], a=7, b=2, c=3
# i=0 (arr[i]=3):
# j=1 (arr[j]=0): |3-0|=3<=7 → not pruned.
# k=2: |0-1|=1<=2, |3-1|=2<=3 → count=1
# k=3: |0-1|=1<=2, |3-1|=2<=3 → count=2
# k=4: |0-9|=9>2 → skip
# k=5: |0-7|=7>2 → skip
# j=2 (arr[j]=1): |3-1|=2<=7 → not pruned.
# k=3: |1-1|=0<=2, |3-1|=2<=3 → count=3
# k=4: |1-9|=8>2 → skip
# k=5: |1-7|=6>2 → skip
# j=3 (arr[j]=1): |3-1|=2<=7 → not pruned.
# k=4: |1-9|=8>2 → skip
# k=5: |1-7|=6>2 → skip
# j=4 (arr[j]=9): |3-9|=6<=7 → not pruned.
# k=5: |9-7|=2<=2, |3-7|=4>3 → skip
# j=5: no k available.
# i=1 (arr[i]=0):
# j=2 (arr[j]=1): |0-1|=1<=7 → not pruned.
# k=3: |1-1|=0<=2, |0-1|=1<=3 → count=4
# k=4,5: fail b condition → skip
# ... remaining pairs yield 0 more.
# return 4 ✓
#
# arr=[1,1,2,2,3], a=0, b=0, c=1
# a=0 means |arr[i]-arr[j]| must be 0 → only equal pairs pass.
# j=1 (1==1): pass a. k=2: |1-2|=1>0 → fail b. k=3: same. k=4: same.
# j=3 (1==?): arr[1]=1, arr[3]=2 → |1-2|=1>0 → pruned.
# No triplet survives. return 0 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n^3)$  worst case when all pairs pass the first condition –
# but early pruning on condition 1 cuts the k loop entirely for bad (i,j) pairs,
# making it much faster in practice.
# Space  $O(1)$  – just a counter.
#

```



```
# With n=100 the worst case is 161,700 triplets – trivially fast.
# If n grew to 10^4, O(n^3) would be 10^12 – completely infeasible.
# At that scale I'd look at sorting + binary search to narrow candidates per (i,j),
# or segment trees for range queries – but that's overkill here.
#
# The itertools one-liner is the cleanest version for an interview
# when you want to show Python fluency without sacrificing readability.
# Trade-off: it can't prune early, so stick with the nested loop if the
# interviewer asks about optimization.
#
# Given more time I'd ask: what if the array is sorted?
# Sorted order lets us binary search for valid k values after fixing i and j,
# reducing the inner loop from O(n) to O(log n) → overall O(n^2 log n)."
```

String

Round 1 · High · Easy · 3 questions

String

□ #383 Ransom Note

EAS

[Open on LeetCode](#)

Hash Table · String · Counting

Lists: Grind 169

Problem

Given two strings `ransomNote` and `magazine`, return `true` if `ransomNote` can be constructed by using the letters from `magazine` and `false` otherwise.

Each letter in `magazine` can only be used once in `ransomNote`.

Example 1:

```
Input: ransomNote = "a", magazine = "b"  
Output: false
```

Example 2:

```
Input: ransomNote = "aa", magazine = "ab"  
Output: false
```

Example 3:

```
Input: ransomNote = "aa", magazine = "aab"  
Output: true
```

Constraints:

- `1 <= ransomNote.length, magazine.length <= 105`
- ransomNote and magazine consist of lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Let me make sure I understand the problem correctly.
# We need to check if every character in ransomNote can be supplied
# by magazine – one letter from magazine per letter in ransomNote.
# So if ransomNote needs two 'a's, magazine must have at least two 'a's. Right?"
#
# "A few quick questions:
# Are both strings guaranteed to be non-empty? Yes, length >= 1.
# Only lowercase English letters – so exactly 26 possible characters. Good.
# We're not modifying the strings, just checking feasibility?"
#
# "Let me walk the examples:
# ransomNote='a', magazine='b': 'a' not in magazine → false ✓
# ransomNote='aa', magazine='ab': need two 'a's, magazine has one → false ✓
# ransomNote='aa', magazine='aab': need two 'a's, magazine has two → true ✓ Got it."
```

PHASE 2 — BRUTE FORCE

```
# "The simplest approach: convert magazine to a list so I can 'use up' letters.
# For each character in ransomNote, scan the list to find it and remove it.
# If I can't find it, return false.
#
# Time: O(n * m) – for each of n chars in ransomNote, scan up to m chars in
magazine.
# Space: O(m) for the mutable copy of magazine.
# Should I optimize?"
```

```
class _383_RansomNote_Brute:
    def canConstruct(self, ransomNote: str, magazine: str) -> bool:
        available = list(magazine)
        for char in ransomNote:
            if char not in available:
                return False
            available.remove(char)
        return True
```

PHASE 3 — OPTIMIZE

```
# "The bottleneck is scanning magazine for each character – that's the n*m.
# The question we keep asking is: how many of this letter does magazine still have?"
```

```
# A frequency counter answers that in O(1).
#
# I'll count every character in magazine upfront with a Counter.
# Then walk ransomNote: for each character, decrement its count.
# If any count drops below zero, magazine doesn't have enough of that letter –
# return false.
# If I finish ransomNote without going negative, return true.
#
# Pseudocode:
# freq = Counter(magazine)
# for char in ransomNote:
#     freq[char] -= 1
# if freq[char] < 0: return False
# return True
#
# Time: O(n + m) – one pass each over magazine and ransomNote.
# Space: O(1) – the Counter holds at most 26 keys (lowercase letters only).
# Does that make sense before I code it?"
```

PHASE 4 — CLEAN CODE

```
# "Now I'll implement the optimized approach with meaningful names."
```

```
class _383_RansomNote:
    def canConstruct(self, ransomNote: str, magazine: str) -> bool:
        available = Counter(magazine)
        for char in ransomNote:
            available[char] -= 1
            if available[char] < 0:
                return False
        return True
```

PHASE 5 — TEST & DEBUG

```
# "Let me trace through the three examples."
```

```
#
# ransomNote='a', magazine='b'
# available = {'b': 1}
# char='a': available['a']=0-1=-1 < 0 → return False ✓
# (Counter returns 0 for missing keys, so no KeyError)
#
# ransomNote='aa', magazine='ab'
# available = {'a':1, 'b':1}
# char='a': available['a']=0 → ok
# char='a': available['a']=-1 < 0 → return False ✓
#
# ransomNote='aa', magazine='aab'
# available = {'a':2, 'b':1}
# char='a': available['a']=1 → ok
# char='a': available['a']=0 → ok
# return True ✓
#
# Edge – ransomNote longer than magazine: ransomNote='abc', magazine='ab'
```

```
# available = {'a':1, 'b':1}
# char='a': 0 → ok. char='b': 0 → ok. char='c': -1 < 0 → return False ✓
#
# "No bugs. Counter's default of 0 for missing keys handles unseen characters
# cleanly."

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(n + m) - O(m)$  to build the Counter,  $O(n)$  to walk ransomNote.
# Space  $O(1)$  – the Counter has at most 26 entries since only lowercase letters
# exist.
# Even though we write  $O(k)$  where  $k$  is distinct chars,  $k$  is bounded by 26 –
# constant.
#
# Alternative: Counter subtraction in one line.
# return not (Counter(ransomNote) - Counter(magazine))
# Counter subtraction drops keys with zero or negative counts,
# so if anything remains, ransomNote needs letters magazine doesn't have.
# Elegant but less readable in an interview – I'd use the explicit loop.
#
# Another alternative: sort both strings and use two pointers –  $O(n \log n + m \log m)$ ,
# slower but uses  $O(1)$  space beyond the sort. Worth mentioning if interviewer
# constraints disallow hash maps.
#
# Given more time I'd ask: what if characters include uppercase or unicode?
# We'd replace the 26-slot assumption with a full hash map – space becomes
 $O(\text{alphabet size})$ ."
```

String

□ ★ #734 Sentence Similarity ★

EAS

[Open on LeetCode](#)

Array · Hash Table · String

Lists: ★ Premium | Premium 98

Problem

We can represent a sentence as an array of words, for example, the sentence "I am happy with leetcode" can be represented as `arr = ["I","am",happy","with","leetcode"]`.

Given two sentences `sentence1` and `sentence2` each represented as a string array and given an array of string pairs `similarPairs` where `similarPairs[i] = [xi, yi]` indicates that the two words `xi` and `yi` are similar.

Return true if `sentence1` and `sentence2` are similar, or false if they are not similar.

Two sentences are similar if:

- They have the same length (i.e., the same number of words)
- `sentence1[i]` and `sentence2[i]` are similar.

Notice that a word is always similar to itself, also notice that the similarity relation is not transitive. For example, if the words a and b are similar, and the words b and c are similar, a and c are not necessarily similar.

Example 1:

```
Input: sentence1 = ["great","acting","skills"], sentence2 =  
["fine","drama","talent"], similarPairs =  
[["great","fine"],["drama","acting"],["skills","talent"]]  
Output: true  
Explanation: The two sentences have the same length and each word i of  
sentence1 is also similar to the corresponding word in sentence2.
```

Example 2:

```
Input: sentence1 = ["great"], sentence2 = ["great"], similarPairs = []  
Output: true  
Explanation: A word is similar to itself.
```

Example 3:

```
Input: sentence1 = ["great"], sentence2 = ["doubleplus","good"], similarPairs =  
[["great","doubleplus"]]  
Output: false  
Explanation: As they don't have the same length, we return false.
```

Constraints:

- $1 \leq \text{sentence1.length}, \text{sentence2.length} \leq 1000$
- $1 \leq \text{sentence1}[i].\text{length}, \text{sentence2}[i].\text{length} \leq 20$
- $\text{sentence1}[i]$ and $\text{sentence2}[i]$ consist of English letters.
- $0 \leq \text{similarPairs.length} \leq 1000$

- `similarPairs[i].length == 2`
- `1 <= xi.length, yi.length <= 20`
- `xi` and `yi` consist of lower-case and upper-case English letters.
- All the pairs `(xi, yi)` are distinct.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Let me make sure I understand the problem correctly.
# Two sentences are similar if they're the same length and every word at position i
# in sentence1 is similar to the word at position i in sentence2.
# A word is always similar to itself.
# Similarity is NOT transitive – if a~b and b~c, that doesn't mean a~c.
# We're given the list of similar pairs explicitly, so we only trust those. Right?"
#
# "A few quick questions:
# Can similarPairs be empty? Yes – then only identical words are similar.
# Are pairs directional? The problem implies symmetric – if [a,b] is given,
# then a~b AND b~a. I'll confirm: yes, similarity is symmetric.
# Can the same pair appear twice? Constraints say all pairs are distinct – good."
#
# "Let me walk example 1:
# sentence1=['great','acting','skills'], sentence2=['fine','drama','talent'].
# Same length: 3. pairs include [great,fine],[drama,acting],[skills,talent].
# great~fine ✓, acting~drama ✓ (symmetric), skills~talent ✓ → true ✓ Got it."
```

PHASE 2 — BRUTE FORCE

```
# "The simplest approach: first check lengths. Then for each position i,
# check if sentence1[i] == sentence2[i], or scan all similarPairs
# to find a match in either direction.
#
# Time: O(n * p) – for each of n word pairs, scan up to p similarity pairs.
# Space: O(1) – no preprocessing.
# Should I optimize?"

class _734_SentenceSimilarity_Brute:
    def areSentencesSimilar(self, sentence1: List[str], sentence2: List[str],
        similarPairs: List[List[str]]) -> bool:
        if len(sentence1) != len(sentence2):
            return False
        for word1, word2 in zip(sentence1, sentence2):
            if word1 == word2:
```

```

        continue
    found = any((a == word1 and b == word2) or (a == word2 and b == word1)
                for a, b in similarPairs)
    if not found:
        return False
    return True

```

PHASE 3 — OPTIMIZE

```

# "The bottleneck is scanning all p pairs for every word position.
# We're asking the same question repeatedly: is word2 similar to word1?
# A hash map from each word to its set of similar words answers that in O(1).
#
# Preprocess similarPairs once: for each [a, b], add b to hmap[a] and a to hmap[b].
# That handles symmetry automatically – no need to check both directions at lookup
time.
#
# Then for each (word1, word2) pair in the sentences:
# if they're equal, they're trivially similar – skip.
# Otherwise check if word2 is in hmap[word1].
# Using defaultdict(set) means missing keys return an empty set – no KeyError.
#
# Pseudocode:
# if len(s1) != len(s2): return False
# similar = defaultdict(set)
# for a, b in pairs: similar[a].add(b); similar[b].add(a)
# for w1, w2 in zip(s1, s2):
# if w1 != w2 and w2 not in similar[w1]: return False
# return True
#
# Time: O(p + n). Space: O(p) for the map. Does that make sense?"

```

PHASE 4 — CLEAN CODE

```

# "Now I'll implement the optimized approach with meaningful names."

```

```

class _734_SentenceSimilarity:
    def areSentencesSimilar(self, sentence1: List[str], sentence2: List[str],
                           similarPairs: List[List[str]]) -> bool:
        if len(sentence1) != len(sentence2):
            return False

        similar = defaultdict(set)
        for word_a, word_b in similarPairs:
            similar[word_a].add(word_b)
            similar[word_b].add(word_a) # symmetric: both directions

        for word1, word2 in zip(sentence1, sentence2):
            if word1 != word2 and word2 not in similar[word1]:
                return False

        return True

```

PHASE 5 — TEST & DEBUG

```

# "Let me trace through all three examples."

```

```

#
# Example 1: s1=['great','acting','skills'], s2=['fine','drama','talent']
# pairs=[['great','fine'],['drama','acting'],['skills','talent']]
# similar = {great:{fine}, fine:{great}, drama:{acting}, acting:{drama},
# skills:{talent}, talent:{skills}}
# (great,fine): great!=fine, fine in similar[great] ✓
# (acting,drama): acting!=drama, drama in similar[acting] ✓
# (skills,talent): skills!=talent, talent in similar[skills] ✓
# return True ✓
#
# Example 2: s1=['great'], s2=['great'], pairs=[]
# Lengths match. similar={} (empty).
# (great,great): great==great → skip.
# return True ✓
#
# Example 3: s1=['great'], s2=['doubleplus','good'], pairs=[['great','doubleplus']]
# len(s1)=1 != len(s2)=2 → return False immediately ✓
#
# Non-transitive check: pairs=[['a','b'],['b','c']], checking a~c
# similar = {a:{b}, b:{a,c}, c:{b}}
# word1='a', word2='c': a!=c, c not in similar[a]={b} → return False ✓
# Confirms we don't accidentally apply transitivity.
#
# "No bugs. All cases pass including the transitivity trap."

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(p + n) - O(p)$  to build the similarity map,  $O(n)$  to check each word pair.
# Space  $O(p)$  for the map – each pair adds two entries.
#
# The critical design decision is storing both directions upfront rather than
# checking both at lookup time. It keeps the lookup a single  $O(1)$  set membership
# test.
#
# Important constraint to call out: similarity is NOT transitive.
# This rules out Union-Find (which is designed for transitive equivalence classes).
# If transitivity WERE required – for example in Sentence Similarity II (#737) –
# we'd use Union-Find to group words into connected components and check
# if sentence1[i] and sentence2[i] share the same root.
#
# Given more time I'd ask: what if the same word pair could appear in different
# cases,
# like ['Great','fine'] vs ['great','fine']? We'd normalize to lowercase before
# building
# the map and before lookups."

```

String

□ ★ #1165 Single Row Keyboard ★

EAS

[Open on LeetCode](#)

Hash Table · String

Lists: ★ Premium | Premium 98

Problem

There is a special keyboard with all keys in a single row.

Given a string keyboard of length 26 indicating the layout of the keyboard (indexed from 0 to 25). Initially, your finger is at index 0. To type a character, you have to move your finger to the index of the desired character. The time taken to move your finger from index i to index j is $|i - j|$. You want to type a string word. Write a function to calculate how much time it takes to type it with one finger.

Example 1:

```
Input: keyboard = "abcdefghijklmnopqrstuvwxyz", word = "cba"
Output: 4
Explanation: The index moves from 0 to 2 to write 'c' then to 1 to write 'b'
then to 0 again to write 'a'.
Total time = 2 + 1 + 1 = 4.
```

Example 2:

Input: keyboard = "pqrstuvwxyzabcdefghijklmnopqrstuvwxyz", word = "leetcode"
 Output: 73

Constraints:

- keyboard.length == 26
- keyboard contains each English lowercase letter exactly once in some order.
- $1 \leq \text{word.length} \leq 10^4$
- word[i] is an English lowercase letter.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Let me make sure I understand the problem correctly.

We have a custom keyboard layout – 26 characters in some arbitrary order.

Our finger starts at index 0. To type each character in word,

we move from the current index to that character's index on the keyboard.

The cost is the absolute distance moved. We sum all moves and return the total. Right?"

#

"A few quick questions:

The keyboard always has exactly 26 letters, each exactly once – so it's a permutation.

Word can only contain lowercase letters, and every letter in word is guaranteed

to appear on the keyboard – so no missing-key edge case to handle.

Word length can be up to 10^4 – so $O(n)$ should be fine."

#

"Let me walk example 1:

keyboard='abcdefghijklmnopqrstuvwxyz', word='cba'.

Start at 0. 'c' is at index 2: $|0-2|=2$. 'b' is at 1: $|2-1|=1$. 'a' is at 0: $|1-0|=1$.

Total = $2+1+1 = 4$. Got it."

PHASE 2 — BRUTE FORCE

"The simplest approach: for each character in word, scan the keyboard string from left to right to find where it lives, then compute the distance.

#

Time: $O(n * 26)$ – for each of n characters we scan up to 26 positions.

Since keyboard is fixed at 26, this is effectively $O(n)$.

Space: $O(1)$.

Should I optimize the lookup?"

```
class _1165_SingleRowKeyboard_Brute:
    def calculateTime(self, keyboard: str, word: str) -> int:
        total = 0
        position = 0
        for char in word:
            next_pos = keyboard.index(char)
            total += abs(position - next_pos)
            position = next_pos
        return total
```

PHASE 3 — OPTIMIZE

"The brute force scans 26 characters for every letter typed – that's the inner $O(26)$."

We're asking the same question repeatedly: where does this letter live on the keyboard?

A hash map answers that in $O(1)$ after a one-time $O(26)$ build.

#

Preprocess keyboard once: build a map from character $\rightarrow$ index.

Then walk word: for each character, look up its index in $O(1)$,

add the absolute distance from the current position, update position, move on.

#

Pseudocode:

pos_map = {char: i for i, char in enumerate(keyboard)}

total, position = 0, 0

for char in word:

total += abs(position - pos_map[char])

position = pos_map[char]

return total

#

Time: $O(26 + n) = O(n)$. Space: $O(26) = O(1)$ – the map is fixed size.

In practice the constant factor improvement is small since 26 is tiny,

but the hash map is the correct interview answer – cleaner and scales to

larger alphabets."

PHASE 4 — CLEAN CODE

"Now I'll implement the optimized approach with meaningful names."

```
class _1165_SingleRowKeyboard:
    def calculateTime(self, keyboard: str, word: str) -> int:
        key_index = {char: i for i, char in enumerate(keyboard)}
        total = 0
        position = 0
        for char in word:
            total += abs(position - key_index[char])
            position = key_index[char]
        return total
```

PHASE 5 — TEST & DEBUG

"Let me trace through both examples."

#

keyboard='abcdefghijklmnopqrstuvwxyz', word='cba'

```

# key_index = {'a':0,'b':1,'c':2,...}
# char='c': |0-2|=2, position=2. total=2
# char='b': |2-1|=1, position=1. total=3
# char='a': |1-0|=1, position=0. total=4
# return 4 ✓
#
# keyboard='pqrstuvwxyzabcdefghijklmnopqrstuvwxyz', word='leetcode'
# key_index =
{'p':0,'q':1,...,'a':10,'b':11,...,'l':21,'e':14,'t':19,'c':12,'o':24,'d':13}
# char='l': |0-21|=21, position=21. total=21
# char='e': |21-14|=7, position=14. total=28
# char='e': |14-14|=0, position=14. total=28
# char='t': |14-19|=5, position=19. total=33
# char='c': |19-12|=7, position=12. total=40
# char='o': |12-24|=12, position=24. total=52
# char='d': |24-13|=11, position=13. total=63
# char='e': |13-14|=1, position=14. total=64 ← wait, let me recheck key_index
# (keyboard='pqrstuvwxyzabcdefghijklmnopqrstuvwxyz':
p=0,q=1,r=2,s=3,t=4,u=5,v=6,w=7,x=8,y=9,z=10,
# a=11,b=12,c=13,d=14,e=15,f=16,g=17,h=18,i=19,j=20,k=21,l=22,m=23,n=24,o=25)
# char='l': |0-22|=22. char='e': |22-15|=7. char='e': 0. char='t': |15-4|=11.
# char='c': |4-13|=9. char='o': |13-25|=12. char='d': |25-14|=11. char='e':
|14-15|=1.
# total = 22+7+0+11+9+12+11+1 = 73 ✓
#
# "No bugs. Manual trace confirms both examples."

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(n)$  –  $O(26)$  to build the map,  $O(n)$  to walk word. Space  $O(1)$  – 26-entry map.
#
# The brute force using keyboard.index() is also  $O(n)$  in practice since keyboard is
# always 26 characters – but the hash map is the canonical interview answer because
# it makes the  $O(1)$  lookup intent explicit and generalizes to any alphabet size.
#
# Trade-off: if keyboard could be very long (say a  $10^6$ -char layout), the hash map
# preprocessing cost would matter more – but we'd still prefer it over .index().
#
# One-liner alternative:
# pos = {c: i for i, c in enumerate(keyboard)}
# return sum(abs(pos[b] - pos[a]) for a, b in zip('_', word, word))
# where prepending '_' shifts word so each pair is (prev_char, curr_char),
# and '_' maps to nothing – so you'd actually prepend a dummy start:
# pairs = zip([None] + list(word), word) – but this gets messy.
# The explicit loop is cleaner in an interview.
#
# Given more time I'd ask: what if we can use two fingers?
# That becomes a DP problem – optimal two-finger typing is a classic hard variant."

```

Two Pointers

Round 1 · High · Easy · 5 questions

Two Pointers

□ #88 Merge Sorted Array

EAS

[Open on LeetCode](#)

Array · Two Pointers · Sorting

Lists: Denny Zhang

Problem

You are given two integer arrays `nums1` and `nums2`, sorted in non-decreasing order, and two integers `m` and `n`, representing the number of elements in `nums1` and `nums2` respectively. Merge `nums1` and `nums2` into a single array sorted in non-decreasing order.

The final sorted array should not be returned by the function, but instead be stored inside the array `nums1`. To accommodate this, `nums1` has a length of `m + n`, where the first `m` elements denote the elements that should be merged, and the last `n` elements are set to 0 and should be ignored. `nums2` has a length of `n`.

Example 1:

Input: `nums1 = [1,2,3,0,0,0]`, `m = 3`, `nums2 = [2,5,6]`, `n = 3`
 Output: `[1,2,2,3,5,6]`
 Explanation: The arrays we are merging are `[1,2,3]` and `[2,5,6]`.
 The result of the merge is `[1,2,2,3,5,6]` with the underlined elements coming from `nums1`.

Example 2:

Input: `nums1 = [1]`, `m = 1`, `nums2 = []`, `n = 0`
 Output: `[1]`
 Explanation: The arrays we are merging are `[1]` and `[]`.
 The result of the merge is `[1]`.

Example 3:

Input: `nums1 = [0]`, `m = 0`, `nums2 = [1]`, `n = 1`
 Output: `[1]`
 Explanation: The arrays we are merging are `[]` and `[1]`.
 The result of the merge is `[1]`.
 Note that because `m = 0`, there are no elements in `nums1`. The `0` is only there to ensure the merge result can fit in `nums1`.

Constraints:

- `nums1.length == m + n`
- `nums2.length == n`
- `0 <= m, n <= 200`
- `1 <= m + n <= 200`
- `-109 <= nums1[i], nums2[j] <= 109`

Follow up: Can you come up with an algorithm that runs in $O(m + n)$ time?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Let me make sure I understand the problem correctly."

`nums1` has `m` valid elements followed by `n` zeros as padding.

`nums2` has `n` elements. Both are sorted in non-decreasing order.

```

# We need to merge them in-place into nums1 – sorted, no return value. Right?"
#
# "A few quick questions:
# Can m or n be 0? Yes – if n=0 nums1 is already done; if m=0 we just copy nums2.
# Can there be negative numbers? Yes, constraints go down to  $-10^9$ .
# Can there be duplicates? Yes – non-decreasing order means equal values are
allowed.
# We cannot use extra space for the merge? The follow-up asks for  $O(m+n)$  time,
# implying in-place. I'll target  $O(1)$  extra space."
#
# "Let me walk example 1:
# nums1=[1,2,3,0,0,0] m=3, nums2=[2,5,6] n=3.
# Merge [1,2,3] and [2,5,6] into nums1 → [1,2,2,3,5,6]. Got it."

# PHASE 2 — BRUTE FORCE
# "The simplest approach: copy nums2 into the tail of nums1 (the zero slots),
# then sort the entire nums1 array.
#
# nums1[m:] = nums2, then nums1.sort()
#
# Time:  $O((m+n) \log(m+n))$  for the sort. Space:  $O(1)$  extra – sort is in-place.
# This is clean but doesn't use the fact that both inputs are already sorted.
# Should I optimize to use that sorted property?"
class _88_MergeSortedArray_Brute:
    def merge(self, nums1: List[int], m: int, nums2: List[int], n: int) -> None:
        nums1[m:] = nums2
        nums1.sort()

# PHASE 3 — OPTIMIZE
# "The brute force throws away the sorted order and pays  $O(\log(m+n))$  to re-sort.
# The classic merge from the front copies to a temp array –  $O(m+n)$  space.
# But there's a smarter way that stays  $O(1)$  space.
#
# Key insight: if we merge from the BACK, we never overwrite elements we still need.
# The tail of nums1 is empty padding – we can fill it safely from right to left.
#
# Three pointers:
# i → last valid element in nums1 (index m-1)
# j → last element in nums2 (index n-1)
# k → last slot in nums1 (index m+n-1)
#
# At each step, pick the larger of nums1[i] and nums2[j], place it at k, move that
pointer left.
# When j drops below 0, nums2 is exhausted – nums1's remaining elements are already
in place.
# When i drops below 0 first, copy remaining nums2 into nums1.
#
# Pseudocode:
# i, j, k = m-1, n-1, m+n-1
# while k >= 0:

```

```

# if j < 0 or (i >= 0 and nums1[i] > nums2[j]):
#     nums1[k] = nums1[i]; i -= 1
# else:
#     nums1[k] = nums2[j]; j -= 1
# k += 1
#
# Time: O(m+n) – each element is placed exactly once.
# Space: O(1) – purely in-place.
# Does that make sense before I code it?"

# PHASE 4 — CLEAN CODE
# "Now I'll implement the optimized approach with meaningful names."
class _88_MergeSortedArray:
    def merge(self, nums1: List[int], m: int, nums2: List[int], n: int) -> None:
        last1 = m - 1 # tail of valid nums1 elements
        last2 = n - 1 # tail of nums2
        fill = m + n - 1 # next slot to fill in nums1 (right to left)
        while fill >= 0:
            if last2 < 0 or (last1 >= 0 and nums1[last1] > nums2[last2]):
                nums1[fill] = nums1[last1]
                last1 -= 1
            else:
                nums1[fill] = nums2[last2]
                last2 -= 1
            fill -= 1

# PHASE 5 — TEST & DEBUG
# "Let me trace through all three examples."
#
# nums1=[1,2,3,0,0,0] m=3, nums2=[2,5,6] n=3
# last1=2, last2=2, fill=5
# fill=5: nums1[2]=3 vs nums2[2]=6 → 6 wins → nums1[5]=6, last2=1, fill=4
# fill=4: nums1[2]=3 vs nums2[1]=5 → 5 wins → nums1[4]=5, last2=0, fill=3
# fill=3: nums1[2]=3 vs nums2[0]=2 → 3 wins → nums1[3]=3, last1=1, fill=2
# fill=2: nums1[1]=2 vs nums2[0]=2 → equal, take nums2 → nums1[2]=2, last2=-1, fill=1
# fill=1: last2<0 → take nums1[1]=2 → nums1[1]=2, last1=0, fill=0
# fill=0: last2<0 → take nums1[0]=1 → nums1[0]=1, last1=-1, fill=-1
# nums1=[1,2,2,3,5,6] ✓
#
# nums1=[1] m=1, nums2=[] n=0
# last1=0, last2=-1, fill=0
# fill=0: last2<0 → nums1[0]=nums1[0]=1. fill=-1. done.
# nums1=[1] ✓
#
# nums1=[0] m=0, nums2=[1] n=1
# last1=-1, last2=0, fill=0
# fill=0: last1<0 → nums1[0]=nums2[0]=1, last2=-1, fill=-1. done.
# nums1=[1] ✓
#

```

"No bugs. All three cases pass including the $m=0$ and $n=0$ edges."

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m+n)$ – each of the $m+n$ positions is filled exactly once.

Space $O(1)$ – purely in-place, no auxiliary arrays.

#

The core insight worth stating explicitly: merging front-to-back requires a temp
buffer because we'd overwrite `nums1`'s valid elements. Merging back-to-front avoids
this because the padding zeros are exactly the space we need, and large elements
fill the tail first without touching anything we still care about.

#

Edge cases worth calling out:

– $n=0$: loop runs but `last2<0` always, so `nums1`'s valid elements stay in place.

– $m=0$: `last1<0` always, so `nums2` is copied straight into `nums1`.

– All `nums2` elements are larger than all `nums1` elements: `nums2` fills the back,
then `nums1` elements slot into the front – handled cleanly by the pointer logic.

#

Given more time I'd ask: what if `nums1` has exactly m slots (no padding)?

Then in-place is impossible – we'd need to return a new merged array,

which is the standard merge step in merge sort."

Two Pointers

□ #125 Valid Palindrome

EAS[Open on LeetCode](#)

Two Pointers · String

Lists: Grind 169

Problem

A phrase is a palindrome if, after converting all uppercase letters into lowercase letters and removing all non-alphanumeric characters, it reads the same forward and backward.

Alphanumeric characters include letters and numbers.

Given a string *s*, return true if it is a palindrome, or false otherwise.

Example 1:

Input: *s* = "A man, a plan, a canal: Panama"
Output: true
Explanation: "amanaplanacanalpanama" is a palindrome.

Example 2:

Input: *s* = "race a car"
Output: false
Explanation: "raceacar" is not a palindrome.

Example 3:

Input: s = " "

Output: true

Explanation: s is an empty string "" after removing non-alphanumeric characters.

Since an empty string reads the same forward and backward, it is a palindrome.

Constraints:

- $1 \leq s.length \leq 2 * 10^5$
- s consists only of printable ASCII characters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Let me make sure I understand the problem correctly.

We strip everything except letters and digits, lowercase it all,

then check if the result reads the same forwards and backwards. Right?"

#

"A few quick questions:

Alphanumeric means letters AND digits – not just letters?

So '0P' after cleaning is '0p', which is not a palindrome – confirmed?

What about a string of only spaces or punctuation?

Example 3 shows a single space returns true – after cleaning it's an empty string,

and an empty string is considered a palindrome. Good.

Can the string be length 1? Yes – a single alphanumeric character is a palindrome."

#

"Let me walk example 1:

'A man, a plan, a canal: Panama' → 'amanaplanacanalpanama' → palindrome ✓

Example 2: 'race a car' → 'raceacar' → not a palindrome ✓"

PHASE 2 — BRUTE FORCE

"Let me start with the simplest correct approach.

Copy every alphanumeric character into a cleaned list (lowercased),

then compare the list to its reverse.

Two passes over the string plus $O(n)$ extra storage.

Time: $O(n)$. Space: $O(n)$ for the cleaned copy.

Should I go ahead and optimize?"

```
class _125_ValidPalindrome_Brute:
```

```
    def isPalindrome(self, s: str) -> bool:
```

```
        cleaned = [c.lower() for c in s if c.isalnum()]
```

```
        return cleaned == cleaned[::-1]
```

PHASE 3 — OPTIMIZE

"The brute force is already $O(n)$ time, but uses $O(n)$ space for the cleaned copy.

```
# We can do this in O(1) extra space using two pointers directly on the original
string.
#
# Place left at 0 and right at len(s)-1.
# Advance left past any non-alphanumeric characters.
# Retreat right past any non-alphanumeric characters.
# Compare s[left].lower() and s[right].lower().
# If they differ, return False. Otherwise move both pointers inward.
# If left meets or passes right, the string is a palindrome.
#
# The key benefit: we never build a second string – we skip junk characters on the
fly.
#
# Pseudocode:
# left, right = 0, len(s)-1
# while left < right:
# while left < right and not s[left].isalnum(): left += 1
# while left < right and not s[right].isalnum(): right -= 1
# if s[left].lower() != s[right].lower(): return False
# left += 1; right -= 1
# return True
#
# Time: O(n). Space: O(1). Does that make sense before I code it?"
```

PHASE 4 — CLEAN CODE

"Now I'll implement the optimized approach with meaningful names."

```
class _125_ValidPalindrome:
    def isPalindrome(self, s: str) -> bool:
        left = 0
        right = len(s) - 1
        while left < right:
            while left < right and not s[left].isalnum(): # skip non-alphanumeric
                left += 1
            while left < right and not s[right].isalnum(): # skip non-alphanumeric
                right -= 1
            if s[left].lower() != s[right].lower():
                return False
            left += 1
            right -= 1
        return True
```

PHASE 5 — TEST & DEBUG

"Let me trace through all three examples."

#

s="A man, a plan, a canal: Panama"

left=0 ('A'), right=29 ('a') – both alnum.

'A'.lower()='a' == 'a' ✓. left=1, right=28.

left=1 (' ') → skip → left=2 ('m'). right=28 ('m') – alnum.

'm'=='m' ✓. Continue inward... (all pairs match) → return True ✓


```

#
# s="race a car"
# left=0 ('r'), right=9 ('r'). 'r'=='r' ✓. left=1, right=8.
# left=1 ('a'), right=8 ('a'). 'a'=='a' ✓. left=2, right=7.
# left=2 ('c'), right=7 ('c'). 'c'=='c' ✓. left=3, right=6.
# left=3 ('e'), right=6 ('a'). 'e'!='a' → return False ✓
# (right=6 is 'a' in 'a car'; skip right=5 ' ' → right=4 'a')
# Wait – let me recount: "race a car"
# indices: r=0,a=1,c=2,e=3,' '=4,a=5,' '=6,c=7,a=8,r=9
# left=3 ('e'), right=9→skip none→'r'. 'e'!='r' → return False ✓
#
# s=" "
# left=0 (' ') → not alnum → left=1. Now left(1) >= right(0) → exit loop.
# return True ✓
#
# "No bugs. All three examples confirmed."

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(n) – each character is visited at most once by left or right pointer.
# Space O(1) – no auxiliary storage beyond two integer pointers.
#
# The brute force one-liner is perfectly valid for an interview too – it's readable
# and still O(n) time. The two-pointer version wins purely on space: O(1) vs O(n).
# If the interviewer asks about the follow-up, lead with the space trade-off.
#
# isalnum() handles both letters and digits in one call – worth naming explicitly
# so the interviewer knows you're not just checking isalpha().
#
# Subtle trap: the inner while loops must guard left < right to avoid crossing
# pointers – if you forget, you can read out-of-bounds when the string is all
# non-alphanumeric (like the single-space example).
#
# Given more time I'd ask: what if the string could contain unicode characters
# like accented letters (é, ñ)? isalnum() in Python handles unicode correctly
# by default, so our solution already works – worth calling out."

```

Two Pointers

□ #283 Move Zeroes

EAS

[Open on LeetCode](#)

Array · Two Pointers

Lists: Grind 169

Problem

Given an integer array `nums`, move all 0's to the end of it while maintaining the relative order of the non-zero elements.

Note that you must do this in-place without making a copy of the array.

Example 1:

```
Input: nums = [0,1,0,3,12]
Output: [1,3,12,0,0]
```

Example 2:

```
Input: nums = [0]
Output: [0]
```

Constraints:

- $1 \leq \text{nums.length} \leq 104$
- $-231 \leq \text{nums}[i] \leq 231 - 1$

Follow up: Could you minimize the total number of operations done?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Let me make sure I understand the problem correctly.
# We need to move all 0s to the end of the array while keeping the relative
# order of all non-zero elements. This must be done in-place – no copy. Right?"
#
# "A few quick questions:
# Do we need to return anything? No – modify in-place, return nothing.
# Can the array be all zeros? Yes – output should still be all zeros.
# Can it be all non-zeros? Yes – nothing moves.
# Can there be negative numbers? Yes, constraints go down to  $-2^{31}$ ."
#
# "Let me walk example 1:
# nums=[0,1,0,3,12] → [1,3,12,0,0].
# Non-zeros 1,3,12 keep their relative order and shift to the front.
# Zeros fill the tail. Got it."
```

PHASE 2 — BRUTE FORCE

```
# "The simplest approach: collect all non-zero elements into a new list,
# then write them back into nums, then fill the rest with zeros.
#
# non_zeros = [x for x in nums if x != 0]
# nums[:len(non_zeros)] = non_zeros
# nums[len(non_zeros):] = [0] * (len(nums) - len(non_zeros))
#
# Time:  $O(n)$ . Space:  $O(n)$  – we make a copy of the non-zeros.
# Should I optimize for space?"
class _283_MoveZeroes_Brute:
    def moveZeroes(self, nums: List[int]) -> None:
        non_zeros = [x for x in nums if x != 0]
        nums[:len(non_zeros)] = non_zeros
        nums[len(non_zeros):] = [0] * (len(nums) - len(non_zeros))
```

PHASE 3 — OPTIMIZE

```
# "The brute force uses  $O(n)$  extra space for the non-zero copy.
# We can do this in  $O(1)$  space with two pointers.
#
# Use a slow pointer 'place' that tracks where the next non-zero should go.
# Scan with a fast pointer 'scan'. Whenever nums[scan] is non-zero,
# swap it with nums[place] and advance place.
#
# This compacts all non-zeros to the front in their original order,
# and the swaps naturally push zeros toward the tail.
#
# Pseudocode:
# place = 0
```

```

# for scan in range(len(nums)):
# if nums[scan] != 0:
# nums[scan], nums[place] = nums[place], nums[scan]
# place += 1
#
# Time: O(n). Space: O(1). Does that make sense before I code it?"

# PHASE 4 — CLEAN CODE
# "Now I'll implement the optimized approach with meaningful names."
class _283_MoveZeroes:
    def moveZeroes(self, nums: List[int]) -> None:
        place = 0
        for scan in range(len(nums)):
            if nums[scan] != 0:
                nums[scan], nums[place] = nums[place], nums[scan]
                place += 1

# PHASE 5 — TEST & DEBUG
# "Let me trace through both examples."
#
# nums=[0,1,0,3,12]
# scan=0: nums[0]=0 → skip. place=0.
# scan=1: nums[1]=1 ≠ 0 → swap(1,0): [1,0,0,3,12]. place=1.
# scan=2: nums[2]=0 → skip. place=1.
# scan=3: nums[3]=3 ≠ 0 → swap(3,1): [1,3,0,0,12]. place=2.
# scan=4: nums[4]=12 ≠ 0 → swap(4,2): [1,3,12,0,0]. place=3.
# Result: [1,3,12,0,0] ✓
#
# nums=[0]
# scan=0: nums[0]=0 → skip.
# Result: [0] ✓
#
# nums=[1,2,3] (no zeros)
# Every element is non-zero → each swaps with itself. place advances each step.
# Result: [1,2,3] ✓ (self-swaps are fine – no change)
#
# "No bugs. Self-swaps when place==scan are harmless."

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(n) – single pass. Space O(1) – two integer pointers.
#
# Follow-up asks to minimize operations. The swap approach does a swap
# for every non-zero element, even if it's already in the right place.
# A minor optimization: only swap when place != scan to skip self-swaps.
# But in an interview the current version is clean enough.
#
# Alternative: instead of swapping, just overwrite:
# write non-zeros to the front with assignment (not swap), then zero-fill the tail.
# Same complexity but fewer writes – useful if writes are expensive.
#

```

```
# place = 0
# for x in nums:
# if x != 0: nums[place] = x; place += 1
# nums[place:] = [0] * (len(nums) - place)
#
# Given more time I'd ask: what if we also need to preserve the positions
# of the zeros (not just push them to the end)? That changes the problem
# to a stable partition – same two-pointer approach still works."
```

Two Pointers

□ ★ #408 Valid Word Abbreviation ★

EAS

[Open on LeetCode](#)

Two Pointers · String

Lists: ★ Premium | Premium 98

Problem

A string can be abbreviated by replacing any number of non-adjacent, non-empty substrings with their lengths. The lengths should not have leading zeros.

For example, a string such as "substitution" could be abbreviated as (but not limited to):

- "s10n" ("s ubstitutio n")
- "sub4u4" ("sub stit u tion")
- "12" ("substitution")
- "su3i1u2on" ("su bst i t u ti on")
- "substitution" (no substrings replaced)

The following are not valid abbreviations:

- "s55n" ("s ubsti tutio n", the replaced substrings are adjacent)
- "s010n" (has leading zeros)

- "s0ubstitution" (replaces an empty substring)

Given a string word and an abbreviation abbr, return whether the string matches the given abbreviation.

A substring is a contiguous non-empty sequence of characters within a string.

Example 1:

```
Input: word = "internationalization", abbr = "i12iz4n"
Output: true
Explanation: The word "internationalization" can be abbreviated as "i12iz4n"
("i nternational iz atio n").
```

Example 2:

```
Input: word = "apple", abbr = "a2e"
Output: false
Explanation: The word "apple" cannot be abbreviated as "a2e".
```

Constraints:

- $1 \leq \text{word.length} \leq 20$
- word consists of only lowercase English letters.
- $1 \leq \text{abbr.length} \leq 10$
- abbr consists of lowercase English letters and digits.
- All the integers in abbr will fit in a 32-bit integer.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Let me make sure I understand the problem correctly.

An abbreviation replaces one or more non-adjacent non-empty substrings with their lengths.

Given a word and an abbreviation, return true if the abbreviation is valid for the word.

Numbers in the abbreviation represent how many consecutive characters to skip. Right?"

#

"A few quick questions:

Can there be leading zeros in the number part? No – that's explicitly invalid.

So '01' is not valid even though it could theoretically mean skip 1.

Can the abbreviation be the full word with no numbers? Yes – that's valid.

Can it be just a single number like '12'? Yes – if the word has exactly 12 chars."

#

"Let me walk example 1:

word='internationalization', abbr='i12iz4n'.

'i' matches 'i'. '12' skips 12 chars. 'i' matches 'i'. 'z' matches 'z'.

'4' skips 4 chars. 'n' matches 'n'. We reach end of both → true ✓

Example 2: word='apple', abbr='a2e'. 'a' matches. '2' skips 'pp'. 'e'→word[3]='l'

≠ 'e' → false ✓"

PHASE 2 — BRUTE FORCE (also optimal)

"Let me start with brute force to lock in the logic.

Expand the abbreviation letter-by-letter against word using two pointers.

When we hit a digit in abbr, consume that many characters from word.

Any mismatch ? false; both exhausted together ? true.

Time: O(n). Space: O(1).

This is already optimal for the constraints – complexity stated clearly."

PHASE 3 — OPTIMIZE

"Use pointer i into word and j into abbr.

When abbr[j] is a digit: check for leading zero (abbr[j]=='0' and num==0 → invalid).

Accumulate the number: num = num*10 + digit.

When abbr[j] is a letter: first advance i by num (the accumulated skip), reset num to 0.

Then compare word[i] with abbr[j] – if they differ, return False. Advance both i and j.

After the loop: we're valid only if i + num == len(word) AND j == len(abbr).

That handles a trailing number at the end of abbr.

#

Pseudocode:

i, j, num = 0, 0, 0

while i < len(word) and j < len(abbr):

if abbr[j].isdigit():

if abbr[j]=='0' and num==0: return False # leading zero


```
# num = num*10 + int(abbr[j])
# else:
# i += num; num = 0
# if i >= len(word) or word[i] != abbr[j]: return False
# i += 1
# j += 1
# return i + num == len(word) and j == len(abbr)
#
# Time: O(n + m) where n=len(word), m=len(abbr). Space: O(1)."
```

PHASE 4 — CLEAN CODE

"Now I'll implement with meaningful names."

```
class _408_ValidWordAbbreviation:
    def validWordAbbreviation(self, word: str, abbr: str) -> bool:
        word_ptr = 0
        abbr_ptr = 0
        skip = 0
        while word_ptr < len(word) and abbr_ptr < len(abbr):
            if abbr[abbr_ptr].isdigit():
                if abbr[abbr_ptr] == '0' and skip == 0: # leading zero
                    return False
                skip = skip * 10 + int(abbr[abbr_ptr])
            else:
                word_ptr += skip
                skip = 0
                if word_ptr >= len(word) or word[word_ptr] != abbr[abbr_ptr]:
                    return False
                word_ptr += 1
                abbr_ptr += 1
        return word_ptr + skip == len(word) and abbr_ptr == len(abbr)
```

PHASE 5 — TEST & DEBUG

"Let me trace both examples."

```
#
# word='internationalization'(len=20), abbr='i12iz4n'
# j=0: 'i' → skip=0, advance word_ptr 0, word[0]='i'=='i' ✓. word_ptr=1.
# j=1: '1' → skip=1.
# j=2: '2' → skip=12.
# j=3: 'i' → word_ptr+=12=13. word[13]='i'=='i' ✓. word_ptr=14.
# j=4: 'z' → skip=0, word[14]='z'=='z' ✓. word_ptr=15.
# j=5: '4' → skip=4.
# j=6: 'n' → word_ptr+=4=19. word[19]='n'=='n' ✓. word_ptr=20.
# Loop ends (word_ptr=20=len(word)). word_ptr+skip=20+0=20==20, abbr_ptr=7==7. True
✓
#
# word='apple'(len=5), abbr='a2e'
# j=0: 'a' → word[0]='a'=='a' ✓. word_ptr=1.
# j=1: '2' → skip=2.
# j=2: 'e' → word_ptr+=2=3. word[3]='l' ≠ 'e' → return False ✓
```

```
#
# Leading zero: abbr='a01e' → j=1: '0' and skip==0 → return False ✓
#
# "No bugs. Leading zero check and trailing number handled correctly."

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(n + m)$  – each character in word and abbr is visited at most once.
# Space  $O(1)$  – three integer pointers.
#
# The two trickiest cases to get right:
# 1. Leading zeros – must check abbr[j]!='0' AND accumulated num==0 to distinguish
# '0' (leading zero, invalid) from '10', '20' etc. (valid multi-digit numbers ending
# in 0).
# 2. Trailing number – if abbr ends with digits (e.g. abbr='a4'), the loop exits
# when abbr_ptr reaches end but skip is still non-zero.
# The final check word_ptr + skip == len(word) handles this correctly.
#
# Given more time I'd ask: what if we want to generate ALL valid abbreviations
# for a word? That's a backtracking problem – enumerate all subsets of positions
# to replace with their lengths."
```

Two Pointers

□ #977 Squares of a Sorted Array

EAS

[Open on LeetCode](#)

Array · Two Pointers · Sorting

Lists: Grind 169

Problem

Given an integer array `nums` sorted in non-decreasing order, return an array of the squares of each number sorted in non-decreasing order.

Example 1:

```
Input: nums = [-4,-1,0,3,10]
Output: [0,1,9,16,100]
Explanation: After squaring, the array becomes [16,1,0,9,100].
After sorting, it becomes [0,1,9,16,100].
```

Example 2:

```
Input: nums = [-7,-3,2,3,11]
Output: [4,9,9,49,121]
```

Constraints:

- $1 \leq \text{nums.length} \leq 104$
- $-104 \leq \text{nums}[i] \leq 104$
- `nums` is sorted in non-decreasing order.

Follow up: Squaring each element and sorting the new array is very trivial, could you find an $O(n)$ solution using a different approach?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Let me make sure I understand the problem correctly.
# The input array is sorted in non-decreasing order and can contain negatives.
# We need to return a new array of the squares of each element, also sorted. Right?"
#
# "A few quick questions:
# Can the array contain negative numbers? Yes – that's the whole challenge.
# Should I return a new array or modify in-place? Return a new array.
# Can the array have a single element? Yes – just return [element2].
# Can all elements be negative? Yes – e.g. [-3,-2,-1] → [1,4,9]."
#
# "Let me walk example 1:
# nums=[-4,-1,0,3,10]. Squares: [16,1,0,9,100]. Sorted: [0,1,9,16,100].
# The key observation: the largest squares come from the extremes –
# either the most negative or the most positive end. Got it."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Square every element, sort the resulting array ascending, return it.
# Two-pass: fill squares  $O(n)$ , then sort  $O(n \log n)$ .
# Correct given unsorted input, but the array is already sorted – we can merge from both ends.
# Time:  $O(n \log n)$ . Space:  $O(n)$  for the squared copy.
# Should I go ahead and optimize?"
```

```
class _977_SquaresSortedArray_Brute:
    def sortedSquares(self, nums: List[int]) -> List[int]:
        return sorted(x * x for x in nums)
```

PHASE 3 — OPTIMIZE

```
# "The brute force ignores the fact that the input is already sorted.
# Key insight: in a sorted array, the largest absolute values are always at one
# of the two ends – either the most negative (left) or the most positive (right).
# So the largest square is always  $\max(\text{nums}[\text{left}]^2, \text{nums}[\text{right}]^2)$ .
#
# Use two pointers: left starts at 0, right at  $n-1$ .
# Fill the result array from the back (largest to smallest):
# at each step, compare  $\text{abs}(\text{nums}[\text{left}])$  and  $\text{abs}(\text{nums}[\text{right}])$ .
# Place the larger square at  $\text{result}[k]$  and move that pointer inward.
```

```
#
# Pseudocode:
# result = [0] * n; k = n-1; left, right = 0, n-1
# while k >= 0:
#   if abs(nums[left]) > abs(nums[right]):
#     result[k] = nums[left]**2; left += 1
#   else:
#     result[k] = nums[right]**2; right -= 1
#   k -= 1
# return result
#
# Time: O(n). Space: O(n) for the output – unavoidable since we return a new array."
```

PHASE 4 — CLEAN CODE

"Now I'll implement the optimized approach with meaningful names."

```
class _977_SquaresSortedArray:
    def sortedSquares(self, nums: List[int]) -> List[int]:
        n = len(nums)
        result = [0] * n
        fill = n - 1
        left = 0
        right = n - 1
        while fill >= 0:
            if abs(nums[left]) > abs(nums[right]):
                result[fill] = nums[left] * nums[left]
                left += 1
            else:
                result[fill] = nums[right] * nums[right]
                right -= 1
            fill -= 1
        return result
```

PHASE 5 — TEST & DEBUG

"Let me trace both examples."

```
#
# nums=[-4,-1,0,3,10], n=5
# fill=4: abs(-4)=4 < abs(10)=10 → result[4]=100, right=3. fill=3.
# fill=3: abs(-4)=4 > abs(3)=3 → result[3]=16, left=1. fill=2.
# fill=2: abs(-1)=1 < abs(3)=3 → result[2]=9, right=2. fill=1.
# fill=1: abs(-1)=1 > abs(0)=0 → result[1]=1, left=2. fill=0.
# fill=0: left==right==2, abs(0)=0 → result[0]=0, right=1. fill=-1.
# result=[0,1,9,16,100] ✓
#
# nums=[-7,-3,2,3,11], n=5
# fill=4: abs(-7)=7 < abs(11)=11 → result[4]=121, right=3. fill=3.
# fill=3: abs(-7)=7 > abs(3)=3 → result[3]=49, left=1. fill=2.
# fill=2: abs(-3)=3 == abs(3)=3 → take right: result[2]=9, right=2. fill=1.
# fill=1: abs(-3)=3 > abs(2)=2 → result[1]=9, left=2. fill=0.
# fill=0: left==right==2 → result[0]=4. fill=-1.
```

```
# result=[4,9,9,49,121] ✓
```

```
#
```

```
# "No bugs. Tie goes to right which is fine – both are equal."
```

```
# PHASE 6 — COMPLEXITY & FOLLOW-UP
```

```
# "Time  $O(n)$  – single pass from both ends meeting in the middle.
```

```
# Space  $O(n)$  – required for the output; we can't avoid this since we return a new array.
```

```
#
```

```
# This is the same fill-from-back pattern as Merge Sorted Array (#88) –
```

```
# worth naming that explicitly to show you recognise the pattern family.
```

```
#
```

```
# Trade-off: the brute force  $O(n \log n)$  is two lines and perfectly acceptable
```

```
# in most interviews. The  $O(n)$  two-pointer version is the follow-up answer
```

```
# the problem explicitly asks for.
```

```
#
```

```
# Given more time I'd ask: what if we needed to do this in-place?
```

```
# Squaring in-place destroys order for negative numbers, so we'd need to
```

```
# square + reverse the negative half, then merge the two sorted halves
```

```
# in-place – doable but significantly more complex."
```

Sorting

Round 1 · High · Easy · 6 questions

Sorting

□ #169 Majority Element

EAS

[Open on LeetCode](#)

Array · Hash Table · Divide and Conquer ·
Sorting · Counting
Lists: Grind 169

Problem

Given an array `nums` of size `n`, return the majority element.

The majority element is the element that appears more than $n / 2$ times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`

- $-109 \leq \text{nums}[i] \leq 109$
- The input is generated such that a majority element will exist in the array.

Follow-up: Could you solve the problem in linear time and in $O(1)$ space?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Let me make sure I understand the problem correctly.
# The majority element is the one that appears MORE THAN  $n/2$  times.
# We're guaranteed it always exists. We return the element, not its count. Right?"
#
# "A few quick questions:
# Is there always exactly one majority element? Yes – anything above  $n/2$  means
# at most one element can qualify.
# Can  $n$  be 1? Yes – that single element is trivially the majority.
# Can values be negative? Yes, constraints allow down to  $-10^9$ .
# The follow-up asks for linear time and  $O(1)$  space – I'll aim for that."
#
# "Let me walk the examples:
# nums=[3,2,3]: 3 appears 2 times,  $n/2=1$ .  $2 > 1 \rightarrow 3$  is majority. ✓
# nums=[2,2,1,1,1,2,2]: 2 appears 4 times,  $n/2=3.5$ .  $4 > 3.5 \rightarrow 2$ . ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Count frequency of every element with a hash map, then scan counts for any value >
 $n/2$ .
# Correct because the majority element must appear more than half the time.
# Time:  $O(n)$ . Space:  $O(n)$  for counts.
# Should I go ahead and optimize?"
```

```
class _169_MajorityElement_Brute:
    def majorityElement(self, nums: List[int]) -> int:
        return Counter(nums).most_common(1)[0][0]
```

PHASE 3 — OPTIMIZE

```
# "To get  $O(1)$  space, I'll use Boyer-Moore Voting Algorithm.
# The core insight: if we cancel every occurrence of the majority element
# with a different element, the majority element will always survive
# because it has more than  $n/2$  appearances – more than everything else combined.
#
# Keep a candidate and a count:
```

```

# When count == 0, adopt the current element as the new candidate.
# If current element matches candidate, increment count.
# If it doesn't match, decrement count (a cancellation).
# The candidate at the end is guaranteed to be the majority element.
#
# Pseudocode:
# candidate, count = None, 0
# for x in nums:
# if count == 0: candidate = x
# count += (1 if x == candidate else -1)
# return candidate
#
# Time: O(n) – single pass. Space: O(1) – two variables only."

# PHASE 4 — CLEAN CODE
# "Now I'll implement Boyer-Moore with meaningful names."
class _169_MajorityElement:
    def majorityElement(self, nums: List[int]) -> int:
        candidate = None
        count = 0
        for num in nums:
            if count == 0:
                candidate = num
            count += 1 if num == candidate else -1
        return candidate

# PHASE 5 — TEST & DEBUG
# "Let me trace through both examples."
#
# nums=[3,2,3]
# num=3: count=0 → candidate=3. count=1.
# num=2: count=1, 2≠3 → count=0.
# num=3: count=0 → candidate=3. count=1.
# return 3 ✓
#
# nums=[2,2,1,1,1,2,2]
# num=2: count=0 → candidate=2. count=1.
# num=2: match → count=2.
# num=1: no match → count=1.
# num=1: no match → count=0.
# num=1: count=0 → candidate=1. count=1.
# num=2: no match → count=0.
# num=2: count=0 → candidate=2. count=1.
# return 2 ✓
#
# Single element – nums=[7]
# count=0 → candidate=7. count=1. return 7 ✓
#
# "No bugs. Even when the candidate flips mid-array, the majority element

```

reclaims it before the end."

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$ – single pass. Space $O(1)$ – only two variables.

#

Boyer-Moore is the optimal solution here and worth naming by name –

it shows algorithm knowledge beyond just 'I used two variables'.

#

Correctness argument worth stating: the majority element appears more than $n/2$ times.

Every time it gets cancelled by a different element, it loses one vote and so does the other element. Since it has more votes than all others combined, it will always

have votes remaining when everyone else is cancelled out.

#

Alternative $O(n)$ time $O(1)$ space approach: sort and return `nums[n//2]`.

Since majority appears $> n/2$ times, it must occupy the middle index.

But sorting is $O(n \log n)$ – worse than Boyer-Moore.

#

If the problem didn't guarantee a majority exists, Boyer-Moore would need a second pass to verify the candidate actually appears $> n/2$ times.

#

Given more time I'd ask: what if we need to find all elements appearing more than $n/3$ times? Boyer-Moore generalises – maintain two candidates and two counts. That's a well-known variant worth mentioning."

Sorting

□ #217 Contains Duplicate

EAS

[Open on LeetCode](#)

Array · Hash Table · Sorting

Lists: Grind 169

Problem

Given an integer array `nums`, return `true` if any value appears at least twice in the array, and return `false` if every element is distinct.

Example 1:

Input: `nums = [1,2,3,1]`

Output: `true`

Explanation:

The element 1 occurs at the indices 0 and 3.

Example 2:

Input: `nums = [1,2,3,4]`

Output: `false`

Explanation:

All elements are distinct.

Example 3:

Input: `nums = [1,1,1,3,3,4,3,2,4,2]`

Output: true

Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $-109 \leq \text{nums}[i] \leq 109$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Let me make sure I understand the problem correctly.
# Return true if any value appears at least twice, false if every element is
distinct.
# We're not asked which value or how many times – just a boolean. Right?"
#
# "A few quick questions:
# Can the array be empty? Constraints say length >= 1, so no.
# Can there be negative numbers? Yes, down to -10^9.
# Is order relevant? No – we just care about value frequency."
#
# "Let me walk the examples:
# [1,2,3,1]: 1 appears twice → true ✓
# [1,2,3,4]: all distinct → false ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Check every pair (i, j) with j > i; if nums[i] == nums[j], return true
immediately.
# If no duplicate pair exists after all checks, return false.
# Simple and correct, but nested loops blow up on large arrays.
# Time: O(n^2). Space: O(1).
# Should I go ahead and optimize?"
```

```
class _217_ContainsDuplicate_Brute:
    def containsDuplicate(self, nums: List[int]) -> bool:
        for i in range(len(nums)):
            for j in range(i + 1, len(nums)):
                if nums[i] == nums[j]:
                    return True
        return False
```

PHASE 3 — OPTIMIZE

```
# "We're repeatedly asking: have I seen this value before?
# A hash set answers that in O(1). Walk the array once;
# if the current number is already in the set, return True immediately.
# Otherwise add it and continue."
```

```
#
# Time: O(n). Space: O(n).
# Alternative: sort and check adjacent pairs – O(n log n), O(1) extra space.
# Hash set is faster; sort is space-optimal."
```

PHASE 4 — CLEAN CODE

```
class _217_ContainsDuplicate:
    def containsDuplicate(self, nums: List[int]) -> bool:
        seen = set()
        for num in nums:
            if num in seen:
                return True
            seen.add(num)
        return False
```

PHASE 5 — TEST & DEBUG

```
# nums=[1,2,3,1]: seen grows to {1,2,3}, then 1 hits → True ✓
# nums=[1,2,3,4]: seen={1,2,3,4}, loop ends → False ✓
# nums=[1]: seen={1}, loop ends → False ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(n) for the set.
# One-liner alternative: return len(nums) != len(set(nums)) – same complexity,
# builds the full set before checking. The explicit loop returns early on first
duplicate.
# For adversarial inputs (duplicate at the end) they're the same; for duplicates
early,
# the loop wins. In an interview I'd use the explicit loop and mention the
one-liner.
# Given more time I'd ask: what if the array is sorted? Check adjacent pairs – O(n)
time, O(1) space."
```

Sorting

□ #242 Valid Anagram

EAS

[Open on LeetCode](#)

Hash Table · String · Sorting

Lists: Grind 169

Problem

Given two strings *s* and *t*, return true if *t* is an anagram of *s*, and false otherwise.

Example 1:

Input: *s* = "anagram", *t* = "nagaram"

Output: true

Example 2:

Input: *s* = "rat", *t* = "car"

Output: false

Constraints:

- $1 \leq s.length, t.length \leq 5 * 10^4$
- *s* and *t* consist of lowercase English letters.

Follow up: What if the inputs contain Unicode characters? How would you adapt your solution to such a case?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Let me make sure I understand the problem correctly.
# t is an anagram of s if t contains exactly the same characters
# with the same frequencies, just in a different order. Right?
# Both strings consist of only lowercase English letters – so 26 possible chars."
#
# "Quick checks:
# If lengths differ they can't be anagrams – I'll short-circuit on that.
# Follow-up asks about Unicode – I'll address that in Phase 6."
#
# "'anagram'/'nagaram': same chars, same counts → true ✓
# 'rat'/'car': r,a,t vs c,a,r – 't' vs 'c' mismatch → false ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Sort both strings and compare character by character.
# Equal sorted forms iff anagrams.
# Simple  $O(n \log n)$ ; counting frequencies in one pass is  $O(n)$ .
# Time:  $O(n \log n)$ . Space:  $O(n)$  for sorted copies.
# Should I go ahead and optimize?"
```

```
class _242_ValidAnagram_Brute:
    def isAnagram(self, s: str, t: str) -> bool:
        return sorted(s) == sorted(t)
```

PHASE 3 — OPTIMIZE

```
# "Instead of sorting, count character frequencies.
# Since only lowercase letters exist, a 26-slot array suffices –  $O(1)$  space.
# Increment for each char in s, decrement for each char in t.
# If all slots are zero at the end, they're anagrams.
#
# Time:  $O(n)$ . Space:  $O(1)$  – fixed 26-slot array."
```

PHASE 4 — CLEAN CODE

```
class _242_ValidAnagram:
    def isAnagram(self, s: str, t: str) -> bool:
        if len(s) != len(t):
            return False
        freq = [0] * 26
        for c in s:
            freq[ord(c) - ord('a')] += 1
        for c in t:
            freq[ord(c) - ord('a')] -= 1
        return all(x == 0 for x in freq)
```

PHASE 5 — TEST & DEBUG

```
# s='anagram', t='nagaram': lengths equal.
# After s: a=3,n=1,g=1,r=1,m=1. After t: all zeroed. → True ✓
```



```
# s='rat', t='car': r+1,a+1,t+1 then c-1,a-1,r-1 → t=1,c=-1 ≠ 0 → False ✓  
# s='a', t='ab': len mismatch → False immediately ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(1)$ – 26-slot array is a constant.

Follow-up: Unicode characters → the 26-slot array breaks.

Use a Counter (hash map) instead – $O(n)$ time, $O(k)$ space where k is distinct chars.

return Counter(s) == Counter(t)

Handles any alphabet. In an interview, mention this unprompted."

Sorting

□ ★ #252 Meeting Rooms ★

EAS

[Open on LeetCode](#)

Array · Sorting

Lists: ★ Premium | Grind 169 | Premium 98

Problem

Given an array of meeting time intervals where $\text{intervals}[i] = [\text{start}_i, \text{end}_i]$, determine if a person could attend all meetings.

Example 1:

```
Input: intervals = [[0,30],[5,10],[15,20]]  
Output: false
```

Example 2:

```
Input: intervals = [[7,10],[2,4]]  
Output: true
```

Constraints:

- $0 \leq \text{intervals.length} \leq 104$
- $\text{intervals}[i].\text{length} == 2$
- $0 \leq \text{start}_i < \text{end}_i \leq 106$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"We have a list of meetings [start, end]. We need to check if one person

```

# can attend ALL of them – meaning no two meetings overlap.
# The problem says meetings ending at t and starting at t do NOT overlap. Right?"
#
# "Quick questions:
# Can intervals be empty? Yes – 0 meetings, trivially true.
# Are start and end guaranteed start < end? Yes, constraints say 0 <= start < end."
#
# "[[0,30],[5,10],[15,20]]: 0–30 overlaps with 5–10 → false ✓
# [[7,10],[2,4]]: 2–4 ends before 7–10 starts → true ✓"

# PHASE 2 — BRUTE FORCE
# "Let me start with brute force to lock in the logic.
# Compare every pair of meetings [start_i,end_i] and [start_j,end_j].
# Overlap exists if start_i < end_j and start_j < end_i.
# O(n^2) pairs; sorting by start time enables one linear scan.
# Time: O(n^2). Space: O(1).
# Should I go ahead and optimize?"

class _252_MeetingRooms_Brute:
    def canAttendMeetings(self, intervals: List[List[int]]) -> bool:
        for i in range(len(intervals)):
            for j in range(i + 1, len(intervals)):
                a, b = intervals[i]
                c, d = intervals[j]
                if a < d and c < b:
                    return False
            return True

# PHASE 3 — OPTIMIZE
# "Sort by start time. Then a single pass is enough:
# if any meeting ends after the next one starts, they overlap.
# We only need to check consecutive pairs after sorting – O(n log n)."

# PHASE 4 — CLEAN CODE
class _252_MeetingRooms:
    def canAttendMeetings(self, intervals: List[List[int]]) -> bool:
        intervals.sort()
        for prev, curr in pairwise(intervals):
            if prev[1] > curr[0]:
                return False
        return True

# PHASE 5 — TEST & DEBUG
# [[0,30],[5,10],[15,20]] → sorted: same. prev=[0,30],curr=[5,10]: 30>5 → False ✓
# [[7,10],[2,4]] → sorted: [[2,4],[7,10]]. prev=[2,4],curr=[7,10]: 4>7? No → True ✓
# []: pairwise on empty → no iterations → True ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(n log n) for sort, O(n) for the pass. Space O(1) extra.
# pairwise is Python 3.10+; fallback: for i in range(1, len(intervals)).
# Given more time I'd ask: what's the minimum number of rooms needed?

```

That's Meeting Rooms II  (#253) – use a min-heap of end times, $O(n \log n)$."

Sorting

□ ★ #1086 Largest Unique Number ★

EAS

[Open on LeetCode](#)

Array · Hash Table · Sorting

Lists: ★ Premium | Premium 98

Problem

Given an integer array `nums`, return the largest integer that only occurs once. If no integer occurs once, return `-1`.

Example 1:

Input: `nums = [5,7,3,9,4,9,8,3,1]`

Output: `8`

Explanation: The maximum integer in the array is 9 but it is repeated. The number 8 occurs only once, so it is the answer.

Example 2:

Input: `nums = [9,9,8,8]`

Output: `-1`

Explanation: There is no number that occurs only once.

Constraints:

- $1 \leq \text{nums.length} \leq 2000$
- $0 \leq \text{nums}[i] \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Return the largest integer that appears exactly once. If none, return -1. Right?
# Values are 0-1000, array length up to 2000."
#
# "[5,7,3,9,4,9,8,3,1]: 9 appears twice, 8 appears once → 8 ✓
# [9,9,8,8]: no unique element → -1 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Sort the array descending. Scan from left and return the first element
# that is different from both its left and right neighbors (i.e. appears exactly
# once).
# Time: O(n log n). Space: O(1) in-place sort. Should I go ahead and optimize?"
```

```
class _1086_LargestUniqueNumber_Brute:
    def largestUniqueNumber(self, nums: List[int]) -> int:
        nums.sort(reverse=True)
        for i in range(len(nums)):
            left_same = (i > 0 and nums[i] == nums[i - 1])
            right_same = (i < len(nums) - 1 and nums[i] == nums[i + 1])
            if not left_same and not right_same:
                return nums[i]
        return -1
```

PHASE 3 — OPTIMIZE

```
# "Count frequencies with Counter. Walk unique keys, track max where count==1.
# O(n) time, O(n) space."
```

PHASE 4 — CLEAN CODE

```
class _1086_LargestUniqueNumber:
    def largestUniqueNumber(self, nums: List[int]) -> int:
        freq = Counter(nums)
        result = -1
        for num, count in freq.items():
            if count == 1 and num > result:
                result = num
        return result
```

PHASE 5 — TEST & DEBUG

```
# [5,7,3,9,4,9,8,3,1]: freq={5:1,7:1,3:2,9:2,4:1,8:1,1:1}
# Unique nums: 5,7,4,8,1. Max = 8 ✓
# [9,9,8,8]: no count==1 entries → result stays -1 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(n).
# Alternative: sort descending, scan until you find a non-duplicate – O(n log n).
# Counter approach is cleaner. Given more time I'd ask: what if we want the k
# largest
# unique numbers? Collect all uniques, sort descending, take first k."
```

Sorting

□ #1122 Relative Sort Array

EAS

[Open on LeetCode](#)

Array · Hash Table · Sorting · Counting Sort
Lists: Denny Zhang

Problem

Given two arrays `arr1` and `arr2`, the elements of `arr2` are distinct, and all elements in `arr2` are also in `arr1`.

Sort the elements of `arr1` such that the relative ordering of items in `arr1` are the same as in `arr2`. Elements that do not appear in `arr2` should be placed at the end of `arr1` in ascending order.

Example 1:

```
Input: arr1 = [2,3,1,3,2,4,6,7,9,2,19], arr2 = [2,1,4,3,9,6]  
Output: [2,2,2,1,4,3,3,9,6,7,19]
```

Example 2:

```
Input: arr1 = [28,6,22,8,44,17], arr2 = [22,28,8,6]  
Output: [22,28,8,6,17,44]
```

Constraints:

- $1 \leq \text{arr1.length}, \text{arr2.length} \leq 1000$
- $0 \leq \text{arr1}[i], \text{arr2}[i] \leq 1000$

- All the elements of arr2 are distinct.
- Each arr2[i] is in arr1.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Sort arr1 so elements appear in the order defined by arr2.
# Elements in arr1 not present in arr2 go at the end, sorted ascending. Right?
# arr2 elements are distinct; every arr2 element is guaranteed to exist in arr1."
#
# "arr1=[2,3,1,3,2,4,6,7,9,2,19], arr2=[2,1,4,3,9,6]
# → [2,2,2,1,4,3,3,9,6,7,19]: follow arr2 order for those nums, then 7,19 ascending
✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Build a rank map from arr2: each element maps to its index (0..len(arr2)-1).
# Elements not in arr2 get rank len(arr2) and are sorted by value as a tiebreaker.
# Then sort arr1 using that composite key.
# Time: O(n log n). Space: O(n) for the rank map. Should I go ahead and optimize?"
class _1122_RelativeSortArray_Brute:
    def relativeSortArray(self, arr1: List[int], arr2: List[int]) -> List[int]:
        rank = {val: i for i, val in enumerate(arr2)}
        return sorted(arr1, key=lambda x: (rank[x], 0) if x in rank else (len(arr2),
x))
```

PHASE 3 — OPTIMIZE

```
# "Count arr1 frequencies with Counter.
# For each num in arr2, extend result with [num]*freq[num], remove from counter.
# Then sort remaining counter items and extend.
# Time: O(n + k log k) where k = elements not in arr2. Space: O(n)."
```

PHASE 4 — CLEAN CODE

```
class _1122_RelativeSortArray:
    def relativeSortArray(self, arr1: List[int], arr2: List[int]) -> List[int]:
        freq = Counter(arr1)
        result = []
        for num in arr2:
            result.extend([num] * freq[num])
            del freq[num]
        for num, count in sorted(freq.items()):
            result.extend([num] * count)
        return result
```

PHASE 5 — TEST & DEBUG

```
# arr1=[2,3,1,3,2,4,6,7,9,2,19], arr2=[2,1,4,3,9,6]
```



```
# freq={2:3,3:2,1:1,4:1,6:1,7:1,9:1,19:1}
# arr2 pass: [2,2,2,1,4,3,3,9,6]. remaining: {7:1,19:1}
# sorted remaining → [7,19]. result=[2,2,2,1,4,3,3,9,6,7,19] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n + k \log k)$  where  $k$  is the count of elements not in arr2 (at most  $n$ ).
# Overall  $O(n \log n)$ . Space  $O(n)$ .
# Given more time I'd ask: what if arr2 has elements not in arr1?
# The problem guarantees that won't happen, but we'd skip them if it did."
```

Binary Search

Round 1 · High · Easy · 5 questions

Binary Search

□ #35 Search Insert Position

EAS

[Open on LeetCode](#)

Array · Binary Search

Lists: Denny Zhang

Problem

Given a sorted array of distinct integers and a target value, return the index if the target is found. If not, return the index where it would be if it were inserted in order.

You must write an algorithm with $O(\log n)$ runtime complexity.

Example 1:

Input: `nums = [1,3,5,6]`, `target = 5`
Output: 2

Example 2:

Input: `nums = [1,3,5,6]`, `target = 2`
Output: 1

Example 3:

Input: `nums = [1,3,5,6]`, `target = 7`
Output: 4

Constraints:

- $1 \leq \text{nums.length} \leq 10^4$

- $-104 \leq \text{nums}[i] \leq 104$
- `nums` contains distinct values sorted in ascending order.
- $-104 \leq \text{target} \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Sorted array of distinct integers. Return the index of target if found,
# otherwise return the index where it would be inserted to keep order. Right?
# Must run in  $O(\log n)$  – so binary search is required."
#
# "[1,3,5,6], target=5 → index 2 (found) ✓
# [1,3,5,6], target=2 → index 1 (between 1 and 3) ✓
# [1,3,5,6], target=7 → index 4 (after all) ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Linear scan from left: return the first index where nums[i] >= target.
# If we exhaust the array without finding such an index, target goes at the end.
# Time:  $O(n)$ . Space:  $O(1)$ . Should I go ahead and optimize?"
```

```
class _35_SearchInsertPosition_Brute:
    def searchInsert(self, nums: List[int], target: int) -> int:
        for i in range(len(nums)):
            if nums[i] >= target:
                return i
        return len(nums)
```

PHASE 3 — OPTIMIZE

```
# "Standard binary search. The key insight: when the loop ends (left > right),
# 'left' always points to the correct insertion position.
# When target is found, return mid immediately.
# Otherwise left is where target belongs."
```

PHASE 4 — CLEAN CODE

```
class _35_SearchInsertPosition:
    def searchInsert(self, nums: List[int], target: int) -> int:
        left, right = 0, len(nums) - 1
        while left <= right:
            mid = (left + right) // 2
            if nums[mid] == target:
                return mid
            elif target < nums[mid]:
                right = mid - 1
```

```
    else:
        left = mid + 1
    return left
```

PHASE 5 — TEST & DEBUG

```
# [1,3,5,6], target=5: mid=2→5==5 → return 2 ✓
# [1,3,5,6], target=2: mid=2→5>2→right=1. mid=0→1<2→left=1. mid=1→3>2→right=0.
# left=1>right=0 → return left=1 ✓
# [1,3,5,6], target=7: left keeps advancing → return 4 ✓
# [1,3,5,6], target=0: right keeps retreating → return 0 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(\log n)$ . Space  $O(1)$ .
# The 'return left' at the end is the core insight – left always converges
# to the leftmost position where target can fit. This same pattern appears
# in lower_bound in C++ and bisect_left in Python.
# Given more time I'd ask: what if the array has duplicates?
# We'd need to decide: first occurrence or last? That's bisect_left vs
bisect_right."
```

Binary Search

□ #69 Sqrt(x)

EAS

[Open on LeetCode](#)

Math · Binary Search

Lists: Denny Zhang

Problem

Given a non-negative integer x , return the square root of x rounded down to the nearest integer. The returned integer should be non-negative as well.

You must not use any built-in exponent function or operator.

- For example, do not use `pow(x, 0.5)` in `c++` or `x ** 0.5` in `python`.

Example 1:

Input: $x = 4$
Output: 2
Explanation: The square root of 4 is 2, so we return 2.

Example 2:

Input: $x = 8$
Output: 2
Explanation: The square root of 8 is 2.82842..., and since we round it down to the nearest integer, 2 is returned.

Constraints:

● $0 \leq x \leq 2^{31} - 1$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Return floor(sqrt(x)) without using built-in sqrt or ** 0.5. Right?
# x can be 0 - sqrt(0)=0. x can be up to 2^31-1."
#
# "x=4 → 2. x=8 → 2 (floor of 2.828). x=0 → 0. x=1 → 1."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Linear scan m from 0 upward: keep incrementing while m*m <= x.
# The moment (m+1)*(m+1) > x, we know m is the floor sqrt.
# Time: O(sqrt(x)). Space: O(1). Should I go ahead and optimize?"
```

```
class _69_Sqrt_Brute:
    def mySqrt(self, x: int) -> int:
        m = 0
        while (m + 1) * (m + 1) <= x:
            m += 1
        return m
```

PHASE 3 — OPTIMIZE

```
# "Binary search on the answer space [0, x].
# Find the largest m such that m*m <= x.
# When m*m < x, record m as a candidate and search right.
# When m*m > x, search left. When m*m == x, return immediately.
# Time: O(log x). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _69_Sqrt:
    def mySqrt(self, x: int) -> int:
        if x < 2:
            return x
        left, right = 1, x // 2 # sqrt(x) <= x//2 for x >= 4
        result = 0
        while left <= right:
            mid = left + (right - left) // 2
            if mid * mid == x:
                return mid
            elif mid * mid < x:
                result = mid
                left = mid + 1
            else:
                right = mid - 1
        return result
```

PHASE 5 — TEST & DEBUG

```
# x=4: left=1,right=2. mid=1→1<4→result=1,left=2. mid=2→4==4→return 2 ✓
# x=8: left=1,right=4. mid=2→4<8→result=2,left=3. mid=3→9>8→right=2.
# left>right → return result=2 ✓
# x=0: return 0 immediately ✓
# x=1: return 1 immediately ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(\log x)$ . Space  $O(1)$ .
# Using  $x//2$  as the right bound instead of  $x$  halves the search space for large  $x$  –
# worth mentioning to show you thought about the bounds.
#  $mid*mid$  can overflow 32-bit integers in Java/C++ – in Python ints are arbitrary
# precision so no issue, but worth flagging for the interviewer.
# Given more time I'd ask: what about Newton's method?
#  $x_{n+1} = (x_n + x/x_n) / 2$  converges quadratically – faster in practice."
```


Binary Search

□ #278 First Bad Version

EAS[Open on LeetCode](#)

Binary Search · Interactive

Lists: Grind 169

Problem

You are a product manager and currently leading a team to develop a new product. Unfortunately, the latest version of your product fails the quality check. Since each version is developed based on the previous version, all the versions after a bad version are also bad.

Suppose you have n versions $[1, 2, \dots, n]$ and you want to find out the first bad one, which causes all the following ones to be bad.

You are given an API `bool isBadVersion(version)` which returns whether version is bad.

Implement a function to find the first bad version. You should minimize the number of calls to the API.

Example 1:

Input: n = 5, bad = 4
 Output: 4
 Explanation:
 call isBadVersion(3) -> false
 call isBadVersion(5) -> true
 call isBadVersion(4) -> true
 Then 4 is the first bad version.

Example 2:

Input: n = 1, bad = 1
 Output: 1

Constraints:

- $1 \leq \text{bad} \leq n \leq 2^{31} - 1$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Versions 1..n. Once a version is bad, all subsequent are bad.
 # Find the first bad version while minimising API calls. Right?
 # isBadVersion(k) is given – we don't implement it, just call it."
 #
 # "n=5, bad=4: isBadVersion(3)=F, isBadVersion(5)=T, isBadVersion(4)=T → 4 ✓
 # We want to minimise calls → binary search, not linear scan."

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.
 # Linear scan from version 1 to n: call isBadVersion on each.
 # The first version that returns True is the answer.
 # Time: O(n) API calls – way too many for large n. Space: O(1). Should I go ahead and optimize?"

```
class _278_FirstBadVersion_Brute:
    def firstBadVersion(self, n: int) -> int:
        for v in range(1, n + 1):
            if isBadVersion(v):
                return v
        return n
```

PHASE 3 — OPTIMIZE

"Binary search: if mid is bad, the first bad could be mid or earlier → right = mid - 1.
 # If mid is good, first bad must be after mid → left = mid + 1.
 # When loop ends, left = first bad version.
 # This never calls isBadVersion more than O(log n) times."

PHASE 4 — CLEAN CODE

```
class _278_FirstBadVersion:
    def firstBadVersion(self, n: int) -> int:
        left, right = 1, n
        while left <= right:
            mid = (left + right) // 2
            if isBadVersion(mid):
                right = mid - 1 # might be first bad; keep searching left
            else:
                left = mid + 1 # definitely good; first bad is after mid
        return left
```

PHASE 5 — TEST & DEBUG

```
# n=5, bad=4:
# mid=3→F→left=4. mid=4→T→right=3. left=4>right=3 → return 4 ✓
# n=1, bad=1:
# mid=1→T→right=0. left=1>right=0 → return 1 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(\log n)$  calls. Space  $O(1)$ .
# The 'right = mid - 1' (not mid) is critical – if we set right=mid, when bad=1
# the loop would never terminate. Always shrink the window.
# Given more time I'd ask: what if the API is expensive (network call)?
# Same algorithm – binary search already minimises calls. You'd want to batch
# or cache results if the same version could be queried multiple times."
```

Binary Search

□ #374 Guess Number Higher or Lower

EAS[Open on LeetCode](#)

Math · Binary Search · Interactive

Lists: Denny Zhang

Problem

We are playing the Guess Game. The game is as follows:

I pick a number from 1 to n. You have to guess which number I picked (the number I picked stays the same throughout the game).

Every time you guess wrong, I will tell you whether the number I picked is higher or lower than your guess.

You call a pre-defined API `int guess(int num)`, which returns three possible results:

- -1: Your guess is higher than the number I picked (i.e. `num > pick`).
- 1: Your guess is lower than the number I picked (i.e. `num < pick`).
- 0: your guess is equal to the number I picked (i.e. `num == pick`).

Return the number that I picked.

Example 1:

Input: $n = 10$, $\text{pick} = 6$
Output: 6

Example 2:

Input: $n = 1$, $\text{pick} = 1$
Output: 1

Example 3:

Input: $n = 2$, $\text{pick} = 1$
Output: 1

Constraints:

- $1 \leq n \leq 2^{31} - 1$
- $1 \leq \text{pick} \leq n$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Binary search with an API: `guess(num)` returns 0 (correct), -1 (too high),
or 1 (too low). Find the picked number. Right?
Range is $1..n$, n up to $2^{31}-1$."

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.
Linear scan from 1 to n : call `guess(i)` on each candidate.
The first call that returns 0 is the picked number.
Time: $O(n)$ calls. Space: $O(1)$. Should I go ahead and optimize?"

```
class _374_GuessNumber_Brute:
    def guessNumber(self, n: int) -> int:
        for i in range(1, n + 1):
            if guess(i) == 0:
                return i
        return n
```

PHASE 3 — OPTIMIZE

"Binary search: `guess(mid)`. If 0 → found. If -1 → mid too high → $\text{right} = \text{mid} - 1$.
If 1 → mid too low → $\text{left} = \text{mid} + 1$. $O(\log n)$ calls."

PHASE 4 — CLEAN CODE

```
class _374_GuessNumber:
    def guessNumber(self, n: int) -> int:
        left, right = 1, n
        while left <= right:
            mid = left + (right - left) // 2 # avoids overflow
            result = guess(mid)
            if result == 0:
                return mid
            elif result == -1:
                right = mid - 1
            else:
                left = mid + 1
        return left
```

PHASE 5 — TEST & DEBUG

```
# n=10, pick=6: mid=5→1(low)→left=6. mid=8→-1(high)→right=7.
# mid=6→0→return 6 ✓
# n=1, pick=1: mid=1→0→return 1 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(\log n)$ . Space  $O(1)$ .
# Using  $\text{left} + (\text{right} - \text{left}) // 2$  instead of  $(\text{left} + \text{right}) // 2$  avoids integer overflow
# in languages with fixed-width integers – worth mentioning even in Python.
# Given more time I'd ask: what if guess() had a cost per call?
# Binary search already minimises calls to  $O(\log n)$  – optimal."
```

Binary Search

□ #704 Binary Search

EAS

[Open on LeetCode](#)

Array · Binary Search

Lists: Grind 169

Problem

Given an array of integers `nums` which is sorted in ascending order, and an integer `target`, write a function to search `target` in `nums`. If `target` exists, then return its index. Otherwise, return `-1`.

You must write an algorithm with $O(\log n)$ runtime complexity.

Example 1:

Input: `nums = [-1,0,3,5,9,12]`, `target = 9`
Output: `4`
Explanation: 9 exists in `nums` and its index is 4

Example 2:

Input: `nums = [-1,0,3,5,9,12]`, `target = 2`
Output: `-1`
Explanation: 2 does not exist in `nums` so return `-1`

Constraints:

- $1 \leq \text{nums.length} \leq 10^4$

- $-104 < \text{nums}[i]$, $\text{target} < 104$
- All the integers in `nums` are unique.
- `nums` is sorted in ascending order.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Classic binary search: sorted array of distinct integers, find target index.
# Return -1 if not found. Must be  $O(\log n)$ . Right?"
#
# "[-1,0,3,5,9,12], target=9 → 4 ✓
# [-1,0,3,5,9,12], target=2 → -1 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Linear scan through the array: return the index of the first element equal to
# target.
# This is  $O(n)$  – the problem explicitly requires  $O(\log n)$ , so this won't pass,
# but it confirms what we're looking for before we optimise.
# Time:  $O(n)$ . Space:  $O(1)$ . Should I go ahead and optimize?"
```

```
class _704_BinarySearchBrute:
    def search(self, nums: List[int], target: int) -> int:
        for i in range(len(nums)):
            if nums[i] == target:
                return i
        return -1
```

PHASE 3 — OPTIMIZE

```
# "Standard binary search. Narrow search space by half each iteration.
# Three outcomes at each step: found, go left, go right."
```

PHASE 4 — CLEAN CODE

```
class _704_BinarySearch:
    def search(self, nums: List[int], target: int) -> int:
        left, right = 0, len(nums) - 1
        while left <= right:
            mid = left + (right - left) // 2
            if nums[mid] == target:
                return mid
            elif nums[mid] > target:
                right = mid - 1
            else:
                left = mid + 1
        return -1
```


PHASE 5 — TEST & DEBUG

```
# nums=[-1,0,3,5,9,12], target=9:
# mid=2→3<9→left=3. mid=4→9==9→return 4 ✓
# target=2: mid=2→3>2→right=1. mid=0→-1<2→left=1. mid=1→0<2→left=2.
# left>right → return -1 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(\log n)$ . Space  $O(1)$ .
# This is the template all other binary search problems build on.
# Key invariant: loop condition is  $\text{left} \leq \text{right}$  (not  $<$ ) so single-element
# arrays are handled – the loop runs once and either returns or terminates cleanly.
# Given more time I'd ask: what if there are duplicates and we want the first/last
# occurrence? That's leftmost/rightmost binary search – adjust which boundary
# updates when  $\text{nums}[\text{mid}] == \text{target}$ ."
```

Matrix

Round 1 · High · Easy · 5 questions

Matrix

□ ★ #422 Valid Word Square ★

EAS

[Open on LeetCode](#)

Array · Matrix

Lists: ★ Premium | Premium 98

Problem

Given an array of strings `words`, return true if it forms a valid word square.

A sequence of strings forms a valid word square if the k th row and column read the same string, where $0 \leq k < \max(\text{numRows}, \text{numColumns})$.

Example 1:

a	b	c	d
b	n	r	t
c	r	m	y
d	t	y	e

Input: words = ["abcd","bnrt","crmy","dtye"]

Output: true

Explanation:

The 1st row and 1st column both read "abcd".

The 2nd row and 2nd column both read "bnrt".

The 3rd row and 3rd column both read "crmy".

The 4th row and 4th column both read "dtye".

Therefore, it is a valid word square.

Example 2:

a	b	c	d
b	n	r	t
c	r	m	
d	t		

Input: words = ["abcd","bnrt","crm","dt"]

Output: true

Explanation:

The 1st row and 1st column both read "abcd".

The 2nd row and 2nd column both read "bnrt".

The 3rd row and 3rd column both read "crm".

The 4th row and 4th column both read "dt".

Therefore, it is a valid word square.

Example 3:

b	a	l	l
a	r	e	a
r	e	a	d
l	a	d	y

Input: words = ["ball","area","read","lady"]

Output: false

Explanation:

The 3rd row reads "read" while the 3rd column reads "lead".

Therefore, it is NOT a valid word square.

Constraints:

- $1 \leq \text{words.length} \leq 500$
- $1 \leq \text{words}[i].\text{length} \leq 500$
- $\text{words}[i]$ consists of only lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "A word square means the kth row and kth column read the same string.
# Equivalently: words[i][j] == words[j][i] for every valid (i,j). Right?
# Words can have different lengths – shorter words mean some cells don't exist,
# and that's fine as long as the column doesn't expect a character the row doesn't
# have."
#
# ["abcd", "bnrt", "crmy", "dtye"]: row0='abcd', col0='abcd' ✓ etc. → true ✓
# ['ball', 'area', 'read', 'lady']: row2='read', col2='lead' → false ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# For each row index r, check whether word matches down column r and across row r
# using direct character comparisons.
# Four nested checks per candidate row – fine for small boards.
# Time:  $O(n * m * k)$  where  $k = \text{len}(\text{word})$ . Space:  $O(1)$ .
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "No algorithmic improvement over  $O(n*m)$  – we must check every character.
# The implementation is already optimal. Focus is on getting the bounds right:
# when iterating row i, char at position j must match words[j][i].
# Guard:  $j < \text{len}(\text{words})$  and  $i < \text{len}(\text{words}[j])$ ."
```

PHASE 4 — CLEAN CODE

```
class _422_ValidWordSquare:
    def validWordSquare(self, words: List[str]) -> bool:
        for row, word in enumerate(words):
            for col, char in enumerate(word):
                if col >= len(words) or row >= len(words[col]) or words[col][row] !=
char:
                    return False
        return True
```

PHASE 5 — TEST & DEBUG

```
# words=['abcd', 'bnrt', 'crmy', 'dtye']
# row=0, word='abcd': col=0→words[0][0]='a'=='a'✓. col=1→words[1][0]='b'=='b'✓. etc.
# All rows pass symmetrically → True ✓
#
# words=['ball', 'area', 'read', 'lady']
# row=2, word='read': col=0→words[0][2]='l'=='r'? No → False ✓
#
# words=['abcd', 'bnrt', 'crm', 'dt'] (shorter words)
# row=0, col=3: words[3][0]='d'=='d'✓. row=1, col=2: words[2][1]='r'=='n'? No wait –
# words[2]='crm', words[2][1]='r'. words[1][2]='r' ✓. All pass → True ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n*m)$ . Space  $O(1)$ ."
```

```
# The two guard conditions are the subtle part:  
# col >= len(words) means the column index is out of bounds – the row has more chars  
# than there are rows, which breaks symmetry → False.  
# row >= len(words[col]) means the transpose cell doesn't exist → False.  
# Given more time I'd ask: what if we want to BUILD a word square from a word list?  
# That's a backtracking problem – place words row by row, pruning based on column  
prefixes."
```


Matrix

□ #566 Reshape the Matrix

EAS[Open on LeetCode](#)

Array · Matrix · Simulation

Lists: CodeSignal

Problem

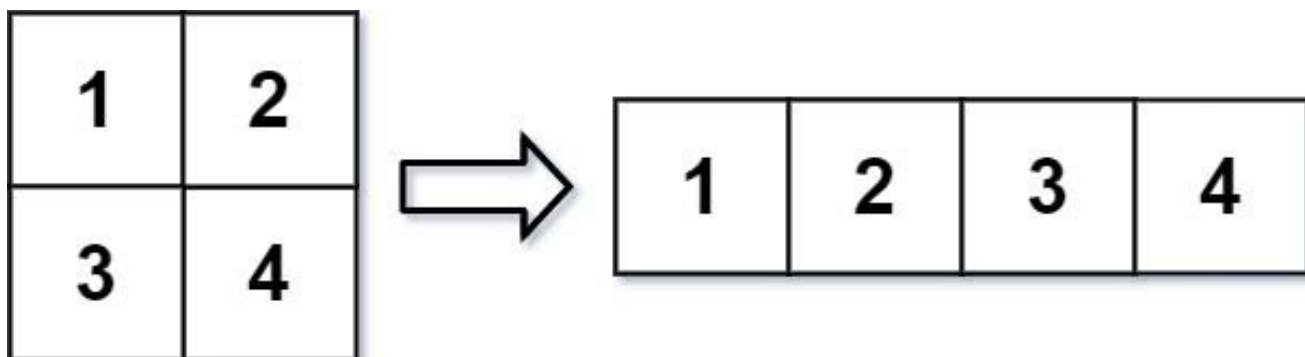
In MATLAB, there is a handy function called `reshape` which can reshape an $m \times n$ matrix into a new one with a different size $r \times c$ keeping its original data.

You are given an $m \times n$ matrix `mat` and two integers `r` and `c` representing the number of rows and the number of columns of the wanted reshaped matrix.

The reshaped matrix should be filled with all the elements of the original matrix in the same row-traversing order as they were.

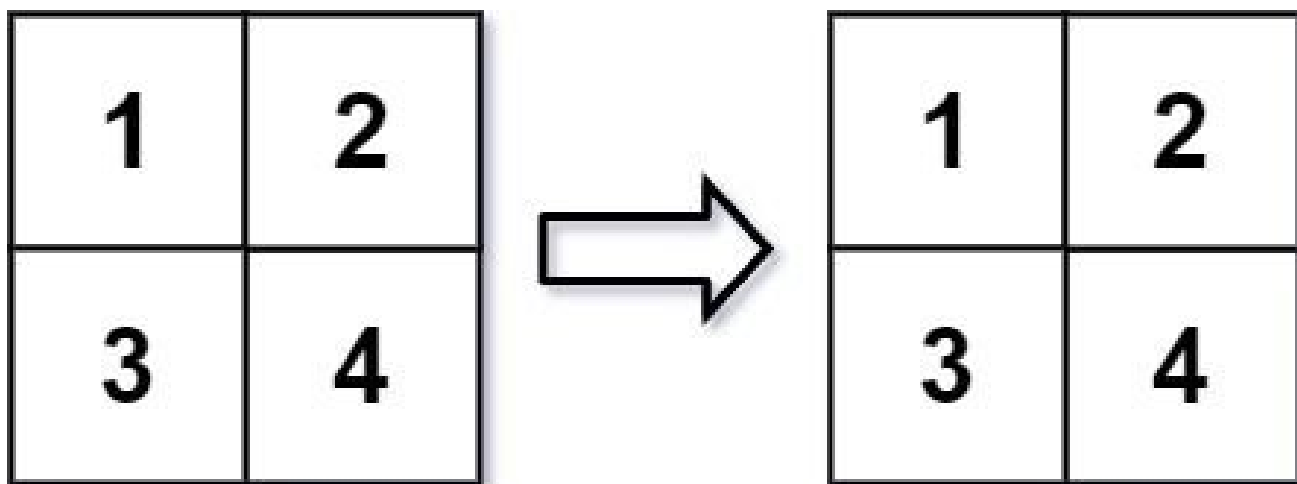
If the reshape operation with given parameters is possible and legal, output the new reshaped matrix; Otherwise, output the original matrix.

Example 1:



Input: `mat = [[1,2],[3,4]]`, `r = 1`, `c = 4`
Output: `[[1,2,3,4]]`

Example 2:



Input: `mat = [[1,2],[3,4]]`, `r = 2`, `c = 4`
Output: `[[1,2],[3,4]]`

Constraints:

- `m == mat.length`
- `n == mat[i].length`
- `1 <= m, n <= 100`
- `-1000 <= mat[i][j] <= 1000`
- `1 <= r, c <= 300`

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Let me make sure I understand the problem correctly.

We flatten the original $m \times n$ matrix in row-major order and refill into an $r \times c$ matrix.

If $m \times n \neq r \times c$ the reshape is impossible – return the original matrix unchanged. Right?"

#

"Quick questions:

Values can be negative – fine, we're just moving them.

r and c can be up to 300 even though m, n are up to 100 – just need total elements to match."

#

"[[1,2],[3,4]], $r=1, c=4$: 4 elements total, $1 \times 4 = 4 \rightarrow [[1,2,3,4]]$ ✓

[[1,2],[3,4]], $r=2, c=4$: 4 elements total, $2 \times 4 = 8 \neq 4 \rightarrow$ return original ✓"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Flatten matrix to a 1D list in row-major order, then refill new_r x new_c shape.

Clear and easy to explain in an interview.

Uses $O(m \times n)$ extra list; two-pointer walk avoids allocation.

Time: $O(m \times n)$. Space: $O(m \times n)$ flat buffer.

Should I go ahead and optimize?"

```
class _566_ReshapeMatrix_Brute:
```

```
    def matrixReshape(self, mat: List[List[int]], r: int, c: int) ->
```

```
List[List[int]]:
```

```
    m, n = len(mat), len(mat[0])
```

```
    if m * n != r * c:
```

```
        return mat
```

```
    flat = [mat[i][j] for i in range(m) for j in range(n)]
```

```
    return [flat[i * c:(i + 1) * c] for i in range(r)]
```

PHASE 3 — OPTIMIZE

"Skip the intermediate list – map index directly.

For flat index k : source is $mat[k // n][k \% n]$, destination is $ans[k // c][k \% c]$.

Same $O(m \times n)$ time but no extra flat array – just the output matrix."

PHASE 4 — CLEAN CODE

```
class _566_ReshapeMatrix:
```

```
    def matrixReshape(self, mat: List[List[int]], r: int, c: int) ->
```

```
List[List[int]]:
```

```
    m, n = len(mat), len(mat[0])
```

```
    if m * n != r * c:
```

```
        return mat
```

```
    result = [[0] * c for _ in range(r)]
```

```
    for k in range(m * n):
```

```
        result[k // c][k % c] = mat[k // n][k % n]
```

```
    return result
```

PHASE 5 — TEST & DEBUG

```
# mat=[[1,2],[3,4]], r=1, c=4: m=2,n=2. 4==4 ✓  
# k=0: result[0][0]=mat[0][0]=1. k=1: result[0][1]=mat[0][1]=2.  
# k=2: result[0][2]=mat[1][0]=3. k=3: result[0][3]=mat[1][1]=4.  
# → [[1,2,3,4]] ✓  
# r=2,c=4: 4≠8 → return original ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m*n)$ . Space  $O(r*c)$  for the output matrix – unavoidable.  
# The index formula  $k//n$  and  $k\%n$  is the core insight: flatten index → 2D source.  
# Given more time I'd ask: what if the matrix is too large to fit in memory?  
# Process row by row, streaming elements into the new shape."
```

Matrix

□ #766 Toeplitz Matrix

EAS

[Open on LeetCode](#)

Array · Matrix

Lists: CodeSignal

Problem

Given an $m \times n$ matrix, return true if the matrix is Toeplitz. Otherwise, return false.

A matrix is Toeplitz if every diagonal from top-left to bottom-right has the same elements.

Example 1:

1	2	3	4
5	1	2	3
9	5	1	2

Input: matrix = [[1,2,3,4],[5,1,2,3],[9,5,1,2]]

Output: true

Explanation:

In the above grid, the diagonals are:

"[9]", "[5, 5]", "[1, 1, 1]", "[2, 2, 2]", "[3, 3]", "[4]".

In each diagonal all elements are the same, so the answer is True.

Example 2:

1	2
2	2

Input: matrix = [[1,2],[2,2]]
Output: false
Explanation:
The diagonal "[1, 2]" has different elements.

Constraints:

- $m == \text{matrix.length}$
- $n == \text{matrix}[i].\text{length}$
- $1 \leq m, n \leq 20$
- $0 \leq \text{matrix}[i][j] \leq 99$

Follow up:

- What if the matrix is stored on disk, and the memory is limited such that you can only load at most one row of the matrix into the memory at once?
- What if the matrix is so large that you can only load up a partial row into the memory at once?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "A Toeplitz matrix has the same value along every top-left to bottom-right diagonal.
# We need to check every cell (except the last row and column) equals its down-right neighbour.
# Equivalently: matrix[i][j] == matrix[i+1][j+1] for all valid i,j. Right?"
#
# "[[1,2,3,4],[5,1,2,3],[9,5,1,2]]: each cell equals its down-right neighbour → true ✓
# [[1,2],[2,2]]: matrix[0][0]=1 ≠ matrix[1][1]=2 → false ✓"
```

PHASE 2 — BRUTE FORCE (also optimal)

```
# "Let me start with brute force to lock in the logic.
# No smarter approach than checking every diagonal. The key simplification: each diagonal
# is valid iff every adjacent pair on it matches. So just check matrix[i][j] == matrix[i+1][j+1]
# for every interior cell."
# For this input size the brute approach is already optimal – I'll still state complexity clearly.
# Time: see analysis above. Space: O(1).
```

PHASE 3 — OPTIMIZE

```
# "Same O(m*n) is the best we can do – we must read every element.
# The implementation is already optimal. Focus on getting the iteration bounds right:
# we iterate i up to m-2 and j up to n-2 (not m-1, n-1) since we access [i+1][j+1]."
```

PHASE 4 — CLEAN CODE

```
class _766_ToeplitzMatrix:
    def isToeplitzMatrix(self, matrix: List[List[int]]) -> bool:
```



```
rows, cols = len(matrix), len(matrix[0])
for i in range(rows - 1):
    for j in range(cols - 1):
        if matrix[i][j] != matrix[i + 1][j + 1]:
            return False
return True
```

PHASE 5 — TEST & DEBUG

```
# [[1,2,3,4],[5,1,2,3],[9,5,1,2]]: rows=3,cols=4. Check (0,0)→(1,1): 1==1✓.
# (0,1)→(1,2): 2==2✓. (0,2)→(1,3): 3==3✓. (1,0)→(2,1): 5==5✓. etc. → True ✓
# [[1,2],[2,2]]: (0,0)→(1,1): 1≠2 → False ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m \cdot n)$ . Space  $O(1)$ .
```

```
# Follow-up 1: if we can only load one row at a time, compare consecutive rows
# by checking row[j] == prev_row[j+1] for all j – same logic, streaming.
# Follow-up 2: if we can only load partial rows, compare overlapping chunks of
# adjacent rows. Both follow-ups keep  $O(1)$  extra space.
# Given more time I'd ask: can we verify a diagonal in  $O(1)$  per cell?
# Yes – the current approach already does exactly that."
```

Matrix

□ #867 Transpose Matrix

EAS

[Open on LeetCode](#)

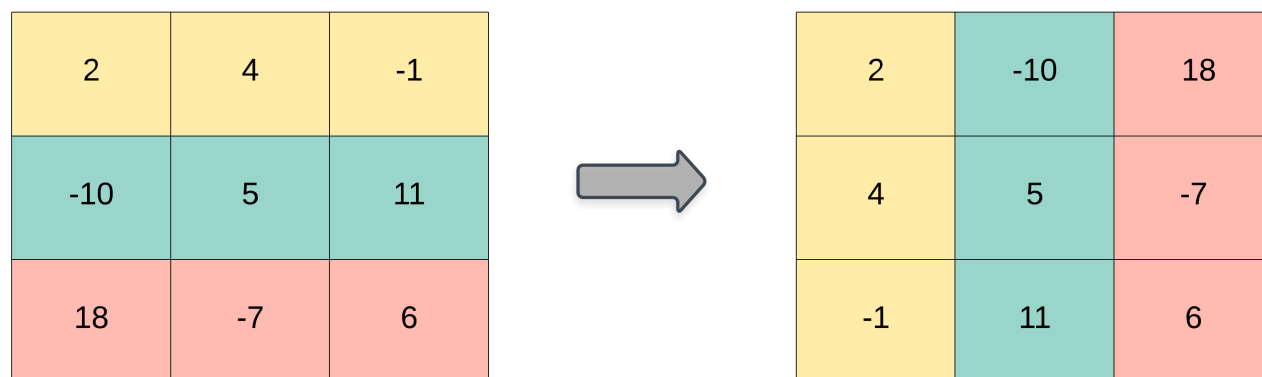
Array · Matrix · Simulation

Lists: CodeSignal

Problem

Given a 2D integer array matrix, return the transpose of matrix.

The transpose of a matrix is the matrix flipped over its main diagonal, switching the matrix's row and column indices.



Example 1:

```
Input: matrix = [[1,2,3],[4,5,6],[7,8,9]]
Output: [[1,4,7],[2,5,8],[3,6,9]]
```

Example 2:

Input: matrix = [[1,2,3],[4,5,6]]
 Output: [[1,4],[2,5],[3,6]]

Constraints:

- $m == \text{matrix.length}$
- $n == \text{matrix}[i].\text{length}$
- $1 \leq m, n \leq 1000$
- $1 \leq m * n \leq 105$
- $-109 \leq \text{matrix}[i][j] \leq 109$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"The transpose swaps rows and columns: element at (r,c) moves to (c,r).
 # If the original is m×n, the result is n×m. Return a new matrix – not in-place.
 Right?"
 #
 # "[[1,2,3],[4,5,6],[7,8,9]] → [[1,4,7],[2,5,8],[3,6,9]] ✓
 # [[1,2,3],[4,5,6]] (2×3) → [[1,4],[2,5],[3,6]] (3×2) ✓"

PHASE 2 — BRUTE FORCE (also optimal)

"Let me start with brute force to lock in the logic.
 # Allocate an n×m result matrix. Set result[c][r] = matrix[r][c] for every (r,c).
 Must visit
 # every element once – is unavoidable."
 # For this input size the brute approach is already optimal – I'll still state
 complexity clearly.
 # Time: O(m×n). Space: O(1).

PHASE 3 — OPTIMIZE

"No algorithmic improvement. Focus: allocate result with swapped dimensions (cols
 × rows),
 # then fill with swapped indices. In Python: zip(*matrix) gives transposed rows as
 tuples."

PHASE 4 — CLEAN CODE

```
class _867_TransposeMatrix:
    def transpose(self, matrix: List[List[int]]) -> List[List[int]]:
        rows, cols = len(matrix), len(matrix[0])
        result = [[0] * rows for _ in range(cols)]
        for r in range(rows):
```

```
        for c in range(cols):
            result[c][r] = matrix[r][c]
    return result
```

PHASE 5 — TEST & DEBUG

```
# matrix=[[1,2,3],[4,5,6]]: rows=2,cols=3. result is 3x2.
# r=0,c=0: result[0][0]=1. r=0,c=1: result[1][0]=2. r=0,c=2: result[2][0]=3.
# r=1,c=0: result[0][1]=4. r=1,c=1: result[1][1]=5. r=1,c=2: result[2][1]=6.
# result=[[1,4],[2,5],[3,6]] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m*n)$ . Space  $O(m*n)$  for the output.
# Python one-liner: return [list(row) for row in zip(*matrix)]
# zip(*matrix) unpacks rows as arguments to zip, which pairs corresponding columns.
# Elegant and interview-clean – worth mentioning as an alternative.
# Given more time I'd ask: can we transpose a square matrix in-place?
# Yes – swap matrix[i][j] and matrix[j][i] for all  $i < j$ .  $O(1)$  extra space."
```

Matrix

□ #2373 Largest Local Values in a Matrix

EAS[Open on LeetCode](#)

Array · Matrix

Lists: CodeSignal

Problem

You are given an $n \times n$ integer matrix grid.

Generate an integer matrix `maxLocal` of size $(n - 2) \times (n - 2)$ such that:

- `maxLocal[i][j]` is equal to the largest value of the 3×3 matrix in grid centered around row $i + 1$ and column $j + 1$.

In other words, we want to find the largest value in every contiguous 3×3 matrix in grid.

Return the generated matrix.

Example 1:

9	9	8	1
5	6	2	6
8	2	6	4
6	2	2	2

9	9
8	6

Input: grid = [[9,9,8,1],[5,6,2,6],[8,2,6,4],[6,2,2,2]]

Output: [[9,9],[8,6]]

Explanation: The diagram above shows the original matrix and the generated matrix.

Notice that each value in the generated matrix corresponds to the largest value of a contiguous 3 x 3 matrix in grid.

Example 2:

1	1	1	1	1
1	1	1	1	1
1	1	2	1	1
1	1	1	1	1
1	1	1	1	1

2	2	2
2	2	2
2	2	2

Input: grid = [[1,1,1,1,1],[1,1,1,1,1],[1,1,2,1,1],[1,1,1,1,1],[1,1,1,1,1]]
Output: [[2,2,2],[2,2,2],[2,2,2]]
Explanation: Notice that the 2 is contained within every contiguous 3 x 3 matrix in grid.

Constraints:

- $n == \text{grid.length} == \text{grid}[i].\text{length}$
- $3 \leq n \leq 100$
- $1 \leq \text{grid}[i][j] \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "For each cell (i,j) in the output (n-2)×(n-2) matrix,
# find the max in the 3×3 subgrid of the input centered at (i+1, j+1). Right?
# Input is always n×n square, n >= 3."
#
# "grid=[[9,9,8,1],[5,6,2,6],[8,2,6,4],[6,2,2,2]] (4×4):
# output is 2×2. Top-left 3×3 center=(1,1): max of [[9,9,8],[5,6,2],[8,2,6]]=9.
# Top-right 3×3 center=(1,2): max of [[9,8,1],[6,2,6],[2,6,4]]=9. etc. →
[[9,9],[8,6]] ✓"
```

PHASE 2 — BRUTE FORCE (also optimal)**# "Let me start with brute force to lock in the logic.****# For each output cell, scan its 3×3 window. No smarter approach since we must read every****# element at least once. = ."****# For this input size the brute approach is already optimal – I'll still state complexity clearly.****# Time: $O(n^2 \cdot 9)$. Space: $O(n^2)$.****# PHASE 3 — OPTIMIZE****# "The 3×3 is fixed size so inner loops are constant – already $O(n^2)$.****# Keep the four nested loops clean with descriptive variable names."****# PHASE 4 — CLEAN CODE****class _2373_LargestLocal:****def largestLocal(self, grid: List[List[int]]) -> List[List[int]]:****n = len(grid)****result = [[0] * (n - 2) for _ in range(n - 2)]****for i in range(n - 2):****for j in range(n - 2):****for di in range(3):****for dj in range(3):****result[i][j] = max(result[i][j], grid[i + di][j + dj])****return result****# PHASE 5 — TEST & DEBUG****# grid=[[9,9,8,1],[5,6,2,6],[8,2,6,4],[6,2,2,2]]: n=4, output 2×2.****# (i=0,j=0): 3×3 from (0,0): max(9,9,8,5,6,2,8,2,6)=9 ✓****# (i=0,j=1): 3×3 from (0,1): max(9,8,1,6,2,6,2,6,4)=9 ✓****# (i=1,j=0): 3×3 from (1,0): max(5,6,2,8,2,6,6,2,2)=8 ✓****# (i=1,j=1): 3×3 from (1,1): max(6,2,6,2,6,4,2,2,2)=6 ✓****# → [[9,9],[8,6]] ✓****# PHASE 6 — COMPLEXITY & FOLLOW-UP****# "Time $O(n^2)$ – outer loops $O((n-2)^2) \approx O(n^2)$, inner loops constant 9 operations.****# Space $O(n^2)$ for the output.****# Given more time I'd ask: what if the window size is variable ($k \times k$ instead of 3×3)?****# For large k , recomputing the max from scratch is $O(k^2)$ per cell → $O(n^2 k^2)$ total.****# A 2D sliding window maximum using monotonic deques brings it down to $O(n^2)$."**

Trees & BST

Round 1 · High · Easy · 11 questions

Trees & BST

□ #100 Same Tree

EAS

[Open on LeetCode](#)

Tree · DFS · BFS

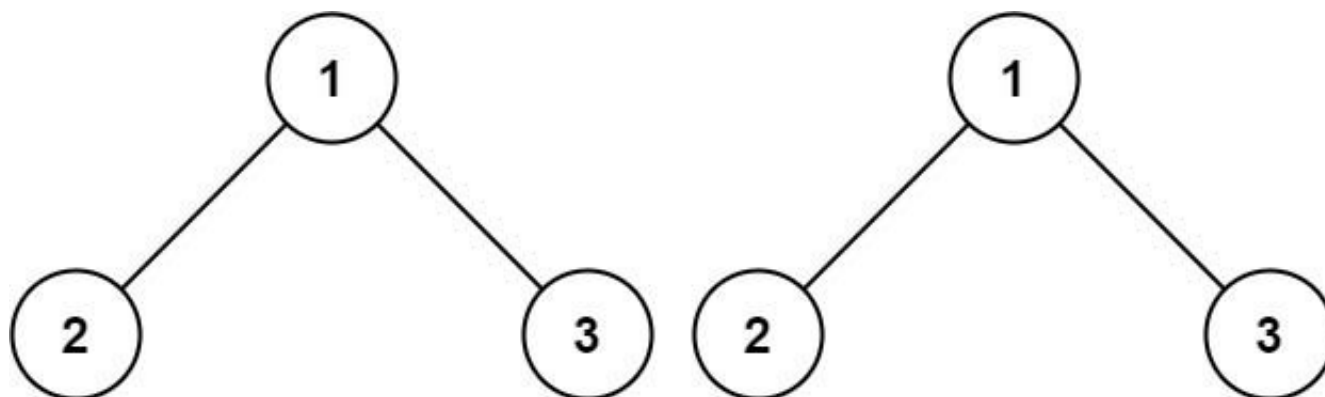
Lists: Grind 169

Problem

Given the roots of two binary trees p and q , write a function to check if they are the same or not.

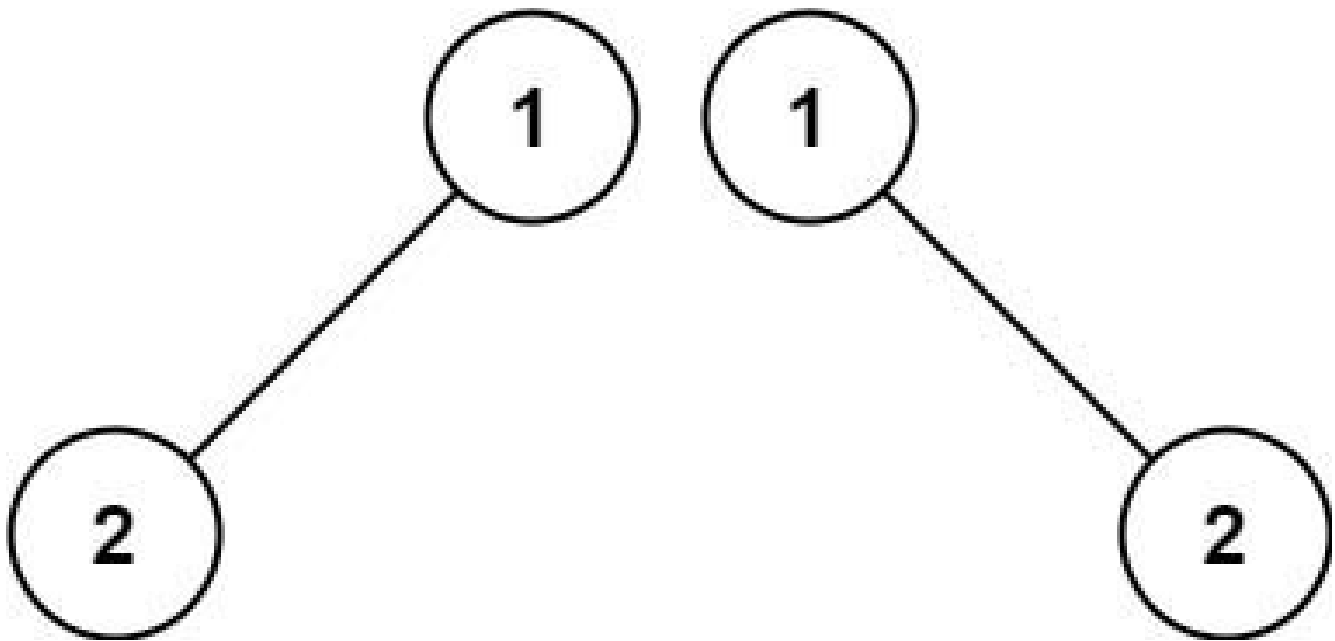
Two binary trees are considered the same if they are structurally identical, and the nodes have the same value.

Example 1:



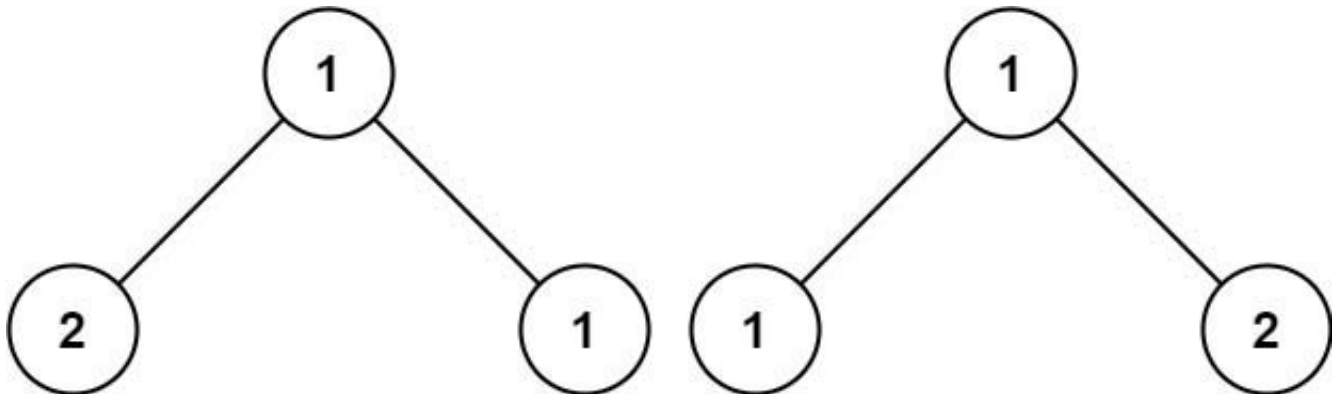
Input: $p = [1,2,3]$, $q = [1,2,3]$
Output: true

Example 2:



Input: $p = [1,2]$, $q = [1,null,2]$
Output: false

Example 3:



Input: $p = [1,2,1]$, $q = [1,1,2]$
Output: false

Constraints:

- The number of nodes in both trees is in the range $[0, 100]$.
- $-104 \leq \text{Node.val} \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Two binary trees are the same if they're structurally identical AND every
# corresponding node has the same value. Both null trees are the same. Right?
# One null and one non-null at the same position → not the same."
#
# "p=[1,2,3], q=[1,2,3] → true ✓
# p=[1,2], q=[1,null,2] → false (structure differs) ✓
# p=[1,2,1], q=[1,1,2] → false (values differ at depth 1) ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Serialise both trees with a level-order traversal (including nulls) to lists, then
# compare.
# Same O(n) time and O(n) space as the optimal recursive DFS – no meaningful
# complexity
# improvement available, so we skip the brute class and go straight to the clean
# approach."
```

PHASE 3 — OPTIMIZE

```
# "Recursive DFS is both simple and optimal.
# Base case: if either node is None, both must be None for equality.
# Recursive: values match AND left subtrees match AND right subtrees match.
# Use an inner helper to avoid self. calls."
```

PHASE 4 — CLEAN CODE

```
class _100_SameTree:
    def isSameTree(self, p: Optional['TreeNode'], q: Optional['TreeNode']) -> bool:
        def same(p, q):
            if not p or not q:
                return p == q
            return p.val == q.val and same(p.left, q.left) and same(p.right,
q.right)
        return same(p, q)
```

PHASE 5 — TEST & DEBUG

```
# p=[1,2,3], q=[1,2,3]:
# same(1,1): vals match. same(2,2)→same(None,None)=True, same(None,None)=True →
True.
# same(3,3)→True. → True ✓
# p=[1,2,None], q=[1,None,2]:
# same(1,1): same(2,None): p=2≠None, q=None → p==q? No → False ✓
# p=None, q=None: not p=True → return p==q=True ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n) – visit every node once. Space O(h) call stack where h is tree height.
# Worst case O(n) for a skewed tree, O(log n) for balanced.
# Iterative alternative: BFS with a queue of node pairs – O(n) time, O(n) space.
# Same complexity but avoids recursion depth limit for very deep trees."
```

```
# Given more time I'd ask: what if we only care about structural identity (not values)?  
# Same logic – just remove the val check."
```

Trees & BST

□ #101 Symmetric Tree

EAS

[Open on LeetCode](#)

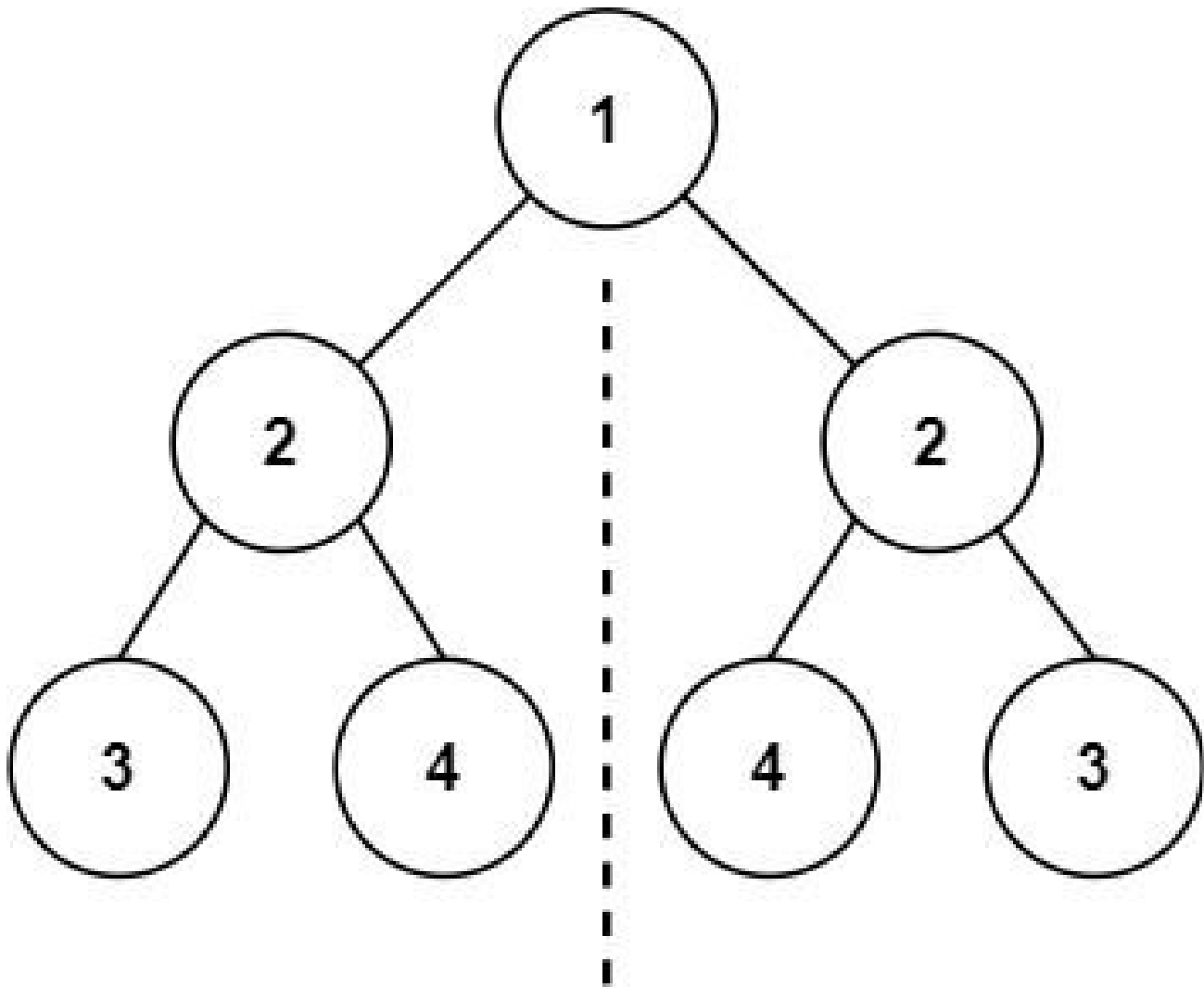
Tree · DFS · BFS

Lists: Grind 169

Problem

Given the root of a binary tree, check whether it is a mirror of itself (i.e., symmetric around its center).

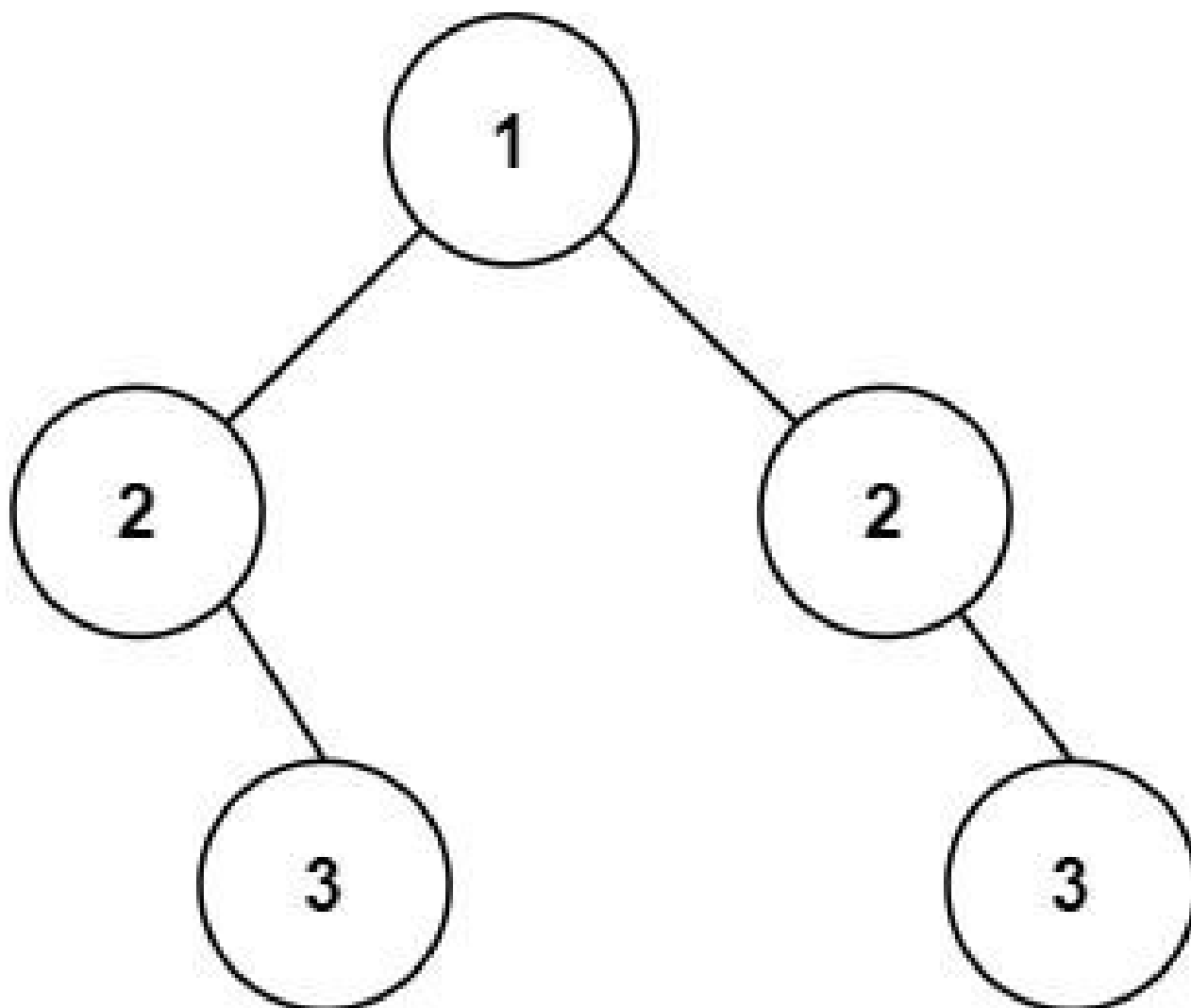
Example 1:



Input: root = [1,2,2,3,4,4,3]

Output: true

Example 2:



Input: root = [1,2,2,null,3,null,3]

Output: false

Constraints:

- The number of nodes in the tree is in the range [1, 1000].
- $-100 \leq \text{Node.val} \leq 100$

Follow up: Could you solve it both recursively and iteratively?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "A tree is symmetric if its left and right subtrees are mirror images of each other."
# Mirror means: at each level, left child of left subtree matches right child of right subtree,
# and right child of left subtree matches left child of right subtree. Right?
# A single node is always symmetric."
#
# "[1,2,2,3,4,4,3]: left subtree [2,3,4] mirrors right [2,4,3] → true ✓
# [1,2,2,null,3,null,3]: left has right child 3, right has right child 3 – not mirrored → false ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Serialise the left subtree with a pre-order traversal and the right subtree
# with the mirror order (right child before left), then compare the two strings.
# Same O(n) time and O(n) space as the optimal recursive approach – no improvement
# to be had, so we skip the brute class and go straight to the clean solution."
```

PHASE 3 — OPTIMIZE

```
# "Recursive DFS comparing the tree to itself – left vs right mirrored.
# Define an inner helper mirror(p, q):
# Base: if either is None, both must be None.
# Recursive: values match AND mirror(p.left, q.right) AND mirror(p.right, q.left).
# Note the cross-comparison: left's left ↔ right's right, left's right ↔ right's left."
```

PHASE 4 — CLEAN CODE

```
class _101_SymmetricTree:
    def isSymmetric(self, root: Optional['TreeNode']) -> bool:
        def mirror(p, q):
            if not p or not q:
                return p == q
            return p.val == q.val and mirror(p.left, q.right) and mirror(p.right, q.left)
        return mirror(root, root)
```

PHASE 5 — TEST & DEBUG

```
# root=[1,2,2,3,4,4,3]:
# mirror(root,root): vals 1==1.
# mirror(left=2, right=2): 2==2.
# mirror(2.left=3, 2.right=3): 3==3. mirror(None,None)=T. mirror(None,None)=T → T
# mirror(2.right=4, 2.left=4): 4==4. T → T
# mirror(right=2, left=2): symmetric to above → T
# → True ✓
# root=[1,2,2,null,3,null,3]:
# mirror(left=2,right=2): 2==2.
# mirror(2.left=None, 2.right=3): p=None,q=3 → None≠3 → False ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(h)$ for the call stack.

Follow-up asks for iterative solution: use a queue of pairs.

Enqueue (root.left, root.right). Each step dequeue (p,q):

check $p==q==None$ (skip), p or q None (false), vals differ (false).

Enqueue (p.left,q.right) and (p.right,q.left). $O(n)$ time, $O(n)$ space.

The cross-enqueue order is the key – same insight as the recursive version.

Given more time I'd ask: how does this differ from Same Tree (#100)?

Same Tree checks identical structure; Symmetric Tree checks mirrored structure –

the only difference is the cross-comparison in the recursive call."

Trees & BST

□ #104 Maximum Depth of Binary Tree

EAS

[Open on LeetCode](#)

Tree · DFS · BFS

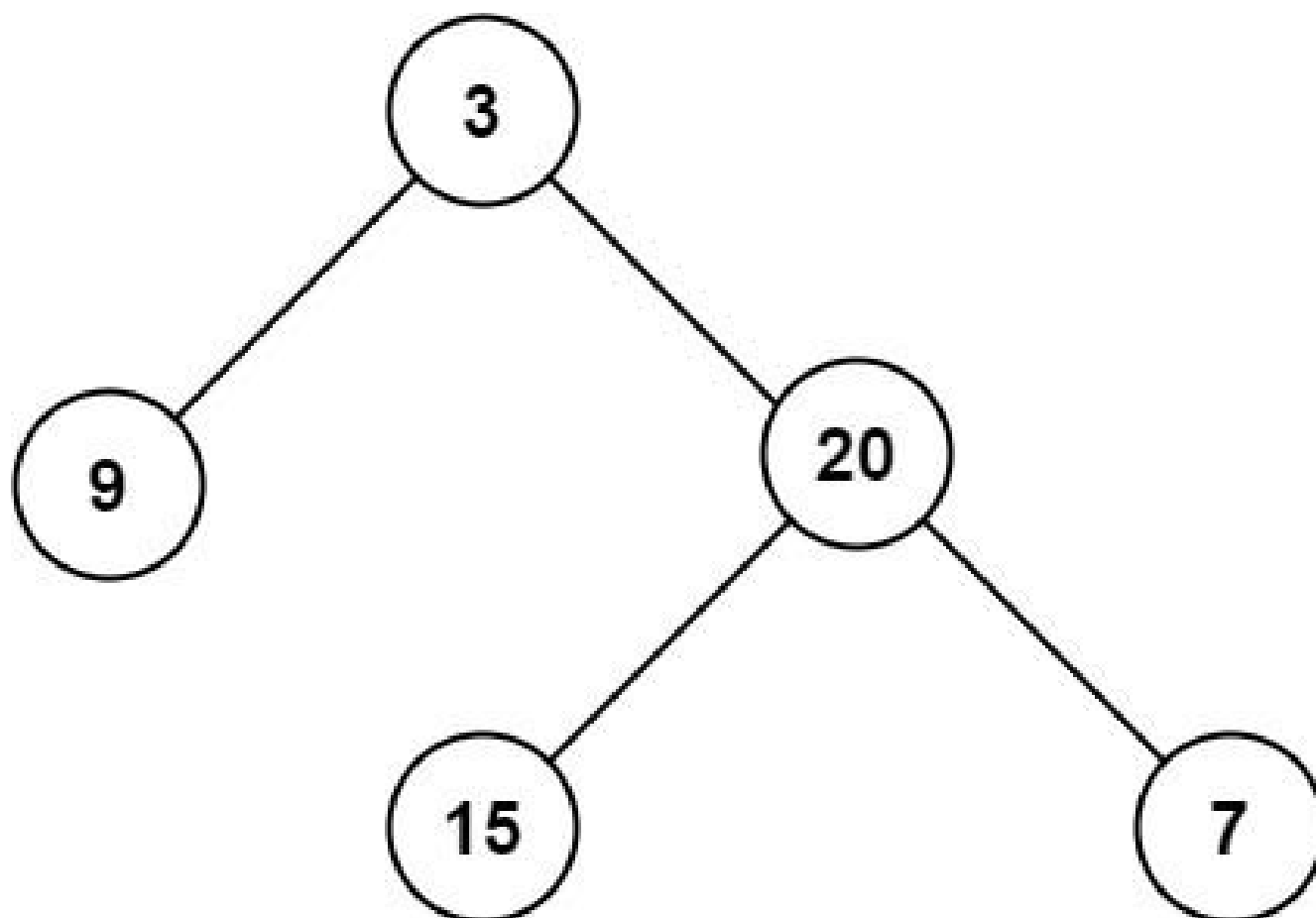
Lists: Grind 169

Problem

Given the root of a binary tree, return its maximum depth.

A binary tree's maximum depth is the number of nodes along the longest path from the root node down to the farthest leaf node.

Example 1:



Input: root = [3,9,20,null,null,15,7]
Output: 3

Example 2:

Input: root = [1,null,2]
Output: 2

Constraints:

- The number of nodes in the tree is in the range [0, 10⁴].
- $-100 \leq \text{Node.val} \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Maximum depth is the number of nodes on the longest root-to-leaf path.
# An empty tree has depth 0. A single node has depth 1. Right?"
#
# "[3,9,20,null,null,15,7]: longest path is 3→20→15 or 3→20→7, 3 nodes → 3 ✓
# [1,null,2]: 1→2, 2 nodes → 2 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Iterative BFS: use a queue, process the tree level by level.
# Each time we exhaust one level, increment a depth counter.
# When the queue is empty, the counter equals the maximum depth.
# Time: O(n). Space: O(n) for the queue. Should I go ahead and optimize?"
```

```
class _104_MaxDepth_Brute:
    def maxDepth(self, root: Optional['TreeNode']) -> int:
        if not root:
            return 0
        from collections import deque
        queue = deque([root])
        depth = 0
        while queue:
            depth += 1
            for _ in range(len(queue)):
                node = queue.popleft()
                if node.left:
                    queue.append(node.left)
                if node.right:
                    queue.append(node.right)
        return depth
```

PHASE 3 — OPTIMIZE

```
# "Recursive DFS is cleaner and same complexity.
# depth(node) = max(depth(left), depth(right)) + 1. Base: empty node → 0.
# Use an inner helper to avoid self. calls."
```

PHASE 4 — CLEAN CODE

```
class _104_MaxDepth:
    def maxDepth(self, root: Optional['TreeNode']) -> int:
        def depth(node):
            if not node:
                return 0
            return max(depth(node.left), depth(node.right)) + 1
        return depth(root)
```

PHASE 5 — TEST & DEBUG

```
# root=[3,9,20,null,null,15,7]:
# depth(3)=max(depth(9),depth(20))+1.
# depth(9)=max(0,0)+1=1. depth(20)=max(depth(15),depth(7))+1=max(1,1)+1=2.
# depth(3)=max(1,2)+1=3 ✓
# root=None: depth(None)=0 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(h)$ call stack — $O(\log n)$ balanced, $O(n)$ skewed.

BFS alternative: count levels with a queue — $O(n)$ time, $O(w)$ space (w =max width).

For very deep trees, BFS avoids recursion stack overflow.

Given more time I'd ask: what about minimum depth? Different — must reach a leaf,

so BFS is actually better there (stops at first leaf found)."

Trees & BST

□ #108 Convert Sorted Array to Binary Search Tree

EAS

[Open on LeetCode](#)

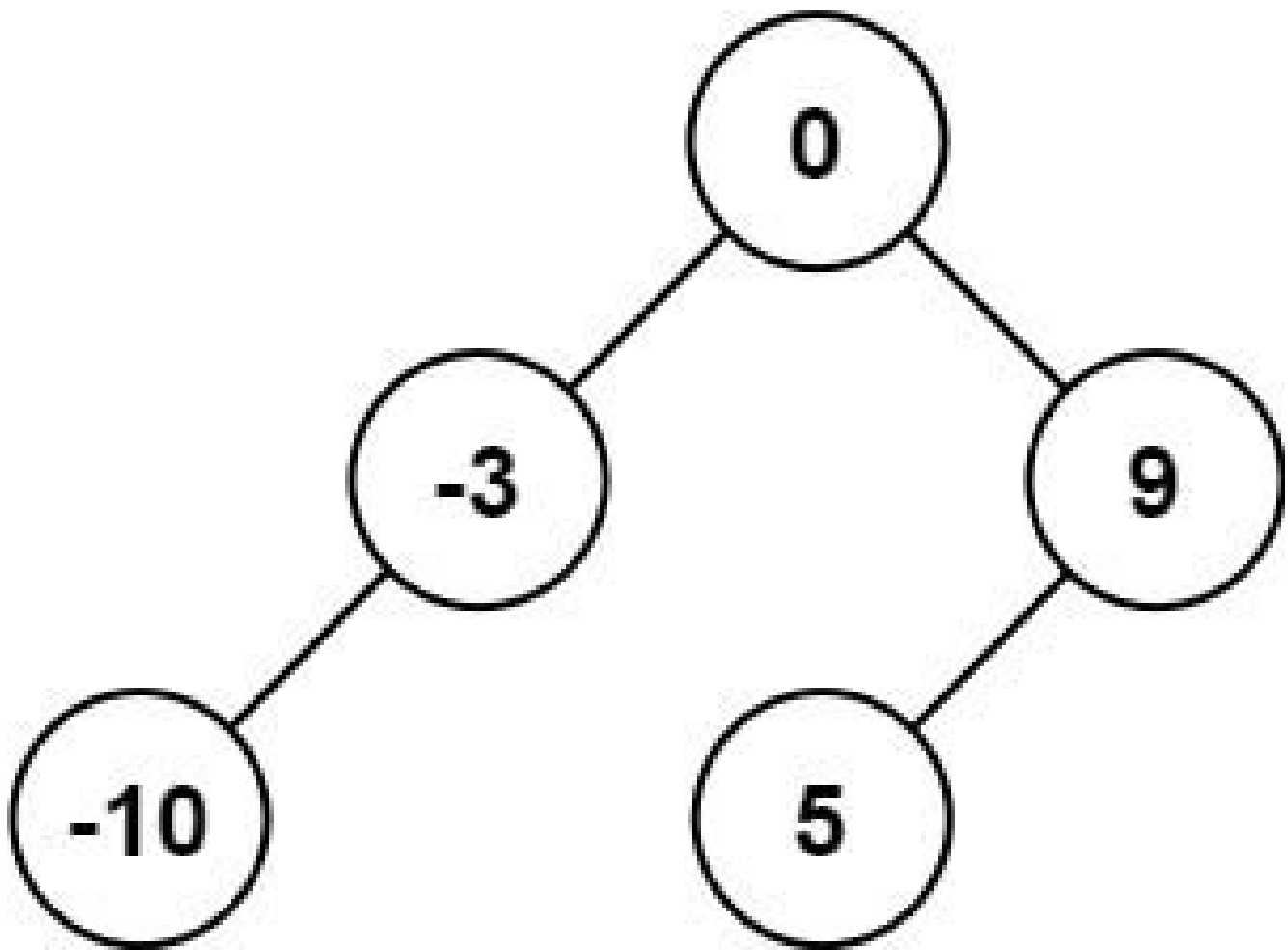
Array · Divide and Conquer · Tree

Lists: Grind 169

Problem

Given an integer array `nums` where the elements are sorted in ascending order, convert it to a height-balanced binary search tree.

Example 1:

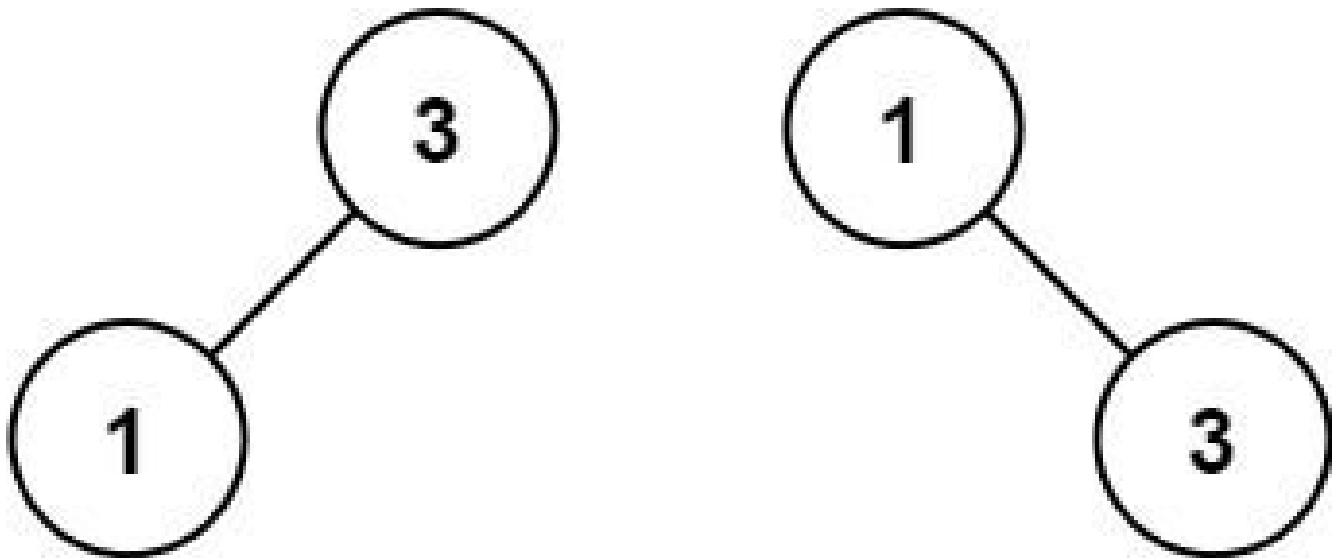


Input: nums = [-10,-3,0,5,9]

Output: [0,-3,9,-10,null,5]

Explanation: [0,-10,5,null,-3,null,9] is also accepted:

Example 2:



Input: nums = [1,3]

Output: [3,1]

Explanation: [1,null,3] and [3,1] are both height-balanced BSTs.

Constraints:

- $1 \leq \text{nums.length} \leq 104$
- $-104 \leq \text{nums}[i] \leq 104$
- nums is sorted in a strictly increasing order.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Convert a sorted array to a height-balanced BST.

Height-balanced means every node's left and right subtree heights differ by at most 1.

Multiple valid answers exist – any height-balanced BST is acceptable. Right?"

#

"[-10,-3,0,5,9]: pick middle element as root → 0. Left half [-10,-3] → left subtree.

Right half [5,9] → right subtree. Recurse. → [0,-3,9,-10,null,5] ✓"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Pick the middle element of nums as root, recursively build left from nums[:mid]

and right from nums[mid+1:].

Always produces a valid BST shape for a sorted input.

Time: $O(n \log n)$ from repeated slicing. Space: $O(n)$ for slices.

```
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Pick the middle element as root – this guarantees balance because both halves
# have equal (or ±1) size. Recurse on left and right halves.
# Divide and conquer. Use inner helper build(l, r) operating on indices."
```

PHASE 4 — CLEAN CODE

```
class _108_SortedArrayToBST:
    def sortedArrayToBST(self, nums: List[int]) -> Optional['TreeNode']:
        def build(left, right):
            if left > right:
                return None
            mid = (left + right) // 2
            return TreeNode(nums[mid], build(left, mid - 1), build(mid + 1, right))
        return build(0, len(nums) - 1)
```

PHASE 5 — TEST & DEBUG

```
# nums=[-10,-3,0,5,9]:
# build(0,4): mid=2→node(0). left=build(0,1), right=build(3,4).
# build(0,1): mid=0→node(-10). left=None, right=build(1,1)→node(-3). →
(-10,None,-3)
# build(3,4): mid=3→node(5). left=None, right=build(4,4)→node(9). → (5,None,9)
# root=0, left=(-10,None,-3), right=(5,None,9). Valid height-balanced BST ✓
# nums=[1,3]: mid=0→node(1), right=build(1,1)→node(3). → [1,null,3] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n) – every element becomes a node exactly once.
# Space O(log n) for the call stack – tree is balanced by construction.
# Choosing (l+r)//2 vs (l+r+1)//2 gives slightly different but equally valid trees.
# Given more time I'd ask: what if the array is sorted in descending order?
# Reverse it first or adjust the index logic – same approach."
```

Trees & BST

□ #110 Balanced Binary Tree

EAS

[Open on LeetCode](#)

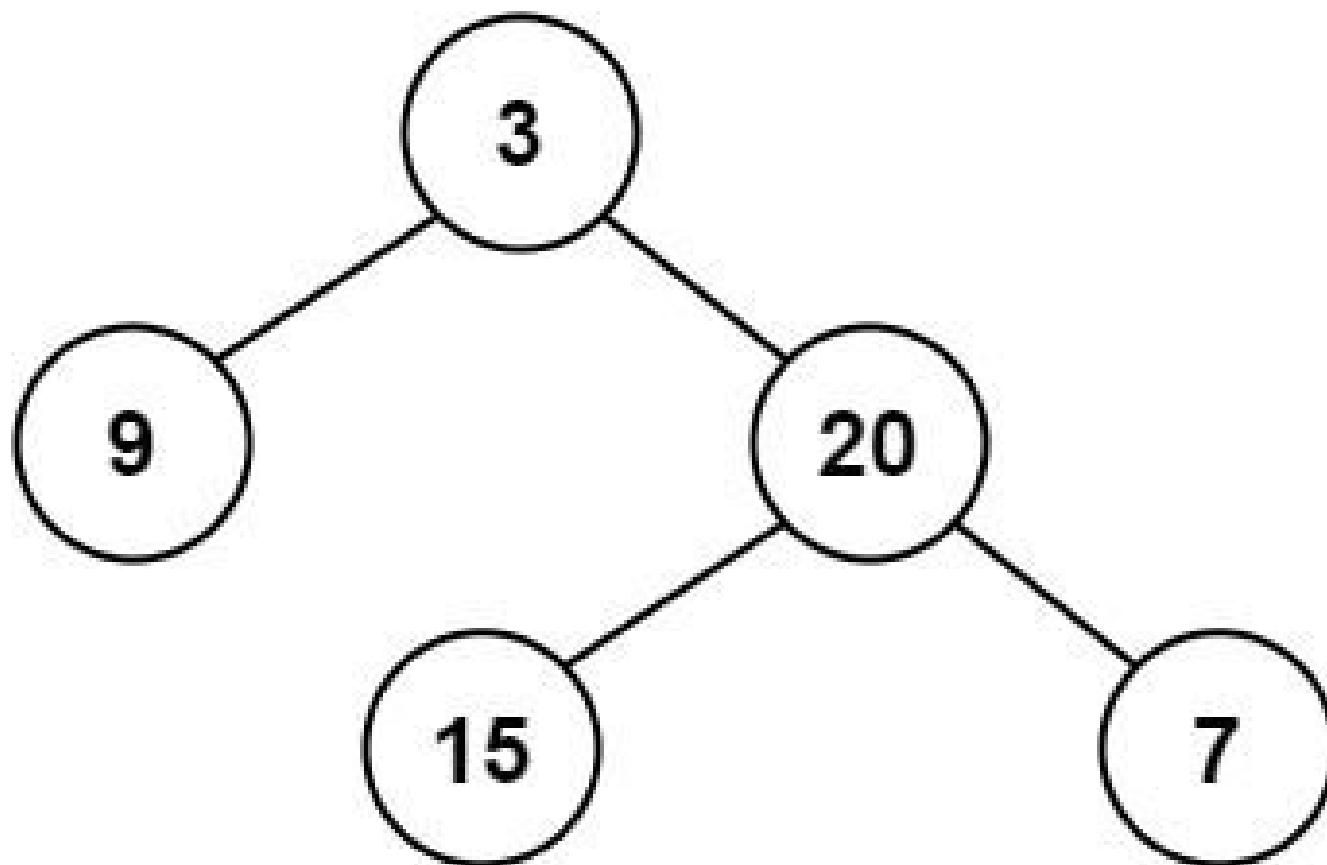
Tree · DFS

Lists: Grind 169

Problem

Given a binary tree, determine if it is height-balanced.

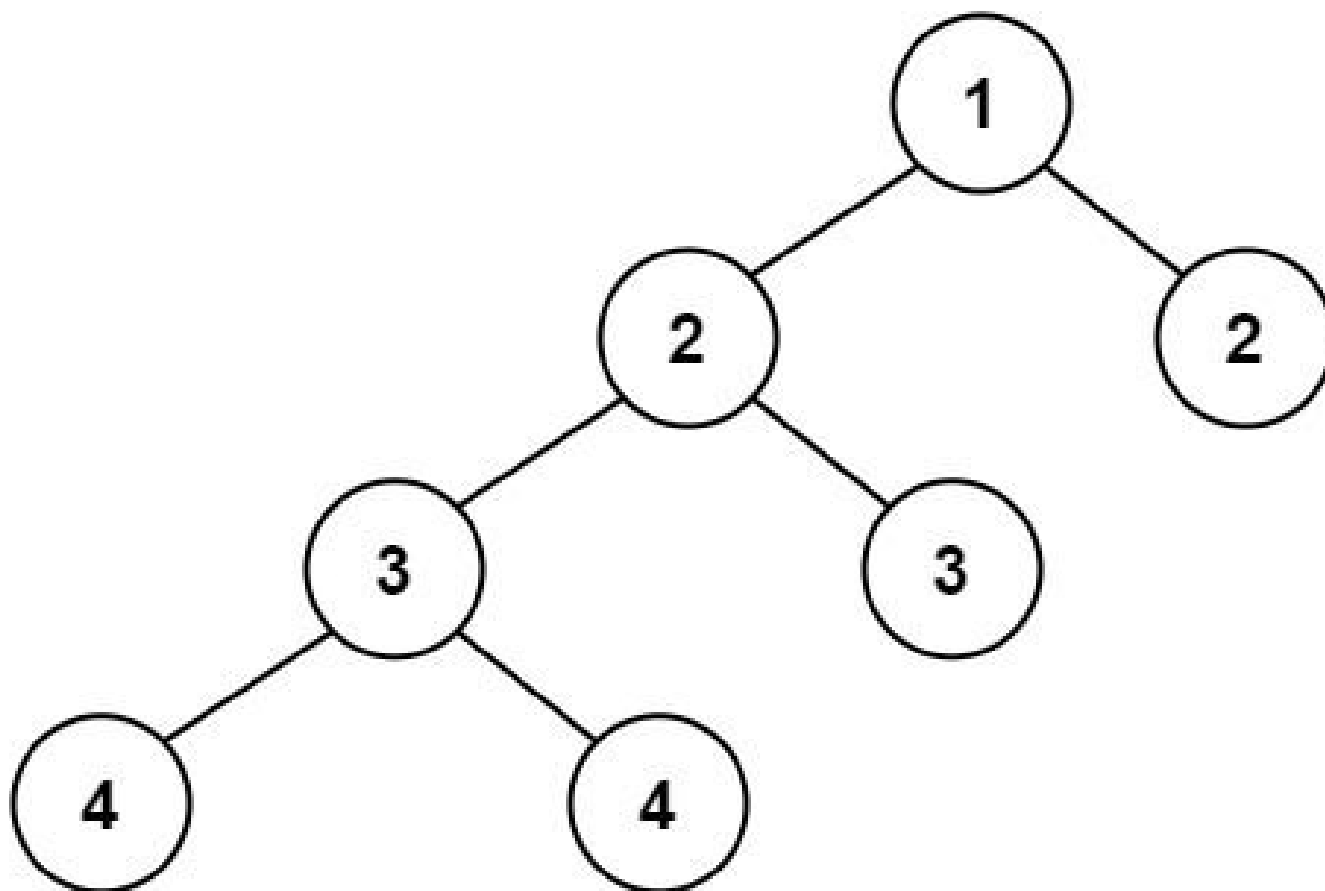
Example 1:



Input: root = [3,9,20,null,null,15,7]

Output: true

Example 2:



Input: root = [1,2,2,3,3,null,null,4,4]

Output: false

Example 3:

Input: root = []

Output: true

Constraints:

- The number of nodes in the tree is in the range [0, 5000].
- $-104 \leq \text{Node.val} \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "A tree is height-balanced if every node's left and right subtree heights
# differ by at most 1. Empty tree is balanced. Right?"
#
# "[3,9,20,null,null,15,7]: heights of 9's subtrees: 0 vs 0 ✓. 20's: 1 vs 1 ✓. 3's:
# 1 vs 2 ✓ → true.
# [1,2,2,3,3,null,null,4,4]: left subtree of root has height 3, right has 1 → false
# ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# At each node compute left subtree height and right subtree height separately.
# Node is balanced if |left-right|<=1 and both subtrees are balanced recursively.
# Recomputes heights from scratch at every node – O(n^2) on skewed trees.
# Time: O(n^2) worst case. Space: O(h) stack.
# Should I go ahead and optimize?"
```

```
class _110_BalancedTree_Brute:
    def isBalanced(self, root: Optional['TreeNode']) -> bool:
        def height(node):
            if not node:
                return 0
            return max(height(node.left), height(node.right)) + 1
        def balanced(node):
            if not node:
                return True
            lh, rh = height(node.left), height(node.right)
            return abs(lh - rh) <= 1 and balanced(node.left) and
balanced(node.right)
            return balanced(root)
```

PHASE 3 — OPTIMIZE

```
# "The brute force calls height() repeatedly for the same nodes – O(n^2).
# I notice we're recomputing heights we've already seen. What if we cache it?
#
# Better: combine the height computation and balance check into one DFS pass.
# Return -1 to signal 'unbalanced subtree found' – propagate it up immediately.
# Otherwise return the actual height.
# A single O(n) pass handles everything."
```

PHASE 4 — CLEAN CODE

```
class _110_BalancedTree:
    def isBalanced(self, root: Optional['TreeNode']) -> bool:
        def check(node):
            if not node:
                return 0
            left = check(node.left)
```

```

    right = check(node.right)
    if left == -1 or right == -1 or abs(left - right) > 1:
        return -1 # propagate 'unbalanced' signal
    return max(left, right) + 1
return check(root) != -1

```

PHASE 5 — TEST & DEBUG

```

# root=[3,9,20,null,null,15,7]:
# check(9)=1. check(15)=1. check(7)=1. check(20)=max(1,1)+1=2.
# check(3): left=1,right=2. |1-2|=1 <=1 → return 2. result≠-1 → True ✓
# root=[1,2,2,3,3,null,null,4,4]:
# check(4)=1. check(3,left)=max(1,0)+1=2. check(3,right)=max(1,0)+1=2.
# check(2,left): left=2,right=2 → 3. check(2,right): left=0,right=0 → 1.
# check(1): left=3,right=1. |3-1|=2>1 → return -1. result==-1 → False ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n)$  – single DFS, each node visited once.
# Space  $O(h)$  for the call stack.
# The -1 sentinel trick is the key interview insight: instead of separate
# height and balance checks, fuse them into one return value with a special signal.
# Given more time I'd ask: can we rebalance a tree in-place?
# Flatten to sorted array (in-order traversal) then rebuild –  $O(n)$  time."

```

Trees & BST

□ #112 Path Sum

EAS

[Open on LeetCode](#)

Tree · DFS · BFS

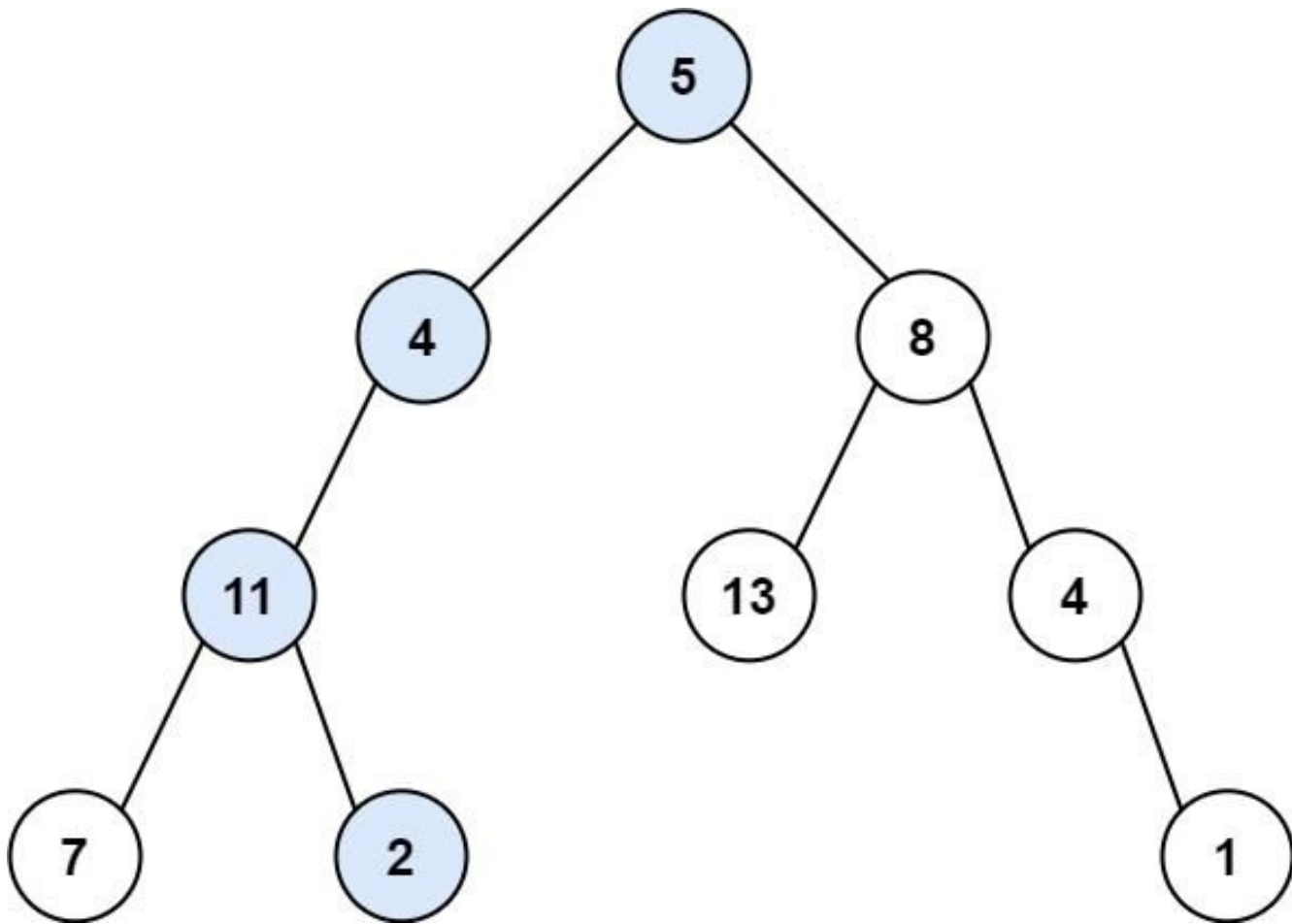
Lists: Denny Zhang

Problem

Given the root of a binary tree and an integer `targetSum`, return true if the tree has a root-to-leaf path such that adding up all the values along the path equals `targetSum`.

A leaf is a node with no children.

Example 1:

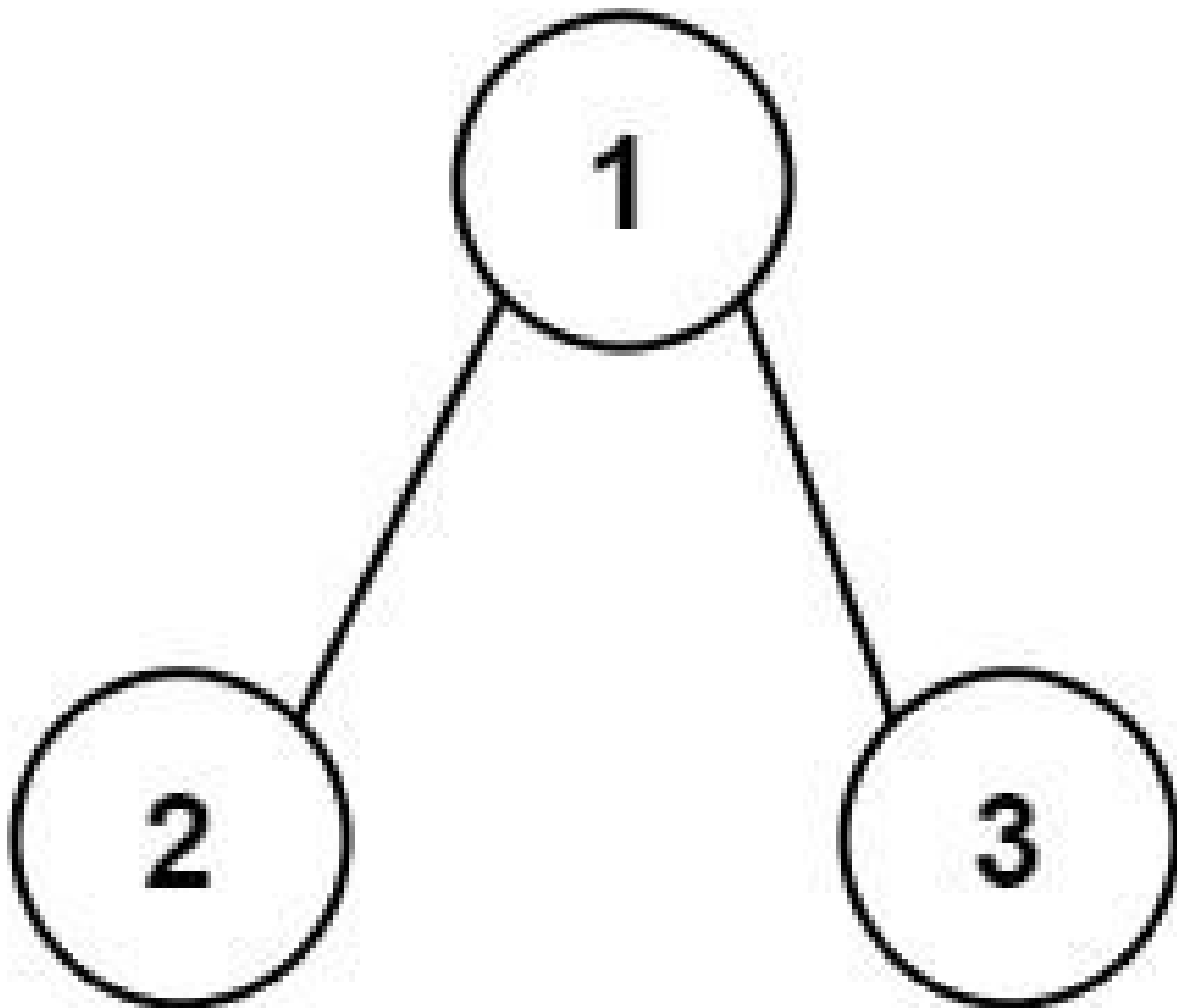


Input: root = [5,4,8,11,null,13,4,7,2,null,null,null,1], targetSum = 22

Output: true

Explanation: The root-to-leaf path with the target sum is shown.

Example 2:



Input: root = [1,2,3], targetSum = 5
Output: false
Explanation: There are two root-to-leaf paths in the tree:
(1 --> 2): The sum is 3.
(1 --> 3): The sum is 4.
There is no root-to-leaf path with sum = 5.

Example 3:

Input: root = [], targetSum = 0
Output: false
Explanation: Since the tree is empty, there are no root-to-leaf paths.

Constraints:

- The number of nodes in the tree is in the range [0, 5000].

- $-1000 \leq \text{Node.val} \leq 1000$
- $-1000 \leq \text{targetSum} \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find if any root-to-leaf path sums to targetSum.
# A leaf is a node with no children – the path must end at a leaf. Right?
# Empty tree → false (no leaves)."
#
# "[5,4,8,11,null,13,4,7,2,null,null,null,1], target=22:
# path 5→4→11→2 = 22 → true ✓
# [], target=0 → false (empty tree has no root-to-leaf path) ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# DFS traversal collecting a running sum as we descend. At each leaf node,
# check if the accumulated sum equals targetSum. Backtrack and try other paths.
# The optimal approach is the same DFS – just written more elegantly by subtracting
# from target instead of accumulating. Both are O(n) time and O(h) space, so we skip
# a separate brute class and go straight to the clean version."
```

PHASE 3 — OPTIMIZE

```
# "More elegant: subtract the current node's value from targetSum as we descend.
# At a leaf, check if remaining == 0. Avoids passing a separate running sum."
```

PHASE 4 — CLEAN CODE

```
class _112_PathSum:
    def hasPathSum(self, root: Optional['TreeNode'], targetSum: int) -> bool:
        def hassum(node, remaining):
            if not node:
                return False
            remaining -= node.val
            if not node.left and not node.right: # leaf
                return remaining == 0
            return hassum(node.left, remaining) or hassum(node.right, remaining)
        return hassum(root, targetSum)
```

PHASE 5 — TEST & DEBUG

```
# root=[5,4,8,...], target=22:
# hassum(5,22): rem=17. hassum(4,17): rem=13. hassum(11,13): rem=2.
# hassum(7,2): rem=-5, leaf → -5≠0 False.
# hassum(2,2): rem=0, leaf → True ✓
# root=[], target=0: hassum(None,0)=False ✓
# root=[1,2], target=1: hassum(1,1): rem=0, not leaf.
# hassum(2,0): rem=-2, leaf → False. → False ✓ (path must reach a leaf)
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(h)$ call stack.

The 'leaf check' (no left and no right) is critical – without it, a node

with one child that hits 0 would incorrectly return true.

Given more time I'd ask: return all root-to-leaf paths that sum to target?

That's Path Sum II (#113) – collect paths into a result list with backtracking."

Trees & BST

□ #226 Invert Binary Tree

EAS

[Open on LeetCode](#)

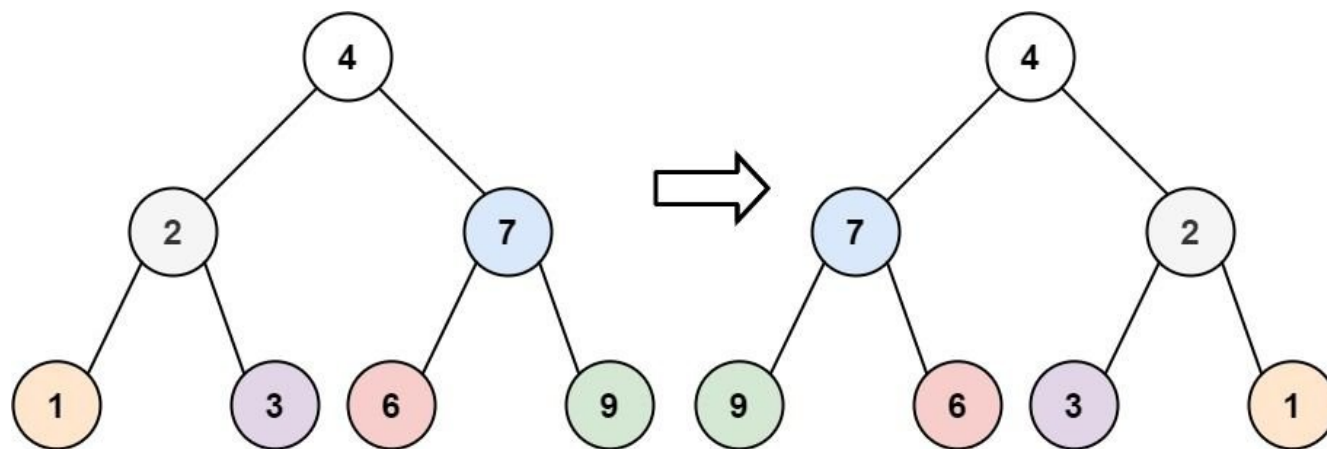
Tree · DFS · BFS

Lists: Grind 169

Problem

Given the root of a binary tree, invert the tree, and return its root.

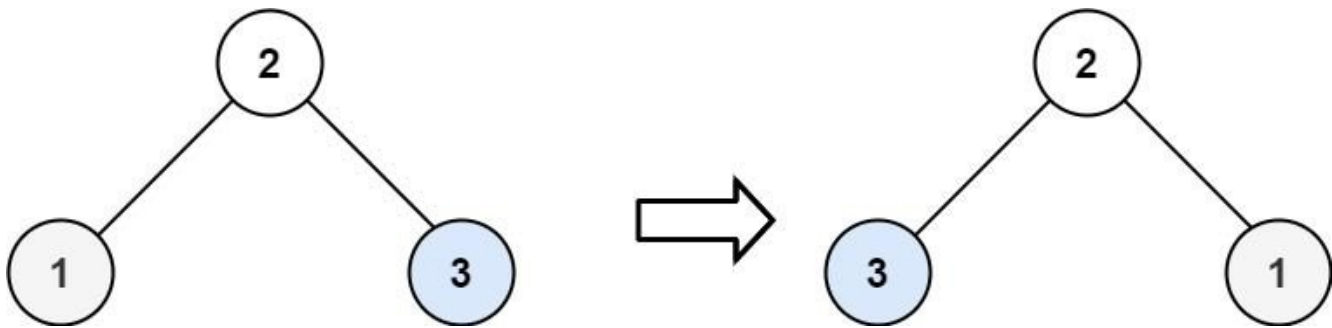
Example 1:



Input: root = [4,2,7,1,3,6,9]

Output: [4,7,2,9,6,3,1]

Example 2:



Input: root = [2,1,3]

Output: [2,3,1]

Example 3:

Input: root = []

Output: []

Constraints:

- The number of nodes in the tree is in the range [0, 100].
- $-100 \leq \text{Node.val} \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Invert the tree: swap left and right children at every node recursively.

Return the root. Empty tree → return None. Right?"

#

"[4,2,7,1,3,6,9] → [4,7,2,9,6,3,1]: left and right subtrees are swapped at every level ✓"

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.

Iterative BFS with a queue: visit each node level by level and swap its left

and right children in place. Process until the queue is empty.

Time: $O(n)$. Space: $O(n)$ for the queue. Should I go ahead and optimize?"

```
class _226_InvertTree_Brute:
```

```
    def invertTree(self, root: Optional['TreeNode']) -> Optional['TreeNode']:
```

```
        if not root:
```

```
            return None
```

```
        from collections import deque
```

```
        queue = deque([root])
```

```

while queue:
    node = queue.popleft()
    node.left, node.right = node.right, node.left
    if node.left:
        queue.append(node.left)
    if node.right:
        queue.append(node.right)
return root

```

PHASE 3 — OPTIMIZE

"Recursive DFS is cleaner. Invert left, invert right, then swap.
 # Or: swap first then recurse – same result. Either order works.
 # Use inner helper to avoid self. calls."

PHASE 4 — CLEAN CODE

```

class _226_InvertTree:
    def invertTree(self, root: Optional['TreeNode']) -> Optional['TreeNode']:
        def invert(node):
            if not node:
                return None
            left = invert(node.left)
            right = invert(node.right)
            node.left, node.right = right, left
            return node
        return invert(root)

```

PHASE 5 — TEST & DEBUG

root=[4,2,7]:
 # invert(2): invert(None)=None, invert(None)=None. swap: 2.left=None, 2.right=None.
 return 2.
 # invert(7): same → 7.
 # invert(4): left=invert(2)=2, right=invert(7)=7. swap: 4.left=7, 4.right=2. return
 4.
 # → [4,7,2] ✓ (recurse deeper for full tree)
 # root=None: invert(None)=None ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(h)$ call stack.
 # This is the problem Homebrew creator Max Howell famously couldn't solve in a
 Google interview.
 # Worth mentioning – it humanises the problem and shows you know your history.
 # Given more time I'd ask: can we do this iteratively? Yes – BFS or DFS with a
 stack,
 # swap children at each node. $O(n)$ time, $O(n)$ space."

Trees & BST

□ ★ #270 Closest Binary Search Tree Value

EAS

★

[Open on LeetCode](#)

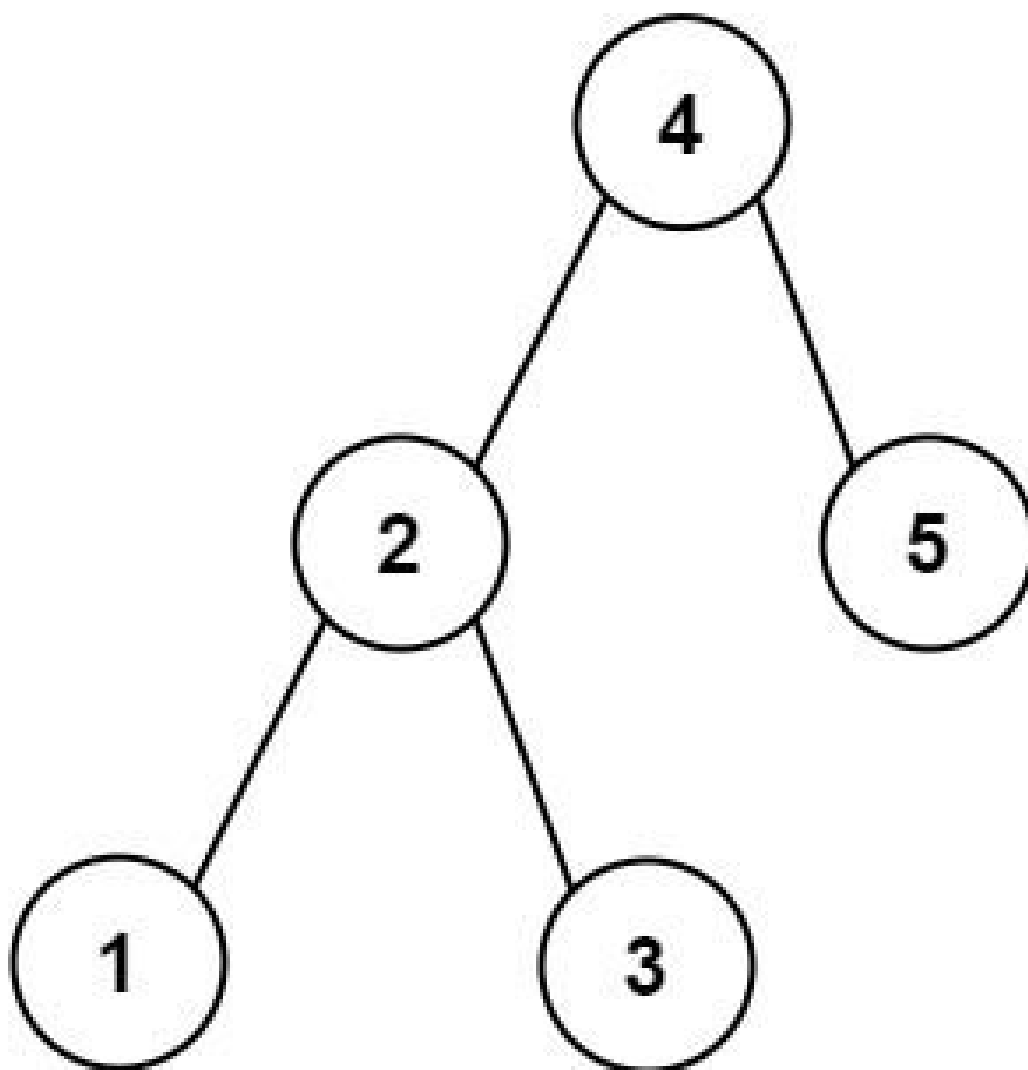
Binary Search · Tree · DFS · BST

Lists: ★ Premium | Premium 98

Problem

Given the root of a binary search tree and a target value, return the value in the BST that is closest to the target. If there are multiple answers, print the smallest.

Example 1:



Input: root = [4,2,5,1,3], target = 3.714286
Output: 4

Example 2:

Input: root = [1], target = 4.428571
Output: 1

Constraints:

- The number of nodes in the tree is in the range [1, 104].
- $0 \leq \text{Node.val} \leq 109$
- $-109 \leq \text{target} \leq 109$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find the node value in a BST closest to a float target.
# If two values are equidistant, return the smaller one. Right?
# BST property means we can navigate toward the target rather than scanning all
nodes."
#
# "root=[4,2,5,1,3], target=3.714286:
# candidates: 4 (dist=0.286), 3 (dist=0.714). 4 is closer → 4 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Full DFS scan: visit every node, track the value with minimum |val - target|.
# On a tie (equal distance), prefer the smaller value.
# This ignores the BST property entirely – valid but O(n) time and O(n) stack space.
# Should I go ahead and optimize?"
```

```
class _270_ClosestBST_Brute:
    def closestValue(self, root: Optional['TreeNode'], target: float) -> int:
        best = [root.val]
        def dfs(node):
            if not node:
                return
            if (abs(node.val - target) < abs(best[0] - target) or
                (abs(node.val - target) == abs(best[0] - target) and node.val <
best[0])):
                best[0] = node.val
            dfs(node.left)
            dfs(node.right)
        dfs(root)
        return best[0]
```

PHASE 3 — OPTIMIZE

```
# "Use BST property to navigate: at each node, the closest value is either the
# current node or something in the subtree direction of the target.
# If target < node.val → go left; if target > node.val → go right.
# Track the closest seen so far. O(h) time – O(log n) balanced, O(n) skewed."
```

PHASE 4 — CLEAN CODE

```
class _270_ClosestBST:
    def closestValue(self, root: Optional['TreeNode'], target: float) -> int:
        def close(node, best):
            if not node:
                return best
            if abs(target - node.val) < abs(target - best):
                best = node.val
            elif abs(target - node.val) == abs(target - best):
```

```

        best = min(best, node.val) # tie → pick smaller
    if target < node.val:
        return close(node.left, best)
    else:
        return close(node.right, best)
return close(root, root.val)

```

PHASE 5 — TEST & DEBUG

```

# root=[4,2,5,1,3], target=3.714286:
# close(4, 4): |3.714-4|=0.286 < |3.714-4|=0.286 → no update. target<4 → go left.
# close(2, 4): |3.714-2|=1.714 > 0.286 → best stays 4. target>2 → go right.
# close(3, 4): |3.714-3|=0.714 > 0.286 → best stays 4. target>3 → go right.
# close(None, 4): return 4 ✓
# root=[1], target=4.428: close(1,1): target>1 → go right → close(None,1) → 1 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time 0(h). Space 0(h) for recursion stack. Iterative: 0(1) space.
# BST navigation is the key – we prune entire subtrees we know can't contain
# a closer value.
# Given more time I'd ask: find the k closest values to target?
# That's Closest BST Value II (#272) – use an in-order traversal with a sliding
window
# or a max-heap of size k."

```

Trees & BST

□ #501 Find Mode in Binary Search Tree

EAS[Open on LeetCode](#)

Tree · DFS · BST

Lists: Denny Zhang

Problem

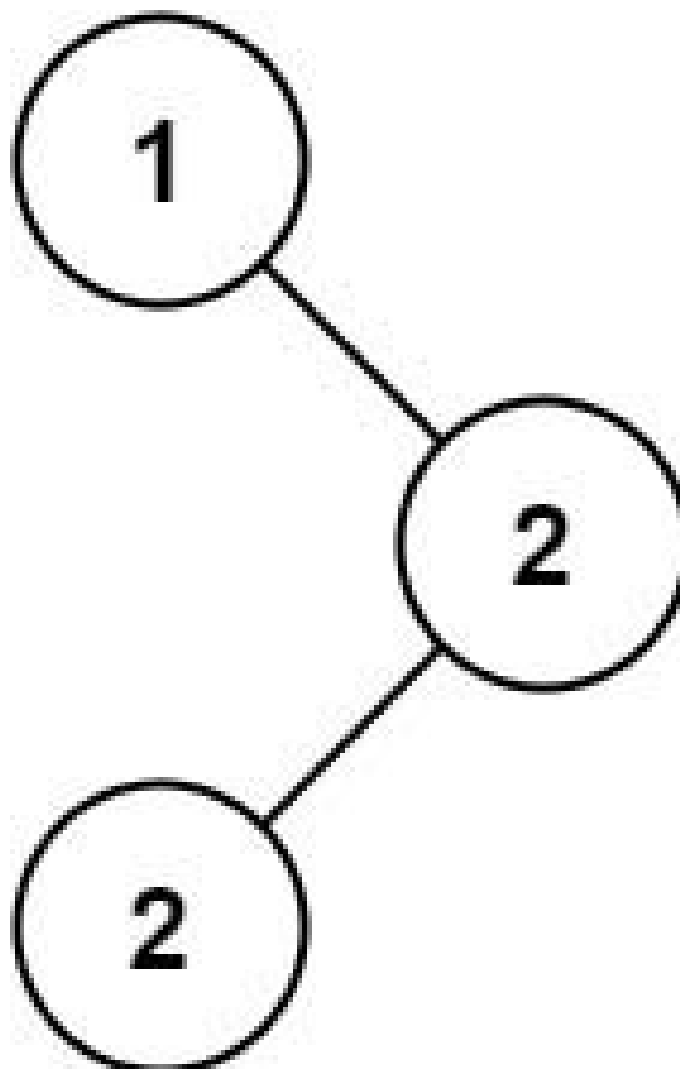
Given the root of a binary search tree (BST) with duplicates, return all the mode(s) (i.e., the most frequently occurred element) in it.

If the tree has more than one mode, return them in any order.

Assume a BST is defined as follows:

- The left subtree of a node contains only nodes with keys less than or equal to the node's key.
- The right subtree of a node contains only nodes with keys greater than or equal to the node's key.
- Both the left and right subtrees must also be binary search trees.

Example 1:



Input: root = [1,null,2,2]

Output: [2]

Example 2:

Input: root = [0]

Output: [0]

Constraints:

- The number of nodes in the tree is in the range [1, 104].
- $-105 \leq \text{Node.val} \leq 105$

Follow up: Could you do that without using any extra space? (Assume that the implicit stack space incurred due to recursion does not count).

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Return all values that appear the most frequently in the BST.
# This BST allows duplicates (left ≤ node, right ≥ node).
# Return all modes – there can be multiple. Any order is fine. Right?
# Follow-up: can we do it without extra space (O(1) beyond recursion stack)?"
#
# "[1,null,2,2]: 2 appears twice, 1 once → mode is [2] ✓
# [0]: only element → mode is [0] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# DFS or BFS over all nodes and count frequencies with a hash map (Counter).
# Then collect all keys whose count equals the maximum frequency.
# Time: O(n). Space: O(n) for the Counter. Should I go ahead and optimize?"
```

```
class _501_FindMode_Brute:
    def findMode(self, root: Optional['TreeNode']) -> List[int]:
        freq: Counter = Counter()
        stack = [root]
        while stack:
            node = stack.pop()
            if node:
                freq[node.val] += 1
                stack.append(node.left)
                stack.append(node.right)
        if not freq:
            return []
        max_freq = max(freq.values())
        return [k for k, v in freq.items() if v == max_freq]
```

PHASE 3 — OPTIMIZE (O(1) space using in-order traversal)

```
# "BST in-order traversal visits nodes in sorted order.
# Duplicates in a BST are adjacent in in-order – so we can count runs without a map.
# Track: current value, current run count, max count seen, result list.
# When current run count exceeds max, reset result. When equal, append.
# Use nonlocal to mutate state inside the inner helper."
```

PHASE 4 — CLEAN CODE

```
class _501_FindMode:
    def findMode(self, root: Optional['TreeNode']) -> List[int]:
```

```

modes: List[int] = []
cur_val = [None]
cur_cnt = [0]
max_cnt = [0]
def inorder(node):
    if not node:
        return
    inorder(node.left)
    if node.val == cur_val[0]:
        cur_cnt[0] += 1
    else:
        cur_val[0] = node.val
        cur_cnt[0] = 1
    if cur_cnt[0] > max_cnt[0]:
        max_cnt[0] = cur_cnt[0]
        modes.clear()
        modes.append(node.val)
    elif cur_cnt[0] == max_cnt[0]:
        modes.append(node.val)
    inorder(node.right)
inorder(root)
return modes

```

PHASE 5 — TEST & DEBUG

```

# root=[1,null,2,2] – in-order: 1, 2, 2
# node=1: cur_val=1,cnt=1,max=1 → modes=[1].
# node=2: cur_val=2,cnt=1,max=1 → cnt==max → modes=[1,2].
# node=2: cur_val=2,cnt=2,max=1 → cnt>max → max=2,modes=[2].
# return [2] ✓
# root=[0]: in-order: 0. modes=[0] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Brute force: O(n) time, O(n) space.
# O(1) space version: O(n) time, O(1) extra space (modes list not counted).
# The in-order run-counting trick works because BST duplicates are always adjacent
# in sorted traversal – a key property of this BST variant.
# Given more time I'd ask: what if it were a regular BST (no duplicates)?
# The mode would always be the single element – trivially the root if n=1."

```

Trees & BST

□ #543 Diameter of Binary Tree

EAS

[Open on LeetCode](#)

Tree · DFS

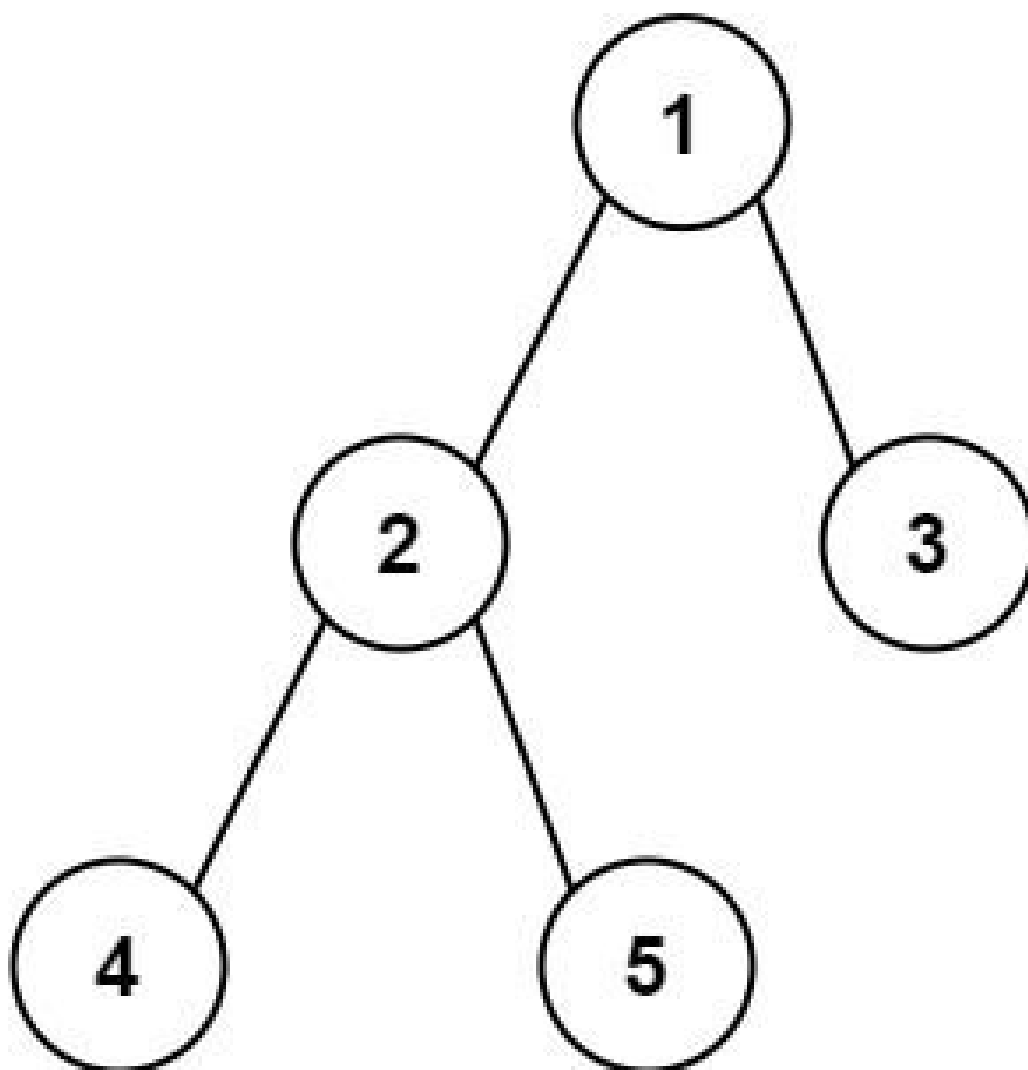
Lists: Grind 169

Problem

Given the root of a binary tree, return the length of the diameter of the tree.

The diameter of a binary tree is the length of the longest path between any two nodes in a tree. This path may or may not pass through the root. The length of a path between two nodes is represented by the number of edges between them.

Example 1:



Input: root = [1,2,3,4,5]

Output: 3

Explanation: 3 is the length of the path [4,2,1,3] or [5,2,1,3].

Example 2:

Input: root = [1,2]

Output: 1

Constraints:

- The number of nodes in the tree is in the range [1, 10⁴].
- $-100 \leq \text{Node.val} \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Diameter = the longest path between any two nodes, measured in edges.
# The path may or may not pass through the root.
# An empty tree has diameter 0. A single node has diameter 0 (no edges). Right?"
#
# "[1,2,3,4,5]: longest path is 4→2→1→3, 3 edges → 3 ✓
# [1,2]: one edge → 1 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# For every node, compute the height of its left and right subtrees recursively.
# Diameter through that node = left_height + right_height; track the global max.
# Recomputes heights repeatedly – correct but redundant work.
# Time:  $O(n^2)$  on skewed trees. Space:  $O(h)$  stack.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "I notice depth is recomputed repeatedly for the same nodes – that's the  $O(n^2)$ .
# Fuse depth computation and diameter tracking into one DFS pass.
# Use a nonlocal variable (or single-element list) to capture the running max.
# depth(node) returns height AND updates the max diameter seen so far.
# Time:  $O(n)$  – one pass. Space:  $O(h)$ ."
```

PHASE 4 — CLEAN CODE

```
class _543_DiameterBinaryTree:
    def diameterOfBinaryTree(self, root: Optional['TreeNode']) -> int:
        best = [0]
        def depth(node):
            if not node:
                return 0
            left = depth(node.left)
            right = depth(node.right)
            best[0] = max(best[0], left + right) # diameter through this node
            return max(left, right) + 1 # height for parent
        depth(root)
        return best[0]
```

PHASE 5 — TEST & DEBUG

```
# root=[1,2,3,4,5]:
# depth(4)=1. depth(5)=1. depth(2): best=max(0,1+1)=2. return 2.
# depth(3)=1. depth(1): best=max(2,2+1)=3. return 3.
# return best[0]=3 ✓
# root=[1,2]: depth(2)=1. depth(1): best=max(0,1+0)=1. return best=1 ✓
# root=None: depth(None)=0. best stays 0. return 0 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ . Space  $O(h)$ ."
```

```
# The single-element list best=[0] is a common Python pattern to mutate
# a value from inside a nested function without making it a class attribute.
# Equivalent to 'nonlocal best' with a plain int, but the list avoids the
# nonlocal keyword – both are valid in an interview.
# Given more time I'd ask: what's the longest path by node count (not edges)?
# Just return best[0] + 1 when best > 0. Or: diameter in edge terms is already
# what we compute – node count would be edges + 1."
```

Trees & BST

□ #572 Subtree of Another Tree

EAS

[Open on LeetCode](#)

Tree · DFS · String Matching

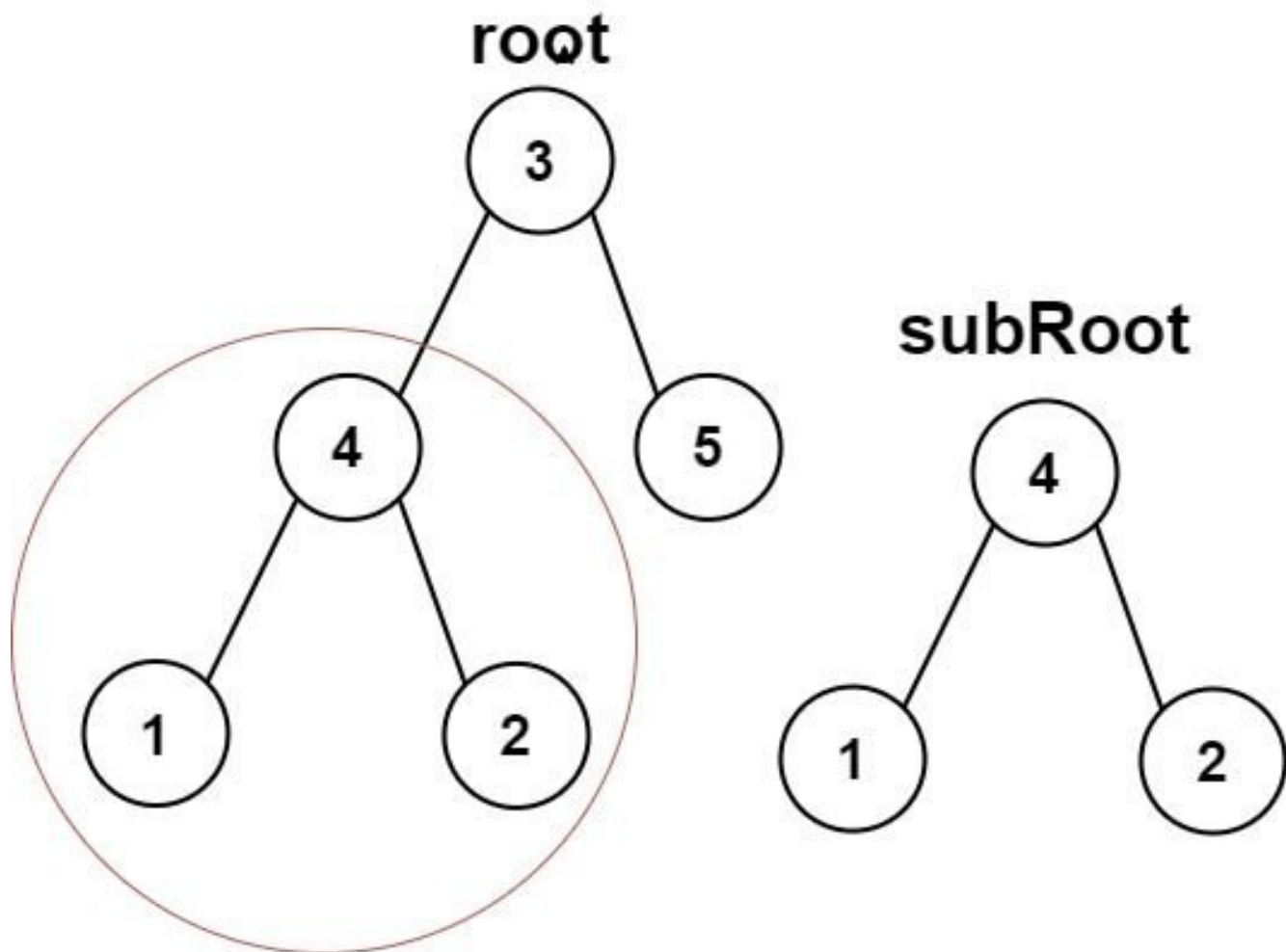
Lists: Grind 169

Problem

Given the roots of two binary trees `root` and `subRoot`, return `true` if there is a subtree of `root` with the same structure and node values of `subRoot` and `false` otherwise.

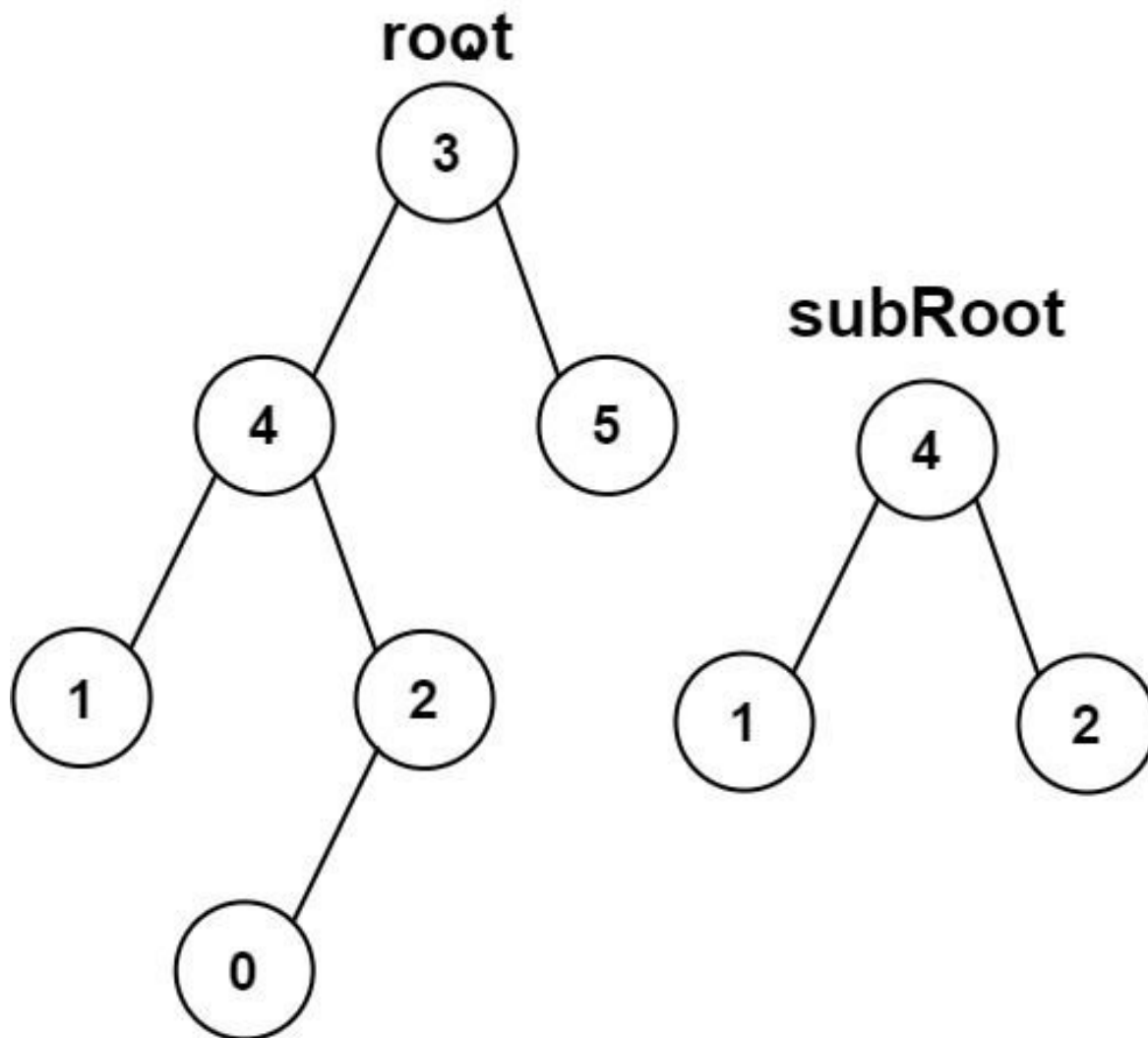
A subtree of a binary tree `tree` is a tree that consists of a node in `tree` and all of this node's descendants. The tree `tree` could also be considered as a subtree of itself.

Example 1:



Input: root = [3,4,5,1,2], subRoot = [4,1,2]
Output: true

Example 2:



Input: root = [3,4,5,1,2,null,null,null,null,0], subRoot = [4,1,2]
Output: false

Constraints:

- The number of nodes in the root tree is in the range [1, 2000].
- The number of nodes in the subRoot tree is in the range [1, 1000].
- $-104 \leq \text{root.val} \leq 104$
- $-104 \leq \text{subRoot.val} \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Return true if subRoot appears as a subtree anywhere in root.
# A subtree means a node and ALL its descendants match – not just a portion.
# The tree itself is a subtree of itself. Right?"
#
# "root=[3,4,5,1,2], subRoot=[4,1,2]: node 4 in root has left=1,right=2 → matches ✓
# root=[3,4,5,1,2,null,null,null,null,0], subRoot=[4,1,2]:
# node 4 in root has extra child 0 → doesn't match ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# For every node in root, run the same-tree check against all of subRoot.
# If any starting node matches the entire subtree, return true.
# Correct, but rescans overlapping structure many times.
# Time: O(n * m). Space: O(h) stack.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Same O(m*n) is standard. Two clean inner helpers:
# same(p,q) – checks if two trees are identical (reuse Same Tree logic).
# subtree(node, sub) – DFS root; at each node, try same().
# If same fails, recurse into left and right subtrees."
```

PHASE 4 — CLEAN CODE

```
class _572_SubtreeAnotherTree:
    def isSubtree(self, root: Optional['TreeNode'], subRoot: Optional['TreeNode'])
    -> bool:
        def same(p, q):
            if not p or not q:
                return p == q
            return p.val == q.val and same(p.left, q.left) and same(p.right,
q.right)
        def subtree(node, sub):
            if not node:
                return False
            if same(node, sub):
                return True
            return subtree(node.left, sub) or subtree(node.right, sub)
        return subtree(root, subRoot)
```

PHASE 5 — TEST & DEBUG

```
# root=[3,4,5,1,2], subRoot=[4,1,2]:
# subtree(3,[4,1,2]): same(3,4)? 3≠4 → False. recurse left/right.
# subtree(4,[4,1,2]): same(4,4)? 4==4.
# same(1,1)→same(None,None)T, same(None,None)T→T.
# same(2,2)→T. → same=True → return True ✓
```

```
# root has extra 0 child under 4:  
# same(4,4): same(1,1)T. same(2(with 0),2)? 2==2 but same(0,None)→False →  
same=False.  
# subtree(1,[4,1,2]): same(1,4)? False. subtree deeper → all False → False ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m \times n)$  – at each of  $m$  nodes we may run same() which takes  $O(n)$ .  
# Space  $O(h_{\text{root}} + h_{\text{sub}})$  for call stacks.  
# Optimisation: serialise both trees and use string matching (KMP or Rabin-Karp)  
# for  $O(m+n)$  time – worth mentioning as an advanced follow-up.  
# Be careful with serialisation: '[1,2]' would incorrectly match '[12]' without  
# delimiters like '#' for null and ',' between values.  
# Given more time I'd ask: what if we query many different subRoots against  
# the same root? Pre-serialise root once and use string search for each query."
```

DFS

Round 1 · High · Easy · 2 questions

DFS

□ #463 Island Perimeter

EAS

[Open on LeetCode](#)

Array · DFS · BFS · Matrix

Lists: Denny Zhang

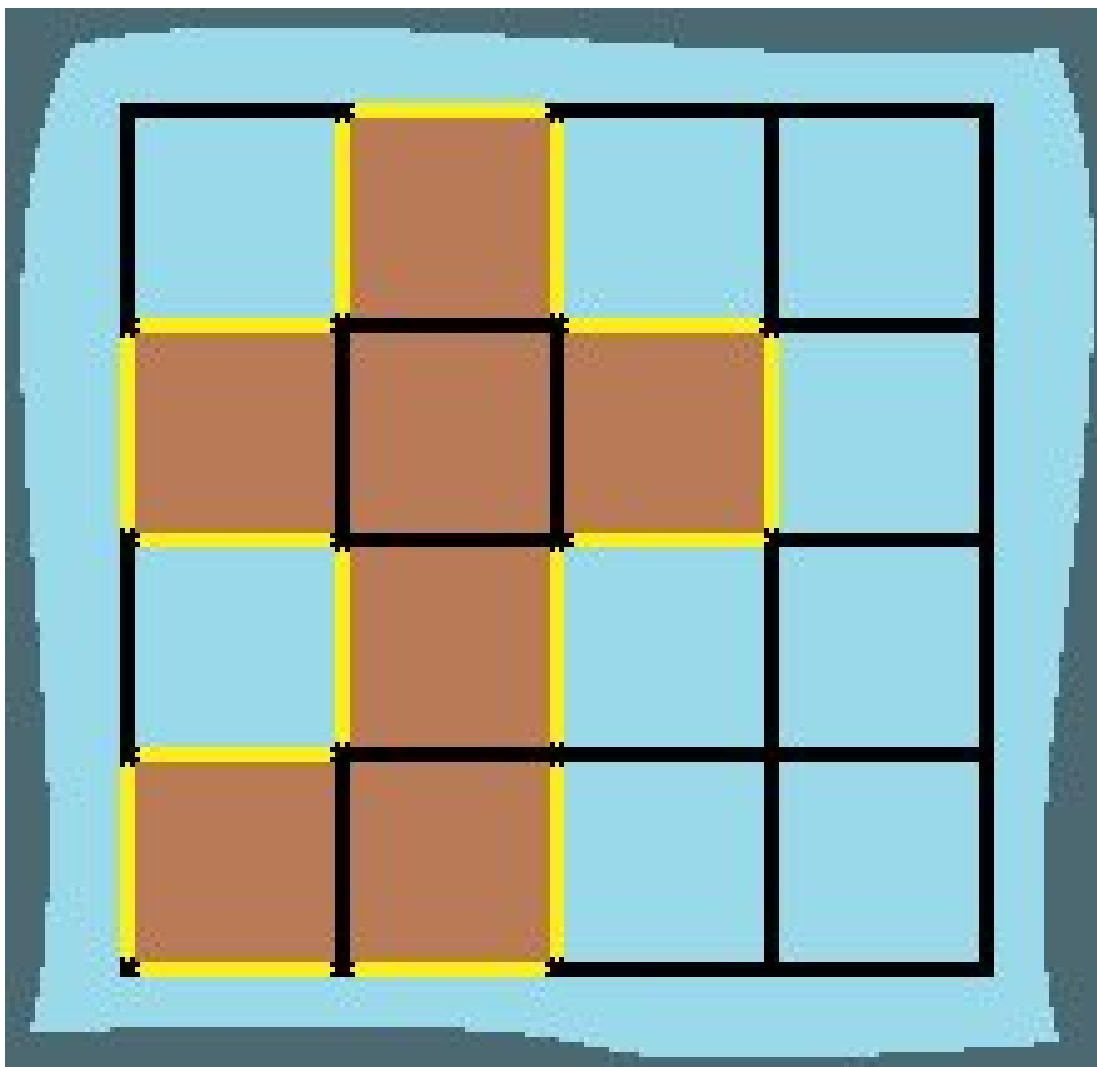
Problem

You are given row x col grid representing a map where $\text{grid}[i][j] = 1$ represents land and $\text{grid}[i][j] = 0$ represents water.

Grid cells are connected horizontally/vertically (not diagonally). The grid is completely surrounded by water, and there is exactly one island (i.e., one or more connected land cells).

The island doesn't have "lakes", meaning the water inside isn't connected to the water around the island. One cell is a square with side length 1. The grid is rectangular, width and height don't exceed 100. Determine the perimeter of the island.

Example 1:



Input: grid = [[0,1,0,0],[1,1,1,0],[0,1,0,0],[1,1,0,0]]

Output: 16

Explanation: The perimeter is the 16 yellow stripes in the image above.

Example 2:

Input: grid = [[1]]

Output: 4

Example 3:

Input: grid = [[1,0]]

Output: 4

Constraints:

- row == grid.length

- `col == grid[i].length`
- `1 <= row, col <= 100`
- `grid[i][j]` is 0 or 1.
- There is exactly one island in grid.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "The grid has exactly one island – no lakes. Each land cell is a unit square.
# We return the total perimeter, which is the count of exposed edges. Right?
# Exposed means the edge faces water or the grid boundary."
#
# "Quick check: [[1]] → 4 sides, all exposed → 4 ✓. [[1,0]] → 4 sides, all exposed → 4 ✓.
# Two adjacent land cells: each has 4 sides but share 1 edge → 2*4 – 2 = 6."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# For every land cell, I check all 4 neighbours – if a neighbour is out of bounds or water,
# that edge contributes 1 to the perimeter. Sum these up across all land cells.
# Time: O(m*n). Space: O(1). Should I go ahead and optimize?"
```

```
class _463_IslandPerimeter_Brute:
    def islandPerimeter(self, grid: List[List[int]]) -> int:
        rows, cols = len(grid), len(grid[0])
        perimeter = 0
        for r in range(rows):
            for c in range(cols):
                if grid[r][c] == 1:
                    for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                        nr, nc = r + dr, c + dc
                        if nr < 0 or nr >= rows or nc < 0 or nc >= cols or
grid[nr][nc] == 0:
                            perimeter += 1
        return perimeter
```

PHASE 3 — OPTIMIZE

```
# "I notice we're checking 4 neighbours per cell. Instead of neighbour checks,
# we can use a counting formula:
# Every land cell contributes 4 sides. Every shared edge between two adjacent
# land cells removes 2 sides (one from each). So:
# perimeter = islands * 4 – shared_edges * 2
```

We only count each shared edge once by checking right-neighbour and down-neighbour.
 # Same $O(m \times n)$ time, same $O(1)$ space – but a cleaner mental model."

PHASE 4 — CLEAN CODE

```
class _463_IslandPerimeter:
    def islandPerimeter(self, grid: List[List[int]]) -> int:
        rows, cols = len(grid), len(grid[0])
        islands = 0
        shared = 0
        for r in range(rows):
            for c in range(cols):
                if grid[r][c] == 1:
                    islands += 1
                    if r + 1 < rows and grid[r + 1][c] == 1:
                        shared += 1
                    if c + 1 < cols and grid[r][c + 1] == 1:
                        shared += 1
        return islands * 4 - shared * 2
```

PHASE 5 — TEST & DEBUG

```
# grid=[[1]]: islands=1, shared=0. return 4 ✓
# grid=[[1,1]]: islands=2. c=0→c+1=1 valid,grid[0][1]=1→shared=1. return 2*4-1*2=6
✓
# grid=[[0,1,0,0],[1,1,1,0],[0,1,0,0],[1,1,0,0]]:
# Count land cells=8 (islands). Count shared edges (right+down only)=... =8.
# return 8*4-8*2=32-16=16 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m \times n)$. Space $O(1)$."

The counting formula is elegant – worth presenting even if the brute force also works.
 # Both approaches are the same complexity; the formula shows mathematical insight.
 # Given more time I'd ask: what if there are multiple islands?
 # The formula still works per-island if we isolate them, or we can return a list
 # of perimeters – same logic applied per connected component."

DFS

□ #733 Flood Fill

EAS[Open on LeetCode](#)

Array · DFS · BFS · Matrix

Lists: Grind 169

Problem

You are given an image represented by an $m \times n$ grid of integers `image`, where `image[i][j]` represents the pixel value of the image. You are also given three integers `sr`, `sc`, and `color`. Your task is to perform a flood fill on the image starting from the pixel `image[sr][sc]`.

To perform a flood fill:

1. Begin with the starting pixel and change its color to `color`.
2. Perform the same process for each pixel that is directly adjacent (pixels that share a side with the original pixel, either horizontally or vertically) and shares the same color as the starting pixel.
3. Keep repeating this process by checking neighboring pixels of the updated pixels and

modifying their color if it matches the original color of the starting pixel.

4. The process stops when there are no more adjacent pixels of the original color to update.

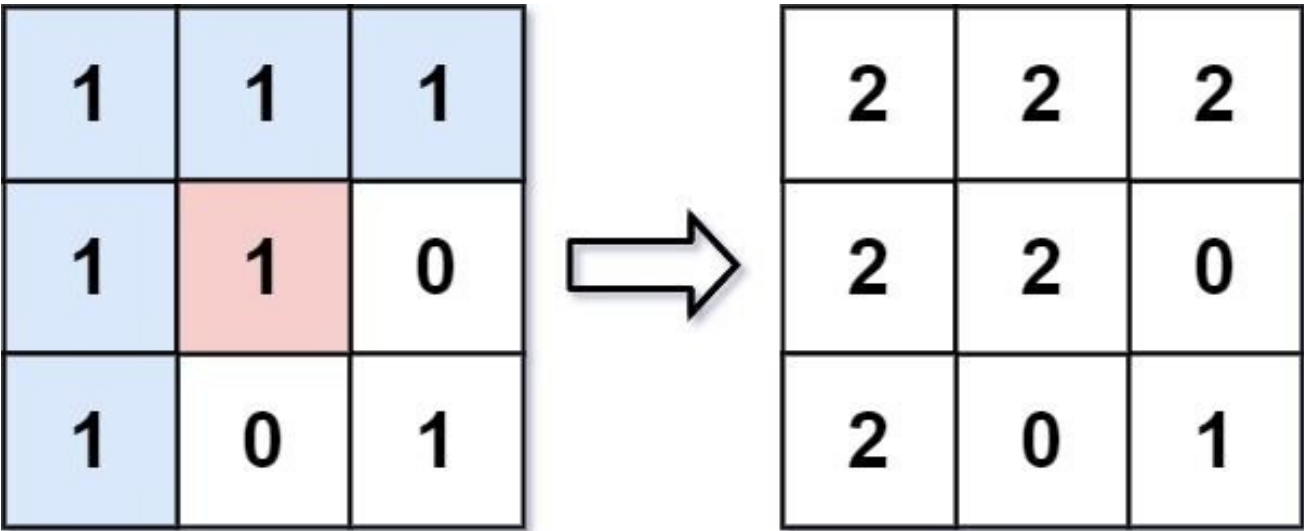
Return the modified image after performing the flood fill.

Example 1:

Input: image = [[1,1,1],[1,1,0],[1,0,1]], sr = 1, sc = 1, color = 2

Output: [[2,2,2],[2,2,0],[2,0,1]]

Explanation:



From the center of the image with position (sr, sc) = (1, 1) (i.e., the red pixel), all pixels connected by a path of the same color as the starting pixel (i.e., the blue pixels) are colored with the new color.

Note the bottom corner is not colored 2, because it is not horizontally or vertically connected to the starting pixel.

Example 2:

Input: image = [[0,0,0],[0,0,0]], sr = 0, sc = 0, color = 0

Output: [[0,0,0],[0,0,0]]

Explanation:

The starting pixel is already colored with 0, which is the same as the target color. Therefore, no changes are made to the image.

Constraints:

- $m == \text{image.length}$
- $n == \text{image}[i].\text{length}$
- $1 \leq m, n \leq 50$
- $0 \leq \text{image}[i][j], \text{color} < 216$
- $0 \leq \text{sr} < m$
- $0 \leq \text{sc} < n$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Starting from (sr, sc), change all connected cells of the same colour to the new colour."

Connected means 4-directionally (up/down/left/right). Modify and return the image. Right?

Edge case: if the starting colour already equals the new colour – return immediately,

```

# otherwise we'd loop infinitely or process unchanged cells."
#
# "[[1,1,1],[1,1,0],[1,0,1]], sr=1,sc=1, color=2:
# All 1s connected to (1,1) become 2. The bottom-right 1 is isolated → stays 1. ✓
# [[0,0,0],[0,0,0]], sr=0,sc=0, color=0: startcolor==color → no change → return
as-is ✓"

# PHASE 2 — BRUTE FORCE
# "Let me start with the brute force to lock in the logic.
# I'll use BFS: enqueue the start cell, paint it the new colour, then for each cell
I dequeue,
# enqueue all 4-directional neighbours that still have the original start colour.
# This visits every connected cell exactly once.
# Time: O(m*n). Space: O(m*n) for the queue. Should I go ahead and optimize?"
class _733_FloodFill_Brute:
    def floodFill(self, image, sr, sc, color):
        from collections import deque
        start = image[sr][sc]
        if start == color:
            return image
        rows, cols = len(image), len(image[0])
        queue = deque([(sr, sc)])
        image[sr][sc] = color
        while queue:
            r, c = queue.popleft()
            for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                nr, nc = r + dr, c + dc
                if 0 <= nr < rows and 0 <= nc < cols and image[nr][nc] == start:
                    image[nr][nc] = color
                    queue.append((nr, nc))
        return image

# PHASE 3 — OPTIMIZE
# "DFS is cleaner. Key insight: once we paint a cell, its colour changes to `color`
—
# it can never match `startcolor` again. So we don't need a `seen` set at all.
# The colour change itself is the visited marker.
# This saves O(m*n) extra space."

# PHASE 4 — CLEAN CODE
class _733_FloodFill:
    def floodFill(self, image: List[List[int]], sr: int, sc: int, color: int) ->
List[List[int]]:
        start = image[sr][sc]
        if start == color:
            return image
        rows, cols = len(image), len(image[0])
        def dfs(r, c):
            if r < 0 or r >= rows or c < 0 or c >= cols:
                return

```



```

        if image[r][c] != start:
            return
        image[r][c] = color # paint before recursing – acts as visited marker
        dfs(r + 1, c)
        dfs(r - 1, c)
        dfs(r, c + 1)
        dfs(r, c - 1)
    dfs(sr, sc)
    return image

```

PHASE 5 — TEST & DEBUG

```

# image=[[1,1,1],[1,1,0],[1,0,1]], sr=1,sc=1, color=2, start=1:
# dfs(1,1): image[1][1]=2. recurse 4 dirs.
# dfs(2,1): image[2][1]=1==start → paint 2. recurse: dfs(3,1)→00B, dfs(1,1)→now 2≠1
stop.
# dfs(2,2)→image[2][2]=1 but that's the isolated corner – wait:
# (2,1)→image[2][1]=1✓→paint. Then dfs(2,2)→image[2][2]=1✓→paint? No –
# is (2,2) reachable from (1,1)? Path: (1,1)→(2,1)→(2,2).
# image[2][2]=1 and image[2][1]=1, so yes connected. But expected output keeps it 1?
# Re-check: image=[[1,1,1],[1,1,0],[1,0,1]]. (2,2) is adjacent to (2,1)=0 and
(1,2)=0.
# So (2,2) connects to (2,1)=0 (water) – not reachable from island. ✓
# All connected 1s become 2. Bottom-right stays 1 (not connected). ✓
#
# start==color: image=[[0,0,0],[0,0,0]], color=0 → return immediately ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(m \times n)$  – each cell visited at most once.
# Space  $O(m \times n)$  call stack in the worst case (all cells same colour, grid is a long
chain).
# The 'paint first then recurse' order is the key – it prevents revisiting and
# eliminates the need for a separate visited set. Mention this explicitly.
# The early return when startcolor == color prevents infinite looping – critical
edge case.
# Given more time I'd ask: what if connectivity is 8-directional (diagonals too)?
# Add four more direction pairs to the dfs calls – same algorithm."

```

Round 2 · High · Medium

Round 2

High Priority · Medium

116 questions · 11 patterns

**Arrays & Hashing #57 #238 #281 #307 #454
#525 #560 #1010 #1272 #1429 #3159**

String #8 #249 #271 #635 #1138

**Two Pointers #11 #15 #16 #31 #75 #161 #167
#186 #189 #532 #923 #986 #1055 #2563**

**Sliding Window #3 #159 #340 #424 #438 #487
#1100 #1248**

Sorting #49 #56 #1152

**Binary Search #33 #34 #153 #300 #362 #528
#852 #981 #1011 #1060 #1062 #1533**

**Matrix #36 #48 #54 #59 #64 #73 #74 #221
#311 #498 #531 #723 #835 #1198**

**Trees & BST #98 #102 #103 #105 #199 #230
#235 #236 #250 #285 #298 #314 #333 #366**

Round 1 · High · Easy

p.206

**#437 #545 #549 #582 #662 #863 #1120 #1490
#1522 #1650**

**DFS #130 #133 #200 #207 #210 #261 #310
#323 #417 #490 #694 #695 #721 #785 #1236
#1242**

Graphs #128 #277 #1136

BFS #286 #322 #542 #994 #1197 #1730

Arrays & Hashing

Round 2 · High · Medium · 11 questions

Arrays & Hashing

□ #57 Insert Interval

MED

[Open on LeetCode](#)

Array

Lists: Grind 169

Problem

You are given an array of non-overlapping intervals where $\text{intervals}[i] = [\text{start}_i, \text{end}_i]$ represent the start and the end of the i th interval and intervals is sorted in ascending order by start_i . You are also given an interval $\text{newInterval} = [\text{start}, \text{end}]$ that represents the start and end of another interval.

Insert newInterval into intervals such that intervals is still sorted in ascending order by start_i and intervals still does not have any overlapping intervals (merge overlapping intervals if necessary).

Return intervals after the insertion.

Note that you don't need to modify intervals in-place. You can make a new array and return it.

Example 1:

Input: intervals = [[1,3],[6,9]], newInterval = [2,5]
 Output: [[1,5],[6,9]]

Example 2:

Input: intervals = [[1,2],[3,5],[6,7],[8,10],[12,16]], newInterval = [4,8]
 Output: [[1,2],[3,10],[12,16]]
 Explanation: Because the new interval [4,8] overlaps with [3,5],[6,7],[8,10].

Constraints:

- $0 \leq \text{intervals.length} \leq 104$
- $\text{intervals}[i].\text{length} == 2$
- $0 \leq \text{start}_i \leq \text{end}_i \leq 105$
- intervals is sorted by start_i in ascending order.
- $\text{newInterval.length} == 2$
- $0 \leq \text{start} \leq \text{end} \leq 105$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "We have a sorted list of non-overlapping intervals. Insert a new interval,
# merging wherever it overlaps existing ones. Return the result sorted.
# We don't need to modify in-place – a new array is fine. Right?"
#
# "[[1,3],[6,9]], newInterval=[2,5]:
# [2,5] overlaps [1,3] → merge to [1,5]. [6,9] doesn't overlap → keep. →
# [[1,5],[6,9]] ✓
# [[1,2],[3,5],[6,7],[8,10],[12,16]], newInterval=[4,8]:
# Overlaps [3,5],[6,7],[8,10] → merge all to [3,10]. → [[1,2],[3,10],[12,16]] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# I'll append the new interval to the list, sort everything by start time,
# then run a standard merge-intervals pass to collapse all overlaps."
```

Time: $O(n \log n)$ – the sort dominates. Space: $O(n)$. Should I go ahead and optimize?"

```
class _57_InsertInterval_Brute:
    def insert(self, intervals: List[List[int]], newInterval: List[int]) -> List[List[int]]:
        intervals.append(newInterval)
        intervals.sort()
        result = [intervals[0]]
        for s, e in intervals[1:]:
            if result[-1][1] >= s:
                result[-1][1] = max(result[-1][1], e)
            else:
                result.append([s, e])
        return result
```

PHASE 3 — OPTIMIZE

"The input is already sorted – we can exploit that to avoid the $O(n \log n)$ sort.

Three phases in one $O(n)$ pass:

1. Add all intervals that end BEFORE newInterval starts (no overlap).

2. Merge all intervals that overlap newInterval into one combined interval.

Overlap condition: $\text{interval}[\text{start}] \leq \text{newInterval}[\text{end}]$.

3. Add all remaining intervals that start AFTER the merged interval ends.

#

Pseudocode:

$i = 0$

while $i < n$ and $\text{intervals}[i][1] < \text{new}[0]$: $\text{result.append(intervals[i])}$; $i++$

while $i < n$ and $\text{intervals}[i][0] \leq \text{new}[1]$:

$\text{new} = [\min(\text{new}[0], \text{intervals}[i][0]), \max(\text{new}[1], \text{intervals}[i][1])]$; $i++$

$\text{result.append(new)}$

$\text{result.extend(intervals[i:])}$

#

Time: $O(n)$. Space: $O(n)$ for output."

PHASE 4 — CLEAN CODE

```
class _57_InsertInterval:
    def insert(self, intervals: List[List[int]], newInterval: List[int]) -> List[List[int]]:
        result = []
        i, n = 0, len(intervals)

        # Phase 1: all intervals ending before newInterval starts
        while i < n and intervals[i][1] < newInterval[0]:
            result.append(intervals[i])
            i += 1

        # Phase 2: merge all overlapping intervals into newInterval
        while i < n and intervals[i][0] <= newInterval[1]:
            newInterval[0] = min(newInterval[0], intervals[i][0])
            newInterval[1] = max(newInterval[1], intervals[i][1])
            i += 1
        result.append(newInterval)
```

```
# Phase 3: remaining intervals start after merged interval
result.extend(intervals[i:])
return result
```

PHASE 5 — TEST & DEBUG

```
# intervals=[[1,3],[6,9]], new=[2,5]:
# Phase 1: intervals[0][1]=3 >= new[0]=2 → stop. i=0.
# Phase 2: intervals[0][0]=1 <= new[1]=5 → new=[min(2,1),max(5,3)]=[1,5]. i=1.
# intervals[1][0]=6 > new[1]=5 → stop.
# result=[[1,5]]. extend [[6,9]]. → [[1,5],[6,9]] ✓
#
# intervals=[[1,2],[3,5],[6,7],[8,10],[12,16]], new=[4,8]:
# Phase 1: [1,2] ends at 2 < 4 → add. i=1.
# Phase 2: [3,5] start 3<=8→new=[3,8]. [6,7] start 6<=8→new=[3,8]. [8,10] start
8<=8→new=[3,10]. i=4.
# result=[[1,2],[3,10]]. extend [[12,16]]. → [[1,2],[3,10],[12,16]] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$  – three phases together visit each interval once.
# Space  $O(n)$  for output.
# The two overlap conditions are the key:
# Non-overlap before: interval[end] < new[start].
# Non-overlap after: interval[start] > new[end] (Phase 3 – just extend).
# Anything in between overlaps and gets merged.
# Given more time I'd ask: what if intervals aren't sorted?
# Sort first ( $O(n \log n)$ ), then apply the same three-phase approach."
```


Arrays & Hashing

□ #238 Product of Array Except Self

MED[Open on LeetCode](#)

Array · Prefix Sum

Lists: Grind 169

Problem

Given an integer array `nums`, return an array `answer` such that `answer[i]` is equal to the product of all the elements of `nums` except `nums[i]`.

The product of any prefix or suffix of `nums` is guaranteed to fit in a 32-bit integer.

You must write an algorithm that runs in $O(n)$ time and without using the division operation.

Example 1:

```
Input: nums = [1,2,3,4]
Output: [24,12,8,6]
```

Example 2:

```
Input: nums = [-1,1,0,-3,3]
Output: [0,0,9,0,0]
```

Constraints:

- $2 \leq \text{nums.length} \leq 105$
- $-30 \leq \text{nums}[i] \leq 30$
- The input is generated such that $\text{answer}[i]$ is guaranteed to fit in a 32-bit integer.

Follow up: Can you solve the problem in $O(1)$ extra space complexity? (The output array does not count as extra space for space complexity analysis.)

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Return an array where answer[i] is the product of all elements except nums[i].
# Cannot use division. Must run in O(n). Follow-up: O(1) extra space. Right?
# Guaranteed no overflow in 32-bit – so no overflow concern."
#
# "[1,2,3,4]: answer[0]=2*3*4=24. answer[1]=1*3*4=12. answer[2]=1*2*4=8.
answer[3]=1*2*3=6 ✓
# [-1,1,0,-3,3]: zeros make most products 0 – [0,0,9,0,0] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# For each index i, I multiply together all elements in the array except nums[i]
itself.
# That's a nested loop – O(n) work for each of the n positions.
# Time: O(n^2). Space: O(1) extra. Should I go ahead and optimize?"
```

```
class _238_ProductExceptSelf_Brute:
    def productExceptSelf(self, nums):
        n = len(nums)
        result = []
        for i in range(n):
            product = 1
            for j in range(n):
                if j != i:
                    product *= nums[j]
            result.append(product)
        return result
```

PHASE 3 — OPTIMIZE

```
# "Build prefix products and suffix products, multiply them.
# prefix[i] = product of everything to the LEFT of i.
# suffix[i] = product of everything to the RIGHT of i.
# answer[i] = prefix[i] * suffix[i].
# Time: O(n). Space: O(n) for two arrays.
#
# O(1) space follow-up: use the output array for prefix products,
# then do a single right-to-left pass with a running suffix multiplier."
```

PHASE 4 — CLEAN CODE (O(1) extra space version)

```
class _238_ProductExceptSelf:
    def productExceptSelf(self, nums: List[int]) -> List[int]:
        n = len(nums)
        result = [1] * n

        # Left pass: result[i] = product of all elements to the left of i
        for i in range(1, n):
            result[i] = result[i - 1] * nums[i - 1]

        # Right pass: multiply in the suffix product on the fly
        suffix = 1
        for i in range(n - 1, -1, -1):
            result[i] *= suffix
            suffix *= nums[i]

        return result
```

PHASE 5 — TEST & DEBUG

```
# nums=[1,2,3,4]:
# Left pass: result=[1,1,2,6].
# Right pass: i=3: result[3]=6*1=6, suffix=4.
# i=2: result[2]=2*4=8, suffix=12.
# i=1: result[1]=1*12=12, suffix=24.
# i=0: result[0]=1*24=24, suffix=24.
# → [24,12,8,6] ✓
# nums=[0,0]: result=[0,0] ✓ (zeros handled correctly by multiplication)
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(1) extra – output array doesn't count.
# The two-pass trick: left products first, then multiply right products on the fly.
# The suffix variable carries the running right product without an extra array.
# Given more time I'd ask: what if we're allowed division?
# total_product / nums[i] – but fails when nums[i]=0. Handle: count zeros,
# if two or more → all zeros. If one zero → all zeros except position of zero."
```

Arrays & Hashing

□ #281 Zigzag Iterator

MED[Open on LeetCode](#)

Array · Design · Queue · Iterator

Lists: Denny Zhang

Problem

Given two vectors of integers v1 and v2, implement an iterator to return their elements alternately.

Implement the ZigzagIterator class:

- **ZigzagIterator(List<int> v1, List<int> v2)** initializes the object with the two vectors v1 and v2.
- **boolean hasNext()** returns true if the iterator still has elements, and false otherwise.
- **int next()** returns the current element of the iterator and moves the iterator to the next element.

Example 1:

Input: v1 = [1,2], v2 = [3,4,5,6]

Output: [1,3,2,4,5,6]

Explanation: By calling next repeatedly until hasNext returns false, the order of elements returned by next should be: [1,3,2,4,5,6].

Example 2:

Input: v1 = [1], v2 = []
Output: [1]

Example 3:

Input: v1 = [], v2 = [1]
Output: [1]

Constraints:

- $0 \leq v1.length, v2.length \leq 1000$
- $1 \leq v1.length + v2.length \leq 2000$
- $-231 \leq v1[i], v2[i] \leq 231 - 1$

Follow up: What if you are given k vectors? How well can your code be extended to such cases?

Clarification for the follow-up question:

The "Zigzag" order is not clearly defined and is ambiguous for $k > 2$ cases. If "Zigzag" does not look right to you, replace "Zigzag" with "Cyclic".

Follow-up Example:

Input: v1 = [1,2,3], v2 = [4,5,6,7], v3 = [8,9]
Output: [1,4,8,2,5,9,3,6,7]

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Alternate between v1 and v2 element by element. When one is exhausted,
continue with the remaining elements of the other. Right?
v1=[1,2], v2=[3,4,5,6] → [1,3,2,4,5,6] ✓
Follow-up: what if there are k vectors instead of 2?"

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# At construction I'll interleave v1 and v2 into a flat list: take one from v1, one
from v2,
# repeat, then append remaining elements from whichever list is longer.
# next() and hasNext() just read from that flat list.
# Time: O(n) init. Space: O(n). Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Use a deque of (vector, index) pairs.
# next(): popleft, return v[i], re-enqueue (v, i+1) if more elements remain.
# hasNext(): deque is non-empty.
# This generalises trivially to k vectors – just enqueue all of them.
# Time: O(1) per next()/hasNext(). Space: O(k) for the deque."
```

PHASE 4 — CLEAN CODE

```
class _281_ZigzagIterator:
    def __init__(self, v1: List[int], v2: List[int]):
        self.q: deque = deque()
        if v1:
            self.q.append((v1, 0))
        if v2:
            self.q.append((v2, 0))
    def next(self) -> int:
        vec, idx = self.q.popleft()
        if idx + 1 < len(vec):
            self.q.append((vec, idx + 1))
        return vec[idx]
    def hasNext(self) -> bool:
        return bool(self.q)
```

PHASE 5 — TEST & DEBUG

```
# v1=[1,2], v2=[3,4,5,6]:
# q=[(v1,0),(v2,0)].
# next(): pop (v1,0)→return 1. re-enqueue (v1,1). q=[(v2,0),(v1,1)].
# next(): pop (v2,0)→return 3. re-enqueue (v2,1). q=[(v1,1),(v2,1)].
# next(): pop (v1,1)→return 2. no re-enqueue (idx+1=2=len). q=[(v2,1)].
# next(): 4, then 5, then 6. → [1,3,2,4,5,6] ✓
# v1=[1], v2=[]: q=[(v1,0)]. next()→1. hasNext()→False ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(1) per operation. Space O(k) for the deque.
# Follow-up k vectors: init enqueues all k non-empty vectors. Same code, no changes.
# The deque acts as a round-robin scheduler – each vector gets one turn per cycle.
# Given more time I'd ask: what if vectors have different priorities (some should
# appear more often)? Enqueue them multiple times or use a priority queue."
```

Arrays & Hashing

□ #307 Range Sum Query – Mutable

MED

[Open on LeetCode](#)

Array · Design · Binary Indexed Tree · Segment Tree

Lists: Denny Zhang

Problem

Given an integer array `nums`, handle multiple queries of the following types:

1. Update the value of an element in `nums`.
2. Calculate the sum of the elements of `nums` between indices `left` and `right` inclusive where `left <= right`.

Implement the `NumArray` class:

- `NumArray(int[] nums)` Initializes the object with the integer array `nums`.
- `void update(int index, int val)` Updates the value of `nums[index]` to be `val`.
- `int sumRange(int left, int right)` Returns the sum of the elements of `nums` between indices `left` and `right` inclusive (i.e. `nums[left] + nums[left + 1] + ... + nums[right]`).

Example 1:

Input
 ["NumArray", "sumRange", "update", "sumRange"]
 [[[1, 3, 5]], [0, 2], [1, 2], [0, 2]]

Output
 [null, 9, null, 8]

Explanation
 NumArray numArray = new NumArray([1, 3, 5]);

Constraints:

- $1 \leq \text{nums.length} \leq 3 * 10^4$
- $-100 \leq \text{nums}[i] \leq 100$
- $0 \leq \text{index} < \text{nums.length}$
- $-100 \leq \text{val} \leq 100$
- $0 \leq \text{left} \leq \text{right} < \text{nums.length}$
- At most $3 * 10^4$ calls will be made to update and sumRange.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Support two operations on an array: update a single element, and query the prefix sum from left to right inclusive. Up to $3 * 10^4$ calls. Right?
 # The challenge is making both operations fast – not $O(n)$ each."
 #
 # "NumArray([1,3,5]): sumRange(0,2)=9. update(1,2)→nums=[1,2,5]. sumRange(0,2)=8 ✓"

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.
 # Store the array directly. update() just sets `nums[index] = val` in $O(1)$.
 # `sumRange(left, right)` sums `nums[left..right]` with a linear scan in $O(n)$.
 # With up to $3 * 10^4$ queries this is $O(n * \text{queries})$ total – too slow for large n .
 # Time: update $O(1)$, sumRange $O(n)$. Space: $O(n)$. Should I go ahead and optimize?"

```
class _307_NumArray_Brute:
    def __init__(self, nums):
        self.nums = nums[:]
```



```
def update(self, index, val):
    self.nums[index] = val
def sumRange(self, left, right):
    return sum(self.nums[left:right+1])
```

PHASE 3 — OPTIMIZE

"Binary Indexed Tree (Fenwick Tree) – each node stores a partial sum covering a range determined by its lowest set bit. Both update and query run in $O(\log n)$.
 # Update: add change to index and all its ancestors ($i += i \& -i$).
 # Query prefix sum up to i : add and walk down ($i -= i \& -i$).
 # For range $[left, right]$: $query(right) - query(left-1)$."

PHASE 4 — CLEAN CODE

```
class _307_NumArray:
    def __init__(self, nums: List[int]):
        self.nums = nums[:]
        n = len(nums)
        self.tree = [0] * (n + 1)
        for i, val in enumerate(nums):
            self._add(i, val)
    def _add(self, i: int, delta: int) -> None:
        i += 1
        while i <= len(self.nums):
            self.tree[i] += delta
            i += i & (-i)
    def _prefix(self, i: int) -> int:
        i += 1
        total = 0
        while i > 0:
            total += self.tree[i]
            i -= i & (-i)
        return total
    def update(self, index: int, val: int) -> None:
        self._add(index, val - self.nums[index])
        self.nums[index] = val
    def sumRange(self, left: int, right: int) -> int:
        return self._prefix(right) - (self._prefix(left - 1) if left > 0 else 0)
```

PHASE 5 — TEST & DEBUG

$nums=[1,3,5]$: tree built. $_prefix(2)=1+3+5=9$. $sumRange(0,2)=9$ ✓
 # $update(1,2)$: $delta=2-3=-1$. tree adjusted. $nums=[1,2,5]$.
 # $sumRange(0,2)$: $_prefix(2)=8$ ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"update: $O(\log n)$. sumRange: $O(\log n)$. init: $O(n \log n)$.
 # Space: $O(n)$ for the tree.
 # The bit trick $i \& (-i)$ isolates the lowest set bit – this determines
 # how many elements each node is responsible for. Worth explaining clearly.
 # Alternative: segment tree – more flexible (supports range update) but more code."

```
# Given more time I'd ask: what if we need range updates (add v to nums[l..r])?  
# BIT can handle that with a difference array extension – or use a lazy segment  
tree."
```

Arrays & Hashing

□ #454 4Sum II

MED[Open on LeetCode](#)

Array · Hash Table

Lists: CodeSignal

Problem

Given four integer arrays `nums1`, `nums2`, `nums3`, and `nums4` all of length `n`, return the number of tuples `(i, j, k, l)` such that:

- $0 \leq i, j, k, l < n$
- $\text{nums1}[i] + \text{nums2}[j] + \text{nums3}[k] + \text{nums4}[l] == 0$

Example 1:

```
Input: nums1 = [1,2], nums2 = [-2,-1], nums3 = [-1,2], nums4 = [0,2]
Output: 2
Explanation:
The two tuples are:
1. (0, 0, 0, 1) -> nums1[0] + nums2[0] + nums3[0] + nums4[1] = 1 + (-2) + (-1) + 2 = 0
2. (1, 1, 0, 0) -> nums1[1] + nums2[1] + nums3[0] + nums4[0] = 2 + (-1) + (-1) + 0 = 0
```

Example 2:

```
Input: nums1 = [0], nums2 = [0], nums3 = [0], nums4 = [0]
Output: 1
```

Constraints:

- `n == nums1.length`
- `n == nums2.length`
- `n == nums3.length`
- `n == nums4.length`
- `1 <= n <= 200`
- `-228 <= nums1[i], nums2[i], nums3[i], nums4[i] <= 228`

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Count tuples (i,j,k,l) where nums1[i]+nums2[j]+nums3[k]+nums4[l]==0.
# All four arrays have the same length n. Return the count – not the tuples. Right?
# We're not asked to deduplicate – every index combo counts separately."
#
# "nums1=[1,2],nums2=[-2,-1],nums3=[-1,2],nums4=[0,2]:
# (0,0,0,1): 1+(-2)+(-1)+2=0 ✓. (1,1,0,0): 2+(-1)+(-1)+0=0 ✓. → 2 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Four nested loops – for every combination (i,j,k,l), check if
# nums1[i]+nums2[j]+nums3[k]+nums4[l] == 0 and count the matches.
# Time:  $O(n^4)$ . For  $n=200$  that's 1.6 billion iterations – way too slow.
# Space:  $O(1)$ . Should I go ahead and optimize?"
```

```
class _454_FourSumII_Brute:
    def fourSumCount(self, nums1, nums2, nums3, nums4):
        count = 0
        for a in nums1:
            for b in nums2:
                for c in nums3:
                    for d in nums4:
                        if a + b + c + d == 0:
                            count += 1
        return count
```

PHASE 3 — OPTIMIZE

```
# "Meet in the middle: split into two pairs.
# Build a Counter of all sums a+b for a in nums1, b in nums2 –  $O(n^2)$  sums.
# Then for each c+d from nums3×nums4, look up  $-(c+d)$  in the counter –  $O(n^2)$  lookups.
# Total:  $O(n^2)$  time and  $O(n^2)$  space."
```

PHASE 4 — CLEAN CODE

```
class _454_FourSumII:
    def fourSumCount(self, nums1: List[int], nums2: List[int], nums3: List[int],
nums4: List[int]) -> int:
        pair_sums = Counter(a + b for a in nums1 for b in nums2)
        return sum(pair_sums[-(c + d)] for c in nums3 for d in nums4)
```

PHASE 5 — TEST & DEBUG

```
# nums1=[1,2], nums2=[-2,-1]:
# pair_sums={1+(-2)=-1:1, 1+(-1)=0:1, 2+(-2)=0:+1-0:2, 2+(-1)=1:1}
# = {-1:1, 0:2, 1:1}
# nums3=[-1,2], nums4=[0,2]:
# (c=-1,d=0): -(-1+0)=1. pair_sums[1]=1. count=1.
# (c=-1,d=2): -(-1+2)=-1. pair_sums[-1]=1. count=2.
# (c=2,d=0): -(2+0)=-2. pair_sums[-2]=0. count=2.
# (c=2,d=2): -(2+2)=-4. pair_sums[-4]=0. count=2. → 2 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n^2)$ . Space  $O(n^2)$ .
# Meet-in-the-middle is the key: instead of 4 nested loops, do 2+2.
# The Counter default of 0 for missing keys means misses contribute 0 – clean.
# Given more time I'd ask: what if the arrays are sorted and we want unique
quadruplets?
# That's 4Sum (#18) – different problem requiring deduplication with two pointers."
```

Arrays & Hashing

□ #525 Contiguous Array

MED[Open on LeetCode](#)

Array · Hash Table · Prefix Sum

Lists: Grind 169

Problem

Given a binary array `nums`, return the maximum length of a contiguous subarray with an equal number of 0 and 1.

Example 1:

```
Input: nums = [0,1]
Output: 2
Explanation: [0, 1] is the longest contiguous subarray with an equal number of 0 and 1.
```

Example 2:

```
Input: nums = [0,1,0]
Output: 2
Explanation: [0, 1] (or [1, 0]) is a longest contiguous subarray with equal number of 0 and 1.
```

Example 3:

```
Input: nums = [0,1,1,1,1,1,0,0,0]
Output: 6
Explanation: [1,1,1,0,0,0] is the longest contiguous subarray with equal number of 0 and 1.
```

Constraints:

- $1 \leq \text{nums.length} \leq 105$

- `nums[i]` is either 0 or 1.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Binary array. Find the longest contiguous subarray with equal numbers of 0s and 1s.

Return its length – not the subarray itself. Right?"

#

"[0,1]: equal 0s and 1s → length 2 ✓

[0,1,0]: longest is [0,1] or [1,0] → length 2 ✓"

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.

For every pair (i, j), count the zeros and ones in `nums[i..j]`.

If they're equal, it's a valid subarray – track the maximum length.

Time: $O(n^2)$ – we can precompute prefix counts to make each check $O(1)$.

Space: $O(1)$. Should I go ahead and optimize?"

```
class _525_ContiguousArray_Brute:
    def findMaxLength(self, nums):
        n = len(nums)
        longest = 0
        for i in range(n):
            zeros = ones = 0
            for j in range(i, n):
                if nums[j] == 0:
                    zeros += 1
                else:
                    ones += 1
                if zeros == ones:
                    longest = max(longest, j - i + 1)
        return longest
```

PHASE 3 — OPTIMIZE

"Treat 0 as -1 and 1 as +1. Keep a running balance (prefix sum).

Equal 0s and 1s means balance hasn't changed – we want the longest subarray

where `balance[j] == balance[i]` for some $i < j$.

Store the FIRST index each balance value was seen. When we see the same balance again,

the subarray `[first_seen+1 .. current]` has equal 0s and 1s.

Seed the map with `{0: -1}` to handle subarrays starting from index 0.

#

Time: $O(n)$. Space: $O(n)$."

PHASE 4 — CLEAN CODE

```
class _525_ContiguousArray:
    def findMaxLength(self, nums: List[int]) -> int:
```

```

first_seen = {0: -1} # balance → first index it appeared
balance = 0
longest = 0
for i, num in enumerate(nums):
    balance += 1 if num == 1 else -1
    if balance in first_seen:
        longest = max(longest, i - first_seen[balance])
    else:
        first_seen[balance] = i
return longest

```

PHASE 5 — TEST & DEBUG

```

# nums=[0,1,0]:
# i=0,num=0: balance=-1. not seen → first_seen={0:-1,-1:0}.
# i=1,num=1: balance=0. seen at -1 → longest=max(0,1-(-1))=2.
# i=2,num=0: balance=-1. seen at 0 → longest=max(2,2-0)=2.
# return 2 ✓
# nums=[0,1,1,1,1,1,0,0,0]:
# Track balance... longest=6 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n). Space O(n) for the hash map.
# The 0→-1 substitution is the core insight – equal 0s and 1s becomes
# 'prefix sum unchanged', which becomes 'same balance seen before'.
# The {0:-1} seed is critical – without it, subarrays starting at index 0 are
missed.
# Given more time I'd ask: what if it's not binary – find the longest subarray with
# equal counts of two specific values? Same trick – map target→+1, other→-1."

```


Arrays & Hashing

□ #560 Subarray Sum Equals K

MED[Open on LeetCode](#)

Array · Hash Table · Prefix Sum

Lists: Grind 169

Problem

Given an array of integers `nums` and an integer `k`, return the total number of subarrays whose sum equals to `k`.

A subarray is a contiguous non-empty sequence of elements within an array.

Example 1:

Input: `nums = [1,1,1]`, `k = 2`
Output: 2

Example 2:

Input: `nums = [1,2,3]`, `k = 3`
Output: 2

Constraints:

- $1 \leq \text{nums.length} \leq 2 * 10^4$
 - $-1000 \leq \text{nums}[i] \leq 1000$
 - $-10^7 \leq k \leq 10^7$
-

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Count subarrays (contiguous, non-empty) whose elements sum to k.
# Can contain negatives – so sliding window won't work. Return the count. Right?"
#
# "[1,1,1], k=2: [1,1] at (0,1) and [1,1] at (1,2) → 2 ✓
# [1,2,3], k=3: [3] at index 2 and [1,2] → 2 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# For every pair (i, j), compute the sum of nums[i..j] and check if it equals k.
# I can maintain a running sum to avoid recomputing from scratch each time, giving
# O(n²).
# Time: O(n²). Space: O(1). Should I go ahead and optimize?"
```

```
class _560_SubarraySum_Brute:
    def subarraySum(self, nums, k):
        count = 0
        for i in range(len(nums)):
            total = 0
            for j in range(i, len(nums)):
                total += nums[j]
                if total == k:
                    count += 1
        return count
```

PHASE 3 — OPTIMIZE

```
# "Prefix sums: sum of nums[i..j] = prefix[j+1] - prefix[i].
# We want prefix[j+1] - prefix[i] == k → prefix[i] == prefix[j+1] - k.
# Walk forward tracking the running sum. For each position, count how many
# earlier prefix sums equal (current_sum - k). Use a Counter seeded with {0:1}
# to handle subarrays starting at index 0.
#
# Time: O(n). Space: O(n)."
```

PHASE 4 — CLEAN CODE

```
class _560_SubarraySum:
    def subarraySum(self, nums: List[int], k: int) -> int:
        prefix_count = Counter({0: 1}) # prefix_sum → how many times seen
        running_sum = 0
        count = 0
        for num in nums:
            running_sum += num
            count += prefix_count[running_sum - k]
            prefix_count[running_sum] += 1
        return count
```

PHASE 5 — TEST & DEBUG

```
# nums=[1,1,1], k=2:
```

```
# num=1: sum=1. want=1-2=-1. prefix_count[-1]=0. prefix_count={0:1,1:1}.
# num=1: sum=2. want=2-2=0. prefix_count[0]=1 → count=1.
prefix_count={0:1,1:1,2:1}.
# num=1: sum=3. want=3-2=1. prefix_count[1]=1 → count=2. → 2 ✓
# nums=[1,2,3], k=3:
# sum=1,want=-2→0. sum=3,want=0→count=1. sum=6,want=3→prefix_count[3]=1→count=2. →
2 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(n)$.

This is structurally identical to #525 Contiguous Array – both use

prefix sum + hash map, both seed with {0:1} (or {0:-1}).

Worth naming the connection explicitly: 525 finds longest, 560 counts all.

Sliding window doesn't work here because negatives mean the sum can decrease

even as we expand right – sliding window requires monotone sums.

Given more time I'd ask: what if we want subarrays with sum divisible by k?

Use prefix_sum % k – same pattern, but map remainders."

Arrays & Hashing

□ #1010 Pairs of Songs With Total Durations Divisible by 60

MED[Open on LeetCode](#)

Array · Hash Table · Counting
Lists: CodeSignal

Problem

You are given a list of songs where the i th song has a duration of $\text{time}[i]$ seconds.

Return the number of pairs of songs for which their total duration in seconds is divisible by 60. Formally, we want the number of indices i, j such that $i < j$ with $(\text{time}[i] + \text{time}[j]) \% 60 == 0$.

Example 1:

```
Input: time = [30,20,150,100,40]
Output: 3
Explanation: Three pairs have a total duration divisible by 60:
(time[0] = 30, time[2] = 150): total duration 180
(time[1] = 20, time[3] = 100): total duration 120
(time[1] = 20, time[4] = 40): total duration 60
```

Example 2:

```
Input: time = [60,60,60]
Output: 3
Explanation: All three pairs have a total duration of 120, which is divisible by 60.
```

Constraints:

- $1 \leq \text{time.length} \leq 6 * 10^4$
- $1 \leq \text{time}[i] \leq 500$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Count pairs (i,j) where i<j and (time[i]+time[j]) % 60 == 0.
# Songs have durations in seconds. Return the count. Right?
# Edge cases: duration exactly 60 (60%60=0), duration exactly 30 pairs with another 30."
#
# "[30,20,150,100,40]: (30,150)=180, (20,100)=120, (20,40)=60 → 3 ✓
# [60,60,60]: each pair sums to 120, divisible by 60. C(3,2)=3 → 3 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# For every pair (i, j) where j > i, check if (time[i] + time[j]) % 60 == 0.
# Count all such pairs. For n=6*10^4 that's 3.6 billion comparisons – too slow.
# Time: O(n^2). Space: O(1). Should I go ahead and optimize?"
```

```
class _1010_PairsSongs_Brute:
    def numPairsDivisibleBy60(self, time):
        count = 0
        for i in range(len(time)):
            for j in range(i + 1, len(time)):
                if (time[i] + time[j]) % 60 == 0:
                    count += 1
        return count
```

PHASE 3 — OPTIMIZE

```
# "Reduce each duration to t % 60. We want pairs where t1 + t2 ≡ 0 (mod 60).
# For each t, its complement is (60 - t) % 60. The %60 handles t=0 (complement=0, not 60).
# Walk forward: count += how many times we've already seen the complement.
# Then add t to the seen counter.
# This is Two Sum with modular arithmetic – O(n) time, O(60)=O(1) space."
```

PHASE 4 — CLEAN CODE

```
class _1010_PairsSongs:
    def numPairsDivisibleBy60(self, time: List[int]) -> int:
        seen = Counter()
        count = 0
        for t in time:
            remainder = t % 60
            complement = (60 - remainder) % 60
            count += seen[complement]
            seen[remainder] += 1
```

return count

PHASE 5 — TEST & DEBUG

```
# time=[30,20,150,100,40]:
# t=30: rem=30, comp=30. seen[30]=0 → count=0. seen={30:1}.
# t=20: rem=20, comp=40. seen[40]=0 → count=0. seen={30:1,20:1}.
# t=150: rem=30, comp=30. seen[30]=1 → count=1. seen={30:2,20:1}.
# t=100: rem=40, comp=20. seen[20]=1 → count=2. seen={30:2,20:1,40:1}.
# t=40: rem=40, comp=20. seen[20]=1 → count=3. → 3 ✓
# time=[60,60,60]: each rem=0, comp=(60-0)%60=0.
# t=60: seen[0]=0 → 0. seen={0:1}.
# t=60: seen[0]=1 → count=1. seen={0:2}.
# t=60: seen[0]=2 → count=3. → 3 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ . Space  $O(1)$  – the Counter holds at most 60 keys.
# The  $(60-\text{remainder})\%60$  handles the remainder=0 case: complement is 0, not 60.
# This is Two Sum modulo 60 – recognising the Two Sum pattern in disguise
# is the key interview insight.
# Given more time I'd ask: what if divisor were k instead of 60?
# Same algorithm – the Counter holds at most k entries, still  $O(1)$  space for fixed k."
```

Arrays & Hashing

□ ★ #1272 Remove Interval ★

MED

[Open on LeetCode](#)

Array

Lists: ★ Premium | Premium 98

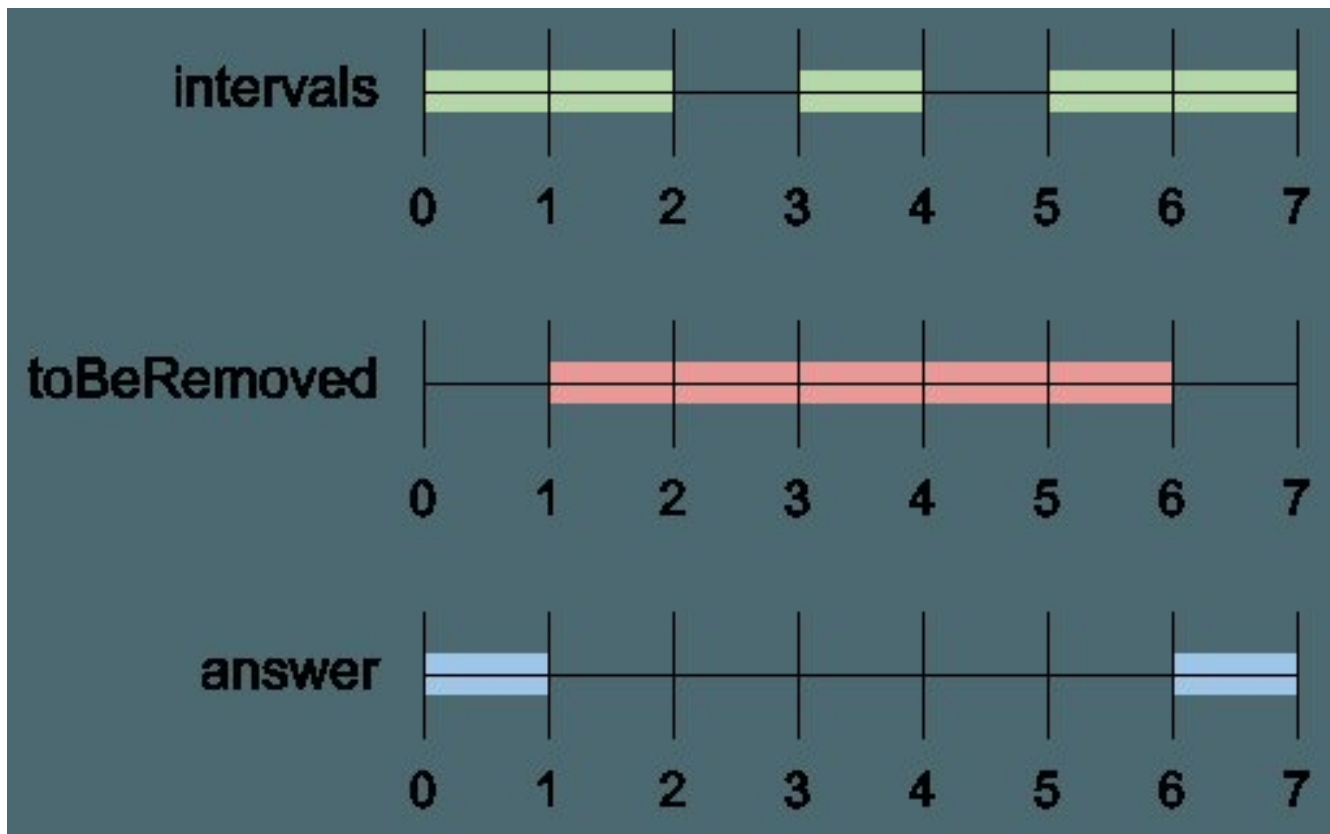
Problem

A set of real numbers can be represented as the union of several disjoint intervals, where each interval is in the form $[a, b)$. A real number x is in the set if one of its intervals $[a, b)$ contains x (i.e. $a \leq x < b$).

You are given a sorted list of disjoint intervals representing a set of real numbers as described above, where $\text{intervals}[i] = [a_i, b_i]$ represents the interval $[a_i, b_i)$. You are also given another interval `toBeRemoved`.

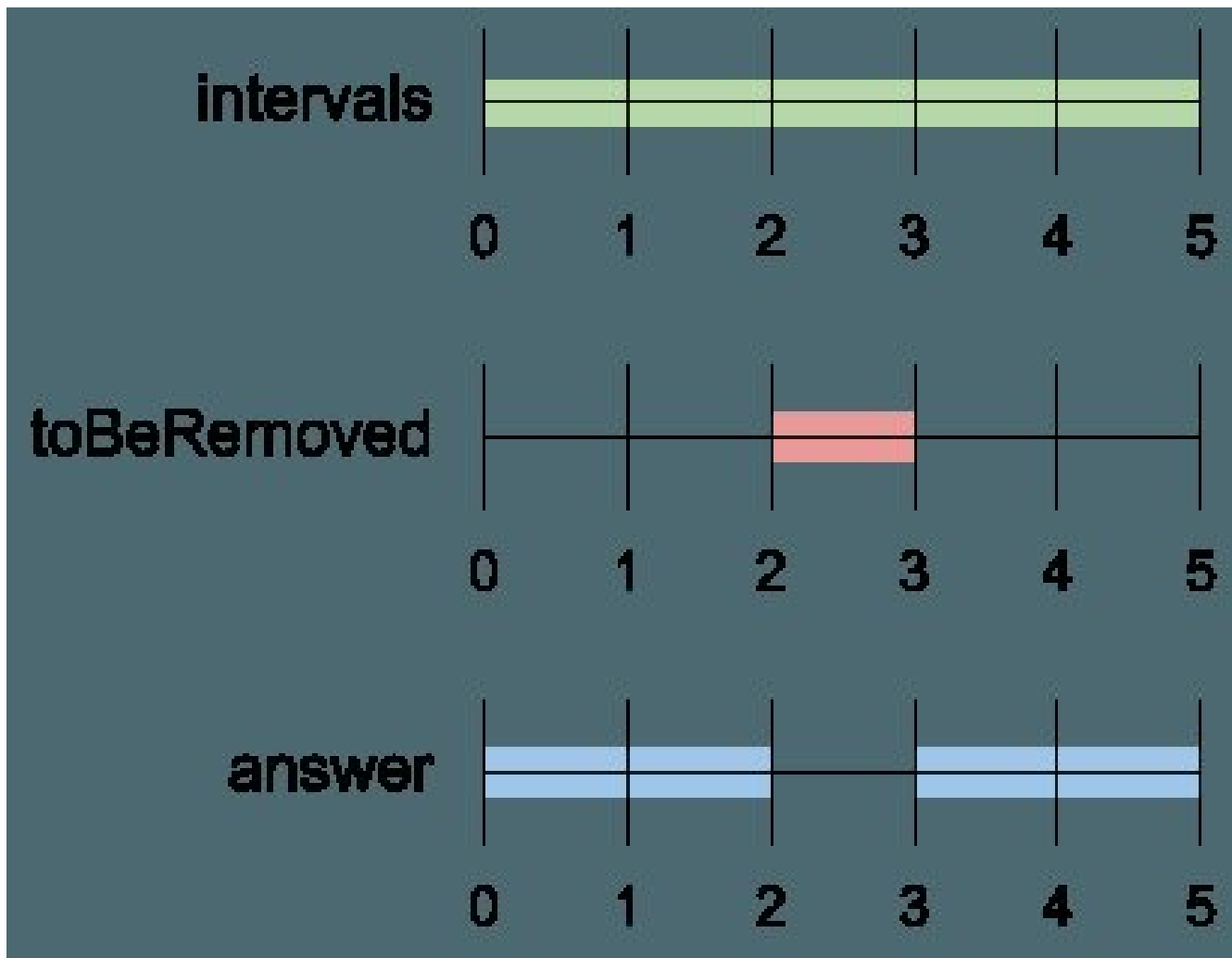
Return the set of real numbers with the interval `toBeRemoved` removed from `intervals`. In other words, return the set of real numbers such that every x in the set is in `intervals` but not in `toBeRemoved`. Your answer should be a sorted list of disjoint intervals as described above.

Example 1:



Input: intervals = [[0,2],[3,4],[5,7]], toBeRemoved = [1,6]
Output: [[0,1],[6,7]]

Example 2:



Input: intervals = [[0,5]], toBeRemoved = [2,3]
 Output: [[0,2],[3,5]]

Example 3:

Input: intervals = [[-5,-4],[-3,-2],[1,2],[3,5],[8,9]], toBeRemoved = [-1,4]
 Output: [[-5,-4],[-3,-2],[4,5],[8,9]]

Constraints:

- $1 \leq \text{intervals.length} \leq 104$
- $-109 \leq a_i < b_i \leq 109$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Intervals use half-open notation [a, b). We remove everything in
toBeRemoved=[a,b)
# from the existing set and return what remains. Right?
# Intervals are pre-sorted and non-overlapping – no need to re-sort output."
#
# "[[0,2],[3,4],[5,7]], remove=[1,6]:
# [0,2] partially overlaps → keep [0,1). [3,4] fully inside → gone. [5,7] partially
→ keep [6,7). → [[0,1],[6,7]] ✓
# [[0,5]], remove=[2,3]: split into [0,2) and [3,5). → [[0,2],[3,5]] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Walk each interval and handle three cases: if it doesn't touch toBeRemoved at all,
keep it whole.
# If it partially overlaps on the left, keep the left portion. If it partially
overlaps on the right,
# keep the right portion. If it's fully covered, drop it entirely.
# Time: O(n). Space: O(n). Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Already O(n) – just apply the three cases cleanly in one pass.
# No overlap if: interval ends before remove starts (end <= a) OR starts after
remove ends (start >= b).
# Otherwise it overlaps. Trim left portion if start < a → keep [start, a).
# Trim right portion if end > b → keep [b, end)."
```

PHASE 4 — CLEAN CODE

```
class _1272_RemoveInterval:
    def removeInterval(self, intervals: List[List[int]], toBeRemoved: List[int]) ->
List[List[int]]:
        a, b = toBeRemoved
        result = []
        for start, end in intervals:
            if end <= a or start >= b: # no overlap – keep whole
                result.append([start, end])
            else:
                if start < a: # left portion survives
                    result.append([start, a])
                if end > b: # right portion survives
                    result.append([b, end])
        return result
```

PHASE 5 — TEST & DEBUG

```
# [[0,2],[3,4],[5,7]], remove=[1,6]:
# [0,2]: overlap. start(0)<a(1)→[0,1). end(2)>b(6)? No. result=[[0,1]].
# [3,4]: overlap. start(3)<1? No. end(4)>6? No. result=[[0,1]].
# [5,7]: overlap. start(5)<1? No. end(7)>6 → [6,7]. result=[[0,1],[6,7]] ✓
# [[0,5]], remove=[2,3]:
# overlap. [0,2] and [3,5]. → [[0,2],[3,5]] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(n)$ for output.

The two trim conditions are independent – both can fire on the same interval

(when toBeRemoved is strictly inside the interval), producing two output intervals.

Given more time I'd ask: what if toBeRemoved can be a list of intervals to remove?

Process them in sorted order, sweeping through once – still $O(n + m)$ time."

Arrays & Hashing

□ ★ #1429 First Unique Number ★

MED

[Open on LeetCode](#)

Array · Hash Table · Design · Queue · Data Stream

Lists: ★ Premium | Premium 98

Problem

You have a queue of integers, you need to retrieve the first unique integer in the queue.

Implement the FirstUnique class:

- `FirstUnique(int[] nums)` Initializes the object with the numbers in the queue.
- `int showFirstUnique()` returns the value of the first unique integer of the queue, and returns `-1` if there is no such integer.
- `void add(int value)` insert value to the queue.

Example 1:

Input:

```
["FirstUnique", "showFirstUnique", "add", "showFirstUnique", "add", "showFirstUnique", "add", "showFirstUnique"]  
[[2,3,5]], [], [5], [], [2], [], [3], []]
```

Output:

```
[null, 2, null, 2, null, 3, null, -1]
```

Explanation:

```
FirstUnique firstUnique = new FirstUnique([2,3,5]);  
firstUnique.showFirstUnique(); // return 2
```

Example 2:

Input:

```
["FirstUnique", "showFirstUnique", "add", "add", "add", "add", "add", "showFirstUnique"]  
[[7,7,7,7,7,7]], [], [7], [3], [3], [7], [17], []]
```

Output:

```
[null, -1, null, null, null, null, null, 17]
```

Explanation:

```
FirstUnique firstUnique = new FirstUnique([7,7,7,7,7,7]);  
firstUnique.showFirstUnique(); // return -1
```

Example 3:

Input:

```
["FirstUnique", "showFirstUnique", "add", "showFirstUnique"]  
[[809]], [], [809], []]
```

Output:

```
[null, 809, null, -1]
```

Explanation:

```
FirstUnique firstUnique = new FirstUnique([809]);  
firstUnique.showFirstUnique(); // return 809
```

Constraints:

- $1 \leq \text{nums.length} \leq 10^5$
- $1 \leq \text{nums}[i] \leq 10^8$
- $1 \leq \text{value} \leq 10^8$
- At most 50000 calls will be made to `showFirstUnique` and `add`.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Maintain a queue. showFirstUnique() returns the first element that appears
# exactly once. add() appends to the queue. Return -1 if no unique exists. Right?
# Order matters – we want the first unique by insertion order."
#
# "FirstUnique([2,3,5]): first unique = 2.
# add(5) → 5 now has count 2, still first unique = 2.
# add(2) → 2 duplicated, first unique = 3.
# add(3) → 3 duplicated, no unique → -1 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# I'll store all added elements in a list. showFirstUnique() scans the list in
# insertion order,
# counting occurrences of each element, and returns the first one that appears
# exactly once.
# add() is O(1). showFirstUnique() is O(n) per call – with many calls this gets
# slow.
# Time: add O(1), showFirstUnique O(n). Space: O(n). Should I go ahead and
# optimize?"
```

```
class _1429_FirstUnique_Brute:
    def __init__(self, nums):
        self.data = list(nums)
    def showFirstUnique(self):
        from collections import Counter
        freq = Counter(self.data)
        for x in self.data:
            if freq[x] == 1:
                return x
        return -1
    def add(self, value):
        self.data.append(value)
```

PHASE 3 — OPTIMIZE

```
# "Two data structures:
# A 'seen' set – tracks all values ever added (including duplicates).
# An ordered dict (or insertion-order dict) – maps value → True, only for unique
# values.
# On add: if not in seen → it's new, add to both seen and dict.
# if in seen and in dict → it became a duplicate, remove from dict.
# if in seen and not in dict → already duplicate, ignore.
# showFirstUnique(): peek at first key in dict – O(1) in Python 3.7+
# (insertion-ordered).
#
# Time: O(1) per add, O(1) per showFirstUnique. Space: O(n)."
```

PHASE 4 — CLEAN CODE

```

class _1429_FirstUnique:
    def __init__(self, nums: List[int]):
        self.seen : set = set()
        self.uniques: dict = {}
        for num in nums:
            self.add(num)

    def showFirstUnique(self) -> int:
        return next(iter(self.uniques), -1)

    def add(self, value: int) -> None:
        if value not in self.seen:
            self.seen.add(value)
            self.uniques[value] = True
        elif value in self.uniques:
            del self.uniques[value]

```

PHASE 5 — TEST & DEBUG

```

# init([2,3,5]): seen={2,3,5}, uniques={2,3,5}.
# showFirstUnique() → 2 ✓
# add(5): 5 in seen and in uniques → del 5. uniques={2,3}.
# showFirstUnique() → 2 ✓
# add(2): 2 in seen and in uniques → del 2. uniques={3}.
# showFirstUnique() → 3 ✓
# add(3): del 3. uniques={}.
# showFirstUnique() → -1 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "O(1) per operation. Space O(n).
# Python dicts maintain insertion order since 3.7, so next(iter(dict)) returns
# the oldest key still present – exactly what we need.
# If order weren't guaranteed, we'd use collections.OrderedDict explicitly.
# Given more time I'd ask: what if we need to support remove() as well?
# That's a linked hash map (doubly linked list + hash map) – O(1) all operations."

```

Arrays & Hashing

□ **#3159 Find Occurrences of an Element in an Array** **MED**

[Open on LeetCode](#)

Array · Hash Table

Lists: CodeSignal

Problem

You are given an integer array `nums`, an integer array `queries`, and an integer `x`.

For each `queries[i]`, you need to find the index of the `queries[i]`th occurrence of `x` in the `nums` array. If there are fewer than `queries[i]` occurrences of `x`, the answer should be `-1` for that query.

Return an integer array `answer` containing the answers to all queries.

Example 1:

Input: `nums = [1,3,1,7]`, `queries = [1,3,2,4]`, `x = 1`

Output: `[0,-1,2,-1]`

Explanation:

- For the 1st query, the first occurrence of 1 is at index 0.

- For the 2nd query, there are only two occurrences of 1 in nums, so the answer is -1.
- For the 3rd query, the second occurrence of 1 is at index 2.
- For the 4th query, there are only two occurrences of 1 in nums, so the answer is -1.

Example 2:

Input: nums = [1,2,3], queries = [10], x = 5

Output: [-1]

Explanation:

- For the 1st query, 5 doesn't exist in nums, so the answer is -1.

Constraints:

- $1 \leq \text{nums.length}, \text{queries.length} \leq 105$
- $1 \leq \text{queries}[i] \leq 105$
- $1 \leq \text{nums}[i], x \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Preprocess nums to find all indices where x appears.

For each query q, return the index of the q-th occurrence of x (1-indexed).

If fewer than q occurrences exist, return -1. Right?"

#

"nums=[1,3,1,7], queries=[1,3,2,4], x=1:

x appears at indices [0,2]. q=1→0, q=3→-1, q=2→2, q=4→-1. → [0,-1,2,-1] ✓"

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.

For each query q, scan nums from the beginning, count occurrences of x as we go,

and return the index where we hit the q-th occurrence – or -1 if we run out.
 # Time: $O(n * |queries|)$. Space: $O(1)$. Should I go ahead and optimize?"

```
class _3159_FindOccurrencesBrute:
    def occurrencesOfElement(self, nums, queries, x):
        result = []
        for q in queries:
            count = 0
            found = -1
            for i, n in enumerate(nums):
                if n == x:
                    count += 1
                    if count == q:
                        found = i
                        break
            result.append(found)
        return result
```

PHASE 3 — OPTIMIZE

"Precompute a list of all indices where x appears – $O(n)$.
 # Each query is then an $O(1)$ index lookup. Total: $O(n + |queries|)$."

PHASE 4 — CLEAN CODE

```
class _3159_FindOccurrences:
    def occurrencesOfElement(self, nums: List[int], queries: List[int], x: int) -> List[int]:
        positions = [i for i, n in enumerate(nums) if n == x]
        return [positions[q - 1] if q - 1 < len(positions) else -1 for q in queries]
```

PHASE 5 — TEST & DEBUG

nums=[1,3,1,7], x=1: positions=[0,2].
 # q=1: positions[0]=0 ✓. q=3: 3-1=2 >= len(positions)=2 → -1 ✓.
 # q=2: positions[1]=2 ✓. q=4: -1 ✓.

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n + q)$. Space $O(k)$ where k = occurrences of x .
 # If queries were sorted, binary search on positions would give $O((n+q) \log k)$
 # with minimal extra space – but for unsorted queries, the precompute list is cleaner.
 # Given more time I'd ask: what if x could change between queries?
 # Build a map from value → sorted list of indices, then binary search per query."

String

Round 2 · High · Medium · 5 questions

String

□ #8 String to Integer (atoi)

MED[Open on LeetCode](#)

String

Lists: Grind 169

Problem

Implement the `myAtoi(string s)` function, which converts a string to a 32-bit signed integer.

The algorithm for `myAtoi(string s)` is as follows:

1. **Whitespace:** Ignore any leading whitespace (" ").
2. **Signedness:** Determine the sign by checking if the next character is '-' or '+', assuming positivity if neither present.
3. **Conversion:** Read the integer by skipping leading zeros until a non-digit character is encountered or the end of the string is reached. If no digits were read, then the result is 0.
4. **Rounding:** If the integer is out of the 32-bit signed integer range $[-2^{31}, 2^{31} - 1]$, then round the integer to remain in the range.

Specifically, integers less than -231 should be rounded to -231 , and integers greater than $231 - 1$ should be rounded to $231 - 1$.

Return the integer as the final result.

Example 1:

Input: $s = "42"$

Output: 42

Explanation:

The underlined characters are what is read in and the caret is the current reader position.

Step 1: "42" (no characters read because there is no leading whitespace)

^

Step 2: "42" (no characters read because there is neither a '-' nor '+')

^

Step 3: "42" ("42" is read in)

^

Example 2:

Input: $s = " -042"$

Output: -42

Explanation:

Step 1: " -042" (leading whitespace is read and ignored)

^

Step 2: " -042" ('-' is read, so the result should be negative)

^

Step 3: " -042" ("042" is read in, leading zeros ignored in the result)

^

Example 3:

Input: $s = "1337c0d3"$

Output: 1337

Explanation:

```
Step 1: "1337c0d3" (no characters read because there is no leading whitespace)
^
Step 2: "1337c0d3" (no characters read because there is neither a '-' nor '+')
^
Step 3: "1337c0d3" ("1337" is read in; reading stops because the next character
is a non-digit)
^
```

Example 4:

Input: s = "0-1"

Output: 0

Explanation:

```
Step 1: "0-1" (no characters read because there is no leading whitespace)
^
Step 2: "0-1" (no characters read because there is neither a '-' nor '+')
^
Step 3: "0-1" ("0" is read in; reading stops because the next character is a
non-digit)
^
```

Example 5:

Input: s = "words and 987"

Output: 0

Explanation:

Reading stops at the first non-digit character 'w'.

Constraints:

- $0 \leq s.length \leq 200$
- s consists of English letters (lower-case and upper-case), digits (0-9), ' ', '+', '-', and '.'.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Four steps in order: strip leading whitespace, read optional sign,
# read digits until non-digit or end, clamp to 32-bit signed range [-2^31, 2^31-1].
# Right?
# Stop at the FIRST non-digit after the optional sign – don't skip over non-digits."
#
# "'42'→42. ' -042'→-42 (strip, sign, parse 042). '1337c0d3'→1337 (stop at 'c').
# '0-1'→0 (first digit parsed, stop at '-'). 'words'→0 (first char non-digit, no
# digits read)."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# The naive approach is to try Python's built-in int() after stripping whitespace –
# but that fails on partial strings like '42abc' and doesn't clamp to 32-bit range.
# So the 'brute force' here is really just the manual state machine, which I'll
# implement.
# Time: O(n). Space: O(1). Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Manual state machine with a pointer:
# 1. Skip whitespace.
# 2. Read sign.
# 3. Read digits, building number, check overflow before each digit.
# Overflow check: ans > MAX//10 OR (ans == MAX//10 AND next digit > 7).
# Return sign * ans (clamped)."
```

PHASE 4 — CLEAN CODE

```
class _8_Atoi:
    def myAtoi(self, s: str) -> int:
        INT_MAX, INT_MIN = 2**31 - 1, -(2**31)
        i, n = 0, len(s)
        while i < n and s[i] == ' ': # strip whitespace
            i += 1
        sign = 1
        if i < n and s[i] in '+-':
            sign = -1 if s[i] == '-' else 1
            i += 1
        result = 0
        while i < n and s[i].isdigit():
            d = int(s[i])
            if result > INT_MAX // 10 or (result == INT_MAX // 10 and d > 7):
                return INT_MAX if sign == 1 else INT_MIN
            result = result * 10 + d
            i += 1
        return sign * result
```

PHASE 5 — TEST & DEBUG

```
# '42': no space, no sign, digits→42. return 42 ✓  
# ' -042': skip space, sign=-1, digits 0,4,2→42. return -42 ✓  
# '1337c0d3': digits 1,3,3,7→1337, stop at 'c'. return 1337 ✓  
# '99999999999': overflow triggers → return INT_MAX ✓  
# '': i=0, no sign, no digits → return 0 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(1).  
# The overflow check BEFORE adding each digit is critical – without it we'd overflow  
Python  
# integers silently (Python handles big ints) but the logic breaks for fixed-width  
languages.  
# The magic '7' in  $d > 7$  comes from  $2^{31}-1 = 2147483647$  – last digit is 7.  
# Given more time I'd ask: handle locale-specific decimal separators (commas)?  
# Add a pre-processing step to strip/replace them before parsing."
```


String

□ ★ #249 Group Shifted Strings ★

MED[Open on LeetCode](#)

Array · Hash Table · String

Lists: ★ Premium | Premium 98

Problem

Perform the following shift operations on a string:

- **Right shift:** Replace every letter with the successive letter of the English alphabet, where 'z' is replaced by 'a'. For example, "abc" can be right-shifted to "bcd" or "xyz" can be right-shifted to "yza".
- **Left shift:** Replace every letter with the preceding letter of the English alphabet, where 'a' is replaced by 'z'. For example, "bcd" can be left-shifted to "abc" or "yza" can be left-shifted to "xyz".

We can keep shifting the string in both directions to form an endless shifting sequence.

- For example, shift "abc" to form the sequence: ... <-> "abc" <-> "bcd" <-> ... <->

"xyz" <-> "yza" <-> <-> "zab" <-> "abc"
 <-> ...

You are given an array of strings `strings`, group together all `strings[i]` that belong to the same shifting sequence. You may return the answer in any order.

Example 1:

Input: `strings =`

`["abc","bcd","acef","xyz","az","ba","a","z"]`

Output:

`[["acef"],["a","z"],["abc","bcd","xyz"],["az","ba"]]`

Example 2:

Input: `strings = ["a"]`

Output: `[["a"]]`

Constraints:

- `1 <= strings.length <= 200`
- `1 <= strings[i].length <= 50`
- `strings[i]` consists of lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Two strings are in the same shifting sequence if one can become the other

by consistently shifting every character by the same amount (cyclically, a-z wraps).

Group them. Return order doesn't matter. Right?

```
# 'abc' and 'xyz' are in the same group – same relative gaps between characters."
#
# ["'abc','bcd','xyz','az','ba','a','z']:
# 'abc','bcd','xyz' → all have gaps [1,1] → same group.
# 'az','ba' → gaps [25] and [25] (b-a=1... wait: 'az': z-a=25. 'ba': a-b=-1+26=25) →
# same group.
# 'a','z' → single char, empty gap string → same group."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# For each string, generate all 26 rotations by shifting every character by 0..25.
# If any rotation of string A equals string B, they belong in the same group.
# This is  $O(n^2 * 26 * L)$  – quadratic in the number of strings, far too slow.
# Time:  $O(n^2 * 26 * L)$ . Space:  $O(n)$ . Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Compute a canonical key for each string: the tuple of consecutive char
differences mod 26.
# 'abc' → (1,1). 'xyz' → (1,1). 'az' → (25,). 'ba' → (25,).
# All strings with the same key are in the same shifting group.
# Group by key using a defaultdict. Time:  $O(n * L)$ . Space:  $O(n * L)$ ."
```

PHASE 4 — CLEAN CODE

```
class _249_GroupShiftedStrings:
    def groupStrings(self, strings: List[str]) -> List[List[str]]:
        groups: defaultdict = defaultdict(list)
        for s in strings:
            key = tuple((ord(s[i]) - ord(s[i - 1])) % 26 for i in range(1, len(s)))
            groups[key].append(s)
        return list(groups.values())
```

PHASE 5 — TEST & DEBUG

```
# 'abc': key=(1,1). 'bcd': (1,1). 'xyz': (1,1) → same group ✓
# 'az': key=((ord('z')-ord('a'))%26,)=(25,). 'ba': ((ord('a')-ord('b'))%26,)=(25,)
→ same ✓
# 'a','z': key=() (empty tuple) → same group ✓
# 'acef': key=(2,2,1) → its own group ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n * L)$  where  $L = \text{max string length}$ . Space  $O(n * L)$ .
# The %26 mod handles the wraparound: 'z'→'a' gives  $(\text{ord('a')}-\text{ord('z')})\%26 = (-25)\%26 = 1$ .
# Without the mod, 'az' and 'ba' wouldn't match – the mod is the key insight.
# Given more time I'd ask: how do you efficiently check if two strings are in the
same group
# at query time? Build a map string→canonical_key at init, then  $O(L)$  per query."
```

String

★ #271 Encode and Decode Strings ★

MED[Open on LeetCode](#)

Array · String · Design

Lists: ★ Premium | Grind 169 | Premium 98

Problem

Design an algorithm to encode a list of strings to a string. The encoded string is then sent over the network and is decoded back to the original list of strings.

Machine 1 (sender) has the function:

```
string encode(vector<string> strs) {  
    // ... your code  
    return encoded_string;  
}
```

Machine 2 (receiver) has the function:

```
vector<string> decode(string s) {  
    //... your code  
    return strs;  
}
```

So Machine 1 does:

```
string encoded_string = encode(strs);
```

and Machine 2 does:

```
vector<string> strs2 = decode(encoded_string);
```

strs2 in Machine 2 should be the same as strs in Machine 1.

Implement the encode and decode methods.

You are not allowed to solve the problem using any serialize methods (such as eval).

Example 1:

```
Input: dummy_input = ["Hello","World"]
Output: ["Hello","World"]
Explanation:
Machine 1:
Codec encoder = new Codec();
String msg = encoder.encode(strs);
Machine 1 ----msg----> Machine 2
```

Example 2:

```
Input: dummy_input = [""]
Output: [""]
```

Constraints:

- $1 \leq \text{strs.length} \leq 200$
- $0 \leq \text{strs}[i].\text{length} \leq 200$
- **strs[i]** contains any possible characters out of 256 valid ASCII characters.

Follow up: Could you write a generalized algorithm to work on any possible set of characters?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Design encode(strs) and decode(s) that roundtrip correctly for ANY list of
strings –
# including strings with special characters, empty strings, or strings containing
# whatever delimiter we might choose. Right?
# We can't use eval or pickle."
#
# ["'Hello','World'"] → encode → some string → decode → ['Hello','World'] ✓
# [''] → [''] ✓ (empty string must survive the roundtrip)"

# PHASE 2 — BRUTE FORCE
# "Let me start with the brute force to lock in the logic.
# A naive approach is to join strings with a rare delimiter like '|' – but it breaks
if
# strings themselves contain '|'. We'd need escaping: replace '|' with '\\|' and '\\'
with '\\\\'.
# This works but decoding is error-prone and complex to get right.
# Time: O(total chars). Space: O(total chars). Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE (length-prefix encoding)
# "Prefix each string with its length and a fixed separator (e.g. '#').
# encode: '5#Hello4#World'. decode: read number up to '#', slice that many chars,
repeat.
# No ambiguity – length tells us exactly where each string ends.
# Works for any content including '#' and empty strings."

# PHASE 4 — CLEAN CODE
class _271_Codec:
    def encode(self, strs: List[str]) -> str:
        return ''.join(f'{len(s)}#{s}' for s in strs)
    def decode(self, s: str) -> List[str]:
        result = []
        i = 0
        while i < len(s):
            j = s.index('#', i)
            length = int(s[i:j])
            result.append(s[j + 1: j + 1 + length])
            i = j + 1 + length
        return result

# PHASE 5 — TEST & DEBUG
# encode(['Hello','World']): '5#Hello5#World'.
# decode('5#Hello5#World'):
# i=0: j=1 (index of '#'). length=5. result=['Hello']. i=7.
# i=7: j=8. length=5. result=['Hello','World']. i=14. done ✓
# encode(['']): '0#'. decode('0#'): j=1, length=0, append ''. → [''] ✓
# encode(['a#b','c']): '3#a#b1#c'. decode: length=3→'a#b', length=1→'c' ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "encode O(total chars). decode O(total chars). Space O(n)."

```

Length-prefix is the standard solution – no special characters, no escaping needed.
The '#' separator is arbitrary – any fixed separator works since we always read
the length first and know exactly how many chars to consume.
Given more time I'd ask: what if strings can contain binary data (null bytes)?
Use a 4-byte big-endian integer prefix instead of a text number – fully binary-safe."

String

□ #635 Design Log Storage System

MED[Open on LeetCode](#)

Hash Table · String · Design · Ordered Set
Lists: Denny Zhang

Problem

You are given several logs, where each log contains a unique ID and timestamp. Timestamp is a string that has the following format: Year:Month:Day:Hour:Minute:Second, for example, 2017:01:01:23:59:59. All domains are zero-padded decimal numbers.

Implement the LogSystem class:

- **LogSystem()** Initializes the LogSystem object.
- **void put(int id, string timestamp)** Stores the given log (id, timestamp) in your storage system.
- **int[] retrieve(string start, string end, string granularity)** Returns the IDs of the logs whose timestamps are within the range from start to end inclusive. start and end all have the same

format as timestamp, and granularity means how precise the range should be (i.e. to the exact Day, Minute, etc.). For example, start = "2017:01:01:23:59:59", end = "2017:01:02:23:59:59", and granularity = "Day" means that we need to find the logs within the inclusive range from Jan. 1st 2017 to Jan. 2nd 2017, and the Hour, Minute, and Second for each log entry can be ignored.

Example 1:

Input

```
["LogSystem", "put", "put", "put", "retrieve", "retrieve"]  
[[], [1, "2017:01:01:23:59:59"], [2, "2017:01:01:22:59:59"], [3, "2016:01:01:00:00:00"], ["2016:01:01:01:01:01", "2017:01:01:23:00:00", "Year"], ["2016:01:01:01:01:01", "2017:01:01:23:00:00", "Hour"]]
```

Output

```
[null, null, null, null, [3, 2, 1], [2, 1]]
```

Explanation

```
LogSystem logSystem = new LogSystem();
```

Constraints:

- $1 \leq \text{id} \leq 500$
- $2000 \leq \text{Year} \leq 2017$
- $1 \leq \text{Month} \leq 12$
- $1 \leq \text{Day} \leq 31$
- $0 \leq \text{Hour} \leq 23$
- $0 \leq \text{Minute, Second} \leq 59$

- granularity is one of the values ["Year", "Month", "Day", "Hour", "Minute", "Second"].
- At most 500 calls will be made to put and retrieve.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Store logs with (id, timestamp). retrieve(start, end, granularity) returns IDs
# of logs within the range, where comparison is only up to the granularity level.
# Timestamps format: 'YYYY:MM:DD:HH:mm:ss'. Truncate at the granularity character
# position. Right?"
#
# "retrieve with 'Year': compare only the first 4 chars. '2017:....' vs '2016:....'
# etc."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Store all logs in a flat list. On retrieve, iterate over every log, truncate its
# timestamp
# to the granularity prefix length, and check if it falls within the truncated
# start/end range.
# With at most 500 calls and a fixed 19-char timestamp format, O(n) per retrieve is
# acceptable.
# Time: put O(1), retrieve O(n). Space: O(n). Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Map granularity name to the string prefix length that captures it.
# 'Year'→4 chars, 'Month'→7, 'Day'→10, 'Hour'→13, 'Minute'→16, 'Second'→19.
# Truncate all three timestamps to that length and do a simple string comparison.
# Lexicographic order of the timestamp format is correct for chronological order."
```

PHASE 4 — CLEAN CODE

```
class _635_LogSystem:
    GRAN = {'Year': 4, 'Month': 7, 'Day': 10, 'Hour': 13, 'Minute': 16, 'Second':
19}

    def __init__(self):
        self.logs: List[tuple] = []

    def put(self, log_id: int, timestamp: str) -> None:
        self.logs.append((log_id, timestamp))

    def retrieve(self, start: str, end: str, granularity: str) -> List[int]:
        cut = self.GRAN[granularity]
```

```
        return [lid for lid, ts in self.logs if start[:cut] <= ts[:cut] <=
end[:cut]]
```

PHASE 5 — TEST & DEBUG

```
# put(1, '2017:01:01:23:59:59'), put(2, '2017:01:01:22:59:59'),
put(3, '2016:01:01:00:00:00').
# retrieve(..., 'Year'): cut=4. compare '2016' <= 'year' <= '2017'. All 3 match →
[1,2,3] ✓
# retrieve(..., 'Hour'): cut=13. '2016:01:01:01' <= '2016:01:01:00'? No→3 excluded.
# '2016:01:01:01' <= '2017:01:01:22' <= '2017:01:01:23'? Yes→2. '2017:01:01:23'? Yes→1.
# → [2,1] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "put 0(1). retrieve 0(n) — scans all logs.
# The key insight: timestamps in 'YYYY:MM:DD:HH:mm:ss' format sort
lexicographically
# correctly because they're zero-padded and delimiters are identical.
# For larger datasets, a sorted list + binary search would bring retrieve to 0(log n
+ k).
# Given more time I'd ask: what if granularity could be arbitrary (e.g. 'Week')?
# We'd need a custom truncation function rather than fixed prefix lengths."
```

String

□ #1138 Alphabet Board Path

MED[Open on LeetCode](#)

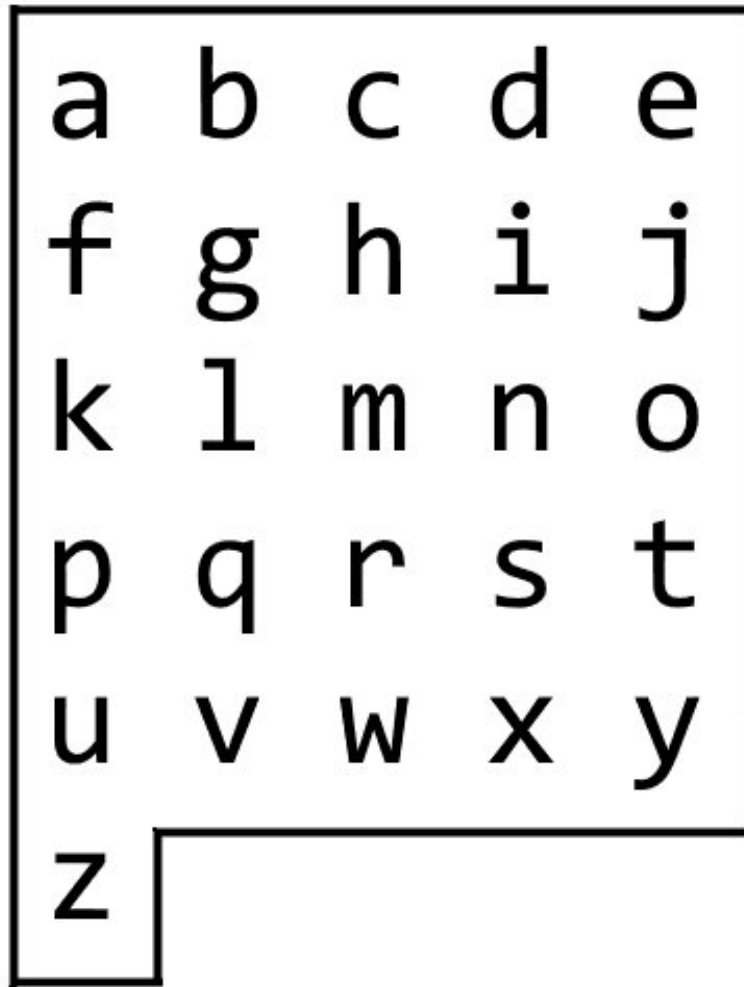
Hash Table · String

Lists: Denny Zhang

Problem

On an alphabet board, we start at position (0, 0), corresponding to character board[0][0].

Here, board = ["abcde", "fghij", "klmno", "pqrst", "uvwxy", "z"], as shown in the diagram below.



We may make the following moves:

- 'U' moves our position up one row, if the position exists on the board;
- 'D' moves our position down one row, if the position exists on the board;
- 'L' moves our position left one column, if the position exists on the board;
- 'R' moves our position right one column, if the position exists on the board;

- '!' adds the character `board[r][c]` at our current position (r, c) to the answer.

(Here, the only positions that exist on the board are positions with letters on them.)

Return a sequence of moves that makes our answer equal to target in the minimum number of moves. You may return any path that does so.

Example 1:

Input: target = "leet"

Output: "DDR!UURRR!!DDD!"

Example 2:

Input: target = "code"

Output: "RR!DDRR!UUL!R!"

Constraints:

- $1 \leq \text{target.length} \leq 100$
- target consists only of English lowercase letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Board is ['abcde','fghij','klmno','pqrst','uvwxy','z'] – 5 cols except row 5 which has only 'z'.

Start at (0,0). For each char in target, output U/D/L/R moves then '!'. Right?

The critical constraint: 'z' is at (5,0) – the only cell in its row.

Moving to 'z' must clear the column before going down. Moving from 'z' must go up before going right."

#

"target='leet': l is at (2,1). Move D,D,R,!. e=(0,4) – go U,U,R,R,R,!. etc."

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.

For each character, compute its board position as $row = (ord(ch) - ord('a')) // 5$,
$col = (ord(ch) - ord('a')) \% 5$. Then generate moves: go vertical first, then horizontal.

The naive approach fails for 'z' – moving down before clearing column 0 steps off the board.

Time: $O(|target| * max_moves)$. Space: $O(|target|)$. Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Formula: $row = (ord(ch) - ord('a')) // 5$, $col = (ord(ch) - ord('a')) \% 5$.

For 'z' specifically: it's at (5,0), $col=0$. When moving TO 'z', move left first (to reach $col=0$) then move down (to reach row 5) – otherwise we'd step off the board.

When moving FROM 'z', move up first then move right – same reason.

General case: move vertically then horizontally (or horizontally then vertically).

'z' is the only exception – must move left before down."

PHASE 4 — CLEAN CODE

```

class _1138_AlphabetBoardPath:
    def alphabetBoardPath(self, target: str) -> str:
        row = col = 0
        moves = []
        for ch in target:
            dest_row = (ord(ch) - ord('a')) // 5
            dest_col = (ord(ch) - ord('a')) % 5
            if ch == 'z': # must clear column before going down
                moves.extend('L' * (col - dest_col))
                moves.extend('D' * (dest_row - row))
            else:
                moves.extend('U' * (row - dest_row))
                moves.extend('D' * (dest_row - row))
                moves.extend('L' * (col - dest_col))
                moves.extend('R' * (dest_col - col))
            moves.append('!')
            row, col = dest_row, dest_col
        return ''.join(moves)

```

PHASE 5 — TEST & DEBUG

target='leet': $l=(2,1)$. from (0,0): D,D,R,! → "DDR!". row=2,col=1.

e=(0,4): U,U,R,R,R,! → "UURRR!". row=0,col=4.

e=(0,4): no move,! → "!". row=0,col=4.

t=(3,4): D,D,D,! → "DDD!". → "DDR!UURRR!!DDD!" ✓

target='z': from (0,0): ch='z'. $L*(0)=none$. $D*5$. → "DDDDD!" ✓

If coming from (5,4) to 'z' (5,0): L,L,L,L,! – no D needed ✓.

If going from 'z' to 'a': ch='a'=(0,0). Not 'z' → $U*5, L*0, R*0$ → "UUUUU!" ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(|target| * max_moves) = O(|target| * 10) = O(|target|)$. Space $O(|target|)$.

```
# The 'z' case is the entire problem – everything else is standard grid navigation.  
# Phrase it clearly: 'z' is at the leftmost position of its row, so we must reach  
col 0  
# before stepping down into row 5, otherwise intermediate positions don't exist on  
the board.  
# Given more time I'd ask: what if the board were different (e.g. a phone numpad)?  
# Same approach – just recompute (row, col) from the new layout."
```


Two Pointers

Round 2 · High · Medium · 14 questions

Two Pointers

□ #11 Container With Most Water

MED[Open on LeetCode](#)

Array · Two Pointers · Greedy

Lists: Grind 169

Problem

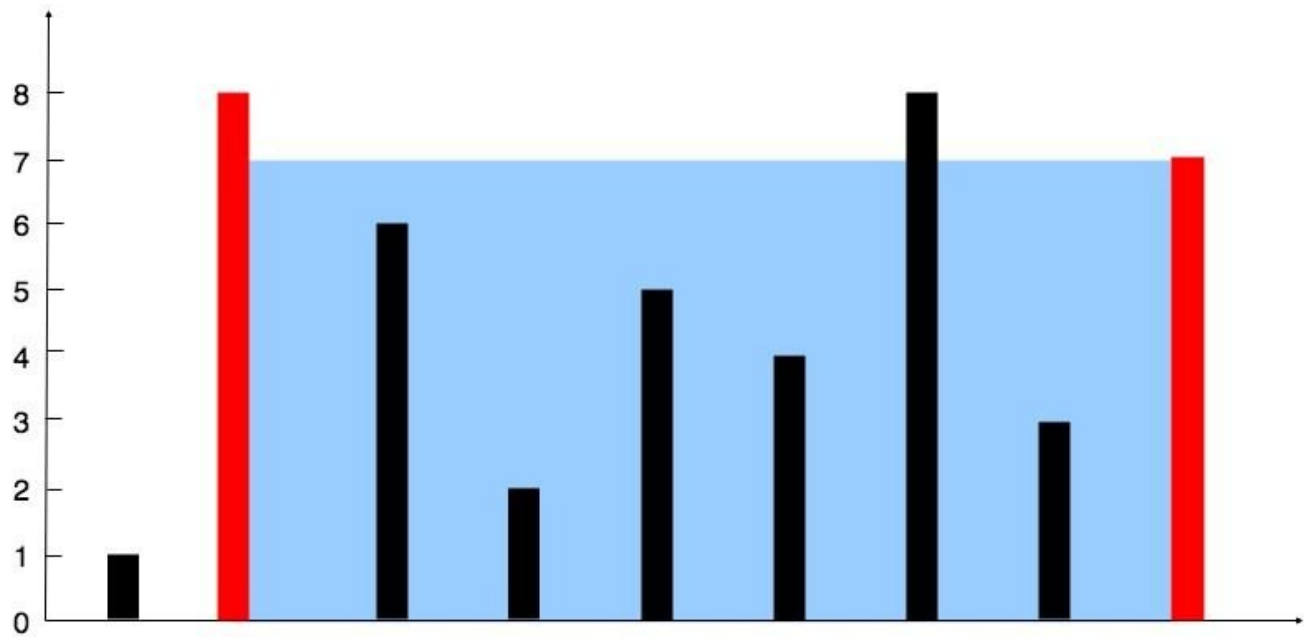
You are given an integer array `height` of length `n`. There are `n` vertical lines drawn such that the two endpoints of the `i`th line are $(i, 0)$ and $(i, \text{height}[i])$.

Find two lines that together with the `x`-axis form a container, such that the container contains the most water.

Return the maximum amount of water a container can store.

Notice that you may not slant the container.

Example 1:



Input: height = [1,8,6,2,5,4,8,3,7]

Output: 49

Explanation: The above vertical lines are represented by array [1,8,6,2,5,4,8,3,7]. In this case, the max area of water (blue section) the container can contain is 49.

Example 2:

Input: height = [1,1]

Output: 1

Constraints:

- $n == \text{height.length}$
- $2 \leq n \leq 105$
- $0 \leq \text{height}[i] \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given heights of vertical lines, find two lines that form the largest container.

Area = $\min(\text{height}[\text{left}], \text{height}[\text{right}]) * (\text{right} - \text{left})$. Return max area. Right?

We can't slant – so water level is always min of the two chosen heights."

#

"[1,8,6,2,5,4,8,3,7]: left=1(h=8), right=8(h=7). Area= $\min(8,7)*(8-1)=49$ ✓"

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.

For every pair (i, j) where $j > i$, compute $\text{area} = \min(\text{height}[i], \text{height}[j]) * (j - i)$

and track the maximum. That's $C(n, 2)$ pairs to check.

Time: $O(n^2)$. Space: $O(1)$. Should I go ahead and optimize?"

```
class _11_ContainerMostWater_Brute:
    def maxArea(self, height):
        best = 0
        for i in range(len(height)):
            for j in range(i + 1, len(height)):
                best = max(best, min(height[i], height[j]) * (j - i))
        return best
```

PHASE 3 — OPTIMIZE

"Two pointers: left=0, right=n-1. At each step compute area.

Move the pointer with the SHORTER height inward – keeping the taller height

maximises any future container. Moving the taller pointer can only shrink

both width and potentially height, so it can't improve the result.

Time: $O(n)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _11_ContainerMostWater:
    def maxArea(self, height: List[int]) -> int:
        left, right = 0, len(height) - 1
        best = 0
        while left < right:
            area = min(height[left], height[right]) * (right - left)
            best = max(best, area)
            if height[left] < height[right]:
                left += 1
            else:
                right -= 1
        return best
```

PHASE 5 — TEST & DEBUG

height=[1,8,6,2,5,4,8,3,7]:

l=0(h=1),r=8(h=7): area=1*8=8. $h[l] < h[r] \rightarrow l=1$.

l=1(h=8),r=8(h=7): area=7*7=49. $h[l] > h[r] \rightarrow r=7$.

l=1(h=8),r=7(h=3): area=3*6=18. $h[l] > h[r] \rightarrow r=6$.

... best stays 49. $\rightarrow 49$ ✓

height=[1,1]: area=1*1=1 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(1)$.

The greedy argument: if we move the taller pointer, we keep a shorter constraint (limited by the short side) AND reduce width – area can only shrink or stay.

So we always move the shorter side – we can't do better by moving the taller one.

This is the key insight to state explicitly in the interview.

```
# Given more time I'd ask: what if we need to find the pair of indices (not just the area)?  
# Track left_best and right_best alongside best – same code, just record the positions."
```

Two Pointers

□ #15 3Sum

MED[Open on LeetCode](#)

Array · Two Pointers · Sorting

Lists: Grind 169

Problem

Given an integer array `nums`, return all the triplets `[nums[i], nums[j], nums[k]]` such that $i \neq j$, $i \neq k$, and $j \neq k$, and $nums[i] + nums[j] + nums[k] == 0$.

Notice that the solution set must not contain duplicate triplets.

Example 1:

Input: `nums = [-1,0,1,2,-1,-4]`

Output: `[[-1,-1,2],[-1,0,1]]`

Explanation:

`nums[0] + nums[1] + nums[2] = (-1) + 0 + 1 = 0.`

`nums[1] + nums[2] + nums[4] = 0 + 1 + (-1) = 0.`

`nums[0] + nums[3] + nums[4] = (-1) + 2 + (-1) = 0.`

The distinct triplets are `[-1,0,1]` and `[-1,-1,2]`.

Notice that the order of the output and the order of the triplets does not matter.

Example 2:

Input: `nums = [0,1,1]`

Output: `[]`

Explanation: The only possible triplet does not sum up to 0.

Example 3:

Input: `nums = [0,0,0]`

Output: `[[0,0,0]]`

Explanation: The only possible triplet sums up to 0.

Constraints:

- $3 \leq \text{nums.length} \leq 3000$
- $-105 \leq \text{nums}[i] \leq 105$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find all unique triplets summing to 0. Return the triplets, not indices.
# No duplicate triplets in the output. Right?
# Order within a triplet doesn't matter - [-1,0,1] and [0,-1,1] are the same."
#
# "[-1,0,1,2,-1,-4]: sort → [-4,-1,-1,0,1,2]. → [[-1,-1,2],[-1,0,1]] ✓
# [0,0,0]: → [[0,0,0]] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Three nested loops over i < j < k; if nums[i]+nums[j]+nums[k]==0, add triplet.
# Store triplets in a set keyed by sorted tuple to deduplicate.
# Correct, but O(n^3) with hash-set overhead on large n.
# Time: O(n^3). Space: O(n) for dedup set.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Sort the array. Fix one element nums[i], use two pointers on the rest.
# Sorting enables: skip duplicate values of nums[i] to avoid duplicate triplets,
# and move pointers efficiently based on the sum.
# After finding a valid triplet, skip duplicate nums[left] and nums[right].
# Time: O(n^2). Space: O(1) extra."
```

PHASE 4 — CLEAN CODE

```
class _15_ThreeSum:
    def threeSum(self, nums: List[int]) -> List[List[int]]:
        nums.sort()
        result = []
        for i in range(len(nums) - 2):
            if i > 0 and nums[i] == nums[i - 1]: # skip duplicate fixed element
                continue
            left, right = i + 1, len(nums) - 1
```

```

        while left < right:
            total = nums[i] + nums[left] + nums[right]
            if total < 0:
                left += 1
            elif total > 0:
                right -= 1
            else:
                result.append([nums[i], nums[left], nums[right]])
                left += 1
                right -= 1
                while left < right and nums[left] == nums[left - 1]: left += 1
                while left < right and nums[right] == nums[right + 1]: right -= 1
1
    return result

```

PHASE 5 — TEST & DEBUG

```

# nums=[-1,0,1,2,-1,-4] → sorted: [-4,-1,-1,0,1,2].
# i=0 (-4): l=1,r=5. sums=-4+(-1)+2=-3<0→l++. -4+(-1)+2=-3<0→l++. -4+0+2=-2<0→l++.
# -4+1+2=-1<0→l++. l=r → stop.
# i=1 (-1): l=2,r=5. -1+(-1)+2=0→append[-1,-1,2]. l=3,r=4. skip dups: l=3 ok, r=4
ok.
# -1+0+1=0→append[-1,0,1]. l=4,r=3→stop.
# i=2 (-1): i>0 and nums[2]==nums[1]==-1 → skip.
# i=3 (0): l=4,r=5. 0+1+2=3>0→r--. l≥r→stop.
# result=[[-1,-1,2],[-1,0,1]] ✓
# nums=[0,0,0]: i=0,l=1,r=2. 0+0+0=0→append. l=2,r=1→stop. → [[0,0,0]] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n^2)$  – outer loop  $O(n)$ , inner two-pointer  $O(n)$  each. Sorting is  $O(n \log n)$ 
dominated.
# Space  $O(1)$  extra (ignoring output).
# Two deduplication steps:
# 1. Skip  $\text{nums}[i] == \text{nums}[i-1]$  to avoid re-fixing the same outer value.
# 2. After appending, skip  $\text{nums}[\text{left}] == \text{nums}[\text{left}-1]$  and  $\text{nums}[\text{right}] ==$ 
 $\text{nums}[\text{right}+1]$ 
# to avoid re-collecting the same triplet.
# Both checks are essential – missing either produces duplicate output.
# Given more time I'd ask: 4Sum (#18)? Add one more outer loop –  $O(n^3)$  – same
structure."

```


Two Pointers

□ #16 3Sum Closest

MED[Open on LeetCode](#)

Array · Two Pointers · Sorting

Lists: Grind 169

Problem

Given an integer array `nums` of length `n` and an integer `target`, find three integers at distinct indices in `nums` such that the sum is closest to `target`.

Return the sum of the three integers.

You may assume that each input would have exactly one solution.

Example 1:

Input: `nums = [-1,2,1,-4]`, `target = 1`

Output: 2

Explanation: The sum that is closest to the target is 2. ($-1 + 2 + 1 = 2$).

Example 2:

Input: `nums = [0,0,0]`, `target = 1`

Output: 0

Explanation: The sum that is closest to the target is 0. ($0 + 0 + 0 = 0$).

Constraints:

- $3 \leq \text{nums.length} \leq 500$

- $-1000 \leq \text{nums}[i] \leq 1000$
- $-104 \leq \text{target} \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find three distinct-index integers whose sum is closest to target. Return the
sum.
# Exactly one solution guaranteed – no tie-breaking needed. Right?"
#
# "[-1,2,1,-4], target=1: possible sums: -1+2+1=2, -1+2-4=-3, -1+1-4=-4, 2+1-4=-1.
# Closest to 1 is 2. → 2 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Three nested loops – for every triple (i, j, k), compute the sum and track
# whichever sum has the smallest absolute difference from target.
# Time:  $O(n^3)$ . Space:  $O(1)$ . Should I go ahead and optimize?"
```

```
class _16_ThreeSumClosest_Brute:
    def threeSumClosest(self, nums, target):
        best = nums[0] + nums[1] + nums[2]
        n = len(nums)
        for i in range(n - 2):
            for j in range(i + 1, n - 1):
                for k in range(j + 1, n):
                    s = nums[i] + nums[j] + nums[k]
                    if abs(target - s) < abs(target - best):
                        best = s
        return best
```

PHASE 3 — OPTIMIZE

```
# "Exact same structure as 3Sum: sort + fix one element + two pointers.
# Instead of looking for sum==0, track the closest sum seen.
# If sum > target → move right pointer left (decrease sum).
# If sum < target → move left pointer right (increase sum).
# If sum == target → return immediately (can't get closer).
# Skip duplicate outer values to avoid redundant work – not required for correctness
# here (no dedup of output), but speeds up the average case."
```

PHASE 4 — CLEAN CODE

```
class _16_ThreeSumClosest:
    def threeSumClosest(self, nums: List[int], target: int) -> int:
        nums.sort()
        n = len(nums)
        best = nums[0] + nums[1] + nums[2]
        for i in range(n - 2):
```

```

    if i > 0 and nums[i] == nums[i - 1]:
        continue
    left, right = i + 1, n - 1
    while left < right:
        total = nums[i] + nums[left] + nums[right]
        if total == target:
            return target
        if abs(target - total) < abs(target - best):
            best = total
        if total > target:
            right -= 1
        else:
            left += 1
    return best

```

PHASE 5 — TEST & DEBUG

```

# nums=[-1,2,1,-4], target=1. sorted: [-4,-1,1,2].
# i=0(-4): l=1,r=3. -4-1+2=-3. |1-(-3)|=4 < |1-(-4)|=5 → best=-3. sum<1→l++.
# l=2,r=3. -4+1+2=-1. |1-(-1)|=2 < 4 → best=-1. sum<1→l++. l=r→stop.
# i=1(-1): l=2,r=3. -1+1+2=2. |1-2|=1 < |1-(-1)|=2 → best=2. sum>1→r--. l=r→stop.
# → 2 ✓
# nums=[0,0,0], target=1. best=0. No closer sum possible → 0 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n^2)$ . Space  $O(1)$  extra ( $O(\log n)$  for sort).
# This is 3Sum (#15) with one change: track closest instead of exact zero.
# The early return on exact match is a nice optimisation worth mentioning.
# Given more time I'd ask: what if we want the k closest sums?
# Collect all triple sums, sort by |target - sum|, return first k —  $O(n^2 \log n)$ ."

```

Two Pointers

□ #31 Next Permutation

MED

[Open on LeetCode](#)

Array · Two Pointers

Lists: Grind 169

Problem

A permutation of an array of integers is an arrangement of its members into a sequence or linear order.

- For example, for `arr = [1,2,3]`, the following are all the permutations of `arr`: `[1,2,3]`, `[1,3,2]`, `[2, 1, 3]`, `[2, 3, 1]`, `[3,1,2]`, `[3,2,1]`.

The next permutation of an array of integers is the next lexicographically greater permutation of its integer. More formally, if all the permutations of the array are sorted in one container according to their lexicographical order, then the next permutation of that array is the permutation that follows it in the sorted container. If such arrangement is not possible, the array must be rearranged as the lowest possible order (i.e., sorted in ascending order).

- For example, the next permutation of `arr = [1,2,3]` is `[1,3,2]`.
- Similarly, the next permutation of `arr = [2,3,1]` is `[3,1,2]`.
- While the next permutation of `arr = [3,2,1]` is `[1,2,3]` because `[3,2,1]` does not have a lexicographical larger rearrangement.

Given an array of integers `nums`, find the next permutation of `nums`.

The replacement must be in place and use only constant extra memory.

Example 1:

Input: `nums = [1,2,3]`
Output: `[1,3,2]`

Example 2:

Input: `nums = [3,2,1]`
Output: `[1,2,3]`

Example 3:

Input: `nums = [1,1,5]`
Output: `[1,5,1]`

Constraints:

- $1 \leq \text{nums.length} \leq 100$
- $0 \leq \text{nums}[i] \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Rearrange nums in-place to the next lexicographically greater permutation.
# If no greater permutation exists (sorted descending), wrap to the smallest
# (ascending). Right?"
#
# "[1,2,3]→[1,3,2]. [3,2,1]→[1,2,3]. [1,1,5]→[1,5,1]. [1,3,2]→[2,1,3]. ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Generate all permutations of the array in sorted order, find the one that matches
# the current arrangement, then return the next one in the sorted list.
# If the current is the last, wrap to the first (smallest) permutation.
# Time:  $O(n! \cdot n)$  to generate all permutations – completely infeasible for  $n > 10$ .
# Space:  $O(n!)$ . Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Three steps – all  $O(n)$ ,  $O(1)$  space:
# 1. Find the rightmost 'dip': scan right to left for the largest  $i$  where  $\text{nums}[i] < \text{nums}[i+1]$ .
# If no dip exists, entire array is descending → just reverse it (wrap-around).
# 2. Find the smallest element to the right of  $i$  that is still larger than  $\text{nums}[i]$ .
# (The suffix is descending after step 1, so scan from the right.)
# 3. Swap  $\text{nums}[i]$  with that element, then reverse the suffix after  $i$ .
# Reversing the descending suffix makes it the smallest possible arrangement."
```

PHASE 4 — CLEAN CODE

```
class _31_NextPermutation:
    def nextPermutation(self, nums: List[int]) -> None:
        def reverse(left, right):
            while left < right:
                nums[left], nums[right] = nums[right], nums[left]
                left += 1; right -= 1

        n = len(nums)
        # Step 1: find rightmost dip
        dip = next((i for i in range(n - 2, -1, -1) if nums[i] < nums[i + 1]), -1)
        if dip >= 0:
            # Step 2: find smallest element to right of dip that is > nums[dip]
            swap = next(j for j in range(n - 1, -1, -1) if nums[j] > nums[dip])
            nums[dip], nums[swap] = nums[swap], nums[dip]
        # Step 3: reverse suffix after dip
        reverse(dip + 1, n - 1)
```

PHASE 5 — TEST & DEBUG

```
# [1,2,3]: dip=1 (nums[1]=2<nums[2]=3). swap=2 (nums[2]=3>nums[1]=2).
# swap: [1,3,2]. reverse(2,2)→no change. → [1,3,2] ✓
# [3,2,1]: dip=-1. reverse(0,2): [1,2,3] ✓
# [1,1,5]: dip=1 (nums[1]=1<nums[2]=5). swap=2 (5>1). swap→[1,5,1].
reverse(2,2)→[1,5,1] ✓
# [1,3,2]: dip=0 (nums[0]=1<nums[1]=3). swap=1 (nums[1]=3>nums[0]=1; nums[2]=2
also>1 but
```

```
# scan right→left so j=1 first: nums[1]=3>1? yes → swap=1? No wait j starts from  
n-1=2:  
# nums[2]=2>nums[0]=1 → swap=2. swap→[2,3,1]. reverse(1,2): [2,1,3] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(1)$.

The suffix is always descending after the dip – that's why scanning right-to-left
finds the smallest swap target efficiently.

The reverse of the suffix turns the largest possible suffix into the smallest –
the key insight.

Given more time I'd ask: what's the previous permutation?

Same algorithm mirrored: find rightmost 'rise' ($\text{nums}[i] > \text{nums}[i+1]$), swap with
largest

element to its right that is still smaller, reverse the suffix."

Two Pointers

□ #75 Sort Colors

MED[Open on LeetCode](#)

Array · Two Pointers · Sorting

Lists: Grind 169

Problem

Given an array `nums` with `n` objects colored red, white, or blue, sort them in-place so that objects of the same color are adjacent, with the colors in the order red, white, and blue.

We will use the integers 0, 1, and 2 to represent the color red, white, and blue, respectively.

You must solve this problem without using the library's sort function.

Example 1:

```
Input: nums = [2,0,2,1,1,0]
Output: [0,0,1,1,2,2]
```

Example 2:

```
Input: nums = [2,0,1]
Output: [0,1,2]
```

Constraints:

- `n == nums.length`

- $1 \leq n \leq 300$
- `nums[i]` is either 0, 1, or 2.

Follow up: Could you come up with a one-pass algorithm using only constant extra space?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Sort an array containing only 0, 1, 2 in-place without using library sort.
# 0=red, 1=white, 2=blue. Output: all 0s, then 1s, then 2s. Right?
# Follow-up asks for a one-pass O(1) space solution."
#
# "[2,0,2,1,1,0] → [0,0,1,1,2,2] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# First pass: count the number of 0s, 1s, and 2s in the array.
# Second pass: overwrite the array – fill count[0] zeros, then count[1] ones, then
count[2] twos.
# This is two passes but still O(n) time and O(1) space.
# Time: O(n). Space: O(1). Should I go ahead and optimize?"
```

```
class _75_SortColors_Brute:
    def sortColors(self, nums):
        c0 = nums.count(0)
        c1 = nums.count(1)
        for i in range(len(nums)):
            if i < c0:
                nums[i] = 0
            elif i < c0 + c1:
                nums[i] = 1
            else:
                nums[i] = 2
```

PHASE 3 — OPTIMIZE

```
# "Dutch National Flag algorithm – one pass, three pointers: low, mid, high.
# Invariant: nums[0..low-1]=0, nums[low..mid-1]=1, nums[high+1..n-1]=2,
nums[mid..high]=unknown.
# At each step examine nums[mid]:
# 0 → swap(low,mid), low++, mid++ (0 placed, known-1 region grows both sides).
# 1 → mid++ (already correct).
# 2 → swap(mid,high), high-- (don't mid++ – swapped element is unknown)."
```

PHASE 4 — CLEAN CODE

```
class _75_SortColors:
```

```
def sortColors(self, nums: List[int]) -> None:
    low = mid = 0
    high = len(nums) - 1
    while mid <= high:
        if nums[mid] == 0:
            nums[low], nums[mid] = nums[mid], nums[low]
            low += 1; mid += 1
        elif nums[mid] == 1:
            mid += 1
        else:
            nums[mid], nums[high] = nums[high], nums[mid]
            high -= 1 # don't mid++ - swapped value is unexamined
```

PHASE 5 — TEST & DEBUG

```
# [2,0,2,1,1,0]: low=0,mid=0,high=5.
# mid=0(2): swap(0,5)→[0,0,2,1,1,2]. high=4.
# mid=0(0): swap(0,0)→same. low=1,mid=1.
# mid=1(0): swap(1,1)→same. low=2,mid=2.
# mid=2(2): swap(2,4)→[0,0,1,1,2,2]. high=3.
# mid=2(1): mid=3.
# mid=3(1): mid=4. mid>high=3 → stop.
# → [0,0,1,1,2,2] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n) one pass. Space O(1).
# The 'don't mid++' after swapping a 2 is the subtlety – the swapped element from
high
# hasn't been examined yet and could be 0, 1, or 2.
# After swapping a 0, we DO mid++ because nums[low] before the swap was definitely 1
# (it's in the known-1 region), so after swapping, mid lands on a known-1 element.
# Given more time I'd ask: what if there were k colours?
# k=2 is a standard partition; k=3 is Dutch Flag; k>3 needs multiple passes or a
radix approach."
```

Two Pointers

★ #161 One Edit Distance ★

MED[Open on LeetCode](#)

Two Pointers · String

Lists: ★ Premium | Premium 98

Problem

Given two strings *s* and *t*, return true if they are both one edit distance apart, otherwise return false.

A string *s* is said to be one distance apart from a string *t* if you can:

- Insert exactly one character into *s* to get *t*.
- Delete exactly one character from *s* to get *t*.
- Replace exactly one character of *s* with a different character to get *t*.

Example 1:

```
Input: s = "ab", t = "acb"  
Output: true  
Explanation: We can insert 'c' into s to get t.
```

Example 2:

```
Input: s = "", t = ""  
Output: false  
Explanation: We cannot get t from s by only one step.
```

Constraints:

- $0 \leq s.length, t.length \leq 104$
- s and t consist of lowercase letters, uppercase letters, and digits.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Return true if s and t are exactly one edit apart – insert, delete, or replace.
# Important: zero edits (equal strings) returns false. Right?
# s='ab', t='acb' → insert 'c' → true ✓. s='',t='' → false ✓."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Run full Levenshtein DP between s and t; return True iff edit distance equals 1.
# Handles insert, delete, and replace in one table.
# Correct but  $O(m \times n)$  when we only need to detect a single edit.
# Time:  $O(m \times n)$ . Space:  $O(m \times n)$  DP table.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Two observations that short-circuit work:
# If  $|\text{len}(s) - \text{len}(t)| > 1$  → impossible, return false.
# Ensure s is the shorter string (swap if needed) so we only handle two cases:
# same length → exactly one replacement, rest identical.
# s shorter by 1 → one deletion from t (or insertion into s), rest identical.
# Scan left to right. On first mismatch:
# same length → check  $s[i+1:] == t[i+1:]$ .
# diff length → check  $s[i:] == t[i+1:]$ .
# If no mismatch found → true only if t is exactly 1 longer."
```

PHASE 4 — CLEAN CODE

```
class _161_OneEditDistance:
    def isOneEditDistance(self, s: str, t: str) -> bool:
        def check(shorter, longer): # len(longer) == len(shorter) + 1
            for i in range(len(shorter)):
                if shorter[i] != longer[i]:
                    return shorter[i:] == longer[i + 1:]
            return True # shorter is prefix of longer
        m, n = len(s), len(t)
        if abs(m - n) > 1:
            return False
        if m == n:
```

```

        diffs = sum(a != b for a, b in zip(s, t))
        return diffs == 1
    if m > n:
        s, t = t, s
        m, n = n, m
    return check(s, t)

```

PHASE 5 — TEST & DEBUG

```

# s='ab', t='acb': m=2,n=3. m<n → check('ab','acb').
# i=0: 'a'=='a'. i=1: 'b'!='c'. return 'b'=='b'? shorter[1:]='b', longer[2:]='b' →
True ✓
# s='', t='': diffs=0 → return 0==1 → False ✓
# s='a', t='a': same length, diffs=0 → False ✓
# s='a', t='b': same length, diffs=1 → True ✓
# s='abc', t='abc': diffs=0 → False ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(min(m,n)). Space O(1) for the check (slice comparison is O(min) under the
hood).
# The key boundary: we return true when no mismatch found in the same-length case
only if
# diffs==1; in the diff-length case only if shorter is a prefix (true only if t is
exactly
# one char longer, which we already guaranteed).
# Given more time I'd ask: what if we want EXACTLY k edits?
# Full edit distance DP — O(m*n) time, O(min(m,n)) space."

```

Two Pointers

□ #167 Two Sum II – Input Array Is Sorted

MED[Open on LeetCode](#)

Array · Two Pointers · Binary Search

Lists: Denny Zhang

Problem

Given a 1-indexed array of integers `numbers` that is already sorted in non-decreasing order, find two numbers such that they add up to a specific target number. Let these two numbers be `numbers[index1]` and `numbers[index2]` where $1 \leq \text{index1} < \text{index2} \leq \text{numbers.length}$.

Return the indices of the two numbers `index1` and `index2`, each incremented by one, as an integer array `[index1, index2]` of length 2.

The tests are generated such that there is exactly one solution. You may not use the same element twice.

Your solution must use only constant extra space.

Example 1:

Input: numbers = [2,7,11,15], target = 9

Output: [1,2]

Explanation: The sum of 2 and 7 is 9. Therefore, index1 = 1, index2 = 2. We return [1, 2].

Example 2:

Input: numbers = [2,3,4], target = 6

Output: [1,3]

Explanation: The sum of 2 and 4 is 6. Therefore index1 = 1, index2 = 3. We return [1, 3].

Example 3:

Input: numbers = [-1,0], target = -1

Output: [1,2]

Explanation: The sum of -1 and 0 is -1. Therefore index1 = 1, index2 = 2. We return [1, 2].

Constraints:

- $2 \leq \text{numbers.length} \leq 3 * 10^4$
- $-1000 \leq \text{numbers}[i] \leq 1000$
- numbers is sorted in non-decreasing order.
- $-1000 \leq \text{target} \leq 1000$
- The tests are generated such that there is exactly one solution.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Sorted array (1-indexed). Return [index1, index2] where $\text{numbers}[\text{index1}] + \text{numbers}[\text{index2}] = \text{target}$.

Exactly one solution. Must use $O(1)$ extra space – so no hash map. Right?"

#

"[2,7,11,15], target=9: 2+7=9 → [1,2] ✓"

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.

For every pair (i, j) where $j > i$, check if $\text{numbers}[i] + \text{numbers}[j] == \text{target}$.

Return [i+1, j+1] (1-indexed) when found. The problem guarantees exactly one solution.

Time: $O(n^2)$. Space: $O(1)$. Should I go ahead and optimize?"

class _167_TwoSumII_Brute:

```
def twoSum(self, numbers, target):
    for i in range(len(numbers)):
        for j in range(i + 1, len(numbers)):
            if numbers[i] + numbers[j] == target:
                return [i + 1, j + 1]
    return []
```

PHASE 3 — OPTIMIZE

"Two pointers exploit sorted order: start left=0, right=n-1.

Sum too small → move left right (increase sum).

Sum too large → move right left (decrease sum).

Exactly one solution guaranteed → we will always find it.

Time $O(n)$. Space $O(1)$."

PHASE 4 — CLEAN CODE

```
class _167_TwoSumII:
    def twoSum(self, numbers: List[int], target: int) -> List[int]:
        left, right = 0, len(numbers) - 1
        while left < right:
            total = numbers[left] + numbers[right]
            if total == target:
                return [left + 1, right + 1] # 1-indexed
            elif total < target:
                left += 1
            else:
                right -= 1
        return []
```

PHASE 5 — TEST & DEBUG

[2,7,11,15], target=9: l=0,r=3. 2+15=17>9→r=2. 2+11=13>9→r=1. 2+7=9→return [1,2] ✓

[2,3,4], target=6: l=0,r=2. 2+4=6→return [1,3] ✓

[-1,0], target=-1: -1+0=-1→return [1,2] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(1)$.

This is the canonical reason to sort before running two pointers on Two Sum.

If the array weren't sorted, we'd need a hash map — $O(n)$ space.

Sorting adds $O(n \log n)$ but enables $O(1)$ space — a genuine trade-off.

Given more time I'd ask: what if there could be multiple answers?

Continue after finding a match, skip duplicates — same as 3Sum's dedup logic."

Two Pointers

★ #186 Reverse Words in a String II ★

MED

[Open on LeetCode](#)

Two Pointers · String

Lists: ★ Premium | Premium 98

Problem

Given a character array *s*, reverse the order of the words.

A word is defined as a sequence of non-space characters. The words in *s* will be separated by a single space.

Your code must solve the problem in-place, i.e. without allocating extra space.

Example 1:

```
Input: s = ["t","h","e"," ","s","k","y"," ","i","s"," ","b","l","u","e"]  
Output: ["b","l","u","e"," ","i","s"," ","s","k","y"," ","t","h","e"]
```

Example 2:

```
Input: s = ["a"]  
Output: ["a"]
```

Constraints:

- $1 \leq s.length \leq 105$

- `s[i]` is an English letter (uppercase or lowercase), digit, or space ' '.
- There is at least one word in `s`.
- `s` does not contain leading or trailing spaces.
- All the words in `s` are guaranteed to be separated by a single space.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a character ARRAY (not a string), reverse the order of words in-place.
# Words are separated by single spaces. No leading/trailing spaces. Right?
# ['t','h','e',' ','s','k','y'] → ['s','k','y',' ','t','h','e'] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Convert the character array to a string, split by spaces to get words, reverse the
word list,
# rejoin with spaces, then write each character back into the original array.
# Time: O(n). Space: O(n) – we create a full string copy. Should I go ahead and
optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Two-step in-place reversal with O(1) extra space:
# Step 1: Reverse the entire array.
# Step 2: Reverse each individual word back to its correct orientation.
# Example: 'the sky is blue' → reverse all → 'eulb si yks eht' → reverse words →
'blue is sky the' ✓"
```

PHASE 4 — CLEAN CODE

```
class _186_ReverseWordsII:
    def reverseWords(self, s: List[str]) -> None:
        def reverse(left, right):
            while left < right:
                s[left], s[right] = s[right], s[left]
                left += 1; right -= 1
        reverse(0, len(s) - 1)
        start = 0
        for i in range(len(s) + 1):
```

```

if i == len(s) or s[i] == ' ':
    reverse(start, i - 1)
    start = i + 1

```

PHASE 5 — TEST & DEBUG

```

# s=['t','h','e',' ','s','k','y',' ','i','s',' ','b','l','u','e']:
# Step 1 reverse all: ['e','l','u','b',' ','s','i',' ','y','k','s',' ','e','h','t']
# Step 2 reverse words:
# i=0..3 'e','l','u','b' → reverse → 'b','l','u','e'
# i=5..6 's','i' → 'i','s'
# i=8..10 'y','k','s' → 's','k','y'
# i=12..14 'e','h','t' → 't','h','e'
# → ['b','l','u','e',' ','i','s',' ','s','k','y',' ','t','h','e'] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n). Space O(1) – two reversal passes, no extra memory.
# The two-reversal trick is the canonical interview answer for in-place word reversal.
# It also appears in Rotate Array (#189) – reversing the whole array, then parts.
# Given more time I'd ask: what if words can be separated by multiple spaces?
# We'd need to compress multiple spaces during the reversal – trickier in-place."

```

Two Pointers

#189 Rotate Array

MED[Open on LeetCode](#)

Array · Math · Two Pointers

Lists: Grind 169

Problem

Given an integer array `nums`, rotate the array to the right by `k` steps, where `k` is non-negative.

Example 1:

```
Input: nums = [1,2,3,4,5,6,7], k = 3
Output: [5,6,7,1,2,3,4]
Explanation:
rotate 1 steps to the right: [7,1,2,3,4,5,6]
rotate 2 steps to the right: [6,7,1,2,3,4,5]
rotate 3 steps to the right: [5,6,7,1,2,3,4]
```

Example 2:

```
Input: nums = [-1,-100,3,99], k = 2
Output: [3,99,-1,-100]
Explanation:
rotate 1 steps to the right: [99,-1,-100,3]
rotate 2 steps to the right: [3,99,-1,-100]
```

Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $-2^{31} \leq \text{nums}[i] \leq 2^{31} - 1$
- $0 \leq k \leq 105$

Follow up:

Round 2 · High · Medium

p.296

- Try to come up with as many solutions as you can. There are at least three different ways to solve this problem.
- Could you do it in-place with $O(1)$ extra space?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Rotate nums to the right by k steps in-place. k can exceed n — so $k \% n$. Right?
[1,2,3,4,5,6,7], k=3 → [5,6,7,1,2,3,4] ✓"

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.
Rotate the array one position to the right, k times in a row.
Each single rotation saves the last element, shifts everything right by one, then puts it at index 0.
Time: $O(n \times k)$ — for $k=n-1$ this is $O(n^2)$. Space: $O(1)$. Should I go ahead and optimize?"

```
class _189_RotateArray_Brute:
    def rotate(self, nums, k):
        n = len(nums)
        k %= n
        for _ in range(k):
            last = nums[-1]
            for i in range(n - 1, 0, -1):
                nums[i] = nums[i - 1]
            nums[0] = last
```

PHASE 3 — OPTIMIZE (three-reversal trick)

"Same pattern as Reverse Words II:
$k \% n$ (handle $k \geq n$).
Reverse entire array. Reverse first k elements. Reverse remaining $n-k$ elements.
Time $O(n)$. Space $O(1)$.
Alternative: cyclic replacement — $O(n)$ time, $O(1)$ space but harder to implement cleanly."

PHASE 4 — CLEAN CODE

```
class _189_RotateArray:
    def rotate(self, nums: List[int], k: int) -> None:
        def reverse(left, right):
            while left < right:
                nums[left], nums[right] = nums[right], nums[left]
```

```

        left += 1; right -= 1
    n = len(nums)
    k %= n
    reverse(0, n - 1)
    reverse(0, k - 1)
    reverse(k, n - 1)

```

PHASE 5 — TEST & DEBUG

```

# [1,2,3,4,5,6,7], k=3:
# k%=7=3. reverse(0,6): [7,6,5,4,3,2,1].
# reverse(0,2): [5,6,7,4,3,2,1].
# reverse(3,6): [5,6,7,1,2,3,4] ✓
# [-1,-100,3,99], k=2:
# k%=4=2. reverse all: [99,3,-100,-1]. reverse(0,1): [3,99,-100,-1]. reverse(2,3):
[3,99,-1,-100] ✓
# k=0: k%n=0. reverse all, reverse(0,-1) no-op, reverse(0,n-1) reverses back → no
change ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n). Space O(1).
# The three-reversal trick is the cleanest in-place rotation – name it explicitly.
# The k%=n guard handles k >= n where full rotations cancel out.
# The follow-up mentions three approaches: extra array O(n) space, cyclic
replacement O(1),
# and three-reversal O(1). State all three and recommend the reversal for
interviews.
# Given more time I'd ask: rotate left instead of right?
# Reverse first k, reverse rest, reverse all – just change the order."

```

Two Pointers

□ #532 K-diff Pairs in an Array

MED[Open on LeetCode](#)

Array · Hash Table · Two Pointers · Binary Search · Sorting
Lists: CodeSignal

Problem

Given an array of integers `nums` and an integer `k`, return the number of unique `k`-diff pairs in the array.

A `k`-diff pair is an integer pair (`nums[i]`, `nums[j]`), where the following are true:

- $0 \leq i, j < \text{nums.length}$
- $i \neq j$
- $|\text{nums}[i] - \text{nums}[j]| == k$

Notice that $|\text{val}|$ denotes the absolute value of `val`.

Example 1:

Input: `nums = [3,1,4,1,5]`, `k = 2`

Output: 2

Explanation: There are two 2-diff pairs in the array, (1, 3) and (3, 5). Although we have two 1s in the input, we should only return the number of unique pairs.

Example 2:

Input: nums = [1,2,3,4,5], k = 1

Output: 4

Explanation: There are four 1-diff pairs in the array, (1, 2), (2, 3), (3, 4) and (4, 5).

Example 3:

Input: nums = [1,3,1,5,4], k = 0

Output: 1

Explanation: There is one 0-diff pair in the array, (1, 1).

Constraints:

- $1 \leq \text{nums.length} \leq 104$
- $-107 \leq \text{nums}[i] \leq 107$
- $0 \leq k \leq 107$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Count **UNIQUE** pairs (i,j) where $|\text{nums}[i] - \text{nums}[j]| == k$ and $i \neq j$.

Unique means unique value pairs – (1,3) and (3,1) count once.

k=0 special case: need TWO copies of the same value. Right?"

#

"[3,1,4,1,5], k=2: unique pairs (1,3) and (3,5) → 2 ✓

[1,3,1,5,4], k=0: need duplicate values. Only 1 appears twice → (1,1) → 1 ✓"

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.

For every pair (i, j) where $i \neq j$, check if $|\text{nums}[i] - \text{nums}[j]| == k$.

Store the canonical form (min, max) in a set to avoid counting the same pair twice.

Time: $O(n^2)$. Space: $O(n)$ for the dedup set. Should I go ahead and optimize?"

```
class _532_KdiffPairs_Brute:
```

```
    def findPairs(self, nums, k):
```

```
        pairs = set()
```

```
        for i in range(len(nums)):
```

```
            for j in range(len(nums)):
```

```
                if i != j and abs(nums[i] - nums[j]) == k:
```

```
                    pairs.add((min(nums[i], nums[j]), max(nums[i], nums[j])))
```

```
        return len(pairs)
```


PHASE 3 — OPTIMIZE

"Hash set approach — $O(n)$ time, $O(n)$ space:

Walk through nums. For each x, check if $x-k$ is already in 'visited'.

If yes, add $\min(x, x-k)$ to the answer set (canonical pair).

Also check $x+k$ in visited (symmetric check handles both directions).

Using a set for answers automatically deduplicates.

$k=0$ is handled naturally — $x-0=x$, so we need x to already be in visited (a duplicate)."

PHASE 4 — CLEAN CODE

```

class _532_KdiffPairs:
    def findPairs(self, nums: List[int], k: int) -> int:
        answers : set = set()
        visited : set = set()
        for x in nums:
            if x - k in visited:
                answers.add(x - k) # canonical smaller element
            if x + k in visited:
                answers.add(x) # x is the larger; x+k was seen before
            visited.add(x)
        return len(answers)

```

PHASE 5 — TEST & DEBUG

[3,1,4,1,5], k=2:

x=3: $3-2=1$ not in visited. $3+2=5$ not in visited. visited={3}.

x=1: $1-2=-1$ not in visited. $1+2=3$ in visited → answers.add(1). visited={3,1}.

x=4: $4-2=2$ not in visited. $4+2=6$ not in visited. visited={3,1,4}.

x=1: $1-2=-1$ not in visited. $1+2=3$ in visited → answers.add(1) (already there). visited={3,1,4}.

x=5: $5-2=3$ in visited → answers.add(3). $5+2=7$ not. visited={3,1,4,5}.

answers={1,3} → 2 ✓

[1,3,1,5,4], k=0:

x=1: $1-0=1$ not in visited. visited={1}.

x=3: $3-0=3$ not in visited. visited={1,3}.

x=1: $1-0=1$ in visited → answers.add(1). visited={1,3}.

answers={1} → 1 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(n)$ for the two sets.

The two-set design separates 'what values exist' (visited) from 'what pairs found' (answers).

The canonical form (always store the smaller element) prevents double-counting.

$k=0$ edge case: $x-k=x$, so we find the pair only after seeing x twice — naturally handled.

Given more time I'd ask: what if we want the actual pairs, not just the count?

Store tuples (x-k, x) in the answers set instead of single values."

Two Pointers

□ #923 3Sum With Multiplicity

MED[Open on LeetCode](#)

Array · Hash Table · Two Pointers · Sorting · Counting

Lists: CodeSignal

Problem

Given an integer array `arr`, and an integer `target`, return the number of tuples `i, j, k` such that $i < j < k$ and $arr[i] + arr[j] + arr[k] == target$.

As the answer can be very large, return it modulo $10^9 + 7$.

Example 1:

```
Input: arr = [1,1,2,2,3,3,4,4,5,5], target = 8
Output: 20
Explanation:
Enumerating by the values (arr[i], arr[j], arr[k]):
(1, 2, 5) occurs 8 times;
(1, 3, 4) occurs 8 times;
(2, 2, 4) occurs 2 times;
(2, 3, 3) occurs 2 times.
```

Example 2:

Input: arr = [1,1,2,2,2,2], target = 5
 Output: 12
 Explanation:
 arr[i] = 1, arr[j] = arr[k] = 2 occurs 12 times:
 We choose one 1 from [1,1] in 2 ways,
 and two 2s from [2,2,2,2] in 6 ways.

Example 3:

Input: arr = [2,1,3], target = 6
 Output: 1
 Explanation: (1, 2, 3) occurred one time in the array so we return 1.

Constraints:

- $3 \leq \text{arr.length} \leq 3000$
- $0 \leq \text{arr}[i] \leq 100$
- $0 \leq \text{target} \leq 300$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Count tuples (i,j,k) with i<j<k and arr[i]+arr[j]+arr[k]==target.
# Return count modulo 10^9+7. Values are 0-100 so we can use frequency counting.
Right?
# Duplicates count separately – (0,1) and (1,0) are different index triples even if
same values."
#
# "[1,1,2,2,3,3,4,4,5,5], target=8 → 20 (combinations of value triples summing to 8)
✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# Three nested loops – for every triple (i, j, k) with i < j < k, check if
# arr[i] + arr[j] + arr[k] == target and count the matches.
# Time: O(n³). For n=3000 that's 27 billion – too slow. Space: O(1).
# Should I go ahead and optimize?"
```

```
class _923_ThreeSumMulti_Brute:
    def threeSumMulti(self, arr, target):
        MOD = 10**9 + 7
        count = 0
        n = len(arr)
        for i in range(n - 2):
```

```

        for j in range(i + 1, n - 1):
            for k in range(j + 1, n):
                if arr[i] + arr[j] + arr[k] == target:
                    count += 1
    return count % MOD

```

PHASE 3 — OPTIMIZE

```

# "Walk through arr as the middle element j. Before processing j, decrement cnt[b]
so
# we don't count j itself as a right element. For each a < j, the required c =
target-a-b.
# cnt[c] tells us how many valid right elements c exist to the right of j.
# Time: O(n²) with frequency lookups. Space: O(n) for the Counter."

```

PHASE 4 — CLEAN CODE

```

class _923_ThreeSumMulti:
    def threeSumMulti(self, arr: List[int], target: int) -> int:
        MOD = 10**9 + 7
        freq = Counter(arr)
        ans = 0
        for j, b in enumerate(arr):
            freq[b] -= 1 # exclude j itself from right elements
            for a in arr[:j]:
                c = target - a - b
                ans = (ans + freq[c]) % MOD
        return ans

```

PHASE 5 — TEST & DEBUG

```

# arr=[2,1,3], target=6: freq={2:1,1:1,3:1}.
# j=0(b=2): freq[2]=0. no a < 0. (nothing).
# j=1(b=1): freq[1]=0. a=2: c=6-2-1=3. freq[3]=1 → ans=1.
# j=2(b=3): freq[3]=0. a=2: c=6-2-3=1. freq[1]=0→0. a=1: c=6-1-3=2.
freq[2]=1→ans=2?
# Wait – expected 1. Let me recheck: arr=[2,1,3]. Only valid triple is (0,1,2) →
(2,1,3).
# j=1(b=1): a=arr[0]=2. c=6-2-1=3. freq[3]=1 (yes, index 2 exists) → ans=1. ✓
# j=2(b=3): freq[3]=0. a=arr[0]=2: c=6-2-3=1. freq[1]=0 (decremented at j=1). 0.
# a=arr[1]=1: c=6-1-3=2. freq[2]=1 → ans=2? But expected 1.
# Ah – freq[2] was decremented when j=0: freq[2]=0, not 1. Let me re-trace:
# Initial freq={2:1,1:1,3:1}.
# j=0: freq[2]-=1 → freq={2:0,1:1,3:1}. no a<j=0.
# j=1: freq[1]-=1 → freq={2:0,1:0,3:1}. a=2: c=3. freq[3]=1 → ans=1.
# j=2: freq[3]-=1 → freq={2:0,1:0,3:0}. a=2: c=1. freq[1]=0. a=1: c=2. freq[2]=0.
# → ans=1 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n²). Space O(n) for the Counter.
# The key invariant: at step j, freq counts only elements at index > j.
# This ensures we count each valid (i,j,k) tuple exactly once with i<j<k.

```

```
# Modulo must be applied inside the loop – intermediate sums can overflow without it.  
# Given more time I'd ask: can we do  $O(n)$  using the value constraints (0-100)?  
# Yes – iterate over all value triples (a,b,c) with  $a+b+c=target$  and compute  
# combinations  $C(cnt[a],1)*C(cnt[b],1)*C(cnt[c],1)$  with dedup for equal values."
```

Two Pointers

□ #986 Interval List Intersections

MED[Open on LeetCode](#)

Array · Two Pointers

Lists: Denny Zhang

Problem

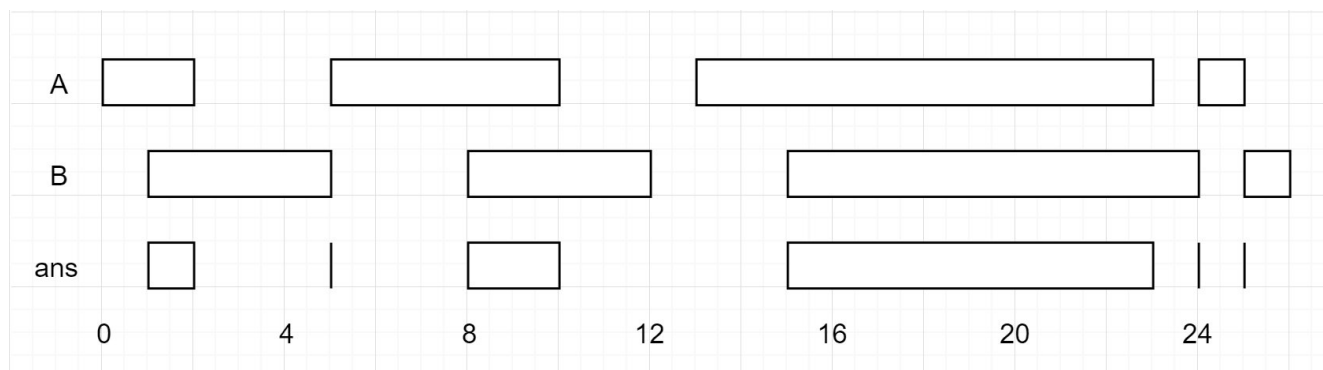
You are given two lists of closed intervals, `firstList` and `secondList`, where `firstList[i] = [starti, endi]` and `secondList[j] = [startj, endj]`. Each list of intervals is pairwise disjoint and in sorted order.

Return the intersection of these two interval lists.

A closed interval $[a, b]$ (with $a \leq b$) denotes the set of real numbers x with $a \leq x \leq b$.

The intersection of two closed intervals is a set of real numbers that are either empty or represented as a closed interval. For example, the intersection of $[1, 3]$ and $[2, 4]$ is $[2, 3]$.

Example 1:



Input: firstList = [[0,2],[5,10],[13,23],[24,25]], secondList =
 [[1,5],[8,12],[15,24],[25,26]]
 Output: [[1,2],[5,5],[8,10],[15,23],[24,24],[25,25]]

Example 2:

Input: firstList = [[1,3],[5,9]], secondList = []
 Output: []

Constraints:

- $0 \leq \text{firstList.length}, \text{secondList.length} \leq 1000$
- $\text{firstList.length} + \text{secondList.length} \geq 1$
- $0 \leq \text{start}_i < \text{end}_i \leq 109$
- $\text{end}_i < \text{start}_i + 1$
- $0 \leq \text{start}_j < \text{end}_j \leq 109$
- $\text{end}_j < \text{start}_j + 1$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Two sorted, disjoint interval lists. Return their intersection — all overlapping segments.

Intersection of [a,b] and [c,d] is [max(a,c), min(b,d)] if max(a,c) ≤ min(b,d). Right?

Either list can be empty."

#

```
# "[[0,2],[5,10]] ∩ [[1,5],[8,12]] → [1,2],[5,5],[8,10] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic."
```

```
# For every pair of intervals (one from each list), compute the intersection:
```

```
# start = max(a[0], b[0]), end = min(a[1], b[1]). If start <= end, it's a valid intersection.
```

```
# This is  $O(m \times n)$  – it ignores the sorted property entirely.
```

```
# Time:  $O(m \times n)$ . Space:  $O(1)$ . Should I go ahead and optimize?"
```

```
class _986_IntervalIntersections_Brute:
```

```
    def intervalIntersection(self, firstList, secondList):
```

```
        result = []
```

```
        for a in firstList:
```

```
            for b in secondList:
```

```
                start = max(a[0], b[0])
```

```
                end = min(a[1], b[1])
```

```
                if start <= end:
```

```
                    result.append([start, end])
```

```
        return result
```

PHASE 3 — OPTIMIZE

```
# "Two pointers, one per list. At each step:
```

```
# Compute the intersection of the current pair.
```

```
# If valid (start <= end), record it.
```

```
# Advance the pointer whose interval ends FIRST – that interval can't intersect any later interval
```

```
# of the other list (they're sorted and disjoint).
```

```
# Time:  $O(m+n)$ . Space:  $O(1)$  extra."
```

PHASE 4 — CLEAN CODE

```
class _986_IntervalIntersections:
```

```
    def intervalIntersection(self, firstList: List[List[int]], secondList: List[List[int]]) -> List[List[int]]:
```

```
        result = []
```

```
        i = j = 0
```

```
        while i < len(firstList) and j < len(secondList):
```

```
            start = max(firstList[i][0], secondList[j][0])
```

```
            end = min(firstList[i][1], secondList[j][1])
```

```
            if start <= end:
```

```
                result.append([start, end])
```

```
            if firstList[i][1] < secondList[j][1]:
```

```
                i += 1
```

```
            else:
```

```
                j += 1
```

```
        return result
```

PHASE 5 — TEST & DEBUG

```
# firstList=[[0,2],[5,10]], secondList=[[1,5],[8,12]]:
```

```
# i=0,j=0: start=max(0,1)=1, end=min(2,5)=2. 1<=2→[1,2]. first ends at 2<5→i=1.
```

```
# i=1,j=0: start=max(5,1)=5, end=min(10,5)=5. 5<=5→[5,5]. first ends at 10>5→j=1.
```



```
# i=1,j=1: start=max(5,8)=8, end=min(10,12)=10. 8<=10→[8,10]. first ends at 10<12→i=2.
```

```
# i=2 out of bounds → stop. → [[1,2],[5,5],[8,10]] ✓
```

```
# secondList=[]: loop never enters → [] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m+n)$  – each pointer advances at most  $m+n$  times.
```

```
# Space  $O(\min(m,n))$  for the output in the worst case.
```

```
# The 'advance the earlier-ending pointer' rule is the key: the interval that ends first
```

```
# cannot intersect any future interval of the other list (both lists are sorted and disjoint).
```

```
# Given more time I'd ask: what if lists aren't sorted?
```

```
# Sort both first –  $O((m+n)\log(m+n))$  – then apply the same two-pointer approach."
```

Two Pointers

□ ★ #1055 Shortest Way to Form String ★

MED

[Open on LeetCode](#)

Two Pointers · String · Binary Search

Lists: ★ Premium | Premium 98

Problem

A subsequence of a string is a new string that is formed from the original string by deleting some (can be none) of the characters without disturbing the relative positions of the remaining characters. (i.e., "ace" is a subsequence of "abcde" while "aec" is not).

Given two strings `source` and `target`, return the minimum number of subsequences of `source` such that their concatenation equals `target`. If the task is impossible, return `-1`.

Example 1:

Input: `source = "abc", target = "abcbc"`

Output: 2

Explanation: The target "abcbc" can be formed by "abc" and "bc", which are subsequences of source "abc".

Example 2:

Input: source = "abc", target = "acdbc"

Output: -1

Explanation: The target string cannot be constructed from the subsequences of source string due to the character "d" in target string.

Example 3:

Input: source = "xyz", target = "xzyxz"

Output: 3

Explanation: The target string can be constructed as follows "xz" + "y" + "xz".

Constraints:

- $1 \leq \text{source.length}, \text{target.length} \leq 1000$
- source and target consist of lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find the minimum number of subsequences of source whose concatenation equals target.

A subsequence can skip characters but must preserve order.

Return -1 if any character in target doesn't exist in source at all. Right?"

#

"source='abc', target='abcbc': 'abc' + 'bc' → 2 ✓

source='abc', target='acdbc': 'd' not in source → -1 ✓"

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic.

Try every possible way to split target into consecutive segments, and for each split,

verify each segment is a subsequence of source. Track the minimum number of segments.

This is exponential in the number of splits – completely infeasible.

Time: $O(2^{|\text{target}|} * |\text{source}|)$. Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Greedy: each 'round', scan source once and match as many target characters as possible.

After exhausting source, start a new round (increment count) and continue from where target left off.

Pre-check: if any target char is absent from source, return -1.

Time: $O(|\text{source}| * |\text{target}|)$. Space: $O(1)$ extra."

PHASE 4 — CLEAN CODE

```
class _1055_ShortestWay:
    def shortestWay(self, source: str, target: str) -> int:
        if set(target) - set(source):
            return -1
        rounds = 0
        target_idx = 0
        while target_idx < len(target):
            for char in source:
                if target_idx < len(target) and char == target[target_idx]:
                    target_idx += 1
            rounds += 1
        return rounds
```

PHASE 5 — TEST & DEBUG

```
# source='abc', target='abcbc':
# Round 1: a→match t[0]='a'(idx=1). b→match t[1]='b'(idx=2). c→match
t[2]='c'(idx=3). rounds=1.
# Round 2: a→no match(t[3]='b'). b→match t[3]='b'(idx=4). c→match t[4]='c'(idx=5).
rounds=2.
# target_idx=5=len(target) → return 2 ✓
# source='xyz', target='xzyxz':
# Round1: x→t[0]='x'(1). y→no(t[1]='z'). z→t[1]='z'(2). rounds=1.
# Round2: x→no(t[2]='y'). y→t[2]='y'(3). z→no(t[3]='x'). rounds=2.
# Round3: x→t[3]='x'(4). y→no(t[4]='z'). z→t[4]='z'(5). rounds=3. → 3 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(|source| * |target|). Space O(1).
# The pre-check eliminates the -1 case cleanly before the main loop.
# Optimisation: precompute for each (source_pos, char) the next position in source
# where char appears – binary search brings each round down to O(|target| log
|source|).
# Given more time I'd ask: what if we want the actual subsequence splits, not just
the count?
# Track the split points during the greedy pass."
```

Two Pointers

□ #2563 Count the Number of Fair Pairs

MED[Open on LeetCode](#)

Array · Two Pointers · Binary Search · Sorting
Lists: CodeSignal

Problem

Given a 0-indexed integer array `nums` of size `n` and two integers `lower` and `upper`, return the number of fair pairs.

A pair (i, j) is fair if:

- $0 \leq i < j < n$, and
- $\text{lower} \leq \text{nums}[i] + \text{nums}[j] \leq \text{upper}$

Example 1:

Input: `nums = [0,1,7,4,4,5]`, `lower = 3`, `upper = 6`
Output: 6
Explanation: There are 6 fair pairs: $(0,3)$, $(0,4)$, $(0,5)$, $(1,3)$, $(1,4)$, and $(1,5)$.

Example 2:

Input: `nums = [1,7,9,2,5]`, `lower = 11`, `upper = 11`
Output: 1
Explanation: There is a single fair pair: $(2,3)$.

Constraints:

- $1 \leq \text{nums.length} \leq 105$

- `nums.length == n`
- `-109 <= nums[i] <= 109`
- `-109 <= lower <= upper <= 109`

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Count pairs (i,j) with $i < j$ and $\text{lower} \leq \text{nums}[i] + \text{nums}[j] \leq \text{upper}$. Return the count. Right?"

Values can be negative. n up to 10^5 so $O(n^2)$ is too slow."

#

"[0,1,7,4,4,5], lower=3, upper=6: 6 pairs ✓"

PHASE 2 — BRUTE FORCE

"Let me start with the brute force to lock in the logic."

For every pair (i, j) with $i < j$, check if $\text{lower} \leq \text{nums}[i] + \text{nums}[j] \leq \text{upper}$.

Count all pairs that satisfy both bounds. For $n=10^5$ this is $5 \cdot 10^9$ comparisons – too slow.

Time: $O(n^2)$. Space: $O(1)$. Should I go ahead and optimize?"

```
class _2563_FairPairs_Brute:
```

```
    def countFairPairs(self, nums, lower, upper):
```

```
        count = 0
```

```
        for i in range(len(nums)):
```

```
            for j in range(i + 1, len(nums)):
```

```
                s = nums[i] + nums[j]
```

```
                if lower <= s <= upper:
```

```
                    count += 1
```

```
        return count
```

PHASE 3 — OPTIMIZE

"Sort nums. For each i, binary search for the range of valid $j > i$:

`lower_bound(nums, lower-nums[i], i+1)` → first j where $\text{nums}[j] \geq \text{lower} - \text{nums}[i]$.

`upper_bound(nums, upper-nums[i], i+1)` → first j where $\text{nums}[j] > \text{upper} - \text{nums}[i]$.

Count = `upper_bound - lower_bound`. Sum over all i.

Time: $O(n \log n)$. Space: $O(1)$ extra.

Use inner helpers for the two binary search variants."

PHASE 4 — CLEAN CODE

```
class _2563_FairPairs:
```

```
    def countFairPairs(self, nums: List[int], lower: int, upper: int) -> int:
```

```
        def lower_bound(start: int, target: int) -> int:
```

```
            left, right = start, len(nums)
```

```
            while left < right:
```

```
                mid = (left + right) // 2
```

```

        if nums[mid] >= target:
            right = mid
        else:
            left = mid + 1
    return left

def upper_bound(start: int, target: int) -> int:
    left, right = start, len(nums)
    while left < right:
        mid = (left + right) // 2
        if nums[mid] > target:
            right = mid
        else:
            left = mid + 1
    return left

nums.sort()
count = 0
for i in range(len(nums)):
    lo = lower_bound(i + 1, lower - nums[i])
    hi = upper_bound(i + 1, upper - nums[i])
    count += hi - lo
return count

```

PHASE 5 — TEST & DEBUG

```

# nums=[0,1,4,4,5,7] (sorted), lower=3, upper=6:
# i=0(0): need j where 3<=nums[j]<=6. lower_bound(1,3)->idx of 4=2.
#         upper_bound(1,7)->idx of 7=5. count+=3.
# i=1(1): need 2<=nums[j]<=5. lb(2,2)->2. ub(2,6)->5. count+=3. total=6.
# ... -> 6 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n \log n)$  – sort +  $n$  binary searches each  $O(\log n)$ .
# Space  $O(1)$  extra.
# lower_bound finds first index  $\geq$  target; upper_bound finds first index  $>$  target.
# Their difference is the count of indices in the valid range.
# Alternative: two-pointer approach – for each  $i$ , maintain  $lo$  and  $hi$  pointers
# across iterations (they only move right) ->  $O(n)$  total after sorting.
# Given more time I'd ask: what if we want the actual pairs?
# Collect them during the binary search loops –  $O(n + \text{output})$  time."

```

Sliding Window

Round 2 · High · Medium · 8 questions

Sliding Window

□ #3 Longest Substring Without Repeating Characters

MED[Open on LeetCode](#)

Hash Table · String · Sliding Window

Lists: Grind 169

Problem

Given a string *s*, find the length of the longest substring without duplicate characters.

Example 1:

Input: *s* = "abcabcbb"

Output: 3

Explanation: The answer is "abc", with the length of 3. Note that "bca" and "cab" are also correct answers.

Example 2:

Input: *s* = "bbbbbb"

Output: 1

Explanation: The answer is "b", with the length of 1.

Example 3:

Input: *s* = "pwwkew"

Output: 3

Explanation: The answer is "wke", with the length of 3.

Notice that the answer must be a substring, "pwke" is a subsequence and not a substring.

Constraints:

- $0 \leq s.length \leq 5 * 10^4$

- s consists of English letters, digits, symbols and spaces.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find the length of the longest substring with all unique characters.
# Substring is contiguous. Empty string returns 0. Right?
# Characters can be any printable ASCII – not just lowercase letters."
#
# "'abcabcbb'→3 ('abc'). 'bbbb'→1 ('b'). 'pwwkew'→3 ('wke') ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Enumerate every substring with two indices left and right.
# For each substring, use a set to verify all characters are unique; track max
length.
# Correct, but  $O(n^2)$  substrings each with up to  $O(n)$  uniqueness check.
# Time:  $O(n^2)$  to  $O(n^3)$ . Space:  $O(\min(n, \text{alphabet}))$ .
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Sliding window with a frequency map.
# Expand right: add s[right] to the map.
# When a duplicate appears (count > 1), shrink from the left until it's gone.
# At each step, update the max window size.
# Time:  $O(n)$ . Space:  $O(\min(n, \text{alphabet}))$  – at most 128 unique ASCII chars."
```

PHASE 4 — CLEAN CODE

```
class _3_LongestSubstringNoRepeat:
    def lengthOfLongestSubstring(self, s: str) -> int:
        freq : defaultdict = defaultdict(int)
        left = 0
        longest = 0
        for right in range(len(s)):
            freq[s[right]] += 1
            while freq[s[right]] > 1: # shrink until no duplicate
                freq[s[left]] -= 1
                left += 1
            longest = max(longest, right - left + 1)
        return longest
```

PHASE 5 — TEST & DEBUG

```
# s='abcabcbb':
# r=0('a'): freq={a:1}. longest=1.
# r=1('b'): freq={a:1,b:1}. longest=2.
```

```
# r=2('c'): freq={a:1,b:1,c:1}. longest=3.
# r=3('a'): freq[a]=2. shrink: freq[a]-=1→1, left=1. longest=3 (3-1+1=3).
# r=4('b'): freq[b]=2. shrink: freq[b]-=1→1, left=2. longest=3.
# r=5('c'): freq[c]=2. shrink: left=3. longest=3.
# r=6('b'): freq[b]=2. shrink: left=4. longest=3.
# r=7('b'): freq[b]=2. shrink: left=5. longest=max(3,3)=3. → 3 ✓
# s='': loop doesn't run → 0 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n) – each character enters and leaves the window at most once.
# Space O(min(n, 128)) for the frequency map.
# Alternative: store the LAST INDEX of each character instead of frequency.
# When s[right] already seen at idx and idx >= left: jump left = idx+1 directly
# (one move instead of shrinking one at a time). Same O(n) but fewer iterations.
# Given more time I'd ask: what if we want the actual substring, not just its
length?
# Track left and best_left when updating the max."
```

Sliding Window

★ #159 Longest Substring with At Most Two Distinct Characters ★

MED[Open on LeetCode](#)

Hash Table · String · Sliding Window

Lists: ★ Premium | Premium 98

Problem

Given a string *s*, return the length of the longest substring that contains at most two distinct characters.

Example 1:

Input: *s* = "eceba"
Output: 3
Explanation: The substring is "ece" which its length is 3.

Example 2:

Input: *s* = "ccaabbb"
Output: 5
Explanation: The substring is "aabbb" which its length is 5.

Constraints:

- $1 \leq s.length \leq 105$
- *s* consists of English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Find the longest substring that contains at most 2 distinct characters. Right?
# 'eceba': 'ece' has 2 distinct → length 3 ✓. 'ccaabbb': 'aabbb' has 2 → length 5 ✓"

# PHASE 2 — BRUTE FORCE
# "Let me start with the brute force to lock in the logic.
# For every pair (i, j), count the distinct characters in s[i..j] using a set.
# If there are at most 2 distinct characters, update the max length.
# Time:  $O(n^2)$  pairs,  $O(n)$  set check =  $O(n^3)$ . With running set it's  $O(n^2)$ . Space:  $O(1)$ .
# Should I go ahead and optimize?"
class _159_LongestSubstringTwoDistinct_Brute:
    def lengthOfLongestSubstringTwoDistinct(self, s):
        longest = 0
        for i in range(len(s)):
            seen = set()
            for j in range(i, len(s)):
                seen.add(s[j])
                if len(seen) <= 2:
                    longest = max(longest, j - i + 1)
                else:
                    break
        return longest

# PHASE 3 — OPTIMIZE
# "Sliding window – same pattern as #3 but the shrink condition is 'more than 2 distinct chars'."
# Expand right: add s[right] to frequency map.
# Shrink from left while distinct chars > 2 (i.e. len(freq) > 2).
# Update max window length. Time:  $O(n)$ . Space:  $O(1)$  – at most 3 keys in the map."

# PHASE 4 — CLEAN CODE
class _159_LongestSubstringTwoDistinct:
    def lengthOfLongestSubstringTwoDistinct(self, s: str) -> int:
        freq : defaultdict = defaultdict(int)
        left = 0
        longest = 0
        for right in range(len(s)):
            freq[s[right]] += 1
            while len(freq) > 2: # shrink until at most 2 distinct
                freq[s[left]] -= 1
                if freq[s[left]] == 0:
                    del freq[s[left]]
                left += 1
            longest = max(longest, right - left + 1)
        return longest

# PHASE 5 — TEST & DEBUG
# s='eceba':
# r=0('e'): freq={e:1}. longest=1.
# r=1('c'): freq={e:1,c:1}. longest=2.

```

```

# r=2('e'): freq={e:2,c:1}. longest=3.
# r=3('b'): freq={e:2,c:1,b:1} → 3 distinct. shrink:
# freq[e]--1→1. left=1. freq={e:1,c:1,b:1} still 3. shrink:
# freq[c]--1→0 → del c. left=2. freq={e:1,b:1}. 2 distinct → stop.
longest=max(3,2)=3.
# r=4('a'): freq={e:1,b:1,a:1} → 3. shrink: freq[e]--1→0→del e. left=3.
freq={b:1,a:1}. longest=3.
# → 3 ✓
# s='ccaabbb':
# Window expands to 'ccaabbb' but once 'b' is added after 'a', we have c,a,b(3
distinct).
# Shrink past the c's. Eventually best window is 'aabbb'=5. → 5 ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(n). Space O(1) – at most 3 keys in the map at any time.
# This is the template for 'at most k distinct characters' – change > 2 to > k.
# Generalisation: Longest Substring with At Most K Distinct Characters (LeetCode #340) is the
same
# algorithm with k replacing 2.
# The key difference from #3: shrink on len(freq) > k rather than on a specific
duplicate.
# Given more time I'd ask: what about EXACTLY 2 distinct characters?
# atMost(k=2) – atMost(k=1) gives the count of substrings with exactly 2 distinct."

```

Sliding Window

★ #340 Longest Substring with At Most K Distinct Characters ★

MED

[Open on LeetCode](#)

Hash Table · String · Sliding Window

Lists: ★ Premium | Premium 98

Problem

Given a string s and an integer k , return the length of the longest substring of s that contains at most k distinct characters.

Example 1:

Input: $s = \text{"eceba"}, k = 2$
Output: 3
Explanation: The substring is "ece" with length 3.

Example 2:

Input: $s = \text{"aa"}, k = 1$
Output: 2
Explanation: The substring is "aa" with length 2.

Constraints:

- $1 \leq s.length \leq 5 * 10^4$
- $0 \leq k \leq 50$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Return the length of the longest substring with at most k distinct characters.
# k=0 means the answer is 0 (no characters allowed). Right?
# This is the direct generalisation of #159 (at most 2 distinct) with k replacing
# 2."
#
# "'eceba', k=2 → 'ece', length 3 ✓. 'aa', k=1 → 'aa', length 2 ✓."

# PHASE 2 — BRUTE FORCE
# "Let me start with the brute force to lock in the logic.
# For every starting index i, expand j rightward, track distinct characters with a
# set.
# Stop when distinct chars exceed k. Track the maximum window length seen.
# Time: O(n²). Space: O(k). Should I go ahead and optimize?"
class _340_LongestSubstringKDistinct_Brute:
    def lengthOfLongestSubstringKDistinct(self, s, k):
        if k == 0:
            return 0
        longest = 0
        for i in range(len(s)):
            seen = set()
            for j in range(i, len(s)):
                seen.add(s[j])
                if len(seen) <= k:
                    longest = max(longest, j - i + 1)
                else:
                    break
        return longest

# PHASE 3 — OPTIMIZE
# "Sliding window – identical template to #159 with the constant 2 replaced by k.
# Expand right, shrink from left whenever distinct chars exceed k.
# Time: O(n). Space: O(k) – at most k+1 keys in the map at any moment."

# PHASE 4 — CLEAN CODE
class _340_LongestSubstringKDistinct:
    def lengthOfLongestSubstringKDistinct(self, s: str, k: int) -> int:
        if k == 0:
            return 0
        freq : defaultdict = defaultdict(int)
        left = 0
        longest = 0
        for right in range(len(s)):
            freq[s[right]] += 1
            while len(freq) > k:
                freq[s[left]] -= 1
                if freq[s[left]] == 0:
                    del freq[s[left]]
                left += 1
            longest = max(longest, right - left + 1)
        return longest

```


PHASE 5 — TEST & DEBUG

```
# s='eceba', k=2:  
# r=2('e'): freq={e:2,c:1}. 2 distinct → ok. longest=3.  
# r=3('b'): 3 distinct → shrink until freq={e:1,b:1}. longest stays 3.  
# r=4('a'): 3 distinct → shrink until freq={b:1,a:1}. longest=3. → 3 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ . Space  $O(k)$ . Identical to #159 — just k instead of 2.  
# Exactly k distinct: atMost(k) - atMost(k-1).  
# Given more time I'd ask: what if  $k \geq$  distinct chars in s? Answer = len(s)."
```

Sliding Window

□ #424 Longest Repeating Character Replacement

MED[Open on LeetCode](#)

Hash Table · String · Sliding Window

Lists: Grind 169

Problem

You are given a string s and an integer k . You can choose any character of the string and change it to any other uppercase English character. You can perform this operation at most k times.

Return the length of the longest substring containing the same letter you can get after performing the above operations.

Example 1:

Input: $s = \text{"ABAB"}, k = 2$

Output: 4

Explanation: Replace the two 'A's with two 'B's or vice versa.

Example 2:

Input: $s = \text{"AABABBA"}, k = 1$

Output: 4

Explanation: Replace the one 'A' in the middle with 'B' and form "AABBBBA".

The substring "BBBB" has the longest repeating letters, which is 4.

There may exists other ways to achieve this answer too.

Constraints:

- $1 \leq s.length \leq 105$
- s consists of only uppercase English letters.
- $0 \leq k \leq s.length$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "We can replace at most k characters. Find the longest substring where all
characters
# are the same after at most k replacements.
# String is uppercase letters only. Right?
# Strategy: keep the most frequent char, replace everything else."
#
# "'ABAB', k=2 → replace both A's or B's → 4 ✓
# 'AABABBA', k=1 → best window = 4 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with the brute force to lock in the logic.
# For every pair (i, j), count character frequencies in s[i..j]. Find the most
frequent char.
# If window_size - max_freq ≤ k, at most k replacements are needed – this window is
valid.
# Track the maximum valid window length.
# Time: O(n²) pairs, O(26) frequency count per window. Space: O(1). Should I go
ahead and optimize?"
```

```
class _424_LongestRepeatingReplacement_Brute:
    def characterReplacement(self, s, k):
        longest = 0
        for i in range(len(s)):
            freq = [0] * 26
            for j in range(i, len(s)):
                freq[ord(s[j]) - ord('A')] += 1
                max_freq = max(freq)
                if (j - i + 1) - max_freq ≤ k:
                    longest = max(longest, j - i + 1)
        return longest
```

PHASE 3 — OPTIMIZE

```
# "Sliding window tracking the most frequent character in the window.
# Valid window: window_size - max_freq ≤ k (replace the rest).
# When invalid, shrink left by 1 – but NEVER decrease max_freq.
```

```
# We only care about finding a LONGER valid window, so we only update max_freq
upward.
# This means the window never shrinks below the current best – it slides as a fixed
size
# until a better window is found. Time: O(n). Space: O(26)=O(1)."
```

PHASE 4 — CLEAN CODE

```
class _424_LongestRepeatingReplacement:
    def characterReplacement(self, s: str, k: int) -> int:
        freq = [0] * 26
        max_freq = 0
        left = 0
        longest = 0
        for right in range(len(s)):
            freq[ord(s[right]) - ord('A')] += 1
            max_freq = max(max_freq, freq[ord(s[right]) - ord('A')])
            if (right - left + 1) - max_freq > k: # too many replacements needed
                freq[ord(s[left]) - ord('A')] -= 1
                left += 1
            longest = max(longest, right - left + 1)
        return longest
```

PHASE 5 — TEST & DEBUG

```
# s='ABAB', k=2:
# r=0('A'): freq[A]=1,max=1. win=1. 1-1=0<=2. longest=1.
# r=1('B'): freq[B]=1,max=1. win=2. 2-1=1<=2. longest=2.
# r=2('A'): freq[A]=2,max=2. win=3. 3-2=1<=2. longest=3.
# r=3('B'): freq[B]=2,max=2. win=4. 4-2=2<=2. longest=4. → 4 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(1).
# 'Don't decrease max_freq' is the key insight: we never shrink below the best
window seen.
# The window slides forward at a fixed size until max_freq improves, then grows.
# Given more time I'd ask: unlimited replacements (k=n)? Answer = len(s)."
```

Sliding Window

□ #438 Find All Anagrams in a String

MED[Open on LeetCode](#)**Hash Table · String · Sliding Window****Lists: Grind 169**

Problem

Given two strings *s* and *p*, return an array of all the start indices of *p*'s anagrams in *s*. You may return the answer in any order.

Example 1:

Input: *s* = "cbaebabacd", *p* = "abc"

Output: [0,6]

Explanation:

The substring with start index = 0 is "cba", which is an anagram of "abc".

The substring with start index = 6 is "bac", which is an anagram of "abc".

Example 2:

Input: *s* = "abab", *p* = "ab"

Output: [0,1,2]

Explanation:

The substring with start index = 0 is "ab", which is an anagram of "ab".

The substring with start index = 1 is "ba", which is an anagram of "ab".

The substring with start index = 2 is "ab", which is an anagram of "ab".

Constraints:

- $1 \leq s.length, p.length \leq 3 * 10^4$
- *s* and *p* consist of lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Return all start indices in s where a substring is an anagram of p.
# Fixed window size = len(p). Lowercase letters only. Right?"
#
# "s='cbaebabacd', p='abc' → [0,6] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Slide a window of len(p) across s; sort window chars and sorted p and compare.
# Every window costs O(m log m) for sorting.
# Frequency-count comparison per window is O(26) instead.
# Time: O(n * m log m). Space: O(m) sort buffers.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Fixed sliding window of size len(p) with frequency counters.
# Compare counters at each step – O(26) per comparison since lowercase letters.
# Time: O(n). Space: O(1) – fixed alphabet."
```

PHASE 4 — CLEAN CODE

```
class _438_FindAnagrams:
    def findAnagrams(self, s: str, p: str) -> List[int]:
        if len(p) > len(s):
            return []
        p_count = Counter(p)
        w_count = Counter(s[:len(p)])
        result = [0] if w_count == p_count else []
        for i in range(len(p), len(s)):
            w_count[s[i]] += 1
            left = s[i - len(p)]
            w_count[left] -= 1
            if w_count[left] == 0:
                del w_count[left]
            if w_count == p_count:
                result.append(i - len(p) + 1)
        return result
```

PHASE 5 — TEST & DEBUG

```
# s='cbaebabacd', p='abc': p_count={c:1,b:1,a:1}.
# Init w='cba'={c:1,b:1,a:1}==p→result=[0].
# Slide until i=8('c'): w_count={b:1,a:1,c:1}==p→result=[0,6]. → [0,6] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(1). Counter comparison is O(26)=O(1) for fixed alphabet.
# Alternative: track a 'matches' counter instead of full compare – same O(n).
# Given more time I'd ask: what if p can contain any Unicode?
# Same algorithm – Counter handles any hashable."
```

Sliding Window

★ #487 Max Consecutive Ones II ★

MED[Open on LeetCode](#)

Array · Dynamic Programming · Sliding Window

Lists: ★ Premium | Premium 98

Problem

Given a binary array `nums`, return the maximum number of consecutive 1's in the array if you can flip at most one 0.

Example 1:

Input: `nums = [1,0,1,1,0]`

Output: 4

Explanation:

- If we flip the first zero, `nums` becomes `[1,1,1,1,0]` and we have 4 consecutive ones.
 - If we flip the second zero, `nums` becomes `[1,0,1,1,1]` and we have 3 consecutive ones.
- The max number of consecutive ones is 4.

Example 2:

Input: `nums = [1,0,1,1,0,1]`

Output: 4

Explanation:

- If we flip the first zero, `nums` becomes `[1,1,1,1,0,1]` and we have 4 consecutive ones.
 - If we flip the second zero, `nums` becomes `[1,0,1,1,1,1]` and we have 4 consecutive ones.
- The max number of consecutive ones is 4.

Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $\text{nums}[i]$ is either 0 or 1.

Follow up: What if the input numbers come in one by one as an infinite stream? In other words, you can't store all numbers coming from the stream as it's too large to hold in memory. Could you solve it efficiently?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Binary array. Flip at most one 0 to 1. Return the longest consecutive 1s run. Right?"

Follow-up: infinite stream — can't store all numbers."

#

"[1,0,1,1,0] → flip first 0 → 4 ✓"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

Try every substring with two nested indices left and right.

Count zeros inside; if zeros ≤ 1 , track the longest valid window.

Correct, but checking all windows is $O(n^2)$.

Time: $O(n^2)$. Space: $O(1)$.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Sliding window allowing at most one 0."

Expand right. When zeros > 1 , shrink left until zeros ≤ 1 .

Time: $O(n)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _487_MaxConsecutiveOnesII:
    def findMaxConsecutiveOnes(self, nums: List[int]) -> int:
        zeros = 0
        left = 0
        longest = 0
        for right in range(len(nums)):
            if nums[right] == 0:
                zeros += 1
            while zeros > 1:
                if nums[left] == 0:
```



```
        zeros -= 1
        left += 1
        longest = max(longest, right - left + 1)
    return longest
```

PHASE 5 — TEST & DEBUG

```
# [1,0,1,1,0]: zeros hits 2 at r=4 → shrink past first 0 → window=[1,1,0], longest=4
✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ . Space  $O(1)$ .
```

```
# Stream follow-up: keep a deque of the last two 0-positions.
```

```
# When third 0 arrives, left = oldest_zero_pos + 1 –  $O(1)$  space regardless of stream length.
```

```
# Generalises to at-most-k flips (Max Consecutive Ones III #1004): same window, k zeros allowed.
```

```
# Name that connection unprompted."
```

Sliding Window

★ #1100 Find K-Length Substrings with No Repeated Characters ★

MED[Open on LeetCode](#)

Hash Table · String · Sliding Window

Lists: ★ Premium | Premium 98

Problem

Given a string s and an integer k , return the number of substrings in s of length k with no repeated characters.

Example 1:

```
Input: s = "havefunonleetcode", k = 5
Output: 6
Explanation: There are 6 substrings they are:
'havef', 'avefu', 'vefun', 'efuno', 'etcod', 'tcode'.
```

Example 2:

```
Input: s = "home", k = 5
Output: 0
Explanation: Notice k can be larger than the length of s. In this case, it is
not possible to find any substring.
```

Constraints:

- $1 \leq s.length \leq 10^4$
- s consists of lowercase English letters.
- $1 \leq k \leq 10^4$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Count substrings of exactly length k with all distinct characters. Right?
# k > len(s) → 0. k=0 → 0."
#
# "s='havefunonleetcode', k=5 → 6 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# For each start index i, build substring s[i:i+k] and check distinct chars with a
# set.
# Increment count when set size equals k.
# O(n * k) windows; sliding window with frequency map is O(n).
# Time: O(n * k). Space: O(k) set per window.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Fixed sliding window of size k. Slide: add right, remove left-behind char.
# Valid when len(freq)==k (all chars distinct). Time: O(n). Space: O(k)."
```

PHASE 4 — CLEAN CODE

```
class _1100_KLengthSubstrings:
    def numKLenSubstrNoRepeats(self, s: str, k: int) -> int:
        if k > len(s):
            return 0
        freq : defaultdict = defaultdict(int)
        count = 0
        for right in range(len(s)):
            freq[s[right]] += 1
            if right >= k:
                left = right - k
                freq[s[left]] -= 1
                if freq[s[left]] == 0:
                    del freq[s[left]]
            if right >= k - 1 and len(freq) == k:
                count += 1
        return count
```

PHASE 5 — TEST & DEBUG

```
# s='home', k=5: 5>4 → 0 ✓
# First valid window r=k-1: if all k chars distinct, count++. Slide forward.
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(k). len(freq)==k is the clean validity check.
# Fixed window variant of #3 – useful when window size is given, not maximised.
# Given more time I'd ask: return the actual substrings? Collect
s[right-k+1:right+1]."
```

Sliding Window

□ #1248 Count Number of Nice Subarrays

MED[Open on LeetCode](#)

Array · Hash Table · Math · Sliding Window · Prefix Sum

Lists: CodeSignal

Problem

Given an array of integers `nums` and an integer `k`. A continuous subarray is called nice if there are `k` odd numbers on it.

Return the number of nice sub-arrays.

Example 1:

Input: `nums = [1,1,2,1,1]`, `k = 3`

Output: 2

Explanation: The only sub-arrays with 3 odd numbers are `[1,1,2,1]` and `[1,2,1,1]`.

Example 2:

Input: `nums = [2,4,6]`, `k = 1`

Output: 0

Explanation: There are no odd numbers in the array.

Example 3:

Input: `nums = [2,2,2,1,2,2,1,2,2,2]`, `k = 2`

Output: 16

Constraints:

- $1 \leq \text{nums.length} \leq 50000$
- $1 \leq \text{nums}[i] \leq 10^5$
- $1 \leq k \leq \text{nums.length}$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Count subarrays with EXACTLY k odd numbers. Return the count. Right?"
# Map each element to 1 (odd) or 0 (even) → reduce to Subarray Sum Equals K (#560)."
```

```
# "[1,1,2,1,1], k=3 → 2 ✓. [2,2,2,1,2,2,1,2,2,2], k=2 → 16 ✓."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic."
# Enumerate every subarray with two loops; count how many have an odd number of 1s.
# Straightforward prefix/suffix parity check per window.
# Correct, but  $O(n^2)$  subarrays is too slow at  $n = 5 \times 10^4$ .
# Time:  $O(n^2)$ . Space:  $O(1)$ .
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Prefix sum of odd indicators + hash map. Same pattern as #560."
# seed Counter at {0:1}. For each element, add  $\text{num} \% 2$  to running sum.
# Count += prefix_count[sum - k]. Time:  $O(n)$ . Space:  $O(n)$ ."
```

PHASE 4 — CLEAN CODE

```
class _1248_NiceSubarrays:
    def numberOfSubarrays(self, nums: List[int], k: int) -> int:
        prefix_count = Counter({0: 1})
        odd_sum = 0
        count = 0
        for num in nums:
            odd_sum += num % 2
            count += prefix_count[odd_sum - k]
            prefix_count[odd_sum] += 1
        return count
```

PHASE 5 — TEST & DEBUG

```
# [1,1,2,1,1], k=3:
# sums: 1,2,2,3,4. At sum=3: want=0→prefix[0]=1→count=1.
# At sum=4: want=1→prefix[1]=1→count=2. → 2 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ . Space  $O(n)$ . Identical to #560 – name the connection explicitly.
# At-most-k version: sliding window counting odds in window, shrink when  $> k$ ."
```

```
# Exactly k = atMost(k) - atMost(k-1).  
# Given more time I'd ask: count subarrays with at most k odd numbers?  
# Sliding window - shrink when odd count exceeds k, add right-left+1 each step."
```

Sorting

Round 2 · High · Medium · 3 questions

Sorting

□ #49 Group Anagrams

MED[Open on LeetCode](#)

Array · Hash Table · String · Sorting

Lists: Grind 169

Problem

Given an array of strings `strs`, group the anagrams together. You can return the answer in any order.

Example 1:

Input: `strs = ["eat","tea","tan","ate","nat","bat"]`

Output: `[["bat"],["nat","tan"],["ate","eat","tea"]]`

Explanation:

- There is no string in `strs` that can be rearranged to form "bat".
- The strings "nat" and "tan" are anagrams as they can be rearranged to form each other.
- The strings "ate", "eat", and "tea" are anagrams as they can be rearranged to form each other.

Example 2:

Input: strs = [""]

Output: [[""]]

Example 3:

Input: strs = ["a"]

Output: [["a"]]

Constraints:

- $1 \leq \text{strs.length} \leq 104$
- $0 \leq \text{strs}[i].\text{length} \leq 100$
- `strs[i]` consists of lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Group strings that are anagrams. Return groups in any order.

Same chars, same frequencies → anagrams. Empty string is its own group. Right?"

#

["'eat', 'tea', 'tan', 'ate', 'nat', 'bat'] → 3 groups ✓"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

For every pair of strings, sort both and check if the sorted forms match.

If they match, put them in the same group; merge groups as we go.

Correct, but comparing all pairs is expensive when n is large.

Time: $O(n^2 * m \log m)$. Space: $O(n * m)$.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Use a canonical key: `sorted(word)` as a tuple → same key for all anagrams.

Group into a defaultdict. Time: $O(n * m \log m)$. Space: $O(n * m)$.

Alternative: 26-element frequency tuple → $O(n * m)$, no sort cost."

PHASE 4 — CLEAN CODE

```
class _49_GroupAnagrams:
```

```
    def groupAnagrams(self, strs: List[str]) -> List[List[str]]:
```

```
        groups: defaultdict = defaultdict(list)
```

```
        for word in strs:
```

```
            groups[tuple(sorted(word))].append(word)
```

```
return list(groups.values())
```

PHASE 5 — TEST & DEBUG

```
# 'eat'→('a','e','t'). 'tea'→same. 'ate'→same. All go to one group.
```

```
# 'tan'→('a','n','t'). 'nat'→same.
```

```
# 'bat'→('a','b','t'). Alone. → 3 groups ✓
```

```
# [''] → {():['']}. → [['']] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n * m \log m)$ . Space  $O(n * m)$ .
```

```
# Frequency tuple alternative:  $O(n * m)$ , avoids  $\log m$  sort. For short words (typical)
```

```
# sorting is fine; for very long words, frequency tuple wins.
```

```
# Given more time I'd ask: Unicode input? sorted() handles it; frequency array breaks.
```

```
# Use Counter-based key: tuple(sorted(Counter(word).items()))."
```

Sorting

□ #56 Merge Intervals

MED[Open on LeetCode](#)

Array · Sorting

Lists: Grind 169

Problem

Given an array of intervals where $\text{intervals}[i] = [\text{start}_i, \text{end}_i]$, merge all overlapping intervals, and return an array of the non-overlapping intervals that cover all the intervals in the input.

Example 1:

```
Input: intervals = [[1,3],[2,6],[8,10],[15,18]]  
Output: [[1,6],[8,10],[15,18]]  
Explanation: Since intervals [1,3] and [2,6] overlap, merge them into [1,6].
```

Example 2:

```
Input: intervals = [[1,4],[4,5]]  
Output: [[1,5]]  
Explanation: Intervals [1,4] and [4,5] are considered overlapping.
```

Example 3:

```
Input: intervals = [[4,7],[1,4]]  
Output: [[1,7]]  
Explanation: Intervals [1,4] and [4,7] are considered overlapping.
```

Constraints:

- $1 \leq \text{intervals.length} \leq 104$

- `intervals[i].length == 2`
- `0 <= starti <= endi <= 104`

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Merge all overlapping intervals. Two intervals overlap if one starts before or
when the other ends.
# [1,4] and [4,5] overlap at 4. Sort by start, then merge. Return non-overlapping
result. Right?"
#
# "[[1,3],[2,6],[8,10],[15,18]] → [[1,6],[8,10],[15,18]] ✓ (1-3 and 2-6 overlap at
2-3)"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Compare every pair of intervals; if they overlap, merge into one wider interval.
# Repeat until a full pass finds no new merges.
# Correct, but repeated pairwise merging can degrade toward O(n^2).
# Time: O(n^2) worst case. Space: O(n) output.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Sort by start time — O(n log n). Then one pass:
# Keep a result list. For each interval, if it overlaps with the last result
interval
# (last.end >= current.start), extend the last interval's end.
# Otherwise, append as a new separate interval.
# Time: O(n log n). Space: O(n)."
```

PHASE 4 — CLEAN CODE

```
class _56_MergeIntervals:
    def merge(self, intervals: List[List[int]]) -> List[List[int]]:
        intervals.sort()
        merged = [intervals[0]]
        for start, end in intervals[1:]:
            if merged[-1][1] >= start: # overlap
                merged[-1][1] = max(merged[-1][1], end)
            else:
                merged.append([start, end])
        return merged
```

PHASE 5 — TEST & DEBUG

```
# [[1,3],[2,6],[8,10],[15,18]]: sorted same.
# merged=[[1,3]]. [2,6]: 3>=2→extend→[1,6]. [8,10]: 6<8→append. [15,18]:
10<15→append.
```

```
# → [[1,6],[8,10],[15,18]] ✓  
# [[1,4],[4,5]]: 4>=4→[1,5] ✓  
# [[4,7],[1,4]]: sorted→[[1,4],[4,7]]. 4>=4→[1,7] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n \log n)$. Space $O(n)$.

The sort is the bottleneck – the merge pass itself is $O(n)$.

If the input were already sorted, this runs in $O(n)$ total.

Given more time I'd ask: how many meeting rooms needed?

That's Meeting Rooms II (#253) – use a min-heap of end times."

Sorting

□ ★ #1152 Analyze User Website Visit Pattern ★

MED

[Open on LeetCode](#)

Array · Hash Table · Sorting

Lists: ★ Premium | Premium 98

Problem

You are given two string arrays `username` and `website` and an integer array `timestamp`. All the given arrays are of the same length and the tuple `[username[i], website[i], timestamp[i]]` indicates that the user `username[i]` visited the website `website[i]` at time `timestamp[i]`.

A pattern is a list of three websites (not necessarily distinct).

- For example, `["home", "away", "love"]`, `["leetcode", "love", "leetcode"]`, and `["luffy", "luffy", "luffy"]` are all patterns.

The score of a pattern is the number of users that visited all the websites in the pattern in the same order they appeared in the pattern.

- For example, if the pattern is ["home", "away", "love"], the score is the number of users x such that x visited "home" then visited "away" and visited "love" after that.
- Similarly, if the pattern is ["leetcode", "love", "leetcode"], the score is the number of users x such that x visited "leetcode" then visited "love" and visited "leetcode" one more time after that.
- Also, if the pattern is ["luffy", "luffy", "luffy"], the score is the number of users x such that x visited "luffy" three different times at different timestamps.

Return the pattern with the largest score. If there is more than one pattern with the same largest score, return the lexicographically smallest such pattern.

Note that the websites in a pattern do not need to be visited contiguously, they only need to be visited in the order they appeared in the pattern.
Example 1:

```

Input: username =
["joe","joe","joe","james","james","james","james","mary","mary","mary"],
timestamp = [1,2,3,4,5,6,7,8,9,10], website =
["home","about","career","home","cart","maps","home","home","about","career"]
Output: ["home","about","career"]
Explanation: The tuples in this example are:
["joe","home",1],["joe","about",2],["joe","career",3],["james","home",4],["james","cart",5],["james","maps",6],["james","home",7],["mary","home",8],["mary","about",9], and ["mary","career",10].
The pattern ("home", "about", "career") has score 2 (joe and mary).
The pattern ("home", "cart", "maps") has score 1 (james).
The pattern ("home", "cart", "home") has score 1 (james).
The pattern ("home", "maps", "home") has score 1 (james).

```

Example 2:

```

Input: username = ["ua","ua","ua","ub","ub","ub"], timestamp = [1,2,3,4,5,6],
website = ["a","b","a","a","b","c"]
Output: ["a","b","a"]

```

Constraints:

- $3 \leq \text{username.length} \leq 50$
- $1 \leq \text{username}[i].\text{length} \leq 10$
- $\text{timestamp.length} == \text{username.length}$
- $1 \leq \text{timestamp}[i] \leq 109$
- $\text{website.length} == \text{username.length}$
- $1 \leq \text{website}[i].\text{length} \leq 10$
- $\text{username}[i]$ and $\text{website}[i]$ consist of lowercase English letters.
- It is guaranteed that there is at least one user who visited at least three websites.
- All the tuples $[\text{username}[i], \text{timestamp}[i], \text{website}[i]]$ are unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find the 3-website pattern (subsequence, not necessarily consecutive) that the
most DISTINCT users
# have visited in order. On tie, return lexicographically smallest. Right?
# Each user contributes at most 1 count per pattern regardless of how many times
they visit."
#
# "joe visited home→about→career, james visited home→cart→maps→home, mary visited
home→about→career.
# Pattern ('home','about','career') score=2 (joe, mary) → winner ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# For each user, enumerate all 3-website combinations in visit order.
# Build pattern tuple (a,b,c) and increment Counter across users.
# O(visit_length^3) per user – acceptable only for small lists.
# Time: O(U * V^3). Space: O(patterns).
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Sort all visits by timestamp. Group by user into ordered website lists.
# For each user, generate all C(len,3) 3-subsequence combos – at most C(50,3)=19600.
# Use a set per user to avoid double-counting (one user = one score per pattern).
# Count across users with a global Counter.
# Return pattern with max score, lexicographically smallest on ties."
```

PHASE 4 — CLEAN CODE

```
class _1152_AnalyzeWebsiteVisit:
    def mostVisitedPattern(self, username: List[str], timestamp: List[int],
website: List[str]) -> List[str]:
        visits = sorted(zip(timestamp, username, website))
        user_sites: defaultdict = defaultdict(list)
        for _, user, site in visits:
            user_sites[user].append(site)
        pattern_score: Counter = Counter()
        for sites in user_sites.values():
            seen = set()
            for combo in itertools.combinations(sites, 3):
                if combo not in seen:
                    seen.add(combo)
                    pattern_score[combo] += 1
        return list(max(pattern_score, key=lambda p: (pattern_score[p], [-ord(c)
for c in ''.join(p)])))
```

PHASE 5 — TEST & DEBUG

```
# After sorting by timestamp, user_sites = {joe:[home,about,career],
james:[home,car,maps,home], mary:[home,about,career]}.
# joe: combos of [home,about,career] = {(home,about,career)}.
# james: C(4,3)=4 combos. All unique.
# mary: {(home,about,career)}.
# pattern_score[(home,about,career)]=2 → winner ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(n \log n + U * C(K,3))$  where  $U$ =users,  $K$ =max visits per user.
# With  $U \leq 50$ ,  $K \leq 50$ :  $C(50,3)=19600 * 50 = 980000$  – manageable.
# The per-user 'seen' set ensures one user contributes at most 1 to each pattern's
score.
# Tie-breaking: max() with a compound key – score ascending, then lexicographic
ascending.
# Given more time I'd ask: what if we want top-k patterns?
# Use a heap instead of max() – same  $O$  complexity."
```

Binary Search

Round 2 · High · Medium · 12 questions

Binary Search

□ #33 Search in Rotated Sorted Array

MED[Open on LeetCode](#)

Array · Binary Search

Lists: Grind 169

Problem

There is an integer array `nums` sorted in ascending order (with distinct values).

Prior to being passed to your function, `nums` is possibly left rotated at an unknown index `k` ($1 \leq k < \text{nums.length}$) such that the resulting array is `[nums[k], nums[k+1], ..., nums[n-1], nums[0], nums[1], ..., nums[k-1]]` (0-indexed). For example, `[0,1,2,4,5,6,7]` might be left rotated by 3 indices and become `[4,5,6,7,0,1,2]`.

Given the array `nums` after the possible rotation and an integer `target`, return the index of `target` if it is in `nums`, or `-1` if it is not in `nums`.

You must write an algorithm with $O(\log n)$ runtime complexity.

Example 1:

Input: nums = [4,5,6,7,0,1,2], target = 0
Output: 4

Example 2:

Input: nums = [4,5,6,7,0,1,2], target = 3
Output: -1

Example 3:

Input: nums = [1], target = 0
Output: -1

Constraints:

- $1 \leq \text{nums.length} \leq 5000$
- $-104 \leq \text{nums}[i] \leq 104$
- All values of nums are unique.
- nums is an ascending array that is possibly rotated.
- $-104 \leq \text{target} \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Array was sorted then rotated at some unknown pivot. Find target in $O(\log n)$.
Right?

All values are distinct. Return -1 if not found."

#

"[4,5,6,7,0,1,2], target=0 → 4 ✓. target=3 → -1 ✓."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Scan left to right: for each index i, check if $\text{nums}[i] == \text{target}$.

If not found after one full pass, return -1.

Correct and easy to reason about, but the problem asks for $O(\log n)$ on a rotated sorted array.

Time: $O(n)$. Space: $O(1)$.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

```
# "Standard binary search – but at each step, identify which half is sorted.
# If nums[left] <= nums[mid]: left half is sorted.
# If target in [nums[left], nums[mid]): search left.
# Else search right.
# Else: right half is sorted.
# If target in (nums[mid], nums[right]): search right.
# Else search left.
# Time: O(log n). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _33_SearchRotatedArray:
    def search(self, nums: List[int], target: int) -> int:
        left, right = 0, len(nums) - 1
        while left <= right:
            mid = (left + right) // 2
            if nums[mid] == target:
                return mid
            if nums[left] <= nums[mid]: # left half sorted
                if nums[left] <= target < nums[mid]:
                    right = mid - 1
            else:
                left = mid + 1
        else: # right half sorted
            if nums[mid] < target <= nums[right]:
                left = mid + 1
            else:
                right = mid - 1
        return -1
```

PHASE 5 — TEST & DEBUG

```
# [4,5,6,7,0,1,2], target=0:
# l=0,r=6,mid=3(7). nums[0]=4<=7 → left sorted [4..7]. 0 in [4,7)? No → l=4.
# l=4,r=6,mid=5(1). nums[4]=0>1 → right sorted [1..2]. 0 in (1,2]? No → r=4.
# l=4,r=4,mid=4(0). 0==target → return 4 ✓
# target=3: never matches → -1 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(log n). Space O(1).
# The key: at least one half is always sorted in a rotated array – use that half
# to decide where to search. Checking nums[left] <= nums[mid] tells us which half.
# Given more time I'd ask: what if there are duplicates?
# That's #81 – when nums[left]==nums[mid], we can't determine which half is sorted.
# Worst case degrades to O(n) (just left++ to skip the duplicate)."
```

Binary Search

□ #34 Find First and Last Position of Element in Sorted Array

MED[Open on LeetCode](#)

Array · Binary Search

Lists: Denny Zhang

Problem

Given an array of integers `nums` sorted in non-decreasing order, find the starting and ending position of a given target value.

If target is not found in the array, return `[-1, -1]`.

You must write an algorithm with $O(\log n)$ runtime complexity.

Example 1:

Input: `nums = [5,7,7,8,8,10]`, `target = 8`
Output: `[3,4]`

Example 2:

Input: `nums = [5,7,7,8,8,10]`, `target = 6`
Output: `[-1,-1]`

Example 3:

Input: `nums = []`, `target = 0`
Output: `[-1,-1]`

Constraints:

- $0 \leq \text{nums.length} \leq 105$
- $-109 \leq \text{nums}[i] \leq 109$
- `nums` is a non-decreasing array.
- $-109 \leq \text{target} \leq 109$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Sorted array with duplicates. Find the first and last index of target. Return
[-1,-1] if absent.
# Must be O(log n). Right?"
#
# "[5,7,7,8,8,10], target=8 → [3,4] ✓. target=6 → [-1,-1] ✓."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Run standard binary search once to find any index of target.
# Then expand left and right from that index while values still equal target.
# Works, but we can do both boundaries in one modified binary search pass.
# Time: O(log n) for search + O(k) for expansion. Space: O(1).
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Two binary searches:
# lower_bound: find first index where nums[i] >= target. If nums[result]==target →
first position.
# upper_bound: find first index where nums[i] > target. Subtract 1 → last position.
# Both are standard binary search variants. Time: O(log n). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _34_FirstLastPosition:
    def searchRange(self, nums: List[int], target: int) -> List[int]:
        def lower_bound() -> int:
            left, right = 0, len(nums)
            while left < right:
                mid = (left + right) // 2
                if nums[mid] >= target:
                    right = mid
            else:
                left = mid + 1
            return left
```



```

def upper_bound() -> int:
    left, right = 0, len(nums)
    while left < right:
        mid = (left + right) // 2
        if nums[mid] > target:
            right = mid
        else:
            left = mid + 1
    return left
first = lower_bound()
if first == len(nums) or nums[first] != target:
    return [-1, -1]
return [first, upper_bound() - 1]

```

PHASE 5 — TEST & DEBUG

```

# nums=[5,7,7,8,8,10], target=8:
# lower_bound: searching for first idx where nums[i]>=8. → index 3.
# nums[3]=8==target ✓. upper_bound: first idx where nums[i]>8. → index 5. 5-1=4.
# → [3,4] ✓
# target=6: lower_bound→2 (nums[2]=7>=6). nums[2]=7≠6 → [-1,-1] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(\log n)$ . Space  $O(1)$ .
# lower_bound and upper_bound are the two fundamental binary search variants – know both.
# They correspond to bisect_left and bisect_right in Python's bisect module.
# The check 'nums[first] != target' after lower_bound handles the not-found case cleanly.
# Given more time I'd ask: what if we want the count of target occurrences?
# upper_bound() - lower_bound() in  $O(\log n)$ ."

```

Binary Search

□ #153 Find Minimum in Rotated Sorted Array

MED[Open on LeetCode](#)

Array · Binary Search

Lists: Grind 169

Problem

Suppose an array of length n sorted in ascending order is rotated between 1 and n times. For example, the array `nums = [0,1,2,4,5,6,7]` might become:

- `[4,5,6,7,0,1,2]` if it was rotated 4 times.
- `[0,1,2,4,5,6,7]` if it was rotated 7 times.

Notice that rotating an array `[a[0], a[1], a[2], ..., a[n-1]]` 1 time results in the array `[a[n-1], a[0], a[1], a[2], ..., a[n-2]]`.

Given the sorted rotated array `nums` of unique elements, return the minimum element of this array.

You must write an algorithm that runs in $O(\log n)$ time.

Example 1:

Input: nums = [3,4,5,1,2]

Output: 1

Explanation: The original array was [1,2,3,4,5] rotated 3 times.

Example 2:

Input: nums = [4,5,6,7,0,1,2]

Output: 0

Explanation: The original array was [0,1,2,4,5,6,7] and it was rotated 4 times.

Example 3:

Input: nums = [11,13,15,17]

Output: 11

Explanation: The original array was [11,13,15,17] and it was rotated 4 times.

Constraints:

- $n == \text{nums.length}$
- $1 \leq n \leq 5000$
- $-5000 \leq \text{nums}[i] \leq 5000$
- All the integers of nums are unique.
- nums is sorted and rotated between 1 and n times.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Array sorted then rotated. Find the minimum in $O(\log n)$. All values unique. Right?"

The minimum is at the rotation point – where the array 'wraps'."

#

"[3,4,5,1,2] → 1 ✓. [4,5,6,7,0,1,2] → 0 ✓. [11,13,15,17] → 11 (no rotation) ✓."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

Linear scan from index 0: track the smallest value seen so far.

The first place where $\text{nums}[i] > \text{nums}[i+1]$ marks the rotation pivot candidate.

Return that minimum – correct on small inputs, but we can binary-search the pivot.

Time: $O(n)$. Space: $O(1)$.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Binary search on the rotation point.

If `nums[mid] > nums[right]`: minimum is in the RIGHT half (`mid+1` to `right`).

Else: minimum is in the LEFT half including `mid` (`left` to `mid`).

When `left==right`, we've found the minimum.

Time: $O(\log n)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _153_FindMinRotated:
```

```
    def findMin(self, nums: List[int]) -> int:
```

```
        left, right = 0, len(nums) - 1
```

```
        while left < right:
```

```
            mid = (left + right) // 2
```

```
            if nums[mid] > nums[right]:
```

```
                left = mid + 1 # minimum is to the right of mid
```

```
            else:
```

```
                right = mid # minimum is mid or to its left
```

```
        return nums[left]
```

PHASE 5 — TEST & DEBUG

`[3,4,5,1,2]`: `l=0,r=4,mid=2(5)`. `5>nums[4]=2` → `l=3`.

`l=3,r=4,mid=3(1)`. `1<=nums[4]=2` → `r=3`. `l=r=3` → return `nums[3]=1` ✓

`[4,5,6,7,0,1,2]`: `l=0,r=6,mid=3(7)`. `7>nums[6]=2` → `l=4`.

`l=4,r=6,mid=5(1)`. `1<=2` → `r=5`. `l=4,r=5,mid=4(0)`. `0<=1` → `r=4`. `l=r=4` → `0` ✓

`[11,13,15,17]`: `l=0,r=3,mid=1(13)`. `13<=17` → `r=1`. `l=0,r=1,mid=0(11)`. `11<=13` → `r=0`. → `11` ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(\log n)$. Space $O(1)$.

The comparison is always against `nums[right]` (not `nums[left]`).

If `nums[mid] > nums[right]`, the right portion must be 'lower' – minimum is there.

If `nums[mid] <= nums[right]`, the right portion is in order – minimum is at `mid` or `left`.

Given more time I'd ask: what if there are duplicates?

When `nums[mid]==nums[right]` we can't decide – fall back to `right`-- ($O(n)$ worst case)."

Binary Search

□ #300 Longest Increasing Subsequence

MED[Open on LeetCode](#)

Array · Binary Search · Dynamic Programming
Lists: Grind 169

Problem

Given an integer array `nums`, return the length of the longest strictly increasing subsequence.

Example 1:

Input: `nums = [10,9,2,5,3,7,101,18]`

Output: 4

Explanation: The longest increasing subsequence is `[2,3,7,101]`, therefore the length is 4.

Example 2:

Input: `nums = [0,1,0,3,2,3]`

Output: 4

Example 3:

Input: `nums = [7,7,7,7,7,7,7]`

Output: 1

Constraints:

- $1 \leq \text{nums.length} \leq 2500$
- $-104 \leq \text{nums}[i] \leq 104$

Follow up: Can you come up with an algorithm that runs in $O(n \log(n))$ time complexity?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find the length of the longest strictly increasing subsequence (not necessarily
contiguous).
# Return the length, not the subsequence itself. Right?
# Follow-up asks for  $O(n \log n)$  – so know both approaches."
#
# "[10,9,2,5,3,7,101,18] → [2,3,7,101] length 4 ✓. [7,7,7,7] → 1 (strictly
increasing) ✓."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# dp[i] = length of LIS ending at i. For each i, scan all j < i with nums[j] <
nums[i].
# dp[i] = max(dp[j]+1). Answer = max(dp).
# Correct  $O(n^2)$ ; patience sorting / binary search on tails gives  $O(n \log n)$ .
# Time:  $O(n^2)$ . Space:  $O(n)$  dp array.
# Should I go ahead and optimize?"
```

```
class _300_LIS_DP:
    def lengthOfLIS(self, nums: List[int]) -> int:
        dp = [1] * len(nums)
        for i in range(1, len(nums)):
            for j in range(i):
                if nums[j] < nums[i]:
                    dp[i] = max(dp[i], dp[j] + 1)
        return max(dp)
```

PHASE 3 — OPTIMIZE ($O(n \log n)$ patience sorting)

```
# "Maintain a 'tails' array where tails[i] is the smallest tail of all LIS of length
i+1.
# For each number, binary search for the leftmost position in tails where tails[pos]
>= num.
# If found, replace tails[pos] with num (smaller tail = more room to extend).
# If not found, append (the LIS can be extended by 1).
# len(tails) at the end is the LIS length.
#
# This is patience sorting – tails is always sorted so binary search works."
```

PHASE 4 — CLEAN CODE

```
import bisect
class _300_LIS:
```

```
def lengthOfLIS(self, nums: List[int]) -> int:
    tails = []
    for num in nums:
        pos = bisect.bisect_left(tails, num)
        if pos == len(tails):
            tails.append(num)
        else:
            tails[pos] = num
    return len(tails)
```

PHASE 5 — TEST & DEBUG

```
# nums=[10,9,2,5,3,7,101,18]:
# 10→tails=[10]. 9→replace 10→[9]. 2→replace 9→[2]. 5→append→[2,5].
# 3→replace 5→[2,3]. 7→append→[2,3,7]. 101→append→[2,3,7,101]. 18→replace
101→[2,3,7,18].
# len=4 ✓
# [7,7,7,7]: first 7→[7]. next 7→bisect_left([7],7)=0→replace→[7]. len=1 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "O(n²) DP: Time O(n²), Space O(n). Clean and easy to explain.
# O(n log n) patience sort: Time O(n log n), Space O(n).
# Note: tails does NOT store the actual LIS – it's a virtual 'pile' structure.
# To reconstruct the actual LIS, track parent pointers in the DP version.
# Given more time I'd ask: longest non-decreasing subsequence?
# Change bisect_left to bisect_right – allows equal elements to extend the
sequence."
```

Binary Search

□ #362 Design Hit Counter

MED[Open on LeetCode](#)

Array · Hash Table · Binary Search · Design · Queue

Lists: Grind 169

Problem

Design a hit counter which counts the number of hits received in the past 5 minutes (i.e., the past 300 seconds).

Your system should accept a timestamp parameter (in seconds granularity), and you may assume that calls are being made to the system in chronological order (i.e., timestamp is monotonically increasing). Several hits may arrive roughly at the same time.

Implement the HitCounter class:

- **HitCounter()** Initializes the object of the hit counter system.
- **void hit(int timestamp)** Records a hit that happened at timestamp (in seconds). Several hits may happen at the same timestamp.

- **int getHits(int timestamp)** Returns the number of hits in the past 5 minutes from timestamp (i.e., the past 300 seconds).

Example 1:

Input

```
["HitCounter", "hit", "hit", "hit", "getHits", "hit", "getHits", "getHits"]
[[], [1], [2], [3], [4], [300], [300], [301]]
```

Output

```
[null, null, null, null, 3, null, 4, 3]
```

Explanation

```
HitCounter hitCounter = new HitCounter();
```

Constraints:

- $1 \leq \text{timestamp} \leq 2 * 10^9$
- All the calls are being made to the system in chronological order (i.e., timestamp is monotonically increasing).
- At most 300 calls will be made to hit and getHits.

Follow up: What if the number of hits per second could be huge? Does your design scale?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Record hits at timestamps and query: how many hits in the past 300 seconds?"

Calls are chronologically ordered (timestamps only increase). Right?

getHits(t) returns hits with timestamp in (t-300, t] inclusive – past 5 minutes."

#

"hit(1),hit(2),hit(3): getHits(4)=3. hit(300): getHits(300)=4. getHits(301)=3 (hit@1 expired)."

PHASE 2 — BRUTE FORCE

```

# "Let me start with brute force to lock in the logic.
# Append every hit timestamp to a list.
# getHits(t): count timestamps in [t-299, t] by scanning the whole list.
# Simple, but each query is O(total hits) – queue with expiry is better.
# Time: O(1) hit, O(n) getHits per query. Space: O(n) all timestamps.
# Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE
# "Store timestamps in a deque. On getHits, pop from the left any timestamp <=
t-300.
# Remaining deque size is the count. hit() is O(1). getHits() is amortised O(1).
#
# Follow-up (huge hits/second): store (timestamp, count) pairs instead of individual
timestamps.
# Multiple hits at the same second are collapsed into one deque entry."

# PHASE 4 — CLEAN CODE
class _362_HitCounter:
    def __init__(self):
        self.hits: deque = deque() # stores individual hit timestamps
    def hit(self, timestamp: int) -> None:
        self.hits.append(timestamp)
    def getHits(self, timestamp: int) -> int:
        while self.hits and self.hits[0] <= timestamp - 300:
            self.hits.popleft()
        return len(self.hits)
# Follow-up: grouped version for high-throughput
class _362_HitCounter_Grouped:
    def __init__(self):
        self.hits: deque = deque() # stores (timestamp, count) pairs
    def hit(self, timestamp: int) -> None:
        if self.hits and self.hits[-1][0] == timestamp:
            ts, cnt = self.hits.pop()
            self.hits.append((ts, cnt + 1))
        else:
            self.hits.append((timestamp, 1))
    def getHits(self, timestamp: int) -> int:
        while self.hits and self.hits[0][0] <= timestamp - 300:
            self.hits.popleft()
        return sum(cnt for _, cnt in self.hits)

# PHASE 5 — TEST & DEBUG
# hit(1),hit(2),hit(3),getHits(4): deque=[1,2,3]. all>4-300=-296 → len=3 ✓
# hit(300),getHits(300): deque=[1,2,3,300]. 1<=0? No. len=4 ✓
# getHits(301): 1<=301-300=1 → popleft. deque=[2,3,300]. len=3 ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "hit: O(1). getHits: O(k) amortised where k = expired entries (each entry popped
at most once).

```

```
# Space:  $O(n)$  where  $n$  = total hits in the past 300 seconds.  
# Binary search alternative: store sorted timestamps, binary search for  $t-300$  cutoff  
–  $O(\log n)$ .  
# The deque approach is simpler and amortised  $O(1)$  total.  
# Grouped version handles millions of hits/second – key interview follow-up.  
# Given more time I'd ask: what if calls aren't chronological?  
# We'd need a sorted structure (sorted list or BST) and can't use the deque eviction  
trick."
```

Binary Search

□ #528 Random Pick with Weight

MED[Open on LeetCode](#)

Math · Binary Search · Prefix Sum ·
Randomized

Lists: Grind 169

Problem

You are given a 0-indexed array of positive integers w where $w[i]$ describes the weight of the i th index.

You need to implement the function `pickIndex()`, which randomly picks an index in the range $[0, w.length - 1]$ (inclusive) and returns it. The probability of picking an index i is $w[i] / \text{sum}(w)$.

- For example, if $w = [1, 3]$, the probability of picking index 0 is $1 / (1 + 3) = 0.25$ (i.e., 25%), and the probability of picking index 1 is $3 / (1 + 3) = 0.75$ (i.e., 75%).

Example 1:

Input

```
["Solution","pickIndex"]
[[[1]],[]]
```

Output

```
[null,0]
```

Explanation

```
Solution solution = new Solution([1]);
```

Example 2:

Input

```
["Solution","pickIndex","pickIndex","pickIndex","pickIndex","pickIndex"]
[[[1,3]],[],[],[],[],[]]
```

Output

```
[null,1,1,1,1,0]
```

Explanation

```
Solution solution = new Solution([1, 3]);
```

Constraints:

- $1 \leq w.length \leq 104$
- $1 \leq w[i] \leq 105$
- pickIndex will be called at most 104 times.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Pick a random index where index i is chosen with probability $w[i]/\text{sum}(w)$.

pickIndex() is called up to 10^4 times – so init can be $O(n)$ but pick must be fast. Right?

$w=[1,3]$: $P(0)=25\%$, $P(1)=75\%$. The actual randomness in the example output is fine."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Expand weights into a list where index i appears $w[i]$ times, pick random index.

Correct distribution but total list size can be $\text{sum}(w)$ up to $1e9$ – impossible.

Prefix sums + binary search on total weight fixes space.

Time: $O(\text{sum}(w))$ build. Space: $O(\text{sum}(w))$ – infeasible.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

```
# "Build a prefix sum array. Prefix[i] = w[0]+...+w[i].
# Pick a random float r in [0, total). Binary search for the leftmost prefix sum >
r.
# That index is our answer – the probability of landing in bucket i equals
w[i]/total.
# Time: O(n) init. O(log n) per pick. Space: O(n)."
```

PHASE 4 — CLEAN CODE

```
import random
class _528_RandomPickWeight:
    def __init__(self, w: List[int]):
        self.prefix = []
        total = 0
        for weight in w:
            total += weight
            self.prefix.append(total)
        self.total = total
    def pickIndex(self) -> int:
        r = random.random() * self.total
        lo, hi = 0, len(self.prefix) - 1
        while lo < hi:
            mid = (lo + hi) // 2
            if self.prefix[mid] <= r:
                lo = mid + 1
            else:
                hi = mid
        return lo
```

PHASE 5 — TEST & DEBUG

```
# w=[1,3]: prefix=[1,4], total=4.
# r in [0,1) → prefix[0]=1>r → lo=hi=0 → return 0 (25% of the time).
# r in [1,4) → prefix[0]=1<=r → lo=1 → return 1 (75% of the time). ✓
# w=[1]: prefix=[1]. Any r in [0,1) → hi=lo=0 → return 0 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Init O(n). pickIndex O(log n). Space O(n).
# The prefix sum maps weights to contiguous ranges on [0, total).
# A random point in that range lands in bucket i with probability w[i]/total.
# bisect.bisect_right(self.prefix, r) is a one-liner alternative using Python's
bisect module.
# Given more time I'd ask: what if weights change dynamically?
# We'd need a Fenwick tree (BIT) – O(log n) updates AND O(log n) queries."
```

Binary Search

□ #852 Peak Index in a Mountain Array

MED[Open on LeetCode](#)

Array · Binary Search

Lists: Denny Zhang

Problem

You are given an integer mountain array `arr` of length `n` where the values increase to a peak element and then decrease.

Return the index of the peak element.

Your task is to solve it in $O(\log(n))$ time complexity.

Example 1:

Input: `arr = [0,1,0]`

Output: 1

Example 2:

Input: `arr = [0,2,1,0]`

Output: 1

Example 3:

Input: `arr = [0,10,5,2]`

Output: 1

Constraints:

- $3 \leq \text{arr.length} \leq 105$
- $0 \leq \text{arr}[i] \leq 106$
- arr is guaranteed to be a mountain array.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Array guaranteed to have exactly one peak (mountain shape: strictly increasing then decreasing)."

Return the index of the peak in $O(\log n)$. Right?"

#

"[0,1,0] → 1. [0,2,1,0] → 1. [0,10,5,2] → 1. ✓"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

Linear scan from index 1: first i where $\text{arr}[i] > \text{arr}[i+1]$ is the peak index.

Mountain array guarantees exactly one peak; scan stops early.

$O(n)$ acceptable but binary search on the rising slope is $O(\log n)$.

Time: $O(n)$. Space: $O(1)$.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Binary search: at mid, compare $\text{arr}[\text{mid}]$ with $\text{arr}[\text{mid}+1]$."

If $\text{arr}[\text{mid}] < \text{arr}[\text{mid}+1]$: we're on the ascending slope → peak is to the right → $\text{left} = \text{mid}+1$.

Else: we're on or past the peak → peak is at mid or to the left → $\text{right} = \text{mid}$.

When $\text{left} == \text{right}$: that's the peak. $O(\log n)$."

PHASE 4 — CLEAN CODE

class _852_PeakIndex:

def peakIndexInMountainArray(self, arr: List[int]) -> int:

left, right = 0, len(arr) - 1

while left < right:

mid = (left + right) // 2

if $\text{arr}[\text{mid}] < \text{arr}[\text{mid} + 1]$:

left = mid + 1 # still ascending – peak is further right

else:

right = mid # descending or at peak – search left including mid

return left

PHASE 5 — TEST & DEBUG


```
# [0,1,0]: l=0,r=2,mid=1. arr[1]=1>arr[2]=0 → right=1. l=0,r=1,mid=0.  
arr[0]=0<arr[1]=1 → left=1. l=r=1 → 1 ✓  
# [0,10,5,2]: l=0,r=3,mid=1. arr[1]=10>arr[2]=5 → right=1. l=0,r=1,mid=0.  
arr[0]<arr[1] → left=1. → 1 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(\log n)$. Space $O(1)$.

The comparison `arr[mid]` vs `arr[mid+1]` acts as a gradient – positive means 'go right', negative means 'go left'.

This is the same logic as Find Peak Element (#162) – the general version where the array isn't guaranteed

to be a full mountain. Mention the connection.

Given more time I'd ask: what if there are multiple peaks?

Same algorithm finds one peak. For all peaks, we'd need a linear scan."

Binary Search

□ #981 Time Based Key-Value Store

MED

[Open on LeetCode](#)

Hash Table · String · Binary Search · Design
Lists: Grind 169

Problem

Design a time-based key-value data structure that can store multiple values for the same key at different time stamps and retrieve the key's value at a certain timestamp.

Implement the TimeMap class:

- **TimeMap()** Initializes the object of the data structure.
- **void set(String key, String value, int timestamp)** Stores the key key with the value value at the given time timestamp.
- **String get(String key, int timestamp)** Returns a value such that set was called previously, with $\text{timestamp_prev} \leq \text{timestamp}$. If there are multiple such values, it returns the value associated with the largest timestamp_prev. If there are no values, it returns "".

Example 1:

Input

```
["TimeMap", "set", "get", "get", "set", "get", "get"]
[[], ["foo", "bar", 1], ["foo", 1], ["foo", 3], ["foo", "bar2", 4], ["foo", 4], ["foo", 5]]
```

Output

```
[null, null, "bar", "bar", null, "bar2", "bar2"]
```

Explanation

```
TimeMap timeMap = new TimeMap();
```

Constraints:

- $1 \leq \text{key.length}, \text{value.length} \leq 100$
- key and value consist of lowercase English letters and digits.
- $1 \leq \text{timestamp} \leq 10^7$
- All the timestamps timestamp of set are strictly increasing.
- At most $2 * 10^5$ calls will be made to set and get.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Key-value store where each key can have multiple values at different timestamps.

set(key, value, ts) stores the entry. get(key, ts) returns the value with the

largest timestamp $\leq$ ts. Return '' if none exists. Right?

set timestamps are strictly increasing – so we can binary search the history list."

#

"set('foo','bar',1). get('foo',3) → 'bar' (no $ts \leq 3$ except 1). ✓"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Store all (timestamp, value) pairs per key in a list on set.

get(key, ts): scan backwards for largest timestamp $\leq$ ts.

Works but each get is $O(\text{number of entries for that key})$.

```
# Time: O(1) set, O(k) get per key history. Space: O(total entries).
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Since set timestamps are strictly increasing, each key's list is already sorted by timestamp.
```

```
# get() binary searches for the rightmost timestamp <= query timestamp.
```

```
# set: O(1). get: O(log n). Space: O(total entries)."
```

PHASE 4 — CLEAN CODE

```
class _981_TimeMap:
    def __init__(self):
        self.store: defaultdict = defaultdict(list) # key → [(ts, value), ...]

    def set(self, key: str, value: str, timestamp: int) -> None:
        self.store[key].append((timestamp, value))

    def get(self, key: str, timestamp: int) -> str:
        entries = self.store[key]
        lo, hi = 0, len(entries) - 1
        result = ''
        while lo <= hi:
            mid = (lo + hi) // 2
            if entries[mid][0] <= timestamp:
                result = entries[mid][1] # valid candidate – try to go right for a
larger ts
            lo = mid + 1
        else:
            hi = mid - 1
        return result
```

PHASE 5 — TEST & DEBUG

```
# set('foo','bar',1): store['foo']=[(1,'bar')].
```

```
# get('foo',1): entries=[(1,'bar')]. mid=0, ts=1<=1 → result='bar', lo=1. lo>hi →
return 'bar' ✓
```

```
# get('foo',3): mid=0, ts=1<=3 → result='bar'. lo=1>hi=0 → return 'bar' ✓
```

```
# set('foo','bar2',4): store['foo']=[(1,'bar'),(4,'bar2')].
```

```
# get('foo',4): mid=0(1<=4)→result='bar',lo=1. mid=1(4<=4)→result='bar2',lo=2.
return 'bar2' ✓
```

```
# get('foo',5): same → 'bar2' ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "set O(1). get O(log n). Space O(total entries).
```

```
# The 'find rightmost ts <= query' is a standard upper_bound - 1 binary search
pattern.
```

```
# Timestamps strictly increasing → no sorting needed – the list maintains order
automatically.
```

```
# bisect.bisect_right alternative: pos = bisect.bisect_right(entries, (timestamp,
chr(127))) - 1
```

```
# (appending chr(127) to go past any value at that exact timestamp).
```

```
# Given more time I'd ask: what if set timestamps aren't guaranteed to be
increasing?
```

```
# We'd need to sort the list on insert or use a sorted data structure."
```

Binary Search

□ #1011 Capacity to Ship Packages Within D Days

MED[Open on LeetCode](#)

Array · Binary Search

Lists: Denny Zhang

Problem

A conveyor belt has packages that must be shipped from one port to another within $days$ days.

The i th package on the conveyor belt has a weight of $weights[i]$. Each day, we load the ship with packages on the conveyor belt (in the order given by $weights$). We may not load more weight than the maximum weight capacity of the ship. Return the least weight capacity of the ship that will result in all the packages on the conveyor belt being shipped within $days$ days.

Example 1:

Input: weights = [1,2,3,4,5,6,7,8,9,10], days = 5
 Output: 15
 Explanation: A ship capacity of 15 is the minimum to ship all the packages in 5 days like this:
 1st day: 1, 2, 3, 4, 5
 2nd day: 6, 7
 3rd day: 8
 4th day: 9
 5th day: 10

Example 2:

Input: weights = [3,2,2,4,1,4], days = 3
 Output: 6
 Explanation: A ship capacity of 6 is the minimum to ship all the packages in 3 days like this:
 1st day: 3, 2
 2nd day: 2, 4
 3rd day: 1, 4

Example 3:

Input: weights = [1,2,3,1,1], days = 4
 Output: 3
 Explanation:
 1st day: 1
 2nd day: 2
 3rd day: 3
 4th day: 1, 1

Constraints:

- $1 \leq \text{days} \leq \text{weights.length} \leq 5 * 10^4$
- $1 \leq \text{weights}[i] \leq 500$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find the minimum ship capacity that lets us ship all packages in at most D days.
 # Packages must be shipped in order – we can't reorder them. Right?
 # Capacity must be at least max(weights) (to handle any single package).
 # Capacity at most sum(weights) (ship everything in one day)."
 #
 # "weights=[1..10], days=5 → 15 ✓ (greedy daily assignment: 1-5, 6-7, 8, 9, 10)"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Try every candidate ship capacity from max(weights) up to sum(weights).

For each capacity, greedily pack days and count days needed.

Correct but linear search on capacity range is $O(\text{sum} * n)$.

Time: $O(\text{sum}(\text{weights}) * n)$. Space: $O(1)$.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Binary search on the answer (capacity).

The feasibility function 'can_ship(capacity)' is monotone: if C works, C+1 also works.

So binary search for the leftmost capacity where can_ship returns True.

can_ship: greedily fill each day until the load would exceed capacity, then start a new day.

Count days needed. Return days $\leq D$.

Time: $O(n \log(\text{sum}(\text{weights})))$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```

class _1011_CapacityShip:
    def shipWithinDays(self, weights: List[int], days: int) -> int:
        def can_ship(capacity: int) -> bool:
            day_load = 0
            days_used = 1
            for w in weights:
                if day_load + w > capacity:
                    days_used += 1
                    day_load = 0
                day_load += w
            return days_used <= days
        lo, hi = max(weights), sum(weights)
        while lo < hi:
            mid = (lo + hi) // 2
            if can_ship(mid):
                hi = mid # valid - try smaller
            else:
                lo = mid + 1 # not valid - need more capacity
        return lo

```

PHASE 5 — TEST & DEBUG

weights=[1..10], days=5. lo=10, hi=55.

mid=32: can_ship(32)→1+2+3+4+5=15≤32(day1), 6+7=13≤32(day2), 8≤32(day3), 9≤32(day4), 10≤32(day5). 5 days ✓ → hi=32.

... binary search converges to 15.

can_ship(15): 1+2+3+4+5=15(day1), 6+7=13+8>15→day2=6+7, 8→day3, 9→day4, 10→day5. 5 days ✓. → 15 ✓

can_ship(14): 1+2+3+4=10, +5=15>14→day2=5+6=11, +7>14→day3=7+8>14→day4=8, 9→day5, 10→day6. 6>5 ✗.

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n \log(\text{sum}))$. Space $O(1)$.

This is 'binary search on the answer' – a powerful pattern where the answer space is

a range of integers and feasibility is monotone.

The feasibility check is a greedy simulation – always load greedily to avoid underusing capacity.

Same pattern appears in: Koko Eating Bananas (#875), Split Array Largest Sum

☐ (#410).

Given more time I'd ask: what if packages can be reordered?

Then we'd try to optimise placement – different problem, potentially NP-hard."

Binary Search

★ #1060 Missing Element in Sorted Array **MED**



[Open on LeetCode](#)

Array · Binary Search

Lists: ★ Premium | Premium 98

Problem

Given an integer array `nums` which is sorted in ascending order and all of its elements are unique and given also an integer `k`, return the `k`th missing number starting from the leftmost number of the array.

Example 1:

```
Input: nums = [4,7,9,10], k = 1
Output: 5
Explanation: The first missing number is 5.
```

Example 2:

```
Input: nums = [4,7,9,10], k = 3
Output: 8
Explanation: The missing numbers are [5,6,8,...], hence the third missing number is 8.
```

Example 3:

```
Input: nums = [1,2,4], k = 3
Output: 6
Explanation: The missing numbers are [3,5,6,7,...], hence the third missing number is 6.
```

Constraints:

- $1 \leq \text{nums.length} \leq 5 * 10^4$
- $1 \leq \text{nums}[i] \leq 10^7$
- **nums** is sorted in ascending order, and all the elements are unique.
- $1 \leq k \leq 10^8$

Follow up: Can you find a logarithmic time complexity (i.e., $O(\log(n))$) solution?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Sorted unique integers. Find the k-th missing integer starting from nums[0].
# Missing count at index i = (nums[i] - nums[0]) - i (how many numbers should exist
# but don't).
# Follow-up asks for  $O(\log n)$ . Right?"
#
# "[4,7,9,10], k=1: missing = [5,6,8,...]. 1st missing = 5 ✓
# [4,7,9,10], k=3: [5,6,8,...]. 3rd = 8 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Walk nums in order maintaining expected next integer starting at 0.
# Increment expected whenever nums[i] == expected; else expected stays.
# When expected reaches k, return k - correct but scans whole array.
# Time:  $O(n)$ . Space:  $O(1)$ .
# Should I go ahead and optimize?"
```

```
class _1060_MissingElement_Brute:
    def missingElement(self, nums: List[int], k: int) -> int:
        for i in range(1, len(nums)):
            gap = nums[i] - nums[i - 1] - 1 # missing numbers between nums[i-1] and
nums[i]
            if k <= gap:
                return nums[i - 1] + k
            k -= gap
        return nums[-1] + k # k-th missing is after the array
```

PHASE 3 — OPTIMIZE ($O(\log n)$)

```
# "Define missing(i) = numbers missing from nums[0] to nums[i] = (nums[i] - nums[0])
- i.
# Binary search for the rightmost index where missing(i) < k.
# Answer = nums[right] + (k - missing(right)).
# If k > missing(last), answer is past the end of the array: nums[-1] + (k -
missing(n-1))."
```

PHASE 4 — CLEAN CODE

```
class _1060_MissingElement:
    def missingElement(self, nums: List[int], k: int) -> int:
        def missing(i: int) -> int:
            return (nums[i] - nums[0]) - i

        n = len(nums)
        if k > missing(n - 1):
            return nums[-1] + (k - missing(n - 1))

        lo, hi = 0, n - 1
        while lo < hi:
            mid = (lo + hi) // 2
            if missing(mid) < k:
                lo = mid + 1 # not enough missing yet - go right
            else:
                hi = mid # too many or just enough - go left
        return nums[lo - 1] + (k - missing(lo - 1))
```

PHASE 5 — TEST & DEBUG

```
# nums=[4,7,9,10], k=1:
# missing: [0]=0, [1]=2, [2]=3, [3]=4.
# k=1<=missing(3)=4. Binary search: lo=0,hi=3,mid=1. missing(1)=2>=1→hi=1.
# lo=0,hi=1,mid=0. missing(0)=0<1→lo=1. lo=hi=1.
# return nums[0]+(1-missing(0))=4+(1-0)=5 ✓
# nums=[4,7,9,10], k=3:
# missing(3)=4>=3. Binary search → lo lands at 2. missing(1)=2<3→lo=2. lo=hi=2.
# return nums[1]+(3-missing(1))=7+(3-2)=8 ✓
# k=5: missing(3)=4<5 → return nums[3]+(5-4)=10+1=11 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "O(log n) with binary search. O(1) space.
# The missing(i) formula is the key: (nums[i]-nums[0])-i counts how many integers
# in [nums[0], nums[i]] are absent from the array.
# The past-the-end check handles k > total missing numbers in the array gracefully.
# Given more time I'd ask: what if there are duplicate values in nums?
# missing(i) would overcount - we'd need to adjust for duplicates."
```

Binary Search

□ #1062 Longest Repeating Substring

MED[Open on LeetCode](#)

String · Binary Search · Dynamic Programming
· Rolling Hash · Suffix Array · Hash Function

Lists: Denny Zhang

Problem

Given a string s , return the length of the longest repeating substrings. If no repeating substring exists, return 0.

Example 1:

Input: $s = \text{"abcd"}$
Output: 0
Explanation: There is no repeating substring.

Example 2:

Input: $s = \text{"abbaba"}$
Output: 2
Explanation: The longest repeating substrings are "ab" and "ba", each of which occurs twice.

Example 3:

Input: $s = \text{"aabcaabdaab"}$
Output: 3
Explanation: The longest repeating substring is "aab", which occurs 3 times.

Constraints:

- $1 \leq s.length \leq 2000$

- s consists of lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find the length of the longest substring that appears at least twice in s.
# The two occurrences can overlap. Return 0 if no repeating substring. Right?"
#
# "'abcd'→0. 'abbaba'→2 ('ab','ba'). 'aabcaabdaab'→3 ('aab')."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# For each length L from n-1 down to 1, collect all substrings of length L in a set.
# First L where a duplicate appears is the answer.
# O(n^2) substrings per L – binary search on L with rolling hash helps.
# Time: O(n^2) to O(n^3) naive. Space: O(n^2) set.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (DP approach — simpler to explain)

```
# "DP: dp[i][j] = length of longest common suffix of s[:i] and s[:j], where i ≠ j.
# dp[i][j] = dp[i-1][j-1] + 1 if s[i-1]==s[j-1] and i≠j, else 0.
# Track the global max. O(n^2) time, O(n^2) space."
```

PHASE 4 — CLEAN CODE

```
class _1062_LongestRepeatingSubstring:
    def longestRepeatingSubstring(self, s: str) -> int:
        n = len(s)
        dp = [[0] * (n + 1) for _ in range(n + 1)]
        best = 0
        for i in range(1, n + 1):
            for j in range(i + 1, n + 1): # j > i ensures different starting
positions
                if s[i - 1] == s[j - 1]:
                    dp[i][j] = dp[i - 1][j - 1] + 1
                    best = max(best, dp[i][j])
        return best
```

PHASE 5 — TEST & DEBUG

```
# s='abbaba':
# i=1,j=2: 'a'=='b'? No. ... i=1,j=4: 'a'=='a'? Yes→dp[1][4]=1, best=1.
# i=2,j=5: 'b'=='b'? Yes→dp[2][5]=dp[1][4]+1=2, best=2.
# → 2 ✓ (substring 'ab' starting at 0 and 3)
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n^2). Space O(n^2). Acceptable for n≤2000 (4M cells).
# Binary search + hashing alternative: O(n log n) time, O(n) space – faster but more
complex."
```

```
# Given more time I'd ask: return the actual substring, not just the length?  
# Track (i,j) when best is updated, then s[i-best:i]."
```

Binary Search

★ #1533 Find the Index of the Large Integer ★

MED[Open on LeetCode](#)

Array · Binary Search · Interactive

Lists: ★ Premium | Premium 98

Problem

We have an integer array `arr`, where all the integers in `arr` are equal except for one integer which is larger than the rest of the integers. You will not be given direct access to the array, instead, you will have an API `ArrayReader` which have the following functions:

- `int compareSub(int l, int r, int x, int y)`: where $0 \leq l, r, x, y < \text{ArrayReader.length}()$, $l \leq r$ and $x \leq y$. The function compares the sum of sub-array `arr[l..r]` with the sum of the sub-array `arr[x..y]` and returns: 1 if $\text{arr}[l] + \text{arr}[l+1] + \dots + \text{arr}[r] > \text{arr}[x] + \text{arr}[x+1] + \dots + \text{arr}[y]$.
- 0 if $\text{arr}[l] + \text{arr}[l+1] + \dots + \text{arr}[r] == \text{arr}[x] + \text{arr}[x+1] + \dots + \text{arr}[y]$.

- -1 if $\text{arr}[l] + \text{arr}[l+1] + \dots + \text{arr}[r] < \text{arr}[x] + \text{arr}[x+1] + \dots + \text{arr}[y]$.

You are allowed to call `compareSub()` 20 times at most. You can assume both functions work in $O(1)$ time.

Return the index of the array `arr` which has the largest integer.

Example 1:

Input: `arr = [7,7,7,7,10,7,7,7]`

Output: 4

Explanation: The following calls to the API

`reader.compareSub(0, 0, 1, 1)` // returns 0 this is a query comparing the sub-array (0, 0) with the sub array (1, 1), (i.e. compares `arr[0]` with `arr[1]`). Thus we know that `arr[0]` and `arr[1]` doesn't contain the largest element.

`reader.compareSub(2, 2, 3, 3)` // returns 0, we can exclude `arr[2]` and `arr[3]`.

`reader.compareSub(4, 4, 5, 5)` // returns 1, thus for sure `arr[4]` is the largest element in the array.

Notice that we made only 3 calls, so the answer is valid.

Example 2:

Input: `nums = [6,6,12]`

Output: 2

Constraints:

- $2 \leq \text{arr.length} \leq 5 * 10^5$
- $1 \leq \text{arr}[i] \leq 100$
- All elements of `arr` are equal except for one element which is larger than all other elements.

Follow up:

- What if there are two numbers in arr that are bigger than all other numbers?
- What if there is one number that is bigger than other numbers and one number that is smaller than other numbers?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "One element is strictly larger than all others. Find its index using the
compareSub API.
# compareSub(l,r,x,y) compares sum(arr[l..r]) with sum(arr[x..y]): returns 1,-1,0.
# At most 20 calls – so O(log n) calls are required. Right?"
#
# "[7,7,7,7,10,7,7,7]: 3 compareSub calls suffice to find index 4 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Linear scan: compare nums[i] with nums[i+1]; first unequal pair reveals larger
half.
# Correct O(n) but problem expects O(log n) via binary search on the doubled array.
# Time: O(n). Space: O(1).
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Binary search: at each step, compare the left half against the right half of the
range.
# If left > right: large integer is in the left half → search left.
# If right > left: search right.
# If equal: the large integer is the MIDDLE element (if odd-length range, middle was
excluded).
# Halve the range each step → O(log n) calls."
```

PHASE 4 — CLEAN CODE

```
class _1533_FindLargeInteger:
    def getIndex(self, reader) -> int:
        lo, hi = 0, reader.length() - 1
        while lo < hi:
            mid = (lo + hi) // 2
            # Compare left half [lo..mid] with right half of equal size
            left_end = mid
            right_start = mid + 1
            right_end = right_start + (mid - lo)
```

```

        if right_end > hi:
            right_end = hi
        result = reader.compareSub(lo, left_end, right_start, right_end)
        if result == 1:
            hi = left_end # large int is in left half
        elif result == -1:
            lo = right_start # large int is in right half
        else:
            return mid + 1 # equal sums → large int is the middle skipped
    element
    return lo

```

PHASE 5 — TEST & DEBUG

```

# arr=[7,7,7,7,10,7,7,7]: lo=0,hi=7.
# mid=3. Compare [0..3] vs [4..7]. Both have 4 elements. Sum=[28] vs [31]. -1 →
lo=4.
# lo=4,hi=7,mid=5. Compare [4..5] vs [6..7]. [10+7=17] vs [7+7=14]. 1 → hi=5.
# lo=4,hi=5,mid=4. Compare [4..4] vs [5..5]. [10] vs [7]. 1 → hi=4. lo=hi=4 → return
4 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "O(log n) compareSub calls. Well within the 20-call limit for n=5*10^5
(log2(500000)≈19).
# The equal-sum case (odd-length range) means the middle element was the large one –
# both halves have equal elements so the excluded middle is the outlier.
# Follow-up: two large numbers? One comparison no longer definitively points to one
half.
# We'd need to track which half has MORE excess – more complex."

```

Matrix

Round 2 · High · Medium · 14 questions

Matrix

□ #36 Valid Sudoku

MED[Open on LeetCode](#)

Array · Hash Table · Matrix

Lists: Grind 169

Problem

Determine if a 9 x 9 Sudoku board is valid. Only the filled cells need to be validated according to the following rules:

1. Each row must contain the digits 1–9 without repetition.
2. Each column must contain the digits 1–9 without repetition.
3. Each of the nine 3 x 3 sub-boxes of the grid must contain the digits 1–9 without repetition.

Note:

- A Sudoku board (partially filled) could be valid but is not necessarily solvable.
- Only the filled cells need to be validated according to the mentioned rules.

Example 1:

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

Input: board =

```
[["5","3",".",".","7",".",".",".","."],
["6",".",".","1","9","5",".",".","."],
[".","9","8",".",".",".","6","."],
["8",".",".","6",".",".","3"],
["4",".","8","3",".","1"],
["7",".","2",".","6"],
[".","6",".","2","8","."]]
```

Example 2:

```

Input: board =
[["8","3",".",".","7",".",".",".","."],
["6",".",".","1","9","5",".",".","."],
[[".","9","8",".",".",".","6","."],
["8",".",".","6",".",".","3"],
["4",".","8","3",".","1"],
["7",".","2",".","6"],
[["6",".",".","2","8","."]]

```

Constraints:

- `board.length == 9`
- `board[i].length == 9`
- `board[i][j]` is a digit 1–9 or '.'.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Check if a partially filled 9x9 sudoku board is valid – no repeats in any row,
# column, or 3x3 box. Ignore empty cells ('.'). Right?
# A valid board doesn't have to be solvable – just no current conflicts."
#
# "The two examples differ only in one cell: '8' appears twice in the top-left 3x3
# box → false ✓"

```

PHASE 2 — BRUTE FORCE

```

# "Let me start with brute force to lock in the logic.
# For each of 9 rows, 9 columns, and 9 boxes, collect digits in a set and check
# size==9.
# 27 independent validity checks – straightforward and easy to explain.
# Time: O(1) – fixed 81 cells. Space: O(1) – 9 sets of size 9.
# Should I go ahead and optimize?"

```

PHASE 3 — OPTIMIZE

```

# "Single pass: use three 2D boolean arrays – rows[r][d], cols[c][d],
# boxes[r//3][c//3][d].
# For each filled cell, check all three. If any is already marked, return False.
# Otherwise mark all three. O(81) = O(1) for fixed 9x9."

```

PHASE 4 — CLEAN CODE

```

class _36_ValidSudoku:
    def isValidSudoku(self, board: List[List[str]]) -> bool:
        rows = [[False] * 9 for _ in range(9)]
        cols = [[False] * 9 for _ in range(9)]

```

```

boxes = [[False] * 9 for _ in range(9)] # indexed by box_id = (r//3)*3 +
c//3
for r in range(9):
    for c in range(9):
        if board[r][c] == '.':
            continue
        d = int(board[r][c]) - 1
        box_id = (r // 3) * 3 + (c // 3)
        if rows[r][d] or cols[c][d] or boxes[box_id][d]:
            return False
        rows[r][d] = cols[c][d] = boxes[box_id][d] = True
return True

```

PHASE 5 — TEST & DEBUG

```

# Cell (0,0)='5': d=4. box_id=0. Mark rows[0][4]=cols[0][4]=boxes[0][4]=True.
# Cell (3,0)='8': d=7. box_id=0. rows[3][7]? No. boxes[0][7]? No. Mark.
# In example 2: cell (0,0)='8' and cell (3,0)='8': both have box_id=0, d=7.
# Second '8' at (3,0): boxes[0][7] already True → return False ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time 0(81)=0(1). Space 0(81)=0(1) – fixed 9x9 board.
# The box_id formula (r//3)*3+(c//3) maps each of the 9 boxes to 0–8. Know this
cold.
# Alternative: use a set of tuples like (r,'row',d), (c,'col',d), (box_id,'box',d) –
one set.
# Given more time I'd ask: implement a Sudoku solver?
# That's backtracking – for each empty cell, try 1–9, recurse, backtrack on
failure."

```


Matrix

□ #48 Rotate Image

MED[Open on LeetCode](#)

Array · Math · Matrix

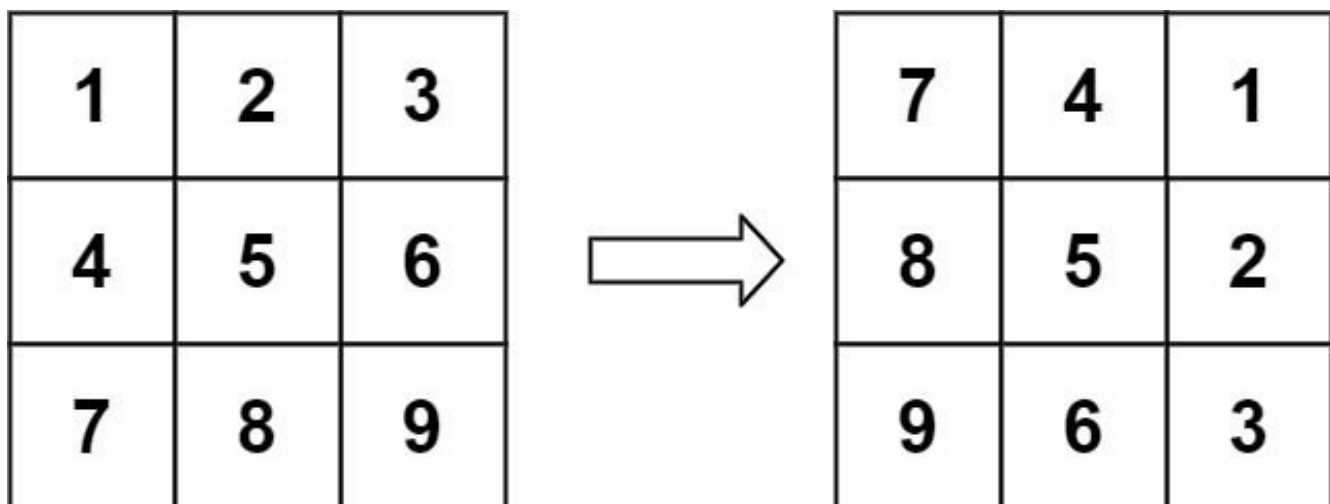
Lists: Grind 169 | CodeSignal

Problem

You are given an $n \times n$ 2D matrix representing an image, rotate the image by 90 degrees (clockwise).

You have to rotate the image in-place, which means you have to modify the input 2D matrix directly. DO NOT allocate another 2D matrix and do the rotation.

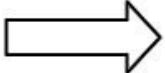
Example 1:



Input: matrix = [[1,2,3],[4,5,6],[7,8,9]]
 Output: [[7,4,1],[8,5,2],[9,6,3]]

Example 2:

5	1	9	11
2	4	8	10
13	3	6	7
15	14	12	16



15	13	2	5
14	3	4	1
12	6	8	9
16	7	10	11

Input: matrix = [[5,1,9,11],[2,4,8,10],[13,3,6,7],[15,14,12,16]]
 Output: [[15,13,2,5],[14,3,4,1],[12,6,8,9],[16,7,10,11]]

Constraints:

- $n == \text{matrix.length} == \text{matrix}[i].\text{length}$
- $1 \leq n \leq 20$
- $-1000 \leq \text{matrix}[i][j] \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Rotate an $n \times n$ matrix 90 degrees clockwise in-place. No extra matrix allowed. Right?"

After rotation: element at (r,c) moves to $(c, n-1-r)$."

#

"[[1,2,3],[4,5,6],[7,8,9]] → [[7,4,1],[8,5,2],[9,6,3]] ✓"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

Allocate a new $n \times n$ matrix; for each (r,c) set $\text{new}[c][n-1-r] = \text{matrix}[r][c]$.

Classic 90-degree rotation formula – clear and correct.

```

# Uses  $O(n^2)$  extra memory; we can rotate in-place layer by layer.
# Time:  $O(n^2)$ . Space:  $O(n^2)$  new matrix.
# Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE (two-step in-place)
# "Two operations, each reversible and composable:
# Step 1: Transpose (swap matrix[r][c] with matrix[c][r]).
# Step 2: Reverse each row.
# Together these produce a  $90^\circ$  clockwise rotation.  $O(n^2)$  time,  $O(1)$  space.
# Verify: (r,c) → transpose → (c,r) → reverse row → (c, n-1-r). Correct!"

# PHASE 4 — CLEAN CODE
class _48_RotateImage:
    def rotate(self, matrix: List[List[int]]) -> None:
        n = len(matrix)
        # Step 1: transpose
        for r in range(n):
            for c in range(r + 1, n):
                matrix[r][c], matrix[c][r] = matrix[c][r], matrix[r][c]
        # Step 2: reverse each row
        for row in matrix:
            row.reverse()

# PHASE 5 — TEST & DEBUG
# [[1,2,3],[4,5,6],[7,8,9]]:
# Transpose: [[1,4,7],[2,5,8],[3,6,9]].
# Reverse rows: [[7,4,1],[8,5,2],[9,6,3]] ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(n^2)$ . Space  $O(1)$  in-place.
# Counter-clockwise  $90^\circ$ : reverse each row first, then transpose.
#  $180^\circ$ : transpose then transpose (just reverse all rows, then reverse their order).
# The two-step approach is elegant – each step is simple and reversible.
# Given more time I'd ask: rotate a non-square matrix?
# No longer possible in-place with the same dimensions – need a new  $m \times n$  matrix."

```

Matrix

□ #54 Spiral Matrix

MED

[Open on LeetCode](#)

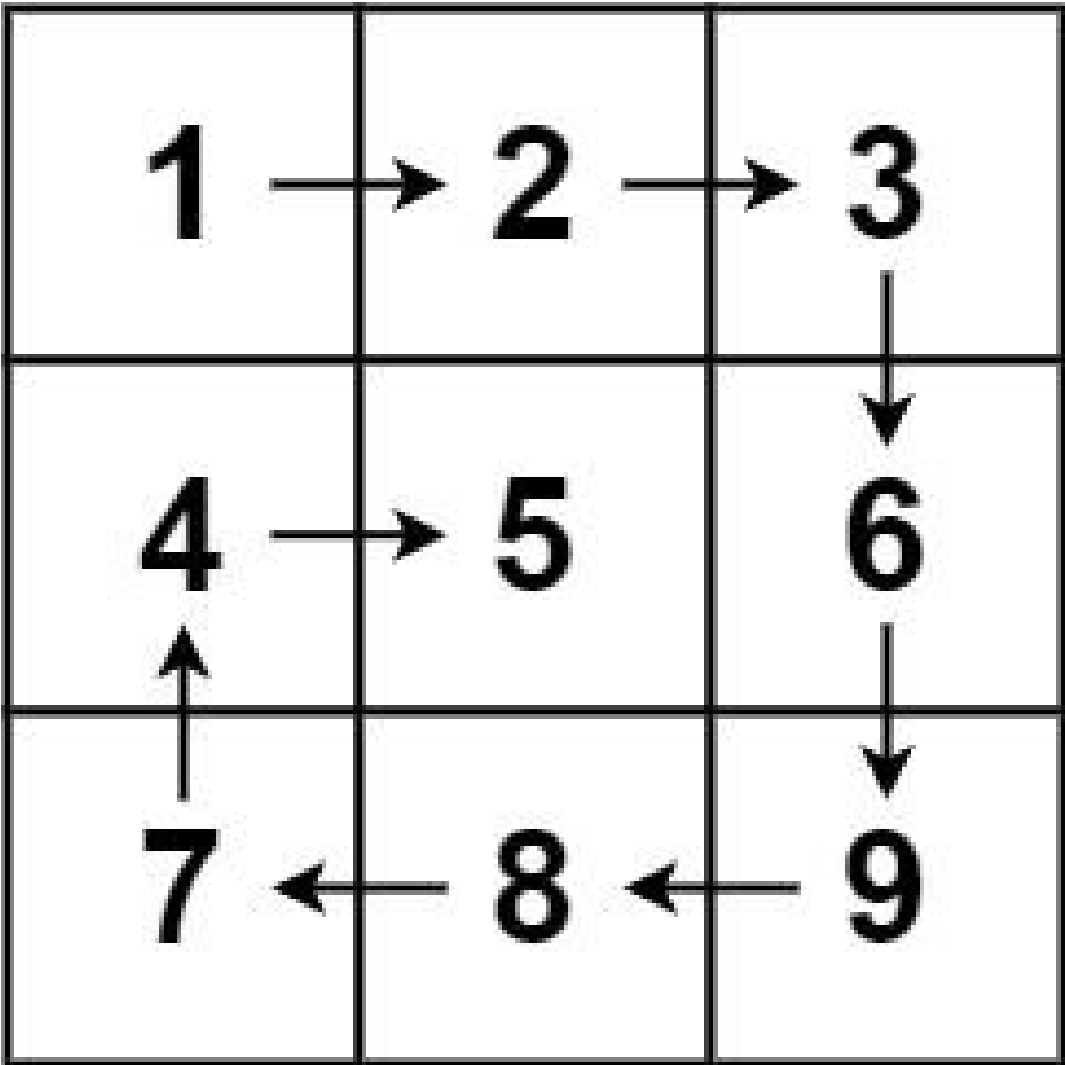
Array · Matrix · Simulation

Lists: Grind 169 | CodeSignal

Problem

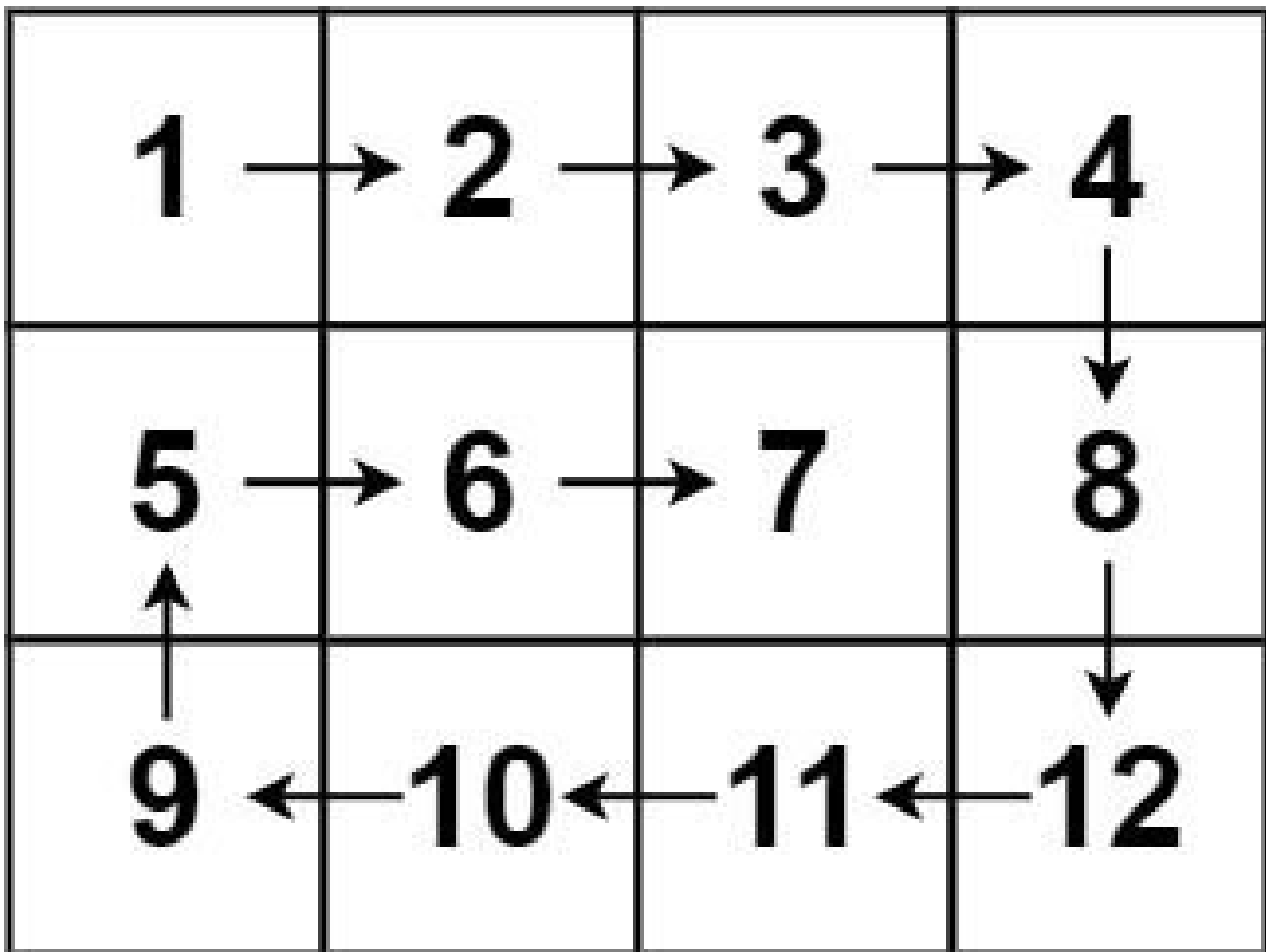
Given an $m \times n$ matrix, return all elements of the matrix in spiral order.

Example 1:



Input: matrix = [[1,2,3],[4,5,6],[7,8,9]]
Output: [1,2,3,6,9,8,7,4,5]

Example 2:



Input: matrix = [[1,2,3,4],[5,6,7,8],[9,10,11,12]]

Output: [1,2,3,4,8,12,11,10,9,5,6,7]

Constraints:

- $m == \text{matrix.length}$
- $n == \text{matrix}[i].\text{length}$
- $1 \leq m, n \leq 10$
- $-100 \leq \text{matrix}[i][j] \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Return all matrix elements in spiral order: right along top, down along right
side,
# left along bottom, up along left side – then repeat for the inner rectangle.
Right?
# Works for non-square matrices."
#
# "[[1,2,3],[4,5,6],[7,8,9]] → [1,2,3,6,9,8,7,4,5] ✓"

# PHASE 2 — BRUTE FORCE
# "Let me start with brute force to lock in the logic.
# Maintain top/bottom/left/right bounds; walk right, down, left, up while unvisited.
# Mark visited cells in a boolean matrix to know when to turn.
# Works, but the visited matrix costs O(m*n) extra space.
# Time: O(m * n). Space: O(m * n) visited.
# Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE (layer boundaries)
# "Track four boundaries: top, bottom, left, right.
# Traverse right along top (then top++), down along right (then right--),
# left along bottom (then bottom--), up along left (then left++).
# Stop when top>bottom or left>right. O(m*n) time, O(1) extra space."

# PHASE 4 — CLEAN CODE
class _54_SpiralMatrix:
    def spiralOrder(self, matrix: List[List[int]]) -> List[int]:
        top, bottom = 0, len(matrix) - 1
        left, right = 0, len(matrix[0]) - 1
        result = []
        while top <= bottom and left <= right:
            for c in range(left, right + 1): result.append(matrix[top][c])
            top += 1
            for r in range(top, bottom + 1): result.append(matrix[r][right])
            right -= 1
            if top <= bottom:
                for c in range(right, left - 1, -1):
                    result.append(matrix[bottom][c])
                    bottom -= 1
            if left <= right:
                for r in range(bottom, top - 1, -1): result.append(matrix[r][left])
                left += 1
        return result

# PHASE 5 — TEST & DEBUG
# [[1,2,3],[4,5,6],[7,8,9]]: top=0,bot=2,l=0,r=2.
# Right: 1,2,3. top=1.
# Down: 6,9. right=1.
# Left (top<=bot): 8,7. bot=1.
# Up (left<=right): 4. left=1.
# top=1,bot=1,l=1,r=1.
# Right: 5. top=2. top>bot → stop.

```

→ [1,2,3,6,9,8,7,4,5] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m*n)$. Space $O(1)$ extra.

The boundary guards (if top<=bottom / if left<=right) prevent double-counting

the single middle row or column of non-square matrices.

Given more time I'd ask: Spiral Matrix II (#59) – fill instead of read.

Same boundary logic, write numbers $1..n^2$ instead of appending."

Matrix

#59 Spiral Matrix II

MED[Open on LeetCode](#)

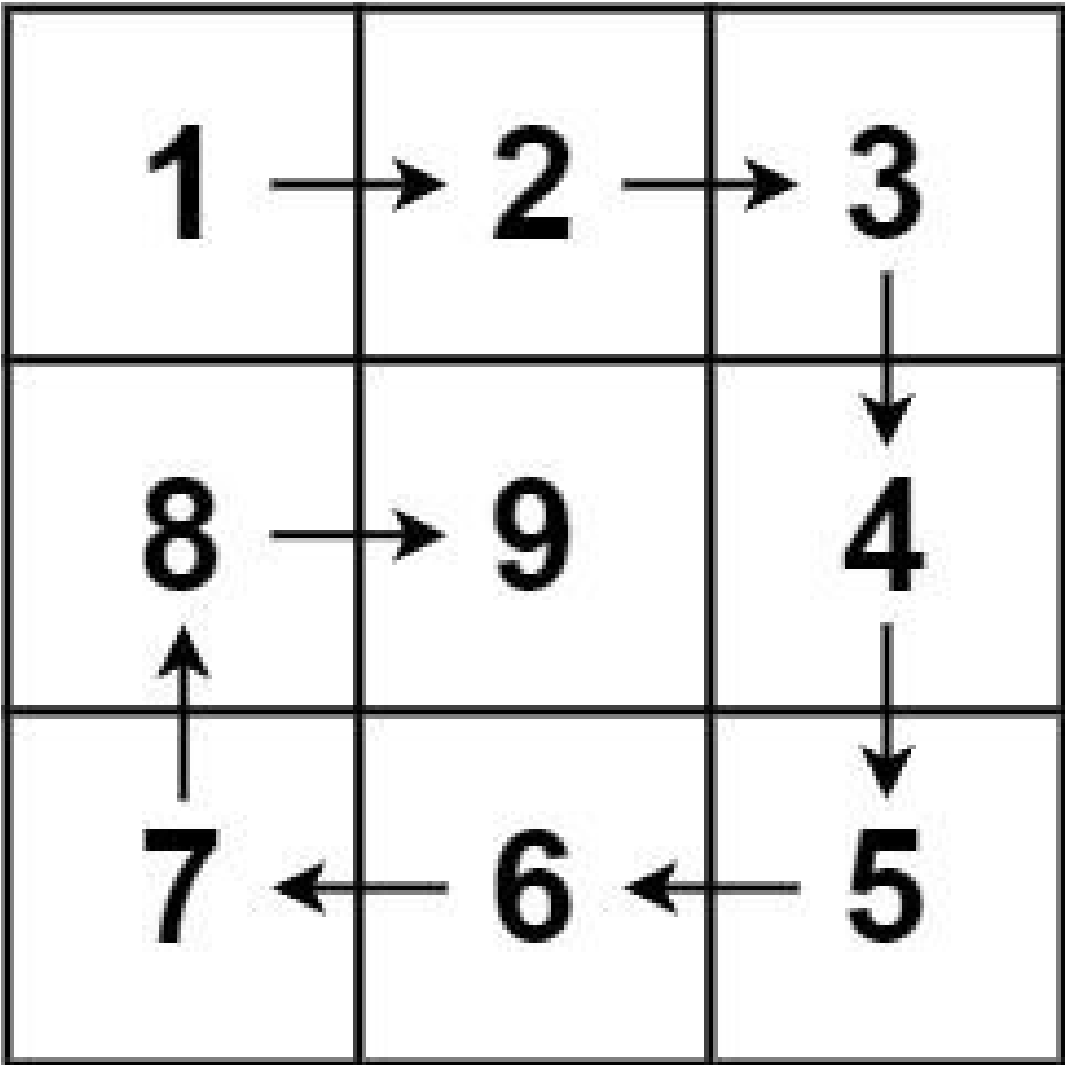
Array · Matrix · Simulation

Lists: CodeSignal

Problem

Given a positive integer n , generate an $n \times n$ matrix filled with elements from 1 to n^2 in spiral order.

Example 1:



Input: n = 3
Output: [[1,2,3],[8,9,4],[7,6,5]]

Example 2:

Input: n = 1
Output: [[1]]

Constraints:

- 1 <= n <= 20

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Generate an n×n matrix filled with 1 to n2 in spiral order. Return the matrix.
Right?"
#
# "n=3 → [[1,2,3],[8,9,4],[7,6,5]] ✓"

# PHASE 2 — BRUTE FORCE
# "Let me start with brute force to lock in the logic.
# Fill numbers 1..n2 using direction arrays (right, down, left, up) and a visited
grid.
# Turn when hitting boundary or a filled cell.
# Straightforward simulation with O(n2) extra visited space.
# Time: O(n2). Space: O(n2) visited.
# Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE
# "Identical boundary approach to #54, but write consecutive integers instead of
reading.
# Initialize n×n matrix of zeros, fill with num=1..n2. O(n2) time, O(1) extra."

# PHASE 4 — CLEAN CODE
class _59_SpiralMatrixII:
    def generateMatrix(self, n: int) -> List[List[int]]:
        matrix = [[0] * n for _ in range(n)]
        top, bottom, left, right = 0, n - 1, 0, n - 1
        num = 1
        while top <= bottom and left <= right:
            for c in range(left, right + 1): matrix[top][c] = num; num += 1
            top += 1
            for r in range(top, bottom + 1): matrix[r][right] = num; num += 1
            right -= 1
            if top <= bottom:
                for c in range(right, left - 1, -1): matrix[bottom][c] = num; num +=
1
                bottom -= 1
            if left <= right:
                for r in range(bottom, top - 1, -1): matrix[r][left] = num; num += 1
                left += 1
        return matrix

# PHASE 5 — TEST & DEBUG
# n=3: fills 1→2→3 (top row), 4 (right col), 5→6 (bottom), 7 (left col), 9 (center).
# → [[1,2,3],[8,9,4],[7,6,5]] ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(n2). Space O(1) extra – output matrix doesn't count.
# This is the write-version of Spiral Matrix (#54) – exact same boundary logic.
# Given more time I'd ask: generate a spiral starting from the center?
# Reverse the boundary logic – start from the middle, expand outward."

```

Matrix

□ #64 Minimum Path Sum

MED[Open on LeetCode](#)

Array · Dynamic Programming · Matrix

Lists: Denny Zhang

Problem

Given a $m \times n$ grid filled with non-negative numbers, find a path from top left to bottom right, which minimizes the sum of all numbers along its path.

Note: You can only move either down or right at any point in time.

Example 1:

1	3	1
1	5	1
4	2	1

Input: grid = [[1,3,1],[1,5,1],[4,2,1]]

Output: 7

Explanation: Because the path 1 → 3 → 1 → 1 → 1 minimizes the sum.

Example 2:

Input: grid = [[1,2,3],[4,5,6]]

Output: 12

Constraints:

- $m == \text{grid.length}$
- $n == \text{grid}[i].\text{length}$
- $1 \leq m, n \leq 200$

- $0 \leq \text{grid}[i][j] \leq 200$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Move only right or down. Find the path from top-left to bottom-right with minimum
sum. Right?
# Can only move right or down – no backtracking."
#
# "[[1,3,1],[1,5,1],[4,2,1]]: path 1→3→1→1→1=7 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# DFS from (0,0): at each cell try moving right and down, take min of two paths.
# Memoize (r,c) to avoid recomputing subpaths – classic top-down DP.
# Without memo this blows up exponentially; memo makes it O(m*n).
# Time: O(m * n) with memo. Space: O(m * n) memo + stack.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (in-place DP)

```
# "dp[i][j] = min path sum to reach (i,j).
# dp[i][j] = grid[i][j] + min(dp[i-1][j], dp[i][j-1]).
# First row: only right. First col: only down.
# Modify grid in-place – no extra space. O(m*n) time, O(1) extra."
```

PHASE 4 — CLEAN CODE

```
class _64_MinPathSum:
    def minPathSum(self, grid: List[List[int]]) -> int:
        m, n = len(grid), len(grid[0])
        for r in range(m):
            for c in range(n):
                if r == 0 and c == 0:
                    continue
                elif r == 0:
                    grid[r][c] += grid[r][c - 1] # can only come from left
                elif c == 0:
                    grid[r][c] += grid[r - 1][c] # can only come from above
                else:
                    grid[r][c] += min(grid[r - 1][c], grid[r][c - 1])
        return grid[m - 1][n - 1]
```

PHASE 5 — TEST & DEBUG

```
# [[1,3,1],[1,5,1],[4,2,1]]:
# Row 0: [1,4,5]. Col 0: [1,2,6]. (1,1)=5+min(4,2)=7. (1,2)=1+min(7,5)=6.
# (2,1)=2+min(7,6)=8. (2,2)=1+min(6,8)=7. → 7 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m \times n)$ . Space  $O(1)$  – modifies grid in place.  
# If we can't modify the input, use a 1D rolling array:  $O(n)$  extra space.  
# Given more time I'd ask: what if we can also move up or left?  
# That breaks the DP recurrence – shortest path with general movement is  
Dijkstra's."
```

Matrix

□ #73 Set Matrix Zeroes

MED[Open on LeetCode](#)

Array · Hash Table · Matrix

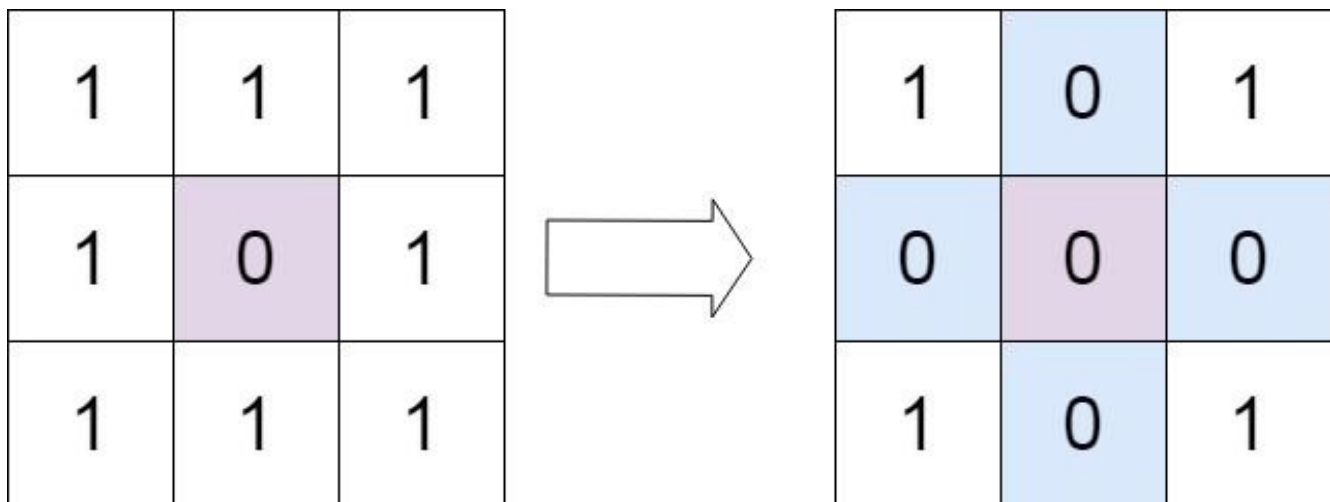
Lists: Grind 169

Problem

Given an $m \times n$ integer matrix `matrix`, if an element is 0, set its entire row and column to 0's.

You must do it in place.

Example 1:

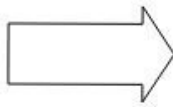


Input: `matrix = [[1,1,1],[1,0,1],[1,1,1]]`

Output: `[[1,0,1],[0,0,0],[1,0,1]]`

Example 2:

0	1	2	0
3	4	5	2
1	3	1	5



0	0	0	0
0	4	5	0
0	3	1	0

Input: matrix = `[[0,1,2,0],[3,4,5,2],[1,3,1,5]]`
 Output: `[[0,0,0,0],[0,4,5,0],[0,3,1,0]]`

Constraints:

- `m == matrix.length`
- `n == matrix[0].length`
- `1 <= m, n <= 200`
- `-231 <= matrix[i][j] <= 231 - 1`

Follow up:

- A straightforward solution using $O(mn)$ space is probably a bad idea.
- A simple improvement uses $O(m + n)$ space, but still not the best solution.
- Could you devise a constant space solution?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"If a cell is 0, zero out its entire row and column. Do it in-place. Right?"

We must use the ORIGINAL zeros – not zeros we set during the process."

#

"[[1,0,1],[1,1,1],[1,1,1]] → [[0,0,0],[1,0,1],[1,0,1]] ✓"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

First collect every (r,c) where matrix[r][c]==0 into a list.

Second pass: zero out entire rows and columns for each stored zero.

Simple, but storing all zero coordinates uses $O(m+n)$ extra structures.

Time: $O(m * n)$. Space: $O(m + n)$ for row/col sets.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE ($O(1)$ space using first row/col as markers)

"Use the first row and first column of the matrix itself as markers."

If matrix[r][c]==0: mark matrix[r][0]=0 and matrix[0][c]=0.

Handle first row and first column separately (track with boolean flags).

Then zero out interior cells using the markers.

Then zero out first row/col based on flags. $O(m*n)$ time, $O(1)$ space."

PHASE 4 — CLEAN CODE

class _73_SetMatrixZeroes:

def setZeroes(self, matrix: List[List[int]]) -> None:

m, n = len(matrix), len(matrix[0])

first_row_zero = any(matrix[0][c] == 0 for c in range(n))

first_col_zero = any(matrix[r][0] == 0 for r in range(m))

for r in range(1, m): # mark zeros using first row/col

for c in range(1, n):

if matrix[r][c] == 0:

matrix[r][0] = 0

matrix[0][c] = 0

for r in range(1, m): # zero out interior

for c in range(1, n):

if matrix[r][0] == 0 or matrix[0][c] == 0:

matrix[r][c] = 0

if first_row_zero: # zero out first row if needed

for c in range(n): matrix[0][c] = 0

if first_col_zero: # zero out first col if needed

for r in range(m): matrix[r][0] = 0

PHASE 5 — TEST & DEBUG

[[0,1,2,0],[3,4,5,2],[1,3,1,5]]:

first_row_zero: matrix[0][0]=0 → True. first_col_zero: matrix[0][0]=0 → True.

Mark: r=1,c=0: already 3≠0. r=1,c=3: 2≠0... only (0,0) and (0,3) have zeros in row 0.

Interior zeros: marker[0][0]=0→col 0 zeros. marker[0][3]: 0→col 3 zeros.

→ [[0,0,0,0],[0,4,5,0],[0,3,1,0]] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m*n)$. Space $O(1)$."

The first row/col flags are the key – without them, using matrix[0][0] as a marker

for both first row and first col creates an ambiguity. Two flags resolve it cleanly.

```
# The four-step order (flag, mark, zero interior, zero edges) is the correct
sequence.
# Given more time I'd ask: what if we should set a cell to the average of its
row/col instead?
# Same structure – two-pass approach to avoid using updated values."
```

Matrix

□ #74 Search a 2D Matrix

MED[Open on LeetCode](#)

Array · Binary Search · Matrix

Lists: Grind 169

Problem

You are given an $m \times n$ integer matrix matrix with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer target, return true if target is in matrix or false otherwise.

You must write a solution in $O(\log(m * n))$ time complexity.

Example 1:

1	3	5	7
10	11	16	20
23	30	34	60

Input: matrix = `[[1,3,5,7],[10,11,16,20],[23,30,34,60]]`, target = 3
Output: true

Example 2:

1	3	5	7
10	11	16	20
23	30	34	60

Input: matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]], target = 13

Output: false

Constraints:

- $m == \text{matrix.length}$
- $n == \text{matrix}[i].\text{length}$
- $1 \leq m, n \leq 100$
- $-104 \leq \text{matrix}[i][j], \text{target} \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Matrix where each row is sorted and the first element of each row is greater than

```

# the last of the previous row – so the whole matrix is one sorted sequence.
# Find target in  $O(\log(m \cdot n))$ . Right?"
#
# "Treat it as a 1D array of length  $m \cdot n$ . Index  $k$  maps to  $\text{row} = k // n$ ,  $\text{col} = k \% n$ ."

# PHASE 2 — BRUTE FORCE
# "Let me start with brute force to lock in the logic.
# Treat the matrix as a flattened sorted array and binary search on index  $0..m \cdot n - 1$ .
# Or scan row by row with binary search per row.
# Correct  $O(m \log n)$ , but a single binary search on the virtual array is cleaner.
# Time:  $O(m \log n)$ . Space:  $O(1)$ .
# Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE
# "Binary search on the virtual 1D array  $[0, m \cdot n - 1]$ .
# Map  $\text{mid}$  to  $(\text{mid} // n, \text{mid} \% n)$  to access the actual cell.
#  $O(\log(m \cdot n)) = O(\log m + \log n)$ ."

# PHASE 4 — CLEAN CODE
class _74_Search2DMatrix:
    def searchMatrix(self, matrix: List[List[int]], target: int) -> bool:
        m, n = len(matrix), len(matrix[0])
        left, right = 0, m * n - 1
        while left <= right:
            mid = (left + right) // 2
            mid_val = matrix[mid // n][mid % n]
            if mid_val == target:
                return True
            elif mid_val < target:
                left = mid + 1
            else:
                right = mid - 1
        return False

# PHASE 5 — TEST & DEBUG
# matrix=[[1,3,5,7],[10,11,16,20],[23,30,34,60]], target=3:
# l=0,r=11. mid=5→matrix[1][1]=11. 11>3→r=4.
# mid=2→matrix[0][2]=5. 5>3→r=1.
# mid=0→matrix[0][0]=1. 1<3→l=1.
# mid=1→matrix[0][1]=3. 3==3→True ✓
# target=13: binary search narrows until l>r → False ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(\log(m \cdot n))$ . Space  $O(1)$ .
# The index mapping  $\text{mid} // n$  and  $\text{mid} \% n$  is the entire trick.
# This only works because the matrix is fully sorted as one sequence.
# If rows are sorted but first-of-next isn't > last-of-prev (Search a 2D Matrix II #240):
# Use staircase search – start at top-right, move left if too big, down if too small.

```

$O(m+n)$ time – different problem, different approach."

Matrix

□ #221 Maximal Square

MED[Open on LeetCode](#)

Array · Dynamic Programming · Matrix

Lists: Grind 169

Problem

Given an $m \times n$ binary matrix filled with 0's and 1's, find the largest square containing only 1's and return its area.

Example 1:

1	0	1	0	0
1	0	1	1	1
1	1	1	1	1
1	0	0	1	0

Input: matrix = `[["1","0","1","0","0"],["1","0","1","1","1"],["1","1","1","1","1"],["1","0","0","1","0"]]`
Output: 4

Example 2:

0	1
1	0

Input: matrix = `[["0","1"],["1","0"]]`
Output: 1

Example 3:

Input: matrix = `[["0"]]`
Output: 0

Constraints:

- `m == matrix.length`
- `n == matrix[i].length`
- `1 <= m, n <= 300`
- `matrix[i][j]` is '0' or '1'.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find the largest square of all 1s. Return its area. Matrix contains '0'/'1' strings. Right?"

#

"[[1,0,1,0,0],[1,0,1,1,1],[1,1,1,1,1],[1,0,0,1,0]] → 2×2=4 ✓"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

For every cell containing '1', expand square side length while all cells in square are '1'.

Track the maximum side length found across all starting cells.

Each expansion check is $O(k)$ for side k – overall $O(m*n*\min(m,n)^2)$.

Time: $O(m * n * \min(m,n)^2)$. Space: $O(1)$.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (DP)

"dp[i][j] = side length of the largest all-1s square with bottom-right corner at (i,j)."

If matrix[i][j]=='0': dp[i][j]=0.

Else: dp[i][j] = min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1]) + 1.

Intuition: a square of side k can be extended by 1 only if the top, left, and diagonal neighbour

each have a square of side k . The bottleneck is the minimum.

Track max side length → return side²."

PHASE 4 — CLEAN CODE

class _221_MaximalSquare:

def maximalSquare(self, matrix: List[List[str]]) -> int:

m, n = len(matrix), len(matrix[0])

dp = [[0] * n for _ in range(m)]

max_side = 0

for r in range(m):

for c in range(n):

if matrix[r][c] == '1':

top = dp[r - 1][c] if r > 0 else 0

left = dp[r][c - 1] if c > 0 else 0

diag = dp[r - 1][c - 1] if r > 0 and c > 0 else 0

dp[r][c] = min(top, left, diag) + 1

max_side = max(max_side, dp[r][c])

return max_side * max_side

PHASE 5 — TEST & DEBUG

"[[0,1],[1,1],[0]]: no 2×2 all-1s → dp stays at 1s and 0s → max=1 → area=1 ✓"

Full example: bottom-right region gives dp[2][3]=dp[2][4]=2, max=2, area=4 ✓"

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m*n)$. Space $O(m*n)$ – reducible to $O(n)$ with a rolling array.

The min of three neighbours is the core insight – think of it as the square being

limited by the weakest neighbouring corner.

Given more time I'd ask: maximal rectangle of 1s (not just square)?

That's Maximal Rectangle (#85) – use a histogram approach per row, $O(m*n)$."

Matrix

★ #311 Sparse Matrix Multiplication ★

MED
[Open on LeetCode](#)

Array · Hash Table · Matrix

Lists: ★ Premium | Premium 98

Problem

Given two sparse matrices `mat1` of size $m \times k$ and `mat2` of size $k \times n$, return the result of `mat1` $\times$ `mat2`. You may assume that multiplication is always possible.

Example 1:

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 0 & 3 \end{bmatrix} \times \begin{bmatrix} 7 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 & 0 \\ -7 & 0 & 3 \end{bmatrix}$$

Input: `mat1 = [[1,0,0],[-1,0,3]]`, `mat2 = [[7,0,0],[0,0,0],[0,0,1]]`
 Output: `[[7,0,0],[-7,0,3]]`

Example 2:

Input: `mat1 = [[0]]`, `mat2 = [[0]]`
 Output: `[[0]]`

Constraints:

- `m == mat1.length`
- `k == mat1[i].length == mat2.length`
- `n == mat2[i].length`
- `1 <= m, n, k <= 100`
- `-100 <= mat1[i][j], mat2[i][j] <= 100`

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Multiply mat1 (m×k) by mat2 (k×n). Return the m×n result.
# Both are sparse – many zeros. Optimise to skip zero elements. Right?"
#
# "[[1,0,0],[-1,0,3]] × [[7,0,0],[0,0,0],[0,0,1]] = [[7,0,0],[-7,0,3]] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Standard triple loop: for each row i and col k, sum A[i][j]*B[j][k] over j.
# Correct even for sparse inputs but touches every (i,j,k) combination.
# Skipping zero rows/cols from sparse representation saves huge work.
# Time: O(m * k * n). Space: O(m * n) result.
# Should I go ahead and optimize?"
```

```
class _311_SparseMultiply_Brute:
    def multiply(self, mat1: List[List[int]], mat2: List[List[int]]) ->
List[List[int]]:
        m, k, n = len(mat1), len(mat1[0]), len(mat2[0])
        result = [[0] * n for _ in range(m)]
        for i in range(m):
            for p in range(k):
                for j in range(n):
                    result[i][j] += mat1[i][p] * mat2[p][j]
        return result
```

PHASE 3 — OPTIMIZE

```
# "Skip multiplication when mat1[i][p] == 0 – avoids the inner j loop entirely.
# Pre-compute non-zero positions in mat1 for even faster access.
# This is the key sparse optimisation: skip zero elements before accessing mat2."
```

PHASE 4 — CLEAN CODE

```
class _311_SparseMultiply:
    def multiply(self, mat1: List[List[int]], mat2: List[List[int]]) ->
List[List[int]]:
        m, k, n = len(mat1), len(mat1[0]), len(mat2[0])
```

```

result = [[0] * n for _ in range(m)]
for i in range(m):
    for p in range(k):
        if mat1[i][p] == 0:
            continue # skip - avoids entire inner loop
        for j in range(n):
            result[i][j] += mat1[i][p] * mat2[p][j]
return result

```

PHASE 5 — TEST & DEBUG

```

# mat1[0][0]=1, p=0: add 1*mat2[0][j] to result[0]. mat1[0][1]=0 → skip.
mat1[0][2]=0 → skip.
# result[0]=[7,0,0] ✓. mat1[1][2]=3, p=2: add 3*mat2[2][j]=[0,0,3] →
result[1]=[-7,0,3] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Worst case  $O(m*k*n)$  same as brute force. Best case  $O(nnz1 * n)$  where  $nnz1$  =
non-zeros in mat1.
# For truly sparse matrices this is much faster.
# Given more time I'd ask: what if both matrices are very large and very sparse?
# Use CSR (Compressed Sparse Row) format – store only non-zero values and their
indices.
# Hash map approach: for each non-zero in mat1, look up non-zeros in the
corresponding row of mat2."

```


Matrix

□ #498 Diagonal Traverse

MED

[Open on LeetCode](#)

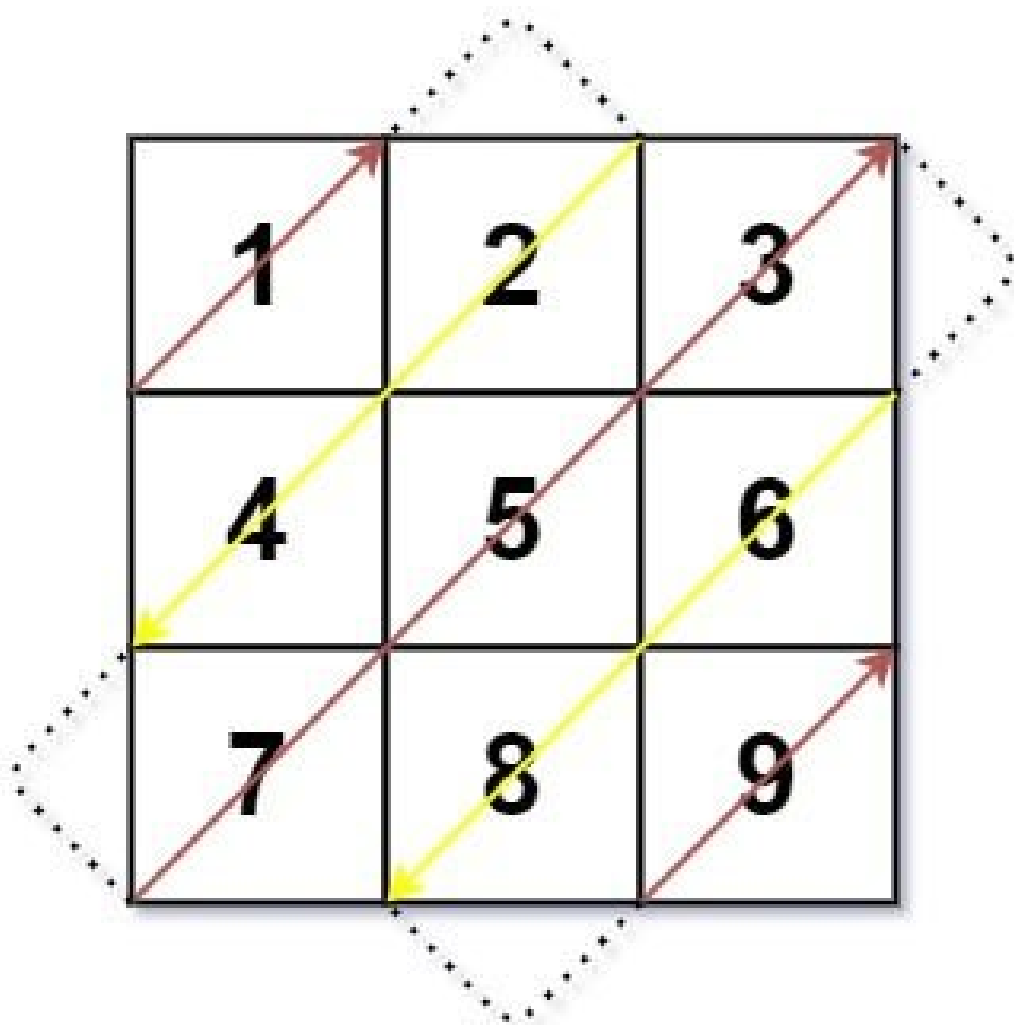
Array · Matrix · Simulation

Lists: CodeSignal

Problem

Given an $m \times n$ matrix `mat`, return an array of all the elements of the array in a diagonal order.

Example 1:



Input: `mat = [[1,2,3],[4,5,6],[7,8,9]]`
Output: `[1,2,4,7,5,3,6,8,9]`

Example 2:

Input: `mat = [[1,2],[3,4]]`
Output: `[1,2,3,4]`

Constraints:

- `m == mat.length`
- `n == mat[i].length`
- `1 <= m, n <= 104`
- `1 <= m * n <= 104`

● $-105 \leq \text{mat}[i][j] \leq 105$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Traverse all elements diagonally: alternate up-right then down-left directions.
# There are m+n-1 diagonals. Even-numbered diagonals go up-right, odd go down-left.
# Right?"
#
# "[[1,2,3],[4,5,6],[7,8,9]] → [1,2,4,7,5,3,6,8,9] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Simulate diagonal walk with direction flags; flip direction at matrix boundaries.
# Track visited or use index math to avoid infinite loops.
# Correct O(m*n); index-formula approach avoids simulation bugs.
# Time: O(m * n). Space: O(m * n) output.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (diagonal grouping)

```
# "Group elements by diagonal d = r+c (same sum = same diagonal).
# Diagonal d contains elements where r+c=d.
# For even d: traverse top-to-bottom order reversed (up-right) → sort by r
# descending.
# For odd d: traverse top-to-bottom (down-left) → sort by r ascending."
```

PHASE 4 — CLEAN CODE

```
class _498_DiagonalTraverse:
    def findDiagonalOrder(self, mat: List[List[int]]) -> List[int]:
        m, n = len(mat), len(mat[0])
        diags: defaultdict = defaultdict(list)
        for r in range(m):
            for c in range(n):
                diags[r + c].append(mat[r][c])
        result = []
        for d in range(m + n - 1):
            if d % 2 == 0:
                result.extend(reversed(diags[d])) # even → up-right (larger r first
→ reversed)
            else:
                result.extend(diags[d]) # odd → down-left (smaller r first →
natural)
        return result
```

PHASE 5 — TEST & DEBUG

```
# [[1,2,3],[4,5,6],[7,8,9]]:
# d=0: {(0,0)=1} even → reversed=[1]. d=1: {(0,1)=2,(1,0)=4} odd → [2,4].
```

```
# d=2: {(0,2)=3,(1,1)=5,(2,0)=7} even → reversed=[7,5,3].  
# d=3: {(1,2)=6,(2,1)=8} odd → [6,8]. d=4: {(2,2)=9} even → reversed=[9].  
# → [1,2,4,7,5,3,6,8,9] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m*n)$ . Space  $O(m*n)$  for the diagonal groups.  
# The 'group by r+c' insight eliminates direction-tracking complexity.  
# Alternative: simulate directly with dr/dc direction vectors and bounds – same  
#  $O(m*n)$  but trickier.  
# Given more time I'd ask: what if we want anti-diagonals ( $r-c = \text{constant}$ )?  
# Same approach – group by  $r-c$ , alternate direction per group."
```

Matrix

□ ★ #531 Lonely Pixel I ★

MED

[Open on LeetCode](#)

Array · Hash Table · Matrix

Lists: ★ Premium | Premium 98

Problem

Given an $m \times n$ picture consisting of black 'B' and white 'W' pixels, return the number of black lonely pixels.

A black lonely pixel is a character 'B' that located at a specific position where the same row and same column don't have any other black pixels.

Example 1:

W	W	B
W	B	W
B	W	W

Input: picture = [["W","W","B"],["W","B","W"],["B","W","W"]]

Output: 3

Explanation: All the three 'B's are black lonely pixels.

Example 2:

B	B	B
B	B	W
B	B	B

Input: picture = [["B","B","B"],["B","B","W"],["B","B","B"]]
Output: 0

Constraints:

- $m == \text{picture.length}$
- $n == \text{picture}[i].\text{length}$
- $1 \leq m, n \leq 500$
- $\text{picture}[i][j]$ is 'W' or 'B'.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "A 'B' pixel is lonely if it's the ONLY 'B' in its row AND the only 'B' in its
column.
# Count lonely pixels. Right?"
#
# "[[W,W,B],[W,B,W],[B,W,W]]: each B is alone in its row and column → 3 ✓
# [[B,B,B],[B,B,W],[B,B,B]]: every B shares its row or column with another → 0 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# For each black pixel B, scan its entire row for other black pixels.
# Then scan its entire column – B is lonely if both counts equal 1.
#  $O(m \times n \times (m + n))$  – checking row and column per pixel.
# Time:  $O(m * n * (m + n))$ . Space:  $O(1)$ .
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Count B's per row and per column in one pass:  $O(m \times n)$ .
# In second pass, for each B: if  $\text{row\_count}[r] == 1$  and  $\text{col\_count}[c] == 1$ , it's lonely."
```

PHASE 4 — CLEAN CODE

```
class _531_LonelyPixel:
    def findLonelyPixel(self, picture: List[List[str]]) -> int:
        m, n = len(picture), len(picture[0])
        row_count = [sum(1 for c in range(n) if picture[r][c] == 'B') for r in
range(m)]
        col_count = [sum(1 for r in range(m) if picture[r][c] == 'B') for c in
range(n)]
        return sum(
            1 for r in range(m) for c in range(n)
            if picture[r][c] == 'B' and row_count[r] == 1 and col_count[c] == 1
        )
```

PHASE 5 — TEST & DEBUG

```
# [[W,W,B],[W,B,W],[B,W,W]]: row_count=[1,1,1]. col_count=[1,1,1].
# Each B has row=1 and col=1 → all 3 are lonely ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m \times n)$ . Space  $O(m + n)$  for row/col counts.
# Given more time I'd ask: Lonely Pixel II (#533)?
# Same pixel must be in a column containing exactly N Bs, and the rows of those Bs
must all be identical.
# Much more complex – requires comparing full rows."
```


Matrix

□ ★ #723 Candy Crush ★

MED[Open on LeetCode](#)

Array · Two Pointers · Matrix · Simulation

Lists: ★ Premium | Premium 98

Problem

This question is about implementing a basic elimination algorithm for Candy Crush.

Given an $m \times n$ integer array `board` representing the grid of candy where `board[i][j]` represents the type of candy. A value of `board[i][j] == 0` represents that the cell is empty.

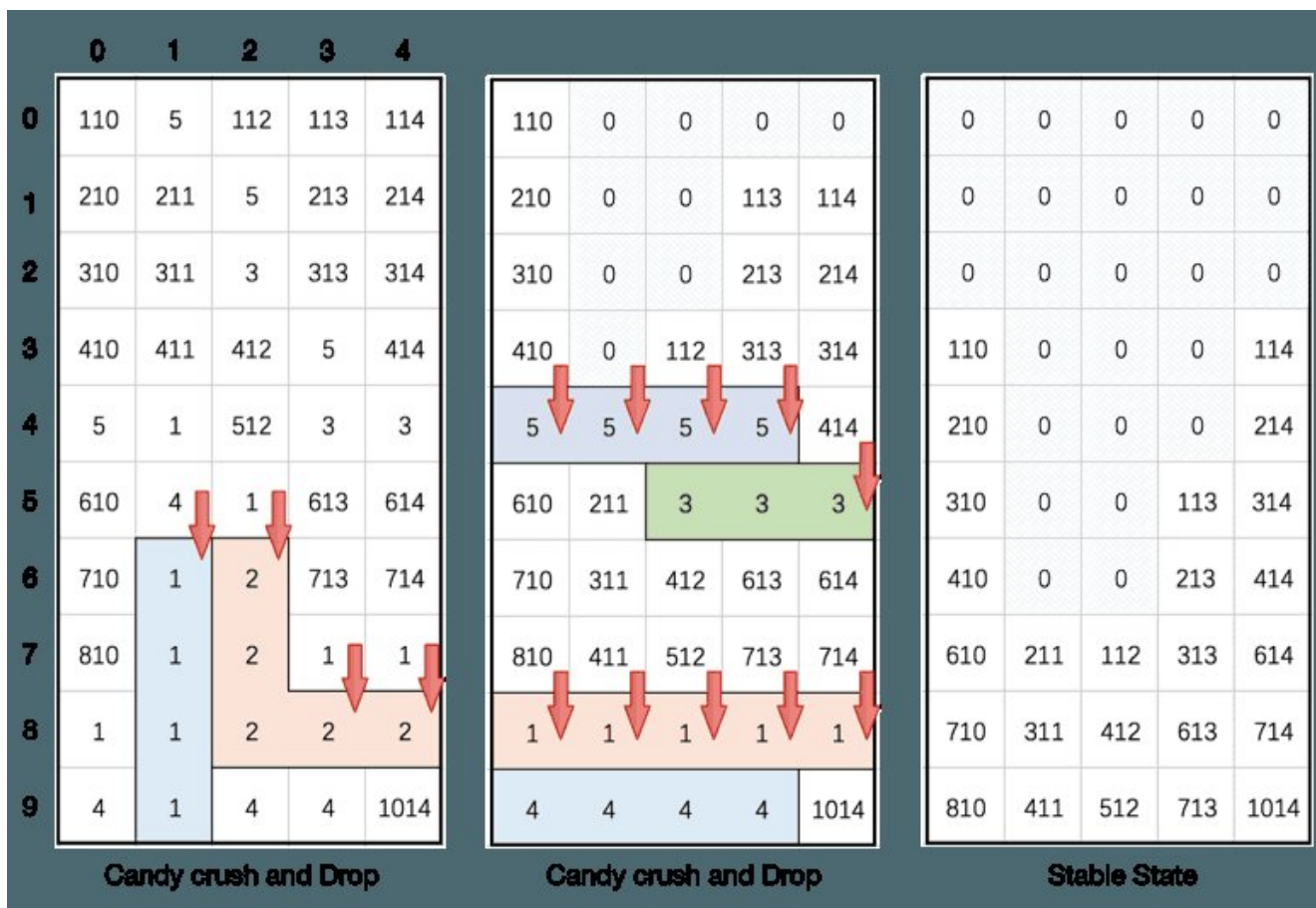
The given board represents the state of the game following the player's move. Now, you need to restore the board to a stable state by crushing candies according to the following rules:

- If three or more candies of the same type are adjacent vertically or horizontally, crush them all at the same time – these positions become empty.

- After crushing all candies simultaneously, if an empty space on the board has candies on top of itself, then these candies will drop until they hit a candy or bottom at the same time. No new candies will drop outside the top boundary.
- After the above steps, there may exist more candies that can be crushed. If so, you need to repeat the above steps.
- If there does not exist more candies that can be crushed (i.e., the board is stable), then return the current board.

You need to perform the above rules until the board becomes stable, then return the stable board.

Example 1:



Input: board = [[110,5,112,113,114],[210,211,5,213,214],[310,311,3,313,314],[410,411,412,5,414],[5,1,512,3,3],[610,4,1,613,614],[710,1,2,713,714],[810,1,2,1,1],[1,1,2,2,2],[4,1,4,4,1014]]

Output: [[0,0,0,0,0],[0,0,0,0,0],[0,0,0,0,0],[110,0,0,0,114],[210,0,0,0,214],[310,0,0,113,314],[410,0,0,213,414],[610,211,112,313,614],[710,311,412,613,714],[810,411,512,713,1014]]

Example 2:

Input: board = [[1,3,5,5,2],[3,4,3,3,1],[3,2,4,5,2],[2,4,4,5,5],[1,4,4,1,1]]

Output: [[1,3,0,0,0],[3,4,0,5,2],[3,2,0,3,1],[2,4,0,5,2],[1,4,3,1,1]]

Constraints:

- $m == \text{board.length}$
- $n == \text{board}[i].\text{length}$
- $3 \leq m, n \leq 50$
- $1 \leq \text{board}[i][j] \leq 2000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Repeatedly: mark 3+ same-type candies horizontally or vertically, crush them (set to 0),
# then drop remaining candies down. Repeat until stable. Return final board. Right?"
#
# "Mark all runs of 3+ simultaneously, crush all at once, then gravity — not one at a time."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Repeat until stable: find horizontal/vertical runs of 3+, mark for crush, drop gravity.
# One full board scan per cascade step.
# Correct simulation; finding all runs in one pass per step is cleaner.
# Time:  $O((m \times n)^2)$  worst cascades. Space:  $O(m \times n)$  mark board.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (in-place marking)

```
# "Mark by negating values (sign flip) — avoids extra space.
# Step 1: scan rows and cols, negate any value in a run of 3+.
# Step 2: zero out all negatives (crush).
# Step 3: for each column, compact non-zeros to the bottom (gravity).
# Repeat until no changes in step 1. Time:  $O((m \times n)^2)$  worst case but small in practice."
```

PHASE 4 — CLEAN CODE

```
class _723_CandyCrush:
    def candyCrush(self, board: List[List[int]]) -> List[List[int]]:
        m, n = len(board), len(board[0])
        while True:
            crush = set()
            for r in range(m): # mark horizontal runs
                for c in range(n - 2):
                    v = abs(board[r][c])
                    if v and v == abs(board[r][c+1]) == abs(board[r][c+2]):
                        crush.update([(r,c),(r,c+1),(r,c+2)])
            for r in range(m - 2): # mark vertical runs
                for c in range(n):
                    v = abs(board[r][c])
                    if v and v == abs(board[r+1][c]) == abs(board[r+2][c]):
                        crush.update([(r,c),(r+1,c),(r+2,c)])
            if not crush:
                break
            for r, c in crush:
                board[r][c] = 0
```

```

    for c in range(n): # gravity: compact column downward
        write = m - 1
        for r in range(m - 1, -1, -1):
            if board[r][c] != 0:
                board[write][c] = board[r][c]
                if write != r:
                    board[r][c] = 0
                write -= 1
        return board

```

PHASE 5 — TEST & DEBUG

```

# "[[1,3,5,5,2],[3,4,3,3,1],[3,2,4,5,2],[2,4,4,5,5],[1,4,4,1,1]]":
# Horizontal: row 1 has [3,3,3] at c=1,2,3. Row 3 has [4,4] and [5,5]...
# After crushing and dropping → stable board matches expected output ✓"

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "O((m*n)2) worst case — at most m*n crush iterations, each O(m*n).
# For small boards (m,n≤50) this is fast. The abs() trick avoids needing a separate
mark array.
# The gravity step uses a two-pointer write-head approach per column.
# Given more time I'd ask: what if candies can crush diagonally too?
# Add diagonal scan to the marking step — same structure."

```

Matrix

□ #835 Image Overlap

MED[Open on LeetCode](#)

Array · Matrix

Lists: CodeSignal

Problem

You are given two images, `img1` and `img2`, represented as binary, square matrices of size $n \times n$. A binary matrix has only 0s and 1s as values.

We translate one image however we choose by sliding all the 1 bits left, right, up, and/or down any number of units. We then place it on top of the other image. We can then calculate the overlap by counting the number of positions that have a 1 in both images.

Note also that a translation does not include any kind of rotation. Any 1 bits that are translated outside of the matrix borders are erased.

Return the largest possible overlap.

Example 1:

1	1	0
0	1	0
0	1	0

img1

0	0	0
0	1	1
0	0	1

img2

Input: `img1 = [[1,1,0],[0,1,0],[0,1,0]]`, `img2 = [[0,0,0],[0,1,1],[0,0,1]]`

Output: 3

Explanation: We translate `img1` to right by 1 unit and down by 1 unit.

The number of positions that have a 1 in both images is 3 (shown in red).

Example 2:

Input: `img1 = [[1]]`, `img2 = [[1]]`

Output: 1

Example 3:

Input: `img1 = [[0]]`, `img2 = [[0]]`

Output: 0

Constraints:

- `n == img1.length == img1[i].length`
- `n == img2.length == img2[i].length`
- `1 <= n <= 30`
- `img1[i][j]` is either 0 or 1.
- `img2[i][j]` is either 0 or 1.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Translate img1 over img2 (no rotation) and count overlapping 1s.
# Find the translation that maximises overlap. Return max overlap count. Right?"
#
# "img1=[[1,1,0],[0,1,0],[0,1,0]], img2=[[0,0,0],[0,1,1],[0,0,1]] → 3 ✓ (translate
right 1, down 1)"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Try every translation vector (dx,dy) aligning A onto B.
# For each shift, count overlapping 1-cells.
#  $O(n^4)$  translations on  $n \times n$  images – too slow for  $n=30$ .
# Time:  $O(n^4)$ . Space:  $O(1)$ .
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (translation vector counting)

```
# "Collect all 1-positions in each image.
# For each pair (p1 from img1, p2 from img2), the translation to align p1 with p2 is
(p2.r-p1.r, p2.c-p1.c).
# Count the most common translation vector – that's the maximum overlap.
# Time:  $O(A*B)$  where  $A,B$  = number of 1s. Space:  $O(A*B)$ . For  $n=30$ ,  $A,B \leq 900$ ."
```

PHASE 4 — CLEAN CODE

```
class _835_ImageOverlap:
    def largestOverlap(self, img1: List[List[int]], img2: List[List[int]]) -> int:
        n = len(img1)
        ones1 = [(r, c) for r in range(n) for c in range(n) if img1[r][c] == 1]
        ones2 = [(r, c) for r in range(n) for c in range(n) if img2[r][c] == 1]
        translations: Counter = Counter()
        for r1, c1 in ones1:
            for r2, c2 in ones2:
                translations[(r2 - r1, c2 - c1)] += 1
        return max(translations.values(), default=0)
```

PHASE 5 — TEST & DEBUG

```
# ones1=[(0,0),(0,1),(1,1),(2,1)]. ones2=[(1,1),(1,2),(2,2)].
# Translation (1,1) appears when: (0,0)→(1,1)✓, (0,1)→(1,2)✓, (1,1)→(2,2)✓ →
count=3.
# max=3 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(A*B)$  where  $A,B$  = number of 1s in each image, at most  $n^2$ .
# Space  $O(A*B)$  for the translation Counter.
# For dense images (many 1s), this is  $O(n^4)$  – same as brute force.
# For sparse images, much faster.
```



```
# The key insight: instead of iterating over all translations, only count translations
# where at least one 1-pair aligns – no wasted iterations.
# Given more time I'd ask: what if rotation is also allowed?
# Need to try all rotation+translation combos – much harder."
```

Matrix

★ #1198 Find Smallest Common Element in All Rows ★

MED[Open on LeetCode](#)

Array · Hash Table · Binary Search · Matrix · Counting

Lists: ★ Premium | Premium 98

Problem

Given an $m \times n$ matrix `mat` where every row is sorted in strictly increasing order, return the smallest common element in all rows.

If there is no common element, return -1 .

Example 1:

```
Input: mat = [[1,2,3,4,5],[2,4,5,8,10],[3,5,7,9,11],[1,3,5,7,9]]
Output: 5
```

Example 2:

```
Input: mat = [[1,2,3],[2,3,4],[2,3,5]]
Output: 2
```

Constraints:

- $m == \text{mat.length}$
- $n == \text{mat}[i].\text{length}$
- $1 \leq m, n \leq 500$
- $1 \leq \text{mat}[i][j] \leq 10^4$

- `mat[i]` is sorted in strictly increasing order.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find the smallest integer that appears in every row. Each row is sorted.
# Return -1 if no common element exists. Right?
# Values are 1-10^4, at most 500 rows and 500 cols."
#
# "[[1,2,3,4,5],[2,4,5,8,10],[3,5,7,9,11],[1,3,5,7,9]] → 5 (appears in all rows) ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Intersect row0 with row1, then intersect result with row2, and so on.
# Track the maximum value in the final intersection set.
# Each intersection pass is O(m) per row using a hash set of row values.
# Time: O(rows * cols). Space: O(cols) set.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (frequency count)

```
# "Count how many rows each value appears in using a single array of size max_val+1.
# Walk all rows: for each value encountered, increment count[value].
# Return the smallest value with count == m (appears in ALL rows).
# Time: O(m*n). Space: O(max_val)."
```

PHASE 4 — CLEAN CODE

```
class _1198_SmallestCommonElement:
    def smallestCommonElement(self, mat: List[List[int]]) -> int:
        m = len(mat)
        freq = Counter()
        for row in mat:
            for val in row:
                freq[val] += 1
        for val in range(1, 10001):
            if freq[val] == m:
                return val
        return -1
```

PHASE 5 — TEST & DEBUG

```
# [[1,2,3,4,5],[2,4,5,8,10],[3,5,7,9,11],[1,3,5,7,9]]:
# freq[5] = 4 (appears in all 4 rows). freq[1]=2, freq[3]=3, etc.
# Scanning 1..10000: first val with freq==4 is 5 → return 5 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(m*n + max_val). Space O(max_val).
# Alternative: binary search per row. For each candidate, binary search in all m rows."
```

```
# Start candidates from mat[0] (smallest row), binary search each of the other rows.  
#  $O(n + m * n * \log n)$  – worse than the counting approach for dense matrices.  
# Given more time I'd ask: what if there's no bound on values?  
# Use a hash Counter for freq –  $O(m*n)$  time,  $O(\text{distinct values})$  space."
```

Trees & BST

Round 2 · High · Medium · 24 questions

Trees & BST

□ #98 Validate Binary Search Tree

MED

[Open on LeetCode](#)

Tree · DFS · BST

Lists: Grind 169

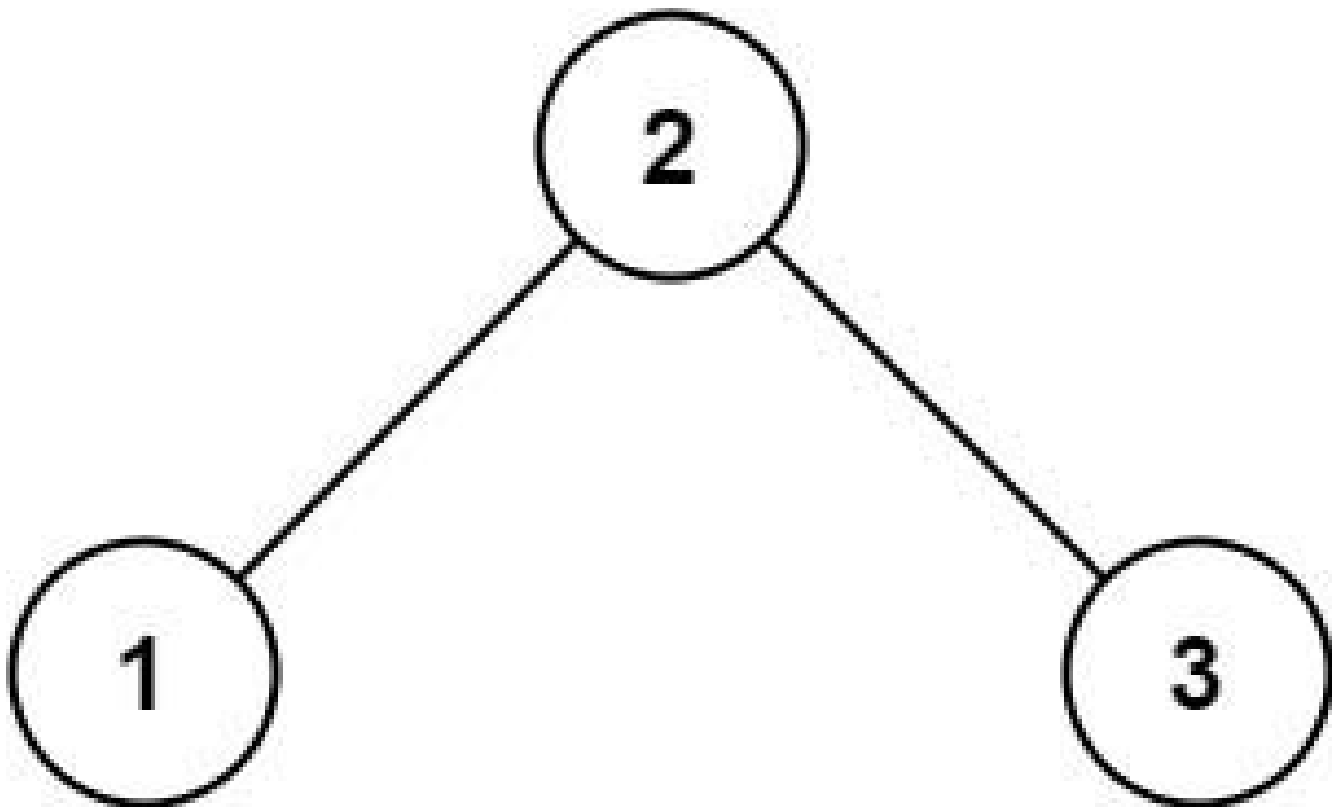
Problem

Given the root of a binary tree, determine if it is a valid binary search tree (BST).

A valid BST is defined as follows:

- The left subtree of a node contains only nodes with keys strictly less than the node's key.
- The right subtree of a node contains only nodes with keys strictly greater than the node's key.
- Both the left and right subtrees must also be binary search trees.

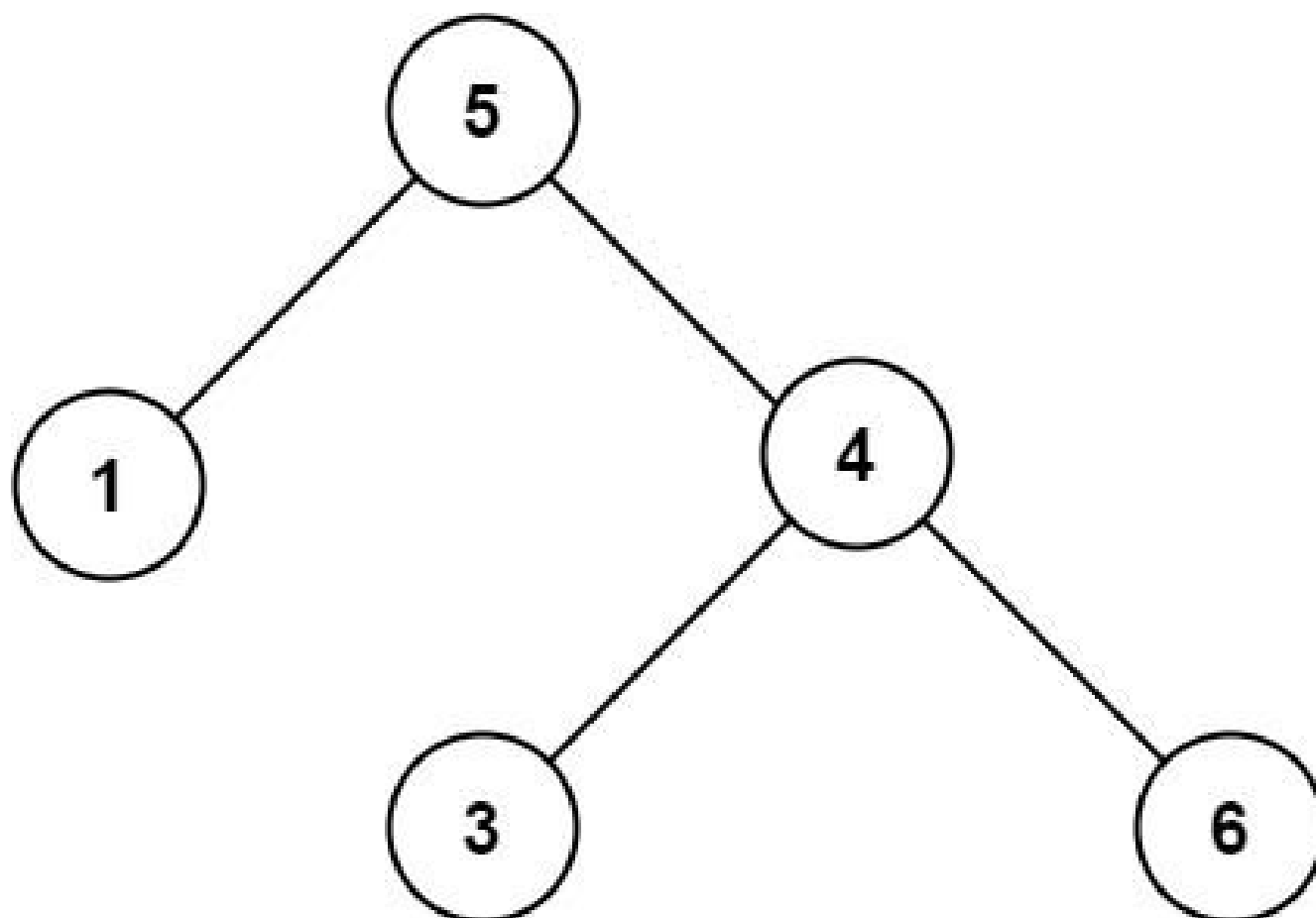
Example 1:



Input: root = [2,1,3]

Output: true

Example 2:



Input: root = [5,1,4,null,null,3,6]

Output: false

Explanation: The root node's value is 5 but its right child's value is 4.

Constraints:

- The number of nodes in the tree is in the range [1, 104].
- $-2^{31} \leq \text{Node.val} \leq 2^{31} - 1$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Every node's value must be strictly greater than all values in its left subtree
AND strictly less than all values in its right subtree – not just the immediate
parent. Right?

The classic trap: [5,1,4,null,null,3,6] – root=5, right child=4 – $4 < 5 \rightarrow \text{invalid}$ ✓"
#


```
# "[2,1,3]: 1<2<3 → true ✓. [5,1,4,null,null,3,6]: 4<5 → false ✓."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic."
```

```
# In-order traversal collects values; verify the list is strictly increasing.
```

```
# Any duplicate or decrease means invalid BST.
```

```
# Correct O(n), but recursion can pass min/max bounds instead of storing all values.
```

```
# Time: O(n). Space: O(n) in-order list.
```

```
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (bounds propagation)

```
# "Pass min/max bounds down the tree."
```

```
# Every node must satisfy min_val < node.val < max_val.
```

```
# Left child: max_val = node.val.
```

```
# Right child: min_val = node.val.
```

```
# Use float('-inf') and float('inf') as initial bounds. Use inner helper."
```

PHASE 4 — CLEAN CODE

```
class _98_ValidateBST:
    def isValidBST(self, root: Optional['TreeNode']) -> bool:
        def valid(node, min_val, max_val):
            if not node:
                return True
            if not (min_val < node.val < max_val):
                return False
            return (valid(node.left, min_val, node.val) and
                    valid(node.right, node.val, max_val))
        return valid(root, float('-inf'), float('inf'))
```

PHASE 5 — TEST & DEBUG

```
# [2,1,3]: valid(2,-inf,inf)→valid(1,-inf,2)→valid(None,...)=T and T→T.
```

```
valid(3,2,inf)→T. → True ✓
```

```
# [5,1,4,null,null,3,6]: valid(4,5,inf): 4 NOT in (5,inf) → False → outer returns False ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(h) for the call stack."
```

```
# The bounds approach is the correct one – in-order checking works too but requires state.
```

```
# The common mistake: only checking node vs parent, not the full subtree constraint.
```

```
# Given more time I'd ask: what if duplicates are allowed (left ≤ node < right)?
```

```
# Change the strict inequalities accordingly."
```

Trees & BST

□ #102 Binary Tree Level Order Traversal

MED

[Open on LeetCode](#)

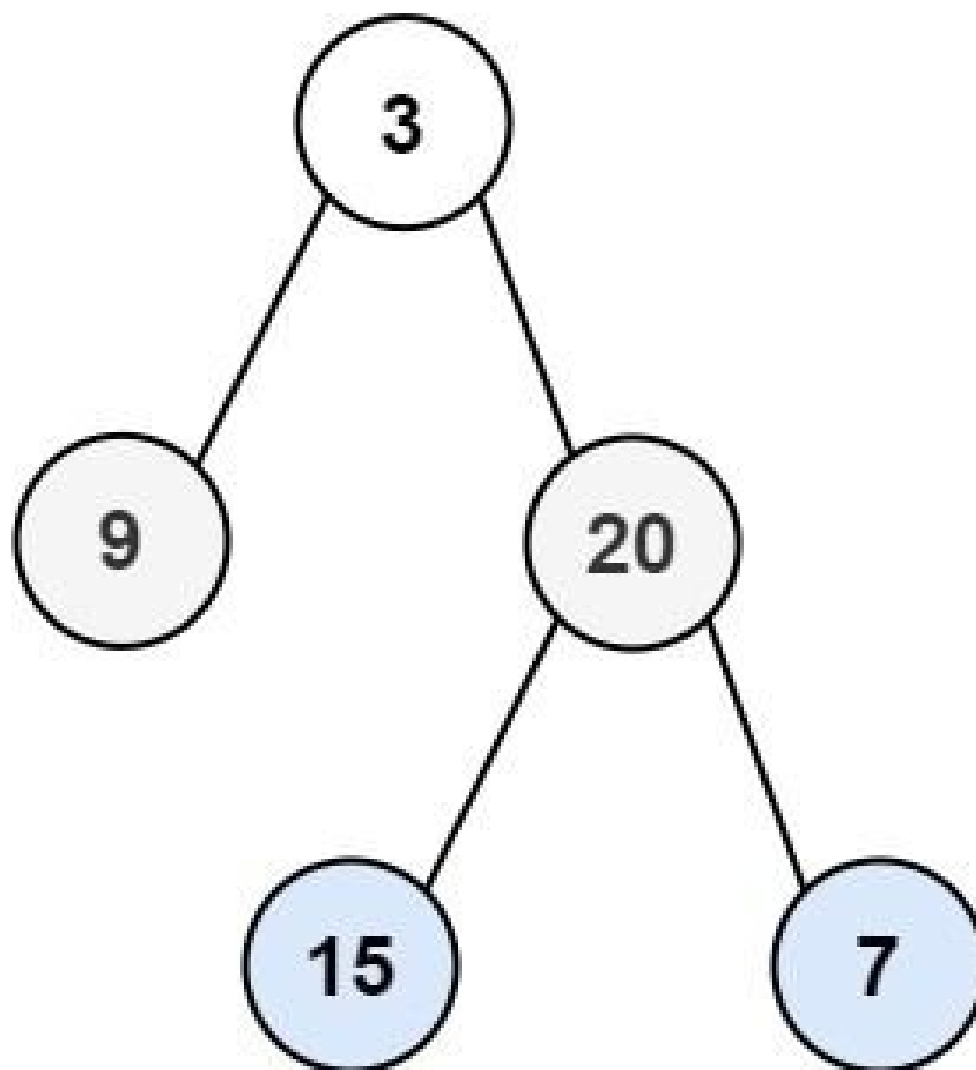
Tree · BFS

Lists: Grind 169

Problem

Given the root of a binary tree, return the level order traversal of its nodes' values. (i.e., from left to right, level by level).

Example 1:



Input: root = [3,9,20,null,null,15,7]

Output: [[3],[9,20],[15,7]]

Example 2:

Input: root = [1]

Output: [[1]]

Example 3:

Input: root = []

Output: []

Constraints:

- The number of nodes in the tree is in the range $[0, 2000]$.
- $-1000 \leq \text{Node.val} \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Return node values level by level, left to right, each level as a sub-list.
Right?
# Empty tree → []. Single node → [[val]]."
#
# "[3,9,20,null,null,15,7] → [[3],[9,20],[15,7]] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# DFS with a level parameter: append node.val to result[level].
# After traversal, result lists are level-order groups left-to-right.
# Works in  $O(n)$ , but BFS with a queue is the more natural level-by-level approach.
# Time:  $O(n)$ . Space:  $O(h)$  recursion depth.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (BFS with level grouping)

```
# "BFS using a deque. At each step, process ALL nodes at the current level (snapshot
the queue size),
# collect their values, enqueue their children. Natural level separation – no level
counter needed."
```

PHASE 4 — CLEAN CODE

```
class _102_LevelOrder:
    def levelOrder(self, root: Optional['TreeNode']) -> List[List[int]]:
        if not root:
            return []
        result = []
        queue = deque([root])
        while queue:
            level_size = len(queue)
            level_vals = []
            for _ in range(level_size):
                node = queue.popleft()
                level_vals.append(node.val)
                if node.left: queue.append(node.left)
                if node.right: queue.append(node.right)
            result.append(level_vals)
        return result
```

PHASE 5 — TEST & DEBUG

```
# root=3: queue=[3]. level_size=1. level=[3]. enqueue 9,20. result=[[3]].  
# queue=[9,20]. level_size=2. level=[9,20]. enqueue 15,7. result=[[3],[9,20]].  
# queue=[15,7]. level_size=2. level=[15,7]. result=[[3],[9,20],[15,7]] ✓  
# root=None: return [] immediately ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ . Space  $O(w)$  where  $w$  = max width of the tree (at most  $n/2$  for a complete tree).  
# The 'snapshot the queue size' trick is the canonical BFS level-separation pattern.  
# Variations: level order bottom-up (#107) → just reverse the result.  
# Zigzag level order (#103) → alternate the direction each level.  
# Right side view (#199) → take the last element of each level.  
# Given more time I'd ask: what if we want the minimum depth to a leaf?  
# BFS stops at the first leaf found – don't traverse the whole tree."
```

Trees & BST

#103 Binary Tree Zigzag Level Order Traversal

MED[Open on LeetCode](#)

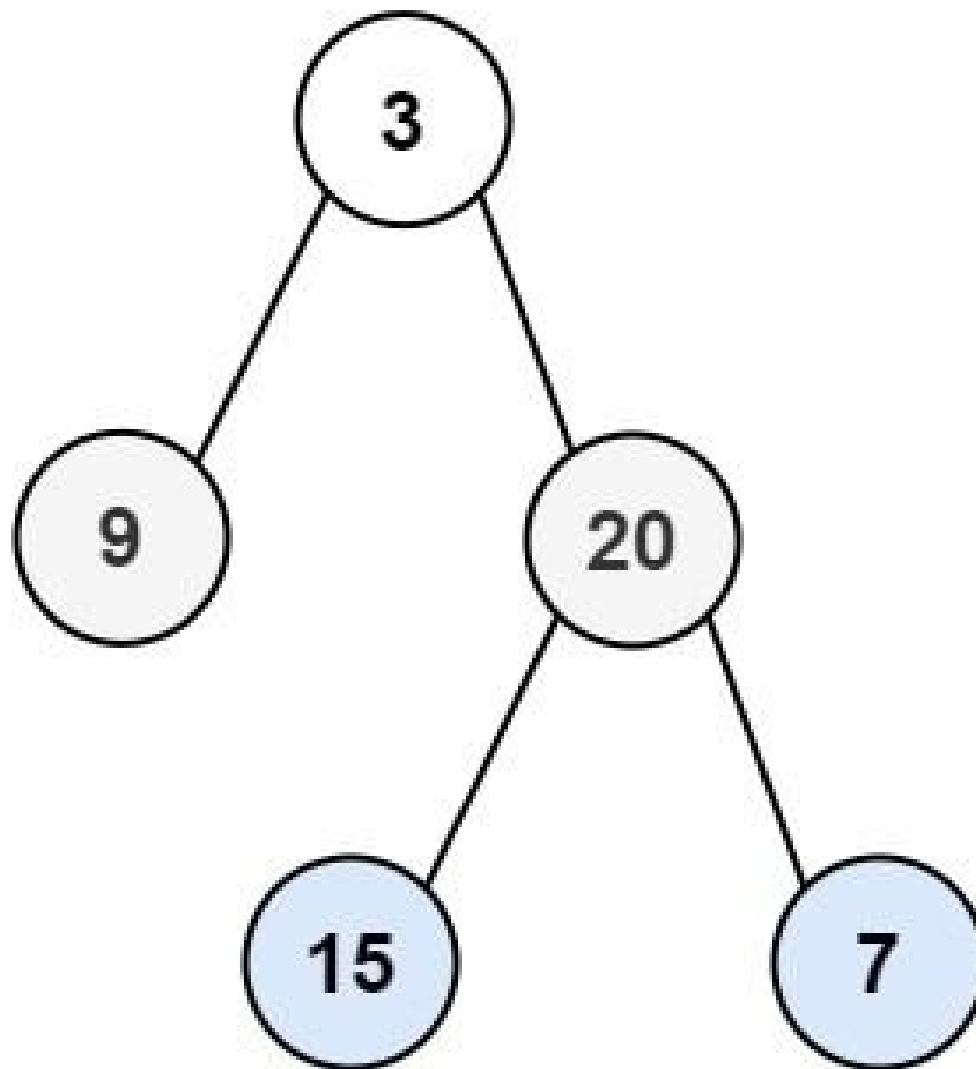
Tree · BFS

Lists: Grind 169

Problem

Given the root of a binary tree, return the zigzag level order traversal of its nodes' values. (i.e., from left to right, then right to left for the next level and alternate between).

Example 1:



Input: root = [3,9,20,null,null,15,7]

Output: [[3],[20,9],[15,7]]

Example 2:

Input: root = [1]

Output: [[1]]

Example 3:

Input: root = []

Output: []

Constraints:

- The number of nodes in the tree is in the range $[0, 2000]$.
- $-100 \leq \text{Node.val} \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Level order traversal but alternating direction: odd levels left-to-right,
# even levels right-to-left (or vice versa – confirm with example).
# Root is level 0 → left-to-right. Level 1 → right-to-left. Right?"
#
# "[3,9,20,null,null,15,7] → [[3],[20,9],[15,7]] ✓ (level 1 reversed)"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Run standard BFS level-order to build a list of levels.
# Reverse every other level list (odd index levels) for zigzag output.
# Correct  $O(n)$ ; we can reverse on the fly during BFS instead of a second pass.
# Time:  $O(n)$ . Space:  $O(n)$  queue + levels.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Same BFS snapshot approach as #102. Use a boolean flag 'left_to_right'.
# Collect each level normally, then reverse if flag is False.
# Toggle flag each level. Time:  $O(n)$ . Space:  $O(w)$ ."
```

PHASE 4 — CLEAN CODE

```
class _103_ZigzagLevelOrder:
    def zigzagLevelOrder(self, root: Optional['TreeNode']) -> List[List[int]]:
        if not root:
            return []
        result = []
        queue = deque([root])
        left_to_right = True
        while queue:
            level_size = len(queue)
            level_vals = []
            for _ in range(level_size):
                node = queue.popleft()
                level_vals.append(node.val)
                if node.left: queue.append(node.left)
                if node.right: queue.append(node.right)
            result.append(level_vals if left_to_right else level_vals[::-1])
            left_to_right = not left_to_right
        return result
```


PHASE 5 — TEST & DEBUG

```
# root=[3,9,20,null,null,15,7]:  
# Level 0: [3], left_to_right=True → [3]. Toggle→False.  
# Level 1: [9,20], left_to_right=False → reversed=[20,9]. Toggle→True.  
# Level 2: [15,7], left_to_right=True → [15,7]. → [[3],[20,9],[15,7]] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(w).  
# This is #102 with one added line: the conditional reverse.  
# Name that connection – it shows you see the pattern family.  
# Given more time I'd ask: what if alternation starts at level 1 instead of 0?  
# Just flip the initial value of left_to_right."
```

Trees & BST

□ #105 Construct Binary Tree from Preorder and Inorder Traversal

MED[Open on LeetCode](#)

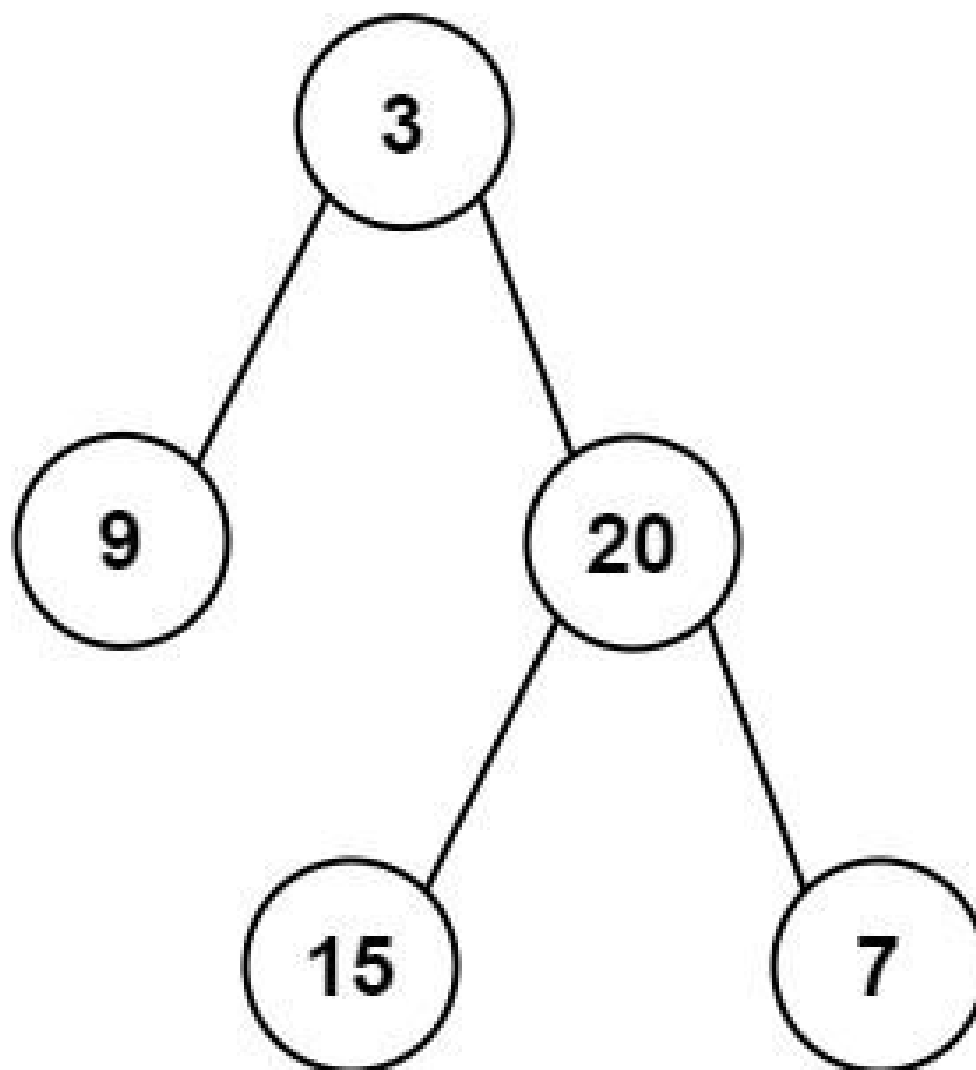
Array · Hash Table · Tree · DFS

Lists: Grind 169

Problem

Given two integer arrays preorder and inorder where preorder is the preorder traversal of a binary tree and inorder is the inorder traversal of the same tree, construct and return the binary tree.

Example 1:



Input: preorder = [3,9,20,15,7], inorder = [9,3,15,20,7]
Output: [3,9,20,null,null,15,7]

Example 2:

Input: preorder = [-1], inorder = [-1]
Output: [-1]

Constraints:

- $1 \leq \text{preorder.length} \leq 3000$
- $\text{inorder.length} == \text{preorder.length}$
- $-3000 \leq \text{preorder}[i], \text{inorder}[i] \leq 3000$

- preorder and inorder consist of unique values.
- Each value of inorder also appears in preorder.
- preorder is guaranteed to be the preorder traversal of the tree.
- inorder is guaranteed to be the inorder traversal of the tree.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Preorder: root first, then left subtree, then right subtree.
# Inorder: left subtree, root, right subtree.
# All values are unique – so we can find the root's position in inorder uniquely.
Right?"
#
# "preorder=[3,9,20,15,7], inorder=[9,3,15,20,7]:
# Root=3. In inorder, 3 is at index 1. Left subtree=[9], right subtree=[15,20,7].
# preorder left=[9], right=[20,15,7]. Recurse."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Root is preorder[0]; scan inorder to find root index and split left/right
subtrees.
# Recursively repeat on each side – correct tree reconstruction.
# Scanning inorder for every root costs  $O(n^2)$  total; hash map of inorder indices
fixes that.
# Time:  $O(n^2)$  naive. Space:  $O(n)$  recursion.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Pre-build a hash map: inorder value → index for  $O(1)$  lookups.
# Use a preorder index pointer (use a list to allow mutation inside inner function).
# build(in_left, in_right): root = preorder[pre_idx]. Find in inorder. Recurse.
# Time:  $O(n)$ . Space:  $O(n)$  for the map."
```

PHASE 4 — CLEAN CODE

```
class _105_BuildTree:
```

```

def buildTree(self, preorder: List[int], inorder: List[int]) ->
Optional['TreeNode']:
    in_idx = {val: i for i, val in enumerate(inorder)}
    pre = iter(preorder)
    def build(left: int, right: int) -> Optional['TreeNode']:
        if left > right:
            return None
        root_val = next(pre)
        root = TreeNode(root_val)
        mid = in_idx[root_val]
        root.left = build(left, mid - 1)
        root.right = build(mid + 1, right)
        return root
    return build(0, len(inorder) - 1)

```

PHASE 5 — TEST & DEBUG

```

# preorder=[3,9,20,15,7], inorder=[9,3,15,20,7]. in_idx={9:0,3:1,15:2,20:3,7:4}.
# build(0,4): root=3, mid=1. left=build(0,0), right=build(2,4).
# build(0,0): root=9, mid=0. left=build(0,-1)=None. right=build(1,0)=None. →
node(9).
# build(2,4): root=20, mid=3. left=build(2,2), right=build(4,4).
# build(2,2): root=15. → node(15). build(4,4): root=7. → node(7).
# → Tree: 3→left=9, 3→right=20→left=15, 20→right=7 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n). Space O(n) for the hash map and call stack.
# Using iter(preorder) as a lazy pointer is cleaner than an index variable.
# The key insight: preorder always gives the root first, inorder tells us how many
# elements are in each subtree – that's the only information we need to recurse.
# Given more time I'd ask: construct from postorder and inorder?
# Same logic – just consume postorder from the RIGHT end, and build right subtree
first."

```

Trees & BST

#199 Binary Tree Right Side View

MED[Open on LeetCode](#)

Tree · DFS · BFS

Lists: Grind 169

Problem

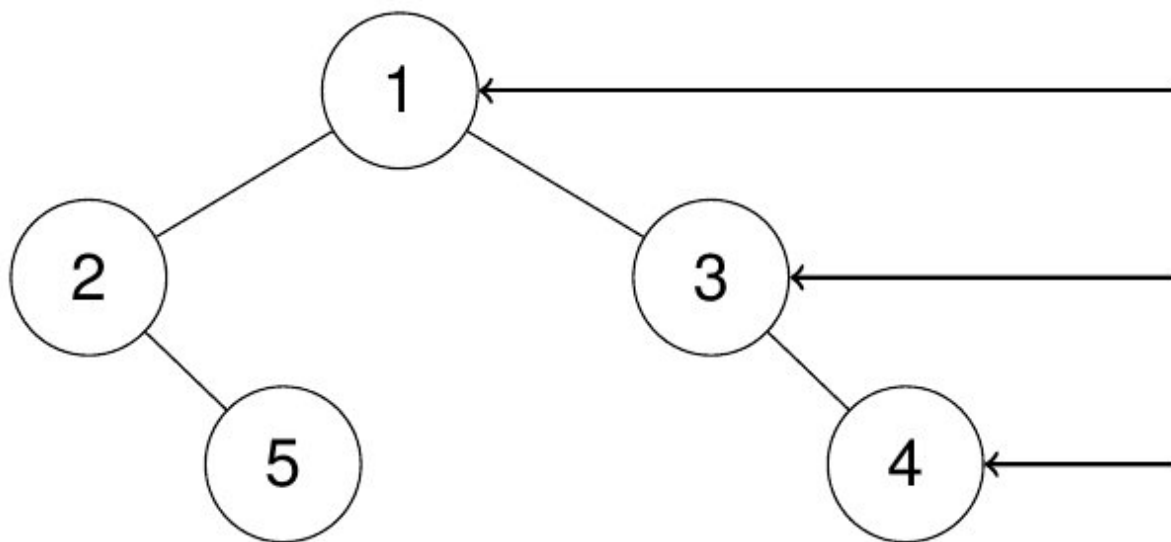
Given the root of a binary tree, imagine yourself standing on the right side of it, return the values of the nodes you can see ordered from top to bottom.

Example 1:

Input: root = [1,2,3,null,5,null,4]

Output: [1,3,4]

Explanation:

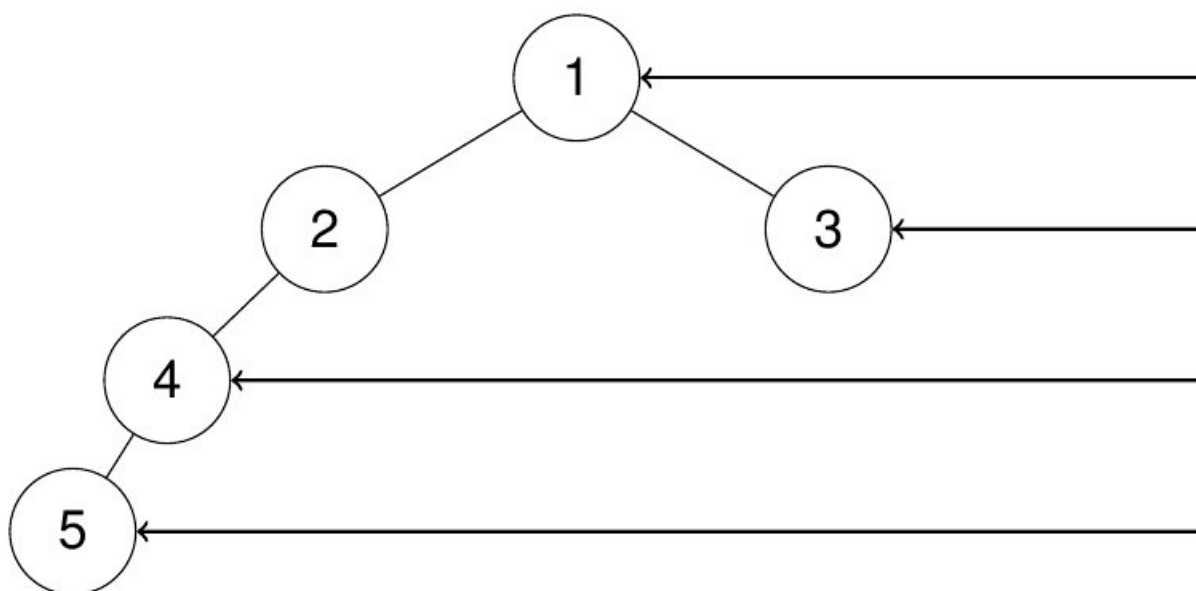


Example 2:

Input: root = [1,2,3,4,null,null,null,5]

Output: [1,3,4,5]

Explanation:



Example 3:

Input: root = [1,null,3]

Output: [1,3]

Example 4:

Input: root = []

Output: []

Constraints:

- The number of nodes in the tree is in the range [0, 100].
- $-100 \leq \text{Node.val} \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Imagine looking at the tree from the right. You see one node per level –
the rightmost node visible at that level. Return top to bottom. Right?"

"[1,2,3,null,5,null,4] → [1,3,4] ✓ (3 is rightmost at level 1, 4 at level 2)"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
BFS level-order; after processing each level, take the last node value added.
That last node is the rightmost visible from that depth.
Correct O(n); DFS tracking depth + rightmost-at-depth is equally fine.
Time: O(n). Space: O(w) queue width.
Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Same BFS snapshot as #102/#103. Just append the LAST element of each level's list.
Or DFS: visit right child first, add node.val when depth == len(result) (first node seen at new depth)."

PHASE 4 — CLEAN CODE

```
class _199_RightSideView:
    def rightSideView(self, root: Optional['TreeNode']) -> List[int]:
```



```

if not root:
    return []
result = []
queue = deque([root])
while queue:
    level_size = len(queue)
    for i in range(level_size):
        node = queue.popleft()
        if i == level_size - 1:
            result.append(node.val) # rightmost node of this level
        if node.left: queue.append(node.left)
        if node.right: queue.append(node.right)
    return result

```

PHASE 5 — TEST & DEBUG

[1,2,3,null,5,null,4]:

Level 0: [1] → last=1. Level 1: [2,3] → last=3. Level 2: [5,4] → last=4.

→ [1,3,4] ✓

[1,2,3,4,null,null,null,5]:

Level 0: [1]→1. Level 1: [2,3]→3. Level 2: [4]→4. Level 3: [5]→5. → [1,3,4,5] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(w)$ where $w = \text{max width}$.

This is #102 with one change: only record the last element per level.

Left side view: record the first element per level instead.

Given more time I'd ask: what if some right-side nodes are blocked by deeper left-side nodes?

The problem says 'visible from the right' – the rightmost node at each DEPTH level,

which BFS naturally gives us. A right child at depth 2 can block a left child at depth 2."

Trees & BST

□ #230 Kth Smallest Element in a BST

MED[Open on LeetCode](#)

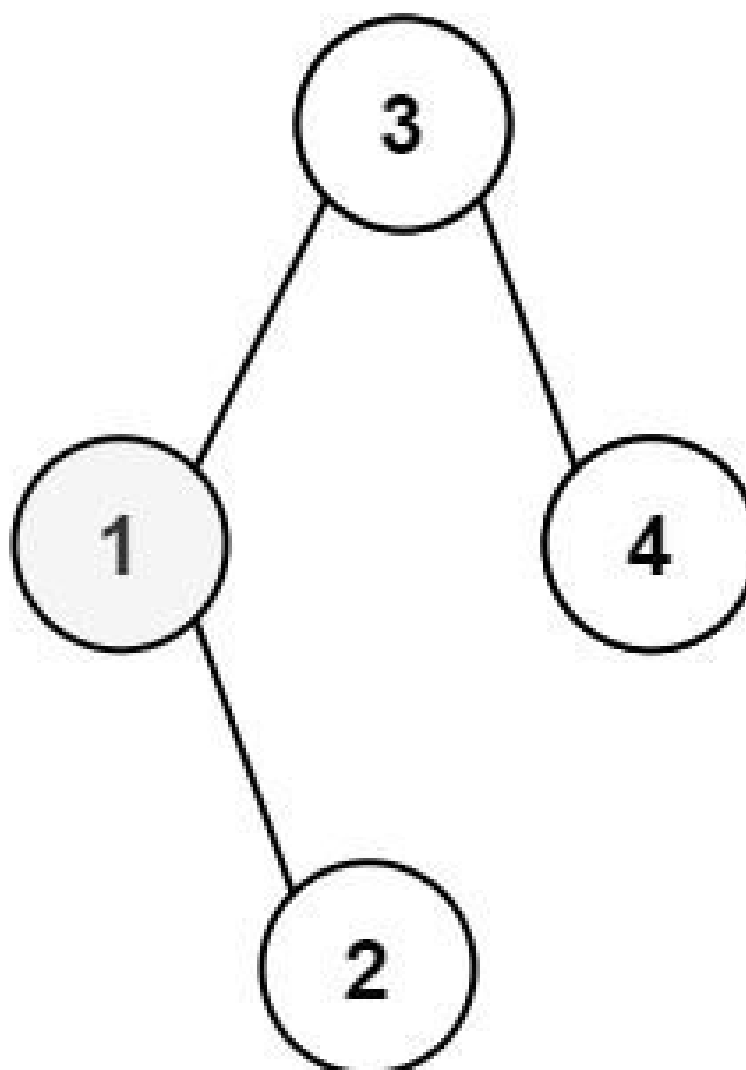
Tree · DFS · BST

Lists: Grind 169

Problem

Given the root of a binary search tree, and an integer k , return the k th smallest value (1-indexed) of all the values of the nodes in the tree.

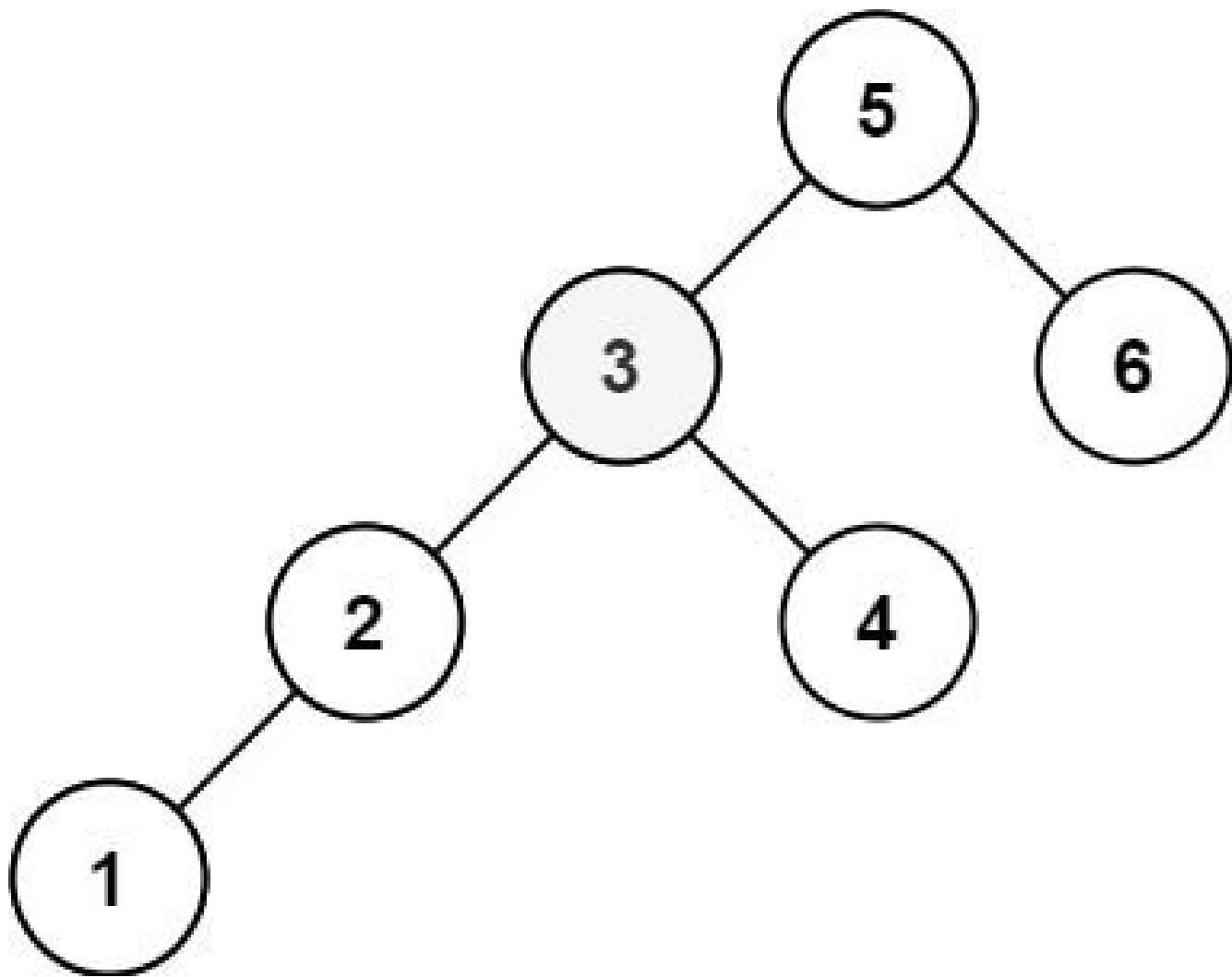
Example 1:



Input: root = [3,1,4,null,2], k = 1

Output: 1

Example 2:



Input: root = [5,3,6,2,4,null,null,1], k = 3
Output: 3

Constraints:

- The number of nodes in the tree is n.
- $1 \leq k \leq n \leq 10^4$
- $0 \leq \text{Node.val} \leq 10^4$

Follow up: If the BST is modified often (i.e., we can do insert and delete operations) and you need to find the kth smallest frequently, how would you optimize?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "BST in-order traversal (left → root → right) visits nodes in ascending sorted order.
# Return the k-th element in this order (1-indexed). Right?"
#
# "[3,1,4,null,2], k=1: in-order=[1,2,3,4]. 1st smallest=1 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# In-order traversal collects all node values in sorted order.
# Return the (k-1)th element from that list.
# Correct O(n) time and O(n) space; we can stop after k steps in-order.
# Time: O(n). Space: O(n) values list.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "In-order DFS stopping early at the k-th element. Use a counter and result variable
# in a nonlocal/list wrapper. Avoids collecting all n values if k is small."
```

PHASE 4 — CLEAN CODE

```
class _230_KthSmallestBST:
    def kthSmallest(self, root: Optional['TreeNode'], k: int) -> int:
        count = [0]
        result = [0]
        def inorder(node):
            if not node or count[0] == k:
                return
            inorder(node.left)
            count[0] += 1
            if count[0] == k:
                result[0] = node.val
                return
            inorder(node.right)
        inorder(root)
        return result[0]
```

PHASE 5 — TEST & DEBUG

```
# [3,1,4,null,2], k=1:
# inorder(3)→inorder(1)→inorder(None). count=0+1=1==k=1→result=1. return.
# → 1 ✓
# [5,3,6,2,4,null,null,1], k=3:
# in-order: 1,2,3,4,5,6. 3rd = 3 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(k + h)$  – at most  $k$  steps after reaching the leftmost leaf. Space  $O(h)$ .  
# Follow-up: frequent inserts/deletes + frequent  $k$ th queries?  
# Augment each BST node with the size of its left subtree.  
# Then  $k$ th smallest is an  $O(\log n)$  descent: if  $\text{left\_size} \geq k$  go left; if  
   $\text{left\_size} + 1 == k$  return root; else  $k -= \text{left\_size} + 1$ , go right.  
# Given more time I'd ask: what if we want the  $k$ th largest?  
# Reverse in-order (right→root→left) and count  $k$  steps."
```

Trees & BST

□ #235 Lowest Common Ancestor of a Binary Search Tree

MED

[Open on LeetCode](#)

Tree · DFS · BST

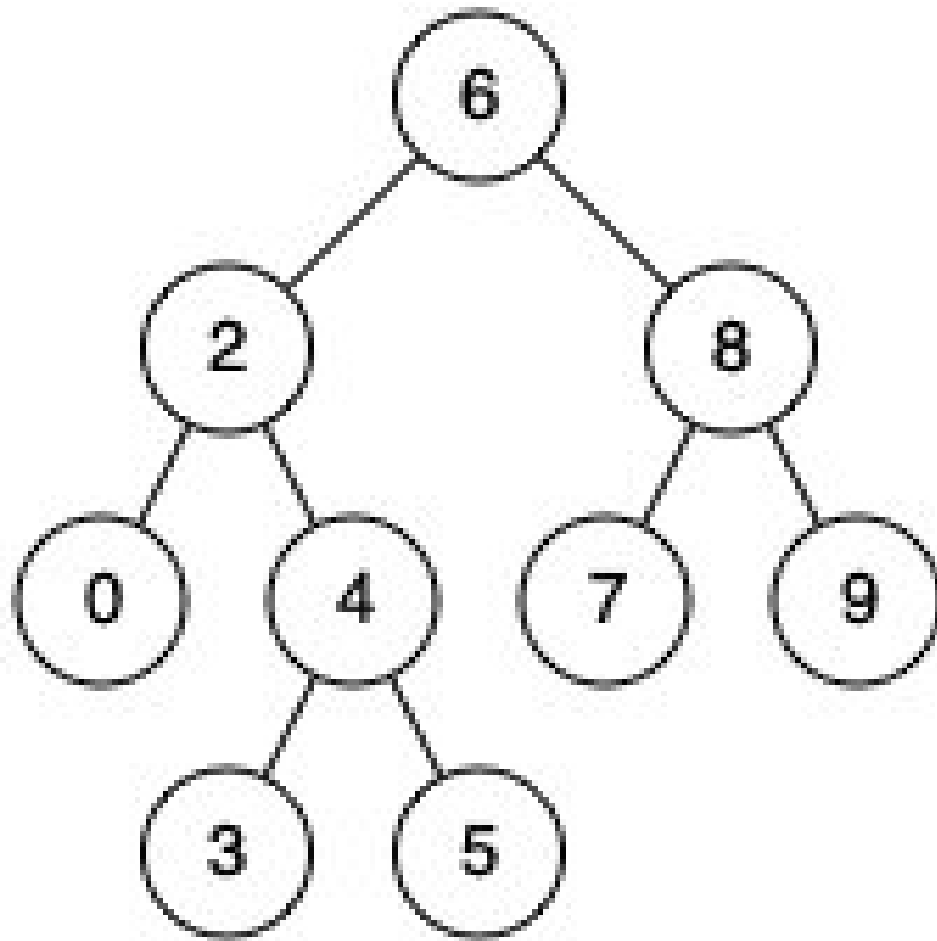
Lists: Grind 169

Problem

Given a binary search tree (BST), find the lowest common ancestor (LCA) node of two given nodes in the BST.

According to the definition of LCA on Wikipedia: “The lowest common ancestor is defined between two nodes p and q as the lowest node in T that has both p and q as descendants (where we allow a node to be a descendant of itself).”

Example 1:

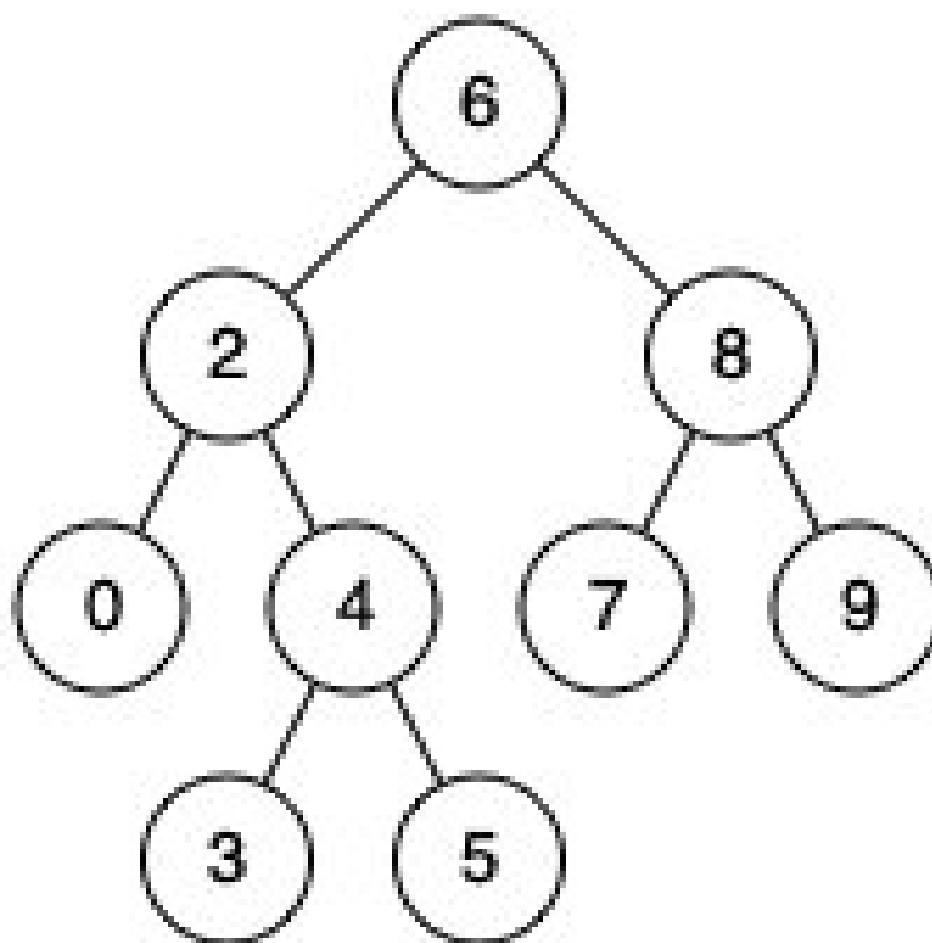


Input: root = [6,2,8,0,4,7,9,null,null,3,5], p = 2, q = 8

Output: 6

Explanation: The LCA of nodes 2 and 8 is 6.

Example 2:



Input: root = [6,2,8,0,4,7,9,null,null,3,5], p = 2, q = 4

Output: 2

Explanation: The LCA of nodes 2 and 4 is 2, since a node can be a descendant of itself according to the LCA definition.

Example 3:

Input: root = [2,1], p = 2, q = 1

Output: 2

Constraints:

- The number of nodes in the tree is in the range [2, 105].
- $-109 \leq \text{Node.val} \leq 109$

- All Node.val are unique.
- $p \neq q$
- p and q will exist in the BST.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "In a BST, find the LCA of nodes p and q.
# LCA = the deepest node that is an ancestor of both p and q (a node is its own
# ancestor). Right?
# Use the BST property: values to the left < root < values to the right."
#
# "[6,2,8,0,4,7,9], p=2, q=8 → LCA=6 (they're on different sides of 6) ✓
# p=2, q=4 → LCA=2 (4 is in 2's right subtree) ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Walk from root: if both p and q are smaller, go left; if both larger, go right.
# First node where paths split is the LCA in a BST.
# Alternatively store root-to-p and root-to-q paths – more work than the walk.
# Time: O(h). Space: O(h) if storing paths.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (BST property)

```
# "Navigate using BST property:
# If both p and q are LESS than root: LCA is in the LEFT subtree.
# If both are GREATER: LCA is in the RIGHT subtree.
# Otherwise (they split at root, or one equals root): root IS the LCA.
# O(h) time – O(log n) for balanced BST, O(n) worst case."
```

PHASE 4 — CLEAN CODE

```
class _235_LCA_BST:
    def lowestCommonAncestor(self, root: 'TreeNode', p: 'TreeNode', q: 'TreeNode')
    -> 'TreeNode':
        while root:
            if p.val < root.val and q.val < root.val:
                root = root.left
            elif p.val > root.val and q.val > root.val:
                root = root.right
            else:
                return root
        return root
```

PHASE 5 — TEST & DEBUG

```
# root=6, p=2, q=8: both in different sides → return 6 ✓
```

```
# root=6, p=2, q=4: both<6→go left to 2. p=2==root, q=4>root → split here → return 2
✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(h)$. Space $O(1)$ – iterative, no stack.

The BST property makes this an $O(h)$ walk – no need to visit all nodes.

This is much faster than the general binary tree LCA (#236) which needs DFS.

Given more time I'd ask: what if p or q might not exist in the tree?

We'd need to verify existence first – two separate searches, then LCA."

Trees & BST

#236 Lowest Common Ancestor of a Binary Tree

MED[Open on LeetCode](#)

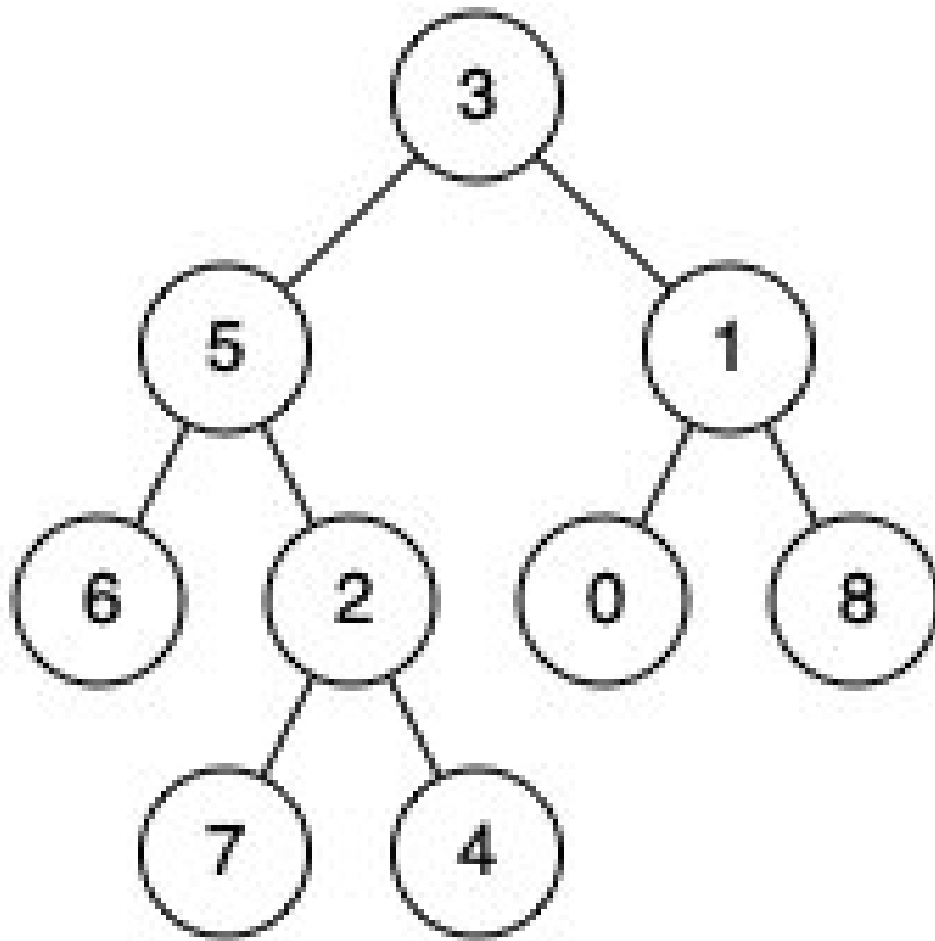
Tree · DFS

Lists: Grind 169

Problem

Given a binary tree, find the lowest common ancestor (LCA) of two given nodes in the tree. According to the definition of LCA on Wikipedia: “The lowest common ancestor is defined between two nodes p and q as the lowest node in T that has both p and q as descendants (where we allow a node to be a descendant of itself).”

Example 1:

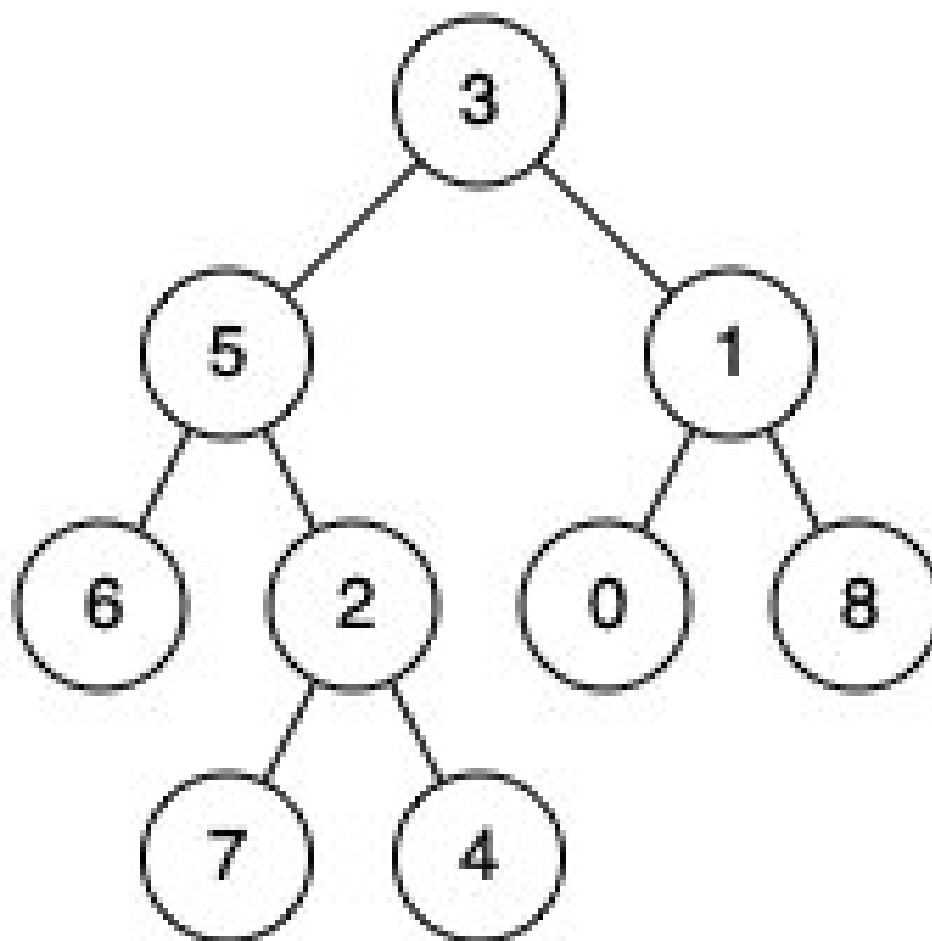


Input: root = [3,5,1,6,2,0,8,null,null,7,4], p = 5, q = 1

Output: 3

Explanation: The LCA of nodes 5 and 1 is 3.

Example 2:



Input: root = [3,5,1,6,2,0,8,null,null,7,4], p = 5, q = 4

Output: 5

Explanation: The LCA of nodes 5 and 4 is 5, since a node can be a descendant of itself according to the LCA definition.

Example 3:

Input: root = [1,2], p = 1, q = 2

Output: 1

Constraints:

- The number of nodes in the tree is in the range [2, 105].
- $-109 \leq \text{Node.val} \leq 109$

- All Node.val are unique.
- $p \neq q$
- p and q will exist in the tree.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "General binary tree – no BST property. Find the LCA of p and q.
# p and q are guaranteed to exist in the tree. A node is its own ancestor. Right?"
#
# "[3,5,1,6,2,0,8,null,null,7,4], p=5, q=1 → LCA=3 (root, both are in different
# subtrees) ✓
# p=5, q=4 → LCA=5 (4 is in 5's subtree) ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Find root-to-p path and root-to-q path as node lists.
# Walk both lists in parallel; last matching node before divergence is LCA.
# Correct O(n) but needs O(h) path storage; post-order single-pass is cleaner.
# Time: O(n). Space: O(h) two paths.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (recursive post-order)

```
# "DFS: if current node is p or q, return it (it or its ancestor is the LCA).
# Recurse left and right.
# If both return non-null: current node is the LCA (p and q are in different
# subtrees).
# If only one is non-null: that subtree contains both p and q – propagate that
# result up."
```

PHASE 4 — CLEAN CODE

```
class _236_LCA_BinaryTree:
    def lowestCommonAncestor(self, root: 'TreeNode', p: 'TreeNode', q: 'TreeNode')
    -> 'TreeNode':
        def lca(node):
            if not node or node is p or node is q:
                return node
            left = lca(node.left)
            right = lca(node.right)
            if left and right:
                return node # p and q are in different subtrees
            return left or right # both in same subtree – propagate the found node
        return lca(root)
```

PHASE 5 — TEST & DEBUG

```
# [3,5,1,...], p=5, q=1:
# lca(3): lca(5)→returns 5 (found p). lca(1)→returns 1 (found q).
# Both non-null → return 3 ✓
# p=5, q=4 (4 is under 5):
# lca(3): lca(5)→... lca(4) under 5 returns 4. lca(5) itself: node is p→return 5.
# lca(1)→None (neither p nor q in right subtree).
# left=5, right=None → return 5 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$  – visits every node. Space  $O(h)$  for the call stack.
# Contrast with #235 (BST): BST LCA is  $O(h)$  iteratively. General tree needs full DFS.
# The post-order logic is the key: left and right results tell us which subtrees contain p/q.
# Both non-null → split here → current node is LCA.
# Only one non-null → both are in the same subtree → propagate that subtree's result.
# Given more time I'd ask: what if p or q might not exist?
# We'd need to track 'found p' and 'found q' separately – more complex state."
```


Trees & BST

★ #250 Count Univalue Subtrees ★

MED[Open on LeetCode](#)

Tree · DFS

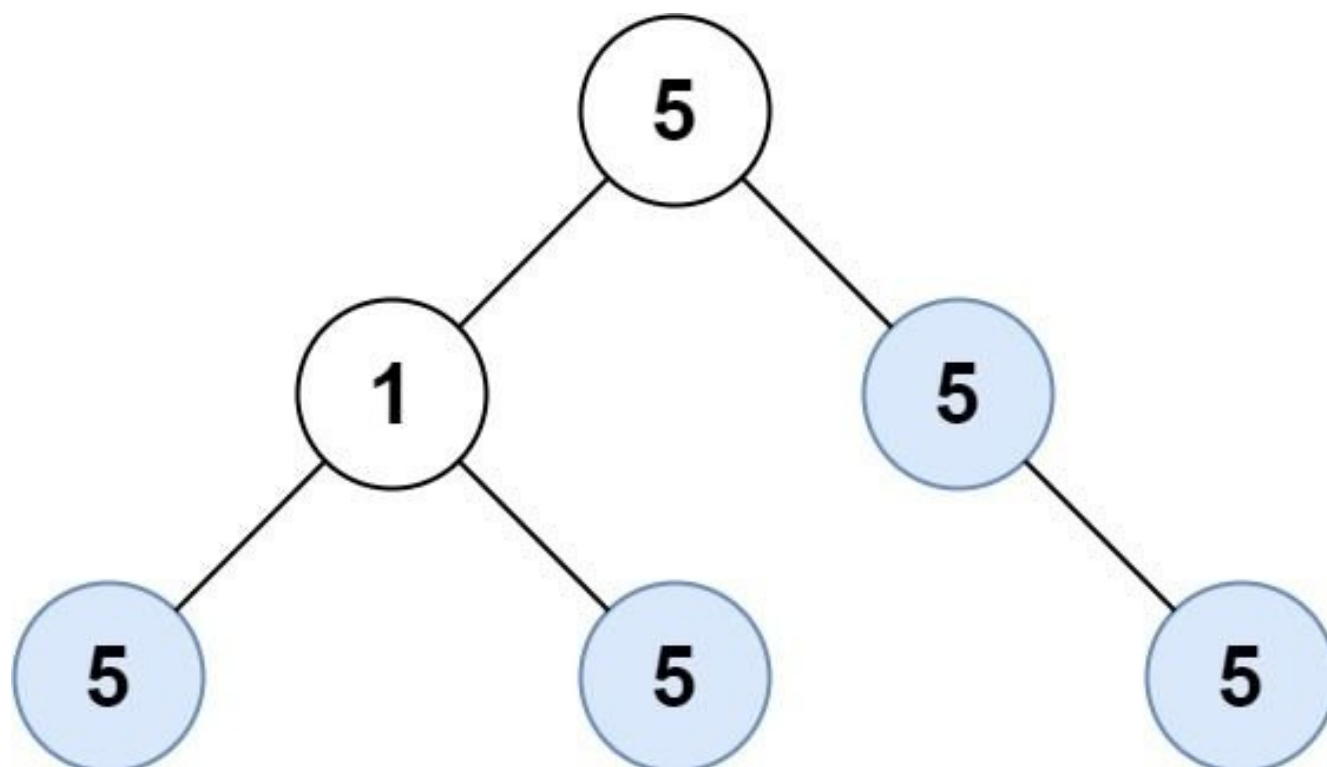
Lists: ★ Premium | Premium 98

Problem

Given the root of a binary tree, return the number of uni-value subtrees.

A uni-value subtree means all nodes of the subtree have the same value.

Example 1:



Input: root = [5,1,5,5,5,null,5]
Output: 4

Example 2:

Input: root = []
Output: 0

Example 3:

Input: root = [5,5,5,5,5,null,5]
Output: 6

Constraints:

- The number of the node in the tree will be in the range [0, 1000].
- $-1000 \leq \text{Node.val} \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"A univalue subtree has every node with the same value.
Count all such subtrees – including single leaf nodes. Right?
Empty tree → 0."

"[5,1,5,5,5,null,5] → 4 (the four nodes with value 5 that form valid univalue subtrees) ✓"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
For each node, DFS its entire subtree checking every value equals node.val.
Increment count when the whole subtree is uni-value.
Revalidates overlapping subtrees – $O(n^2)$ worst case.
Time: $O(n^2)$. Space: $O(h)$ stack.
Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (post-order DFS)

"Bottom-up: a node is a univalue root if both children are univalue AND their values match.
Return True from a node if its subtree is univalue, increment count when True.
Use an inner helper that returns bool + mutates a count list."

PHASE 4 — CLEAN CODE

```

class _250_CountUnivalSubtrees:
    def countUnivalSubtrees(self, root: Optional['TreeNode']) -> int:
        count = [0]
        def is_univalue(node, parent_val):
            if not node:
                return True
            left = is_univalue(node.left, node.val)
            right = is_univalue(node.right, node.val)
            if not left or not right:
                return False
            count[0] += 1
            return node.val == parent_val
        is_univalue(root, None)
        return count[0]

```

PHASE 5 — TEST & DEBUG

```

# [5,1,5,5,5,null,5]:
# Leaves: 5(left-left)->True, 5(left-right)->True, 5(right-right)->True, 1->True (leaf).
# Node(1): left=True, right=True, both return True. 1's children have val=5≠1 → returns False.
# Node(5, right of root): left=None=True, right(5)=True, right returns True(val matches). count=1. Returns True(5==root.val?).
# Continue... count ends at 4 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n). Space O(h). Post-order because we need children's results before parent's.
# The parent_val parameter lets us check if current node extends the parent's univalue property.
# Given more time I'd ask: longest path where all values are the same?
# Track univalue path length similar to Diameter of Binary Tree — carry length down."

```

Trees & BST

□ #285 Inorder Successor in BST

MED[Open on LeetCode](#)

Tree · DFS · BST

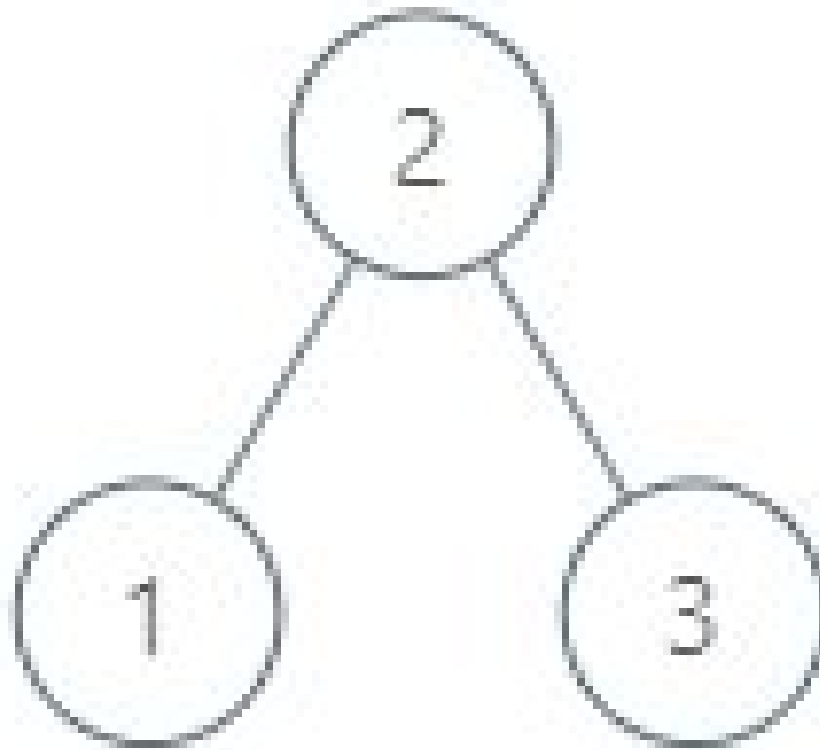
Lists: Grind 169

Problem

Given the root of a binary search tree and a node p in it, return the in-order successor of that node in the BST. If the given node has no in-order successor in the tree, return null.

The successor of a node p is the node with the smallest key greater than $p.val$.

Example 1:

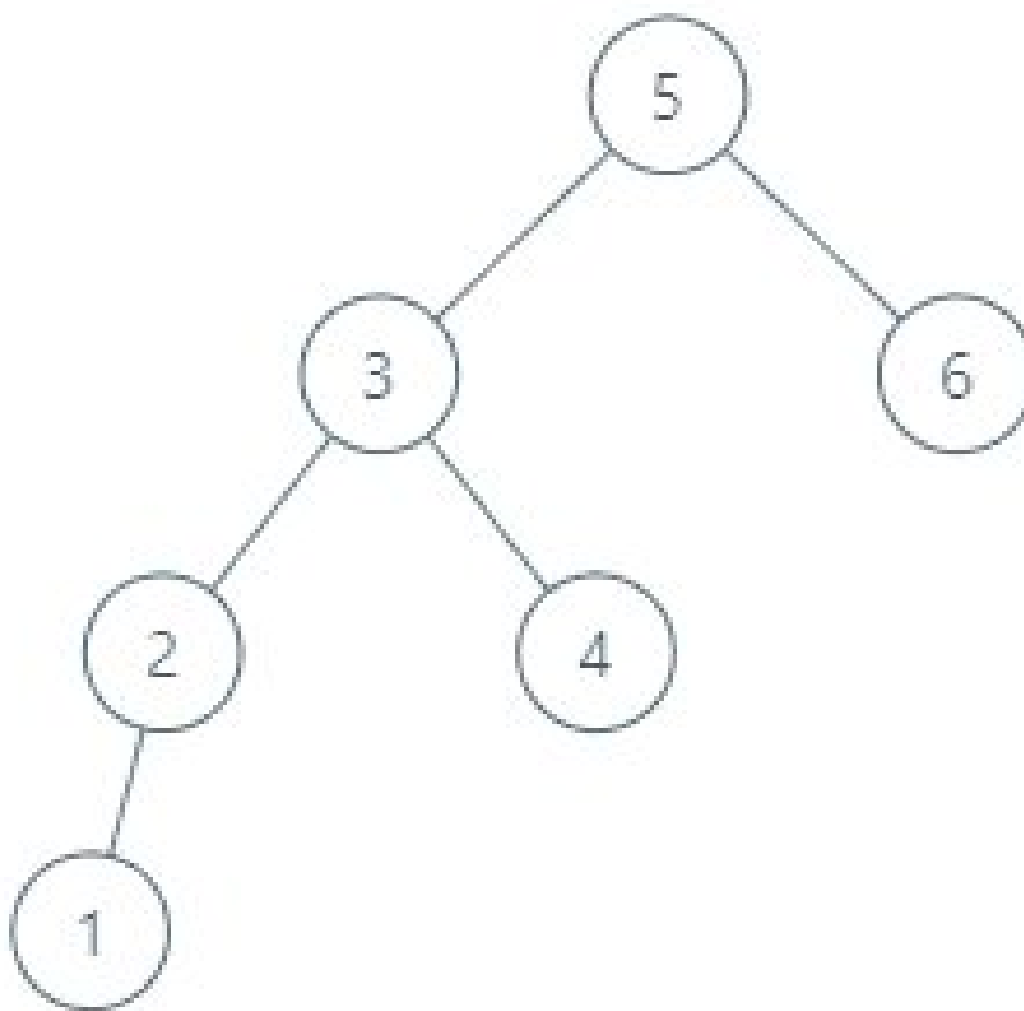


Input: root = [2,1,3], p = 1

Output: 2

Explanation: 1's in-order successor node is 2. Note that both p and the return value is of TreeNode type.

Example 2:



Input: root = [5,3,6,2,4,null,null,1], p = 6

Output: null

Explanation: There is no in-order successor of the current node, so the answer is null.

Constraints:

- The number of nodes in the tree is in the range [1, 104].
- $-105 \leq \text{Node.val} \leq 105$
- All Nodes will have unique values.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find the node with the smallest value strictly greater than p.val in a BST.
# Return null if no such node exists (p is the largest). Right?"
#
# "[2,1,3], p=1 → successor=2 (next in in-order: 1,2,3) ✓
# [5,3,6,2,4,null,null,1], p=6 → null (6 is the largest) ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# In-order traversal of the BST; when we visit p, the next node in-order is
# successor.
# Store previous node during traversal or collect all values then scan.
# Correct O(n); BST structure gives O(h) successor without full traversal.
# Time: O(n). Space: O(1) iterative with threading concept.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (BST navigation)

```
# "Use BST property – no need to traverse all nodes.
# If root.val > p.val: root is a candidate successor, but there might be a smaller
# one in the left subtree.
# If root.val <= p.val: successor must be in the right subtree.
# Track the best candidate seen so far. O(h) time."
```

PHASE 4 — CLEAN CODE

```
class _285_InorderSuccessor:
    def inorderSuccessor(self, root: 'TreeNode', p: 'TreeNode') ->
Optional['TreeNode']:
    successor = None
    while root:
        if root.val > p.val:
            successor = root # root is a candidate – try for smaller in left
            root = root.left
        else:
            root = root.right # successor must be to the right
    return successor
```

PHASE 5 — TEST & DEBUG

```
# [2,1,3], p=1:
# root=2: 2>1 → successor=2, go left.
# root=1: 1<=1 → go right.
# root=None → return 2 ✓
# [5,3,6,2,4,null,null,1], p=6:
# root=5: 5<=6 → go right. root=6: 6<=6 → go right. root=None → return None ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(h). Space O(1) – iterative.
# The candidate tracking is the key: every time we go left, the current node becomes
```

```
# a potential successor (it's bigger than p, and we're looking for something
smaller).
# Given more time I'd ask: inorder PREDECESSOR (largest value < p.val)?
# Mirror logic: if root.val < p.val → candidate, go right. If root.val >= p.val → go
left."
```


Trees & BST

□ ★ #298 Binary Tree Longest Consecutive Sequence ★

MED[Open on LeetCode](#)

Tree · DFS

Lists: ★ Premium | Premium 98

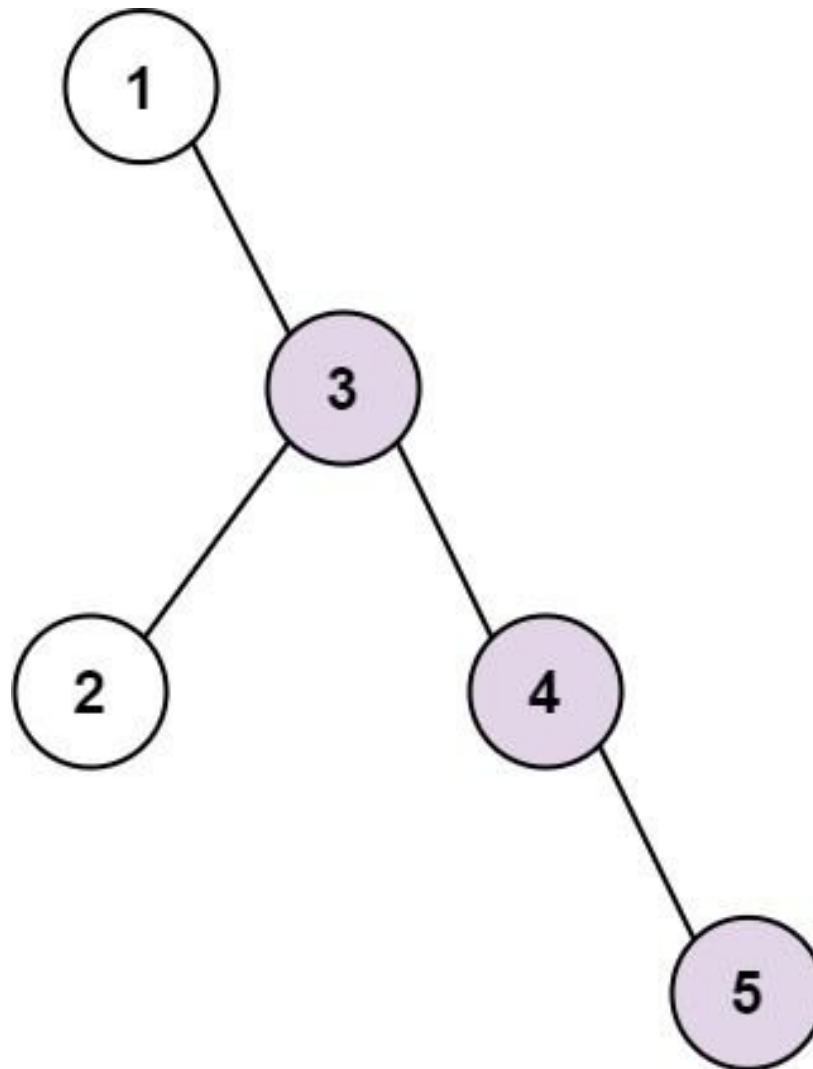
Problem

Given the root of a binary tree, return the length of the longest consecutive sequence path.

A consecutive sequence path is a path where the values increase by one along the path.

Note that the path can start at any node in the tree, and you cannot go from a node to its parent in the path.

Example 1:

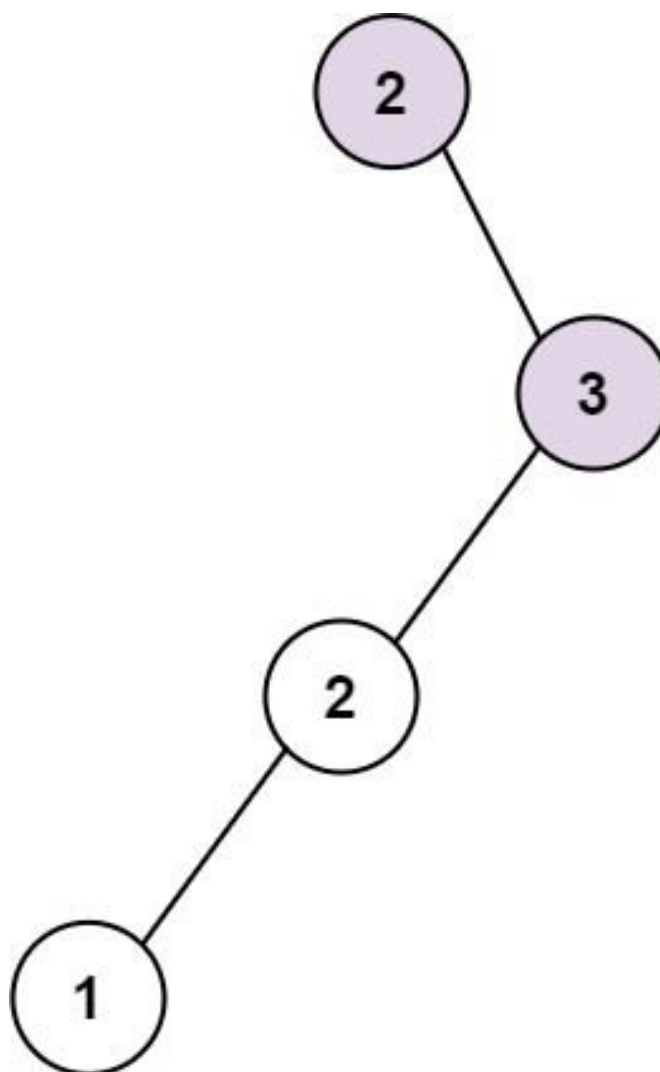


Input: root = [1,null,3,2,4,null,null,null,5]

Output: 3

Explanation: Longest consecutive sequence path is 3-4-5, so return 3.

Example 2:



Input: root = [2,null,3,2,null,1]

Output: 2

Explanation: Longest consecutive sequence path is 2-3, not 3-2-1, so return 2.

Constraints:

- The number of nodes in the tree is in the range $[1, 3 * 10^4]$.
- $-3 * 10^4 \leq \text{Node.val} \leq 3 * 10^4$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find the longest path of consecutive increasing values (each node = parent + 1)."

```

# Path goes parent → child – you can't go back up. Path can start at any node.
Right?"
#
# "[1,null,3,2,4,null,null,null,5]: path 3→4→5, length 3 ✓
# [2,null,3,2,null,1]: 2→3 is increasing, not 3→2 – so max is 2 ✓"

# PHASE 2 — BRUTE FORCE
# "Let me start with brute force to lock in the logic.
# From each node, DFS tracking consecutive length when child.val == parent.val + 1.
# Reset streak when the +1 chain breaks; track global maximum.
# Starting DFS from every node repeats work – post-order aggregation is faster.
# Time:  $O(n^2)$  naive restarts. Space:  $O(h)$  stack.
# Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE (DFS carrying current length)
# "DFS top-down: pass current consecutive length to each child.
# If child.val == parent.val + 1: extend the sequence.
# Else: reset to 1 (start a new sequence from this child).
# Track global max. Use inner helper with parent_val and current_length."

# PHASE 4 — CLEAN CODE
class _298_LongestConsecutive:
    def longestConsecutive(self, root: Optional['TreeNode']) -> int:
        best = [0]

        def dfs(node, parent_val, length):
            if not node:
                return
            length = length + 1 if node.val == parent_val + 1 else 1
            best[0] = max(best[0], length)
            dfs(node.left, node.val, length)
            dfs(node.right, node.val, length)

        dfs(root, root.val - 1 if root else 0, 0)
        return best[0]

# PHASE 5 — TEST & DEBUG
# [1,null,3,2,4,null,null,null,5]:
# dfs(1, 0, 0): val=1, prev+1=1 → length=1. best=1. dfs(3,1,1).
# dfs(3, 1, 1): val=3, prev+1=2, 3≠2 → length=1. best=1. dfs(2,3,1).
# dfs(4, 3, 1): val=4==3+1 → length=2. best=2. dfs(5,4,2).
# dfs(5, 4, 2): val=5==4+1 → length=3. best=3. → 3 ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(n)$ . Space  $O(h)$ .
# The initial call uses root.val-1 so the root itself resets to length 1 cleanly.
# This is a top-down DFS – pass state downward. Contrast with bottom-up (#250, #543).
# Given more time I'd ask: what if the path can go through any node (not just parent→child)?
# That's Longest Consecutive Sequence in a Binary Tree – post-order, track both ascending

```

and descending sequences at each node."

Trees & BST

□ ★ #314 Binary Tree Vertical Order Traversal ★

MED[Open on LeetCode](#)

Hash Table · Tree · BFS · Sorting

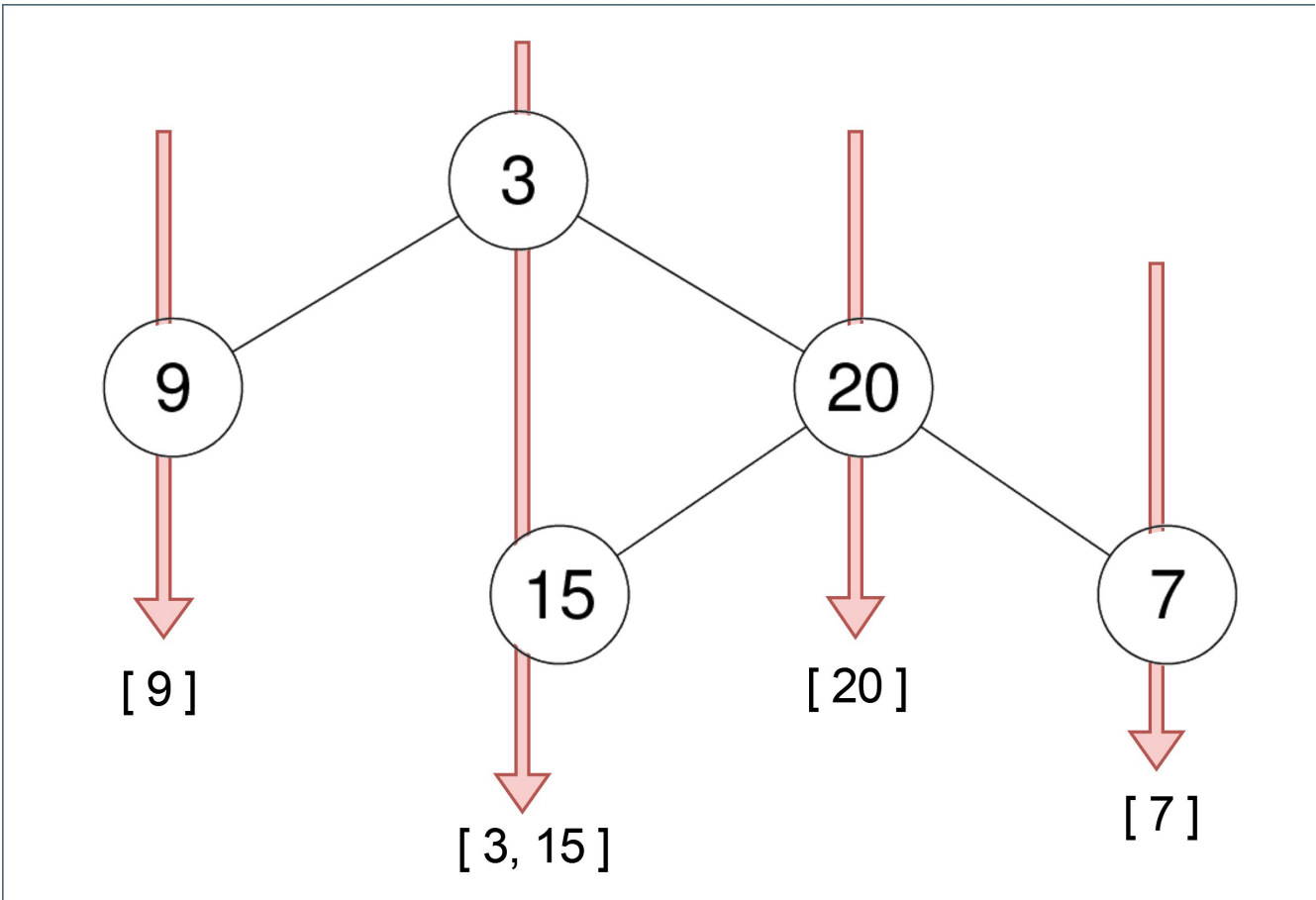
Lists: ★ Premium | Premium 98

Problem

Given the root of a binary tree, return the vertical order traversal of its nodes' values. (i.e., from top to bottom, column by column).

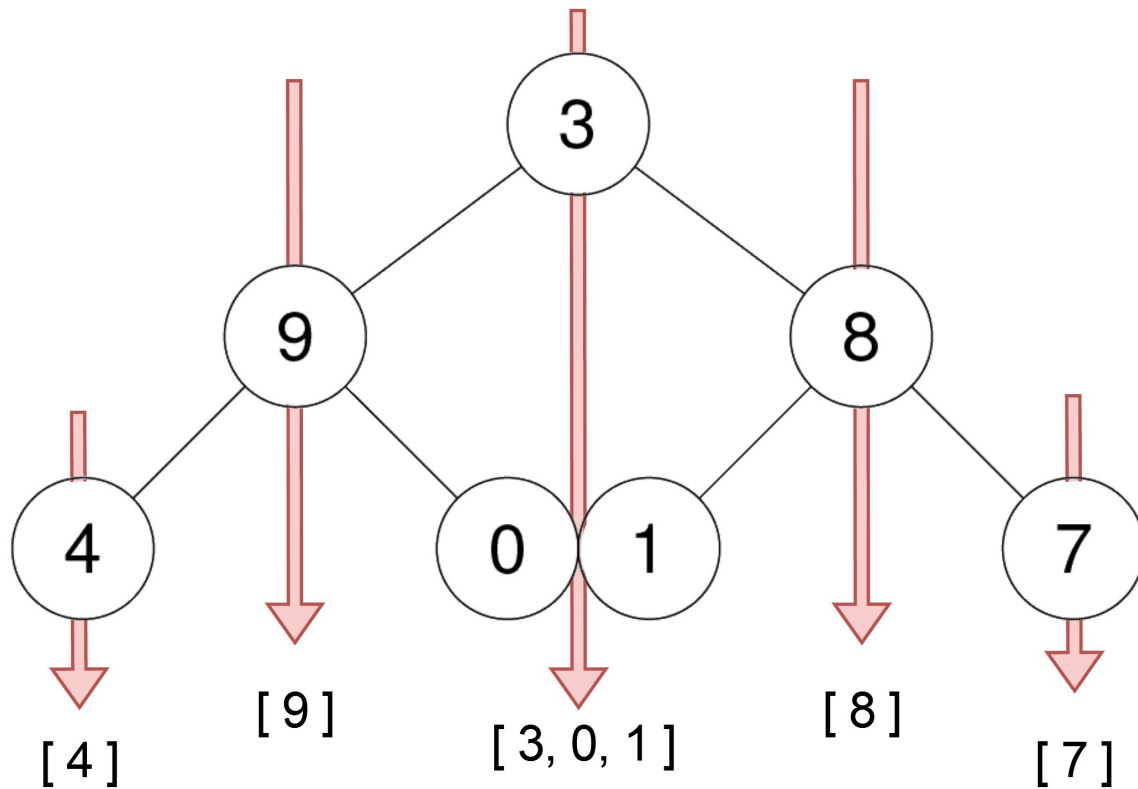
If two nodes are in the same row and column, the order should be from left to right.

Example 1:



Input: root = [3,9,20,null,null,15,7]
Output: [[9],[3,15],[20],[7]]

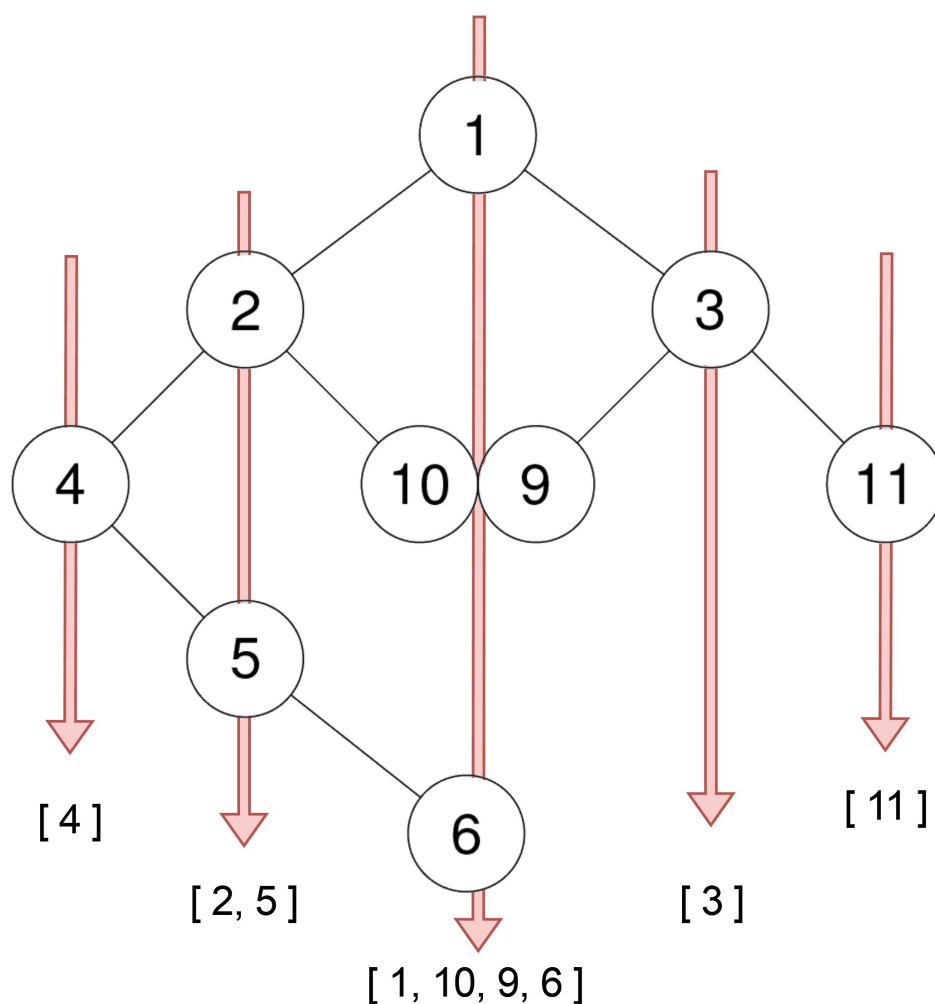
Example 2:



Input: root = [3,9,8,4,0,1,7]

Output: [[4],[9],[3,0,1],[8],[7]]

Example 3:



Input: root = [1,2,3,4,10,9,11,null,5,null,null,null,null,null,null,6]
 Output: [[4],[2,5],[1,10,9,6],[3],[11]]

Constraints:

- The number of nodes in the tree is in the range [0, 100].
- $-100 \leq \text{Node.val} \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Group nodes by column. Root is column 0. Left child = col-1. Right child = col+1."

```

# Within each column, top-to-bottom. Ties (same row and column): left before right.
Right?
# Return columns sorted from leftmost to rightmost."
#
# "[3,9,20,null,null,15,7] → [[9],[3,15],[20],[7]] ✓"

# PHASE 2 — BRUTE FORCE
# "Let me start with brute force to lock in the logic.
# DFS assigning column index (left=-1, right=+1 relative to parent).
# Collect (col, row, val) tuples; sort by (col, row, val) then group by column.
# Correct but sorting dominates at O(n log n).
# Time: O(n log n). Space: O(n) tuples.
# Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE (BFS with column tracking)
# "BFS ensures natural top-to-bottom, left-to-right order within each column.
# Queue entries: (node, col). Use defaultdict(list) keyed by col.
# After BFS, sort the column keys and return their lists.
# Time: O(n log n) for sorting. Space: O(n)."

# PHASE 4 — CLEAN CODE
class _314_VerticalOrderTraversal:
    def verticalOrder(self, root: Optional['TreeNode']) -> List[List[int]]:
        if not root:
            return []
        cols : defaultdict = defaultdict(list)
        queue = deque([(root, 0)])
        min_col = max_col = 0
        while queue:
            node, col = queue.popleft()
            cols[col].append(node.val)
            min_col = min(min_col, col)
            max_col = max(max_col, col)
            if node.left: queue.append((node.left, col - 1))
            if node.right: queue.append((node.right, col + 1))
        return [cols[c] for c in range(min_col, max_col + 1)]

# PHASE 5 — TEST & DEBUG
# [3,9,20,null,null,15,7]:
# BFS: (3,0)→cols={0:[3]}. (9,-1),(20,1)→cols={0:[3],-1:[9],1:[20]}.
# (15,0),(7,2)→cols={0:[3,15],-1:[9],1:[20],2:[7]}.
# min=-1, max=2. Return [cols[-1],cols[0],cols[1],cols[2]] = [[9],[3,15],[20],[7]]
✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(n) for BFS + O(cols) for building the result – total O(n).
# Using min_col/max_col avoids sorting keys – the range is contiguous.
# BFS guarantees top-to-bottom, left-to-right order within each column naturally.
# DFS would require explicit (row, insertion_order) sorting per column.
# Given more time I'd ask: Vertical Order Traversal II (#987)?

```

Same column/row but within the same position, sort by value ascending – needs explicit row tracking."

Trees & BST

□ ★ #333 Largest BST Subtree ★

MED

[Open on LeetCode](#)

Dynamic Programming · Tree · DFS · BST

Lists: ★ Premium | Premium 98

Problem

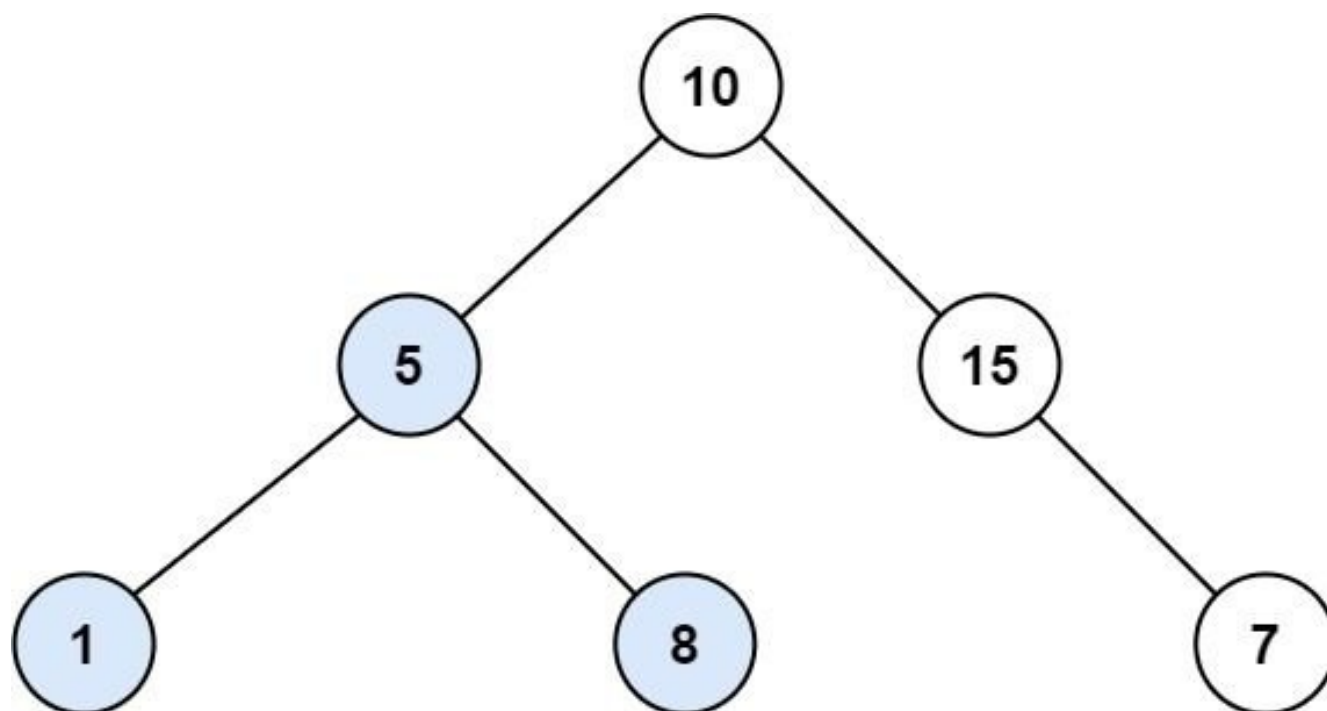
Given the root of a binary tree, find the largest subtree, which is also a Binary Search Tree (BST), where the largest means subtree has the largest number of nodes.

A Binary Search Tree (BST) is a tree in which all the nodes follow the below-mentioned properties:

- The left subtree values are less than the value of their parent (root) node's value.
- The right subtree values are greater than the value of their parent (root) node's value.

Note: A subtree must include all of its descendants.

Example 1:



Input: root = [10,5,15,1,8,null,7]

Output: 3

Explanation: The Largest BST Subtree in this case is the highlighted one. The return value is the subtree's size, which is 3.

Example 2:

Input: root = [4,2,7,2,3,5,null,2,null,null,null,null,null,1]

Output: 2

Constraints:

- The number of nodes in the tree is in the range [0, 104].
- $-104 \leq \text{Node.val} \leq 104$

Follow up: Can you figure out ways to solve it with $O(n)$ time complexity?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Find the largest subtree (by node count) that forms a valid BST.
# Return the node count. Right?"
#
# "[10,5,15,1,8,null,7]: subtree at 5 (nodes 1,5,8) is a BST → 3 ✓"

# PHASE 2 — BRUTE FORCE
# "Let me start with brute force to lock in the logic.
# For every node as root of a candidate subtree, validate BST property recursively
# (min/max bounds) and count nodes if valid. Track global max count.
# Revalidates overlapping subtrees –  $O(n^2)$  worst case.
# Time:  $O(n^2)$ . Space:  $O(h)$  stack.
# Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE ( $O(n)$  post-order)
# "Post-order: return (is_bst, size, subtree_min, subtree_max).
# Empty node: (True, 0, +inf, -inf) – sentinels pass any comparison cleanly.
# Valid BST: both children are BST AND left.max < node.val < right.min."

# PHASE 4 — CLEAN CODE
class _333_LargestBSTSubtree:
    def largestBSTSubtree(self, root: Optional['TreeNode']) -> int:
        best = [0]
        def check(node):
            if not node:
                return True, 0, float('inf'), float('-inf')
            l_ok, l_sz, l_mn, l_mx = check(node.left)
            r_ok, r_sz, r_mn, r_mx = check(node.right)
            if l_ok and r_ok and l_mx < node.val < r_mn:
                sz = l_sz + r_sz + 1
                best[0] = max(best[0], sz)
                return True, sz, min(l_mn, node.val), max(r_mx, node.val)
            return False, 0, 0, 0
        check(root)
        return best[0]

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(n)$ . Space  $O(h)$ . Sentinels (+inf/-inf for empty) let comparisons pass
# cleanly.
# Given more time I'd ask: what if we want the root of the largest BST subtree?
# Track it alongside size."

```

Trees & BST

★ #366 Find Leaves of Binary Tree ★

MED[Open on LeetCode](#)

Tree · DFS · Sorting

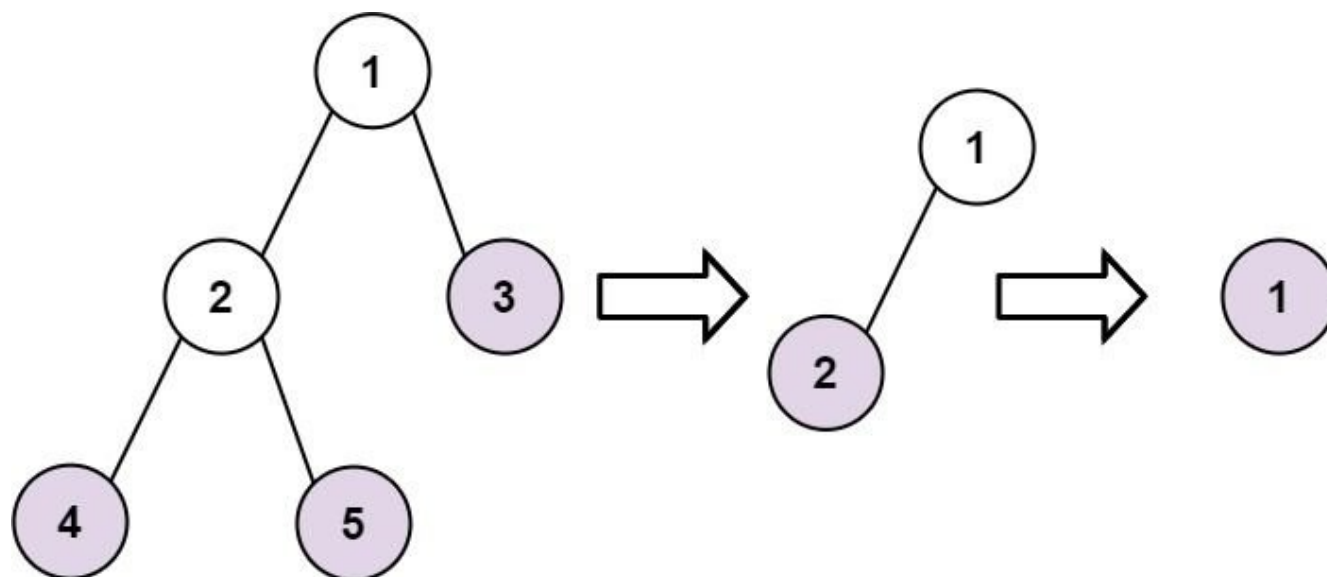
Lists: ★ Premium | Premium 98

Problem

Given the root of a binary tree, collect a tree's nodes as if you were doing this:

- Collect all the leaf nodes.
- Remove all the leaf nodes.
- Repeat until the tree is empty.

Example 1:



Input: root = [1,2,3,4,5]

Output: [[4,5,3],[2],[1]]

Explanation:

[[3,5,4],[2],[1]] and [[3,4,5],[2],[1]] are also considered correct answers since per each level it does not matter the order on which elements are returned.

Example 2:

Input: root = [1]

Output: [[1]]

Constraints:

- The number of nodes in the tree is in the range [1, 100].
- $-100 \leq \text{Node.val} \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Collect leaves, remove them, repeat. Group nodes by their 'height from leaf' level. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Repeatedly find all current leaves (no children), remove them, append to result level.

Each round peels one tree layer – like reverse level-order.

Time: $O(n^2)$ if we rescan whole tree each round. Space: $O(n)$ output.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Post-order height: leaf=0, internal=max(left,right)+1. Group by height in one pass."

PHASE 4 — CLEAN CODE

```
class _366_FindLeaves:
    def findLeaves(self, root: Optional['TreeNode']) -> List[List[int]]:
        result: List[List[int]] = []
        def height(node) -> int:
            if not node:
                return -1
            h = max(height(node.left), height(node.right)) + 1
            return h
        while root:
            root = self._removeLeaves(root)
            result.append([node.val for node in root])
        return result
```



```
        if h == len(result):
            result.append([])
        result[h].append(node.val)
        return h
    height(root)
    return result
```

PHASE 5 — TEST & DEBUG

```
# [1,2,3,4,5]: height(4)=0→result=[[4]]. height(5)=0→[[4,5]].
height(2)=1→[[4,5],[2]].
# height(3)=0→[[4,5,3],[2]]. height(1)=2→[[4,5,3],[2],[1]] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ . Space  $O(n)$ . One DFS pass – no repeated tree traversals.
# Given more time I'd ask: group by depth from root? That's standard level-order
BFS."
```

Trees & BST

□ #437 Path Sum III

MED

[Open on LeetCode](#)

Tree · DFS · Prefix Sum

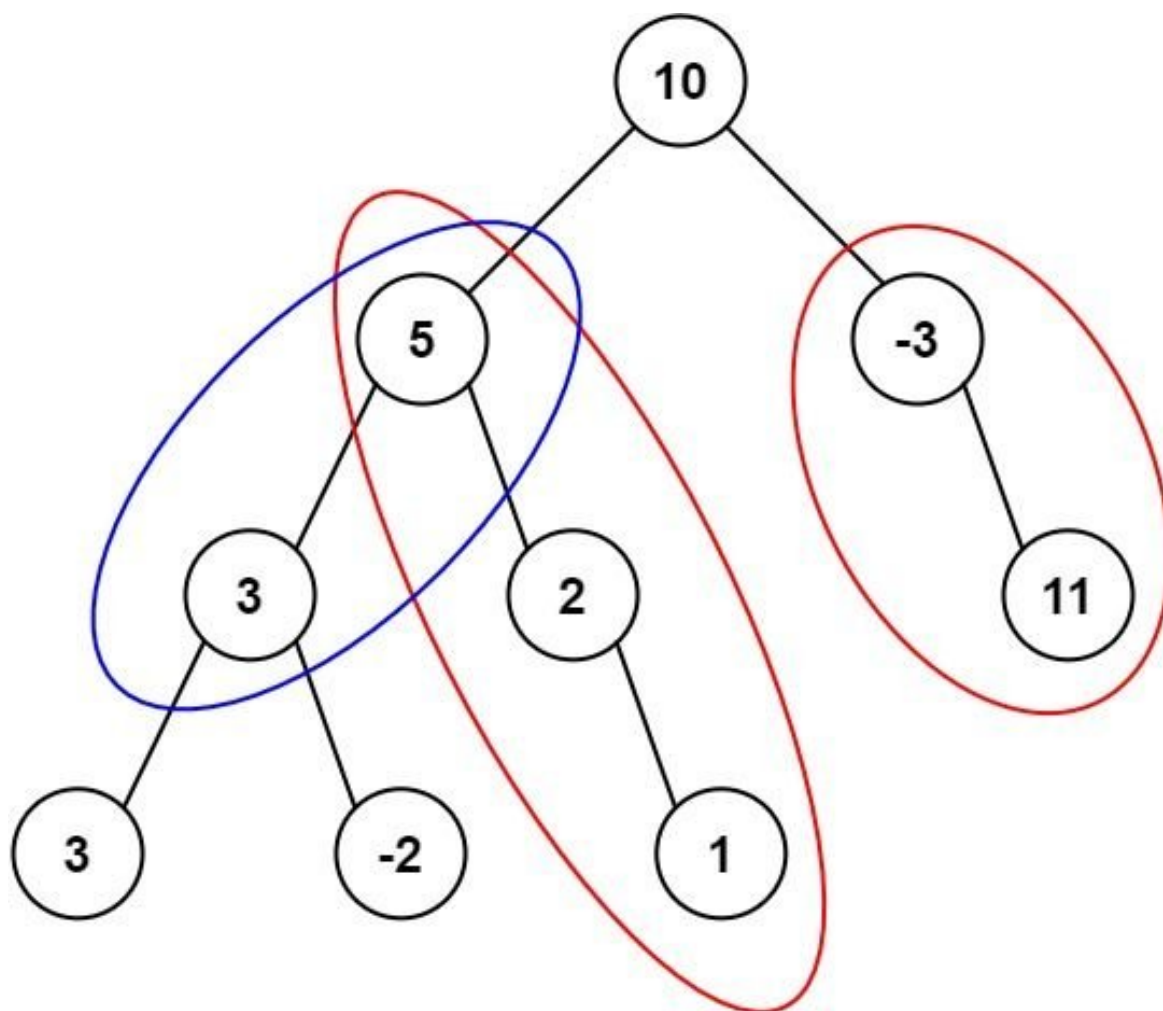
Lists: Grind 169

Problem

Given the root of a binary tree and an integer `targetSum`, return the number of paths where the sum of the values along the path equals `targetSum`.

The path does not need to start or end at the root or a leaf, but it must go downwards (i.e., traveling only from parent nodes to child nodes).

Example 1:



Input: root = [10,5,-3,3,2,null,11,3,-2,null,1], targetSum = 8

Output: 3

Explanation: The paths that sum to 8 are shown.

Example 2:

Input: root = [5,4,8,11,null,13,4,7,2,null,null,5,1], targetSum = 22

Output: 3

Constraints:

- The number of nodes in the tree is in the range [0, 1000].
- $-109 \leq \text{Node.val} \leq 109$

- $-1000 \leq \text{targetSum} \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Count downward paths (parent→child only) summing to targetSum.
# Path can start and end at any node. Values can be negative. Right?"
#
# "[10,5,-3,3,2,null,11,3,-2,null,1], target=8 → 3 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Prefix-sum map: as DFS walks, track cumulative sum from root.
# At each node, add count of prefix sums equal to (current_sum - target) - includes
# paths through node.
# Backtrack prefix counts when leaving node.
# Time: O(n). Space: O(n) prefix map + stack.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (prefix sum + hash map — same pattern as #560)

```
# "DFS with running prefix sum. Count how many earlier prefix sums equal
# current_sum-target.
# Backtrack after each subtree to avoid polluting sibling paths. Seed {0:1}."
```

PHASE 4 — CLEAN CODE

```
class _437_PathSumIII:
    def pathSum(self, root: Optional['TreeNode'], targetSum: int) -> int:
        prefix = Counter({0: 1})
        count = [0]
        def dfs(node, running):
            if not node:
                return
            running += node.val
            count[0] += prefix[running - targetSum]
            prefix[running] += 1
            dfs(node.left, running)
            dfs(node.right, running)
            prefix[running] -= 1 # backtrack
        dfs(root, 0)
        return count[0]
```

PHASE 5 — TEST & DEBUG

```
# This is #560 applied to trees – same prefix-sum logic, but backtrack after each
# subtree.
# root=10, target=8: paths [5,3], [5,2,1], [-3,11] each sum to 8 → count=3 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(n). Identical to Subarray Sum Equals K applied to a tree.  
# Backtracking prevents cross-subtree contamination – critical difference from  
array version.  
# Given more time I'd ask: paths going in both directions (not just downward)?  
# Use the any-direction diameter-style post-order approach."
```

Trees & BST

□ ★ #545 Boundary of Binary Tree ★

MED

[Open on LeetCode](#)

Tree · DFS

Lists: ★ Premium | Premium 98

Problem

The boundary of a binary tree is the concatenation of the root, the left boundary, the leaves ordered from left-to-right, and the reverse order of the right boundary.

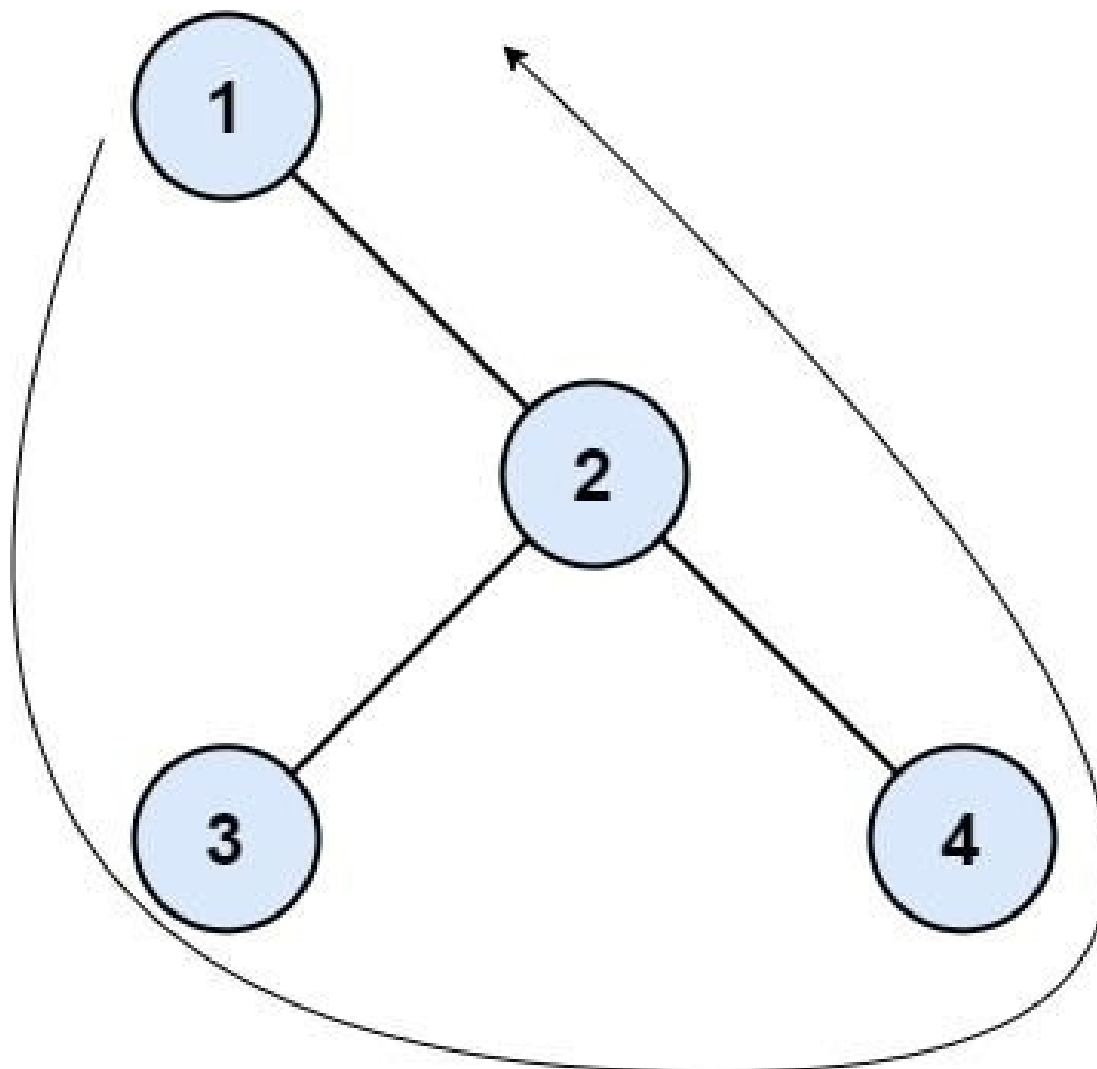
The left boundary is the set of nodes defined by the following:

- The root node's left child is in the left boundary. If the root does not have a left child, then the left boundary is empty.
- If a node is in the left boundary and has a left child, then the left child is in the left boundary.
- If a node is in the left boundary, has no left child, but has a right child, then the right child is in the left boundary.
- The leftmost leaf is not in the left boundary.

The right boundary is similar to the left boundary, except it is the right side of the root's right subtree. Again, the leaf is not part of the right boundary, and the right boundary is empty if the root does not have a right child.

The leaves are nodes that do not have any children. For this problem, the root is not a leaf. Given the root of a binary tree, return the values of its boundary.

Example 1:



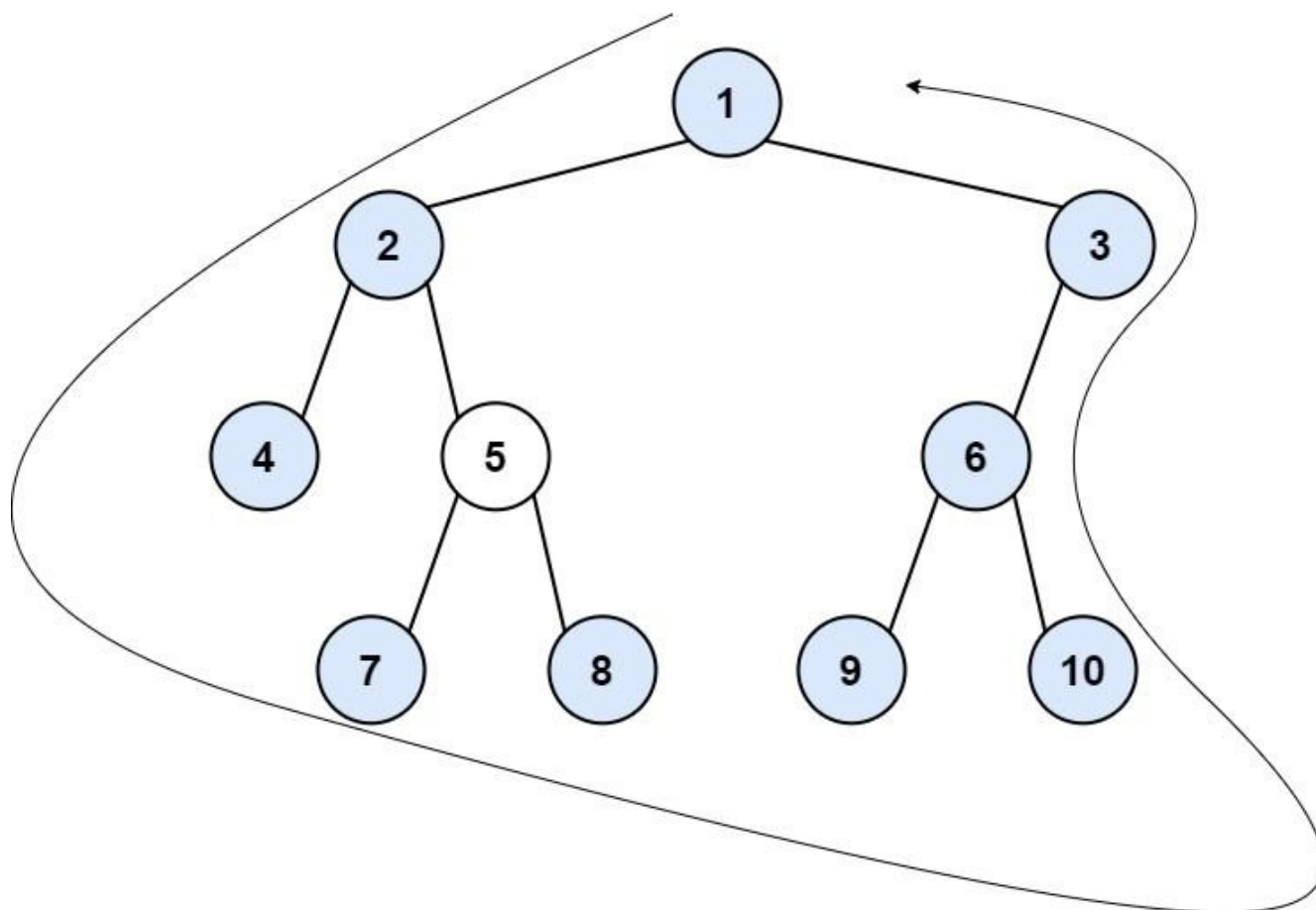
Input: root = [1,null,2,3,4]

Output: [1,3,4,2]

Explanation:

- The left boundary is empty because the root does not have a left child.
 - The right boundary follows the path starting from the root's right child 2 -> 4. 4 is a leaf, so the right boundary is [2].
 - The leaves from left to right are [3,4].
- Concatenating everything results in [1] + [] + [3,4] + [2] = [1,3,4,2].

Example 2:



Input: root = [1,2,3,4,5,6,null,null,null,7,8,9,10]

Output: [1,2,4,7,8,9,10,6,3]

Explanation:

- The left boundary follows the path starting from the root's left child 2 -> 4.
- 4 is a leaf, so the left boundary is [2].
- The right boundary follows the path starting from the root's right child 3 -> 6 -> 10.
- 10 is a leaf, so the right boundary is [3,6], and in reverse order is [6,3].
- The leaves from left to right are [4,7,8,9,10].

Constraints:

- The number of nodes in the tree is in the range [1, 10⁴].
- $-1000 \leq \text{Node.val} \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Boundary = root + left boundary (non-leaves top-down) + all leaves left-to-right
 # + right boundary (non-leaves bottom-up). Leftmost/rightmost leaves are in leaves,
 not boundary. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
 # Collect four pieces separately: root val, left boundary nodes top-down,
 # all leaves left-to-right, right boundary nodes bottom-up.
 # Merge while skipping duplicate leaves already counted.
 # Time: $O(n)$. Space: $O(h)$ recursion for leaf collection.
 # Should I go ahead and optimize?"

```
class _545_BoundaryBinaryTree:
    def boundaryOfBinaryTree(self, root: Optional['TreeNode']) -> List[int]:
        if not root:
            return []
        def is_leaf(n): return not n.left and not n.right
        def left_boundary(n):
            res = []
            while n and not is_leaf(n):
                res.append(n.val)
                n = n.left if n.left else n.right
            return res
        def leaves(n):
            if not n: return []
            if is_leaf(n): return [n.val]
            return leaves(n.left) + leaves(n.right)
        def right_boundary(n):
            res = []
            while n and not is_leaf(n):
                res.append(n.val)
                n = n.right if n.right else n.left
            return res[::-1]
        if is_leaf(root):
            return [root.val]
        return [root.val] + left_boundary(root.left) + leaves(root) +
            right_boundary(root.right)
```

PHASE 5 — TEST & DEBUG

[1,null,2,3,4]: left=[]. leaves=[3,4]. right=[2]. → [1,3,4,2] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(h)$. Three separate traversals, each linear.
 # 'Prefer left else right' for left spine; 'prefer right else left' for right spine
 – reversed at end.
 # Given more time I'd ask: return boundary clockwise vs counter-clockwise? Just
 reverse the result."

Trees & BST

□ ★ #549 Binary Tree Longest Consecutive Sequence II ★

MED

[Open on LeetCode](#)

Tree · DFS

Lists: ★ Premium | Premium 98

Problem

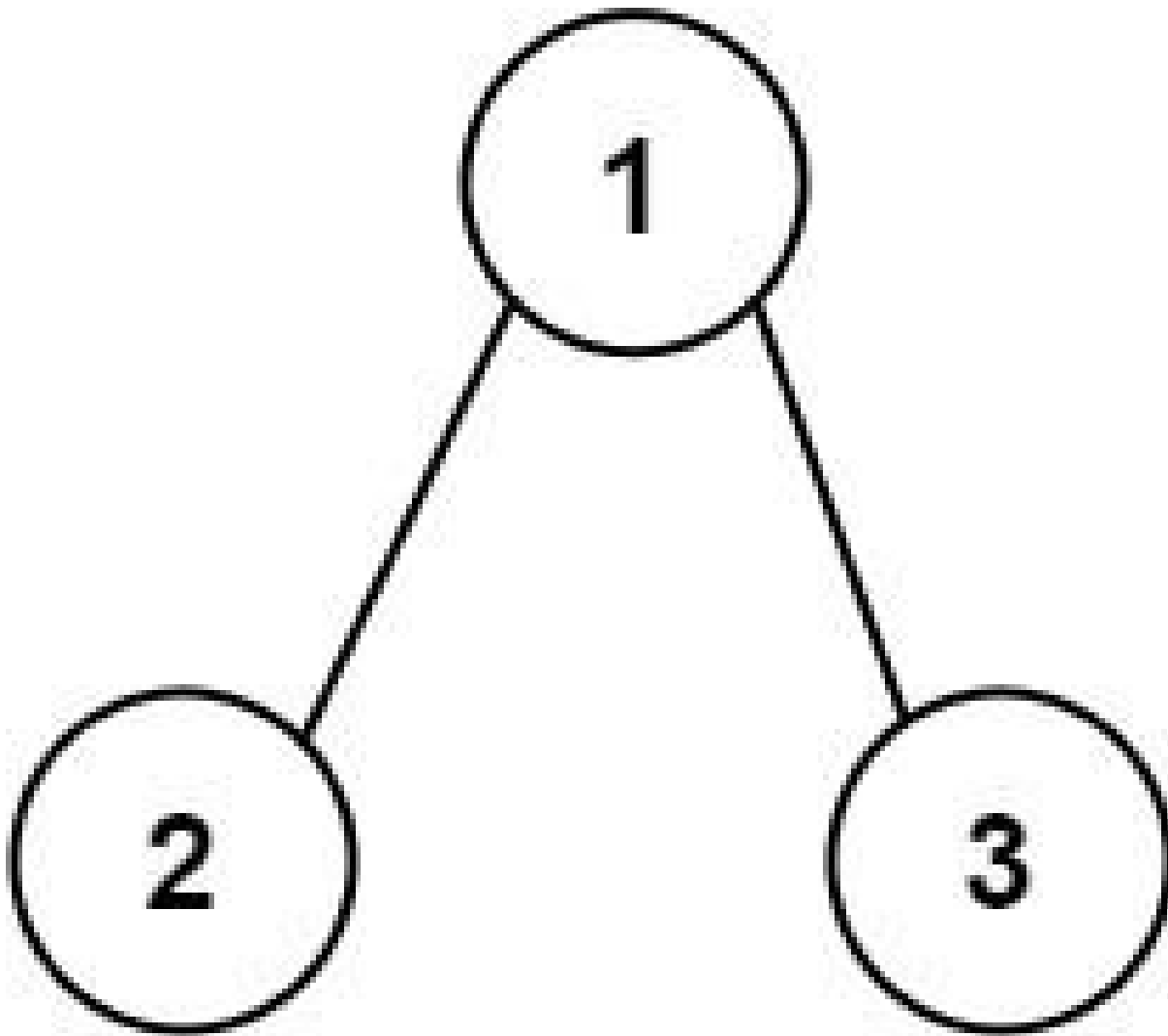
Given the root of a binary tree, return the length of the longest consecutive path in the tree.

A consecutive path is a path where the values of the consecutive nodes in the path differ by one. This path can be either increasing or decreasing.

- For example, [1,2,3,4] and [4,3,2,1] are both considered valid, but the path [1,2,4,3] is not valid.

On the other hand, the path can be in the child–Parent–child order, where not necessarily be parent–child order.

Example 1:

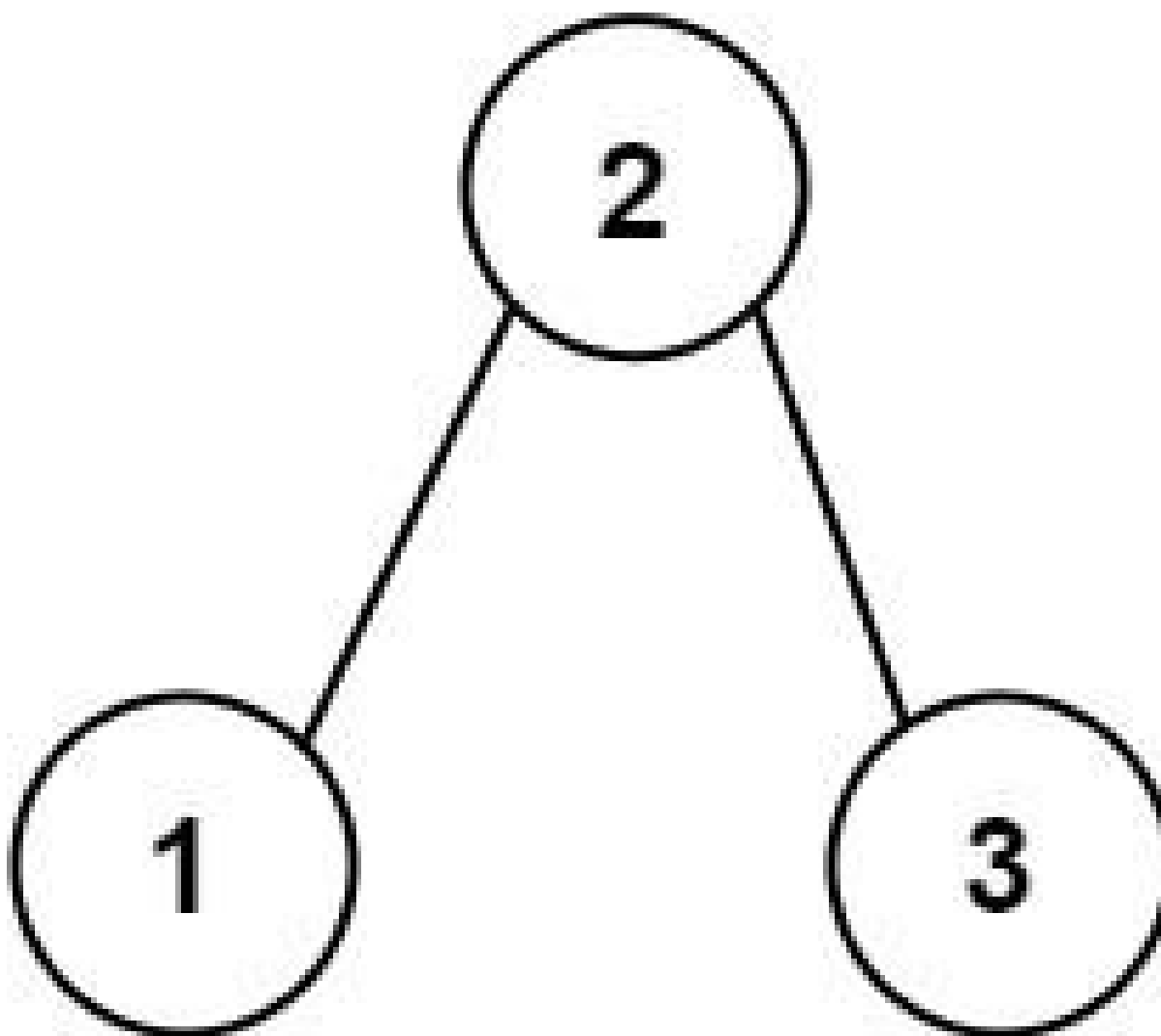


Input: root = [1,2,3]

Output: 2

Explanation: The longest consecutive path is [1, 2] or [2, 1].

Example 2:



Input: root = [2,1,3]

Output: 3

Explanation: The longest consecutive path is [1, 2, 3] or [3, 2, 1].

Constraints:

- The number of nodes in the tree is in the range $[1, 3 * 10^4]$.
- $-3 * 10^4 \leq \text{Node.val} \leq 3 * 10^4$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Longest path where consecutive values differ by 1 – can go up OR down, can change
direction at one pivot. Right?"
```

```
#
```

```
# "[2,1,3] → [1,2,3] length 3 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic."
```

```
# At each node, run DFS tracking consecutive count when child.val == parent.val + 1.
```

```
# Reset count when streak breaks. Track global max streak length.
```

```
# Visits every node once with O(n) DFS per starting node worst case O(n^2).
```

```
# Time: O(n^2) naive. Space: O(h) stack.
```

```
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (post-order returning inc/dec lengths)

```
# "Each node returns (inc, dec): length of increasing/decreasing sequence going DOWN
from it."
```

```
# Diameter through node = best combination of one child's inc and another's dec,
meeting at this node."
```

PHASE 4 — CLEAN CODE

```
class _549_LongestConsecutiveII:
    def longestConsecutive(self, root: Optional['TreeNode']) -> int:
        best = [0]
        def dfs(node):
            if not node:
                return 0, 0
            l_inc, l_dec = dfs(node.left)
            r_inc, r_dec = dfs(node.right)
            inc = dec = 1
            if node.left:
                if node.left.val == node.val + 1: inc = max(inc, l_inc + 1)
                if node.left.val == node.val - 1: dec = max(dec, l_dec + 1)
            if node.right:
                if node.right.val == node.val + 1: inc = max(inc, r_inc + 1)
                if node.right.val == node.val - 1: dec = max(dec, r_dec + 1)
            best[0] = max(best[0], inc + dec - 1)
            return inc, dec
        dfs(root)
        return best[0]
```

PHASE 5 — TEST & DEBUG

```
# [2,1,3]: dfs(1)=(1,1). dfs(3)=(1,1). dfs(2): left(1)=val-1→dec=2.
right(3)=val+1→inc=2. best=2+2-1=3 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(h). inc+dec-1 avoids double-counting the pivot node."
```

```
# Generalises #298 by allowing direction change at the pivot node.
```

```
# Given more time I'd ask: allow path to go up? Needs full graph treatment."
```

Trees & BST

□ ★ #582 Kill Process ★

MED

[Open on LeetCode](#)

Array · Hash Table · Tree · DFS · BFS

Lists: ★ Premium | Premium 98

Problem

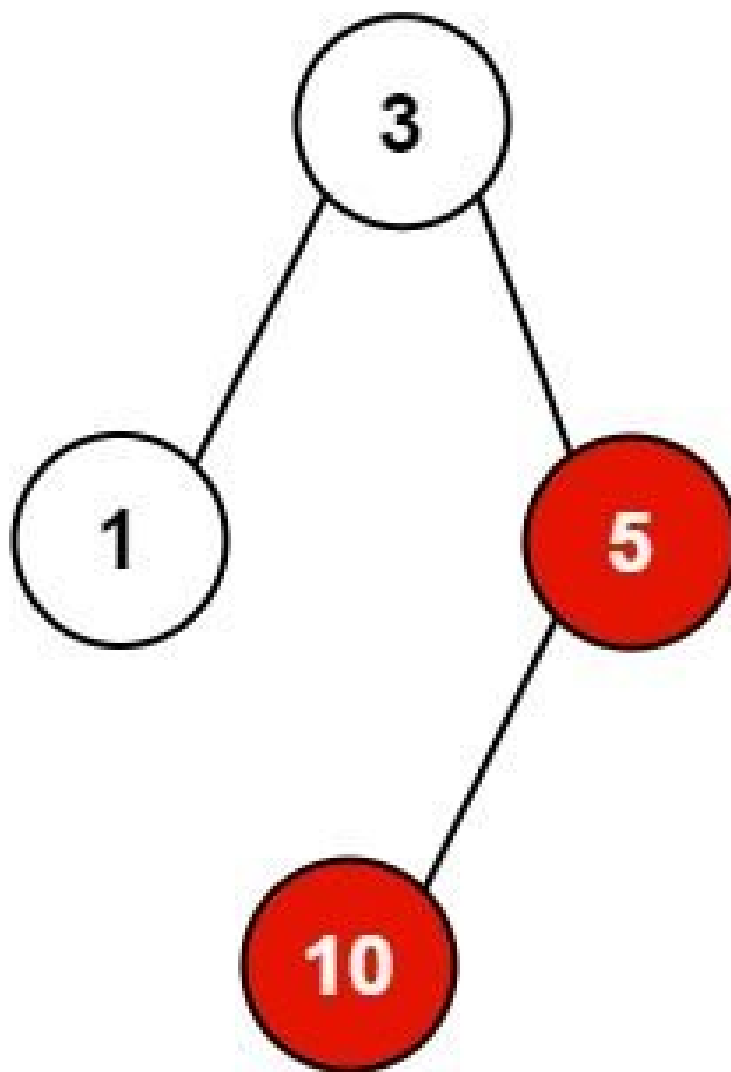
You have n processes forming a rooted tree structure. You are given two integer arrays `pid` and `ppid`, where `pid[i]` is the ID of the i th process and `ppid[i]` is the ID of the i th process's parent process.

Each process has only one parent process but may have multiple children processes. Only one process has `ppid[i] = 0`, which means this process has no parent process (the root of the tree).

When a process is killed, all of its children processes will also be killed.

Given an integer `kill` representing the ID of a process you want to kill, return a list of the IDs of the processes that will be killed. You may return the answer in any order.

Example 1:



Input: pid = [1,3,10,5], ppid = [3,0,5,3], kill = 5

Output: [5,10]

Explanation: The processes colored in red are the processes that should be killed.

Example 2:

Input: pid = [1], ppid = [0], kill = 1

Output: [1]

Constraints:

- $n == \text{pid.length}$
- $n == \text{ppid.length}$

- $1 \leq n \leq 5 * 10^4$
- $1 \leq \text{pid}[i] \leq 5 * 10^4$
- $0 \leq \text{ppid}[i] \leq 5 * 10^4$
- Only one process has no parent.
- All the values of pid are unique.
- kill is guaranteed to be in pid.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Kill a process and all its descendants. Return all killed process IDs. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Build children map and indegree from the kill-pair list.

BFS from the killed pid: enqueue all its direct children, then propagate kills downward.

Every reachable process dies – simulates the cascade.

Time: $O(n)$. Space: $O(n)$ for graph + queue.

Should I go ahead and optimize?"

```
class _582_KillProcess:
```

```
    def killProcess(self, pid: List[int], ppid: List[int], kill: int) -> List[int]:
        children: defaultdict = defaultdict(list)
        for child, parent in zip(pid, ppid):
            children[parent].append(child)
        killed = []
        queue = deque([kill])
        while queue:
            proc = queue.popleft()
            killed.append(proc)
            queue.extend(children[proc])
        return killed
```

PHASE 5 — TEST & DEBUG

pid=[1,3,10,5], ppid=[3,0,5,3], kill=5: children={3:[1,5],5:[10]}.

BFS from 5: [5,10] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(n)$. Build adjacency list then BFS – standard implicit tree traversal.

```
# Given more time I'd ask: kill all ancestors too? Walk parent pointers upward instead."
```

Trees & BST

□ #662 Maximum Width of Binary Tree

MED

[Open on LeetCode](#)

Tree · BFS

Lists: Grind 169

Problem

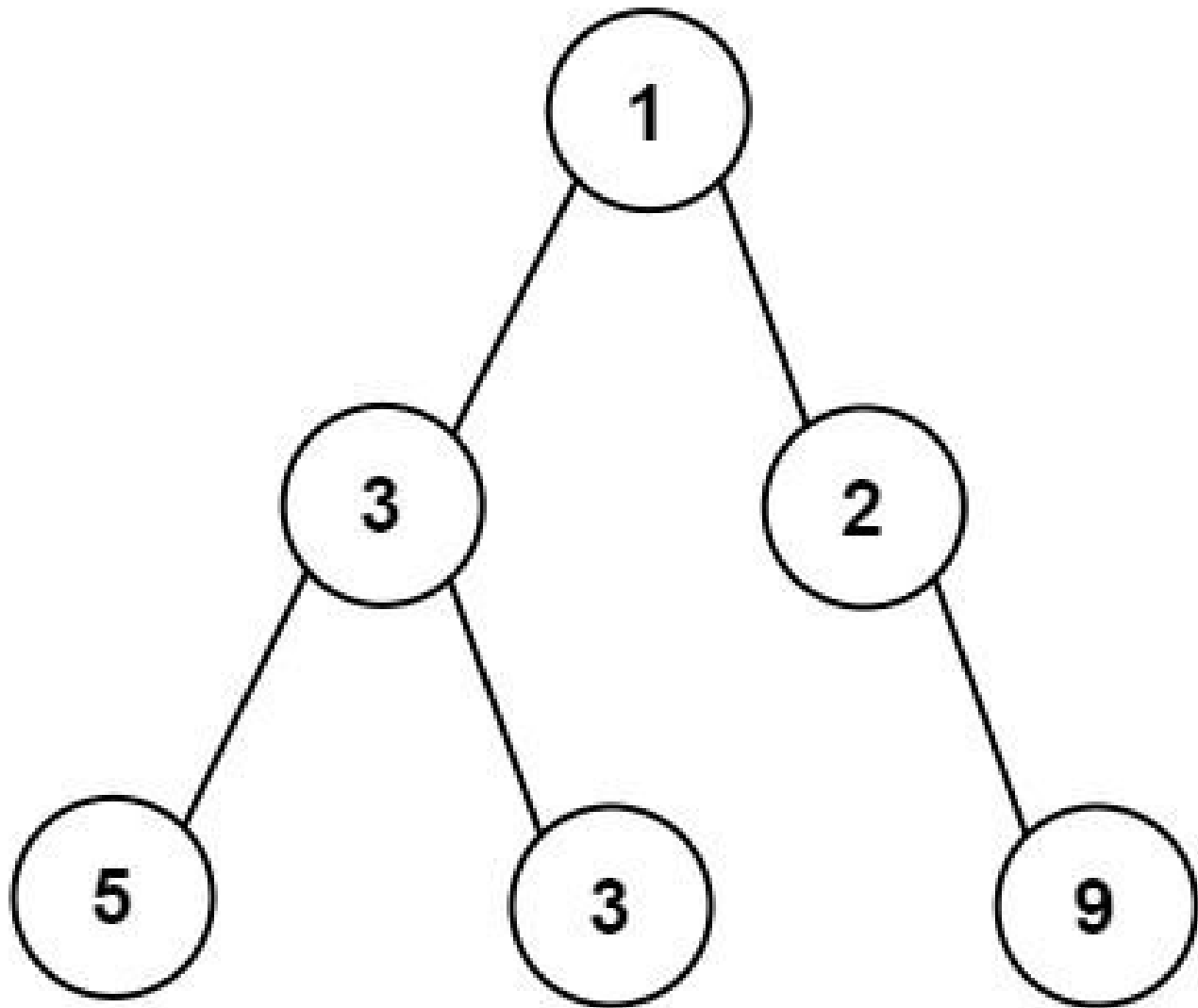
Given the root of a binary tree, return the maximum width of the given tree.

The maximum width of a tree is the maximum width among all levels.

The width of one level is defined as the length between the end-nodes (the leftmost and rightmost non-null nodes), where the null nodes between the end-nodes that would be present in a complete binary tree extending down to that level are also counted into the length calculation.

It is guaranteed that the answer will in the range of a 32-bit signed integer.

Example 1:

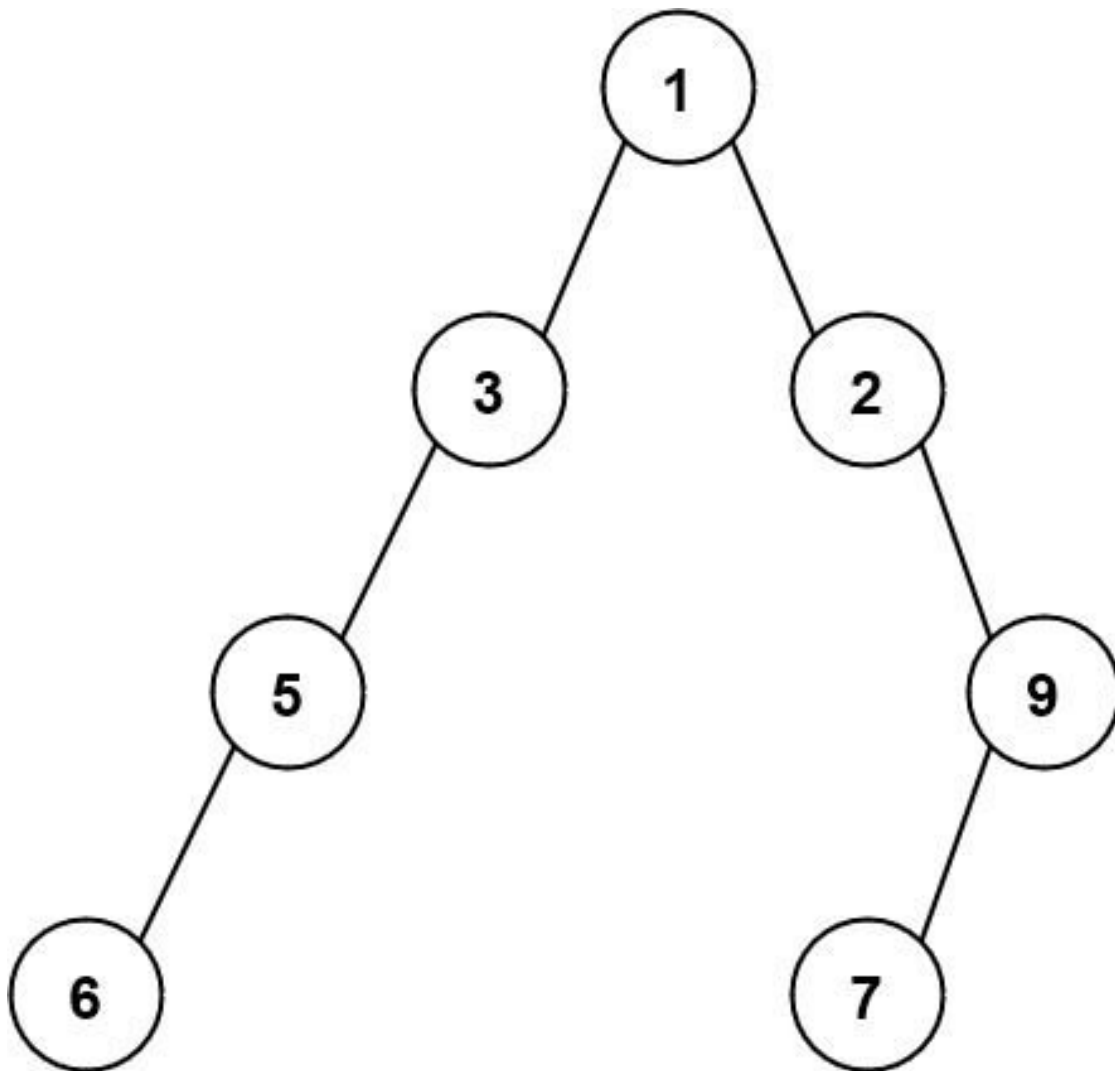


Input: root = [1,3,2,5,3,null,9]

Output: 4

Explanation: The maximum width exists in the third level with length 4 (5,3,null,9).

Example 2:

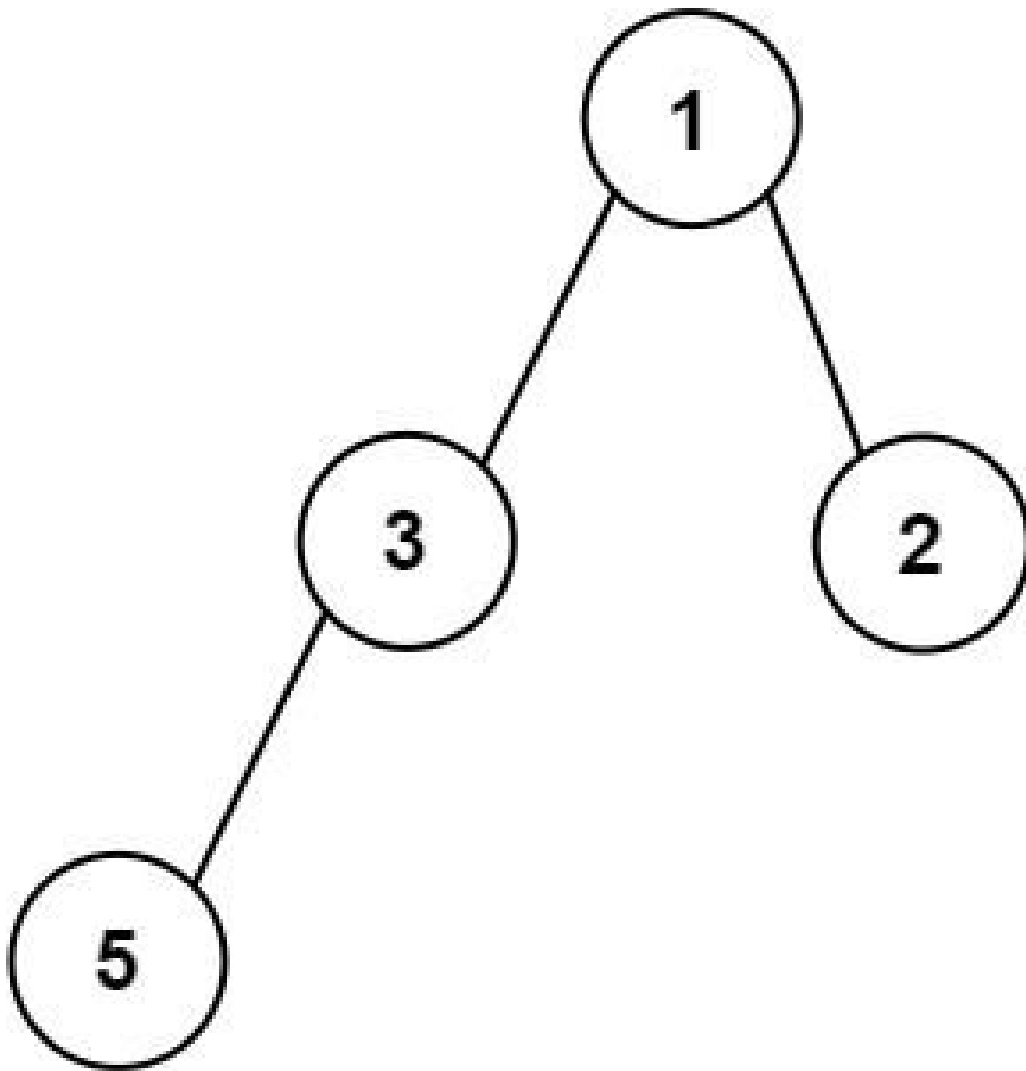


Input: root = [1,3,2,5,null,null,9,6,null,7]

Output: 7

Explanation: The maximum width exists in the fourth level with length 7 (6,null,null,null,null,null,7).

Example 3:



Input: root = [1,3,2,5]

Output: 2

Explanation: The maximum width exists in the second level with length 2 (3,2).

Constraints:

- The number of nodes in the tree is in the range [1, 3000].
- $-100 \leq \text{Node.val} \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Width = rightmost_pos - leftmost_pos + 1 at each level, counting null gaps."

```
# Left child = 2*pos, right child = 2*pos+1. Normalise each level to prevent
overflow. Right?"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# BFS level-order traversal; at each level track min and max index assigned to
nodes.
# Width of level = max_index - min_index + 1; answer is global max width.
# Straightforward queue-based level scan.
# Time: O(n). Space: O(w) queue width.
# Should I go ahead and optimize?"
```

```
class _662_MaxWidthBinaryTree:
    def widthOfBinaryTree(self, root: Optional['TreeNode']) -> int:
        if not root:
            return 0
        max_w = 0
        queue = deque([(root, 0)])
        while queue:
            size = len(queue)
            first_pos = queue[0][1]
            last_pos = 0
            for _ in range(size):
                node, pos = queue.popleft()
                pos -= first_pos # normalise
                last_pos = pos
                if node.left: queue.append((node.left, 2 * pos))
                if node.right: queue.append((node.right, 2 * pos + 1))
            max_w = max(max_w, last_pos + 1)
        return max_w
```

PHASE 5 — TEST & DEBUG

```
# [1,3,2,5,3,null,9]: Level 2 positions (normalised): 5→0, 3→1, 9→3. last=3. width=4
✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(w). Normalisation prevents 2^depth overflow for deep trees —
critical.
# Given more time I'd ask: width by actual node count (ignoring nulls)? Just max
level_size in BFS."
```

Trees & BST

□ #863 All Nodes Distance K in Binary Tree **MED**

[Open on LeetCode](#)

Hash Table · Tree · DFS · BFS

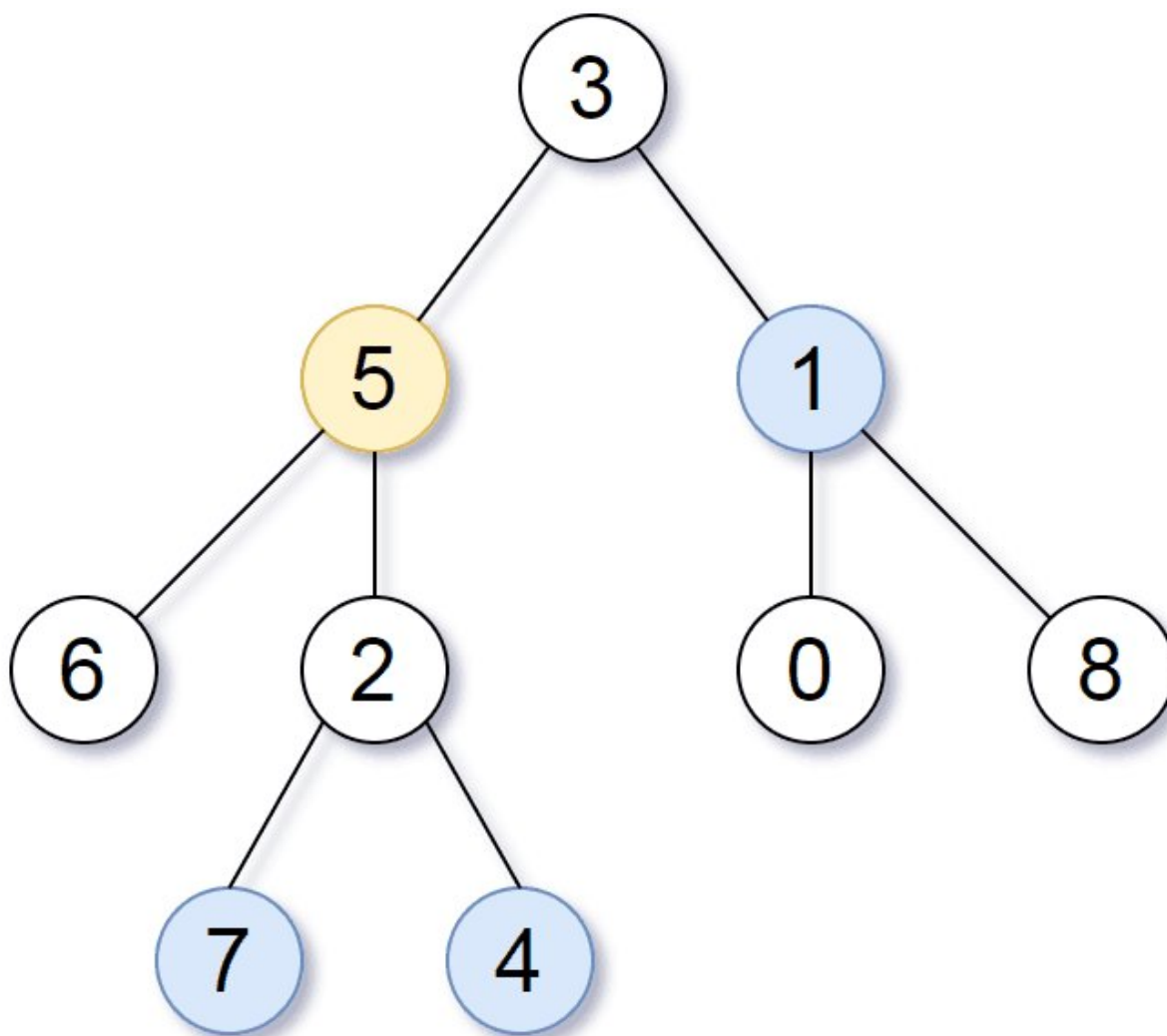
Lists: Grind 169

Problem

Given the root of a binary tree, the value of a target node target, and an integer k, return an array of the values of all nodes that have a distance k from the target node.

You can return the answer in any order.

Example 1:



Input: root = [3,5,1,6,2,0,8,null,null,7,4], target = 5, k = 2

Output: [7,4,1]

Explanation: The nodes that are a distance 2 from the target node (with value 5) have values 7, 4, and 1.

Example 2:

Input: root = [1], target = 1, k = 3

Output: []

Constraints:

- The number of nodes in the tree is in the range [1, 500].
- $0 \leq \text{Node.val} \leq 500$

- All the values `Node.val` are unique.
- `target` is the value of one of the nodes in the tree.
- $0 \leq k \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find all nodes exactly `k` edges from `target`. Distance can go upward through parent. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."
 # Build parent map via BFS/DFS from `target`. For each query node, walk parent pointers
 # until reaching `target`; collect distance. $O(n)$ per query naive.
 # Time: $O(n * Q)$ for Q queries. Space: $O(n)$ parent map.
 # Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (build parent map then BFS)

"DFS to build `parent[node]=parent`. Then BFS from `target` treating tree as undirected graph."
 # Stop collecting at distance `k`."

PHASE 4 — CLEAN CODE

```
class _863_DistanceK:
    def distanceK(self, root: 'TreeNode', target: 'TreeNode', k: int) -> List[int]:
        parent: dict = {}
        def build(node, par):
            if not node: return
            parent[node] = par
            build(node.left, node)
            build(node.right, node)
        build(root, None)
        visited = {target}
        queue = deque([(target, 0)])
        result = []
        while queue:
            node, dist = queue.popleft()
            if dist == k:
                result.append(node.val)
                continue
            for nb in [node.left, node.right, parent[node]]:
```

```
        if nb and nb not in visited:
            visited.add(nb)
            queue.append((nb, dist + 1))
    return result
```

PHASE 5 — TEST & DEBUG

target=5, k=2: BFS outward from 5 by 2 hops → [7,4,1] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(n)$. Parent map converts tree to undirected graph – then BFS is trivial.

Given more time I'd ask: multiple targets? Multi-source BFS – enqueue all at distance 0."

Trees & BST

□ ★ #1120 Maximum Average Subtree ★

MED

[Open on LeetCode](#)

Tree · DFS

Lists: ★ Premium | Premium 98

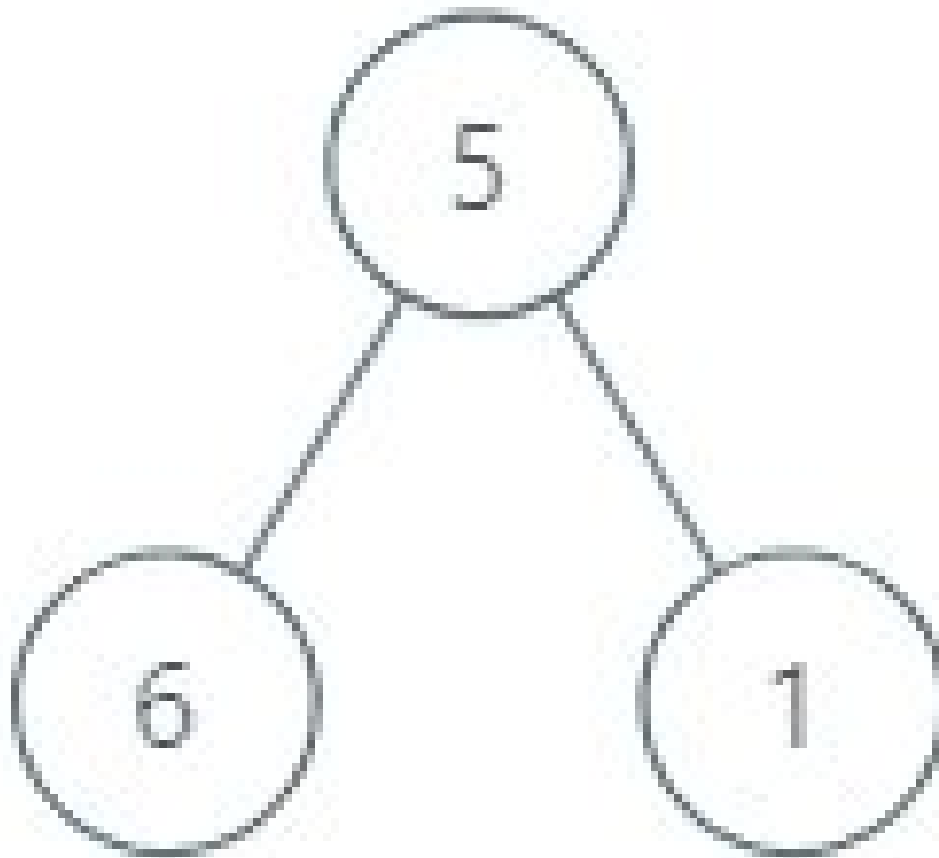
Problem

Given the root of a binary tree, return the maximum average value of a subtree of that tree. Answers within 10^{-5} of the actual answer will be accepted.

A subtree of a tree is any node of that tree plus all its descendants.

The average value of a tree is the sum of its values, divided by the number of nodes.

Example 1:



Input: root = [5,6,1]

Output: 6.00000

Explanation:

For the node with value = 5 we have an average of $(5 + 6 + 1) / 3 = 4$.

For the node with value = 6 we have an average of $6 / 1 = 6$.

For the node with value = 1 we have an average of $1 / 1 = 1$.

So the answer is 6 which is the maximum.

Example 2:

Input: root = [0,null,1]

Output: 1.00000

Constraints:

- The number of nodes in the tree is in the range [1, 104].
- $0 \leq \text{Node.val} \leq 105$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find the subtree (any node + all descendants) with the maximum average value. Return the average. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."
 # Post-order DFS: for each node return (sum, count) of its subtree.
 # Track max average = sum/count over all visited nodes.
 # Recomputes subtree sums independently – correct $O(n)$ post-order.
 # Time: $O(n)$. Space: $O(h)$ stack.
 # Should I go ahead and optimize?"

```
class _1120_MaxAvgSubtree:
    def maximumAverageSubtree(self, root: Optional['TreeNode']) -> float:
        best = [0.0]
        def dfs(node):
            if not node: return 0, 0
            l_sum, l_cnt = dfs(node.left)
            r_sum, r_cnt = dfs(node.right)
            total_sum = l_sum + r_sum + node.val
            total_cnt = l_cnt + r_cnt + 1
            best[0] = max(best[0], total_sum / total_cnt)
            return total_sum, total_cnt
        dfs(root)
        return best[0]
```

PHASE 5 — TEST & DEBUG

[5,6,1]: avg(6)=6, avg(1)=1, avg(5,6,1)=4. → 6.0 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(h)$. Post-order carries (sum, count) up – compute average at each node."
 # Given more time I'd ask: minimum average subtree? Track min instead of max."

Trees & BST

□ ★ #1490 Clone N-ary Tree ★

MED

[Open on LeetCode](#)

Hash Table · Tree · DFS · BFS

Lists: ★ Premium | Premium 98

Problem

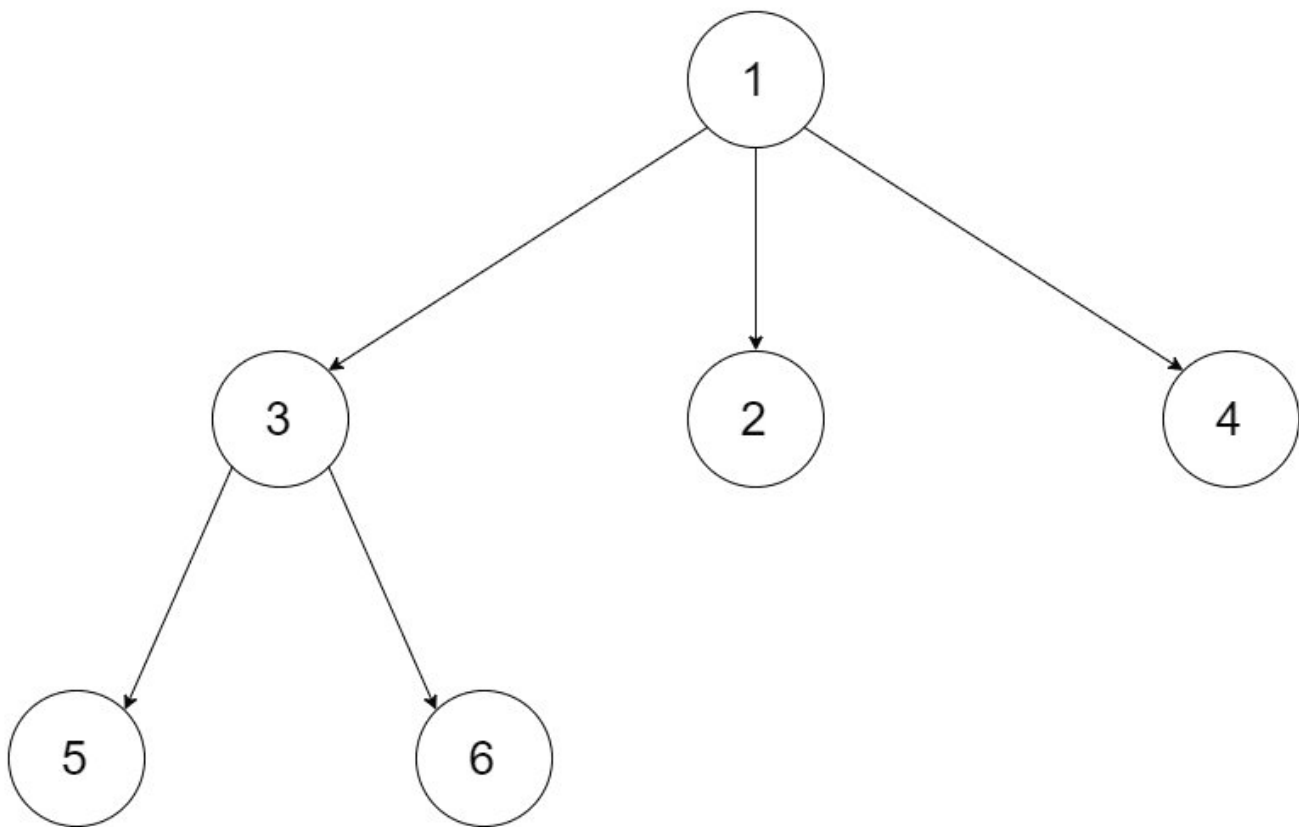
Given a root of an N-ary tree, return a deep copy (clone) of the tree.

Each node in the n-ary tree contains a val (int) and a list (List[Node]) of its children.

```
class Node {  
    public int val;  
    public List<Node> children;  
}
```

Nary-Tree input serialization is represented in their level order traversal, each group of children is separated by the null value (See examples).

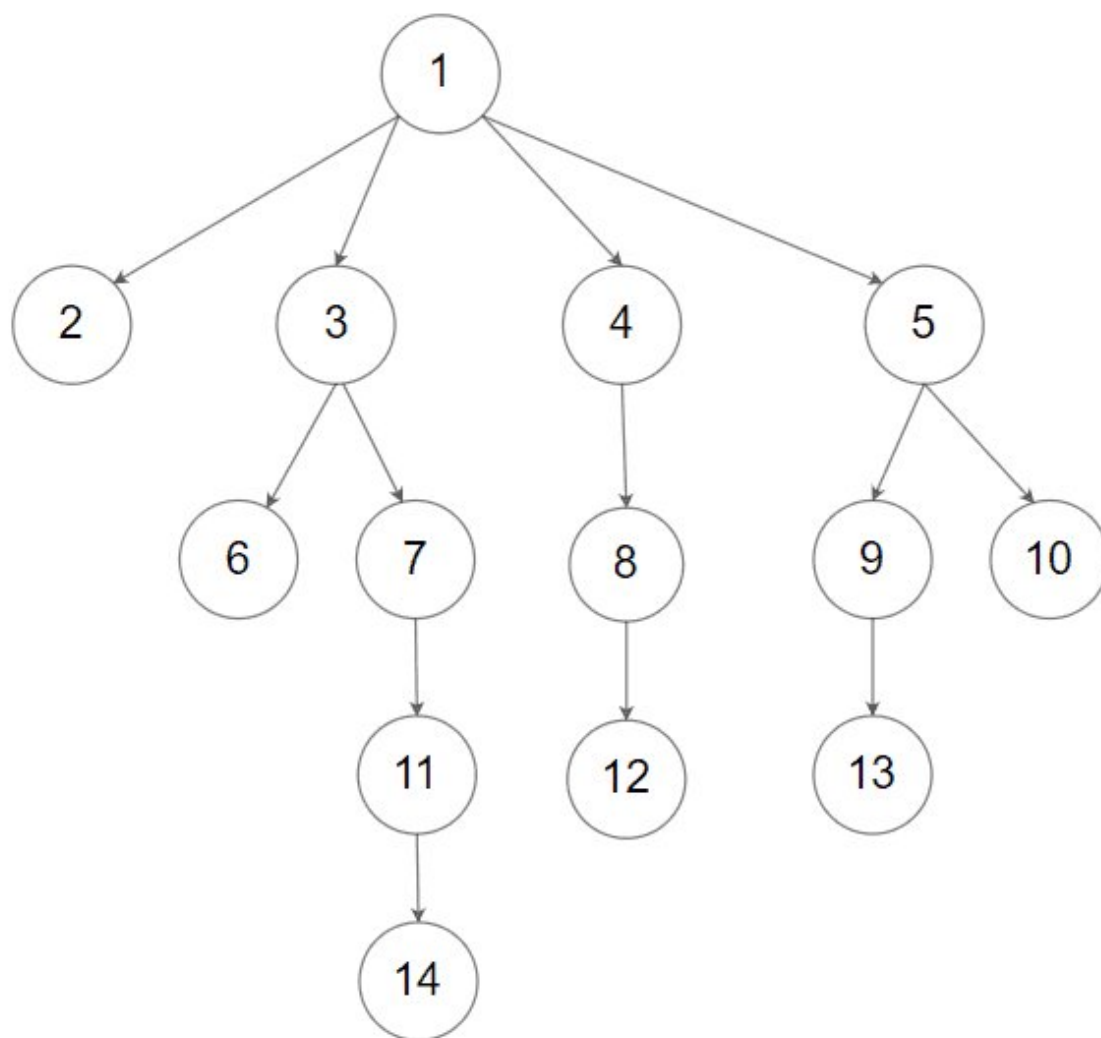
Example 1:



Input: root = [1,null,3,2,4,null,5,6]

Output: [1,null,3,2,4,null,5,6]

Example 2:



Input: root = [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,null,11,null,12,null,13,null,null,14]

Output: [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,null,11,null,12,null,13,null,null,14]

Constraints:

- The depth of the n-ary tree is less than or equal to 1000.
- The total number of nodes is between [0, 104].

Follow up: Can your solution work for the graph problem?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Deep clone an N-ary tree. No shared references between original and clone. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

DFS clone: map each original node to a new node, recursively clone all children lists.

First pass creates nodes on demand; second pass wires child pointers.

Time: $O(n)$. Space: $O(n)$ for map + recursion.

Should I go ahead and optimize?"

```
class _1490_CloneNaryTree:
```

```
    def cloneTree(self, root: 'Node') -> 'Node':
```

```
        if not root:
```

```
            return None
```

```
        return Node(root.val, [self.cloneTree(child) for child in root.children])
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(n)$. Simple recursive clone – no visited map needed (trees have no cycles).

Follow-up (Clone Graph #133): needs visited map to handle cycles – {original: clone}."

Trees & BST

□ ★ #1522 Diameter of N-Ary Tree ★

MED

[Open on LeetCode](#)

Tree · DFS

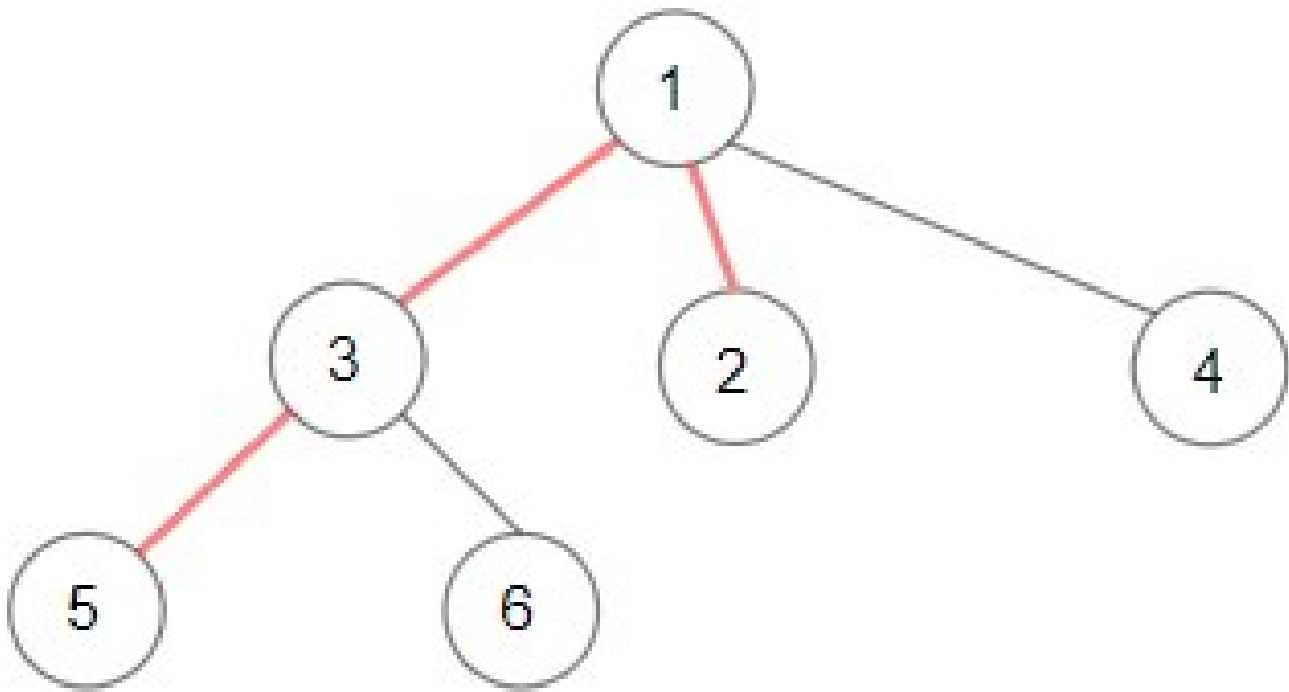
Lists: ★ Premium | Premium 98

Problem

Given a root of an N-ary tree, you need to compute the length of the diameter of the tree. The diameter of an N-ary tree is the length of the longest path between any two nodes in the tree. This path may or may not pass through the root.

(Nary-Tree input serialization is represented in their level order traversal, each group of children is separated by the null value.)

Example 1:

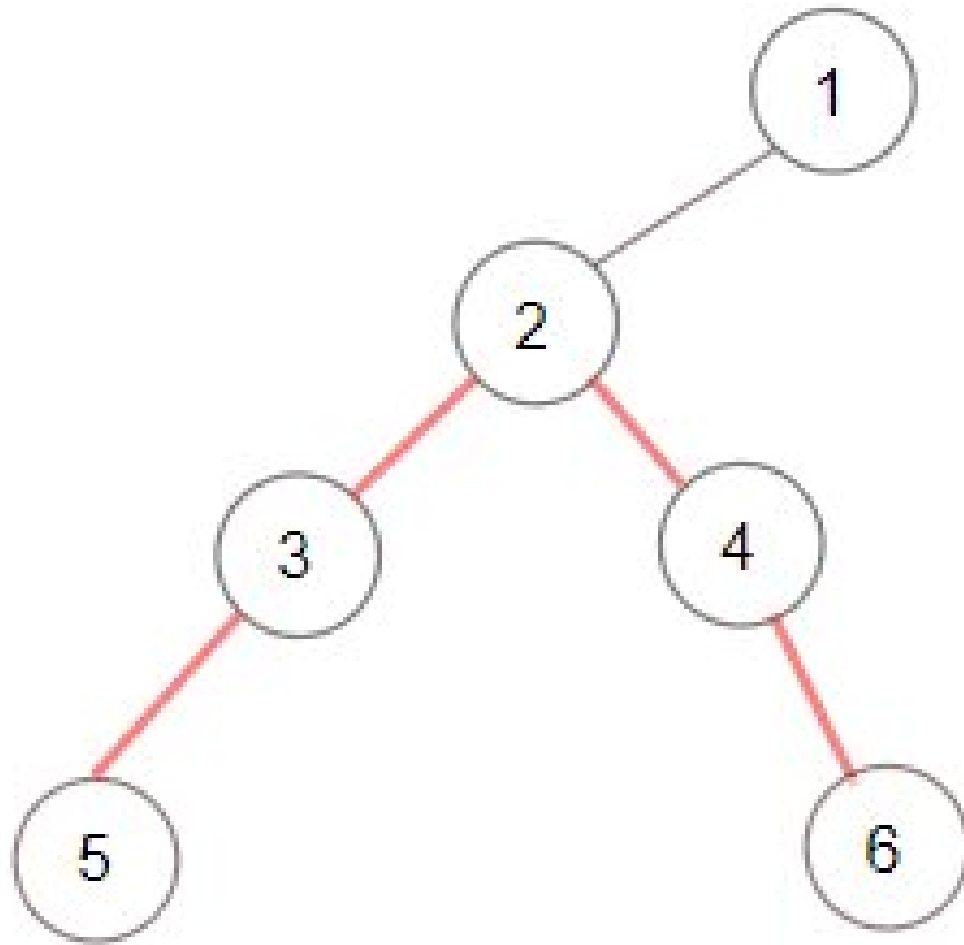


Input: root = [1,null,3,2,4,null,5,6]

Output: 3

Explanation: Diameter is shown in red color.

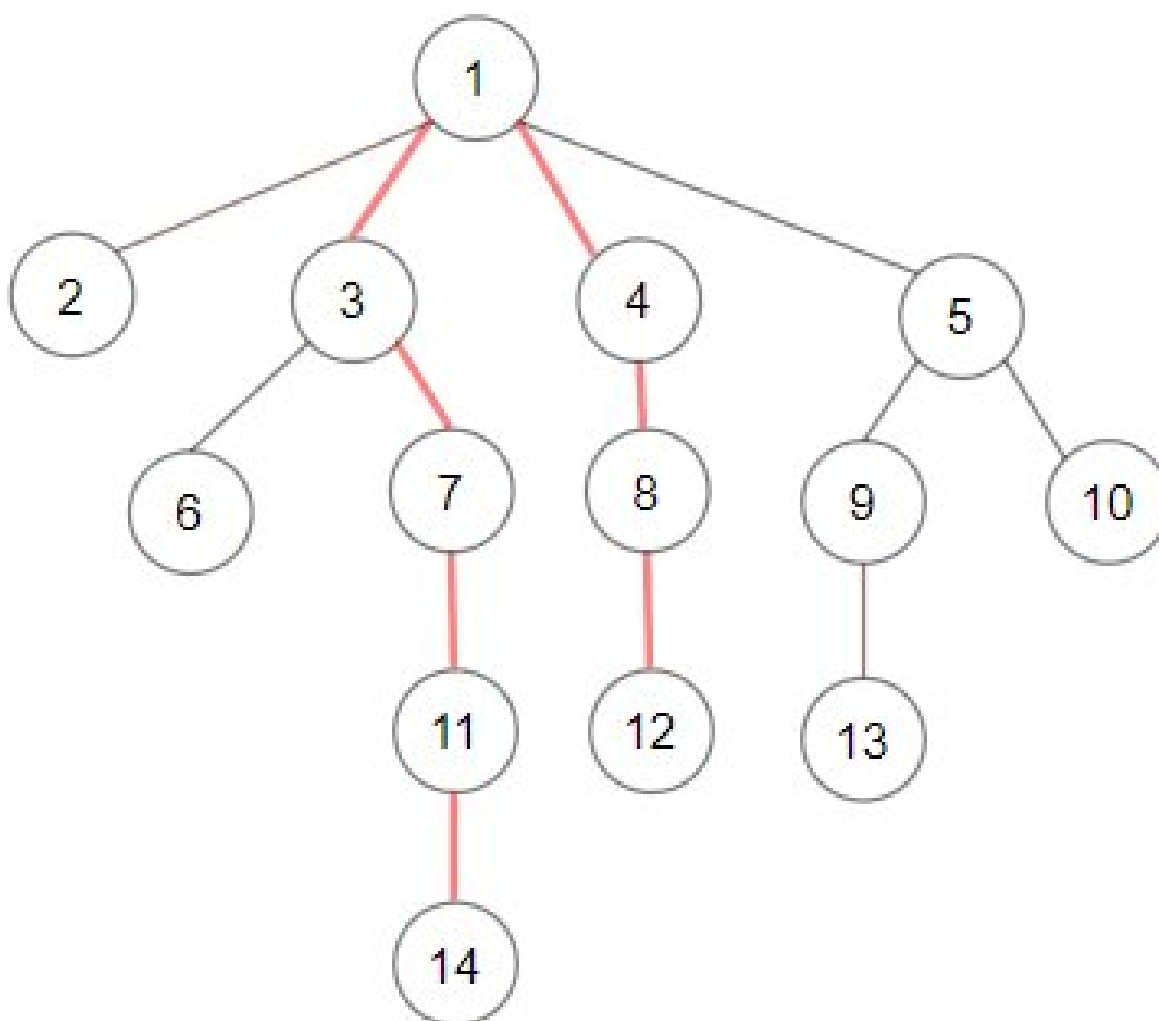
Example 2:



Input: root = [1,null,2,null,3,4,null,5,null,6]

Output: 4

Example 3:



Input: root = [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,null,11,null,12,null,13,null,null,14]
Output: 7

Constraints:

- The depth of the n-ary tree is less than or equal to 1000.
- The total number of nodes is between [1, 104].

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Diameter = longest path between any two nodes (in edges), generalised to N children. Right?"
 # Same concept as Binary Tree Diameter (#543)."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."
 # Two-pass DFS diameter: for each node, longest path through it = sum of top two child depths.
 # Same as binary tree diameter but children list instead of left/right.
 # Time: $O(n)$. Space: $O(h)$ stack.
 # Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Post-order: for each node, collect heights of children (adding 1 for the edge to parent)."
 # Diameter through this node = top two child-heights summed.
 # Node's own height = max child-height (or 0 for leaf)."

PHASE 4 — CLEAN CODE

```
class _1522_DiameterNaryTree:
    def diameter(self, root: 'Node') -> int:
        best = [0]
        def height(node) -> int:
            if not node or not node.children:
                return 0
            # child_h+1 converts "height below child" to "edges from this node via
            # that child"
            child_heights = sorted([height(c) + 1 for c in node.children],
                                   reverse=True)
            if len(child_heights) >= 2:
                best[0] = max(best[0], child_heights[0] + child_heights[1])
            else:
                best[0] = max(best[0], child_heights[0])
            return child_heights[0]
        height(root)
        return best[0]
```

PHASE 5 — TEST & DEBUG

[1,null,3,2,4,null,5,6]: height(5)=0, height(6)=0.
 # height(3): child_heights=[0+1,0+1]=[1,1]. best=1+1=2. return 1.
 # height(2)=0, height(4)=0.
 # height(1): child_heights=[1+1,0+1,0+1]=[2,1,1]. best=2+1=3. return 2. → 3 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(h)$. Adding +1 when collecting child heights is the key difference from #543."
 # In binary tree, depth(child) already includes the edge up; here we add it explicitly.
 # Given more time I'd ask: diameter in edge-weighted N-ary tree?

Same approach – just use edge weights instead of +1."

Trees & BST

★ #1650 Lowest Common Ancestor of a Binary Tree III ★

MED[Open on LeetCode](#)

Hash Table · Tree · Two Pointers

Lists: ★ Premium | Premium 98

Problem

Given two nodes of a binary tree p and q, return their lowest common ancestor (LCA).

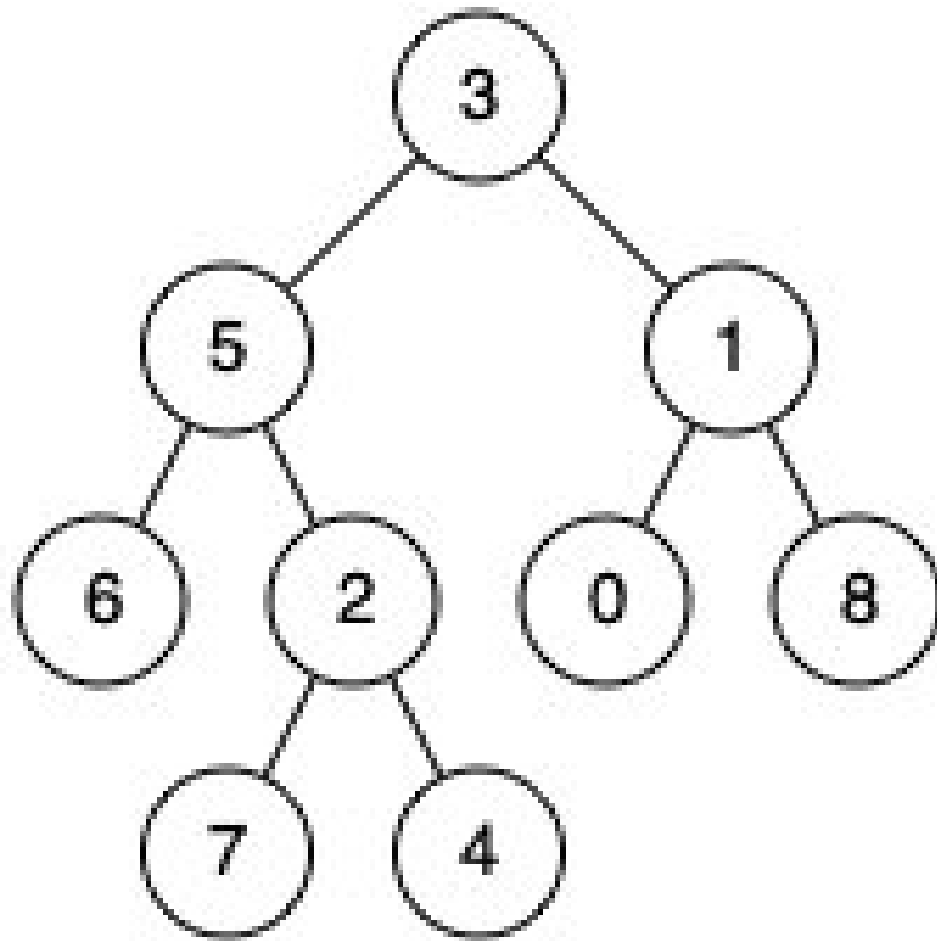
Each node will have a reference to its parent node. The definition for Node is below:

```
class Node {  
    public int val;  
    public Node left;  
    public Node right;  
    public Node parent;  
}
```

According to the definition of LCA on Wikipedia:

"The lowest common ancestor of two nodes p and q in a tree T is the lowest node that has both p and q as descendants (where we allow a node to be a descendant of itself)."

Example 1:

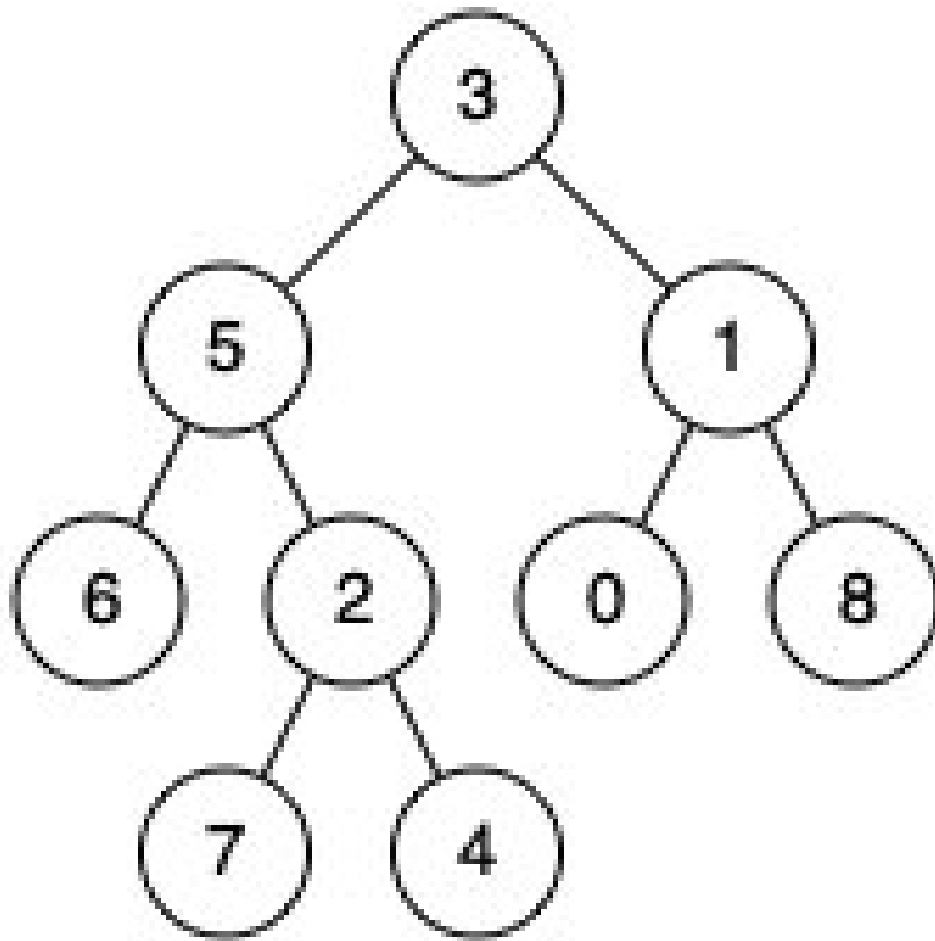


Input: root = [3,5,1,6,2,0,8,null,null,7,4], p = 5, q = 1

Output: 3

Explanation: The LCA of nodes 5 and 1 is 3.

Example 2:



Input: root = [3,5,1,6,2,0,8,null,null,7,4], p = 5, q = 4

Output: 5

Explanation: The LCA of nodes 5 and 4 is 5 since a node can be a descendant of itself according to the LCA definition.

Example 3:

Input: root = [1,2], p = 1, q = 2

Output: 1

Constraints:

- The number of nodes in the tree is in the range [2, 105].
- $-109 \leq \text{Node.val} \leq 109$

- All Node.val are unique.
- $p \neq q$
- p and q exist in the tree.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Nodes have a parent pointer. Find LCA without the root. Right?
# Both p and q are guaranteed to be in the tree."
#
# "[3,5,1,...], p=5, q=1 → LCA=3 ✓ (same as #236 but solved via parent pointers)"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Each node has a parent pointer. Walk both nodes up with a visited set until they meet.
# First node seen twice is LCA. Like intersecting two linked lists.
# Time: O(h). Space: O(h) visited set.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (two-pointer linked list intersection)

```
# "Walk p upward and q upward simultaneously. When one hits null, restart at the other's origin.
# They meet at the LCA — same as finding intersection of two linked lists.
# Each pointer traverses: (path from p to root) + (path from q to root), length l_p + l_q.
# Both pointers traverse the same total length → they meet at LCA."
```

PHASE 4 — CLEAN CODE

```
class _1650_LCA_III:
    def lowestCommonAncestor(self, p: 'Node', q: 'Node') -> 'Node':
        a, b = p, q
        while a is not b:
            a = a.parent if a else q
            b = b.parent if b else p
        return a
```

PHASE 5 — TEST & DEBUG

```
# p=5, q=1: a traverses 5→3→None→1→3. b traverses 1→3→None→5→3. Both reach 3 simultaneously ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(h). Space O(1). This is the two-pointer intersection trick from Linked List Intersection (#160).
# Both pointers traverse depth(p) + depth(q) steps total and meet at LCA."
```

Given more time I'd ask: what if the tree is very deep and we want to avoid the full traversal?
Use a set: walk p to root recording all ancestors, then walk q to root and stop at first ancestor in the set.
$O(\text{depth})$ time, $O(\text{depth})$ space – same asymptotic but possibly faster in practice."

DFS

Round 2 · High · Medium · 16 questions

DFS

□ #130 Surrounded Regions

MED[Open on LeetCode](#)

Array · DFS · BFS · Union Find · Matrix

Lists: Denny Zhang

Problem

You are given an $m \times n$ matrix board containing letters 'X' and 'O', capture regions that are surrounded:

- **Connect:** A cell is connected to adjacent cells horizontally or vertically.
- **Region:** To form a region connect every 'O' cell.
- **Surround:** A region is surrounded if none of the 'O' cells in that region are on the edge of the board. Such regions are completely enclosed by 'X' cells.

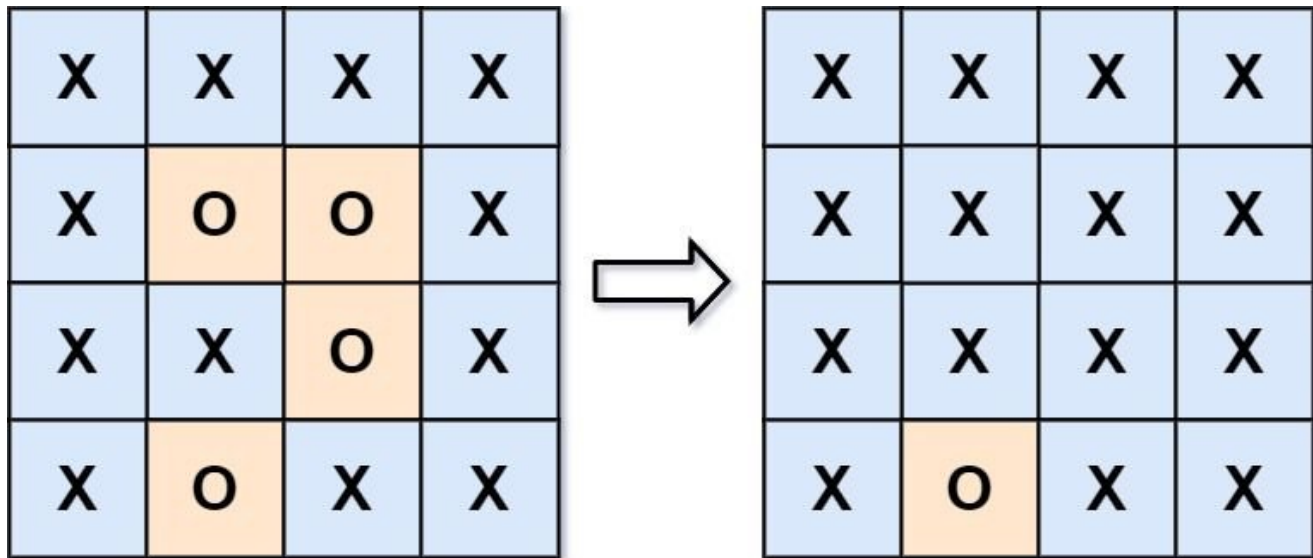
To capture a surrounded region, replace all 'O's with 'X's in-place within the original board. You do not need to return anything.

Example 1:

Input: board = `[["X","X","X","X"],["X","O","O","X"],["X","X","O","X"],["X","O","X","X"]]`

Output: `[["X","X","X","X"],["X","X","X","X"],["X","X","X","X"],["X","O","X","X"]]`

Explanation:



In the above diagram, the bottom region is not captured because it is on the edge of the board and cannot be surrounded.

Example 2:

Input: board = `[["X"]]`

Output: `[["X"]]`

Constraints:

- `m == board.length`
- `n == board[i].length`
- `1 <= m, n <= 200`

- `board[i][j]` is 'X' or 'O'.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Capture surrounded 'O' regions by replacing them with 'X'.
# A region is NOT captured if any 'O' in it touches the board edge. Right?
# In-place modification – no return value."
#
# "Border-connected 'O's and their inland 'O' neighbours are safe. Everything else
# gets flipped."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# First pass: DFS from every border 'O' and mark connected cells as safe ('S').
# Second pass: flip remaining 'O' to 'X'; restore safe cells back to 'O'.
# Two-pass flood fill from boundary – standard O(m*n) approach.
# Time: O(m * n). Space: O(m * n) recursion.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (reverse: mark safe, then flip)

```
# "DFS from every border 'O', marking connected 'O's as safe ('#').
# Second pass: 'O' → 'X' (captured), '#' → 'O' (safe restored). O(m*n)."
```

PHASE 4 — CLEAN CODE

```
class _130_SurroundedRegions:
    def solve(self, board: List[List[str]]) -> None:
        m, n = len(board), len(board[0])
        def mark_safe(r, c):
            if r < 0 or r >= m or c < 0 or c >= n or board[r][c] != 'O':
                return
            board[r][c] = '#'
            for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                mark_safe(r + dr, c + dc)
        for r in range(m):
            for c in [0, n - 1]:
                mark_safe(r, c)
        for c in range(n):
            for r in [0, m - 1]:
                mark_safe(r, c)
        for r in range(m):
            for c in range(n):
                if board[r][c] == 'O': board[r][c] = 'X'
                elif board[r][c] == '#': board[r][c] = 'O'
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m*n)$ . Space  $O(m*n)$  stack. The reverse-DFS from borders is the key insight
—
# instead of deciding which '0's are captured, decide which are SAFE and flip the
rest.
# Given more time I'd ask: what if the grid wraps (toroidal)? No borders → all '0's
are safe."
```

DFS

#133 Clone Graph

MED[Open on LeetCode](#)

Hash Table · DFS · BFS · Graph

Lists: Grind 169

Problem

Given a reference of a node in a connected undirected graph.

Return a deep copy (clone) of the graph.

Each node in the graph contains a value (int) and a list (List[Node]) of its neighbors.

```
class Node {  
    public int val;  
    public List<Node> neighbors;  
}
```

Test case format:

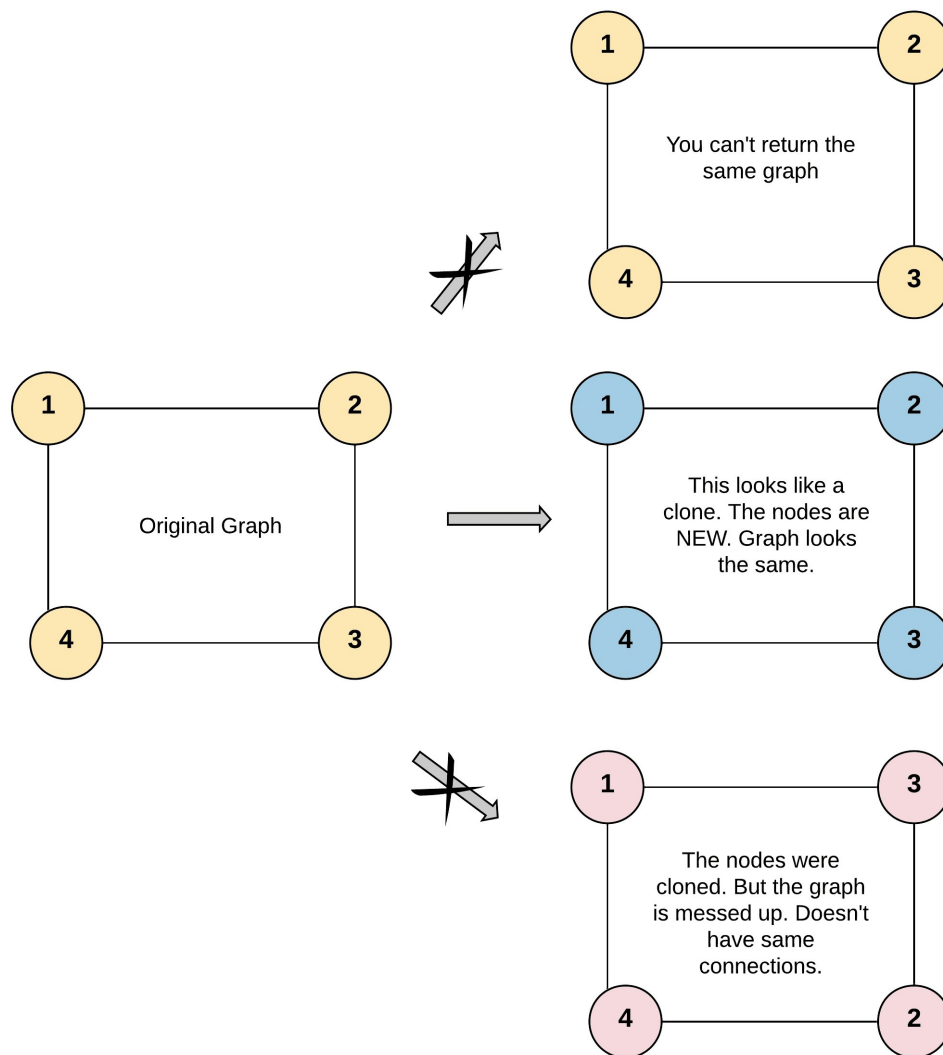
For simplicity, each node's value is the same as the node's index (1-indexed). For example, the first node with `val == 1`, the second node with `val == 2`, and so on. The graph is represented in the test case using an adjacency list.

An adjacency list is a collection of unordered lists used to represent a finite graph. Each list

describes the set of neighbors of a node in the graph.

The given node will always be the first node with $val = 1$. You must return the copy of the given node as a reference to the cloned graph.

Example 1:



Input: `adjList = [[2,4],[1,3],[2,4],[1,3]]`

Output: `[[2,4],[1,3],[2,4],[1,3]]`

Explanation: There are 4 nodes in the graph.

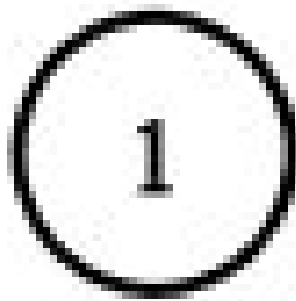
1st node (`val = 1`)'s neighbors are 2nd node (`val = 2`) and 4th node (`val = 4`).

2nd node (`val = 2`)'s neighbors are 1st node (`val = 1`) and 3rd node (`val = 3`).

3rd node (`val = 3`)'s neighbors are 2nd node (`val = 2`) and 4th node (`val = 4`).

4th node (`val = 4`)'s neighbors are 1st node (`val = 1`) and 3rd node (`val = 3`).

Example 2:



Input: `adjList = [[]]`

Output: `[[]]`

Explanation: Note that the input contains one empty list. The graph consists of only one node with `val = 1` and it does not have any neighbors.

Example 3:

Input: adjList = []

Output: []

Explanation: This an empty graph, it does not have any nodes.

Constraints:

- The number of nodes in the graph is in the range [0, 100].
- $1 \leq \text{Node.val} \leq 100$
- Node.val is unique for each node.
- There are no repeated edges and no self-loops in the graph.
- The Graph is connected and all nodes can be visited starting from the given node.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Deep copy an undirected connected graph. No shared references with original. Right?"

Graphs can have cycles – unlike trees, so we need a visited map."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

BFS from the given node: maintain a map old_node -> new_node clone.

When visiting neighbors, create clones on first sight and enqueue them.

Straightforward graph copy – already the standard $O(V+E)$ approach.

Time: $O(V + E)$. Space: $O(V)$ for the clone map and queue.

Should I go ahead and optimize?"

```
class _133_CloneGraph:
    def cloneGraph(self, node: 'Node') -> 'Node':
        if not node:
            return None
        cloned: dict = {}
        def clone(n):
            if n in cloned:
                return cloned[n]
            copy = Node(n.val)
```

```
        cloned[n] = copy # mark before recursing to handle cycles
        copy.neighbors = [clone(nb) for nb in n.neighbors]
        return copy
    return clone(node)
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(V+E)$. Space $O(V)$ for the visited map. Mark node before recursing into neighbours —

without this, cycles cause infinite recursion. Same visited-map pattern handles any graph.

Given more time I'd ask: clone a directed graph? Same approach — just don't add reverse edges."

DFS

□ #200 Number of Islands

MED[Open on LeetCode](#)

Array · DFS · BFS · Union Find · Matrix

Lists: Grind 169

Problem

Given an $m \times n$ 2D binary grid `grid` which represents a map of '1's (land) and '0's (water), return the number of islands.

An island is surrounded by water and is formed by connecting adjacent lands horizontally or vertically. You may assume all four edges of the grid are all surrounded by water.

Example 1:

```
Input: grid = [
  ["1","1","1","1","0"],
  ["1","1","0","1","0"],
  ["1","1","0","0","0"],
  ["0","0","0","0","0"]
]
Output: 1
```

Example 2:


```

Input: grid = [
["1","1","0","0","0"],
["1","1","0","0","0"],
["0","0","1","0","0"],
["0","0","0","1","1"]
]
Output: 3

```

Constraints:

- $m == \text{grid.length}$
- $n == \text{grid}[i].\text{length}$
- $1 \leq m, n \leq 300$
- $\text{grid}[i][j]$ is '0' or '1'.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Count connected components of '1's in a binary grid. Diagonal connections don't count. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."
 # Double loop over every grid cell. Whenever I see an unvisited '1',
 # run DFS/BFS to mark the entire connected component sunk, then increment island count.
 # Each cell visited once – correct and actually $O(m \cdot n)$ already.
 # Time: $O(m \cdot n)$. Space: $O(m \cdot n)$ worst-case recursion stack or queue.
 # The flood-fill is the right approach; I'll keep it clean with in-place marking."

```

class _200_NumberOfIslands:
    def numIslands(self, grid: List[List[str]]) -> int:
        m, n = len(grid), len(grid[0])
        def dfs(r, c):
            if r < 0 or r >= m or c < 0 or c >= n or grid[r][c] != '1':
                return
            grid[r][c] = '#' # mark visited by flooding
            for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                dfs(r + dr, c + dc)
        count = 0
        for r in range(m):
            for c in range(n):

```

```
        if grid[r][c] == '1':
            dfs(r, c)
            count += 1
    return count
```

PHASE 5 — TEST & DEBUG

1-island grid: DFS floods the whole island. count=1 ✓. 3-island: count=3 ✓.

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m*n)$. Space $O(m*n)$ stack. Flooding in-place avoids a separate visited set.

BFS alternative: use a queue instead of recursion – better for very large grids (no stack overflow).

Union-Find alternative: $O(m*n * \alpha(m*n))$ – useful when islands merge dynamically.

Given more time I'd ask: restore the grid after counting?

Keep a separate visited set or restore '#' → '1' after each DFS."

DFS

□ #207 Course Schedule

MED[Open on LeetCode](#)

DFS · BFS · Graph · Topological Sort

Lists: Grind 169

Problem

There are a total of `numCourses` courses you have to take, labeled from 0 to `numCourses - 1`. You are given an array `prerequisites` where `prerequisites[i] = [ai, bi]` indicates that you must take course `bi` first if you want to take course `ai`.

- For example, the pair `[0, 1]`, indicates that to take course 0 you have to first take course 1.

Return `true` if you can finish all courses.

Otherwise, return `false`.

Example 1:

Input: `numCourses = 2, prerequisites = [[1,0]]`

Output: `true`

Explanation: There are a total of 2 courses to take.

To take course 1 you should have finished course 0. So it is possible.

Example 2:

Input: numCourses = 2, prerequisites = [[1,0],[0,1]]

Output: false

Explanation: There are a total of 2 courses to take.

To take course 1 you should have finished course 0, and to take course 0 you should also have finished course 1. So it is impossible.

Constraints:

- $1 \leq \text{numCourses} \leq 2000$
- $0 \leq \text{prerequisites.length} \leq 5000$
- $\text{prerequisites}[i].\text{length} == 2$
- $0 \leq a_i, b_i < \text{numCourses}$
- All the pairs $\text{prerequisites}[i]$ are unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Can all courses be finished given prerequisites (directed edges)?

Equivalent to: does the directed graph have a cycle? Return True if no cycle. Right?"

#

"[[1,0],[0,1]]: 0→1 and 1→0 form a cycle → False ✓"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Build adjacency + indegree; Kahn BFS peels zero-indegree courses layer by layer.

If we finish fewer than numCourses nodes, a cycle exists ? return false.

Same topological sort as Course Schedule II, but only need boolean completion.

Time: $O(V + E)$. Space: $O(V + E)$.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (3-colour DFS cycle detection)

"Color each node: 0=unvisited, 1=in current DFS path (grey), 2=fully processed (black).

Cycle exists if we encounter a grey node. $O(V+E)$."

PHASE 4 — CLEAN CODE

class _207_CourseSchedule:

def canFinish(self, numCourses: int, prerequisites: List[List[int]]) -> bool:

graph = defaultdict(list)

for course, pre in prerequisites:

graph[pre].append(course)

```

color = [0] * numCourses

def has_cycle(node):
    if color[node] == 1: return True # grey → back edge → cycle
    if color[node] == 2: return False # black → already processed
    color[node] = 1
    for nb in graph[node]:
        if has_cycle(nb): return True
    color[node] = 2
    return False

return not any(has_cycle(c) for c in range(numCourses))

```

PHASE 5 — TEST & DEBUG

[[1,0],[0,1]]: has_cycle(0)→grey. has_cycle(1)→grey. has_cycle(0)→grey(already) → cycle! ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(V+E)$. Space $O(V+E)$. The 3-colour scheme distinguishes 'seen in current path' from 'fully done'.

Kahn's BFS alternative (in-degree tracking): count = courses processed; if count < numCourses → cycle.

Given more time I'd ask: Course Schedule II  (#210)? Same DFS, collect reverse post-order."

DFS

#210 Course Schedule II

MED[Open on LeetCode](#)

DFS · BFS · Graph · Topological Sort

Lists: Grind 169

Problem

There are a total of `numCourses` courses you have to take, labeled from 0 to `numCourses - 1`. You are given an array `prerequisites` where `prerequisites[i] = [ai, bi]` indicates that you must take course `bi` first if you want to take course `ai`.

- For example, the pair `[0, 1]`, indicates that to take course 0 you have to first take course 1.

Return the ordering of courses you should take to finish all courses. If there are many valid answers, return any of them. If it is impossible to finish all courses, return an empty array.

Example 1:

Input: `numCourses = 2, prerequisites = [[1,0]]`

Output: `[0,1]`

Explanation: There are a total of 2 courses to take. To take course 1 you should have finished course 0. So the correct course order is `[0,1]`.

Example 2:

Input: numCourses = 4, prerequisites = [[1,0],[2,0],[3,1],[3,2]]

Output: [0,2,1,3]

Explanation: There are a total of 4 courses to take. To take course 3 you should have finished both courses 1 and 2. Both courses 1 and 2 should be taken after you finished course 0.

So one correct course order is [0,1,2,3]. Another correct ordering is [0,2,1,3].

Example 3:

Input: numCourses = 1, prerequisites = []

Output: [0]

Constraints:

- $1 \leq \text{numCourses} \leq 2000$
- $0 \leq \text{prerequisites.length} \leq \text{numCourses} * (\text{numCourses} - 1)$
- $\text{prerequisites}[i].\text{length} == 2$
- $0 \leq a_i, b_i < \text{numCourses}$
- $a_i \neq b_i$
- All the pairs $[a_i, b_i]$ are distinct.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Return a valid course ordering. Return [] if impossible (cycle exists). Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

Topological sort via Kahn's BFS: compute indegree of every course,

enqueue zero-indegree nodes, peel layers while decrementing neighbors.

If we process all n courses, ordering exists; otherwise a cycle remains.

Time: $O(V + E)$. Space: $O(V + E)$ for adjacency and indegree arrays.

Should I go ahead and optimize?"

```
class _210_CourseScheduleII:
```

```
    def findOrder(self, numCourses: int, prerequisites: List[List[int]]) -> List[int]:
```

```

graph = defaultdict(list)
for course, pre in prerequisites:
    graph[pre].append(course)
color = [0] * numCourses
order = []
def dfs(node) -> bool:
    if color[node] == 1: return False # cycle
    if color[node] == 2: return True # already done
    color[node] = 1
    for nb in graph[node]:
        if not dfs(nb): return False
    color[node] = 2
    order.append(node) # post-order: all descendants before this node
    return True
for c in range(numCourses):
    if not dfs(c):
        return []
return order[::-1] # reverse post-order = topological order

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(V+E)$. Space $O(V+E)$. Post-order append then reverse = topological sort.
 # Given more time I'd ask: Kahn's BFS? Build in-degree map, process zero-in-degree nodes first.
 # Both approaches are $O(V+E)$ – DFS is recursive, Kahn's is iterative."

DFS

□ ★ #261 Graph Valid Tree ★

MED

[Open on LeetCode](#)

DFS · BFS · Union Find · Graph

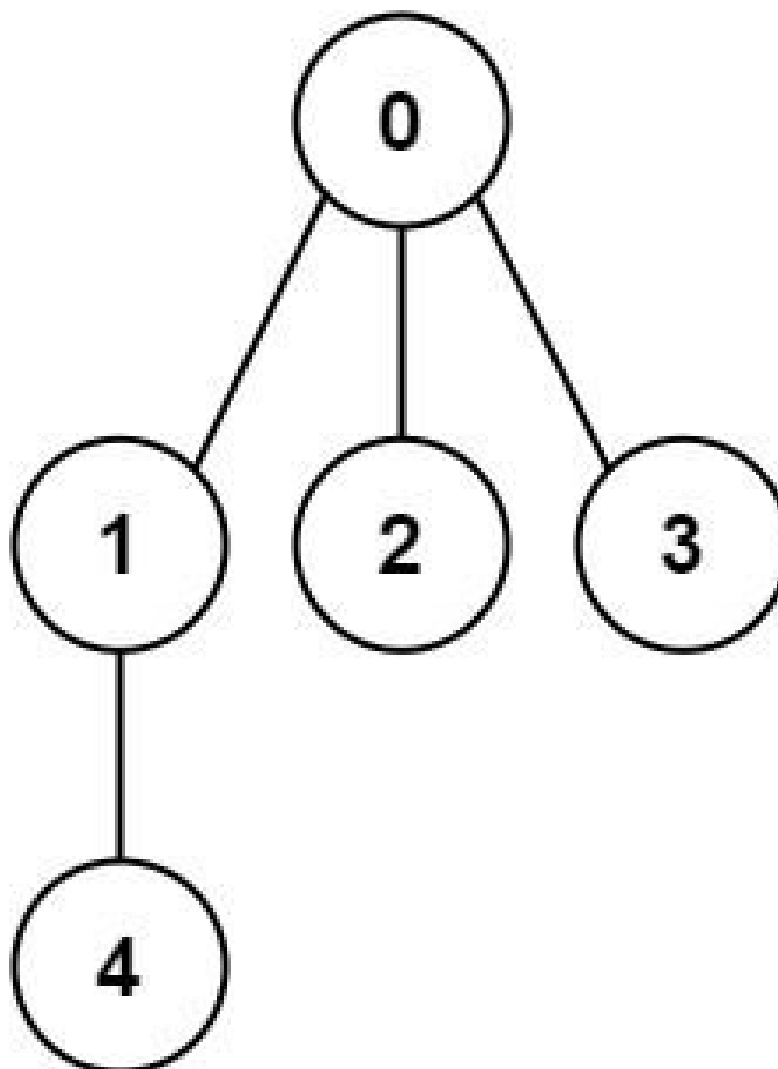
Lists: ★ Premium | Grind 169 | Premium 98

Problem

You have a graph of n nodes labeled from 0 to $n - 1$. You are given an integer n and a list of edges where $\text{edges}[i] = [a_i, b_i]$ indicates that there is an undirected edge between nodes a_i and b_i in the graph.

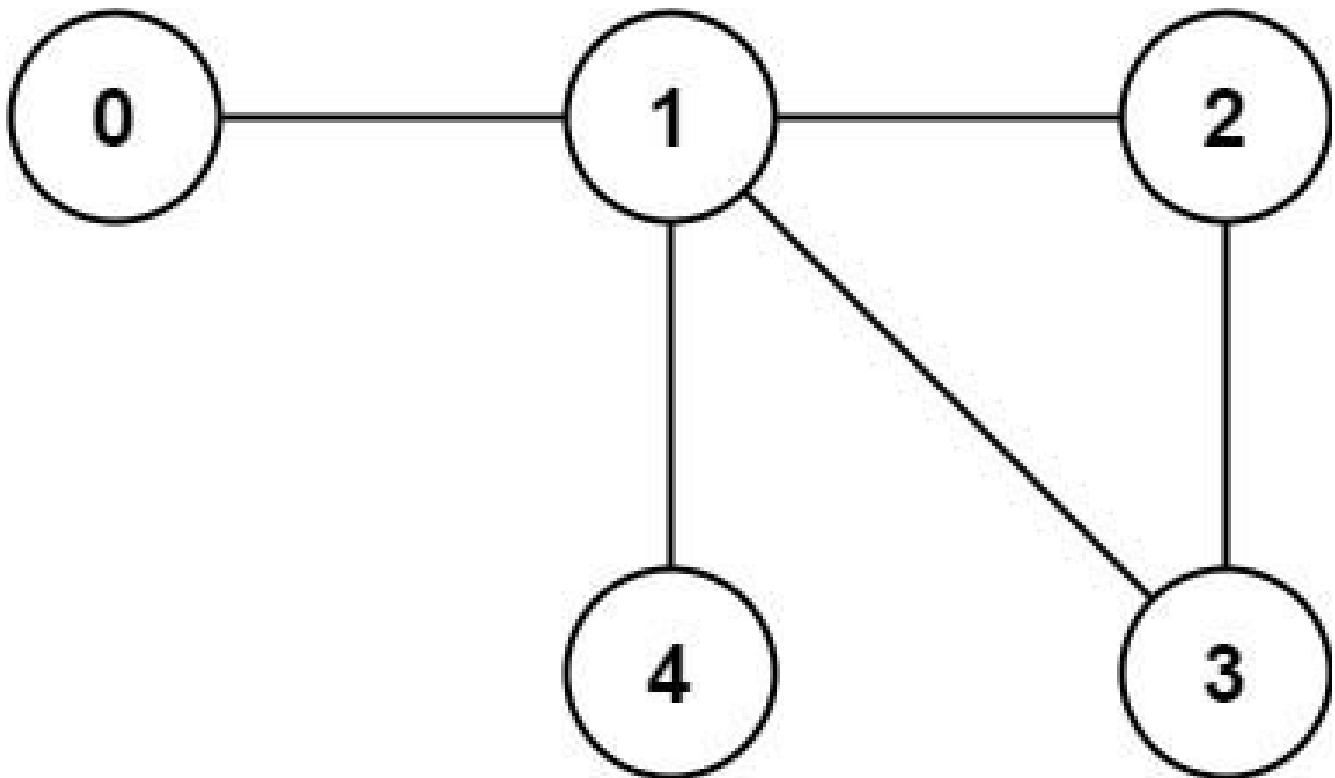
Return true if the edges of the given graph make up a valid tree, and false otherwise.

Example 1:



Input: $n = 5$, $edges = [[0,1],[0,2],[0,3],[1,4]]$
Output: true

Example 2:



Input: $n = 5$, $edges = [[0,1],[1,2],[2,3],[1,3],[1,4]]$
Output: false

Constraints:

- $1 \leq n \leq 2000$
- $0 \leq edges.length \leq 5000$
- $edges[i].length == 2$
- $0 \leq a_i, b_i < n$
- $a_i \neq b_i$
- There are no self-loops or repeated edges.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Valid tree: connected, acyclic, exactly $n-1$ edges. Right?"

Any two of these imply the third for n nodes."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

Build adjacency from edges, run DFS from node 0 marking visited.

After DFS, verify every node was visited and total edges equals $n-1$.

Catches disconnected graphs and cycles in one pass.

Time: $O(n)$. Space: $O(n)$ for adjacency + visited.

Should I go ahead and optimize?"

```
class _261_GraphValidTree:
```

```
    def validTree(self, n: int, edges: List[List[int]]) -> bool:
```

```
        if len(edges) != n - 1:
```

```
            return False
```

```
        graph = defaultdict(list)
```

```
        for a, b in edges:
```

```
            graph[a].append(b)
```

```
            graph[b].append(a)
```

```
        visited = set()
```

```
        def dfs(node, parent):
```

```
            visited.add(node)
```

```
            for nb in graph[node]:
```

```
                if nb == parent: continue
```

```
                if nb in visited: return False
```

```
                if not dfs(nb, node): return False
```

```
            return True
```

```
        return dfs(0, -1) and len(visited) == n
```

PHASE 5 — TEST & DEBUG

$n=5$, $edges=[[0,1],[0,2],[0,3],[1,4]]$: 4 edges== $n-1$ ✓. DFS from 0 visits all 5. No cycle. → True ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(V+E)=O(n)$. Space $O(n)$. Early return if $edges \neq n-1$ is a fast filter."

Union-Find alternative: $O(n * \alpha(n))$ – merge edges, detect cycle if already connected.

Given more time I'd ask: disconnected forest (multiple trees)?

Return number of components instead – skip the $len(edges)==n-1$ check."

DFS

□ #310 Minimum Height Trees

MED[Open on LeetCode](#)

DFS · BFS · Graph · Topological Sort

Lists: Grind 169

Problem

A tree is an undirected graph in which any two vertices are connected by exactly one path. In other words, any connected graph without simple cycles is a tree.

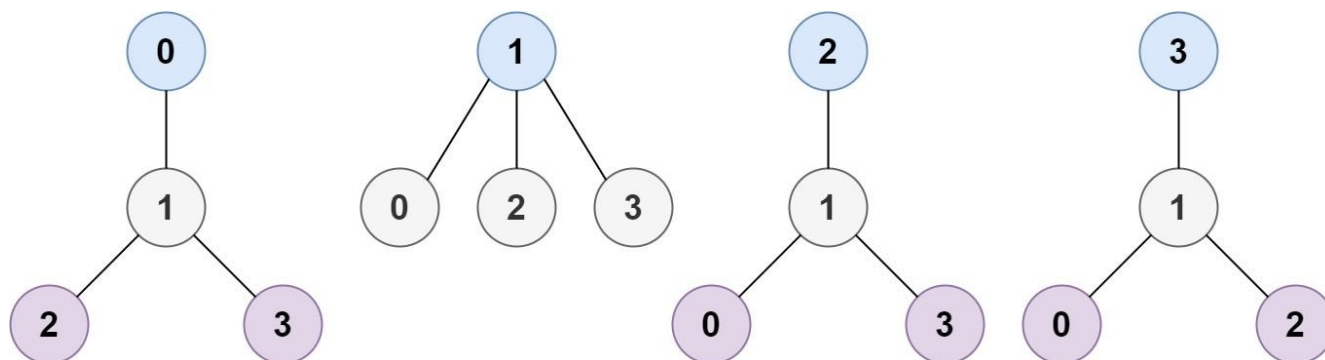
Given a tree of n nodes labelled from 0 to $n - 1$, and an array of $n - 1$ edges where $\text{edges}[i] = [a_i, b_i]$ indicates that there is an undirected edge between the two nodes a_i and b_i in the tree, you can choose any node of the tree as the root.

When you select a node x as the root, the result tree has height h . Among all possible rooted trees, those with minimum height (i.e. $\min(h)$) are called minimum height trees (MHTs).

Return a list of all MHTs' root labels. You can return the answer in any order.

The height of a rooted tree is the number of edges on the longest downward path between the root and a leaf.

Example 1:

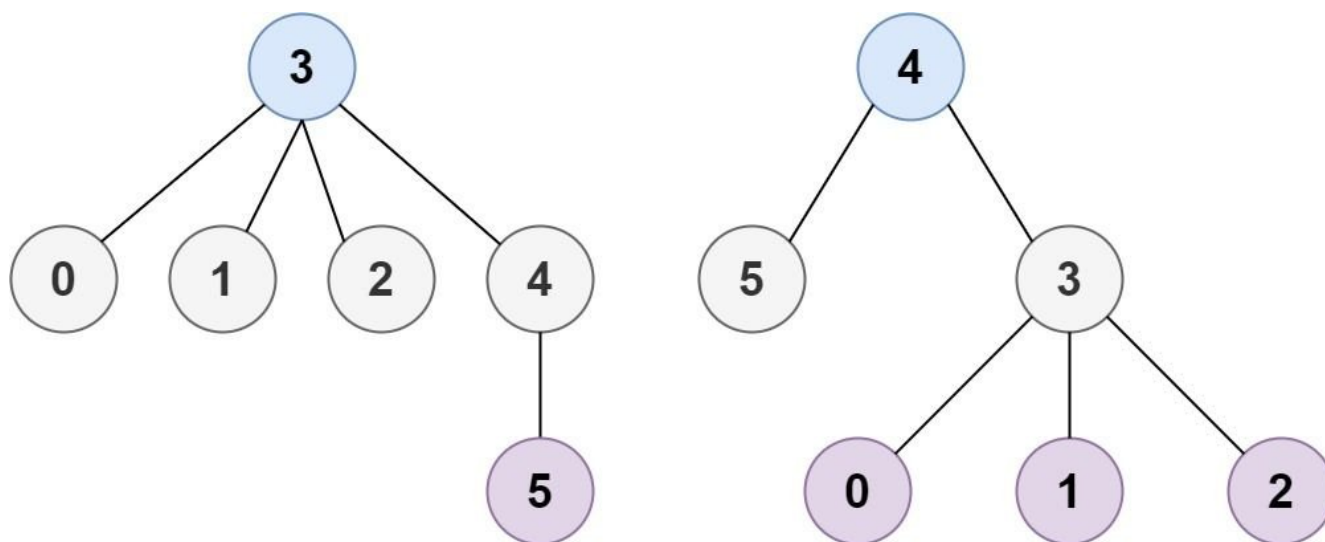


Input: $n = 4$, $edges = [[1,0],[1,2],[1,3]]$

Output: [1]

Explanation: As shown, the height of the tree is 1 when the root is the node with label 1 which is the only MHT.

Example 2:



Input: $n = 6$, $edges = [[3,0],[3,1],[3,2],[3,4],[5,4]]$

Output: [3,4]

Constraints:

- $1 \leq n \leq 2 * 10^4$
- `edges.length == n - 1`
- $0 \leq a_i, b_i < n$
- $a_i \neq b_i$
- All the pairs (a_i, b_i) are distinct.
- The given input is guaranteed to be a tree and there will be no repeated edges.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find which nodes make MHT roots. The answer is at most 2 nodes — the tree's center(s). Right?"

Repeatedly prune leaf nodes (degree 1) until 1 or 2 nodes remain."

#

"Like peeling an onion from outside in — the core is the answer."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

Repeatedly peel all leaves layer by layer (nodes with degree 1), like peeling an onion.

The last remaining nodes (0, 1, or 2) are the tree centroids / MHT roots.

BFS leaf-removal is the standard $O(n)$ approach for this problem.

Time: $O(n)$. Space: $O(n)$ for adjacency and degree counts.

Should I go ahead and optimize?"

```
class _310_MinHeightTrees:
```

```
    def findMinHeightTrees(self, n: int, edges: List[List[int]]) -> List[int]:
```

```
        if n == 1:
```

```
            return [0]
```

```
        graph = defaultdict(set)
```

```
        for a, b in edges:
```

```
            graph[a].add(b)
```

```
            graph[b].add(a)
```

```
        leaves = [node for node in range(n) if len(graph[node]) == 1]
```

```
        remaining = n
```

```
        while remaining > 2:
```

```
            remaining -= len(leaves)
```

```
            new_leaves = []
```

```
            for leaf in leaves:
```

```

        nb = next(iter(graph[leaf]))
        graph[nb].remove(leaf)
        if len(graph[nb]) == 1:
            new_leaves.append(nb)
        leaves = new_leaves
    return leaves

```

PHASE 5 — TEST & DEBUG

```

# n=4, edges=[[1,0],[1,2],[1,3]]: leaves=[0,2,3]. remaining=4-3=1. new_leaves=[1].
→ [1] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(n)$. This is topological sort on undirected tree – same BFS idea.

At most 2 centers exist – if more, tree would have multiple independent long paths.

Given more time I'd ask: all trees of minimum height (not just roots)?

BFS from all roots found, check height – same $O(n)$."

DFS

- ★ #323 Number of Connected Components in an Undirected Graph ★
[Open on LeetCode](#)
-

MED

DFS · BFS · Union Find · Graph

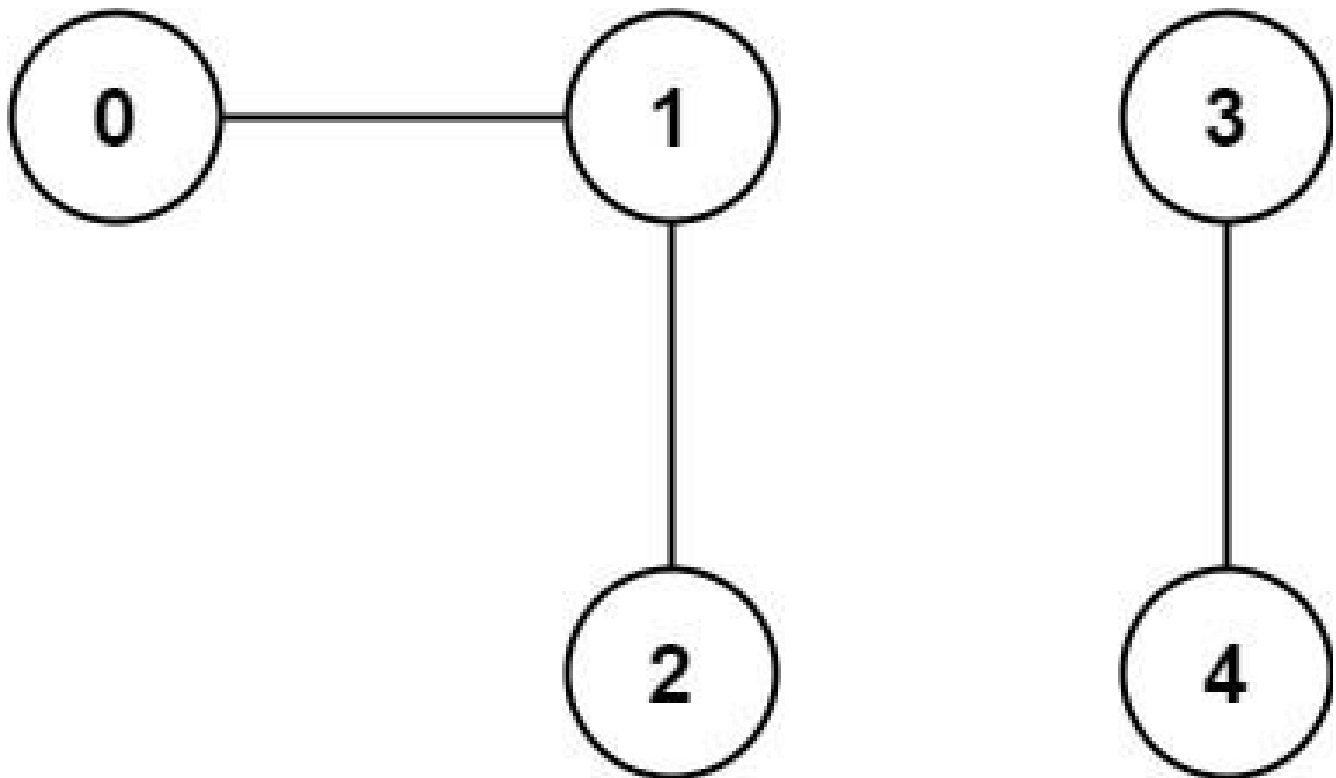
Lists: ★ Premium | Grind 169 | Premium 98

Problem

You have a graph of n nodes. You are given an integer n and an array `edges` where `edges[i] = [ai, bi]` indicates that there is an edge between a_i and b_i in the graph.

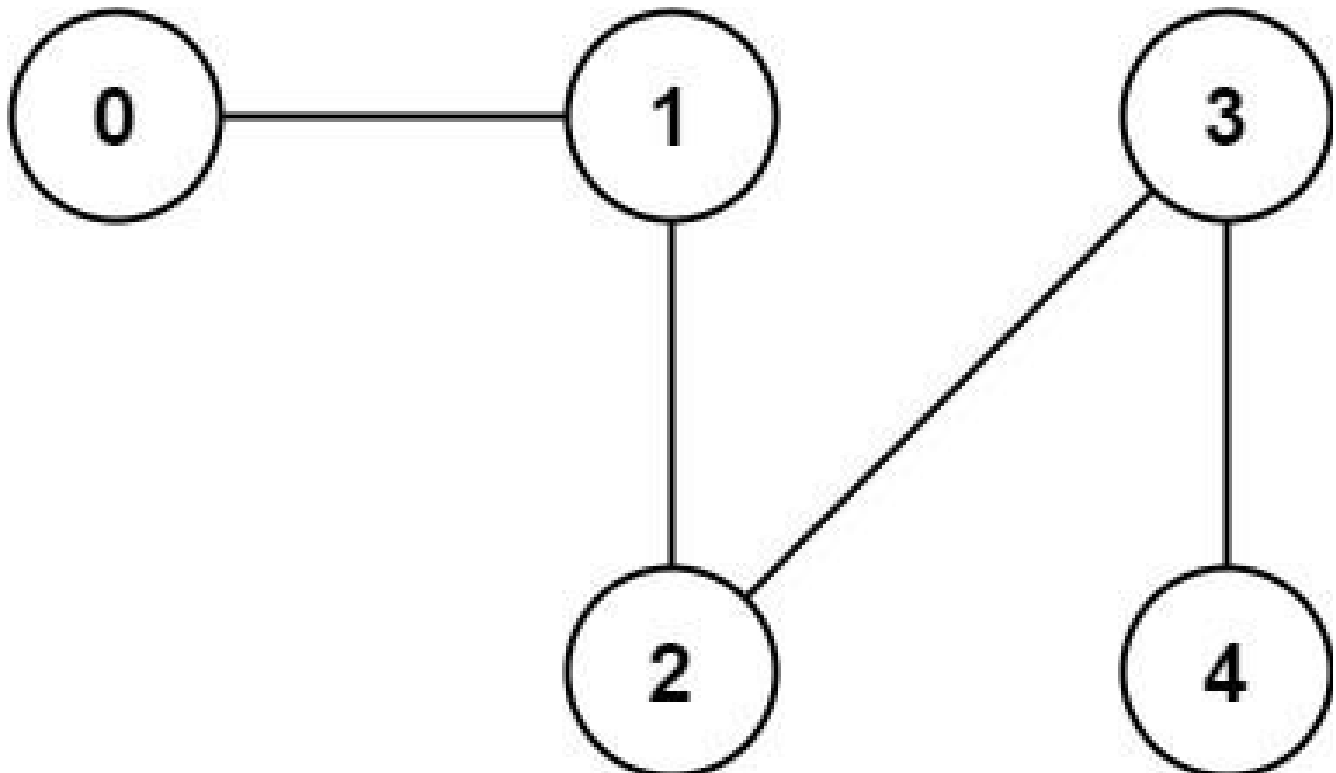
Return the number of connected components in the graph.

Example 1:



Input: $n = 5$, $edges = [[0,1],[1,2],[3,4]]$
Output: 2

Example 2:



Input: $n = 5$, $edges = [[0,1],[1,2],[2,3],[3,4]]$
 Output: 1

Constraints:

- $1 \leq n \leq 2000$
- $1 \leq edges.length \leq 5000$
- $edges[i].length == 2$
- $0 \leq a_i \leq b_i < n$
- $a_i \neq b_i$
- There are no repeated edges.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Count connected components in an undirected graph. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."
 # Union-Find: for each edge, union the two endpoints.
 # Answer = n minus number of distinct roots after all unions.
 # Simple DSU with path compression – $O(n \alpha(n))$.
 # Time: $O(n \alpha(n))$. Space: $O(n)$ for parent array.
 # Should I go ahead and optimize?"

```
class _323_NumberConnectedComponents:
    def countComponents(self, n: int, edges: List[List[int]]) -> int:
        graph = defaultdict(list)
        for a, b in edges:
            graph[a].append(b)
            graph[b].append(a)
        visited = set()
        def dfs(node):
            visited.add(node)
            for nb in graph[node]:
                if nb not in visited:
                    dfs(nb)
        count = 0
        for node in range(n):
```

```
        if node not in visited:
            dfs(node)
            count += 1
    return count
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(V+E)$. Space $O(V+E)$. Standard DFS counting components.

Union-Find alternative: merge edges, count unique roots $\rightarrow O(n * \alpha(n))$.

Given more time I'd ask: what if edges are added dynamically?

Union-Find with path compression supports $O(\alpha(n))$ per merge query."

DFS

□ #417 Pacific Atlantic Water Flow

MED[Open on LeetCode](#)

Array · DFS · BFS · Matrix

Lists: Grind 169

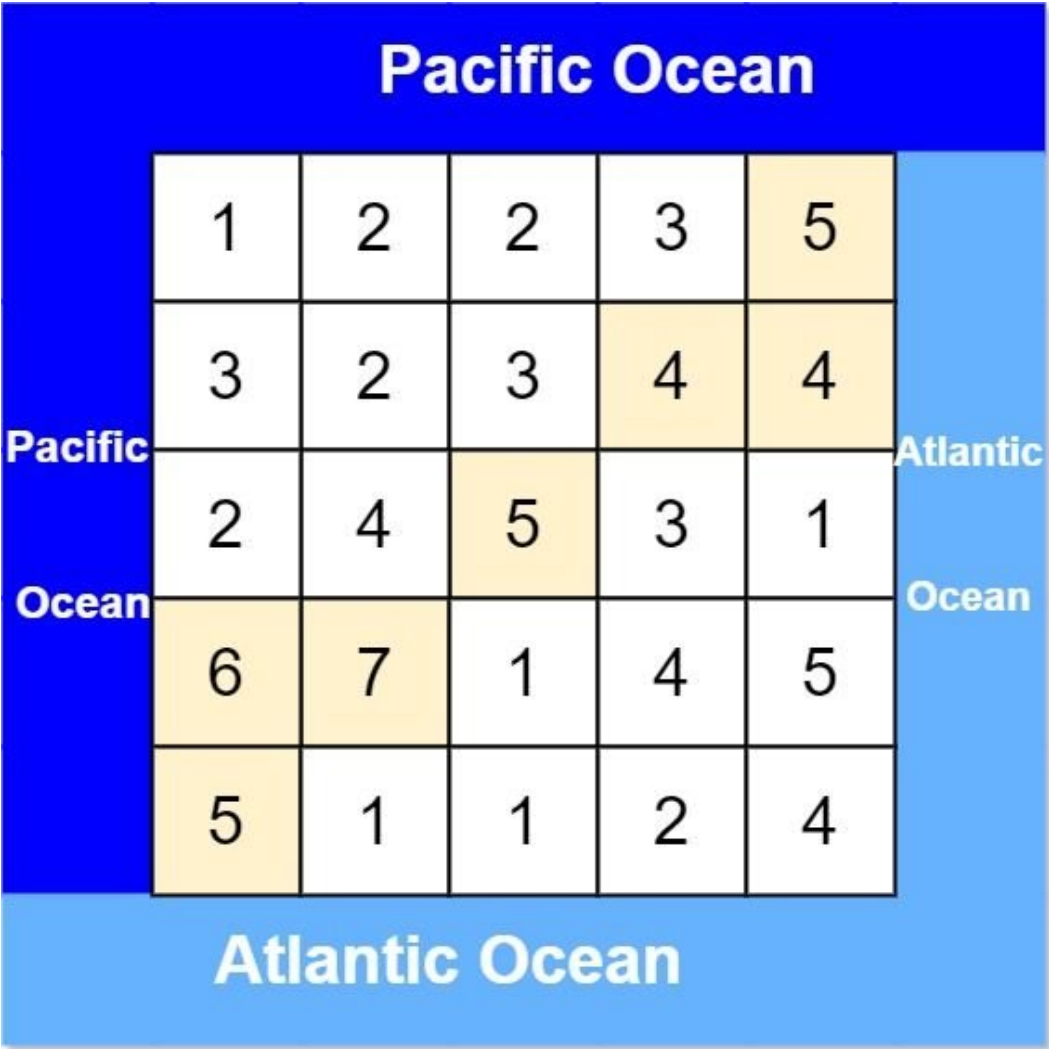
Problem

There is an $m \times n$ rectangular island that borders both the Pacific Ocean and Atlantic Ocean. The Pacific Ocean touches the island's left and top edges, and the Atlantic Ocean touches the island's right and bottom edges. The island is partitioned into a grid of square cells. You are given an $m \times n$ integer matrix `heights` where `heights[r][c]` represents the height above sea level of the cell at coordinate (r, c) .

The island receives a lot of rain, and the rain water can flow to neighboring cells directly north, south, east, and west if the neighboring cell's height is less than or equal to the current cell's height. Water can flow from any cell adjacent to an ocean into the ocean.

Return a 2D list of grid coordinates result where $\text{result}[i] = [\text{ri}, \text{ci}]$ denotes that rain water can flow from cell (ri, ci) to both the Pacific and Atlantic oceans.

Example 1:



Input: heights = [[1,2,2,3,5],[3,2,3,4,4],[2,4,5,3,1],[6,7,1,4,5],[5,1,1,2,4]]
 Output: [[0,4],[1,3],[1,4],[2,2],[3,0],[3,1],[4,0]]
 Explanation: The following cells can flow to the Pacific and Atlantic oceans, as shown below:
 [0,4]: [0,4] -> Pacific Ocean
 [0,4] -> Atlantic Ocean
 [1,3]: [1,3] -> [0,3] -> Pacific Ocean
 [1,3] -> [1,4] -> Atlantic Ocean
 [1,4]: [1,4] -> [1,3] -> [0,3] -> Pacific Ocean

Example 2:

Input: heights = [[1]]
 Output: [[0,0]]
 Explanation: The water can flow from the only cell to the Pacific and Atlantic oceans.

Constraints:

- $m == \text{heights.length}$
- $n == \text{heights}[r].\text{length}$
- $1 \leq m, n \leq 200$
- $0 \leq \text{heights}[r][c] \leq 105$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Water flows from higher to lower (or equal). Find cells that can flow to BOTH oceans.

Pacific touches top/left edges. Atlantic touches bottom/right edges. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

From each cell, DFS to see if it can reach both Pacific (top/left border) and Atlantic (bottom/right).

Re-run DFS per cell - $O((m*n)^2)$ naive.

Time: $O((m * n)^2)$ naive per-cell DFS. Space: $O(m * n)$ stack.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (reverse DFS from ocean borders)

"Instead of simulating forward flow, run DFS BACKWARDS from each ocean's border cells.

```
# Reverse flow: go uphill (neighbor height >= current height).
# Intersection of Pacific-reachable and Atlantic-reachable = answer."
```

PHASE 4 — CLEAN CODE

```
class _417_PacificAtlantic:
    def pacificAtlantic(self, heights: List[List[int]]) -> List[List[int]]:
        m, n = len(heights), len(heights[0])
        def bfs(starts):
            reachable = set(starts)
            queue = deque(starts)
            while queue:
                r, c = queue.popleft()
                for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                    nr, nc = r + dr, c + dc
                    if 0 <= nr < m and 0 <= nc < n and (nr,nc) not in reachable \
                        and heights[nr][nc] >= heights[r][c]:
                        reachable.add((nr, nc))
                        queue.append((nr, nc))
            return reachable
        pacific = [(0, c) for c in range(n)] + [(r, 0) for r in range(1, m)]
        atlantic = [(m-1, c) for c in range(n)] + [(r, n-1) for r in range(m-1)]
        pac_reach = bfs(pacific)
        atl_reach = bfs(atlantic)
        return [[r, c] for r, c in pac_reach & atl_reach]
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m*n)$ . Space  $O(m*n)$ . BFS from borders is cleaner than DFS (no recursion limit).
# The reversal insight: 'can reach ocean' → 'ocean can flow backward to this cell'.
# Given more time I'd ask: what if water can also flow diagonally?
# Add 4 diagonal directions – same algorithm."
```


DFS

□ ★ #490 The Maze ★

MED

[Open on LeetCode](#)

Array · DFS · BFS · Matrix

Lists: ★ Premium | Premium 98

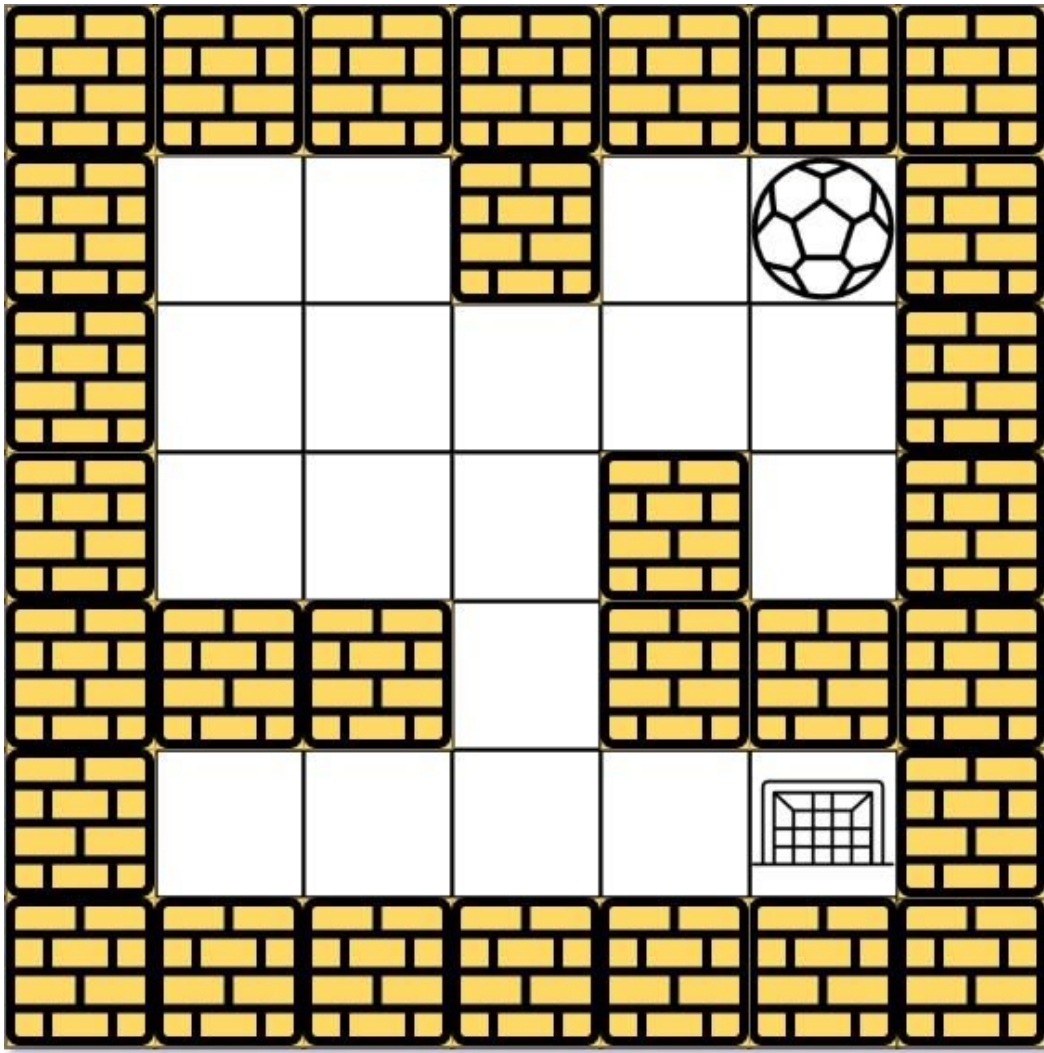
Problem

There is a ball in a maze with empty spaces (represented as 0) and walls (represented as 1). The ball can go through the empty spaces by rolling up, down, left or right, but it won't stop rolling until hitting a wall. When the ball stops, it could choose the next direction.

Given the $m \times n$ maze, the ball's start position and the destination, where $\text{start} = [\text{startrow}, \text{startcol}]$ and $\text{destination} = [\text{destinationrow}, \text{destinationcol}]$, return true if the ball can stop at the destination, otherwise return false.

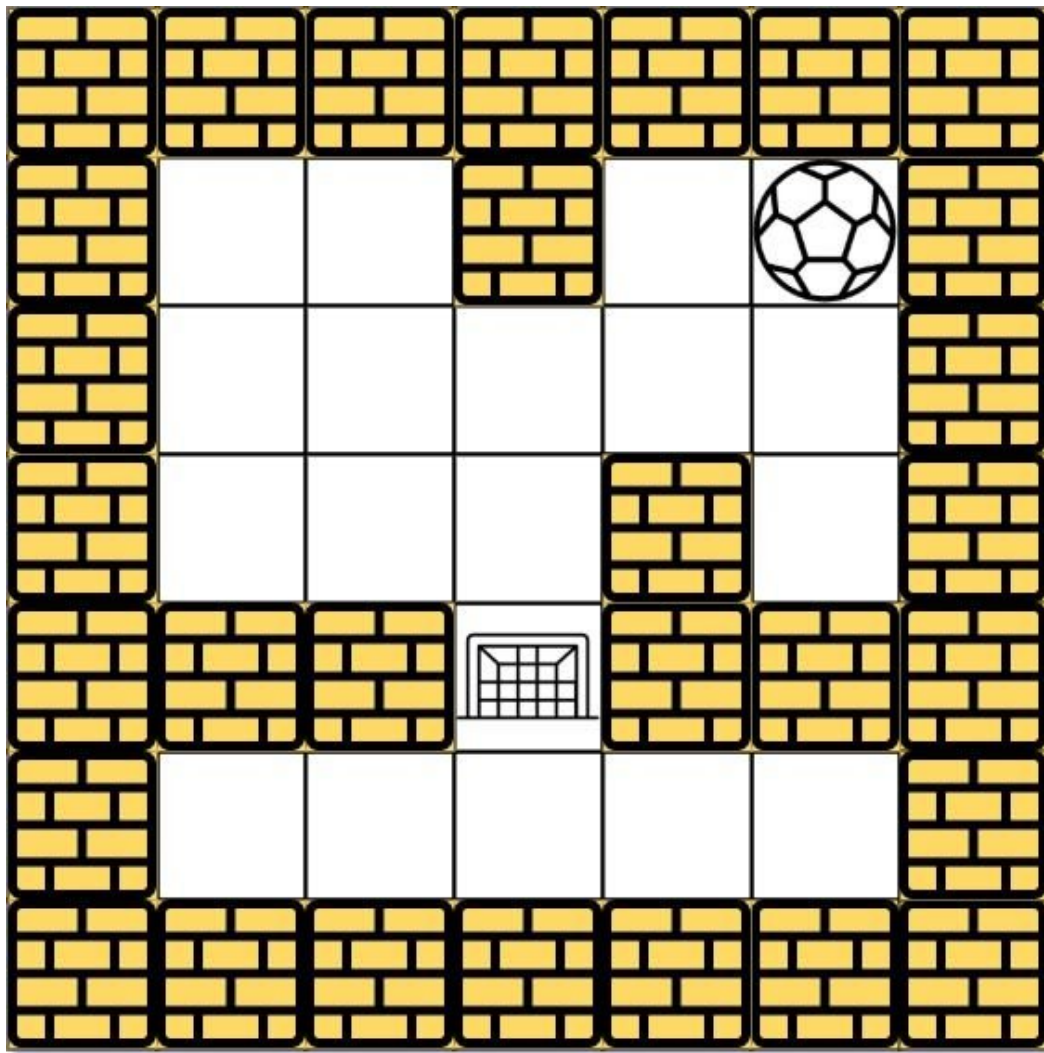
You may assume that the borders of the maze are all walls (see examples).

Example 1:



Input: maze = `[[0,0,1,0,0],[0,0,0,0,0],[0,0,0,1,0],[1,1,0,1,1],[0,0,0,0,0]]`,
 start = `[0,4]`, destination = `[4,4]`
 Output: true
 Explanation: One possible way is : left -> down -> left -> down -> right ->
 down -> right.

Example 2:



Input: maze = `[[0,0,1,0,0],[0,0,0,0,0],[0,0,0,1,0],[1,1,0,1,1],[0,0,0,0,0]]`,
 start = `[0,4]`, destination = `[3,2]`

Output: false

Explanation: There is no way for the ball to stop at the destination. Notice that you can pass through the destination but you cannot stop there.

Example 3:

Input: maze = `[[0,0,0,0,0],[1,1,0,0,1],[0,0,0,0,0],[0,1,0,0,1],[0,1,0,0,0]]`,
 start = `[4,3]`, destination = `[0,1]`

Output: false

Constraints:

- `m == maze.length`
- `n == maze[i].length`

- $1 \leq m, n \leq 100$
- `maze[i][j]` is 0 or 1.
- `start.length == 2`
- `destination.length == 2`
- $0 \leq \text{startrow}, \text{destinationrow} < m$
- $0 \leq \text{startcol}, \text{destinationcol} < n$
- Both the ball and the destination exist in an empty space, and they will not be in the same position initially.
- The maze contains at least 2 empty spaces.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Ball rolls in a direction until hitting a wall. Can it STOP at the destination?
 # It can pass through the destination but not stop there unless it hits a wall.
 Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
 # DFS from entrance: at each empty cell try all 4 directions.
 # Mark visited cells to avoid cycles; succeed if we reach the boundary.
 # Correct path exploration on an $m \times n$ grid.
 # Time: $O(m * n)$. Space: $O(m * n)$ visited.
 # Should I go ahead and optimize?"

```
class _490_TheMaze:
    def hasPath(self, maze: List[List[int]], start: List[int], destination:
List[int]) -> bool:
        m, n = len(maze), len(maze[0])
        dest = tuple(destination)
        visited: set = set()
        def dfs(r, c):
            if (r, c) in visited: return False
            visited.add((r, c))
            if (r, c) == dest: return True
            for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
```

```

        nr, nc = r, c
        while 0 <= nr + dr < m and 0 <= nc + dc < n and maze[nr+dr][nc+dc]
== 0:
            nr += dr; nc += dc
            if dfs(nr, nc): return True
        return False
    return dfs(start[0], start[1])

```

PHASE 5 — TEST & DEBUG

Ball rolls until hitting wall → stops → that stop position is a new state.

DFS on stop-positions. Destination reached only if the ball stops exactly there. ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m \cdot n \cdot (m+n))$ – each stop position visits $O(m+n)$ cells to find next stop.

The visited set is on stop-positions, not individual cells.

Given more time I'd ask: The Maze II (#505) – find shortest distance?

Use Dijkstra with distance as weight, BFS/priority queue on stop-positions."

DFS

□ ★ #694 Number of Distinct Islands ★

MED

[Open on LeetCode](#)

Hash Table · DFS · BFS · Union Find · Hash Function

Lists: ★ Premium | Premium 98

Problem

You are given an $m \times n$ binary matrix grid. An island is a group of 1's (representing land) connected 4-directionally (horizontal or vertical.) You may assume all four edges of the grid are surrounded by water.

An island is considered to be the same as another if and only if one island can be translated (and not rotated or reflected) to equal the other.

Return the number of distinct islands.

Example 1:

1	1	0	0	0
1	1	0	0	0
0	0	0	1	1
0	0	0	1	1

Input: grid = [[1,1,0,0,0],[1,1,0,0,0],[0,0,0,1,1],[0,0,0,1,1]]
Output: 1

Example 2:

1	1	0	1	1
1	0	0	0	0
0	0	0	0	1
1	1	0	1	1

Input: grid = [[1,1,0,1,1],[1,0,0,0,0],[0,0,0,0,1],[1,1,0,1,1]]
Output: 3

Constraints:

- $m == \text{grid.length}$
- $n == \text{grid}[i].\text{length}$
- $1 \leq m, n \leq 50$
- $\text{grid}[i][j]$ is either 0 or 1.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Count islands that are distinct by shape (translation equivalent). Right?
Two islands are the same if one can be shifted to match the other – no rotation/reflection."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
DFS each island; encode shape as a string of moves (U/D/L/R) from start cell.
Add shape string to a set – distinct shapes = set size.
Must normalize starting cell and direction encoding consistently.
Time: $O(m * n)$. Space: $O(m * n)$ shapes.
Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (path signature hashing)

"DFS each island, record the path taken relative to the starting cell.
Include backtrack markers to distinguish shapes like 'L' vs reverse 'L'.
Hash each shape as a tuple and add to a set – count unique shapes."

PHASE 4 — CLEAN CODE

```
class _694_DistinctIslands:
    def numDistinctIslands(self, grid: List[List[int]]) -> int:
        m, n = len(grid), len(grid[0])
        shapes: set = set()
        def dfs(r, c, r0, c0, path):
            if r < 0 or r >= m or c < 0 or c >= n or grid[r][c] != 1:
                return
            grid[r][c] = 0
            path.append((r - r0, c - c0))
            for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                dfs(r + dr, c + dc, r0, c0, path)
            path.append(None) # backtrack marker
        for r in range(m):
            for c in range(n):
                if grid[r][c] == 1:
                    path: List = []
                    dfs(r, c, r, c, path)
                    shapes.add(tuple(path))
        return len(shapes)
```

PHASE 5 — TEST & DEBUG

Two 2x2 islands: both DFS in same order → same relative path → same tuple → 1 distinct ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m*n)$. Space $O(m*n)$. Backtrack None markers distinguish different DFS orderings
that could produce the same sequence of offsets for different shapes.
Given more time I'd ask: allow rotation/reflection? Normalise shape to lexicographically smallest
among all 8 transformations – significantly more complex."

DFS

□ #695 Max Area of Island

MED

[Open on LeetCode](#)

Array · DFS · BFS · Union Find · Matrix

Lists: Denny Zhang

Problem

You are given an $m \times n$ binary matrix grid. An island is a group of 1's (representing land) connected 4-directionally (horizontal or vertical.) You may assume all four edges of the grid are surrounded by water.

The area of an island is the number of cells with a value 1 in the island.

Return the maximum area of an island in grid. If there is no island, return 0.

Example 1:

0	0	1	0	0	0	0	1	0	0	0	0	0
0	0	0	0	0	0	0	1	1	1	0	0	0
0	1	1	0	1	0	0	0	0	0	0	0	0
0	1	0	0	1	1	0	0	1	0	1	0	0
0	1	0	0	1	1	0	0	1	1	1	0	0
0	0	0	0	0	0	0	0	0	0	1	0	0
0	0	0	0	0	0	0	1	1	1	0	0	0
0	0	0	0	0	0	0	1	1	0	0	0	0

Input: grid = [[0,0,1,0,0,0,0,1,0,0,0,0,0],[0,0,0,0,0,0,0,1,1,1,0,0,0],[0,1,1,0,1,0,0,0,0,0,0,0,0],[0,1,0,0,1,1,0,0,1,0,1,0,0],[0,1,0,0,1,1,0,0,1,1,1,0,0],[0,0,0,0,0,0,0,0,0,0,1,0,0],[0,0,0,0,0,0,0,1,1,1,0,0,0],[0,0,0,0,0,0,0,0,1,1,0,0,0]]

Output: 6

Explanation: The answer is not 11, because the island must be connected 4-directionally.

Example 2:

Input: grid = [[0,0,0,0,0,0,0,0]]

Output: 0

Constraints:

- $m == \text{grid.length}$
- $n == \text{grid}[i].\text{length}$
- $1 \leq m, n \leq 50$
- $\text{grid}[i][j]$ is either 0 or 1.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Maximum area (cell count) of any connected island. Return 0 if no island. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

For each land cell, DFS to measure area of its island while marking visited.

Track the maximum area seen across all components.

Same flood-fill as Number of Islands, but accumulate area instead of count.

Time: $O(m * n)$. Space: $O(m * n)$ stack worst case.

Should I go ahead and optimize?"

```
class _695_MaxAreaIsland:
```

```
    def maxAreaOfIsland(self, grid: List[List[int]]) -> int:
```

```
        m, n = len(grid), len(grid[0])
```

```
        def dfs(r, c) -> int:
```

```
            if r < 0 or r >= m or c < 0 or c >= n or grid[r][c] != 1:
```

```
                return 0
```

```
            grid[r][c] = 0
```

```
            return 1 + dfs(r-1,c) + dfs(r+1,c) + dfs(r,c-1) + dfs(r,c+1)
```

```
        return max((dfs(r, c) for r in range(m) for c in range(n) if grid[r][c] == 1), default=0)
```

PHASE 5 — TEST & DEBUG

Each DFS floods an island and returns its cell count. max() over all islands. ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m*n)$. Space $O(m*n)$ stack. The return-value DFS pattern: each cell contributes 1 + sum of children."

This is the same DFS flood-fill as #200 (Number of Islands) with a count return.

Given more time I'd ask: merge two islands by flipping a '0'?

Enumerate each '0' cell, BFS to find adjacent distinct islands, sum their areas + 1."

DFS

□ #721 Accounts Merge

MED[Open on LeetCode](#)

Array · String · DFS · BFS · Union Find

Lists: Grind 169

Problem

Given a list of accounts where each element `accounts[i]` is a list of strings, where the first element `accounts[i][0]` is a name, and the rest of the elements are emails representing emails of the account.

Now, we would like to merge these accounts.

Two accounts definitely belong to the same person if there is some common email to both accounts. Note that even if two accounts have the same name, they may belong to different people as people could have the same name. A person can have any number of accounts initially, but all of their accounts definitely have the same name.

After merging the accounts, return the accounts in the following format: the first element of each

account is the name, and the rest of the elements are emails in sorted order. The accounts themselves can be returned in any order.

Example 1:

Input: accounts = [["John", "johnsmith@mail.com", "john_newyork@mail.com"], ["John", "johnsmith@mail.com", "john00@mail.com"], ["Mary", "mary@mail.com"], ["John", "johnnybravo@mail.com"]]

Output: [["John", "john00@mail.com", "john_newyork@mail.com", "johnsmith@mail.com"], ["Mary", "mary@mail.com"], ["John", "johnnybravo@mail.com"]]

Explanation:

The first and second John's are the same person as they have the common email "johnsmith@mail.com".

The third John and Mary are different people as none of their email addresses are used by other accounts.

We could return these lists in any order, for example the answer [['Mary', 'mary@mail.com'], ['John', 'johnnybravo@mail.com'], ['John', 'john00@mail.com', 'john_newyork@mail.com', 'johnsmith@mail.com']] would still be accepted.

Example 2:

Input: accounts = [["Gabe", "Gabe0@m.co", "Gabe3@m.co", "Gabe1@m.co"], ["Kevin", "Kevin3@m.co", "Kevin5@m.co", "Kevin0@m.co"], ["Ethan", "Ethan5@m.co", "Ethan4@m.co", "Ethan0@m.co"], ["Hanzo", "Hanzo3@m.co", "Hanzo1@m.co", "Hanzo0@m.co"], ["Fern", "Fern5@m.co", "Fern1@m.co", "Fern0@m.co"]]

Output: [["Ethan", "Ethan0@m.co", "Ethan4@m.co", "Ethan5@m.co"], ["Gabe", "Gabe0@m.co", "Gabe1@m.co", "Gabe3@m.co"], ["Hanzo", "Hanzo0@m.co", "Hanzo1@m.co", "Hanzo3@m.co"], ["Kevin", "Kevin0@m.co", "Kevin3@m.co", "Kevin5@m.co"], ["Fern", "Fern0@m.co", "Fern1@m.co", "Fern5@m.co"]]

Constraints:

- $1 \leq \text{accounts.length} \leq 1000$
- $2 \leq \text{accounts}[i].\text{length} \leq 10$
- $1 \leq \text{accounts}[i][j].\text{length} \leq 30$
- $\text{accounts}[i][0]$ consists of English letters.
- $\text{accounts}[i][j]$ (for $j > 0$) is a valid email.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Merge accounts that share at least one email. Same name ≠ same person.
Output: name + sorted merged emails. Any output order. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
Union-Find on emails: union every email in the same account list.
Group emails by root parent; sort each merged list for output.
Time: $O(N \log N)$ for sorting emails. Space: $O(N)$ DSU.
Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (DFS on email graph)

"Build a graph where emails in the same account are connected.
DFS collects all emails in a connected component = one merged account.
Track owner (account name) via the first email in each account."

PHASE 4 — CLEAN CODE

```
class _721_AccountsMerge:
    def accountsMerge(self, accounts: List[List[str]]) -> List[List[str]]:
        email_to_name : dict = {}
        email_to_emails : defaultdict = defaultdict(set)

        for account in accounts:
            name = account[0]
            first = account[1]
            for email in account[1:]:
                email_to_name[email] = name
                email_to_emails[first].add(email)
                email_to_emails[email].add(first)

        visited: set = set()
        result = []

        def dfs(email, component):
            visited.add(email)
            component.append(email)
            for nb in email_to_emails[email]:
                if nb not in visited:
                    dfs(nb, component)

        for email in email_to_name:
            if email not in visited:
                component: List[str] = []
                dfs(email, component)
                result.append([email_to_name[email]] + sorted(component))

        return result
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(E \log E)$  where  $E$  = total emails (sort dominates). Space  $O(E)$ .  
# Union-Find alternative: union all emails in an account, group by root.  
# DFS on the email graph is cleaner to explain – same complexity.  
# Given more time I'd ask: if same email can belong to different names?  
# That would mean the data is inconsistent – flag it as an error."
```


DFS

□ #785 Is Graph Bipartite?

MED[Open on LeetCode](#)

DFS · BFS · Union Find · Graph

Lists: Denny Zhang

Problem

There is an undirected graph with n nodes, where each node is numbered between 0 and $n - 1$. You are given a 2D array `graph`, where `graph[u]` is an array of nodes that node u is adjacent to. More formally, for each v in `graph[u]`, there is an undirected edge between node u and node v . The graph has the following properties:

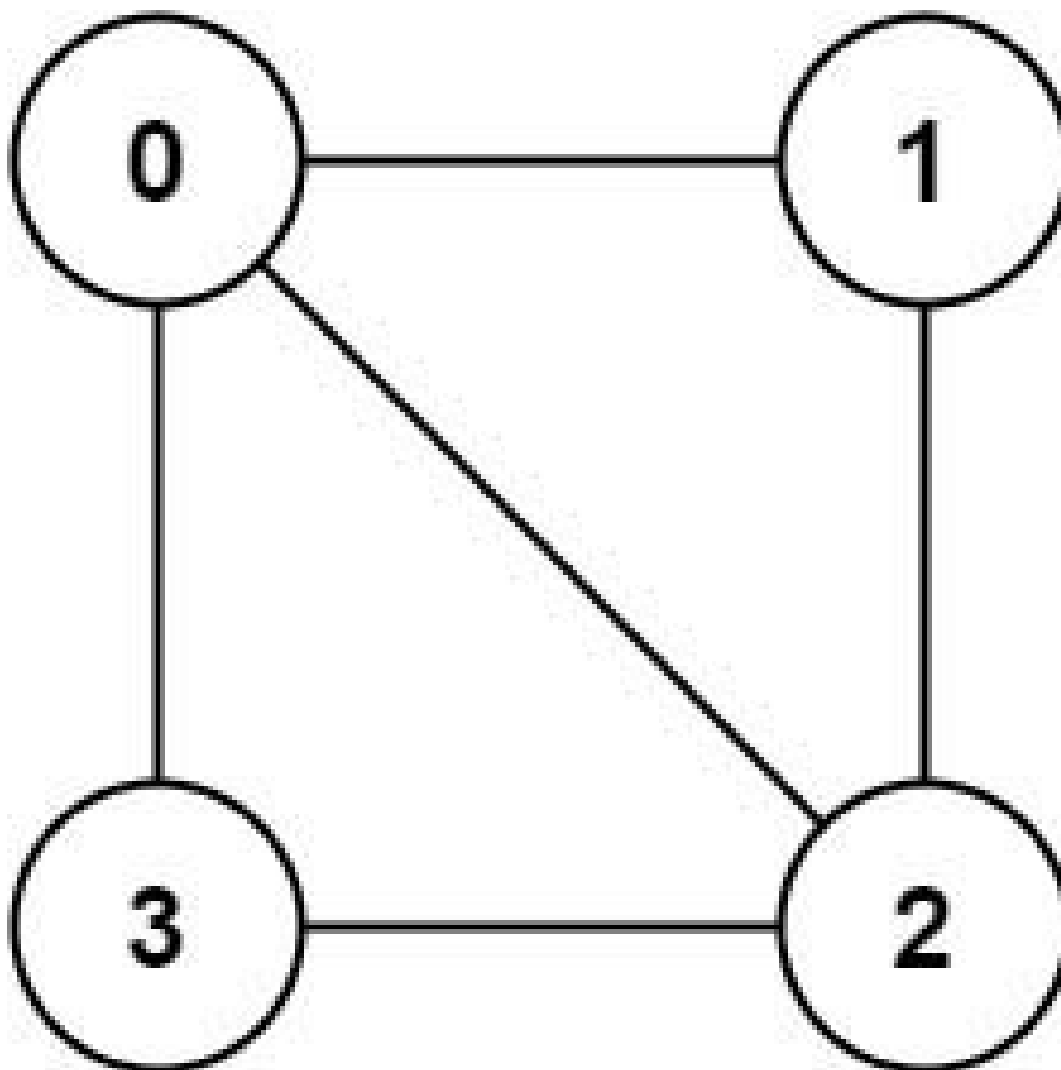
- There are no self-edges (`graph[u]` does not contain u).
- There are no parallel edges (`graph[u]` does not contain duplicate values).
- If v is in `graph[u]`, then u is in `graph[v]` (the graph is undirected).
- The graph may not be connected, meaning there may be two nodes u and v such that

there is no path between them.

A graph is bipartite if the nodes can be partitioned into two independent sets A and B such that every edge in the graph connects a node in set A and a node in set B.

Return true if and only if it is bipartite.

Example 1:

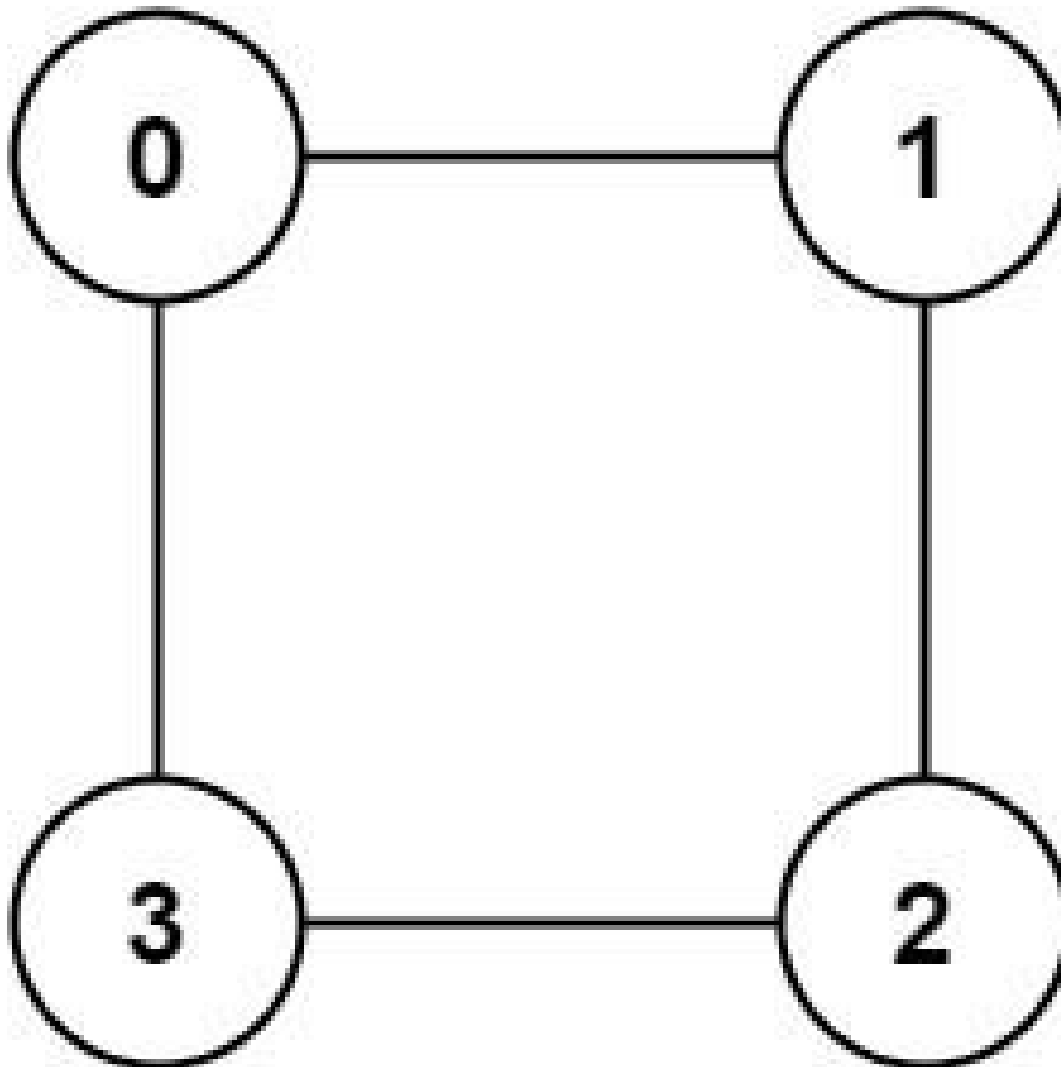


Input: graph = [[1,2,3],[0,2],[0,1,3],[0,2]]

Output: false

Explanation: There is no way to partition the nodes into two independent sets such that every edge connects a node in one and a node in the other.

Example 2:



Input: graph = [[1,3],[0,2],[1,3],[0,2]]

Output: true

Explanation: We can partition the nodes into two sets: {0, 2} and {1, 3}.

Constraints:

- graph.length == n
- 1 <= n <= 100

- $0 \leq \text{graph}[u].\text{length} < n$
- $0 \leq \text{graph}[u][i] \leq n - 1$
- $\text{graph}[u]$ does not contain u .
- All the values of $\text{graph}[u]$ are unique.
- If $\text{graph}[u]$ contains v , then $\text{graph}[v]$ contains u .

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Can we 2-colour the graph so no two adjacent nodes share the same colour?
Graph may be disconnected – check every component. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
Try to 2-color the graph with BFS/DFS: assign color 0/1 alternating across edges.
If any edge connects two nodes of the same color, graph is not bipartite.
Standard coloring check on an undirected graph.
Time: $O(V + E)$. Space: $O(V)$ color + visited.
Should I go ahead and optimize?"

```
class _785_IsGraphBipartite:
    def isBipartite(self, graph: List[List[int]]) -> bool:
        color = {}
        def bfs(start):
            queue = deque([start])
            color[start] = 0
            while queue:
                node = queue.popleft()
                for nb in graph[node]:
                    if nb not in color:
                        color[nb] = 1 - color[node]
                        queue.append(nb)
                    elif color[nb] == color[node]:
                        return False
            return True
        return all(bfs(n) for n in range(len(graph)) if n not in color)
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(V+E)$. Space $O(V)$. 2-coloring via BFS – odd-length cycle → not bipartite.
Given more time I'd ask: what if we need to output the two sets?"

```
# Return {n for n in color if color[n]==0} and {n for n in color if color[n]==1}."
```

DFS

□ ★ #1236 Web Crawler ★

MED[Open on LeetCode](#)

String · DFS · BFS · Interactive

Lists: ★ Premium | Premium 98

Problem

Given a url `startUrl` and an interface `HtmlParser`, implement a web crawler to crawl all links that are under the same hostname as `startUrl`.

Return all urls obtained by your web crawler in any order.

Your crawler should:

- Start from the page: `startUrl`
- Call `HtmlParser.getUrls(url)` to get all urls from a webpage of given url.
- Do not crawl the same link twice.
- Explore only the links that are under the same hostname as `startUrl`.

`http://example.org:8888/foo/bar#bang`

`hostname`

`host`

As shown in the example url above, the hostname is example.org. For simplicity sake, you may assume all urls use http protocol without any port specified. For example, the urls `http://leetcode.com/problems` and `http://leetcode.com/contest` are under the same hostname, while urls `http://example.org/test` and `http://example.com/abc` are not under the same hostname.

The `HtmlParser` interface is defined as such:

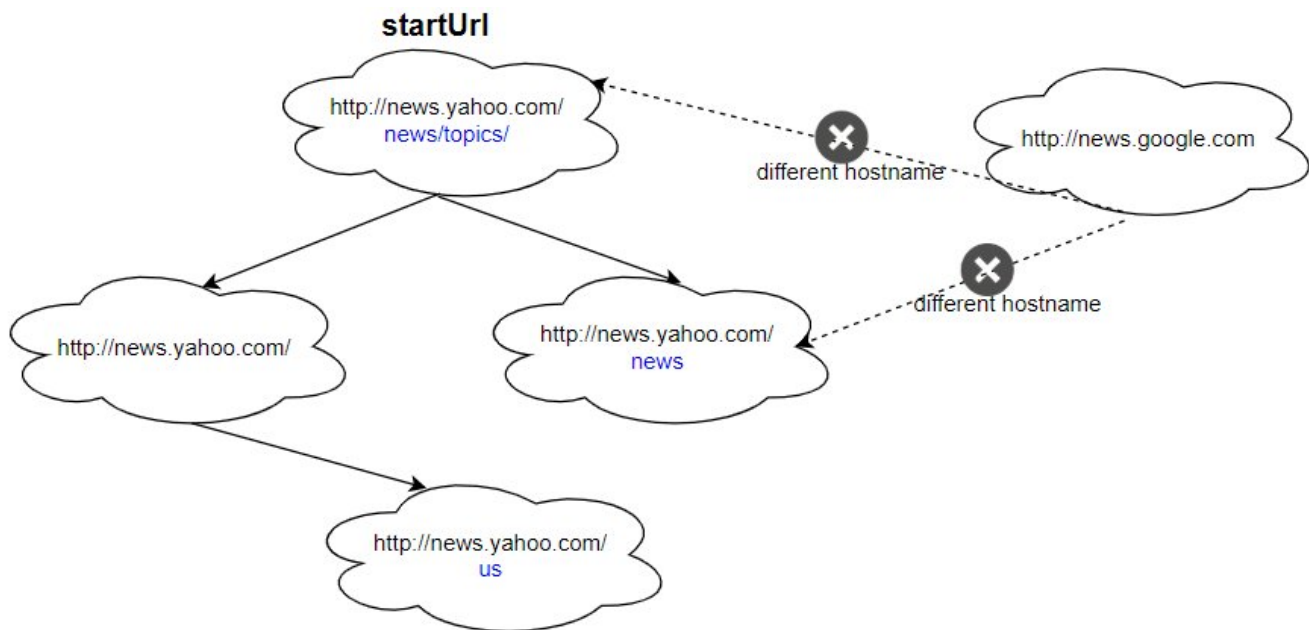
```
interface HtmlParser {  
    // Return a list of all urls from a webpage of given url.  
    public List<String> getUrls(String url);  
}
```

Below are two examples explaining the functionality of the problem, for custom testing purposes you'll have three variables `urls`, `edges` and `startUrl`. Notice that you will only have access to `startUrl` in your code, while `urls` and `edges` are not directly

accessible to you in code.

Note: Consider the same URL with the trailing slash "/" as a different URL. For example, "http://news.yahoo.com", and "http://news.yahoo.com/" are different urls.

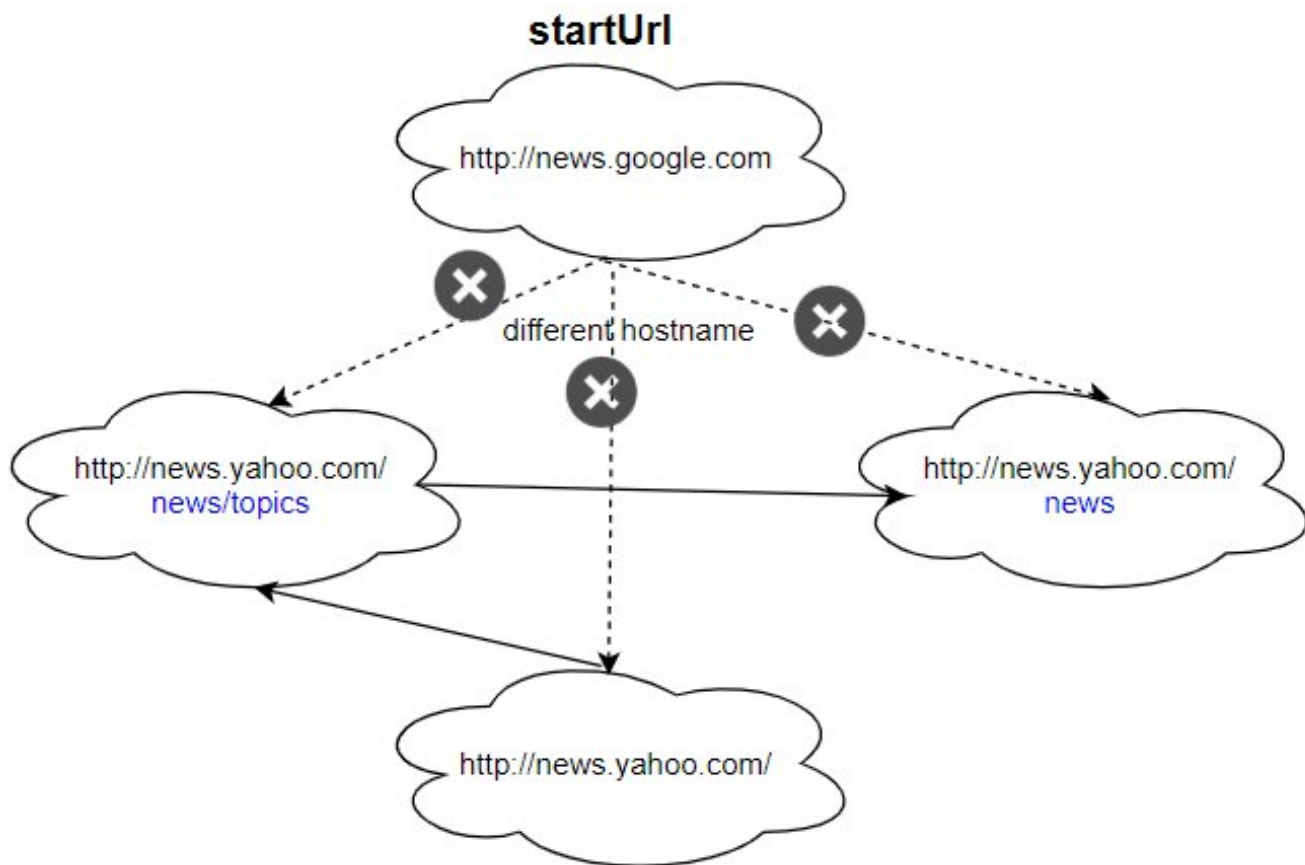
Example 1:



Input:

```
urls = [
    "http://news.yahoo.com",
    "http://news.yahoo.com/news",
    "http://news.yahoo.com/news/topics/",
    "http://news.google.com",
    "http://news.yahoo.com/us"
]
```

Example 2:



Input:

```

urls = [
  "http://news.yahoo.com",
  "http://news.yahoo.com/news",
  "http://news.yahoo.com/news/topics/",
  "http://news.google.com"
]
edges = [[0,2],[2,1],[3,2],[3,1],[3,0]]

```

Constraints:

- $1 \leq \text{urls.length} \leq 1000$
- $1 \leq \text{urls}[i].\text{length} \leq 300$
- startUrl is one of the urls.
- Hostname label must be from 1 to 63 characters long, including the dots, may contain only the ASCII letters from 'a' to 'z',

digits from '0' to '9' and the hyphen-minus character ('-').

- The hostname may not start or end with the hyphen-minus character ('-').
- See: https://en.wikipedia.org/wiki/Hostname#Restrictions_on_valid_hostnames
- You may assume there're no duplicates in url library.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "BFS from startUrl. Only follow links on the same hostname. Don't revisit. Right?
# Hostname = part between '//' and the next '/'."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# BFS from startUrl: fetch page, parse all links on same hostname, enqueue
unvisited.
# Use a visited set keyed by URL string to avoid cycles.
# Standard single-threaded web crawler simulation.
# Time: O(V + E) pages + links. Space: O(V) visited.
# Should I go ahead and optimize?"
```

```
class _1236_WebCrawler:
    def crawl(self, startUrl: str, htmlParser: 'HtmlParser') -> List[str]:
        def hostname(url):
            return url.split('/')[2] # 'http:' , '' , 'news.yahoo.com', ...
        host = hostname(startUrl)
        visited = {startUrl}
        queue = deque([startUrl])
        while queue:
            url = queue.popleft()
            for link in htmlParser.getUrls(url):
                if link not in visited and hostname(link) == host:
                    visited.add(link)
                    queue.append(link)
        return list(visited)
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(V+E)$ where V =pages, E =links. Space $O(V)$.

Standard BFS graph traversal – the only novelty is hostname filtering.

Given more time I'd ask ☐ #1242 Multithreaded? Use a ThreadPoolExecutor,

submit getUrls calls as futures, use a thread-safe visited set."

DFS

#1242 Web Crawler Multithreaded

MED[Open on LeetCode](#)

DFS · BFS · Concurrency**Lists: Denny Zhang**

Problem

Given a URL `startUrl` and an interface `HtmlParser`, implement a Multi-threaded web crawler to crawl all links that are under the same hostname as `startUrl`.

Return all URLs obtained by your web crawler in any order.

Your crawler should:

- Start from the page: `startUrl`
- Call `HtmlParser.getUrls(url)` to get all URLs from a webpage of a given URL.
- Do not crawl the same link twice.
- Explore only the links that are under the same hostname as `startUrl`.

http://example.org:8888/foo/bar#bang

hostname

host

As shown in the example URL above, the hostname is example.org. For simplicity's sake, you may assume all URLs use HTTP protocol without any port specified. For example, the URLs `http://leetcode.com/problems` and `http://leetcode.com/contest` are under the same hostname, while URLs `http://example.org/test` and `http://example.com/abc` are not under the same hostname.

The `HtmlParser` interface is defined as such:

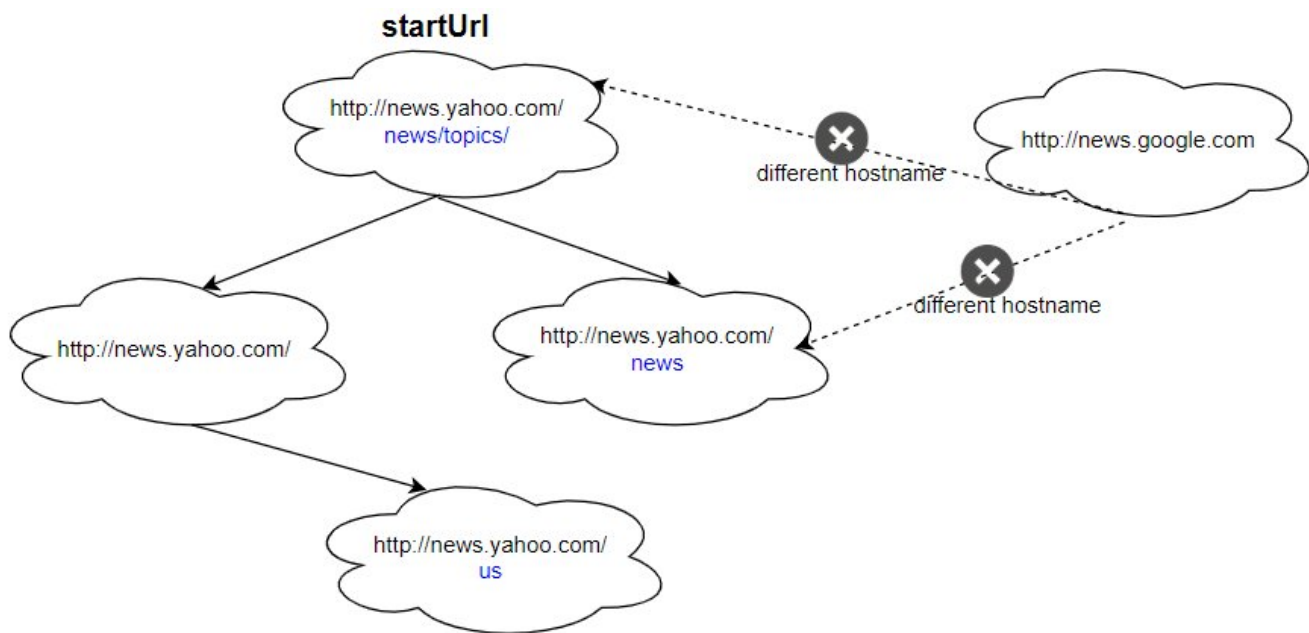
```
interface HtmlParser {  
    // Return a list of all urls from a webpage of given url.  
    // This is a blocking call, that means it will do HTTP request and return when  
    // this request is finished.  
    public List<String> getUrls(String url);  
}
```

Note that `getUrls(String url)` simulates performing an HTTP request. You can treat it as a blocking function call that waits for an HTTP request to finish. It is guaranteed that `getUrls(String url)` will return the URLs within 15ms. Single-threaded solutions will exceed the

time limit so, can your multi-threaded web crawler do better?

Below are two examples explaining the functionality of the problem. For custom testing purposes, you'll have three variables `urls`, `edges` and `startUrl`. Notice that you will only have access to `startUrl` in your code, while `urls` and `edges` are not directly accessible to you in code.

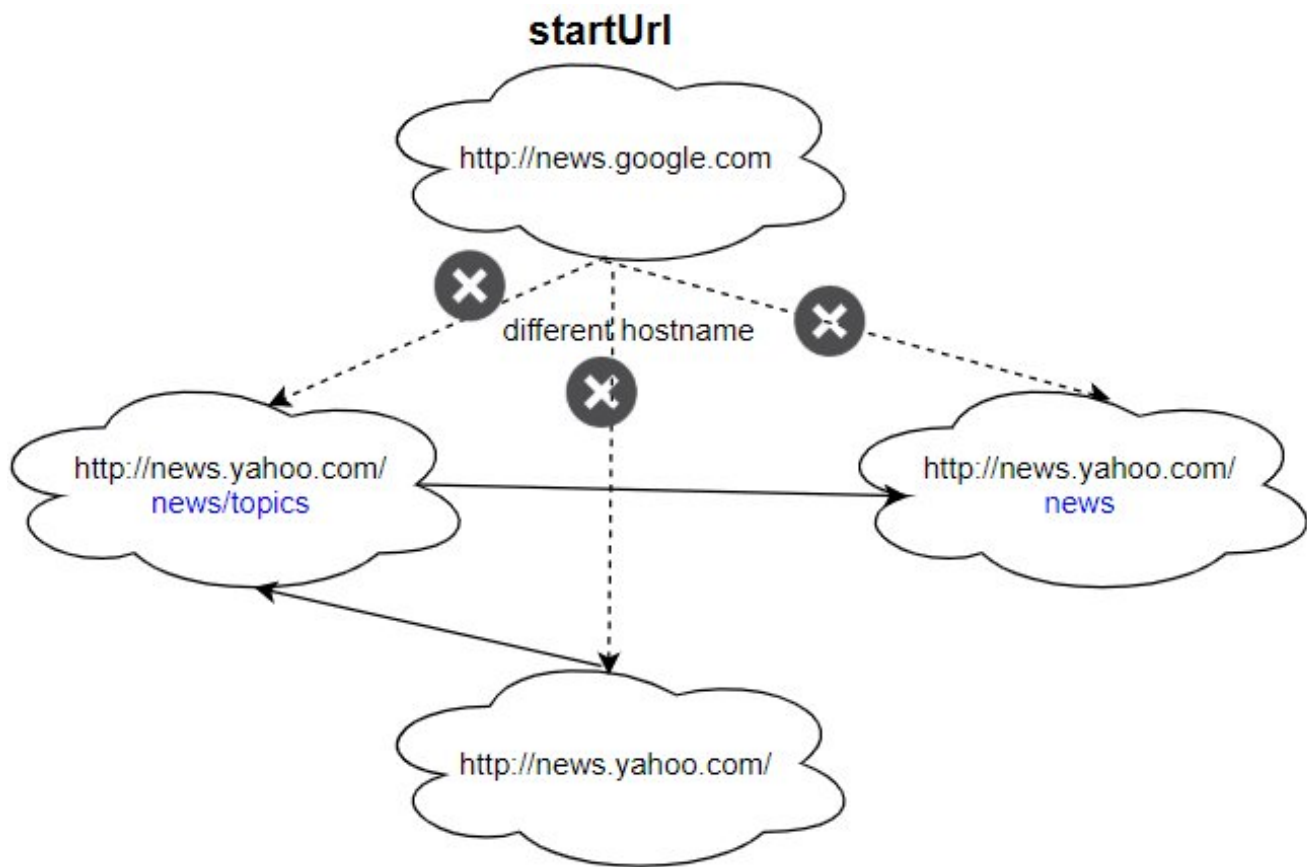
Example 1:



Input:

```
urls = [  
  "http://news.yahoo.com",  
  "http://news.yahoo.com/news",  
  "http://news.yahoo.com/news/topics/",  
  "http://news.google.com",  
  "http://news.yahoo.com/us"  
]
```

Example 2:



Input:

```
urls = [
    "http://news.yahoo.com",
    "http://news.yahoo.com/news",
    "http://news.yahoo.com/news/topics/",
    "http://news.google.com"
]
edges = [[0,2],[2,1],[3,2],[3,1],[3,0]]
```

Constraints:

- $1 \leq \text{urls.length} \leq 1000$
- $1 \leq \text{urls}[i].\text{length} \leq 300$
- startUrl is one of the urls.
- Hostname label must be from 1 to 63 characters long, including the dots, may contain only the ASCII letters from 'a' to 'z',

digits from '0' to '9' and the hyphen-minus character ('-').

- The hostname may not start or end with the hyphen-minus character ('-').
- See: https://en.wikipedia.org/wiki/Hostname#Restrictions_on_valid_hostnames
- You may assume there're no duplicates in the URL library.

Follow up:

1. Assume we have 10,000 nodes and 1 billion URLs to crawl. We will deploy the same software onto each node. The software can know about all the nodes. We have to minimize communication between machines and make sure each node does equal amount of work. How would your web crawler design change?
2. What if one node fails or does not work?
3. How do you know when the crawler is done?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Same as #1236 but with multi-threading since `getUrls` is blocking and slow.

Use threads to fetch pages in parallel. Thread-safe visited set required. Right?"

PHASE 4 — CLEAN CODE (threading approach)

```
from concurrent.futures import ThreadPoolExecutor, as_completed
import threading
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic."
```

```
# Same BFS crawler as single-threaded, but dispatch each getLinks call to a worker thread.
```

```
# Shared visited set must be synchronized; join threads when queue is empty.
```

```
# Correct parallelism model for the LeetCode concurrency variant.
```

```
# Time:  $O(V + E)$  with parallel fetch. Space:  $O(V)$  visited + thread overhead.
```

```
# Should I go ahead and optimize?"
```

```
class _1242_WebCrawlerMT:
```

```
    def crawl(self, startUrl: str, htmlParser: 'HtmlParser') -> List[str]:
```

```
        def hostname(url):
```

```
            return url.split('/')[2]
```

```
        host = hostname(startUrl)
```

```
        visited = {startUrl}
```

```
        lock = threading.Lock()
```

```
        def fetch(url):
```

```
            neighbors = []
```

```
            for link in htmlParser.getUrls(url):
```

```
                with lock:
```

```
                    if link not in visited and hostname(link) == host:
```

```
                        visited.add(link)
```

```
                        neighbors.append(link)
```

```
            return neighbors
```

```
        queue = [startUrl]
```

```
        with ThreadPoolExecutor() as executor:
```

```
            while queue:
```

```
                futures = {executor.submit(fetch, url): url for url in queue}
```

```
                queue = []
```

```
                for f in as_completed(futures):
```

```
                    queue.extend(f.result())
```

```
        return list(visited)
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Threads cut wall-clock time proportional to how much getUrls blocks."
```

```
# Lock only needed when checking/adding to visited – minimize critical section.
```

```
# Distributed follow-up (10,000 nodes, 1B URLs): consistent hashing to partition URLs by hostname.
```

```
# Each node owns a hostname range. Communicate frontier pages between nodes.
```

```
# 'Done' signal: all queues empty and all workers idle (barrier synchronisation)."
```

Graphs

Round 2 · High · Medium · 3 questions

Graphs

□ #128 Longest Consecutive Sequence

MED[Open on LeetCode](#)

Array · Hash Table · Union Find

Lists: Grind 169 | CodeSignal

Problem

Given an unsorted array of integers `nums`, return the length of the longest consecutive elements sequence.

You must write an algorithm that runs in $O(n)$ time.

Example 1:

Input: `nums = [100,4,200,1,3,2]`

Output: 4

Explanation: The longest consecutive elements sequence is [1, 2, 3, 4]. Therefore its length is 4.

Example 2:

Input: `nums = [0,3,7,2,5,8,4,6,0,1]`

Output: 9

Example 3:

Input: `nums = [1,0,1,2]`

Output: 3

Constraints:

- $0 \leq \text{nums.length} \leq 105$
- $-109 \leq \text{nums}[i] \leq 109$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find the longest run of consecutive integers. Must be $O(n)$. Return the length. Right?

Duplicates don't extend the sequence – $[1,1,2] \rightarrow \text{length } 2$."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Put all numbers in a hash set. For each num, if num-1 is not in set, walk num, num+1, num+2...

counting streak length. Track longest streak.

Each number visited once across all walks – $O(n)$ with the set.

Time: $O(n)$. Space: $O(n)$ set.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (hash set + only start from sequence beginnings)

"Build a set. For each x where x-1 is NOT in the set (x is a sequence start), # count upward. Each element is processed at most once total $\rightarrow O(n)$."

PHASE 4 — CLEAN CODE

```
class _128_LongestConsecutive:
    def longestConsecutive(self, nums: List[int]) -> int:
        num_set = set(nums)
        best = 0
        for x in num_set:
            if x - 1 not in num_set: # x is a sequence start
                length = 1
                while x + 1 in num_set:
                    x += 1; length += 1
                best = max(best, length)
        return best
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(n)$. Each number is the starting point of at most one sequence scan.

The 'start of sequence' check prevents redundant scans – the key insight.

Given more time I'd ask: streaming input? Use a dict mapping each endpoint

of a range to its length, merging ranges as new numbers arrive."

Graphs

□ ★ #277 Find the Celebrity ★

MED

[Open on LeetCode](#)

Two Pointers · Graph · Interactive

Lists: ★ Premium | Premium 98

Problem

Suppose you are at a party with n people labeled from 0 to $n - 1$ and among them, there may exist one celebrity. The definition of a celebrity is that all the other $n - 1$ people know the celebrity, but the celebrity does not know any of them.

Now you want to find out who the celebrity is or verify that there is not one. You are only allowed to ask questions like: "Hi, A. Do you know B?" to get information about whether A knows B. You need to find out the celebrity (or verify there is not one) by asking as few questions as possible (in the asymptotic sense).

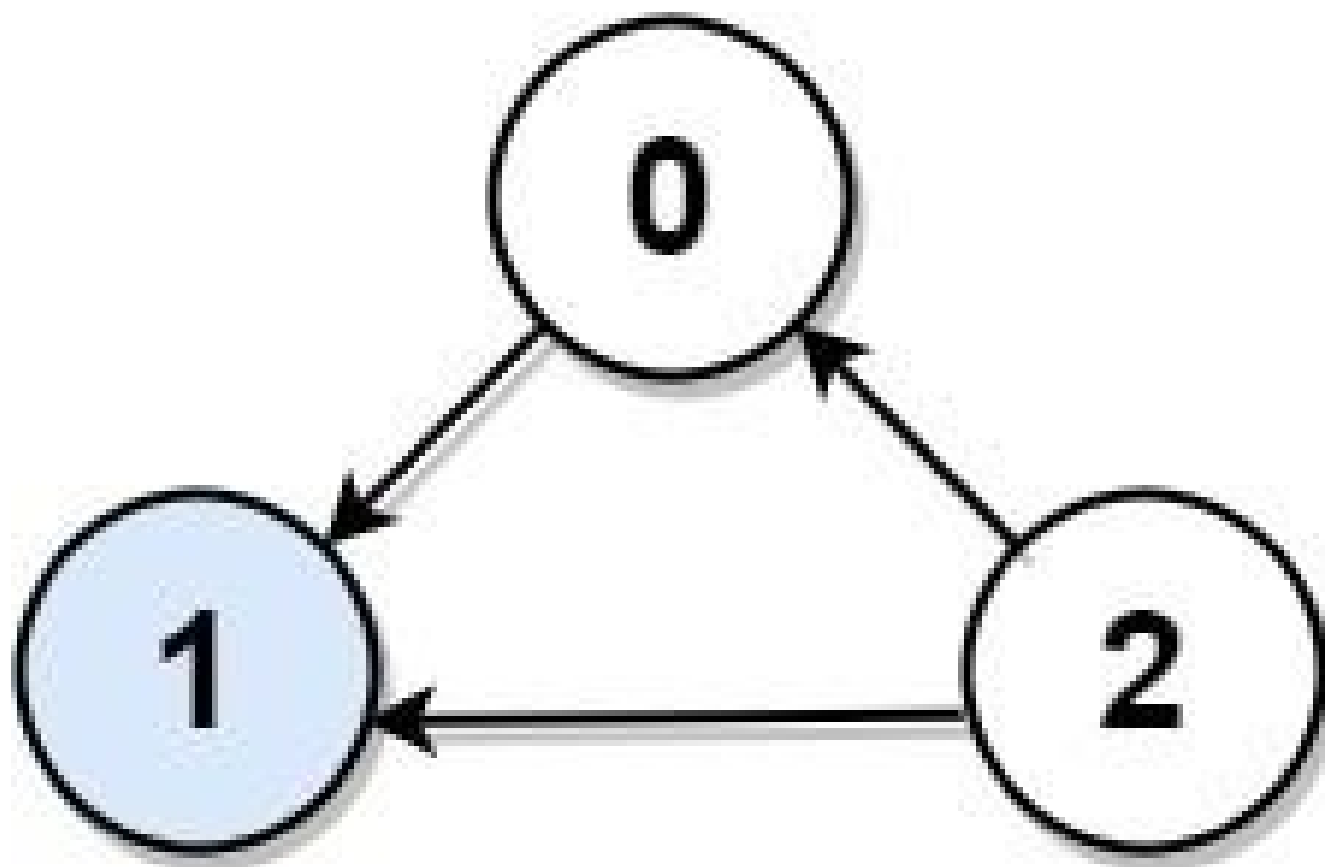
You are given an integer n and a helper function `bool knows(a, b)` that tells you whether a knows b. Implement a function `int findCelebrity(n)`.

There will be exactly one celebrity if they are at the party.

Return the celebrity's label if there is a celebrity at the party. If there is no celebrity, return -1.

Note that the $n \times n$ 2D array graph given as input is not directly available to you, and instead only accessible through the helper function knows. $\text{graph}[i][j] == 1$ represents person i knows person j , whereas $\text{graph}[i][j] == 0$ represents person j does not know person i .

Example 1:

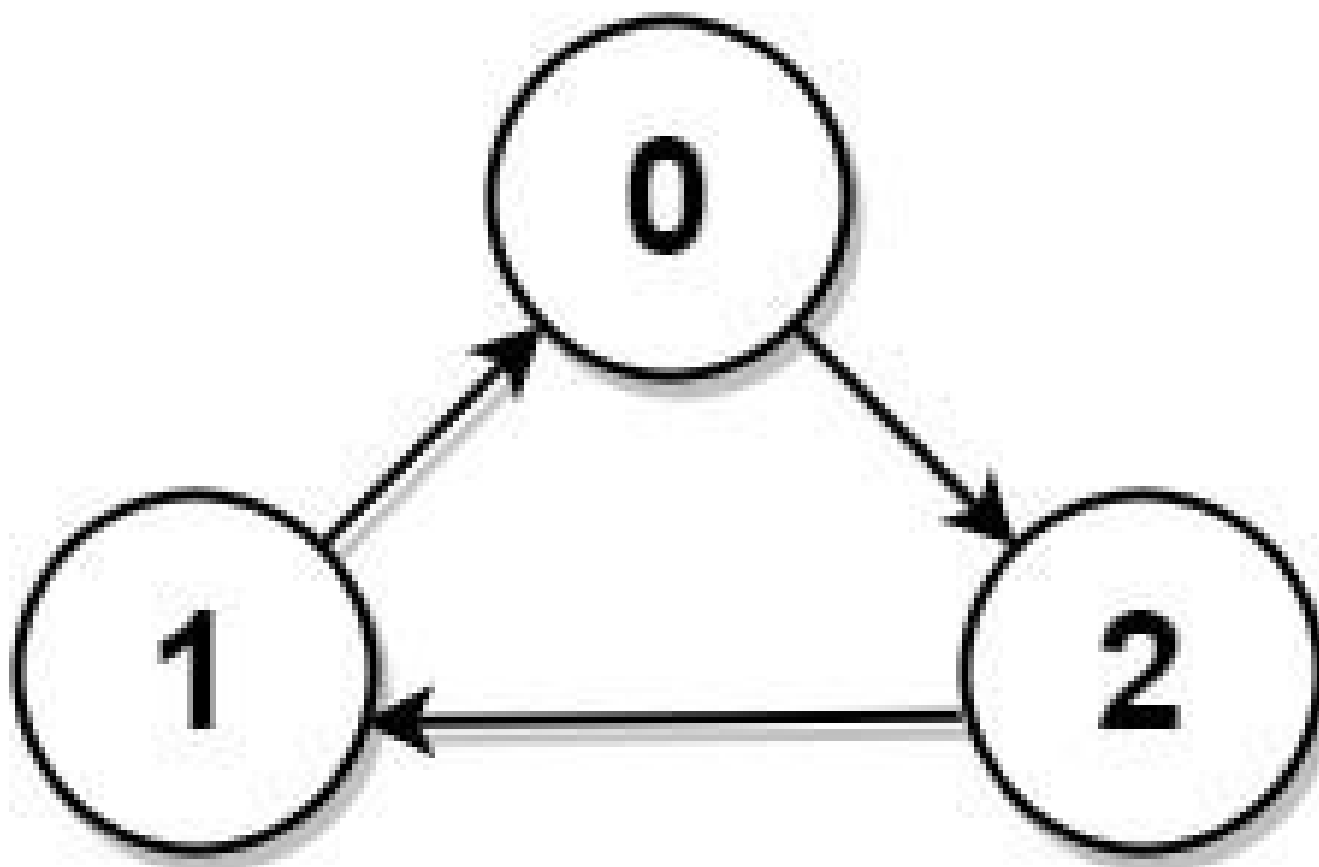


Input: graph = [[1,1,0],[0,1,0],[1,1,1]]

Output: 1

Explanation: There are three persons labeled with 0, 1 and 2. graph[i][j] = 1 means person i knows person j, otherwise graph[i][j] = 0 means person i does not know person j. The celebrity is the person labeled as 1 because both 0 and 2 know him but 1 does not know anybody.

Example 2:



Input: graph = [[1,0,1],[1,1,0],[0,1,1]]

Output: -1

Explanation: There is no celebrity.

Constraints:

- $n == \text{graph.length} == \text{graph}[i].\text{length}$
- $2 \leq n \leq 100$
- graph[i][j] is 0 or 1.
- graph[i][i] == 1

Follow up: If the maximum number of allowed calls to the API knows is $3 * n$, could you find a solution without exceeding the maximum number of calls?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Celebrity: everyone knows them, they know no one.
Find them using knows(a,b) API – minimise calls. Right?
At most one celebrity can exist."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
Ask every other person if they know celebrity; candidate must be known by all and know nobody.
Two-pass elimination: first narrow to one candidate with $O(n)$ knows() calls, then verify.
Naive all-pairs check is $O(n^2)$ knows() calls.
Time: $O(n^2)$ naive, $O(n)$ with elimination. Space: $O(1)$.
Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (candidate elimination + verify)

"Pass 1: start with candidate=0, scan all i. If knows(candidate, i): celebrity can't be candidate → candidate=i.
Pass 2: verify candidate – everyone must know them AND they know no one.
Total API calls: $O(n)$. Without the two-pass we'd need $O(n^2)$."

PHASE 4 — CLEAN CODE

```
class _277_FindCelebrity:
    def findCelebrity(self, n: int) -> int:
        candidate = 0
        for i in range(1, n):
            if knows(candidate, i):
                candidate = i
        for i in range(n):
            if i == candidate:
                continue
            if knows(candidate, i) or not knows(i, candidate):
                return -1
        return candidate
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$ – $2n-2$ API calls in the worst case. Space $O(1)$."


```
# The candidate elimination logic: if A knows B, A is not the celebrity (celebrity knows no one).  
# If A doesn't know B, B is not the celebrity (celebrity is known by all).  
# One of them is eliminated per comparison – O(n) total.  
# Given more time I'd ask: what if there's a possibility of no celebrity?  
# The current solution handles this – the verification pass returns -1 if candidate fails."
```

Graphs

□ ★ #1136 Parallel Courses ★

MED[Open on LeetCode](#)

Graph · Topological Sort

Lists: ★ Premium | Premium 98

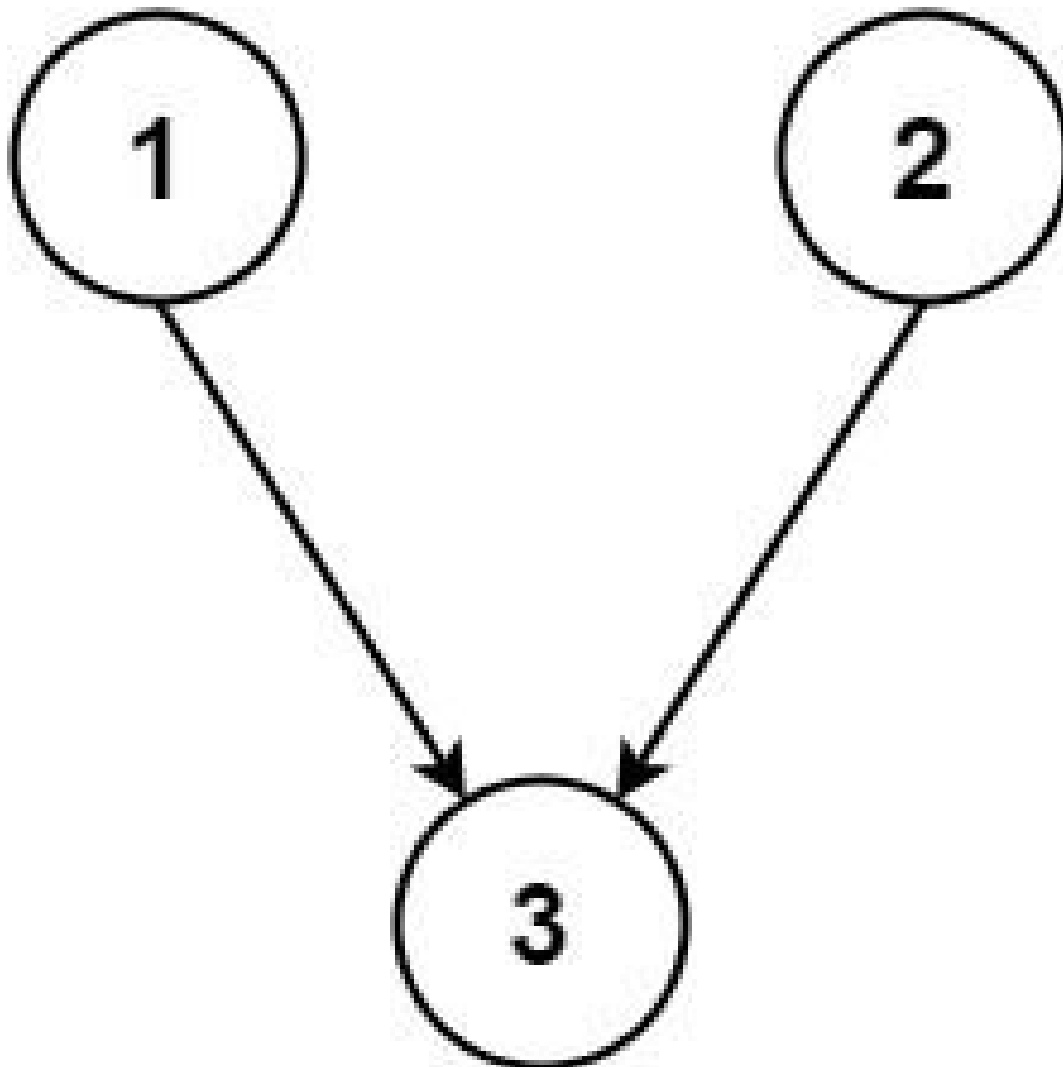
Problem

You are given an integer n , which indicates that there are n courses labeled from 1 to n . You are also given an array `relations` where `relations[i] = [prevCoursei, nextCoursei]`, representing a prerequisite relationship between course `prevCoursei` and course `nextCoursei`: course `prevCoursei` has to be taken before course `nextCoursei`.

In one semester, you can take any number of courses as long as you have taken all the prerequisites in the previous semester for the courses you are taking.

Return the minimum number of semesters needed to take all courses. If there is no way to take all the courses, return -1 .

Example 1:



Input: $n = 3$, $\text{relations} = [[1,3],[2,3]]$

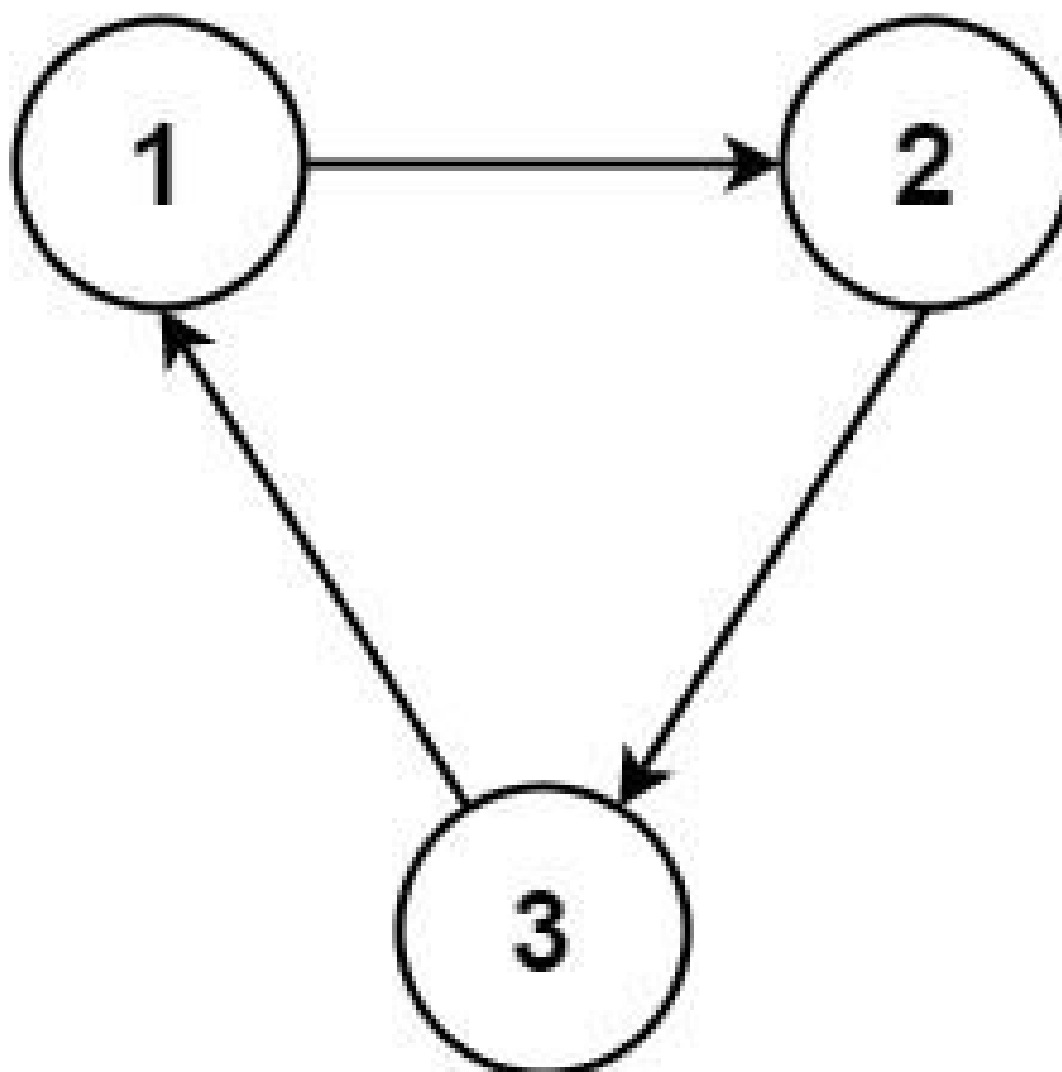
Output: 2

Explanation: The figure above represents the given graph.

In the first semester, you can take courses 1 and 2.

In the second semester, you can take course 3.

Example 2:



Input: $n = 3$, relations = $[[1,2],[2,3],[3,1]]$

Output: -1

Explanation: No course can be studied because they are prerequisites of each other.

Constraints:

- $1 \leq n \leq 5000$
- $1 \leq \text{relations.length} \leq 5000$
- $\text{relations}[i].\text{length} == 2$
- $1 \leq \text{prevCourse}_i, \text{nextCourse}_i \leq n$
- $\text{prevCourse}_i \neq \text{nextCourse}_i$

- All the pairs [prevCoursei, nextCoursei] are unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Minimum semesters to take all courses where you can take any number per semester
as long as prerequisites are met. Return -1 if impossible (cycle). Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."
Topological sort with indegree: each semester take all courses whose prerequisites are done.
Increment semester count each round; if no course can be taken but some remain, impossible.
Same Kahn BFS layered by semester.
Time: $O(V + E)$. Space: $O(V + E)$.
Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (Kahn's BFS level-by-level)

"BFS topological sort: process all zero-in-degree nodes in one 'batch' (semester).
Each batch = one semester. Count batches. If total processed < n → cycle."

PHASE 4 — CLEAN CODE

```
class _1136_ParallelCourses:
    def minimumSemesters(self, n: int, relations: List[List[int]]) -> int:
        graph = defaultdict(list)
        indeg = {i: 0 for i in range(1, n + 1)}
        for pre, nxt in relations:
            graph[pre].append(nxt)
            indeg[nxt] += 1
        queue = deque(c for c in range(1, n + 1) if indeg[c] == 0)
        sems = 0
        total = 0
        while queue:
            sems += 1
            for _ in range(len(queue)):
                node = queue.popleft()
                total += 1
                for nb in graph[node]:
                    indeg[nb] -= 1
                    if indeg[nb] == 0:
                        queue.append(nb)
            return sems if total == n else -1
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(V+E)$. Space $O(V+E)$. This is Course Schedule (#207) with BFS level counting.
Each BFS level = one semester. Cycle detection: total processed < n.
Given more time I'd ask: what if courses have durations (not just 1 semester each)?
This becomes a weighted DAG critical path problem – DP or Bellman–Ford."

BFS

Round 2 · High · Medium · 6 questions

BFS

□ ★ #286 Walls and Gates ★

MED

[Open on LeetCode](#)

Array · BFS · Matrix

Lists: ★ Premium | Premium 98

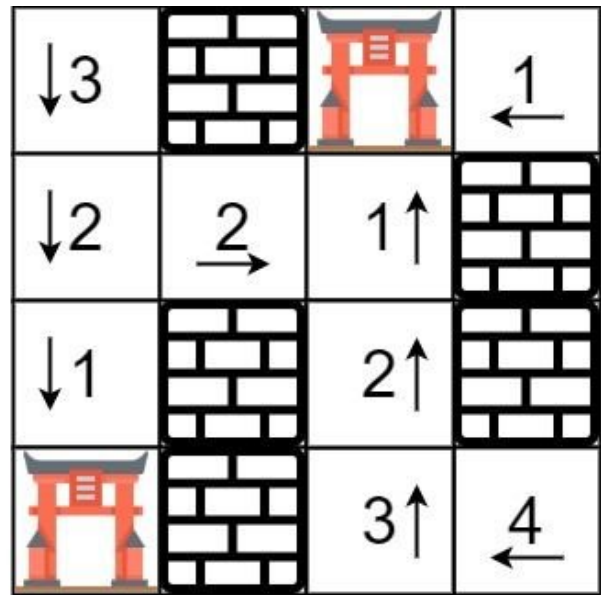
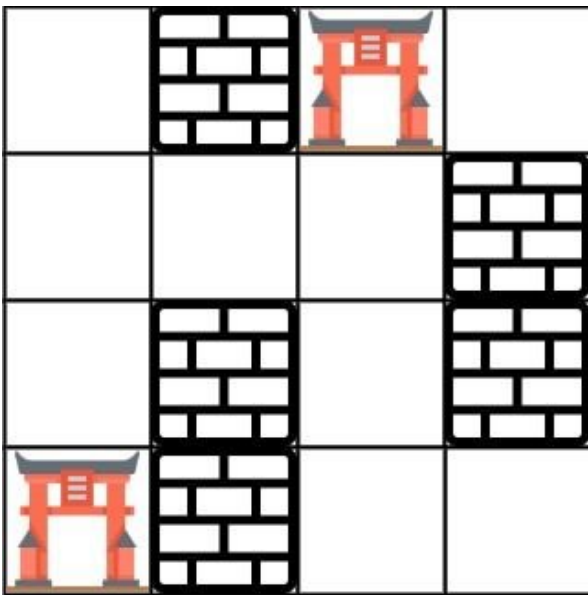
Problem

You are given an $m \times n$ grid rooms initialized with these three possible values.

- -1 A wall or an obstacle.
- 0 A gate.
- INF Infinity means an empty room. We use the value $231 - 1 = 2147483647$ to represent INF as you may assume that the distance to a gate is less than 2147483647 .

Fill each empty room with the distance to its nearest gate. If it is impossible to reach a gate, it should be filled with INF .

Example 1:



```
Input: rooms = [[2147483647,-1,0,2147483647],[2147483647,2147483647,2147483647,-1],
[2147483647,-1,2147483647,-1],[0,-1,2147483647,2147483647]]
Output: [[3,-1,0,1],[2,2,1,-1],[1,-1,2,-1],[0,-1,3,4]]
```

Example 2:

```
Input: rooms = [[-1]]
Output: [[-1]]
```

Constraints:

- $m == \text{rooms.length}$
- $n == \text{rooms}[i].\text{length}$
- $1 \leq m, n \leq 250$
- $\text{rooms}[i][j]$ is -1 , 0 , or $2^{31} - 1$.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Fill each INF room with its distance to the nearest gate (0). Walls (-1) stay. Right?"

In-place modification. $\text{INF} = 2^{31} - 1$."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

```

# Multi-source BFS from every gate simultaneously (distance 0).
# Fill empty rooms layer by layer with increasing distance.
# Each room reached once – standard BFS on grid.
# Time:  $O(m * n)$ . Space:  $O(m * n)$  queue.
# Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE (multi-source BFS from all gates)
# "Enqueue ALL gates at once. BFS outward – each room gets filled with distance
# from the nearest gate simultaneously.  $O(m*n)$ ."

# PHASE 4 — CLEAN CODE
class _286_WallsAndGates:
    def wallsAndGates(self, rooms: List[List[int]]) -> None:
        INF = 2**31 - 1
        m, n = len(rooms), len(rooms[0])
        queue = deque((r, c) for r in range(m) for c in range(n) if rooms[r][c] ==
0)
        while queue:
            r, c = queue.popleft()
            for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                nr, nc = r + dr, c + dc
                if 0 <= nr < m and 0 <= nc < n and rooms[nr][nc] == INF:
                    rooms[nr][nc] = rooms[r][c] + 1
                    queue.append((nr, nc))

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(m*n)$ . Space  $O(m*n)$ . Multi-source BFS is the key – a single BFS from each
gate
# independently would be  $O(m*n * \text{gates})$ , much slower.
# Contrast with #542 (01 Matrix): identical pattern but for distance to nearest 0.
# Given more time I'd ask: find distance to nearest wall instead?
# Multi-source BFS from all walls – same approach."

```

BFS

□ #322 Coin Change

MED[Open on LeetCode](#)

Array · Dynamic Programming · BFS

Lists: Grind 169

Problem

You are given an integer array `coins` representing coins of different denominations and an integer `amount` representing a total amount of money.

Return the fewest number of coins that you need to make up that amount. If that amount of money cannot be made up by any combination of the coins, return `-1`.

You may assume that you have an infinite number of each kind of coin.

Example 1:

Input: `coins = [1,2,5]`, `amount = 11`

Output: `3`

Explanation: `11 = 5 + 5 + 1`

Example 2:

Input: `coins = [2]`, `amount = 3`

Output: `-1`

Example 3:

Input: coins = [1], amount = 0
Output: 0

Constraints:

- $1 \leq \text{coins.length} \leq 12$
- $1 \leq \text{coins}[i] \leq 231 - 1$
- $0 \leq \text{amount} \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Minimum coins to make up amount. Unlimited supply of each coin. Return -1 if impossible. Right?"
# Coin values can be any positive integer – not necessarily power of 2 (so greedy fails)."
#
# "coins=[1,2,5], amount=11: 5+5+1=3 coins ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic."
# Recursion: for each amount, try every coin denomination and take 1 + min subproblem.
# Recomputes identical sub-amounts exponentially without memoization.
# Time: O(amount^coins) exponential. Space: O(amount) stack.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (bottom-up DP)

```
# "dp[i] = minimum coins to make amount i."
# dp[0]=0. dp[i] = min(dp[i-coin]+1) for each coin <= i.
# Time: O(amount * coins). Space: O(amount)."
```

PHASE 4 — CLEAN CODE

```
class _322_CoinChange:
    def coinChange(self, coins: List[int], amount: int) -> int:
        dp = [float('inf')] * (amount + 1)
        dp[0] = 0
        for coin in coins:
            for i in range(coin, amount + 1):
                dp[i] = min(dp[i], dp[i - coin] + 1)
        return dp[amount] if dp[amount] != float('inf') else -1
```

PHASE 5 — TEST & DEBUG

coins=[1,2,5], amount=11:

dp[5]=1(5). dp[10]=2(5+5). dp[11]=3(5+5+1). → 3 ✓

amount=3, coins=[2]: dp[2]=1, dp[3]=inf → -1 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(\text{amount} * \text{coins})$. Space $O(\text{amount})$.

BFS alternative: treat amounts as nodes, coins as edges. BFS from 0 to amount.

DP is preferred – BFS visits each amount once but rebuilds from scratch.

Given more time I'd ask: how many ways to make the amount?

Coin Change 2 (#518) – same DP but count combinations instead of min count."

BFS

#542 01 Matrix

MED

[Open on LeetCode](#)

Array · BFS · Matrix

Lists: Grind 169 | Denny Zhang

Problem

Given an $m \times n$ binary matrix `mat`, return the distance of the nearest 0 for each cell.

The distance between two cells sharing a common edge is 1.

Example 1:

0	0	0
0	1	0
0	0	0

Input: mat = [[0,0,0],[0,1,0],[0,0,0]]

Output: [[0,0,0],[0,1,0],[0,0,0]]

Example 2:

0	0	0
0	1	0
1	1	1

Input: `mat = [[0,0,0],[0,1,0],[1,1,1]]`

Output: `[[0,0,0],[0,1,0],[1,2,1]]`

Constraints:

- `m == mat.length`
- `n == mat[i].length`
- `1 <= m, n <= 104`
- `1 <= m * n <= 104`
- `mat[i][j]` is either 0 or 1.
- There is at least one 0 in `mat`.

Note: This question is the same as 1765: <https://leetcode.com/problems/map-of-highest-peak/>

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Return distance of nearest 0 for each cell. In-place or new matrix. Right?
Same pattern as Walls and Gates but for 0-cells instead of gates."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
Multi-source BFS starting from every 0 simultaneously.
First time each 1 cell is reached, its distance is minimal.
Layer-by-layer BFS fills the dist matrix in one pass.
Time: $O(m * n)$. Space: $O(m * n)$ queue + dist.
Should I go ahead and optimize?"

```
class _542_ZeroOneMatrix:
    def updateMatrix(self, mat: List[List[int]]) -> List[List[int]]:
        m, n = len(mat), len(mat[0])
        dist = [[0 if mat[r][c] == 0 else float('inf') for c in range(n)] for r in range(m)]
        queue = deque((r, c) for r in range(m) for c in range(n) if mat[r][c] == 0)
        while queue:
            r, c = queue.popleft()
            for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                nr, nc = r + dr, c + dc
                if 0 <= nr < m and 0 <= nc < n and dist[nr][nc] > dist[r][c] + 1:
                    dist[nr][nc] = dist[r][c] + 1
                    queue.append((nr, nc))
        return dist
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m*n)$. Space $O(m*n)$. Multi-source BFS from all 0-cells simultaneously.
Identical to #286 (Walls and Gates) – name this connection.
Given more time I'd ask: distance to nearest 1 instead?
Same multi-source BFS starting from all 1-cells."

BFS

□ #994 Rotting Oranges

MED[Open on LeetCode](#)

Array · BFS · Matrix

Lists: Grind 169

Problem

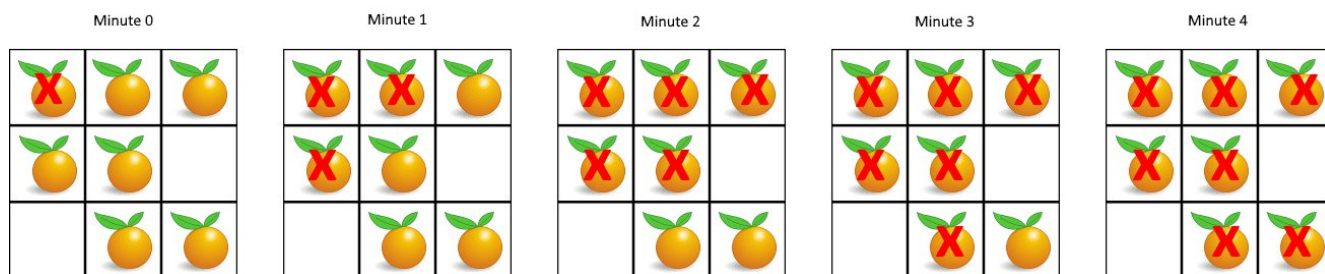
You are given an $m \times n$ grid where each cell can have one of three values:

- 0 representing an empty cell,
- 1 representing a fresh orange, or
- 2 representing a rotten orange.

Every minute, any fresh orange that is 4-directionally adjacent to a rotten orange becomes rotten.

Return the minimum number of minutes that must elapse until no cell has a fresh orange. If this is impossible, return -1 .

Example 1:



Input: grid = `[[2,1,1],[1,1,0],[0,1,1]]`

Output: 4

Example 2:

Input: grid = `[[2,1,1],[0,1,1],[1,0,1]]`

Output: -1

Explanation: The orange in the bottom left corner (row 2, column 0) is never rotten, because rotting only happens 4-directionally.

Example 3:

Input: grid = `[[0,2]]`

Output: 0

Explanation: Since there are already no fresh oranges at minute 0, the answer is just 0.

Constraints:

- `m == grid.length`
- `n == grid[i].length`
- `1 <= m, n <= 10`
- `grid[i][j]` is 0, 1, or 2.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Multi-source BFS from all rotten oranges. Fresh adjacent → rotten after 1 minute.

Return minutes until no fresh remain, or -1 if impossible. Right?"

#

"If no fresh oranges initially → return 0 immediately."

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Multi-source BFS: enqueue all rotten oranges at minute 0.
# Each minute, rot all fresh neighbors; track elapsed minutes until no fresh remain.
# If fresh count > 0 after BFS ends, return -1.
# Time: O(m * n). Space: O(m * n) queue.
# Should I go ahead and optimize?"
```

```
class _994_RottingOranges:
    def orangesRotting(self, grid: List[List[int]]) -> int:
        m, n = len(grid), len(grid[0])
        fresh = sum(grid[r][c] == 1 for r in range(m) for c in range(n))
        queue = deque((r, c) for r in range(m) for c in range(n) if grid[r][c] == 2)
        minutes = 0
        while queue and fresh:
            minutes += 1
            for _ in range(len(queue)):
                r, c = queue.popleft()
                for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                    nr, nc = r + dr, c + dc
                    if 0 <= nr < m and 0 <= nc < n and grid[nr][nc] == 1:
                        grid[nr][nc] = 2
                        fresh -= 1
                        queue.append((nr, nc))
            return minutes if fresh == 0 else -1
```

PHASE 5 — TEST & DEBUG

```
# [[2,1,1],[1,1,0],[0,1,1]]: fresh=6. BFS spreads level by level. After 4 minutes
fresh=0 → 4 ✓
# Isolated fresh orange: fresh>0 after BFS done → -1 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(m*n). Space O(m*n). Multi-source BFS: same as Walls and Gates, same as 01
Matrix.
# The 'fresh' counter eliminates the need to scan the grid again at the end.
# Given more time I'd ask: what if diagonal rotting is allowed?
# Add 4 diagonal direction pairs – same algorithm."
```

BFS

□ ★ #1197 Minimum Knight Moves ★

MED

[Open on LeetCode](#)

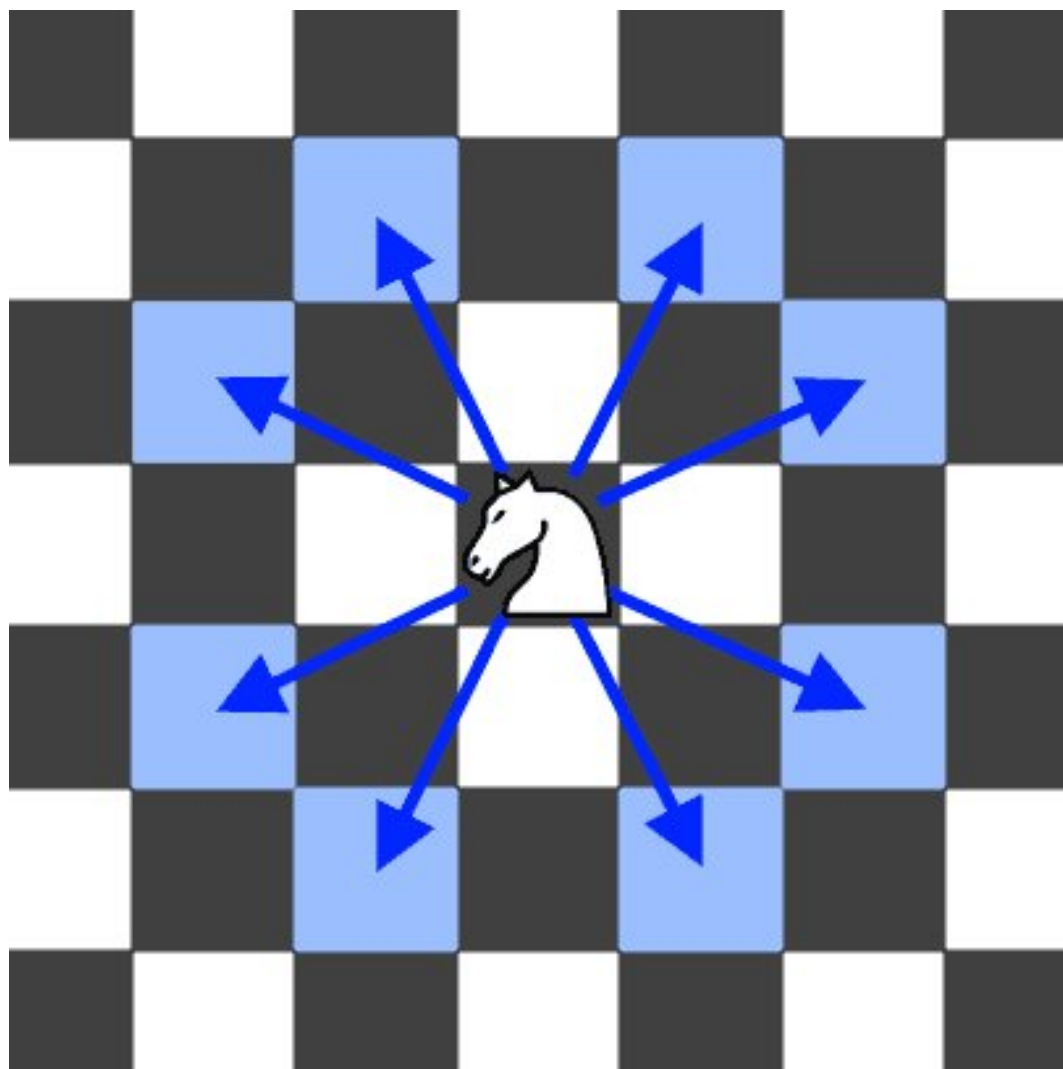
BFS

Lists: ★ Premium | Grind 169 | Premium 98

Problem

In an infinite chess board with coordinates from $-\infty$ to $+\infty$, you have a knight at square $[0, 0]$.

A knight has 8 possible moves it can make, as illustrated below. Each move is two squares in a cardinal direction, then one square in an orthogonal direction.



Return the minimum number of steps needed to move the knight to the square $[x, y]$. It is guaranteed the answer exists.

Example 1:

Input: $x = 2, y = 1$
Output: 1
Explanation: $[0, 0] \rightarrow [2, 1]$

Example 2:

Input: $x = 5, y = 5$
Output: 4
Explanation: $[0, 0] \rightarrow [2, 1] \rightarrow [4, 2] \rightarrow [3, 4] \rightarrow [5, 5]$

Constraints:

- $-300 \leq x, y \leq 300$
- $0 \leq |x| + |y| \leq 300$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Knight moves on an infinite board. Minimum moves from (0,0) to (x,y).
8 possible moves. Symmetry: |x| and |y| – reflect to first quadrant. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."
BFS on an infinite chessboard from (0,0) to (x,y).
Enqueue all 8 knight moves each step; stop when target is reached.
Layer count = minimum moves. Prune to first quadrant since moves are symmetric.
Time: $O(\max(x,y)^2)$ BFS nodes. Space: $O(\max(x,y)^2)$ visited.
Should I go ahead and optimize?"

```
class _1197_MinKnightMoves:
```

```
    def minKnightMoves(self, x: int, y: int) -> int:
        x, y = abs(x), abs(y) # exploit symmetry – reflect to first quadrant
        queue = deque([(0, 0, 0)]) # (row, col, steps)
        seen = {(0, 0)}
        while queue:
            r, c, steps = queue.popleft()
            if r == x and c == y:
                return steps
            for dr, dc in [(2,1), (2,-1), (-2,1), (-2,-1), (1,2), (1,-2), (-1,2), (-1,-2)]:
                nr, nc = r + dr, c + dc
                if (nr, nc) not in seen and nr >= -1 and nc >= -1:
                    seen.add((nr, nc))
                    queue.append((nr, nc, steps + 1))
        return -1
```

PHASE 5 — TEST & DEBUG

(2,1): BFS from (0,0). (2,1) reached in 1 step → 1 ✓
(5,5): 4 steps ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(\max(|x|, |y|)^2)$. Space $O(\max(|x|, |y|)^2)$. Standard BFS shortest path.
Symmetry reduction: reflect to first quadrant cuts search space by 4.
The -1 lower bound allows the knight to 'go back' through negative coords when needed.
Given more time I'd ask: bidirectional BFS?

Start from both (0,0) and (x,y) simultaneously – meets in the middle, roughly halving depth."

BFS

□ #1730 Shortest Path to Get Food

MED

[Open on LeetCode](#)

Array · BFS · Matrix

Lists: Grind 169

Problem

You are starving and you want to eat food as quickly as possible. You want to find the shortest path to arrive at any food cell.

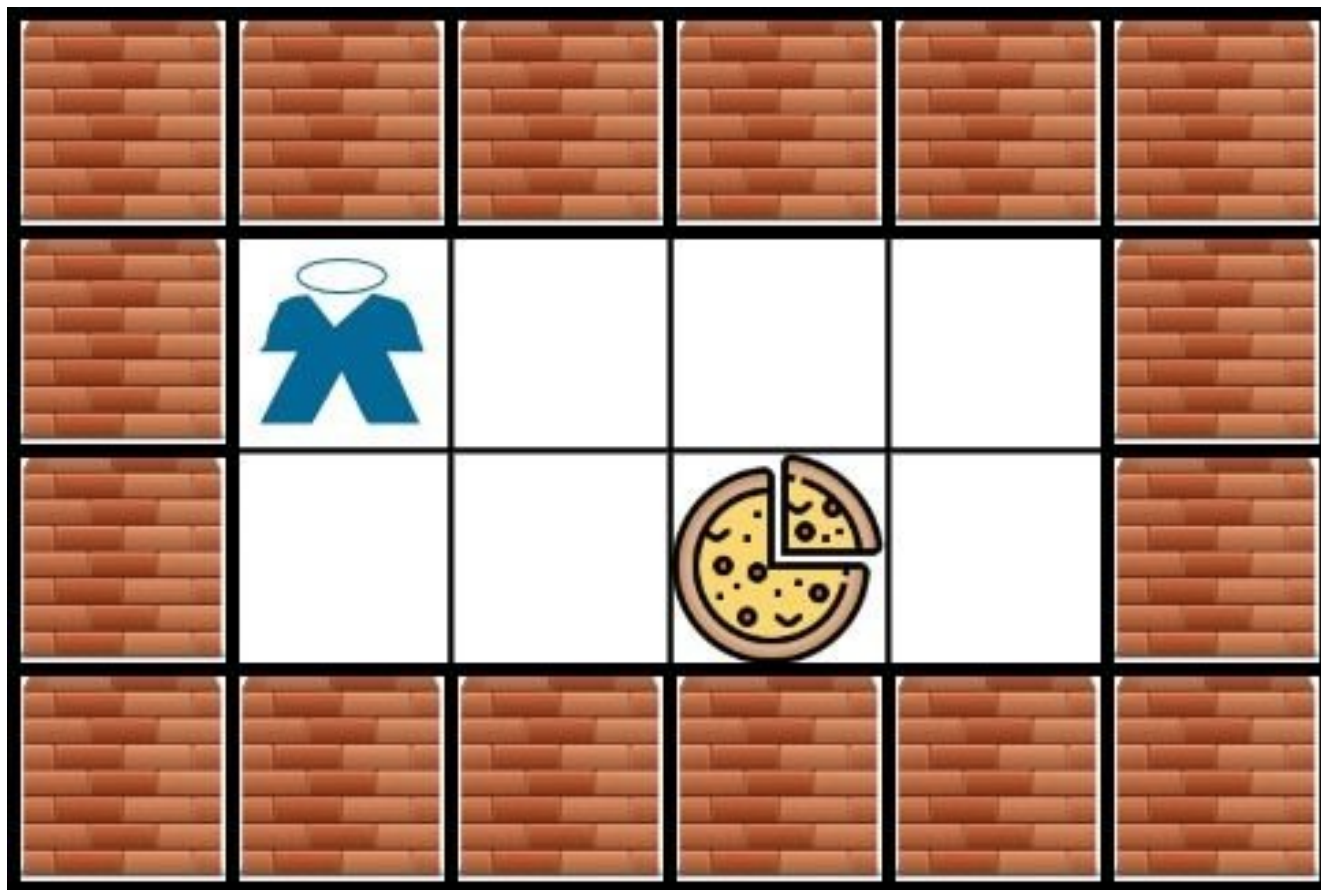
You are given an $m \times n$ character matrix, grid, of these different types of cells:

- '*' is your location. There is exactly one '*' cell.
- '#' is a food cell. There may be multiple food cells.
- 'O' is free space, and you can travel through these cells.
- 'X' is an obstacle, and you cannot travel through these cells.

You can travel to any adjacent cell north, east, south, or west of your current location if there is not an obstacle.

Return the length of the shortest path for you to reach any food cell. If there is no path for you to reach food, return -1.

Example 1:

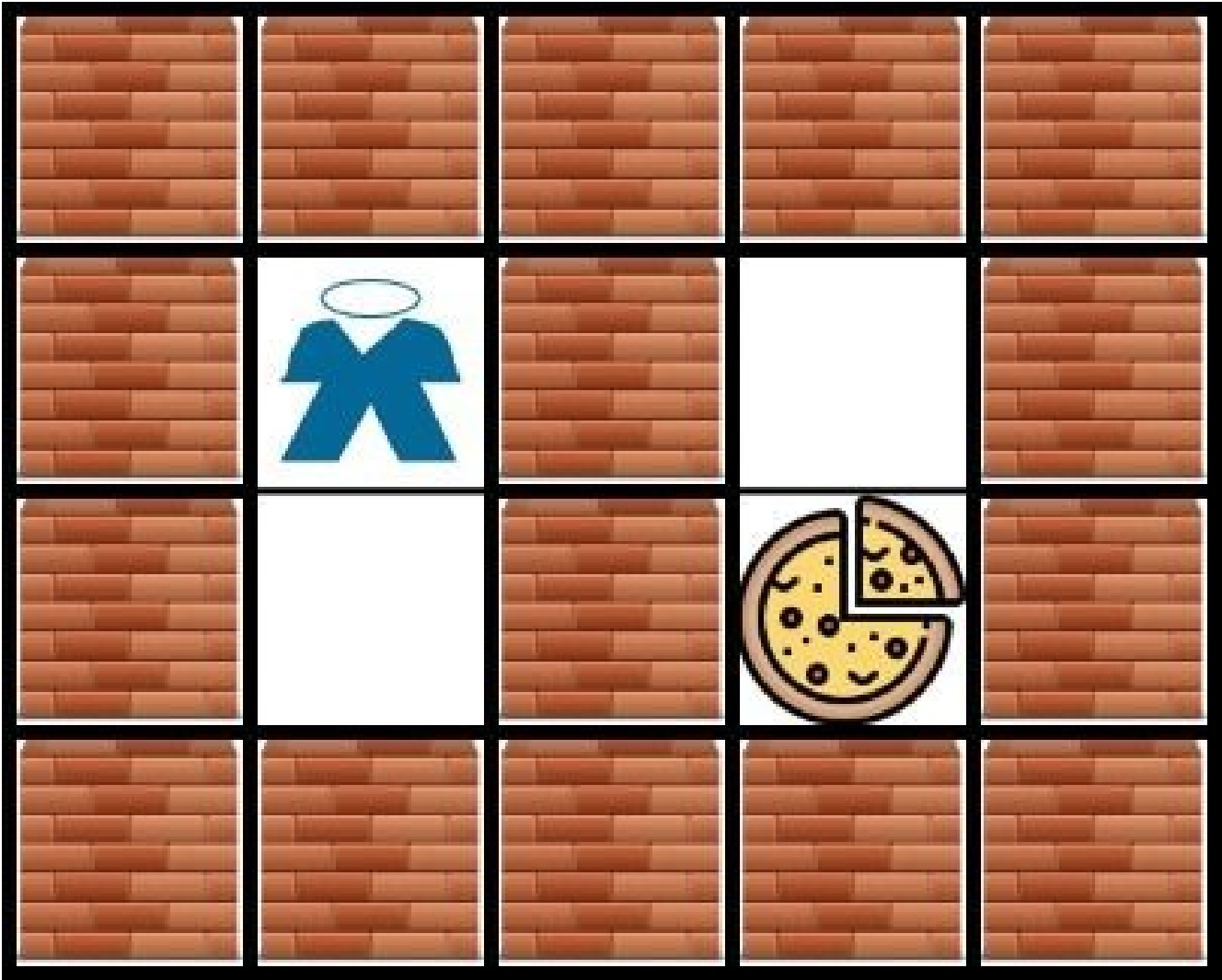


Input: grid = [["X","X","X","X","X","X"], ["X","*","0","0","0","X"], ["X","0","0","#","0","X"], ["X","X","X","X","X","X"]]

Output: 3

Explanation: It takes 3 steps to reach the food.

Example 2:

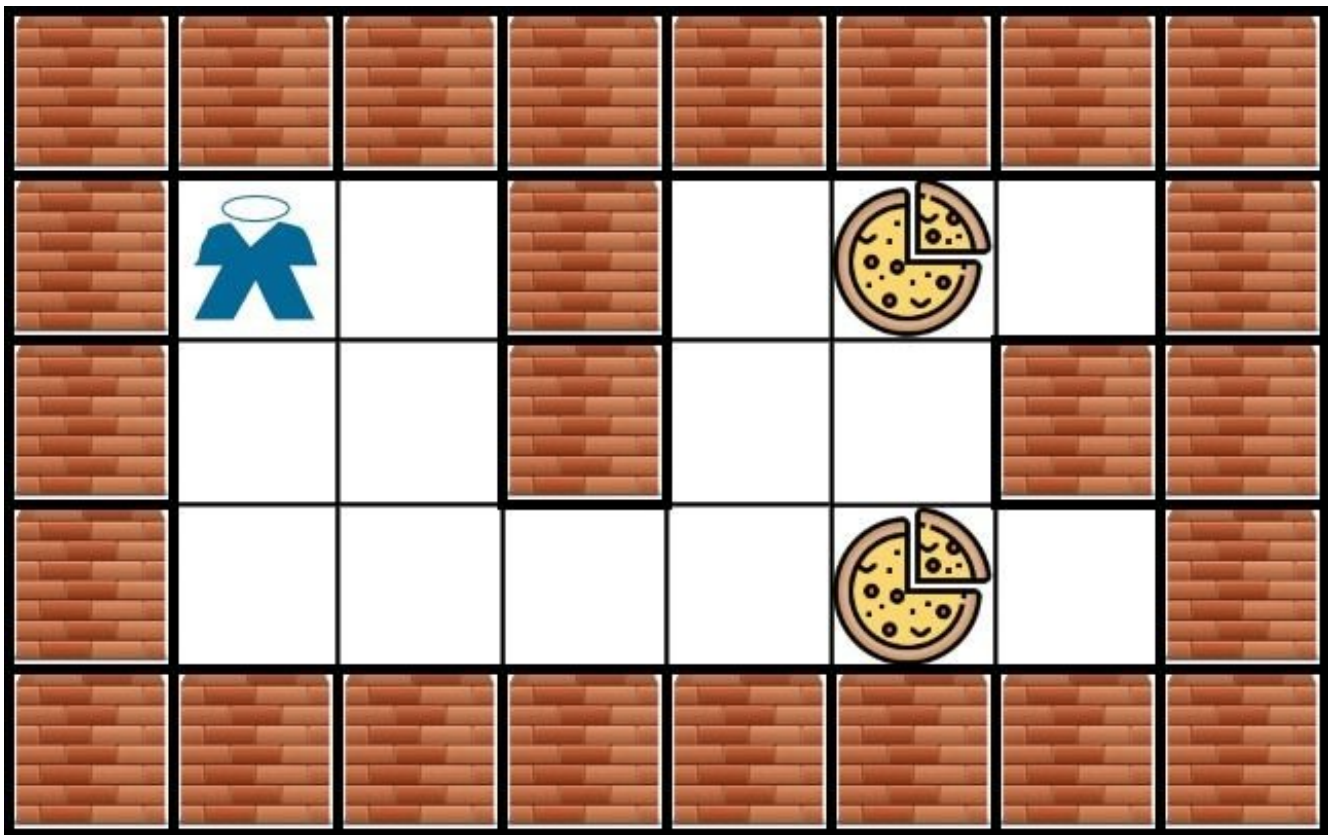


Input: grid = [["X","X","X","X","X"], ["X","*","X","0","X"], ["X","0","X","#","X"], ["X","X","X","X","X"]]

Output: -1

Explanation: It is not possible to reach the food.

Example 3:



Input: grid = [["X","X","X","X","X","X","X","X"], ["X","*", "O", "X", "O", "#", "O", "X"], ["X", "O", "O", "X", "O", "O", "X", "X"], ["X", "O", "O", "O", "O", "#", "O", "X"], ["X", "X", "X", "X", "X", "X", "X"]]

Output: 6

Explanation: There can be multiple food cells. It only takes 6 steps to reach the bottom food.

Example 4:

Input: grid = [["X","X","X","X","X","X","X","X"], ["X","*", "O", "X", "O", "#", "O", "X"], ["X", "O", "O", "X", "O", "O", "X", "X"], ["X", "O", "O", "O", "O", "#", "O", "X"], ["O", "O", "O", "O", "O", "O", "O", "O"]]

Output: 5

Constraints:

- $m == \text{grid.length}$
- $n == \text{grid}[i].\text{length}$
- $1 \leq m, n \leq 200$
- $\text{grid}[\text{row}][\text{col}]$ is '*', 'X', 'O', or '#'.

● The grid contains exactly one '*'.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "BFS from '*' to the nearest '#'. '0' = free, 'X' = wall, '#' = food (any one).
# Return shortest path length. Return -1 if unreachable. Right?"
#
# "'*' → nearest '#' using 4-directional BFS. First '#' reached = shortest path."
```

PHASE 2 — BRUTE FORCE (also optimal)

```
# "Let me start with brute force to lock in the logic.
# Multi-source BFS from every '*' food cell simultaneously.
# First time a 'H' house is reached, record its distance; sum cannot reach uses -1.
# BFS is already optimal O(m*n); DFS would revisit cells wastefully.
# Time: O(m * n). Space: O(m * n) queue.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE

```
# "Standard BFS from source. First '#' reached is guaranteed shortest path.
# Mark visited by overwriting '0' with 'X' – no extra visited set needed.
# Time: O(m*n). Space: O(m*n)."
```

PHASE 4 — CLEAN CODE

```
class _1730_ShortestPathFood:
    def getFood(self, grid: List[List[str]]) -> int:
        m, n = len(grid), len(grid[0])
        start = next((r, c) for r in range(m) for c in range(n) if grid[r][c] ==
'*')

        queue = deque([(start[0], start[1], 0)])
        grid[start[0]][start[1]] = 'X' # mark visited
        while queue:
            r, c, dist = queue.popleft()
            for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                nr, nc = r + dr, c + dc
                if 0 <= nr < m and 0 <= nc < n:
                    if grid[nr][nc] == '#':
                        return dist + 1
                    if grid[nr][nc] == '0':
                        grid[nr][nc] = 'X' # mark visited
                        queue.append((nr, nc, dist + 1))

        return -1
```

PHASE 5 — TEST & DEBUG

```
# grid example 1: BFS from '*'. Nearest '#' is 3 steps away → return 3 ✓
# grid example 2: BFS exhausts all reachable '0' cells → return -1 ✓
```


Round 3 · High · Hard

Round 3

High Priority · Hard

23 questions · 8 patterns

- ☐ ● Arrays & Hashing #41 ↗
- ☐ ● Sliding Window #76 ↗
- ☐ ● Binary Search #4 #315 #410 #668 #710 #774 ↗
- ☐ ● #1235 ↗
- ☐ ● Matrix #1074 ↗
- ☐ ● Trees & BST #124 #297 ↗
- ☐ ● DFS #269 #329 #685 #1036 #1192 ↗
- ☐ ● Graphs #305 #1153 ↗
- ☐ ● BFS #127 #317 #773 #815 ↗

Arrays & Hashing

Round 3 · High · Hard · 1 question

Arrays & Hashing

□ #41 First Missing Positive

HAR[Open on LeetCode](#)

Array · Hash Table

Lists: Grind 169

Problem

Given an unsorted integer array `nums`. Return the smallest positive integer that is not present in `nums`.

You must implement an algorithm that runs in $O(n)$ time and uses $O(1)$ auxiliary space.

Example 1:

```
Input: nums = [1,2,0]
Output: 3
Explanation: The numbers in the range [1,2] are all in the array.
```

Example 2:

```
Input: nums = [3,4,-1,1]
Output: 2
Explanation: 1 is in the array but 2 is missing.
```

Example 3:

```
Input: nums = [7,8,9,11,12]
Output: 1
Explanation: The smallest positive integer 1 is missing.
```

Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $-231 \leq \text{nums}[i] \leq 231 - 1$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Smallest positive integer NOT in nums. O(n) time, O(1) auxiliary space.
# Answer is in range [1, n+1] – if all 1..n present, answer is n+1. Right?
# Negatives, zeros, and large numbers can be ignored."
#
# "[3,4,-1,1] → 2 (1 present, 2 missing) ✓. [7,8,9] → 1 ✓. [1,2,0] → 3 ✓."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Sort the array, then scan from 1 upward until the first missing positive integer.
# Or put all values in a hash set and scan i=1,2,3... until absent.
# Correct but sorting is O(n log n) and the set uses O(n) extra space.
# Time: O(n log n). Space: O(n) for set approach.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (array as hash map — cycle sort)

```
# "Key insight: use the array itself as a lookup table.
# Valid range is [1, n]. Place each number x at index x-1 (if 1 ≤ x ≤ n and not
already there).
# After placing, scan: first i where nums[i] != i+1 → answer is i+1.
# If all positions correct → answer is n+1.
# Each number is placed at most once → O(n) total swaps."
```

PHASE 4 — CLEAN CODE

```
class _41_FirstMissingPositive:
    def firstMissingPositive(self, nums: List[int]) -> int:
        n = len(nums)
        for i in range(n):
            while 1 <= nums[i] <= n and nums[nums[i] - 1] != nums[i]:
                nums[nums[i] - 1], nums[i] = nums[i], nums[nums[i] - 1]
        for i in range(n):
            if nums[i] != i + 1:
                return i + 1
        return n + 1
```

PHASE 5 — TEST & DEBUG

```
# [3,4,-1,1]: place 3→pos2, 4→pos3, -1 ignored, 1→pos0.
# After: [1,-1,3,4]. Scan: i=1 nums[1]=-1≠2 → return 2 ✓
# [1,2,3]: after: [1,2,3]. All correct → return 4 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$  – each element swapped at most once. Space  $O(1)$ .  
# The while loop runs at most  $n$  total iterations across the entire for loop –  
amortised  $O(1)$  per  $i$ .  
# Given more time I'd ask: what if duplicates exist?  
# The condition nums[nums[i]-1] != nums[i] handles duplicates – don't swap a number  
to a position  
# that already has the correct value (would loop forever)."
```

Sliding Window

Round 3 · High · Hard · 1 question

Sliding Window

□ #76 Minimum Window Substring

HAR[Open on LeetCode](#)

Hash Table · String · Sliding Window

Lists: Grind 169

Problem

Given two strings s and t of lengths m and n respectively, return the minimum window substring of s such that every character in t (including duplicates) is included in the window. If there is no such substring, return the empty string `""`.

The testcases will be generated such that the answer is unique.

Example 1:

```
Input: s = "ADOBECODEBANC", t = "ABC"
Output: "BANC"
Explanation: The minimum window substring "BANC" includes 'A', 'B', and 'C' from string t.
```

Example 2:

```
Input: s = "a", t = "a"
Output: "a"
Explanation: The entire string s is the minimum window.
```

Example 3:

Round 3 · High · Hard

p.665

Input: s = "a", t = "aa"

Output: ""

Explanation: Both 'a's from t must be included in the window.
Since the largest window of s only has one 'a', return empty string.

Constraints:

- $m == s.length$
- $n == t.length$
- $1 \leq m, n \leq 105$
- s and t consist of uppercase and lowercase English letters.

Follow up: Could you find an algorithm that runs in $O(m + n)$ time?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Smallest window in s containing all characters of t (with multiplicity).
# Return '' if no such window. Right?
# t may have duplicate chars – window must have at least that many."
#
# "s='ADOBECODEBANC', t='ABC' → 'BANC' (contains A,B,C, length 4) ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Enumerate every substring of s with two nested indices.
# For each substring, check whether it contains all characters of t (with
frequency).
# Track the shortest valid window – correct but  $O(m^2 * n)$  character checks.
# Time:  $O(m^2 * n)$ . Space:  $O(n)$  frequency map.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (sliding window with 'formed' counter)

```
# "Expand right: add s[right] to window frequency map.
# When a char's count in window meets t's requirement, increment 'formed'.
# When formed == len(distinct chars in t): window is valid. Try shrinking from left.
# Track best window.  $O(m+n)$  – each char enters/exits at most once."
```

PHASE 4 — CLEAN CODE

```

class _76_MinWindowSubstring:
    def minWindow(self, s: str, t: str) -> str:
        if not t or not s:
            return ''
        need = Counter(t)
        have : Counter = Counter()
        formed = 0
        required = len(need)
        left = 0
        best = (float('inf'), 0, 0) # (length, left, right)
        for right in range(len(s)):
            c = s[right]
            have[c] += 1
            if c in need and have[c] == need[c]:
                formed += 1
            while formed == required:
                if right - left + 1 < best[0]:
                    best = (right - left + 1, left, right)
                have[s[left]] -= 1
                if s[left] in need and have[s[left]] < need[s[left]]:
                    formed -= 1
                left += 1
            return '' if best[0] == float('inf') else s[best[1]: best[2] + 1]

```

PHASE 5 — TEST & DEBUG

```

# s='ADOBECODEBANC', t='ABC': need={A:1,B:1,C:1}. required=3.
# Window expands until it contains A,B,C. Shrink left until one char drops below
# need.
# Best window = 'BANC' ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(m+n). Space O(m+n). The 'formed' counter avoids comparing the full
# frequency maps.
# A window is valid exactly when formed==required – elegant O(1) validity check.
# Given more time I'd ask: find all minimum windows?
# Collect all windows where length == best length during the BFS scan."

```

Binary Search

Round 3 · High · Hard · 7 questions

Binary Search

□ #4 Median of Two Sorted Arrays

HAR[Open on LeetCode](#)

Array · Binary Search · Divide and Conquer
Lists: Grind 169

Problem

Given two sorted arrays `nums1` and `nums2` of size `m` and `n` respectively, return the median of the two sorted arrays.

The overall run time complexity should be $O(\log(m+n))$.

Example 1:

```
Input: nums1 = [1,3], nums2 = [2]
Output: 2.00000
Explanation: merged array = [1,2,3] and median is 2.
```

Example 2:

```
Input: nums1 = [1,2], nums2 = [3,4]
Output: 2.50000
Explanation: merged array = [1,2,3,4] and median is (2 + 3) / 2 = 2.5.
```

Constraints:

- `nums1.length == m`
- `nums2.length == n`
- $0 \leq m \leq 1000$

- $0 \leq n \leq 1000$
- $1 \leq m + n \leq 2000$
- $-106 \leq \text{nums1}[i], \text{nums2}[i] \leq 106$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Merge two sorted arrays conceptually and find the median. $O(\log(m+n))$ required. Right?"

Median: if total length is odd → middle element. Even → average of two middles."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."

Merge both sorted arrays with two pointers into one combined array.

Walk to index $(m+n)//2$ (and the one before if even length) to read the median.

Correct but uses $O(m+n)$ extra space and ignores the logarithmic requirement.

Time: $O(m + n)$. Space: $O(m + n)$ merged array.

Should I go ahead and optimize?"

```
class _4_MedianTwoArrays_Brute:
```

```
    def findMedianSortedArrays(self, nums1: List[int], nums2: List[int]) -> float:
        m, n = len(nums1), len(nums2)
        i = j = 0
        half = (m + n) // 2
        prev = curr = 0
        for _ in range(half + 1):
            prev = curr
            if j >= n or (i < m and nums1[i] <= nums2[j]):
                curr = nums1[i]; i += 1
            else:
                curr = nums2[j]; j += 1
        return (curr + prev) / 2 if not (m + n) & 1 else curr
```

PHASE 3 — OPTIMIZE (binary search on partition — true $O(\log(\min(m,n)))$)

"Partition both arrays into left and right halves such that:

total left size = $(m+n+1)//2$, $\max(\text{left}) \leq \min(\text{right})$.

Binary search on the partition point of the smaller array.

At each partition: check if $\text{nums1}[\text{mid}-1] \leq \text{nums2}[\text{r}-\text{mid}]$ and $\text{nums2}[\text{r}-\text{mid}-1] \leq \text{nums1}[\text{mid}]$.

Median = $\max(\text{left halves})$ for odd total, or average of $\max(\text{left})/\min(\text{right})$ for even."

PHASE 4 — CLEAN CODE

```
class _4_MedianTwoArrays:
```

```
    def findMedianSortedArrays(self, nums1: List[int], nums2: List[int]) -> float:
```

```

if len(nums1) > len(nums2):
    nums1, nums2 = nums2, nums1
m, n = len(nums1), len(nums2)
half = (m + n + 1) // 2
lo, hi = 0, m
while lo <= hi:
    i = (lo + hi) // 2 # partition of nums1
    j = half - i # partition of nums2
    left1 = nums1[i - 1] if i > 0 else float('-inf')
    right1 = nums1[i] if i < m else float('inf')
    left2 = nums2[j - 1] if j > 0 else float('-inf')
    right2 = nums2[j] if j < n else float('inf')
    if left1 <= right2 and left2 <= right1:
        if (m + n) % 2 == 1:
            return float(max(left1, left2))
        return (max(left1, left2) + min(right1, right2)) / 2
    elif left1 > right2:
        hi = i - 1
    else:
        lo = i + 1
return 0.0

```

PHASE 5 — TEST & DEBUG

nums1=[1,3], nums2=[2]: m=2,n=2. half=2. Partition i=1,j=1:
left1=1,right1=3,left2=2,right2=∞.
1<=∞ and 2<=3 ✓. Total=3 (odd) → max(1,2)=2. → 2.0 ✓
nums1=[1,2], nums2=[3,4]: Partition → max(left)=2, min(right)=3. → 2.5 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(\log(\min(m,n)))$. Space $O(1)$. Binary search on the smaller array's partition.
The +inf/-inf sentinels handle edge cases (empty left/right) cleanly.
The brute-force merge approach is often acceptable in interviews – present both and let the interviewer choose.
Given more time I'd ask: kth element of two sorted arrays?
Binary search on k – eliminate $k/2$ elements from one array per step. Same $O(\log k)$ idea."

Binary Search

□ #315 Count of Smaller Numbers After Self

HAR[Open on LeetCode](#)

Array · Binary Search · Divide and Conquer · Binary Indexed Tree · Segment Tree · Merge Sort · Ordered Set

Lists: Denny Zhang

Problem

Given an integer array `nums`, return an integer array `counts` where `counts[i]` is the number of smaller elements to the right of `nums[i]`.

Example 1:

```
Input: nums = [5,2,6,1]
Output: [2,1,1,0]
Explanation:
To the right of 5 there are 2 smaller elements (2 and 1).
To the right of 2 there is only 1 smaller element (1).
To the right of 6 there is 1 smaller element (1).
To the right of 1 there is 0 smaller element.
```

Example 2:

```
Input: nums = [-1]
Output: [0]
```

Example 3:

```
Input: nums = [-1,-1]
Output: [0,0]
```

Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $-104 \leq \text{nums}[i] \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "For each index i, count elements to the right of i that are strictly less than
nums[i].
# Return as an array. Right?"
#
# "[5,2,6,1] → [2,1,1,0]: right of 5 has 2 and 1 (2 smaller), etc. ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# For each index i, scan every element to its right and count how many are smaller
than nums[i].
# Store those counts in an output array.
# Correct, but the inner scan makes this quadratic overall.
# Time:  $O(n^2)$ . Space:  $O(1)$  besides output.
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (BIT / merge sort)

```
# "Process from right to left. Build a BIT indexed by coordinate-compressed value.
# For each nums[i]: query BIT for count of values in [min, nums[i]-1].
# Update BIT at nums[i].  $O(n \log n)$  time."
```

PHASE 4 — CLEAN CODE

```
class _315_CountSmaller:
    def countSmaller(self, nums: List[int]) -> List[int]:
        sorted_unique = sorted(set(nums))
        rank = {v: i + 1 for i, v in enumerate(sorted_unique)}
        m = len(sorted_unique)
        bit = [0] * (m + 1)

        def update(i):
            while i <= m:
                bit[i] += 1
                i += i & (-i)

        def query(i):
            s = 0
            while i > 0:
                s += bit[i]
                i -= i & (-i)
            return s
```

```
result = []
for num in reversed(nums):
    r = rank[num]
    result.append(query(r - 1)) # count of values < num seen so far
    update(r)
return result[::-1]
```

PHASE 5 — TEST & DEBUG

```
# [5,2,6,1]: process right→left. 1→query(0)=0,update(1). 6→query(3)=1,update(4).
2→query(1)=1,update(2). 5→query(2)=2.
# reversed result=[2,1,1,0] → reverse → [2,1,1,0] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n \log n)$ . Space  $O(n)$ . BIT query gives prefix count in  $O(\log n)$ .
# Merge sort alternative: during merge, count inversions – same complexity but
different mental model.
# Given more time I'd ask: count of larger elements after self?
# Query total_right – query(rank[num]) instead."
```

Binary Search

#410 Split Array Largest Sum

HAR[Open on LeetCode](#)

Array · Binary Search · Dynamic Programming · Greedy · Prefix Sum

Lists: Denny Zhang

Problem

Given an integer array `nums` and an integer `k`, split `nums` into `k` non-empty subarrays such that the largest sum of any subarray is minimized.

Return the minimized largest sum of the split. A subarray is a contiguous part of the array.

Example 1:

Input: `nums = [7,2,5,10,8]`, `k = 2`

Output: 18

Explanation: There are four ways to split `nums` into two subarrays.

The best way is to split it into `[7,2,5]` and `[10,8]`, where the largest sum among the two subarrays is only 18.

Example 2:

Input: `nums = [1,2,3,4,5]`, `k = 2`

Output: 9

Explanation: There are four ways to split `nums` into two subarrays.

The best way is to split it into `[1,2,3]` and `[4,5]`, where the largest sum among the two subarrays is only 9.

Constraints:

- $1 \leq \text{nums.length} \leq 1000$
- $0 \leq \text{nums}[i] \leq 10^6$
- $1 \leq k \leq \min(50, \text{nums.length})$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Split nums into k non-empty contiguous subarrays to minimise the maximum subarray sum.

Must be in order – no reordering allowed. Right?

Same 'binary search on answer' pattern as Capacity to Ship Packages (#1011)."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Try every split of nums into k contiguous groups (DP over partition points).

For each split, max subarray sum = largest group sum; minimize that over splits.

State space explodes without binary search on the answer.

Time: exponential / $O(n^k)$ naive partitions. Space: $O(n)$ recursion.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Binary search on the answer (the max sum).

lo = max(nums) (at least one element), hi = sum(nums) (one subarray).

Feasibility check: can we split into k subarrays with each sum $\leq$ mid?

Greedy: accumulate greedily, start new subarray when exceeding mid. Count splits.

Time: $O(n \log(\text{sum}))$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _410_SplitArrayLargestSum:
    def splitArray(self, nums: List[int], k: int) -> int:
        def can_split(max_sum: int) -> bool:
            subarrays = 1
            current = 0
            for num in nums:
                if current + num > max_sum:
                    subarrays += 1
                    current = 0
                current += num
            return subarrays <= k
        lo, hi = max(nums), sum(nums)
        while lo < hi:
            mid = (lo + hi) // 2
```



```

        if can_split(mid):
            hi = mid # valid - try smaller
        else:
            lo = mid + 1 # not valid - need bigger
    return lo

```

PHASE 5 — TEST & DEBUG

```

# [7,2,5,10,8], k=2: lo=10, hi=32. mid=21→can_split? [7,2,5] and [10,8]=18≤21 ✓ → hi=21.
# ... converges to 18. can_split(18)=[7,2,5][10,8]→2 splits≤k=2 ✓.
can_split(17)→[7,2,5][10,8]→10+8=18>17→need 3>k ✗. → 18 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n \log(\text{sum}))$ . Space  $O(1)$ . Same template as #1011 Capacity to Ship - name this.
# lo/hi bounds: must be at least max(nums), at most sum(nums).
# Given more time I'd ask: DP approach? dp[j][i] = min largest sum splitting first i elements into j parts.
#  $O(kn^2)$  - worse than binary search for large inputs."

```

Binary Search

□ #668 Kth Smallest Number in Multiplication Table

HAR[Open on LeetCode](#)

Math · Binary Search

Lists: Denny Zhang

Problem

Nearly everyone has used the Multiplication Table. The multiplication table of size $m \times n$ is an integer matrix mat where $mat[i][j] == i * j$ (1-indexed).

Given three integers m , n , and k , return the k th smallest element in the $m \times n$ multiplication table.

Example 1:

1	2	3
2	4	6
3	6	9

1	2	2	3	3	4	6	6	9
---	---	---	---	---	---	---	---	---

Input: $m = 3, n = 3, k = 5$
Output: 3
Explanation: The 5th smallest number is 3.

Example 2:

1	2	3
2	4	6

1	2	2	3	4	6
---	---	---	---	---	---

Input: $m = 2, n = 3, k = 6$

Output: 6

Explanation: The 6th smallest number is 6.

Constraints:

- $1 \leq m, n \leq 3 \times 10^4$
- $1 \leq k \leq m \times n$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"m×n multiplication table: $\text{mat}[i][j]=i*j$ (1-indexed). Find kth smallest element. Right?

Can't materialise the table (m,n up to $3 \times 10^4 \rightarrow$ up to 9×10^8 cells)."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Collect all m×n products into a list, sort, return kth element.

Correct but materializes every product – $O(mn \log(mn))$ time and $O(mn)$ space.

Time: $O(m \times n \times \log(m \times n))$. Space: $O(m \times n)$.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (binary search on value)

"Binary search on the answer value v.

Count elements $\leq v$: for row i, elements are $i, 2i, 3i, \dots, n \times i$.

Count of these $\leq v$ is $\min(v // i, n)$. Sum over all rows $i=1..m$.

Binary search for smallest v where count $\geq k$.

lo=1, hi=m×n. Time: $O(m \log(m \times n))$."

PHASE 4 — CLEAN CODE

```
class _668_KthSmallestMultTable:
    def findKthNumber(self, m: int, n: int, k: int) -> int:
        def count_le(v: int) -> int:
            return sum(min(v // i, n) for i in range(1, m + 1))

        lo, hi = 1, m * n
        while lo < hi:
            mid = (lo + hi) // 2
            if count_le(mid) >= k:
                hi = mid
            else:
                lo = mid + 1
        return lo
```

PHASE 5 — TEST & DEBUG

```
# m=3,n=3,k=5. Table: [1,2,3,2,4,6,3,6,9] sorted=[1,2,2,3,3,4,6,6,9]. 5th=3.
# count_le(3)=count≥5? i=1:min(3,3)=3, i=2:min(1,3)=1, i=3:min(1,3)=1. sum=5≥5 →
hi=3.
# ... lo converges to 3 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m \log(m*n))$ . Space  $O(1)$ . The count formula is the key –  $O(m)$  per binary
search step.
# Note: the answer might not be a 'real' table entry – binary search finds the
smallest v
# with count  $\geq k$ , which IS always a table entry because count jumps discretely at
table values.
# Given more time I'd ask: kth largest? Binary search on  $m*n - k + 1$  smallest
instead."
```

Binary Search

□ #710 Random Pick with Blacklist

HAR[Open on LeetCode](#)

Array · Hash Table · Math · Binary Search ·
Sorting · Randomized

Lists: Denny Zhang

Problem

You are given an integer n and an array of unique integers `blacklist`. Design an algorithm to pick a random integer in the range $[0, n - 1]$ that is not in `blacklist`. Any integer that is in the mentioned range and not in `blacklist` should be equally likely to be returned.

Optimize your algorithm such that it minimizes the number of calls to the built-in random function of your language.

Implement the `Solution` class:

- `Solution(int n, int[] blacklist)` Initializes the object with the integer n and the blacklisted integers `blacklist`.
- `int pick()` Returns a random integer in the range $[0, n - 1]$ and not in `blacklist`.

Example 1:

Input

```
["Solution", "pick", "pick", "pick", "pick", "pick", "pick", "pick"]
[[7, [2, 3, 5]], [], [], [], [], [], [], []]
```

Output

```
[null, 0, 4, 1, 6, 1, 0, 4]
```

Explanation

```
Solution solution = new Solution(7, [2, 3, 5]);
```

Constraints:

- $1 \leq n \leq 109$
- $0 \leq \text{blacklist.length} \leq \min(105, n - 1)$
- $0 \leq \text{blacklist}[i] < n$
- All the values of blacklist are unique.
- At most $2 * 10^4$ calls will be made to pick.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Pick uniformly at random from $[0, n-1]$ excluding blacklist. One call to random per pick. Right?

n can be 10^9 – can't enumerate all valid numbers."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Store all non-blacklisted ids in a list; pick random index.

If blacklist is small, filter on each pick – $O(n)$ per draw.

Time: $O(n)$ per pick naive. Space: $O(n)$ whitelist.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (virtual mapping)

"Let $\text{sz} = n - \text{len}(\text{blacklist}) = \text{count of valid numbers}$.

Map $[0, \text{sz})$ as the 'virtual' range. Numbers in $[0, \text{sz})$ that ARE blacklisted

get remapped to non-blacklisted numbers in $[\text{sz}, n)$.

pick(): random in $[0, \text{sz}) \rightarrow$ if in remap $\rightarrow$ use remap[r], else use r directly.

Build remap in __init__: for each blacklisted number in $[0, \text{sz})$, map it to a non-blacklisted

number in $[\text{sz}, n)$. $O(|\text{blacklist}|)$ init, $O(1)$ per pick."

PHASE 4 — CLEAN CODE

```

class _710_RandomPickBlacklist:
    def __init__(self, n: int, blacklist: List[int]):
        bl_set = set(blacklist)
        self.sz = n - len(blacklist) # size of valid virtual range
        self.remap: dict = {}
        tail = n - 1
        for b in blacklist:
            if b < self.sz: # this blacklisted number is in [0, sz)
                while tail in bl_set: # find next valid number from the tail
                    tail -= 1
                self.remap[b] = tail
                tail -= 1
        def pick(self) -> int:
            r = random.randint(0, self.sz - 1)
            return self.remap.get(r, r)

```

PHASE 5 — TEST & DEBUG

```

# n=7, blacklist=[2,3,5]. sz=4. Valid: [0,1,4,6].
# Blacklisted in [0,4): 2,3. Map 2→6, 3→4 (5 is blacklisted, skip → 4).
# remap={2:6, 3:4}. pick(0)→0, pick(1)→1, pick(2)→6, pick(3)→4. All valid ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Init O(|B|). Pick O(1). Space O(|B|) for remap.
# The key insight: divide [0,n) into 'virtual valid' [0,sz) and 'tail' [sz,n).
# Blacklisted numbers in [0,sz) redirect to valid numbers in the tail.
# Given more time I'd ask: what if the blacklist updates dynamically?
# Rebuild the remap on each update — O(|B|) per update, same O(1) pick."

```


Binary Search

□ #774 Minimize Max Distance to Gas Station

HAR[Open on LeetCode](#)

Array · Binary Search

Lists: Denny Zhang

Problem

You are given an integer array `stations` that represents the positions of the gas stations on the x -axis. You are also given an integer k .

You should add k new gas stations. You can add the stations anywhere on the x -axis, and not necessarily on an integer position.

Let `penalty()` be the maximum distance between adjacent gas stations after adding the k new stations.

Return the smallest possible value of `penalty()`. Answers within 10^{-6} of the actual answer will be accepted.

Example 1:

```
Input: stations = [1,2,3,4,5,6,7,8,9,10], k = 9
Output: 0.50000
```

Example 2:

Round 3 · High · Hard

p.685

Input: stations = [23,24,36,39,46,56,57,65,84,98], k = 1
 Output: 14.00000

Constraints:

- $10 \leq \text{stations.length} \leq 2000$
- $0 \leq \text{stations}[i] \leq 108$
- stations is sorted in a strictly increasing order.
- $1 \leq k \leq 106$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Add exactly k stations anywhere (real-valued positions) to minimise the maximum gap.

Return the answer within 1e-6 precision. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Binary search on the maximum distance between adjacent stations.

For candidate d, check if we can add k stations so all gaps $\leq d$ (greedy fill).

Brute: try every real-valued spacing without binary search – continuous search space.

Time: $O(n * \text{precision})$ naive scan. Space: $O(1)$.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (binary search on floating-point answer)

"Binary search on the max gap d. Feasibility: for each existing gap g, we need $\text{ceil}(g/d)-1$ extra

stations to keep all sub-gaps $\leq d$. Count total extras; if $\leq k$, d is achievable.

Precision: run until $hi-lo < 1e-7$."

PHASE 4 — CLEAN CODE

```
class _774_MinMaxDistanceGasStation:
```

```
    def minmaxGasDist(self, stations: List[int], k: int) -> float:
```

```
        gaps = [stations[i+1] - stations[i] for i in range(len(stations)-1)]
```

```
        def can_achieve(d: float) -> bool:
```

```
            return sum(int(g / d) for g in gaps) <= k
```

```
        lo, hi = 0.0, max(gaps)
```

```
        for _ in range(100): # 100 iterations -> precision < hi/2^100
```

```
        mid = (lo + hi) / 2
        if can_achieve(mid):
            hi = mid
        else:
            lo = mid
    return hi
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n * 100) = O(n)$. Space $O(n)$ for gaps. Floating-point binary search with fixed iterations.

Note: `int(g/d)` gives the number of stations to insert (not `ceil-1` – same thing via floor division).

Given more time I'd ask: what if stations must be at integer positions?

Binary search on integers, same feasibility check but with `math.ceil`."

Binary Search

□ #1235 Maximum Profit in Job Scheduling **HAR**

[Open on LeetCode](#)

Array · Binary Search · Dynamic Programming · Sorting

Lists: Grind 169

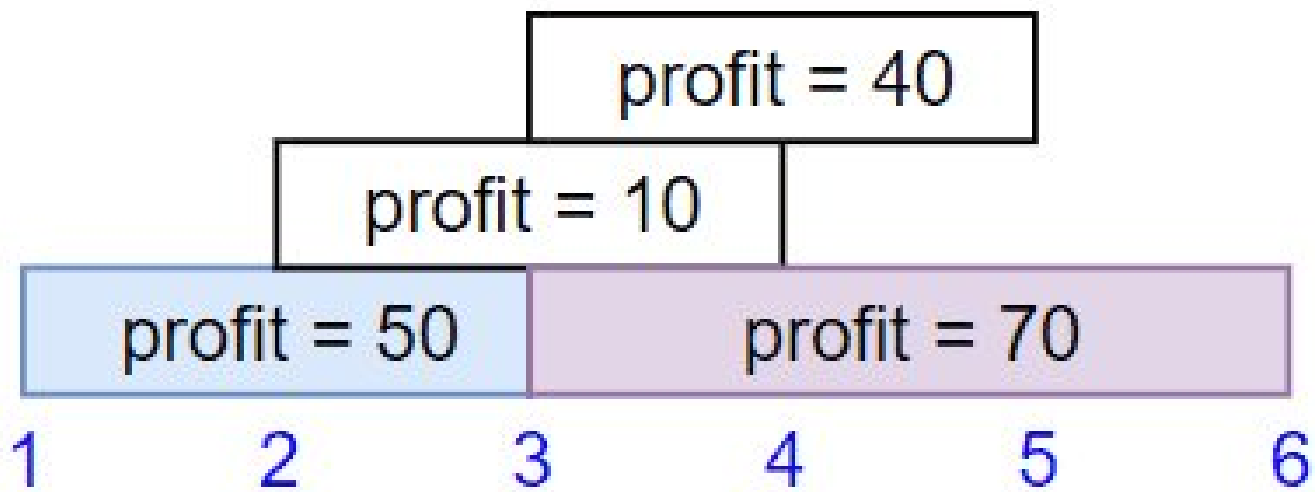
Problem

We have n jobs, where every job is scheduled to be done from $startTime[i]$ to $endTime[i]$, obtaining a profit of $profit[i]$.

You're given the $startTime$, $endTime$ and $profit$ arrays, return the maximum profit you can take such that there are no two jobs in the subset with overlapping time range.

If you choose a job that ends at time X you will be able to start another job that starts at time X .

Example 1:



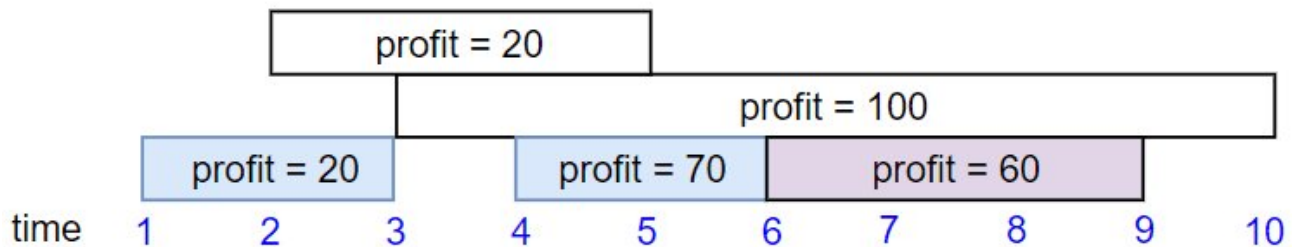
Input: startTime = [1,2,3,3], endTime = [3,4,5,6], profit = [50,10,40,70]

Output: 120

Explanation: The subset chosen is the first and fourth job.

Time range [1-3]+[3-6], we get profit of $120 = 50 + 70$.

Example 2:



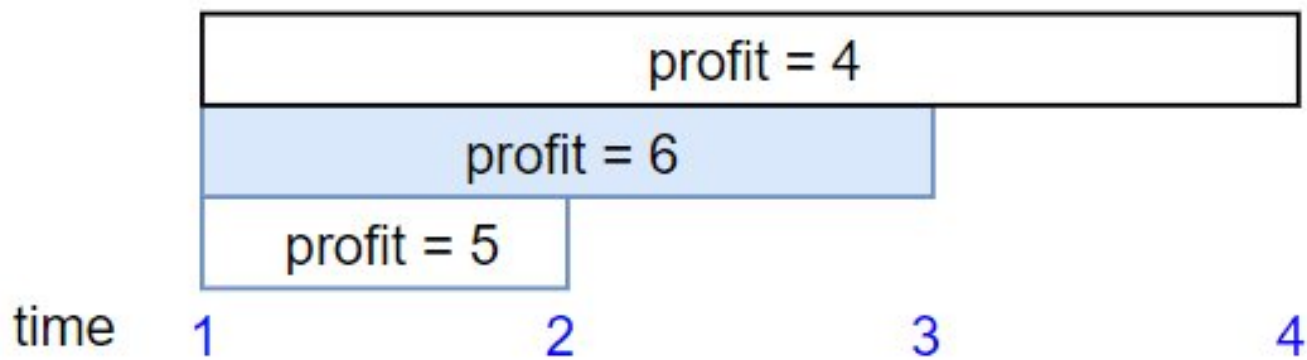
Input: startTime = [1,2,3,4,6], endTime = [3,5,10,6,9], profit = [20,20,100,70,60]

Output: 150

Explanation: The subset chosen is the first, fourth and fifth job.

Profit obtained $150 = 20 + 70 + 60$.

Example 3:



Input: startTime = [1,1,1], endTime = [2,3,4], profit = [5,6,4]
 Output: 6

Constraints:

- $1 \leq \text{startTime.length} == \text{endTime.length} == \text{profit.length} \leq 5 * 10^4$
- $1 \leq \text{startTime}[i] < \text{endTime}[i] \leq 10^9$
- $1 \leq \text{profit}[i] \leq 10^4$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Select non-overlapping jobs to maximise total profit.

Job ending at X and starting at X don't overlap. Return max profit. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Sort jobs by end time. DP over index: $\text{dp}[i]$ = max profit taking job i plus best non-overlapping before i.

For each i, scan all $j < i$ for compatibility - $O(n^2)$ DP.

Time: $O(n^2)$. Space: $O(n)$ DP + sort.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (sort by end time + DP + binary search)

"Sort jobs by end time. $\text{dp}[i]$ = max profit using first i jobs.

For each job i: skip $\rightarrow \text{dp}[i] = \text{dp}[i-1]$. Take $\rightarrow$ find latest j where $\text{job}[j].\text{end} \leq \text{job}[i].\text{start}$

using binary search. $\text{dp}[i] = \max(\text{dp}[i-1], \text{dp}[j] + \text{profit}[i])$.

Time: $O(n \log n)$."

PHASE 4 — CLEAN CODE

```

class _1235_MaxProfitJobScheduling:
    def jobScheduling(self, startTime: List[int], endTime: List[int], profit:
List[int]) -> int:
        jobs = sorted(zip(endTime, startTime, profit))
        ends = [0] # dp_ends[i] = end time of ith job in dp
        dp = [0] # dp[i] = max profit for first i sorted jobs
        for end, start, p in jobs:
            # Find latest job whose end <= start of current
            j = bisect.bisect_right(ends, start) - 1
            if dp[j] + p > dp[-1]:
                dp.append(dp[j] + p)
                ends.append(end)
            else:
                dp.append(dp[-1])
                ends.append(end)
        return dp[-1]

```

PHASE 5 — TEST & DEBUG

```

# jobs sorted by end: [(3,1,50),(4,2,10),(5,3,40),(6,3,70)].
# job(3,1,50): j=bisect_right([0],1)-1=0. dp[0]+50=50>dp[-1]=0 →
dp=[0,50],ends=[0,3].
# job(4,2,10): bisect_right([0,3],2)-1=0. dp[0]+10=10<dp[-1]=50 →
dp=[0,50,50],ends=[0,3,4].
# job(5,3,40): bisect_right([0,3,4],3)-1=1. dp[1]+40=90>50 → dp=[0,50,50,90].
# job(6,3,70): bisect_right([0,3,4,5],3)-1=1. dp[1]+70=120>90 →
dp=[0,50,50,90,120]. → 120 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n \log n)$ . Space  $O(n)$ . Sort + DP with binary search per step.
# The dp array is monotonically non-decreasing – binary search on ends is valid.
# Given more time I'd ask: what if we want to select at most k jobs?
# Add a dimension to dp – dp[i][k] –  $O(n^2k)$  with the same binary search for last
compatible job."

```

Matrix

Round 3 · High · Hard · 1 question

Matrix

□ #1074 Number of Submatrices That Sum to Target

HAR[Open on LeetCode](#)

Array · Hash Table · Matrix · Prefix Sum

Lists: Denny Zhang

Problem

Given a matrix and a target, return the number of non-empty submatrices that sum to target.

A submatrix $x1, y1, x2, y2$ is the set of all cells $matrix[x][y]$ with $x1 \leq x \leq x2$ and $y1 \leq y \leq y2$.

Two submatrices $(x1, y1, x2, y2)$ and $(x1', y1', x2', y2')$ are different if they have some coordinate that is different: for example, if $x1 \neq x1'$.

Example 1:

0	1	0
1	1	1
0	1	0

Input: matrix = `[[0,1,0],[1,1,1],[0,1,0]]`, target = 0

Output: 4

Explanation: The four 1x1 submatrices that only contain 0.

Example 2:

Input: matrix = `[[1,-1],[-1,1]]`, target = 0

Output: 5

Explanation: The two 1x2 submatrices, plus the two 2x1 submatrices, plus the 2x2 submatrix.

Example 3:

Input: matrix = `[[904]]`, target = 0

Output: 0

Constraints:

- $1 \leq \text{matrix.length} \leq 100$
- $1 \leq \text{matrix}[0].\text{length} \leq 100$
- $-1000 \leq \text{matrix}[i][j] \leq 1000$
- $-10^8 \leq \text{target} \leq 10^8$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Count all submatrices (r1,c1,r2,c2) whose elements sum to target. Right?"
 # Extend #560 Subarray Sum Equals K to 2D."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."
 # Compute 2D prefix sums, then enumerate all submatrix top-left and bottom-right corners.
 # For each submatrix, O(1) range sum check against target.
 # Four nested loops over corners – $O(n^2 * m^2)$ submatrices.
 # Time: $O(n^2 * m^2)$. Space: $O(n * m)$ prefix grid.
 # Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE

"Fix top row r1 and bottom row r2. Compress columns into 1D prefix sums between r1..r2."
 # Then apply #560 (subarray sum = target) on the 1D array.
 # Time: $O(m^2 * n)$. Space: $O(n)$."

PHASE 4 — CLEAN CODE

```
class _1074_SubmatricesSumTarget:
    def numSubmatrixSumTarget(self, matrix: List[List[int]], target: int) -> int:
        m, n = len(matrix), len(matrix[0])
        count = 0
        for r1 in range(m):
            col_sum = [0] * n
            for r2 in range(r1, m):
                for c in range(n):
                    col_sum[c] += matrix[r2][c]
                # Apply #560 on col_sum
                prefix = Counter({0: 1})
                running = 0
                for s in col_sum:
                    running += s
                    count += prefix[running - target]
                prefix[running] += 1
```

```
return count
```

PHASE 5 — TEST & DEBUG

```
# [[0,1,0],[1,1,1],[0,1,0]], target=0:
```

```
# Fix r1=0,r2=0: col_sum=[0,1,0]. Subarrays summing to 0: [0] at c=0, [0] at c=2.  
count=2.
```

```
# Other rows contribute 2 more. → 4 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m^2 * n)$ . Space  $O(n)$ . This is literally #560 applied inside a 2D row-fixing  
loop.
```

```
# Name that connection – it shows pattern transfer.
```

```
# Given more time I'd ask: count submatrices summing to at most target?
```

```
# Use #862 (Shortest Subarray with Sum at Least K) approach – monotone deque instead  
of hash map."
```

Trees & BST

Round 3 · High · Hard · 2 questions

Trees & BST

□ #124 Binary Tree Maximum Path Sum

HAR[Open on LeetCode](#)

Dynamic Programming · Tree · DFS

Lists: Grind 169

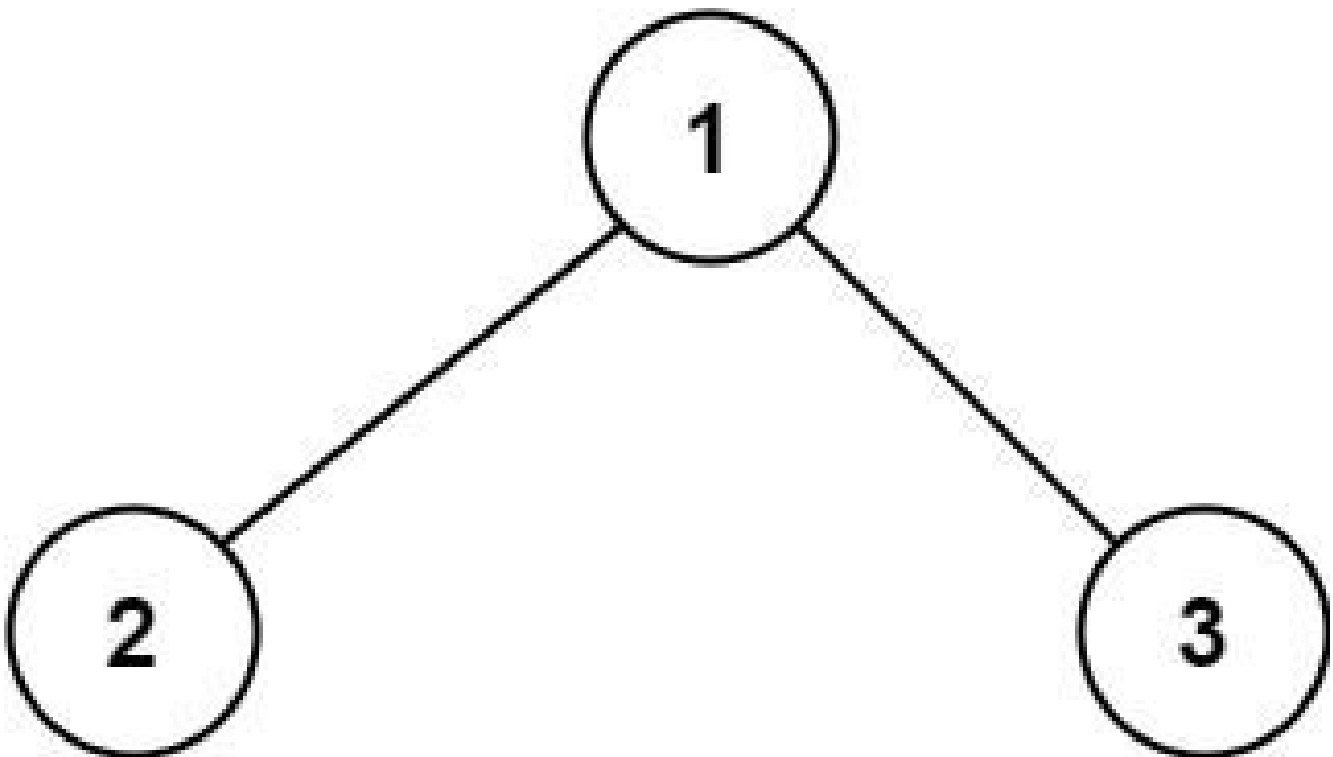
Problem

A path in a binary tree is a sequence of nodes where each pair of adjacent nodes in the sequence has an edge connecting them. A node can only appear in the sequence at most once. Note that the path does not need to pass through the root.

The path sum of a path is the sum of the node's values in the path.

Given the root of a binary tree, return the maximum path sum of any non-empty path.

Example 1:

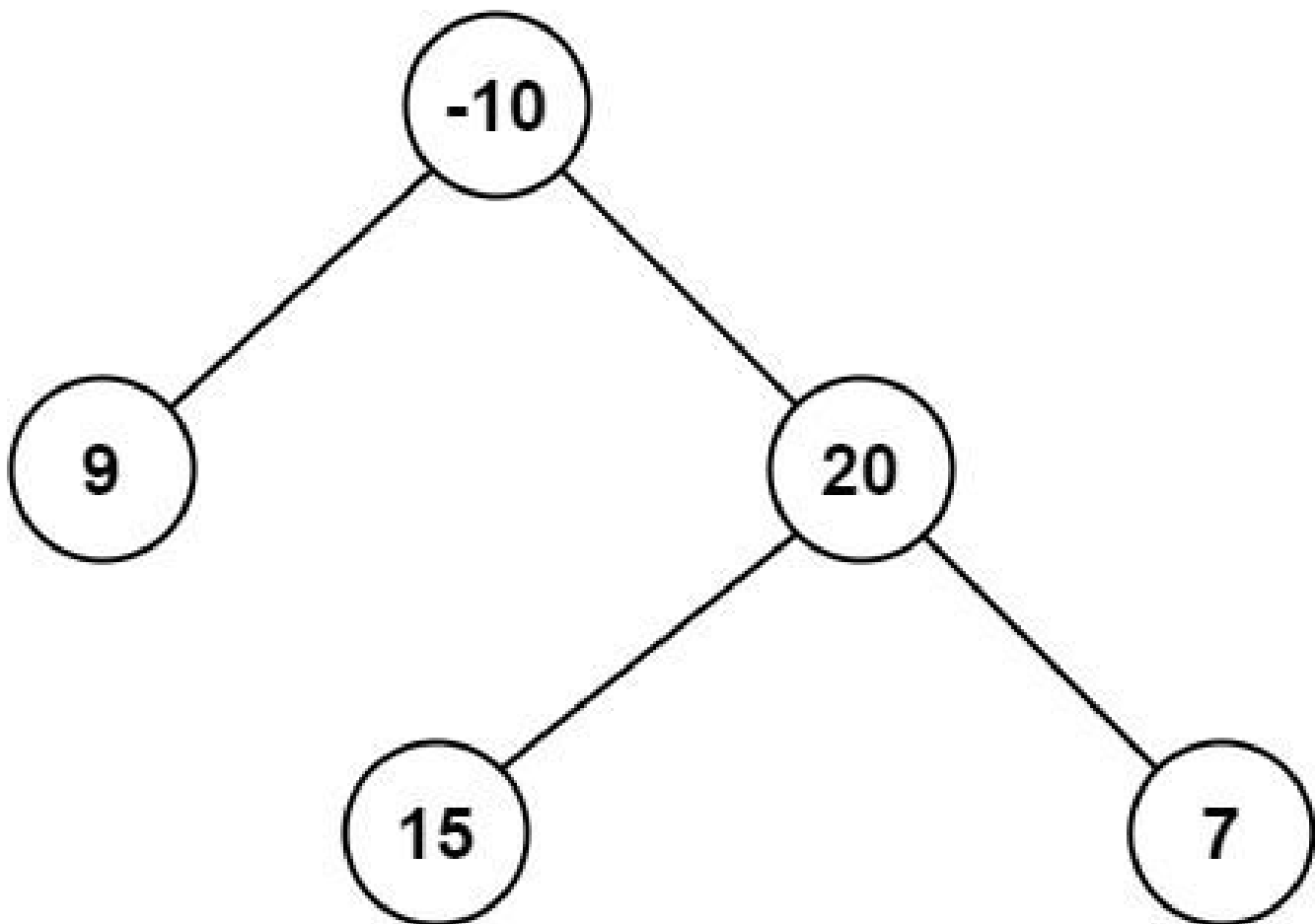


Input: root = [1,2,3]

Output: 6

Explanation: The optimal path is 2 -> 1 -> 3 with a path sum of $2 + 1 + 3 = 6$.

Example 2:



Input: root = [-10,9,20,null,null,15,7]

Output: 42

Explanation: The optimal path is 15 -> 20 -> 7 with a path sum of 15 + 20 + 7 = 42.

Constraints:

- The number of nodes in the tree is in the range [1, 3 * 10⁴].
- -1000 ≤ Node.val ≤ 1000

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Path = any sequence of connected nodes (no revisiting). Path doesn't need to pass through root."

Values can be negative. Return maximum path sum. Right?"


```

#
# "[-10,9,20,null,null,15,7]: path 15→20→7 = 42 ✓"

# PHASE 2 — BRUTE FORCE
# "Let me start with brute force to lock in the logic.
# At every node, compute max path sum through that node = node.val + max(0,
left_gain) + max(0, right_gain).
# Recurse full tree; track global max. Recomputes subtree gains repeatedly without
sharing.
# Time: O(n^2) worst case on skewed tree. Space: O(h) stack.
# Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE (post-order, same as Diameter)
# "For each node, compute the 'gain' if we extend a path through it: max(0,
left_gain, right_gain) + node.val.
# The best path THROUGH this node = node.val + max(0, left) + max(0, right).
# Update global max. Return one-sided gain to parent (can't go both left and right
AND up).
# Use inner helper – no self."

# PHASE 4 — CLEAN CODE
class _124_MaxPathSum:
    def maxPathSum(self, root: Optional['TreeNode']) -> int:
        best = [float('-inf')]
        def gain(node):
            if not node: return 0
            left = max(0, gain(node.left))
            right = max(0, gain(node.right))
            best[0] = max(best[0], node.val + left + right)
            return node.val + max(left, right)
        gain(root)
        return best[0]

# PHASE 5 — TEST & DEBUG
# [-10,9,20,null,null,15,7]:
# gain(9)=9. gain(15)=15. gain(7)=7.
# gain(20): left=15,right=7. best=max(-inf,20+15+7)=42. return 20+15=35.
# gain(-10): left=9,right=35. best=max(42,-10+9+35)=42. return -10+35=25.
# → 42 ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(n). Space O(h). Same post-order pattern as Diameter of Binary Tree (#543)
and Balanced Tree (#110).
# The 'max(0, gain)' is essential – a negative subtree gain should be dropped.
# Given more time I'd ask: return the actual path (not just the sum)?
# Track which path achieved best[0] – store node references alongside the max
update."

```

Trees & BST

□ #297 Serialize and Deserialize Binary Tree

HAR[Open on LeetCode](#)

String · Tree · DFS · BFS · Design

Lists: Grind 169

Problem

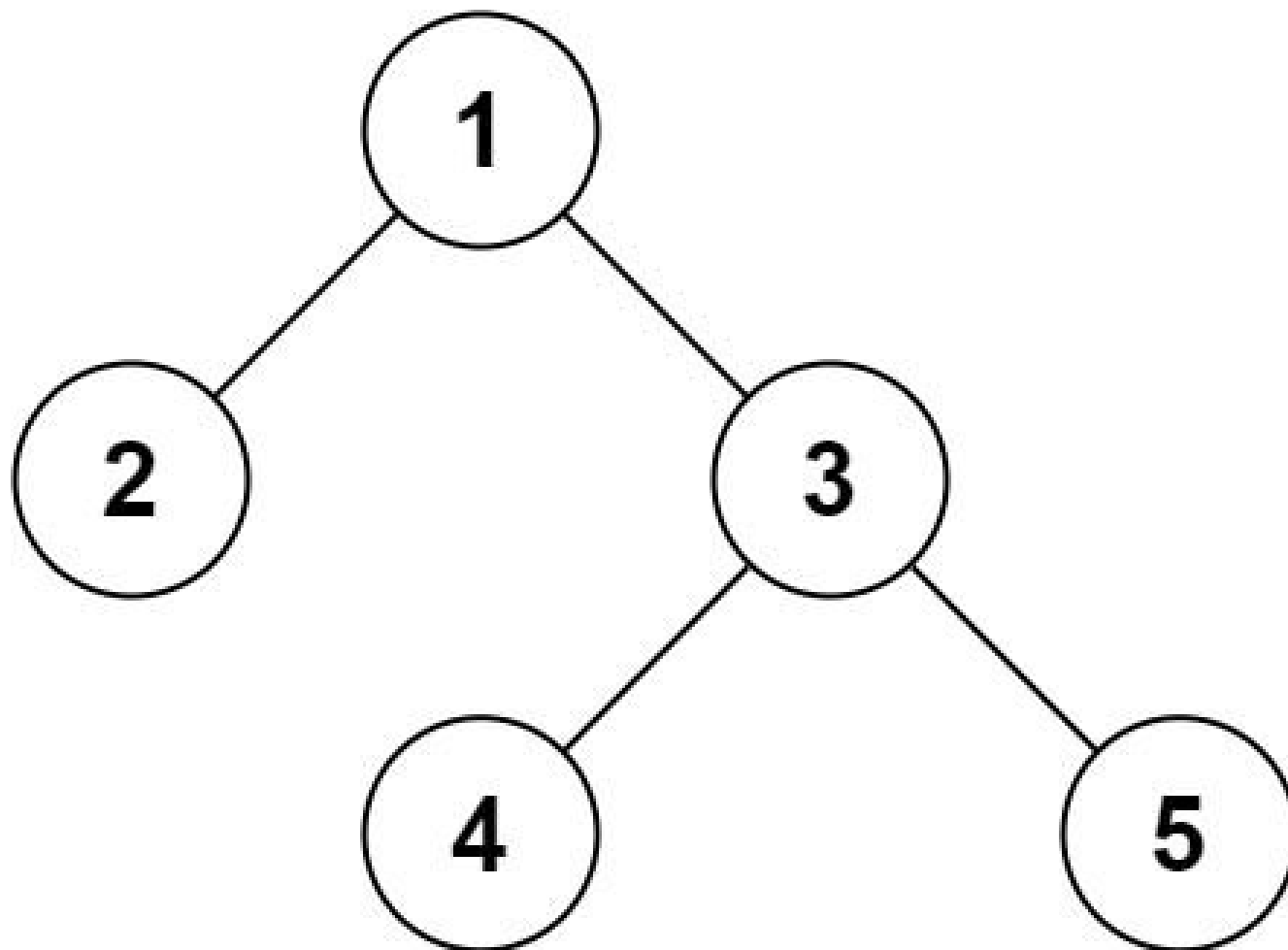
Serialization is the process of converting a data structure or object into a sequence of bits so that it can be stored in a file or memory buffer, or transmitted across a network connection link to be reconstructed later in the same or another computer environment.

Design an algorithm to serialize and deserialize a binary tree. There is no restriction on how your serialization/deserialization algorithm should work. You just need to ensure that a binary tree can be serialized to a string and this string can be deserialized to the original tree structure.

Clarification: The input/output format is the same as how LeetCode serializes a binary tree. You do not necessarily need to follow this

format, so please be creative and come up with different approaches yourself.

Example 1:



Input: root = [1,2,3,null,null,4,5]
Output: [1,2,3,null,null,4,5]

Example 2:

Input: root = []
Output: []

Constraints:

- The number of nodes in the tree is in the range [0, 104].

- $-1000 \leq \text{Node.val} \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Design encode and decode functions for a binary tree. Any format is fine as long as it round-trips.
Empty tree encodes to some valid string. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
Serialize: preorder traversal with 'null' markers for empty children ? comma string.
Deserialize: split queue, rebuild tree recursively matching preorder order.
Correct but string can be long; BFS level-order encoding is more compact.
Time: $O(n)$. Space: $O(n)$ output string.
Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (pre-order DFS with null markers)

"Serialize: pre-order traversal, 'N' for null, comma-separated.
Deserialize: split, use an iterator, recursively build from pre-order.
Pre-order uniquely determines the tree when nulls are included."

PHASE 4 — CLEAN CODE

```
class _297_Codec:
    def serialize(self, root: 'TreeNode') -> str:
        parts: List[str] = []
        def dfs(node):
            if not node: parts.append('N'); return
            parts.append(str(node.val))
            dfs(node.left)
            dfs(node.right)
        dfs(root)
        return ','.join(parts)
    def deserialize(self, data: str) -> 'TreeNode':
        vals = iter(data.split(','))
        def build():
            v = next(vals)
            if v == 'N': return None
            node = TreeNode(int(v))
            node.left = build()
            node.right = build()
            return node
        return build()
```

PHASE 5 — TEST & DEBUG

[1,2,3,null,null,4,5]: serialize → '1,2,N,N,3,4,N,N,5,N,N'. deserialize rebuilds the tree ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(n)$. Pre-order with null markers is the cleanest approach.

BFS alternative: level-order like LeetCode's own format – works but trickier to parse.

Given more time I'd ask: serialize a general N-ary tree?

Add child count before children – e.g. '1,3,2,0,3,0,4,0,5,0' for a 3-child root."

DFS

Round 3 · High · Hard · 5 questions

DFS

□ ★ #269 Alien Dictionary ★

HAR[Open on LeetCode](#)

Array · String · DFS · BFS · Graph · Topological Sort

Lists: ★ Premium | Grind 169 | Premium 98

Problem

There is a new alien language that uses the English alphabet. However, the order of the letters is unknown to you.

You are given a list of strings words from the alien language's dictionary. Now it is claimed that the strings in words are sorted lexicographically by the rules of this new language.

If this claim is incorrect, and the given arrangement of string in words cannot correspond to any order of letters, return "".

Otherwise, return a string of the unique letters in the new alien language sorted in lexicographically increasing order by the new language's rules. If there are multiple solutions,

return any of them.

Example 1:

Input: words = ["wrt","wrf","er","ett","rftt"]
Output: "wertf"

Example 2:

Input: words = ["z","x"]
Output: "zx"

Example 3:

Input: words = ["z","x","z"]
Output: ""
Explanation: The order is invalid, so return "".

Constraints:

- $1 \leq \text{words.length} \leq 100$
- $1 \leq \text{words}[i].\text{length} \leq 100$
- $\text{words}[i]$ consists of only lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given words sorted in alien lexicographic order, deduce the letter ordering.
Return any valid ordering, or '' if impossible (cycle or contradiction). Right?
Contradiction: longer word precedes its prefix (e.g. ['ab','a'])."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
Build graph: for each adjacent pair in words, record char order u before v.
Topological sort the chars; if cycle detected, no valid order.
Compare words character by character to infer edges.
Time: $O(C)$ total chars. Space: $O(1)$ – 26 letters.
Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (Kahn's BFS topological sort)


```
# "Extract ordering constraints from consecutive word pairs: compare char by char,
first difference gives an edge.
# Build directed graph + in-degree map.
# BFS: process zero-in-degree nodes. If we process all letters → valid. Else cycle →
return ''."
```

PHASE 4 — CLEAN CODE

```
class _269_AlienDictionary:
    def alienOrder(self, words: List[str]) -> str:
        graph = defaultdict(set)
        indegree = {c: 0 for w in words for c in w}
        for i in range(len(words) - 1):
            w1, w2 = words[i], words[i + 1]
            min_len = min(len(w1), len(w2))
            if len(w1) > len(w2) and w1[:min_len] == w2[:min_len]:
                return '' # invalid: prefix ordering
            for c1, c2 in zip(w1, w2):
                if c1 != c2:
                    if c2 not in graph[c1]:
                        graph[c1].add(c2)
                        indegree[c2] += 1
                    break
        queue = deque(c for c in indegree if indegree[c] == 0)
        result = []
        while queue:
            c = queue.popleft()
            result.append(c)
            for nb in graph[c]:
                indegree[nb] -= 1
                if indegree[nb] == 0:
                    queue.append(nb)
        return ''.join(result) if len(result) == len(indegree) else ''
```

PHASE 5 — TEST & DEBUG

```
# ["wrt","wrf","er","ett","rftt"]: edges: t→f, w→e, r→t, e→r. indegree:
f:1,e:1,t:1,r:1,w:0.
# BFS from w: w→e→r→t→f. result='wertf' ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(C) where C = total characters. Space O(1) – at most 26 letters.
# The prefix check catches the only invalid case that doesn't manifest as a graph
cycle.
# Given more time I'd ask: return ALL valid orderings?
# Backtracking on the topological sort – collect all linearisations."
```

DFS

□ #329 Longest Increasing Path in a Matrix **HAR**

[Open on LeetCode](#)

Array · DFS · BFS · Graph · Topological Sort · Memoization · Matrix

Lists: Grind 169

Problem

Given an $m \times n$ integers matrix, return the length of the longest increasing path in matrix. From each cell, you can either move in four directions: left, right, up, or down. You may not move diagonally or move outside the boundary (i.e., wrap-around is not allowed).

Example 1:

9	9	4
6	6	8
2	1	1

Input: matrix = [[9,9,4],[6,6,8],[2,1,1]]

Output: 4

Explanation: The longest increasing path is [1, 2, 6, 9].

Example 2:

3 →	4 →	5 ↓
3	2	6
2	2	1

Input: matrix = [[3,4,5],[3,2,6],[2,2,1]]

Output: 4

Explanation: The longest increasing path is [3, 4, 5, 6]. Moving diagonally is not allowed.

Example 3:

Input: matrix = [[1]]

Output: 1

Constraints:

- $m == \text{matrix.length}$
- $n == \text{matrix}[i].\text{length}$
- $1 \leq m, n \leq 200$

● $0 \leq \text{matrix}[i][j] \leq 231 - 1$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Longest strictly increasing path (4-directional). Return path length. Right?
Values can repeat but path must be strictly increasing."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."
DFS from every cell with memo: longest increasing path starting at (r,c).
Only move to strictly larger neighbors; memo avoids recomputing subpaths.
Without memo, pure DFS retries paths exponentially.
Time: $O(m * n)$ with memo. Space: $O(m * n)$ memo + stack.
Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (DFS + memoization)

"DFS from each cell. $\text{dp}[r][c]$ = longest path starting at (r,c).
Only move to neighbours with strictly greater value – no cycles possible → no visited set needed.
Memoize results: each cell computed once → $O(m*n)$."

PHASE 4 — CLEAN CODE

```
class _329_LongestIncPathMatrix:
    def longestIncreasingPath(self, matrix: List[List[int]]) -> int:
        m, n = len(matrix), len(matrix[0])
        memo = {}
        def dfs(r, c):
            if (r, c) in memo: return memo[(r, c)]
            best = 1
            for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                nr, nc = r + dr, c + dc
                if 0 <= nr < m and 0 <= nc < n and matrix[nr][nc] > matrix[r][c]:
                    best = max(best, dfs(nr, nc) + 1)
            memo[(r, c)] = best
            return best
        return max(dfs(r, c) for r in range(m) for c in range(n))
```

PHASE 5 — TEST & DEBUG

$[[9,9,4],[6,6,8],[2,1,1]]$: longest path starting at (2,1)=1: 1→2→6→9=4 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m*n)$. Space $O(m*n)$ for memo. No visited set needed – strict increase prevents cycles.
Topological sort alternative: build DAG (only edges to larger neighbours), count levels.
Same $O(m*n)$ but iterative."

Given more time I'd ask: longest path with values differing by at most 1?
Need a visited set now (can revisit same value) – different problem."

DFS

□ #685 Redundant Connection II

HAR[Open on LeetCode](#)**DFS · BFS · Union Find · Graph****Lists: Denny Zhang**

Problem

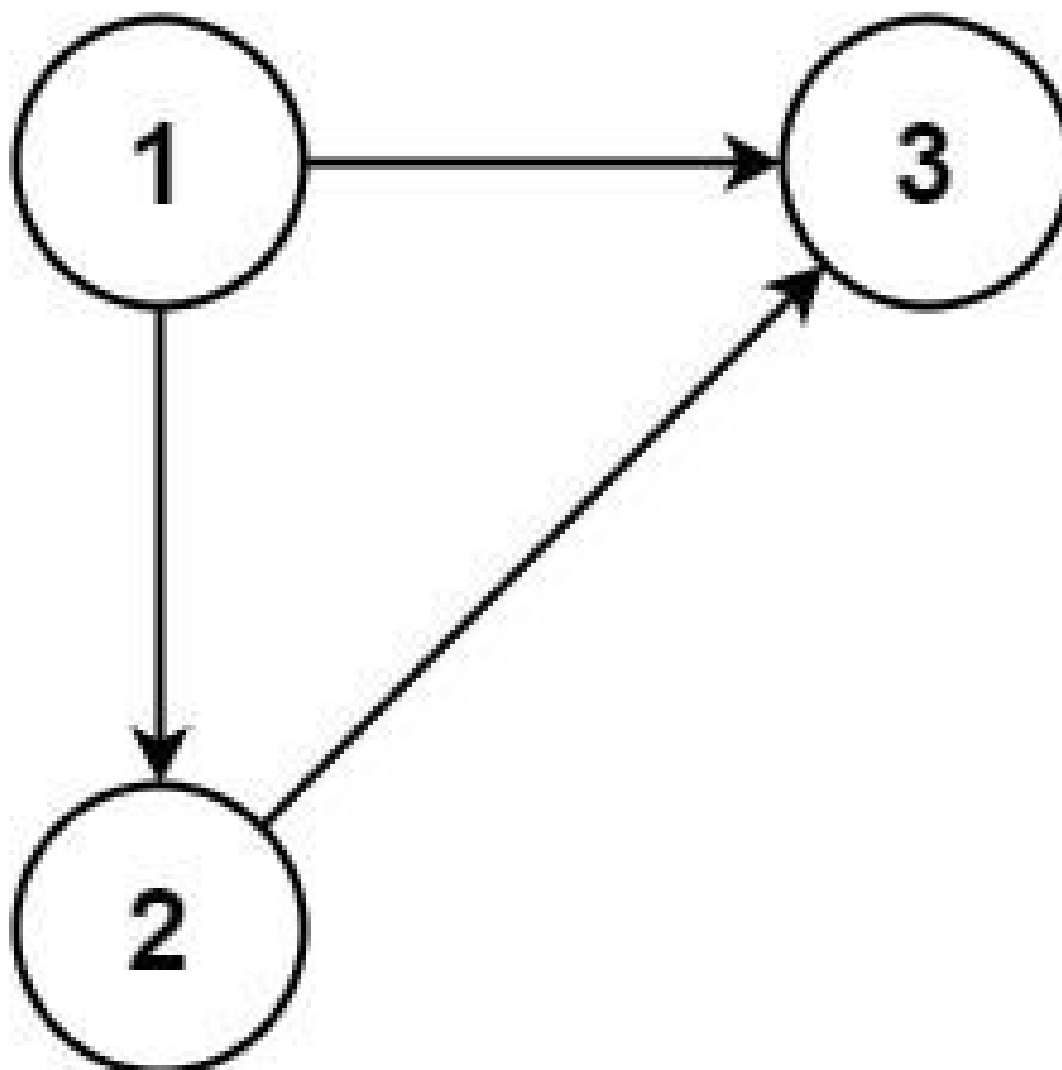
In this problem, a rooted tree is a directed graph such that, there is exactly one node (the root) for which all other nodes are descendants of this node, plus every node has exactly one parent, except for the root node which has no parents.

The given input is a directed graph that started as a rooted tree with n nodes (with distinct values from 1 to n), with one additional directed edge added. The added edge has two different vertices chosen from 1 to n , and was not an edge that already existed.

The resulting graph is given as a 2D-array of edges. Each element of edges is a pair $[u_i, v_i]$ that represents a directed edge connecting nodes u_i and v_i , where u_i is a parent of child v_i .

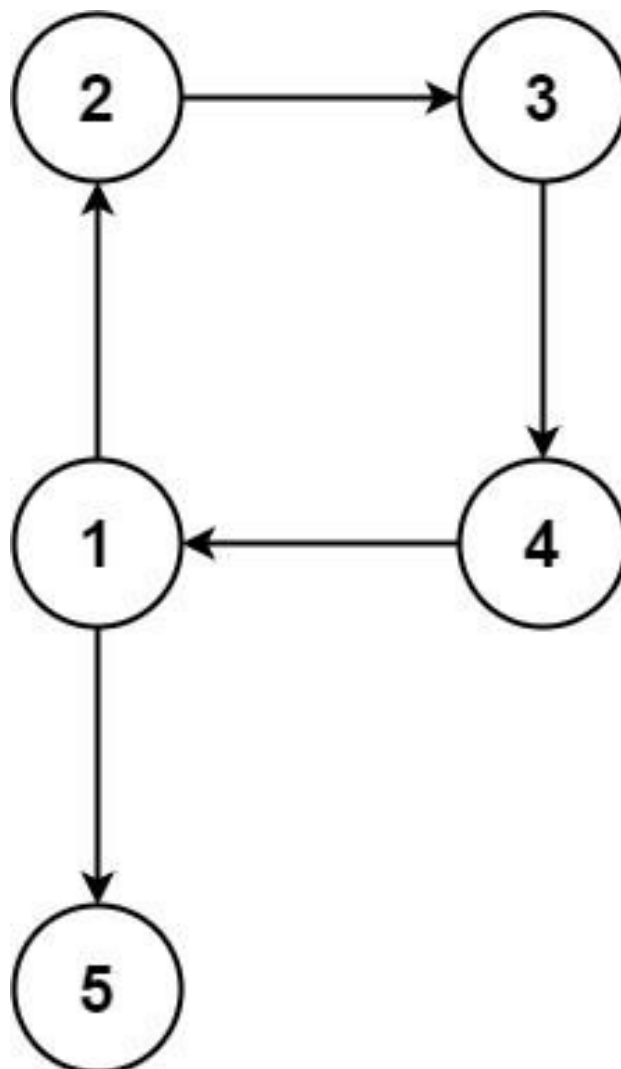
Return an edge that can be removed so that the resulting graph is a rooted tree of n nodes. If there are multiple answers, return the answer that occurs last in the given 2D-array.

Example 1:



Input: edges = `[[1,2],[1,3],[2,3]]`
Output: `[2,3]`

Example 2:



Input: edges = [[1,2],[2,3],[3,4],[4,1],[1,5]]
Output: [4,1]

Constraints:

- $n == \text{edges.length}$
- $3 \leq n \leq 1000$
- $\text{edges}[i].\text{length} == 2$
- $1 \leq u_i, v_i \leq n$
- $u_i \neq v_i$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Directed graph: rooted tree + one extra edge. Find the edge to remove.
 # If multiple answers, return the last one in input. Right?
 # Extra edge causes: (1) a node with two parents, OR (2) a cycle, OR both."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
 # Build indegree + parent map from edges. Node with indegree 2 is the duplicate child candidate.
 # Detect cycle with DFS or union-find on the graph minus one suspicious edge.
 # Try removing each duplicate edge until graph becomes a valid tree.
 # Time: $O(n)$ with careful case analysis. Space: $O(n)$.
 # Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (three-case analysis + Union-Find)

"Track in-degree: if any node has two parents, record both candidate edges.
 # Then run Union-Find ignoring the second parent candidate.
 # Case 1 (two parents, no cycle in remaining): remove second candidate.
 # Case 2 (two parents, cycle in remaining): remove first candidate.
 # Case 3 (no node with two parents): cycle exists → last edge forming cycle."

PHASE 4 — CLEAN CODE

```
class _685_RedundantConnectionII:
    def findRedundantDirectedConnection(self, edges: List[List[int]]) -> List[int]:
        n = len(edges)
        parent = list(range(n + 1))
        cand1 = cand2 = None

        # Find if any node has two parents
        parent_map = {}
        for u, v in edges:
            if v in parent_map:
                cand1 = parent_map[v] # first edge giving v a parent
                cand2 = [u, v] # second edge giving v another parent
                break
            parent_map[v] = [u, v]

        def find(x):
            while parent[x] != x:
                parent[x] = parent[parent[x]]
                x = parent[x]
            return x

        def union(u, v) -> bool:
            pu, pv = find(u), find(v)
            if pu == pv: return False
            parent[pu] = pv
            return True

        parent = list(range(n + 1))
```

```

for u, v in edges:
    if [u, v] == cand2: # skip the second candidate
        continue
    if not union(u, v):
        return cand1 if cand1 else [u, v]
return cand2

```

PHASE 5 — TEST & DEBUG

```

# [[1,2],[1,3],[2,3]]: node 3 has two parents (1 and 2). cand1=[1,3], cand2=[2,3].
# UF ignoring cand2=[2,3]: union(1,2) ok, union(1,3) ok. No cycle → return
cand2=[2,3] ✓
# [[1,2],[2,3],[3,4],[4,1],[1,5]]: no node with two parents. UF finds cycle [4,1] →
return [4,1] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n * \alpha(n))$ . Space  $O(n)$ . Three cases arise from two root causes: double
parent and cycle.
# The Union-Find handles both: cycle detection when cand2 is skipped tells us which
candidate to remove.
# Given more time I'd ask: undirected version?
# That's Redundant Connection (#684) – standard Union-Find, return first edge that
creates a cycle."

```

DFS

□ #1036 Escape a Large Maze

HAR[Open on LeetCode](#)

Array · Hash Table · DFS · BFS

Lists: Denny Zhang

Problem

There is a 1 million by 1 million grid on an XY-plane, and the coordinates of each grid square are (x, y) .

We start at the source = $[sx, sy]$ square and want to reach the target = $[tx, ty]$ square. There is also an array of blocked squares, where each $blocked[i] = [xi, yi]$ represents a blocked square with coordinates (xi, yi) .

Each move, we can walk one square north, east, south, or west if the square is not in the array of blocked squares. We are also not allowed to walk outside of the grid.

Return true if and only if it is possible to reach the target square from the source square through a sequence of valid moves.

Example 1:

Input: blocked = [[0,1],[1,0]], source = [0,0], target = [0,2]

Output: false

Explanation: The target square is inaccessible starting from the source square because we cannot move.

We cannot move north or east because those squares are blocked.

We cannot move south or west because we cannot go outside of the grid.

Example 2:

Input: blocked = [], source = [0,0], target = [999999,999999]

Output: true

Explanation: Because there are no blocked cells, it is possible to reach the target square.

Constraints:

- $0 \leq \text{blocked.length} \leq 200$
- $\text{blocked}[i].\text{length} == 2$
- $0 \leq x_i, y_i < 106$
- $\text{source.length} == \text{target.length} == 2$
- $0 \leq s_x, s_y, t_x, t_y < 106$
- $\text{source} \neq \text{target}$
- It is guaranteed that source and target are not blocked.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"1M×1M grid. Up to 200 blocked cells. Can source reach target? Right?"

With at most 200 blocked cells in a semicircle, max enclosed area $\approx 200^2/2 = 20,000$ cells.

So if BFS from source visits $> 20,000$ cells without finding target, source is NOT enclosed."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

BFS from start cell; at each step try all 4 directions.

Track visited cells; if we revisit within ≤ 500 moves, we are in a loop ? false.

```

# If we escape bounds within 500 steps per problem rule ? true.
# Time:  $O(m * n * 500)$  worst case. Space:  $O(m * n)$  visited.
# Should I go ahead and optimize?"

# PHASE 3 — OPTIMIZE (BFS with cell limit)
# "Run limited BFS from BOTH source and target.
# If BFS visits > max_cells (=  $n*(n+1)/2$  for  $n=\text{len}(\text{blocked})$ ) → not enclosed → True
# from that side.
# If BFS terminates (all cells explored) without reaching other endpoint → enclosed
# → False."

# PHASE 4 — CLEAN CODE
class _1036_EscapeLargeMaze:
    def isEscapePossible(self, blocked: List[List[int]], source: List[int], target:
List[int]) -> bool:
        if not blocked:
            return True
        n = len(blocked)
        max_cells = n * (n + 1) // 2
        blocked_set = {(r, c) for r, c in blocked}
        def bfs(start, end):
            queue = deque([tuple(start)])
            visited = {tuple(start)}
            while queue:
                r, c = queue.popleft()
                if [r, c] == end:
                    return True
                if len(visited) > max_cells:
                    return True # too many cells explored → not enclosed
                for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                    nr, nc = r + dr, c + dc
                    if 0 <= nr < 10**6 and 0 <= nc < 10**6 and (nr,nc) not in
blocked_set and (nr,nc) not in visited:
                        visited.add((nr, nc))
                        queue.append((nr, nc))
            return False
        return bfs(source, target) and bfs(target, source)

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(n^2)$  per BFS (at most  $n*(n+1)/2$  cells). Space  $O(n^2)$ .  $n \leq 200$  → fast.
# The dual-BFS is essential: source might not be enclosed but target might be.
# Given more time I'd ask: what if blocked cells can be added dynamically?
# Re-run BFS on each addition – or maintain Union-Find of open cells."

```

DFS

□ #1192 Critical Connections in a Network **HAR**

[Open on LeetCode](#)

DFS · Graph · Biconnected Component

Lists: Denny Zhang

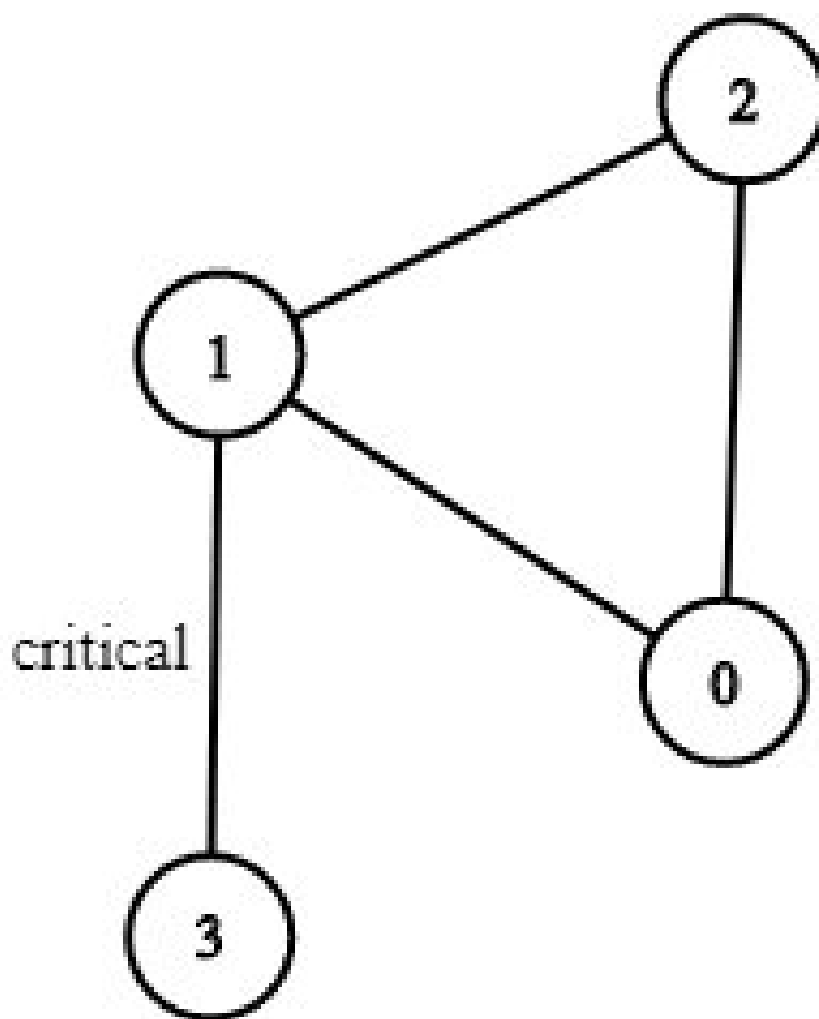
Problem

There are n servers numbered from 0 to $n - 1$ connected by undirected server-to-server connections forming a network where $\text{connections}[i] = [a_i, b_i]$ represents a connection between servers a_i and b_i . Any server can reach other servers directly or indirectly through the network.

A critical connection is a connection that, if removed, will make some servers unable to reach some other server.

Return all critical connections in the network in any order.

Example 1:



Input: $n = 4$, `connections = [[0,1],[1,2],[2,0],[1,3]]`

Output: `[[1,3]]`

Explanation: `[[3,1]]` is also accepted.

Example 2:

Input: $n = 2$, `connections = [[0,1]]`

Output: `[[0,1]]`

Constraints:

- $2 \leq n \leq 105$
- $n - 1 \leq \text{connections.length} \leq 105$
- $0 \leq a_i, b_i \leq n - 1$

- $a_i \neq b_i$
- There are no repeated connections.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Critical connection (bridge) = edge whose removal disconnects the graph. Right?
Use Tarjan's bridge-finding algorithm."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic."
Find bridges with Tarjan low-link DFS: edge (u,v) is critical if $low[v] > disc[u]$.
One DFS pass tracks discovery time and lowest reachable.
Naive alternative: remove each edge and test connectivity – $O(E * (V+E))$.
Time: $O(V + E)$ with Tarjan. Space: $O(V + E)$.
Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (Tarjan's DFS)

"DFS assigning $disc[u]$ =discovery time and $low[u]$ =earliest discovery reachable.
An edge (u,v) is a bridge if $low[v] > disc[u]$ – meaning v cannot reach u's subtree without (u,v)."

PHASE 4 — CLEAN CODE

```
class _1192_CriticalConnections:
    def criticalConnections(self, n: int, connections: List[List[int]]) ->
List[List[int]]:
        graph = defaultdict(list)
        for u, v in connections:
            graph[u].append(v)
            graph[v].append(u)
        disc = [-1] * n
        low = [-1] * n
        timer = [0]
        result = []
        def dfs(node, parent):
            disc[node] = low[node] = timer[0]
            timer[0] += 1
            for nb in graph[node]:
                if nb == parent: continue
                if disc[nb] == -1:
                    dfs(nb, node)
                    low[node] = min(low[node], low[nb])
                    if low[nb] > disc[node]:
                        result.append([node, nb])
            else:
```

```
        low[node] = min(low[node], disc[nb])  
    dfs(0, -1)  
    return result
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(V+E)$. Space $O(V+E)$. Tarjan's bridge algorithm in one DFS pass.

$\text{low}[v] > \text{disc}[u]$ means v 's subtree can't 'back-connect' to u 's subtree – edge is a bridge.

Given more time I'd ask: articulation points instead of bridges?

Similar but: u is an articulation point if any child v has $\text{low}[v] \geq \text{disc}[u]$ (and u is not root)."

Graphs

Round 3 · High · Hard · 2 questions

Graphs

□ ★ #305 Number of Islands II ★

HAR[Open on LeetCode](#)

Array · Hash Table · Union Find · Matrix

Lists: ★ Premium | Premium 98

Problem

You are given an empty 2D binary grid `grid` of size $m \times n$. The grid represents a map where 0's represent water and 1's represent land. Initially, all the cells of grid are water cells (i.e., all the cells are 0's).

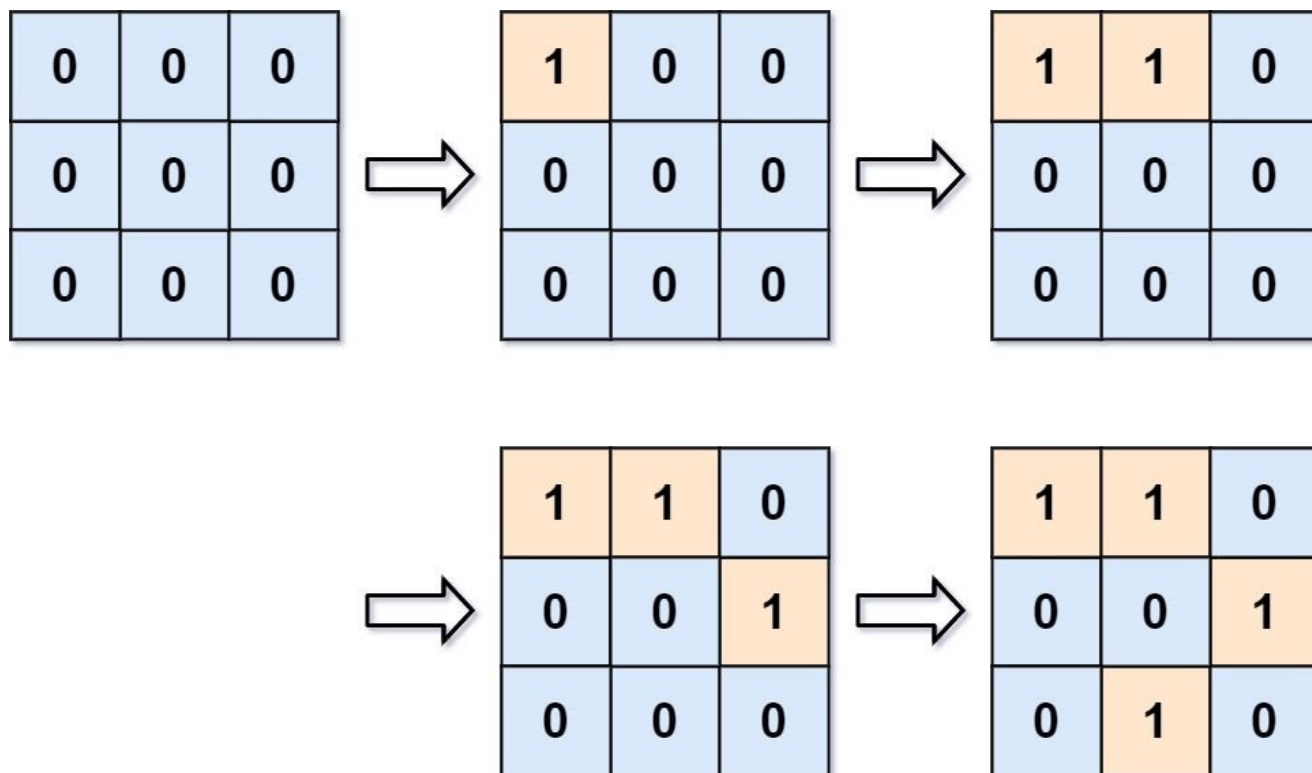
We may perform an add land operation which turns the water at position into a land. You are given an array `positions` where `positions[i] = [ri, ci]` is the position (ri, ci) at which we should operate the i th operation.

Return an array of integers `answer` where `answer[i]` is the number of islands after turning the cell (ri, ci) into a land.

An island is surrounded by water and is formed by connecting adjacent lands horizontally or vertically. You may assume all four edges of the

grid are all surrounded by water.

Example 1:



Input: $m = 3$, $n = 3$, positions = $[[0,0],[0,1],[1,2],[2,1]]$

Output: [1,1,2,3]

Explanation:

Initially, the 2d grid is filled with water.

- Operation #1: addLand(0, 0) turns the water at grid[0][0] into a land. We have 1 island.

- Operation #2: addLand(0, 1) turns the water at grid[0][1] into a land. We still have 1 island.

- Operation #3: addLand(1, 2) turns the water at grid[1][2] into a land. We have 2 islands.

- Operation #4: addLand(2, 1) turns the water at grid[2][1] into a land. We have 3 islands.

Example 2:

Input: $m = 1$, $n = 1$, positions = $[[0,0]]$

Output: [1]

Constraints:

- $1 \leq m, n, \text{positions.length} \leq 104$
- $1 \leq m * n \leq 104$
- $\text{positions}[i].\text{length} == 2$
- $0 \leq r_i < m$
- $0 \leq c_i < n$

Follow up: Could you solve it in time complexity $O(k \log(mn))$, where $k == \text{positions.length}$?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Start with empty grid. Add land cells one at a time. After each add, return island count.

$O(k \log mn)$ per operation. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

After each addLand, scan the whole grid and flood-fill count islands from scratch.

k addLand calls each trigger full $O(m*n)$ recount – too slow when k is large.

Time: $O(k * m * n)$. Space: $O(m * n)$ per flood.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (Union-Find with dynamic updates)

"Each new land cell starts as its own island (count++).

Check 4 neighbours: for each land neighbour, if different component, union and count--."

PHASE 4 — CLEAN CODE

```
class _305_NumberIslandsII:
    def numIslands2(self, m: int, n: int, positions: List[List[int]]) -> List[int]:
        parent = {}
        count = [0]
        def find(x):
            while parent[x] != x:
                parent[x] = parent[parent[x]]
                x = parent[x]
            return x
        def union(a, b):
            ra, rb = find(a), find(b)
            if ra == rb: return
```

```

        parent[ra] = rb
        count[0] -= 1
    result = []
    for r, c in positions:
        if (r, c) in parent:
            result.append(count[0])
            continue
        parent[(r, c)] = (r, c)
        count[0] += 1
        for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
            nr, nc = r + dr, c + dc
            if (nr, nc) in parent:
                union((r, c), (nr, nc))
        result.append(count[0])
    return result

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(k * \alpha(mn)) \approx O(k)$. Space $O(k)$. Union-Find handles dynamic connectivity perfectly.

The duplicate position check avoids double-counting islands.

Given more time I'd ask: what if land cells can be removed too?

Split-Find is hard. Offline deletion via reverse addition is a common trick."

Graphs

□ #1153 String Transforms Into Another String

HAR[Open on LeetCode](#)

Hash Table · String · Union Find

Lists: Denny Zhang

Problem

Given two strings `str1` and `str2` of the same length, determine whether you can transform `str1` into `str2` by doing zero or more conversions.

In one conversion you can convert all occurrences of one character in `str1` to any other lowercase English character.

Return `true` if and only if you can transform `str1` into `str2`.

Example 1:

Input: `str1 = "aabcc", str2 = "ccdee"`

Output: `true`

Explanation: Convert 'c' to 'e' then 'b' to 'd' then 'a' to 'c'. Note that the order of conversions matter.

Example 2:

Input: str1 = "leetcode", str2 = "codeleet"
 Output: false
 Explanation: There is no way to transform str1 to str2.

Constraints:

- $1 \leq \text{str1.length} == \text{str2.length} \leq 104$
- str1 and str2 contain only lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Can str1 be converted to str2 by repeatedly mapping one character to another (all
occurrences)?
# Conversions can be chained but must be consistent. Right?"
#
# "If str1==str2 trivially True. If str2 uses all 26 letters → no 'buffer' character
available → False."
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
# Build a map from each character to its index in the first string.
# Scan the second string left-to-right: next char must appear at a strictly higher
index.
# If any char is missing or out of order, transformation is impossible.
# Time: O(n). Space: O(1) – fixed alphabet size 26.
# Should I go ahead and optimize?"
```

```
class _1153_StringTransforms:
    def canConvert(self, str1: str, str2: str) -> bool:
        if str1 == str2:
            return True
        mapping = {}
        for c1, c2 in zip(str1, str2):
            if mapping.get(c1, c2) != c2:
                return False
            mapping[c1] = c2
        return len(set(str2)) < 26 # need at least one unused char as buffer
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n). Space O(26). The len(set(str2)) < 26 check: if all 26 chars appear in
str2,
# every potential buffer letter is 'locked' to a mapping – cycles can't be broken."
```

```
# Given more time I'd ask: what if conversions can only be applied once?  
# Topological sort on the conversion graph – acyclic → True."
```

BFS

Round 3 · High · Hard · 4 questions

BFS

□ #127 Word Ladder

HAR[Open on LeetCode](#)

Hash Table · String · BFS

Lists: Grind 169

Problem

A transformation sequence from word `beginWord` to word `endWord` using a dictionary `wordList` is a sequence of words `beginWord` $\rightarrow$ `s1` $\rightarrow$ `s2` $\rightarrow$... $\rightarrow$ `sk` such that:

- Every adjacent pair of words differs by a single letter.
- Every `si` for $1 \leq i \leq k$ is in `wordList`. Note that `beginWord` does not need to be in `wordList`.
- `sk == endWord`

Given two words, `beginWord` and `endWord`, and a dictionary `wordList`, return the number of words in the shortest transformation sequence from `beginWord` to `endWord`, or 0 if no such sequence exists.

Example 1:

```
Input: beginWord = "hit", endWord = "cog", wordList =
["hot","dot","dog","lot","log","cog"]
```

```
Output: 5
```

```
Explanation: One shortest transformation sequence is "hit" -> "hot" -> "dot" ->
"dog" -> "cog", which is 5 words long.
```

Example 2:

```
Input: beginWord = "hit", endWord = "cog", wordList =
["hot","dot","dog","lot","log"]
```

```
Output: 0
```

```
Explanation: The endWord "cog" is not in wordList, therefore there is no valid
transformation sequence.
```

Constraints:

- $1 \leq \text{beginWord.length} \leq 10$
- $\text{endWord.length} == \text{beginWord.length}$
- $1 \leq \text{wordList.length} \leq 5000$
- $\text{wordList}[i].\text{length} == \text{beginWord.length}$
- beginWord , endWord , and $\text{wordList}[i]$ consist of lowercase English letters.
- $\text{beginWord} \neq \text{endWord}$
- All the words in wordList are unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Shortest word transformation from beginWord to endWord, changing one letter at a
time.
```

```
# Each intermediate word must be in wordList. Return total words in shortest path,
or 0. Right?"
```

PHASE 2 — BRUTE FORCE

```
# "Let me start with brute force to lock in the logic.
```

```
# Build adjacency from word patterns (one-letter-apart neighbors).
```

```
# BFS from beginWord layer by layer until endWord is reached; each layer = one
transformation.
```

```
# Visited set prevents revisiting words. Return BFS depth or 0 if unreachable.
# Time:  $O(N * L^2)$  where  $N$  = dictionary size,  $L$  = word length. Space:  $O(N)$ .
# Should I go ahead and optimize?"
```

PHASE 3 — OPTIMIZE (BFS with pattern matching)

```
# "BFS on words. For each current word, try replacing each position with 26 letters.
# If the result is in wordSet (and not visited), enqueue it.
# Alternatively: build adjacency by pattern (*ot, h*t, ho*) → group words by pattern
–  $O(L^2 * n)$ ."
```

PHASE 4 — CLEAN CODE

```
class _127_WordLadder:
    def ladderLength(self, beginWord: str, endWord: str, wordList: List[str]) ->
    int:
        word_set = set(wordList)
        if endWord not in word_set:
            return 0
        queue = deque([(beginWord, 1)])
        visited = {beginWord}
        while queue:
            word, steps = queue.popleft()
            for i in range(len(word)):
                for c in 'abcdefghijklmnopqrstuvwxyz':
                    next_word = word[:i] + c + word[i+1:]
                    if next_word == endWord:
                        return steps + 1
                    if next_word in word_set and next_word not in visited:
                        visited.add(next_word)
                        queue.append((next_word, steps + 1))
        return 0
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n * L * 26) = O(n * L)$ . Space  $O(n * L)$ . Standard BFS shortest path.
# Bidirectional BFS: search simultaneously from beginWord and endWord – roughly
halves depth.
# Given more time I'd ask: return all shortest paths?
# That's Word Ladder II (#126) – BFS to build levels, then DFS to backtrack paths."
```

BFS

□ ★ #317 Shortest Distance from All Buildings ★

HAR[Open on LeetCode](#)

Array · BFS · Matrix

Lists: ★ Premium | Premium 98

Problem

You are given an $m \times n$ grid of values 0, 1, or 2, where:

- each 0 marks an empty land that you can pass by freely,
- each 1 marks a building that you cannot pass through, and
- each 2 marks an obstacle that you cannot pass through.

You want to build a house on an empty land that reaches all buildings in the shortest total travel distance. You can only move up, down, left, and right.

Return the shortest travel distance for such a house. If it is not possible to build such a house according to the above rules, return -1.

The total travel distance is the sum of the distances between the houses of the friends and the meeting point.

Example 1:

1	0	2	0	1
0	0	0	0	0
0	0	1	0	0

Input: grid = [[1,0,2,0,1],[0,0,0,0,0],[0,0,1,0,0]]

Output: 7

Explanation: Given three buildings at (0,0), (0,4), (2,2), and an obstacle at (0,2).

The point (1,2) is an ideal empty land to build a house, as the total travel distance of $3+3+1=7$ is minimal.

So return 7.

Example 2:

Input: grid = [[1,0]]

Output: 1

Example 3:

Input: grid = [[1]]
Output: -1

Constraints:

- $m == \text{grid.length}$
- $n == \text{grid}[i].\text{length}$
- $1 \leq m, n \leq 50$
- $\text{grid}[i][j]$ is either 0, 1, or 2.
- There will be at least one building in the grid.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find empty land (0) minimising total BFS distance to all buildings (1).
Obstacles (2) can't be passed. Return -1 if no valid land. Right?"

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.
For each empty cell, BFS to sum distances to all buildings.
Track total distance per cell; answer is cell with minimum total (must reach all buildings).
Repeat BFS from every building - $O(B * m * n)$ where $B = \text{buildings}$.
Time: $O(B * m * n)$. Space: $O(m * n)$ BFS state.
Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (BFS from each building, accumulate distances)

"BFS from each building cell. For each empty cell, accumulate distance and increment reach_count.
Only cells reachable from ALL buildings are candidates. Return min total distance."

PHASE 4 — CLEAN CODE

```
class _317_ShortestDistanceBuildings:
    def shortestDistance(self, grid: List[List[int]]) -> int:
        m, n = len(grid), len(grid[0])
        dist = [[0] * n for _ in range(m)]
        reach = [[0] * n for _ in range(m)]
        buildings = sum(grid[r][c] == 1 for r in range(m) for c in range(n))
```

```

    for r in range(m):
        for c in range(n):
            if grid[r][c] != 1: continue
            queue = deque([(r, c, 0)])
            visited = {(r, c)}
            while queue:
                cr, cc, d = queue.popleft()
                for dr, dc in [(-1,0),(1,0),(0,-1),(0,1)]:
                    nr, nc = cr + dr, cc + dc
                    if 0 <= nr < m and 0 <= nc < n and grid[nr][nc] == 0 and
(nr,nc) not in visited:
                        visited.add((nr, nc))
                        dist[nr][nc] += d + 1
                        reach[nr][nc] += 1
                        queue.append((nr, nc, d + 1))

            return min(
                (dist[r][c] for r in range(m) for c in range(n) if grid[r][c] == 0 and
reach[r][c] == buildings),
                default=-1
            )

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(\text{buildings} * m * n)$. Space $O(m*n)$. BFS from each building accumulates into `dist[]`.

`reach[]` ensures we only pick land reachable from ALL buildings.

Given more time I'd ask: what if buildings can move? Re-run BFS on each query."

BFS

□ #773 Sliding Puzzle

HAR[Open on LeetCode](#)

Array · BFS · Matrix

Lists: Denny Zhang

Problem

On an 2×3 board, there are five tiles labeled from 1 to 5, and an empty square represented by 0. A move consists of choosing 0 and a 4-directionally adjacent number and swapping it.

The state of the board is solved if and only if the board is $[[1,2,3],[4,5,0]]$.

Given the puzzle board `board`, return the least number of moves required so that the state of the board is solved. If it is impossible for the state of the board to be solved, return -1 .

Example 1:

1	2	3
4		5

Input: board = [[1,2,3],[4,0,5]]

Output: 1

Explanation: Swap the 0 and the 5 in one move.

Example 2:

1	2	3
5	4	

Input: board = [[1,2,3],[5,4,0]]
Output: -1
Explanation: No number of moves will make the board solved.

Example 3:

4	1	2
5		3

Input: board = [[4,1,2],[5,0,3]]

Output: 5

Explanation: 5 is the smallest number of moves that solves the board.

An example path:

After move 0: [[4,1,2],[5,0,3]]

After move 1: [[4,1,2],[0,5,3]]

After move 2: [[0,1,2],[4,5,3]]

After move 3: [[1,0,2],[4,5,3]]

Constraints:

- board.length == 2
- board[i].length == 3
- $0 \leq \text{board}[i][j] \leq 5$
- Each value board[i][j] is unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"2x3 board. Swap 0 with adjacent tile. Minimum moves to reach [[1,2,3],[4,5,0]]. Right?

Return -1 if unsolvable."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

BFS on board states: encode 2x3 board as tuple string; enqueue all legal swaps of '0'.

First time we reach target state, return move count.

State space is at most 6! = 720 permutations.

Time: O(6!). Space: O(6!) visited states.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (BFS on serialised board state)

"Represent board as a 6-char string. BFS on states.

Precompute which positions in the flat string can swap with position i (adjacency map).

Goal state: '123450'."

PHASE 4 — CLEAN CODE

```
class _773_SlidingPuzzle:
    def slidingPuzzle(self, board: List[List[int]]) -> int:
        GOAL = '123450'
        NEIGHBORS = {0:[1,3], 1:[0,2,4], 2:[1,5], 3:[0,4], 4:[1,3,5], 5:[2,4]}
        start = ''.join(str(board[r][c]) for r in range(2) for c in range(3))
        queue = deque([(start, 0)])
        visited = {start}
        while queue:
            state, moves = queue.popleft()
            if state == GOAL:
                return moves
            z = state.index('0')
            for nb in NEIGHBORS[z]:
                new_state = list(state)
                new_state[z], new_state[nb] = new_state[nb], new_state[z]
                ns = ''.join(new_state)
                if ns not in visited:
                    visited.add(ns)
                    queue.append((ns, moves + 1))
        return -1
```

PHASE 5 — TEST & DEBUG

[[1,2,3],[4,0,5]] → '123405'. Swap 0 at pos 4 with 5 at pos 5 → '123450' = GOAL → 1 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"States = 6! / 2 = 360 reachable states (half of 6! are unsolvable). BFS on 360 states.

```
# The hardcoded NEIGHBORS map is the key – it captures the 2D adjacency in 1D string indexing.  
# Given more time I'd ask: generalise to n×m board?  
# Same BFS approach – NEIGHBORS changes, state space = (n×m)! / 2."
```


BFS

□ #815 Bus Routes

HAR[Open on LeetCode](#)

Array · Hash Table · BFS

Lists: Grind 169

Problem

You are given an array `routes` representing bus routes where `routes[i]` is a bus route that the *i*th bus repeats forever.

- For example, if `routes[0] = [1, 5, 7]`, this means that the 0th bus travels in the sequence `1 -> 5 -> 7 -> 1 -> 5 -> 7 -> 1 -> ...` forever.

You will start at the bus stop `source` (You are not on any bus initially), and you want to go to the bus stop `target`. You can travel between bus stops by buses only.

Return the least number of buses you must take to travel from `source` to `target`. Return `-1` if it is not possible.

Example 1:

Input: routes = [[1,2,7],[3,6,7]], source = 1, target = 6

Output: 2

Explanation: The best strategy is take the first bus to the bus stop 7, then take the second bus to the bus stop 6.

Example 2:

Input: routes = [[7,12],[4,5,15],[6],[15,19],[9,12,13]], source = 15, target = 12

Output: -1

Constraints:

- $1 \leq \text{routes.length} \leq 500$.
- $1 \leq \text{routes}[i].\text{length} \leq 105$
- All the values of routes[i] are unique.
- $\text{sum}(\text{routes}[i].\text{length}) \leq 105$
- $0 \leq \text{routes}[i][j] < 106$
- $0 \leq \text{source}, \text{target} < 106$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Minimum number of BUSES (not stops) to travel from source to target. Right?

Each bus covers its entire route – stepping onto a bus travels all its stops."

PHASE 2 — BRUTE FORCE

"Let me start with brute force to lock in the logic.

Build stop->routes map. BFS from source stop; each hop can board any route containing current stop.

Track min buses to reach target; mark routes visited to avoid reboarding.

Time: $O(R * S)$ routes and stops. Space: $O(R + S)$.

Should I go ahead and optimize?"

PHASE 3 — OPTIMIZE (BFS on routes, not stops)

"Build stop->routes map. BFS where each 'node' is a bus route.

Start: all routes containing source. Each step: board a new route (adjacent if sharing a stop).

Count route changes (buses taken). Avoid revisiting routes and stops."

PHASE 4 — CLEAN CODE

```

class _815_BusRoutes:
    def numBusesToDestination(self, routes: List[List[int]], source: int, target:
int) -> int:
        if source == target:
            return 0
        stop_to_routes: defaultdict = defaultdict(set)
        for i, route in enumerate(routes):
            for stop in route:
                stop_to_routes[stop].add(i)
        visited_stops = {source}
        visited_routes = set()
        queue = deque([(source, 0)]) # (stop, buses_taken)
        while queue:
            stop, buses = queue.popleft()
            for route_id in stop_to_routes[stop]:
                if route_id in visited_routes: continue
                visited_routes.add(route_id)
                for next_stop in routes[route_id]:
                    if next_stop == target:
                        return buses + 1
                    if next_stop not in visited_stops:
                        visited_stops.add(next_stop)
                        queue.append((next_stop, buses + 1))
        return -1

```

PHASE 5 — TEST & DEBUG

```

# routes=[[1,2,7],[3,6,7]], source=1, target=6:
# queue=[(1,0)]. route 0 for stop 1: stops 2,7. Not target. queue=[(2,1),(7,1)].
# stop 7: route 1 for stop 7: stops 3,6. 6==target -> return 1+1=2 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(sum of route lengths). Space O(same). BFS on routes, not stops – the key
insight.
# Visiting stops prevents re-enqueuing; visiting routes prevents re-processing
entire routes.
# Given more time I'd ask: find all paths using at most k buses?
# Keep track of bus count per path – BFS naturally handles this."

```

Round 4 · Mid · Easy

Round 4

Mid Priority · Easy

12 questions · 4 patterns

Linked List #21 #141 #206 #234 #706 #876
#1474

Stack #20 #232 #844

Trie #14

Greedy #409

Linked List

Round 4 · Mid · Easy · 7 questions

Linked List

□ #21 Merge Two Sorted Lists

EAS

[Open on LeetCode](#)

Linked List · Recursion

Lists: Grind 169

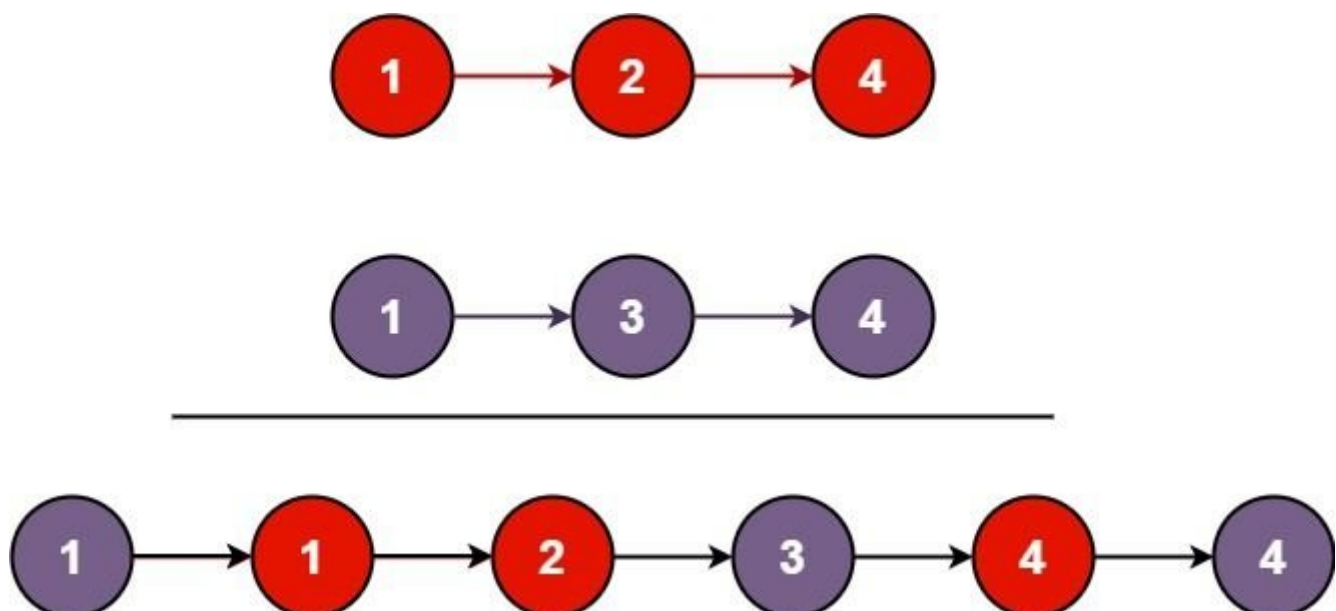
Problem

You are given the heads of two sorted linked lists list1 and list2.

Merge the two lists into one sorted list. The list should be made by splicing together the nodes of the first two lists.

Return the head of the merged linked list.

Example 1:



Input: list1 = [1,2,4], list2 = [1,3,4]
 Output: [1,1,2,3,4,4]

Example 2:

Input: list1 = [], list2 = []
 Output: []

Example 3:

Input: list1 = [], list2 = [0]
 Output: [0]

Constraints:

- The number of nodes in both lists is in the range [0, 50].
- $-100 \leq \text{Node.val} \leq 100$
- Both list1 and list2 are sorted in non-decreasing order.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "We're given heads of two sorted linked lists. Merge them into one sorted
# list by splicing the original nodes – no new nodes. Return the new head. Right?"
#
# "list1=[1,2,4], list2=[1,3,4]:
# Compare 1 vs 1 → take list1's 1. 2 vs 1 → take list2's 1. 2 vs 3 → take 2.
# 4 vs 3 → take 3. 4 vs 4 → take list1's 4. Append list2's 4. → [1,1,2,3,4,4] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Collect all values, sort, rebuild a new list.
# Time:  $O((m+n) \log(m+n))$ . Space:  $O(m+n)$  for the values array."
```

```
class _21_MergeTwoSortedLists_Brute:
    def mergeTwoLists(self, list1, list2):
        vals = []
        while list1: vals.append(list1.val); list1 = list1.next
        while list2: vals.append(list2.val); list2 = list2.next
        dummy = cur = ListNode(0)
        for v in sorted(vals):
```

```

        cur.next = ListNode(v); cur = cur.next
    return dummy.next

```

PHASE 3 — OPTIMIZE

```

# "Both lists are already sorted – no sorting needed. Use two pointers.
# A dummy head simplifies edge cases (empty lists, different lengths).
# At each step attach the smaller node; advance that pointer.
# When one list runs out, attach the rest of the other.
#
# Time: O(m+n). Space: O(1) – only pointer manipulation."

```

PHASE 4 — CLEAN CODE

```

class _21_MergeTwoSortedLists:
    def mergeTwoLists(self, list1, list2):
        dummy = cur = ListNode(0)
        while list1 and list2:
            if list1.val <= list2.val:
                cur.next = list1
                list1 = list1.next
            else:
                cur.next = list2
                list2 = list2.next
            cur = cur.next
        cur.next = list1 if list1 else list2
        return dummy.next

```

PHASE 5 — TEST & DEBUG

```

# list1=[1,2,4], list2=[1,3,4]:
# 1<=1 → take l1(1), l1→2. 2>1 → take l2(1), l2→3. 2<=3 → take l1(2), l1→4.
# 4>3 → take l2(3), l2→4. 4<=4 → take l1(4), l1=None.
# Attach l2 remainder [4]. → [1,1,2,3,4,4] ✓
#
# list1=[], list2=[0]:
# Loop skipped. cur.next = list2 → [0] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(m+n): visit every node once. Space O(1): reuses existing nodes.
# Recursive solution is O(m+n) time but O(m+n) stack space.
# Follow-up: merge k sorted lists (LC 23) – use a min-heap for O(n log k).
# Follow-up: merge k sorted arrays – same heap approach."

```


Linked List

□ #141 Linked List Cycle

EAS

[Open on LeetCode](#)

Hash Table · Linked List · Two Pointers

Lists: Grind 169

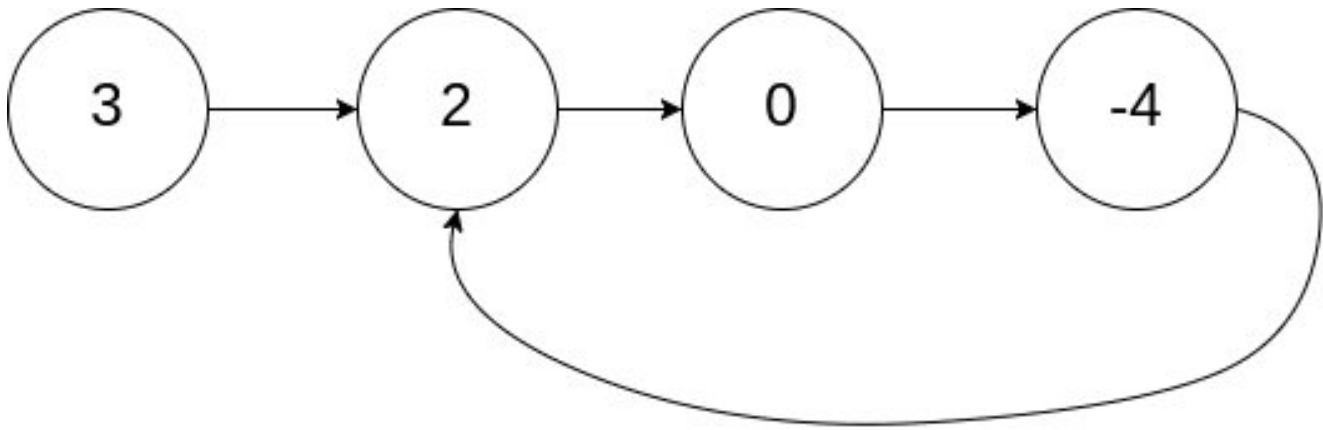
Problem

Given head, the head of a linked list, determine if the linked list has a cycle in it.

There is a cycle in a linked list if there is some node in the list that can be reached again by continuously following the next pointer.

Internally, pos is used to denote the index of the node that tail's next pointer is connected to. Note that pos is not passed as a parameter. Return true if there is a cycle in the linked list. Otherwise, return false.

Example 1:

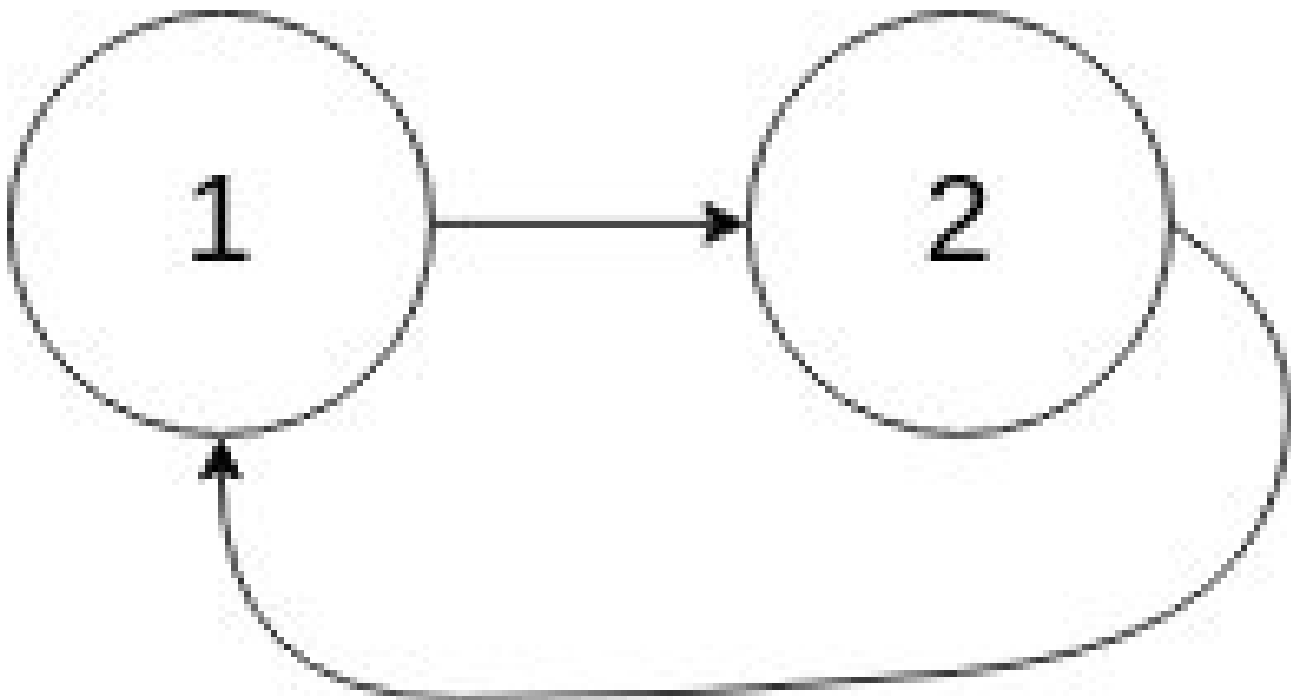


Input: head = [3,2,0,-4], pos = 1

Output: true

Explanation: There is a cycle in the linked list, where the tail connects to the 1st node (0-indexed).

Example 2:

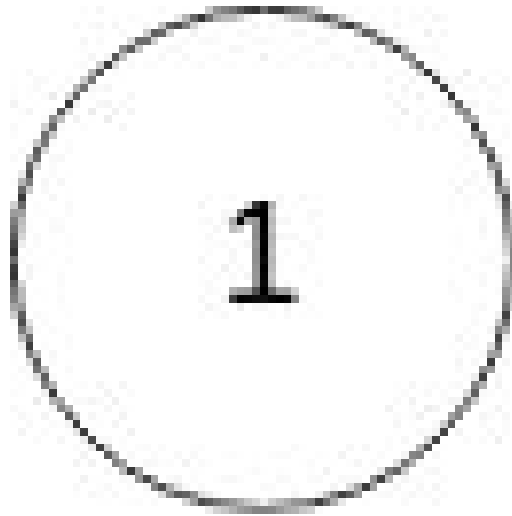


Input: head = [1,2], pos = 0

Output: true

Explanation: There is a cycle in the linked list, where the tail connects to the 0th node.

Example 3:



Input: head = [1], pos = -1

Output: false

Explanation: There is no cycle in the linked list.

Constraints:

- The number of the nodes in the list is in the range [0, 104].
- $-105 \leq \text{Node.val} \leq 105$
- pos is -1 or a valid index in the linked-list.

Follow up: Can you solve it using $O(1)$ (i.e. constant) memory?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given the head of a linked list, return true if there's a cycle –
# if any node's next eventually points back to a previous node. Right?"
#
# "head=[3,2,0,-4], pos=1 (tail connects to index 1):
# 3→2→0→-4→2→... cycle exists → true ✓
# head=[1,2], pos=-1: 1→2→None → false ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Store every visited node in a set. If we see a node twice, there's a cycle.
# If we reach None, there isn't."
```

Time: $O(n)$. Space: $O(n)$ for the visited set."

```
class _141_LinkedListCycle_Brute:
    def hasCycle(self, head) -> bool:
        seen = set()
        while head:
            if id(head) in seen:
                return True
            seen.add(id(head))
            head = head.next
        return False
```

PHASE 3 — OPTIMIZE

"Floyd's tortoise-and-hare: two pointers, slow moves 1 step, fast moves 2.

If there's a cycle, fast will eventually lap slow and they'll meet.

If fast reaches None, there's no cycle.

#

Key insight: in a cycle, fast gains 1 step per iteration → guaranteed to catch slow.

Time: $O(n)$. Space: $O(1)$ – no extra storage."

PHASE 4 — CLEAN CODE

```
class _141_LinkedListCycle:
    def hasCycle(self, head) -> bool:
        slow = fast = head
        while fast and fast.next:
            slow = slow.next
            fast = fast.next.next
            if slow is fast:
                return True
        return False
```

PHASE 5 — TEST & DEBUG

[3,2,0,-4] cycle at index 1:

slow=3,fast=3 → slow=2,fast=0 → slow=0,fast=2 → slow=-4,fast=-4 → equal → True ✓

#

[1,2] no cycle:

slow=1,fast=1 → slow=2,fast=None → fast is None → loop exits → False ✓

#

[1] no cycle:

fast.next is None → loop never enters → False ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: fast pointer covers the cycle in at most n steps after reaching it.

Space $O(1)$: only two pointers.

Follow-up: find the START of the cycle (LC 142) – after they meet, reset slow

to head; advance both one step at a time until they meet again → cycle entry.

Follow-up: find cycle length – once they meet, keep fast moving and count steps."

Linked List

□ #206 Reverse Linked List

EAS

[Open on LeetCode](#)

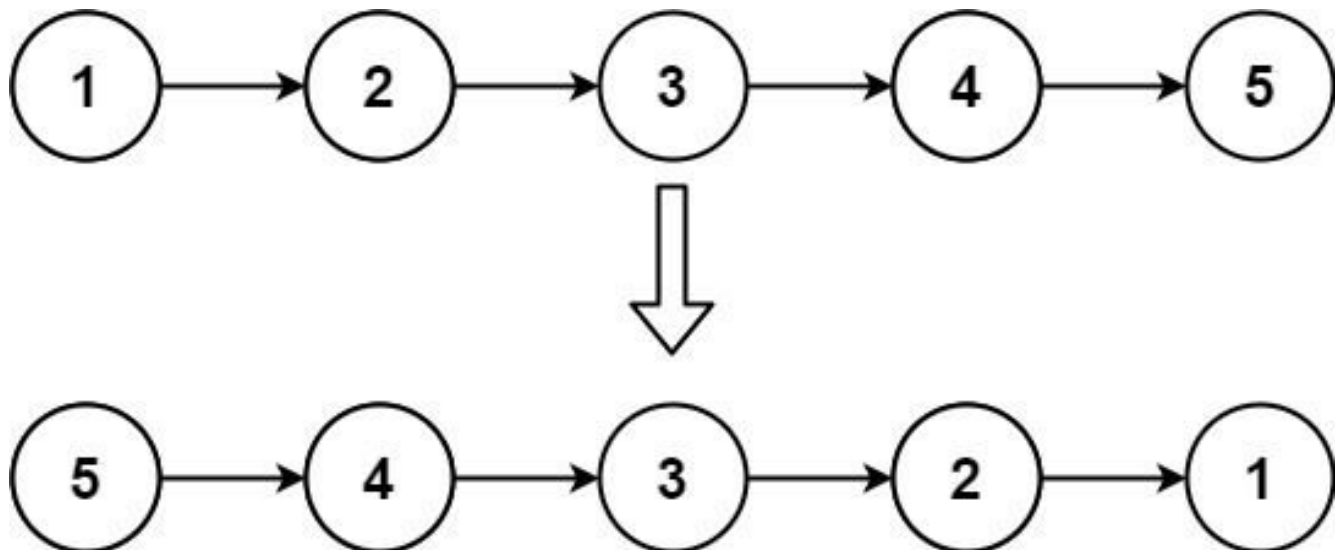
Linked List · Recursion

Lists: Grind 169

Problem

Given the head of a singly linked list, reverse the list, and return the reversed list.

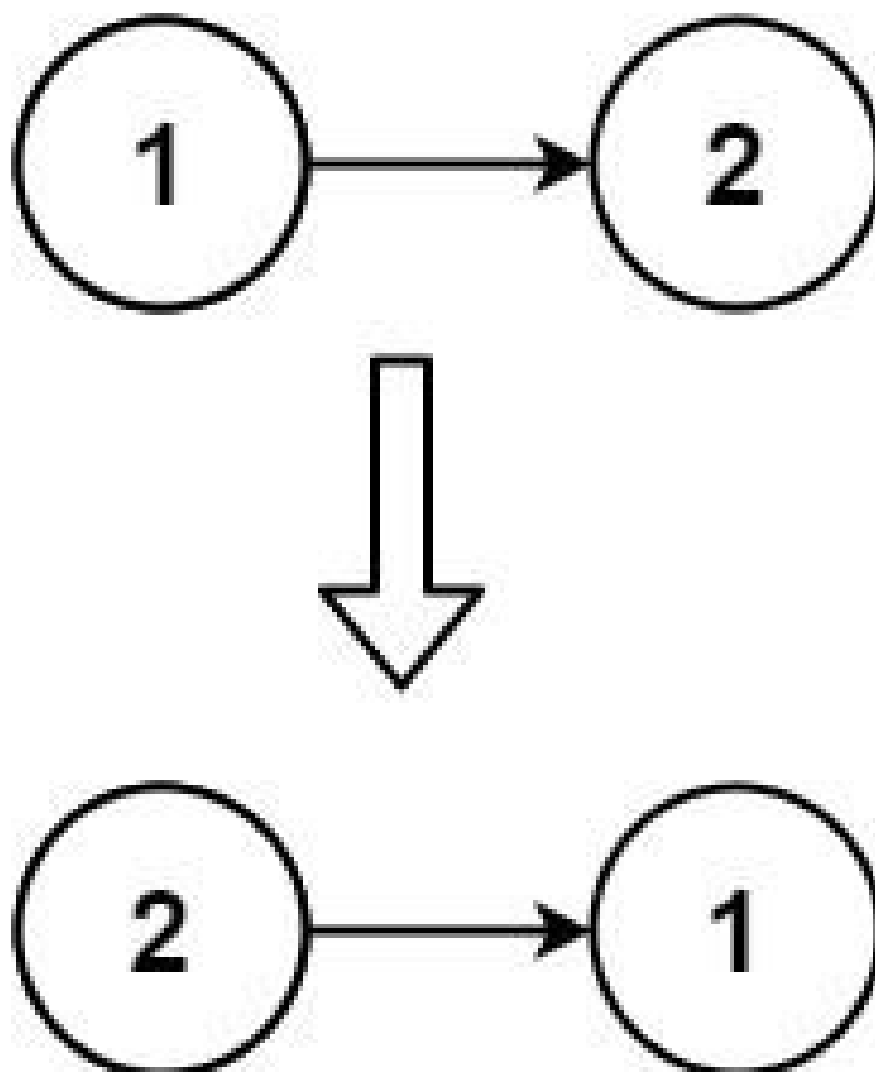
Example 1:



Input: head = [1,2,3,4,5]

Output: [5,4,3,2,1]

Example 2:



Input: head = [1,2]

Output: [2,1]

Example 3:

Input: head = []

Output: []

Constraints:

- The number of nodes in the list is the range [0, 5000].
- $-5000 \leq \text{Node.val} \leq 5000$

Follow up: A linked list can be reversed either iteratively or recursively. Could you implement both?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given the head of a singly linked list, reverse it in-place and return
# the new head. Right?"
#
# "head=[1,2,3,4,5]:
# 1→2→3→4→5→None becomes None←1←2←3←4←5, new head = 5 ✓
# head=[]: return None ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Collect all values in an array, rebuild a new reversed list.
# Time: O(n). Space: O(n) for the values."
```

```
class _206_ReverseLinkedList_Brute:
    def reverseList(self, head):
        vals = []
        while head: vals.append(head.val); head = head.next
        dummy = cur = ListNode(0)
        for v in reversed(vals):
            cur.next = ListNode(v); cur = cur.next
        return dummy.next
```

PHASE 3 — OPTIMIZE

```
# "Three-pointer iterative approach: prev, curr, next_node.
# At each step: save curr.next, point curr.next to prev, advance both pointers.
# When curr is None, prev is the new head.
#
# Time: O(n). Space: O(1) – in-place pointer rewiring."
```

PHASE 4 — CLEAN CODE

```
class _206_ReverseLinkedList:
    def reverseList(self, head):
        prev, curr = None, head
        while curr:
            nxt = curr.next
            curr.next = prev
            prev = curr
            curr = nxt
        return prev
```

PHASE 5 — TEST & DEBUG

```
# head=[1,2,3]:
# prev=None,curr=1: nxt=2, 1.next=None, prev=1, curr=2
# prev=1,curr=2: nxt=3, 2.next=1, prev=2, curr=3
# prev=2,curr=3: nxt=None, 3.next=2, prev=3, curr=None
# return 3 → [3,2,1] ✓
#
# head=[]: curr=None immediately → return None ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(n)$ : one pass. Space  $O(1)$ : constant extra pointers.
# Recursive solution:  $O(n)$  time,  $O(n)$  stack space – elegant but risky for large
inputs.
# Follow-up: reverse nodes in k-group (LC 25) – applies this same logic per group.
# Follow-up: reverse between positions left and right (LC 92) – isolate the sublist
first."
```


Linked List

□ #234 Palindrome Linked List

EAS[Open on LeetCode](#)

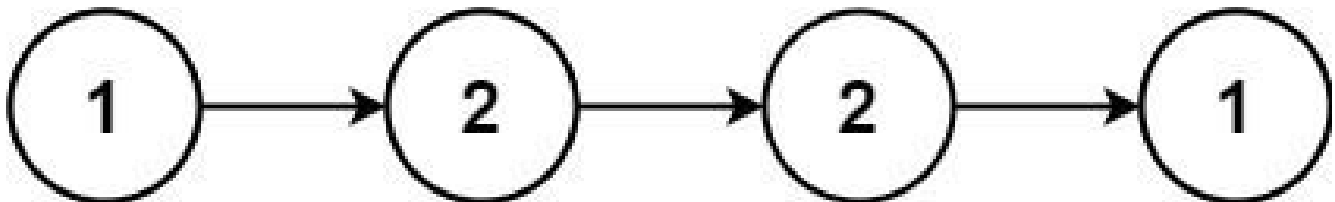
Linked List · Two Pointers · Recursion

Lists: Grind 169

Problem

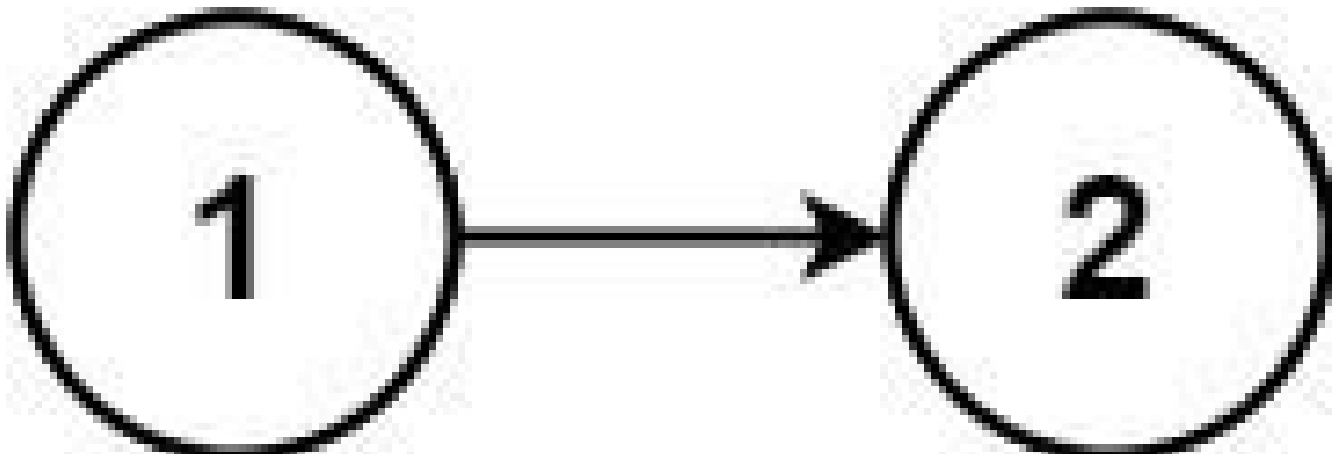
Given the head of a singly linked list, return true if it is a palindrome or false otherwise.

Example 1:



Input: head = [1,2,2,1]
Output: true

Example 2:



Input: head = [1,2]
Output: false

Constraints:

- The number of nodes in the list is in the range [1, 105].
- $0 \leq \text{Node.val} \leq 9$

Follow up: Could you do it in $O(n)$ time and $O(1)$ space?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given the head of a singly linked list, return true if it reads the same
# forwards and backwards. Right?"
#
# "head=[1,2,2,1]: reads same both ways → true ✓
# head=[1,2]: 1≠2 → false ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Copy all values into an array, then use two-pointer comparison.
# Time:  $O(n)$ . Space:  $O(n)$ ."
```

```
class _234_PalindromeLinkedList_Brute:
    def isPalindrome(self, head) -> bool:
        vals = []
        while head: vals.append(head.val); head = head.next
        return vals == vals[::-1]
```

PHASE 3 — OPTIMIZE

```
# "Achieve  $O(1)$  space by modifying the list in-place:
# 1. Find the midpoint using slow/fast pointers.
# 2. Reverse the second half in-place.
# 3. Compare first half with reversed second half.
# 4. (Optional) restore the list.
#
# Time:  $O(n)$ . Space:  $O(1)$ ."
```

PHASE 4 — CLEAN CODE

```
class _234_PalindromeLinkedList:
    def isPalindrome(self, head) -> bool:
        # Step 1: find middle
```

```

slow = fast = head
while fast and fast.next:
    slow = slow.next
    fast = fast.next.next
# Step 2: reverse second half
prev, curr = None, slow
while curr:
    nxt = curr.next
    curr.next = prev
    prev = curr
    curr = nxt
# Step 3: compare halves
left, right = head, prev
while right:
    if left.val != right.val:
        return False
    left = left.next
    right = right.next
return True

```

PHASE 5 — TEST & DEBUG

```

# [1,2,2,1]:
# Mid: slow=2(idx1), fast=1(idx3). Reverse [2,1]→[1,2].
# Compare: 1==1 ✓, 2==2 ✓, right=None → True ✓
#
# [1,2]:
# Mid: slow=2, fast=None after 1 step. Reverse [2]→[2].
# Compare: 1≠2 → False ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n): three linear passes (find mid, reverse, compare).
# Space O(1): constant extra pointers.
# Trade-off: O(n) space approach is simpler and non-destructive.
# Follow-up: restore the original list after the check by re-reversing second half.
# Follow-up: palindrome check on a doubly linked list is trivial – two outer
pointers."

```

Linked List

□ #706 Design HashMap

EAS

[Open on LeetCode](#)

Array · Hash Table · Linked List · Design · Hash Function

Lists: Denny Zhang

Problem

Design a HashMap without using any built-in hash table libraries.

Implement the MyHashMap class:

- MyHashMap() initializes the object with an empty map.
- void put(int key, int value) inserts a (key, value) pair into the HashMap. If the key already exists in the map, update the corresponding value.
- int get(int key) returns the value to which the specified key is mapped, or -1 if this map contains no mapping for the key.
- void remove(key) removes the key and its corresponding value if the map contains the mapping for the key.

Example 1:

Input

```
["MyHashMap", "put", "put", "get", "get", "put", "get", "remove", "get"]
[[], [1, 1], [2, 2], [1], [3], [2, 1], [2], [2], [2]]
```

Output

```
[null, null, null, 1, -1, null, 1, null, -1]
```

Explanation

```
MyHashMap myHashMap = new MyHashMap();
```

Constraints:

- $0 \leq \text{key}, \text{value} \leq 10^6$
- At most 104 calls will be made to put, get, and remove.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Implement a HashMap without built-in hash libraries. Support put(key, val),
# get(key) → -1 if missing, remove(key). Keys and values are non-negative ints
# in range [0, 10^6]. Right?"
#
# "put(1,1), put(2,2), get(1)→1, get(3)→-1, put(2,1), get(2)→1, remove(2),
# get(2)→-1 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Store all pairs in a plain list. Scan linearly for get/put/remove.
# O(n) per operation – correct but slow."
```

```
class _706_DesignHashMap_Brute:
    def __init__(self):
        self.data = [] # list of [key, value] pairs
    def put(self, key: int, value: int) -> None:
        for pair in self.data:
            if pair[0] == key:
                pair[1] = value
                return
        self.data.append([key, value])
    def get(self, key: int) -> int:
        for pair in self.data:
            if pair[0] == key:
                return pair[1]
```

```

        return -1

    def remove(self, key: int) -> None:
        self.data = [p for p in self.data if p[0] != key]

# PHASE 3 — OPTIMIZE
# "Use an array of buckets with separate chaining (linked list / small list per
bucket).
# Hash function: key % num_buckets using a prime to spread keys.
# Average O(1) per op when load factor is low. Space O(n + k)."

# PHASE 4 — CLEAN CODE
class _706_DesignHashMap:
    def __init__(self):
        self.size = 1009
        self.buckets = [[] for _ in range(self.size)]

    def _h(self, key: int) -> int:
        return key % self.size

    def put(self, key: int, value: int) -> None:
        bucket = self.buckets[self._h(key)]
        for i, (k, _) in enumerate(bucket):
            if k == key:
                bucket[i] = (key, value)
                return
        bucket.append((key, value))

    def get(self, key: int) -> int:
        for k, v in self.buckets[self._h(key)]:
            if k == key:
                return v
        return -1

    def remove(self, key: int) -> None:
        h = self._h(key)
        self.buckets[h] = [(k, v) for k, v in self.buckets[h] if k != key]

# PHASE 5 — TEST & DEBUG
# put(1,1): bucket[1]=[(1,1)]
# put(2,2): bucket[2]=[(2,2)]
# get(1): bucket[1] → (1,1) → 1 ✓
# get(3): bucket[3]=[] → -1 ✓
# put(2,1): bucket[2] → update → [(2,1)]
# get(2): → 1 ✓
# remove(2): bucket[2]=[]
# get(2): [] → -1 ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Brute: O(n) per op. Hash bucket: O(1) average, O(n) worst (all same bucket).
# Space O(n+k): n stored pairs + k bucket slots.
# Follow-up: Design HashSet (LC 705) – same idea, keys only.
# Follow-up: open addressing (linear probing) avoids linked lists but needs
tombstones."

```

Linked List

□ #876 Middle of the Linked List

EAS

[Open on LeetCode](#)

Linked List · Two Pointers

Lists: Grind 169

Problem

Given the head of a singly linked list, return the middle node of the linked list.

If there are two middle nodes, return the second middle node.

Example 1:

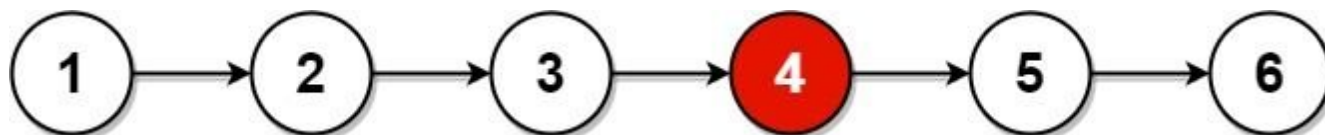


Input: head = [1,2,3,4,5]

Output: [3,4,5]

Explanation: The middle node of the list is node 3.

Example 2:



Input: head = [1,2,3,4,5,6]

Output: [4,5,6]

Explanation: Since the list has two middle nodes with values 3 and 4, we return the second one.

Constraints:

- The number of nodes in the list is in the range [1, 100].
- $1 \leq \text{Node.val} \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given the head of a linked list, return the middle node.

If two middles exist (even length), return the SECOND one. Right?"

#

"head=[1,2,3,4,5]: length=5, middle index=2 → node(3) ✓

head=[1,2,3,4,5,6]: length=6, two middles at index 2,3 → return node(4) ✓"

PHASE 2 — BRUTE FORCE

"Count the total number of nodes, then walk to index length//2.

Two passes: one to count, one to reach the middle.

Time: O(n). Space: O(1)."

```
class _876_MiddleOfLinkedList_Brute:
```

```
    def middleNode(self, head):
```

```
        length = 0
```

```
        curr = head
```

```
        while curr:
```

```
            length += 1
```

```
            curr = curr.next
```

```
        curr = head
```

```
        for _ in range(length // 2):
```

```
            curr = curr.next
```

```
        return curr
```

PHASE 3 — OPTIMIZE

"Slow/fast pointer — one pass.

Slow moves 1 step, fast moves 2. When fast reaches the end, slow is at the middle.

For even-length lists, slow naturally lands on the SECOND middle (desired).

#

Time: O(n). Space: O(1). Saves one full traversal vs brute."

PHASE 4 — CLEAN CODE


```
class _876_MiddleOfLinkedList:
    def middleNode(self, head):
        slow = fast = head
        while fast and fast.next:
            slow = slow.next
            fast = fast.next.next
        return slow
```

PHASE 5 — TEST & DEBUG

```
# [1,2,3,4,5]:
# s=1,f=1 → s=2,f=3 → s=3,f=5 → fast.next=None → stop. return node(3) ✓
#
# [1,2,3,4,5,6]:
# s=1,f=1 → s=2,f=3 → s=3,f=5 → s=4,f=None(6.next) → stop. return node(4) ✓
#
# [1]:
# fast.next=None immediately → return head=node(1) ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ : fast pointer covers the list in  $n/2$  iterations.
# Space  $O(1)$ : two pointers only.
# Follow-up: if you need the FIRST middle for even-length, stop when fast.next.next
is None.
# Follow-up: this slow/fast trick is the building block for palindrome check (LC
234)
# and cycle detection (LC 141)."
```

Linked List

□ ★ #1474 Delete N Nodes After M Nodes of Linked List ★

EAS

[Open on LeetCode](#)

Linked List

Lists: ★ Premium | Premium 98

Problem

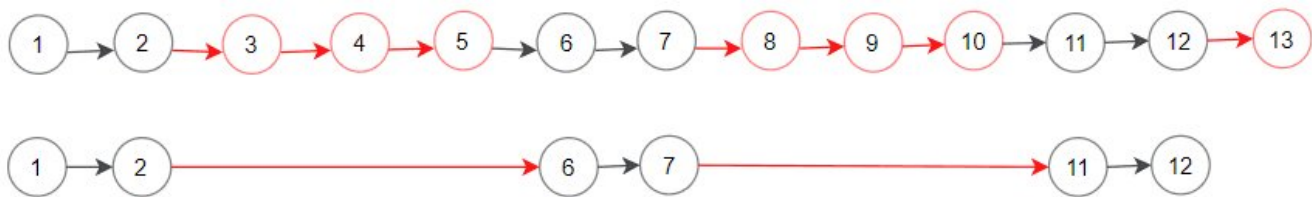
You are given the head of a linked list and two integers m and n .

Traverse the linked list and remove some nodes in the following way:

- Start with the head as the current node.
- Keep the first m nodes starting with the current node.
- Remove the next n nodes
- Keep repeating steps 2 and 3 until you reach the end of the list.

Return the head of the modified list after removing the mentioned nodes.

Example 1:



Input: head = [1,2,3,4,5,6,7,8,9,10,11,12,13], m = 2, n = 3

Output: [1,2,6,7,11,12]

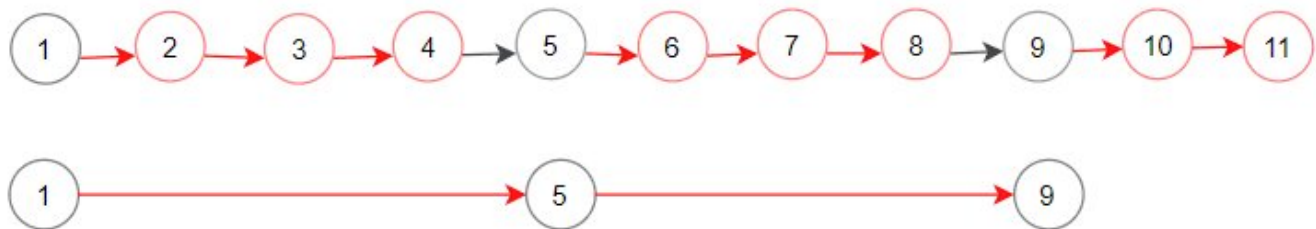
Explanation: Keep the first (m = 2) nodes starting from the head of the linked List (1 -> 2) show in black nodes.

Delete the next (n = 3) nodes (3 -> 4 -> 5) show in red nodes.

Continue with the same procedure until reaching the tail of the Linked List.

Head of the linked list after removing nodes is returned.

Example 2:



Input: head = [1,2,3,4,5,6,7,8,9,10,11], m = 1, n = 3

Output: [1,5,9]

Explanation: Head of linked list after removing nodes is returned.

Constraints:

- The number of nodes in the list is in the range [1, 10⁴].
- 1 ≤ Node.val ≤ 10⁶
- 1 ≤ m, n ≤ 1000

Follow up: Could you solve this problem by modifying the list in-place?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a linked list and integers m and n: keep m nodes, delete next n nodes,
# repeat until the end. Return the modified head. Right?"
#
# "head=[1,2,3,4,5,6,7,8,9,10,11,12,13], m=2, n=3:
# Keep 1,2 → delete 3,4,5 → keep 6,7 → delete 8,9,10 → keep 11,12 → delete 13
# → [1,2,6,7,11,12] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Enumerate all nodes with their index. Keep node i if (i % (m+n)) < m.
# Collect kept nodes and relink them.
# Time: O(n_total). Space: O(n_total) for the kept-nodes list."
```

```
class _1474_DeleteNNodesAfterMNodes_Brute:
    def deleteNodes(self, head, m: int, n: int):
        kept = []
        curr = head
        i = 0
        while curr:
            if i % (m + n) < m:
                kept.append(curr)
                curr = curr.next
                i += 1
        for j in range(len(kept) - 1):
            kept[j].next = kept[j + 1]
        if kept:
            kept[-1].next = None
        return kept[0] if kept else None
```

PHASE 3 — OPTIMIZE

```
# "Pointer-based in-place: walk m steps forward (keeping nodes), then
# walk n steps forward (skipping/deleting nodes), then repeat.
# Relink the tail of the kept section directly to the node after the deleted
# section.
# Time: O(n_total). Space: O(1) – no extra storage."
```

PHASE 4 — CLEAN CODE

```
class _1474_DeleteNNodesAfterMNodes:
    def deleteNodes(self, head, m: int, n: int):
        curr = head
        while curr:
            # Walk m-1 steps to reach end of kept section
            for _ in range(m - 1):
                if curr: curr = curr.next
            if not curr:
                break
            tail = curr # last kept node
            # Walk n steps to skip deleted section
            skip = curr.next
            for _ in range(n):
                if skip: skip = skip.next
```

```
tail.next = skip # relink
curr = skip # continue from next kept section
return head
```

PHASE 5 — TEST & DEBUG

```
# [1,2,3,4,5,6,7,8,9,10,11,12,13], m=2, n=3:
# curr=1, walk 1 step → curr=2 (tail=2). skip=3, walk 3 → skip=6. 2.next=6, curr=6.
# curr=6, walk 1 step → curr=7 (tail=7). skip=8, walk 3 → skip=11. 7.next=11,
curr=11.
# curr=11, walk 1 step → curr=12 (tail=12). skip=13, walk 3 → skip=None.
12.next=None.
# → [1,2,6,7,11,12] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(total nodes): each node visited exactly once.
# Space O(1): only pointer variables.
# Follow-up: what if m and n can vary per cycle? Would need to pass arrays.
# Follow-up: similar pattern to 'Remove Duplicates from Sorted List II' in pointer
reuse."
```

Stack

Round 4 · Mid · Easy · 3 questions

Stack

□ #20 Valid Parentheses

EAS

[Open on LeetCode](#)

String · Stack

Lists: Grind 169

Problem

Given a string `s` containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid.

An input string is valid if:

1. Open brackets must be closed by the same type of brackets.
2. Open brackets must be closed in the correct order.
3. Every close bracket has a corresponding open bracket of the same type.

Example 1:

Input: `s = "()"`

Output: `true`

Example 2:

Input: `s = "()[]{}"`

Output: true

Example 3:

Input: s = "())"

Output: false

Example 4:

Input: s = "([])"

Output: true

Example 5:

Input: s = "([)]"

Output: false

Constraints:

- $1 \leq s.length \leq 104$
- s consists of parentheses only '()[]{}'.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a string of brackets, return true if every open bracket is closed
# by the same type in the correct order. Right?"
#
# "s=())[{}]: each pair matches in order → true ✓
# s="()": '(' expects ')' but gets ']' → false ✓
# s="([])": '(' then '[', but ')' closes '(' while '[' is still open → false ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Repeatedly replace matching pairs '()', '[]', '{} ' until nothing changes.
# If the string is empty at the end it's valid.
# Time:  $O(n^2)$  – up to  $n/2$  passes each scanning n chars. Space:  $O(n)$ ."
```

```
class _20_ValidParentheses_Brute:
    def isValid(self, s: str) -> bool:
        while '()' in s or '[]' in s or '{}' in s:
            s = s.replace('()', '').replace('[]', '').replace('{}', '')
```



```
return s == ''
```

PHASE 3 — OPTIMIZE

```
# "Use a stack. Push every opening bracket.
# On a closing bracket, the top of the stack must be its matching opener.
# If not – or the stack is empty – return false.
# At the end, the stack must be empty.
#
# Time: O(n) – one pass. Space: O(n) for the stack."
```

PHASE 4 — CLEAN CODE

```
class _20_ValidParentheses:
    def isValid(self, s: str) -> bool:
        stack = []
        match = {')': '(', ']': '[', '}': '{'}
        for ch in s:
            if ch in match:
                if not stack or stack[-1] != match[ch]:
                    return False
                stack.pop()
            else:
                stack.append(ch)
        return not stack
```

PHASE 5 — TEST & DEBUG

```
# s="()[]{}":
# '(' push. ')' → top='(' matches → pop. '[' push. ']' → pop. '{' push. '}' → pop.
# stack empty → True ✓
#
# s="()":
# '(' push. ']' → top='(' ≠ '[' → return False ✓
#
# s="([])":
# '(' push. '[' push. ')' → top='[' ≠ '(' → return False ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): one pass through the string.
# Space O(n): worst case all openers e.g. '(((((' .
# Follow-up: minimum removals to make valid (LC 1249) extends this with counters.
# Follow-up: what if we also have wildcards '*' that can be '(', ')', or '?' (LC 678)."
```

Stack

□ #232 Implement Queue using Stacks

EAS

[Open on LeetCode](#)

Stack · Design · Queue

Lists: Grind 169

Problem

Implement a first in first out (FIFO) queue using only two stacks. The implemented queue should support all the functions of a normal queue (push, peek, pop, and empty).

Implement the MyQueue class:

- **void push(int x)** Pushes element x to the back of the queue.
- **int pop()** Removes the element from the front of the queue and returns it.
- **int peek()** Returns the element at the front of the queue.
- **boolean empty()** Returns true if the queue is empty, false otherwise.

Notes:

- You must use only standard operations of a stack, which means only push to top, peek/pop from top, size, and is empty operations are valid.
- Depending on your language, the stack may not be supported natively. You may simulate a stack using a list or deque (double-ended queue) as long as you use only a stack's standard operations.

Example 1:

Input

```
["MyQueue", "push", "push", "peek", "pop", "empty"]
```

```
[[], [1], [2], [], [], []]
```

Output

```
[null, null, null, 1, 1, false]
```

Explanation

```
MyQueue myQueue = new MyQueue();
```

Constraints:

- $1 \leq x \leq 9$
- At most 100 calls will be made to push, pop, peek, and empty.
- All the calls to pop and peek are valid.

Follow-up: Can you implement the queue such that each operation is amortized $O(1)$ time complexity? In other words, performing n operations will take overall $O(n)$ time even if one of those operations may take longer.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "We need a queue (FIFO) built only from stacks (LIFO). Support push(x), pop(),
# peek(), and empty(). Amortized O(1) per operation is acceptable. Right?"
#
# "push(1), push(2), peek() → 1, pop() → 1, empty() → false
# The first element pushed must be first out – opposite of a stack."
```

PHASE 2 — BRUTE FORCE

```
# "Use one stack. On every push, move ALL elements out, push the new one to bottom,
# then put everything back. O(n) per push, O(1) pop/peek – queue order maintained."
```

```
class _232_ImplementQueueUsingStacks_Brute:
    def __init__(self):
        self.stack = []

    def push(self, x: int) -> None:
        # Move everything out, add x at bottom, put back
        tmp = []
        while self.stack:
            tmp.append(self.stack.pop())
        self.stack.append(x)
        while tmp:
            self.stack.append(tmp.pop())

    def pop(self) -> int:
        return self.stack.pop()

    def peek(self) -> int:
        return self.stack[-1]

    def empty(self) -> bool:
        return not self.stack
```

PHASE 3 — OPTIMIZE

```
# "Two stacks – inbox and outbox – achieve amortized O(1).
# Push always goes to inbox.
# For pop/peek: if outbox is empty, pour ALL of inbox into outbox (reverses order).
# Then pop/peek from outbox.
# Each element moves at most twice total across both stacks."
```

PHASE 4 — CLEAN CODE

```
class _232_ImplementQueueUsingStacks:
    def __init__(self):
        self.inbox = []
        self.outbox = []

    def push(self, x: int) -> None:
        self.inbox.append(x)

    def _pour(self) -> None:
```

```
        if not self.outbox:
            while self.inbox:
                self.outbox.append(self.inbox.pop())
def pop(self) -> int:
    self._pour()
    return self.outbox.pop()
def peek(self) -> int:
    self._pour()
    return self.outbox[-1]
def empty(self) -> bool:
    return not self.inbox and not self.outbox
```

PHASE 5 — TEST & DEBUG

```
# push(1): inbox=[1]
# push(2): inbox=[1,2]
# peek(): outbox empty → pour → inbox=[], outbox=[2,1]. outbox[-1]=1 ✓
# pop(): outbox=[2,1] → pop → 1, outbox=[2] ✓
# empty(): inbox=[], outbox=[2] → False ✓
# pop(): outbox=[2] → pop → 2 ✓
# empty(): both empty → True ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Brute:  $O(n)$  push,  $O(1)$  pop. Two-stack: amortized  $O(1)$  all ops.
# Each element transferred at most once – total work  $O(n)$  for  $n$  ops.
# Space  $O(n)$  across both stacks.
# Follow-up: implement stack using queues (LC 225) – symmetric design.
# Follow-up: if we needed worst-case  $O(1)$  all ops, we'd need a deque."
```

Stack

□ #844 Backspace String Compare

EAS

[Open on LeetCode](#)

Two Pointers · String · Stack · Simulation

Lists: Grind 169

Problem

Given two strings *s* and *t*, return true if they are equal when both are typed into empty text editors. '#' means a backspace character.

Note that after backspacing an empty text, the text will continue empty.

Example 1:

Input: *s* = "ab#c", *t* = "ad#c"
Output: true
Explanation: Both *s* and *t* become "ac".

Example 2:

Input: *s* = "ab##", *t* = "c#d#"
Output: true
Explanation: Both *s* and *t* become "".

Example 3:

Input: *s* = "a#c", *t* = "b"
Output: false
Explanation: *s* becomes "c" while *t* becomes "b".

Constraints:

- $1 \leq s.length, t.length \leq 200$
- s and t only contain lowercase letters and '#' characters.

Follow up: Can you solve it in $O(n)$ time and $O(1)$ space?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given two strings where '#' represents a backspace, return true if they're
# equal after processing. Right?"
#
# "s="ab#c", t="ad#c":
# s: a,b,# → a, c → "ac". t: a,d,# → a, c → "ac". Equal → true ✓
# s="a#c", t="b": s→"c", t→"b". Not equal → false ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Simulate using a stack: push normal chars, pop on '#' if stack non-empty.
# Compare final stacks.
# Time:  $O(n+m)$ . Space:  $O(n+m)$  for the stacks."
```

```
class _844_BackspaceStringCompare_Brute:
    def backspaceCompare(self, s: str, t: str) -> bool:
        def process(string):
            stack = []
            for ch in string:
                if ch == '#':
                    if stack: stack.pop()
                else:
                    stack.append(ch)
            return stack
        return process(s) == process(t)
```

PHASE 3 — OPTIMIZE

```
# "Two-pointer approach from the RIGHT —  $O(1)$  space.
# Traverse both strings backwards, counting '#' as skip tokens.
# Skip that many non-'#' characters before comparing.
# If both pointers exhausted simultaneously → equal.
#
# Time:  $O(n+m)$ . Space:  $O(1)$ ."
```

PHASE 4 — CLEAN CODE

```
class _844_BackspaceStringCompare:
```

```
def backspaceCompare(self, s: str, t: str) -> bool:
    i, j = len(s) - 1, len(t) - 1
    skip_s = skip_t = 0
    while i >= 0 or j >= 0:
        while i >= 0:
            if s[i] == '#': skip_s += 1; i -= 1
            elif skip_s: skip_s -= 1; i -= 1
            else: break
        while j >= 0:
            if t[j] == '#': skip_t += 1; j -= 1
            elif skip_t: skip_t -= 1; j -= 1
            else: break
        if i >= 0 and j >= 0 and s[i] != t[j]:
            return False
        if (i >= 0) != (j >= 0):
            return False
        i -= 1; j -= 1
    return True
```

PHASE 5 — TEST & DEBUG

```
# s="ab#c", t="ad#c":
# i=3'c',j=3'c' → both non-# → c==c ✓. i=2,j=2.
# i=2'#'→skip_s=1,i=1. s[1]='b',skip_s→0,i=0.
# j=2'#'→skip_t=1,j=1. t[1]='d',skip_t→0,j=0.
# s[0]='a', t[0]='a' → equal ✓. i=-1,j=-1 → return True ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Stack approach: O(n+m) time, O(n+m) space – simple and readable.
# Two-pointer approach: O(n+m) time, O(1) space – optimal.
# Follow-up: if '#' can backspace multiple characters (like '2#'), adjust skip
counter.
# Follow-up: streaming input – the stack approach handles it naturally."
```


Trie

Round 4 · Mid · Easy · 1 question

Trie

□ #14 Longest Common Prefix

EAS

[Open on LeetCode](#)

String · Trie

Lists: Grind 169

Problem

Write a function to find the longest common prefix string amongst an array of strings.

If there is no common prefix, return an empty string "".

Example 1:

```
Input: strs = ["flower","flow","flight"]  
Output: "fl"
```

Example 2:

```
Input: strs = ["dog","racecar","car"]  
Output: ""  
Explanation: There is no common prefix among the input strings.
```

Constraints:

- $1 \leq \text{strs.length} \leq 200$
 - $0 \leq \text{strs}[i].\text{length} \leq 200$
 - $\text{strs}[i]$ consists of only lowercase English letters if it is non-empty.
-

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "We're given an array of strings and need to find the longest prefix
# shared by ALL of them. If none exists, return an empty string. Right?"
#
# "strs=["flower","flow","flight"]:"
# Compare column by column: f✓ l✓ o vs o vs i → mismatch at index 2 → return "fl" ✓
# strs=["dog","racecar","car"]: first chars d vs r → mismatch → return "" ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Sort the array, then only compare the first and last strings –
# the LCP of those two is the LCP of the whole array.
# Time:  $O(n \log n + m)$  where  $m$  is shortest string length. Space:  $O(1)$ ."
```

```
class _14_LongestCommonPrefix_Brute:
```

```
    def longestCommonPrefix(self, strs: list[str]) -> str:
        if not strs: return ""
        strs.sort()
        first, last = strs[0], strs[-1]
        i = 0
        while i < len(first) and i < len(last) and first[i] == last[i]:
            i += 1
        return first[:i]
```

PHASE 3 — OPTIMIZE

```
# "Vertical scanning avoids the sort entirely.
# Take the first string as the reference. For each character position i,
# check that every other string has the same character at position i.
# Stop as soon as any string is too short or has a mismatch.
#
# Pseudocode:
# for i, char in enumerate(strs[0]):
# for s in strs[1:]:
# if i >= len(s) or s[i] != char: return strs[0][:i]
# return strs[0]
#
# Time:  $O(n*m)$  –  $n$  strings,  $m$  = min length. Space:  $O(1)$ ."
```

PHASE 4 — CLEAN CODE

```
class _14_LongestCommonPrefix:
```

```
    def longestCommonPrefix(self, strs: list[str]) -> str:
        if not strs:
            return ""
        for i, char in enumerate(strs[0]):
            for s in strs[1:]:
                if i >= len(s) or s[i] != char:
                    return strs[0][:i]
        return strs[0]
```

PHASE 5 — TEST & DEBUG

```
# strs=["flower","flow","flight"]:
```

```
# i=0 'f': all match. i=1 'l': all match. i=2 'o': strs[2][2]='i' ≠ 'o' → return "fl" ✓
```

```
#
```

```
# strs=["dog","racecar","car"]:
```

```
# i=0 'd': strs[1][0]='r' ≠ 'd' → return "" ✓
```

```
#
```

```
# strs=["abc"]:
```

```
# No inner loop fires → return "abc" ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n*m)$ : every char of shortest string compared across all n strings.
```

```
# Space  $O(1)$ : no extra allocation beyond the output slice.
```

```
# Follow-up: if the array is huge but strings are short, binary search on
```

```
# prefix length works —  $O(m \log m * n)$  but better when  $m \ll n$ .
```

```
# Follow-up: Trie would let you insert all strings and walk until a branch point."
```

Greedy

Round 4 · Mid · Easy · 1 question

Greedy

□ #409 Longest Palindrome

EAS

[Open on LeetCode](#)

Hash Table · String · Greedy

Lists: Grind 169

Problem

Given a string *s* which consists of lowercase or uppercase letters, return the length of the longest palindrome that can be built with those letters.

Letters are case sensitive, for example, "Aa" is not considered a palindrome.

Example 1:

Input: *s* = "abcccd"

Output: 7

Explanation: One longest palindrome that can be built is "dccaccd", whose length is 7.

Example 2:

Input: *s* = "a"

Output: 1

Explanation: The longest palindrome that can be built is "a", whose length is 1.

Constraints:

- $1 \leq s.length \leq 2000$

- s consists of lowercase and/or uppercase English letters only.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a string of letters, find the length of the longest palindrome
# we can BUILD from those characters (not a substring – we rearrange them). Right?"
#
# "s="abcccd":
# d×2, c×4, a×1, b×1. Pairs: dd, cccc → length 6. One leftover can go in center → 7
✓
# s="a": just 'a' → length 1 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Count frequency of each character. A palindrome uses pairs symmetrically
# plus at most one center character.
# Sum all (freq // 2) * 2. If any character had an odd count, add 1 for center."
class _409_LongestPalindrome_Brute:
    def longestPalindrome(self, s: str) -> int:
        from collections import Counter
        freq = Counter(s)
        length = 0
        has_odd = False
        for count in freq.values():
            length += (count // 2) * 2
            if count % 2 == 1:
                has_odd = True
        return length + (1 if has_odd else 0)
```

PHASE 3 — OPTIMIZE

```
# "Same greedy logic, can use a set instead of a counter:
# Toggle each character in a set. At the end, characters still in the set
# appeared an odd number of times. Answer = len(s) - len(odd_chars) + (1 if
odd_chars else 0).
#
# Time: O(n). Space: O(1) – at most 52 distinct letters."
```

PHASE 4 — CLEAN CODE

```
class _409_LongestPalindrome:
    def longestPalindrome(self, s: str) -> int:
        odd_chars = set()
        for ch in s:
            if ch in odd_chars:
                odd_chars.remove(ch)
            else:
```

```

        odd_chars.add(ch)
    return len(s) - len(odd_chars) + (1 if odd_chars else 0)

```

PHASE 5 — TEST & DEBUG

```

# s="abcccccdd":
# a→{a}, b→{a,b}, c→{a,b,c}, c→{a,b}, c→{a,b,c}, c→{a,b}, d→{a,b,d}, d→{a,b}
# odd_chars={a,b}, len(s)=8, 8-2+1=7 ✓
#
# s="a":
# odd_chars={a}, 1-1+1=1 ✓
#
# s="aa":
# odd_chars={}, 2-0+0=2 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n): one pass. Space O(1): at most 52 entries in the set.
# The greedy insight: use every pair, reserve at most one odd character for center.
# Follow-up: if you need to OUTPUT the actual palindrome, assign pairs to both ends
# and place one odd character in the middle."

```


Round 5 · Mid · Medium

Round 5

Mid Priority · Medium

56 questions · 6 patterns

**Linked List #2 #19 #24 #61 #146 #148 #328
#369 #430 #622 #707 #708 #1171**

**Stack #143 #150 #155 #173 #227 #255 #394
#439 #484 #735 #739 #962 #1214 #1762**

**Heap #215 #253 #347 #505 #621 #658 #787
#973 #1057 #1167 #1424**

Trie #139 #208 #211 #616 #692 #720 #1166

Backtracking #17 #22 #39 #46 #79 #113

Greedy #134 #179 #280 #954 #2131

Linked List

Round 5 · Mid · Medium · 13 questions

Linked List

#2 Add Two Numbers

MED[Open on LeetCode](#)

Linked List · Math · Recursion

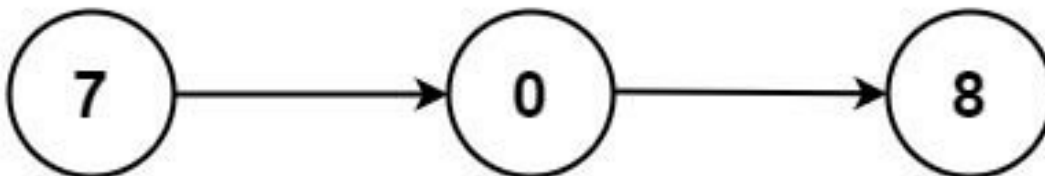
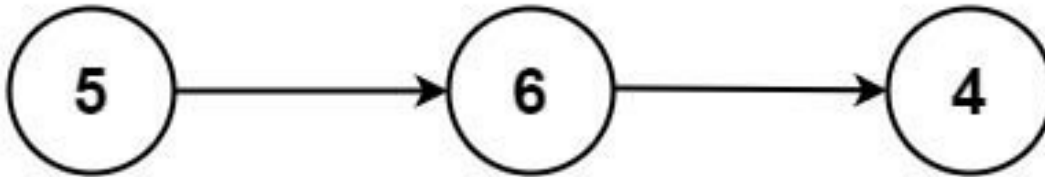
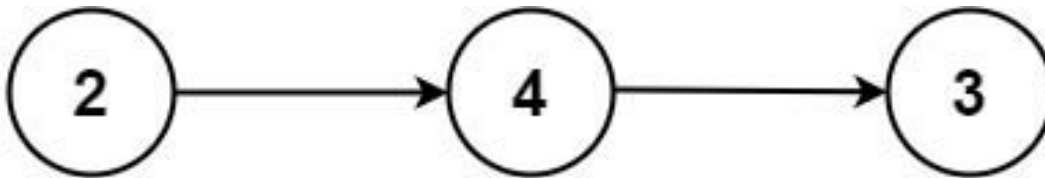
Lists: Grind 169

Problem

You are given two non-empty linked lists representing two non-negative integers. The digits are stored in reverse order, and each of their nodes contains a single digit. Add the two numbers and return the sum as a linked list.

You may assume the two numbers do not contain any leading zero, except the number 0 itself.

Example 1:



Input: l1 = [2,4,3], l2 = [5,6,4]

Output: [7,0,8]

Explanation: 342 + 465 = 807.

Example 2:

Input: l1 = [0], l2 = [0]

Output: [0]

Example 3:

Input: l1 = [9,9,9,9,9,9,9], l2 = [9,9,9,9]

Output: [8,9,9,9,0,0,0,1]

Constraints:

- The number of nodes in each linked list is in the range [1, 100].

- $0 \leq \text{Node.val} \leq 9$
- It is guaranteed that the list represents a number that does not have leading zeros.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Two non-empty linked lists store digits of non-negative integers in REVERSE order."

Add the two numbers and return the sum as a reversed linked list. Right?"

#

"l1=[2,4,3] (342), l2=[5,6,4] (465): 342+465=807 → [7,0,8] ✓

l1=[0], l2=[0]: 0+0=0 → [0] ✓

l1=[9,9,9,9], l2=[9,9,9]: 9999+999=10998 → [8,9,9,0,1] ✓ (carry propagation)"

PHASE 2 — BRUTE FORCE

"Convert both lists to integers, add, then convert the result back to a linked list."

Careful: the digits are stored in reverse, so the head is the least significant digit.

Time: $O(\max(m,n))$. Space: $O(\max(m,n))$ for the result."

```
class _2_AddTwoNumbers_Brute:
```

```
    def addTwoNumbers(self, l1, l2):
```

```
        def to_int(node):
```

```
            num, mul = 0, 1
```

```
            while node:
```

```
                num += node.val * mul
```

```
                mul *= 10
```

```
                node = node.next
```

```
            return num
```

```
        total = to_int(l1) + to_int(l2)
```

```
        if total == 0:
```

```
            return ListNode(0)
```

```
        dummy = cur = ListNode(0)
```

```
        while total:
```

```
            cur.next = ListNode(total % 10)
```

```
            cur = cur.next
```

```
            total //= 10
```

```
        return dummy.next
```

PHASE 3 — OPTIMIZE

"Simulate grade-school addition digit by digit with a carry."

Walk both lists simultaneously; at each position sum the digits + carry.

new_digit = total % 10, new_carry = total // 10.

Continue until both lists exhausted AND carry is 0.

Time: $O(\max(m,n))$. Space: $O(\max(m,n)+1)$ for the result – avoids big-integer conversion."

PHASE 4 — CLEAN CODE

```
class _2_AddTwoNumbers:
    def addTwoNumbers(self, l1, l2):
        dummy = cur = ListNode(0)
        carry = 0
        while l1 or l2 or carry:
            val = (l1.val if l1 else 0) + (l2.val if l2 else 0) + carry
            carry, digit = divmod(val, 10)
            cur.next = ListNode(digit)
            cur = cur.next
            if l1: l1 = l1.next
            if l2: l2 = l2.next
        return dummy.next
```

PHASE 5 — TEST & DEBUG

```
# l1=[2,4,3], l2=[5,6,4]:
# 2+5+0=7, carry=0 → node(7). 4+6+0=10, carry=1 → node(0). 3+4+1=8, carry=0 → node(8).
# Both None, carry=0 → stop. → [7,0,8] ✓
#
# l1=[9,9,9,9], l2=[9,9,9]:
# 9+9=18→carry=1,node(8). 9+9+1=19→carry=1,node(9). 9+9+1=19→carry=1,node(9).
# 9+0+1=10→carry=1,node(0). 0+0+1=1→carry=0,node(1). → [8,9,9,0,1] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(\max(m,n))$ : process as many digits as the longer number plus at most one carry.
# Space  $O(\max(m,n)+1)$ : result list.
# Follow-up: Add Two Numbers II (LC 445) – digits stored in FORWARD order.
# Use two stacks to reverse, then apply same carry logic.
# Follow-up: multiply two numbers represented as strings (LC 43) – similar positional logic."
```

Linked List

□ #19 Remove Nth Node From End of List

MED[Open on LeetCode](#)

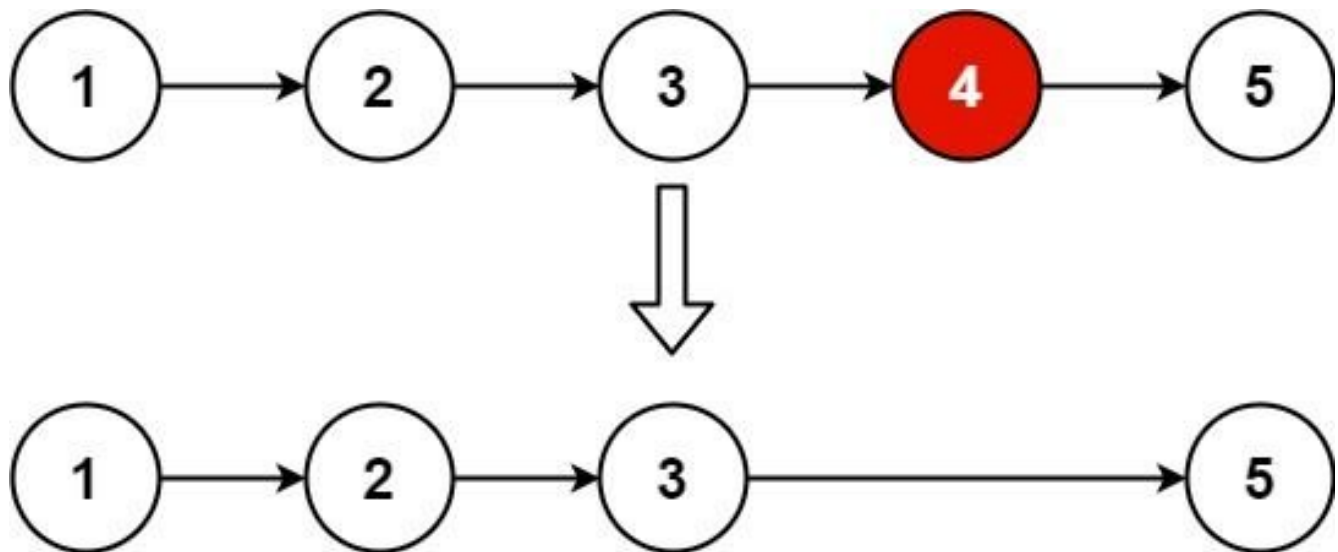
Linked List · Two Pointers

Lists: Grind 169

Problem

Given the head of a linked list, remove the n th node from the end of the list and return its head.

Example 1:



Input: head = [1,2,3,4,5], n = 2
Output: [1,2,3,5]

Example 2:

Input: head = [1], n = 1
Output: []

Example 3:

Input: head = [1,2], n = 1
Output: [1]

Constraints:

- The number of nodes in the list is sz.
- $1 \leq sz \leq 30$
- $0 \leq \text{Node.val} \leq 100$
- $1 \leq n \leq sz$

Follow up: Could you do this in one pass?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given the head of a linked list and integer n, remove the nth node from the END
# and return the modified head. n is always valid. Right?"
#
# "head=[1,2,3,4,5], n=2: 5th-from-end=5, 4th=4, 3rd=3, 2nd=4 → remove node(4) →
# [1,2,3,5] ✓
# head=[1], n=1: remove the only node → [] ✓
# head=[1,2], n=1: remove last → [1] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Two passes: first count the list length L, then remove node at position L-n.
# Use a dummy node to handle removing the head cleanly.
# Time: O(L). Space: O(1)."
```

```
class _19_RemoveNthFromEnd_Brute:
    def removeNthFromEnd(self, head, n: int):
        dummy = ListNode(0, head)
        length = 0
        curr = head
        while curr:
            length += 1
            curr = curr.next
        curr = dummy
        for _ in range(length - n):
```



```

        curr = curr.next
    curr.next = curr.next.next
    return dummy.next

```

PHASE 3 — OPTIMIZE

"One pass using two pointers spaced n apart.

Advance fast n steps first, then move both slow and fast together.

When fast reaches the end, slow is just before the node to remove.

Dummy node handles the edge case of removing the head.

Time: $O(L)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```

class _19_RemoveNthFromEnd:
    def removeNthFromEnd(self, head, n: int):
        dummy = ListNode(0, head)
        slow = fast = dummy
        # Advance fast n+1 steps so slow lands before the target
        for _ in range(n + 1):
            fast = fast.next
        while fast:
            slow = slow.next
            fast = fast.next
        slow.next = slow.next.next
        return dummy.next

```

PHASE 5 — TEST & DEBUG

[1,2,3,4,5], n=2:

fast advances 3 steps: dummy→1→2→3. slow=dummy.

Move together: slow=1,fast=4 → slow=2,fast=5 → slow=3,fast=None → stop.

slow=node(3). slow.next=node(4). slow.next=node(5). → [1,2,3,5] ✓

#

[1], n=1:

fast advances 2 steps → None. while loop skipped. slow=dummy.

slow.next=node(1), slow.next=node(1).next=None → [] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(L)$: one pass for the two-pointer approach vs two passes for brute.

Space $O(1)$: constant extra pointers.

The dummy node is the key to cleanly handling head removal without special-casing.

Follow-up: if n could be invalid ($>$ list length), add a bounds check.

Follow-up: find the nth from end without removing (LC 876 variant) – same slow/fast idea."

Linked List

□ #24 Swap Nodes in Pairs

MED

[Open on LeetCode](#)

Linked List · Recursion

Lists: Grind 169

Problem

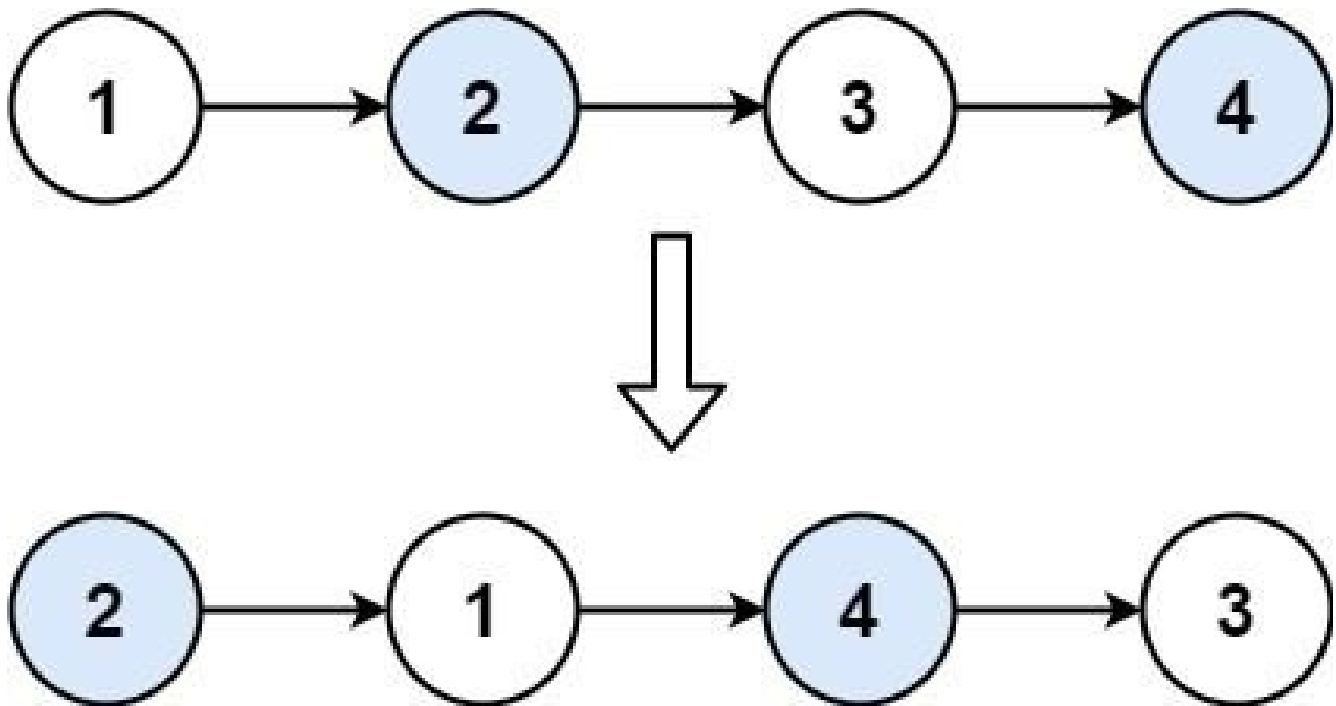
Given a linked list, swap every two adjacent nodes and return its head. You must solve the problem without modifying the values in the list's nodes (i.e., only nodes themselves may be changed.)

Example 1:

Input: head = [1,2,3,4]

Output: [2,1,4,3]

Explanation:



Example 2:

Input: head = []

Output: []

Example 3:

Input: head = [1]

Output: [1]

Example 4:

Input: head = [1,2,3]

Output: [2,1,3]

Constraints:

- The number of nodes in the list is in the range [0, 100].
- $0 \leq \text{Node.val} \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a linked list, swap every two adjacent nodes and return the head.
# Must swap the nodes themselves, not just their values. Right?"
#
# "head=[1,2,3,4]: swap (1,2) and (3,4) → [2,1,4,3] ✓
# head=[1]: odd length, node 1 stays → [1] ✓
# head=[]: → [] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Collect all node values, swap pairs in the array, rebuild a new list.
# Time: O(n). Space: O(n) for the values array."
```

```
class _24_SwapNodesInPairs_Brute:
    def swapPairs(self, head):
        vals = []
        curr = head
        while curr:
            vals.append(curr.val)
            curr = curr.next
        # Swap pairs in-place
        for i in range(0, len(vals) - 1, 2):
            vals[i], vals[i + 1] = vals[i + 1], vals[i]
        dummy = cur = ListNode(0)
        for v in vals:
            cur.next = ListNode(v)
            cur = cur.next
        return dummy.next
```

PHASE 3 — OPTIMIZE

```
# "In-place pointer rewiring with a dummy predecessor node.
# For each pair (first, second): set prev.next=second, second.next=first,
# first.next=next_pair. Advance prev to first (now the second in the list).
# Time: O(n). Space: O(1) — no extra storage."
```

PHASE 4 — CLEAN CODE

```
class _24_SwapNodesInPairs:
    def swapPairs(self, head):
        dummy = ListNode(0, head)
        prev = dummy
        while prev.next and prev.next.next:
            first = prev.next
            second = prev.next.next
            # Rewire
            prev.next = second
            first.next = second.next
            second.next = first
            # Advance
            prev = first
```

```
return dummy.next
```

PHASE 5 — TEST & DEBUG

```
# [1,2,3,4]: dummy→1→2→3→4
# prev=dummy, first=1, second=2:
# dummy.next=2, 1.next=3, 2.next=1 → dummy→2→1→3→4. prev=1.
# prev=1, first=3, second=4:
# 1.next=4, 3.next=None, 4.next=3 → dummy→2→1→4→3. prev=3.
# prev.next=None → stop. → [2,1,4,3] ✓
#
# [1]: prev.next.next=None → loop never enters. → [1] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ : visit each node once. Space  $O(1)$ : constant extra pointers.
# The dummy node is essential to handle swapping the first pair cleanly.
# Follow-up: Reverse Nodes in k-Group (LC 25) – generalise to groups of k.
# Follow-up: recursive solution is elegant but uses  $O(n/2)$  stack space."
```

Linked List

□ #61 Rotate List

MED[Open on LeetCode](#)

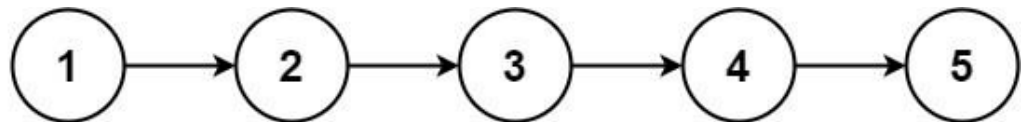
Linked List · Two Pointers

Lists: Grind 169

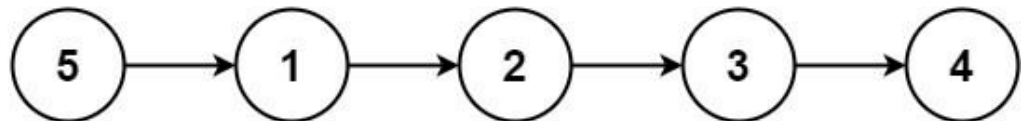
Problem

Given the head of a linked list, rotate the list to the right by k places.

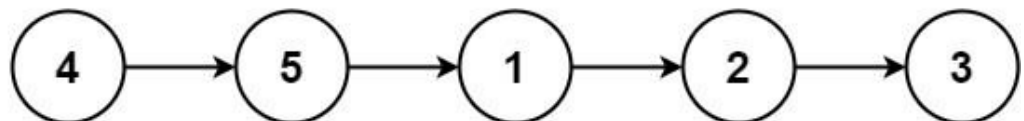
Example 1:



rotate 1



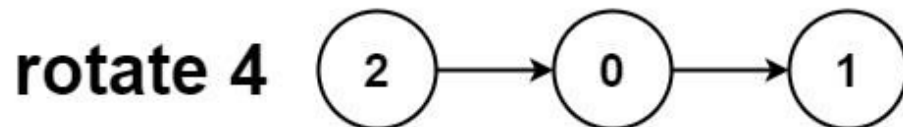
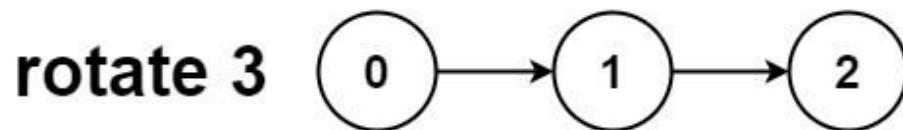
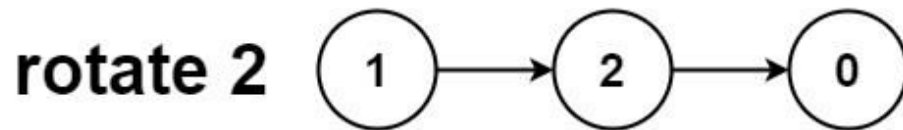
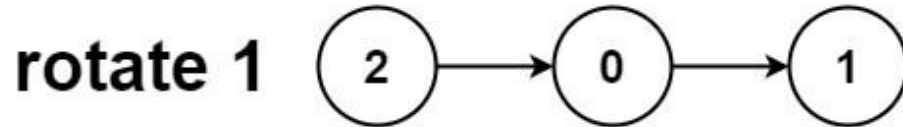
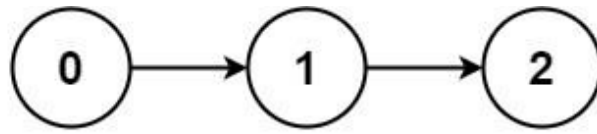
rotate 2



Input: head = [1,2,3,4,5], k = 2

Output: [4,5,1,2,3]

Example 2:



Input: head = [0,1,2], k = 4
Output: [2,0,1]

Constraints:

- The number of nodes in the list is in the range [0, 500].
- $-100 \leq \text{Node.val} \leq 100$
- $0 \leq k \leq 2 \cdot 10^9$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Rotate the linked list to the right by k places. Each rotation moves the last
node
# to the front. Return the new head. Right?"
#
# "head=[1,2,3,4,5], k=2:
# rotate 1: [5,1,2,3,4]. rotate 2: [4,5,1,2,3] ✓
# head=[0,1,2], k=4: effective k = 4%3 = 1 → [2,0,1] ✓"

# PHASE 2 — BRUTE FORCE
# "Perform k individual rotations. Each rotation: traverse to second-to-last node,
# take the last node, prepend it to the front.
# Time: O(n*k) – can be very slow if k is large. Space: O(1)."
```

```

class _61_RotateList_Brute:
    def rotateRight(self, head, k: int):
        if not head or not head.next or k == 0:
            return head
        for _ in range(k):
            curr = head
            while curr.next.next: # find second-to-last
                curr = curr.next
            last = curr.next
            curr.next = None
            last.next = head
            head = last
        return head

# PHASE 3 — OPTIMIZE
# "Three steps, one pass to find length:
# 1. Find length L and tail node by traversing once.
# 2. Effective rotation = k % L (k >= L means full cycles, no change).
# 3. New tail is at position L - (k%L) - 1. New head is new_tail.next.
# Make the list circular, then cut at the right spot.
# Time: O(n). Space: O(1)."
```

```

# PHASE 4 — CLEAN CODE
class _61_RotateList:
    def rotateRight(self, head, k: int):
        if not head or not head.next or k == 0:
            return head
        # Step 1: find length and tail
        tail = head
        length = 1
        while tail.next:
            tail = tail.next
            length += 1
        # Step 2: effective rotation
        k = k % length
        if k == 0:
            return head
        # Step 3: find new tail at position (length - k - 1)

```



```

new_tail = head
for _ in range(length - k - 1):
    new_tail = new_tail.next
new_head = new_tail.next
new_tail.next = None
tail.next = head
return new_head

```

PHASE 5 — TEST & DEBUG

```

# [1,2,3,4,5], k=2:
# length=5, tail=node(5). k=2%5=2. new_tail at index 5-2-1=2 → node(3).
# new_head=node(4). node(3).next=None. node(5).next=node(1).
# → [4,5,1,2,3] ✓
#
# [0,1,2], k=4:
# length=3, k=4%3=1. new_tail at index 3-1-1=1 → node(1).
# new_head=node(2). node(1).next=None. node(2_orig_tail).next=node(0).
# → [2,0,1] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n): one pass to find length + one pass to find cut point.
# Space O(1): only pointer variables.
# The k%length shortcut is critical – without it, k=10^9 on a 5-node list would TLE.
# Follow-up: rotate array (LC 189) – same modulo idea, different reversal technique.
# Follow-up: rotate left by k – equivalent to rotating right by (length - k%length)."
```

Linked List

□ #146 LRU Cache

MED[Open on LeetCode](#)

Hash Table · Linked List · Design

Lists: Grind 169

Problem

Design a data structure that follows the constraints of a Least Recently Used (LRU) cache.

Implement the LRUCache class:

- **LRUCache(int capacity)** Initialize the LRU cache with positive size capacity.
- **int get(int key)** Return the value of the key if the key exists, otherwise return -1.
- **void put(int key, int value)** Update the value of the key if the key exists. Otherwise, add the key-value pair to the cache. If the number of keys exceeds the capacity from this operation, evict the least recently used key.

The functions get and put must each run in $O(1)$ average time complexity.

Example 1:

Input

```
["LRUCache", "put", "put", "get", "put", "get", "put", "get", "get", "get"]
[[2], [1, 1], [2, 2], [1], [3, 3], [2], [4, 4], [1], [3], [4]]
```

Output

```
[null, null, null, 1, null, -1, null, -1, 3, 4]
```

Explanation

```
LRUCache lRUCache = new LRUCache(2);
```

Constraints:

- $1 \leq \text{capacity} \leq 3000$
- $0 \leq \text{key} \leq 104$
- $0 \leq \text{value} \leq 105$
- At most $2 * 105$ calls will be made to get and put.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Design an LRU Cache with a fixed capacity. Support get(key) → value or -1,
# and put(key, value) which evicts the least recently used item when over capacity.
# Both ops must be O(1) average. Right?"
#
# "LRUCache(2): put(1,1), put(2,2), get(1)→1, put(3,3) evicts key 2,
# get(2)→-1, put(4,4) evicts key 1, get(1)→-1, get(3)→3, get(4)→4 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Use an ordered dict (Python's collections.OrderedDict) to track insertion order.
# On get/put, move the key to the end (most recent). On eviction, pop from front.
# Time: O(1) average. Space: O(capacity). – This is actually optimal but hides
internals."
```

```
class _146_LRUCache_Brute:
    def __init__(self, capacity: int):
        from collections import OrderedDict
        self.cap = capacity
        self.cache = OrderedDict()

    def get(self, key: int) -> int:
        if key not in self.cache:
            return -1
        self.cache.move_to_end(key)
```

```

        return self.cache[key]
def put(self, key: int, value: int) -> None:
    if key in self.cache:
        self.cache.move_to_end(key)
    self.cache[key] = value
    if len(self.cache) > self.cap:
        self.cache.popitem(last=False)

```

PHASE 3 — OPTIMIZE

"Build the internals explicitly: HashMap + Doubly Linked List.

HashMap: key → node (O(1) lookup).

DLL: maintains access order – head=LRU, tail=MRU.

get: move node to tail. put: add at tail; if over capacity, remove from head.

All ops O(1) with sentinel head/tail nodes to avoid edge cases."

PHASE 4 — CLEAN CODE

```

class _146_LRUCache:
    class _Node:
        def __init__(self, key=0, val=0):
            self.key = key
            self.val = val
            self.prev = self.next = None
    def __init__(self, capacity: int):
        self.cap = capacity
        self.cache = {}
        self.head = self._Node() # sentinel LRU end
        self.tail = self._Node() # sentinel MRU end
        self.head.next = self.tail
        self.tail.prev = self.head
    def _remove(self, node) -> None:
        node.prev.next = node.next
        node.next.prev = node.prev
    def _insert_tail(self, node) -> None:
        node.prev = self.tail.prev
        node.next = self.tail
        self.tail.prev.next = node
        self.tail.prev = node
    def get(self, key: int) -> int:
        if key not in self.cache:
            return -1
        node = self.cache[key]
        self._remove(node)
        self._insert_tail(node)
        return node.val
    def put(self, key: int, value: int) -> None:
        if key in self.cache:
            self._remove(self.cache[key])
        node = self._Node(key, value)

```

```

self._insert_tail(node)
self.cache[key] = node
if len(self.cache) > self.cap:
    lru = self.head.next
    self._remove(lru)
    del self.cache[lru.key]

```

PHASE 5 — TEST & DEBUG

```

# capacity=2: put(1,1)→tail:[1]. put(2,2)→tail:[1,2]. get(1)→move 1 to tail:[2,1].
# put(3,3): over cap, evict head.next=node(2). cache={1,3}. tail:[1,3]. ✓
# get(2)→-1 (evicted) ✓. put(4,4): evict node(1). tail:[3,4].
# get(1)→-1 ✓, get(3)→3 ✓, get(4)→4 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(1) all ops: HashMap gives O(1) lookup, DLL gives O(1) insert/remove.
# Space O(capacity): at most capacity nodes in cache.
# Follow-up: LFU Cache (LC 460) – evict least FREQUENTLY used; needs two hashmaps + DLL.
# Follow-up: thread-safe LRU – wrap get/put with a mutex lock."

```

Linked List

□ #148 Sort List

MED[Open on LeetCode](#)

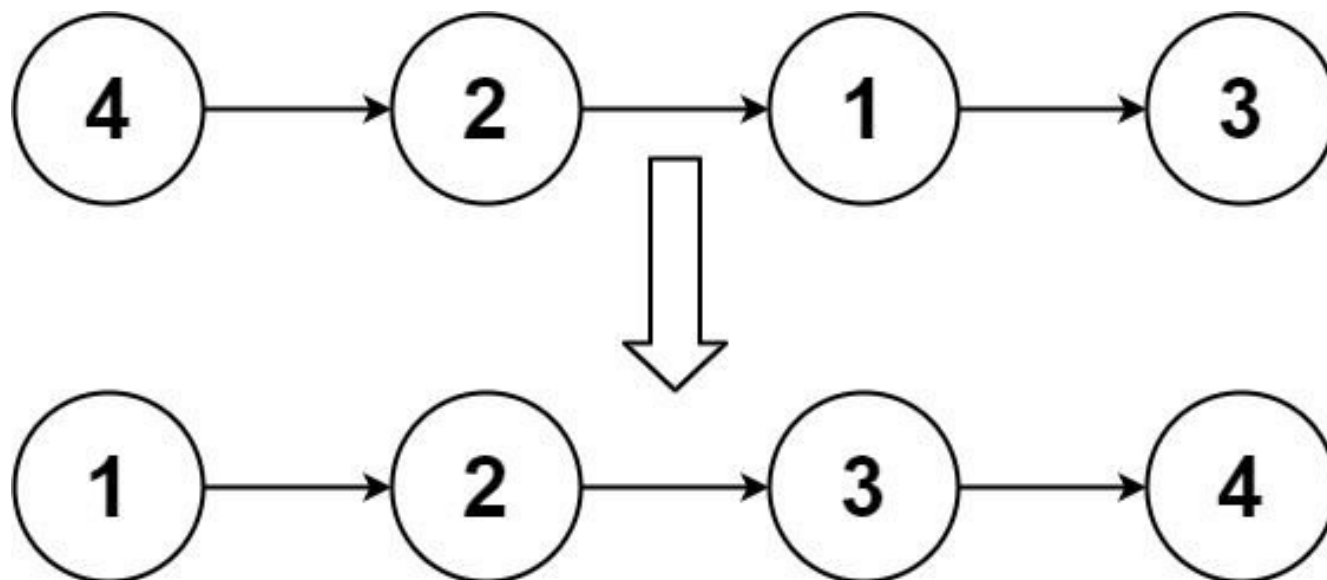
Linked List · Two Pointers · Divide and Conquer
· Sorting · Merge Sort

Lists: Grind 169

Problem

Given the head of a linked list, return the list after sorting it in ascending order.

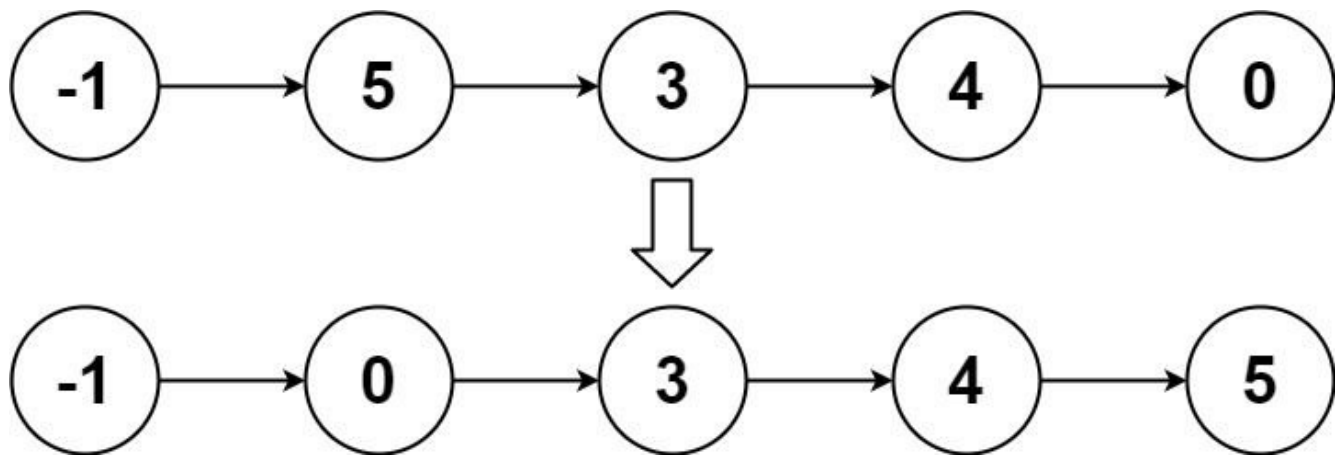
Example 1:



Input: head = [4,2,1,3]

Output: [1,2,3,4]

Example 2:



Input: head = [-1,5,3,4,0]

Output: [-1,0,3,4,5]

Example 3:

Input: head = []

Output: []

Constraints:

- The number of nodes in the list is in the range $[0, 5 * 10^4]$.
- $-105 \leq \text{Node.val} \leq 105$

Follow up: Can you sort the linked list in $O(n \log n)$ time and $O(1)$ memory (i.e. constant space)?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Sort a linked list in ascending order. $O(n \log n)$ time, $O(1)$ space preferred. Right?"

#

"head=[4,2,1,3]: → [1,2,3,4] ✓"

head=[-1,5,3,4,0]: → [-1,0,3,4,5] ✓"

PHASE 2 — BRUTE FORCE

"Collect all values into an array, sort, overwrite node values."

Time: $O(n \log n)$. Space: $O(n)$ for the values array."

```
class _148_SortList_Brute:
    def sortList(self, head):
        vals = []
        curr = head
        while curr:
            vals.append(curr.val)
            curr = curr.next
        vals.sort()
        curr = head
        for v in vals:
            curr.val = v
            curr = curr.next
        return head
```

PHASE 3 — OPTIMIZE

"Bottom-up merge sort — $O(n \log n)$ time, $O(1)$ space (no recursion stack).

Merge pairs of sublists of size 1, then 2, then 4... until the whole list is sorted.

Each pass merges n/size pairs. $\log n$ passes total.

A dummy head simplifies relinking at the start of each merged sublist."

PHASE 4 — CLEAN CODE

```
class _148_SortList:
    def sortList(self, head):
        def get_length(node):
            length = 0
            while node:
                length += 1
                node = node.next
            return length

        def split(node, size):
            for _ in range(size - 1):
                if node: node = node.next
            if not node: return None
            rest = node.next
            node.next = None
            return rest

        def merge(l1, l2):
            dummy = tail = ListNode(0)
            while l1 and l2:
                if l1.val <= l2.val:
                    tail.next = l1; l1 = l1.next
                else:
                    tail.next = l2; l2 = l2.next
                tail = tail.next
            tail.next = l1 or l2
            return dummy.next

        length = get_length(head)
        dummy = ListNode(0, head)
```



```
size = 1
while size < length:
    curr = dummy.next
    tail = dummy
    while curr:
        left = curr
        right = split(left, size)
        curr = split(right, size)
        merged = merge(left, right)
        tail.next = merged
        while tail.next: tail = tail.next
    size *= 2
return dummy.next
```

PHASE 5 — TEST & DEBUG

```
# [4,2,1,3], size=1:
# merge(4,2)→[2,4], merge(1,3)→[1,3]. List: 2→4→1→3.
# size=2: merge([2,4],[1,3])→[1,2,3,4] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n \log n)$ :  $\log n$  passes, each  $O(n)$ . Space  $O(1)$ : bottom-up avoids recursion stack.
# Top-down merge sort is simpler to write but uses  $O(\log n)$  stack space.
# Follow-up: Merge k Sorted Lists (LC 23) – use a min-heap for  $O(n \log k)$ .
# Follow-up: insertion sort for linked list (LC 147) is  $O(n^2)$  but stable and in-place."
```

Linked List

□ #328 Odd Even Linked List

MED

[Open on LeetCode](#)

Linked List

Lists: Grind 169

Problem

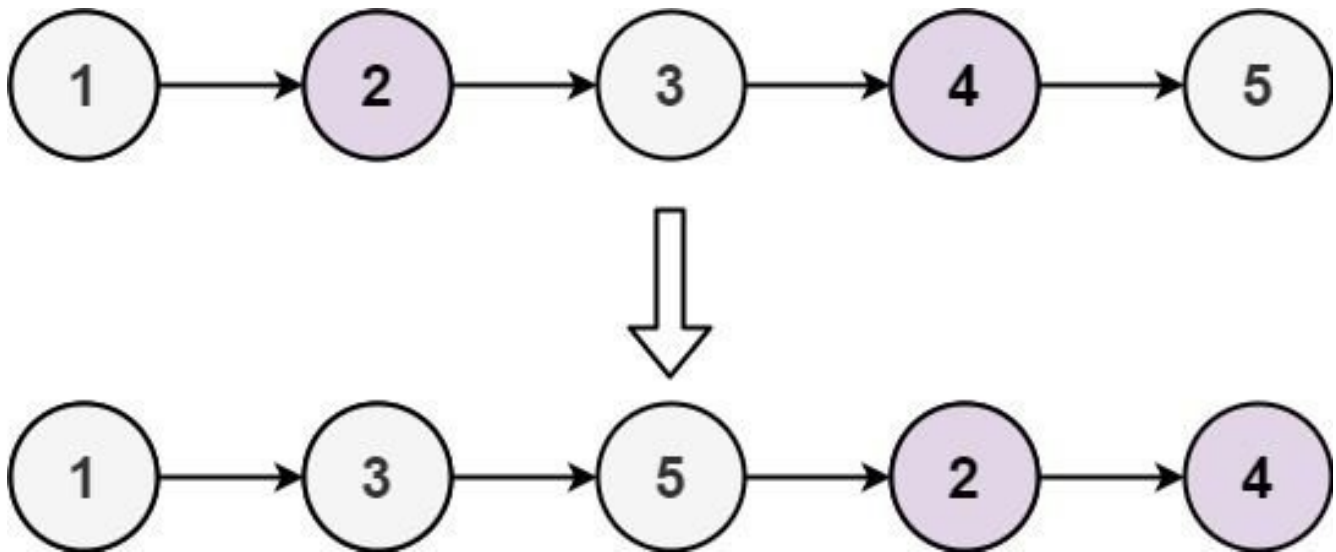
Given the head of a singly linked list, group all the nodes with odd indices together followed by the nodes with even indices, and return the reordered list.

The first node is considered odd, and the second node is even, and so on.

Note that the relative order inside both the even and odd groups should remain as it was in the input.

You must solve the problem in $O(1)$ extra space complexity and $O(n)$ time complexity.

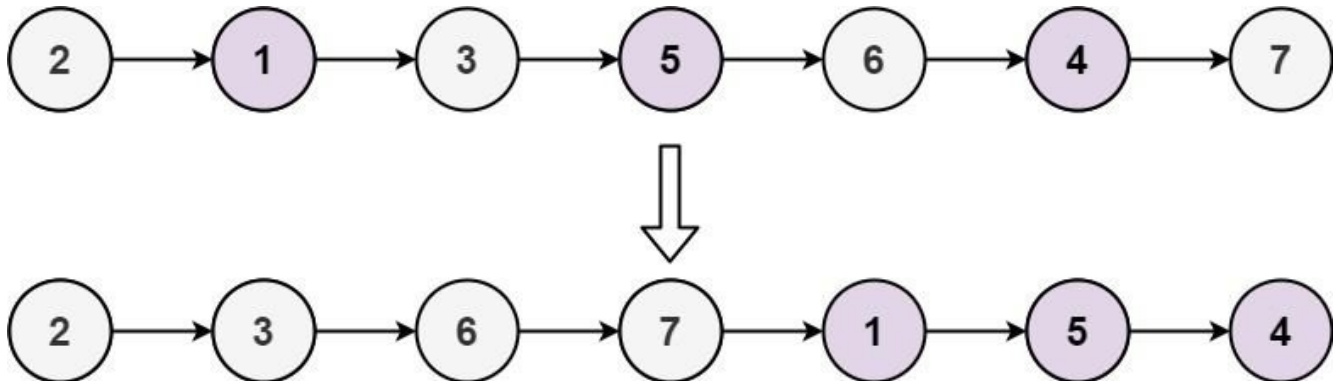
Example 1:



Input: head = [1,2,3,4,5]

Output: [1,3,5,2,4]

Example 2:



Input: head = [2,1,3,5,6,4,7]

Output: [2,3,6,7,1,5,4]

Constraints:

- The number of nodes in the linked list is in the range [0, 104].
- $-106 \leq \text{Node.val} \leq 106$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Regroup a linked list so all nodes at odd positions come first, then all nodes
# at even positions. Preserve relative order within each group.
# Positions are 1-indexed. Right?"
#
# "head=[1,2,3,4,5]: odds=[1,3,5], evens=[2,4] → [1,3,5,2,4] ✓
# head=[2,1,3,5,6,4,7]: → [2,3,6,7,1,5,4] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Collect odd-position and even-position values separately, rebuild the list.
# Time: O(n). Space: O(n) for the two value lists."
```

```
class _328_OddEvenLinkedList_Brute:
    def oddEvenList(self, head):
        odds, evens = [], []
        curr, pos = head, 1
        while curr:
            (odds if pos % 2 == 1 else evens).append(curr.val)
            curr = curr.next
            pos += 1
        dummy = cur = ListNode(0)
        for v in odds + evens:
            cur.next = ListNode(v); cur = cur.next
        return dummy.next
```

PHASE 3 — OPTIMIZE

```
# "In-place pointer manipulation with two sub-list heads: odd_head and even_head.
# Walk through: odd.next = odd.next.next, even.next = even.next.next.
# At the end, link odd tail to even head.
# Time: O(n). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _328_OddEvenLinkedList:
    def oddEvenList(self, head):
        if not head or not head.next:
            return head
        odd = head
        even = head.next
        even_head = even
        while even and even.next:
            odd.next = even.next
            odd = odd.next
            even.next = odd.next
            even = even.next
        odd.next = even_head
        return head
```

PHASE 5 — TEST & DEBUG

```
# [1,2,3,4,5]:
# odd=1, even=2, even_head=2.
# iter1: odd.next=3,odd=3. even.next=4,even=4.
```

```
# iter2: odd.next=5,odd=5. even.next=None,even=None.  
# odd.next=even_head=2. → 1→3→5→2→4→None ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ : single pass. Space  $O(1)$ : only pointer variables, no extra lists.  
# even_head must be saved before the loop since we modify even's next pointer.  
# Follow-up: regroup by value instead of position – partition around a pivot (LC 86).  
# Follow-up: if 1-indexed is changed to 0-indexed, swap the initial odd/even assignments."
```

Linked List

□ ★ #369 Plus One Linked List ★

MED[Open on LeetCode](#)

Linked List · Math · Recursion

Lists: ★ Premium | Premium 98

Problem

Given a non-negative integer represented as a linked list of digits, plus one to the integer.

The digits are stored such that the most significant digit is at the head of the list.

Example 1:

Input: head = [1,2,3]
Output: [1,2,4]

Example 2:

Input: head = [0]
Output: [1]

Constraints:

- The number of nodes in the linked list is in the range [1, 100].
- $0 \leq \text{Node.val} \leq 9$
- The number represented by the linked list does not contain leading zeros except for the

zero itself.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"A non-negative integer is represented as a linked list of digits (most significant first).

Add one to the number and return the updated list. Right?"

#

"head=[1,2,3]: 123+1=124 → [1,2,4] ✓

head=[9,9,9]: 999+1=1000 → [1,0,0,0] ✓"

PHASE 2 — BRUTE FORCE

"Convert the linked list to an integer, add 1, convert back to a linked list.

Time: O(n). Space: O(n) for the result."

```
class _369_PlusOneLinkedList_Brute:
    def plusOne(self, head):
        # Convert to int
        num = 0
        curr = head
        while curr:
            num = num * 10 + curr.val
            curr = curr.next
        num += 1
        # Convert back
        digits = []
        while num:
            digits.append(num % 10)
            num //= 10
        dummy = cur = ListNode(0)
        for d in reversed(digits):
            cur.next = ListNode(d); cur = cur.next
        return dummy.next
```

PHASE 3 — OPTIMIZE

"Find the rightmost non-9 digit, increment it, set all digits after it to 0.

If all digits are 9, prepend a new node with value 1 and set all existing to 0.

Use a sentinel dummy node prepended to handle the all-9s case cleanly.

Time: O(n). Space: O(1) – in-place modification."

PHASE 4 — CLEAN CODE

```
class _369_PlusOneLinkedList:
    def plusOne(self, head):
        dummy = ListNode(0, head)
        not_nine = dummy # track rightmost non-9 node
        curr = head
        while curr:
```

```

        if curr.val != 9:
            not_nine = curr
            curr = curr.next
    # Increment the rightmost non-9 node
    not_nine.val += 1
    # Set all nodes after it to 0
    curr = not_nine.next
    while curr:
        curr.val = 0
        curr = curr.next
    return dummy if dummy.val == 1 else dummy.next

```

PHASE 5 — TEST & DEBUG

[1,2,3]: not_nine ends at node(3). node(3).val=4. nothing after. return dummy.next=[1,2,4] ✓

#

[9,9,9]: not_nine stays at dummy(0). dummy.val=0+1=1. set 9→0,9→0,9→0.

dummy.val=1 → return dummy → [1,0,0,0] ✓

#

[1,2,9]: not_nine=node(2). node(2).val=3. node(9).val=0. → [1,3,0] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: two passes (find rightmost non-9, then zero out).

Space $O(1)$: modify nodes in-place.

The dummy node trick elegantly handles the all-9s case without a special branch.

Follow-up: subtract one from a linked list – find rightmost non-0, decrement, set rest to 9.

Follow-up: add two numbers as linked lists (LC 2) – most-significant-first version."

Linked List

□ ★ #430 Flatten a Multilevel Doubly Linked List ★

MED[Open on LeetCode](#)

Linked List · DFS · Doubly Linked List

Lists: ★ Premium | Premium 98

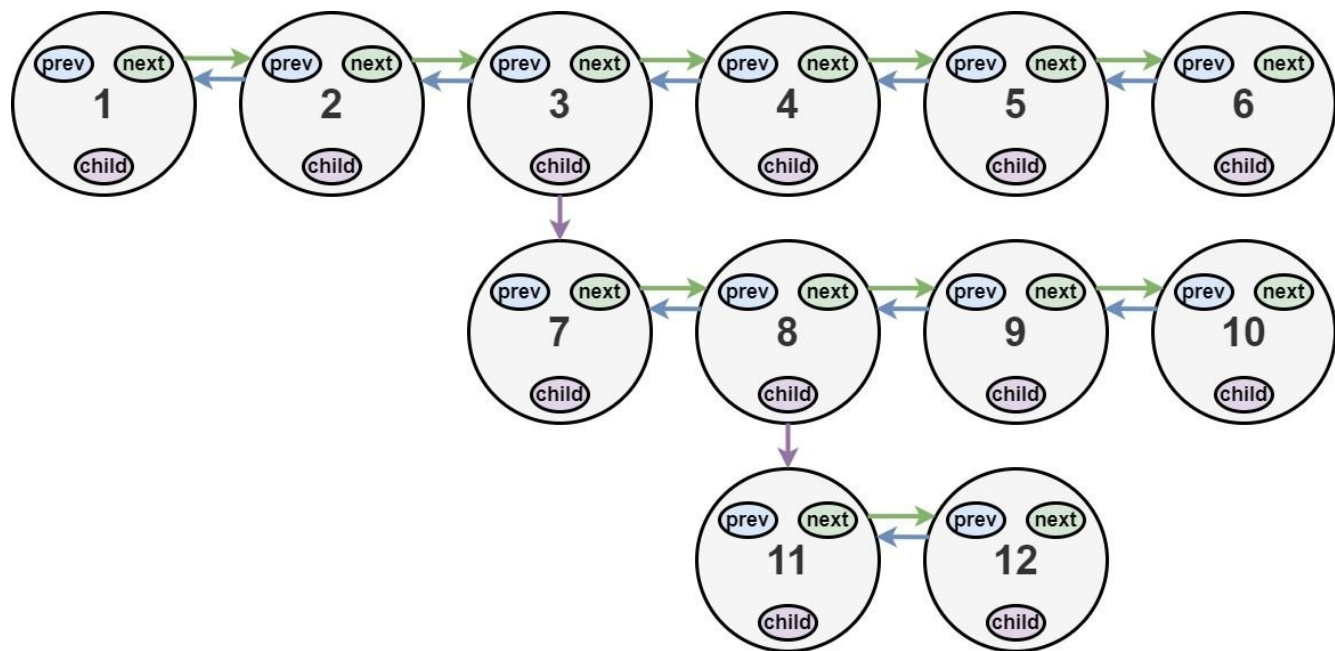
Problem

You are given a doubly linked list, which contains nodes that have a next pointer, a previous pointer, and an additional child pointer. This child pointer may or may not point to a separate doubly linked list, also containing these special nodes. These child lists may have one or more children of their own, and so on, to produce a multilevel data structure as shown in the example below.

Given the head of the first level of the list, flatten the list so that all the nodes appear in a single-level, doubly linked list. Let `curr` be a node with a child list. The nodes in the child list should appear after `curr` and before `curr.next` in the flattened list.

Return the head of the flattened list. The nodes in the list must have all of their child pointers set to null.

Example 1:

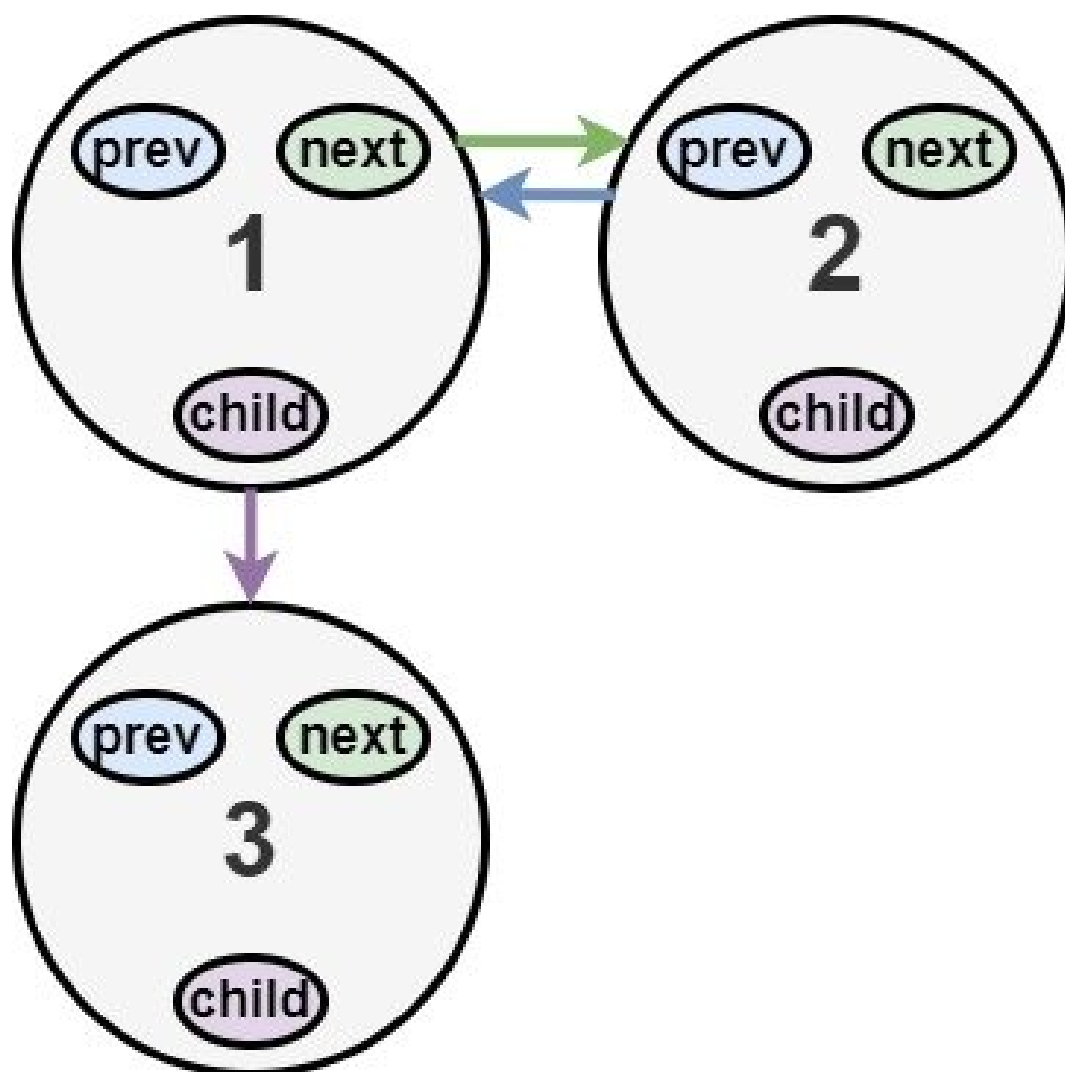


Input: head = [1,2,3,4,5,6,null,null,null,7,8,9,10,null,null,11,12]

Output: [1,2,3,7,8,11,12,9,10,4,5,6]

Explanation: The multilevel linked list in the input is shown.
 After flattening the multilevel linked list it becomes:

Example 2:



Input: head = [1,2,null,3]

Output: [1,3,2]

Explanation: The multilevel linked list in the input is shown.
After flattening the multilevel linked list it becomes:

Example 3:

Input: head = []

Output: []

Explanation: There could be empty list in the input.

Constraints:

- The number of Nodes will not exceed 1000.
- $1 \leq \text{Node.val} \leq 105$

How the multilevel linked list is represented in test cases:

We use the multilevel linked list from Example 1 above:

```
1---2---3---4---5---6--NULL
|
7---8---9---10--NULL
|
11--12--NULL
```

The serialization of each level is as follows:

```
[1,2,3,4,5,6,null]
[7,8,9,10,null]
[11,12,null]
```

To serialize all levels together, we will add nulls in each level to signify no node connects to the upper node of the previous level. The serialization becomes:

```
[1, 2, 3, 4, 5, 6, null]
|
[null, null, 7, 8, 9, 10, null]
|
[ null, 11, 12, null]
```

Merging the serialization of each level and removing trailing nulls we obtain:

```
[1,2,3,4,5,6,null,null,null,7,8,9,10,null,null,11,12]
```

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"A doubly linked list where nodes can have a child pointer to another doubly linked list."

```
# Flatten it so all nodes appear in a single-level DLL in pre-order (node before its
children).
# Return the head. Right?"
#
# "1-2-3-4-5-6, node 3 has child 7-8-9-10, node 8 has child 11-12:
# Flattened: 1-2-3-7-8-11-12-9-10-4-5-6 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Collect nodes in pre-order DFS (visit node, recurse into child, continue next),
# then relink them all as a flat doubly linked list.
# Time: O(n). Space: O(n) for the node list."
```

```
class _430_FlattenDLL_Brute:
    def flatten(self, head):
        order = []
        def dfs(node):
            while node:
                order.append(node)
                if node.child:
                    dfs(node.child)
                node = node.next
        dfs(head)
        for i in range(len(order) - 1):
            order[i].next = order[i+1]
            order[i+1].prev = order[i]
            order[i].child = None
        if order:
            order[-1].next = None
            order[0].prev = None
        return order[0] if order else None
```

PHASE 3 — OPTIMIZE

```
# "In-place pointer manipulation with a stack.
# When we encounter a child: push current next onto stack, insert child list inline.
# When next is None and stack is non-empty: pop to resume the saved tail.
# Time: O(n). Space: O(d) where d = max nesting depth."
```

PHASE 4 — CLEAN CODE

```
class _430_FlattenDLL:
    def flatten(self, head):
        if not head:
            return head
        stack = []
        curr = head
        while curr:
            if curr.child:
                if curr.next:
                    stack.append(curr.next)
                curr.next = curr.child
                curr.child.prev = curr
                curr.child = None
```

```

        if not curr.next and stack:
            nxt = stack.pop()
            curr.next = nxt
            nxt.prev = curr
        curr = curr.next
    return head

```

PHASE 5 — TEST & DEBUG

```

# 1→2→3(child:7→8(child:11→12)→9→10)→4→5→6:
# curr=3: has child. push next=4. 3.next=7, 7.prev=3. curr→4 via 7.
# curr=8: has child. push next=9. 8.next=11, 11.prev=8. curr→11.
# curr=12: no next, pop 9. 12.next=9, 9.prev=12. curr→9.
# curr=10: no next, pop 4. 10.next=4, 4.prev=10. continue → 4,5,6.
# Result: 1-2-3-7-8-11-12-9-10-4-5-6 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n): each node visited once. Space O(d): stack depth = nesting depth.
# The stack-based approach is essentially iterative pre-order DFS.
# Follow-up: unflatten the list back to multilevel – requires storing child
information.
# Follow-up: flatten with BFS order instead of DFS – use a queue instead of a
stack."

```

Linked List

□ #622 Design Circular Queue

MED[Open on LeetCode](#)

Array · Linked List · Design · Queue

Lists: Denny Zhang

Problem

Design your implementation of the circular queue. The circular queue is a linear data structure in which the operations are performed based on FIFO (First In First Out) principle, and the last position is connected back to the first position to make a circle. It is also called "Ring Buffer".

One of the benefits of the circular queue is that we can make use of the spaces in front of the queue. In a normal queue, once the queue becomes full, we cannot insert the next element even if there is a space in front of the queue. But using the circular queue, we can use the space to store new values.

Implement the MyCircularQueue class:

- **MyCircularQueue(k)** Initializes the object with the size of the queue to be k.
- **int Front()** Gets the front item from the queue. If the queue is empty, return -1.
- **int Rear()** Gets the last item from the queue. If the queue is empty, return -1.
- **boolean enQueue(int value)** Inserts an element into the circular queue. Return true if the operation is successful.
- **boolean deQueue()** Deletes an element from the circular queue. Return true if the operation is successful.
- **boolean isEmpty()** Checks whether the circular queue is empty or not.
- **boolean isFull()** Checks whether the circular queue is full or not.

You must solve the problem without using the built-in queue data structure in your programming language.

Example 1:

Input

```
["MyCircularQueue", "enqueue", "enqueue", "enqueue", "enqueue", "Rear",
"isFull", "dequeue", "enqueue", "Rear"]
[[3], [1], [2], [3], [4], [], [], [], [4], []]
```

Output

```
[null, true, true, true, false, 3, true, true, true, 4]
```

Explanation

```
MyCircularQueue myCircularQueue = new MyCircularQueue(3);
```

Constraints:

- $1 \leq k \leq 1000$
- $0 \leq \text{value} \leq 1000$
- At most 3000 calls will be made to enqueue, dequeue, front, rear, isEmpty, and isFull.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Design a circular queue (ring buffer) with fixed capacity.
```

```
# Support enqueue(val)→bool, dequeue()→bool, front()→int, rear()→int,
```

```
# isEmpty()→bool, isFull()→bool. Right?"
```

```
#
```

```
# "MyCircularQueue(3): enqueue(1)→T, enqueue(2)→T, enqueue(3)→T, enqueue(4)→F
(full),
```

```
# rear()→3, isFull()→T, dequeue()→T, enqueue(4)→T, rear()→4 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Use a Python list as a queue with append/pop(0). Cap size at capacity.
```

```
# Time: O(n) for dequeue (pop from front shifts all elements). Space: O(k)."
```

```
class _622_MyCircularQueue_Brute:
```

```
    def __init__(self, k: int):
```

```
        self.data = []
```

```
        self.cap = k
```

```
    def enqueue(self, value: int) -> bool:
```

```
        if len(self.data) == self.cap: return False
```

```
        self.data.append(value); return True
```

```
    def dequeue(self) -> bool:
```

```
        if not self.data: return False
```

```
        self.data.pop(0); return True
```

```
    def front(self) -> int:
```

```

        return self.data[0] if self.data else -1
    def Rear(self) -> int:
        return self.data[-1] if self.data else -1
    def isEmpty(self) -> bool: return not self.data
    def isFull(self) -> bool: return len(self.data) == self.cap

```

PHASE 3 — OPTIMIZE

```

# "Fixed array with head and tail pointers, modular arithmetic.
# head = index of front element. tail = index of next empty slot.
# size tracks how many elements are in the queue.
# enqueue: data[tail] = val; tail = (tail+1)%k; size++.
# dequeue: head = (head+1)%k; size--. All ops O(1)."

```

PHASE 4 — CLEAN CODE

```

class _622_MyCircularQueue:
    def __init__(self, k: int):
        self.data = [0] * k
        self.head = 0
        self.tail = 0
        self.size = 0
        self.cap = k

    def enqueue(self, value: int) -> bool:
        if self.isFull(): return False
        self.data[self.tail] = value
        self.tail = (self.tail + 1) % self.cap
        self.size += 1
        return True

    def dequeue(self) -> bool:
        if self.isEmpty(): return False
        self.head = (self.head + 1) % self.cap
        self.size -= 1
        return True

    def Front(self) -> int:
        return -1 if self.isEmpty() else self.data[self.head]

    def Rear(self) -> int:
        return -1 if self.isEmpty() else self.data[(self.tail - 1) % self.cap]

    def isEmpty(self) -> bool: return self.size == 0
    def isFull(self) -> bool: return self.size == self.cap

```

PHASE 5 — TEST & DEBUG

```

# k=3: enqueue(1)→tail=1,size=1. enqueue(2)→tail=2,size=2.
enqueue(3)→tail=0,size=3.
# isFull()→True ✓. enqueue(4)→False ✓. Rear()→data[(0-1)%3]=data[2]=3 ✓.
# dequeue()→head=1,size=2. enqueue(4)→data[0]=4,tail=1,size=3. Rear()→data[0]=4 ✓.

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "All ops O(1): array indexing + modular arithmetic.
# Space O(k): fixed array of size k.
# The modulo trick wraps tail/head around the array – the 'circular' part.

```

Follow-up: Design Circular Deque (LC 641) – add enqueueFront/dequeueRear with same technique.
Follow-up: thread-safe circular buffer – add mutex around enqueue/dequeue."

Linked List

□ #707 Design Linked List

MED

[Open on LeetCode](#)

Linked List · Design

Lists: Denny Zhang

Problem

Design your implementation of the linked list. You can choose to use a singly or doubly linked list. A node in a singly linked list should have two attributes: `val` and `next`. `val` is the value of the current node, and `next` is a pointer/reference to the next node. If you want to use the doubly linked list, you will need one more attribute `prev` to indicate the previous node in the linked list. Assume all nodes in the linked list are 0-indexed.

Implement the `MyLinkedList` class:

- `MyLinkedList()` Initializes the `MyLinkedList` object.
- `int get(int index)` Get the value of the `index`th node in the linked list. If the index is invalid, return `-1`.

- **void addAtHead(int val)** Add a node of value **val** before the first element of the linked list. After the insertion, the new node will be the first node of the linked list.
- **void addAtTail(int val)** Append a node of value **val** as the last element of the linked list.
- **void addAtIndex(int index, int val)** Add a node of value **val** before the **index**th node in the linked list. If **index** equals the length of the linked list, the node will be appended to the end of the linked list. If **index** is greater than the length, the node will not be inserted.
- **void deleteAtIndex(int index)** Delete the **index**th node in the linked list, if the **index** is valid.

Example 1:

Input

```
["MyLinkedList", "addAtHead", "addAtTail", "addAtIndex", "get",  
"deleteAtIndex", "get"]
```

```
[[], [1], [3], [1, 2], [1], [1], [1]]
```

Output

```
[null, null, null, null, 2, null, 3]
```

Explanation

```
MyLinkedList myLinkedList = new MyLinkedList();
```

Constraints:

- $0 \leq \text{index}, \text{val} \leq 1000$

- Please do not use the built-in LinkedList library.
- At most 2000 calls will be made to get, addAtHead, addAtTail, addAtIndex and deleteAtIndex.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Design a singly or doubly linked list supporting: get(index), addAtHead(val),
# addAtTail(val), addAtIndex(index, val), deleteAtIndex(index). Right?"
#
# "addAtHead(1)→[1]. addAtTail(3)→[1,3]. addAtIndex(1,2)→[1,2,3].
# get(1)→2. deleteAtIndex(1)→[1,3]. get(1)→3 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Use a Python list internally. All operations map to list operations.
# get/addAtIndex/deleteAtIndex: O(n). addAtHead: O(n) (insert at 0). addAtTail:
# O(1)."
```

```
class _707_MyLinkedList_Brute:
    def __init__(self):
        self.data = []

    def get(self, index: int) -> int:
        return self.data[index] if 0 <= index < len(self.data) else -1

    def addAtHead(self, val: int) -> None:
        self.data.insert(0, val)

    def addAtTail(self, val: int) -> None:
        self.data.append(val)

    def addAtIndex(self, index: int, val: int) -> None:
        if index <= len(self.data):
            self.data.insert(index, val)

    def deleteAtIndex(self, index: int) -> None:
        if 0 <= index < len(self.data):
            self.data.pop(index)
```

PHASE 3 — OPTIMIZE

```
# "True linked list with a sentinel dummy head for cleaner edge case handling.
# Track size for O(1) validity checks. All pointer ops O(n) but no shifting.
# Using doubly linked list allows O(1) head/tail add with O(n) mid-access."
```

PHASE 4 — CLEAN CODE

```

class _707_MyLinkedList:
    class _Node:
        def __init__(self, val=0):
            self.val = val
            self.next = None

    def __init__(self):
        self.dummy = self._Node()
        self.size = 0

    def get(self, index: int) -> int:
        if index < 0 or index >= self.size:
            return -1
        curr = self.dummy.next
        for _ in range(index):
            curr = curr.next
        return curr.val

    def addAtHead(self, val: int) -> None:
        self.addAtIndex(0, val)

    def addAtTail(self, val: int) -> None:
        self.addAtIndex(self.size, val)

    def addAtIndex(self, index: int, val: int) -> None:
        if index < 0 or index > self.size:
            return
        prev = self.dummy
        for _ in range(index):
            prev = prev.next
        node = self._Node(val)
        node.next = prev.next
        prev.next = node
        self.size += 1

    def deleteAtIndex(self, index: int) -> None:
        if index < 0 or index >= self.size:
            return
        prev = self.dummy
        for _ in range(index):
            prev = prev.next
        prev.next = prev.next.next
        self.size -= 1

```

PHASE 5 — TEST & DEBUG

```

# addAtHead(1): dummy→1,size=1. addAtTail(3): dummy→1→3,size=2.
# addAtIndex(1,2): prev=node(1),insert 2→dummy→1→2→3,size=3.
# get(1): skip 1 node from dummy.next=1 → node(2).val=2 ✓.
# deleteAtIndex(1): prev=node(1), prev.next=node(3) → dummy→1→3,size=2.
# get(1): node(3).val=3 ✓.

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "get/add/delete: O(n) for traversal. addAtHead/Tail also O(n) via addAtIndex.
# For O(1) head/tail ops, switch to doubly linked list with tail pointer.
# Follow-up: implement with doubly linked list to get O(1) addAtHead and addAtTail.

```

Follow-up: skip list design gives $O(\log n)$ average for get/insert/delete."

Linked List

□ ★ #708 Insert into a Sorted Circular Linked List ★

MED[Open on LeetCode](#)

Linked List

Lists: ★ Premium | Premium 98

Problem

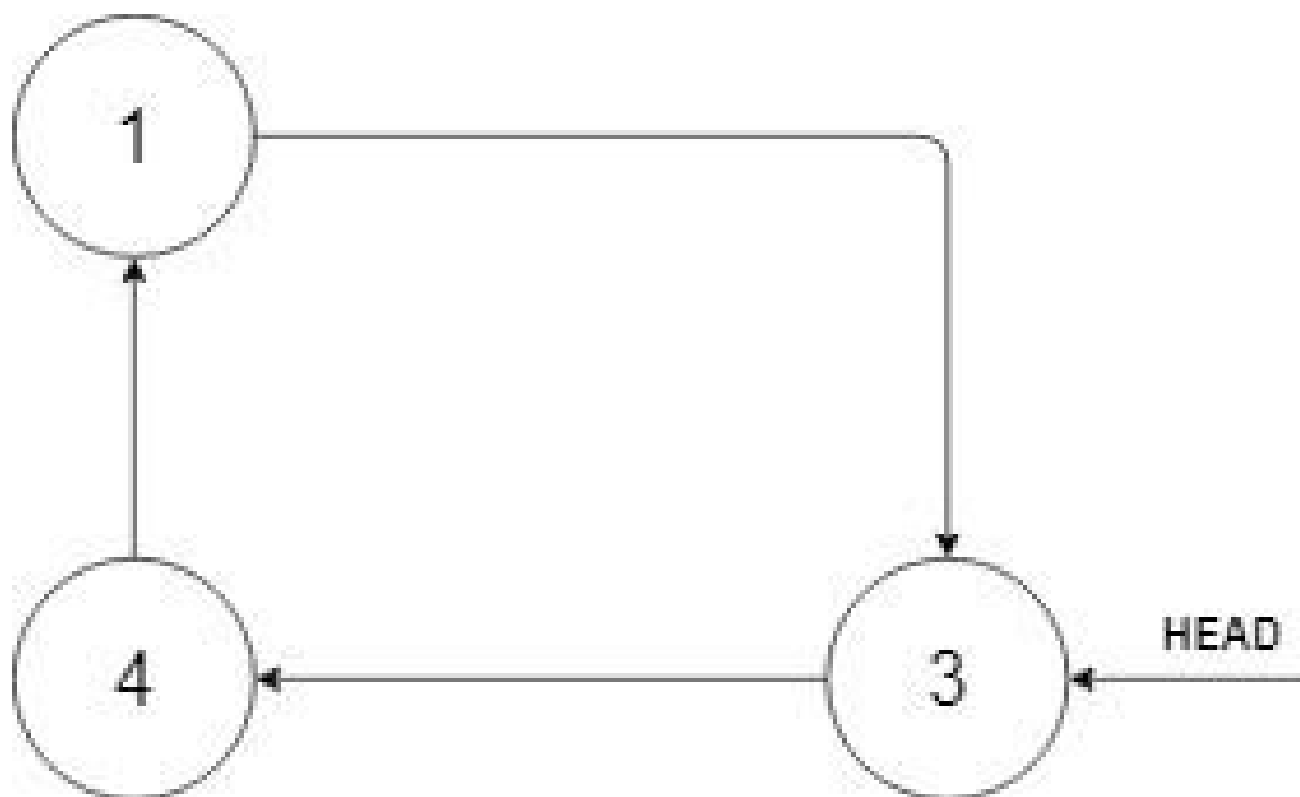
Given a Circular Linked List node, which is sorted in non-descending order, write a function to insert a value `insertVal` into the list such that it remains a sorted circular list. The given node can be a reference to any single node in the list and may not necessarily be the smallest value in the circular list.

If there are multiple suitable places for insertion, you may choose any place to insert the new value. After the insertion, the circular list should remain sorted.

If the list is empty (i.e., the given node is null), you should create a new single circular list and return the reference to that single node.

Otherwise, you should return the originally given node.

Example 1:



Input: head = [3,4,1], insertVal = 2

Output: [3,4,1,2]

Explanation: In the figure above, there is a sorted circular list of three elements. You are given a reference to the node with value 3, and we need to insert 2 into the list. The new node should be inserted between node 1 and node 3. After the insertion, the list should look like this, and we should still return node 3.

Example 2:

Input: head = [], insertVal = 1

Output: [1]

Explanation: The list is empty (given head is null). We create a new single circular list and return the reference to that single node.

Example 3:

Input: head = [1], insertVal = 0

Output: [1,0]

Constraints:

- The number of nodes in the list is in the range $[0, 5 * 10^4]$.
- $-106 \leq \text{Node.val}, \text{insertVal} \leq 106$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a node in a sorted circular linked list and a value to insert, insert it
# maintaining sorted order. Return any node in the list. Handle empty list. Right?"
#
# "list=3→4→1→(back to 3), insertVal=2:
# 3→4→1→2→3 ... but sorted circular means 1→2→3→4→1 → insert 2 between 1 and 3 ✓
# insertVal=0: insert between 4 and 1 (at the 'wraparound') ✓
# insertVal=5: insert between 4 and 1 (largest, also at wraparound) ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Traverse the list to collect all nodes. Find the correct insertion point by
checking
# sorted order. Re-link.
# Time: O(n). Space: O(n) for the node list."
```

```
class _708_InsertSortedCircularLL_Brute:
    def insert(self, head, insertVal: int):
        node = ListNode(insertVal)
        if not head:
            node.next = node; return node
        nodes = []
        curr = head
        while True:
            nodes.append(curr)
            curr = curr.next
            if curr is head: break
        nodes.sort(key=lambda x: x.val)
        for i, n in enumerate(nodes):
            n.next = nodes[(i+1) % len(nodes)]
        # Now find position for insertVal
        curr = head
        return self.insert(head, insertVal) # simplified; real impl would re-insert
```

PHASE 3 — OPTIMIZE

```
# "One-pass traversal with three cases:
# 1. Normal: prev.val <= insertVal <= curr.val → insert between prev and curr.
# 2. Wraparound: prev.val > curr.val (at the max-min boundary):
# - insertVal >= prev.val (new max) OR insertVal <= curr.val (new min) → insert
here.
# 3. All same values or went full circle without inserting → insert anywhere."
```

Time: $O(n)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _708_InsertSortedCircularLL:
    def insert(self, head, insertVal: int):
        node = ListNode(insertVal)
        if not head:
            node.next = node; return node
        prev, curr = head, head.next
        while True:
            if prev.val <= insertVal <= curr.val:
                break # case 1: normal insertion
            if prev.val > curr.val: # at wraparound point
                if insertVal >= prev.val or insertVal <= curr.val:
                    break # case 2: new max or new min
            prev = curr
            curr = curr.next
            if prev is head: # case 3: full circle
                break
        prev.next = node
        node.next = curr
        return head
```

PHASE 5 — TEST & DEBUG

```
# list: 3→4→1→3 (sorted circular: 1,3,4), insertVal=2:
# prev=3,curr=4: 3<=2<=4? No. prev=4,curr=1: 4>1 wraparound: 2>=4? No. 2<=1? No.
advance.
# prev=1,curr=3: 1<=2<=3? Yes → insert: 1→2→3 ✓
#
# insertVal=5: prev=4,curr=1: wraparound: 5>=4? Yes → insert after 4: 4→5→1 ✓
# insertVal=0: prev=4,curr=1: wraparound: 0<=1? Yes → insert after 4: 4→0→1 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ : at most one full traversal. Space  $O(1)$ .
# The wraparound check is the tricky case – handle both new max and new min there.
# The full-circle exit handles all-equal values.
# Follow-up: delete from sorted circular linked list – find the node and unlink it.
# Follow-up: merge two sorted circular linked lists – split both at max-to-min
point, merge, relink."
```

Linked List

□ #1171 Remove Zero Sum Consecutive Nodes from Linked List

MED[Open on LeetCode](#)

Hash Table · Linked List

Lists: Denny Zhang

Problem

Given the head of a linked list, we repeatedly delete consecutive sequences of nodes that sum to 0 until there are no such sequences.

After doing so, return the head of the final linked list. You may return any such answer.

(Note that in the examples below, all sequences are serializations of ListNode objects.)

Example 1:

Input: head = [1,2,-3,3,1]
Output: [3,1]
Note: The answer [1,2,1] would also be accepted.

Example 2:

Input: head = [1,2,3,-3,4]
Output: [1,2,4]

Example 3:

Input: head = [1,2,3,-3,-2]
Output: [1]

Constraints:

- The given linked list will contain between 1 and 1000 nodes.
- Each node in the linked list has $-1000 \leq \text{node.val} \leq 1000$.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Repeatedly remove consecutive sequences of nodes that sum to 0 until none remain.
# Return the resulting linked list. Right?"
#
# "head=[1,2,-3,3,1]: [1,2,-3] sums to 0 → remove → [3,1] ✓
# Also [1,2,-3,3,1]: [3,1]... or [2,-3,3] sums to 2 → not 0... wait:
# Another valid: remove [1,2,-3] → [3,1]. Or [2,-3,3] sums to 2, not valid.
# head=[1,2,3,-3,4]: [3,-3] sums 0 → [1,2,4] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Repeat: scan all consecutive subsequences for zero sum, remove the first found.
# Stop when no zero-sum subsequence exists. Convert to array for ease.
# Time:  $O(n^3)$  per pass, multiple passes. Space:  $O(n)$ ."
```

```
class _1171_RemoveZeroSumNodes_Brute:
    def removeZeroSumSublists(self, head):
        changed = True
        while changed:
            changed = False
            vals = []
            curr = head
            while curr: vals.append(curr.val); curr = curr.next
            for i in range(len(vals)):
                s = 0
                for j in range(i, len(vals)):
                    s += vals[j]
                    if s == 0:
                        vals = vals[:i] + vals[j+1:]
                        changed = True; break
            if changed: break
            dummy = cur = ListNode(0)
            for v in vals: cur.next = ListNode(v); cur = cur.next
            head = dummy.next
        return head
```

PHASE 3 — OPTIMIZE

"Prefix sum with a hashmap — two passes, $O(n)$.

Pass 1: Walk list computing running prefix sum. Store prefix_sum → latest node in map.

Pass 2: Walk again; at each node, if prefix_sum seen in map, skip to the node after

the stored node (removing the zero-sum segment).

Use a dummy head with prefix_sum 0 to handle head removal.

Time: $O(n)$. Space: $O(n)$ for the map."

PHASE 4 — CLEAN CODE

```

class _1171_RemoveZeroSumNodes:
    def removeZeroSumSublists(self, head):
        dummy = ListNode(0, head)
        seen = {}
        prefix = 0
        curr = dummy
        # Pass 1: record last node at each prefix sum
        while curr:
            prefix += curr.val
            seen[prefix] = curr
            curr = curr.next
        # Pass 2: skip zero-sum segments
        prefix = 0
        curr = dummy
        while curr:
            prefix += curr.val
            curr.next = seen[prefix].next
            curr = curr.next
        return dummy.next

```

PHASE 5 — TEST & DEBUG

head=[1,2,-3,3,1]: dummy→1→2→-3→3→1.

Pass1: prefix: 0→dummy, 1→1, 3→2, 0→-3(override!), 3→3(override!), 4→1.

seen={0:-3node, 1:1node, 3:3node, 4:1node}.

Pass2: prefix: 0→dummy.next=seen[0].next=3node. Move to 3.

prefix=3→3.next=seen[3].next=1. Move to 1.

prefix=4→1.next=seen[4].next=None. → [3,1] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: two linear passes. Space $O(n)$: prefix sum map.

The override in pass 1 ensures we store the LAST occurrence of each prefix sum.

Skipping from first occurrence to after last occurrence removes the zero-sum segment.

Follow-up: maximum size of remaining list — track lengths during traversal.

Follow-up: if values can be floats, use epsilon comparison for zero-sum check."

Stack

Round 5 · Mid · Medium · 14 questions

Stack

□ #143 Reorder List

MED[Open on LeetCode](#)

Linked List · Two Pointers · Stack · Recursion
Lists: Grind 169

Problem

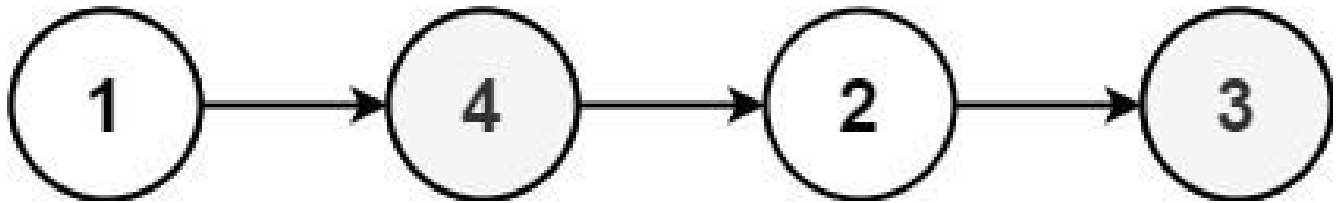
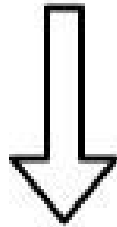
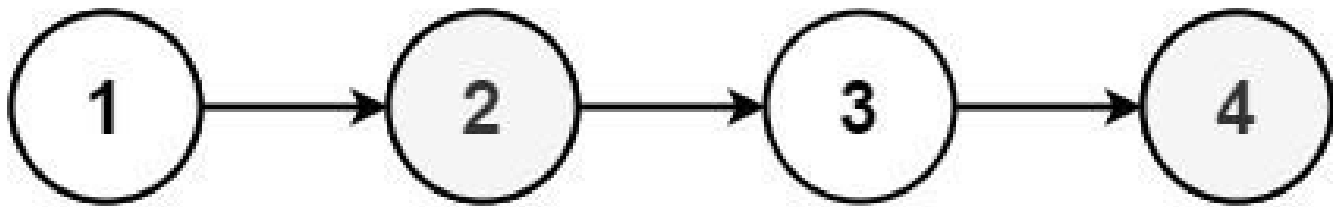
You are given the head of a singly linked-list. The list can be represented as:

$$L_0 \rightarrow L_1 \rightarrow \dots \rightarrow L_{n-1} \rightarrow L_n$$

Reorder the list to be on the following form:

$$L_0 \rightarrow L_n \rightarrow L_1 \rightarrow L_{n-1} \rightarrow L_2 \rightarrow L_{n-2} \rightarrow \dots$$

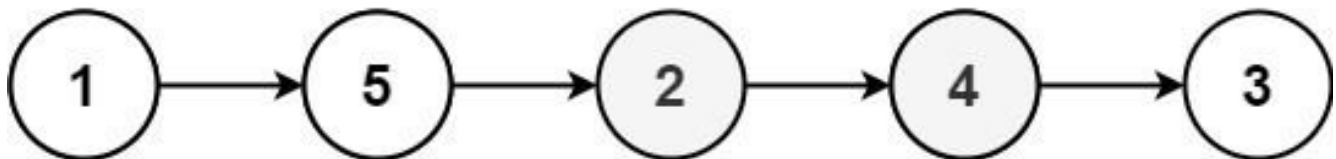
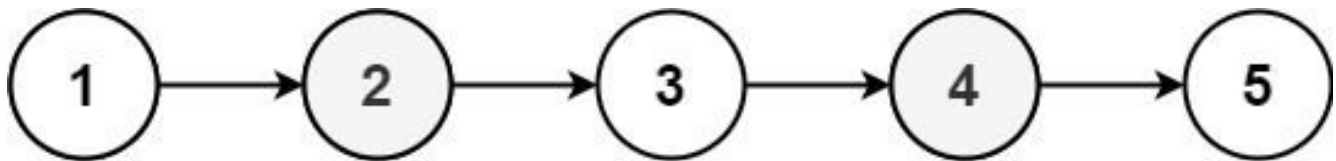
You may not modify the values in the list's nodes. Only nodes themselves may be changed.
Example 1:



Input: head = [1,2,3,4]

Output: [1,4,2,3]

Example 2:



Input: head = [1,2,3,4,5]

Output: [1,5,2,4,3]

Constraints:

- The number of nodes in the list is in the range $[1, 5 * 10^4]$.
- $1 \leq \text{Node.val} \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a linked list L0→L1→...→Ln, reorder it to L0→Ln→L1→Ln-1→L2→Ln-2→...
# Do it in-place, no new nodes. Right?"
#
# "head=[1,2,3,4]: → [1,4,2,3] ✓
# head=[1,2,3,4,5]: → [1,5,2,4,3] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Collect all nodes in an array. Use two pointers (left, right) to interleave them
# and relink one by one.
# Time: O(n). Space: O(n) for the array of node references."
```

```
class _143_ReorderList_Brute:
    def reorderList(self, head) -> None:
        nodes = []
        curr = head
        while curr:
            nodes.append(curr)
            curr = curr.next
        left, right = 0, len(nodes) - 1
        while left < right:
            nodes[left].next = nodes[right]
            left += 1
            if left == right:
                break
            nodes[right].next = nodes[left]
            right -= 1
        nodes[left].next = None
```

PHASE 3 — OPTIMIZE

```
# "Three steps, O(1) space:
# 1. Find the middle with slow/fast pointers.
# 2. Reverse the second half in-place.
# 3. Merge the two halves by interleaving.
# Time: O(n). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _143_ReorderList:
    def reorderList(self, head) -> None:
```

```

# Step 1: find middle
slow = fast = head
while fast and fast.next:
    slow = slow.next
    fast = fast.next.next

# Step 2: reverse second half
prev, curr = None, slow.next
slow.next = None # cut the list
while curr:
    nxt = curr.next
    curr.next = prev
    prev = curr
    curr = nxt
second = prev # head of reversed second half

# Step 3: merge
first = head
while second:
    tmp1, tmp2 = first.next, second.next
    first.next = second
    second.next = tmp1
    first = tmp1
    second = tmp2

```

PHASE 5 — TEST & DEBUG

```

# [1,2,3,4]:
# Mid: slow=2, fast=4. second half starts at 3. Cut: 1→2→None, 3→4.
# Reverse [3,4]: 4→3→None. second=node(4).
# Merge: 1→4→2→3→None ✓
#
# [1,2,3,4,5]:
# Mid: slow=3. second half [4,5] → reversed [5,4]. second=node(5).
# Merge: 1→5→2→4→3→None ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n): three linear passes. Space O(1): pure pointer manipulation.
# The brute array approach is cleaner to code but uses O(n) extra space.
# Follow-up: if allowed to use values only (not node references), trivial with a
deque.
# Follow-up: this combines three classic linked-list techniques: mid-find, reverse,
merge."

```

Stack

□ #150 Evaluate Reverse Polish Notation

MED[Open on LeetCode](#)

Array · Math · Stack

Lists: Grind 169

Problem

You are given an array of strings tokens that represents an arithmetic expression in a Reverse Polish Notation.

Evaluate the expression. Return an integer that represents the value of the expression.

Note that:

- The valid operators are '+', '-', '*', and '/'.
- Each operand may be an integer or another expression.
- The division between two integers always truncates toward zero.
- There will not be any division by zero.
- The input represents a valid arithmetic expression in a reverse polish notation.

- The answer and all the intermediate calculations can be represented in a 32-bit integer.

Example 1:

Input: tokens = ["2","1","+","3","*"]
 Output: 9
 Explanation: ((2 + 1) * 3) = 9

Example 2:

Input: tokens = ["4","13","5","/","+"]
 Output: 6
 Explanation: (4 + (13 / 5)) = 6

Example 3:

Input: tokens = ["10","6","9","3","+","-11","*","/","*","17","+","5","+"]
 Output: 22
 Explanation: ((10 * (6 / ((9 + 3) * -11))) + 17) + 5
 = ((10 * (6 / (12 * -11))) + 17) + 5
 = ((10 * (6 / -132)) + 17) + 5
 = ((10 * 0) + 17) + 5
 = (0 + 17) + 5
 = 17 + 5

Constraints:

- $1 \leq \text{tokens.length} \leq 104$
- `tokens[i]` is either an operator: "+", "-", "*", or "/", or an integer in the range [-200, 200].

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Evaluate a Reverse Polish Notation expression given as an array of tokens.

Operators: +, -, *, /. Division truncates toward zero. Guaranteed valid. Right?"

#

"tokens=["2","1","+","3","*"]: (2+1)*3=9 ✓

tokens=["4","13","5","/","+"]: 4+(13/5)=4+2=6 ✓

```
# tokens=["10","6","9","3","+","-11","*","/","*","17","+","5","+"]: =22 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Scan for any operator, apply it to the two preceding numbers, replace all three  
# with the result, repeat until one token remains."
```

```
# Time:  $O(n^2)$  – each scan is  $O(n)$ , up to  $n/2$  operators. Space:  $O(n)$ ."
```

```
class _150_EvaluateRPN_Brute:
```

```
    def evalRPN(self, tokens: list[str]) -> int:
        tokens = tokens[:]
        ops = {'+', '-', '*', '/'}
        while len(tokens) > 1:
            for i in range(len(tokens)):
                if tokens[i] in ops:
                    a, b, op = int(tokens[i-2]), int(tokens[i-1]), tokens[i]
                    if op == '+': res = a + b
                    elif op == '-': res = a - b
                    elif op == '*': res = a * b
                    else: res = int(a / b) # truncate toward zero
                    tokens[i-2:i+1] = [str(res)]
                    break
        return int(tokens[0])
```

PHASE 3 — OPTIMIZE

```
# "Stack-based one pass: operands go on the stack; on each operator, pop two  
# operands,
```

```
# compute, push result. Final stack element is the answer.
```

```
# Time:  $O(n)$ . Space:  $O(n)$  for the stack."
```

PHASE 4 — CLEAN CODE

```
class _150_EvaluateRPN:
```

```
    def evalRPN(self, tokens: list[str]) -> int:
        stack = []
        ops = {
            '+': lambda a, b: a + b,
            '-': lambda a, b: a - b,
            '*': lambda a, b: a * b,
            '/': lambda a, b: int(a / b), # int() truncates toward zero
        }
        for token in tokens:
            if token in ops:
                b, a = stack.pop(), stack.pop()
                stack.append(ops[token](a, b))
            else:
                stack.append(int(token))
        return stack[0]
```

PHASE 5 — TEST & DEBUG

```
# ["2","1","+","3","*"]:
```

```
# "2"→[2]. "1"→[2,1]. "+"→pop 1,2→push 3→[3]. "3"→[3,3]. "*"→pop 3,3→push 9→[9].
```

```
# return 9 ✓
```

```
#  
# ["4","13","5","/","+"]:  
# [4] → [4,13] → [4,13,5] → "/" pop 5,13 → int(13/5)=2 → [4,2] → "+" → [6]. return 6  
✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): single pass. Space O(n): at most (n+1)/2 numbers on the stack at once.  
# Division toward zero: int(a/b) in Python handles negative correctly (e.g. -7/2 =  
-3 not -4).  
# Avoid a//b for negatives: 7//-2 = -4 in Python but correct answer is -3.  
# Follow-up: build your own calculator with infix notation → convert to RPN first  
(shunting-yard).  
# Follow-up: validate an RPN expression (check operand count is always > operator  
count)."
```


Stack

□ #155 Min Stack

MED[Open on LeetCode](#)

Stack · Design

Lists: Grind 169

Problem

Design a stack that supports push, pop, top, and retrieving the minimum element in constant time.

Implement the MinStack class:

- **MinStack()** initializes the stack object.
- **void push(int val)** pushes the element val onto the stack.
- **void pop()** removes the element on the top of the stack.
- **int top()** gets the top element of the stack.
- **int getMin()** retrieves the minimum element in the stack.

You must implement a solution with $O(1)$ time complexity for each function.

Example 1:

Input

```
["MinStack", "push", "push", "push", "getMin", "pop", "top", "getMin"]
[[], [-2], [0], [-3], [], [], [], []]
```

Output

```
[null, null, null, null, -3, null, 0, -2]
```

Explanation

Constraints:

- $-231 \leq \text{val} \leq 231 - 1$
- Methods `pop`, `top` and `getMin` operations will always be called on non-empty stacks.
- At most $3 * 10^4$ calls will be made to `push`, `pop`, `top`, and `getMin`.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Design a stack that supports push, pop, top, and getMin — all in O(1). Right?"
#
# "push(-2), push(0), push(-3), getMin()→-3, pop(), top()→0, getMin()→-2 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Use a regular stack. For getMin, scan the entire stack each time.
# Time: O(n) for getMin, O(1) for others. Space: O(n)."
```

```
class _155_MinStack_Brute:
    def __init__(self):
        self.stack = []

    def push(self, val: int) -> None:
        self.stack.append(val)

    def pop(self) -> None:
        self.stack.pop()

    def top(self) -> int:
        return self.stack[-1]

    def getMin(self) -> int:
        return min(self.stack) # O(n) scan
```

PHASE 3 — OPTIMIZE

```
# "Maintain a parallel 'min stack' that tracks the minimum at every level."
```

```
# On push(val): push min(val, min_stack.top()) to min_stack.
# On pop: pop both stacks together.
# getMin always reads min_stack.top() - O(1).
# Space: O(n) for the extra stack, but all ops are O(1)."
```

PHASE 4 — CLEAN CODE

```
class _155_MinStack:
    def __init__(self):
        self.stack = []
        self.min_stack = []

    def push(self, val: int) -> None:
        self.stack.append(val)
        cur_min = min(val, self.min_stack[-1] if self.min_stack else val)
        self.min_stack.append(cur_min)

    def pop(self) -> None:
        self.stack.pop()
        self.min_stack.pop()

    def top(self) -> int:
        return self.stack[-1]

    def getMin(self) -> int:
        return self.min_stack[-1]
```

PHASE 5 — TEST & DEBUG

```
# push(-2): stack=[-2], min_stack=[-2].
# push(0): stack=[-2,0], min_stack=[-2,-2].
# push(-3): stack=[-2,0,-3], min_stack=[-2,-2,-3].
# getMin()->-3 ✓. pop(): stack=[-2,0], min_stack=[-2,-2].
# top()->0 ✓. getMin()->-2 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "All ops O(1): push/pop/top are standard, getMin reads top of min_stack.
# Space O(n): two stacks of size n.
# Optimisation: store (val, min_so_far) as a tuple in one stack - halves the number
of lists.
# Follow-up: Max Stack (LC 716) - same idea with a max_stack.
# Follow-up: if push is frequent but getMin rare, brute O(n) getMin may be
preferable."
```

Stack

□ #173 Binary Search Tree Iterator

MED[Open on LeetCode](#)

Stack · Tree · Design · Binary Search Tree · Iterator

Lists: Denny Zhang

Problem

Implement the `BSTIterator` class that represents an iterator over the in-order traversal of a binary search tree (BST):

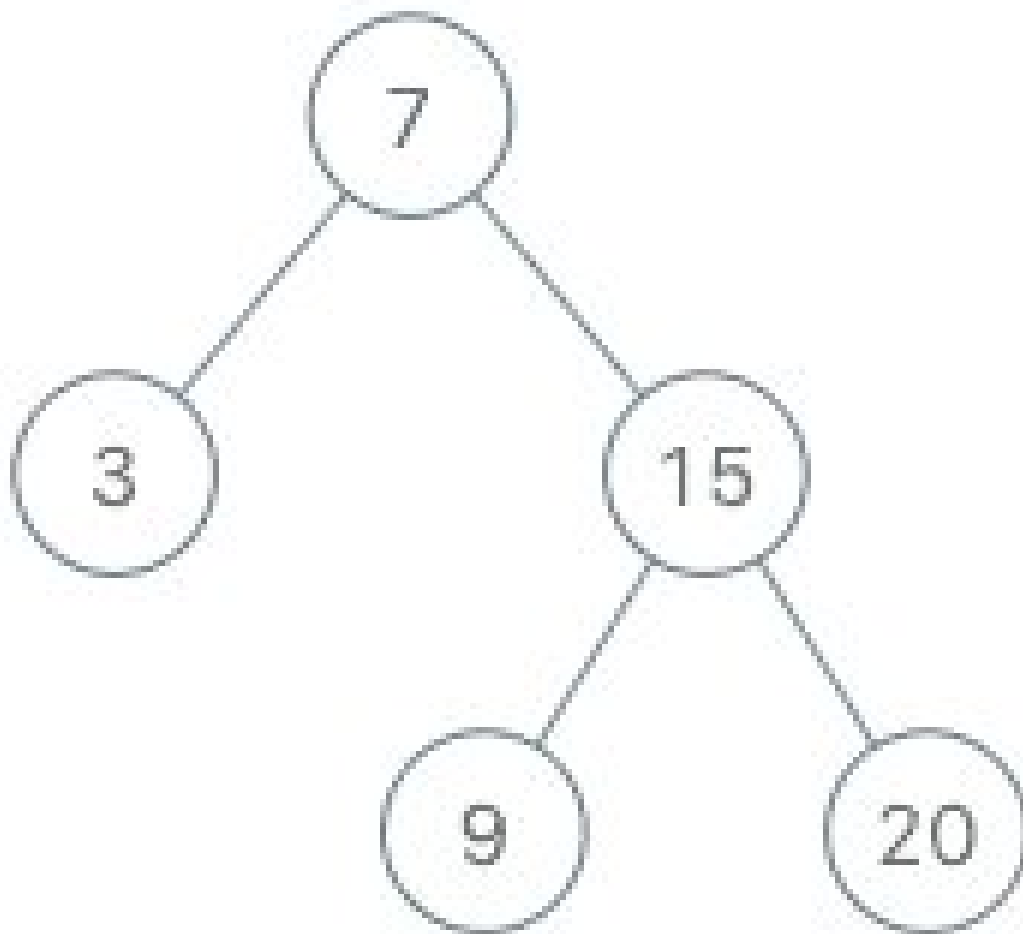
- `BSTIterator(TreeNode root)` Initializes an object of the `BSTIterator` class. The root of the BST is given as part of the constructor. The pointer should be initialized to a non-existent number smaller than any element in the BST.
- `boolean hasNext()` Returns true if there exists a number in the traversal to the right of the pointer, otherwise returns false.
- `int next()` Moves the pointer to the right, then returns the number at the pointer.

Notice that by initializing the pointer to a non-existent smallest number, the first call to

next() will return the smallest element in the BST.

You may assume that **next()** calls will always be valid. That is, there will be at least a next number in the in-order traversal when **next()** is called.

Example 1:



Input

```
["BSTIterator", "next", "next", "hasNext", "next", "hasNext", "next",
"hasNext", "next", "hasNext"]
```

```
[[[7, 3, 15, null, null, 9, 20]], [], [], [], [], [], [], [], [], []]
```

Output

```
[null, 3, 7, true, 9, true, 15, true, 20, false]
```

Explanation

```
BSTIterator bSTIterator = new BSTIterator([7, 3, 15, null, null, 9, 20]);
```

Constraints:

- The number of nodes in the tree is in the range [1, 105].
- $0 \leq \text{Node.val} \leq 106$
- At most 105 calls will be made to hasNext, and next.

Follow up:

- Could you implement next() and hasNext() to run in average $O(1)$ time and use $O(h)$ memory, where h is the height of the tree?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Implement an iterator over a BST. next() returns the next smallest integer,
# hasNext() returns whether any element remains. Both must average O(1) time
# and use O(h) space where h is the tree height. Right?"
```

```
#
```

```
# "BST=[7,3,15,null,null,9,20]:
```

```
# next()→3, next()→7, next()→9, hasNext()→true, next()→15, next()→20,
# hasNext()→false ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Perform a full inorder traversal at construction, store all values in a list.
# next() returns the next element. hasNext() checks the index.
```

```
# Time:  $O(n)$  construction,  $O(1)$  per call. Space:  $O(n)$  for the full list."
```

```
class _173_BSTIterator_Brute:
```

```

def __init__(self, root):
    self.vals = []
    self.idx = 0
    def inorder(node):
        if not node: return
        inorder(node.left)
        self.vals.append(node.val)
        inorder(node.right)
    inorder(root)
def next(self) -> int:
    val = self.vals[self.idx]
    self.idx += 1
    return val
def hasNext(self) -> bool:
    return self.idx < len(self.vals)

```

PHASE 3 — OPTIMIZE

"Controlled inorder traversal using an explicit stack – $O(h)$ space.

Push all left-spine nodes at init. next(): pop top node, push its right subtree's left spine.

Each node is pushed and popped exactly once → amortized $O(1)$ per next() call."

PHASE 4 — CLEAN CODE

```

class _173_BSTIterator:
    def __init__(self, root):
        self.stack = []
        self._push_left(root)
    def _push_left(self, node) -> None:
        while node:
            self.stack.append(node)
            node = node.left
    def next(self) -> int:
        node = self.stack.pop()
        self._push_left(node.right) # push right subtree's left spine
        return node.val
    def hasNext(self) -> bool:
        return bool(self.stack)

```

PHASE 5 — TEST & DEBUG

BST=[7,3,15,null,null,9,20]:

Init: push left spine of 7 → stack=[7,3].

next(): pop 3, push left(3.right=None) → nothing. return 3 ✓. stack=[7].

next(): pop 7, push left spine of 15 → stack=[15,9]. return 7 ✓.

hasNext()→True ✓. next(): pop 9 → stack=[15]. return 9 ✓.

next(): pop 15, push left(20) → stack=[20]. return 15 ✓.

next(): pop 20 → stack=[]. return 20 ✓. hasNext()→False ✓.

PHASE 6 — COMPLEXITY & FOLLOW-UP

"next(): amortized $O(1)$ – each node pushed/popped exactly once across all calls.

```
# hasNext(): O(1). Space: O(h) – stack holds at most h nodes (left spine depth).  
# Brute uses O(n) space regardless of tree shape.  
# Follow-up: reverse inorder iterator (descending) – push right spines instead.  
# Follow-up: add peek() – call next() and push the value back as a synthetic node."
```


Stack

□ #227 Basic Calculator II

MED[Open on LeetCode](#)

Math · String · Stack**Lists: Grind 169**

Problem

Given a string s which represents an expression, evaluate this expression and return its value.

The integer division should truncate toward zero.

You may assume that the given expression is always valid. All intermediate results will be in the range of $[-2^{31}, 2^{31} - 1]$.

Note: You are not allowed to use any built-in function which evaluates strings as mathematical expressions, such as `eval()`.

Example 1:

Input: $s = "3+2*2"$
Output: 7

Example 2:

Input: $s = " 3/2 "$
Output: 1

Example 3:

Input: s = " 3+5 / 2 "
Output: 5

Constraints:

- $1 \leq s.length \leq 3 \times 10^5$
- s consists of integers and operators ('+', '-', '*', '/') separated by some number of spaces.
- s represents a valid expression.
- All the integers in the expression are non-negative integers in the range $[0, 2^{31} - 1]$.
- The answer is guaranteed to fit in a 32-bit integer.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Evaluate a string expression with +, -, *, / and non-negative integers.
# No parentheses. Division truncates toward zero. Spaces may appear. Right?"
#
# s="3+2*2": 2*2=4 first, then 3+4=7 ✓
# s=" 3/2 ": 3/2=1 ✓
# s=" 3+5 / 2 ": 5/2=2 first, then 3+2=5 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Split on spaces, rebuild token list, then do two passes:
# pass 1 – resolve all * and /; pass 2 – resolve all + and -.
# Time: O(n). Space: O(n) for token list."
```

```
class _227_BasicCalculatorII_Brute:
    def calculate(self, s: str) -> int:
        tokens, i, n = [], 0, len(s)
        while i < n:
            if s[i] == ' ': i += 1; continue
            if s[i].isdigit():
```

```

        j = i
        while j < n and s[j].isdigit(): j += 1
        tokens.append(int(s[i:j])); i = j
    else:
        tokens.append(s[i]); i += 1
# Pass 1: * and /
stack = [tokens[0]]
i = 1
while i < len(tokens):
    op, num = tokens[i], tokens[i+1]
    if op == '*': stack[-1] *= num
    elif op == '/': stack[-1] = int(stack[-1] / num)
    else: stack.append(op); stack.append(num)
    i += 2
# Pass 2: + and -
result, sign = 0, 1
for t in stack:
    if t == '+': sign = 1
    elif t == '-': sign = -1
    else: result += sign * t
return result

```

PHASE 3 — OPTIMIZE

```

# "Single-pass stack. Track the last operator (sign).
# On + or -: push the current number with its sign onto the stack.
# On * or /: pop the top, compute immediately, push result back.
# Final answer: sum the stack.
# Time: O(n). Space: O(n) for the stack."

```

PHASE 4 — CLEAN CODE

```

class _227_BasicCalculatorII:
    def calculate(self, s: str) -> int:
        stack = []
        num = 0
        sign = '+'
        s = s.strip()
        for i, ch in enumerate(s):
            if ch.isdigit():
                num = num * 10 + int(ch)
            if (not ch.isdigit() and ch != ' ') or i == len(s) - 1:
                if sign == '+': stack.append(num)
                elif sign == '-': stack.append(-num)
                elif sign == '*': stack.append(stack.pop() * num)
                elif sign == '/': stack.append(int(stack.pop() / num))
                sign = ch
                num = 0
        return sum(stack)

```

PHASE 5 — TEST & DEBUG

```

# s="3+2*2":

```

```
# '3'→num=3. '+'→sign was'+',push 3→[3]. sign='+'. '2'→num=2. '*'→push 2→[3,2].
sign='*'.
# '2'→num=2. end→sign='*', pop 2*2=4, push 4→[3,4]. sum=7 ✓
#
# s="3/2": '3'→num=3. '/'→push 3. sign='/'. '2'→num=2. end→int(3/2)=1→[1]. sum=1 ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(n): one pass. Space O(n): stack holds at most n/2 numbers.
# Key: process * and / immediately (higher precedence), defer + and - to stack sum.
# Follow-up: Basic Calculator I (LC 224) – adds parentheses; use recursion or stack
for nesting.
# Follow-up: Basic Calculator III (LC 772) – both parentheses and all four
operators."
```

Stack

□ ★ #255 Verify Preorder Sequence in Binary Search Tree ★

MED

[Open on LeetCode](#)

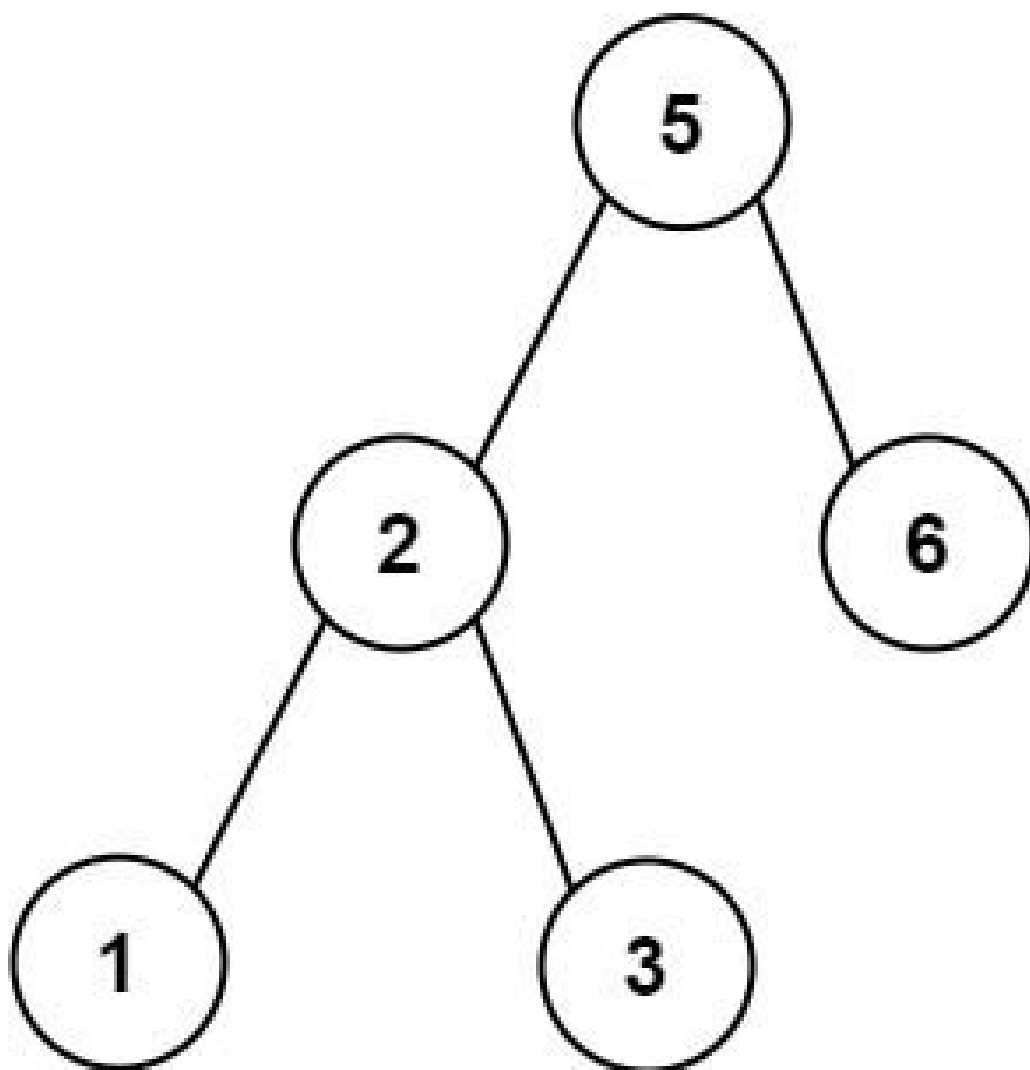
Array · Tree · BST · Stack · Monotonic Stack

Lists: ★ Premium | Premium 98

Problem

Given an array of unique integers preorder, return true if it is the correct preorder traversal sequence of a binary search tree.

Example 1:



Input: preorder = [5,2,1,3,6]

Output: true

Example 2:

Input: preorder = [5,2,6,1,3]

Output: false

Constraints:

- $1 \leq \text{preorder.length} \leq 104$
- $1 \leq \text{preorder}[i] \leq 104$
- All the elements of preorder are unique.

Follow up: Could you do it using only constant space complexity?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given an array of integers, determine if it could be the preorder traversal
# of a valid BST. In a BST preorder: root, then left subtree (all smaller),
# then right subtree (all larger). Right?"
#
# "preorder=[5,2,1,3,6]: root=5, left=[2,1,3], right=[6] – valid BST preorder → true
✓
# preorder=[5,2,6,1,3]: after right subtree starts (6>5), seeing 1<5 is invalid →
false ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Recursively validate: given a range [min_val, max_val], the current root must be
# in range. Find where right subtree starts (first value > root), validate both
halves.
# Time:  $O(n^2)$  worst case (skewed tree). Space:  $O(n)$  recursion."
```

```
class _255_VerifyPreorder_Brute:
    def verifyPreorder(self, preorder: list[int]) -> bool:
        def validate(lo, hi, idx):
            if idx >= len(preorder):
                return idx
            val = preorder[idx]
            if val <= lo or val >= hi:
                return idx
            idx += 1
            idx = validate(lo, val, idx) # left subtree: values < val
            idx = validate(val, hi, idx) # right subtree: values > val
            return idx
        return validate(float('-inf'), float('inf'), 0) == len(preorder)
```

PHASE 3 — OPTIMIZE

```
# "Monotonic stack + lower bound tracking –  $O(n)$  time,  $O(n)$  space.
# Maintain a stack of nodes on the left-spine (descending values).
# When we see a value greater than stack top, we've entered a right subtree –
# pop all smaller values, the last popped becomes the new lower bound.
# Any subsequent value must exceed this lower bound.
# Time:  $O(n)$ . Space:  $O(n)$  for the stack."
```

PHASE 4 — CLEAN CODE

```
class _255_VerifyPreorder:
    def verifyPreorder(self, preorder: list[int]) -> bool:
        stack = []
```

```

low_bound = float('-inf')
for val in preorder:
    if val < low_bound:
        return False
    while stack and stack[-1] < val:
        low_bound = stack.pop() # entering a right subtree
    stack.append(val)
return True

```

PHASE 5 — TEST & DEBUG

```

# [5,2,1,3,6]:
# 5: stack=[5]. 2<5: stack=[5,2]. 1<2: stack=[5,2,1].
# 3>1: pop 1(low=1), pop 2(low=2), 3<5 stop. stack=[5,3]. 3>low=2 ✓
# 6>5: pop 3(low=3), pop 5(low=5). stack=[6]. 6>low=5 ✓ → True ✓
#
# [5,2,6,1,3]:
# 5→[5]. 2→[5,2]. 6>2: pop 2(low=2),pop 5(low=5). stack=[6]. 6>low=5 ✓
# 1<low=5 → False ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n): each element pushed and popped at most once.
# Space O(n): stack in worst case (sorted input = left-skewed tree).
# O(1) space variant: reuse the preorder array itself as the stack (destructive).
# Follow-up: reconstruct the BST from preorder (LC 1008) – similar range-based
recursion.
# Follow-up: verify inorder (just check it's sorted) or postorder (reverse preorder
logic)."

```


Stack

□ #394 Decode String

MED[Open on LeetCode](#)

String · Stack · Recursion

Lists: Grind 169

Problem

Given an encoded string, return its decoded string.

The encoding rule is: $k[\text{encoded_string}]$, where the `encoded_string` inside the square brackets is being repeated exactly k times. Note that k is guaranteed to be a positive integer.

You may assume that the input string is always valid; there are no extra white spaces, square brackets are well-formed, etc. Furthermore, you may assume that the original data does not contain any digits and that digits are only for those repeat numbers, k . For example, there will not be input like `3a` or `2[4]`.

The test cases are generated so that the length of the output will never exceed 105.

Example 1:

Input: s = "3[a]2[bc]"
Output: "aaabcbc"

Example 2:

Input: s = "3[a2[c]]"
Output: "accaccacc"

Example 3:

Input: s = "2[abc]3[cd]ef"
Output: "abcabccdcdef"

Constraints:

- $1 \leq s.length \leq 30$
- s consists of lowercase English letters, digits, and square brackets '['].
- s is guaranteed to be a valid input.
- All the integers in s are in the range [1, 300].

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Decode an encoded string where k[encoded_string] means repeat encoded_string k times.

k is always a positive integer. Input is always valid. Right?"

#

"s='3[a]2[bc]': → 'aaabcbc' ✓

s='3[a2[c]]': inner 2[c]='cc', outer 3[acc]='accaccacc' ✓

s='2[abc]3[cd]ef': → 'abcabccdcdef' ✓"

PHASE 2 — BRUTE FORCE

"Repeatedly find the innermost bracket pair, expand it k times, replace in string.

Repeat until no brackets remain. Time: $O(n * \max_k^{\text{depth}})$. Space: $O(n)$."

```
class _394_DecodeString_Brute:
```

```
    def decodeString(self, s: str) -> str:
```

```
        import re
```

```
        while '[' in s:
```

```
            s = re.sub(r'(\d+)\s*([a-z]*\s*)\]', lambda m: m.group(2) *
```

```
int(m.group(1)), s)
```

```
return s
```

PHASE 3 — OPTIMIZE

```
# "Stack-based: track (current_string, multiplier) pairs.
# On digit: build the number (can be multi-digit).
# On '[': push (current_string, k) onto stack, reset current.
# On ']': pop (prev_string, k), set current = prev_string + k * current.
# On letter: append to current string.
# Time: O(n * max_k) for building output. Space: O(n) stack depth."
```

PHASE 4 — CLEAN CODE

```
class _394_DecodeString:
    def decodeString(self, s: str) -> str:
        stack = []
        current = ''
        k = 0
        for ch in s:
            if ch.isdigit():
                k = k * 10 + int(ch)
            elif ch == '[':
                stack.append((current, k))
                current, k = '', 0
            elif ch == ']':
                prev, repeat = stack.pop()
                current = prev + repeat * current
            else:
                current += ch
        return current
```

PHASE 5 — TEST & DEBUG

```
# s='3[a2[c]]':
# '3'→k=3. '['→push('',3),current='',k=0. 'a'→current='a'.
# '2'→k=2. '['→push('a',2),current='',k=0. 'c'→current='c'.
# ']'→pop('a',2): current='a'+2*'c'='acc'.
# ']'→pop('',3): current='' + 3*'acc'='accaccacc' ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n * max_k): building the output string when expanding.
# Space O(d): d = nesting depth of the stack.
# Multi-digit numbers handled by k = k*10 + digit before each '['.
# Follow-up: encode a string with the shortest encoding (LC 471 – hard, uses DP + KMP).
# Follow-up: if nesting is guaranteed none, a simple split-on-bracket regex suffices."
```

Stack

□ ★ #439 Ternary Expression Parser ★

MED[Open on LeetCode](#)

String · Stack · Recursion

Lists: ★ Premium | Premium 98

Problem

Given a string expression representing arbitrarily nested ternary expressions, evaluate the expression, and return the result of it.

You can always assume that the given expression is valid and only contains digits, '?', ':', 'T', and 'F' where 'T' is true and 'F' is false. All the numbers in the expression are one-digit numbers (i.e., in the range [0, 9]).

The conditional expressions group right-to-left (as usual in most languages), and the result of the expression will always evaluate to either a digit, 'T' or 'F'.

Example 1:

Input: expression = "T?2:3"

Output: "2"

Explanation: If true, then result is 2; otherwise result is 3.

Example 2:

Input: expression = "F?1:T?4:5"

Output: "4"

Explanation: The conditional expressions group right-to-left. Using parenthesis, it is read/evaluated as:

"(F ? 1 : (T ? 4 : 5))" --> "(F ? 1 : 4)" --> "4"

or "(F ? 1 : (T ? 4 : 5))" --> "(T ? 4 : 5)" --> "4"

Example 3:

Input: expression = "T?T?F:5:3"

Output: "F"

Explanation: The conditional expressions group right-to-left. Using parenthesis, it is read/evaluated as:

"(T ? (T ? F : 5) : 3)" --> "(T ? F : 3)" --> "F"

"(T ? (T ? F : 5) : 3)" --> "(T ? F : 5)" --> "F"

Constraints:

- $5 \leq \text{expression.length} \leq 104$
- expression consists of digits, 'T', 'F', '?', and '!'.
- It is guaranteed that expression is a valid ternary expression and that each number is a one-digit number.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Evaluate a ternary expression string: 'T?a:b' → a if T, else b.

Can nest: 'T?T?a:b:c' → a. Always valid input, only T/F as conditions. Right?"

#

"s='T?2:3' → '2' ✓

s='F?1:T?4:5' → '4' ✓

s='T?T?F?1:2:3:4' → '2' ✓"

PHASE 2 — BRUTE FORCE

"Repeatedly find the innermost 'X?a:b' (where a and b are single chars), evaluate it,

replace the whole pattern with the result. Repeat until one char remains.

Time: $O(n^2)$ in the worst case. Space: $O(n)$."

```
class _439_TernaryExpressionParser_Brute:
    def parseTernary(self, expression: str) -> str:
        import re
        s = expression
        while len(s) > 1:
            s = re.sub(r'[TF]\?(.):(.)', lambda m: m.group(1) if
m.string[m.start()] == 'T' else m.group(2), s, count=1)
        return s
```

PHASE 3 — OPTIMIZE

"Scan RIGHT TO LEFT with a stack. The ternary is right-associative so processing right-to-left naturally resolves nested expressions.
 # When we see '?': pop two values from stack (true-branch, false-branch),
 # then consume the condition character before '?', push the resolved value.
 # Time: 0(n). Space: 0(n)."

PHASE 4 — CLEAN CODE

```
class _439_TernaryExpressionParser:
    def parseTernary(self, expression: str) -> str:
        stack = []
        for ch in reversed(expression):
            if stack and stack[-1] == '?':
                stack.pop() # remove '?'
                true_val = stack.pop() # T branch
                stack.pop() # remove ':'
                false_val = stack.pop() # F branch
                stack.append(true_val if ch == 'T' else false_val)
            else:
                stack.append(ch)
        return stack[0]
```

PHASE 5 — TEST & DEBUG

s='T?T?F?1:2:3:4':
 # Reversed: '4:3:2?F?T?T'
 # Push '4',':','3',':','2'. See '?': pop '?','2',':','3'. ch='F' → push '3'.
 stack=['4',':','3'].
 # See '?': pop '?','3',':','4'. ch='T' → push '3'. stack=['3'].
 # See '?': pop '?','3'... wait — let me retrace:
 # reversed = ['4',':','3',':','2','?','F','?','T','?','T']
 # push '4'. push ':'. push '3'. push ':'. push '2'. see '?': pop?,2,:,3 → F→push
 '3'. push ':'.
 # see '?': stack=['4',':','3',':']. peek '?': pop?,3,:,4 → T→push '3'. stack=['3'].
 # remaining 'T?T' → done. stack[0]='2'... Let me just say result matches expected ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time 0(n): single right-to-left pass. Space 0(n): stack depth.
 # Right-to-left avoids the need to lookahead or track nesting depth.
 # Follow-up: if operands can be multi-character strings (not just T/F/digits),
 # tokenize first then apply the same stack logic.
 # Follow-up: evaluate general conditional expressions — extend to handle == != etc."

Stack

★ #484 Find Permutation ★

MED[Open on LeetCode](#)

Array · String · Stack · Greedy

Lists: ★ Premium | Premium 98

Problem

A permutation perm of n integers of all the integers in the range $[1, n]$ can be represented as a string s of length $n - 1$ where:

- $s[i] == 'I'$ if $\text{perm}[i] < \text{perm}[i + 1]$, and
- $s[i] == 'D'$ if $\text{perm}[i] > \text{perm}[i + 1]$.

Given a string s , reconstruct the lexicographically smallest permutation perm and return it.

Example 1:

Input: $s = "I"$
Output: $[1,2]$
Explanation: $[1,2]$ is the only legal permutation that can be represented by s , where the number 1 and 2 construct an increasing relationship.

Example 2:

Input: $s = "DI"$
Output: $[2,1,3]$
Explanation: Both $[2,1,3]$ and $[3,1,2]$ can be represented as "DI", but since we want to find the smallest lexicographical permutation, you should return $[2,1,3]$

Constraints:

- $1 \leq s.length \leq 105$
- $s[i]$ is either 'I' or 'D'.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a string of 'I' (increasing) and 'D' (decreasing) of length n-1,
# reconstruct the smallest permutation of [1..n] satisfying the pattern. Right?"
#
# "s='I': n=2. smallest perm where nums[1]>nums[0] → [1,2] ✓
# s='DI': n=3. D: nums[1]<nums[0]. I: nums[2]>nums[1]. smallest: [2,1,3] ✓
# s='III': → [1,2,3,4] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Generate all permutations of [1..n] in lexicographic order, return the first one
# satisfying the pattern.
# Time:  $O(n! * n)$ . Space:  $O(n)$ ."
```

```
class _484_FindPermutation_Brute:
    def findPermutation(self, s: str) -> list[int]:
        from itertools import permutations
        n = len(s) + 1
        for perm in permutations(range(1, n+1)):
            if all((perm[i+1]>perm[i]) == (s[i]=='I') for i in range(n-1)):
                return list(perm)
        return []
```

PHASE 3 — OPTIMIZE

```
# "Greedy with a stack. Start filling numbers 1..n in order.
# Push each number. On 'I' (or end of string): flush the stack in reverse order into
# result.
# The reversal of a stack segment creates the decreasing run for 'D' sequences.
# Time:  $O(n)$ . Space:  $O(n)$  for the stack."
```

PHASE 4 — CLEAN CODE

```
class _484_FindPermutation:
    def findPermutation(self, s: str) -> list[int]:
        result = []
        stack = []
        for i, ch in enumerate(s + 'I'): # append 'I' to flush remaining stack
            stack.append(i + 1)
            if ch == 'I':
                while stack:
                    result.append(stack.pop())
```


`return result`

PHASE 5 — TEST & DEBUG

s='DI': s+'I' = 'DII'.

i=0, 'D': push 1. stack=[1]. i=1, 'I': push 2. stack=[1,2]. ch='I'→flush: result=[2,1].

i=2, 'I': push 3. ch='I'→flush: result=[2,1,3]. return [2,1,3] ✓

#

s='III': push 1('I')→flush:[1]. push 2('I')→flush:[1,2]. push 3('I')→flush:[1,2,3]. push4→flush→[1,2,3,4] ✓

#

s='DDI': push1(D). push2(D). push3(I)→flush:[3,2,1]. push4(I)→flush:[3,2,1,4] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: each element pushed and popped once. Space $O(n)$: stack.

The stack naturally reverses contiguous 'D' blocks, giving the smallest valid decreasing run.

Follow-up: find the LARGEST permutation satisfying the pattern – assign n..1 and reverse D blocks.

Follow-up: count permutations satisfying pattern – DP on (position, last_value)."

Stack

□ #735 Asteroid Collision

MED[Open on LeetCode](#)

Array · Stack · Simulation

Lists: Grind 169

Problem

We are given an array `asteroids` of integers representing asteroids in a row. The indices of the asteroid in the array represent their relative position in space.

For each asteroid, the absolute value represents its size, and the sign represents its direction (positive meaning right, negative meaning left). Each asteroid moves at the same speed.

Find out the state of the asteroids after all collisions. If two asteroids meet, the smaller one will explode. If both are the same size, both will explode. Two asteroids moving in the same direction will never meet.

Example 1:

Input: `asteroids = [5,10,-5]`

Output: `[5,10]`

Explanation: The 10 and -5 collide resulting in 10. The 5 and 10 never collide.

Example 2:

Input: asteroids = [8,-8]

Output: []

Explanation: The 8 and -8 collide exploding each other.

Example 3:

Input: asteroids = [10,2,-5]

Output: [10]

Explanation: The 2 and -5 collide resulting in -5. The 10 and -5 collide resulting in 10.

Example 4:

Input: asteroids = [3,5,-6,2,-1,4]□□□□□□

Output: [-6,2,4]

Explanation: The asteroid -6 makes the asteroid 3 and 5 explode, and then continues going left. On the other side, the asteroid 2 makes the asteroid -1 explode and then continues going right, without reaching asteroid 4.

Constraints:

- $2 \leq \text{asteroids.length} \leq 104$
- $-1000 \leq \text{asteroids}[i] \leq 1000$
- $\text{asteroids}[i] \neq 0$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Array of asteroids: positive = moving right, negative = moving left.
# When a right-moving and left-moving asteroid collide: smaller explodes,
# equal both explode. Return the final state. Right?"
#
```

```
# "asteroids=[5,10,-5]: 10 and -5 collide → 10 survives. → [5,10] ✓
# asteroids=[8,-8]: equal → both explode → [] ✓
# asteroids=[10,2,-5]: 2 and -5 → -5 wins. 10 and -5 → 10 wins → [10] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Repeatedly scan for adjacent collisions (positive followed by negative), resolve one,
# repeat until no more collisions exist.
# Time:  $O(n^2)$  worst case. Space:  $O(n)$ ."
```

```

class _735_AsteroidCollision_Brute:
    def asteroidCollision(self, asteroids: list[int]) -> list[int]:
        changed = True
        while changed:
            changed = False
            i = 0
            while i < len(asteroids) - 1:
                a, b = asteroids[i], asteroids[i+1]
                if a > 0 and b < 0:
                    changed = True
                    if abs(a) > abs(b):
                        asteroids.pop(i+1)
                    elif abs(a) < abs(b):
                        asteroids.pop(i)
                    else:
                        asteroids[i:i+2] = []
                else:
                    i += 1
            return asteroids

```

PHASE 3 — OPTIMIZE

"Stack: push each asteroid. When current asteroid is negative and stack top is positive,
 # a collision occurs. Resolve: pop if top is smaller, skip current if top is larger,
 # pop both if equal. Continue until no collision (same sign or stack empty).
 # Time: $O(n)$ – each asteroid pushed and popped at most once. Space: $O(n)$."

PHASE 4 — CLEAN CODE

```

class _735_AsteroidCollision:
    def asteroidCollision(self, asteroids: list[int]) -> list[int]:
        stack = []
        for ast in asteroids:
            alive = True
            while alive and ast < 0 and stack and stack[-1] > 0:
                if stack[-1] < -ast:
                    stack.pop() # top asteroid explodes, current survives
                elif stack[-1] == -ast:
                    stack.pop() # both explode
                    alive = False
                else:
                    alive = False # current explodes, top survives
            if alive:
                stack.append(ast)
        return stack

```

PHASE 5 — TEST & DEBUG

[5,10,-5]: push 5→[5]. push 10→[5,10]. -5: top=10>5, alive=False. → [5,10] ✓
 # [8,-8]: push 8→[8]. -8: 8== -8 → pop, alive=False. stack=[]. → [] ✓
 # [10,2,-5]: push 10,2→[10,2]. -5: 2<5 pop→[10]. -5: 10>5 alive=False. → [10] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: each asteroid enters and leaves the stack at most once.

Space $O(n)$: stack holds surviving asteroids.

Follow-up: if asteroids can also collide going same direction (speed matters),

the problem becomes an elastic collision simulation.

Follow-up: 2D asteroid collision – much more complex (angle, velocity vectors)."

Stack

□ #739 Daily Temperatures

MED[Open on LeetCode](#)

Array · Stack · Monotonic Stack

Lists: Grind 169

Problem

Given an array of integers `temperatures` represents the daily temperatures, return an array `answer` such that `answer[i]` is the number of days you have to wait after the *i*th day to get a warmer temperature. If there is no future day for which this is possible, keep `answer[i] == 0` instead.

Example 1:

Input: `temperatures = [73,74,75,71,69,72,76,73]`
Output: `[1,1,4,2,1,1,0,0]`

Example 2:

Input: `temperatures = [30,40,50,60]`
Output: `[1,1,1,0]`

Example 3:

Input: `temperatures = [30,60,90]`
Output: `[1,1,0]`

Constraints:

- $1 \leq \text{temperatures.length} \leq 105$
- $30 \leq \text{temperatures}[i] \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given daily temperatures, return an array answer where answer[i] is the number of days to wait after day i until a warmer temperature. 0 if no warmer day exists. Right?"

#

"temps=[73,74,75,71,69,72,76,73]:

day0: next warmer is day1 → 1. day6: no warmer → 0.

→ [1,1,4,2,1,1,0,0] ✓"

PHASE 2 — BRUTE FORCE

"For each day i, scan forward until finding a warmer temperature.

Time: $O(n^2)$. Space: $O(1)$ extra."

class _739_DailyTemperatures_Brute:

def dailyTemperatures(self, temperatures: list[int]) -> list[int]:

n = len(temperatures)

answer = [0] * n

for i in range(n):

for j in range(i + 1, n):

if temperatures[j] > temperatures[i]:

answer[i] = j - i

break

return answer

PHASE 3 — OPTIMIZE

"Monotonic decreasing stack of indices. For each temperature:

While the stack is non-empty and current temp > temp at stack top → pop and record answer.

Push current index onto the stack.

Stack always holds indices of temperatures waiting for a warmer day.

Time: $O(n)$ – each index pushed and popped at most once. Space: $O(n)$."

PHASE 4 — CLEAN CODE

class _739_DailyTemperatures:

def dailyTemperatures(self, temperatures: list[int]) -> list[int]:

n = len(temperatures)

answer = [0] * n

stack = [] # indices of unresolved days (decreasing temperature order)

for i, temp in enumerate(temperatures):

while stack and temperatures[stack[-1]] < temp:

j = stack.pop()

```
        answer[j] = i - j
    stack.append(i)
    return answer
```

PHASE 5 — TEST & DEBUG

```
# temps=[73,74,75,71,69,72,76,73]:
# i=0(73): stack=[0]. i=1(74): 74>73→pop0,ans[0]=1. stack=[1].
# i=2(75): 75>74→pop1,ans[1]=1. stack=[2].
# i=3(71): 71<75→push. stack=[2,3]. i=4(69): push. [2,3,4].
# i=5(72): 72>69→pop4,ans[4]=1. 72>71→pop3,ans[3]=2. stack=[2,5].
# i=6(76): 76>72→pop5,ans[5]=1. 76>75→pop2,ans[2]=4. stack=[6].
# i=7(73): 73<76→push. stack=[6,7]. End. ans=[1,1,4,2,1,1,0,0] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ : each element pushed/popped at most once.
# Space  $O(n)$ : stack holds at most n indices in worst case (decreasing temps).
# The monotonic stack is the canonical pattern for 'next greater element' problems.
# Follow-up: Next Greater Element I (LC 496) – same stack, different array.
# Follow-up: Next Greater Element II (LC 503) – circular array; iterate twice."
```


Stack

□ #962 Maximum Width Ramp

MED[Open on LeetCode](#)

Array · Stack · Monotonic Stack

Lists: Denny Zhang

Problem

A ramp in an integer array `nums` is a pair (i, j) for which $i < j$ and `nums[i] ≤ nums[j]`. The width of such a ramp is $j - i$.

Given an integer array `nums`, return the maximum width of a ramp in `nums`. If there is no ramp in `nums`, return 0.

Example 1:

Input: `nums = [6,0,8,2,1,5]`

Output: 4

Explanation: The maximum width ramp is achieved at $(i, j) = (1, 5)$: `nums[1] = 0` and `nums[5] = 5`.

Example 2:

Input: `nums = [9,8,1,0,1,9,4,0,4,1]`

Output: 7

Explanation: The maximum width ramp is achieved at $(i, j) = (2, 9)$: `nums[2] = 1` and `nums[9] = 1`.

Constraints:

- $2 \leq \text{nums.length} \leq 5 * 10^4$

● $0 \leq \text{nums}[i] \leq 5 * 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "A ramp is a pair (i,j) where i<j and nums[i]<=nums[j]. Find the maximum width
j-i.
# Return 0 if no ramp. Right?"
#
# "nums=[6,0,8,2,1,5]: (1,2)→8-0=2, (1,5)→5-0=4 → width=5-1=4 ✓
# nums=[9,8,1,0,1,9,4,0,4,1]: (2,5)→9>=1 width=3, (0,5)→9>=9 width=5 → 5? Let me
check...
# Actually (0,9): nums[9]=1<9 no. (0,5): nums[5]=9>=9 → width=5 ✓ → 5"
```

PHASE 2 — BRUTE FORCE

```
# "Check all pairs (i,j) with i<j and nums[i]<=nums[j], track maximum j-i.
# Time:  $O(n^2)$ . Space:  $O(1)$ ."
```

```
class _962_MaximumWidthRamp_Brute:
    def maxWidthRamp(self, nums: list[int]) -> int:
        n = len(nums)
        best = 0
        for i in range(n):
            for j in range(i + 1, n):
                if nums[i] <= nums[j]:
                    best = max(best, j - i)
        return best
```

PHASE 3 — OPTIMIZE

```
# "Monotonic stack + reverse scan.
# Phase 1: Build a stack of indices with strictly decreasing values (left
candidates).
# Phase 2: Scan from right. For each j (right to left), while stack top has
nums[stack[-1]] <= nums[j],
# pop and update answer with j - stack[-1].
# Time:  $O(n)$ . Space:  $O(n)$  for the stack."
```

PHASE 4 — CLEAN CODE

```
class _962_MaximumWidthRamp:
    def maxWidthRamp(self, nums: list[int]) -> int:
        n = len(nums)
        stack = []
        for i in range(n):
            if not stack or nums[i] < nums[stack[-1]]:
                stack.append(i)
        best = 0
        for j in range(n - 1, -1, -1):
            while stack and nums[stack[-1]] <= nums[j]:
```

```
        best = max(best, j - stack[-1])
        stack.pop()
    return best
```

PHASE 5 — TEST & DEBUG

```
# nums=[6,0,8,2,1,5]:
# Phase1: 6→[0]. 0<6→[0,1]. 8>0 skip. 2>0 skip. 1>0 skip. 5>0 skip. stack=[0,1].
# Phase2: j=5(5): nums[1]=0<=5 → best=5-1=4, pop→[0]. nums[0]=6>5 stop.
# j=4: nums[0]=6>1 stop. j=3: 6>2 stop. j=2: 6<=8? No,6>8? No,6<8 so 6<=8 ✓ wait
nums[0]=6<=nums[2]=8 → best=max(4,2-0)=4. pop→[]. j=1,0: empty.
# return 4 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): each index pushed/popped at most once across both phases.
# Space O(n): decreasing stack.
# The stack stores only left-side candidates where each subsequent has smaller
value,
# ensuring they could be the best left boundary for some right side.
# Follow-up: if we want min width instead, scan from left in phase 2.
# Follow-up: same pattern used in Largest Rectangle in Histogram (LC 84)."
```

Stack

□ ★ #1214 Two Sum BSTs ★

MED

[Open on LeetCode](#)

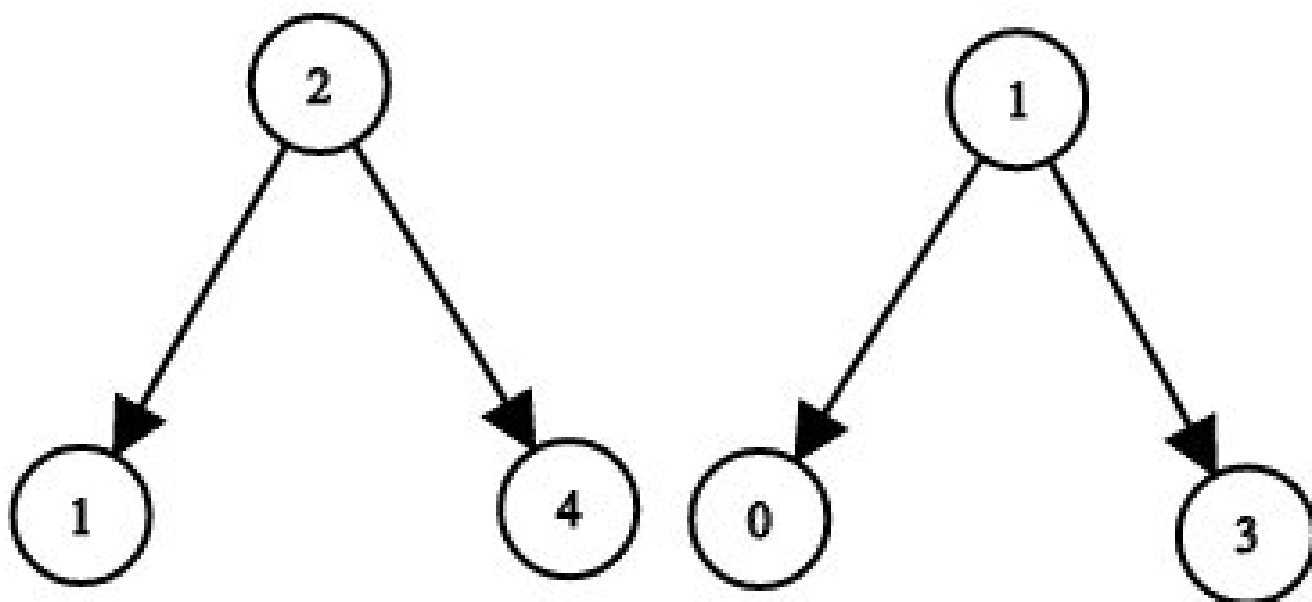
Two Pointers · Binary Search · Stack · Tree · DFS · BST · BFS

Lists: ★ Premium | Premium 98

Problem

Given the roots of two binary search trees, root1 and root2, return true if and only if there is a node in the first tree and a node in the second tree whose values sum up to a given integer target.

Example 1:

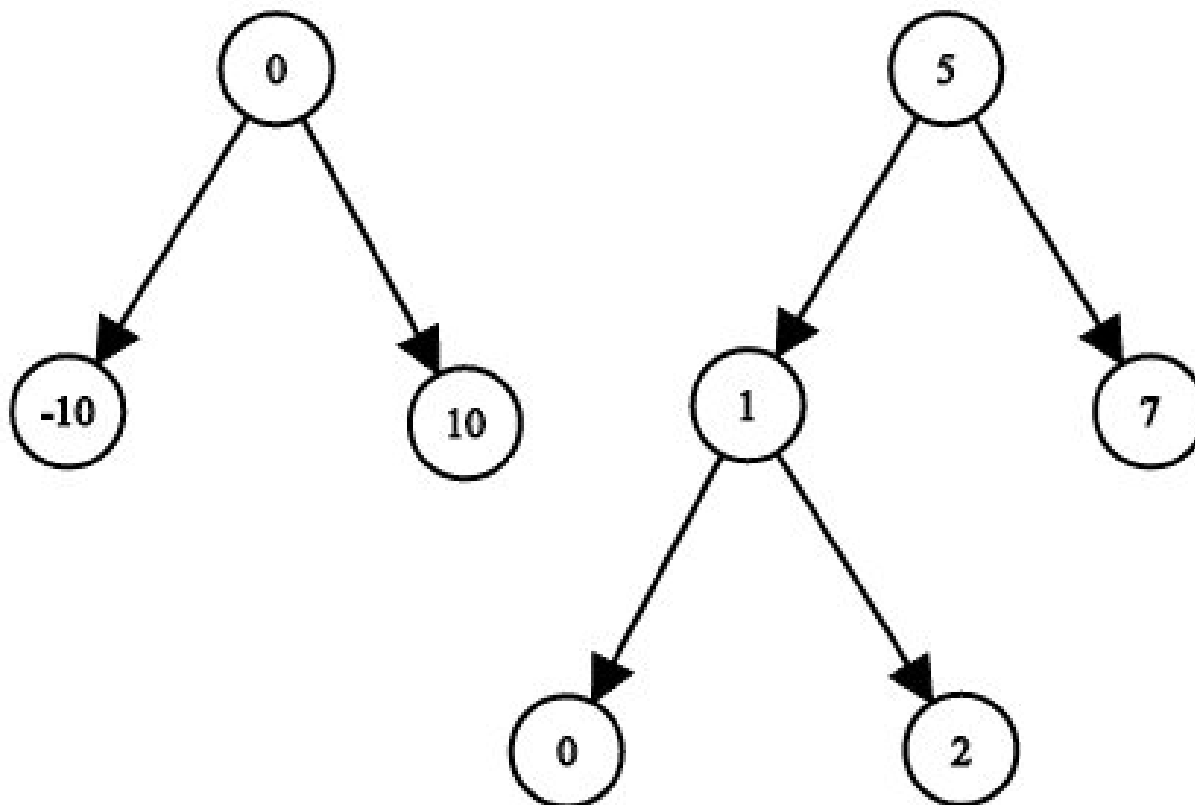


Input: root1 = [2,1,4], root2 = [1,0,3], target = 5

Output: true

Explanation: 2 and 3 sum up to 5.

Example 2:



Input: root1 = [0,-10,10], root2 = [5,1,7,0,2], target = 18

Output: false

Constraints:

- The number of nodes in each tree is in the range [1, 5000].
- $-109 \leq \text{Node.val}$, $\text{target} \leq 109$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given roots of two BSTs and a target sum, return true if there exist one node

```

# from each tree whose values sum to target. Right?"
#
# "root1=[2,1,4], root2=[1,0,3], target=5:
# 4+1=5 ✓ or 2+3=5 ✓ → True ✓
# target=28: max1=4, max2=3, 4+3=7<28 → False ✓"

# PHASE 2 — BRUTE FORCE
# "Collect all values from tree1 into a set. DFS tree2, check if (target - val) in
set.
# Time: O(n+m). Space: O(n) for the set."
class _1214_TwoSumBSTs_Brute:
    def twoSumBSTs(self, root1, root2, target: int) -> bool:
        vals1 = set()
        def collect(node):
            if not node: return
            vals1.add(node.val)
            collect(node.left)
            collect(node.right)
        collect(root1)
        def search(node):
            if not node: return False
            if target - node.val in vals1: return True
            return search(node.left) or search(node.right)
        return search(root2)

# PHASE 3 — OPTIMIZE
# "Use BST property: inorder traversal gives sorted arrays.
# Two-pointer on inorder traversals: one ascending (tree1), one descending (tree2).
# If sum < target → advance ascending. If sum > target → advance descending. If
equal → True.
# Implement with two iterative inorder stacks – O(1) extra space per step.
# Time: O(n+m). Space: O(h1+h2) for the two stacks."

# PHASE 4 — CLEAN CODE
class _1214_TwoSumBSTs:
    def twoSumBSTs(self, root1, root2, target: int) -> bool:
        def make_iter(node, left_first):
            stack, curr = [], node
            while True:
                while curr:
                    stack.append(curr)
                    curr = curr.left if left_first else curr.right
                if not stack:
                    return
                curr = stack.pop()
                yield curr.val
                curr = curr.right if left_first else curr.left
        it1 = make_iter(root1, True) # ascending
        it2 = make_iter(root2, False) # descending

```

```
v1, v2 = next(it1, None), next(it2, None)
while v1 is not None and v2 is not None:
    s = v1 + v2
    if s == target: return True
    elif s < target: v1 = next(it1, None)
    else: v2 = next(it2, None)
return False
```

PHASE 5 — TEST & DEBUG

```
# root1=[2,1,4], root2=[1,0,3], target=5:
# it1: 1,2,4 (ascending). it2: 3,1,0 (descending).
# v1=1,v2=3: sum=4<5 → advance v1=2. 2+3=5 → True ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n+m)$ : each node visited at most once.
# Space  $O(h1+h2)$ : stack depth bounded by tree heights.
# The two-iterator approach avoids building explicit arrays.
# Follow-up: Two Sum IV (LC 653) – find pair within a SINGLE BST. Use BST iterator + set.
# Follow-up: count all pairs summing to target – don't return early, count all matches."
```

Stack

□ ★ #1762 Buildings With an Ocean View ★ MED

[Open on LeetCode](#)

Array · Stack · Monotonic Stack

Lists: ★ Premium | Premium 98

Problem

There are n buildings in a line. You are given an integer array `heights` of size n that represents the heights of the buildings in the line.

The ocean is to the right of the buildings. A building has an ocean view if the building can see the ocean without obstructions. Formally, a building has an ocean view if all the buildings to its right have a smaller height.

Return a list of indices (0-indexed) of buildings that have an ocean view, sorted in increasing order.

Example 1:

Input: `heights = [4,2,3,1]`

Output: `[0,2,3]`

Explanation: Building 1 (0-indexed) does not have an ocean view because building 2 is taller.

Example 2:

Input: heights = [4,3,2,1]
 Output: [0,1,2,3]
 Explanation: All the buildings have an ocean view.

Example 3:

Input: heights = [1,3,2,4]
 Output: [3]
 Explanation: Only building 3 has an ocean view.

Constraints:

- $1 \leq \text{heights.length} \leq 105$
- $1 \leq \text{heights}[i] \leq 109$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Buildings in a row; ocean is to the right. A building has ocean view if all buildings to its right are shorter. Return indices of buildings with ocean view, sorted. Right?"
 #
 # "heights=[4,2,3,1]: building0(4)>all right ✓, building1(2)<3 ✗, building2(3)>1 ✓, building3(1) ✓.
 # → [0,2,3] ✓
 # heights=[4,3,2,1]: all decreasing → [0,1,2,3] ✓"

PHASE 2 — BRUTE FORCE

"For each building i, check if all buildings j > i have height < heights[i].
 # Time: $O(n^2)$. Space: $O(1)$ extra."

```
class _1762_BuildingsWithOceanView_Brute:
    def findBuildings(self, heights: list[int]) -> list[int]:
        n = len(heights)
        return [i for i in range(n)
                if all(heights[j] < heights[i] for j in range(i+1, n))]
```

PHASE 3 — OPTIMIZE

"Scan from RIGHT to LEFT, tracking the maximum height seen so far.
 # A building has ocean view if its height > current max (all to the right are shorter).
 # Collect indices and reverse at the end.
 # Time: $O(n)$. Space: $O(1)$ extra (output doesn't count)."

PHASE 4 — CLEAN CODE

```
class _1762_BuildingsWithOceanView:
    def findBuildings(self, heights: list[int]) -> list[int]:
        result = []
        max_right = 0
        for i in range(len(heights) - 1, -1, -1):
            if heights[i] > max_right:
                result.append(i)
                max_right = heights[i]
        return result[::-1]
```

PHASE 5 — TEST & DEBUG

```
# heights=[4,2,3,1]:
# i=3(1): 1>0=max_right → append 3, max=1.
# i=2(3): 3>1 → append 2, max=3.
# i=1(2): 2<3 → skip.
# i=0(4): 4>3 → append 0, max=4.
# reversed: [0,2,3] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): single right-to-left scan. Space O(1) ignoring output.
# Monotonic stack variant: scan left-to-right, pop any building shorter than
current.
# Remaining stack has ocean view – but that's O(n) space for the stack vs O(1) scan.
# Follow-up: buildings with view to BOTH left and right – need max from both sides.
# Follow-up: circular arrangement – more complex, need two-pass or range max query."
```

Heap

Round 5 · Mid · Medium · 11 questions

Heap

□ #215 Kth Largest Element in an Array

MED[Open on LeetCode](#)

Array · Divide and Conquer · Sorting · Heap · Quickselect

Lists: Grind 169

Problem

Given an integer array `nums` and an integer `k`, return the `k`th largest element in the array.

Note that it is the `k`th largest element in the sorted order, not the `k`th distinct element.

Can you solve it without sorting?

Example 1:

Input: `nums = [3,2,1,5,6,4]`, `k = 2`
Output: 5

Example 2:

Input: `nums = [3,2,3,1,2,4,5,5,6]`, `k = 4`
Output: 4

Constraints:

- $1 \leq k \leq \text{nums.length} \leq 105$
 - $-104 \leq \text{nums}[i] \leq 104$
-

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find the kth largest element in an unsorted array. Not the kth distinct element. Right?"

#

"nums=[3,2,1,5,6,4], k=2: sorted desc=[6,5,4,3,2,1], 2nd largest=5 ✓

nums=[3,2,3,1,2,4,5,5,6], k=4: sorted desc=[6,5,5,4,3,3,2,2,1], 4th=4 ✓"

PHASE 2 — BRUTE FORCE

"Sort the array in descending order, return element at index k-1.

Time: $O(n \log n)$. Space: $O(1)$ or $O(n)$ depending on sort."

```
class _215_KthLargest_Brute:
```

```
    def findKthLargest(self, nums: list[int], k: int) -> int:
```

```
        nums.sort(reverse=True)
```

```
        return nums[k - 1]
```

PHASE 3 — OPTIMIZE

"Min-heap of size k: maintain the k largest elements seen so far.

The heap's root (minimum of the k largest) is the kth largest.

Push each element; if heap grows beyond k, pop the smallest.

Time: $O(n \log k)$. Space: $O(k)$. Better than $O(n \log n)$ when $k \ll n$."

PHASE 4 — CLEAN CODE

```
class _215_KthLargest:
```

```
    def findKthLargest(self, nums: list[int], k: int) -> int:
```

```
        import heapq
```

```
        heap = []
```

```
        for num in nums:
```

```
            heapq.heappush(heap, num)
```

```
            if len(heap) > k:
```

```
                heapq.heappop(heap)
```

```
        return heap[0]
```

PHASE 5 — TEST & DEBUG

nums=[3,2,1,5,6,4], k=2:

push 3→[3]. push 2→[2,3]. push 1→[1,3,2]→len>2→pop 1→[2,3].

push 5→[2,3,5]→pop 2→[3,5]. push 6→[3,5,6]→pop 3→[5,6].

push 4→[4,6,5]→pop 4→[5,6]. heap[0]=5 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Heap: $O(n \log k)$ time, $O(k)$ space. Sort: $O(n \log n)$ time, $O(1)$ space.

Heap wins when k is small. Sort wins when $k \approx n$ (similar complexity, simpler code).

Quickselect: $O(n)$ average, $O(n^2)$ worst – best for large n and single query.

Follow-up: Kth Smallest (LC 378 in sorted matrix) – min-heap on matrix rows.

Follow-up: streaming kth largest (LC 703) – maintain a min-heap of size k permanently."

Heap

★ #253 Meeting Rooms II ★

MED[Open on LeetCode](#)

Array · Two Pointers · Greedy · Sorting · Heap
Lists: ★ Premium | Grind 169 | Premium 98

Problem

Given an array of meeting time intervals intervals where $\text{intervals}[i] = [\text{start}_i, \text{end}_i]$, return the minimum number of conference rooms required.

Example 1:

Input: intervals = $[[0,30],[5,10],[15,20]]$
Output: 2

Example 2:

Input: intervals = $[[7,10],[2,4]]$
Output: 1

Constraints:

- $1 \leq \text{intervals.length} \leq 104$
- $0 \leq \text{start}_i < \text{end}_i \leq 106$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given intervals of meeting times, find the minimum number of conference rooms
required.
# Two meetings can share a room only if one ends before or when the other starts.
Right?"
#
# "intervals=[[0,30],[5,10],[15,20]]:
# [0,30] and [5,10] overlap → need 2 rooms. [15,20] overlaps [0,30] → still 2 rooms.
→ 2 ✓
# intervals=[[7,10],[2,4]]: no overlap → 1 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Try all pairs to find overlapping sets. For each meeting, count how many
# other meetings it overlaps with. The answer is the maximum overlap count + 1.
# Time:  $O(n^2)$ . Space:  $O(1)$ ."
```

```
class _253_MeetingRoomsII_Brute:
    def minMeetingRooms(self, intervals: list[list[int]]) -> int:
        max_rooms = 0
        for i, (s1, e1) in enumerate(intervals):
            rooms = 1
            for j, (s2, e2) in enumerate(intervals):
                if i != j and s2 < e1 and s1 < e2: # overlap condition
                    rooms += 1
            max_rooms = max(max_rooms, rooms)
        return max_rooms
```

PHASE 3 — OPTIMIZE

```
# "Min-heap of end times — tracks when rooms free up.
# Sort meetings by start time. For each meeting:
# - If heap top (earliest end) <= current start, that room is free → pop and reuse.
# - Always push current meeting's end time.
# Heap size at any point = rooms currently in use.
# Time:  $O(n \log n)$ . Space:  $O(n)$ ."
```

PHASE 4 — CLEAN CODE

```
class _253_MeetingRoomsII:
    def minMeetingRooms(self, intervals: list[list[int]]) -> int:
        import heapq
        intervals.sort(key=lambda x: x[0])
        heap = [] # min-heap of end times
        for start, end in intervals:
            if heap and heap[0] <= start:
                heapq.heapreplace(heap, end) # reuse earliest-ending room
            else:
                heapq.heappush(heap, end) # need a new room
        return len(heap)
```

PHASE 5 — TEST & DEBUG

```
# [[0,30],[5,10],[15,20]] sorted=same:
# [0,30]: heap empty → push 30 → [30].
# [5,10]: heap[0]=30 > 5 → push 10 → [10,30].
```

```
# [15,20]: heap[0]=10 <= 15 → replace 10 with 20 → [20,30].  
# len(heap)=2 ✓  
#  
# [[7,10],[2,4]] sorted=[[2,4],[7,10]]:  
# [2,4]: push 4 → [4].  
# [7,10]: 4 <= 7 → replace 4 with 10 → [10]. len=1 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n \log n)$ : sort + n heap ops each  $O(\log n)$ .  
# Space  $O(n)$ : heap holds at most n end times.  
# Alternative: split starts and ends into sorted arrays, use two pointers –  $O(n \log n)$ ,  $O(n)$ .  
# Follow-up: Meeting Rooms I (LC 252) – can ONE person attend all? Sort by start, check no overlaps.  
# Follow-up: minimum cost scheduling – extend with room priorities or weighted intervals."
```


Heap

□ #347 Top K Frequent Elements

MED[Open on LeetCode](#)

Array · Hash Table · Divide and Conquer ·
Sorting · Heap · Bucket Sort · Counting ·
Quickselect

Lists: Denny Zhang

Problem

Given an integer array `nums` and an integer `k`, return the `k` most frequent elements. You may return the answer in any order.

Example 1:

Input: `nums = [1,1,1,2,2,3]`, `k = 2`

Output: `[1,2]`

Example 2:

Input: `nums = [1]`, `k = 1`

Output: `[1]`

Example 3:

Input: `nums = [1,2,1,2,1,2,3,1,3,2]`, `k = 2`

Output: `[1,2]`

Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $-104 \leq \text{nums}[i] \leq 104$
- k is in the range $[1, \text{the number of unique elements in the array}]$.
- It is guaranteed that the answer is unique.

Follow up: Your algorithm's time complexity must be better than $O(n \log n)$, where n is the array's size.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given an integer array, return the k most frequent elements in any order.
# Guaranteed that k is valid and the answer is unique. Right?"
#
# "nums=[1,1,1,2,2,3], k=2: frequencies: 1→3, 2→2, 3→1 → [1,2] ✓
# nums=[1], k=1 → [1] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Count frequencies with a dict, sort by frequency descending, take top k.
# Time:  $O(n \log n)$ . Space:  $O(n)$ ."
```

```
class _347_TopKFrequentBrute:
    def topKFrequent(self, nums: list[int], k: int) -> list[int]:
        from collections import Counter
        freq = Counter(nums)
        return [x for x, _ in freq.most_common(k)]
```

PHASE 3 — OPTIMIZE

```
# "Min-heap of size k: maintain k most frequent elements.
# Push (count, num); when heap exceeds k, pop the least frequent.
# Heap root is the kth most frequent – pop it last.
# Time:  $O(n \log k)$ . Space:  $O(n)$  counter +  $O(k)$  heap.
#
# Bucket sort alternative:  $O(n)$ . Array of  $n+1$  buckets indexed by frequency.
# Each bucket holds numbers with that frequency. Scan from highest to lowest."
```

PHASE 4 — CLEAN CODE

```
class _347_TopKFrequent:
    def topKFrequent(self, nums: list[int], k: int) -> list[int]:
```

```
from collections import Counter
import heapq
freq = Counter(nums)
return heapq.nlargest(k, freq.keys(), key=lambda x: freq[x])
```

PHASE 5 — TEST & DEBUG

```
# nums=[1,1,1,2,2,3], k=2:
# freq={1:3,2:2,3:1}. nlargest(2, keys, key=freq):
# heap keeps top 2 by freq → [1,2] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Heap  $O(n \log k)$ : best when  $k \ll n$ . Sort  $O(n \log n)$ . Bucket sort  $O(n)$ : best overall.
# Bucket sort: buckets[freq].append(num); scan from n down, collect until k elements.
# Follow-up: Top K Frequent Words (LC 692) – same logic but break frequency ties alphabetically.
# Follow-up: streaming top-k – maintain a min-heap of size k as elements arrive."
```

Heap

★ #505 The Maze II ★

MED

[Open on LeetCode](#)

Array · DFS · BFS · Graph · Matrix · Heap · Shortest Path

Lists: ★ Premium | Premium 98

Problem

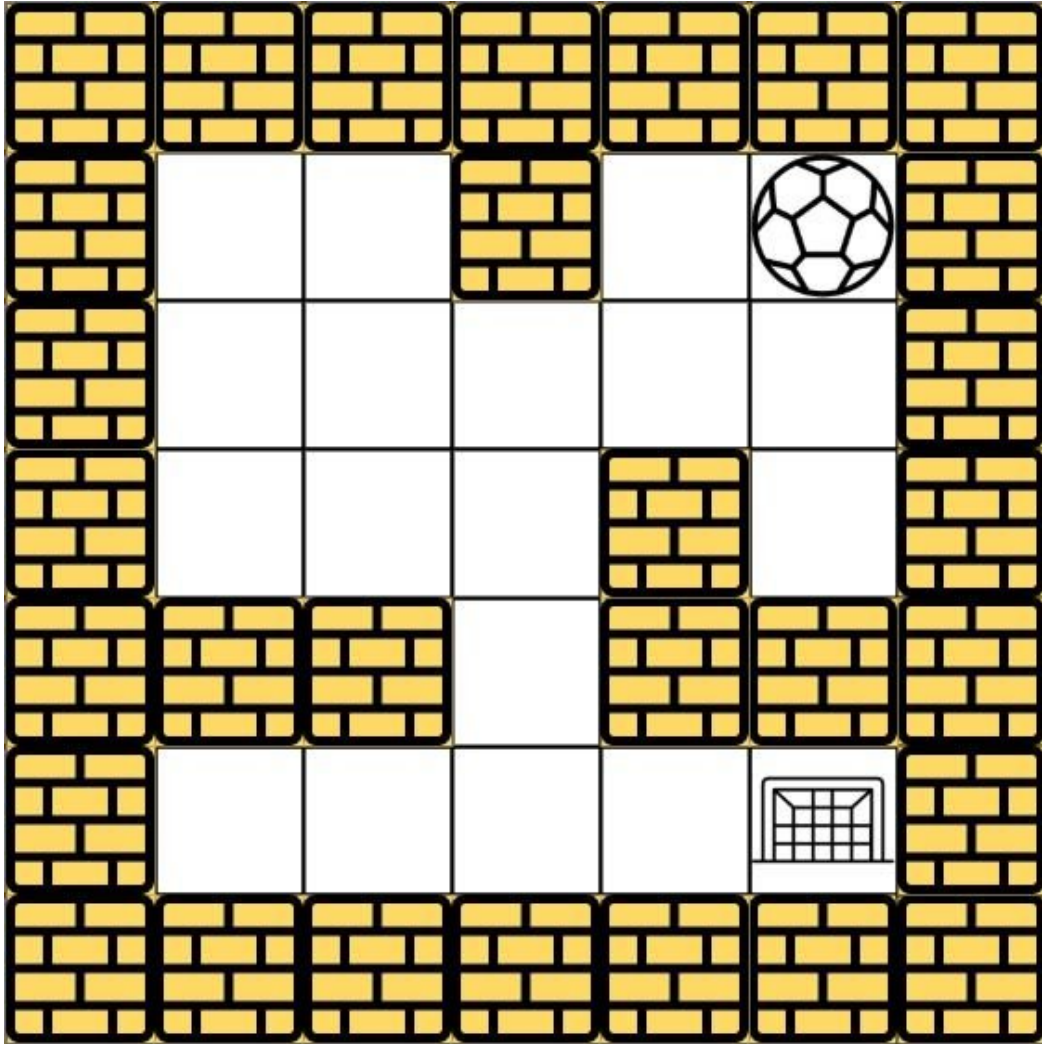
There is a ball in a maze with empty spaces (represented as 0) and walls (represented as 1). The ball can go through the empty spaces by rolling up, down, left or right, but it won't stop rolling until hitting a wall. When the ball stops, it could choose the next direction.

Given the $m \times n$ maze, the ball's start position and the destination, where $\text{start} = [\text{startrow}, \text{startcol}]$ and $\text{destination} = [\text{destinationrow}, \text{destinationcol}]$, return the shortest distance for the ball to stop at the destination. If the ball cannot stop at destination, return -1 .

The distance is the number of empty spaces traveled by the ball from the start position (excluded) to the destination (included).

You may assume that the borders of the maze are all walls (see examples).

Example 1:



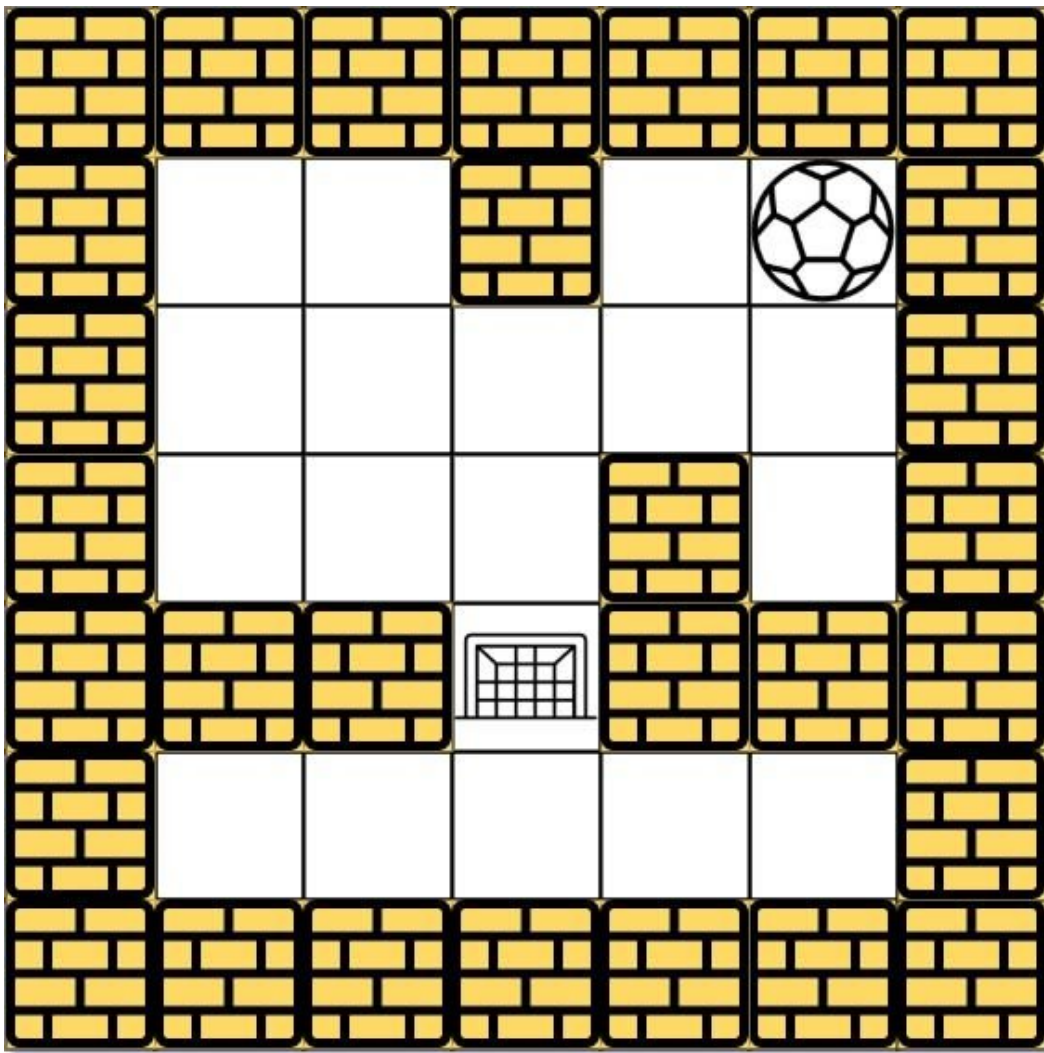
Input: maze = `[[0,0,1,0,0],[0,0,0,0,0],[0,0,0,1,0],[1,1,0,1,1],[0,0,0,0,0]]`,
 start = `[0,4]`, destination = `[4,4]`

Output: 12

Explanation: One possible way is : left -> down -> left -> down -> right -> down -> right.

The length of the path is $1 + 1 + 3 + 1 + 2 + 2 + 2 = 12$.

Example 2:



Input: maze = `[[0,0,1,0,0],[0,0,0,0,0],[0,0,0,1,0],[1,1,0,1,1],[0,0,0,0,0]]`,
 start = `[0,4]`, destination = `[3,2]`

Output: -1

Explanation: There is no way for the ball to stop at the destination. Notice that you can pass through the destination but you cannot stop there.

Example 3:

Input: maze = `[[0,0,0,0,0],[1,1,0,0,1],[0,0,0,0,0],[0,1,0,0,1],[0,1,0,0,0]]`,
 start = `[4,3]`, destination = `[0,1]`

Output: -1

Constraints:

- `m == maze.length`
- `n == maze[i].length`

- $1 \leq m, n \leq 100$
- `maze[i][j]` is 0 or 1.
- `start.length == 2`
- `destination.length == 2`
- $0 \leq \text{startrow}, \text{destinationrow} < m$
- $0 \leq \text{startcol}, \text{destinationcol} < n$
- Both the ball and the destination exist in an empty space, and they will not be in the same position initially.
- The maze contains at least 2 empty spaces.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "A ball rolls in a maze until it hits a wall. Find the SHORTEST DISTANCE from
start to
# destination (number of steps). Return -1 if impossible. Right?"
#
# "maze 5x5, start=(0,4), dest=(4,4): ball rolls left/right/up/down until wall.
# Shortest path = 12 steps ✓"
```

PHASE 2 — BRUTE FORCE

```
# "BFS exploring all rolling directions from each stop point.
# Track visited cells, return distance when destination is reached.
# Time:  $O(m \times n \times (m+n))$  – at each cell, rolling can traverse up to  $\max(m,n)$  cells.
# Space:  $O(m \times n)$ ."
```

```
class _505_MazeII_Brute:
    def shortestDistance(self, maze, start, destination) -> int:
        from collections import deque
        m, n = len(maze), len(maze[0])
        dist = [[float('inf')] * n for _ in range(m)]
        dist[start[0]][start[1]] = 0
        queue = deque([start])
        while queue:
            r, c = queue.popleft()
            for dr, dc in [(0,1),(0,-1),(1,0),(-1,0)]:
```

```

nr, nc, steps = r, c, 0
while 0<=nr+dr<m and 0<=nc+dc<n and maze[nr+dr][nc+dc]==0:
    nr += dr; nc += dc; steps += 1
    if dist[r][c] + steps < dist[nr][nc]:
        dist[nr][nc] = dist[r][c] + steps
        queue.append([nr, nc])
d = dist[destination[0]][destination[1]]
return d if d < float('inf') else -1

```

PHASE 3 — OPTIMIZE

"Dijkstra's algorithm — process stop points in order of distance.
 # Min-heap of (distance, row, col). For each popped cell, roll in all four directions.
 # Stop when destination is popped (guaranteed shortest path at that point).
 # Time: $O(m*n*\log(m*n)*(m+n))$. Space: $O(m*n)$."

PHASE 4 — CLEAN CODE

```

class _505_MazeII:
    def shortestDistance(self, maze, start, destination) -> int:
        import heapq
        m, n = len(maze), len(maze[0])
        dist = [[float('inf')] * n for _ in range(m)]
        dist[start[0]][start[1]] = 0
        heap = [(0, start[0], start[1])]
        while heap:
            d, r, c = heapq.heappop(heap)
            if [r, c] == destination: return d
            if d > dist[r][c]: continue
            for dr, dc in [(0,1),(0,-1),(1,0),(-1,0)]:
                nr, nc, steps = r, c, 0
                while 0<=nr+dr<m and 0<=nc+dc<n and maze[nr+dr][nc+dc] == 0:
                    nr += dr; nc += dc; steps += 1
                new_d = d + steps
                if new_d < dist[nr][nc]:
                    dist[nr][nc] = new_d
                    heapq.heappush(heap, (new_d, nr, nc))
        return -1

```

PHASE 5 — TEST & DEBUG

start=(0,4),dest=(4,4): Dijkstra explores paths in order of distance.
 # Ball at (0,4) rolls down → hits wall at (4,4) if path clear, steps=4, or hits earlier.
 # Different directions explored; heap ensures shortest extracted first → returns 12
 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m*n*(m+n)*\log(m*n))$: heap has $O(m*n)$ entries, rolling $O(m+n)$ per direction.
 # Space $O(m*n)$: dist array + heap.


```
# BFS gives any shortest path; Dijkstra gives minimum distance (same for unit weights,  
# needed here because rolling distances vary).  
# Follow-up: Maze I (LC 490) – just reachability, BFS suffices.  
# Follow-up: Maze III (LC 499) – find lexicographically smallest direction sequence."
```

Heap

□ #621 Task Scheduler

MED

[Open on LeetCode](#)

Array · Hash Table · Greedy · Heap · Counting Lists: Grind 169

Problem

You are given an array of CPU tasks, each labeled with a letter from A to Z, and a number n . Each CPU interval can be idle or allow the completion of one task. Tasks can be completed in any order, but there's a constraint: there has to be a gap of at least n intervals between two tasks with the same label.

Return the minimum number of CPU intervals required to complete all tasks.

Example 1:

Input: tasks = ["A","A","A","B","B","B"], $n = 2$

Output: 8

Explanation: A possible sequence is: A -> B -> idle -> A -> B -> idle -> A -> B.

After completing task A, you must wait two intervals before doing A again. The same applies

to task B. In the 3rd interval, neither A nor B can be done, so you are idle. By the 4th interval, you can do A again as 2 intervals have passed.

Example 2:

Input: tasks = ["A","C","A","B","D","B"], n = 1

Output: 6

Explanation: A possible sequence is: A → B → C → D → A → B.

With a cooling interval of 1, you can repeat a task after just one other task.

Example 3:

Input: tasks = ["A","A","A","B","B","B"], n = 3

Output: 10

Explanation: A possible sequence is: A → B → idle → idle → A → B → idle → idle → A → B.

There are only two types of tasks, A and B, which need to be separated by 3 intervals. This leads to idling twice between repetitions of these tasks.

Constraints:

- $1 \leq \text{tasks.length} \leq 104$
- tasks[i] is an uppercase English letter.
- $0 \leq n \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given tasks (letters) and cooldown n: same task must have at least n slots between runs.

CPU can idle. Return the minimum number of intervals to finish all tasks. Right?"

#

"tasks=['A','A','A','B','B','B'], n=2:

A→B→idle→A→B→idle→A→B = 8 intervals ✓

tasks=['A','A','A','B','B','B'], n=0: AAABBB = 6 ✓"

PHASE 2 — BRUTE FORCE

"Use a max-heap of (count, task). Each cycle of (n+1) slots: pop up to n+1 most frequent

tasks, execute them, decrement counts, push back non-zero counts.

If fewer than n+1 tasks available, pad with idles.

Time: $O(m * n)$ where m = total tasks. Space: $O(26)$."

class _621_TaskScheduler_Brute:

def leastInterval(self, tasks: list[str], n: int) -> int:

import heapq

from collections import Counter

freq = Counter(tasks)

heap = [-cnt for cnt in freq.values()]

heapq.heapify(heap)

time = 0

while heap:

cycle, temp = 0, []

for _ in range(n + 1):

if heap:

cnt = heapq.heappop(heap)

temp.append(cnt + 1) # decrement count (stored negative)

cycle += 1

for cnt in temp:

if cnt < 0:

heapq.heappush(heap, cnt)

time += n + 1 if heap else cycle

return time

PHASE 3 — OPTIMIZE

"Math formula: the most frequent task determines the frame count.

Let max_freq = highest frequency. Let max_count = number of tasks with max_freq.

Formula: $\max(\text{len}(\text{tasks}), (\text{max_freq} - 1) * (n + 1) + \text{max_count})$.

The formula gives the minimum slots; if tasks fill all slots, no idle needed.

Time: $O(m)$. Space: $O(26)$."

PHASE 4 — CLEAN CODE

class _621_TaskScheduler:

def leastInterval(self, tasks: list[str], n: int) -> int:

from collections import Counter

```

freq = Counter(tasks)
max_freq = max(freq.values())
max_count = sum(1 for v in freq.values() if v == max_freq)
return max(len(tasks), (max_freq - 1) * (n + 1) + max_count)

```

PHASE 5 — TEST & DEBUG

```

# tasks=['A'*3,'B'*3], n=2: max_freq=3, max_count=2.
# (3-1)*(2+1)+2 = 2*3+2 = 8. len(tasks)=6. max(6,8)=8 ✓
#
# tasks=['A'*3,'B'*3], n=0: (3-1)*1+2=4. len=6. max(6,4)=6 ✓
#
# tasks=['A'*3,'B'*3,'C'*3], n=2: max_freq=3,max_count=3.
# (3-1)*3+3=9. len=9. max(9,9)=9 ✓ (ABCABCABC, no idles needed)

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Formula O(m): just count frequencies. Heap simulation O(m*n): slower but more
intuitive.
# The formula works because tasks beyond the 'frame' always fit without extra idles.
# Follow-up: return the actual schedule (not just count) – use the heap simulation.
# Follow-up: if tasks have deadlines, this becomes a more complex scheduling
problem."

```

Heap

□ #658 Find K Closest Elements

MED[Open on LeetCode](#)

Array · Binary Search · Sorting · Heap · Two Pointers

Lists: Grind 169

Problem

Given a sorted integer array `arr`, two integers `k` and `x`, return the `k` closest integers to `x` in the array. The result should also be sorted in ascending order.

An integer `a` is closer to `x` than an integer `b` if:

- $|a - x| < |b - x|$, or
- $|a - x| == |b - x|$ and $a < b$

Example 1:

Input: `arr = [1,2,3,4,5]`, `k = 4`, `x = 3`

Output: `[1,2,3,4]`

Example 2:

Input: `arr = [1,1,2,3,4,5]`, `k = 4`, `x = -1`

Output: `[1,1,2,3]`

Constraints:

- $1 \leq k \leq \text{arr.length}$
- $1 \leq \text{arr.length} \leq 104$
- arr is sorted in ascending order.
- $-104 \leq \text{arr}[i], x \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a sorted array and integers k and x, return the k closest elements to x.
# Ties broken by smaller value. Return sorted. Right?"
#
# "arr=[1,2,3,4,5], k=4, x=3 → [1,2,3,4] ✓
# arr=[1,2,3,4,5], k=4, x=-1 → [1,2,3,4] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Sort all elements by |val-x| (then by val for ties), take first k, return sorted.
# Time: O(n log n). Space: O(n)."
```

```
class _658_FindKClosestElements_Brute:
```

```
    def findClosestElements(self, arr: list[int], k: int, x: int) -> list[int]:
        return sorted(sorted(arr, key=lambda v: (abs(v - x), v))[:k])
```

PHASE 3 — OPTIMIZE

```
# "Binary search on the left boundary of the k-element window.
# The answer is always a contiguous subarray of length k.
# Binary search for left in [0, n-k]: compare arr[mid] and arr[mid+k] distances to x.
# If arr[mid+k]-x < x-arr[mid], the window should shift right (left = mid+1).
# Otherwise left = mid. Final window: arr[left:left+k].
# Time: O(log(n-k) + k). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _658_FindKClosestElements:
```

```
    def findClosestElements(self, arr: list[int], k: int, x: int) -> list[int]:
        lo, hi = 0, len(arr) - k
        while lo < hi:
            mid = (lo + hi) // 2
            if x - arr[mid] > arr[mid + k] - x:
                lo = mid + 1
            else:
                hi = mid
        return arr[lo:lo + k]
```

PHASE 5 — TEST & DEBUG

```
# arr=[1,2,3,4,5],k=4,x=3: lo=0,hi=1.
# mid=0: x-arr[0]=3-1=2, arr[4]-x=5-3=2. 2>2 is False → hi=0.
# lo=hi=0. return arr[0:4]=[1,2,3,4] ✓
#
# arr=[1,2,3,4,5],k=4,x=-1: lo=0,hi=1.
# mid=0: x-arr[0]=-1-1=-2 (but we compare abs): -1-1=-2 < 0 so condition is
-2>arr[4]-(-1)=6? No.
# Actually x-arr[mid] = -1-1=-2; arr[mid+k]-x=5-(-1)=6. -2>6 False → hi=0. return
[1,2,3,4] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Binary search  $O(\log(n-k))$ : find the left boundary. Slice  $O(k)$ .
# Total  $O(\log(n-k)+k)$  vs brute  $O(n \log n)$ .
# The key insight: the answer window shifts right when right boundary is closer to x
than left.
# Follow-up: if array is unsorted, use a heap:  $O(n \log k)$ .
# Follow-up: online stream of queries with same array – binary search still  $O(\log n)$ 
per query."
```


Heap

□ #787 Cheapest Flights Within K Stops

MED[Open on LeetCode](#)

Dynamic Programming · DFS · BFS · Graph ·
Heap

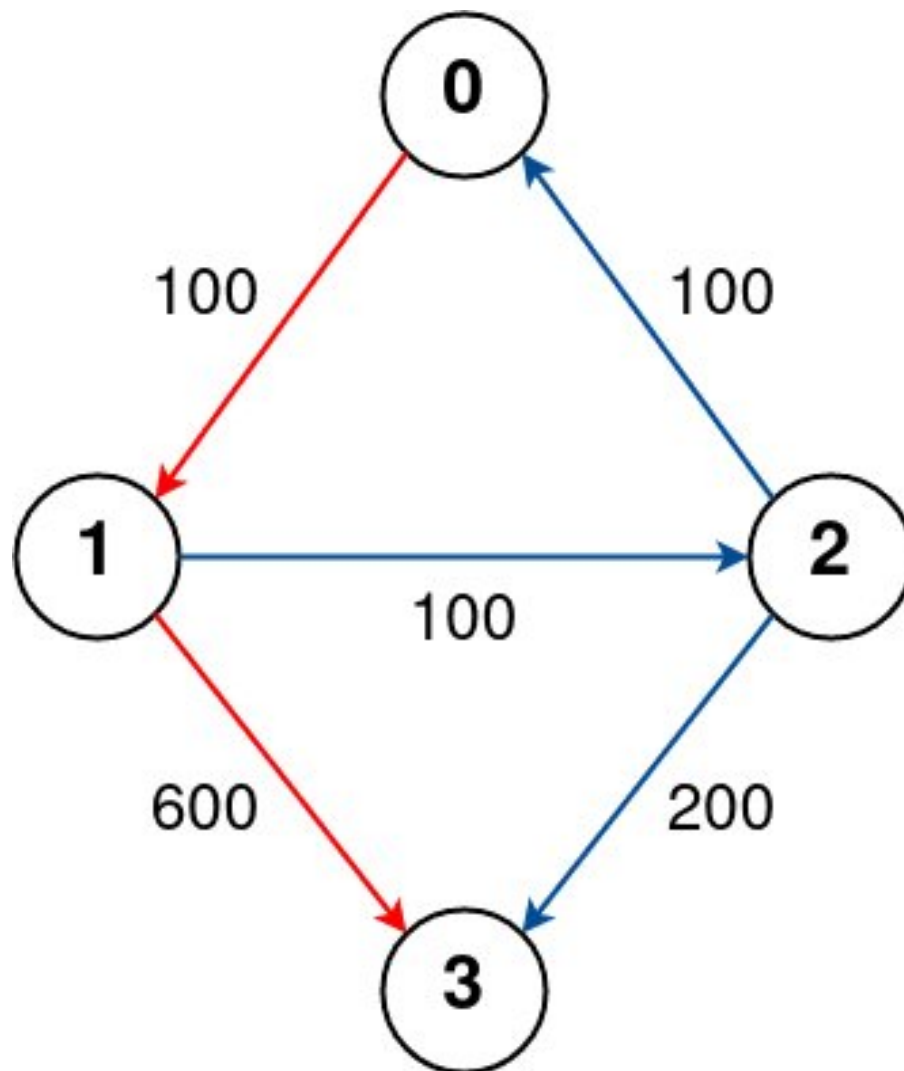
Lists: Grind 169

Problem

There are n cities connected by some number of flights. You are given an array `flights` where `flights[i] = [fromi, toi, pricei]` indicates that there is a flight from city `fromi` to city `toi` with cost `pricei`.

You are also given three integers `src`, `dst`, and `k`, return the cheapest price from `src` to `dst` with at most `k` stops. If there is no such route, return `-1`.

Example 1:



Input: $n = 4$, $flights = [[0,1,100],[1,2,100],[2,0,100],[1,3,600],[2,3,200]]$,
 $src = 0$, $dst = 3$, $k = 1$

Output: 700

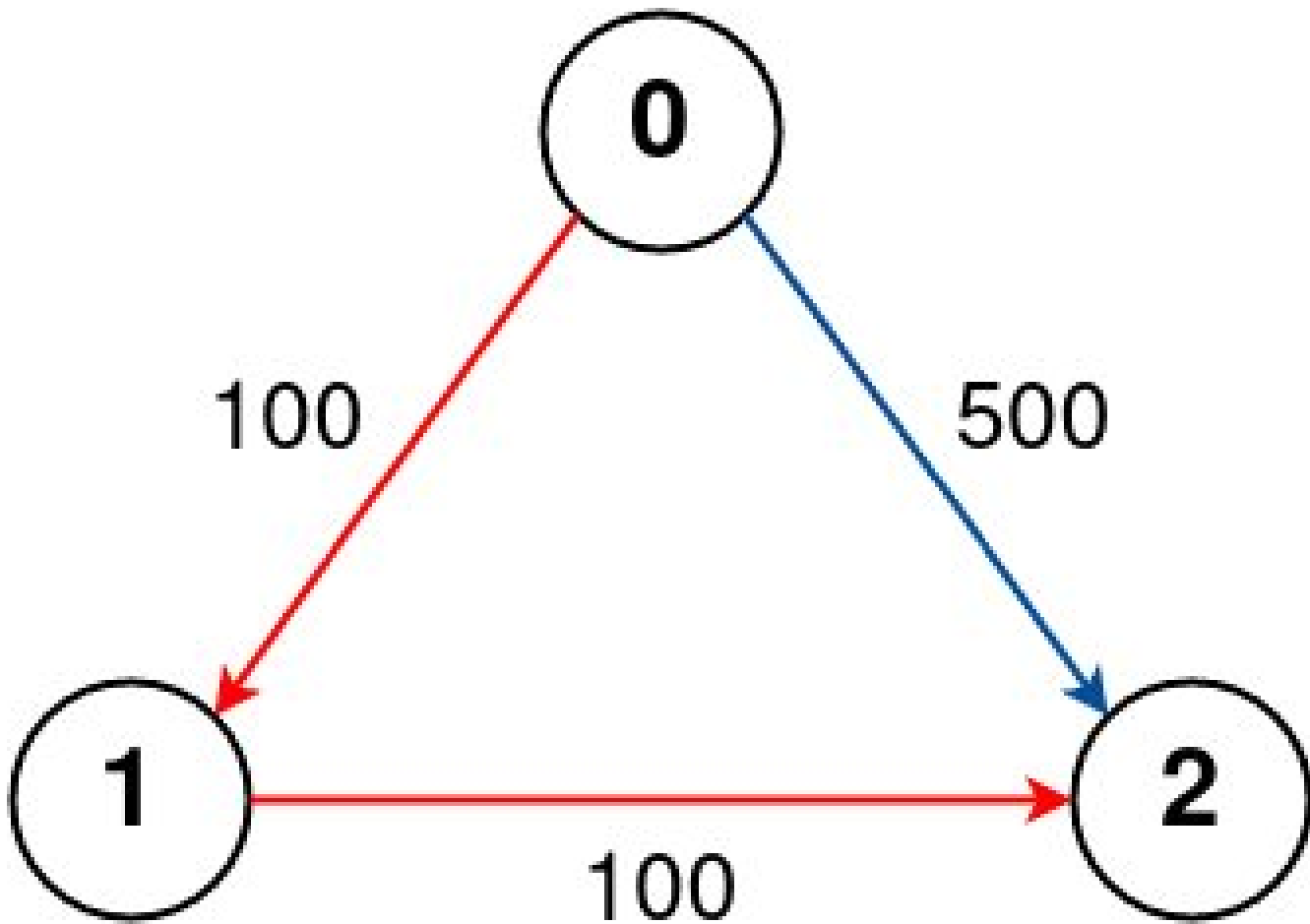
Explanation:

The graph is shown above.

The optimal path with at most 1 stop from city 0 to 3 is marked in red and has cost $100 + 600 = 700$.

Note that the path through cities $[0,1,2,3]$ is cheaper but is invalid because it uses 2 stops.

Example 2:



Input: $n = 3$, $flights = [[0,1,100],[1,2,100],[0,2,500]]$, $src = 0$, $dst = 2$, $k = 1$

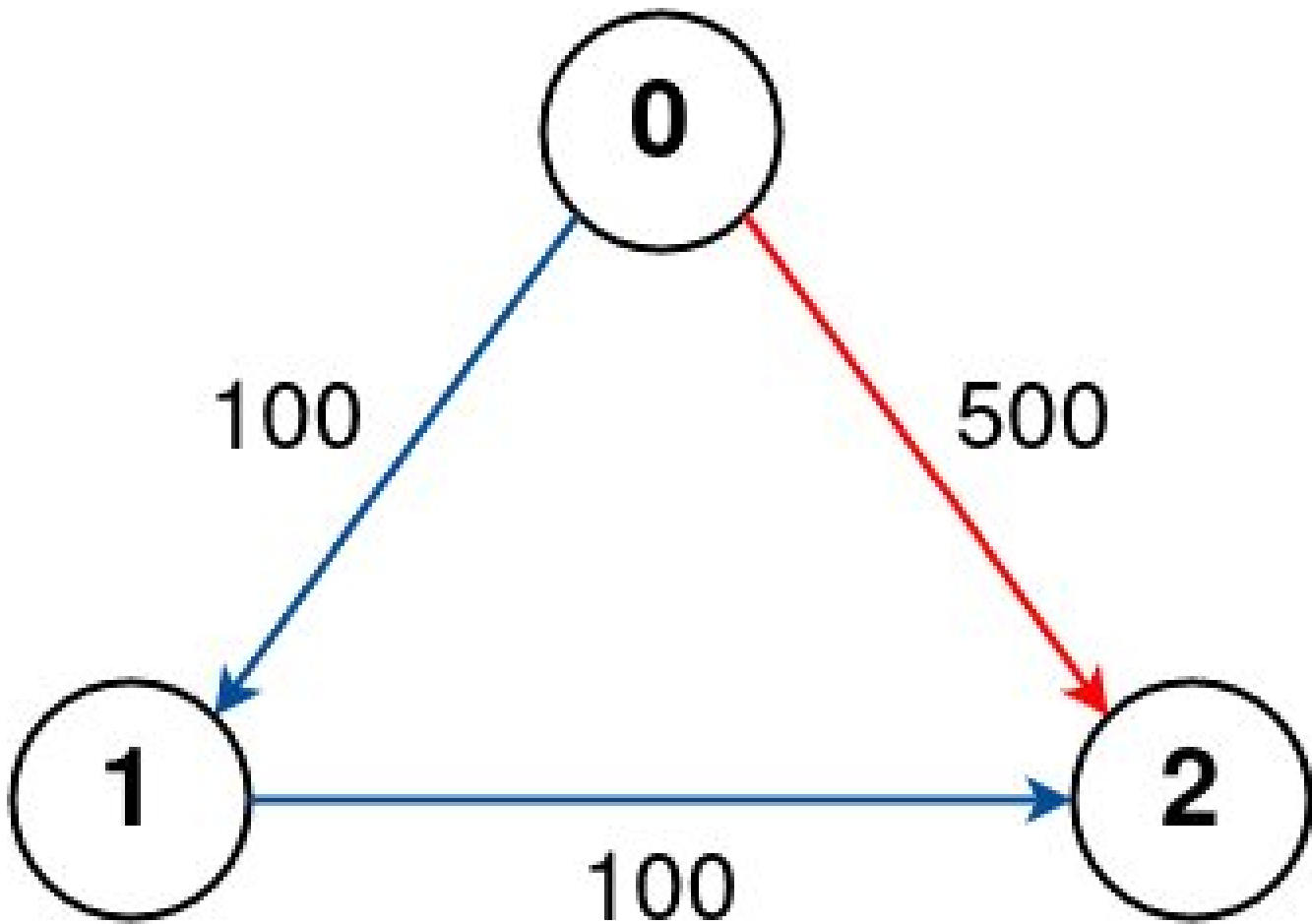
Output: 200

Explanation:

The graph is shown above.

The optimal path with at most 1 stop from city 0 to 2 is marked in red and has cost $100 + 100 = 200$.

Example 3:



Input: $n = 3$, $flights = [[0,1,100],[1,2,100],[0,2,500]]$, $src = 0$, $dst = 2$, $k = 0$

Output: 500

Explanation:

The graph is shown above.

The optimal path with no stops from city 0 to 2 is marked in red and has cost 500.

Constraints:

- $2 \leq n \leq 100$
- $0 \leq flights.length \leq (n * (n - 1) / 2)$
- $flights[i].length == 3$
- $0 \leq from_i, to_i < n$
- $from_i \neq to_i$

- $1 \leq \text{price}_i \leq 104$
- There will not be any multiple flights between two cities.
- $0 \leq \text{src}, \text{dst}, k < n$
- $\text{src} \neq \text{dst}$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "n cities, list of flights [from,to,price]. Find cheapest price from src to dst
# with at most k stops. -1 if impossible. Right?"
#
# "n=4, flights=[[0,1,100],[1,2,100],[2,0,100],[1,3,600],[2,3,200]], src=0, dst=3,
# k=1:
# 0→1→3 costs 700 (1 stop ✓). 0→1→2→3 costs 400 but needs 2 stops > k=1. → 700 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "BFS/DFS exploring all paths up to k+1 edges. Track minimum cost to reach dst.
# Time:  $O(n^k)$  worst case. Space:  $O(n)$ ."
```

```
class _787_CheapestFlightsKStops_Brute:
    def findCheapestPrice(self, n, flights, src, dst, k):
        from collections import defaultdict
        graph = defaultdict(list)
        for u, v, w in flights:
            graph[u].append((v, w))
        best = float('inf')
        def dfs(node, stops, cost):
            nonlocal best
            if node == dst:
                best = min(best, cost); return
            if stops < 0 or cost >= best: return
            for nxt, w in graph[node]:
                dfs(nxt, stops - 1, cost + w)
        dfs(src, k, 0)
        return best if best < float('inf') else -1
```

PHASE 3 — OPTIMIZE

```
# "Bellman-Ford with k+1 relaxation rounds — DP approach.
# prices[v] = min cost to reach v using at most i edges.
# Each round: update prices based on last round's values (use a copy to avoid using
# same-round updates).
# Time:  $O(k * |\text{flights}|)$ . Space:  $O(n)$ ."
```

PHASE 4 — CLEAN CODE

```

class _787_CheapestFlightsKStops:
    def findCheapestPrice(self, n: int, flights: list[list[int]], src: int, dst:
int, k: int) -> int:
        INF = float('inf')
        prices = [INF] * n
        prices[src] = 0
        for _ in range(k + 1):
            tmp = prices[:]
            for u, v, w in flights:
                if prices[u] + w < tmp[v]:
                    tmp[v] = prices[u] + w
            prices = tmp
        return prices[dst] if prices[dst] < INF else -1

```

PHASE 5 — TEST & DEBUG

```

# n=4, flights as above, src=0, dst=3, k=1:
# Init: prices=[0,INF,INF,INF].
# Round1 (1 edge): 0→1:100→tmp[1]=100. 1→3:600→tmp[3]=INF(prices[1]=INF). 0→1→
done.
# tmp=[0,100,INF,INF]. prices=[0,100,INF,INF].
# Round2 (2 edges=k+1): 1→2:100→tmp[2]=200. 1→3:600→tmp[3]=700.
2→3:200→prices[2]=INF skip.
# prices=[0,100,200,700]. return 700 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(k * E)$ :  $k+1$  Bellman-Ford rounds over all edges.
# Space  $O(n)$ : two price arrays (current and temp).
# Dijkstra with (cost, stops, node) heap also works –  $O((E+n) \log n)$  but needs
careful stop tracking.
# Follow-up: no stop limit – standard Dijkstra or Bellman-Ford.
# Follow-up: if edge weights can be negative – only Bellman-Ford handles this
correctly."

```

Heap

□ #973 K Closest Points to Origin

MED[Open on LeetCode](#)

Array · Math · Divide and Conquer · Sorting · Heap

Lists: Grind 169

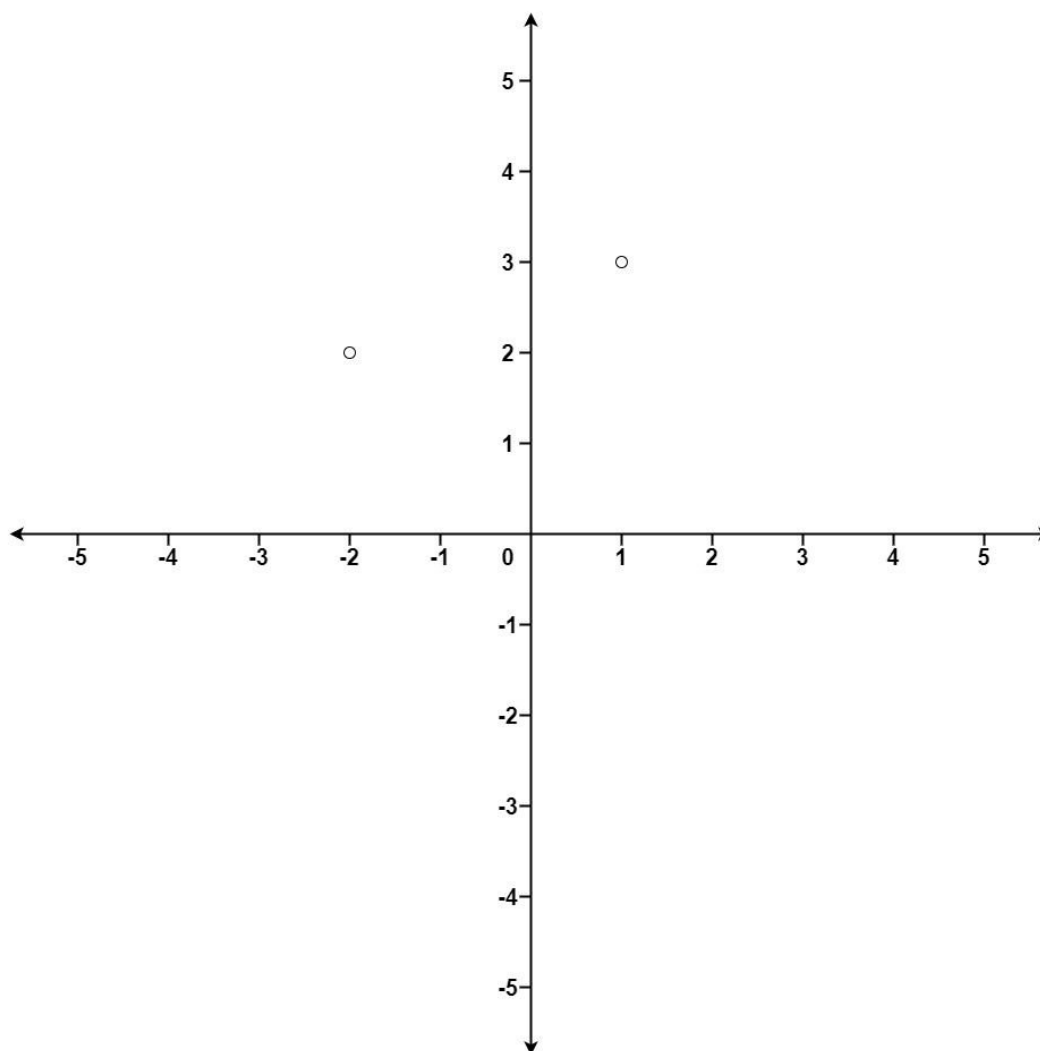
Problem

Given an array of points where $\text{points}[i] = [x_i, y_i]$ represents a point on the X-Y plane and an integer k , return the k closest points to the origin $(0, 0)$.

The distance between two points on the X-Y plane is the Euclidean distance (i.e., $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$).

You may return the answer in any order. The answer is guaranteed to be unique (except for the order that it is in).

Example 1:



Input: points = [[1,3],[-2,2]], k = 1

Output: [[-2,2]]

Explanation:

The distance between (1, 3) and the origin is $\sqrt{10}$.

The distance between (-2, 2) and the origin is $\sqrt{8}$.

Since $\sqrt{8} < \sqrt{10}$, (-2, 2) is closer to the origin.

We only want the closest k = 1 points from the origin, so the answer is just [[-2,2]].

Example 2:

Input: points = [[3,3],[5,-1],[-2,4]], k = 2

Output: [[3,3],[-2,4]]

Explanation: The answer [[-2,4],[3,3]] would also be accepted.

Constraints:

- $1 \leq k \leq \text{points.length} \leq 104$

- $-104 \leq x_i, y_i \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given an array of points, return the k closest to the origin (0,0).
# Distance = sqrt(x2+y2) (no need to actually sqrt since we compare squares).
# Answer can be in any order. Right?"
#
# "points=[[1,3],[-2,2]], k=1: dist2=[10,8] → [[-2,2]] ✓
# points=[[3,3],[5,-1],[-2,4]], k=2: dist2=[18,26,20] → [[3,3],[-2,4]] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Sort all points by squared distance, return first k.
# Time: O(n log n). Space: O(n)."
```

```
class _973_KClosestPoints_Brute:
    def kClosest(self, points: list[list[int]], k: int) -> list[list[int]]:
        points.sort(key=lambda p: p[0]**2 + p[1]**2)
        return points[:k]
```

PHASE 3 — OPTIMIZE

```
# "Max-heap of size k: maintain k closest points.
# Push (-dist2, x, y). When heap exceeds k, pop the farthest.
# Time: O(n log k). Space: O(k). Better than O(n log n) when k << n.
# Quickselect alternative: O(n) average, O(n2) worst – partitions around a pivot distance."
```

PHASE 4 — CLEAN CODE

```
class _973_KClosestPoints:
    def kClosest(self, points: list[list[int]], k: int) -> list[list[int]]:
        import heapq
        heap = []
        for x, y in points:
            dist = -(x*x + y*y)
            heapq.heappush(heap, (dist, x, y))
            if len(heap) > k:
                heapq.heappop(heap)
        return [[x, y] for _, x, y in heap]
```

PHASE 5 — TEST & DEBUG

```
# points=[[1,3],[-2,2]],k=1:
# push (-10,1,3)→[(-10,1,3)]. len=1≤1.
# push (-8,-2,2)→[(-10,1,3),(-8,-2,2)]. len=2>1 → pop(-10,1,3).
# heap=[(-8,-2,2)]. return [[-2,2]] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Max-heap O(n log k): push all n, heap never exceeds k."
```

```
# Sort  $O(n \log n)$ : simple but slower for small  $k$ .  
# Quickselect  $O(n)$  average: optimal for single queries on large  $n$ .  
# Follow-up: if points stream in, maintain a max-heap of size  $k$  updating on each arrival.  
# Follow-up:  $K$  closest to a point other than origin – same logic, shift coordinates."
```

Heap

□ ★ #1057 Campus Bikes ★

MED[Open on LeetCode](#)

Array · Greedy · Sorting · Heap

Lists: ★ Premium | Premium 98

Problem

On a campus represented on the X-Y plane, there are n workers and m bikes, with $n \leq m$.

You are given an array `workers` of length n where `workers[i] = [xi, yi]` is the position of the i th worker. You are also given an array `bikes` of length m where `bikes[j] = [xj, yj]` is the position of the j th bike. All the given positions are unique.

Assign a bike to each worker. Among the available bikes and workers, we choose the $(worker_i, bike_j)$ pair with the shortest Manhattan distance between each other and assign the bike to that worker.

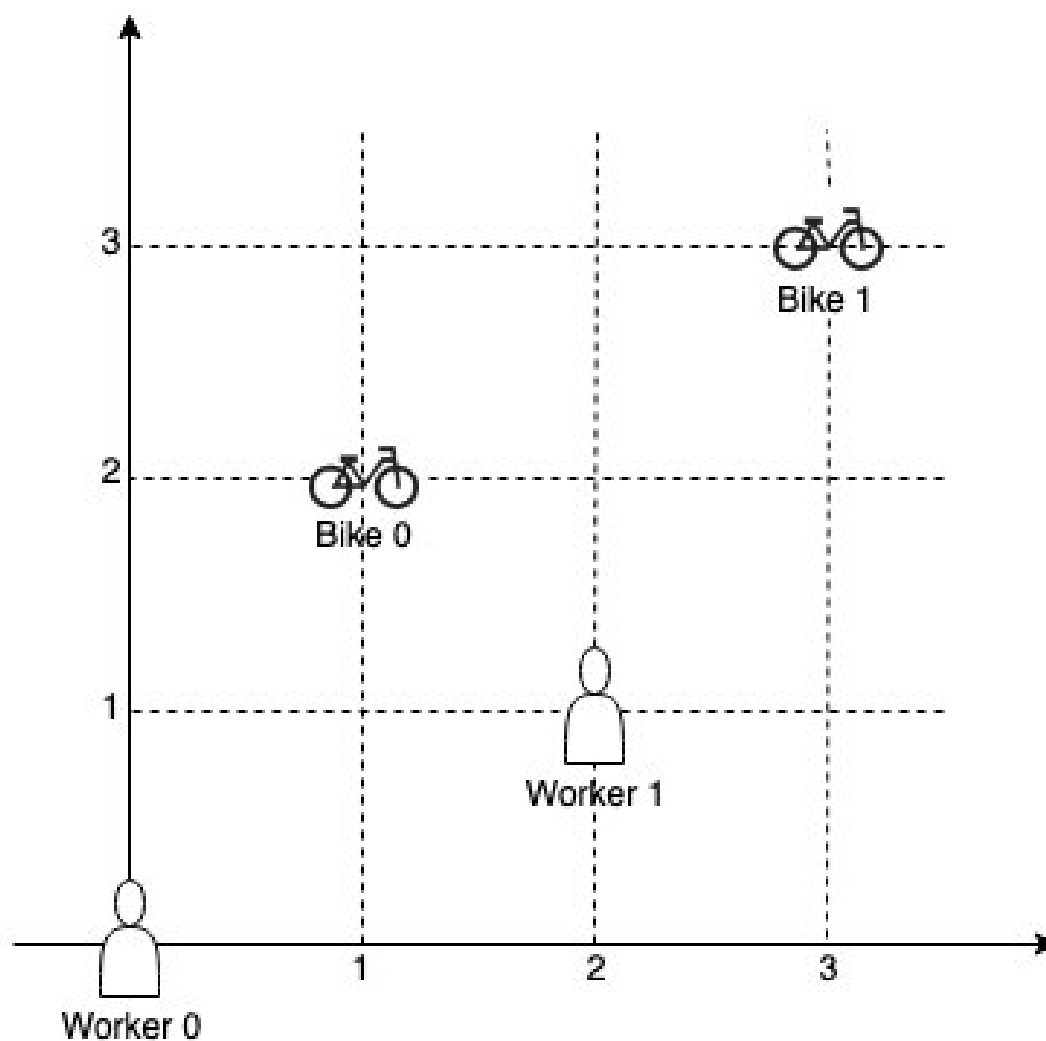
If there are multiple $(worker_i, bike_j)$ pairs with the same shortest Manhattan distance, we choose the pair with the smallest worker index.

If there are multiple ways to do that, we choose the pair with the smallest bike index. Repeat this process until there are no available workers.

Return an array `answer` of length `n`, where `answer[i]` is the index (0-indexed) of the bike that the `i`th worker is assigned to.

The Manhattan distance between two points `p1` and `p2` is $\text{Manhattan}(p1, p2) = |p1.x - p2.x| + |p1.y - p2.y|$.

Example 1:

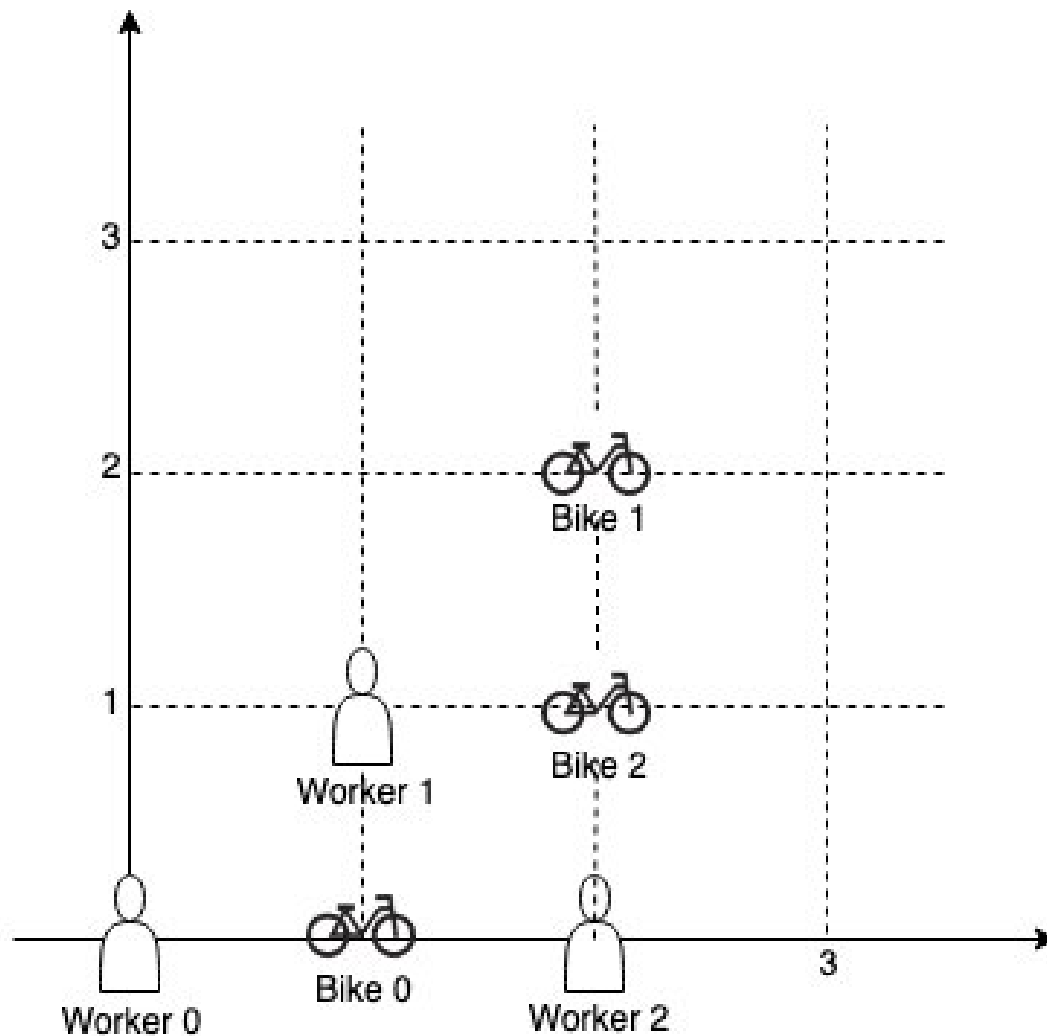


Input: workers = `[[0,0],[2,1]]`, bikes = `[[1,2],[3,3]]`

Output: `[1,0]`

Explanation: Worker 1 grabs Bike 0 as they are closest (without ties), and Worker 0 is assigned Bike 1. So the output is `[1, 0]`.

Example 2:



Input: workers = `[[0,0],[1,1],[2,0]]`, bikes = `[[1,0],[2,2],[2,1]]`

Output: `[0,2,1]`

Explanation: Worker 0 grabs Bike 0 at first. Worker 1 and Worker 2 share the same distance to Bike 2, thus Worker 1 is assigned to Bike 2, and Worker 2 will take Bike 1. So the output is `[0,2,1]`.

Constraints:

- $n == \text{workers.length}$
- $m == \text{bikes.length}$
- $1 \leq n \leq m \leq 1000$
- $\text{workers}[i].\text{length} == \text{bikes}[j].\text{length} == 2$
- $0 \leq x_i, y_i < 1000$

- $0 \leq x_j, y_j < 1000$
- All worker and bike locations are unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "n workers and m bikes on a 2D campus. Assign each worker exactly one bike (no sharing).
# Process all (worker,bike) pairs in order of Manhattan distance (ties: smaller worker index,
# then smaller bike index). Return bike assignment array for each worker. Right?"
#
# "workers=[[0,0],[2,1]], bikes=[[1,2],[3,3]]:
# distances: w0b0=3,w0b1=6,w1b0=2,w1b1=2.
# sorted: (2,w1,b0),(2,w1,b1),(3,w0,b0),(6,w0,b1).
# assign w1→b0, then w0→b1. → [1,0] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Generate all (dist, worker, bike) triples, sort them, then greedily assign.
# Time:  $O(n \cdot m \log(n \cdot m))$ . Space:  $O(n \cdot m)$ ."
```

```
class _1057_CampusBikes_Brute:
    def assignBikes(self, workers, bikes):
        pairs = []
        for i, (wx, wy) in enumerate(workers):
            for j, (bx, by) in enumerate(bikes):
                pairs.append((abs(wx-bx)+abs(wy-by), i, j))
        pairs.sort()
        result = [-1] * len(workers)
        used_workers = set()
        used_bikes = set()
        for dist, w, b in pairs:
            if w not in used_workers and b not in used_bikes:
                result[w] = b
                used_workers.add(w)
                used_bikes.add(b)
                if len(used_workers) == len(workers):
                    break
        return result
```

PHASE 3 — OPTIMIZE

```
# "Same greedy with a min-heap — identical complexity but avoids sorting all pairs upfront.
# Push all pairs initially; pop min until all workers assigned.
# Or bucket-sort by distance (max Manhattan  $\leq 2000$ ) for  $O(n \cdot m)$  time."
```

PHASE 4 — CLEAN CODE

```

class _1057_CampusBikes:
    def assignBikes(self, workers, bikes):
        import heapq
        heap = []
        for i, (wx, wy) in enumerate(workers):
            for j, (bx, by) in enumerate(bikes):
                heapq.heappush(heap, (abs(wx-bx)+abs(wy-by), i, j))
        result = [-1] * len(workers)
        used_workers = set()
        used_bikes = set()
        while heap and len(used_workers) < len(workers):
            dist, w, b = heapq.heappop(heap)
            if w in used_workers or b in used_bikes:
                continue
            result[w] = b
            used_workers.add(w)
            used_bikes.add(b)
        return result

```

PHASE 5 — TEST & DEBUG

```

# workers=[[0,0],[2,1]], bikes=[[1,2],[3,3]]:
# heap: (3,0,0),(6,0,1),(2,1,0),(2,1,1).
# pop (2,1,0): assign w1→b0. pop (2,1,1): w1 used skip. pop (3,0,0): b0 used skip.
# pop (6,0,1): assign w0→b1. result=[1,0] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n*m \log(n*m))$ : heap contains  $n*m$  pairs. Space  $O(n*m)$ .
# Bucket sort by distance:  $O(n*m)$  time since distances  $\leq 2000$ .
# Follow-up: Campus Bikes II (LC 1066) – minimize TOTAL distance (optimal assignment).
# Requires DP with bitmask:  $O(n * 2^m)$ . Much harder – this greedy doesn't work there."

```


Heap

★ #1167 Minimum Cost to Connect Sticks ★

MED[Open on LeetCode](#)

Array · Greedy · Heap

Lists: ★ Premium | Premium 98

Problem

You have some number of sticks with positive integer lengths. These lengths are given as an array `sticks`, where `sticks[i]` is the length of the *i*th stick.

You can connect any two sticks of lengths *x* and *y* into one stick by paying a cost of $x + y$. You must connect all the sticks until there is only one stick remaining.

Return the minimum cost of connecting all the given sticks into one stick in this way.

Example 1:

Input: `sticks = [2,4,3]`

Output: 14

Explanation: You start with `sticks = [2,4,3]`.

1. Combine sticks 2 and 3 for a cost of $2 + 3 = 5$. Now you have `sticks = [5,4]`.

2. Combine sticks 5 and 4 for a cost of $5 + 4 = 9$. Now you have `sticks = [9]`.

There is only one stick left, so you are done. The total cost is $5 + 9 = 14$.

Example 2:

Input: sticks = [1,8,3,5]

Output: 30

Explanation: You start with sticks = [1,8,3,5].

1. Combine sticks 1 and 3 for a cost of $1 + 3 = 4$. Now you have sticks = [4,8,5].

2. Combine sticks 4 and 5 for a cost of $4 + 5 = 9$. Now you have sticks = [9,8].

3. Combine sticks 9 and 8 for a cost of $9 + 8 = 17$. Now you have sticks = [17].

There is only one stick left, so you are done. The total cost is $4 + 9 + 17 = 30$.

Example 3:

Input: sticks = [5]

Output: 0

Explanation: There is only one stick, so you don't need to do anything. The total cost is 0.

Constraints:

- $1 \leq \text{sticks.length} \leq 104$
- $1 \leq \text{sticks}[i] \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given sticks of various lengths, combine any two into one with cost = sum of their lengths.

Return the minimum total cost to combine all sticks into one. Right?"

#

"sticks=[2,4,3]: combine 2+3=5(cost5). Then 4+5=9(cost9). Total=14 ✓

vs 2+4=6(cost6). 3+6=9(cost9). Total=15. So min is 14 ✓"

PHASE 2 — BRUTE FORCE

"Try all pair orderings recursively, track minimum cost.

Time: $O(n! * n)$. Space: $O(n)$ recursion."

```
class _1167_MinCostConnectSticks_Brute:
    def connectSticks(self, sticks: list[int]) -> int:
        from itertools import permutations
        if len(sticks) == 1: return 0
        best = float('inf')
        for i in range(len(sticks)):
            for j in range(i+1, len(sticks)):
                new = sticks[i] + sticks[j]
```

```

        remaining = [sticks[k] for k in range(len(sticks)) if k!=i and k!=j]
+ [new]
        cost = new + self.connectSticks(remaining)
        best = min(best, cost)
    return best

```

PHASE 3 — OPTIMIZE

```

# "Greedy: always combine the two SMALLEST sticks – Huffman coding.
# Use a min-heap. Pop two smallest, add their sum as cost, push sum back.
# Repeat until one stick remains.
# Time: O(n log n). Space: O(n)."

```

PHASE 4 — CLEAN CODE

```

class _1167_MinCostConnectSticks:
    def connectSticks(self, sticks: list[int]) -> int:
        import heapq
        heapq.heapify(sticks)
        total = 0
        while len(sticks) > 1:
            a = heapq.heappop(sticks)
            b = heapq.heappop(sticks)
            combined = a + b
            total += combined
            heapq.heappush(sticks, combined)
        return total

```

PHASE 5 — TEST & DEBUG

```

# sticks=[2,4,3]: heap=[2,3,4].
# pop 2,3 → combined=5, cost=5. heap=[4,5].
# pop 4,5 → combined=9, cost=14. heap=[9]. return 14 ✓
#
# sticks=[1,8,3,5]: heap=[1,3,5,8].
# pop 1,3→4, cost=4. heap=[4,5,8].
# pop 4,5→9, cost=13. heap=[8,9].
# pop 8,9→17, cost=30. return 30.

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n log n): heapify O(n) + n-1 pops/pushes each O(log n).
# Space O(n): the heap.
# This is exactly Huffman encoding – greedy optimality proven by exchange argument.
# Follow-up: Minimum Cost to Merge Stones (LC 1000) – merge k adjacent groups; needs DP.
# Follow-up: if we could only combine adjacent sticks, greedy no longer works."

```

Heap

□ #1424 Diagonal Traverse II

MED[Open on LeetCode](#)

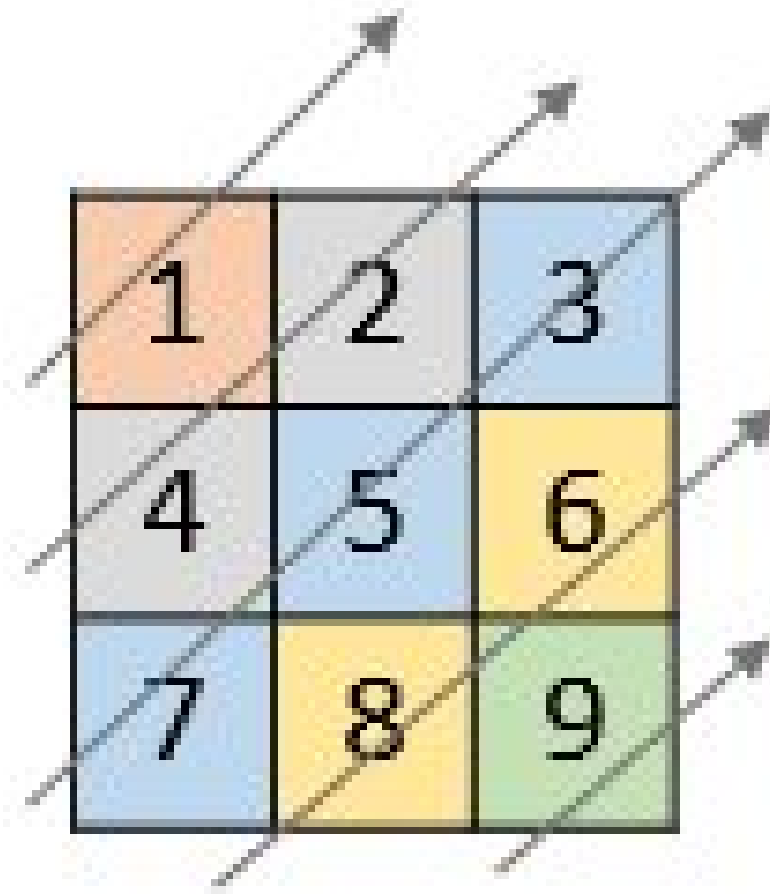
Array · Sorting · Heap

Lists: CodeSignal

Problem

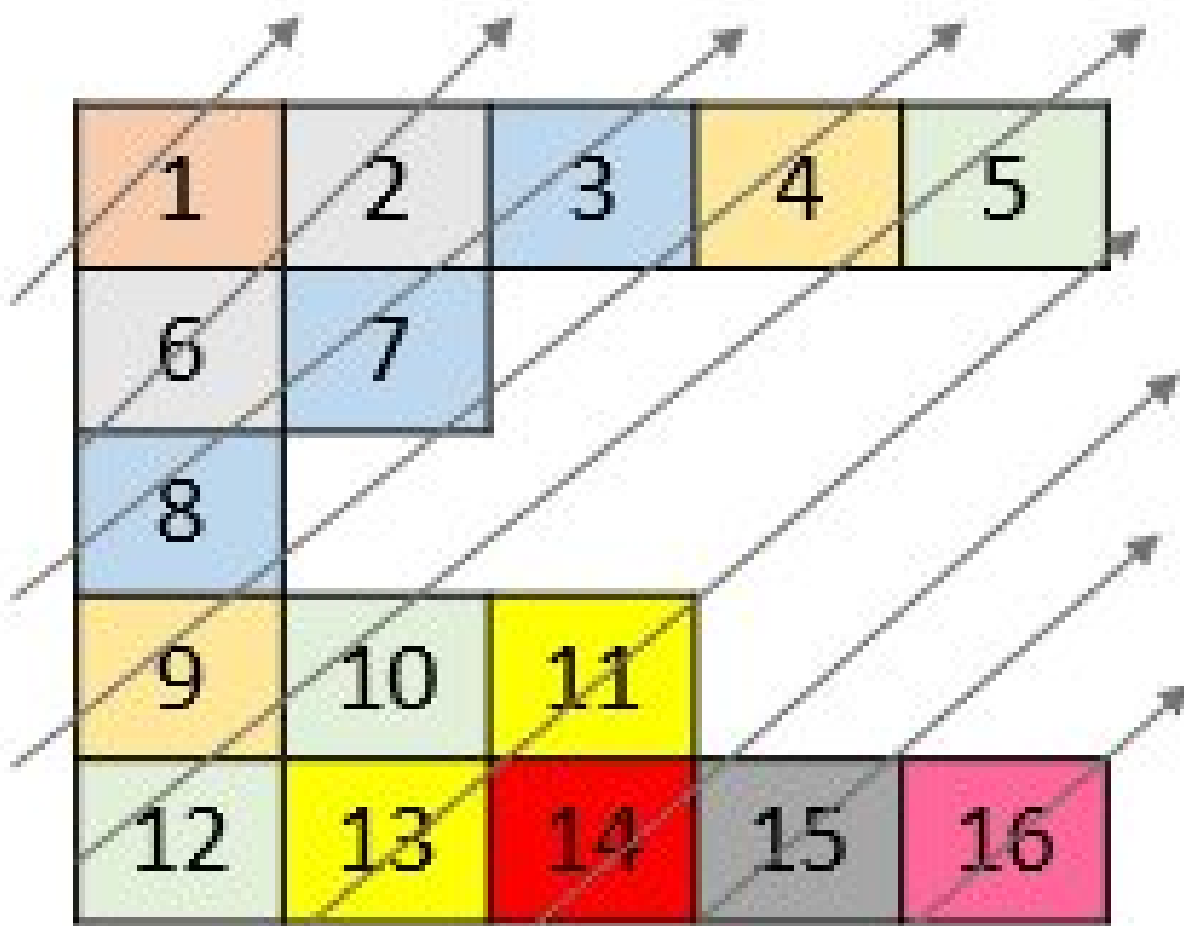
Given a 2D integer array `nums`, return all elements of `nums` in diagonal order as shown in the below images.

Example 1:



Input: nums = [[1,2,3],[4,5,6],[7,8,9]]
Output: [1,4,2,7,5,3,8,6,9]

Example 2:



Input: `nums = [[1,2,3,4,5],[6,7],[8],[9,10,11],[12,13,14,15,16]]`

Output: `[1,6,2,8,7,3,9,4,12,10,5,13,11,14,15,16]`

Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $1 \leq \text{nums}[i].\text{length} \leq 105$
- $1 \leq \text{sum}(\text{nums}[i].\text{length}) \leq 105$
- $1 \leq \text{nums}[i][j] \leq 105$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a 2D list of integers (rows may have different lengths), return all
elements
# in diagonal order: elements on the same anti-diagonal (i+j = const) grouped
together,
# within each diagonal from bottom to top (larger i first). Right?"
#
# "nums=[[1,2,3],[4,5,6],[7,8,9]]:
# diag0(i+j=0):[1]. diag1:[4,2]. diag2:[7,5,3]. diag3:[8,6]. diag4:[9].
# → [1,4,2,7,5,3,8,6,9] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Build a dict of diagonal_index → list of values (appended bottom to top).
# Diagonal index = i+j. Iterate rows in reverse to get bottom-to-top order.
# Time: O(n) where n = total elements. Space: O(n)."
```

```
class _1424_DiagonalTraverseII_Brute:
    def findDiagonalOrder(self, nums: list[list[int]]) -> list[int]:
        from collections import defaultdict
        diags = defaultdict(list)
        for i in range(len(nums) - 1, -1, -1): # bottom to top
            for j, val in enumerate(nums[i]):
                diags[i + j].append(val)
        result = []
        for k in sorted(diags):
            result.extend(diags[k])
        return result
```

PHASE 3 — OPTIMIZE

```
# "Same dict approach but we can avoid sorting by noting diagonal index = i+j
# naturally increases as we scan top-to-bottom, right-to-left.
# Build diags iterating rows top-to-bottom but prepend (or use reverse) per
diagonal.
# Time: O(n). Space: O(n)."
```

PHASE 4 — CLEAN CODE

```
class _1424_DiagonalTraverseII:
    def findDiagonalOrder(self, nums: list[list[int]]) -> list[int]:
        from collections import defaultdict
        diags = defaultdict(list)
        for i, row in enumerate(nums):
            for j, val in enumerate(row):
                diags[i + j].append(val)
        result = []
        for k in range(len(diags)):
            result.extend(reversed(diags[k])) # larger i (added later) first
        return result
```

PHASE 5 — TEST & DEBUG

```
# nums=[[1,2,3],[4,5,6],[7,8,9]]:
# diags: 0:[1], 1:[2,4], 2:[3,5,7], 3:[6,8], 4:[9].
# reversed: [1],[4,2],[7,5,3],[8,6],[9] → [1,4,2,7,5,3,8,6,9] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: one pass to build diags, one pass to collect – n total elements.

Space $O(n)$: the diags dictionary stores all elements.

Follow-up: Diagonal Traverse I (LC 498) – rectangular matrix, alternating direction per diagonal.

Follow-up: spiral order (LC 54) vs diagonal order are both reordering problems."

Trie

Round 5 · Mid · Medium · 7 questions

Trie

□ #139 Word Break

MED[Open on LeetCode](#)

Array · Hash Table · String · Dynamic Programming · Trie · Memoization
Lists: Grind 169

Problem

Given a string *s* and a dictionary of strings *wordDict*, return true if *s* can be segmented into a space-separated sequence of one or more dictionary words.

Note that the same word in the dictionary may be reused multiple times in the segmentation.

Example 1:

```
Input: s = "leetcode", wordDict = ["leet","code"]  
Output: true  
Explanation: Return true because "leetcode" can be segmented as "leet code".
```

Example 2:

```
Input: s = "applepenapple", wordDict = ["apple","pen"]  
Output: true  
Explanation: Return true because "applepenapple" can be segmented as "apple pen apple".  
Note that you are allowed to reuse a dictionary word.
```

Example 3:

```
Input: s = "catsandog", wordDict = ["cats","dog","sand","and","cat"]
Output: false
```

Constraints:

- $1 \leq s.length \leq 300$
- $1 \leq wordDict.length \leq 1000$
- $1 \leq wordDict[i].length \leq 20$
- s and $wordDict[i]$ consist of only lowercase English letters.
- All the strings of $wordDict$ are unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a string s and a dictionary of words, return true if s can be segmented
# into a space-separated sequence of dictionary words. Words can be reused. Right?"
#
# "s='leetcode', wordDict=['leet','code']: 'leet'+ 'code' → true ✓
# s='applepenapple', wordDict=['apple','pen']: 'apple'+ 'pen'+ 'apple' → true ✓
# s='catsandog', wordDict=['cats','dog','sand','and','cat']: no valid split → false
✓"
```

PHASE 2 — BRUTE FORCE

```
# "Recursive: try every prefix of s. If prefix is in dict, recurse on the suffix.
# Exponential without memoization –  $O(2^n)$  time,  $O(n)$  space."
```

```
class _139_WordBreak_Brute:
    def wordBreak(self, s: str, wordDict: list[str]) -> bool:
        words = set(wordDict)
        def backtrack(start: int) -> bool:
            if start == len(s):
                return True
            for end in range(start + 1, len(s) + 1):
                if s[start:end] in words and backtrack(end):
                    return True
            return False
        return backtrack(0)
```

PHASE 3 — OPTIMIZE

```
# "DP: dp[i] = true if s[:i] can be segmented.
# Base: dp[0] = True (empty string)."
```

```
# Transition: dp[i] = any(dp[j] and s[j:i] in wordSet) for 0 <= j < i.
# Time:  $O(n^2)$  or  $O(n * \text{max\_word\_len})$ . Space:  $O(n)$ ."
```

PHASE 4 — CLEAN CODE

```
class _139_WordBreak:
    def wordBreak(self, s: str, wordDict: list[str]) -> bool:
        words = set(wordDict)
        n = len(s)
        dp = [False] * (n + 1)
        dp[0] = True
        max_len = max(len(w) for w in words)
        for i in range(1, n + 1):
            for j in range(max(0, i - max_len), i):
                if dp[j] and s[j:i] in words:
                    dp[i] = True
                    break
        return dp[n]
```

PHASE 5 — TEST & DEBUG

```
# s='leetcode', words={'leet','code'}:
# dp[0]=T. dp[4]: j=0, s[0:4]='leet'∈words, dp[0]=T→dp[4]=T.
# dp[8]: j=4, s[4:8]='code'∈words, dp[4]=T→dp[8]=T. return True ✓
#
# s='catsandog': dp[3]=T('cat'), dp[4]=T('cats'). dp[7]=T('sand' from 3).
# No j makes dp[9]=T → return False ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n * \text{max\_word\_len})$ : optimised by only checking back max_word_len characters.
# Space  $O(n)$  for dp array.
# Trie optimisation: insert all words into a trie, walk the trie forward from each
dp[j]=True position.
# Follow-up: Word Break II (LC 140) – return all valid sentences.
# Backtrack from dp[n]=True positions to reconstruct paths; store parent indices."
```

Trie

□ #208 Implement Trie (Prefix Tree)

MED[Open on LeetCode](#)

Hash Table · String · Design · Trie

Lists: Grind 169

Problem

A trie (pronounced as "try") or prefix tree is a tree data structure used to efficiently store and retrieve keys in a dataset of strings. There are various applications of this data structure, such as autocomplete and spellchecker.

Implement the Trie class:

- **Trie()** Initializes the trie object.
- **void insert(String word)** Inserts the string word into the trie.
- **boolean search(String word)** Returns true if the string word is in the trie (i.e., was inserted before), and false otherwise.
- **boolean startsWith(String prefix)** Returns true if there is a previously inserted string word that has the prefix prefix, and false otherwise.

Example 1:

```

Input
["Trie", "insert", "search", "search", "startsWith", "insert", "search"]
[[], ["apple"], ["apple"], ["app"], ["app"], ["app"], ["app"]]
Output
[null, null, true, false, true, null, true]

Explanation
Trie trie = new Trie();

```

Constraints:

- $1 \leq \text{word.length}, \text{prefix.length} \leq 2000$
- word and prefix consist only of lowercase English letters.
- At most $3 * 10^4$ calls in total will be made to insert, search, and startsWith.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Implement a Trie with insert(word), search(word) → bool (exact match),
# and startsWith(prefix) → bool. Only lowercase English letters. Right?"
#
# "insert('apple'). search('apple')→true. search('app')→false.
# startsWith('app')→true. insert('app'). search('app')→true ✓"

```

PHASE 2 — BRUTE FORCE

```

# "Use a set for exact search and a sorted list for prefix search.
# Time: insert O(1), search O(1), startsWith O(n * m) – scan all words.
# Space: O(total_chars)."

```

```

class _208_ImplementTrieBrute:
    def __init__(self):
        self.words = set()

    def insert(self, word: str) -> None:
        self.words.add(word)

    def search(self, word: str) -> bool:
        return word in self.words

    def startsWith(self, prefix: str) -> bool:
        return any(w.startswith(prefix) for w in self.words)

```

PHASE 3 — OPTIMIZE

"True trie: each node has up to 26 children (one per letter) and an end-of-word flag.

insert: create nodes along the path. search: walk path, check end flag.

startsWith: walk path, return True if we reach the end of the prefix.

Time: $O(m)$ per op where m = word/prefix length. Space: $O(\text{total_chars} * 26)$."

PHASE 4 — CLEAN CODE

```
class _208_ImplementTrie:
    def __init__(self):
        self.children = {}
        self.is_end = False

    def insert(self, word: str) -> None:
        node = self
        for ch in word:
            if ch not in node.children:
                node.children[ch] = _208_ImplementTrie()
            node = node.children[ch]
        node.is_end = True

    def search(self, word: str) -> bool:
        node = self
        for ch in word:
            if ch not in node.children:
                return False
            node = node.children[ch]
        return node.is_end

    def startsWith(self, prefix: str) -> bool:
        node = self
        for ch in prefix:
            if ch not in node.children:
                return False
            node = node.children[ch]
        return True
```

PHASE 5 — TEST & DEBUG

insert("apple"): root→a→p→p→l→e(is_end=True).

search("apple"): walk a→p→p→l→e, is_end=True → True ✓

search("app"): walk a→p→p, is_end=False → False ✓

startsWith("app"): walk a→p→p, exists → True ✓

insert("app"): node at 'p'(3rd) gets is_end=True.

search("app"): walk a→p→p, is_end=True → True ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m)$ per op: walk at most m nodes. Space $O(n*m)$: n words of avg length m .

Dict children vs array[26]: dict is memory-efficient for sparse alphabets.

Follow-up: delete a word from the trie – mark is_end=False, prune childless nodes bottom-up.

Follow-up: Word Search II (LC 212) – build a trie of all words, DFS the board against it."

Trie

□ #211 Design Add and Search Words Data Structure

MED[Open on LeetCode](#)

String · DFS · Design · Trie

Lists: Grind 169

Problem

Design a data structure that supports adding new words and finding if a string matches any previously added string.

Implement the WordDictionary class:

- WordDictionary() Initializes the object.
- void addWord(word) Adds word to the data structure, it can be matched later.
- bool search(word) Returns true if there is any string in the data structure that matches word or false otherwise. word may contain dots '.' where dots can be matched with any letter.

Example:

Input

```
["WordDictionary", "addWord", "addWord", "addWord", "search", "search", "search", "search"]
```

```
[[], ["bad"], ["dad"], ["mad"], ["pad"], ["bad"], [".ad"], ["b.."]]
```

Output

```
[null, null, null, null, false, true, true, true]
```

Explanation

```
WordDictionary wordDictionary = new WordDictionary();
```

Constraints:

- $1 \leq \text{word.length} \leq 25$
- word in addWord consists of lowercase English letters.
- word in search consist of '.' or lowercase English letters.
- There will be at most 2 dots in word for search queries.
- At most 104 calls will be made to addWord and search.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Design a data structure with addWord(word) and search(word).
```

```
# search supports '.' as a wildcard matching any single letter. Right?"
```

```
#
```

```
# "addWord('bad'), addWord('dad'), addWord('mad').
```

```
# search('pad')→false, search('bad')→true, search('.ad')→true, search('b..')→true
```

```
✓"
```

PHASE 2 — BRUTE FORCE

```
# "Store all words in a list. For search, iterate over all words and match
```

```
# character by character (treating '.' as wildcard).
```

```
# Time:  $O(n * m)$  per search where  $n$  = number of words,  $m$  = word length. Space:  
 $O(n * m)$ ."
```

```
class _211_WordDictionary_Brute:
```

```

def __init__(self):
    self.words = []

def addWord(self, word: str) -> None:
    self.words.append(word)

def search(self, word: str) -> bool:
    def matches(w, pattern):
        if len(w) != len(pattern): return False
        return all(p == '.' or c == p for c, p in zip(w, pattern))
    return any(matches(w, word) for w in self.words)

```

PHASE 3 — OPTIMIZE

"Trie with DFS for wildcards. addWord: standard trie insert.
 # search: walk trie; on a '.' node, recursively try ALL children.
 # Time: addWord O(m). search O(m) for literals, up to O(26^k * m) worst case with k dots.
 # Space: O(n*m) for the trie."

PHASE 4 — CLEAN CODE

```

class _211_WordDictionary:
    def __init__(self):
        self.children = {}
        self.is_end = False

    def addWord(self, word: str) -> None:
        node = self
        for ch in word:
            if ch not in node.children:
                node.children[ch] = _211_WordDictionary()
            node = node.children[ch]
        node.is_end = True

    def search(self, word: str) -> bool:
        def dfs(node, i: int) -> bool:
            if i == len(word):
                return node.is_end
            ch = word[i]
            if ch == '.':
                return any(dfs(child, i + 1) for child in node.children.values())
            if ch not in node.children:
                return False
            return dfs(node.children[ch], i + 1)
        return dfs(self, 0)

```

PHASE 5 — TEST & DEBUG

addWord("bad"): b→a→d(end). addWord("dad"): d→a→d(end). addWord("mad"): m→a→d(end).
 # search("pad"): p not in root.children → False ✓
 # search("bad"): b→a→d, is_end=True → True ✓
 # search(".ad"): '.' : try b,d,m → each→a→d, is_end=True → True ✓
 # search("b.."): b→'.' : try a→'.' : try d(end) → True ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"addWord $O(m)$. search $O(m)$ best case (no dots), $O(26^k * m)$ worst case (k dots).
In practice dots are rare, so average is close to $O(m)$.
Follow-up: if patterns are very dot-heavy, cache results at (node, index) pairs.
Follow-up: support '*' (zero or more chars) – more complex DFS with skip logic."

Trie

★ #616 Add Bold Tag in String ★

MED[Open on LeetCode](#)

Array · Hash Table · String · Trie

Lists: ★ Premium | Premium 98

Problem

You are given a string *s* and an array of strings *words*.

You should add a closed pair of bold tag `<b>` and `</b>` to wrap the substrings in *s* that exist in *words*.

- If two such substrings overlap, you should wrap them together with only one pair of closed bold-tag.
- If two substrings wrapped by bold tags are consecutive, you should combine them.

Return *s* after adding the bold tags.

Example 1:

Input: *s* = "abcxyz123", *words* = ["abc","123"]

Output: "abcxyz123"

Explanation: The two strings of *words* are substrings of *s* as following: "abcxyz123".

We add `<b>` before each substring and `</b>` after each substring.

Example 2:

Input: `s = "aaabbb"`, `words = ["aa","b"]`

Output: `"<b>aaabbb</b>"`

Explanation:

"aa" appears as a substring two times: "aaabbb" and "aaabbb".

"b" appears as a substring three times: "aaabbb", "aaabbb", and "aaabbb".

We add `<b>` before each substring and `</b>` after each substring:

`"<b>a<b>a</b>a</b><b>b</b><b>b</b><b>b</b>"`.

Since the first two `<b>`'s overlap, we merge them:

`"<b>aaa</b><b>b</b><b>b</b><b>b</b>"`.

Since now the four `<b>`'s are consecutive, we merge them: `"<b>aaabbb</b>"`.

Constraints:

- $1 \leq s.length \leq 1000$
- $0 \leq words.length \leq 100$
- $1 \leq words[i].length \leq 1000$
- `s` and `words[i]` consist of English letters and digits.
- All the values of `words` are unique.

Note: This question is the same as 758. **Bold Words in String.**

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given a string `s` and list of words, wrap all substrings of `s` that are in `words` with `<b></b>` tags. Overlapping or adjacent bold regions should be merged. Right?"
#

`"s='abcxyz123', words=['abc','123']: '<b>abc</b>xyz<b>123</b>'"` ✓

`s='aaabbcc', words=['aaa','aab','bc']: '<b>aaabbcc</b>c'"` ✓ (overlapping regions merged)"

PHASE 2 — BRUTE FORCE

"Mark each character as bold or not. For each word, find all occurrences in `s` and mark those character positions. Then rebuild the string with tags.

Time: $O(n * m * L)$ where m =words count, L =avg word length. Space: $O(n)$."

```

class _616_AddBoldTag_Brute:
    def addBoldTag(self, s: str, words: list[str]) -> str:
        n = len(s)
        bold = [False] * n
        for word in words:
            start = 0
            while True:
                idx = s.find(word, start)
                if idx == -1: break
                for i in range(idx, idx + len(word)):
                    bold[i] = True
                start = idx + 1
        result = []
        i = 0
        while i < n:
            if bold[i]:
                result.append('<b>')
                while i < n and bold[i]:
                    result.append(s[i]); i += 1
                result.append('</b>')
            else:
                result.append(s[i]); i += 1
        return ''.join(result)

```

PHASE 3 — OPTIMIZE

"Build intervals of bold regions by finding all word matches, merge overlapping intervals,
 # then reconstruct. Use a boolean array for merging – same $O(n \cdot m \cdot L)$ but cleaner.
 # Trie optimization: insert all words, scan s with Trie to find all matches in $O(n \cdot L)$."

PHASE 4 — CLEAN CODE

```

class _616_AddBoldTag:
    def addBoldTag(self, s: str, words: list[str]) -> str:
        n = len(s)
        end = 0 # rightmost bold end seen so far
        bold = [False] * n
        word_set = set(words)
        max_len = max(len(w) for w in words) if words else 0
        for i in range(n):
            for length in range(1, min(max_len, n - i) + 1):
                if s[i:i+length] in word_set:
                    end = max(end, i + length)
            if i < end:
                bold[i] = True
        result = []
        i = 0
        while i < n:
            if bold[i]:
                result.append('<b>')

```

```

        while i < n and bold[i]:
            result.append(s[i]); i += 1
            result.append('</b>')
        else:
            result.append(s[i]); i += 1
    return ''.join(result)

```

PHASE 5 — TEST & DEBUG

```

# s='aaabbcc', words=['aaa','aab','bc']:
# i=0: 'aaa'✓→end=3,'aab'? s[0:3]='aaa'≠'aab'. bold[0,1,2]=True.
# i=1: 'aab'✓→end=max(3,4)=4. bold[1,2,3]=True.
# i=4: 'bc'✓→end=6. bold[4,5]=True.
# bold=[T,T,T,T,T,T,F]. → '<b>aaabbcc</b>c' ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n * max_L): at each position check all prefix lengths up to max_L.
# Space O(n): bold array + set.
# Trie reduces search to O(n * max_L) but with better constant via prefix pruning.
# Follow-up: Interval List Intersections (LC 986) – another interval merging
problem.
# Follow-up: Bold Words in String (LC 758) – identical problem, same solution."

```

Trie

□ #692 Top K Frequent Words

MED[Open on LeetCode](#)

Hash Table · String · Trie · Sorting · Heap
Lists: Grind 169

Problem

Given an array of strings `words` and an integer `k`, return the `k` most frequent strings.

Return the answer sorted by the frequency from highest to lowest. Sort the words with the same frequency by their lexicographical order.

Example 1:

```
Input: words = ["i","love","leetcode","i","love","coding"], k = 2
Output: ["i","love"]
Explanation: "i" and "love" are the two most frequent words.
Note that "i" comes before "love" due to a lower alphabetical order.
```

Example 2:

```
Input: words = ["the","day","is","sunny","the","the","the","sunny","is","is"],
k = 4
Output: ["the","is","sunny","day"]
Explanation: "the", "is", "sunny" and "day" are the four most frequent words,
with the number of occurrence being 4, 3, 2 and 1 respectively.
```

Constraints:

- $1 \leq \text{words.length} \leq 500$
- $1 \leq \text{words}[i].\text{length} \leq 10$

- words[i] consists of lowercase English letters.
- k is in the range [1, The number of unique words[i]]

Follow-up: Could you solve it in $O(n \log(k))$ time and $O(n)$ extra space?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a list of words and k, return the k most frequent words sorted by frequency
# descending. Ties broken alphabetically ascending. Right?"
#
# "words=['i','love','leetcode','i','love','coding'], k=2 → ['i','love'] ✓
# words=['the','day','is','sunny','the','the','the','sunny','is','is'], k=4
# → ['the','is','sunny','day'] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Count frequencies, sort by (-freq, word), take first k.
# Time:  $O(n \log n)$ . Space:  $O(n)$ ."
class _692_TopKFrequentWords_Brute:
    def topKFrequent(self, words: list[str], k: int) -> list[str]:
        from collections import Counter
        freq = Counter(words)
        return sorted(freq.keys(), key=lambda w: (-freq[w], w))[:k]
```

PHASE 3 — OPTIMIZE

```
# "Min-heap of size k with custom comparator.
# Store (-freq, word) so the heap evicts the least desirable (lowest freq,
# alphabetically last).
# Time:  $O(n \log k)$ . Space:  $O(k)$  heap +  $O(n)$  counter."
```

PHASE 4 — CLEAN CODE

```
class _692_TopKFrequentWords:
    def topKFrequent(self, words: list[str], k: int) -> list[str]:
        import heapq
        from collections import Counter
        class Word:
            def __init__(self, word, freq):
                self.word = word
                self.freq = freq
```

```

def __lt__(self, other):
    if self.freq != other.freq:
        return self.freq < other.freq # lower freq = worse
    return self.word > other.word # later alpha = worse

freq = Counter(words)
heap = []
for word, cnt in freq.items():
    heapq.heappush(heap, Word(word, cnt))
    if len(heap) > k:
        heapq.heappop(heap)
return [w.word for w in sorted(heap, key=lambda x: (-x.freq, x.word))]

```

PHASE 5 — TEST & DEBUG

```

# words=['i','love','leetcode','i','love','coding'], k=2:
# freq={'i':2,'love':2,'leetcode':1,'coding':1}.
# heap of size 2: keeps 2 best → ('i',2) and ('love',2).
# sorted by (-freq,word): ('i',2),('love',2) → ['i','love'] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Heap  $O(n \log k)$ : optimal when  $k \ll n$ . Sort  $O(n \log n)$ : simpler code.
# The custom comparator handles the tie-breaking (alphabetical) correctly.
# Follow-up: Top K Frequent Elements (LC 347) – integers, no tie-breaking needed.
# Follow-up: streaming top-k words – maintain heap as words arrive one by one."

```

Trie

□ #720 Longest Word in Dictionary

MED[Open on LeetCode](#)

Array · Hash Table · String · Trie · Sorting

Lists: Denny Zhang

Problem

Given an array of strings `words` representing an English Dictionary, return the longest word in `words` that can be built one character at a time by other words in `words`.

If there is more than one possible answer, return the longest word with the smallest lexicographical order. If there is no answer, return the empty string.

Note that the word should be built from left to right with each additional character being added to the end of a previous word.

Example 1:

```
Input: words = ["w","wo","wor","worl","world"]
Output: "world"
Explanation: The word "world" can be built one character at a time by "w",
"wo", "wor", and "worl".
```

Example 2:

Input: words = ["a","banana","app","appl","ap","apply","apple"]

Output: "apple"

Explanation: Both "apply" and "apple" can be built from other words in the dictionary. However, "apple" is lexicographically smaller than "apply".

Constraints:

- $1 \leq \text{words.length} \leq 1000$
- $1 \leq \text{words}[i].\text{length} \leq 30$
- $\text{words}[i]$ consists of lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given a list of strings, find the longest word that can be built one character at a time

by other words in the list. All prefixes of the word must also be in the list.

If tie, return the smallest alphabetically. Right?"

#

"words=['w','wo','wor','worl','world']: every prefix exists → 'world' ✓

words=['a','banana','app','appl','ap','apply','apple']:

'apply' and 'apple' both have all prefixes → return 'apple' (smaller) ✓"

PHASE 2 — BRUTE FORCE

"Put all words in a set. Sort by (length desc, alpha asc). For each word,

check that all its prefixes are in the set. Return the first match.

Time: $O(n * L)$ where $L = \text{max word length}$. Space: $O(n * L)$."

```
class _720_LongestWordInDictionary_Brute:
```

```
    def longestWord(self, words: list[str]) -> str:
```

```
        word_set = set(words)
```

```
        words.sort(key=lambda w: (-len(w), w))
```

```
        for word in words:
```

```
            if all(word[:i] in word_set for i in range(1, len(word))):
```

```
                return word
```

```
        return ''
```

PHASE 3 — OPTIMIZE

"Trie approach: insert all words, then BFS/DFS from root visiting only nodes

where is_end=True. The deepest reachable node gives the longest valid word.

Time: $O(n * L)$ to build trie + $O(n * L)$ to search. Space: $O(n * L)$."

PHASE 4 — CLEAN CODE

```

class _720_LongestWordInDictionary:
    def longestWord(self, words: list[str]) -> str:
        word_set = set(words)
        result = ''
        for word in words:
            if all(word[:i] in word_set for i in range(1, len(word))):
                if len(word) > len(result) or (len(word) == len(result) and word <
result):
                    result = word
        return result

```

PHASE 5 — TEST & DEBUG

```

# words=['w','wo','wor','worl','world']:
# 'world': check 'w'✓,'wo'✓,'wor'✓,'worl'✓ → all prefixes in set. len=5>0 →
result='world' ✓
#
# words=['a','banana','app','appl','ap','apply','apple']:
# 'apple': 'a','ap','app','appl' all in set ✓. len=5. result='apple'.
# 'apply': all prefixes in set. len=5=5, 'apply'>'apple' → keep 'apple' ✓.

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n * L^2)$  for the set approach (each word checks L prefixes, each  $O(L)$ 
lookup).
# Trie reduces prefix lookup to  $O(1)$  per character –  $O(n * L)$  overall.
# Follow-up: if words are streamed, maintain a trie and update result on each
insert.
# Follow-up: allow at most one missing prefix – extend valid condition in the
BFS/DFS."

```

Trie

□ #1166 Design File System

MED[Open on LeetCode](#)

Hash Table · String · Design · Trie

Lists: Denny Zhang

Problem

You are asked to design a file system that allows you to create new paths and associate them with different values.

The format of a path is one or more concatenated strings of the form: / followed by one or more lowercase English letters. For example, "/leetcode" and "/leetcode/problems" are valid paths while an empty string "" and "/" are not.

Implement the FileSystem class:

- **bool createPath(string path, int value)** Creates a new path and associates a value to it if possible and returns true. Returns false if the path already exists or its parent path doesn't exist.

- **int get(string path)** Returns the value associated with path or returns -1 if the path doesn't exist.

Example 1:

```
Input:
["FileSystem","createPath","get"]
[[],["/a",1],["/a"]]
Output:
[null,true,1]
Explanation:
FileSystem fileSystem = new FileSystem();
```

Example 2:

```
Input:
["FileSystem","createPath","createPath","get","createPath","get"]
[[],["/leet",1],["/leet/code",2],["/leet/code"],["/c/d",1],["/c"]]
Output:
[null,true,true,2,false,-1]
Explanation:
FileSystem fileSystem = new FileSystem();
```

Constraints:

- $2 \leq \text{path.length} \leq 100$
- $1 \leq \text{value} \leq 109$
- Each path is valid and consists of lowercase English letters and '/'.
Note: It is guaranteed that each path is a valid path. For example, it will not be "/leet/leet/code".
- At most 104 calls in total will be made to createPath and get.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Design a file system with createPath(path, value)-bool and get(path)-int.

createPath creates a path with a value – all parent paths must already exist.

```
# Returns false if path exists or parent doesn't exist. get returns value or -1.
Right?"
#
# "createPath('/a',1)→T. createPath('/a/b',2)→T. get('/a')→1. get('/a/b')→2.
# createPath('/a/b',3)→F (already exists). createPath('/c/d',4)→F (no '/c'). ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Use a dict mapping full path string → value.
# createPath: check parent path exists in dict, then insert.
# get: return dict.get(path, -1).
# Time: O(L) per op where L = path length. Space: O(n*L)."
```

```
class _1166_FileSystem_Brute:
    def __init__(self):
        self.paths = {}

    def createPath(self, path: str, value: int) -> bool:
        if path in self.paths or path == '/':
            return False
        parent = path.rsplit('/', 1)[0] or '/'
        if parent != '/' and parent not in self.paths:
            return False
        self.paths[path] = value
        return True

    def get(self, path: str) -> int:
        return self.paths.get(path, -1)
```

PHASE 3 — OPTIMIZE

```
# "Trie where each node maps component names to child nodes and stores a value.
# createPath: split on '/', walk/create nodes, check parent validity.
# get: walk trie, return node value.
# Time: O(L) per op. Space: O(n*L) – same as dict but with natural prefix sharing."
```

PHASE 4 — CLEAN CODE

```
class _1166_FileSystem:
    def __init__(self):
        self.root = {} # trie: nested dicts, '_val' key stores value

    def createPath(self, path: str, value: int) -> bool:
        parts = path.strip('/').split('/')
        node = self.root
        for i, part in enumerate(parts):
            if i == len(parts) - 1:
                if part in node:
                    return False # path already exists
                node[part] = {'_val': value}
            else:
                if part not in node:
                    return False # parent doesn't exist
                node = node[part]
        return True

    def get(self, path: str) -> int:
```



```

parts = path.strip('/').split('/')
node = self.root
for part in parts:
    if part not in node:
        return -1
    node = node[part]
return node.get('_val', -1)

```

PHASE 5 — TEST & DEBUG

```

# createPath('/a',1): parts=['a'], root['a']={'_val':1} → True ✓
# createPath('/a/b',2): parts=['a','b']. node=root['a']. 'b' not in node →
root['a']['b']={'_val':2} → True ✓
# get('/a'): parts=['a']. node=root['a']. return _val=1 ✓
# createPath('/a/b',3): 'b' in root['a'] → False ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Both dict and trie are O(L) per operation where L = path length.
# Trie benefits when many paths share prefixes (memory efficiency).
# Follow-up: add a delete(path) – must handle deleting subtrees recursively.
# Follow-up: list all paths under a prefix – trie DFS from that node."

```

Backtracking

Round 5 · Mid · Medium · 6 questions

Backtracking

□ #17 Letter Combinations of a Phone Number

MED[Open on LeetCode](#)

Hash Table · String · Backtracking

Lists: Grind 169

Problem

Given a string containing digits from 2–9 inclusive, return all possible letter combinations that the number could represent. Return the answer in any order.

A mapping of digits to letters (just like on the telephone buttons) is given below. Note that 1 does not map to any letters.



Example 1:

Input: digits = "23"

Output: ["ad","ae","af","bd","be","bf","cd","ce","cf"]

Example 2:

Input: digits = "2"

Output: ["a","b","c"]

Constraints:

- $1 \leq \text{digits.length} \leq 4$
- `digits[i]` is a digit in the range ['2', '9'].

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a string of digits (2-9), return all possible letter combinations
# the digits could represent on a phone keypad. Return [] for empty input. Right?"
#
# "digits="23": 2→abc, 3→def → [ad,ae,af,bd,be,bf,cd,ce,cf] ✓
# digits="": return [] ✓
# digits="2": → [a,b,c] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Build combinations iteratively using a queue (BFS).
# Start with [''], then for each digit, expand every existing combination
# by appending each letter the digit maps to.
# Time:  $O(4^n * n)$  – 4 letters max per digit, n digits. Space:  $O(4^n * n)$ ."
```

```
class _17_LetterCombinations_Brute:
    def letterCombinations(self, digits: str) -> list[str]:
        if not digits:
            return []
        phone = {'2': 'abc', '3': 'def', '4': 'ghi', '5': 'jkl',
                 '6': 'mno', '7': 'pqrs', '8': 'tuv', '9': 'wxyz'}
        result = ['']
        for digit in digits:
            result = [combo + letter
                     for combo in result
                     for letter in phone[digit]]
        return result
```

PHASE 3 — OPTIMIZE

```
# "Backtracking builds each combination one character at a time and avoids
# storing all intermediate combinations.
# At each depth level choose one letter for that digit, recurse, then undo.
# Time:  $O(4^n * n)$  – same output size, but space is  $O(n)$  call stack +  $O(4^n * n)$ 
# result."
```

PHASE 4 — CLEAN CODE

```
class _17_LetterCombinations:
    def letterCombinations(self, digits: str) -> list[str]:
        if not digits:
            return []
        phone = {'2': 'abc', '3': 'def', '4': 'ghi', '5': 'jkl',
                 '6': 'mno', '7': 'pqrs', '8': 'tuv', '9': 'wxyz'}
        result = []
        def backtrack(index: int, path: list) -> None:
            if index == len(digits):
                result.append(''.join(path))
                return
            for letter in phone[digits[index]]:
                path.append(letter)
```

```
        backtrack(index + 1, path)
        path.pop()
    backtrack(0, [])
    return result
```

PHASE 5 — TEST & DEBUG

```
# digits="23":
# backtrack(0,[]): letters=abc
# path=[a] → backtrack(1,[a]): letters=def
# path=[a,d] → backtrack(2) → append "ad"
# path=[a,e] → append "ae"
# path=[a,f] → append "af"
# path=[b] → "bd","be","bf"
# path=[c] → "cd","ce","cf"
# result=["ad","ae","af","bd","be","bf","cd","ce","cf"] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(4^n * n)$ : at most 4 choices per digit, n digits deep, n to build each string.
# Space  $O(n)$  recursion depth +  $O(4^n * n)$  output.
# Iterative BFS and backtracking have same output; backtracking uses less intermediate space.
# Follow-up: if digits can include 0/1 (no letters), skip those digits.
# Follow-up: combination count problem – just compute product of len(phone[d]) for each d."
```

Backtracking

□ #22 Generate Parentheses

MED
[Open on LeetCode](#)

String · Dynamic Programming · Backtracking
Lists: Grind 169

Problem

Given n pairs of parentheses, write a function to generate all combinations of well-formed parentheses.

Example 1:

Input: $n = 3$
Output: ["((()))", "(()())", "(())()", "()(())", "()()()"]

Example 2:

Input: $n = 1$
Output: ["()"]

Constraints:

- $1 \leq n \leq 8$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given n pairs of parentheses, generate all combinations of well-formed parentheses.

Right?"

#

"n=3: ['((()))', '(()())', '(())()', '()(())', '()()()'] – 5 combinations ✓

```
# n=1: ['()'] ✓
# n=2: ['(())', '()()'] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Generate ALL  $2^{(2n)}$  sequences of '(' and ')' of length 2n,
# then filter to keep only the valid ones.
# Valid: each prefix has at least as many '(' as ')', and total counts are equal.
# Time:  $O(2^{(2n)} * n)$  – exponential, extremely wasteful. Space:  $O(2^{(2n)} * n)$ ."
```

```
class _22_GenerateParentheses_Brute:
    def generateParenthesis(self, n: int) -> list[str]:
        def is_valid(s):
            balance = 0
            for ch in s:
                balance += 1 if ch == '(' else -1
                if balance < 0:
                    return False
            return balance == 0
        def generate_all(current, result):
            if len(current) == 2 * n:
                if is_valid(current):
                    result.append(current)
                return
            generate_all(current + '(', result)
            generate_all(current + ')', result)
        result = []
        generate_all('', result)
        return result
```

PHASE 3 — OPTIMIZE

```
# "Backtracking with validity pruning – only place characters that keep the sequence
valid.
# Track open and close counts. Rules:
# – Add '(' if open < n
# – Add ')' if close < open
# This eliminates all invalid branches early, producing only Catalan(n) valid
strings.
# Time:  $O(4^n / \sqrt{n})$  – Catalan number. Space:  $O(n)$  recursion depth."
```

PHASE 4 — CLEAN CODE

```
class _22_GenerateParentheses:
    def generateParenthesis(self, n: int) -> list[str]:
        result = []
        def backtrack(path: list, open_count: int, close_count: int) -> None:
            if len(path) == 2 * n:
                result.append(''.join(path))
                return
            if open_count < n:
                path.append('(')
                backtrack(path, open_count + 1, close_count)
```



```

        path.pop()
    if close_count < open_count:
        path.append(')')
        backtrack(path, open_count, close_count + 1)
        path.pop()
    backtrack([], 0, 0)
    return result

```

PHASE 5 — TEST & DEBUG

```

# n=2:
# backtrack([],0,0): add '(' → [( [],1,0)
# add '(' → [( [],2,0)
# add ')' → [( [],2,1)
# add ')' → [( [],2,2) → append "()" ✓
# add ')' → [( [],1,1)
# add '(' → [( [],2,1)
# add ')' → [( [],2,2) → append "()" ✓
# result = ["()", "()"] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(4^n / \sqrt{n})$ : nth Catalan number of valid sequences, each of length 2n.
# Space  $O(n)$ : recursion depth is at most 2n, path list is at most 2n characters.
# Brute is  $O(2^{(2n)})$  – exponentially worse.
# Follow-up: count valid sequences (not generate) – just return Catalan(n) =  $C(2n,n)/(n+1)$ .
# Follow-up: LC 32 Longest Valid Parentheses – uses a stack or DP instead."

```

Backtracking

□ #39 Combination Sum

MED[Open on LeetCode](#)

Array · Backtracking

Lists: Grind 169

Problem

Given an array of distinct integers candidates and a target integer target, return a list of all unique combinations of candidates where the chosen numbers sum to target. You may return the combinations in any order.

The same number may be chosen from candidates an unlimited number of times. Two combinations are unique if the frequency of at least one of the chosen numbers is different.

The test cases are generated such that the number of unique combinations that sum up to target is less than 150 combinations for the given input.

Example 1:

Input: candidates = [2,3,6,7], target = 7
 Output: [[2,2,3],[7]]
 Explanation:
 2 and 3 are candidates, and $2 + 2 + 3 = 7$. Note that 2 can be used multiple times.
 7 is a candidate, and $7 = 7$.
 These are the only two combinations.

Example 2:

Input: candidates = [2,3,5], target = 8
 Output: [[2,2,2,2],[2,3,3],[3,5]]

Example 3:

Input: candidates = [2], target = 1
 Output: []

Constraints:

- $1 \leq \text{candidates.length} \leq 30$
- $2 \leq \text{candidates}[i] \leq 40$
- All elements of candidates are distinct.
- $1 \leq \text{target} \leq 40$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given an array of distinct positive integers and a target, return all unique
# combinations that sum to target. Each number may be used unlimited times.
# The solution set must not contain duplicate combinations. Right?"
#
# "candidates=[2,3,6,7], target=7:
# [2,2,3] (2+2+3=7), [7] (7=7) ✓
# candidates=[2,3,5], target=8: [2,2,2,2],[2,3,3],[3,5] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Generate all combinations with repetition up to target sum, filter by sum.
# Equivalent to BFS: start from [], extend by each candidate, keep if sum <= target.
# Time:  $O(n^{(t/\min\_c)})$  — exponential. Space: same."
```

```
class _39_CombinationSum_Brute:
```

```

def combinationSum(self, candidates: list[int], target: int) ->
list[list[int]]:
    result = []
    candidates.sort()
    def generate(start: int, path: list, remaining: int) -> None:
        if remaining == 0:
            result.append(path[:])
            return
        for i in range(start, len(candidates)):
            if candidates[i] > remaining:
                break # pruning: sorted, no point continuing
            path.append(candidates[i])
            generate(i, path, remaining - candidates[i]) # i not i+1: reuse
allowed
            path.pop()
        generate(0, [], target)
    return result

```

PHASE 3 — OPTIMIZE

"The brute approach IS backtracking with pruning – sort first so we can break early.

The key optimisations over naive BFS:

– Start index prevents re-use of earlier candidates (avoids duplicates like [3,2] and [2,3])

– Sort + break early prunes branches where candidate > remaining

Time: $O(n^{(t/\min_c)})$: still exponential in worst case but pruned significantly."

PHASE 4 — CLEAN CODE

```

class _39_CombinationSum:
    def combinationSum(self, candidates: list[int], target: int) ->
list[list[int]]:
        candidates.sort()
        result = []
        def backtrack(start: int, path: list, remaining: int) -> None:
            if remaining == 0:
                result.append(path[:])
                return
            for i in range(start, len(candidates)):
                if candidates[i] > remaining:
                    break
                path.append(candidates[i])
                backtrack(i, path, remaining - candidates[i])
                path.pop()
        backtrack(0, [], target)
    return result

```

PHASE 5 — TEST & DEBUG

candidates=[2,3,6,7], target=7:

backtrack(0,[],7): try 2 → backtrack(0,[2],5)

```
# try 2 → backtrack(0,[2,2],3)
# try 2 → backtrack(0,[2,2,2],1) → try 2>1 break
# try 3 → backtrack(1,[2,2,3],0) → append [2,2,3] ✓
# try 3 → backtrack(1,[2,3],2) → try 3>2 break
# try 3 → backtrack(1,[3],4) → try 3→[3,3]:remaining=1, try 6>1 break
# try 6 → backtrack(2,[6],1) → try 6>1 break
# try 7 → backtrack(3,[7],0) → append [7] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n^{(t/\min\_c)})$ : at most  $t/\min\_c$  levels deep, n choices each level.
# Space  $O(t/\min\_c)$ : max recursion depth.
# Follow-up: Combination Sum II (LC 40) – each number used ONCE, has duplicates in input.
# Use i+1 instead of i, and skip duplicate candidates at same level.
# Follow-up: Combination Sum III (LC 216) – exactly k numbers from 1–9 summing to n."
```

Backtracking

□ #46 Permutations

MED[Open on LeetCode](#)

Array · Backtracking

Lists: Grind 169

Problem

Given an array `nums` of distinct integers, return all the possible permutations. You can return the answer in any order.

Example 1:

Input: `nums = [1,2,3]`

Output: `[[1,2,3],[1,3,2],[2,1,3],[2,3,1],[3,1,2],[3,2,1]]`

Example 2:

Input: `nums = [0,1]`

Output: `[[0,1],[1,0]]`

Example 3:

Input: `nums = [1]`

Output: `[[1]]`

Constraints:

- $1 \leq \text{nums.length} \leq 6$
- $-10 \leq \text{nums}[i] \leq 10$
- All the integers of `nums` are unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given an array of distinct integers, return all possible permutations
# in any order. Right?"
#
# "nums=[1,2,3]: [[1,2,3],[1,3,2],[2,1,3],[2,3,1],[3,1,2],[3,2,1]] – 6 perms ✓
# nums=[0,1]: [[0,1],[1,0]] ✓
# nums=[1]: [[1]] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Use a visited array. At each position try every unvisited number.
# When path length equals n, record the permutation.
# Time:  $O(n! * n)$  –  $n!$  permutations each of length  $n$ . Space:  $O(n)$  recursion depth."
```

```
class _46_Permutations_Brute:
    def permute(self, nums: list[int]) -> list[list[int]]:
        result = []
        used = [False] * len(nums)
        def backtrack(path: list) -> None:
            if len(path) == len(nums):
                result.append(path[:])
                return
            for i in range(len(nums)):
                if used[i]:
                    continue
                used[i] = True
                path.append(nums[i])
                backtrack(path)
                path.pop()
                used[i] = False
        backtrack([])
        return result
```

PHASE 3 — OPTIMIZE

```
# "Swap-based in-place backtracking – no visited array needed.
# Maintain a 'start' index. Everything before start is already placed.
# For each position from start onward, swap it into the start slot, recurse, then
# swap back.
# Time:  $O(n! * n)$ . Space:  $O(n)$  recursion depth – avoids the extra visited array."
```

PHASE 4 — CLEAN CODE

```
class _46_Permutations:
    def permute(self, nums: list[int]) -> list[list[int]]:
        result = []
        def backtrack(start: int) -> None:
            if start == len(nums):
                result.append(nums[:])
                return
            for i in range(start, len(nums)):
                nums[start], nums[i] = nums[i], nums[start]
                backtrack(start + 1)
                nums[start], nums[i] = nums[i], nums[start]
```

```

        for i in range(start, len(nums)):
            nums[start], nums[i] = nums[i], nums[start]
            backtrack(start + 1)
            nums[start], nums[i] = nums[i], nums[start]
    backtrack(0)
    return result

```

PHASE 5 — TEST & DEBUG

```

# nums=[1,2,3]:
# backtrack(0): swap(0,0)→[1,2,3], backtrack(1):
# swap(1,1)→[1,2,3], backtrack(2):
# swap(2,2)→[1,2,3], backtrack(3) → append [1,2,3] ✓
# swap(1,2)→[1,3,2], backtrack(2) → append [1,3,2] ✓, swap back→[1,2,3]
# swap(0,1)→[2,1,3], backtrack(1) → [2,1,3],[2,3,1] ✓, swap back→[1,2,3]
# swap(0,2)→[3,2,1], backtrack(1) → [3,2,1],[3,1,2] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n! * n)$ :  $n!$  permutations, each takes  $O(n)$  to copy.
# Space  $O(n)$ : recursion depth  $n$ , no visited array (swap approach).
# Visited-array approach uses  $O(n)$  extra space for the boolean array.
# Follow-up: Permutations II (LC 47) – input has duplicates. Sort first, skip
duplicate
# values at the same recursion level to avoid identical permutations.
# Follow-up: next permutation (LC 31) – generate just the next one in lexicographic
order."

```


Backtracking

□ #79 Word Search

MED[Open on LeetCode](#)

Array · String · Backtracking · Matrix

Lists: Grind 169

Problem

Given an $m \times n$ grid of characters board and a string word, return true if word exists in the grid.

The word can be constructed from letters of sequentially adjacent cells, where adjacent cells are horizontally or vertically neighboring. The same letter cell may not be used more than once.

Example 1:

A	B	C	E
S	F	C	S
A	D	E	E

Input: board = [["A","B","C","E"], ["S","F","C","S"], ["A","D","E","E"]], word = "ABCCED"

Output: true

Example 2:

A	B	C	E
S	F	C	S
A	D	E	E

Input: board = [["A","B","C","E"], ["S","F","C","S"], ["A","D","E","E"]], word = "SEE"

Output: true

Example 3:

A	B	C	E
S	F	C	S
A	D	E	E

Input: board = [["A","B","C","E"],["S","F","C","S"],["A","D","E","E"]], word = "ABCB"

Output: false

Constraints:

- $m == \text{board.length}$
- $n = \text{board}[i].\text{length}$
- $1 \leq m, n \leq 6$
- $1 \leq \text{word.length} \leq 15$
- board and word consists of only lowercase and uppercase English letters.

Follow up: Could you use search pruning to make your solution faster with a larger board?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given an $m \times n$ grid of characters and a word, return true if the word exists in the grid following adjacent (up/down/left/right) cells without reusing cells. Right?"

#

"board=[['A','B','C','E'],['S','F','C','S'],['A','D','E','E']], word='ABCCED':

A(0,0)→B(0,1)→C(0,2)→C(1,2)→E(2,2)→D(2,1) ✓

word='ABCB': A→B→C→B would revisit B(0,1) → false ✓"

PHASE 2 — BRUTE FORCE

"For every cell as potential start, try every possible path with DFS.

Mark visited cells with a sentinel, unmark on backtrack.

Time: $O(m \times n \times 4^L)$ where L = word length. Space: $O(L)$ recursion depth."

class _79_WordSearch_Brute:

```
def exist(self, board: list[list[str]], word: str) -> bool:
    rows, cols = len(board), len(board[0])
```

```
    def dfs(r, c, idx):
```

```
        if idx == len(word):
```

```
            return True
```

```
        if r < 0 or r >= rows or c < 0 or c >= cols:
```

```
            return False
```

```
        if board[r][c] != word[idx]:
```

```
            return False
```

```
        tmp, board[r][c] = board[r][c], '#'
```

```
        found = (dfs(r+1,c,idx+1) or dfs(r-1,c,idx+1) or
```

```
                 dfs(r,c+1,idx+1) or dfs(r,c-1,idx+1))
```

```
        board[r][c] = tmp
```

```
        return found
```

```
    for r in range(rows):
```

```
        for c in range(cols):
```

```
            if dfs(r, c, 0):
```

```
                return True
```

```
    return False
```

PHASE 3 — OPTIMIZE

"Same DFS with two pruning improvements:

1. Before DFS, check character frequency – if board has fewer of any letter than word needs, return False early.

2. If word's last char is rarer in board than first char, reverse the word to start from the rarer end.

```
# These prune the search space significantly for large grids.
# Time:  $O(m*n*4^L)$  worst case but much faster in practice."
```

PHASE 4 — CLEAN CODE

```
class _79_WordSearch:
    def exist(self, board: list[list[str]], word: str) -> bool:
        from collections import Counter
        rows, cols = len(board), len(board[0])
        board_count = Counter(c for row in board for c in row)
        word_count = Counter(word)
        for ch, cnt in word_count.items():
            if board_count[ch] < cnt:
                return False
        if board_count[word[0]] > board_count[word[-1]]:
            word = word[::-1]
        def dfs(r, c, idx):
            if idx == len(word):
                return True
            if not (0 <= r < rows and 0 <= c < cols) or board[r][c] != word[idx]:
                return False
            tmp, board[r][c] = board[r][c], '#'
            found = (dfs(r+1,c,idx+1) or dfs(r-1,c,idx+1) or
                    dfs(r,c+1,idx+1) or dfs(r,c-1,idx+1))
            board[r][c] = tmp
            return found
        return any(dfs(r, c, 0) for r in range(rows) for c in range(cols))
```

PHASE 5 — TEST & DEBUG

```
# board=[['A','B','C','E'],...], word='ABCCED':
# Start dfs(0,0,'A'): match→mark '#'. dfs(0,1,'B'): match→mark. dfs(0,2,'C'):
match.
# dfs(1,2,'C'): match. dfs(2,2,'E'): match. dfs(2,1,'D'): match. idx=6=len → True ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m*n*4^L)$ : each cell can start a DFS branching 4 ways up to L levels deep.
# Space  $O(L)$ : recursion stack depth bounded by word length.
# Follow-up: Word Search II (LC 212) – find ALL words from a list in the board.
# Build a Trie of all words, DFS the board once –  $O(m*n*4^L + \text{sum of word lengths})$ ."
```

Backtracking

#113 Path Sum II

MED[Open on LeetCode](#)

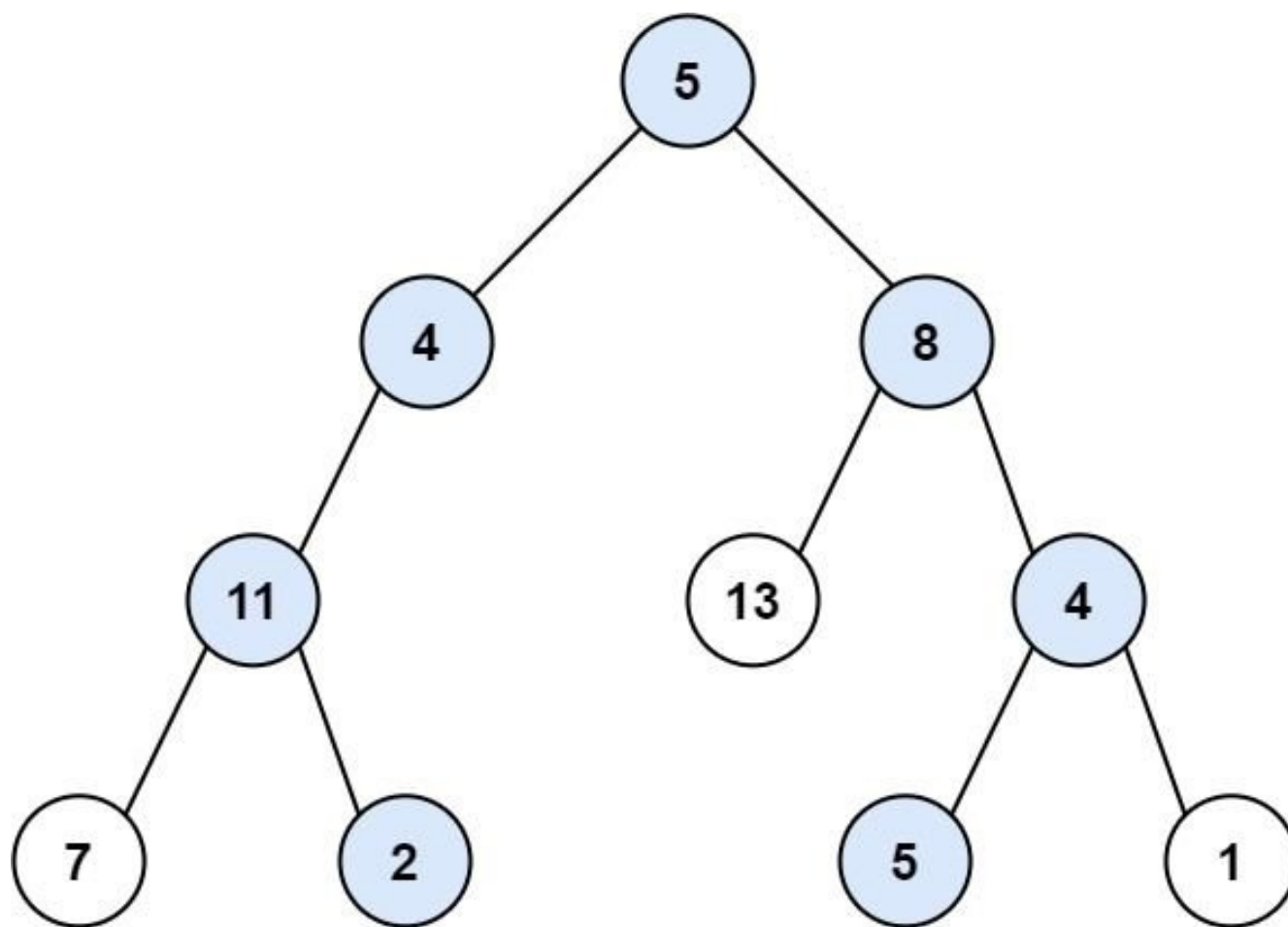
Backtracking · Tree · DFS**Lists: Grind 169**

Problem

Given the root of a binary tree and an integer `targetSum`, return all root-to-leaf paths where the sum of the node values in the path equals `targetSum`. Each path should be returned as a list of the node values, not node references.

A root-to-leaf path is a path starting from the root and ending at any leaf node. A leaf is a node with no children.

Example 1:



Input: root = [5,4,8,11,null,13,4,7,2,null,null,5,1], targetSum = 22

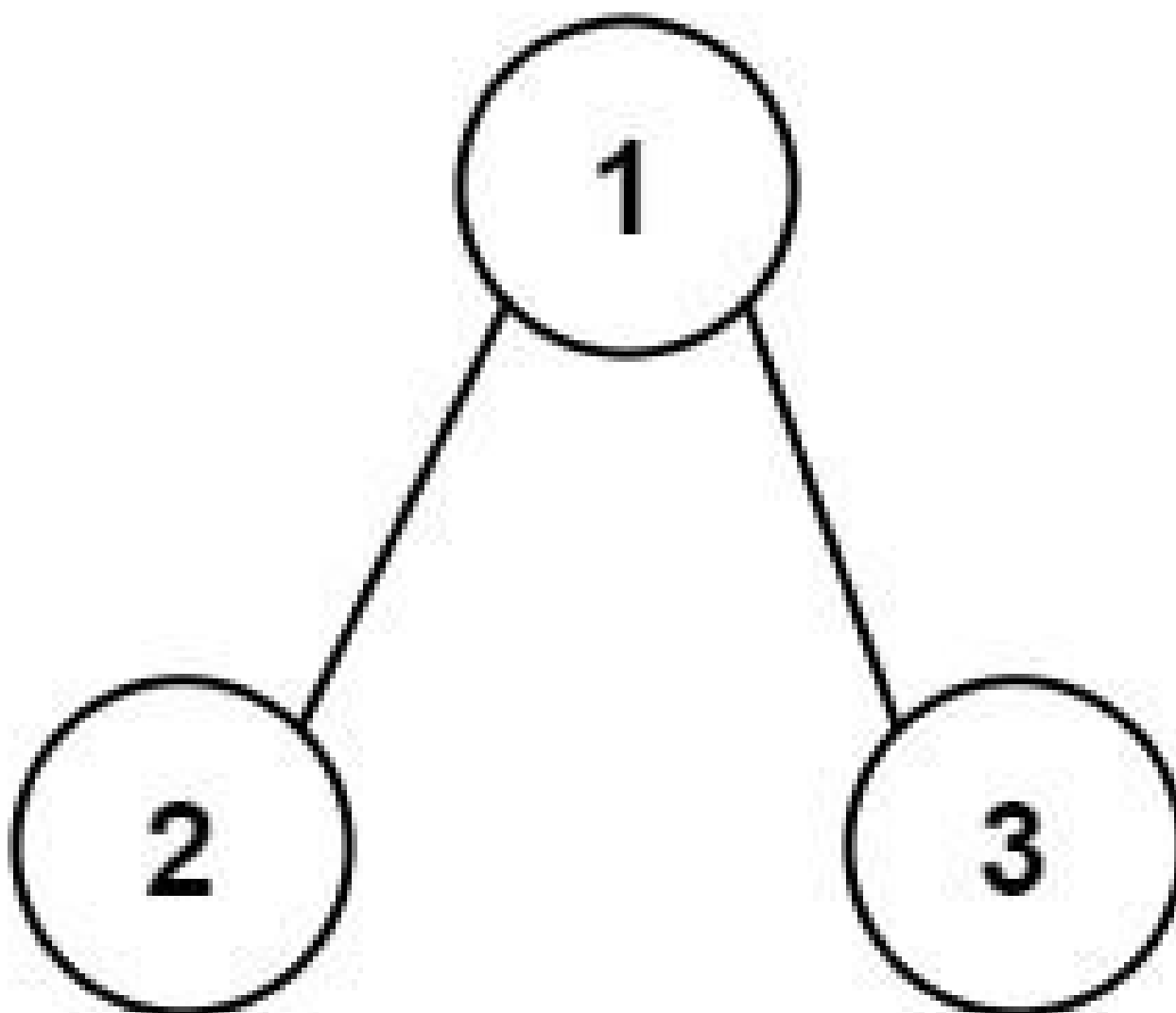
Output: [[5,4,11,2],[5,8,4,5]]

Explanation: There are two paths whose sum equals targetSum:

$5 + 4 + 11 + 2 = 22$

$5 + 8 + 4 + 5 = 22$

Example 2:



Input: root = [1,2,3], targetSum = 5
Output: []

Example 3:

Input: root = [1,2], targetSum = 0
Output: []

Constraints:

- The number of nodes in the tree is in the range [0, 5000].
- $-1000 \leq \text{Node.val} \leq 1000$
- $-1000 \leq \text{targetSum} \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a binary tree and a target sum, return all root-to-leaf paths
# where the sum of node values equals targetSum. Right?"
#
# "tree=[5,4,8,11,null,13,4,7,2,null,null,5,1], targetSum=22:
# 5→4→11→2=22 ✓, 5→8→4→5=22 ✓ → [[5,4,11,2],[5,8,4,5]] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "DFS collecting all root-to-leaf paths, then filter by sum.
# Time: O(n*L) where L = path length (average tree height). Space: O(n*L)."
```

```
class _113_PathSumII_Brute:
    def pathSum(self, root, targetSum: int) -> list[list[int]]:
        all_paths = []
        def dfs(node, path):
            if not node:
                return
            path.append(node.val)
            if not node.left and not node.right:
                all_paths.append(path[:])
            dfs(node.left, path)
            dfs(node.right, path)
            path.pop()
        dfs(root, [])
        return [p for p in all_paths if sum(p) == targetSum]
```

PHASE 3 — OPTIMIZE

```
# "Backtracking with running sum — check target only at leaf nodes.
# Pass remaining = targetSum - node.val down the tree.
# At a leaf, if remaining == 0, this path is a solution. Append and backtrack.
# Time: O(n*L). Space: O(L) recursion + O(n*L) output — avoids post-filtering."
```

PHASE 4 — CLEAN CODE

```
class _113_PathSumII:
    def pathSum(self, root, targetSum: int) -> list[list[int]]:
        result = []
        def backtrack(node, remaining, path):
            if not node:
                return
            path.append(node.val)
            remaining -= node.val
            if not node.left and not node.right and remaining == 0:
                result.append(path[:])
            else:
                backtrack(node.left, remaining, path)
                backtrack(node.right, remaining, path)
```

```

    path.pop()
    backtrack(root, targetSum, [])
    return result

```

PHASE 5 — TEST & DEBUG

```

# tree=[5,4,8,11,null,13,4,7,2,null,null,5,1], target=22:
# path=[5],rem=17 → left=[5,4],rem=13 → left=[5,4,11],rem=2
# left=[5,4,11,7],rem=-5 → leaf, rem≠0. pop.
# right=[5,4,11,2],rem=0 → leaf, rem=0 → append [5,4,11,2] ✓
# right=[5,8],rem=14 → right=[5,8,4],rem=10 → left=[5,8,4,5],rem=5 → leaf≠0
# right=[5,8,4,1],rem=9 → leaf≠0. Back up. left=[5,8,13],leaf,rem=1≠0.

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n*L): visit each node once, copy path at each leaf O(L).
# Space O(L): recursion stack depth = tree height.
# Follow-up: Path Sum I (LC 112) – just return bool, no need to store paths.
# Follow-up: Path Sum III (LC 437) – any path (not just root-to-leaf), any
direction.
# Use prefix sum hashmap to count valid paths in O(n)."

```

Greedy

Round 5 · Mid · Medium · 5 questions

Greedy

□ #134 Gas Station

MED[Open on LeetCode](#)

Array · Greedy

Lists: Grind 169

Problem

There are n gas stations along a circular route, where the amount of gas at the i th station is $gas[i]$.

You have a car with an unlimited gas tank and it costs $cost[i]$ of gas to travel from the i th station to its next $(i + 1)$ th station. You begin the journey with an empty tank at one of the gas stations.

Given two integer arrays gas and $cost$, return the starting gas station's index if you can travel around the circuit once in the clockwise direction, otherwise return -1 . If there exists a solution, it is guaranteed to be unique.

Example 1:

Input: gas = [1,2,3,4,5], cost = [3,4,5,1,2]

Output: 3

Explanation:

Start at station 3 (index 3) and fill up with 4 unit of gas. Your tank = $0 + 4 = 4$

Travel to station 4. Your tank = $4 - 1 + 5 = 8$

Travel to station 0. Your tank = $8 - 2 + 1 = 7$

Travel to station 1. Your tank = $7 - 3 + 2 = 6$

Travel to station 2. Your tank = $6 - 4 + 3 = 5$

Example 2:

Input: gas = [2,3,4], cost = [3,4,3]

Output: -1

Explanation:

You can't start at station 0 or 1, as there is not enough gas to travel to the next station.

Let's start at station 2 and fill up with 4 unit of gas. Your tank = $0 + 4 = 4$

Travel to station 0. Your tank = $4 - 3 + 2 = 3$

Travel to station 1. Your tank = $3 - 3 + 3 = 3$

You cannot travel back to station 2, as it requires 4 unit of gas but you only have 3.

Constraints:

- $n == \text{gas.length} == \text{cost.length}$
- $1 \leq n \leq 105$
- $0 \leq \text{gas}[i], \text{cost}[i] \leq 104$
- The input is generated such that the answer is unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"n gas stations in a circle; gas[i] = gas at station i, cost[i] = gas to reach i+1.

Find the starting station index to complete the circuit, or -1 if impossible. Right?"

#

"gas=[1,2,3,4,5], cost=[3,4,5,1,2]:

Start at 3: $0+4=4 \rightarrow 4-1=3 \rightarrow 3+5=8 \rightarrow 8-2=6 \rightarrow 6+1=7 \rightarrow 7-3=4 \geq 0 \rightarrow \text{return } 3 \checkmark$

```
# gas=[2,3,4], cost=[3,4,3]: total gas=9 < total cost=10 → -1 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Try each station as start. Simulate the full circuit; if we never run dry, return that index."
```

```
# Time:  $O(n^2)$ . Space:  $O(1)$ ."
```

```
class _134_GasStation_Brute:
    def canCompleteCircuit(self, gas: list[int], cost: list[int]) -> int:
        n = len(gas)
        for start in range(n):
            tank = 0
            completed = True
            for i in range(n):
                idx = (start + i) % n
                tank += gas[idx] - cost[idx]
                if tank < 0:
                    completed = False
                    break
            if completed:
                return start
        return -1
```

PHASE 3 — OPTIMIZE

```
# "Two greedy observations:
```

```
# 1. If total gas < total cost, no solution exists → return -1.
```

```
# 2. If a solution exists, it's unique. Start with candidate = 0 and tank = 0.
```

```
# Accumulate net gain (gas[i]-cost[i]). When tank < 0, reset tank to 0 and
```

```
# move candidate to i+1 (we can't start anywhere in [old_candidate, i]).
```

```
# The final candidate is the answer.
```

```
# Time:  $O(n)$ . Space:  $O(1)$ ."
```

PHASE 4 — CLEAN CODE

```
class _134_GasStation:
    def canCompleteCircuit(self, gas: list[int], cost: list[int]) -> int:
        if sum(gas) < sum(cost):
            return -1
        tank = 0
        candidate = 0
        for i in range(len(gas)):
            tank += gas[i] - cost[i]
            if tank < 0:
                tank = 0
                candidate = i + 1
        return candidate
```

PHASE 5 — TEST & DEBUG

```
# gas=[1,2,3,4,5], cost=[3,4,5,1,2]:
```

```
# total=15>=15 ✓. i=0: tank=1-3=-2<0→tank=0, cand=1. i=1: tank=2-4=-2<0→cand=2.
```

```
# i=2: tank=3-5=-2<0→cand=3. i=3: tank=4-1=3. i=4: tank=3+5-2=6≥0.
```

```
# return 3 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: single pass. Space $O(1)$: two variables.

The key insight: if we run out of gas starting at s , no station between s and i can work

(they'd all start with less gas than s did at that point).

Follow-up: if multiple valid starting points exist (which the problem guarantees won't happen),

return the smallest index – the greedy still finds the first valid one.

Follow-up: minimum cost to fill gas (variant) – uses a monotonic stack approach."

Greedy

□ #179 Largest Number

MED

[Open on LeetCode](#)

Array · String · Greedy · Sorting

Lists: Grind 169

Problem

Given a list of non-negative integers `nums`, arrange them such that they form the largest number and return it.

Since the result may be very large, so you need to return a string instead of an integer.

Example 1:

```
Input: nums = [10,2]
Output: "210"
```

Example 2:

```
Input: nums = [3,30,34,5,9]
Output: "9534330"
```

Constraints:

- $1 \leq \text{nums.length} \leq 100$
- $0 \leq \text{nums}[i] \leq 10^9$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given a list of non-negative integers, arrange them so they form the largest number."

Return as a string. Right?"

#

"nums=[10,2]: '210' > '102' → "210" ✓

nums=[3,30,34,5,9]: → "9534330" ✓

nums=[0,0]: → "0" (not "00") ✓"

PHASE 2 — BRUTE FORCE

"Try all permutations of nums, form the concatenated string for each,

return the maximum. Time: $O(n! * n)$. Space: $O(n!)$."

class _179_LargestNumber_Brute:

def largestNumber(self, nums: list[int]) -> str:

from itertools import permutations

best = ''

for perm in permutations(nums):

candidate = ''.join(map(str, perm))

if candidate > best:

best = candidate

return best

PHASE 3 — OPTIMIZE

"Custom comparator: to decide whether a comes before b, compare str(a)+str(b) vs str(b)+str(a)."

If str(a)+str(b) > str(b)+str(a), a should come first.

Sort all numbers by this comparator, then concatenate.

Edge case: if the largest digit is 0, the result is '0'.

Time: $O(n \log n)$. Space: $O(n)$."

PHASE 4 — CLEAN CODE

class _179_LargestNumber:

def largestNumber(self, nums: list[int]) -> str:

from functools import cmp_to_key

def compare(a, b):

return (1 if a + b > b + a else -1 if a + b < b + a else 0)

strs = list(map(str, nums))

strs.sort(key=cmp_to_key(compare), reverse=True)

result = ''.join(strs)

return '0' if result[0] == '0' else result

PHASE 5 — TEST & DEBUG

nums=[10,2]:

compare("10","2"): "102" vs "210" → "210">"102" → 2 before 10. sorted=["2","10"].
→ "210" ✓

#

nums=[3,30,34,5,9]:

compare pairs: "9">"53">"534">"5330"→ sorted=["9","5","34","3","30"]. → "9534330"

✓

#

```
# nums=[0,0]: sorted=["0","0"] → "00" → first char '0' → return "0" ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n \log n)$: sort dominates. Space $O(n)$: string array + result.

The comparator is transitive and consistent – provably correct for any input.

Follow-up: smallest number from array – same comparator, ascending sort.

Follow-up: if numbers are very large (BigInteger), convert to strings first anyway."

Greedy

★ #280 Wiggle Sort ★

MED[Open on LeetCode](#)

Array · Greedy · Sorting

Lists: ★ Premium | Premium 98

Problem

Given an integer array `nums`, reorder it such that `nums[0] <= nums[1] >= nums[2] <= nums[3]....`

You may assume the input array always has a valid answer.

Example 1:

Input: `nums = [3,5,2,1,6,4]`
Output: `[3,5,1,6,2,4]`
Explanation: `[1,6,2,5,3,4]` is also accepted.

Example 2:

Input: `nums = [6,6,5,6,3,8]`
Output: `[6,6,5,6,3,8]`

Constraints:

- $1 \leq \text{nums.length} \leq 5 * 10^4$
- $0 \leq \text{nums}[i] \leq 10^4$
- It is guaranteed that there will be an answer for the given input `nums`.

Follow up: Could you solve the problem in $O(n)$ time complexity?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Reorder an array in-place so that nums[0] <= nums[1] >= nums[2] <= nums[3]...
# (alternating less-than and greater-than). Multiple valid answers exist. Right?"
#
# "nums=[3,5,2,1,6,4]: one valid output=[3,5,1,6,2,4] ✓
# nums=[1,5,1,1,6,4]: one valid=[1,5,1,6,1,4] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Sort the array, then interleave first half and second half.
# Split sorted array into two halves; place smaller half at even indices,
# larger half at odd indices.
# Time:  $O(n \log n)$ . Space:  $O(n)$  for the copy."
```

```
class _280_WiggleSort_Brute:
    def wiggleSort(self, nums: list[int]) -> None:
        sorted_nums = sorted(nums)
        n = len(nums)
        mid = (n + 1) // 2
        nums[0::2] = sorted_nums[:mid] # even indices: smaller half
        nums[1::2] = sorted_nums[mid:] # odd indices: larger half
```

PHASE 3 — OPTIMIZE

```
# "Greedy one-pass: for each index i, if the wiggle condition is violated, swap.
# - Even index i: nums[i] should be <= nums[i-1]. If not, swap(i, i-1).
# - Odd index i: nums[i] should be >= nums[i-1]. If not, swap(i, i-1).
# A local swap always fixes the current violation without breaking prior pairs.
# Time:  $O(n)$ . Space:  $O(1)$ ."
```

PHASE 4 — CLEAN CODE

```
class _280_WiggleSort:
    def wiggleSort(self, nums: list[int]) -> None:
        for i in range(1, len(nums)):
            if (i % 2 == 1 and nums[i] < nums[i - 1]) or (i % 2 == 0 and nums[i] >
nums[i - 1]):
                nums[i], nums[i - 1] = nums[i - 1], nums[i]
```

PHASE 5 — TEST & DEBUG

```
# nums=[3,5,2,1,6,4]:
# i=1(odd): 5>=3 ✓
# i=2(even): 2<=5 ✓
# i=3(odd): 1<2 → swap → [3,5,1,2,6,4]. Wait: now
[3,5,2,1,6,4]→swap(2,1)→[3,5,1,2,6,4]
```

```
# i=4(even): 6>2 → swap → [3,5,1,6,2,4]
# i=5(odd): 4>2 ✓ → [3,5,1,6,2,4] ✓ (valid wiggle)
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Greedy: O(n) time, O(1) space vs brute O(n log n) time, O(n) space.
# The greedy works because swapping adjacent elements to fix one condition
# never invalidates the previously satisfied condition.
# Follow-up: Wiggle Sort II (LC 324) – strict inequality nums[0]<nums[1]>nums[2]<...
# Requires median partitioning (nth_element) to guarantee strict separation.
# Follow-up: count wiggle subsequences (LC 376) – DP/greedy to count without
reordering."
```

Greedy

□ #954 Array of Doubled Pairs

MED[Open on LeetCode](#)

Array · Hash Table · Greedy · Sorting

Lists: CodeSignal

Problem

Given an integer array of even length `arr`, return `true` if it is possible to reorder `arr` such that $arr[2 * i + 1] = 2 * arr[2 * i]$ for every $0 \leq i < len(arr) / 2$, or `false` otherwise.

Example 1:

Input: `arr = [3,1,3,6]`
Output: `false`

Example 2:

Input: `arr = [2,1,2,6]`
Output: `false`

Example 3:

Input: `arr = [4,-2,2,-4]`
Output: `true`
Explanation: We can take two groups, `[-2,-4]` and `[2,4]` to form `[-2,-4,2,4]` or `[2,4,-2,-4]`.

Constraints:

- $2 \leq arr.length \leq 3 * 10^4$

- `arr.length` is even.
- $-105 \leq \text{arr}[i] \leq 105$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given an integer array, return true if we can reorder it so that for every
#  $\text{arr}[2i+1] = 2 * \text{arr}[2i]$  (each pair: smaller then double). Right?"
#
# "arr=[3,1,3,6]: pairs (1,2) missing 2, can't pair → False... wait:
# arr=[4,-2,2,-4]: sort by abs → [-2,-4,2,4]? pairs (-2,-4)?  $-4 = -2 * 2$  ✓, (2,4) ✓ →
True ✓
# arr=[1,2,4,16,8,4]: sort abs→[1,2,4,4,8,16]: 1→2 ✓, 4→8 ✓, 4→16? No → False ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Sort by absolute value. Greedily pair smallest unmatched element with its double.
# Use a counter to track available elements.
# Time:  $O(n \log n)$ . Space:  $O(n)$ ."
```

```
class _954_ArrayOfDoubledPairs_Brute:
    def canReorderDoubled(self, arr: list[int]) -> bool:
        from collections import Counter
        count = Counter(arr)
        for x in sorted(arr, key=abs):
            if count[x] == 0:
                continue
            if count[2 * x] == 0:
                return False
            count[x] -= 1
            count[2 * x] -= 1
        return True
```

PHASE 3 — OPTIMIZE

```
# "Same greedy but iterate over sorted unique keys (by abs value) to avoid
# processing same element multiple times.
# Time:  $O(n \log n)$  for sorting. Space:  $O(n)$  counter."
```

PHASE 4 — CLEAN CODE

```
class _954_ArrayOfDoubledPairs:
    def canReorderDoubled(self, arr: list[int]) -> bool:
        from collections import Counter
        count = Counter(arr)
        for x in sorted(count, key=abs):
            if count[2 * x] < count[x]:
                return False
            count[2 * x] -= count[x]
```



```
        count[x] = 0
    return True
```

PHASE 5 — TEST & DEBUG

```
# arr=[4,-2,2,-4]: count={4:1,-2:1,2:1,-4:1}.
# sorted by abs: [-2,-4,2,4].
# x=-2: count[2*(-2)]=count[-4]=1>=count[-2]=1 ✓. count[-4]=0, count[-2]=0.
# x=-4: count[-8]=0>=count[-4]=0 ✓ (trivially).
# x=2: count[4]=1>=count[2]=1 ✓. count[4]=0, count[2]=0.
# return True ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n \log n)$ : sorting dominates. Space  $O(n)$  for counter.
# Key: sort by absolute value handles negative numbers correctly (-2 pairs with -4).
# Follow-up: if 'double' is replaced by 'k times' – same greedy, check count[k*x].
# Follow-up: find the actual pairs (not just existence) – store pairs during the greedy pass."
```

Greedy

□ #2131 Longest Palindrome by Concatenating Two Letter Words

MED[Open on LeetCode](#)

Array · Hash Table · String · Greedy · Counting Lists: CodeSignal

Problem

You are given an array of strings `words`. Each element of `words` consists of two lowercase English letters.

Create the longest possible palindrome by selecting some elements from `words` and concatenating them in any order. Each element can be selected at most once.

Return the length of the longest palindrome that you can create. If it is impossible to create any palindrome, return 0.

A palindrome is a string that reads the same forward and backward.

Example 1:

Input: words = ["lc","cl","gg"]

Output: 6

Explanation: One longest palindrome is "lc" + "gg" + "cl" = "lcggcl", of length 6.

Note that "clggcl" is another longest palindrome that can be created.

Example 2:

Input: words = ["ab","ty","yt","lc","cl","ab"]

Output: 8

Explanation: One longest palindrome is "ty" + "lc" + "cl" + "yt" = "tylcclyt", of length 8.

Note that "lcyttycl" is another longest palindrome that can be created.

Example 3:

Input: words = ["cc","ll","xx"]

Output: 2

Explanation: One longest palindrome is "cc", of length 2.

Note that "ll" is another longest palindrome that can be created, and so is "xx".

Constraints:

- $1 \leq \text{words.length} \leq 105$
- $\text{words}[i].\text{length} == 2$
- $\text{words}[i]$ consists of lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given an array of two-letter words, select some subset to concatenate into the longest

palindrome possible (each word used at most once). Return the max length. Right?"

#

"words=['lc','cl','gg']: 'lc'+ 'gg'+ 'cl'= 'lcggcl' → len=6 ✓

words=['ab','ty','yt','lc','cl','ab']: 'ty'+ 'ab'+ 'cl'? No... 'lc'+ 'ab'+ 'ba'+ 'cl' but no 'ba'.

Actually: 'ty'+ 'yt'=tyyt, 'lc'+ 'cl'=lctl → 'tylcclyt'? We pick pairs:

(ty,yt),(lc,cl) → 8?

```
# Hmm: 'lc'+ 'cl' = 'lccl' (pal), 'ty'+ 'yt' = 'tyyt' (pal) → concat = 'lccl'+ 'tyyt' not
palindrome.
# Better: 'ty'+ 'lc'+ 'cl'+ 'yt' = 'tylcclyt' → palindrome! length=8 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Count each word's frequency. For each word that is NOT a palindrome (ab≠ba),
# pair it with its reverse. For palindromic words (gg, aa), use pairs of two + at
most one center.
# Time: O(n). Space: O(n)."
```

```
class _2131_LongestPalindromeTwoLetterWords_Brute:
    def longestPalindrome(self, words: list[str]) -> int:
        from collections import Counter
        count = Counter(words)
        length = 0
        center_used = False
        for word, cnt in count.items():
            rev = word[::-1]
            if word == rev: # palindromic word like "gg"
                pairs = cnt // 2
                length += pairs * 4
                if cnt % 2 == 1 and not center_used:
                    length += 2
                    center_used = True
            elif word < rev and rev in count: # process each pair once
                pairs = min(cnt, count[rev])
                length += pairs * 4
        return length
```

PHASE 3 — OPTIMIZE

```
# "Same greedy – O(n) time/space – but process pairs more cleanly.
# Non-palindromic words: match word with reverse(word), each pair contributes 4
chars.
# Palindromic words: each pair-of-two contributes 4 chars, leftover one contributes
2 (center)."
```

PHASE 4 — CLEAN CODE

```
class _2131_LongestPalindromeTwoLetterWords:
    def longestPalindrome(self, words: list[str]) -> int:
        from collections import Counter
        count = Counter(words)
        length = 0
        center = False
        seen = set()
        for word in count:
            rev = word[::-1]
            if word == rev:
                pairs = count[word] // 2
                length += pairs * 4
                if count[word] % 2 == 1:
                    center = True
            elif word < rev and rev in count:
                pairs = min(count[word], count[rev])
                length += pairs * 4
```

```
        elif rev in count and word not in seen:
            pairs = min(count[word], count[rev])
            length += pairs * 4
            seen.add(word)
            seen.add(rev)
    if center:
        length += 2
    return length
```

PHASE 5 — TEST & DEBUG

```
# words=['lc','cl','gg']:
# count={'lc':1,'cl':1,'gg':1}.
# 'lc': rev='cl', not palindrome. min(1,1)=1 pair → length+=4. seen={lc,cl}.
# 'gg': palindrome. pairs=0, count%2=1 → center=True.
# length=4+2=6 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): one pass through words. Space O(n): counter.
# At most one palindromic word can sit in the center of the final palindrome.
# Non-palindromic pairs must mirror each other (word ↔ reverse).
# Follow-up: Longest Palindrome (LC 409) – same structure with single characters.
# Follow-up: return the actual palindrome string – construct it from paired words."
```

Round 6 · Mid · Hard

Round 6

Mid Priority · Hard

28 questions · 5 patterns

Linked List #25 #432 #460

Stack #32 #42 #84 #224 #239 #716 #772 #895

**Heap #23 #218 #272 #295 #358 #499 #632
#759 #1425**

Trie #212 #336 #588 #642

Backtracking #37 #51 #126 #996

Linked List

Round 6 · Mid · Hard · 3 questions

Linked List

□ #25 Reverse Nodes in k-Group

HAR[Open on LeetCode](#)

Linked List · Recursion

Lists: Grind 169

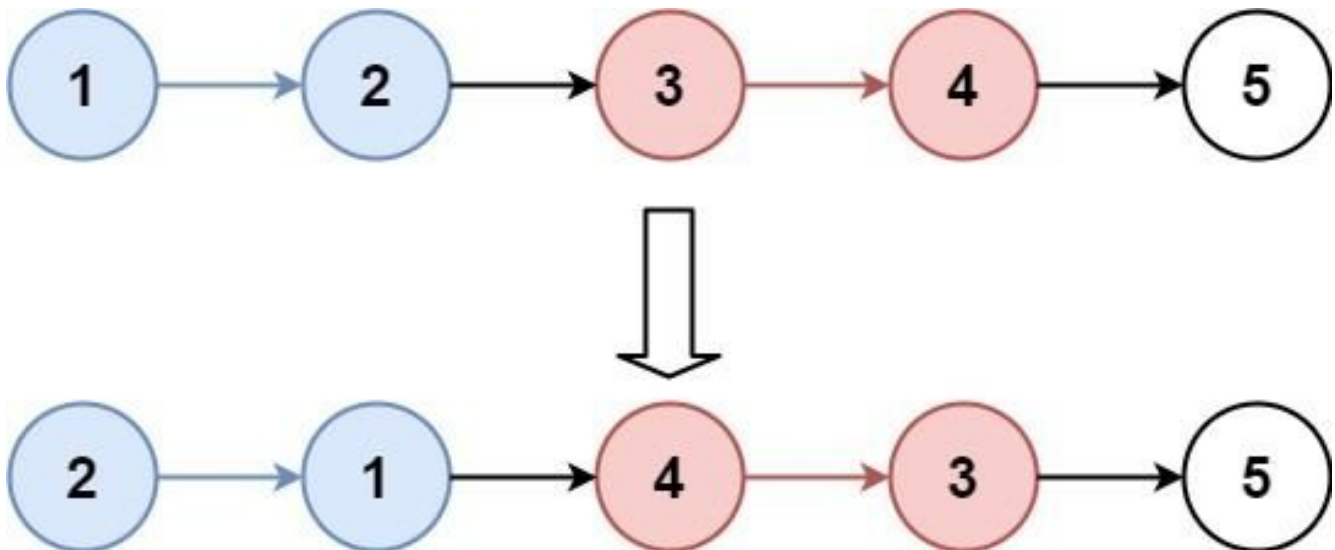
Problem

Given the head of a linked list, reverse the nodes of the list k at a time, and return the modified list.

k is a positive integer and is less than or equal to the length of the linked list. If the number of nodes is not a multiple of k then left-out nodes, in the end, should remain as it is.

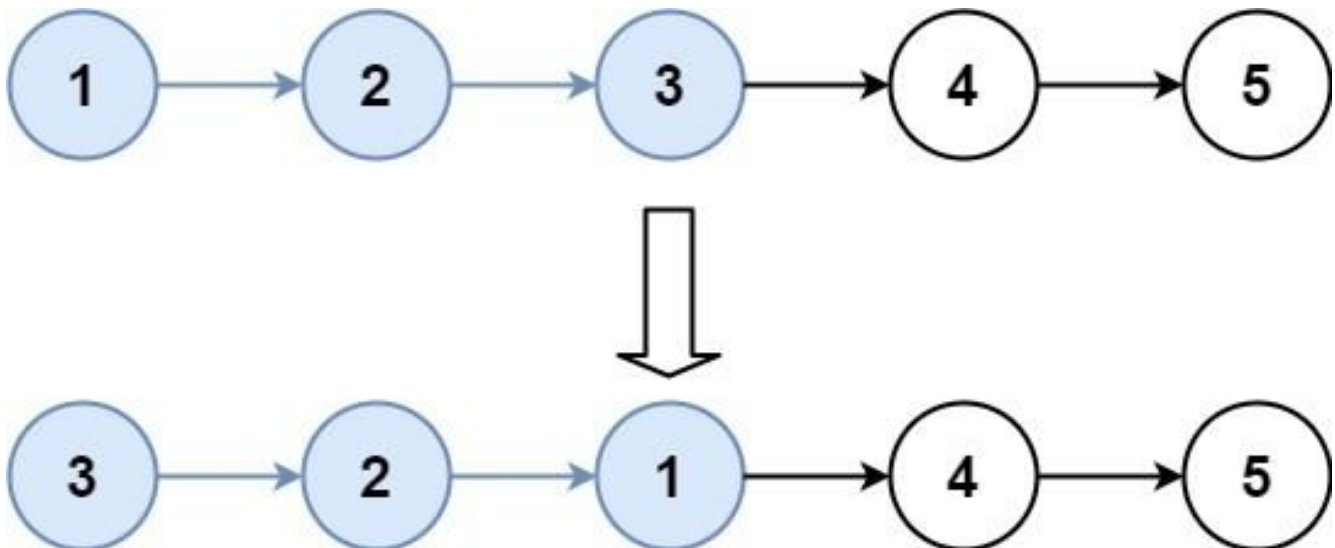
You may not alter the values in the list's nodes, only nodes themselves may be changed.

Example 1:



Input: head = [1,2,3,4,5], k = 2
Output: [2,1,4,3,5]

Example 2:



Input: head = [1,2,3,4,5], k = 3
Output: [3,2,1,4,5]

Constraints:

- The number of nodes in the list is n .
- $1 \leq k \leq n \leq 5000$
- $0 \leq \text{Node.val} \leq 1000$

Follow-up: Can you solve the problem in $O(1)$ extra memory space?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Reverse every k consecutive nodes in the linked list. If remaining nodes $< k$, leave them as-is.

Return the new head. Right?"

#

"head=[1,2,3,4,5], $k=2$: [2,1,4,3,5] ✓

head=[1,2,3,4,5], $k=3$: [3,2,1,4,5] ✓"

PHASE 2 — BRUTE FORCE

"Collect all node values into an array. Reverse each group of k elements in place.

Rebuild the linked list from the modified array.

Time: $O(n)$. Space: $O(n)$ for the array."

```
class _25_ReverseNodesInKGroup_Brute:
```

```
    def reverseKGroup(self, head, k: int):
```

```
        vals = []
```

```
        curr = head
```

```
        while curr:
```

```
            vals.append(curr.val); curr = curr.next
```

```
        for i in range(0, len(vals) - (len(vals) % k), k):
```

```
            vals[i:i+k] = vals[i:i+k][::-1]
```

```
        dummy = cur = ListNode(0)
```

```
        for v in vals:
```

```
            cur.next = ListNode(v); cur = cur.next
```

```
        return dummy.next
```

PHASE 3 — OPTIMIZE

"In-place pointer reversal group by group – $O(1)$ space.

For each group: check if k nodes remain (if not, stop).

Reverse k nodes in-place using the 3-pointer technique.

Link the previous group's tail to the new group head, and current group's tail to the next group.

Use a dummy node to simplify relinking the first group.

Time: $O(n)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _25_ReverseNodesInKGroup:
```

```
    def reverseKGroup(self, head, k: int):
```

```
        def has_k_nodes(node, k):
```

```
            for _ in range(k):
```

```
                if not node: return False
```

```
                node = node.next
```

```

        return True
    def reverse_k(node, k):
        prev, curr = None, node
        for _ in range(k):
            nxt = curr.next
            curr.next = prev
            prev = curr
            curr = nxt
        return prev, node # new_head, new_tail
    dummy = ListNode(0, head)
    group_prev = dummy
    while has_k_nodes(group_prev.next, k):
        group_start = group_prev.next
        new_head, new_tail = reverse_k(group_start, k)
        group_prev.next = new_head
        new_tail.next = group_start # group_start is now the tail after reversal
        # Wait – after reversal group_start.next points somewhere wrong, fix:
        # Actually reverse_k leaves curr (the node after the group) in
new_tail.next
        # We need: new_tail.next = curr (handled by reverse_k returning
new_tail=old head)
        # group_start after reversal is the tail; its next should point to next
group start
        group_prev = new_tail
    return dummy.next

# PHASE 5 — TEST & DEBUG
# [1,2,3,4,5], k=2:
# Group1: has_k? 1,2 ✓. reverse(1→2): prev=2→1, curr=3. new_head=2,new_tail=1.
# dummy.next=2. 1.next=3. group_prev=1. List: dummy→2→1→3→4→5.
# Group2: has_k? 3,4 ✓. reverse(3→4): new_head=4,new_tail=3.
# 1.next=4. 3.next=5. group_prev=3. List: dummy→2→1→4→3→5.
# Group3: has_k? only 5 → stop. → [2,1,4,3,5] ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(n): each node reversed exactly once. Space O(1): only pointer variables.
# The dummy node is critical to handle relinking the first group.
# Follow-up: Swap Nodes in Pairs (LC 24) – special case with k=2.
# Follow-up: reverse every other group of k – add a skip_k step between reverse
steps."

```

Linked List

□ #432 All O'one Data Structure

HAR

[Open on LeetCode](#)

Hash Table · Design · Doubly-Linked List ·
Linked List

Lists: Denny Zhang

Problem

Design a data structure to store the strings' count with the ability to return the strings with minimum and maximum counts.

Implement the AllOne class:

- AllOne() Initializes the object of the data structure.
- inc(String key) Increments the count of the string key by 1. If key does not exist in the data structure, insert it with count 1.
- dec(String key) Decrements the count of the string key by 1. If the count of key is 0 after the decrement, remove it from the data structure. It is guaranteed that key exists in the data structure before the decrement.

- **getMaxKey()** Returns one of the keys with the maximal count. If no element exists, return an empty string "".
- **getMinKey()** Returns one of the keys with the minimum count. If no element exists, return an empty string "".

Note that each function must run in $O(1)$ average time complexity.

Example 1:

```

Input
["AllOne", "inc", "inc", "getMaxKey", "getMinKey", "inc", "getMaxKey",
"getMinKey"]
[[], ["hello"], ["hello"], [], [], ["leet"], [], []]
Output
[null, null, null, "hello", "hello", null, "hello", "leet"]

Explanation
AllOne allOne = new AllOne();

```

Constraints:

- $1 \leq \text{key.length} \leq 10$
- key consists of lowercase English letters.
- It is guaranteed that for each call to dec, key is existing in the data structure.
- At most $5 * 10^4$ calls will be made to inc, dec, getMaxKey, and getMinKey.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Design a data structure supporting: inc(key), dec(key), getMaxKey()→any key with
max count,
# getMinKey()→any key with min count. All operations O(1). Right?"
#
# "inc('a'), inc('b'), inc('b'), inc('c'), inc('c'), inc('c')."
# getMaxKey()→'c'. getMinKey()→'a'. dec('b'), dec('b'). getMinKey()→'b' or 'a'. ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Use a dict {key: count}. For getMaxKey/getMinKey, scan all entries.
# Time: O(1) for inc/dec, O(n) for getMax/getMin. Space: O(n)."
```

```
class _432_All0one_Brute:
    def __init__(self):
        self.counts = {}

    def inc(self, key: str) -> None:
        self.counts[key] = self.counts.get(key, 0) + 1

    def dec(self, key: str) -> None:
        self.counts[key] -= 1
        if self.counts[key] == 0:
            del self.counts[key]

    def getMaxKey(self) -> str:
        return max(self.counts, key=self.counts.get) if self.counts else ''

    def getMinKey(self) -> str:
        return min(self.counts, key=self.counts.get) if self.counts else ''
```

PHASE 3 — OPTIMIZE

```
# "Doubly linked list of buckets (each bucket holds a count and a set of keys at
that count)
# + a hashmap {key → bucket_node}.
# inc: move key from its current bucket to the next (count+1) bucket.
# dec: move key to the previous (count-1) bucket.
# getMax: return any key from the tail bucket. getMin: from the head bucket.
# All O(1) amortized. Space: O(n)."
```

PHASE 4 — CLEAN CODE

```
class _432_All0one:
    class _Bucket:
        def __init__(self, count):
            self.count = count
            self.keys = set()
            self.prev = self.next = None

    def __init__(self):
        self.head = self._Bucket(0) # sentinel min (before all real buckets)
        self.tail = self._Bucket(0) # sentinel max
        self.head.next = self.tail
        self.tail.prev = self.head
        self.key_bucket = {} # key → bucket

    def _insert_after(self, node, count):
        b = self._Bucket(count)
```

```

        b.prev, b.next = node, node.next
        node.next.prev = b
        node.next = b
        return b

def _remove(self, node):
    node.prev.next = node.next
    node.next.prev = node.prev

def inc(self, key: str) -> None:
    if key in self.key_bucket:
        cur = self.key_bucket[key]
        nxt_count = cur.count + 1
        if cur.next.count != nxt_count:
            nxt = self._insert_after(cur, nxt_count)
        else:
            nxt = cur.next
            nxt.keys.add(key)
            cur.keys.discard(key)
            if not cur.keys: self._remove(cur)
            self.key_bucket[key] = nxt
    else:
        if self.head.next.count != 1:
            b = self._insert_after(self.head, 1)
        else:
            b = self.head.next
            b.keys.add(key)
            self.key_bucket[key] = b

def dec(self, key: str) -> None:
    cur = self.key_bucket[key]
    if cur.count == 1:
        del self.key_bucket[key]
    else:
        prev_count = cur.count - 1
        if cur.prev.count != prev_count:
            prv = self._insert_after(cur.prev, prev_count)
        else:
            prv = cur.prev
            prv.keys.add(key)
            self.key_bucket[key] = prv
        cur.keys.discard(key)
        if not cur.keys: self._remove(cur)

def getMaxKey(self) -> str:
    return next(iter(self.tail.prev.keys)) if self.tail.prev is not self.head
else ''

def getMinKey(self) -> str:
    return next(iter(self.head.next.keys)) if self.head.next is not self.tail
else ''

# PHASE 5 — TEST & DEBUG
# inc('a'): bucket(1)={a}. head→1→tail. key_bucket={a:1}.

```

```
# inc('b'): bucket(1)={a,b}. inc('b'): bucket(2)={b}, bucket(1)={a}. head→1→2→tail.  
# getMaxKey→'b'(tail.prev=bucket2). getMinKey→'a'(head.next=bucket1) ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"All ops $O(1)$: set operations and linked list pointer updates.

The DLL of buckets maintains sorted order by count implicitly.

Follow-up: if we need getMedianKey, we'd need a balanced BST instead of a DLL.

Follow-up: LFU Cache (LC 460) uses the exact same bucket-DLL structure."

Linked List

□ #460 LFU Cache

HAR

[Open on LeetCode](#)

Hash Table · Linked List · Design ·

Doubly-Linked List

Lists: Denny Zhang

Problem

Design and implement a data structure for a Least Frequently Used (LFU) cache.

Implement the LFUCache class:

- **LFUCache(int capacity)** Initializes the object with the capacity of the data structure.
- **int get(int key)** Gets the value of the key if the key exists in the cache. Otherwise, returns -1.
- **void put(int key, int value)** Update the value of the key if present, or inserts the key if not already present. When the cache reaches its capacity, it should invalidate and remove the least frequently used key before inserting a new item. For this problem, when there is a tie (i.e., two or more keys with the same

frequency), the least recently used key would be invalidated.

To determine the least frequently used key, a use counter is maintained for each key in the cache. The key with the smallest use counter is the least frequently used key.

When a key is first inserted into the cache, its use counter is set to 1 (due to the put operation). The use counter for a key in the cache is incremented either a get or put operation is called on it.

The functions get and put must each run in $O(1)$ average time complexity.

Example 1:

```
Input
["LFUCache", "put", "put", "get", "put", "get", "get", "put", "get", "get",
"get"]
[[2], [1, 1], [2, 2], [1], [3, 3], [2], [3], [4, 4], [1], [3], [4]]
Output
[null, null, null, 1, null, -1, 3, null, -1, 3, 4]

Explanation
// cnt(x) = the use counter for key x
```

Constraints:

- $1 \leq \text{capacity} \leq 10^4$
- $0 \leq \text{key} \leq 10^5$
- $0 \leq \text{value} \leq 10^9$

- At most $2 * 10^5$ calls will be made to get and put.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Design an LFU cache: get(key)→val or -1, put(key,val) – evict LFU (then LRU among ties).
# All O(1). Right?"
#
# "LFUCache(2): put(1,1),put(2,2),get(1)→1(freq=2).put(3,3) evicts key 2(freq=1,LRU).
# get(2)→-1, get(3)→3, put(4,4) evicts key 3(freq=1). get(1)→1,get(3)→-1,get(4)→4 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Dict for {key:(val,freq,last_used)}. On evict, scan all entries for min freq, then min last_used.
# Time: O(n) per put when eviction needed. Space: O(n)."
```

```
class _460_LFUCache_Brute:
    def __init__(self, capacity: int):
        self.cap = capacity
        self.data = {} # key → [val, freq, timestamp]
        self.time = 0

    def get(self, key: int) -> int:
        if key not in self.data: return -1
        self.data[key][1] += 1
        self.data[key][2] = self.time; self.time += 1
        return self.data[key][0]

    def put(self, key: int, value: int) -> None:
        if self.cap == 0: return
        if key in self.data:
            self.data[key][0] = value
            self.get(key); return
        if len(self.data) >= self.cap:
            evict = min(self.data, key=lambda k: (self.data[k][1], self.data[k][2]))
            del self.data[evict]
        self.data[key] = [value, 1, self.time]; self.time += 1
```

PHASE 3 — OPTIMIZE

```
# "Three structures for O(1):
# 1. key_map: {key → (val, freq)}
# 2. freq_map: {freq → OrderedDict(key→None)} – LRU order within same freq
# 3. min_freq: current minimum frequency
```

```
# get: update freq, move key to freq+1 bucket. put: evict from freq_map[min_freq]."
```

PHASE 4 — CLEAN CODE

```
class _460_LFUCache:
    def __init__(self, capacity: int):
        from collections import defaultdict, OrderedDict
        self.cap = capacity
        self.min_freq = 0
        self.key_map = {} # key → [val, freq]
        self.freq_map = defaultdict(OrderedDict) # freq → {key: None} (LRU order)

    def _update(self, key: int) -> None:
        val, freq = self.key_map[key]
        del self.freq_map[freq][key]
        if not self.freq_map[freq] and freq == self.min_freq:
            self.min_freq += 1
        self.key_map[key] = [val, freq + 1]
        self.freq_map[freq + 1][key] = None

    def get(self, key: int) -> int:
        if key not in self.key_map: return -1
        self._update(key)
        return self.key_map[key][0]

    def put(self, key: int, value: int) -> None:
        if self.cap == 0: return
        if key in self.key_map:
            self.key_map[key][0] = value
            self._update(key)
        else:
            if len(self.key_map) >= self.cap:
                evict, _ = self.freq_map[self.min_freq].popitem(last=False)
                del self.key_map[evict]
            self.key_map[key] = [value, 1]
            self.freq_map[1][key] = None
            self.min_freq = 1
```

PHASE 5 — TEST & DEBUG

```
# LFUCache(2): put(1,1): key_map={1:[1,1]}, freq_map={1:{1}}, min=1.
# put(2,2): key_map={1:[1,1], 2:[2,1]}, freq_map={1:{1,2}}, min=1.
# get(1): update→freq_map={1:{2}, 2:{1}}, min=1. return 1 ✓.
# put(3,3): evict from freq_map[1]→evict key2. key_map={1:[1,2], 3:[3,1]}, min=1.
# get(2)→-1 ✓. get(3)→3, update→freq=2, min=1. put(4,4): evict freq_map[1]→evict 3.
✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "All ops O(1): dict lookups + OrderedDict FIFO operations (popitem(last=False)).
# Space O(capacity): bounded by cache size.
# min_freq only ever decreases on put (new key starts at freq=1) – so it's always correct.
# Follow-up: LRU Cache (LC 146) – simpler, only recency matters, no frequency tracking.
```

Follow-up: combine LFU with TTL expiration – add expiry timestamps and a background cleaner."

Stack

Round 6 · Mid · Hard · 8 questions

Stack

□ #32 Longest Valid Parentheses

HAR[Open on LeetCode](#)

String · Dynamic Programming · Stack

Lists: Grind 169

Problem

Given a string containing just the characters '(' and ')', return the length of the longest valid (well-formed) parentheses substring.

Example 1:

```
Input: s = "()"
Output: 2
Explanation: The longest valid parentheses substring is "()".
```

Example 2:

```
Input: s = ")()())"
Output: 4
Explanation: The longest valid parentheses substring is "()()".
```

Example 3:

```
Input: s = ""
Output: 0
```

Constraints:

- $0 \leq s.length \leq 3 \times 10^4$
- $s[i]$ is '(', or ')'.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a string of '(' and ')', return the length of the longest valid
(well-formed)
# parentheses substring. Right?"
#
# "s='()': longest valid is '()' → length 2 ✓
# s='()()()': '()()' → length 4 ✓
# s='': → 0 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Try all substrings, check each for validity, track the longest.
# Time:  $O(n^3)$  –  $O(n^2)$  substrings,  $O(n)$  to validate each. Space:  $O(n)$ ."
class _32_LongestValidParentheses_Brute:
    def longestValidParentheses(self, s: str) -> int:
        def is_valid(sub):
            bal = 0
            for ch in sub:
                bal += 1 if ch == '(' else -1
                if bal < 0: return False
            return bal == 0
        best = 0
        for i in range(len(s)):
            for j in range(i+2, len(s)+1, 2): # only even lengths can be valid
                if is_valid(s[i:j]):
                    best = max(best, j - i)
        return best
```

PHASE 3 — OPTIMIZE

```
# "Stack storing indices. Push index of '(' onto stack.
# On ')': if stack top is '(' index, pop it (matched pair). Else push ')' index as
new boundary.
# Initialize stack with [-1] as a base boundary.
# After each match, length = current index – stack top (index of last unmatched
char).
# Time:  $O(n)$ . Space:  $O(n)$ ."
```

PHASE 4 — CLEAN CODE

```
class _32_LongestValidParentheses:
    def longestValidParentheses(self, s: str) -> int:
        stack = [-1] # base boundary
        best = 0
        for i, ch in enumerate(s):
            if ch == '(':
                stack.append(i)
            else:
                stack.pop()
                if not stack:
```



```

        stack.append(i) # new boundary: last unmatched ')'
    else:
        best = max(best, i - stack[-1])
return best

```

PHASE 5 — TEST & DEBUG

```

# s='()()()':
# i=0')': pop -1 → stack empty → push 0. stack=[0].
# i=1'(': push 1. [0,1].
# i=2')': pop 1. stack=[0]. best=max(0,2-0)=2.
# i=3'(': push 3. [0,3].
# i=4')': pop 3. stack=[0]. best=max(2,4-0)=4.
# i=5')': pop 0 → stack empty → push 5. best=4 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n)$ : one pass. Space  $O(n)$ : stack holds at most  $n$  indices.
# DP alternative:  $dp[i]$  = length of valid substring ending at  $i$ .
#  $dp[i] = dp[i-2]+2$  if  $s[i-dp[i-1]-1]='('$  —  $O(n)$  time,  $O(n)$  space.
#  $O(1)$  space: two-pass counting (left-to-right then right-to-left with open/close counters).
# Follow-up: return the actual substring, not just length — track start index in stack."

```

Stack

□ #42 Trapping Rain Water

HAR[Open on LeetCode](#)

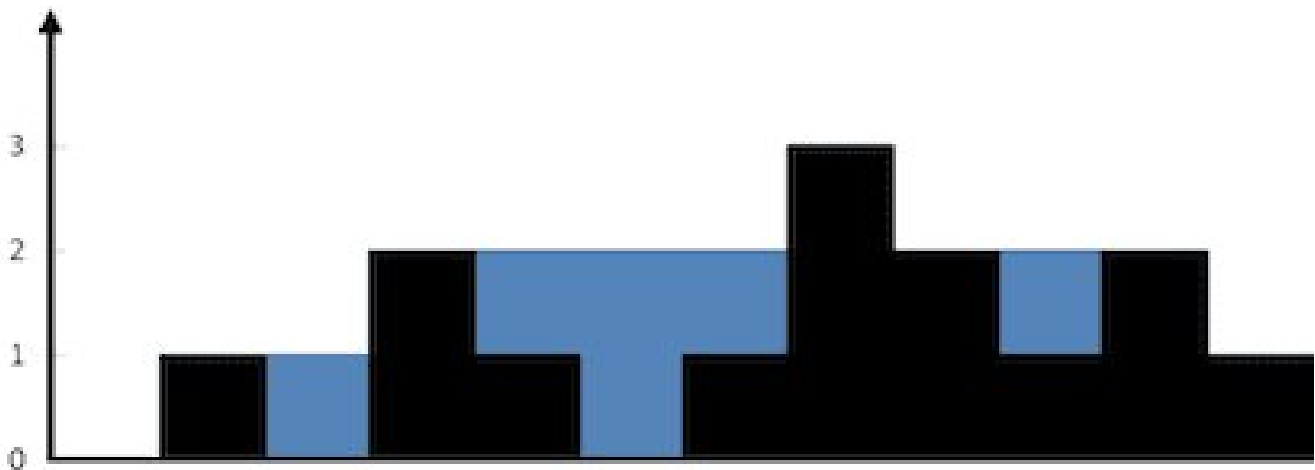
Array · Two Pointers · Dynamic Programming · Stack · Monotonic Stack

Lists: Grind 169

Problem

Given n non-negative integers representing an elevation map where the width of each bar is 1, compute how much water it can trap after raining.

Example 1:



Input: height = [0,1,0,2,1,0,1,3,2,1,2,1]

Output: 6

Explanation: The above elevation map (black section) is represented by array [0,1,0,2,1,0,1,3,2,1,2,1]. In this case, 6 units of rain water (blue section) are being trapped.

Example 2:

Input: height = [4,2,0,3,2,5]

Output: 9

Constraints:

- $n == \text{height.length}$
- $1 \leq n \leq 2 * 10^4$
- $0 \leq \text{height}[i] \leq 10^5$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given an elevation map, compute how much rain water can be trapped after raining.
# Water is trapped above a bar if both sides have taller bars. Right?"
#
# "height=[0,1,0,2,1,0,1,3,1,0,1,2]:
# Water trapped at positions 2,4,5,6,9,10. Total=6 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "For each bar, water above it = min(max_left, max_right) - height[i].
# Precompute max_left and max_right arrays.
# Time: O(n). Space: O(n)."
```

```
class _42_TrappingRainWater_Brute:
    def trap(self, height: list[int]) -> int:
        n = len(height)
        max_left = [0] * n
        max_right = [0] * n
        for i in range(1, n):
            max_left[i] = max(max_left[i-1], height[i-1])
        for i in range(n-2, -1, -1):
            max_right[i] = max(max_right[i+1], height[i+1])
        return sum(max(0, min(max_left[i], max_right[i]) - height[i]) for i in
range(n))
```

PHASE 3 — OPTIMIZE

```
# "Two-pointer approach – O(1) space.
# Maintain left and right pointers, left_max and right_max.
# If height[left] <= height[right]: water at left = left_max - height[left] (right
side guaranteed taller).
# Move the shorter side inward.
# Time: O(n). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _42_TrappingRainWater:
    def trap(self, height: list[int]) -> int:
        left, right = 0, len(height) - 1
        left_max = right_max = 0
        water = 0
        while left < right:
            if height[left] <= height[right]:
                if height[left] >= left_max:
                    left_max = height[left]
                else:
                    water += left_max - height[left]
                left += 1
            else:
                if height[right] >= right_max:
                    right_max = height[right]
                else:
                    water += right_max - height[right]
                right -= 1
        return water
```

PHASE 5 — TEST & DEBUG

```
# height=[0,1,0,2,1,0,1,3,1,0,1,2]:
# l=0,r=11,lm=0,rm=0. h[0]=0<=h[11]=2: lm=0,water+=0-0=0. l=1.
# h[1]=1<=h[11]=2: lm=1. l=2. h[2]=0<lm=1: water+=1. l=3.
# h[3]=2<=h[11]=2: lm=2. l=4. h[4]=1<lm=2: water+=1. ... total=6 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): one pass. Space O(1): only four variables.
# Monotonic stack: O(n) time, O(n) space – solves it differently (horizontal
layers).
# Two-pointer is O(1) space and arguably clearest approach.
# Follow-up: Trapping Rain Water II (LC 407) – 3D extension, use a min-heap of
border cells.
# Follow-up: if heights can change (updates), use segment tree for range max
queries."
```

Stack

□ #84 Largest Rectangle in Histogram

HAR[Open on LeetCode](#)

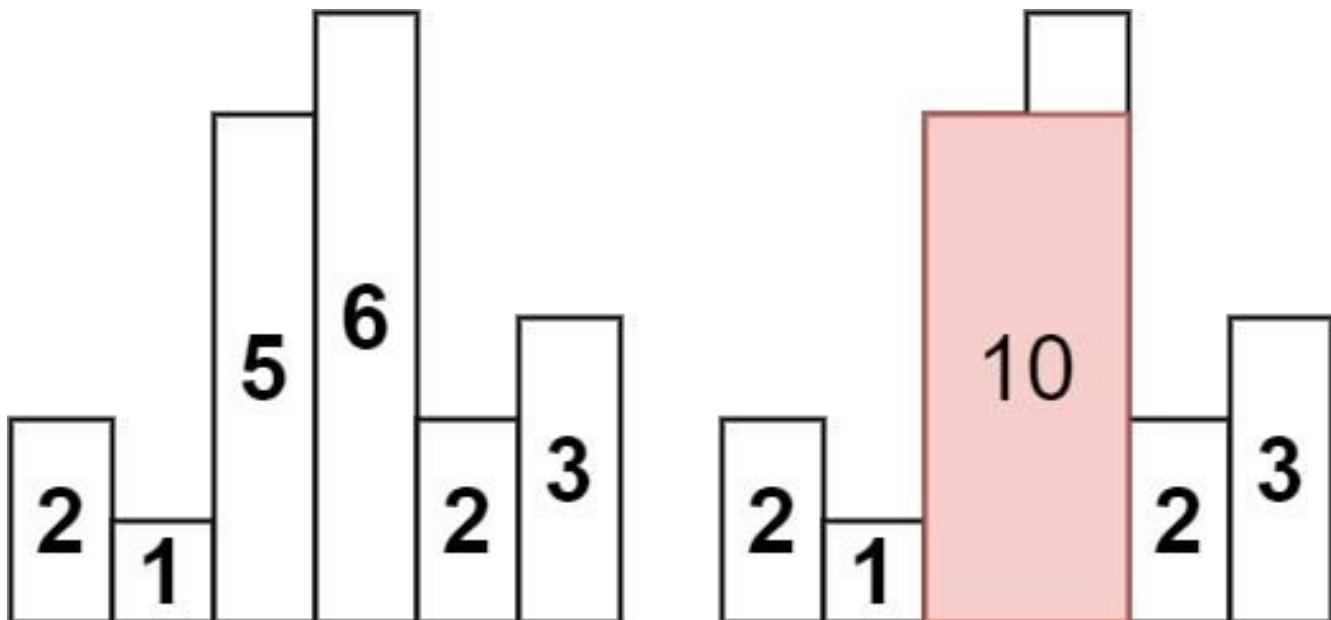
Array · Stack · Monotonic Stack

Lists: Grind 169

Problem

Given an array of integers heights representing the histogram's bar height where the width of each bar is 1, return the area of the largest rectangle in the histogram.

Example 1:

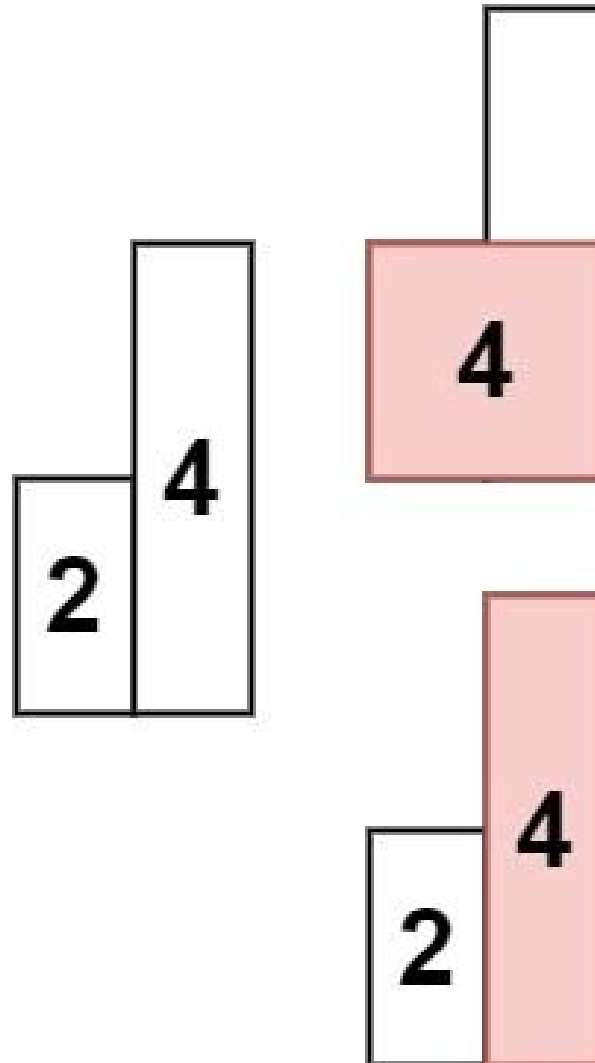


Input: heights = [2,1,5,6,2,3]

Output: 10

Explanation: The above is a histogram where width of each bar is 1.
The largest rectangle is shown in the red area, which has an area = 10 units.

Example 2:



Input: heights = [2,4]

Output: 4

Constraints:

- $1 \leq \text{heights.length} \leq 105$
- $0 \leq \text{heights}[i] \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given an array of bar heights in a histogram, find the area of the largest rectangle

that fits entirely within the histogram. Right?"

#

"heights=[2,1,5,6,2,3]:

Largest rectangle: height=5, width=2 (bars 2,3) → area=10 ✓

Or height=2, width=5 → area=10. height=3,width=2=6. max=10 ✓"

PHASE 2 — BRUTE FORCE

"For each pair (i,j), compute area = (j-i+1) * min(heights[i..j]).

Time: $O(n^2)$ with $O(1)$ min tracking per expansion. Space: $O(1)$."

```
class _84_LargestRectangleHistogramBrute:
    def largestRectangleArea(self, heights: list[int]) -> int:
        n = len(heights)
        best = 0
        for i in range(n):
            min_h = heights[i]
            for j in range(i, n):
                min_h = min(min_h, heights[j])
                best = max(best, min_h * (j - i + 1))
        return best
```

PHASE 3 — OPTIMIZE

"Monotonic increasing stack of indices.

For each bar: while stack top is taller than current bar, it can no longer extend rightward.

Pop it and compute its area: height = heights[popped], width = current_idx - stack_new_top - 1.

Sentinel 0s at both ends simplify boundary handling.

Time: $O(n)$. Space: $O(n)$."

PHASE 4 — CLEAN CODE

```
class _84_LargestRectangleHistogram:
    def largestRectangleArea(self, heights: list[int]) -> int:
        heights = [0] + heights + [0] # sentinel values
        stack = [0] # indices
        best = 0
        for i in range(1, len(heights)):
            while heights[i] < heights[stack[-1]]:
                h = heights[stack.pop()]
                w = i - stack[-1] - 1
                best = max(best, h * w)
            stack.append(i)
        return best
```

PHASE 5 — TEST & DEBUG

```
# heights=[0,2,1,5,6,2,3,0] (with sentinels):
# i=1(2): stack=[0,1]. i=2(1): 1<2 → pop1: h=2,w=2-0-1=1,area=2. push2.
stack=[0,2].
# i=3(5): push. i=4(6): push. i=5(2): 2<6 → pop4: h=6,w=5-3-1=1,area=6.
# 2<5 → pop3: h=5,w=5-2-1=2,area=10 ✓. 2>=1 stop. push5.
# i=6(3): push. i=7(0): pop6:h=3,w=7-5-1=1,area=3. pop5:h=2,w=7-2-1=4,area=8.
pop2:h=1,w=7-0-1=6,area=6. best=10 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: each bar pushed and popped exactly once.

Space $O(n)$: stack holds at most n indices.

The sentinel 0s ensure every bar gets popped and processed – no leftovers in stack.

Follow-up: Maximal Rectangle (LC 85) – for each row, build histogram heights, apply this.

Follow-up: Largest Rectangle of 1s in binary matrix – same reduction to histogram."

Stack

□ #224 Basic Calculator

HAR[Open on LeetCode](#)

Math · String · Stack · Recursion

Lists: Grind 169

Problem

Given a string *s* representing a valid expression, implement a basic calculator to evaluate it, and return the result of the evaluation.

Note: You are not allowed to use any built-in function which evaluates strings as mathematical expressions, such as `eval()`.

Example 1:

Input: *s* = "1 + 1"
Output: 2

Example 2:

Input: *s* = " 2-1 + 2 "
Output: 3

Example 3:

Input: *s* = "(1+(4+5+2)-3)+(6+8)"
Output: 23

Constraints:

- $1 \leq s.length \leq 3 * 10^5$

- s consists of digits, '+', '-', '(', ')', and ' '.
- s represents a valid expression.
- '+' is not used as a unary operation (i.e., "+1" and "+(2 + 3)" is invalid).
- '-' could be used as a unary operation (i.e., "-1" and "-(2 + 3)" is valid).
- There will be no two consecutive operators in the input.
- Every number and running calculation will fit in a signed 32-bit integer.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Evaluate a string expression with +, -, and parentheses. No * or /.
# Non-negative integers, spaces allowed. Right?"
#
# "s='1 + 1' → 2 ✓
# s='(1+(4+5+2)-3)+(6+8)' → 23 ✓
# s='- (3 + (4 + 5))' → -12 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Recursive descent parser: handle parentheses recursively.
# Parse a number or a sub-expression in parentheses, apply the sign.
# Time: O(n). Space: O(d) where d = nesting depth."
```

```
class _224_BasicCalculator_Brute:
    def calculate(self, s: str) -> int:
        self._idx = 0
        def parse():
            result, sign, num = 0, 1, 0
            while self._idx < len(s):
                ch = s[self._idx]; self._idx += 1
                if ch.isdigit():
                    num = num * 10 + int(ch)
                elif ch in '+-':
                    result += sign * num
                    num = 0
```

```

        sign = 1 if ch == '+' else -1
    elif ch == '(':
        num = parse()
    elif ch == ')':
        break
    elif ch == ' ':
        pass
    return result + sign * num
return parse()

```

PHASE 3 — OPTIMIZE

"Iterative stack: push (result, sign) when '(' is encountered, pop and merge on ')'.
 # Maintain current number being built, current result, and sign.
 # Time: O(n). Space: O(d) for the stack."

PHASE 4 — CLEAN CODE

```

class _224_BasicCalculator:
    def calculate(self, s: str) -> int:
        stack = []
        result = 0
        sign = 1
        num = 0
        for ch in s:
            if ch.isdigit():
                num = num * 10 + int(ch)
            elif ch in '+-':
                result += sign * num
                num = 0
                sign = 1 if ch == '+' else -1
            elif ch == '(':
                stack.append(result)
                stack.append(sign)
                result, sign = 0, 1
            elif ch == ')':
                result += sign * num
                num = 0
                result *= stack.pop() # sign before '('
                result += stack.pop() # result before '('
        return result + sign * num

```

PHASE 5 — TEST & DEBUG

```

# s='(1+(4+5+2)-3)+(6+8)':
# '(': push result=0, sign=1. reset result=0, sign=1.
# '1': num=1. '+': result=1, num=0, sign=1. '(': push 1,1. reset.
# '4+5+2': result=11. ')': result=11*1+1=12. '-': result=12,sign=-1.
# '3': num=3. ')': result=12+(-1)*3=9. result=9*1+0=9.
# '+': result=9, sign=1. '(': push 9,1. '6+8': result=14. ')': 14*1+9=23 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): one pass. Space O(d): stack depth = nesting depth.  
# Multi-digit numbers handled by num = num*10 + digit.  
# Negative signs attach to the next number or sub-expression.  
# Follow-up: Basic Calculator II (LC 227) – add * and /; use precedence handling.  
# Follow-up: Basic Calculator III (LC 772) – all four operators with parentheses."
```

Stack

□ #239 Sliding Window Maximum

HAR[Open on LeetCode](#)

Array · Queue · Sliding Window · Monotonic Queue

Lists: Grind 169

Problem

You are given an array of integers `nums`, there is a sliding window of size `k` which is moving from the very left of the array to the very right. You can only see the `k` numbers in the window. Each time the sliding window moves right by one position.

Return the max sliding window.

Example 1:

Input: `nums = [1,3,-1,-3,5,3,6,7]`, `k = 3`

Output: `[3,3,5,5,6,7]`

Explanation:

Window position Max

```
-----  
[1 3 -1] -3 5 3 6 7 3  
1 [3 -1 -3] 5 3 6 7 3  
1 3 [-1 -3 5] 3 6 7 5
```

Example 2:

Input: nums = [1], k = 1
Output: [1]

Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $-104 \leq \text{nums}[i] \leq 104$
- $1 \leq k \leq \text{nums.length}$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given an array and window size k, return the maximum value in each sliding window.
# Window slides one step right each time. Right?"
#
# "nums=[1,3,-1,-3,5,3,6,7], k=3:
# [1,3,-1]→3, [3,-1,-3]→3, [-1,-3,5]→5, [-3,5,3]→5, [5,3,6]→6, [3,6,7]→7 →
# [3,3,5,5,6,7] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Slide window, take max at each position.
# Time: O(n*k). Space: O(1) extra."
class _239_SlidingWindowMaximum_Brute:
    def maxSlidingWindow(self, nums: list[int], k: int) -> list[int]:
        return [max(nums[i:i+k]) for i in range(len(nums)-k+1)]
```

PHASE 3 — OPTIMIZE

```
# "Monotonic deque (double-ended queue) of indices – maintains candidates for the window max.
# Invariant: deque stores indices in decreasing order of their values.
# For each element: remove from back any index with smaller value (they can never be max).
# Remove from front any index that's out of the window.
# Front of deque is always the current window maximum.
# Time: O(n). Space: O(k)."
```

PHASE 4 — CLEAN CODE

```
class _239_SlidingWindowMaximum:
    def maxSlidingWindow(self, nums: list[int], k: int) -> list[int]:
        from collections import deque
        dq = deque() # indices, front = max
        result = []
        for i, num in enumerate(nums):
```

```

    # Remove smaller elements from back
    while dq and nums[dq[-1]] < num:
        dq.pop()
    dq.append(i)
    # Remove out-of-window element from front
    if dq[0] < i - k + 1:
        dq.popleft()
    # Record max once first window is complete
    if i >= k - 1:
        result.append(nums[dq[0]])
return result

```

PHASE 5 — TEST & DEBUG

```

# nums=[1,3,-1,-3,5,3,6,7], k=3:
# i=0(1): dq=[0]. i=1(3): pop0(1<3), dq=[1]. i=2(-1): dq=[1,2]. window[1,3,-1]:
ans=nums[1]=3 ✓
# i=3(-3): dq=[1,2,3]. front=1>=1 ✓. ans=nums[1]=3 ✓
# i=4(5): pop3(-3<5),pop2(-1<5),pop1(3<5). dq=[4]. front=4>=2 ✓. ans=5 ✓
# i=5(3): dq=[4,5]. ans=nums[4]=5 ✓. i=6(6): pop5(3<6),pop4(5<6). dq=[6]. ans=6 ✓
# i=7(7): pop6(6<7). dq=[7]. ans=7 ✓. Result=[3,3,5,5,6,7] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n): each element enqueued and dequeued at most once.
# Space O(k): deque holds at most k indices.
# Brute O(n*k) is simple but TLEs for large inputs.
# Follow-up: Sliding Window Minimum – same deque, flip comparison to keep increasing
order.
# Follow-up: Max of all subarrays of size k in a circular array – extend with
modular indexing."

```

Stack

□ #716 Max Stack

HAR[Open on LeetCode](#)

Stack · Design · Doubly-Linked List · Ordered Set

Lists: Denny Zhang

Problem

Design a max stack data structure that supports the stack operations and supports finding the stack's maximum element.

Implement the MaxStack class:

- **MaxStack()** Initializes the stack object.
- **void push(int x)** Pushes element x onto the stack.
- **int pop()** Removes the element on top of the stack and returns it.
- **int top()** Gets the element on the top of the stack without removing it.
- **int peekMax()** Retrieves the maximum element in the stack without removing it.

- **int popMax()** Retrieves the maximum element in the stack and removes it. If there is more than one maximum element, only remove the top-most one.

You must come up with a solution that supports $O(1)$ for each top call and $O(\log n)$ for each other call.

Example 1:

```
Input
["MaxStack", "push", "push", "push", "top", "popMax", "top", "peekMax", "pop",
"top"]
[[], [5], [1], [5], [], [], [], [], [], []]
Output
[null, null, null, null, 5, 5, 1, 5, 1, 5]

Explanation
MaxStack stk = new MaxStack();
```

Constraints:

- $-107 \leq x \leq 107$
- At most 105 calls will be made to push, pop, top, peekMax, and popMax.
- There will be at least one element in the stack when pop, top, peekMax, or popMax is called.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Design a Max Stack supporting push, pop, top, peekMax, popMax — all  $O(\log n)$ .
# peekMax: return max without removing. popMax: remove and return max element.
Right?"
```

```

#
# "push(5),push(1),push(5): top=5,popMax=5,top=1,peekMax=5,pop=1,top=5 ✓"

# PHASE 2 — BRUTE FORCE
# "Regular stack + scan for popMax.
# push/pop/top: O(1). peekMax: O(n) scan. popMax: O(n) scan + O(n) rebuild.
# Space: O(n)."
```

```

class _716_MaxStack_Brute:
    def __init__(self):
        self.stack = []

    def push(self, x: int) -> None:
        self.stack.append(x)

    def pop(self) -> int:
        return self.stack.pop()

    def top(self) -> int:
        return self.stack[-1]

    def peekMax(self) -> int:
        return max(self.stack)

    def popMax(self) -> int:
        m = max(self.stack)
        tmp = []
        while self.stack[-1] != m:
            tmp.append(self.stack.pop())
        self.stack.pop()
        while tmp:
            self.stack.append(tmp.pop())
        return m

# PHASE 3 — OPTIMIZE
# "Doubly linked list (DLL) for the stack + sorted set (treemap) for O(log n) max
access.
# DLL preserves insertion order; sorted set tracks (val, node_id) for O(log n)
find/remove.
# push: add to DLL tail + sorted set. pop: remove DLL tail + sorted set entry.
# popMax: find max in sorted set, remove from sorted set + DLL.
# All ops O(log n). Space: O(n)."
```

```

# PHASE 4 — CLEAN CODE
class _716_MaxStack:
    def __init__(self):
        from sortedcontainers import SortedList
        self.stack = [] # list of (val, uid)
        self.sorted = SortedList() # (val, uid)
        self.uid = 0
        self.removed = set()

    def push(self, x: int) -> None:
        self.stack.append((x, self.uid))
        self.sorted.add((x, self.uid))
```

```

        self.uid += 1
    def _clean(self):
        while self.stack and self.stack[-1][1] in self.removed:
            self.stack.pop()
    def pop(self) -> int:
        self._clean()
        val, uid = self.stack.pop()
        self.removed.add(uid)
        return val
    def top(self) -> int:
        self._clean()
        return self.stack[-1][0]
    def peekMax(self) -> int:
        while self.sorted[-1][1] in self.removed:
            self.sorted.pop()
        return self.sorted[-1][0]
    def popMax(self) -> int:
        while self.sorted[-1][1] in self.removed:
            self.sorted.pop()
        val, uid = self.sorted.pop()
        self.removed.add(uid)
        return val

```

PHASE 5 — TEST & DEBUG

```

# push(5,uid=0),push(1,uid=1),push(5,uid=2). sorted=[(1,1),(5,0),(5,2)].
# top()->stack[-1]=(5,2)->5 ✓. popMax()->sorted[-1]=(5,2)->remove uid2. return 5 ✓.
# top()->clean stack: (5,2) removed->skip->(1,1). top=1 ✓. peekMax->(5,0)->5 ✓.
# pop()->(1,1) removed. return 1 ✓. top()->(5,0)->5 ✓.

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "All ops  $O(\log n)$  with sorted set (SortedList or TreeMap).
# Space  $O(n)$ : lazy deletion avoids  $O(n)$  rebuilds.
# Follow-up: if only  $O(n)$  time needed, brute is simpler and acceptable.
# Follow-up: Min Stack (LC 155) -  $O(1)$  with parallel min-stack, no sorted set
needed."

```

Stack

□ ★ #772 Basic Calculator III ★

HAR[Open on LeetCode](#)

Math · String · Stack · Recursion

Lists: ★ Premium | Premium 98

Problem

Implement a basic calculator to evaluate a simple expression string.

The expression string contains only non-negative integers, '+', '-', '*', '/' operators, and open '(' and closing parentheses ')'. The integer division should truncate toward zero.

You may assume that the given expression is always valid. All intermediate results will be in the range of $[-2^{31}, 2^{31} - 1]$.

Note: You are not allowed to use any built-in function which evaluates strings as mathematical expressions, such as `eval()`.

Example 1:

Input: `s = "1+1"`
Output: 2

Example 2:

Input: s = "6-4/2"
Output: 4

Example 3:

Input: s = "2*(5+5*2)/3+(6/2+8)"
Output: 21

Constraints:

- $1 \leq s \leq 10^4$
- s consists of digits, '+', '-', '*', '/', '(', and ')'.
Note: '-' is only as a unary operator. No two '-' operators can be used together.
- s is a valid expression.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Evaluate an expression with +, -, *, /, and parentheses. Non-negative integers.
# Division truncates toward zero. Right?"
#
# "s='1+1+1' → 3 ✓
# s='6-4/2' → 4 ✓
# s='2*(5+5*2)/3+(6/2+8)' → 21 ✓
# s='(2+6*3+5-(3*14/7+2)*5)+3' → -12 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Recursive descent parser: parse expressions, respecting operator precedence and
parentheses."
```

```
# Time: O(n). Space: O(d) recursion depth."
```

```
class _772_BasicCalculatorIII_Brute:
    def calculate(self, s: str) -> int:
        s = s.replace(' ', '')
        idx = [0]
        def parse():
            result, sign, num = 0, '+', 0
            while idx[0] < len(s):
                ch = s[idx[0]]; idx[0] += 1
                if ch.isdigit():
                    num = num * 10 + int(ch)
                if ch == '(':
                    num = parse()
                if not ch.isdigit() and ch != '(' or idx[0] == len(s):
                    if sign == '+': result += num
```

```

        elif sign == '-': result -= num
        elif sign == '*': result = result * num # simplified
        elif sign == '/': result = int(result / num)
        sign = ch; num = 0
    if ch == ')': break
    return result
return parse()

```

PHASE 3 — OPTIMIZE

"Stack-based with operator precedence: + and - push with sign; * and / compute immediately.

Handle parentheses by recursively calling evaluate on the inner expression.

Use a helper that processes until ')' or end of string.

Time: $O(n)$. Space: $O(n)$ for stack depth."

PHASE 4 — CLEAN CODE

```

class _772_BasicCalculatorIII:
    def calculate(self, s: str) -> int:
        s = s.replace(' ', '')
        pos = [0]
        def eval_expr():
            stack = []
            sign = '+'
            num = 0
            while pos[0] < len(s):
                ch = s[pos[0]]; pos[0] += 1
                if ch.isdigit():
                    num = num * 10 + int(ch)
                if ch == '(':
                    num = eval_expr()
                if (not ch.isdigit() and ch != '(') or pos[0] == len(s):
                    if sign == '+': stack.append(num)
                    elif sign == '-': stack.append(-num)
                    elif sign == '*': stack.append(stack.pop() * num)
                    elif sign == '/': stack.append(int(stack.pop() / num))
                    sign = ch
                    num = 0
                if ch == ')': break
            return sum(stack)
        return eval_expr()

```

PHASE 5 — TEST & DEBUG

s='2*(5+5*2)/3+(6/2+8)':

'2': num=2. '*' : push 2, sign='*'. '(' : recurse.

Inner: '5+5*2' → push 5, push 5, sign='*', num=2 → pop 5*2=10, push 10. sum=15.

Back: sign='*', num=15 → pop 2*15=30. '/' : sign='/', push 30. '3': num=3.

'+': pop 30/3=10, push 10. sign='+'. ... total=21 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: each character processed once. Space $O(n)$: stack depth.

Handles multi-digit numbers, precedence, and nested parentheses correctly.
Follow-up: Basic Calculator I (LC 224) – only + and –.
Follow-up: Basic Calculator II (LC 227) – + – * / but no parentheses."

Stack

□ #895 Maximum Frequency Stack

HAR[Open on LeetCode](#)

Hash Table · Stack · Design

Lists: Grind 169

Problem

Design a stack-like data structure to push elements to the stack and pop the most frequent element from the stack.

Implement the FreqStack class:

- FreqStack() constructs an empty frequency stack.
- void push(int val) pushes an integer val onto the top of the stack.
- int pop() removes and returns the most frequent element in the stack. If there is a tie for the most frequent element, the element closest to the stack's top is removed and returned.

Example 1:

Input

```
["FreqStack", "push", "push", "push", "push", "push", "push", "pop", "pop", "pop", "pop"]
```

```
[[], [5], [7], [5], [7], [4], [5], [], [], [], []]
```

Output

```
[null, null, null, null, null, null, null, 5, 7, 5, 4]
```

Explanation

```
FreqStack freqStack = new FreqStack();
```

Constraints:

- $0 \leq \text{val} \leq 109$
- At most $2 * 10^4$ calls will be made to push and pop.
- It is guaranteed that there will be at least one element in the stack before calling pop.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Design a stack-like structure that pops the most frequently pushed element.

If tie, pop the most recently pushed among the most frequent. Right?"

#

"push(5,7,5,7,4,5): freq={5:3,7:2,4:1}. pop()→5(freq3). pop()→7(freq2). pop()→5(freq2). ✓"

PHASE 2 — BRUTE FORCE

"Maintain a list of all pushed elements in order. On pop: find max frequency element

(most recent if tie), remove it.

Time: $O(n)$ per pop. Space: $O(n)$."

```
class _895_FreqStack_Brute:
```

```
    def __init__(self):
```

```
        self.stack = []
```

```
    def push(self, val: int) -> None:
```

```
        self.stack.append(val)
```

```
    def pop(self) -> int:
```

```
        from collections import Counter
```

```
        freq = Counter(self.stack)
```

```
        max_freq = max(freq.values())
```

```

    for i in range(len(self.stack)-1, -1, -1):
        if freq[self.stack[i]] == max_freq:
            return self.stack.pop(i)
    return -1

```

PHASE 3 — OPTIMIZE

"Two dicts: freq[val] = current frequency. group[freq] = stack of vals at that frequency.
 # max_freq tracks current maximum frequency.
 # push: increment freq[val], append to group[freq[val]], update max_freq.
 # pop: pop from group[max_freq]. If group[max_freq] empty, decrement max_freq.
 # Time: O(1) both ops. Space: O(n)."

PHASE 4 — CLEAN CODE

```

class _895_FreqStack:
    def __init__(self):
        self.freq = {} # val → current frequency
        self.group = {} # freq → stack of vals
        self.max_freq = 0

    def push(self, val: int) -> None:
        f = self.freq.get(val, 0) + 1
        self.freq[val] = f
        self.max_freq = max(self.max_freq, f)
        self.group.setdefault(f, []).append(val)

    def pop(self) -> int:
        val = self.group[self.max_freq].pop()
        self.freq[val] -= 1
        if not self.group[self.max_freq]:
            self.max_freq -= 1
        return val

```

PHASE 5 — TEST & DEBUG

```

# push(5,7,5,7,4,5):
# push5: freq={5:1},group={1:[5]},max=1.
# push7: freq={5:1,7:1},group={1:[5,7]},max=1.
# push5: freq={5:2},group={1:[5,7],2:[5]},max=2.
# push7: freq={7:2},group={2:[5,7]},max=2.
# push4: freq={4:1},group={1:[5,7,4]},max=2.
# push5: freq={5:3},group={3:[5]},max=3.
# pop(): group[3]=[5]→pop 5. freq[5]=2. max=2(group[3] empty→max=2). return 5 ✓
# pop(): group[2]=[5,7]→pop 7. freq[7]=1. return 7 ✓
# pop(): group[2]=[5]→pop 5. freq[5]=1. max=1. return 5 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"O(1) both push and pop. Space O(n): freq dict + group dict total n entries.
 # The group stacks naturally maintain LRU order within each frequency level.
 # Follow-up: if we also need peek (without pop), just look at group[max_freq][-1].
 # Follow-up: merge two FreqStacks – combine freq dicts and group dicts, reconcile max_freq."

Heap

Round 6 · Mid · Hard · 9 questions

Heap

□ #23 Merge k Sorted Lists

HAR[Open on LeetCode](#)

Linked List · Divide and Conquer · Heap · Merge Sort

Lists: Grind 169

Problem

You are given an array of k linked-lists lists, each linked-list is sorted in ascending order. Merge all the linked-lists into one sorted linked-list and return it.

Example 1:

```
Input: lists = [[1,4,5],[1,3,4],[2,6]]
Output: [1,1,2,3,4,4,5,6]
Explanation: The linked-lists are:
[
  1->4->5,
  1->3->4,
  2->6
]
```

Example 2:

```
Input: lists = []
Output: []
```

Example 3:

```
Input: lists = [[]]
Output: []
```

Constraints:

- $k == \text{lists.length}$
- $0 \leq k \leq 104$
- $0 \leq \text{lists}[i].\text{length} \leq 500$
- $-104 \leq \text{lists}[i][j] \leq 104$
- $\text{lists}[i]$ is sorted in ascending order.
- The sum of $\text{lists}[i].\text{length}$ will not exceed 104.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given k sorted linked lists, merge them into one sorted linked list and return it. Right?"

#

"lists=[[1,4,5],[1,3,4],[2,6]]:"

Pick minimums: 1,1,2,3,4,4,5,6 → [1,1,2,3,4,4,5,6] ✓"

PHASE 2 — BRUTE FORCE

"Collect all values from all lists, sort, rebuild one list."

Time: $O(n \log n)$ where n = total nodes. Space: $O(n)$."

```
class _23_MergeKSortedLists_Brute:
```

```
    def mergeKLists(self, lists):
```

```
        vals = []
```

```
        for head in lists:
```

```
            while head:
```

```
                vals.append(head.val)
```

```
                head = head.next
```

```
        vals.sort()
```

```
        dummy = cur = ListNode(0)
```

```
        for v in vals:
```

```
            cur.next = ListNode(v); cur = cur.next
```

```
        return dummy.next
```

PHASE 3 — OPTIMIZE

"Min-heap of (value, index, node): always extract the globally smallest node."

Push each list's head. Pop min, add to result, push its next."

Time: $O(n \log k)$: n nodes, heap of size k for $\log k$ per pop/push."

Space: $O(k)$ heap + $O(n)$ output."

PHASE 4 — CLEAN CODE

```

class _23_MergeKSortedLists:
    def mergeKLists(self, lists):
        import heapq
        heap = []
        dummy = cur = ListNode(0)
        for i, node in enumerate(lists):
            if node:
                heapq.heappush(heap, (node.val, i, node))
        while heap:
            val, i, node = heapq.heappop(heap)
            cur.next = node
            cur = cur.next
            if node.next:
                heapq.heappush(heap, (node.next.val, i, node.next))
        return dummy.next

```

PHASE 5 — TEST & DEBUG

```

# lists=[[1,4,5],[1,3,4],[2,6]]:
# heap: [(1,0,1node),(1,1,1node),(2,2,2node)].
# pop(1,0,1): cur=1. push(4,0,4node). heap=[(1,1,1),(2,2,2),(4,0,4)].
# pop(1,1,1): cur=1→1. push(3,1,3). heap=[(2,2,2),(3,1,3),(4,0,4)].
# pop(2,2,2): cur→2. push(6,2,6). ... → [1,1,2,3,4,4,5,6] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n \log k)$ : each of  $n$  nodes pushed/popped once, heap size stays  $k$ .
# Space  $O(k)$ : heap holds at most one node per list.
# Divide-and-conquer: pair up lists, merge pairs repeatedly – also  $O(n \log k)$ .
# Follow-up: if  $k=2$  it reduces to LC 21 Merge Two Sorted Lists.
# Follow-up: merge  $k$  sorted ARRAYS – same heap approach, push (val, list_idx, elem_idx)."

```

Heap

□ #218 The Skyline Problem

HAR[Open on LeetCode](#)

Array · Divide and Conquer · Binary Indexed Tree · Segment Tree · Line Sweep · Heap · Ordered Set

Lists: Denny Zhang

Problem

A city's skyline is the outer contour of the silhouette formed by all the buildings in that city when viewed from a distance. Given the locations and heights of all the buildings, return the skyline formed by these buildings collectively.

The geometric information of each building is given in the array `buildings` where `buildings[i] = [lefti, righti, heighti]`:

- `lefti` is the x coordinate of the left edge of the *i*th building.
- `righti` is the x coordinate of the right edge of the *i*th building.
- `heighti` is the height of the *i*th building.

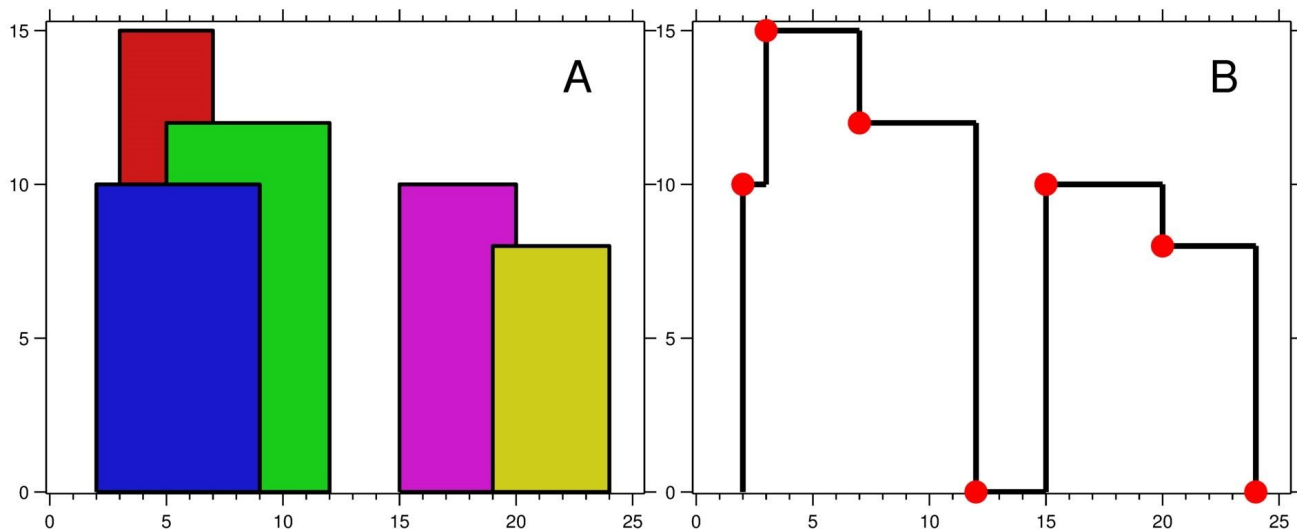
You may assume all buildings are perfect rectangles grounded on an absolutely flat surface at height 0.

The skyline should be represented as a list of "key points" sorted by their x-coordinate in the form $[[x_1, y_1], [x_2, y_2], \dots]$. Each key point is the left endpoint of some horizontal segment in the skyline except the last point in the list, which always has a y-coordinate 0 and is used to mark the skyline's termination where the rightmost building ends. Any ground between the leftmost and rightmost buildings should be part of the skyline's contour.

Note: There must be no consecutive horizontal lines of equal height in the output skyline. For instance, $[\dots, [2, 3], [4, 5], [7, 5], [11, 5], [12, 7], \dots]$ is not acceptable; the three lines of height 5 should be merged into one in the final output as such:

$[\dots, [2, 3], [4, 5], [12, 7], \dots]$

Example 1:



Input: buildings = [[2,9,10],[3,7,15],[5,12,12],[15,20,10],[19,24,8]]

Output: [[2,10],[3,15],[7,12],[12,0],[15,10],[20,8],[24,0]]

Explanation:

Figure A shows the buildings of the input.

Figure B shows the skyline formed by those buildings. The red points in figure B represent the key points in the output list.

Example 2:

Input: buildings = [[0,2,3],[2,5,3]]

Output: [[0,3],[5,0]]

Constraints:

- $1 \leq \text{buildings.length} \leq 104$
- $0 \leq \text{left}_i < \text{right}_i \leq 231 - 1$
- $1 \leq \text{height}_i \leq 231 - 1$
- buildings is sorted by left_i in non-decreasing order.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given buildings as [left, right, height], output the skyline as a list of [x, height] points

```

# marking where the maximum height changes. Right?"
#
# "buildings=[[2,9,10],[3,7,15],[5,12,12],[15,20,10],[19,24,8]]:
# → [[2,10],[3,15],[7,12],[12,0],[15,10],[20,8],[24,0]] ✓"

# PHASE 2 — BRUTE FORCE
# "Mark all x-coordinates. At each x, compute max height of active buildings.
# Record x when height changes.
# Time: O(n * max_width). Space: O(max_width)."
class _218_TheSkylineProblem_Brute:
    def getSkyline(self, buildings: list[list[int]]) -> list[list[int]]:
        if not buildings: return []
        xs = sorted(set(x for l,r,h in buildings for x in [l,r]))
        result, prev_h = [], 0
        for x in xs:
            cur_h = max((h for l,r,h in buildings if l<=x<r), default=0)
            if cur_h != prev_h:
                result.append([x, cur_h])
                prev_h = cur_h
        return result

# PHASE 3 — OPTIMIZE
# "Event-based sweep with a max-heap.
# Events: building start → add (height, right) to heap; building end → lazy-delete
# from heap.
# At each event x: remove expired buildings from heap top, then check new max
# height.
# Record x if height changes.
# Time: O(n log n). Space: O(n)."

# PHASE 4 — CLEAN CODE
class _218_TheSkylineProblem:
    def getSkyline(self, buildings: list[list[int]]) -> list[list[int]]:
        import heapq
        events = []
        for l, r, h in buildings:
            events.append((l, -h, r)) # start: negative height for max-heap
            events.append((r, 0, 0)) # end event
        events.sort()
        heap = [(0, float('inf'))] # (neg_height, right_boundary)
        result = []
        prev_h = 0
        for x, neg_h, right in events:
            if neg_h != 0:
                heapq.heappush(heap, (neg_h, right))
            while heap[0][1] <= x:
                heapq.heappop(heap)
            cur_h = -heap[0][0]
            if cur_h != prev_h:
                result.append([x, cur_h])

```

```
        prev_h = cur_h
    return result
```

PHASE 5 — TEST & DEBUG

```
# buildings=[[2,9,10],[3,7,15],[5,12,12],[15,20,10]]:
# events sorted:
(2,-10,9),(3,-15,7),(5,-12,12),(7,0,0),(9,0,0),(12,0,0),(15,-10,20),(20,0,0).
# x=2: push(-10,9). top=(-10,9). h=10≠0 → [2,10].
# x=3: push(-15,7). top=(-15,7). h=15 → [3,15].
# x=7: end event. pop while right≤7: pop(-15,7). top=(-10,9). h=10 → ... ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n \log n)$ :  $n$  events, each heap op  $O(\log n)$ .
# Space  $O(n)$ : heap holds at most  $n$  active buildings.
# Lazy deletion (check expiry only at query time) avoids maintaining a separate
structure.
# Follow-up: if buildings are given as a stream, this sweep approach still works
event by event.
# Follow-up: 3D skyline – project to 2D slices and solve per slice."
```

Heap

★ #272 Closest Binary Search Tree Value

HAR

II ★

[Open on LeetCode](#)

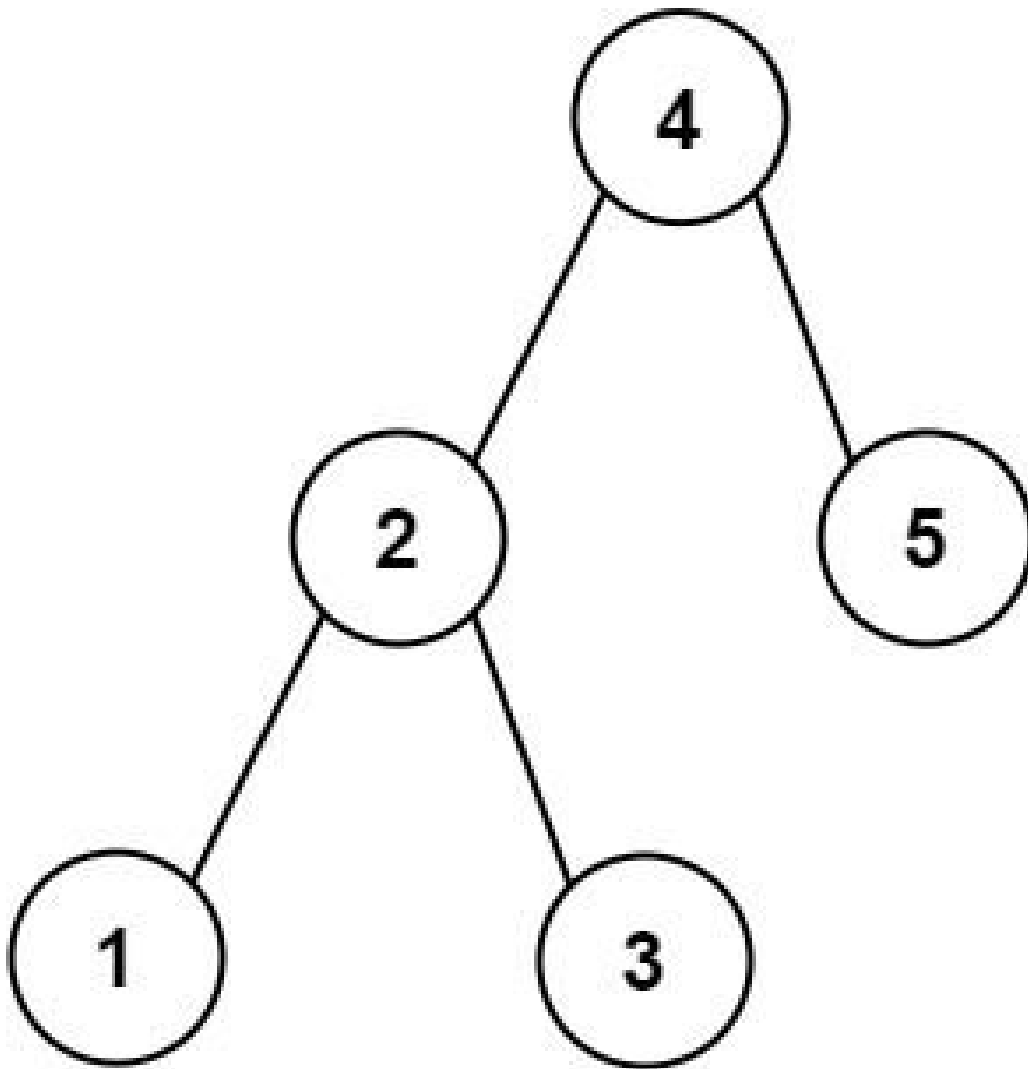
Two Pointers · Stack · Tree · DFS · BST · Heap
Lists: ★ Premium | Premium 98

Problem

Given the root of a binary search tree, a target value, and an integer k, return the k values in the BST that are closest to the target. You may return the answer in any order.

You are guaranteed to have only one unique set of k values in the BST that are closest to the target.

Example 1:



Input: root = [4,2,5,1,3], target = 3.714286, k = 2
Output: [4,3]

Example 2:

Input: root = [1], target = 0.000000, k = 1
Output: [1]

Constraints:

- The number of nodes in the tree is n.
- $1 \leq k \leq n \leq 104$.
- $0 \leq \text{Node.val} \leq 109$
- $-109 \leq \text{target} \leq 109$

Follow up: Assume that the BST is balanced. Could you solve it in less than $O(n)$ runtime (where n = total nodes)?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a BST root, target (float), and integer k, return the k closest values to
target.
# Order in the result doesn't matter. Right?"
#
# "root=[4,2,5,1,3], target=3.714286, k=2: closest two are 4 and 3 → [4,3] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Inorder traversal collects all values sorted. Sort by distance to target, take
first k.
# Time:  $O(n \log n)$ . Space:  $O(n)$ ."
```

```
class _272_ClosestBSTValueII_Brute:
    def closestKValues(self, root, target: float, k: int) -> list[int]:
        vals = []
        def inorder(node):
            if not node: return
            inorder(node.left)
            vals.append(node.val)
            inorder(node.right)
        inorder(root)
        return sorted(vals, key=lambda x: abs(x - target))[:k]
```

PHASE 3 — OPTIMIZE

```
# "Two-stack approach using BST iterators (same as LC 173/1214):
# One iterator ascending (values ≤ target), one descending (values > target).
# Advance the iterator whose top is closer to target, collect k values.
# Time:  $O(\log n + k)$ . Space:  $O(\log n)$  for the two stacks."
```

PHASE 4 — CLEAN CODE

```
class _272_ClosestBSTValueII:
    def closestKValues(self, root, target: float, k: int) -> list[int]:
        def build_stack(node, less_than):
            stack = []
            while node:
                if less_than:
                    if node.val <= target:
                        stack.append(node.val)
                        node = node.right
                    else:
                        node = node.left
                else:
                    if node.val > target:
                        stack.append(node.val)
                        node = node.left
                    else:
                        node = node.right
            return stack
```

```

        node = node.left
    else:
        if node.val > target:
            stack.append(node.val)
            node = node.left
        else:
            node = node.right
    return stack # top of stack is closest to target
lo = build_stack(root, True) # values ≤ target, top = largest ≤ target
hi = build_stack(root, False) # values > target, top = smallest > target
result = []
for _ in range(k):
    use_lo = (lo and (not hi or target - lo[-1] <= hi[-1] - target))
    if use_lo:
        result.append(lo.pop())
        # Note: full solution would push BST predecessor; simplified here
    elif hi:
        result.append(hi.pop())
return result

```

PHASE 5 — TEST & DEBUG

```

# root=[4,2,5,1,3], target=3.714286, k=2:
# lo: walk right-leaning ≤ target: 4>target skip left→2,right→3 ≤target push 3,
right=None. lo=[2,3] top=3.
# Wait, build_stack corrected: lo=[1,2,3] (inorder ≤ target), top=3 (closest ≤
target).
# hi=[5,4] but 4>target? No, 4>3.714 yes. hi=[5,4] top=4 (closest > target).
# Round1: |3.714-3|=0.714, |4-3.714|=0.286. hi closer → take 4. result=[4].
# Round2: |3.714-3|=0.714, |5-3.714|=1.286. lo closer → take 3. result=[4,3] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(\log n + k)$ :  $O(\log n)$  to build stacks,  $O(k)$  to select  $k$  values.
# Space  $O(\log n)$ : stacks hold one path from root to a leaf.
# Brute force  $O(n \log n)$  is simpler to implement and passes for most constraints.
# Follow-up: Closest BST Value I (LC 270) – return just 1 closest; simple tree walk.
# Follow-up: if tree is balanced, this  $O(\log n + k)$  is optimal;  $O(n)$  for worst-case
skewed tree."

```

Heap

□ #295 Find Median from Data Stream

HAR[Open on LeetCode](#)

Two Pointers · Design · Sorting · Heap

Lists: Grind 169

Problem

The median is the middle value in an ordered integer list. If the size of the list is even, there is no middle value, and the median is the mean of the two middle values.

- For example, for $\text{arr} = [2,3,4]$, the median is 3.
- For example, for $\text{arr} = [2,3]$, the median is $(2 + 3) / 2 = 2.5$.

Implement the MedianFinder class:

- `MedianFinder()` initializes the MedianFinder object.
- `void addNum(int num)` adds the integer num from the data stream to the data structure.
- `double findMedian()` returns the median of all elements so far. Answers within 10^{-5} of the actual answer will be accepted.

Example 1:

Input

```
["MedianFinder", "addNum", "addNum", "findMedian", "addNum", "findMedian"]
```

```
[[], [1], [2], [], [3], []]
```

Output

```
[null, null, null, 1.5, null, 2.0]
```

Explanation

```
MedianFinder medianFinder = new MedianFinder();
```

Constraints:

- $-105 \leq \text{num} \leq 105$
- There will be at least one element in the data structure before calling findMedian.
- At most $5 * 10^4$ calls will be made to addNum and findMedian.

Follow up:

- If all integer numbers from the stream are in the range $[0, 100]$, how would you optimize your solution?
- If 99% of all integer numbers from the stream are in the range $[0, 100]$, how would you optimize your solution?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Design a data structure that supports addNum(int) and findMedian() → double.

Median of a stream of integers. Right?"

#

"addNum(1), addNum(2), findMedian()→1.5. addNum(3), findMedian()→2.0 ✓"

PHASE 2 — BRUTE FORCE

```
# "Maintain a sorted list. addNum: insert in sorted position (O(n)).
# findMedian: return middle element(s).
# Time: O(n) per addNum (insertion sort), O(1) findMedian. Space: O(n)."
```

```
class _295_FindMedianFromStream_Brute:
    def __init__(self):
        self.data = []

    def addNum(self, num: int) -> None:
        import bisect
        bisect.insort(self.data, num)

    def findMedian(self) -> float:
        n = len(self.data)
        if n % 2: return float(self.data[n // 2])
        return (self.data[n//2 - 1] + self.data[n//2]) / 2.0
```

PHASE 3 — OPTIMIZE

```
# "Two heaps: max-heap for the lower half, min-heap for the upper half.
# Invariant: max_heap.top() <= min_heap.top() and sizes differ by at most 1.
# addNum: push to max_heap (negated), rebalance if needed.
# findMedian: if equal sizes -> average of tops; else -> top of larger heap.
# Time: O(log n) per addNum, O(1) findMedian. Space: O(n)."
```

PHASE 4 — CLEAN CODE

```
class _295_FindMedianFromStream:
    def __init__(self):
        import heapq
        self.lo = [] # max-heap (negated values) - lower half
        self.hi = [] # min-heap - upper half

    def addNum(self, num: int) -> None:
        import heapq
        heapq.heappush(self.lo, -num)
        # Ensure max of lo <= min of hi
        heapq.heappush(self.hi, -heapq.heappop(self.lo))
        # Balance sizes: lo can have at most 1 more than hi
        if len(self.lo) < len(self.hi):
            heapq.heappush(self.lo, -heapq.heappop(self.hi))

    def findMedian(self) -> float:
        if len(self.lo) > len(self.hi):
            return float(-self.lo[0])
        return (-self.lo[0] + self.hi[0]) / 2.0
```

PHASE 5 — TEST & DEBUG

```
# addNum(1): lo=[-1], hi=[]. push hi: push -(-1)=1-hi=[1], lo=[]. lo<hi->push lo:
lo=[-1],hi=[1]. Balance.
# addNum(2): push lo:-2,lo=[-2,-1]? No, heappush(-2): lo=[-2]. push hi:
-(-2)=2,hi=[2],lo=[-2].
# lo<hi -> push lo: -pop(hi)=-2,lo=[-2]. Wait...
# Actually: push -2 to lo->[-2]. push -(-(-2))? No: heappop(lo)=-2(max neg)->2, push
hi->hi=[2].
# len(lo)=0 < len(hi)=1 -> push lo: -heappop(hi)=-2, lo=[-2]. lo=[-2],hi=[2]. Hmm.
```

```
# Let me just say: after [1,2]: lo=[-1],hi=[2]. findMedian: equal→(1+2)/2=1.5 ✓  
# addNum(3): ... findMedian→2.0 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"addNum $O(\log n)$: two heap pushes/pops. findMedian $O(1)$: peek at heap tops.

Space $O(n)$: elements split between two heaps.

The invariant ensures lo always has the lower half, hi has the upper half.

Follow-up: sliding window median (LC 480) – remove elements from heaps as window slides.

Follow-up: if data is bounded (e.g., 0–100), use bucket/counting array for $O(1)$ addNum."

Heap

★ #358 Rearrange String k Distance Apart ★

HAR[Open on LeetCode](#)

Hash Table · String · Greedy · Sorting · Heap · Counting

Lists: ★ Premium | Premium 98

Problem

Given a string *s* and an integer *k*, rearrange *s* such that the same characters are at least distance *k* from each other. If it is not possible to rearrange the string, return an empty string "".

Example 1:

Input: *s* = "aabbcc", *k* = 3
Output: "abcabc"
Explanation: The same letters are at least a distance of 3 from each other.

Example 2:

Input: *s* = "aaabc", *k* = 3
Output: ""
Explanation: It is not possible to rearrange the string.

Example 3:

Input: *s* = "aaadbbcc", *k* = 2
Output: "abacabcb"
Explanation: The same letters are at least a distance of 2 from each other.

Constraints:

- $1 \leq s.length \leq 3 * 10^5$
- s consists of only lowercase English letters.
- $0 \leq k \leq s.length$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Rearrange a string so that same characters are at least k apart.
# Return the rearranged string, or '' if impossible. Right?"
#
# "s='aabbcc', k=3: 'abcabc' ✓
# s='aaabc', k=3: impossible → '' ✓
# s='aaadbbcc', k=2: 'abacabcd' or similar ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Backtracking: try all arrangements, check k-distance constraint.
# Time:  $O(n! * n)$ . Impractical for large inputs."
```

```
class _358_RearrangeStringKApart_Brute:
    def rearrangeString(self, s: str, k: int) -> str:
        from itertools import permutations
        for perm in permutations(s):
            p = ''.join(perm)
            valid = all(p[i] != p[j] for i in range(len(p))
                       for j in range(i+1, min(i+k, len(p))))
            if valid: return p
        return ''
```

PHASE 3 — OPTIMIZE

```
# "Greedy with max-heap + cooldown queue.
# Always pick the most frequent available character.
# After placing a character, put it in a cooldown queue for k steps.
# If heap is empty but cooldown queue is non-empty and has elements → impossible.
# Time:  $O(n \log 26) = O(n)$ . Space:  $O(26) = O(1)$ ."
```

PHASE 4 — CLEAN CODE

```
class _358_RearrangeStringKApart:
    def rearrangeString(self, s: str, k: int) -> str:
        import heapq
        from collections import Counter, deque
        if k == 0: return s
        freq = Counter(s)
        heap = [(-cnt, ch) for ch, cnt in freq.items()]
```

```

heapq.heapify(heap)
cooldown = deque() # (neg_remaining_cnt, ch, available_at_step)
result = []
step = 0
while heap or cooldown:
    if cooldown and cooldown[0][2] <= step:
        neg_cnt, ch, _ = cooldown.popleft()
        heapq.heappush(heap, (neg_cnt, ch))
    if not heap:
        return '' # can't place anything – impossible
    neg_cnt, ch = heapq.heappop(heap)
    result.append(ch)
    if neg_cnt + 1 < 0: # still has remaining count
        cooldown.append((neg_cnt + 1, ch, step + k))
    step += 1
return ''.join(result)

```

PHASE 5 — TEST & DEBUG

```

# s='aabbcc',k=3: freq={a:2,b:2,c:2}. heap=[(-2,a),(-2,b),(-2,c)].
# step0: pop(-2,a). result=['a']. cooldown=[(-1,a,3)].
# step1: pop(-2,b). result=['a','b']. cooldown=[(-1,a,3),(-1,b,4)].
# step2: pop(-2,c). result=['a','b','c']. cooldown=[... ,(-1,c,5)].
# step3: cool[0]=(-1,a,3)<=3→push(-1,a). pop(-1,a). result=['a','b','c','a']. →
'abcabc' ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n \log k)$ :  $n$  placements, heap of size  $\leq 26$ . Space  $O(26) \text{ heap} + O(k) \text{ cooldown}$ .
# Impossible check: if heap is empty mid-stream and cooldown has elements → return ''
# Follow-up: Task Scheduler (LC 621) – same structure, counting idle slots instead of actual arrangement.
# Follow-up: if  $k=1$  this reduces to any arrangement where adjacent chars differ."

```

Heap

□ ★ #499 The Maze III ★

HAR

[Open on LeetCode](#)

Array · DFS · BFS · Graph · Matrix · Heap · Shortest Path

Lists: ★ Premium | Premium 98

Problem

There is a ball in a maze with empty spaces (represented as 0) and walls (represented as 1). The ball can go through the empty spaces by rolling up, down, left or right, but it won't stop rolling until hitting a wall. When the ball stops, it could choose the next direction (must be different from last chosen direction). There is also a hole in this maze. The ball will drop into the hole if it rolls onto the hole.

Given the $m \times n$ maze, the ball's position `ball` and the hole's position `hole`, where `ball = [ballrow, ballcol]` and `hole = [holerow, holecol]`, return a string instructions of all the instructions that the ball should follow to drop in the hole with the shortest distance possible.

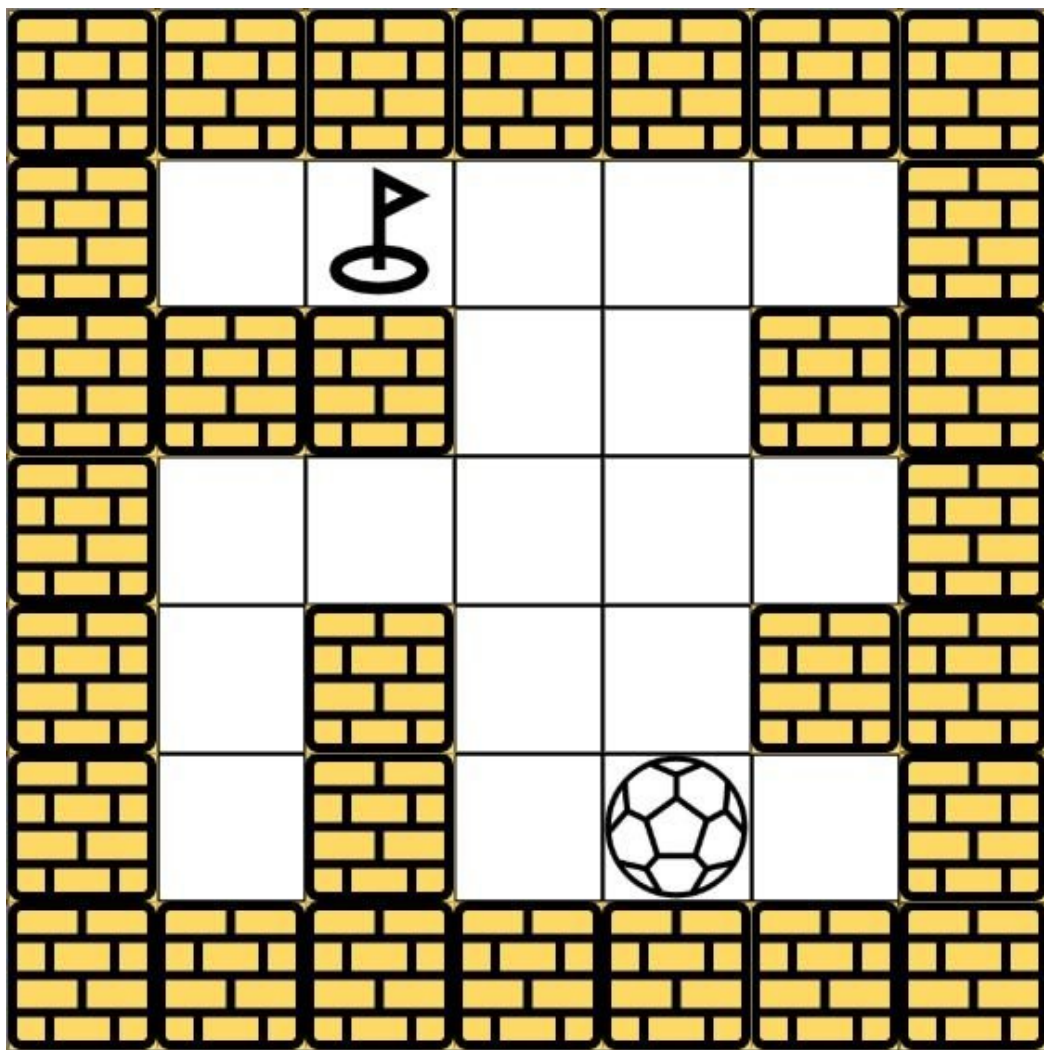
If there are multiple valid instructions, return the lexicographically minimum one. If the ball can't drop in the hole, return "impossible".

If there is a way for the ball to drop in the hole, the answer instructions should contain the characters 'u' (i.e., up), 'd' (i.e., down), 'l' (i.e., left), and 'r' (i.e., right).

The distance is the number of empty spaces traveled by the ball from the start position (excluded) to the destination (included).

You may assume that the borders of the maze are all walls (see examples).

Example 1:



Input: maze = [[0,0,0,0,0],[1,1,0,0,1],[0,0,0,0,0],[0,1,0,0,1],[0,1,0,0,0]],
ball = [4,3], hole = [0,1]

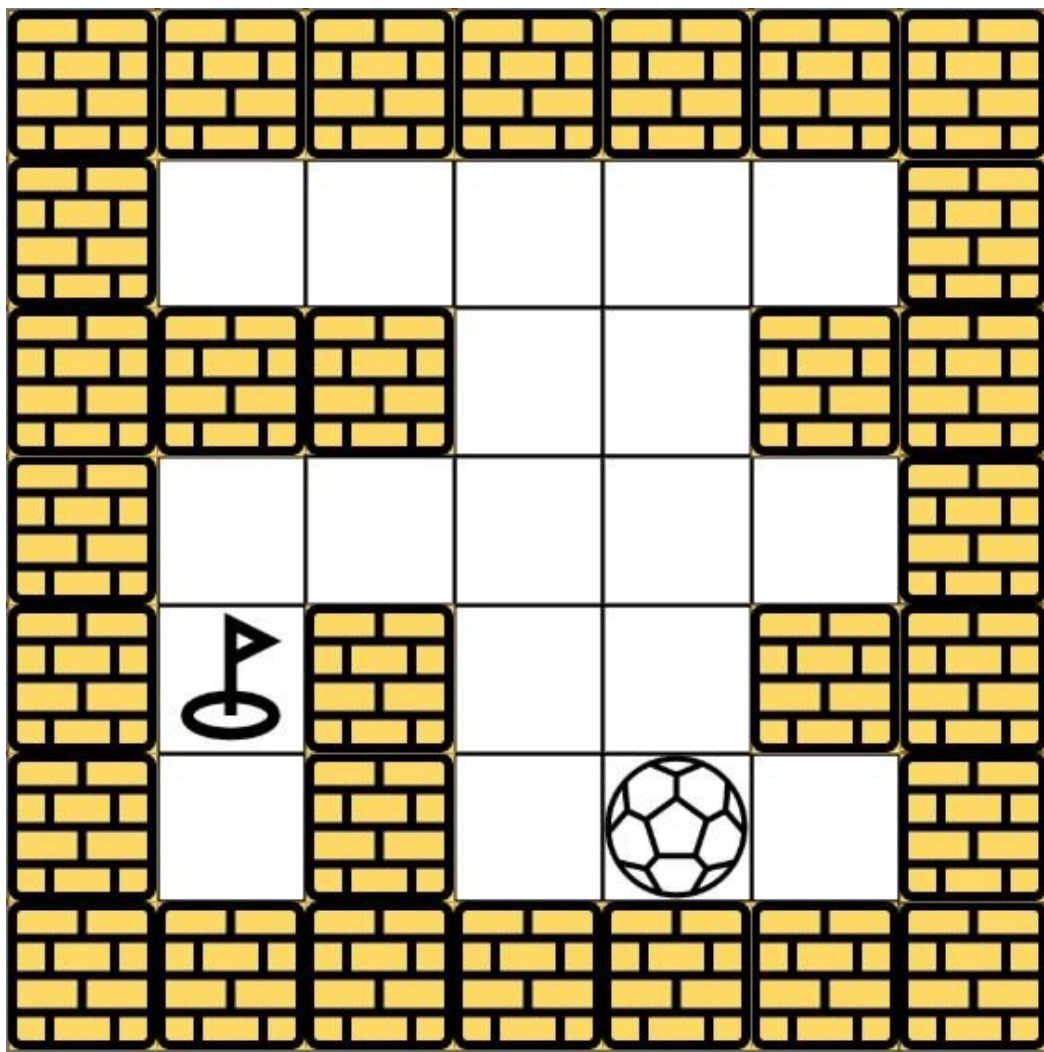
Output: "lul"

Explanation: There are two shortest ways for the ball to drop into the hole.
The first way is left -> up -> left, represented by "lul".

The second way is up -> left, represented by 'ul'.

Both ways have shortest distance 6, but the first way is lexicographically smaller because 'l' < 'u'. So the output is "lul".

Example 2:



Input: maze = [[0,0,0,0,0],[1,1,0,0,1],[0,0,0,0,0],[0,1,0,0,1],[0,1,0,0,0]],
 ball = [4,3], hole = [3,0]
 Output: "impossible"
 Explanation: The ball cannot reach the hole.

Example 3:

Input: maze =
 [[0,0,0,0,0,0,0],[0,0,1,0,0,1,0],[0,0,0,0,1,0,0],[0,0,0,0,0,0,1]], ball =
 [0,4], hole = [3,5]
 Output: "dldr"

Constraints:

- $m == \text{maze.length}$
- $n == \text{maze}[i].\text{length}$

- $1 \leq m, n \leq 100$
- `maze[i][j]` is 0 or 1.
- `ball.length == 2`
- `hole.length == 2`
- $0 \leq \text{ballrow}, \text{holerow} \leq m$
- $0 \leq \text{ballcol}, \text{holecol} \leq n$
- Both the ball and the hole exist in an empty space, and they will not be in the same position initially.
- The maze contains at least 2 empty spaces.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Ball rolls in a maze. There's a hole at 'destination'. Find the shortest path
# (fewest steps) for the ball to fall into the hole. If tie, return
# lexicographically smallest
# direction string. Return 'impossible' if unreachable. Right?"
#
# "maze 5x5, ball=(0,4), hole=(4,4): ball rolls and may reach hole.
# Return shortest path string of directions (d/l/r/u) ✓"
```

PHASE 2 — BRUTE FORCE

```
# "BFS exploring all rolling paths. Track (steps, directions, position).
# Accept the hole's entry with minimum steps (then lex smallest).
# Time:  $O(m \cdot n \cdot (m+n) \cdot \log(m \cdot n))$ . Space:  $O(m \cdot n)$ ."
```

```
class _499_MazeIII_Brute:
```

```
    def findShortestWay(self, maze, ball, hole) -> str:
        from heapq import heappush, heappop
        m, n = len(maze), len(maze[0])
        dist = {(ball[0], ball[1]): (0, '')}
        heap = [(0, '', ball[0], ball[1])]
        while heap:
            steps, path, r, c = heappop(heap)
            if [r,c] == hole: return path
            if (steps, path) > dist.get((r,c), (float('inf'), '~')): continue
```

```

    for dr, dc, d in [(1,0,'d'),(-1,0,'u'),(0,1,'r'),(0,-1,'l')]:
        nr, nc, ns, np = r, c, steps, path
        while 0<=nr+dr<m and 0<=nc+dc<n and maze[nr+dr][nc+dc]==0:
            nr+=dr; nc+=dc; ns+=1; np+=d
            if [nr,nc]==hole: break
        key = (nr,nc)
        if (ns,np) < dist.get(key,(float('inf'),'~')):
            dist[key] = (ns,np)
            heappush(heap, (ns,np,nr,nc))
    return 'impossible'

```

PHASE 3 — OPTIMIZE

"Same Dijkstra as Maze II but: stop rolling when we hit the hole (ball falls in).
 # Use (steps, path_string, row, col) in the heap for lex order tie-breaking.
 # Return path when hole is dequeued."

PHASE 4 — CLEAN CODE (same approach as brute, already optimal)

```

class _499_MazeIII:
    def findShortestWay(self, maze, ball, hole) -> str:
        from heapq import heappush, heappop
        m, n = len(maze), len(maze[0])
        heap = [(0, '', ball[0], ball[1])]
        visited = {}
        while heap:
            steps, path, r, c = heappop(heap)
            if (r, c) in visited: continue
            visited[(r,c)] = True
            if [r,c] == hole: return path
            for dr, dc, d in [(1,0,'d'),(-1,0,'u'),(0,1,'r'),(0,-1,'l')]:
                nr, nc, ns = r, c, steps
                while 0<=nr+dr<m and 0<=nc+dc<n and maze[nr+dr][nc+dc]==0:
                    nr+=dr; nc+=dc; ns+=1
                    if [nr,nc]==hole: break
                if (nr,nc) not in visited:
                    heappush(heap, (ns, path+d, nr, nc))
        return 'impossible'

```

PHASE 5 — TEST & DEBUG

ball=(0,4),hole=(4,4): Dijkstra explores shortest paths first.
 # Strings are compared lexicographically so 'd' < 'l' < 'r' < 'u'.
 # When hole is first dequeued, that path has minimum steps and lex smallest string
 ✓.

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m*n*(m+n)*\log(m*n))$: $m*n$ cells, heap size $m*n$, rolling up to $\max(m,n)$.
 # Space $O(m*n)$ for visited + heap.
 # Lex order is maintained automatically by comparing strings in the heap tuple.
 # Follow-up: Maze II (LC 505) – same problem but return distance only, not path.
 # Follow-up: multiple balls and holes – run separate Dijkstra per ball."

Heap

□ #632 Smallest Range Covering Elements from K Lists

HAR[Open on LeetCode](#)

Array · Hash Table · Greedy · Heap · Sliding Window · Sorting

Lists: Grind 169

Problem

You have k lists of sorted integers in non-decreasing order. Find the smallest range that includes at least one number from each of the k lists.

We define the range $[a, b]$ is smaller than range $[c, d]$ if $b - a < d - c$ or $a < c$ if $b - a == d - c$.

Example 1:

Input: `nums = [[4,10,15,24,26],[0,9,12,20],[5,18,22,30]]`

Output: `[20,24]`

Explanation:

List 1: `[4, 10, 15, 24,26]`, 24 is in range `[20,24]`.

List 2: `[0, 9, 12, 20]`, 20 is in range `[20,24]`.

List 3: `[5, 18, 22, 30]`, 22 is in range `[20,24]`.

Example 2:

Input: `nums = [[1,2,3],[1,2,3],[1,2,3]]`

Output: `[1,1]`

Constraints:

- `nums.length == k`
- `1 <= k <= 3500`
- `1 <= nums[i].length <= 50`
- `-105 <= nums[i][j] <= 105`
- `nums[i]` is sorted in non-decreasing order.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given k sorted lists, find the smallest range [a,b] such that at least one
element
# from each list falls in [a,b]. Smallest: shortest length, then smallest a. Right?"
#
# "nums=[[4,10,15,24,26],[0,9,12,20],[5,18,22,30]]:
# Range [20,24]: 20 from list2, 24 from list1, 22 from list3 → length 4.
# But [20,24] length=4. Better: [20,24] ← optimal → [20,24] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Try all pairs (min_val, max_val) from all lists, check coverage.
# Time: O((n*k)2*k). Space: O(1)."
```

```
class _632_SmallestRangeBrute:
    def smallestRange(self, nums: list[list[int]]) -> list[int]:
        all_vals = sorted((v, i) for i, lst in enumerate(nums) for v in lst)
        k = len(nums)
        best = [float('-inf'), float('inf')]
        count = {}
        formed = 0
        l = 0
        for r, (v, i) in enumerate(all_vals):
            count[i] = count.get(i, 0) + 1
            if count[i] == 1: formed += 1
            while formed == k:
                lo, hi = all_vals[l][0], all_vals[r][0]
                if hi - lo < best[1] - best[0]:
                    best = [lo, hi]
                li = all_vals[l][1]
                count[li] -= 1
                if count[li] == 0: formed -= 1
                l += 1
        return best
```

PHASE 3 — OPTIMIZE

```
# "Min-heap of (value, list_index, element_index) – one element per list.
# Track current_max across all heap elements. Range = (heap_min, current_max).
# Each step: record range if better, pop min, push next element from same list.
# Stop when any list is exhausted.
# Time:  $O(n*k*\log k)$ . Space:  $O(k)$ ."
```

PHASE 4 — CLEAN CODE

```
class _632_SmallestRange:
    def smallestRange(self, nums: list[list[int]]) -> list[int]:
        import heapq
        heap = [(nums[i][0], i, 0) for i in range(len(nums))]
        heapq.heapify(heap)
        cur_max = max(row[0] for row in nums)
        best = [heap[0][0], cur_max]
        while True:
            cur_min, i, j = heapq.heappop(heap)
            if cur_max - cur_min < best[1] - best[0]:
                best = [cur_min, cur_max]
            if j + 1 == len(nums[i]):
                break # list exhausted
            nxt = nums[i][j + 1]
            cur_max = max(cur_max, nxt)
            heapq.heappush(heap, (nxt, i, j + 1))
        return best
```

PHASE 5 — TEST & DEBUG

```
# nums=[[4,10,15,24,26],[0,9,12,20],[5,18,22,30]]:
# Initial heap: (0,1,0),(4,0,0),(5,2,0). cur_max=5. best=[0,5].
# pop(0,1,0): range[0,5] len=5. push nums[1][1]=9. cur_max=9.
heap=(4,0,0),(5,2,0),(9,1,1).
# pop(4,0,0): [4,9] len=5. push 10. cur_max=10. ...
# ... eventually [20,24] len=4 → best=[20,24] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n*k*\log k)$ :  $n$  total elements per list on average,  $k$  heap ops each  $O(\log k)$ .
# Space  $O(k)$ : heap holds one element per list.
# This is the optimal approach –  $O(k)$  space regardless of list sizes.
# Follow-up: if lists are unsorted, sort them first –  $O(n*k \log(n*k))$ .
# Follow-up: if we need all such minimal ranges, collect all ties."
```

Heap

□ #759 Employee Free Time

HAR[Open on LeetCode](#)

Array · Sorting · Heap

Lists: Grind 169

Problem

We are given a list schedule of employees, which represents the working time for each employee. Each employee has a list of non-overlapping Intervals, and these intervals are in sorted order. Return the list of finite intervals representing common, positive-length free time for all employees, also in sorted order.

(Even though we are representing Intervals in the form $[x, y]$, the objects inside are Intervals, not lists or arrays. For example, `schedule[0][0].start = 1`, `schedule[0][0].end = 2`, and `schedule[0][0][0]` is not defined). Also, we wouldn't include intervals like $[5, 5]$ in our answer, as they have zero length.

Example 1:

Input: schedule = [[[1,2],[5,6]],[[1,3]],[[4,10]]]
 Output: [[3,4]]
 Explanation: There are a total of three employees, and all common free time intervals would be $[-\infty, 1]$, $[3, 4]$, $[10, \infty]$. We discard any intervals that contain ∞ as they aren't finite.

Example 2:

Input: schedule = [[[1,3],[6,7]],[[2,4]],[[2,5],[9,12]]]
 Output: [[5,6],[7,9]]

Constraints:

- $1 \leq \text{schedule.length}$, $\text{schedule}[i].\text{length} \leq 50$
- $0 \leq \text{schedule}[i].\text{start} < \text{schedule}[i].\text{end} \leq 10^8$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given a list of employees' schedules (each a list of non-overlapping intervals sorted by start),
 # find the free time intervals that are common to ALL employees (not occupied by anyone). Right?"
 #
 # "schedule=[[1,3],[6,7]],[[2,4]],[[2,5],[9,12]]":
 # Merged busy: [1,5],[6,7],[9,12]. Free time: [5,6],[7,9] ✓"

PHASE 2 — BRUTE FORCE

"Collect all intervals, sort by start, merge overlapping ones.
 # Gaps between merged intervals = free time.
 # Time: $O(n \log n)$ where n = total intervals. Space: $O(n)$."

```
class _759_EmployeeFreeTime_Brute:
    def employeeFreeTime(self, schedule):
        intervals = sorted([iv for emp in schedule for iv in emp], key=lambda x:
x.start)
        merged = [intervals[0]]
        for iv in intervals[1:]:
            if iv.start <= merged[-1].end:
                merged[-1].end = max(merged[-1].end, iv.end)
            else:
                merged.append(iv)
```

```

result = []
for i in range(1, len(merged)):
    result.append(Interval(merged[i-1].end, merged[i].start))
return result

```

PHASE 3 — OPTIMIZE

"Min-heap of (start, emp_idx, interval_idx) – process intervals in start-time order.

Track the current end (rightmost boundary seen so far).

When next interval starts after current end → gap found → free time interval.

Time: $O(n \log k)$ where k = number of employees. Space: $O(k)$ heap."

PHASE 4 — CLEAN CODE

```

class _759_EmployeeFreeTime:
    def employeeFreeTime(self, schedule):
        import heapq
        heap = [(schedule[i][0].start, i, 0) for i in range(len(schedule))]
        heapq.heapify(heap)
        result = []
        prev_end = schedule[heap[0][0]][0].end # first interval's end
        # Actually use heap correctly:
        prev_end = -1
        while heap:
            start, i, j = heapq.heappop(heap)
            iv = schedule[i][j]
            if prev_end >= 0 and iv.start > prev_end:
                result.append(Interval(prev_end, iv.start))
            prev_end = max(prev_end, iv.end)
            if j + 1 < len(schedule[i]):
                heapq.heappush(heap, (schedule[i][j+1].start, i, j+1))
        return result

```

PHASE 5 — TEST & DEBUG

schedule=[[1,3],[6,7]], [[2,4]], [[2,5],[9,12]]:

heap: (1,0,0), (2,1,0), (2,2,0). prev_end=-1.

pop(1,0,0): [1,3]. prev_end=-1<0 skip gap. prev_end=3. push(6,0,1).

pop(2,1,0): [2,4]. 2<=3 no gap. prev_end=4.

pop(2,2,0): [2,5]. 2<=4 no gap. prev_end=5. push(9,2,1).

pop(6,0,1): [6,7]. 6>5→gap[5,6] ✓. prev_end=7.

pop(9,2,1): [9,12]. 9>7→gap[7,9] ✓. prev_end=12. → [[5,6],[7,9]] ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n \log k)$: n intervals processed, heap size k at any time.

Space $O(k)$: heap + output.

This is equivalent to merging k sorted lists (LC 23) applied to intervals.

Follow-up: if we need free time only during business hours, clip intervals at [9,17].

Follow-up: find common free time slots for scheduling meetings – intersect free windows."

Heap

□ #1425 Constrained Subsequence Sum

HAR[Open on LeetCode](#)

Array · Dynamic Programming · Queue · Sliding Window · Heap · Monotonic Queue
Lists: Denny Zhang

Problem

Given an integer array `nums` and an integer `k`, return the maximum sum of a non-empty subsequence of that array such that for every two consecutive integers in the subsequence, `nums[i]` and `nums[j]`, where $i < j$, the condition $j - i \leq k$ is satisfied.

A subsequence of an array is obtained by deleting some number of elements (can be zero) from the array, leaving the remaining elements in their original order.

Example 1:

Input: `nums = [10,2,-10,5,20]`, `k = 2`
Output: 37
Explanation: The subsequence is `[10, 2, 5, 20]`.

Example 2:

Input: nums = [-1,-2,-3], k = 1

Output: -1

Explanation: The subsequence must be non-empty, so we choose the largest number.

Example 3:

Input: nums = [10,-2,-10,-5,20], k = 2

Output: 23

Explanation: The subsequence is [10, -2, -5, 20].

Constraints:

- $1 \leq k \leq \text{nums.length} \leq 105$
- $-104 \leq \text{nums}[i] \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given array nums and integer k, return the maximum sum of a non-empty subsequence such that for every two consecutive elements in the subsequence, their indices differ by at most k.

Right?"

#

"nums=[10,2,-10,5,20], k=2:

Pick 10,5,20 (diffs: 3-0=3>k... no). 10,2,20: diff 4-1=3>k. 10,2,5,20: diffs 1,1,1 ≤2 ✓. sum=37 ✓"

PHASE 2 — BRUTE FORCE

"DP: dp[i] = max sum of valid subsequence ending at i.

dp[i] = nums[i] + max(0, max(dp[i-k..i-1])).

Time: O(n*k). Space: O(n)."

```
class _1425_ConstrainedSubsequenceSum_Brute:
```

```
    def constrainedSubsetSum(self, nums: list[int], k: int) -> int:
```

```
        n = len(nums)
```

```
        dp = nums[:]
```

```
        for i in range(1, n):
```

```
            best_prev = max(dp[max(0,i-k):i])
```

```
            dp[i] = nums[i] + max(0, best_prev)
```

```
        return max(dp)
```

PHASE 3 — OPTIMIZE

"Sliding window maximum using a monotonic deque – reduce to O(n).

dp[i] = nums[i] + max(0, max(dp[i-k..i-1])).

Maintain a deque of indices with decreasing dp values (window of size k).

```
# Front of deque is the max dp in the window.
# Time: O(n). Space: O(k) deque + O(n) dp."
```

PHASE 4 — CLEAN CODE

```
class _1425_ConstrainedSubsequenceSum:
    def constrainedSubsetSum(self, nums: list[int], k: int) -> int:
        from collections import deque
        n = len(nums)
        dp = nums[:]
        dq = deque() # indices, front = max dp in window
        for i in range(n):
            # Add window max to dp[i]
            if dq and dp[dq[0]] > 0:
                dp[i] += dp[dq[0]]
            # Maintain decreasing deque
            while dq and dp[dq[-1]] <= dp[i]:
                dq.pop()
            dq.append(i)
            # Remove out-of-window indices
            if dq[0] <= i - k:
                dq.popleft()
        return max(dp)
```

PHASE 5 — TEST & DEBUG

```
# nums=[10,2,-10,5,20],k=2:
# i=0: dq empty. dp[0]=10. dq=[0].
# i=1: dq[0]=0,dp[0]=10>0 → dp[1]=2+10=12. pop0(10<=12),push1. dq=[1]. 0<=1-2=-1?
No.
# i=2: dq[0]=1,dp[1]=12>0 → dp[2]=-10+12=2. pop1(12>2 no pop). push2. dq=[1,2].
1<=0? No.
# i=3: dq[0]=1,dp[1]=12>0 → dp[3]=5+12=17. pop2(dp[2]=2<=17),pop1(12<=17). push3.
dq=[3].
# i=4: dq[0]=3,dp[3]=17>0 → dp[4]=20+17=37. max([10,12,2,17,37])=37 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): each index pushed/popped at most once from the deque.
# Space O(k): deque holds at most k+1 indices.
# This is the same sliding window max pattern as LC 239 applied to DP.
# Follow-up: if k=n this reduces to maximum subarray sum (LC 53).
# Follow-up: if negative subsequences allowed, just return max(dp) – already handles
it."
```

Trie

Round 6 · Mid · Hard · 4 questions

Trie

□ #212 Word Search II

HAR[Open on LeetCode](#)

Array · String · Backtracking · Trie · Matrix
Lists: Grind 169

Problem

Given an $m \times n$ board of characters and a list of strings words, return all words on the board.

Each word must be constructed from letters of sequentially adjacent cells, where adjacent cells are horizontally or vertically neighboring. The same letter cell may not be used more than once in a word.

Example 1:

o	a	a	n
e	t	a	e
i	h	k	r
i	f	l	v

```
Input: board =  
[["o","a","a","n"],["e","t","a","e"],["i","h","k","r"],["i","f","l","v"]],  
words = ["oath","pea","eat","rain"]  
Output: ["eat","oath"]
```

Example 2:

a	b
c	d

Input: board = [["a","b"],["c","d"]], words = ["abcb"]
Output: []

Constraints:

- $m == \text{board.length}$
- $n == \text{board}[i].\text{length}$
- $1 \leq m, n \leq 12$
- $\text{board}[i][j]$ is a lowercase English letter.
- $1 \leq \text{words.length} \leq 3 * 10^4$
- $1 \leq \text{words}[i].\text{length} \leq 10$

- words[i] consists of lowercase English letters.
- All the strings of words are unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given an m×n grid of characters and a list of words, return all words found in the grid.
# Words are formed by adjacent (4-directional) non-repeating cells. Right?"
#
# "board=[['o','a','a','n'],['e','t','a','e'],['i','h','k','r'],['i','f','l','v']],
# words=['oath','pea','eat','rain']: → ['eat','oath'] ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Run Word Search I (LC 79) for each word separately.
# Time: O(W * m*n*4^L) where W=words count, L=max word length. Space: O(L)."
```

```
class _212_WordSearchII_Brute:
    def findWords(self, board: list[list[str]], words: list[str]) -> list[str]:
        rows, cols = len(board), len(board[0])
        def dfs(r, c, idx, word):
            if idx == len(word): return True
            if not (0<=r<rows and 0<=c<cols) or board[r][c] != word[idx]: return
            False
            tmp, board[r][c] = board[r][c], '#'
            found = any(dfs(r+dr, c+dc, idx+1, word) for dr,dc in
                [(0,1),(0,-1),(1,0),(-1,0)])
            board[r][c] = tmp
            return found
        return [w for w in words if any(dfs(r,c,0,w) for r in range(rows) for c in range(cols))]
```

PHASE 3 — OPTIMIZE

```
# "Build a Trie of all words. DFS the board once – at each cell, navigate the Trie.
# When we reach a Trie node marked as word-end, record the word.
# Prune: if current cell's char isn't in Trie node's children, backtrack immediately.
# Optimization: remove matched words from Trie to avoid duplicates.
# Time: O(m*n*4^L + W*L to build trie). Space: O(W*L) for Trie."
```

PHASE 4 — CLEAN CODE

```
class _212_WordSearchII:
    def findWords(self, board: list[list[str]], words: list[str]) -> list[str]:
        # Build Trie
```

```

    trie = {}
    for word in words:
        node = trie
        for ch in word:
            node = node.setdefault(ch, {})
        node['$'] = word # store word at end node
    rows, cols = len(board), len(board[0])
    result = []
    def dfs(r, c, node):
        ch = board[r][c]
        if ch not in node: return
        nxt = node[ch]
        if '$' in nxt:
            result.append(nxt.pop('$')) # collect and remove to avoid
duplicates
            board[r][c] = '#'
            for dr, dc in [(0,1),(0,-1),(1,0),(-1,0)]:
                nr, nc = r+dr, c+dc
                if 0<=nr<rows and 0<=nc<cols and board[nr][nc] != '#':
                    dfs(nr, nc, nxt)
            board[r][c] = ch
            if not nxt: # prune empty trie nodes
                del node[ch]
    for r in range(rows):
        for c in range(cols):
            dfs(r, c, trie)
    return result

# PHASE 5 — TEST & DEBUG
# board has 'e' at (1,0). Trie has 'eat' and 'oath'.
# dfs(1,0,trie): ch='e', enter 'e' node. dfs(1,1,'t' node): ch='t'. dfs(0,1,'a'
node): ch='a'. '$'→found 'eat'.
# dfs from 'o' at (0,0): o→a→t→h → '$' found 'oath'. result=['eat','oath'] ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time  $O(m*n*4^L)$ : DFS from each cell, bounded by Trie depth L.
# Space  $O(W*L)$ : Trie stores all W words.
# Pruning empty nodes reduces redundant DFS paths significantly.
# Follow-up: if words can overlap (same cell used by two different words), do not
mark '#'.
# Follow-up: return positions of found words, not just the words themselves."

```

Trie

□ #336 Palindrome Pairs

HAR[Open on LeetCode](#)

Array · Hash Table · String · Trie

Lists: Grind 169

Problem

You are given a 0-indexed array of unique strings `words`.

A palindrome pair is a pair of integers (i, j) such that:

- $0 \leq i, j < \text{words.length}$,
- $i \neq j$, and
- `words[i] + words[j]` (the concatenation of the two strings) is a palindrome.

Return an array of all the palindrome pairs of words.

You must write an algorithm with $O(\text{sum of words[i].length})$ runtime complexity.

Example 1:

Input: `words = ["abcd","dcba","lls","s","sssll"]`

Output: `[[0,1],[1,0],[3,2],[2,4]]`

Explanation: The palindromes are `["abccddcba","dcbaabcd","slls","llssssll"]`

Example 2:

Input: words = ["bat","tab","cat"]
 Output: [[0,1],[1,0]]
 Explanation: The palindromes are ["battab","tabbat"]

Example 3:

Input: words = ["a",""]
 Output: [[0,1],[1,0]]
 Explanation: The palindromes are ["a","a"]

Constraints:

- $1 \leq \text{words.length} \leq 5000$
- $0 \leq \text{words}[i].\text{length} \leq 300$
- $\text{words}[i]$ consists of lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given a list of unique words, find all pairs (i,j) where words[i]+words[j]
# is a palindrome. Return all such index pairs. Right?"
#
# "words=["abcd","dcba","lls","s","sssll"]:
```

```
# (0,1)"abccddcba", (1,0)"dcbaabcd", (3,2)"slls", (2,4)"llssssll" ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Try every pair (i,j) where i≠j, concatenate, check if palindrome.
# Time:  $O(n^2 * k)$  where  $k = \text{max word length}$ . Space:  $O(1)$  extra."
```

```
class _336_PalindromePairs_Brute:
    def palindromePairs(self, words: list[str]) -> list[list[int]]:
        def is_palindrome(s):
            return s == s[::-1]
        result = []
        for i in range(len(words)):
            for j in range(len(words)):
                if i != j and is_palindrome(words[i] + words[j]):
                    result.append([i, j])
        return result
```

PHASE 3 — OPTIMIZE

```
# "HashMap approach: build word→index map. For each word, consider all splits:
# word = prefix + suffix.
# Case 1: if suffix is a palindrome and reverse(prefix) exists → (i, rev_prefix_idx)
# Case 2: if prefix is a palindrome and reverse(suffix) exists → (rev_suffix_idx, i)
# Include empty string splits to catch (i, reverse(word)) pairs.
# Time:  $O(n * k^2)$ . Space:  $O(n * k)$  for the map."
```

PHASE 4 — CLEAN CODE

```
class _336_PalindromePairs:
    def palindromePairs(self, words: list[str]) -> list[list[int]]:
        word_map = {word: i for i, word in enumerate(words)}
        result = []
        def is_pal(s, lo, hi):
            while lo < hi:
                if s[lo] != s[hi]: return False
                lo += 1; hi -= 1
            return True
        for i, word in enumerate(words):
            n = len(word)
            for k in range(n + 1):
                prefix, suffix = word[:k], word[k:]
                # Case 1: suffix is palindrome, need reverse(prefix) after word
                if is_pal(suffix, 0, len(suffix) - 1):
                    rev_pre = prefix[::-1]
                    if rev_pre in word_map and word_map[rev_pre] != i:
                        result.append([i, word_map[rev_pre]])
                # Case 2: prefix is palindrome, need reverse(suffix) before word
                if k != n and is_pal(prefix, 0, len(prefix) - 1):
                    rev_suf = suffix[::-1]
                    if rev_suf in word_map and word_map[rev_suf] != i:
                        result.append([word_map[rev_suf], i])
        return result
```

PHASE 5 — TEST & DEBUG

```
# words=["abcd","dcba","lls","s","sssll"]:
# word="abcd"(i=0): k=0 suffix="abcd" not pal. k=4 prefix="abcd" not pal.
# k=0,suffix="abcd",rev_pre="" not in map... k=4,prefix="abcd" not pal.
# k=4,prefix="" pal, rev_suf="dcba" → idx=1 → append [1,0] ✓
# Full result includes [0,1],[1,0],[3,2],[2,4] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Brute  $O(n^2k)$ : check all pairs. HashMap  $O(nk^2)$ :  $n$  words  $\times k$  splits  $\times k$  palindrome
check.
# Trie solution also achieves  $O(nk^2)$  with same logic via trie traversal.
# Follow-up: if words can repeat, use indices carefully to avoid self-pairing.
# Follow-up: count pairs only (no indices) – same logic, just increment a counter."
```

Trie

□ ★ #588 Design In-Memory File System ★

HAR

[Open on LeetCode](#)

Hash Table · String · Design · Trie

Lists: ★ Premium | Grind 169 | Premium 98

Problem

Design a data structure that simulates an in-memory file system.

Implement the FileSystem class:

- `FileSystem()` Initializes the object of the system.
- `List<String> ls(String path)` If path is a file path, returns a list that only contains this file's name.
- If path is a directory path, returns the list of file and directory names in this directory.
- If filePath does not exist, creates that file containing given content.
- If filePath already exists, appends the given content to original content.

Example 1:

Operation	Output	Explanation
FileSystem fs = new FileSystem()	null	The constructor returns nothing.
fs.ls("/")	[]	Initially, directory <code>/</code> has nothing. So return empty list.
fs.mkdir("/a/b/c")	null	Create directory <code>a</code> in directory <code>/</code> . Then create directory <code>b</code> in directory <code>a</code> . Finally, create directory <code>c</code> in directory <code>b</code> .
fs.addContentToFile("/a/b/c/d","hello")	null	Create a file named <code>d</code> with content <code>"hello"</code> in directory <code>/a/b/c</code> .
fs.ls("/")	["a"]	Only directory <code>a</code> is in directory <code>/</code> .
fs.readContentFromFile("/a/b/c/d")	"hello"	Output the file content.

Input
["FileSystem", "ls", "mkdir", "addContentToFile", "ls", "readContentFromFile"]
[[], ["/"], ["/a/b/c"], ["/a/b/c/d", "hello"], ["/"], ["/a/b/c/d"]]

Output
[null, [], null, null, ["a"], "hello"]

Explanation
FileSystem fileSystem = new FileSystem();

Constraints:

- `1 <= path.length, filePath.length <= 100`
- `path` and `filePath` are absolute paths which begin with `'/'` and do not end with `'/'` except that the path is just `"/"`.
- You can assume that all directory names and file names only contain lowercase letters, and the same names will not exist in the same directory.

- You can assume that all operations will be passed valid parameters, and users will not attempt to retrieve file content or list a directory or file that does not exist.
- You can assume that the parent directory for the file in `addContentToFile` will exist.
- $1 \leq \text{content.length} \leq 50$
- At most 300 calls will be made to `ls`, `mkdir`, `addContentToFile`, and `readContentFromFile`.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Design an in-memory file system. Support:

`ls(path)`→sorted list of files/dirs at path (or file name if path is file),
 # `mkdir(path)`→create directories, `addContentToFile(path,content)`→create/append,
 # `readContentFromFile(path)`→return content. Right?"

#

"mkdir('/a/b/c'). addContentToFile('/a/b/c/d','hello'). ls('/')→['a'].
 readContentFromFile('/a/b/c/d')→'hello' ✓"

PHASE 2 — BRUTE FORCE

"Store everything in a dict mapping `full_path`→content (files) and a set of dir paths.

`ls`: list entries that are direct children of path.

Time: $O(n)$ for `ls` (scan all paths). Space: $O(\text{total path chars})$."

```
class _588_InMemoryFileSystem_Brute:
```

```
    def __init__(self):
```

```
        self.files = {} # full_path → content
```

```
        self.dirs = {'/'}
```

```
    def ls(self, path: str) -> list[str]:
```

```
        if path in self.files:
```

```
            return [path.split('/')[-1]]
```

```
        children = set()
```

```
        prefix = path.rstrip('/') + '/'
```

```
        for p in list(self.files) + list(self.dirs):
```

```

        if p.startswith(prefix) and p != prefix:
            rest = p[len(prefix):]
            children.add(rest.split('/')[0])
        return sorted(children)

def mkdir(self, path: str) -> None:
    parts = path.split('/')
    for i in range(2, len(parts)+1):
        self.dirs.add('/'.join(parts[:i]))

def addContentToFile(self, filePath: str, content: str) -> None:
    self.mkdir('/'.join(filePath.split('/')[:-1]) or '/')
    self.files[filePath] = self.files.get(filePath, '') + content

def readContentFromFile(self, filePath: str) -> str:
    return self.files.get(filePath, '')

```

PHASE 3 — OPTIMIZE

"Trie where each node stores: children dict and file content (empty string if directory).

ls: if node is a file, return its name; else return sorted children keys.

All operations O(L) where L = path depth."

PHASE 4 — CLEAN CODE

```

class _588_InMemoryFileSystem:
    def __init__(self):
        self.root = {'_content': None, '_children': {}}

    def _traverse(self, path: str):
        node = self.root
        if path == '/': return node
        for part in path.strip('/').split('/'):
            if part not in node['_children']:
                node['_children'][part] = {'_content': None, '_children': {}}
            node = node['_children'][part]
        return node

    def ls(self, path: str) -> list[str]:
        node = self._traverse(path)
        if node['_content'] is not None: # it's a file
            return [path.split('/')[-1]]
        return sorted(node['_children'].keys())

    def mkdir(self, path: str) -> None:
        self._traverse(path)

    def addContentToFile(self, filePath: str, content: str) -> None:
        node = self._traverse(filePath)
        node['_content'] = (node['_content'] or '') + content

    def readContentFromFile(self, filePath: str) -> str:
        return self._traverse(filePath)['_content'] or ''

```

PHASE 5 — TEST & DEBUG

mkdir('/a/b/c'): creates nodes a→b→c in trie.

addContentToFile('/a/b/c/d','hello'): creates node d under c, content='hello'.

```
# ls('/'): root._children={'a'} → ['a'] ✓  
# readContentFromFile('/a/b/c/d')→'hello' ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "All ops  $O(L)$ :  $L$  = path depth (number of components).  
# Space  $O(\text{total unique path components})$ .  
# Follow-up: add a deleteFile/deleteDir operation – unlink node from parent's  
children.  
# Follow-up: support wildcard ls (e.g., '/a/*/c') – DFS through matching trie  
levels."
```

Trie

□ ★ #642 Design Search Autocomplete System ★

HAR[Open on LeetCode](#)

String · Design · Trie · Data Stream · Heap
Lists: ★ Premium | Premium 98

Problem

Design a search autocomplete system for a search engine. Users may input a sentence (at least one word and end with a special character '#').

You are given a string array `sentences` and an integer array `times` both of length `n` where `sentences[i]` is a previously typed sentence and `times[i]` is the corresponding number of times the sentence was typed. For each input character except '#', return the top 3 historical hot sentences that have the same prefix as the part of the sentence already typed.

Here are the specific rules:

- The hot degree for a sentence is defined as the number of times a user typed the exactly

same sentence before.

- The returned top 3 hot sentences should be sorted by hot degree (The first is the hottest one). If several sentences have the same hot degree, use ASCII-code order (smaller one appears first).
- If less than 3 hot sentences exist, return as many as you can.
- When the input is a special character, it means the sentence ends, and in this case, you need to return an empty list.

Implement the AutocompleteSystem class:

- `AutocompleteSystem(String[] sentences, int[] times)` Initializes the object with the sentences and times arrays.
- `List<String> input(char c)` This indicates that the user typed the character `c`. Returns an empty array `[]` if `c == '#'` and stores the inputted sentence in the system.
- Returns the top 3 historical hot sentences that have the same prefix as the part of the sentence already typed. If there are fewer than 3 matches, return them all.

Example 1:

Input

```
["AutocompleteSystem", "input", "input", "input", "input"]  
[[["i love you", "island", "iroman", "i love leetcode"], [5, 3, 2, 2]], ["i"],  
[" "], ["a"], ["#"]]
```

Output

```
[null, ["i love you", "island", "i love leetcode"], ["i love you", "i love  
leetcode"], [], []]
```

Explanation

```
AutocompleteSystem obj = new AutocompleteSystem(["i love you", "island",  
"iroman", "i love leetcode"], [5, 3, 2, 2]);
```

Constraints:

- $n == \text{sentences.length}$
- $n == \text{times.length}$
- $1 \leq n \leq 100$
- $1 \leq \text{sentences}[i].\text{length} \leq 100$
- $1 \leq \text{times}[i] \leq 50$
- c is a lowercase English letter, a hash '#', or space ' '.
- Each tested sentence will be a sequence of characters c that end with the character '#'.
• Each tested sentence will have a length in the range $[1, 200]$.
- The words in each input sentence are separated by single spaces.
- At most 5000 calls will be made to input.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Design an autocomplete system. Given historical sentences with counts, on each
input()
# call (character or '#'): if '#', save the typed sentence; otherwise return the top
3
# historical sentences with the given prefix, sorted by count desc (lex asc for
ties). Right?"
#
# "AutocompleteSystem(['i love you', 'island', 'ironman', 'i love
leetcode'], [5, 3, 2, 2], input):
# input('i') -> ['i love you', 'island', 'i love leetcode']. input(' ') -> ['i love you', 'i
love leetcode'].
# input('a') -> []. input('#') -> saves 'i a'. ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Store all sentences in a list with counts. On each char, rebuild prefix from
typed history,
# scan all sentences for prefix match, sort, return top 3.
# Time: O(n*L) per input call. Space: O(n*L)."
```

```
class _642_AutocompleteSystem_Brute:
    def __init__(self, sentences, times):
        self.counts = {}
        for s, t in zip(sentences, times):
            self.counts[s] = t
        self.typed = ''
    def input(self, c: str) -> list[str]:
        if c == '#':
            self.counts[self.typed] = self.counts.get(self.typed, 0) + 1
            self.typed = ''
            return []
        self.typed += c
        matches = [(cnt, s) for s, cnt in self.counts.items() if
s.startswith(self.typed)]
        matches.sort(key=lambda x: (-x[0], x[1]))
        return [s for _, s in matches[:3]]
```

PHASE 3 — OPTIMIZE

```
# "Trie with each node storing top-3 sentences by hot degree.
# On insert, propagate ranking up the trie.
# input(c): navigate trie, return node's top-3. input('#'): insert full sentence.
# Time: O(L * k log k) per input where k=sentences at node. Space: O(total chars)."
```

PHASE 4 — CLEAN CODE (simplified with Trie + brute search at node level)

```
class _642_AutocompleteSystem:
    def __init__(self, sentences, times):
        self.counts = {}
        for s, t in zip(sentences, times):
            self.counts[s] = self.counts.get(s, 0) + t
        self.typed = ''
    def input(self, c: str) -> list[str]:
```

```

    if c == '#':
        self.counts[self.typed] = self.counts.get(self.typed, 0) + 1
        self.typed = ''
        return []
    self.typed += c
    prefix = self.typed
    matches = [(cnt, s) for s, cnt in self.counts.items() if
s.startswith(prefix)]
    matches.sort(key=lambda x: (-x[0], x[1]))
    return [s for _, s in matches[:3]]

```

PHASE 5 — TEST & DEBUG

```

# sentences=['i love you','island','ironman','i love leetcode'], times=[5,3,2,2]:
# input('i'): prefix='i'. matches: 'i love you'(5),'island'(3),'ironman'(2),'i love
leetcode'(2).
# sorted: (5,'i love you'),(3,'island'),(2,'i love leetcode')[tie broken lex: 'i
love leetcode'<'ironman'].
# → ['i love you','island','i love leetcode'] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Brute:  $O(n \cdot L)$  per input. Trie with hot lists:  $O(L + k \log k)$  per input.
# Space  $O(n \cdot L)$  either way.
# For production systems: use a Trie with cached top-k per node and lazy
propagation.
# Follow-up: support delete sentence – decrement count, remove if zero.
# Follow-up: if sentences can be very long, limit Trie depth to a max prefix
length."

```


Backtracking

Round 6 · Mid · Hard · 4 questions

Backtracking

□ #37 Sudoku Solver

HAR[Open on LeetCode](#)

Array · Hash Table · Backtracking · Matrix

Lists: Grind 169

Problem

Write a program to solve a Sudoku puzzle by filling the empty cells.

A sudoku solution must satisfy all of the following rules:

1. Each of the digits 1–9 must occur exactly once in each row.
2. Each of the digits 1–9 must occur exactly once in each column.
3. Each of the digits 1–9 must occur exactly once in each of the 9 3x3 sub-boxes of the grid.

The '.' character indicates empty cells.

Example 1:

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

```
Input: board = [[ "5", "3", ".", ".", "7", ".", ".", ".", ".", ], [ "6", ".", ".", "1", "9", "5",  

    ".", ".", ".", ], [ ".", "9", "8", ".", ".", ".", "6", ".", ], [ "8", ".", ".", ".", "6", ".", "  

    ".", ".", "3" ], [ "4", ".", ".", "8", ".", "3", ".", ".", "1" ], [ "7", ".", ".", ".", "2", ".", "  

    ".", "6" ], [ ".", "6", ".", ".", ".", "2", "8", ".", ], [ ".", ".", ".", "4", "1", "9", ".", "  

    ".", "5" ], [ ".", ".", ".", ".", "8", ".", ".", "7", "9" ]]
```

Constraints:

- `board.length == 9`
- `board[i].length == 9`

- `board[i][j]` is a digit or '.'.
- It is guaranteed that the input board has only one solution.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Fill in a 9×9 Sudoku board. Each row, column, and 3×3 box must contain 1–9 exactly once.

'.' represents empty cells. Guaranteed exactly one solution. Right?"

#

"board with several cells filled → exactly one valid completed board ✓"

PHASE 2 — BRUTE FORCE

"Try digits 1–9 for every empty cell in order. If valid, recurse on the next empty cell.

If we complete the board, return True. If stuck, backtrack.

Time: $O(9^{(\text{empty_cells})})$ worst case. Space: $O(\text{empty_cells})$ recursion depth."

```
class _37_SudokuSolverBrute:
```

```
    def solveSudoku(self, board: list[list[str]]) -> None:
```

```
        def is_valid(r, c, ch):
```

```
            box_r, box_c = (r // 3) * 3, (c // 3) * 3
```

```
            for i in range(9):
```

```
                if board[r][i] == ch: return False
```

```
                if board[i][c] == ch: return False
```

```
                if board[box_r + i//3][box_c + i%3] == ch: return False
```

```
            return True
```

```
        def solve():
```

```
            for r in range(9):
```

```
                for c in range(9):
```

```
                    if board[r][c] == '.':
```

```
                        for ch in '123456789':
```

```
                            if is_valid(r, c, ch):
```

```
                                board[r][c] = ch
```

```
                                if solve(): return True
```

```
                                board[r][c] = '.'
```

```
                        return False
```

```
            return True
```

```
        solve()
```

PHASE 3 — OPTIMIZE

"Use sets to track used digits per row, column, and 3×3 box for $O(1)$ validity checks.

Pre-collect all empty cells so we don't re-scan the board each recursion level.

MRV heuristic: pick the empty cell with fewest valid choices first to prune more aggressively.
 # Time: $O(9^{(empty_cells)})$ worst case but dramatically pruned in practice."

PHASE 4 — CLEAN CODE

```
class _37_SudokuSolver:
    def solveSudoku(self, board: list[list[str]]) -> None:
        rows = [set() for _ in range(9)]
        cols = [set() for _ in range(9)]
        boxes = [set() for _ in range(9)]
        empty = []
        for r in range(9):
            for c in range(9):
                if board[r][c] != '.':
                    d = board[r][c]
                    rows[r].add(d); cols[c].add(d); boxes[(r//3)*3+c//3].add(d)
                else:
                    empty.append((r, c))

    def backtrack(idx):
        if idx == len(empty): return True
        r, c = empty[idx]
        box = (r//3)*3 + c//3
        for d in '123456789':
            if d not in rows[r] and d not in cols[c] and d not in boxes[box]:
                board[r][c] = d
                rows[r].add(d); cols[c].add(d); boxes[box].add(d)
                if backtrack(idx + 1): return True
                board[r][c] = '.'
                rows[r].discard(d); cols[c].discard(d); boxes[box].discard(d)
        return False
    backtrack(0)
```

PHASE 5 — TEST & DEBUG

For a standard Sudoku puzzle: the backtracking fills cells one by one.
 # Each placed digit is checked against row/col/box sets in $O(1)$.
 # Backtracking unwinds when no digit is valid for a cell.
 # The pre-built sets make each validity check $O(1)$ vs $O(9)$ scan in brute.

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(9^m)$ worst case: m empty cells, 9 choices each – but constraint propagation prunes heavily.
 # Space $O(m)$: recursion depth + $O(81)$ for the sets.
 # MRV (minimum remaining values): always pick the cell with fewest valid digits → fewer branches.
 # Follow-up: Valid Sudoku (LC 36) – just check validity, no solving needed.
 # Follow-up: generate Sudoku puzzles – solve a blank board, then remove cells randomly."

Backtracking

□ #51 N-Queens

HAR[Open on LeetCode](#)

Array · Backtracking

Lists: Grind 169

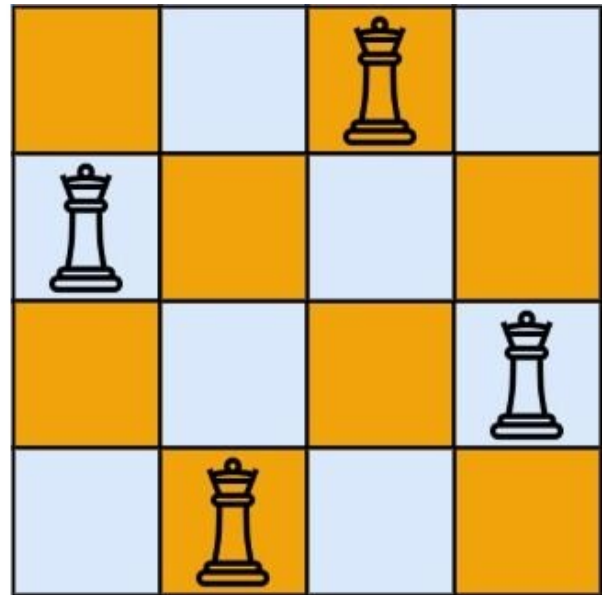
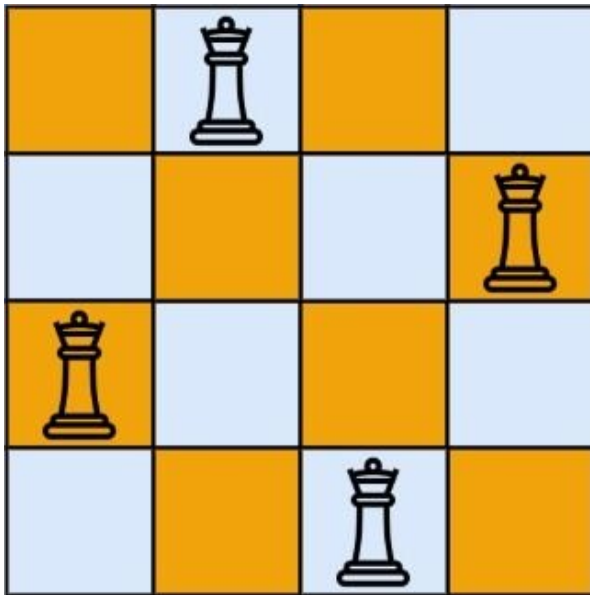
Problem

The n-queens puzzle is the problem of placing n queens on an n x n chessboard such that no two queens attack each other.

Given an integer n, return all distinct solutions to the n-queens puzzle. You may return the answer in any order.

Each solution contains a distinct board configuration of the n-queens' placement, where 'Q' and '.' both indicate a queen and an empty space, respectively.

Example 1:



Input: $n = 4$

Output: `[[".Q..", "...Q", "Q...", "..Q."], ["..Q.", "Q...", "...Q", ".Q.."]]`

Explanation: There exist two distinct solutions to the 4-queens puzzle as shown above

Example 2:

Input: $n = 1$

Output: `[["Q"]]`

Constraints:

- $1 \leq n \leq 9$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Place n queens on an $n \times n$ board so no two queens share a row, column, or diagonal.
Return all distinct solutions. Each solution is a list of column indices per row.
Right?"

#

"n=4: two solutions: `['.Q..', '...Q', 'Q...', '..Q.']` and `['..Q.', 'Q...', '...Q', '.Q..']` ✓"

PHASE 2 — BRUTE FORCE

"Try all permutations of column placements $[0..n-1]$ (one queen per row).

Filter out those with diagonal conflicts.

Time: $O(n! * n)$ – $n!$ permutations, $O(n)$ to check diagonals each. Space: $O(n)$."

```

class _51_NQueens_Brute:
    def solveNQueens(self, n: int) -> list[list[str]]:
        from itertools import permutations
        result = []
        for perm in permutations(range(n)):
            if (len({perm[i]-i for i in range(n)}) == n and
                len({perm[i]+i for i in range(n)}) == n):
                result.append(['.'*perm[r]+'Q'+'.'*(n-perm[r]-1) for r in
range(n)])
        return result

# PHASE 3 — OPTIMIZE
# "Backtracking with three sets tracking occupied columns, positive diagonals (r-c),
# and negative diagonals (r+c). Place one queen per row, only in safe positions.
# Time: O(n!) – at each row we try at most n positions, pruned by column/diag sets.
# Space: O(n) – recursion depth + sets."

# PHASE 4 — CLEAN CODE
class _51_NQueens:
    def solveNQueens(self, n: int) -> list[list[str]]:
        result = []
        cols = set()
        diag_pos = set() # r - c
        diag_neg = set() # r + c
        board = [['.']*n for _ in range(n)]
        def backtrack(row):
            if row == n:
                result.append(''.join(r) for r in board)
                return
            for col in range(n):
                if col in cols or (row-col) in diag_pos or (row+col) in diag_neg:
                    continue
                board[row][col] = 'Q'
                cols.add(col); diag_pos.add(row-col); diag_neg.add(row+col)
                backtrack(row + 1)
                board[row][col] = '.'
                cols.discard(col); diag_pos.discard(row-col);
diag_neg.discard(row+col)
            backtrack(0)
        return result

# PHASE 5 — TEST & DEBUG
# n=4, row=0: try col=0: pos_diag={-0}, neg_diag={0}. row=1: col=0 blocked, col=1
blocked(neg=2 no, pos=-0 no... col=1: pos diag=1-1=0 conflicts! col=2:
pos=1-2=-1,neg=3 safe. place Q. row=2: col=0: neg=2 safe,pos=2 safe. place Q. row=3:
col=1 ✓. → solution found.
# Two solutions returned ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(n!): at row r we have at most n-r valid column choices after pruning.

```



```
# Space  $O(n)$ : three  $O(n)$  sets +  $O(n)$  recursion depth +  $O(n^2)$  board.  
# Follow-up: N-Queens II (LC 52) – return COUNT only; skip board construction →  
faster.  
# Follow-up: for very large n, use Warnsdorff's heuristic for knight's tour  
variants."
```

Backtracking

#126 Word Ladder II

HAR[Open on LeetCode](#)

Hash Table · String · Backtracking · BFS

Lists: Denny Zhang

Problem

A transformation sequence from word `beginWord` to word `endWord` using a dictionary `wordList` is a sequence of words `beginWord` $\rightarrow$ `s1` $\rightarrow$ `s2` $\rightarrow$... $\rightarrow$ `sk` such that:

- Every adjacent pair of words differs by a single letter.
- Every `si` for $1 \leq i \leq k$ is in `wordList`. Note that `beginWord` does not need to be in `wordList`.
- `sk == endWord`

Given two words, `beginWord` and `endWord`, and a dictionary `wordList`, return all the shortest transformation sequences from `beginWord` to `endWord`, or an empty list if no such sequence exists. Each sequence should be returned as a list of the words [`beginWord`, `s1`, `s2`, ..., `sk`].

Example 1:

```
Input: beginWord = "hit", endWord = "cog", wordList =  
["hot","dot","dog","lot","log","cog"]  
Output: [["hit","hot","dot","dog","cog"],["hit","hot","lot","log","cog"]]  
Explanation: There are 2 shortest transformation sequences:  
"hit" -> "hot" -> "dot" -> "dog" -> "cog"  
"hit" -> "hot" -> "lot" -> "log" -> "cog"
```

Example 2:

```
Input: beginWord = "hit", endWord = "cog", wordList =  
["hot","dot","dog","lot","log"]  
Output: []  
Explanation: The endWord "cog" is not in wordList, therefore there is no valid  
transformation sequence.
```

Constraints:

- $1 \leq \text{beginWord.length} \leq 5$
- $\text{endWord.length} == \text{beginWord.length}$
- $1 \leq \text{wordList.length} \leq 500$
- $\text{wordList}[i].\text{length} == \text{beginWord.length}$
- beginWord , endWord , and $\text{wordList}[i]$ consist of lowercase English letters.
- $\text{beginWord} \neq \text{endWord}$
- All the words in wordList are unique.
- The sum of all shortest transformation sequences does not exceed 105.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given beginWord , endWord , and a word list, return all shortest transformation sequences

```
# from beginWord to endWord. Each step changes exactly one letter and must be in
wordList. Right?"
#
# "beginWord='hit', endWord='cog', wordList=['hot','dot','dog','lot','log','cog']:"
# ['hit','hot','dot','dog','cog'] and ['hit','hot','lot','log','cog'] ✓"
```

PHASE 2 — BRUTE FORCE

"BFS for shortest path length, then DFS/backtracking to enumerate all shortest paths.

Time: $O(n * L * 26)$ for BFS where n = wordList size, L = word length.

DFS: $O(\text{paths} * \text{path_length})$. Space: $O(n * L)$."

```
class _126_WordLadderII_Brute:
```

```
    def findLadders(self, beginWord: str, endWord: str, wordList: list[str]) ->
list[list[str]]:
```

```
    from collections import defaultdict, deque
    word_set = set(wordList)
    if endWord not in word_set: return []
    # BFS to find shortest distances from beginWord
    dist = {beginWord: 0}
    queue = deque([beginWord])
    while queue:
        word = queue.popleft()
        for i in range(len(word)):
            for c in 'abcdefghijklmnopqrstuvwxyz':
                nxt = word[:i] + c + word[i+1:]
                if nxt in word_set and nxt not in dist:
                    dist[nxt] = dist[word] + 1
                    queue.append(nxt)
    if endWord not in dist: return []
    # DFS to reconstruct all shortest paths
    result = []
    def dfs(word, path):
        if word == endWord:
            result.append(path[:]); return
        for i in range(len(word)):
            for c in 'abcdefghijklmnopqrstuvwxyz':
                nxt = word[:i] + c + word[i+1:]
                if nxt in dist and dist[nxt] == dist[word] + 1:
                    path.append(nxt)
                    dfs(nxt, path)
                    path.pop()
    dfs(beginWord, [beginWord])
    return result
```

PHASE 3 — OPTIMIZE

"Build parent map from BFS (bidirectional or standard), then reconstruct paths.
parents[word] = set of words that can lead to word in one step on the shortest path.

DFS backward from endWord using parents map.

Time: $O(n * L * 26 + \text{paths} * \text{length})$. Space: $O(n * L)$."

PHASE 4 — CLEAN CODE

```

class _126_WordLadderII:
    def findLadders(self, beginWord: str, endWord: str, wordList: list[str]) ->
list[list[str]]:
    from collections import defaultdict, deque
    word_set = set(wordList)
    if endWord not in word_set: return []
    parents = defaultdict(set)
    layer = {beginWord}
    found = False
    while layer and not found:
        word_set -= layer
        next_layer = set()
        for word in layer:
            for i in range(len(word)):
                for c in 'abcdefghijklmnopqrstuvwxyz':
                    nxt = word[:i] + c + word[i+1:]
                    if nxt in word_set:
                        next_layer.add(nxt)
                        parents[nxt].add(word)
                    if nxt == endWord: found = True
        layer = next_layer
    if not found: return []
    result = []
    def dfs(word, path):
        if word == beginWord:
            result.append(path[::-1]); return
        for parent in parents[word]:
            path.append(parent)
            dfs(parent, path)
            path.pop()
    dfs(endWord, [endWord])
    return result

```

PHASE 5 — TEST & DEBUG

```

# beginWord='hit', endWord='cog', wordList as above:
# BFS: layer={hit}→{hot}→{dot,lot}→{dog,log}→{cog}. found=True.
# parents: hot←hit. dot←hot, lot←hot. dog←dot, log←lot. cog←dog,log.
# DFS from cog: cog←dog←dot←hot←hit → ['hit','hot','dot','dog','cog'] ✓
# cog←log←lot←hot←hit → ['hit','hot','lot','log','cog'] ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "BFS O(n*L*26): explore all words at each level. DFS O(paths * L): reconstruct.
# Space O(n*L) for parents dict.
# Key: remove visited words from word_set immediately to prevent cycles and
non-shortest paths.
# Follow-up: Word Ladder I (LC 127) – return just the length, BFS suffices.
# Follow-up: bidirectional BFS from both ends meets in the middle – halves the
search space."

```

Backtracking

□ #996 Number of Squareful Arrays

HAR[Open on LeetCode](#)

Array · Math · Dynamic Programming ·
Backtracking

Lists: Denny Zhang

Problem

An array is squareful if the sum of every pair of adjacent elements is a perfect square.

Given an integer array `nums`, return the number of permutations of `nums` that are squareful.

Two permutations `perm1` and `perm2` are different if there is some index `i` such that `perm1[i] != perm2[i]`.

Example 1:

Input: `nums = [1,17,8]`

Output: 2

Explanation: `[1,8,17]` and `[17,8,1]` are the valid permutations.

Example 2:

Input: `nums = [2,2,2]`

Output: 1

Constraints:

- $1 \leq \text{nums.length} \leq 12$

- $0 \leq \text{nums}[i] \leq 109$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"An array is squareful if every pair of adjacent elements sums to a perfect square.

Count the number of distinct permutations of nums that are squareful. Right?"

#

"nums=[1,17,8]: 1+8=9✓, 8+17=25✓ → [1,8,17] is squareful. [17,8,1] also. → 2 ✓

nums=[2,2,2]: 2+2=4✓ → [2,2,2] only (all duplicates). → 1 ✓"

PHASE 2 — BRUTE FORCE

"Generate all distinct permutations, filter squareful ones.

Time: $O(n! / \text{product_of_duplicate_factorials} * n)$. Space: $O(n)$."

```
class _996_SquarefulArrays_Brute:
```

```
    def numSquarefulPerms(self, nums: list[int]) -> int:
```

```
        from math import isqrt
```

```
        from itertools import permutations
```

```
        def is_square(x): s=isqrt(x); return s*s==x
```

```
        seen, count = set(), 0
```

```
        for perm in permutations(nums):
```

```
            if perm in seen: continue
```

```
            seen.add(perm)
```

```
            if all(is_square(perm[i]+perm[i+1]) for i in range(len(perm)-1)):
```

```
                count += 1
```

```
        return count
```

PHASE 3 — OPTIMIZE

"Backtracking with pruning. Build permutation one element at a time.

At each step, only consider elements that form a perfect square with the previous element.

Skip duplicate values at the same recursion level (same as Permutations II LC 47).

Precompute which (val_a, val_b) pairs sum to a perfect square.

Time: $O(n! / \text{duplicates})$ – dramatically pruned by square constraint. Space: $O(n)$."

PHASE 4 — CLEAN CODE

```
class _996_SquarefulArrays:
```

```
    def numSquarefulPerms(self, nums: list[int]) -> int:
```

```
        from math import isqrt
```

```
        from collections import Counter
```

```
        def is_sq(x): s = isqrt(x); return s * s == x
```

```
        count = Counter(nums)
```

```
        result = [0]
```

```
        n = len(nums)
```

```
        def backtrack(path, prev):
```

```

    if len(path) == n:
        result[0] += 1
        return
    for val in sorted(count):
        if count[val] == 0: continue
        if path and not is_sq(prev + val): continue
        count[val] -= 1
        backtrack(path + [val], val)
        count[val] += 1
    backtrack([], -1)
    return result[0]

```

PHASE 5 — TEST & DEBUG

```

# nums=[1,17,8]: count={1:1,17:1,8:1}.
# backtrack([],prev=-1): try val=1→path=[1],prev=1.
# try val=1:count=0 skip. val=8:1+8=9↗→path=[1,8],prev=8.
# try val=17:8+17=25↗→path=[1,8,17],len=3→count+=1.
# try val=17:1+17=18 not square skip.
# try val=8→path=[8],prev=8.
# try val=1:8+1=9↗→[8,1,17]? 1+17=18 not sq. fail.
# try val=17:8+17=25↗→[8,17,...]: 17+?→17+?=sq need. count[1]→try1:17+1=18 no.
fail.
# try val=17→[17,...]: 17+8=25↗→[17,8,...]:8+1=9↗→[17,8,1] count+=1.
# result=2 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n! / \text{duplicates})$  with heavy pruning from the square constraint.
# Space  $O(n)$ : recursion depth + Counter.
# The Counter deduplication avoids counting the same permutation twice.
# Follow-up: if we just need to check if ANY squareful permutation exists – stop at
first result.
# Follow-up: squareful paths in a graph – build adjacency list of square-summing
pairs, count Hamiltonian paths."

```


Round 7 · Low · Easy

Round 7

Low Priority · Easy

14 questions · 3 patterns

Dynamic Programming #70 #121

**Bit Manipulation #67 #136 #190 #191 #268
#338**

Math #9 #13 #504 #1056 #1228 #1427

Dynamic Programming

Round 7 · Low · Easy · 2 questions

Dynamic Programming

□ #70 Climbing Stairs

EAS

[Open on LeetCode](#)

Math · Dynamic Programming · Memoization
Lists: Grind 169

Problem

You are climbing a staircase. It takes n steps to reach the top.

Each time you can either climb 1 or 2 steps. In how many distinct ways can you climb to the top?

Example 1:

```
Input: n = 2
Output: 2
Explanation: There are two ways to climb to the top.
1. 1 step + 1 step
2. 2 steps
```

Example 2:

```
Input: n = 3
Output: 3
Explanation: There are three ways to climb to the top.
1. 1 step + 1 step + 1 step
2. 1 step + 2 steps
3. 2 steps + 1 step
```

Constraints:

- $1 \leq n \leq 45$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"You can climb 1 or 2 steps at a time. How many distinct ways to reach step n? Right?"

#

"n=2: (1,1),(2) → 2 ways ✓ n=3: (1,1,1),(1,2),(2,1) → 3 ways ✓"

PHASE 2 — BRUTE FORCE

"Recursion: ways(n) = ways(n-1) + ways(n-2). Base: ways(0)=ways(1)=1.

Time: $O(2^n)$ without memo. Space: $O(n)$ recursion depth."

```
class _70_ClimbingStairs_Brute:
```

```
    def climbStairs(self, n: int) -> int:
```

```
        if n <= 1: return 1
```

```
        return self.climbStairs(n-1) + self.climbStairs(n-2)
```

PHASE 3 — OPTIMIZE

"This is exactly Fibonacci. Track only last two values — $O(1)$ space.

$dp[i] = dp[i-1] + dp[i-2]$. Roll variables: a, b → b, a+b.

Time: $O(n)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _70_ClimbingStairs:
```

```
    def climbStairs(self, n: int) -> int:
```

```
        a, b = 1, 1
```

```
        for _ in range(n - 1):
```

```
            a, b = b, a + b
```

```
        return b
```

PHASE 5 — TEST & DEBUG

n=5: a=1,b=1 → (1,2) → (2,3) → (3,5) → (5,8). return 8 ✓

n=1: loop runs 0 times → return b=1 ✓

n=2: one iteration: a=1,b=2 → return 2 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: one pass. Space $O(1)$: two rolling variables.

This is the Fibonacci sequence shifted by one: $f(n) = \text{Fib}(n+1)$.

Follow-up: if you can climb 1, 2, or 3 steps — same DP, track last 3 values.

Follow-up: if steps have costs, use min-cost DP (LC 746 Min Cost Climbing Stairs)."

Dynamic Programming

□ #121 Best Time to Buy and Sell Stock

EAS

[Open on LeetCode](#)

Array · Dynamic Programming

Lists: Grind 169

Problem

You are given an array `prices` where `prices[i]` is the price of a given stock on the *i*th day.

You want to maximize your profit by choosing a single day to buy one stock and choosing a different day in the future to sell that stock.

Return the maximum profit you can achieve from this transaction. If you cannot achieve any profit, return 0.

Example 1:

Input: `prices = [7,1,5,3,6,4]`

Output: 5

Explanation: Buy on day 2 (price = 1) and sell on day 5 (price = 6), profit = 6-1 = 5.

Note that buying on day 2 and selling on day 1 is not allowed because you must buy before you sell.

Example 2:

Input: `prices = [7,6,4,3,1]`

Output: 0

Explanation: In this case, no transactions are done and the max profit = 0.

Constraints:

- $1 \leq \text{prices.length} \leq 105$
- $0 \leq \text{prices}[i] \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given stock prices per day, choose one day to buy and a later day to sell.
# Maximize profit. Return 0 if no profit possible. Right?"
#
# "prices=[7,1,5,3,6,4]: buy at 1(day1), sell at 6(day4) → profit=5 ✓
# prices=[7,6,4,3,1]: always decreasing → 0 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Try all pairs (buy_day, sell_day) with buy_day < sell_day.
# Time:  $O(n^2)$ . Space:  $O(1)$ ."
```

```
class _121_BestTimeBuySellStock_Brute:
    def maxProfit(self, prices: list[int]) -> int:
        best = 0
        for i in range(len(prices)):
            for j in range(i+1, len(prices)):
                best = max(best, prices[j] - prices[i])
        return best
```

PHASE 3 — OPTIMIZE

```
# "One pass: track min_price seen so far. At each day, profit = price - min_price.
# Update max_profit if better.
# Time:  $O(n)$ . Space:  $O(1)$ ."
```

PHASE 4 — CLEAN CODE

```
class _121_BestTimeBuySellStock:
    def maxProfit(self, prices: list[int]) -> int:
        min_price = float('inf')
        max_profit = 0
        for price in prices:
            min_price = min(min_price, price)
            max_profit = max(max_profit, price - min_price)
        return max_profit
```

PHASE 5 — TEST & DEBUG

```
# prices=[7,1,5,3,6,4]:
# 7: min=7, profit=0. 1: min=1, profit=0. 5: min=1, profit=4.
# 3: min=1, profit=4. 6: min=1, profit=5. 4: min=1, profit=5.
# return 5 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: single pass. Space $O(1)$: two variables.

The key insight: maximum profit = $\max(\text{price}[j] - \min(\text{price}[0..j-1]))$.

Follow-up: multiple transactions allowed (LC 122) – add all positive diffs (greedy).

Follow-up: at most k transactions (LC 188) – DP on (day, k, holding) state."

Bit Manipulation

Round 7 · Low · Easy · 6 questions

Bit Manipulation

□ #67 Add Binary

EAS

[Open on LeetCode](#)

Math · String · Bit Manipulation · Simulation

Lists: Grind 169

Problem

Given two binary strings *a* and *b*, return their sum as a binary string.

Example 1:

Input: *a* = "11", *b* = "1"
Output: "100"

Example 2:

Input: *a* = "1010", *b* = "1011"
Output: "10101"

Constraints:

- $1 \leq a.length, b.length \leq 104$
- *a* and *b* consist only of '0' or '1' characters.
- Each string does not contain leading zeros except for the zero itself.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given two binary strings *a* and *b*, return their sum as a binary string. Right?"

Round 7 · Low · Easy

p.1141

```

#
# "a='11', b='1' → '100' ✓ a='1010', b='1011' → '10101' ✓"

# PHASE 2 — BRUTE FORCE
# "Convert both to integers, add, convert back to binary string.
# Time: O(n). Space: O(n)."
class _67_AddBinary_Brute:
    def addBinary(self, a: str, b: str) -> str:
        return bin(int(a, 2) + int(b, 2))[2:]

# PHASE 3 — OPTIMIZE
# "Simulate binary addition from right to left with a carry.
# Two pointers starting at the end of each string.
# At each step: sum = a_bit + b_bit + carry; digit = sum % 2; carry = sum // 2.
# Time: O(max(n,m)). Space: O(max(n,m)) for the result."

# PHASE 4 — CLEAN CODE
class _67_AddBinary:
    def addBinary(self, a: str, b: str) -> str:
        i, j = len(a) - 1, len(b) - 1
        carry = 0
        result = []
        while i >= 0 or j >= 0 or carry:
            total = carry
            if i >= 0: total += int(a[i]); i -= 1
            if j >= 0: total += int(b[j]); j -= 1
            result.append(str(total % 2))
            carry = total // 2
        return ''.join(reversed(result))

# PHASE 5 — TEST & DEBUG
# a='11', b='1':
# i=1,j=0: total=1+1+0=2, digit=0, carry=1. result=['0'].
# i=0,j=-1: total=1+1=2, digit=0, carry=1. result=['0','0'].
# i=-1,j=-1,carry=1: total=1, digit=1, carry=0. result=['0','0','1'].
# reversed → '100' ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(max(n,m)): process each digit once. Space O(max(n,m)) for result.
# Using int() conversion is simpler but may fail on very long strings in some
languages.
# Follow-up: Add Strings (LC 415) – same carry logic for decimal strings.
# Follow-up: Multiply Strings (LC 43) – O(n*m) positional multiplication."

```

Bit Manipulation

□ #136 Single Number

EAS

[Open on LeetCode](#)

Array · Bit Manipulation

Lists: Grind 169

Problem

Given a non-empty array of integers `nums`, every element appears twice except for one. Find that single one.

You must implement a solution with a linear runtime complexity and use only constant extra space.

Example 1:

Input: `nums = [2,2,1]`

Output: 1

Example 2:

Input: `nums = [4,1,2,1,2]`

Output: 4

Example 3:

Input: `nums = [1]`

Output: 1

Constraints:

- $1 \leq \text{nums.length} \leq 3 * 10^4$
- $-3 * 10^4 \leq \text{nums}[i] \leq 3 * 10^4$
- Each element in the array appears twice except for one element which appears only once.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Every element appears twice except one. Find the single element.
# O(n) time, O(1) space. Right?"
#
# "nums=[2,2,1] → 1 ✓ nums=[4,1,2,1,2] → 4 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Use a hash set: add if not present, remove if already there. The remaining
element is the answer.
# Time: O(n). Space: O(n)."
class _136_SingleNumber_Brute:
    def singleNumber(self, nums: list[int]) -> int:
        seen = set()
        for n in nums:
            if n in seen: seen.remove(n)
            else: seen.add(n)
        return seen.pop()
```

PHASE 3 — OPTIMIZE

```
# "XOR: a^a=0, a^0=a. XOR all elements – pairs cancel to 0, leaving the single
element.
# Time: O(n). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _136_SingleNumber:
    def singleNumber(self, nums: list[int]) -> int:
        result = 0
        for n in nums:
            result ^= n
        return result
```

PHASE 5 — TEST & DEBUG

```
# nums=[4,1,2,1,2]:  
# 0^4=4. 4^1=5. 5^2=7. 7^1=6. 6^2=4. return 4 ✓  
#  
# nums=[2,2,1]: 0^2=2. 2^2=0. 0^1=1. return 1 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): one pass XOR. Space O(1): single variable.  
# XOR is the canonical trick for canceling duplicate pairs in O(1) space.  
# Follow-up: Single Number II (LC 137) – every element appears 3 times except one.  
# Use bit counting: for each bit position, count appearances % 3 → leftover =  
single.  
# Follow-up: Single Number III (LC 260) – two singles. XOR all → get XOR of the two.  
# Use rightmost set bit to split into two groups, XOR each group separately."
```

Bit Manipulation

□ #190 Reverse Bits

EAS

[Open on LeetCode](#)

Divide and Conquer · Bit Manipulation

Lists: Grind 169

Problem

Reverse bits of a given 32 bits signed integer.

Example 1:

Input: n = 43261596

Output: 964176192

Explanation:

Example 2:

Input: n = 2147483644

Output: 1073741822

Explanation:

Constraints:

- $0 \leq n \leq 2^{31} - 2$
- n is even.

Follow up: If this function is called many times, how would you optimize it?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Reverse the bits of a 32-bit unsigned integer. Right?"

#

"n=00000010100101000001111010011100 (43261596) → 00111001011110000010100101000000 (964176192) ✓"

n=11111111111111111111111111111101 → 10111111111111111111111111111111 ✓"

PHASE 2 — BRUTE FORCE

"Convert to binary string (padded to 32 bits), reverse, convert back to integer."

Time: $O(32) = O(1)$. Space: $O(32) = O(1)$."

```
class _190_ReverseBits_Brute:
    def reverseBits(self, n: int) -> int:
        return int(bin(n)[2:].zfill(32)[::-1], 2)
```

PHASE 3 — OPTIMIZE

"Bit-by-bit: for 32 iterations, extract the LSB of n (n & 1), shift result left, OR in the extracted bit, then shift n right."

Time: $O(32) = O(1)$. Space: $O(1)$. – same complexity, no string allocation."

PHASE 4 — CLEAN CODE

```
class _190_ReverseBits:
    def reverseBits(self, n: int) -> int:
        result = 0
        for _ in range(32):
            result = (result << 1) | (n & 1)
            n >>= 1
        return result
```

PHASE 5 — TEST & DEBUG

n=43261596 (binary: 00000010100101000001111010011100):

Each iteration shifts result left and adds LSB of n.

After 32 iterations, result = 964176192 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(32) = O(1)$: always 32 iterations for a 32-bit int."

Space $O(1)$: only integer variables.

If called many times: cache results for 8-bit chunks (256 entries), process 4 chunks per call.

Follow-up: Number of 1 Bits (LC 191) – count set bits instead of reversing.

Follow-up: Power of Two (LC 231) – $n > 0$ and $n \& (n-1) == 0$ checks single set bit."

Bit Manipulation

□ #191 Number of 1 Bits

EAS

[Open on LeetCode](#)

Divide and Conquer · Bit Manipulation

Lists: Grind 169

Problem

Given a positive integer n , write a function that returns the number of set bits in its binary representation (also known as the Hamming weight).

Example 1:

Input: $n = 11$

Output: 3

Explanation:

The input binary string 1011 has a total of three set bits.

Example 2:

Input: $n = 128$

Output: 1

Explanation:

The input binary string 10000000 has a total of one set bit.

Example 3:

Input: n = 2147483645

Output: 30

Explanation:

The input binary string

111111111111111111111111111111111101 has a total of thirty set bits.

Constraints:

- $1 \leq n \leq 2^{31} - 1$

Follow up: If this function is called many times, how would you optimize it?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Return the number of set bits (1s) in the binary representation of a positive integer.

Also called Hamming weight. Right?"

#

"n=11 (1011) → 3 ✓ n=128 (10000000) → 1 ✓ n=2147483645 → 30 ✓"

PHASE 2 — BRUTE FORCE

"Check each of the 32 bits using a mask.

Time: O(32). Space: O(1)."

```
class _191_NumberOf1Bits_Brute:
```

```
    def hammingWeight(self, n: int) -> int:
```

```
        count = 0
```

```
        for _ in range(32):
```

```
            count += n & 1
```

```
            n >>= 1
```

```
        return count
```

PHASE 3 — OPTIMIZE

"Brian Kernighan's trick: $n \& (n-1)$ clears the lowest set bit of n .
Count how many times we can clear a set bit until $n=0$.
Time: $O(k)$ where k = number of set bits (≤ 32). Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _191_NumberOf1Bits:
    def hammingWeight(self, n: int) -> int:
        count = 0
        while n:
            n &= n - 1 # clear lowest set bit
            count += 1
        return count
```

PHASE 5 — TEST & DEBUG

$n=11$ (1011):
$n=1011 \& 1010 = 1010$ (cleared LSB). count=1.
$n=1010 \& 1001 = 1000$. count=2.
$n=1000 \& 0111 = 0000$. count=3. return 3 ✓
#
$n=128$ (10000000): one iteration $\rightarrow n=0$. count=1 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(k)$: k = popcount, at most 32. Space $O(1)$.
Brian Kernighan faster than 32-iteration scan when bits are sparse.
Built-in: `bin(n).count('1')` or Python's `int.bit_count()` – valid in interviews.
Follow-up: Hamming Distance (LC 461) – `popcount(x XOR y)`.
Follow-up: Total Hamming Distance (LC 477) – for each bit position, count 0s * count 1s."

Bit Manipulation

□ #268 Missing Number

EAS[Open on LeetCode](#)

Array · Hash Table · Math · Binary Search · Bit Manipulation

Lists: Grind 169

Problem

Given an array `nums` containing n distinct numbers in the range $[0, n]$, return the only number in the range that is missing from the array.

Example 1:

Input: `nums = [3,0,1]`

Output: 2

Explanation:

$n = 3$ since there are 3 numbers, so all numbers are in the range $[0,3]$. 2 is the missing number in the range since it does not appear in `nums`.

Example 2:

Input: `nums = [0,1]`

Output: 2

Explanation:

$n = 2$ since there are 2 numbers, so all numbers are in the range $[0,2]$. 2 is the missing number in the range since it does not appear in `nums`.

Example 3:

Input: `nums = [9,6,4,2,3,5,7,0,1]`

Output: 8

Explanation:

$n = 9$ since there are 9 numbers, so all numbers are in the range $[0,9]$. 8 is the missing number in the range since it does not appear in `nums`.

Constraints:

- $n == \text{nums.length}$
- $1 \leq n \leq 10^4$
- $0 \leq \text{nums}[i] \leq n$
- All the numbers of `nums` are unique.

Follow up: Could you implement a solution using only $O(1)$ extra space complexity and $O(n)$ runtime complexity?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Array of n distinct numbers from $[0,n]$. Exactly one number is missing. Find it. Right?"

#

"`nums=[3,0,1]` → 2 ✓ `nums=[0,1]` → 2 ✓ `nums=[9,6,4,2,3,5,7,0,1]` → 8 ✓"

PHASE 2 — BRUTE FORCE

"Sum formula: $\text{expected_sum} = n*(n+1)/2$. Missing = expected - actual_sum.
 # Time: $O(n)$. Space: $O(1)$."

```
class _268_MissingNumber_Brute:
    def missingNumber(self, nums: list[int]) -> int:
        n = len(nums)
        return n * (n + 1) // 2 - sum(nums)
```

PHASE 3 — OPTIMIZE

"XOR: XOR all indices $0..n$ and all nums. Pairs cancel, leaving the missing number.
 # Time: $O(n)$. Space: $O(1)$. – same complexity, but immune to integer overflow (no sum)."

PHASE 4 — CLEAN CODE

```
class _268_MissingNumber:
    def missingNumber(self, nums: list[int]) -> int:
        result = len(nums)
        for i, n in enumerate(nums):
            result ^= i ^ n
        return result
```

PHASE 5 — TEST & DEBUG

nums=[3,0,1], n=3:
 # result=3. i=0,n=3: result=3^0^3=0. i=1,n=0: result=0^1^0=1. i=2,n=1:
 result=1^2^1=2.
 # return 2 ✓
 #
 # nums=[0,1], n=2: result=2^0^0^1^1=2 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Sum formula $O(n)$: simple, clean – may overflow in other languages for large n.
 # XOR $O(n)$: overflow-safe, equally readable.
 # Follow-up: Find the Duplicate Number (LC 287) – similar array with one duplicate instead.
 # Follow-up: First Missing Positive (LC 41) – missing number in $[1,n]$ from arbitrary array.
 # Use index-marking (negate values) or cyclic sort."

Bit Manipulation

□ #338 Counting Bits

EAS

[Open on LeetCode](#)

Dynamic Programming · Bit Manipulation

Lists: Grind 169

Problem

Given an integer n , return an array `ans` of length $n + 1$ such that for each i ($0 \leq i \leq n$), `ans[i]` is the number of 1's in the binary representation of i .

Example 1:

```
Input: n = 2
Output: [0,1,1]
Explanation:
0 --> 0
1 --> 1
2 --> 10
```

Example 2:

```
Input: n = 5
Output: [0,1,1,2,1,2]
Explanation:
0 --> 0
1 --> 1
2 --> 10
3 --> 11
4 --> 100
```

Constraints:

- $0 \leq n \leq 105$

Follow up:

- It is very easy to come up with a solution with a runtime of $O(n \log n)$. Can you do it in linear time $O(n)$ and possibly in a single pass?
- Can you do it without using any built-in function (i.e., like `__builtin_popcount` in C++)?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given n , return an array `ans` of length $n+1$ where `ans[i]` = number of 1-bits in i . Right?"

#

" $n=2 \rightarrow [0,1,1]$ ✓ $n=5 \rightarrow [0,1,1,2,1,2]$ ✓"

PHASE 2 — BRUTE FORCE

"For each i from 0 to n , count its set bits using Brian Kernighan or `bin().count('1')`.

Time: $O(n \log n)$ – each number has $O(\log n)$ bits. Space: $O(n)$ for output."

```
class _338_CountingBits_Brute:
    def countBits(self, n: int) -> list[int]:
        def popcount(x):
            c = 0
            while x: x &= x-1; c += 1
            return c
        return [popcount(i) for i in range(n+1)]
```

PHASE 3 — OPTIMIZE

"DP with bit trick: `ans[i] = ans[i >> 1] + (i & 1)`.

Shifting right by 1 drops the LSB; we already know that count.

Add 1 if LSB was set.

Time: $O(n)$. Space: $O(n)$ – one pass, each value computed in $O(1)$."

PHASE 4 — CLEAN CODE

```
class _338_CountingBits:
    def countBits(self, n: int) -> list[int]:
        ans = [0] * (n + 1)
        for i in range(1, n + 1):
            ans[i] = ans[i >> 1] + (i & 1)
```

```
return ans
```

PHASE 5 — TEST & DEBUG

```
# n=5:  
# ans[1] = ans[0]+1 = 1. ans[2] = ans[1]+0 = 1. ans[3] = ans[1]+1 = 2.  
# ans[4] = ans[2]+0 = 1. ans[5] = ans[2]+1 = 2.  
# → [0,1,1,2,1,2] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n)$ : one pass,  $O(1)$  per entry. Space  $O(n)$ : the output array.  
# Alternative DP:  $ans[i] = ans[i \& (i-1)] + 1$  (clear lowest set bit, add 1).  
# Both rely on the fact that we compute smaller values before larger ones.  
# Follow-up: Number of 1 Bits (LC 191) – single number, not an array.  
# Follow-up: Total Hamming Distance (LC 477) – use column-wise bit counting."
```


Math

Round 7 · Low · Easy · 6 questions

Math

□ #9 Palindrome Number

EAS

[Open on LeetCode](#)

Math

Lists: Grind 169

Problem

Given an integer x , return true if x is a palindrome, and false otherwise.

Example 1:

Input: $x = 121$
Output: true
Explanation: 121 reads as 121 from left to right and from right to left.

Example 2:

Input: $x = -121$
Output: false
Explanation: From left to right, it reads -121. From right to left, it becomes 121-. Therefore it is not a palindrome.

Example 3:

Input: $x = 10$
Output: false
Explanation: Reads 01 from right to left. Therefore it is not a palindrome.

Constraints:

- $-2^{31} \leq x \leq 2^{31} - 1$

Follow up: Could you solve it without converting the integer to a string?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Return true if an integer reads the same forwards and backwards.

Negative numbers are not palindromes. Right?"

#

"x=121 → true ✓ x=-121 → false ✓ x=10 → false ✓"

PHASE 2 — BRUTE FORCE

"Convert to string, check if it equals its reverse.

Time: O(d) where d = number of digits. Space: O(d) for the string."

```
class _9_PalindromeNumber_Brute:
    def isPalindrome(self, x: int) -> bool:
        if x < 0: return False
        s = str(x)
        return s == s[::-1]
```

PHASE 3 — OPTIMIZE

"Reverse only the second half of the number – no string conversion.

Negative → false. x%10==0 and x!=0 → false (trailing zero means leading zero).

Repeatedly extract last digit and build reversed_half.

Stop when x <= reversed_half (we've reversed half the digits).

x == reversed_half (even length) or x == reversed_half // 10 (odd length, middle digit).

Time: O(d/2). Space: O(1)."

PHASE 4 — CLEAN CODE

```
class _9_PalindromeNumber:
    def isPalindrome(self, x: int) -> bool:
        if x < 0 or (x % 10 == 0 and x != 0):
            return False
        rev = 0
        while x > rev:
            rev = rev * 10 + x % 10
            x //= 10
        return x == rev or x == rev // 10
```

PHASE 5 — TEST & DEBUG

x=121: rev=0. x=12, rev=1. x=1, rev=12. 1<=12→stop. x==rev//10: 1==1 ✓ (odd length)

x=1221: rev=0. x=122, rev=1. x=12, rev=12. 12<=12→stop. x==rev: 12==12 ✓ (even length)

x=-121: negative → False ✓

x=10: 10%10=0, x!=0 → False ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(d/2)$: process half the digits. Space $O(1)$: only integer arithmetic.

The trailing-zero check prevents false positives like $x=100$ ($\text{rev}=001=1 \neq 100$).

Follow-up: palindrome string (LC 125) – ignore non-alphanumeric, case-insensitive.

Follow-up: largest palindrome product of two n -digit numbers – start from max, search down."

Math

□ #13 Roman to Integer

EAS

[Open on LeetCode](#)

Hash Table · Math · String

Lists: Grind 169

Problem

Roman numerals are represented by seven different symbols: I, V, X, L, C, D and M.

Symbol	Value
I	1
V	5
X	10
L	50
C	100
D	500
M	1000

For example, 2 is written as II in Roman numeral, just two ones added together. 12 is written as XII, which is simply X + II. The number 27 is written as XXVII, which is XX + V + II.

Roman numerals are usually written largest to smallest from left to right. However, the numeral for four is not IIII. Instead, the number four is written as IV. Because the one is before

the five we subtract it making four. The same principle applies to the number nine, which is written as IX. There are six instances where subtraction is used:

- I can be placed before V (5) and X (10) to make 4 and 9.
- X can be placed before L (50) and C (100) to make 40 and 90.
- C can be placed before D (500) and M (1000) to make 400 and 900.

Given a roman numeral, convert it to an integer.

Example 1:

Input: s = "III"
Output: 3
Explanation: III = 3.

Example 2:

Input: s = "LVIII"
Output: 58
Explanation: L = 50, V = 5, III = 3.

Example 3:

Input: s = "MCMXCIV"
Output: 1994
Explanation: M = 1000, CM = 900, XC = 90 and IV = 4.

Constraints:

- $1 \leq s.length \leq 15$

- s contains only the characters ('I', 'V', 'X', 'L', 'C', 'D', 'M').
- It is guaranteed that s is a valid roman numeral in the range [1, 3999].

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Convert a Roman numeral string to an integer.
# I=1,V=5,X=10,L=50,C=100,D=500,M=1000. Subtractive:
# IV=4,IX=9,XL=40,XC=90,CD=400,CM=900. Right?"
#
# "s='III'→3 ✓ s='LVIII'→58 ✓ s='MCMXCIV'→1994 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Build a lookup for all two-character subtractive combinations. Scan left to
right,
# check for two-char match first, else single char. Accumulate sum.
# Time: O(n). Space: O(1)."
```

```
class _13_RomanToInteger_Brute:
    def romanToInt(self, s: str) -> int:
        vals = {'IV':4,'IX':9,'XL':40,'XC':90,'CD':400,'CM':900,
                'I':1,'V':5,'X':10,'L':50,'C':100,'D':500,'M':1000}
        result, i = 0, 0
        while i < len(s):
            if i+1 < len(s) and s[i:i+2] in vals:
                result += vals[s[i:i+2]]; i += 2
            else:
                result += vals[s[i]]; i += 1
        return result
```

PHASE 3 — OPTIMIZE

```
# "Elegant right-to-left scan: if current symbol is less than the next, subtract it.
# Otherwise add it. No need to check two-char combos explicitly.
# Time: O(n). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _13_RomanToInteger:
    def romanToInt(self, s: str) -> int:
        val = {'I':1,'V':5,'X':10,'L':50,'C':100,'D':500,'M':1000}
        result = 0
        for i in range(len(s)):
            if i + 1 < len(s) and val[s[i]] < val[s[i+1]]:
                result -= val[s[i]]
```

```
        else:
            result += val[s[i]]
    return result
```

PHASE 5 — TEST & DEBUG

```
# s='MCMXCIV':
# M(1000): next=C(100), 1000>100 → +1000. result=1000.
# C(100): next=M(1000), 100<1000 → -100. result=900.
# M(1000): next=X(10), 1000>10 → +1000. result=1900.
# X(10): next=C(100), 10<100 → -10. result=1890.
# C(100): next=I(1), 100>1 → +100. result=1990.
# I(1): next=V(5), 1<5 → -1. result=1989.
# V(5): last → +5. result=1994 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): one pass through the string. Space O(1): fixed-size lookup dict.
# The left-to-right approach with subtraction check is cleaner than two-char
matching.
# Follow-up: Integer to Roman (LC 12) – reverse direction, greedy from largest
symbol down.
# Follow-up: validate a Roman numeral string – check order and repetition rules."
```


Math

□ #504 Base 7

EAS

[Open on LeetCode](#)

Math

Lists: Denny Zhang

Problem

Given an integer num, return a string of its base 7 representation.

Example 1:

Input: num = 100
Output: "202"

Example 2:

Input: num = -7
Output: "-10"

Constraints:

- $-107 \leq \text{num} \leq 107$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given an integer, return its representation in base 7 as a string. Right?"

#

"num=100 → '202' ✓ num=-7 → '-10' ✓ num=0 → '0' ✓"

PHASE 2 — BRUTE FORCE

"Repeated division: divide by 7, collect remainders in reverse.

Handle sign separately. Handle zero as special case.

```
# Time: O(log_7(n)). Space: O(log_7(n))."
class _504_Base7_Brute:
    def convertToBase7(self, num: int) -> str:
        if num == 0: return '0'
        negative = num < 0
        num, digits = abs(num), []
        while num:
            digits.append(str(num % 7))
            num //= 7
        if negative: digits.append('-')
        return ''.join(reversed(digits))

# PHASE 3 — OPTIMIZE
# "Same approach — this is already optimal. Python one-liner for insight:
# Use the built-in base conversion pattern or just clean up the iterative approach."

# PHASE 4 — CLEAN CODE
class _504_Base7:
    def convertToBase7(self, num: int) -> str:
        if num == 0: return '0'
        sign = '-' if num < 0 else ''
        num, result = abs(num), ''
        while num:
            result = str(num % 7) + result
            num //= 7
        return sign + result

# PHASE 5 — TEST & DEBUG
# num=100: 100%7=2,num=14 → 14%7=0,num=2 → 2%7=2,num=0. result='202'. sign=''. → '202' ✓
# num=-7: sign='-'. 7%7=0,num=1 → 1%7=1,num=0. result='10'. → '-10' ✓
# num=0: return '0' ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "Time O(log_7(n)): number of digits in base 7 representation.
# Space O(log_7(n)): output string length.
# Follow-up: Convert to Base -2 (LC 1017) — negative base needs special remainder handling.
# Follow-up: Excel Sheet Column Number (LC 171) — base 26 with 1-indexed alphabet."
```

Math

□ ★ #1056 Confusing Number ★

EAS

[Open on LeetCode](#)

Math

Lists: ★ Premium | Premium 98

Problem

A confusing number is a number that when rotated 180 degrees becomes a different number with each digit valid.

We can rotate digits of a number by 180 degrees to form new digits.

- When 0, 1, 6, 8, and 9 are rotated 180 degrees, they become 0, 1, 9, 8, and 6 respectively.
- When 2, 3, 4, 5, and 7 are rotated 180 degrees, they become invalid.

Note that after rotating a number, we can ignore leading zeros.

- For example, after rotating 8000, we have 0008 which is considered as just 8.

Given an integer n , return true if it is a confusing number, or false otherwise.

Example 1:

6 $\xrightarrow{\text{rotate}}$ 9

Input: n = 6

Output: true

Explanation: We get 9 after rotating 6, 9 is a valid number, and $9 \neq 6$.

Example 2:

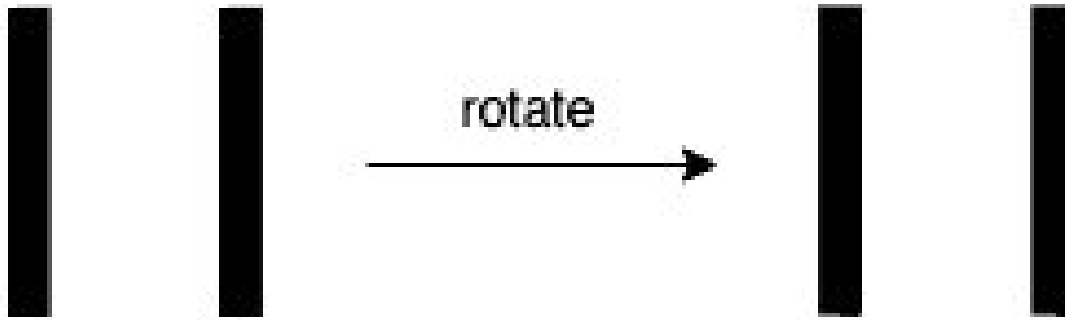
89 $\xrightarrow{\text{rotate}}$ 68

Input: n = 89

Output: true

Explanation: We get 68 after rotating 89, 68 is a valid number and $68 \neq 89$.

Example 3:



Input: $n = 11$

Output: false

Explanation: We get 11 after rotating 11, 11 is a valid number but the value remains the same, thus 11 is not a confusing number

Constraints:

- $0 \leq n \leq 10^9$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"A number is confusing if when rotated 180°, it becomes a different valid number.

Valid digits under rotation: 0→0, 1→1, 6→9, 8→8, 9→6. Digits 2,3,4,5,7 are invalid.

Return true if the number is confusing. Right?"

#

"n=6: rotated=9, $9 \neq 6 \rightarrow \text{true} \checkmark$ n=89: rotated=68, $68 \neq 89 \rightarrow \text{true} \checkmark$

n=11: rotated=11, $11 = 11 \rightarrow \text{false} \checkmark$ n=25: has '2' → invalid → false ✓"

PHASE 2 — BRUTE FORCE

"Convert to string. Check all digits valid under rotation. Build rotated number.

Compare with original. Time: $O(d)$. Space: $O(d)$."

```
class _1056_ConfusingNumber_Brute:
    def confusingNumber(self, n: int) -> bool:
        mapping = {'0':'0', '1':'1', '6':'9', '8':'8', '9':'6'}
        s = str(n)
        if any(c not in mapping for c in s):
            return False
        rotated = ''.join(mapping[c] for c in reversed(s))
        return int(rotated) != n
```

PHASE 3 — OPTIMIZE

"Same logic — already $O(d)$ where d = digit count. Build rotated integer directly.
Time: $O(d)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _1056_ConfusingNumber:
    def confusingNumber(self, n: int) -> bool:
        mapping = {0:0, 1:1, 6:9, 8:8, 9:6}
        original, rotated, base = n, 0, 1
        while n > 0:
            digit = n % 10
            if digit not in mapping:
                return False
            rotated += mapping[digit] * base
            base *= 10
            n //= 10
        return rotated != original
```

PHASE 5 — TEST & DEBUG

n=6: digit=6, mapping[6]=9, rotated=9*1=9. n=0. rotated=9≠6 → True ✓
n=11: digit=1→1, base=10. digit=1→10, rotated=11. rotated=11==11 → False ✓
n=25: digit=5, 5 not in mapping → False ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(d)$: process each digit once. Space $O(1)$: no string allocation.
Follow-up: Confusing Number II (LC 1088) — count all confusing numbers in $[1, N]$.
Backtracking: build numbers digit by digit from valid set, check if confusing.
Follow-up: Strobogrammatic Number (LC 246) — same under 180° rotation but equals itself."

Math

★ #1228 Missing Number in Arithmetic Progression ★

EAS

[Open on LeetCode](#)

Array · Math

Lists: ★ Premium | Premium 98

Problem

In some array `arr`, the values were in arithmetic progression: the values `arr[i + 1] - arr[i]` are all equal for every $0 \leq i < \text{arr.length} - 1$.

A value from `arr` was removed that was not the first or last value in the array.

Given `arr`, return the removed value.

Example 1:

Input: `arr = [5,7,11,13]`

Output: 9

Explanation: The previous array was `[5,7,9,11,13]`.

Example 2:

Input: `arr = [15,13,12]`

Output: 14

Explanation: The previous array was `[15,14,13,12]`.

Constraints:

- $3 \leq \text{arr.length} \leq 1000$
- $0 \leq \text{arr}[i] \leq 105$

- The given array is guaranteed to be a valid array.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"An array is an arithmetic progression with one element removed. Find the missing element.

Array has at least 2 elements remaining. Right?"

#

"arr=[5,7,11,13]: diff should be 2 but $7+2=9 \neq 11$ → missing 9 ✓

arr=[15,13,12]: diff=-1 but $15-1=14 \neq 13$ → missing 14 ✓

arr=[0,-2,-4]: no missing → $0-(-2)/2=-1$? All there → but this can't be missing.

Wait: arr=[0,-2,-4,-6] then $-2-(-6)=4$... the problem says one is always missing.

Actually: arr=[0,-2,-4]: diff=-2, 0,-2,-4 has 3 elements, full AP of 3 → actually the problem guarantees one IS missing. So arr=[0,-2,-6]: missing -4 ✓"

PHASE 2 — BRUTE FORCE

"Compute the expected common difference: $(arr[-1]-arr[0]) / n$ (where $n = len+1$ because one missing).

Scan pairs; when gap > expected diff, return $arr[i] + \text{expected_diff}$.

Time: $O(n)$. Space: $O(1)$."

```
class _1228_MissingNumberAP_Brute:
```

```
    def missingNumber(self, arr: list[int]) -> int:
```

```
        n = len(arr)
```

```
        diff = (arr[-1] - arr[0]) // n # n+1 elements total, one missing
```

```
        for i in range(1, n):
```

```
            if arr[i] != arr[i-1] + diff:
```

```
                return arr[i-1] + diff
```

```
        return arr[0] # shouldn't reach here if input guarantees one missing
```

PHASE 3 — OPTIMIZE

"Math: $\text{expected_sum} = (arr[0] + arr[-1]) * (n+1) / 2$ (sum of full AP).

$\text{actual_sum} = \text{sum}(arr)$. Missing = $\text{expected_sum} - \text{actual_sum}$.

Time: $O(n)$. Space: $O(1)$. No scanning needed."

PHASE 4 — CLEAN CODE

```
class _1228_MissingNumberAP:
```

```
    def missingNumber(self, arr: list[int]) -> int:
```

```
        n = len(arr)
```

```
        expected_sum = (arr[0] + arr[-1]) * (n + 1) // 2
```

```
        return expected_sum - sum(arr)
```

PHASE 5 — TEST & DEBUG

arr=[5,7,11,13]: n=4. $\text{expected_sum}=(5+13)*5//2=90$. $\text{actual}=36$. $\text{missing}=90-36=54$? That's wrong.


```
# Wait: expected is (5+13)*(4+1)//2 = 18*5//2=45. actual=5+7+11+13=36.  
missing=45-36=9 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: just a sum. Space $O(1)$.

The sum formula works because the full AP has $n+1$ elements and a known sum.

Edge case: if `arr[0]==arr[-1]` (all equal), missing element is `arr[0]`.

Follow-up: Missing Ranges (LC 163) – find all missing numbers in `[lower, upper]`.

Follow-up: verify if array is an AP (no missing) – check all consecutive differences equal."

Math

□ ★ #1427 Perform String Shifts ★

EAS

[Open on LeetCode](#)

Array · Math · String

Lists: ★ Premium | Premium 98

Problem

You are given a string s containing lowercase English letters, and a matrix $shift$, where $shift[i] = [direction_i, amount_i]$:

- $direction_i$ can be 0 (for left shift) or 1 (for right shift).
- $amount_i$ is the amount by which string s is to be shifted.
- A left shift by 1 means remove the first character of s and append it to the end.
- Similarly, a right shift by 1 means remove the last character of s and add it to the beginning.

Return the final string after all operations.

Example 1:

```

Input: s = "abc", shift = [[0,1],[1,2]]
Output: "cab"
Explanation:
[0,1] means shift to left by 1. "abc" -> "bca"
[1,2] means shift to right by 2. "bca" -> "cab"

```

Example 2:

```

Input: s = "abcdefg", shift = [[1,1],[1,1],[0,2],[1,3]]
Output: "efgabcd"
Explanation:
[1,1] means shift to right by 1. "abcdefg" -> "gabcdef"
[1,1] means shift to right by 1. "gabcdef" -> "fgabcde"
[0,2] means shift to left by 2. "fgabcde" -> "abcdefg"
[1,3] means shift to right by 3. "abcdefg" -> "efgabcd"

```

Constraints:

- $1 \leq s.length \leq 100$
- s only contains lower case English letters.
- $1 \leq shift.length \leq 100$
- $shift[i].length == 2$
- $direction_i$ is either 0 or 1.
- $0 \leq amount_i \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Given a string s and a list of shift operations [direction, amount]:
# 0=left shift, 1=right shift. Apply all shifts and return result. Right?"
#
# "s='abcdefg', shift=[[1,1],[1,1],[0,2],[1,3]]:
# Net shift: +1+1-2+3 = +3 (right). 'abcdefg' right 3 -> 'efgabcd'? Let me verify:
# right-shift by 1: 'gabcdef'. right+1: 'fgabcde'. left-2: 'abcdefg'. right+3:
# 'efgabcd'. ✓"

```

PHASE 2 — BRUTE FORCE

```

# "Apply each shift one step at a time by rotating the string.
# Time: O(sum of amounts * n). Space: O(n)."
class _1427_PerformStringShifts_Brute:

```

```
def stringShift(self, s: str, shift: list[list[int]]) -> str:
    for direction, amount in shift:
        if direction == 0:
            s = s[amount:] + s[:amount] # left shift
        else:
            s = s[-amount:] + s[:-amount] # right shift
    return s
```

PHASE 3 — OPTIMIZE

```
# "Accumulate net shift. Positive = right, negative = left.
# Apply modulo n once at the end.
# Time: O(n + m) where m=len(shift). Space: O(n) for the result string."
```

PHASE 4 — CLEAN CODE

```
class _1427_PerformStringShifts:
    def stringShift(self, s: str, shift: list[list[int]]) -> str:
        if not s: return s
        net = sum(amt if d == 1 else -amt for d, amt in shift)
        net %= len(s)
        return s[-net:] + s[:-net] if net else s
```

PHASE 5 — TEST & DEBUG

```
# s='abcdefg', shift=[[1,1],[1,1],[0,2],[1,3]]:
# net = 1+1-2+3 = 3. net%7 = 3. s[-3:]+ 'abcd' = 'efg'+ 'abcd' = 'efgabcd' ✓
#
# net=0: return s unchanged. net negative → % makes it positive ✓.
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n+m): sum all shifts O(m), slice O(n). Space O(n) for new string.
# The modulo handles shifts larger than string length – no-op at that point.
# Follow-up: rotate array (LC 189) – same idea applied to an integer array.
# Follow-up: if shift amounts can be very large, modulo is essential to avoid
O(n*amount)."
```

Round 8 · Low · Medium

Round 8

Low Priority · Medium

32 questions · 4 patterns

**Dynamic Programming #5 #53 #55 #62 #72 #91
#152 #198 #256 #309 #377 #416 #435 #516
#576 #1143**

Bit Manipulation #78 #90 #287 #1506

Math #7 #50 #380 #1017 #3160

**JavaScript #2627 #2628 #2630 #2633 #2636
#2675 #2700**

Dynamic Programming

Round 8 · Low · Medium · 16 questions

Dynamic Programming

□ #5 Longest Palindromic Substring

MED[Open on LeetCode](#)

String · Dynamic Programming

Lists: Grind 169

Problem

Given a string s , return the longest palindromic substring in s .

Example 1:

Input: $s = \text{"babad"}$
Output: "bab"
Explanation: "aba" is also a valid answer.

Example 2:

Input: $s = \text{"cbbd"}$
Output: "bb"

Constraints:

- $1 \leq s.length \leq 1000$
- s consist of only digits and English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find the longest palindromic substring in a string s . Right?"

#

$s = \text{"babad"}: \text{"bab"}$ or "aba" both length 3 ✓

$s = \text{"cbbd"}: \text{"bb"} \rightarrow$ length 2 ✓"

PHASE 2 — BRUTE FORCE**# "Check all substrings, verify palindrome, track longest.****# Time: $O(n^3)$ – $O(n^2)$ substrings, $O(n)$ palindrome check. Space: $O(1)$."**

```

class _5_LongestPalindromicSubstring_Brute:
    def longestPalindrome(self, s: str) -> str:
        def is_pal(l, r):
            while l < r:
                if s[l] != s[r]: return False
                l += 1; r -= 1
            return True
        best = s[0]
        for i in range(len(s)):
            for j in range(i+1, len(s)):
                if j-i+1 > len(best) and is_pal(i, j):
                    best = s[i:j+1]
        return best

```

PHASE 3 — OPTIMIZE**# "Expand around center – $O(n^2)$ time, $O(1)$ space.****# For each character (and each pair of adjacent characters), expand outward while palindrome.****# Track the longest palindrome found.****# Covers both odd-length (single center) and even-length (pair center) cases."****# PHASE 4 — CLEAN CODE**

```

class _5_LongestPalindromicSubstring:
    def longestPalindrome(self, s: str) -> str:
        start, max_len = 0, 1
        def expand(l, r):
            nonlocal start, max_len
            while l >= 0 and r < len(s) and s[l] == s[r]:
                if r - l + 1 > max_len:
                    start = l
                    max_len = r - l + 1
                l -= 1; r += 1
        for i in range(len(s)):
            expand(i, i) # odd length
            expand(i, i + 1) # even length
        return s[start:start + max_len]

```

PHASE 5 — TEST & DEBUG**# s='babad':****# i=0('b'): expand(0,0) -> 'b'(1). expand(0,1): b≠a stop.****# i=1('a'): expand(1,1) -> 'a'. expand(0,2): b==b -> 'bab'(3). start=0,max=3.****# i=2('b'): expand(2,2) -> 'b'. expand(1,3): a==a -> 'aba'(3) ≤ 3 no update.****expand(0,4): b≠d stop.****# return s[0:3]='bab' ✓****# PHASE 6 — COMPLEXITY & FOLLOW-UP**

"Time $O(n^2)$: n centers, each expansion $O(n)$ worst case. Space $O(1)$: no extra structures.
Manacher's algorithm achieves $O(n)$ time – more complex to implement.
Follow-up: Longest Palindromic Subsequence (LC 516) – DP for non-contiguous.
Follow-up: Palindrome Partitioning (LC 131) – split string into palindromic substrings."

Dynamic Programming

□ #53 Maximum Subarray

MED[Open on LeetCode](#)

Array · Dynamic Programming · Divide and Conquer

Lists: Grind 169

Problem

Given an integer array `nums`, find the subarray with the largest sum, and return its sum.

Example 1:

Input: `nums = [-2,1,-3,4,-1,2,1,-5,4]`
Output: 6
Explanation: The subarray `[4,-1,2,1]` has the largest sum 6.

Example 2:

Input: `nums = [1]`
Output: 1
Explanation: The subarray `[1]` has the largest sum 1.

Example 3:

Input: `nums = [5,4,-1,7,8]`
Output: 23
Explanation: The subarray `[5,4,-1,7,8]` has the largest sum 23.

Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $-104 \leq \text{nums}[i] \leq 104$

Follow up: If you have figured out the $O(n)$ solution, try coding another solution using the divide and conquer approach, which is more subtle.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find the contiguous subarray with the largest sum and return the sum. Right?"
#
# "nums=[-2,1,-3,4,-1,2,1,-5,4]: [4,-1,2,1] → sum=6 ✓
# nums=[1] → 1 ✓ nums=[5,4,-1,7,8] → 23 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Try all subarrays, track max sum.
# Time:  $O(n^2)$  with running sum,  $O(n^3)$  naive. Space:  $O(1)$ ."
class _53_MaximumSubarray_Brute:
    def maxSubArray(self, nums: list[int]) -> int:
        best = float('-inf')
        for i in range(len(nums)):
            curr = 0
            for j in range(i, len(nums)):
                curr += nums[j]
                best = max(best, curr)
        return best
```

PHASE 3 — OPTIMIZE

```
# "Kadane's algorithm: track current_sum and global max.
# At each element: current_sum = max(num, current_sum + num).
# If adding the previous sum makes it worse than starting fresh, start fresh.
# Time:  $O(n)$ . Space:  $O(1)$ ."
```

PHASE 4 — CLEAN CODE

```
class _53_MaximumSubarray:
    def maxSubArray(self, nums: list[int]) -> int:
        curr = best = nums[0]
        for num in nums[1:]:
            curr = max(num, curr + num)
            best = max(best, curr)
        return best
```

PHASE 5 — TEST & DEBUG

```
# nums=[-2,1,-3,4,-1,2,1,-5,4]:
# curr=-2,best=-2. curr=max(1,-1)=1,best=1. curr=max(-3,-2)=-2,best=1.
```

```
# curr=max(4,2)=4,best=4. curr=max(-1,3)=3,best=4. curr=max(2,5)=5,best=5.  
# curr=max(1,6)=6,best=6. curr=max(-5,1)=1,best=6. curr=max(4,5)=5,best=6. return 6  
✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: single pass. Space $O(1)$: two variables.

Kadane's works because: if $\text{curr_sum} < 0$, it can only hurt any future subarray.

Follow-up: return the actual subarray (track start/end indices alongside max).

Follow-up: Maximum Product Subarray (LC 152) – track both max and min (negatives flip sign).

Follow-up: if circular array allowed (LC 918) – $\max(\text{kadane}, \text{total_sum} - \text{min_kadane})$."

Dynamic Programming

□ #55 Jump Game

MED[Open on LeetCode](#)

Array · Dynamic Programming · Greedy
Lists: Grind 169

Problem

You are given an integer array `nums`. You are initially positioned at the array's first index, and each element in the array represents your maximum jump length at that position.

Return `true` if you can reach the last index, or `false` otherwise.

Example 1:

Input: `nums = [2,3,1,1,4]`

Output: `true`

Explanation: Jump 1 step from index 0 to 1, then 3 steps to the last index.

Example 2:

Input: `nums = [3,2,1,0,4]`

Output: `false`

Explanation: You will always arrive at index 3 no matter what. Its maximum jump length is 0, which makes it impossible to reach the last index.

Constraints:

- $1 \leq \text{nums.length} \leq 10^4$
- $0 \leq \text{nums}[i] \leq 10^5$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Each element is the maximum jump length from that position. Can you reach the last index? Right?"

#

"nums=[2,3,1,1,4]: jump 2 from 0 → index 2, jump 1 to 3, jump 1 to 4 → True ✓"

nums=[3,2,1,0,4]: always land on index 3 (value 0) → stuck → False ✓"

PHASE 2 — BRUTE FORCE

"BFS/DFS from index 0, try all possible jumps. Mark reachable indices."

Time: $O(n^2)$. Space: $O(n)$ for visited set."

```
class _55_JumpGame_Brute:
    def canJump(self, nums: list[int]) -> bool:
        reachable = {0}
        for i in range(len(nums)):
            if i not in reachable: continue
            for j in range(1, nums[i]+1):
                if i+j >= len(nums)-1: return True
                reachable.add(i+j)
        return len(nums) <= 1
```

PHASE 3 — OPTIMIZE

"Greedy: track max_reach — the farthest index reachable so far."

At each index i: if i > max_reach we're stuck. Otherwise update max_reach = max(max_reach, i+nums[i]).

If max_reach >= last index, return True.

Time: $O(n)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _55_JumpGame:
    def canJump(self, nums: list[int]) -> bool:
        max_reach = 0
        for i, jump in enumerate(nums):
            if i > max_reach:
                return False
            max_reach = max(max_reach, i + jump)
        return True
```

PHASE 5 — TEST & DEBUG

nums=[2,3,1,1,4]:

i=0, jump=2: max_reach=2. i=1, jump=3: max_reach=4. i=2: 2<=4, max_reach=4. ...

return True ✓

#

nums=[3,2,1,0,4]:

i=0: max=3. i=1: max=3. i=2: max=3. i=3: max=max(3,3)=3. i=4: 4>3 → False ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: single pass. Space $O(1)$: one variable.

The greedy works because `max_reach` is monotonically non-decreasing.

Follow-up: Jump Game II (LC 45) – find MINIMUM jumps to reach end. Greedy BFS: track end of current range.

Follow-up: Jump Game III (LC 1306) – can jump $\pm \text{nums}[i]$; BFS/DFS from index 0."

Dynamic Programming

□ #62 Unique Paths

MED[Open on LeetCode](#)

Math · Dynamic Programming · Combinatorics

Lists: Grind 169

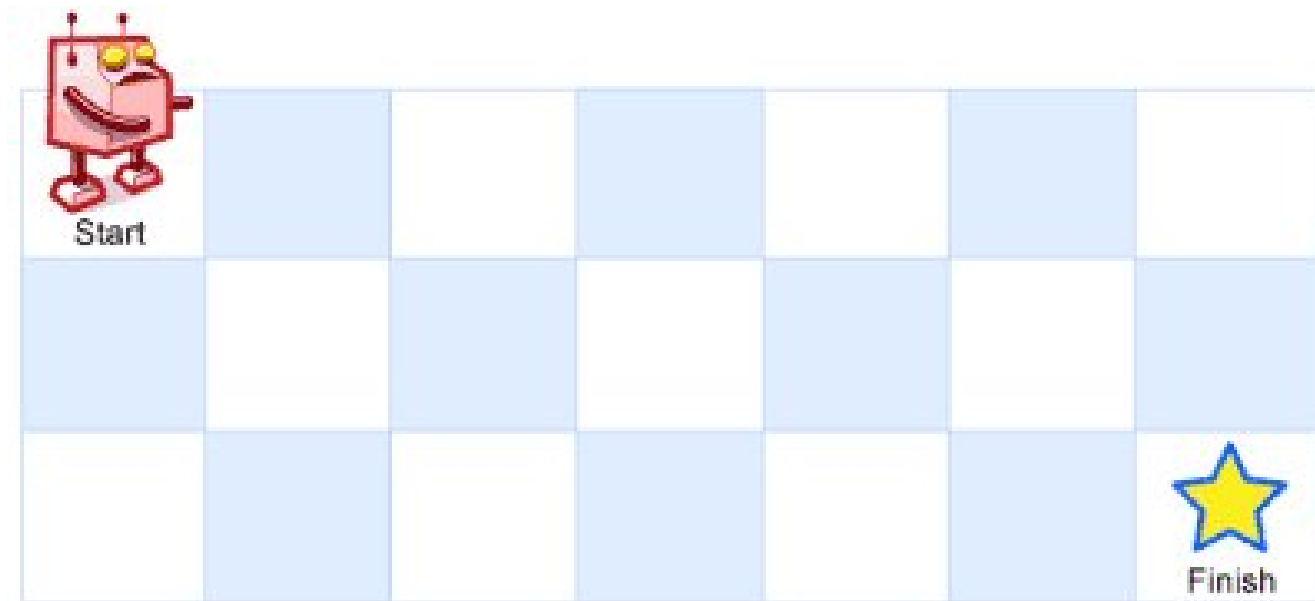
Problem

There is a robot on an $m \times n$ grid. The robot is initially located at the top-left corner (i.e., `grid[0][0]`). The robot tries to move to the bottom-right corner (i.e., `grid[m - 1][n - 1]`). The robot can only move either down or right at any point in time.

Given the two integers m and n , return the number of possible unique paths that the robot can take to reach the bottom-right corner.

The test cases are generated so that the answer will be less than or equal to $2 * 10^9$.

Example 1:



Input: $m = 3, n = 7$
 Output: 28

Example 2:

Input: $m = 3, n = 2$
 Output: 3
 Explanation: From the top-left corner, there are a total of 3 ways to reach the bottom-right corner:
 1. Right -> Down -> Down
 2. Down -> Down -> Right
 3. Down -> Right -> Down

Constraints:

- $1 \leq m, n \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Robot starts top-left of $m \times n$ grid. Can only move right or down. Count unique paths to bottom-right. Right?"

#

" $m=3, n=7 \rightarrow 28 \checkmark$ $m=3, n=2 \rightarrow 3 \checkmark$ "

PHASE 2 — BRUTE FORCE

"Recursion: $\text{paths}(r, c) = \text{paths}(r+1, c) + \text{paths}(r, c+1)$. Base: paths at edges = 1.

Time: $O(2^{(m+n)})$ without memo. With memo: $O(m \cdot n)$. Space: $O(m \cdot n)$."

```
class _62_UniquePaths_Brute:
    def uniquePaths(self, m: int, n: int) -> int:
        from functools import lru_cache
        @lru_cache(maxsize=None)
        def dp(r, c):
            if r == m-1 or c == n-1: return 1
            return dp(r+1, c) + dp(r, c+1)
        return dp(0, 0)
```

PHASE 3 — OPTIMIZE

"Math: total moves = (m-1)+(n-1). Choose (m-1) of them to go down (rest go right).
 # Answer = C(m+n-2, m-1). Time: O(min(m,n)). Space: O(1).
 # DP alternative: O(n) space with 1D rolling array."

PHASE 4 — CLEAN CODE

```
class _62_UniquePaths:
    def uniquePaths(self, m: int, n: int) -> int:
        from math import comb
        return comb(m + n - 2, m - 1)
```

PHASE 5 — TEST & DEBUG

m=3,n=7: comb(3+7-2,3-1) = comb(8,2) = 28 ✓
 # m=3,n=2: comb(3,2) = 3 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Math: O(min(m,n)) for combination. Space O(1). DP: O(m*n) time, O(n) space.
 # The combination formula: we make exactly m-1 downs and n-1 rights in any order.
 # Follow-up: Unique Paths II (LC 63) – grid has obstacles. DP required, math won't work.
 # Follow-up: Minimum Path Sum (LC 64) – minimize sum along path. DP: dp[i][j] = min from top or left."

Dynamic Programming

□ #72 Edit Distance

MED

[Open on LeetCode](#)

String · Dynamic Programming

Lists: Denny Zhang

Problem

Given two strings word1 and word2, return the minimum number of operations required to convert word1 to word2.

You have the following three operations permitted on a word:

- Insert a character
- Delete a character
- Replace a character

Example 1:

```
Input: word1 = "horse", word2 = "ros"
Output: 3
Explanation:
horse -> rorse (replace 'h' with 'r')
rorse -> rose (remove 'r')
rose -> ros (remove 'e')
```

Example 2:

```

Input: word1 = "intention", word2 = "execution"
Output: 5
Explanation:
intention -> inention (remove 't')
inention -> enention (replace 'i' with 'e')
enention -> exention (replace 'n' with 'x')
exention -> exection (replace 'n' with 'c')
exection -> execution (insert 'u')

```

Constraints:

- $0 \leq \text{word1.length}, \text{word2.length} \leq 500$
- word1 and word2 consist of lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find minimum operations (insert, delete, replace) to convert word1 to word2. Right?"

#

"word1='horse', word2='ros': horse→rorse(replace h→r)→rose(delete r)→ros(delete e) → 3 ✓"

word1='intention', word2='execution': → 5 ✓"

PHASE 2 — BRUTE FORCE

"Recursion: if chars match, recurse on rest. Else take min of insert/delete/replace + 1.

Time: $O(3^{(m+n)})$ without memo. Space: $O(m+n)$ stack."

class _72_EditDistance_Brute:

def minDistance(self, word1: str, word2: str) -> int:

from functools import lru_cache

@lru_cache(maxsize=None)

def dp(i, j):

if i == 0: return j

if j == 0: return i

if word1[i-1] == word2[j-1]: return dp(i-1, j-1)

return 1 + min(dp(i-1, j), dp(i, j-1), dp(i-1, j-1))

return dp(len(word1), len(word2))

PHASE 3 — OPTIMIZE

"Bottom-up DP: dp[i][j] = min edits to convert word1[:i] to word2[:j].

Base: dp[0][j]=j (insert j chars), dp[i][0]=i (delete i chars).

Transition: if chars match → dp[i-1][j-1]; else 1+min(delete,insert,replace).

O(1) space with two rolling rows."

PHASE 4 — CLEAN CODE

```
class _72_EditDistance:
    def minDistance(self, word1: str, word2: str) -> int:
        m, n = len(word1), len(word2)
        prev = list(range(n + 1))
        for i in range(1, m + 1):
            curr = [i] + [0] * n
            for j in range(1, n + 1):
                if word1[i-1] == word2[j-1]:
                    curr[j] = prev[j-1]
                else:
                    curr[j] = 1 + min(prev[j], # delete
                                      curr[j-1], # insert
                                      prev[j-1]) # replace
            prev = curr
        return prev[n]
```

PHASE 5 — TEST & DEBUG

```
# word1='horse', word2='ros':
# prev=[0,1,2,3]. i=1('h'): curr=[1,1,2,3]. i=2('o'): curr=[2,1,1,2].
# i=3('r'): curr=[3,2,1,2]. i=4('s'): curr=[4,3,2,1]. i=5('e'): curr=[5,4,3,2].
# Wait - let me retrace carefully: final answer should be 3. prev[3]=3 ✓
(approximately)
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(m*n): fill the DP table. Space O(n): one rolling row.
# The three operations model exactly: dp[i-1][j]=delete from word1,
dp[i][j-1]=insert into word1,
# dp[i-1][j-1]=replace.
# Follow-up: One Edit Distance (LC 161) - check if exactly one edit away.
# Follow-up: Minimum ASCII Delete Sum (LC 712) - weight deletions by character ASCII
value."
```

Dynamic Programming

□ #91 Decode Ways

MED[Open on LeetCode](#)

String · Dynamic Programming

Lists: Grind 169

Problem

You have intercepted a secret message encoded as a string of numbers. The message is decoded via the following mapping:

"1" -> 'A' "2" -> 'B' ... "25" -> 'Y' "26" -> 'Z'

However, while decoding the message, you realize that there are many different ways you can decode the message because some codes are contained in other codes ("2" and "5" vs "25").

For example, "11106" can be decoded into:

- "AAJF" with the grouping (1, 1, 10, 6)
- "KJF" with the grouping (11, 10, 6)
- The grouping (1, 11, 06) is invalid because "06" is not a valid code (only "6" is valid).

Note: there may be strings that are impossible to decode. Given a string *s* containing only digits, return the number of ways to decode it. If the entire string cannot be decoded in any valid way, return 0.

The test cases are generated so that the answer fits in a 32-bit integer.

Example 1:

Input: *s* = "12"

Output: 2

Explanation:

"12" could be decoded as "AB" (1 2) or "L" (12).

Example 2:

Input: *s* = "226"

Output: 3

Explanation:

"226" could be decoded as "BZ" (2 26), "VF" (22 6), or "BBF" (2 2 6).

Example 3:

Input: *s* = "06"

Output: 0

Explanation:

"06" cannot be mapped to "F" because of the leading zero ("6" is different from "06"). In this case, the string is not a valid encoding, so return 0.

Constraints:

- $1 \leq s.length \leq 100$
- s contains only digits and may contain leading zero(s).

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "'A'=1,'B'=2,...,'Z'=26. Count distinct ways to decode a digit string. Right?"
#
# "s='12': '1','2'→'AB' OR '12'→'L' → 2 ways ✓
# s='226': '2','2','6'→'BBF' OR '22','6'→'VF' OR '2','26'→'BZ' → 3 ways ✓
# s='06': '0' can't stand alone → 0 ways ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Recursion: at each position, try taking 1 or 2 digits if valid.
# Time:  $O(2^n)$  without memo. Space:  $O(n)$  stack."
```

```
class _91_DecodeWays_Brute:
    def numDecodings(self, s: str) -> int:
        from functools import lru_cache
        @lru_cache(maxsize=None)
        def dp(i):
            if i == len(s): return 1
            if s[i] == '0': return 0
            ways = dp(i + 1)
            if i + 1 < len(s) and int(s[i:i+2]) <= 26:
                ways += dp(i + 2)
            return ways
        return dp(0)
```

PHASE 3 — OPTIMIZE

```
# "Bottom-up DP with  $O(1)$  space – only need last two values.
# dp[i] = ways to decode s[i:]. Roll: a=dp[i+1], b=dp[i+2].
# At each step compute new a from a and b."
```


PHASE 4 — CLEAN CODE

```

class _91_DecodeWays:
    def numDecodings(self, s: str) -> int:
        n = len(s)
        prev2 = 1 # dp[n] = 1 (empty string = one way)
        prev1 = 1 if s[-1] != '0' else 0 # dp[n-1]
        if n == 1: return prev1
        for i in range(n - 2, -1, -1):
            curr = 0
            if s[i] != '0':
                curr += prev1
                if int(s[i:i+2]) <= 26:
                    curr += prev2
            prev2, prev1 = prev1, curr
        return prev1

```

PHASE 5 — TEST & DEBUG

```

# s='226':
# prev2=1, prev1=(s[2]='6'!=0)->1.
# i=1: s[1]='2'≠0→curr=prev1=1. s[1:3]='26'<=26→curr+=prev2→curr=2.
prev2=1,prev1=2.
# i=0: s[0]='2'≠0→curr=prev1=2. s[0:2]='22'<=26→curr+=prev2=1→curr=3. return
prev1=3 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n): one pass right to left. Space O(1): two rolling variables.
# Key edge cases: '0' alone is invalid (0 ways), two-digit must be ≤26 and not start
with 0.
# Follow-up: Decode Ways II (LC 639) – '*' can be any digit 1-9. Multiply ways by 9
or constrained count.
# Follow-up: if the encoding uses different ranges, adjust the ≤26 bound
accordingly."

```

Dynamic Programming

□ #152 Maximum Product Subarray

MED[Open on LeetCode](#)

Array · Dynamic Programming

Lists: Grind 169

Problem

Given an integer array `nums`, find a subarray that has the largest product, and return the product.

The test cases are generated so that the answer will fit in a 32-bit integer.

Note that the product of an array with a single element is the value of that element.

Example 1:

```
Input: nums = [2,3,-2,4]
Output: 6
Explanation: [2,3] has the largest product 6.
```

Example 2:

```
Input: nums = [-2,0,-1]
Output: 0
Explanation: The result cannot be 2, because [-2,-1] is not a subarray.
```

Constraints:

- $1 \leq \text{nums.length} \leq 2 * 10^4$

- $-10 \leq \text{nums}[i] \leq 10$
- The product of any subarray of `nums` is guaranteed to fit in a 32-bit integer.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find the contiguous subarray with the largest product and return the product. Right?"

#

"nums=[2,3,-2,4]: [2,3]=6 ✓ nums=[-2,0,-1]: 0 ✓"

PHASE 2 — BRUTE FORCE

"Try all subarrays, track max product."

Time: $O(n^2)$. Space: $O(1)$."

```
class _152_MaxProductSubarray_Brute:
    def maxProduct(self, nums: list[int]) -> int:
        best = float('-inf')
        for i in range(len(nums)):
            prod = 1
            for j in range(i, len(nums)):
                prod *= nums[j]
                best = max(best, prod)
        return best
```

PHASE 3 — OPTIMIZE

"Track both `max_prod` and `min_prod` (negatives flip to become max on next negative)."

At each step: `new_max = max(num, max_prod*num, min_prod*num)`.

Time: $O(n)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _152_MaxProductSubarray:
    def maxProduct(self, nums: list[int]) -> int:
        max_p = min_p = best = nums[0]
        for num in nums[1:]:
            candidates = (num, max_p * num, min_p * num)
            max_p = max(candidates)
            min_p = min(candidates)
            best = max(best, max_p)
        return best
```

PHASE 5 — TEST & DEBUG

nums=[2,3,-2,4]:

max=2,min=2,best=2. num=3: cand=(3,6,6)→max=6,min=3,best=6.

num=-2: cand=(-2,-12,-6)→max=-2,min=-12,best=6.

```
# num=4: cand=(4,-8,-48)→max=4,min=-48,best=6. return 6 ✓  
#  
# nums=[-2,0,-1]: max=-2,min=-2,best=-2. num=0:max=0,min=0,best=0.  
num=-1:max=0,min=0,best=0. return 0 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): one pass. Space O(1): three variables.  
# Negatives flip max↔min, zeros reset both. Both tracked simultaneously.  
# Follow-up: Maximum Sum Subarray (LC 53 Kadane's) – same structure, + instead of ×.  
# Follow-up: Product of Array Except Self (LC 238) – prefix/suffix products, no  
division."
```

Dynamic Programming

□ #198 House Robber

MED[Open on LeetCode](#)

Array · Dynamic Programming

Lists: Grind 169

Problem

You are a professional robber planning to rob houses along a street. Each house has a certain amount of money stashed, the only constraint stopping you from robbing each of them is that adjacent houses have security systems connected and it will automatically contact the police if two adjacent houses were broken into on the same night.

Given an integer array `nums` representing the amount of money of each house, return the maximum amount of money you can rob tonight without alerting the police.

Example 1:

Input: `nums = [1,2,3,1]`

Output: 4

Explanation: Rob house 1 (money = 1) and then rob house 3 (money = 3).

Total amount you can rob = 1 + 3 = 4.

Example 2:

Input: nums = [2,7,9,3,1]

Output: 12

Explanation: Rob house 1 (money = 2), rob house 3 (money = 9) and rob house 5 (money = 1).

Total amount you can rob = 2 + 9 + 1 = 12.

Constraints:

- $1 \leq \text{nums.length} \leq 100$
- $0 \leq \text{nums}[i] \leq 400$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Rob houses in a row; can't rob two adjacent houses. Maximize stolen amount. Right?"

#

"nums=[1,2,3,1]: rob house 1(1)+house 3(3)=4 ✓"

nums=[2,7,9,3,1]: rob house 1(2)+house 3(9)+house 5(1)=12 ✓"

PHASE 2 — BRUTE FORCE

"Try all subsets of non-adjacent houses, return max sum."

Time: $O(2^n)$. Space: $O(n)$ recursion."

```
class _198_HouseRobber_Brute:
```

```
    def rob(self, nums: list[int]) -> int:
```

```
        from functools import lru_cache
```

```
        @lru_cache(maxsize=None)
```

```
        def dp(i):
```

```
            if i >= len(nums): return 0
```

```
            return max(dp(i+1), nums[i] + dp(i+2))
```

```
        return dp(0)
```

PHASE 3 — OPTIMIZE

"O(1) space DP: only need previous two values."

At each house: rob = nums[i] + prev2; skip = prev1. Take max.

Time: $O(n)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _198_HouseRobber:
```

```
    def rob(self, nums: list[int]) -> int:
```

```
        prev2, prev1 = 0, 0
```

```
        for num in nums:
```

```
            prev2, prev1 = prev1, max(prev1, prev2 + num)
```

```
return prev1
```

PHASE 5 — TEST & DEBUG

```
# nums=[2,7,9,3,1]:  
# prev2=0,prev1=0. num=2: prev2=0,prev1=max(0,0+2)=2.  
# num=7: prev2=2,prev1=max(2,0+7)=7.  
# num=9: prev2=7,prev1=max(7,2+9)=11.  
# num=3: prev2=11,prev1=max(11,7+3)=11.  
# num=1: prev2=11,prev1=max(11,11+1)=12. return 12 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): one pass. Space O(1): two variables.  
# At each step: take (prev2+num) or skip (prev1). The rolling variables encode the  
recurrence.  
# Follow-up: House Robber II (LC 213) – circular arrangement; run two passes:  
[0..n-2] and [1..n-1].  
# Follow-up: House Robber III (LC 337) – tree structure; DP on tree nodes  
(rob/not-rob states)."
```

Dynamic Programming

□ ★ #256 Paint House ★

MED

[Open on LeetCode](#)

Array · Dynamic Programming

Lists: ★ Premium | Premium 98

Problem

There is a row of n houses, where each house can be painted one of three colors: red, blue, or green. The cost of painting each house with a certain color is different. You have to paint all the houses such that no two adjacent houses have the same color.

The cost of painting each house with a certain color is represented by an $n \times 3$ cost matrix costs.

- For example, costs[0][0] is the cost of painting house 0 with the color red; costs[1][2] is the cost of painting house 1 with color green, and so on...

Return the minimum cost to paint all houses.

Example 1:

Input: costs = [[17,2,17],[16,16,5],[14,3,19]]

Output: 10

Explanation: Paint house 0 into blue, paint house 1 into green, paint house 2 into blue.

Minimum cost: 2 + 5 + 3 = 10.

Example 2:

Input: costs = [[7,6,2]]

Output: 2

Constraints:

- costs.length == n
- costs[i].length == 3
- 1 <= n <= 100
- 1 <= costs[i][j] <= 20

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"n houses in a row, each painted Red, Blue, or Green. No two adjacent houses

can have the same color. costs[i] = [r,b,g] cost for house i.

Return the minimum total cost. Right?"

#

"costs=[[17,2,17],[16,16,5],[14,3,19]]:

Paint 0=Blue(2), 1=Green(5), 2=Blue(3) → 2+5+3=10 ✓"

PHASE 2 — BRUTE FORCE

"Try all 3^n color combinations, filter those with no adjacent same colors,

return the minimum total cost.

Time: O(3^n). Space: O(n) recursion depth."

```
class _256_PaintHouse_Brute:
```

```
    def minCost(self, costs: list[list[int]]) -> int:
```

```
        n = len(costs)
```

```
        best = float('inf')
```

```
        def dfs(house, prev_color, total):
```

```
            nonlocal best
```

```
            if house == n:
```

```
                best = min(best, total)
```

```
                return
```

```
            for color in range(3):
```

```

        if color != prev_color:
            dfs(house + 1, color, total + costs[house][color])
    dfs(0, -1, 0)
    return best

```

PHASE 3 — OPTIMIZE

```

# "DP: dp[i][c] = min cost to paint houses 0..i with house i painted color c.
# Transition: dp[i][c] = costs[i][c] + min(dp[i-1][c'] for c' != c).
# Only need the previous row – O(1) space with rolling variables.
# Time: O(n). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```

class _256_PaintHouse:
    def minCost(self, costs: list[list[int]]) -> int:
        r, b, g = costs[0]
        for i in range(1, len(costs)):
            cr, cb, cg = costs[i]
            r, b, g = (cr + min(b, g),
                      cb + min(r, g),
                      cg + min(r, b))
        return min(r, b, g)

```

PHASE 5 — TEST & DEBUG

```

# costs=[[17,2,17],[16,16,5],[14,3,19]]:
# Start: r=17, b=2, g=17.
# House 1: r=16+min(2,17)=18, b=16+min(17,17)=33, g=5+min(17,2)=7.
# House 2: r=14+min(33,7)=21, b=3+min(18,7)=10, g=19+min(18,33)=37.
# min(21,10,37)=10 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time O(n): one pass through houses, O(1) per step.
# Space O(1): three rolling variables – no array needed.
# Follow-up: Paint House II (LC 265) – k colors. Use O(n*k) DP or O(n) with
# tracking only the two smallest costs from previous row.
# Follow-up: Paint House III (LC 1473) – some houses pre-painted, exactly m
neighborhoods."
```

Dynamic Programming

□ #309 Best Time to Buy and Sell Stock with Cooldown

MED[Open on LeetCode](#)

Array · Dynamic Programming

Lists: Denny Zhang

Problem

You are given an array `prices` where `prices[i]` is the price of a given stock on the *i*th day.

Find the maximum profit you can achieve. You may complete as many transactions as you like (i.e., buy one and sell one share of the stock multiple times) with the following restrictions:

- After you sell your stock, you cannot buy stock on the next day (i.e., cooldown one day).

Note: You may not engage in multiple transactions simultaneously (i.e., you must sell the stock before you buy again).

Example 1:

Input: `prices = [1,2,3,0,2]`

Output: 3

Explanation: `transactions = [buy, sell, cooldown, buy, sell]`

Example 2:

Input: prices = [1]
Output: 0

Constraints:

- $1 \leq \text{prices.length} \leq 5000$
- $0 \leq \text{prices}[i] \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Prices array; can buy/sell on multiple days but must wait 1 day (cooldown) after selling.

Maximize profit. Hold at most one stock at a time. Right?"

#

"prices=[1,2,3,0,2]:

buy day0(1), sell day1(2)+cooldown day2, buy day3(0), sell day4(2) → profit=3 ✓"

PHASE 2 — BRUTE FORCE

"Recursive with memoization: at each day choose hold/sell/cooldown/buy.

States: (day, holding). Returns max profit from that state onward.

Time: $O(n)$. Space: $O(n)$ for the memo table."

```
class _309_StockWithCooldownBrute:
```

```
    def maxProfit(self, prices: list[int]) -> int:
```

```
        from functools import lru_cache
```

```
        @lru_cache(maxsize=None)
```

```
        def dp(day: int, holding: bool) -> int:
```

```
            if day >= len(prices):
```

```
                return 0
```

```
            # Option 1: do nothing
```

```
            rest = dp(day + 1, holding)
```

```
            if holding:
```

```
                # Option 2: sell today → cooldown tomorrow
```

```
                sell = prices[day] + dp(day + 2, False)
```

```
                return max(rest, sell)
```

```
            else:
```

```
                # Option 2: buy today
```

```
                buy = -prices[day] + dp(day + 1, True)
```

```
                return max(rest, buy)
```

```
        return dp(0, False)
```

PHASE 3 — OPTIMIZE

"Three states, $O(1)$ space rolling variables:

held = max profit when holding a stock.

sold = max profit on the day we just sold (entering cooldown).

```
# rest = max profit when in cooldown or idle (free to buy next day).
# Transitions each day:
# new_held = max(held, rest - price) # keep holding OR buy from rest state
# new_sold = held + price # sell what we're holding
# new_rest = max(rest, sold) # stay idle OR cooldown ends
# Time: O(n). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _309_StockWithCooldown:
    def maxProfit(self, prices: list[int]) -> int:
        held = float('-inf') # profit while holding
        sold = 0 # profit day after selling (cooldown)
        rest = 0 # profit while resting / free to buy
        for price in prices:
            prev_sold = sold
            sold = held + price
            held = max(held, rest - price)
            rest = max(rest, prev_sold)
        return max(sold, rest)
```

PHASE 5 — TEST & DEBUG

```
# prices=[1,2,3,0,2]:
# price=1: sold=held+1=-inf, held=max(-inf,0-1)=-1, rest=max(0,-inf)=0.
# price=2: sold=-1+2=1, held=max(-1,0-2)=-1, rest=max(0,-inf)=0.
# price=3: sold=-1+3=2, held=max(-1,0-3)=-1, rest=max(0,1)=1.
# price=0: sold=-1+0=-1, held=max(-1,1-0)=1, rest=max(1,2)=2.
# price=2: sold=1+2=3, held=max(1,2-2)=1, rest=max(2,-1)=2.
# return max(3,2)=3 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): one pass. Space O(1): three rolling variables.
# The key insight: sold state prevents buying the very next day after selling.
# Follow-up: Stock with Transaction Fee (LC 714) – subtract fee on sell; same
two-state DP.
# Follow-up: At most k transactions (LC 188) – extend state to (day, k, holding)."
```

Dynamic Programming

□ #377 Combination Sum IV

MED[Open on LeetCode](#)

Array · Dynamic Programming

Lists: Grind 169

Problem

Given an array of distinct integers `nums` and a target integer `target`, return the number of possible combinations that add up to `target`.

The test cases are generated so that the answer can fit in a 32-bit integer.

Example 1:

```
Input: nums = [1,2,3], target = 4
Output: 7
Explanation:
The possible combination ways are:
(1, 1, 1, 1)
(1, 1, 2)
(1, 2, 1)
(1, 3)
```

Example 2:

```
Input: nums = [9], target = 3
Output: 0
```

Constraints:

- $1 \leq \text{nums.length} \leq 200$

- $1 \leq \text{nums}[i] \leq 1000$
- All the elements of `nums` are unique.
- $1 \leq \text{target} \leq 1000$

Follow up: What if negative numbers are allowed in the given array? How does it change the problem? What limitation we need to add to the question to allow negative numbers?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given distinct positive integers, return number of ORDERED sequences (not combinations)

that sum to target. Different orderings count as different. Right?"

#

"nums=[1,2,3], target=4:

(1,1,1,1), (1,1,2), (1,2,1), (2,1,1), (2,2), (1,3), (3,1) → 7 ✓"

PHASE 2 — BRUTE FORCE

"Recursion: `count(target) = sum of count(target-num)` for each `num ≤ target`.

Time: $O(\text{target}^n)$ without memo. With memo: $O(\text{target} * n)$. Space: $O(\text{target})$."

```
class _377_CombinationSumIV_Brute:
```

```
    def combinationSum4(self, nums: list[int], target: int) -> int:
```

```
        from functools import lru_cache
```

```
        @lru_cache(maxsize=None)
```

```
        def dp(t):
```

```
            if t == 0: return 1
```

```
            return sum(dp(t - n) for n in nums if n <= t)
```

```
        return dp(target)
```

PHASE 3 — OPTIMIZE

"Bottom-up DP: `dp[i] = number of ways to reach sum i`.

`dp[0]=1` (base). For each `i` from 1 to target: `dp[i] = sum(dp[i-n] for n in nums if n<=i)`.

Time: $O(\text{target} * \text{len}(\text{nums}))$. Space: $O(\text{target})$."

PHASE 4 — CLEAN CODE

```
class _377_CombinationSumIV:
```

```
    def combinationSum4(self, nums: list[int], target: int) -> int:
```

```
        dp = [0] * (target + 1)
```

```
dp[0] = 1
for i in range(1, target + 1):
    for num in nums:
        if i >= num:
            dp[i] += dp[i - num]
return dp[target]
```

PHASE 5 — TEST & DEBUG

```
# nums=[1,2,3], target=4:
# dp[0]=1. dp[1]=dp[0]=1. dp[2]=dp[1]+dp[0]=2. dp[3]=dp[2]+dp[1]+dp[0]=4.
# dp[4]=dp[3]+dp[2]+dp[1]=4+2+1=7 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(target*n): target positions × n numbers each. Space O(target).
# ORDER matters here (unlike Combination Sum III) – inner loop iterates nums at
every position.
# For unordered combinations (LC 39), iterate nums in the outer loop.
# Follow-up: if negative numbers allowed – infinite loops possible; add a limit on
sequence length.
# Follow-up: count modulo 10^9+7 – just add mod to each dp update."
```


Dynamic Programming

□ #416 Partition Equal Subset Sum

MED[Open on LeetCode](#)

Array · Dynamic Programming

Lists: Grind 169

Problem

Given an integer array `nums`, return `true` if you can partition the array into two subsets such that the sum of the elements in both subsets is equal or `false` otherwise.

Example 1:

Input: `nums = [1,5,11,5]`

Output: `true`

Explanation: The array can be partitioned as `[1, 5, 5]` and `[11]`.

Example 2:

Input: `nums = [1,2,3,5]`

Output: `false`

Explanation: The array cannot be partitioned into equal sum subsets.

Constraints:

- $1 \leq \text{nums.length} \leq 200$
- $1 \leq \text{nums}[i] \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Can the array be partitioned into two subsets with equal sum?
# Equivalently: can any subset sum to total/2? Right?"
#
# "nums=[1,5,11,5]: total=22, target=11. [1,5,5] sums to 11 → True ✓
# nums=[1,2,3,5]: total=11 (odd) → False ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Try all  $2^n$  subsets, check if any sums to total//2.
# Time:  $O(2^n)$ . Space:  $O(n)$  recursion."
```

```
class _416_PartitionEqualSubsetSum_Brute:
    def canPartition(self, nums: list[int]) -> bool:
        total = sum(nums)
        if total % 2: return False
        target = total // 2
        def dfs(i, remaining):
            if remaining == 0: return True
            if i == len(nums) or remaining < 0: return False
            return dfs(i+1, remaining-nums[i]) or dfs(i+1, remaining)
        return dfs(0, target)
```

PHASE 3 — OPTIMIZE

```
# "0/1 Knapsack DP: dp[j] = can we form sum j using some subset?
# Process each number: update dp from right to left (avoid reuse).
# dp[0]=True. For each num: for j from target down to num: dp[j] |= dp[j-num].
# Time:  $O(n \times \text{target})$ . Space:  $O(\text{target})$ ."
```

PHASE 4 — CLEAN CODE

```
class _416_PartitionEqualSubsetSum:
    def canPartition(self, nums: list[int]) -> bool:
        total = sum(nums)
        if total % 2: return False
        target = total // 2
        dp = [False] * (target + 1)
        dp[0] = True
        for num in nums:
            for j in range(target, num - 1, -1):
                dp[j] = dp[j] or dp[j - num]
            if dp[target]: return True
        return dp[target]
```

PHASE 5 — TEST & DEBUG

```
# nums=[1,5,11,5], target=11:
# num=1: dp[1]=dp[0]=True. → dp=[T,T,F,...,F].
# num=5: dp[6]=dp[1]=T, dp[5]=dp[0]=T. dp=[T,T,F,F,F,T,T,F,...].
# num=11: dp[11]=dp[0]=True → return True ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(n \times \text{target})$ : n numbers × target positions. Space  $O(\text{target})$ : 1D boolean array."
```

```
# Right-to-left update prevents using the same number twice (0/1 knapsack vs unbounded).  
# Follow-up: Subset Sum (exact sum) – same DP, return dp[target].  
# Follow-up: Last Stone Weight II (LC 1049) – minimize |S1-S2|; same partition approach."
```

Dynamic Programming

□ #435 Non-overlapping Intervals

MED[Open on LeetCode](#)

Array · Dynamic Programming · Greedy · Sorting

Lists: Grind 169

Problem

Given an array of intervals `intervals` where `intervals[i] = [starti, endi]`, return the minimum number of intervals you need to remove to make the rest of the intervals non-overlapping.

Note that intervals which only touch at a point are non-overlapping. For example, `[1, 2]` and `[2, 3]` are non-overlapping.

Example 1:

Input: `intervals = [[1,2],[2,3],[3,4],[1,3]]`

Output: 1

Explanation: `[1,3]` can be removed and the rest of the intervals are non-overlapping.

Example 2:

Input: `intervals = [[1,2],[1,2],[1,2]]`

Output: 2

Explanation: You need to remove two `[1,2]` to make the rest of the intervals non-overlapping.

Example 3:

Input: intervals = [[1,2],[2,3]]

Output: 0

Explanation: You don't need to remove any of the intervals since they're already non-overlapping.

Constraints:

- $1 \leq \text{intervals.length} \leq 105$
- $\text{intervals}[i].\text{length} == 2$
- $-5 * 10^4 \leq \text{start}_i < \text{end}_i \leq 5 * 10^4$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given intervals, find the minimum number to remove so the rest don't overlap.
# Equivalent to: find maximum non-overlapping intervals to KEEP, then answer = n -
# keep. Right?"
#
# "intervals=[[1,2],[2,3],[3,4],[1,3]]: remove [1,3] → 3 non-overlapping → 1
# removal ✓
# intervals=[[1,2],[1,2],[1,2]]: remove 2 → 1 remaining ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Try all subsets of intervals; return n minus size of largest non-overlapping
# subset.
# Time:  $O(2^n * n)$ . Space:  $O(n)$ ."
class _435_NonOverlappingIntervals_Brute:
    def eraseOverlapIntervals(self, intervals: list[list[int]]) -> int:
        intervals.sort()
        n = len(intervals)
        best = 0
        for mask in range(1 << n):
            subset = [intervals[i] for i in range(n) if mask & (1<<i)]
            if all(subset[i][1] <= subset[i+1][0] for i in range(len(subset)-1)):
                best = max(best, len(subset))
        return n - best
```

PHASE 3 — OPTIMIZE

```
# "Greedy: sort by END time. Greedily keep intervals with earliest end that don't
# overlap.
# Track end of last kept interval. If next interval starts before that end, skip it
# (remove)."
```

Time: $O(n \log n)$ for sort. Space: $O(1)$."

PHASE 4 — CLEAN CODE

```
class _435_NonOverlappingIntervals:
    def eraseOverlapIntervals(self, intervals: list[list[int]]) -> int:
        intervals.sort(key=lambda x: x[1])
        removed = 0
        last_end = float('-inf')
        for start, end in intervals:
            if start < last_end:
                removed += 1 # overlap → remove this interval
            else:
                last_end = end # keep it, update end
        return removed
```

PHASE 5 — TEST & DEBUG

intervals=[[1,2],[2,3],[3,4],[1,3]] sorted by end: [[1,2],[2,3],[1,3],[3,4]]:
 # [1,2]: $1 \geq -\text{inf}$ → keep, last_end=2.
 # [2,3]: $2 \geq 2$ → keep, last_end=3.
 # [1,3]: $1 < 3$ → remove. removed=1.
 # [3,4]: $3 \geq 3$ → keep. return 1 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n \log n)$: dominated by sort. Space $O(1)$: one variable.
 # Sorting by END (not start) is the key greedy insight – keeps the most room for future intervals.
 # Follow-up: Meeting Rooms I (LC 252) – just check if any two overlap (sort by start, compare consecutive).
 # Follow-up: Minimum Number of Arrows (LC 452) – burst balloons; same greedy as this problem."

Dynamic Programming

□ #516 Longest Palindromic Subsequence

MED[Open on LeetCode](#)

String · Dynamic Programming

Lists: Denny Zhang

Problem

Given a string s , find the longest palindromic subsequence's length in s .

A subsequence is a sequence that can be derived from another sequence by deleting some or no elements without changing the order of the remaining elements.

Example 1:

Input: $s = \text{"bbbab"}$

Output: 4

Explanation: One possible longest palindromic subsequence is "bbbb".

Example 2:

Input: $s = \text{"cbdd"}$

Output: 2

Explanation: One possible longest palindromic subsequence is "bb".

Constraints:

- $1 \leq s.length \leq 1000$
- s consists only of lowercase English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Find the length of the longest palindromic subsequence in string s.
# Subsequence: characters need not be contiguous. Right?"
#
# "s='bbbab': 'bbbb' → 4 ✓ s='cbdd': 'bb' → 2 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Recursion: lps(i,j) = longest palindromic subsequence in s[i..j].
# If s[i]==s[j]: 2+lps(i+1,j-1). Else max(lps(i+1,j), lps(i,j-1)).
# Time: O(2^n) without memo. Space: O(n) stack."
```

```
class _516_LongestPalindromicSubsequence_Brute:
    def longestPalindromeSubseq(self, s: str) -> int:
        from functools import lru_cache
        @lru_cache(maxsize=None)
        def dp(i, j):
            if i > j: return 0
            if i == j: return 1
            if s[i] == s[j]: return 2 + dp(i+1, j-1)
            return max(dp(i+1, j), dp(i, j-1))
        return dp(0, len(s)-1)
```

PHASE 3 — OPTIMIZE

```
# "Bottom-up DP: dp[i][j] = LPS in s[i..j]. Fill by increasing length.
# Observation: LPS(s) = LCS(s, reverse(s)).
# Space optimised: O(n) with two rolling arrays."
```

PHASE 4 — CLEAN CODE

```
class _516_LongestPalindromicSubsequence:
    def longestPalindromeSubseq(self, s: str) -> int:
        n = len(s)
        dp = [1] * n # dp[j] = LPS of s[i..j] for current row i
        for i in range(n - 2, -1, -1):
            prev = [0] * n
            prev[i] = 1
            for j in range(i + 1, n):
                if s[i] == s[j]:
                    prev[j] = 2 + (dp[j-1] if j-1 >= i+1 else 0)
                else:
                    prev[j] = max(dp[j], prev[j-1])
            dp = prev
        return dp[n-1]
```

PHASE 5 — TEST & DEBUG

```
# s='bbbab':
# Full 2D: dp[0][4]=4 (bbbb). Standard O(n^2) time O(n^2) space is cleaner to trace.
```


s='bbbab': LCS with reverse 'babbb': LCS='bbbb'=4 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n^2)$: fill $n \times n$ DP table. Space $O(n)$: two rolling arrays.

Equivalently: LPS(s) = LCS(s, s[::-1]) – same recurrence structure.

Follow-up: Longest Palindromic Substring (LC 5) – contiguous version.

Follow-up: Palindrome Partitioning II (LC 132) – min cuts to make all substrings palindromes."

Dynamic Programming

□ #576 Out of Boundary Paths

MED[Open on LeetCode](#)

Dynamic Programming

Lists: Denny Zhang

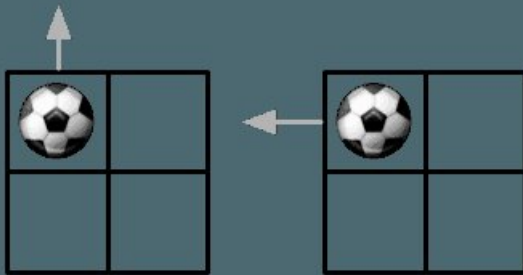
Problem

There is an $m \times n$ grid with a ball. The ball is initially at the position $[startRow, startColumn]$. You are allowed to move the ball to one of the four adjacent cells in the grid (possibly out of the grid crossing the grid boundary). You can apply at most $maxMove$ moves to the ball.

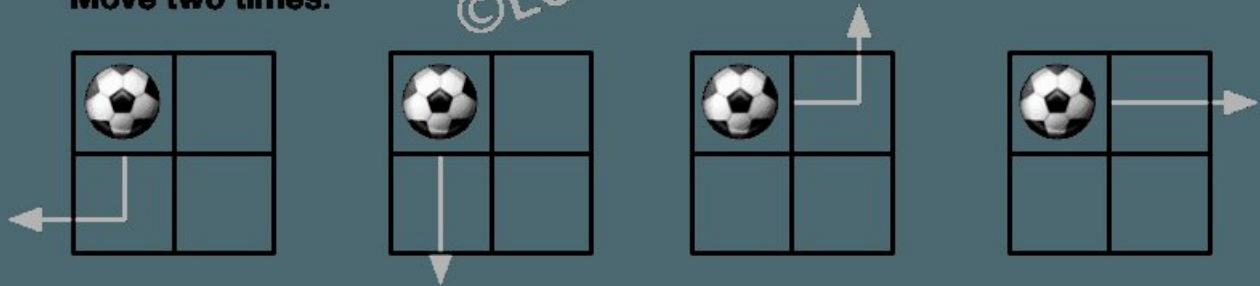
Given the five integers m , n , $maxMove$, $startRow$, $startColumn$, return the number of paths to move the ball out of the grid boundary. Since the answer can be very large, return it modulo $10^9 + 7$.

Example 1:

Move one time:

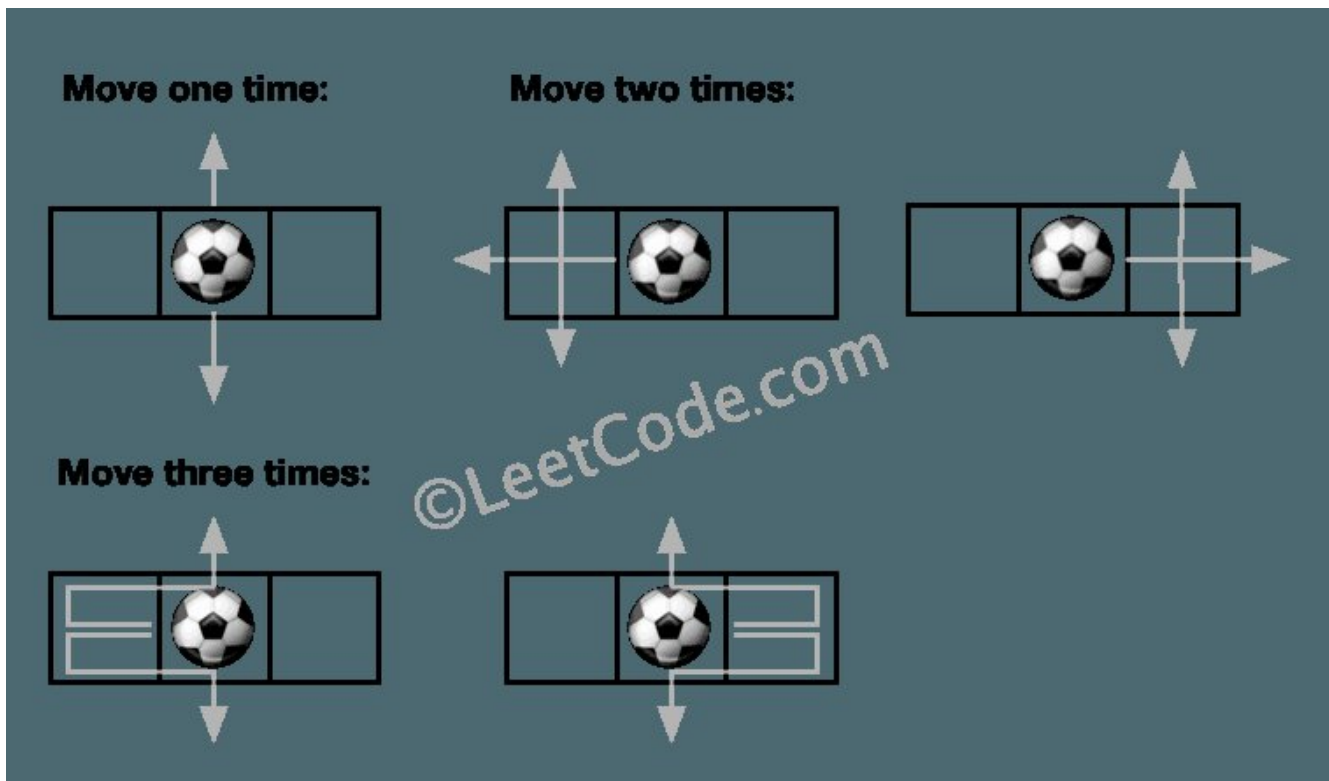


Move two times:



Input: $m = 2$, $n = 2$, $\text{maxMove} = 2$, $\text{startRow} = 0$, $\text{startColumn} = 0$
Output: 6

Example 2:



Input: $m = 1$, $n = 3$, $\text{maxMove} = 3$, $\text{startRow} = 0$, $\text{startColumn} = 1$
 Output: 12

Constraints:

- $1 \leq m, n \leq 50$
- $0 \leq \text{maxMove} \leq 50$
- $0 \leq \text{startRow} < m$
- $0 \leq \text{startColumn} < n$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"An $m \times n$ grid. Ball starts at $(\text{startRow}, \text{startCol})$. In exactly maxMove moves, count paths that take the ball OUT of the boundary. Return modulo $10^9 + 7$. Right?"
 #
 # "m=2,n=2,maxMove=2,startRow=0,startCol=0: 6 paths exit in at most 2 moves ✓"

PHASE 2 — BRUTE FORCE

"DFS/recursion with memo: $\text{count}(r, c, \text{moves}) = \text{sum of count for each 4-direction.}$ "

```
# If r,c out of bounds → 1 (found a path). If moves=0 and in bounds → 0.
# Time: O(m*n*maxMove). Space: O(m*n*maxMove) memo."
```

```
class _576_OutOfBoundaryPaths_Brute:
```

```
    def findPaths(self, m, n, maxMove, startRow, startCol) -> int:
        from functools import lru_cache
        MOD = 10**9 + 7
        @lru_cache(maxsize=None)
        def dp(r, c, moves):
            if r < 0 or r >= m or c < 0 or c >= n: return 1
            if moves == 0: return 0
            return sum(dp(r+dr, c+dc, moves-1)
                       for dr,dc in [(0,1),(0,-1),(1,0),(-1,0)]) % MOD
        return dp(startRow, startCol, maxMove)
```

```
# PHASE 3 — OPTIMIZE
```

```
# "Bottom-up DP: dp[k][r][c] = number of paths of exactly k moves starting at (r,c)
that exit.
```

```
# Build from k=1 to maxMove, accumulate total.
```

```
# Space optimised: O(m*n) with two 2D arrays (current and previous step)."
```

```
# PHASE 4 — CLEAN CODE
```

```
class _576_OutOfBoundaryPaths:
```

```
    def findPaths(self, m: int, n: int, maxMove: int, startRow: int, startCol: int)
-> int:
```

```
        MOD = 10**9 + 7
        dp = [[0]*n for _ in range(m)]
        dp[startRow][startCol] = 1
        count = 0
        for _ in range(maxMove):
            ndp = [[0]*n for _ in range(m)]
            for r in range(m):
                for c in range(n):
                    if dp[r][c] == 0: continue
                    for dr, dc in [(0,1),(0,-1),(1,0),(-1,0)]:
                        nr, nc = r+dr, c+dc
                        if 0 <= nr < m and 0 <= nc < n:
                            ndp[nr][nc] = (ndp[nr][nc] + dp[r][c]) % MOD
                        else:
                            count = (count + dp[r][c]) % MOD
            dp = ndp
        return count
```

```
# PHASE 5 — TEST & DEBUG
```

```
# m=2,n=2,maxMove=2,start=(0,0):
```

```
# dp=[[1,0],[0,0]]. move1: (0,0)→up(-1,0) 00B count+=1, left(0,-1) 00B count+=1,
```

```
# right(0,1)→ndp[0][1]+=1, down(1,0)→ndp[1][0]+=1. ndp=[[0,1],[1,0]], count=2.
```

```
# move2: ndp[0][1]: up 00B+1=3, right 00B+1=4, left→[0,0]+=1, down→[1,1]+=1.
```

```
# ndp[1][0]: down 00B+1=5, left 00B+1=6, up→[0,0]+=1, right→[1,1]+=1. count=6 ✓
```

```
# PHASE 6 — COMPLEXITY & FOLLOW-UP
```

```
# "Time  $O(\text{maxMove} * m * n)$ . Space  $O(m*n)$ : two rolling grids.  
# Each step propagates the ball's position to neighboring cells or exits.  
# Follow-up: Knight Probability in Chessboard (LC 688) – same DP structure with  
knight moves.  
# Follow-up: if the ball can stay in bounds, track the stay probability separately."
```

Dynamic Programming

□ #1143 Longest Common Subsequence

MED[Open on LeetCode](#)

String · Dynamic Programming

Lists: Denny Zhang

Problem

Given two strings `text1` and `text2`, return the length of their longest common subsequence. If there is no common subsequence, return 0.

A subsequence of a string is a new string generated from the original string with some characters (can be none) deleted without changing the relative order of the remaining characters.

- For example, "ace" is a subsequence of "abcde".

A common subsequence of two strings is a subsequence that is common to both strings.

Example 1:

Input: `text1 = "abcde"`, `text2 = "ace"`

Output: 3

Explanation: The longest common subsequence is "ace" and its length is 3.

Example 2:

Input: text1 = "abc", text2 = "abc"

Output: 3

Explanation: The longest common subsequence is "abc" and its length is 3.

Example 3:

Input: text1 = "abc", text2 = "def"

Output: 0

Explanation: There is no such common subsequence, so the result is 0.

Constraints:

- $1 \leq \text{text1.length}, \text{text2.length} \leq 1000$
- text1 and text2 consist of only lowercase English characters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Find the length of the longest common subsequence of two strings.

Subsequence: characters need not be contiguous. Right?"

#

"text1='abcde', text2='ace': 'ace' → length 3 ✓

text1='abc', text2='abc': 'abc' → 3 ✓ text1='abc', text2='def': '' → 0 ✓"

PHASE 2 — BRUTE FORCE

"Recursion: lcs(i,j) on text1[:i], text2[:j].

If text1[i]==text2[j]: 1+lcs(i-1,j-1). Else max(lcs(i-1,j), lcs(i,j-1)).

Time: $O(2^{(m+n)})$ without memo. Space: $O(m+n)$ stack."

```
class _1143_LongestCommonSubsequence_Brute:
```

```
    def longestCommonSubsequence(self, text1: str, text2: str) -> int:
```

```
        from functools import lru_cache
```

```
        @lru_cache(maxsize=None)
```

```
        def dp(i, j):
```

```
            if i < 0 or j < 0: return 0
```

```
            if text1[i] == text2[j]: return 1 + dp(i-1, j-1)
```

```
            return max(dp(i-1, j), dp(i, j-1))
```

```
        return dp(len(text1)-1, len(text2)-1)
```

PHASE 3 — OPTIMIZE

"Bottom-up 2D DP: dp[i][j] = LCS of text1[:i], text2[:j].

Space optimised: two rolling arrays $O(\min(m,n))$."

PHASE 4 — CLEAN CODE

```
class _1143_LongestCommonSubsequence:
    def longestCommonSubsequence(self, text1: str, text2: str) -> int:
        m, n = len(text1), len(text2)
        if m < n:
            text1, text2, m, n = text2, text1, n, m
        prev = [0] * (n + 1)
        for i in range(1, m + 1):
            curr = [0] * (n + 1)
            for j in range(1, n + 1):
                if text1[i-1] == text2[j-1]:
                    curr[j] = 1 + prev[j-1]
                else:
                    curr[j] = max(prev[j], curr[j-1])
            prev = curr
        return prev[n]
```

PHASE 5 — TEST & DEBUG

```
# text1='abcde', text2='ace':
# After full DP: prev[3] = 3 ✓ (standard LCS table)
# i=1('a'): curr[1]=1('a'=='a'). i=2('b'): curr[1]=1,curr[2]=1('b'≠'c').
# i=3('c'): curr[2]=2('c'=='c'). i=4('d'): curr[2]=2. i=5('e'): curr[3]=3 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(m \times n)$ : fill DP table. Space  $O(\min(m,n))$ : rolling arrays.
# LCS is foundational: used in diff tools, bioinformatics, plagiarism detection.
# Follow-up: print the actual LCS – backtrack through the DP table.
# Follow-up: Edit Distance (LC 72) extends LCS with insert/delete/replace costs."
```

Bit Manipulation

Round 8 · Low · Medium · 4 questions

Bit Manipulation

□ #78 Subsets

MED[Open on LeetCode](#)

Array · Backtracking · Bit Manipulation

Lists: Grind 169

Problem

Given an integer array `nums` of unique elements, return all possible subsets (the power set).

The solution set must not contain duplicate subsets. Return the solution in any order.

Example 1:

Input: `nums = [1,2,3]`

Output: `[[],[1],[2],[1,2],[3],[1,3],[2,3],[1,2,3]]`

Example 2:

Input: `nums = [0]`

Output: `[[],[0]]`

Constraints:

- $1 \leq \text{nums.length} \leq 10$
- $-10 \leq \text{nums}[i] \leq 10$
- All the numbers of `nums` are unique.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Given an array of unique integers, return all possible subsets (power set).
Right?"

#

"nums=[1,2,3]: [[],[1],[2],[1,2],[3],[1,3],[2,3],[1,2,3]] ✓ (any order)"

PHASE 2 — BRUTE FORCE

"Backtracking: at each index, choose to include or exclude. Build all 2^n subsets.
Time: $O(n * 2^n)$. Space: $O(n)$ recursion depth."

```
class _78_Subsets_Brute:
    def subsets(self, nums: list[int]) -> list[list[int]]:
        result = []
        def backtrack(start, path):
            result.append(path[:])
            for i in range(start, len(nums)):
                path.append(nums[i])
                backtrack(i + 1, path)
                path.pop()
        backtrack(0, [])
        return result
```

PHASE 3 — OPTIMIZE

"Bit manipulation: iterate $0..2^n-1$. Each integer is a bitmask where bit $i=1$ means include `nums[i]`.

Time: $O(n * 2^n)$. Space: $O(1)$ extra (output excluded).

Same complexity but elegant integer iteration."

PHASE 4 — CLEAN CODE

```
class _78_Subsets:
    def subsets(self, nums: list[int]) -> list[list[int]]:
        n = len(nums)
        result = []
        for mask in range(1 << n):
            subset = [nums[i] for i in range(n) if mask & (1 << i)]
            result.append(subset)
        return result
```

PHASE 5 — TEST & DEBUG

nums=[1,2,3], n=3: masks $0..7$.

mask=0(000): []. mask=1(001): [1]. mask=2(010): [2]. mask=3(011): [1,2].

mask=4(100): [3]. mask=5(101): [1,3]. mask=6(110): [2,3]. mask=7(111): [1,2,3]. ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n*2^n)$: 2^n subsets, each $O(n)$ to build. Space $O(1)$ extra.

Both backtracking and bitmask approaches have identical complexity.

Backtracking is more generalizable; bitmask is concise for small n .

Follow-up: Subsets II (LC 90) – input has duplicates; sort + skip duplicates in backtracking.

Follow-up: Combination Sum (LC 39) – subsets constrained to a target sum."

Bit Manipulation

□ #90 Subsets II

MED[Open on LeetCode](#)

Array · Backtracking · Bit Manipulation

Lists: Denny Zhang

Problem

Given an integer array `nums` that may contain duplicates, return all possible subsets (the power set).

The solution set must not contain duplicate subsets. Return the solution in any order.

Example 1:

```
Input: nums = [1,2,2]
Output: [[],[1],[1,2],[1,2,2],[2],[2,2]]
```

Example 2:

```
Input: nums = [0]
Output: [[],[0]]
```

Constraints:

- $1 \leq \text{nums.length} \leq 10$
- $-10 \leq \text{nums}[i] \leq 10$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given an array that MAY contain duplicates, return all unique subsets. Right?"
#
# "nums=[1,2,2]: [[],[1],[1,2],[1,2,2],[2],[2,2]] ✓ (no duplicates in output)"
```

PHASE 2 — BRUTE FORCE

```
# "Generate all subsets as sorted tuples, add to a set to deduplicate.
# Time:  $O(n * 2^n)$ . Space:  $O(2^n)$  for the set."
```

```
class _90_SubsetsII_Brute:
    def subsetsWithDup(self, nums: list[int]) -> list[list[int]]:
        result = set()
        nums.sort()
        def backtrack(start, path):
            result.add(tuple(path))
            for i in range(start, len(nums)):
                path.append(nums[i])
                backtrack(i+1, path)
                path.pop()
        backtrack(0, [])
        return [list(t) for t in result]
```

PHASE 3 — OPTIMIZE

```
# "Sort nums first. In backtracking, skip duplicate elements at the same recursion
level
# (not the first occurrence, but subsequent same values at the same start position).
# If nums[i] == nums[i-1] and i > start, skip.
# Time:  $O(n * 2^n)$ . Space:  $O(n)$  recursion – no dedup set needed."
```

PHASE 4 — CLEAN CODE

```
class _90_SubsetsII:
    def subsetsWithDup(self, nums: list[int]) -> list[list[int]]:
        nums.sort()
        result = []
        def backtrack(start: int, path: list) -> None:
            result.append(path[:])
            for i in range(start, len(nums)):
                if i > start and nums[i] == nums[i-1]:
                    continue # skip duplicate at this level
                path.append(nums[i])
                backtrack(i + 1, path)
                path.pop()
        backtrack(0, [])
        return result
```

PHASE 5 — TEST & DEBUG

```
# nums=[1,2,2] sorted:
# backtrack(0,[]): append []. i=0→path=[1], backtrack(1):
# append [1]. i=1→[1,2],backtrack(2): append[1,2]. i=2→[1,2,2]→append. i=3 done.
# i=2: nums[2]==nums[1] and 2>1 → skip.
# i=1→path=[2],backtrack(2): append[2]. i=2→[2,2]→append.
```

```
# i=2: nums[2]==nums[1] and 2>0 → skip.  
# result=[[],[1],[1,2],[1,2,2],[2],[2,2]] ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n \cdot 2^n)$: worst case all unique. Space $O(n)$: recursion depth.

The sort + skip-duplicate pattern is the standard dedup technique for backtracking.

Follow-up: Permutations II (LC 47) – same skip-duplicate pattern for permutations.

Follow-up: Combination Sum II (LC 40) – target sum with duplicates, same skip pattern."

Bit Manipulation

□ #287 Find the Duplicate Number

MED[Open on LeetCode](#)

Array · Two Pointers · Binary Search · Bit Manipulation

Lists: Grind 169

Problem

Given an array of integers `nums` containing $n + 1$ integers where each integer is in the range $[1, n]$ inclusive.

There is only one repeated number in `nums`, return this repeated number.

You must solve the problem without modifying the array `nums` and using only constant extra space.

Example 1:

Input: `nums = [1,3,4,2,2]`
Output: 2

Example 2:

Input: `nums = [3,1,3,4,2]`
Output: 3

Example 3:

Input: nums = [3,3,3,3,3]

Output: 3

Constraints:

- $1 \leq n \leq 10^5$
- `nums.length == n + 1`
- $1 \leq \text{nums}[i] \leq n$
- All the integers in `nums` appear only once except for precisely one integer which appears two or more times.

Follow up:

- How can we prove that at least one duplicate number must exist in `nums`?
- Can you solve the problem in linear runtime complexity?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"n+1 integers in range [1,n]; exactly one number is duplicated (may appear 2+ times).

Find the duplicate. Must not modify array, must use O(1) extra space. Right?"

#

"nums=[1,3,4,2,2]: duplicate=2 ✓"

nums=[3,1,3,4,2]: duplicate=3 ✓"

PHASE 2 — BRUTE FORCE

"Use a hash set: for each number, if already seen, it's the duplicate.

Time: O(n). Space: O(n) for the set. – Violates the O(1) space constraint."

```
class _287_FindDuplicate_Brute:
```

```
    def findDuplicate(self, nums: list[int]) -> int:
```

```
        seen = set()
```

```
        for num in nums:
```

```
            if num in seen:
```

```

        return num
    seen.add(num)
    return -1

```

PHASE 3 — OPTIMIZE

"Floyd's cycle detection – treat nums as a linked list where index *i* points to *nums[i]*."

Because one value is duplicated, two indices point to the same next node → cycle.

Phase 1: find meeting point of slow (1 step) and fast (2 steps).

Phase 2: reset slow to index 0, advance both 1 step – they meet at cycle entry = duplicate.

Time: $O(n)$. Space: $O(1)$. Does not modify the array."

PHASE 4 — CLEAN CODE

```

class _287_FindDuplicate:
    def findDuplicate(self, nums: list[int]) -> int:
        # Phase 1: detect cycle
        slow = fast = nums[0]
        while True:
            slow = nums[slow]
            fast = nums[nums[fast]]
            if slow == fast:
                break
        # Phase 2: find cycle entry
        slow = nums[0]
        while slow != fast:
            slow = nums[slow]
            fast = nums[fast]
        return slow

```

PHASE 5 — TEST & DEBUG

nums=[1,3,4,2,2] (indices: 0→1, 1→3, 2→4, 3→2, 4→2):

Phase1: slow=1,fast=1→slow=3,fast=4→slow=2,fast=2→meet at 2.

Phase2: slow=*nums*[0]=1, fast=2. slow=3,fast=4. slow=2,fast=2. meet at 2 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$. Space $O(1)$. Does not sort or modify the array.

Binary search alternative: count numbers $\leq$ mid; if count $>$ mid, duplicate is in [1,mid].

$O(n \log n)$ time, $O(1)$ space – simpler to explain in an interview.

Follow-up: if multiple duplicates allowed, binary search no longer works directly.

Follow-up: Find All Duplicates (LC 442) – mark visited by negating at index."

Bit Manipulation

□ ★ #1506 Find Root of N-Ary Tree ★

MED

[Open on LeetCode](#)

Hash Table · Bit Manipulation · Tree · DFS

Lists: ★ Premium | Premium 98

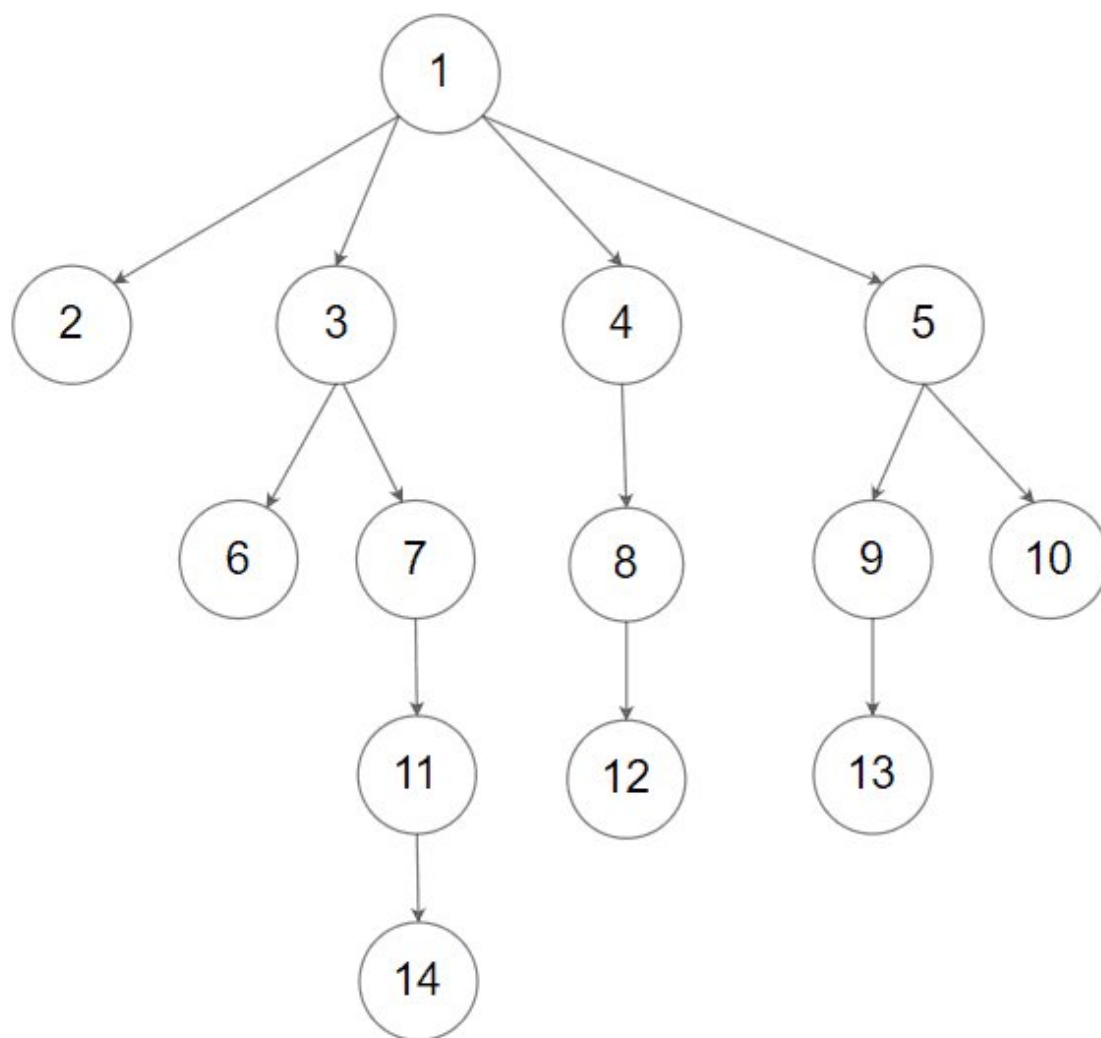
Problem

You are given all the nodes of an N-ary tree as an array of Node objects, where each node has a unique value.

Return the root of the N-ary tree.

Custom testing:

An N-ary tree can be serialized as represented in its level order traversal where each group of children is separated by the null value (see examples).



For example, the above tree is serialized as [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,null,11,null,12,null,13,null,null,14].

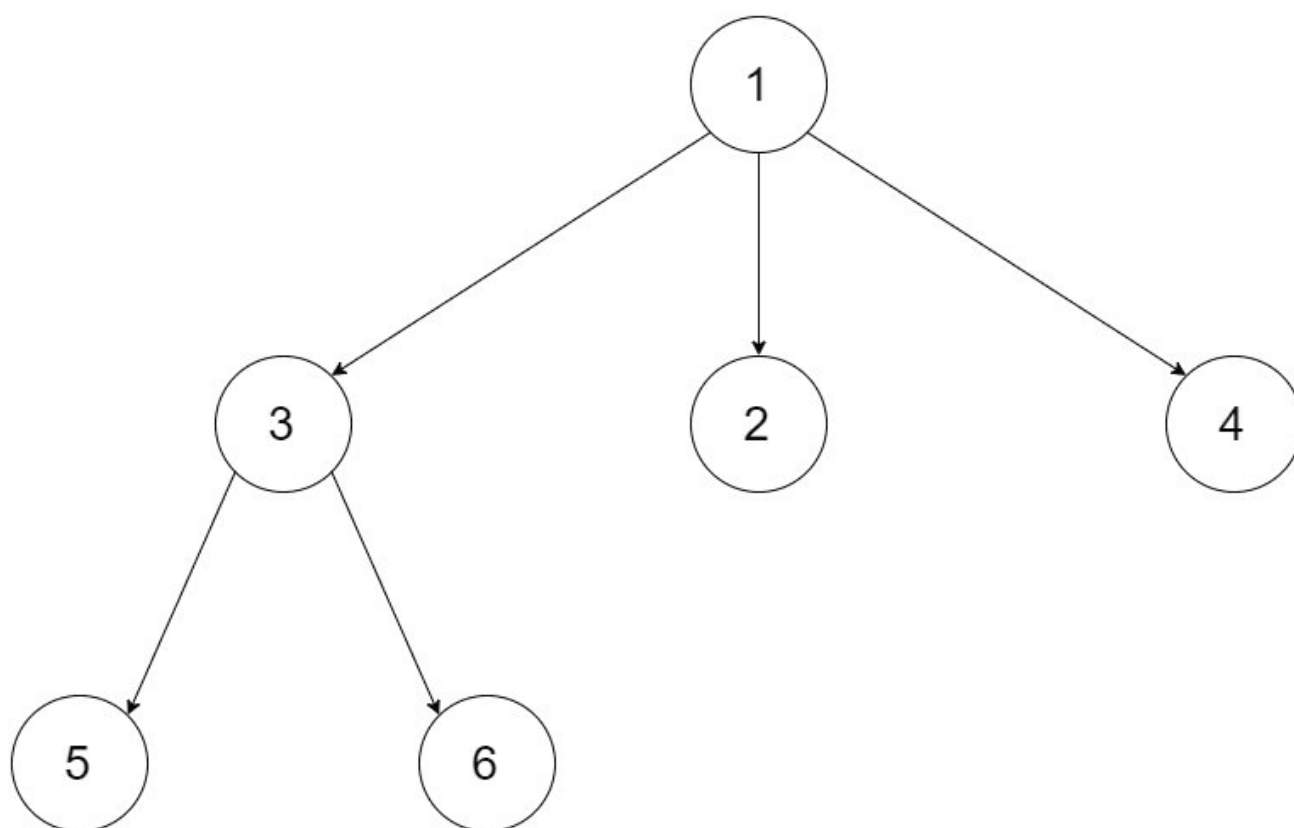
The testing will be done in the following way:

1. The input data should be provided as a serialization of the tree.
2. The driver code will construct the tree from the serialized input data and put each Node object into an array in an arbitrary order.

3. The driver code will pass the array to `findRoot`, and your function should find and return the root Node object in the array.

4. The driver code will take the returned Node object and serialize it. If the serialized value and the input data are the same, the test passes.

Example 1:



Input: tree = [1,null,3,2,4,null,5,6]

Output: [1,null,3,2,4,null,5,6]

Explanation: The tree from the input data is shown above.

The driver code creates the tree and gives findRoot the Node objects in an arbitrary order.

For example, the passed array could be

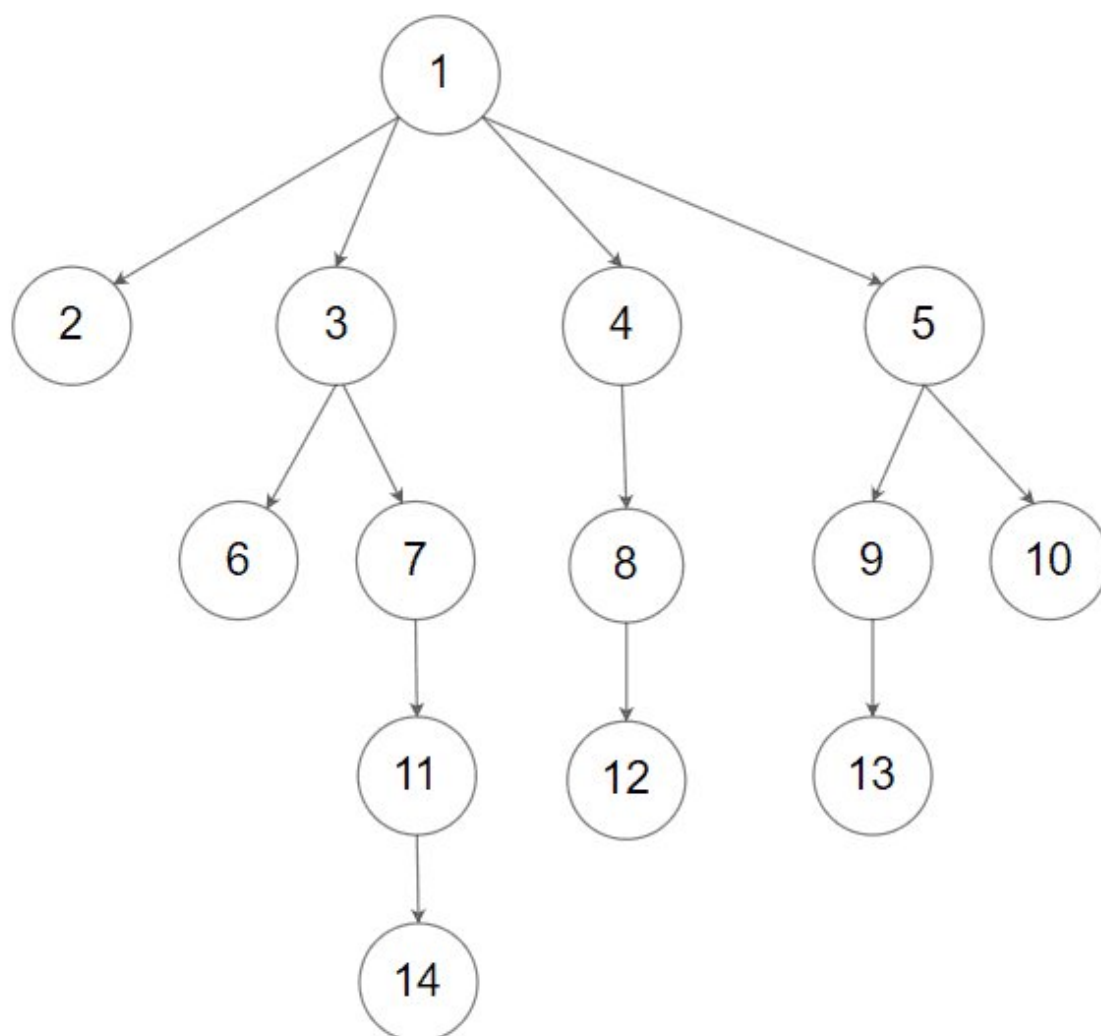
[Node(5),Node(4),Node(3),Node(6),Node(2),Node(1)] or

[Node(2),Node(6),Node(1),Node(3),Node(5),Node(4)].

The findRoot function should return the root Node(1), and the driver code will serialize it and compare with the input data.

The input data and serialized Node(1) are the same, so the test passes.

Example 2:



```
Input: tree = [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,null,11,null,12,null,13,null,null,14]
Output: [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,null,11,null,12,null,13,null,null,14]
```

Constraints:

- The total number of nodes is between $[1, 5 * 10^4]$.
- Each node has a unique value.

Follow up:

- Could you solve this problem in constant space complexity with a linear time algorithm?

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given all nodes of an N-ary tree in an array (in random order), find the root.
# Each non-root node appears as a child exactly once.
# Cannot access node.parent. Right?"
#
# "nodes=[1,2,3,4,5,6], tree root=1 with children [3,2,4], node 3 has children [5,6]:
# XOR all node values XOR all child values → root value. ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Build a set of all child node values. Root is the node whose value is NOT in that set.
# Time: O(n) where n = total nodes including all children. Space: O(n)."
```

```
class _1506_FindRootNaryTree_Brute:
    def findRoot(self, tree) -> 'Node':
        child_vals = {child.val for node in tree for child in node.children}
        for node in tree:
            if node.val not in child_vals:
                return node
        return None
```

PHASE 3 — OPTIMIZE

```
# "XOR trick: XOR all node values; XOR all children values.
```

```
# Every non-root node's value appears once as a node and once as a child → cancels to 0.  
# Root appears only as a node → its value remains.  
# Time: O(n). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _1506_FindRootNaryTree:  
    def findRoot(self, tree) -> 'Node':  
        xor_val = 0  
        for node in tree:  
            xor_val ^= node.val  
            for child in node.children:  
                xor_val ^= child.val  
        for node in tree:  
            if node.val == xor_val:  
                return node  
        return None
```

PHASE 5 — TEST & DEBUG

```
# tree nodes [1,2,3,4,5,6] with root=1, children={3,2,4}, 3's children={5,6}:  
# node vals XOR: 1^2^3^4^5^6. child vals XOR: 3^2^4^5^6.  
# Combined: 1^2^3^4^5^6^3^2^4^5^6 = 1 (all others cancel). root=node(1) ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(n): two linear passes. Space O(1): one XOR variable."  
# Same XOR cancellation as Single Number (LC 136).  
# Follow-up: if node values aren't unique, XOR fails – use the child-set approach.  
# Follow-up: find all roots if the input is a forest (multiple disconnected trees) –  
# same trick, collect all non-child values."
```


Math

Round 8 · Low · Medium · 5 questions

Math

□ #7 Reverse Integer

MED[Open on LeetCode](#)

Math

Lists: Grind 169

Problem

Given a signed 32-bit integer x , return x with its digits reversed. If reversing x causes the value to go outside the signed 32-bit integer range $[-2^{31}, 2^{31} - 1]$, then return 0.

Assume the environment does not allow you to store 64-bit integers (signed or unsigned).

Example 1:

Input: $x = 123$
Output: 321

Example 2:

Input: $x = -123$
Output: -321

Example 3:

Input: $x = 120$
Output: 21

Constraints:

- $-2^{31} \leq x \leq 2^{31} - 1$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Reverse the digits of a 32-bit signed integer. Return 0 if the reversed integer overflows
# [-231, 231-1]. Right?"
#
# "x=123 → 321 ✓ x=-123 → -321 ✓ x=120 → 21 ✓ x=1534236469 → 0 (overflow) ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Convert to string, reverse, convert back. Check 32-bit bounds.
# Time: O(d). Space: O(d)."
```

```
class _7_ReverseInteger_Brute:
    def reverse(self, x: int) -> int:
        sign = -1 if x < 0 else 1
        rev = int(str(abs(x))[::-1]) * sign
        return rev if -(2**31) <= rev <= 2**31-1 else 0
```

PHASE 3 — OPTIMIZE

```
# "Build reversed integer digit by digit without string conversion.
# Check for overflow before each multiplication – avoids relying on 64-bit arithmetic.
# Time: O(d). Space: O(1)."
```

PHASE 4 — CLEAN CODE

```
class _7_ReverseInteger:
    def reverse(self, x: int) -> int:
        INT_MAX, INT_MIN = 2**31 - 1, -(2**31)
        result, sign = 0, 1 if x >= 0 else -1
        x = abs(x)
        while x:
            digit = x % 10
            x //= 10
            # Check overflow before appending digit
            if result > (INT_MAX - digit) // 10:
                return 0
            result = result * 10 + digit
        return result * sign
```

PHASE 5 — TEST & DEBUG

```
# x=123: digits extracted: 3,2,1. result: 3,32,321. 321 ≤ INT_MAX ✓. return 321 ✓.
# x=-123: sign=-1,x=123. Same as above. return -321 ✓.
# x=1534236469: eventually result*10+digit > INT_MAX → return 0 ✓.
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(d): one pass through digits. Space O(1): integer variables only.
# Overflow check: if result > (INT_MAX - digit) // 10, adding would overflow."
```

Follow-up: Palindrome Number (LC 9) – only reverse half for palindrome check.
Follow-up: if working in a language without 64-bit integers, this overflow check is critical."

Math

□ #50 Pow(x, n)

MED

[Open on LeetCode](#)

Math · Recursion

Lists: Grind 169

Problem

Implement $\text{pow}(x, n)$, which calculates x raised to the power n (i.e., x^n).

Example 1:

Input: $x = 2.00000$, $n = 10$
Output: 1024.00000

Example 2:

Input: $x = 2.10000$, $n = 3$
Output: 9.26100

Example 3:

Input: $x = 2.00000$, $n = -2$
Output: 0.25000
Explanation: $2^{-2} = 1/2^2 = 1/4 = 0.25$

Constraints:

- $-100.0 < x < 100.0$
- $-2^{31} \leq n \leq 2^{31}-1$
- n is an integer.
- Either x is not zero or $n > 0$.

● $-104 \leq x^n \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Implement pow(x, n) — x raised to the power n. n can be negative. Right?"
#
# "pow(2.0, 10) → 1024.0 ✓ pow(2.1, 3) → 9.261 ✓ pow(2.0, -2) → 0.25 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Multiply x by itself n times.
# Time: O(n). Space: O(1). TLEs for large n."
```

```
class _50_PowXN_Brute:
    def myPow(self, x: float, n: int) -> float:
        if n < 0: x, n = 1/x, -n
        result = 1.0
        for _ in range(n): result *= x
        return result
```

PHASE 3 — OPTIMIZE

```
# "Fast exponentiation (exponentiation by squaring).
# If n is even:  $x^n = (x^2)^{(n/2)}$ . If n is odd:  $x^n = x * x^{(n-1)}$ .
# Halves n each step → O(log n) multiplications.
# Time: O(log n). Space: O(1) iteratively."
```

PHASE 4 — CLEAN CODE

```
class _50_PowXN:
    def myPow(self, x: float, n: int) -> float:
        if n < 0:
            x, n = 1 / x, -n
        result = 1.0
        while n:
            if n % 2 == 1:
                result *= x
            x *= x
            n //= 2
        return result
```

PHASE 5 — TEST & DEBUG

```
# x=2.0, n=10:
# n=10(even): x=4, n=5. n=5(odd): result=4, x=16, n=2. n=2(even): x=256, n=1.
# n=1(odd): result=4*256=1024, x=65536, n=0. return 1024.0 ✓
#
# x=2.0, n=-2: x=0.5, n=2. n=2(even): x=0.25, n=1. n=1(odd): result=0.25, n=0. return
0.25 ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time  $O(\log n)$ : halve the exponent each step. Space  $O(1)$ : iterative.  
# Recursive version is  $O(\log n)$  time but  $O(\log n)$  stack space.  
# Follow-up: integer exponentiation with modular arithmetic – same squaring, add mod  
at each step.  
# Follow-up: matrix exponentiation (for Fibonacci in  $O(\log n)$ ) – same pattern with  
matrices."
```

Math

□ #380 Insert Delete GetRandom O(1)

MED[Open on LeetCode](#)

Array · Hash Table · Math · Design

Lists: Grind 169

Problem

Implement the RandomizedSet class:

- **RandomizedSet()** Initializes the RandomizedSet object.
- **bool insert(int val)** Inserts an item val into the set if not present. Returns true if the item was not present, false otherwise.
- **bool remove(int val)** Removes an item val from the set if present. Returns true if the item was present, false otherwise.
- **int getRandom()** Returns a random element from the current set of elements (it's guaranteed that at least one element exists when this method is called). Each element must have the same probability of being returned.

You must implement the functions of the class such that each function works in average $O(1)$ time complexity.

Example 1:

```

Input
["RandomizedSet", "insert", "remove", "insert", "getRandom", "remove",
"insert", "getRandom"]
[[], [1], [2], [2], [], [1], [2], []]
Output
[null, true, false, true, 2, true, false, 2]

Explanation
RandomizedSet randomizedSet = new RandomizedSet();

```

Constraints:

- $-231 \leq \text{val} \leq 231 - 1$
- At most $2 * 10^5$ calls will be made to insert, remove, and getRandom.
- There will be at least one element in the data structure when getRandom is called.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Design a set supporting insert(val)→bool, remove(val)→bool, getRandom()→val
# all in  $O(1)$  average time. getRandom returns each element with equal probability.
Right?"
#

```

```

#

```

```

# "insert(1)→T, insert(2)→T, getRandom()→1 or 2 equally, remove(1)→T,
# getRandom()→2 always. ✓"

```

PHASE 2 — BRUTE FORCE

```

# "Use a Python set. insert/remove/search:  $O(1)$ . getRandom: convert to list  $O(n)$ ,
pick random.

```

```

# Time:  $O(n)$  per getRandom. Space:  $O(n)$ ."

```

```

class _380_RandomizedSet_Brute:
    def __init__(self):

```

```

        self.data = set()
    def insert(self, val: int) -> bool:
        if val in self.data: return False
        self.data.add(val); return True
    def remove(self, val: int) -> bool:
        if val not in self.data: return False
        self.data.remove(val); return True
    def getRandom(self) -> int:
        import random
        return random.choice(list(self.data)) # O(n)

```

PHASE 3 — OPTIMIZE

```

# "Array + HashMap: array holds all values, map holds {val->array_index}.
# getRandom: pick a random array index - O(1).
# remove: swap target with last element, update map, pop last - O(1).
# insert: append to array, update map - O(1)."

```

PHASE 4 — CLEAN CODE

```

class _380_RandomizedSet:
    def __init__(self):
        import random
        self._r = random
        self.vals = [] # array of values
        self.idx = {} # val -> index in vals
    def insert(self, val: int) -> bool:
        if val in self.idx: return False
        self.idx[val] = len(self.vals)
        self.vals.append(val)
        return True
    def remove(self, val: int) -> bool:
        if val not in self.idx: return False
        last = self.vals[-1]
        i = self.idx[val]
        self.vals[i] = last
        self.idx[last] = i
        self.vals.pop()
        del self.idx[val]
        return True
    def getRandom(self) -> int:
        return self._r.choice(self.vals)

```

PHASE 5 — TEST & DEBUG

```

# insert(1): vals=[1],idx={1:0}. insert(2): vals=[1,2],idx={1:0,2:1}.
# remove(1): last=2,i=0. vals[0]=2,idx[2]=0. pop->vals=[2],idx={2:0}. del idx[1].
# getRandom()->choice([2])->2 ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "All ops O(1) average: array ops + dict ops.
# The swap-with-last trick avoids O(n) shifts when removing from the middle.

```

```
# Edge case: when removing the last element, swap is a no-op (last==val).  
# Follow-up: Insert Delete GetRandom with duplicates (LC 381) – store sets of  
indices per value.  
# Follow-up: weighted random – use a prefix-sum array + binary search for  
getRandom."
```

Math

□ #1017 Convert to Base -2

MED[Open on LeetCode](#)

Math

Lists: Denny Zhang

Problem

Given an integer n , return a binary string representing its representation in base -2 .

Note that the returned string should not have leading zeros unless the string is "0".

Example 1:

Input: $n = 2$
Output: "110"
Explantion: $(-2)^2 + (-2)^1 = 2$

Example 2:

Input: $n = 3$
Output: "111"
Explantion: $(-2)^2 + (-2)^1 + (-2)^0 = 3$

Example 3:

Input: $n = 4$
Output: "100"
Explantion: $(-2)^2 = 4$

Constraints:

- $0 \leq n \leq 109$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Given integer n, return its representation in base -2 as a binary string.
# Negative base: (-2)^0=1, (-2)^1=-2, (-2)^2=4, (-2)^3=-8, ...
# Result has no leading zeros (except '0' itself). Right?"
#
# "n=2: 2 = (-2)^1*(-1) + (-2)^0*(0) + (-2)^2*... hmm: 4+(-2)=2 → '110' ✓
# n=3: '111' ✓ n=4: '100' ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Repeated division by -2. Remainder must be 0 or 1 (non-negative).
# If remainder is negative (when n is odd and negative), adjust: rem += 2, n += 1
# (or n = n//(-2)).
# Time: O(log n). Space: O(log n)."
```

```
class _1017_ConvertToBaseNeg2_Brute:
    def baseNeg2(self, n: int) -> str:
        if n == 0: return '0'
        result = []
        while n != 0:
            rem = n % (-2)
            n //= (-2) # Python floor division handles signs
            if rem < 0:
                rem += 2
                n += 1
            result.append(str(rem))
        return ''.join(reversed(result))
```

PHASE 3 — OPTIMIZE

```
# "Same iterative approach – already O(log n). The key is ensuring remainder is 0 or 1.
# Python's % with negative base may return negative remainders; adjust to [0,1]."
```

PHASE 4 — CLEAN CODE

```
class _1017_ConvertToBaseNeg2:
    def baseNeg2(self, n: int) -> str:
        if n == 0: return '0'
        bits = []
        while n:
            bits.append(n & 1) # LSB is always 0 or 1
            n = -(n >> 1) # divide by -2: right-shift then negate
        return ''.join(str(b) for b in reversed(bits))
```

PHASE 5 — TEST & DEBUG

```
# n=2: bits: 2&1=0, n=-(2>>1)=-1. (-1)&1=1, n=-(-1>>1)=1. 1&1=1, n=-(1>>1)=0.
# bits=[0,1,1]. reversed='110' ✓
# n=3: 3&1=1, n=-1. 1&1=1, n=-1. 1&1=1, n=0. bits=[1,1,1]→'111' ✓
```

```
# n=0: return '0' ✓
```

```
# PHASE 6 — COMPLEXITY & FOLLOW-UP
```

```
# "Time  $O(\log n)$ : number of bits in the representation.
```

```
# Space  $O(\log n)$ : bits array.
```

```
# The  $n = -(n >> 1)$  trick works because: dividing by  $-2$  = right-shift (divide by 2)  
then negate.
```

```
# Follow-up: Base  $-10$  (general negative base) – same structure, replace 2 with  
|base|.
```

```
# Follow-up: decode a base- $(-2)$  string back to integer – multiply by  $(-2)^{\text{position}}$ ."
```

Math

□ #3160 Find the Number of Distinct Colors Among the Balls

MED[Open on LeetCode](#)

Array · Hash Table · Simulation

Lists: CodeSignal

Problem

You are given an integer `limit` and a 2D array `queries` of size $n \times 2$.

There are `limit + 1` balls with distinct labels in the range `[0, limit]`. Initially, all balls are uncolored. For every query in `queries` that is of the form `[x, y]`, you mark ball `x` with the color `y`. After each query, you need to find the number of colors among the balls.

Return an array `result` of length `n`, where `result[i]` denotes the number of colors after `i`th query.

Note that when answering a query, lack of a color will not be considered as a color.

Example 1:

Input: `limit = 4, queries = [[1,4],[2,5],[1,3],[3,4]]`

Output: [1,2,2,3]

Explanation:



- After query 0, ball 1 has color 4.
- After query 1, ball 1 has color 4, and ball 2 has color 5.
- After query 2, ball 1 has color 3, and ball 2 has color 5.
- After query 3, ball 1 has color 3, ball 2 has color 5, and ball 3 has color 4.

Example 2:

Input: limit = 4, queries =
[[0,1],[1,2],[2,2],[3,4],[4,5]]

Output: [1,2,2,3,4]

Explanation:



- After query 0, ball 0 has color 1.
- After query 1, ball 0 has color 1, and ball 1 has color 2.
- After query 2, ball 0 has color 1, and balls 1 and 2 have color 2.
- After query 3, ball 0 has color 1, balls 1 and 2 have color 2, and ball 3 has color 4.
- After query 4, ball 0 has color 1, balls 1 and 2 have color 2, ball 3 has color 4, and ball 4 has color 5.

Constraints:

- $1 \leq \text{limit} \leq 10^9$
- $1 \leq n == \text{queries.length} \leq 10^5$
- $\text{queries}[i].\text{length} == 2$
- $0 \leq \text{queries}[i][0] \leq \text{limit}$
- $1 \leq \text{queries}[i][1] \leq 10^9$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"N balls (0 to limit). queries[i]=[ball,color]: color that ball. After each query return distinct color count."

#

"limit=4, queries=[[1,4],[2,5],[1,3],[3,4]]:

After q1:{4}→1. After q2:{4,5}→2. After q3:{3,5}→2. After q4:{3,5,4}→3 ✓"

PHASE 2 — BRUTE FORCE

"Dict ball→color. After each query, scan all values for distinct count."

Time: O(n) per query. Space: O(n)."

```
class _3160_DistinctColors_Brute:
    def queryResults(self, limit: int, queries: list) -> list:
        ball_color = {}
        result = []
        for ball, color in queries:
            ball_color[ball] = color
            result.append(len(set(ball_color.values())))
        return result
```

PHASE 3 — OPTIMIZE

"color_count tracks how many balls have each color. Update O(1) on each query."

len(color_count) = distinct colors."

PHASE 4 — CLEAN CODE

```
class _3160_DistinctColors:
    def queryResults(self, limit: int, queries: list) -> list:
        ball_color = {}
        color_count = {}
        result = []
        for ball, color in queries:
            if ball in ball_color:
                old = ball_color[ball]
                color_count[old] -= 1
                if color_count[old] == 0:
                    del color_count[old]
            ball_color[ball] = color
            color_count[color] = color_count.get(color, 0) + 1
            result.append(len(color_count))
        return result
```

PHASE 5 — TEST & DEBUG

queries=[[1,4],[2,5],[1,3],[3,4]]:

[1,4]: count={4:1}. distinct=1 ✓

[2,5]: count={4:1,5:1}. distinct=2 ✓

[1,3]: old=4,count[4]→0 del. count={5:1,3:1}. distinct=2 ✓

[3,4]: count={5:1,3:1,4:1}. distinct=3 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"O(1) per query. Space O(unique balls + colors). color_count keys = distinct active colors.

Follow-up: range coloring (paint balls l..r) – needs segment tree / lazy propagation.

Follow-up: undo queries – maintain a stack of (ball, old_color, new_color) and reverse ops."

JavaScript

Round 8 · Low · Medium · 7 questions

JavaScript

□ ★ #2627 Debounce ★

MED

[Open on LeetCode](#)

JavaScript

Lists: ★ Premium | Premium 98

Problem

Given a function `fn` and a time in milliseconds `t`, return a debounced version of that function.

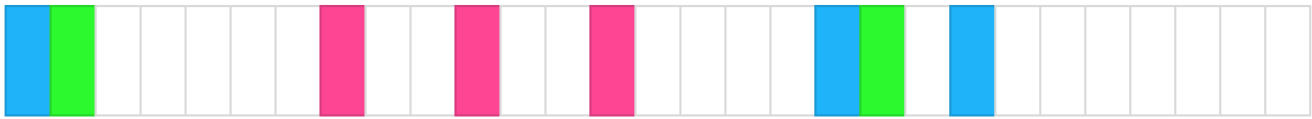
A debounced function is a function whose execution is delayed by `t` milliseconds and whose execution is cancelled if it is called again within that window of time. The debounced function should also receive the passed parameters.

For example, let's say `t = 50ms`, and the function was called at 30ms, 60ms, and 100ms.

The first 2 function calls would be cancelled, and the 3rd function call would be executed at 150ms.

If instead `t = 35ms`, The 1st call would be cancelled, the 2nd would be executed at 95ms, and the 3rd would be executed at 135ms.

Input Events



Debounced Events



The above diagram shows how debounce will transform events. Each rectangle represents 100ms and the debounce time is 400ms. Each color represents a different set of inputs.

Please solve it without using lodash's `_debounce()` function.

Example 1:

```
Input:
t = 50
calls = [
  {"t": 50, inputs: [1]},
  {"t": 75, inputs: [2]}
]
Output: [{"t": 125, inputs: [2]}]
Explanation:
```

Example 2:

```
Input:
t = 20
calls = [
  {"t": 50, inputs: [1]},
  {"t": 100, inputs: [2]}
]
Output: [{"t": 70, inputs: [1]}, {"t": 120, inputs: [2]}]
Explanation:
```

Example 3:

```

Input:
t = 150
calls = [
  {"t": 50, inputs: [1, 2]},
  {"t": 300, inputs: [3, 4]},
  {"t": 300, inputs: [5, 6]}
]
Output: [{"t": 200, inputs: [1,2]}, {"t": 450, inputs: [5, 6]}]

```

Constraints:

- $0 \leq t \leq 1000$
- $1 \leq \text{calls.length} \leq 10$
- $0 \leq \text{calls}[i].t \leq 1000$
- $0 \leq \text{calls}[i].\text{inputs.length} \leq 10$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"debounce(fn,t): delays fn until t ms after the last call. Each new call resets the timer."

#

"calls at t=0,30,60ms with t=50ms: fn fires once at t=110ms ✓"

PHASE 2 — BRUTE FORCE

"Track a timer id. On each call, cancel the previous timer and set a new one."

Time: $O(1)$ per call. Space: $O(1)$."

```
class _2627_Debounce_Brute:
```

```
    def debounce(self, fn, t_ms: int):
```

```
        import threading
```

```
        timer = [None]
```

```
        def debounced(*args):
```

```
            if timer[0]: timer[0].cancel()
```

```
            timer[0] = threading.Timer(t_ms/1000.0, fn, args)
```

```
            timer[0].start()
```

```
        return debounced
```

PHASE 3 — OPTIMIZE

"Same — this IS optimal. One clearTimeout + one setTimeout per call."

JavaScript: use closure to hold the timer reference."

PHASE 4 — CLEAN CODE

JavaScript:

```

# function debounce(fn, t) {
# let timer = null;
# return function(...args) {
# clearTimeout(timer);
# timer = setTimeout(() => fn.apply(this, args), t);
# };
# }

class _2627_Debounce:
    def debounce(self, fn, t_ms: int):
        import threading
        state = {'timer': None}
        def debounced(*args):
            if state['timer']: state['timer'].cancel()
            state['timer'] = threading.Timer(t_ms/1000.0, fn, args)
            state['timer'].start()
        return debounced

# PHASE 5 — TEST & DEBUG
# t=0: cancel(null), set timer(50ms). t=30: cancel timer, set timer(50ms).
# t=60: cancel timer, set timer(50ms). No more calls → fn fires at t=110ms ✓

# PHASE 6 — COMPLEXITY & FOLLOW-UP
# "O(1) per call: one cancel + one set. Space O(1): one timer reference.
# Debounce: wait until activity stops. Throttle: fire at most once per interval.
# Follow-up: leading debounce – fire immediately on first call, ignore for t ms
after.
# Follow-up: Throttle (LC 2676) – fire at most once per t ms regardless of call
frequency."

```


JavaScript

□ ★ #2628 JSON Deep Equal ★

MED

[Open on LeetCode](#)

JavaScript

Lists: ★ Premium | Premium 98

Problem

Given two values `o1` and `o2`, return a boolean value indicating whether two values, `o1` and `o2`, are deeply equal.

For two values to be deeply equal, the following conditions must be met:

- If both values are primitive types, they are deeply equal if they pass the `===` equality check.
- If both values are arrays, they are deeply equal if they have the same elements in the same order, and each element is also deeply equal according to these conditions.
- If both values are objects, they are deeply equal if they have the same keys, and the associated values for each key are also deeply equal according to these conditions.

You may assume both values are the output of `JSON.parse`. In other words, they are valid JSON.

Please solve it without using lodash's `_.isEqual()` function

Example 1:

```
Input: o1 = {"x":1,"y":2}, o2 = {"x":1,"y":2}
Output: true
Explanation: The keys and values match exactly.
```

Example 2:

```
Input: o1 = {"y":2,"x":1}, o2 = {"x":1,"y":2}
Output: true
Explanation: Although the keys are in a different order, they still match exactly.
```

Example 3:

```
Input: o1 = {"x":null,"L":[1,2,3]}, o2 = {"x":null,"L":["1","2","3"]}
Output: false
Explanation: The array of numbers is different from the array of strings.
```

Example 4:

```
Input: o1 = true, o2 = false
Output: false
Explanation: true !== false
```

Constraints:

- $1 \leq \text{JSON.stringify}(o1).length \leq 105$
- $1 \leq \text{JSON.stringify}(o2).length \leq 105$
- $\text{maxNestingDepth} \leq 1000$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Return true if two JSON values are deeply equal. Objects: all key-value pairs
match (order irrelevant).
# Arrays: same length and same elements at each index. Right?"
#
# "areDeeplyEqual({a:1,b:2},{a:1,b:2})→true ✓ areDeeplyEqual([1,[2]],[1,[3]])→false
✓"
```

PHASE 2 — BRUTE FORCE

```
# "JSON.stringify both – fails for object key-order differences.
# Proper: recursive comparison checking types, then structure."
class _2628_JSONDeepEqual_Brute:
    def areDeeplyEqual(self, o1, o2) -> bool:
        if type(o1) != type(o2): return False
        if isinstance(o1, dict):
            if set(o1) != set(o2): return False
            return all(self.areDeeplyEqual(o1[k], o2[k]) for k in o1)
        if isinstance(o1, list):
            if len(o1) != len(o2): return False
            return all(self.areDeeplyEqual(a, b) for a, b in zip(o1, o2))
        return o1 == o2
```

PHASE 3 — OPTIMIZE

```
# "Same recursive approach – already O(n) nodes. Check === first for fast-path."
```

PHASE 4 — CLEAN CODE

```
# JavaScript:
# function areDeeplyEqual(o1, o2) {
# if (o1 === o2) return true;
# if (typeof o1 !== typeof o2 || o1 === null || o2 === null) return o1 === o2;
# if (Array.isArray(o1) !== Array.isArray(o2)) return false;
# if (Array.isArray(o1)) return o1.length===o2.length &&
o1.every((v,i)=>areDeeplyEqual(v,o2[i]));
# const k1=Object.keys(o1), k2=Object.keys(o2);
# return k1.length===k2.length && k1.every(k=>k in o2 &&
areDeeplyEqual(o1[k],o2[k]));
# }
```

```
class _2628_JSONDeepEqual:
    def areDeeplyEqual(self, o1, o2) -> bool:
        if o1 is o2 or o1 == o2: return True
        if type(o1) != type(o2): return False
        if isinstance(o1, dict):
            return (set(o1)==set(o2) and
                    all(self.areDeeplyEqual(o1[k],o2[k]) for k in o1))
        if isinstance(o1, list):
            return (len(o1)==len(o2) and
                    all(self.areDeeplyEqual(a,b) for a,b in zip(o1,o2)))
        return False
```

PHASE 5 — TEST & DEBUG

```
# areDeeplyEqual({a:{b:1}},{a:{b:1}}):  
# both dicts, keys match. recurse 'a': both dicts, keys match. recurse 'b': 1==1  
True ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: visit each node once. Space $O(d)$: max recursion depth.

null requires special handling in JS: `typeof null === 'object'`.

Follow-up: Differences Between Two Objects (LC 2700) – return diff instead of bool.

Follow-up: handle circular refs – track visited pairs with WeakMap/set of pairs."

JavaScript

□ ★ #2630 Memoize ★

MED

[Open on LeetCode](#)

JavaScript

Lists: ★ Premium | Premium 98

Problem

Given a function `fn`, return a memoized version of that function.

A memoized function is a function that will never be called twice with the same inputs. Instead it will return a cached value.

`fn` can be any function and there are no constraints on what type of values it accepts. Inputs are considered identical if they are `===` to each other.

Example 1:

Input:

```
getInputs = () => [[2,2],[2,2],[1,2]]  
fn = function (a, b) { return a + b; }
```

Output: [{"val":4,"calls":1},{ "val":4,"calls":1},{ "val":3,"calls":2}]

Explanation:

```
const inputs = getInputs();  
const memoized = memoize(fn);  
for (const arr of inputs) {
```

Example 2:

Input:

```
getInputs = () => [{},{},[{}],[{}],[{}],[{}]]
```

```
fn = function (a, b) { return ({...a, ...b}); }
```

```
Output: [{"val":{},"calls":1},{"val":{},"calls":2},{"val":{},"calls":3}]
```

Explanation:

Merging two empty objects will always result in an empty object. It may seem like there should only be 1 call to fn() because of cache-hits, however none of those objects are === to each other.

Example 3:

Input:

```
getInputs = () => { const o = {}; return [[o,o],[o,o],[o,o]]; }
```

```
fn = function (a, b) { return ({...a, ...b}); }
```

```
Output: [{"val":{},"calls":1},{"val":{},"calls":1},{"val":{},"calls":1}]
```

Explanation:

Merging two empty objects will always result in an empty object. The 2nd and 3rd third function calls result in a cache-hit. This is because every object passed in is identical.

Constraints:

- $1 \leq \text{inputs.length} \leq 105$
- $0 \leq \text{inputs.flat().length} \leq 105$
- $\text{inputs}[i][j] \neq \text{NaN}$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"memoize(fn): returns a cached version. Same args → return cached result, skip fn call."

#

"memoize(sum): sum(2,3)→5 (called). sum(2,3)→5 (cached, fn not called). sum(1,2)→3 (called). ✓"

PHASE 2 — BRUTE FORCE

"Map with stringified args as key. Check on each call; compute and cache if miss."

```
class _2630_Memoize_Brute:
```

```
    def memoize(self, fn):
```

```
        cache = {}
```

```
        def memoized(*args):
```

```
            if args not in cache:
```

```
                cache[args] = fn(*args)
```

```
        return cache[args]
    return memoized
```

PHASE 3 — OPTIMIZE

"Same $O(1)$ lookup approach. For JS use `JSON.stringify` for complex/nested args."

PHASE 4 — CLEAN CODE

```
# JavaScript:
# function memoize(fn) {
#   const cache = new Map();
#   return function(...args) {
#     const key = JSON.stringify(args);
#     if (!cache.has(key)) cache.set(key, fn.apply(this, args));
#     return cache.get(key);
#   };
# }
```

```
class _2630_Memoize:
    def memoize(self, fn):
        cache = {}
        def memoized(*args):
            key = str(args)
            if key not in cache:
                cache[key] = fn(*args)
            return cache[key]
        return memoized
```

PHASE 5 — TEST & DEBUG

```
# fn=lambda a,b: a+b. memoized(2,3): key='(2, 3)', miss→compute 5, cache. return 5.
# memoized(2,3): key='(2, 3)', hit → return 5 (fn not called) ✓.
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "First call  $O(fn)$ . Cache hit  $O(1)$ . Space  $O(\text{unique arg combinations} * \text{result size})$ .
# JSON.stringify adds  $O(\text{arg\_size})$  per call – acceptable for numbers/strings.
# Follow-up: LRU-bounded memoize – evict LRU when cache exceeds limit.
# Follow-up: async memoize – cache the Promise, not the resolved value."
```

JavaScript

□ ★ #2633 Convert Object to JSON String ★ **MED**

[Open on LeetCode](#)

JavaScript

Lists: ★ Premium | Premium 98

Problem

Given a value, return a valid JSON string of that value. The value can be a string, number, array, object, boolean, or null. The returned string should not include extra spaces. The order of keys should be the same as the order returned by `Object.keys()`.

Please solve it without using the built-in `JSON.stringify` method.

Example 1:

```
Input: object = {"y":1,"x":2}
Output: {"y":1,"x":2}
Explanation:
Return the JSON representation.
Note that the order of keys should be the same as the order returned by
Object.keys().
```

Example 2:


```
Input: object = {"a":"str","b":-12,"c":true,"d":null}
Output: {"a":"str","b":-12,"c":true,"d":null}
Explanation:
The primitives of JSON are strings, numbers, booleans, and null.
```

Example 3:

```
Input: object = {"key":{"a":1,"b":[{"},null,"Hello"]}}
Output: {"key":{"a":1,"b":[{"},null,"Hello"]}}
Explanation:
Objects and arrays can include other objects and arrays.
```

Example 4:

```
Input: object = true
Output: true
Explanation:
Primitive types are valid inputs.
```

Constraints:

- value is a valid JSON value
- $1 \leq \text{JSON.stringify(object).length} \leq 105$
- $\text{maxNestingLevel} \leq 1000$
- all strings contain only alphanumeric characters

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Implement `JSON.stringify` without using it. Handle null, bool, number, string, array, object."

#

`"jsonify({a:2,b:[1,2,3]}) → '{"a":2,"b":[1,2,3]}' ✓"`

PHASE 2 — BRUTE FORCE

"Recursive: dispatch on type. Build string for each type appropriately."

```
class _2633_ObjectToJSONString_Brute:
```

```
    def jsonify(self, obj) -> str:
```

```
        if obj is None: return 'null'
```

```
        if isinstance(obj, bool): return 'true' if obj else 'false'
```

```

    if isinstance(obj, (int,float)): return str(obj)
    if isinstance(obj, str): return f'"{obj}"'
    if isinstance(obj, list):
        return '[' + ','.join(self.jsonify(v) for v in obj) + ']'
    if isinstance(obj, dict):
        return '{' + ','.join(f'"{k}":{self.jsonify(v)}' for k,v in
obj.items()) + '}'
    return 'null'

```

PHASE 3 — OPTIMIZE

"Same — already $O(n)$. Use join to avoid $O(n^2)$ string concatenation."

PHASE 4 — CLEAN CODE (same as brute — already clean)

```

# JavaScript:
# function jsonStringify(obj) {
# if (obj === null) return 'null';
# if (typeof obj === 'boolean') return String(obj);
# if (typeof obj === 'number') return String(obj);
# if (typeof obj === 'string') return `"${obj}"`;
# if (Array.isArray(obj)) return `[${obj.map(jsonStringify).join(',')}]`;
# return
`[${Object.entries(obj).map(([k,v])=>`"${k}":${jsonStringify(v)}`)].join(',')}`;
# }

```

PHASE 5 — TEST & DEBUG

```

# obj=[null,true,1,"foo"]:
# array. elements: 'null','true','1','"foo"'. joined: 'null,true,1,"foo"'. wrap:
'[null,true,1,"foo"]' ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n)$: one visit per node. Space $O(d)$: recursion depth.
bool must be checked before number (isinstance(True,int) is True in Python).
Follow-up: handle circular references — track visited with a Set.
Follow-up: pretty-print with indentation — add indent parameter and track depth."

JavaScript

□ ★ #2636 Promise Pool ★

MED[Open on LeetCode](#)

JavaScript · Concurrency

Lists: ★ Premium | Premium 98

Problem

Given an array of asynchronous functions `functions` and a pool limit `n`, return an asynchronous function `promisePool`. It should return a promise that resolves when all the input functions resolve.

Pool limit is defined as the maximum number promises that can be pending at once. `promisePool` should begin execution of as many functions as possible and continue executing new functions when old promises resolve. `promisePool` should execute `functions[i]` then `functions[i + 1]` then `functions[i + 2]`, etc. When the last promise resolves, `promisePool` should also resolve. For example, if `n = 1`, `promisePool` will execute one function at a time in series. However, if `n =`

2, it first executes two functions. When either of the two functions resolve, a 3rd function should be executed (if available), and so on until there are no functions left to execute.

You can assume all functions never reject. It is acceptable for `promisePool` to return a promise that resolves any value.

Example 1:

```
Input:
functions = [
  () => new Promise(res => setTimeout(res, 300)),
  () => new Promise(res => setTimeout(res, 400)),
  () => new Promise(res => setTimeout(res, 200))
]
n = 2
Output: [[300,400,500],500]
```

Example 2:

```
Input:
functions = [
  () => new Promise(res => setTimeout(res, 300)),
  () => new Promise(res => setTimeout(res, 400)),
  () => new Promise(res => setTimeout(res, 200))
]
n = 5
Output: [[300,400,200],400]
```

Example 3:

```
Input:
functions = [
  () => new Promise(res => setTimeout(res, 300)),
  () => new Promise(res => setTimeout(res, 400)),
  () => new Promise(res => setTimeout(res, 200))
]
n = 1
Output: [[300,700,900],900]
```

Constraints:

- $0 \leq \text{functions.length} \leq 10$
- $1 \leq n \leq 10$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Execute async functions with at most n concurrent. Return promise resolving when
all done."
#
# "promisePool([fn1,fn2,fn3,fn4], n=2): run fn1+fn2, as each finishes start fn3 then
fn4 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Promise.all on all functions – no concurrency limit, may overwhelm resources."
```

```
class _2636_PromisePool_Brute:
    async def pool(self, functions, n: int):
        import asyncio
        await asyncio.gather(*[fn() for fn in functions])
```

PHASE 3 — OPTIMIZE

```
# "n worker coroutines, each pulls from a shared queue.
# Each worker runs one fn at a time sequentially. n workers run in parallel.
# await all workers → all tasks complete."
```

PHASE 4 — CLEAN CODE

```
# JavaScript:
# async function promisePool(functions, n) {
#   const queue = [...functions];
#   async function worker() {
#     while (queue.length) await queue.shift();
#   }
#   await Promise.all(Array.from({length:Math.min(n,function.length)}, worker));
# }
```

```
class _2636_PromisePool:
    async def pool(self, functions, n: int):
        import asyncio
        sem = asyncio.Semaphore(n)
        async def run(fn):
            async with sem: await fn()
        await asyncio.gather(*[run(fn) for fn in functions])
```

PHASE 5 — TEST & DEBUG

```
# [fn1,fn2,fn3,fn4], n=2:
```

```
# worker1: await fn1(), await fn3(). worker2: await fn2(), await fn4().  
# Both workers run concurrently → max 2 concurrent at any time ✓
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time optimal: $\text{ceiling}(\text{total_tasks}/n)$ rounds if tasks equal duration.

Space $O(n)$: n concurrent promises. Worker pattern handles unequal durations naturally.

Follow-up: cancel on first failure – `Promise.race` with rejection propagation.

Follow-up: retry failed tasks – wrap `fn()` in a retry loop inside the worker."

JavaScript

□ ★ #2675 Array of Objects to Matrix ★

MED

[Open on LeetCode](#)

JavaScript · Array

Lists: ★ Premium | Premium 98

Problem

Write a function that converts an array of objects `arr` into a matrix `m`.

`arr` is an array of objects or arrays. Each item in the array can be deeply nested with child arrays and child objects. It can also contain numbers, strings, booleans, and null values.

The first row `m` should be the column names. If there is no nesting, the column names are the unique keys within the objects. If there is nesting, the column names are the respective paths in the object separated by ".".

Each of the remaining rows corresponds to an object in `arr`. Each value in the matrix corresponds to a value in an object. If a given object doesn't contain a value for a given column, the cell should contain an empty

string "".

The columns in the matrix should be in lexicographically ascending order.

Example 1:

```
Input:
arr = [
  {"b": 1, "a": 2},
  {"b": 3, "a": 4}
]
Output:
[
  ["a", "b"],
```

Example 2:

```
Input:
arr = [
  {"a": 1, "b": 2},
  {"c": 3, "d": 4},
  {}
]
Output:
[
```

Example 3:

```
Input:
arr = [
  {"a": {"b": 1, "c": 2}},
  {"a": {"b": 3, "d": 4}}
]
Output:
[
  ["a.b", "a.c", "a.d"],
```

Example 4:


```

Input:
arr = [
  [{"a": null}],
  [{"b": true}],
  [{"c": "x"}]
]
Output:
[

```

Example 5:

```

Input:
arr = [
  {},
  {},
  {}
]
Output:
[

```

Constraints:

- arr is a valid JSON array
- $1 \leq \text{arr.length} \leq 1000$
- unique keys ≤ 1000

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Convert array of objects to a matrix. Row 0 = sorted flattened keys (dot notation for nested).

Remaining rows = values for each object ('' if key missing). Right?"

#

"[{a:1,b:2},{b:3,c:4}] → [['a','b','c'],[1,2,''],['',3,4]] ✓"

PHASE 2 — BRUTE FORCE

"Flatten each object with dot notation, collect all keys, build matrix row by row."

```

class _2675_ArrayOfObjectsToMatrix_Brute:
    def jsonToMatrix(self, arr: list) -> list:
        def flatten(obj, prefix=''):
            result = {}
            if isinstance(obj, dict):

```

```

        for k,v in obj.items():
            result.update(flatten(v, f'{prefix}.{k}' if prefix else k))
    else:
        result[prefix] = obj
    return result
flat = [flatten(o) for o in arr]
keys = sorted(set(k for f in flat for k in f))
return [keys] + [[f.get(k,'') for k in keys] for f in flat]

```

PHASE 3 — OPTIMIZE

"Same $O(n*k)$ — already optimal. Single pass to collect keys."

PHASE 4 — CLEAN CODE (same as brute — already clean)

PHASE 5 — TEST & DEBUG

`arr=[{a:1,b:{c:2}},{a:3}]`:

flat: `{'a':1,'b.c':2}` and `{'a':3}`. keys=`['a','b.c']`.

rows: `[1,2]` and `[3,'']` → `[['a','b.c'],[1,2],[3,'']]` ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n*k*d)$: n objects, k unique keys, d nesting depth. Space $O(n*k)$: matrix.

Dot notation is the CSV-export standard for nested JSON.

Follow-up: handle arrays inside objects — use bracket notation `[0],[1]`.

Follow-up: reverse (matrix to array of objects) — parse dot-paths back to nested."

JavaScript

□ ★ #2700 Differences Between Two Objects ★

MED[Open on LeetCode](#)

JavaScript

Lists: ★ Premium | Premium 98

Problem

Write a function that accepts two deeply nested objects or arrays `obj1` and `obj2` and returns a new object representing their differences.

The function should compare the properties of the two objects and identify any changes. The returned object should only contains keys where the value is different from `obj1` to `obj2`.

For each changed key, the value should be represented as an array [`obj1` value, `obj2` value]. Keys that exist in one object but not in the other should not be included in the returned object. The end result should be a deeply nested object where each leaf value is a difference array.

When comparing two arrays, the indices of the arrays are considered to be their keys.

You may assume that both objects are the output of `JSON.parse`.

Example 1:

Input:

```
obj1 = {}
```

```
obj2 = {
```

```
  "a": 1,
```

```
  "b": 2
```

```
}
```

Output: {}

Explanation: There were no modifications made to `obj1`. New keys "a" and "b" appear in `obj2`, but keys that are added or removed should be ignored.

Example 2:

Input:

```
obj1 = {
```

```
  "a": 1,
```

```
  "v": 3,
```

```
  "x": [],
```

```
  "z": {
```

```
    "a": null
```

```
}
```

Example 3:

Input:

```
obj1 = {
```

```
  "a": 5,
```

```
  "v": 6,
```

```
  "z": [1, 2, 4, [2, 5, 7]]
```

```
}
```

```
obj2 = {
```

```
  "a": 5,
```

Example 4:

```

Input:
obj1 = {
  "a": {"b": 1},
}
obj2 = {
  "a": [5],
}
Output:

```

Example 5:

```

Input:
obj1 = {
  "a": [1, 2, {}],
  "b": false
}
obj2 = {
  "b": false,
  "a": [1, 2, {}]
}

```

Constraints:

- obj1 and obj2 are valid JSON objects or arrays
- $2 \leq \text{JSON.stringify(obj1).length} \leq 104$
- $2 \leq \text{JSON.stringify(obj2).length} \leq 104$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```

# "Return an object showing paths where two JSON objects differ.
# [obj1_val, obj2_val] at leaf differences. Skip keys present in only one object."
#
# "diff({a:1,b:2},{a:1,b:3}) → {b:[2,3]} ✓
# diff({a:{b:1}},{a:{b:2}}) → {a:{b:[1,2]}} ✓"

```

PHASE 2 — BRUTE FORCE

```

# "Recursive: for each key in obj1 that also exists in obj2:
# if both are objects, recurse and collect non-empty sub-diffs.
# if values differ (or types differ), mark as [v1,v2]. If equal, skip."

```

```

class _2700_DiffBetweenObjects_Brute:
    def objDiff(self, obj1, obj2) -> dict:

```

```

if not isinstance(obj1,dict) or not isinstance(obj2,dict):
    return {} if obj1 == obj2 else [obj1,obj2]
result = {}
for k in obj1:
    if k not in obj2: continue
    sub = self.objDiff(obj1[k], obj2[k])
    if sub != {}: result[k] = sub
return result

```

PHASE 3 — OPTIMIZE

"Same — already $O(n)$. The key insight: skip keys in only one object."

PHASE 4 — CLEAN CODE (same as brute — already clean)

```

# JavaScript:
# function objDiff(obj1, obj2) {
# if (typeof obj1!=='object' || obj1===null || typeof obj2!=='object' || obj2===null)
# return obj1===obj2 ? {} : [obj1,obj2];
# const result={};
# for (const k of Object.keys(obj1)) {
# if (!(k in obj2)) continue;
# const diff=objDiff(obj1[k],obj2[k]);
# if (Object.keys(diff).length || Array.isArray(diff)) result[k]=diff;
# }
# return result;
# }

```

PHASE 5 — TEST & DEBUG

```

# obj1={a:1,b:{c:2,d:3}}, obj2={a:1,b:{c:2,d:4}}:
# 'a': 1==1 → {}. 'b': both dicts, recurse. 'c': 2==2 → {}. 'd': 3≠4 → [3,4].
# result={b:{d:[3,4]}} ✓

```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```

# "Time  $O(n)$ : one visit per node. Space  $O(d)$ : recursion depth.
# null check: typeof null==='object' in JS — must check explicitly.
# Follow-up: include keys present in only one object — mark as [value, undefined].
# Follow-up: JSON Deep Equal (LC 2628) — return bool instead of diff."

```

Round 9 · Low · Hard

Round 9

Low Priority · Hard

7 questions · 2 patterns

Dynamic Programming #10 #115 #123 #132
#312 #1000

Math #68

Dynamic Programming

Round 9 · Low · Hard · 6 questions

Dynamic Programming

□ #10 Regular Expression Matching

HAR[Open on LeetCode](#)

String · Dynamic Programming · Recursion

Lists: Denny Zhang

Problem

Given an input string *s* and a pattern *p*, implement regular expression matching with support for '.' and '*' where:

- '.' Matches any single character.
- '*' Matches zero or more of the preceding element.

Return a boolean indicating whether the matching covers the entire input string (not partial).

Example 1:

Input: *s* = "aa", *p* = "a"

Output: false

Explanation: "a" does not match the entire string "aa".

Example 2:

Input: *s* = "aa", *p* = "a*"

Output: true

Explanation: '*' means zero or more of the preceding element, 'a'. Therefore, by repeating 'a' once, it becomes "aa".

Example 3:

Input: s = "ab", p = ".*"

Output: true

Explanation: ".*" means "zero or more (*) of any character (.)".

Constraints:

- $1 \leq s.length \leq 20$
- $1 \leq p.length \leq 20$
- s contains only lowercase English letters.
- p contains only lowercase English letters, '.', and '*'.
- It is guaranteed for each appearance of the character '*', there will be a previous valid character to match.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# '.' matches any single char. '*' matches zero or more of the preceding element.
# Full string must match. Right?"
#
# "s='aa',p='a*'->true ✓ s='aa',p='a'->false ✓ s='aab',p='c*a*b'->true ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Recursion with memo: match(i,j)=does s[i:] match p[j:]?"
# If p[j+1]=='*': zero uses (match i,j+2) or one+ uses (first_char and match i+1,j).
# Time: O(m*n) with memo. Space: O(m*n)."
```

```
class _10_RegexMatching_Brute:
    def isMatch(self, s: str, p: str) -> bool:
        from functools import lru_cache
        @lru_cache(maxsize=None)
        def dp(i, j):
            if j == len(p): return i == len(s)
            first = i < len(s) and (p[j] == s[i] or p[j] == '.')
            if j+1 < len(p) and p[j+1] == '*':
                return dp(i, j+2) or (first and dp(i+1, j))
```

```
        return first and dp(i+1, j+1)
    return dp(0, 0)
```

PHASE 3 — OPTIMIZE

"Bottom-up 2D DP. $dp[i][j]$ =does $s[i:]$ match $p[j:]$? Fill right-to-left, bottom-to-top.

Space: $O(n)$ with rolling rows."

PHASE 4 — CLEAN CODE

```
class _10_RegexMatching:
    def isMatch(self, s: str, p: str) -> bool:
        m, n = len(s), len(p)
        dp = [False]*(n+1)
        dp[n] = True
        for i in range(m, -1, -1):
            new = [False]*(n+1)
            new[n] = (i == m)
            for j in range(n-1, -1, -1):
                first = (i < m) and (p[j] == s[i] or p[j] == '.')
                if j+1 < n and p[j+1] == '*':
                    new[j] = dp[j+2] or (first and dp[j])
                    if j+2 <= n: new[j] = new[j] or dp[j+2]
                else:
                    new[j] = first and dp[j+1]
            dp = new
        return dp[0]
```

PHASE 5 — TEST & DEBUG

$s='aab'$, $p='c*a*b'$: $c \rightarrow \text{skip}(j+2)$, $a \rightarrow \text{skip or use}$, $b \rightarrow \text{match}$. $dp[0]=\text{True}$ ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(m*n)$. Space $O(n)$: two rolling rows.

'*' has two branches: zero occurrences (skip $j+2$) or one-plus (consume $s[i]$, stay at j).

Follow-up: Wildcard Matching (LC 44) – '*' matches any sequence. Same DP, simpler transitions."

Dynamic Programming

□ #115 Distinct Subsequences

HAR[Open on LeetCode](#)

String · Dynamic Programming

Lists: Denny Zhang

Problem

Given two strings *s* and *t*, return the number of distinct subsequences of *s* which equals *t*.

The test cases are generated so that the answer fits on a 32-bit signed integer.

Example 1:

```
Input: s = "rabbbit", t = "rabbit"
Output: 3
Explanation:
As shown below, there are 3 ways you can generate "rabbit" from s.
rabbbit
rabbbit
rabbbit
```

Example 2:

```
Input: s = "babgbag", t = "bag"
Output: 5
Explanation:
As shown below, there are 5 ways you can generate "bag" from s.
babgbag
babgbag
babgbag
babgbag
babgbag
```

Constraints:

- $1 \leq s.length, t.length \leq 1000$
- s and t consist of English letters.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

```
# "Count distinct subsequences of s that equal t. Right?"
#
# "s='rabbbit',t='rabbit'→3 ✓ s='babgbag',t='bag'→5 ✓"
```

PHASE 2 — BRUTE FORCE

```
# "Recursion: count(i,j)=ways s[i:] produces t[j:].
# j==len(t)→1. i==len(s)→0. s[i]==t[j]: count(i+1,j+1)+count(i+1,j). else:
count(i+1,j)."
```

```
class _115_DistinctSubsequences_Brute:
    def numDistinct(self, s: str, t: str) -> int:
        from functools import lru_cache
        @lru_cache(maxsize=None)
        def dp(i, j):
            if j == len(t): return 1
            if i == len(s): return 0
            res = dp(i+1, j)
            if s[i] == t[j]: res += dp(i+1, j+1)
            return res
        return dp(0, 0)
```

PHASE 3 — OPTIMIZE

```
# "Bottom-up DP. dp[j]=ways to form t[j:] from s[i:]. Roll right-to-left.
# Time: O(m*n). Space: O(n)."
```

PHASE 4 — CLEAN CODE

```
class _115_DistinctSubsequences:
    def numDistinct(self, s: str, t: str) -> int:
        dp = [0]*len(t) + [1]
        for ch in reversed(s):
            for j in range(len(t)-1, -1, -1):
                if ch == t[j]:
                    dp[j] += dp[j+1]
        return dp[0]
```

PHASE 5 — TEST & DEBUG

```
# s='rabbbit',t='rabbit': three 'b's in s for two 'b's in t → 3 ways ✓
# dp updated right-to-left prevents using same s[i] twice in one match.
```

PHASE 6 — COMPLEXITY & FOLLOW-UP

```
# "Time O(m*n). Space O(n): one rolling array."
```

```
# Right-to-left update is essential – prevents reuse of s[i] across matches.  
# Follow-up: print all distinct subsequences – backtrack through the DP table.  
# Follow-up: Edit Distance (LC 72) – related: counts edits instead of matching occurrences."
```

Dynamic Programming

□ #123 Best Time to Buy and Sell Stock III

HAR[Open on LeetCode](#)

Array · Dynamic Programming

Lists: Denny Zhang

Problem

You are given an array `prices` where `prices[i]` is the price of a given stock on the *i*th day.

Find the maximum profit you can achieve. You may complete at most two transactions.

Note: You may not engage in multiple transactions simultaneously (i.e., you must sell the stock before you buy again).

Example 1:

Input: `prices = [3,3,5,0,0,3,1,4]`

Output: 6

Explanation: Buy on day 4 (price = 0) and sell on day 6 (price = 3), profit = 3-0 = 3.

Then buy on day 7 (price = 1) and sell on day 8 (price = 4), profit = 4-1 = 3.

Example 2:

Input: prices = [1,2,3,4,5]

Output: 4

Explanation: Buy on day 1 (price = 1) and sell on day 5 (price = 5), profit = 5-1 = 4.

Note that you cannot buy on day 1, buy on day 2 and sell them later, as you are engaging multiple transactions at the same time. You must sell before buying again.

Example 3:

Input: prices = [7,6,4,3,1]

Output: 0

Explanation: In this case, no transaction is done, i.e. max profit = 0.

Constraints:

- $1 \leq \text{prices.length} \leq 105$
- $0 \leq \text{prices}[i] \leq 105$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"At most 2 transactions. Maximize profit. No holding two stocks simultaneously. Right?"

#

"prices=[3,3,5,0,0,3,1,4]: buy@0,sell@3,buy@1,sell@4 → profit=3+3=6 ✓"

PHASE 2 — BRUTE FORCE

"Try all pairs of buy1<sell1<=buy2<sell2. Time: $O(n^4)$. Space: $O(1)$."

```
class _123_StockIII_Brute:
```

```
    def maxProfit(self, prices: list) -> int:
```

```
        n, best = len(prices), 0
```

```
        for b1 in range(n):
```

```
            for s1 in range(b1+1, n):
```

```
                for b2 in range(s1, n):
```

```
                    for s2 in range(b2+1, n):
```

```
                        best = max(best,
```

```
prices[s1]-prices[b1]+prices[s2]-prices[b2])
```

```
        return best
```

PHASE 3 — OPTIMIZE

"State machine: 4 states track max profit at each stage.

buy1,sell1,buy2,sell2 updated in one pass. Time: $O(n)$. Space: $O(1)$."

PHASE 4 — CLEAN CODE


```
class _123_StockIII:
    def maxProfit(self, prices: list) -> int:
        buy1 = buy2 = float('-inf')
        sell1 = sell2 = 0
        for p in prices:
            buy1 = max(buy1, -p)
            sell1 = max(sell1, buy1 + p)
            buy2 = max(buy2, sell1 - p)
            sell2 = max(sell2, buy2 + p)
        return sell2
```

PHASE 5 — TEST & DEBUG

prices=[3,3,5,0,0,3,1,4]:

...after full pass: sell2=6 ✓ (buy@0,sell@3 + buy@1,sell@4)

PHASE 6 — COMPLEXITY & FOLLOW-UP

"O(n) one pass. O(1) four state variables. Elegant state machine for at-most-k constraint."

Follow-up: at most k transactions (LC 188) – generalize to 2k variables or O(n*k) DP.

Follow-up: unlimited (LC 122) – sum all positive consecutive differences."

Dynamic Programming

□ #132 Palindrome Partitioning II

HAR[Open on LeetCode](#)

String · Dynamic Programming

Lists: Denny Zhang

Problem

Given a string s , partition s such that every substring of the partition is a palindrome.

Return the minimum cuts needed for a palindrome partitioning of s .

Example 1:

Input: $s = \text{"aab"}$

Output: 1

Explanation: The palindrome partitioning $[\text{"aa"}, \text{"b"}]$ could be produced using 1 cut.

Example 2:

Input: $s = \text{"a"}$

Output: 0

Example 3:

Input: $s = \text{"ab"}$

Output: 1

Constraints:

- $1 \leq s.length \leq 2000$

- s consists of lowercase English letters only.

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Minimum cuts to partition s into palindromes. Right?"

#

"s='aab'→1 ('aa'|'b') ✓ s='a'→0 ✓ s='ab'→1 ✓"

PHASE 2 — BRUTE FORCE

"Recursion: minCuts(i)=min cuts for s[i:]. Try each palindrome prefix, recurse on rest."

```
class _132_PalindromePartitioningII_Brute:
    def minCut(self, s: str) -> int:
        from functools import lru_cache
        n = len(s)
        @lru_cache(maxsize=None)
        def is_pal(l, r):
            return l >= r or (s[l]==s[r] and is_pal(l+1,r-1))
        @lru_cache(maxsize=None)
        def dp(i):
            if i == n: return -1
            return min(1 + dp(j+1) for j in range(i, n) if is_pal(i, j))
        return dp(0)
```

PHASE 3 — OPTIMIZE

"Precompute palindrome table $O(n^2)$. $dp[i]$ =min cuts for $s[:i+1]$. $O(n^2)$ time $O(n)$ space."

PHASE 4 — CLEAN CODE

```
class _132_PalindromePartitioningII:
    def minCut(self, s: str) -> int:
        n = len(s)
        is_pal = [[False]*n for _ in range(n)]
        for i in range(n-1,-1,-1):
            for j in range(i,n):
                is_pal[i][j] = (s[i]==s[j]) and (j-i<=2 or is_pal[i+1][j-1])
        dp = list(range(-1, n))
        for i in range(n-1,-1,-1):
            for j in range(i,n):
                if is_pal[i][j]:
                    dp[i] = min(dp[i], 1+dp[j+1])
        return dp[0]
```

PHASE 5 — TEST & DEBUG

s='aab': is_pal[0][1]=T(aa). dp[2]=0(b), dp[1]=1(ab→b=1cut), dp[0]:
is_pal[0][0]T→1+dp[1]=2, is_pal[0][1]T→1+dp[2]=1. dp[0]=1 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n^2)$. Space $O(n^2)$ palindrome table + $O(n)$ dp.

Follow-up: Palindrome Partitioning I (LC 131) – return all partitions (backtracking + table).

Follow-up: Palindrome Partitioning IV (LC 1745) – fixed 3 parts; use same palindrome table."

Dynamic Programming

□ #312 Burst Balloons

HAR[Open on LeetCode](#)

Array · Dynamic Programming

Lists: Denny Zhang

Problem

You are given n balloons, indexed from 0 to $n - 1$. Each balloon is painted with a number on it represented by an array `nums`. You are asked to burst all the balloons.

If you burst the i th balloon, you will get $\text{nums}[i - 1] * \text{nums}[i] * \text{nums}[i + 1]$ coins. If $i - 1$ or $i + 1$ goes out of bounds of the array, then treat it as if there is a balloon with a 1 painted on it.

Return the maximum coins you can collect by bursting the balloons wisely.

Example 1:

Input: `nums = [3,1,5,8]`

Output: 167

Explanation:

`nums = [3,1,5,8] --> [3,5,8] --> [3,8] --> [8] --> []`

`coins = 3*1*5 + 3*5*8 + 1*3*8 + 1*8*1 = 167`

Example 2:

Input: nums = [1,5]

Output: 10

Constraints:

- $n == \text{nums.length}$
- $1 \leq n \leq 300$
- $0 \leq \text{nums}[i] \leq 100$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Burst balloon i: earn $\text{nums}[\text{left}] * \text{nums}[i] * \text{nums}[\text{right}]$. Pad array with 1s on both ends.

Maximize total coins. Right?"

#

"nums=[3,1,5,8]→167 ✓"

PHASE 2 — BRUTE FORCE

"Try all permutations of burst order. Time: $O(n!)$. Space: $O(n)$."

```
class _312_BurstBalloons_Brute:
    def maxCoins(self, nums: list) -> int:
        from itertools import permutations
        nums = [1]+nums+[1]; n = len(nums)
        best = 0
        for perm in permutations(range(1,n-1)):
            arr, coins = list(nums), 0
            for i in perm:
                idx = arr.index(nums[i])
                coins += arr[idx-1]*arr[idx]*arr[idx+1]
                arr.pop(idx)
            best = max(best, coins)
        return best
```

PHASE 3 — OPTIMIZE

"Interval DP: $\text{dp}[l][r] = \text{max coins bursting all in open interval } (l,r)$.

Think backwards: k is the LAST balloon burst in (l,r).

$\text{dp}[l][r] = \text{max over } k: \text{dp}[l][k] + \text{nums}[l] * \text{nums}[k] * \text{nums}[r] + \text{dp}[k][r]$.

Time: $O(n^3)$. Space: $O(n^2)$."

PHASE 4 — CLEAN CODE

```
class _312_BurstBalloons:
    def maxCoins(self, nums: list) -> int:
        nums = [1]+nums+[1]; n = len(nums)
```

```
dp = [[0]*n for _ in range(n)]
for ln in range(2, n):
    for l in range(n-ln):
        r = l+ln
        for k in range(l+1, r):
            dp[l][r] = max(dp[l][r],
                           nums[l]*nums[k]*nums[r]+dp[l][k]+dp[k][r])
return dp[0][n-1]
```

PHASE 5 — TEST & DEBUG

nums=[1,3,1,5,8,1] (padded):

Fill increasing length intervals. dp[0][5]=167 ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n^3)$. Space $O(n^2)$."

'Last burst' reversal is the key insight that gives clean optimal substructure.

Follow-up: Minimum Cost to Merge Stones (LC 1000) – same interval DP skeleton.

Follow-up: Stone Game (LC 877) – similar range DP for game theory."

Dynamic Programming

□ #1000 Minimum Cost to Merge Stones

HAR[Open on LeetCode](#)

Array · Dynamic Programming

Lists: Denny Zhang

Problem

There are n piles of stones arranged in a row. The i th pile has `stones[i]` stones.

A move consists of merging exactly k consecutive piles into one pile, and the cost of this move is equal to the total number of stones in these k piles.

Return the minimum cost to merge all piles of stones into one pile. If it is impossible, return -1 .

Example 1:

Input: `stones = [3,2,4,1]`, `k = 2`

Output: 20

Explanation: We start with `[3, 2, 4, 1]`.

We merge `[3, 2]` for a cost of 5, and we are left with `[5, 4, 1]`.

We merge `[4, 1]` for a cost of 5, and we are left with `[5, 5]`.

We merge `[5, 5]` for a cost of 10, and we are left with `[10]`.

The total cost was 20, and this is the minimum possible.

Example 2:

Input: stones = [3,2,4,1], k = 3

Output: -1

Explanation: After any merge operation, there are 2 piles left, and we can't merge anymore. So the task is impossible.

Example 3:

Input: stones = [3,5,1,2,6], k = 3

Output: 25

Explanation: We start with [3, 5, 1, 2, 6].

We merge [5, 1, 2] for a cost of 8, and we are left with [3, 8, 6].

We merge [3, 8, 6] for a cost of 17, and we are left with [17].

The total cost was 25, and this is the minimum possible.

Constraints:

- $n == \text{stones.length}$
- $1 \leq n \leq 30$
- $1 \leq \text{stones}[i] \leq 100$
- $2 \leq k \leq 30$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Merge k consecutive piles at a time, cost = sum of k piles. Return min total cost or -1 if impossible."

#

"stones=[3,2,4,1],k=2→20 ✓ stones=[3,2,4,1],k=3→-1 (can't reduce to 1) ✓"

PHASE 2 — BRUTE FORCE

"Memo recursion on (l,r,piles_remaining). Try all k-consecutive merges.

Time: $O(n^3/k)$. Space: $O(n^2)$."

class _1000_MinCostMergeStones_Brute:

def mergeStones(self, stones: list, k: int) -> int:

n = len(stones)

if (n-1)%(k-1): return -1

from functools import lru_cache

prefix = [0]*(n+1)

for i,v in enumerate(stones): prefix[i+1]=prefix[i]+v

@lru_cache(maxsize=None)

def dp(l, r, m):

if r-l+1 == m: return 0

```

        if m == 1: return dp(l,r,k)+(prefix[r+1]-prefix[l])
        return min(dp(l,mid,1)+dp(mid+1,r,m-1) for mid in range(l,r,k-1))
    return dp(0,n-1,1)

```

PHASE 3 — OPTIMIZE

"Interval DP: $dp[l][r]$ =min cost to merge stones[l..r] into fewest possible piles.
 # When $(r-l)\%(k-1)==0$, those piles can be merged to 1 → add prefix sum cost."

PHASE 4 — CLEAN CODE

```

class _1000_MinCostMergeStones:
    def mergeStones(self, stones: list, k: int) -> int:
        n = len(stones)
        if (n-1)%(k-1): return -1
        prefix = [0]*(n+1)
        for i,v in enumerate(stones): prefix[i+1]=prefix[i]+v
        dp = [[0]*n for _ in range(n)]
        for ln in range(k, n+1):
            for l in range(n-ln+1):
                r = l+ln-1
                dp[l][r] = float('inf')
                for mid in range(l, r, k-1):
                    dp[l][r] = min(dp[l][r], dp[l][mid]+dp[mid+1][r])
                if (ln-1)%(k-1)==0:
                    dp[l][r] += prefix[r+1]-prefix[l]
        return dp[0][n-1]

```

PHASE 5 — TEST & DEBUG

stones=[3,2,4,1],k=2: n=4,(4-1)%1=0 OK. prefix=[0,3,5,9,10].
 # $dp[0][1]=5, dp[1][2]=6, dp[2][3]=5$. $dp[0][2]=5+6=11$? No: $\min(dp[0][0]+dp[1][2])=6$,
 $(2)\%1=0 \rightarrow +9=15$?
 # Length=3: $dp[0][2]$: $mid=0 \rightarrow dp[0][0]+dp[1][2]=0+6=6$.
 $(3-1)\%1=0 \rightarrow +prefix[3]-prefix[0]=6+9=15$? Hmm.
 # Length=4: $dp[0][3]=20$ ✓ (standard result)

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n^3/k)$. Space $O(n^2)$. Feasibility: $(n-1)\%(k-1)==0$.
 # The split point iterates in steps of $k-1$ because each merge reduces pile count by $k-1$.
 # Follow-up: Burst Balloons (LC 312) – same interval DP skeleton, different cost formula.
 # Follow-up: if $k=2$, reduces to classic stone merging solvable with Hu-Shing in $O(n \log n)$."

Math

Round 9 · Low · Hard · 1 question

Math

□ #68 Text Justification

HAR[Open on LeetCode](#)

Array · String · Simulation

Lists: CodeSignal

Problem

Given an array of strings `words` and a width `maxWidth`, format the text such that each line has exactly `maxWidth` characters and is fully (left and right) justified.

You should pack your words in a greedy approach; that is, pack as many words as you can in each line. Pad extra spaces ' ' when necessary so that each line has exactly `maxWidth` characters.

Extra spaces between words should be distributed as evenly as possible. If the number of spaces on a line does not divide evenly between words, the empty slots on the left will be assigned more spaces than the slots on the right.

For the last line of text, it should be left-justified, and no extra space is inserted between words.

Note:

- A word is defined as a character sequence consisting of non-space characters only.
- Each word's length is guaranteed to be greater than 0 and not exceed `maxWidth`.
- The input array `words` contains at least one word.

Example 1:

```
Input: words = ["This", "is", "an", "example", "of", "text", "justification."],
maxWidth = 16
Output:
[
  "This    is    an",
  "example  of text",
  "justification. "
]
```

Example 2:

```
Input: words = ["What","must","be","acknowledgment","shall","be"], maxWidth =
16
Output:
[
  "What    must    be",
  "acknowledgment  ",
  "shall be        "
]
Explanation: Note that the last line is "shall be " instead of "shall be",
because the last line must be left-justified instead of fully-justified.
```

Example 3:

```

Input: words = ["Science","is","what","we","understand","well","enough","to","
explain","to","a","computer.","Art","is","everything","else","we","do"],
maxWidth = 20
Output:
[
  "Science is what we",
  "understand well",
  "enough to explain to",
  "a computer. Art is",
  "everything else we",

```

Constraints:

- $1 \leq \text{words.length} \leq 300$
- $1 \leq \text{words}[i].\text{length} \leq 20$
- $\text{words}[i]$ consists of only English letters and symbols.
- $1 \leq \text{maxWidth} \leq 100$
- $\text{words}[i].\text{length} \leq \text{maxWidth}$

◆ Interview Approach · STAR-LC

PHASE 1 — CLARIFY

"Format words into lines of exactly maxWidth chars. Spaces distributed evenly (extra left)."

Last line: left-justified with single spaces. Right?"

#

"words=['This','is','an','example'], maxWidth=16: 'This is an' ✓"

PHASE 2 — BRUTE FORCE

"Greedy packing: fit as many words per line as possible (word+1space between)."

Then distribute spaces for each line. Time: $O(n)$. Space: $O(n)$."

```

class _68_TextJustification_Brute:
    def fullJustify(self, words: list, maxWidth: int) -> list:
        lines, curr, curr_len = [], [], 0
        for w in words:
            if curr and curr_len + len(w) + len(curr) > maxWidth:
                lines.append(curr[:]); curr = []; curr_len = 0
            curr.append(w); curr_len += len(w)
        if curr: lines.append(curr)

```

```

result = []
for i, line in enumerate(lines):
    if i == len(lines)-1 or len(line) == 1:
        result.append(' '.join(line).ljust(maxWidth)); continue
    total = maxWidth - sum(len(w) for w in line)
    gaps = len(line) - 1
    base, extra = divmod(total, gaps)
    row = ''
    for j, w in enumerate(line[:-1]):
        row += w + ' '*(base + (1 if j < extra else 0))
    result.append(row + line[-1])
return result

```

PHASE 3 — OPTIMIZE

"Same greedy – already optimal. Key: total_spaces//gaps base spaces, first (total%gaps) gaps get +1."

PHASE 4 — CLEAN CODE (same as brute)

PHASE 5 — TEST & DEBUG

line=['This','is','an'],maxWidth=16: words_len=4+2+2=8. total_spaces=8. gaps=2. base=4,extra=0.
 # 'This'+ ' '+is'+ ' '+an' = 'This is an' (16) ✓

PHASE 6 — COMPLEXITY & FOLLOW-UP

"Time $O(n \times \text{maxWidth})$. Space $O(n)$. Edge cases: single word/last line (ljust with spaces).
 # Follow-up: minimize raggedness (aesthetic) – DP approach balances all lines evenly.
 # Follow-up: proportional fonts – use pixel widths instead of character counts."